HL7 FHIR R4 — Chapter 2: Foundation Module (sections 2 – 2.48.10)

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2 Foundation Module

URL: https://hl7.org/fhir/R4/foundation-module.html
Work Group FHIR Infrastructure | Standards Status: Informative

2.0 Foundation Module

The Foundation Module is responsible for the overall infrastructure of the FHIR specification. Every implementer works with the content in the foundation module whichever way they use FHIR.

The Foundation Module maintains most of the basic documentation for the FHIR specification. In addition, the Foundation Module includes the following resources:

Foundation Framework Resource DomainResource Basic Binary Bundle | Content Management Resources. Questionnaire QuestionnaireResponse List Composition DocumentReference DocumentManifest | Data Exchange Resources. OperationOutcome Parameters Subscription MessageHeader MessageDefinition

2.0.1 Relationships with other modules

• All the other modules depend on the foundation module

• The Exchange module builds on the foundation model by defining the recognized methods for exchange of resources

• The Terminology module provides the formal basis for using Concepts defined in Code Systems in the definitions

• The Conformance module provides the basis for extending the foundation for national and local use

• The Security & Privacy provides the linking framework to external standards for security and privacy

• The Implementation Support module builds on the foundation to provide testing and reference implementations

2.0.2 Developmental Roadmap

Several components of the foundation module have now reached normative status. The focus over the next 18-24 months as the 5th release of FHIR is prepared is to focus on some of the non-normative elements and move them towards normative status, such as Questionnaire, List, DocumentReference and Subscription. Exactly which resources will be candidates for normative release will be driven, in part, by the degree of implementation - and whether that implementation is communicated back to HL7.

2.1 Conformance

URL: https://hl7.org/fhir/R4/conformance-rules.html

2.1.0 Conformance

FHIR Infrastructure Work Group | Maturity Level: Normative | Standards Status: Normative

| This page has been approved as part of an ANSI standard. See the Infrastructure Package for further details.

The FHIR specification describes a set of resources and several different frameworks for exchanging resources between different systems. Because of its general nature and wide applicability, the rules made in this specification are fairly loose. As a consequence, this specification allows that different applications might not be able to interoperate because of how they use optional features. Applications claiming conformance to this specification make the claim in respect of a specific exchange framework, and in regard to specific details about their usage of those frameworks and resource contents.

Applications claim conformance to one (or more) of the following exchange frameworks:

• "RESTful FHIR": the RESTful API

• "FHIR messaging": Message based exchange

• "FHIR documents": Document based exchange

To provide details about specific usage of the frameworks and resource contents, FHIR provides a conformance layer that implementers, national/regional programs, and other profiling organizations such as IHE, can use to provide a computable statement about how the resources and their exchange paradigms are used to solve particular use cases. The conformance layer itself is implemented using the following key resources:

Value Set | Defines a set of coded values (see "Using Codes" for more details)

StructureDefinition | Makes rules about how a resource (or type) and its data elements are used in a particular context, including defining how extensions are used. A structure definition references value sets for the coded elements in a resource
CapabilityStatement | A statement of the kinds of resources and operations provided and/or consumed by an application. The Capability Statement references profiles to describe specific use of resources by the application
Implementation Guide | A single coherent collection of capability statements, profiles, extensions, value sets, and documentation describing a set of interoperable applications

The specification also provides a number of tools that can assist with enforcing technical conformance to this base specification.

FHIR is a closed specification. The specification defines what is possible and required for those portions of an API or exchange framework that wish to declare themselves as FHIR conformant, and that includes exchange resources that conform to the documented requirements of this specification. Systems may choose to support capabilities beyond those defined by FHIR (e.g. by adding additional end points/services with other names), but those portions of their interfaces that extend beyond what FHIR explicitly allows cannot be considered or described as "FHIR conformant".

Conformance with this specification does not provide any guarantee of patient or data safety. However, choosing to not conform to this specification carries additional risk in two ways:

• FHIR has been subject to a level of review and vetting unlikely to be received by any non-conformant variation; variations may result in introduction of undetected risks

• FHIR-like solutions (based on FHIR, but not conformant) may set expectations by trading partners which are not met due to the non-conformance of the system and these un-met expectations may also result in risk

Systems can only claim FHIR Conformance for functionality described in the applicable CapabilityStatement.

2.1.0.1 Conformance Language

This specification uses the conformance verbs SHALL, SHOULD, and MAY as defined in RFC 2119. Unlike RFC 2119, however, this specification allows that different applications might not be able to interoperate because of how they use optional features. In particular:

• SHALL: an absolute requirement for all implementations

• SHALL NOT: an absolute prohibition against inclusion for all implementations

• SHOULD/SHOULD NOT: A best practice or recommendation to be considered by implementers within the context of their particular implementation; there may be valid reasons to ignore an item, but the full implications must be understood and carefully weighed before choosing a different course

• MAY: This is truly optional language for an implementation; can be included or omitted as the implementer decides with no implications

2.1.0.2 Base Conformance Rules

The contents of a resource and the formats used to represent resources SHALL conform to the rules described in this specification, as defined in the narrative of the specification, and as controlled by the conformance properties defined below.

Data elements defined in resources and data types have 3 properties that are directly related to conformance: Cardinality, Is-Modifier, and MustSupport. These interact to place conformance requirements on implementations.

2.1.0.3 Cardinality

All attributes defined in FHIR have cardinality as part of their definition - a minimum number of required appearances and a maximum number. These numbers specify the number of times the attribute may appear in any instance of the resource type. This specification only defines the following cardinalities: 0..1, 0..*, 1..1, and 1..*. Profiles that describe specific use cases may use other values for cardinality within the limits of the cardinality defined by the base resource.

Note that when present, elements cannot be empty - they SHALL have a value attribute, child elements, or extensions. This means that setting an element to a minimum cardinality of 1 does not ensure that valid data will be present; specific FHIRPath constraints are required to ensure that the required data will be present.

In this specification, very few elements have a minimum cardinality of 1. Resources are used in many contexts, often quite removed from their primary use case, and sometimes even basic information is quite incomplete. For this reason, the only elements that have a minimum cardinality of 1 are those where they are necessary to any understanding of the resource or element that contains them. The minimum cardinalities should not be taken as a guide to what elements are expected to be present in any particular use of the resource, including their normal/primary usage purpose. In some cases, this specification publishes additional profiles that define which elements are required in particular situations. Similar profiles are published by jurisdictions, vendors, profiling organizations, or projects.

For elements that have cardinality > 1, the order in which they appear may have meaning. Unless the element definition (either in this specification or the extension) defines a meaning to the order explicitly (using ElementDefinition.orderMeaning), the meaning of the order is not defined, and implementations are allowed to reorder the elements. Note that it is not possible to define a meaning for the order of the elements

in a profile using a StructureDefinition. When there is no definition of the meaning of the order, implementations that need to choose a single element from a list of elements for some use SHALL do so based on the semantics of the content of the element that repeats. Profiles and Implementation guides may often make rules about this selection process.

Clients should not depend on servers maintaining ordering of elements, unless the retrieved resource conforms to a profile which mandates maintenance of ordering. If a server cannot maintain ordering, it must strip off known profile tags which require maintenance of ordering, and strip off unknown profiles (since they might require maintenance of ordering).

2.1.0.4 Is-modifier

Is-Modifier is a boolean property that is assigned when an element is defined, either as part of the base resource contents in this specification, or when extensions are defined.

An element is a modifier if and only if it cannot be safely ignored because its value, or its meaning if missing, may cause the interpretation of the containing element or one of its descendants to no longer conform to the stated definition for the element. Typical examples of elements that are labeled "Is-Modifier" are elements such as "status", "active", "refuted", or "certainty" that invert or negate the meaning of the resource (or element) that contains them.

Modifier is not an indication of the degree of importance for a particular piece of information or whether the element ought to be ignored when the resource is used for common use-cases. It is expected that if you ignore an element you may miss an important piece of computable meaning.

For example, consider Observation:

• Observation.status allows the entered-in-error code, which indicates that no actual Observation occurred at all. The definition of Observation indicates that it is a measurement that has been made - and doesn't allow for the possibility of a measurement that wasn't made. As a result, if the status element were ignored and an Observation were interpreted at face value based on its definition, a system or user would infer that an Observation had occurred, which would be false

• If an application ignores the Observation.subject element, it wouldn't know who or what was observed, which would make the remaining information largely useless for most usage. However, any system that ignores the subject would not have a false understanding of the Observation as it's defined. The system would understand what type of observation was made, when and what was found, just not who it was about

• Even something as critical as Observation.code is not a modifier element. While the Observation would have little utility without the code, the understanding when ignoring the element that “some” Observation had been made on the specified subject at the specified date and time would still be true when the element was reintroduced. The only exception would be if a value expressed in the Observation.code element could somehow convey that the Observation had not occurred or otherwise cause the instance to diverge from the defined meaning of Observation when ignoring the element

The definition of is-modifier has a corollary: any element that meets the requirement that it could cause the interpretation of the containing element or its descendants to diverge from their definition SHALL explicitly declare how such divergence could occur and must be marked as a modifier element. Any element not marked is-modifier and without that explanation SHALL NOT be used by an implementer in such a manner as to make the element behave as a modifier. For example, using a special "name" on a patient to indicate that the subject isn’t a real patient, but is instead an artificial structure used for non-patient tests would be non-conformant with FHIR because Patient.name is not a modifier element

Whether an element is a modifier cannot be changed when element usage is described in a constraining Structure Definition. When an element is labeled as Is-Modifier, the documentation must be clear about why it is a modifier.

A typical example of a modifier element is one that negates the element that contains it. For instance, in the following fragment of a resource definition:

Name | Flags | Card. | Type | Description & Constraints
AllergyIntolerance | | | DomainResource | Allergy or Intolerance (generally: Risk of Adverse reaction to a substance)
onset | Σ | 0..1 | dateTime | Date(/time) when manifestations showed
patient | Σ | 1..1 | Reference(Patient) | Who the sensitivity is for
verificationStatus | ?! Σ | 0..1 | CodeableConcept | unconfirmed | confirmed | refuted | entered-in-error AllergyIntoleranceVerificationStatus (Required)
criticality | Σ | 0..1 | code | CRITL | CRITH | CRITU AllergyIntoleranceCriticality (Required)

The definition of an AllergyIntolerance is that it contains information about "Risk of harmful or undesirable, physiological response which is unique to an individual and associated with exposure to a substance". If the value of the 'verificationStatus' element is set to entered-in-error, the entire resource does not actually contain valid information about any risk of exposure, and it is not safe for applications to ignore this element. As a consequence, it is labeled as 'is modifier = true'. In this tabular representation of the resource, this shows as the flag '?!'. The JSON and XML representations of a resource definition have their own representation of 'is modifier = true' status, and it is defined directly in a ElementDefinition.

If a narrative summary is present, and the status is generated, Is-Modifier elements SHALL be represented in the narrative summary of the resource. If Narrative is present with some other status Is-modifier elements SHOULD be represented.

If the value of a modifier element is not explicit in the instance, or known by the context, the resource might not be able to be safely understood. Wherever possible, elements labeled "Is-Modifier = true" also have a minimum cardinality of 1, in order to introduce certainty in their handling. However, sometimes this is not possible - much legacy data is not well described. Implementations producing resources SHOULD ensure that appropriate values for isModifier elements are provided at all times.

Implementations processing the data in resources SHALL understand the impact of the element when using the data. Implementations are not required to "support" the element in any meaningful way - they may achieve this understanding by rejecting instances that contain values outside those they support (for instance, an application may refuse to accept observations with a reliability other than "ok"). Alternatively, implementations may be able to be sure that, due to their implementation environment, such values will never occur. However applications SHOULD always check the value irrespective of this.

Note that processing the data of a resource typically means copying or filtering data out of a resource for use in another context (display to a human, decision support, exchange in another format where not all information is included or storing it for this kind of use). Servers and background processes that simply move whole resources around unchanged are not "processing the data of the resource", and therefore these applications are not required to check Is-Modifier elements.

Every element in the base resource has a value of "true" or "false" for the Is-Modifier flag. The value of the flag cannot be changed by profiles on the resource, in either direction. When a StructureDefinition defines an extension, it labels the extension with the Is-Modifier flag, and this cannot be changed in other profiles. Note that extensions that have is-Modifier = true are represented differently in resource instances ("modifierExtension" instead of "extension"), and there are additional rules about how they are handled.

Most status elements are marked as modifiers because of the presence of the entered-in-error status. Other modifiers are defined:

- Address.use
- ContactPoint.use
- HumanName.use
- Identifier.use
- Quantity.comparator
- BackboneElement.modifierExtension
- DomainResource.modifierExtension
- Resource.implicitRules
- ActivityDefinition.doNotPerform
- AdverseEvent.actuality
- AllergyIntolerance.clinicalStatus
- AllergyIntolerance.verificationStatus
- AppointmentResponse.participantStatus
- Basic.code
- BodyStructure.active
- CarePlan.intent
- CarePlan.activity.detail.doNotPerform
- CommunicationRequest.doNotPerform
- ConceptMap.group.element.target.equivalence
- Condition.clinicalStatus
- Condition.verificationStatus
- Consent.scope
- Contract.term.action.doNotPerform
- DeviceRequest.intent
- Goal.lifecycleStatus
- HealthcareService.active
- Immunization.isSubpotent
- ImmunizationRecommendation.recommendation.forecastStatus
- List.mode
- List.entry.deleted

- MeasureReport.improvementNotation
- MedicationRequest.intent
- MedicationRequest.doNotPerform
- MedicinalProductIngredient.specifiedSubstance.strength.presentationLowLimit
- MedicinalProductIngredient.specifiedSubstance.strength.concentrationLowLimit
- NutritionOrder.intent
- Organization.active
- Patient.active
- Patient.deceased[x]
- Patient.link
- Person.active
- Questionnaire.item.enableWhen
- RelatedPerson.active
- RequestGroup.intent
- Schedule.active
- ServiceRequest.intent
- ServiceRequest.doNotPerform

2.1.0.5 MustSupport

Labeling an element MustSupport means that implementations that produce or consume resources SHALL provide "support" for the element in some meaningful way. Because the base FHIR specification is intended to be independent of any particular implementation context, no elements are flagged as mustSupport=true as part of the base specification. This flag is intended for use in profiles that have a defined implementation context.

For this reason, the specification itself never labels any elements as MustSupport. This is done in StructureDefinitions, where the profile labels an element as mustSupport=true. When a profile does this, it SHALL also make clear exactly what kind of "support" is required, as this could involve expectations around what a system must store, display, allow data capture of, include in decision logic, pass on to other data consumers, etc.

Note that an element that has the property IsModifier is not necessarily a "key" element (e.g. one of the important elements to make use of the resource), nor is it automatically mustSupport - however both of these things are more likely to be true for IsModifier elements than for other elements.

2.1.0.6 Constraints

All elements may have constraints attached to them (also known as 'invariants'). Constraints defined on an element have the following properties:

Key | Identifies the constraint uniquely amongst all the constraints in the context - typically, this is used to refer to the constraint in an error message
Requirements | An explanation of why the constraint has been applied - what harmful conditions are being avoided
Severity | The severity of the invariant - see below
Human Description | A human description of the rule intended to be shown as the explanation for a message when the constraint is not met
Expression | A FHIRPath expression that must evaluate to true when run on the element
XPath | An XPath expression that must evaluate to true when run on the element in the XML representation

Many constraints are defined in the base specification. In addition, additional constraints may be defined in profiles that apply to resources. Systems are not required to evaluate the constraints, just as they are not required to check for conformance, or schema validity. However, systems SHOULD always ensure that all resources are valid against all applicable constraints.

Constraints have a severity level:

Error (rule) | A rule that all resources must conform to. Validators should report it as an error if the rule is violated, and applications processing the content can reject it as an invalid resource
Warning | Report this as a warning that there may be a problem with the resource, but it is considered valid and can be processed normally
Guideline | A warning marked with an extension (http://hl7.org/fhir/StructureDefinition/elementdefinition-bestpractice) that indicates that it should be a treated as an error if the implementation context asks a validator to enforce best practice rules. See Best Practices for a full list

Elements can also be explicitly associated with constraints defined elsewhere. This is a notification to implementers that the element is affected by the constraint. It has no meaning when the constraints are evaluated.

Profiles may define additional constraints that apply to an element, but they cannot alter or remove constraints that are already applied.

2.1.0.7 Other Metadata

In addition to the conformance metadata, each element has other metadata properties defined in the ElementDefinition:

• isSummary - How the element behaves in Summary Searches

• meaningWhenMissing - What it means when an element is missing

• maxLength - How long the representation of the primitive value is allowed to be (not including XML or JSON escaping)

2.1.0.8 Examples and Reference Implementations

This specification includes many examples. While every effort has been made to ensure that the examples are fully conformant to the specification, if the examples disagree with the specification, the specification is considered correct and normative, not the examples. This same rule applies to the reference implementations.

The examples reflected in this specification do *not* represent actual people. Any resemblance to real people - alive or dead - is entirely coincidental. In some cases, examples may be drawn from real clinical data. However, if this has occurred, the content has been scrubbed to remove any identifying information.

2.1.1 Best Practices

URL: https://hl7.org/fhir/R4/best-practices.html

2.1.1 Best Practices

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative

The base FHIR specification is used across the world in many different contexts, with a great variety of use cases. There are many existing restrictions from common practice and regulation that constrain the agreements that the specification represents. In particular there are large amounts of existing data in legacy record stores that need to be represented and exchanged using FHIR resources.

The requirements analysis that occurs during the design of the FHIR process often leads to a clear understanding of how information should be represented, but also can make clear that for a variety of reasons, such best practices cannot be imposed as standards requirements in all contexts. However, it is useful for the FHIR standard to be able to document what are known best practices.

Committees can document best practice by one of two different ways:

• defining invariants and marking them as best practice rules

• adding narrative that defines best practices

When using the FHIR validator, implementers are able to ask for best practice rules to be enforced if they wish.

This page indexes the best practices documented in this specification:

DomainResource

dom-6 | A resource should have narrative for robust management | When a resource has no narrative, only systems that fully understand the data can display the resource to a human safely. Including a human readable representation in the resource makes for a much more robust eco-system and cheaper handling of resources by intermediary systems. Some ecosystems restrict distribution of resources to only those systems that do fully understand the resources, and as a consequence implementers may believe that the narrative is superfluous. However experience shows that such eco-systems often open up to new participants over time.

Condition

con-3 | Condition.clinicalStatus SHALL be present if verificationStatus is not entered-in-error and category is problem-list-item | Most systems will expect a clinicalStatus to be valued for problem-list-items that are managed over time, but might not need a clinicalStatus for point in time encounter-diagnosis.

2.2 FHIR Life Cycle Page

URL: https://hl7.org/fhir/R4/lifecycle.html

2.2 FHIR Life Cycle Page

FHIR Infrastructure Work Group | Maturity Level: 3 | Standards Status: Trial Use

This page describes several issues around lifecycle management for the resources and the content they contain. Specifically, this page describes:

• Resource Status: how resource status codes work

• Current List: issues associated with retrieving "current X list" of resources

• Entered in Error: information about how erroneous entry is handled for the resources

2.2.1 Resource Status

Many FHIR resources have a status element that represents the lifecycle state of the resource or the clinical process represented by the resource. Work groups can specify status values appropriate to the individual resource. Although consistency between resources is not the primary objective, it is helpful to users and developers to have well-crafted value sets that cover all possible states (since the value sets are typically required and non-extensible).

To understand existing status elements, and to help create extensions and resources involving resource states, we note that status value sets follow one of the following life cycles:

• Clinical workflow process life cycle

• Request/Order life cycle

• Entity status life cycle

• Clinical status life cycle

For additional information about managing resource life cycles, see:

• Resource Identity

• Technical vs Business Versions

• Managing Resource Identity (including "Consistent Resource Identification")

2.2.2 Clinical Workflow Process Life Cycle

Describes the lifecycle states of complex activities common in healthcare. Typically, these states follow a chronological life cycle that leads from initiation to the conclusion of the action. A characteristic (but non-exhaustive) set of states for the clinical workflow process life cycle include:

• planned - resources for the activity are being allocated but the activity has not begun

• cancelled - the planned activity did not start and will not take place

• in-progress - the activity has begun

• on-hold (suspended) - the activity has been temporarily interrupted

• stopped (aborted, failed) - the activity has not been completed but no future action is planned

• completed (finished) - the activity has been completed

Examples of the clinical workflow life cycle:

• Communication.status: preparation | in-progress | not-done | on-hold | stopped | completed | entered-in-error | unknown

• Encounter.status: planned | arrived | triaged | in-progress | onleave | finished | cancelled | entered-in-error | unknown

• Goal.lifecycleStatus: proposed | planned | accepted | cancelled | entered-in-error | rejected

• MedicationAdministration.status: in-progress | not-done | on-hold | completed | entered-in-error | stopped | unknown

• MedicationDispense.status: preparation | in-progress | cancelled | on-hold | completed | entered-in-error | stopped | declined | unknown

• Procedure.status: preparation | in-progress | not-done | on-hold | stopped | completed | entered-in-error | unknown

2.2.3 Request/Order Life Cycle

Some resources in FHIR represent orders or requests. The request lifecycle can be generalized in terms of four stages: creating the request, sending the request, receiving acceptance or refusal of the request, and fulfillment of the request. A characteristic (but non-exhaustive) set of states for the request/order pattern include:

• proposed: An actor (e.g. a clinical decision support system) has proposed an action to be requested

• draft: The request is in preliminary form, prior to being requested

• requested: The request has been made

• rejected: The request receiver has declined the request

• accepted: The request receiver has accepted the request

• in-progress: Work to fulfill the request has begun

• on-hold (suspended): Work on the request has been interrupted

• stopped (aborted): The activity has not been completed but no future action is planned

• completed: Work on the requested task has been completed, and no further action is required

• cancelled: The request has been withdrawn

Examples of the request/order life cycle:

• CommunicationRequest.status: draft | active | on-hold | revoked | completed | entered-in-error | unknown

• DeviceRequest.status: draft | active | on-hold | revoked | completed | entered-in-error | unknown

• MedicationRequest.status: active | on-hold | cancelled | completed | entered-in-error | stopped | draft | unknown

• ServiceRequest.status: draft | active | on-hold | revoked | completed | entered-in-error | unknown

2.2.4 Entity Availability Life Cycle

The entity availability life cycle indicates if the resource, or the entity described by the resource, is ready for use, not yet ready for use, or has been retired from use. A characteristic (but non-exhaustive) set of states for the entity availability life cycle include:

• draft: The entity is being prepared but is not yet in use

• active: The entity is in use

• suspended: The entity is not in use at the moment, but may return to active status

• amended: The entity has undergone a revision but is still active

• retired (superseded): The entity is no longer in use.

Examples of the entity availability life cycle:

• DiagnosticReport.status: registered | partial | final | amended | cancelled | entered-in-error | unknown

• MedicationStatement.status: active | completed | entered-in-error | intended | stopped | on-hold | unknown | not-taken. (note: in-progress and completed are states reflecting the administration of the medication)

• DocumentManifest.status: current | superseded | entered-in-error

• CapabilityStatement.status: draft | active | retired | unknown

• StructureDefinition.status: draft | active | retired | unknown

• Questionnaire.status: draft | active | retired | unknown

• DocumentReference.status: current | superseded | entered-in-error

• QuestionnaireResponse.status: in-progress | completed | amended | entered-in-error | stopped

• Flag.status: active | inactive | entered-in-error

• Location.status: active | suspended | inactive

• Organization.active: true | false

• Patient.active: true | false

2.2.5 Clinical Status Life Cycle

Clinical status is somewhat different than the previous status values, since it does not deal with workflow or lifecycle. Instead, it indicates how evidence is affecting a clinical interpretation. Here are two examples:

• AllergyIntolerance.clinicalStatus: active | inactive

• Condition.clinicalStatus: active | inactive

2.2.6 Current Resource Lists

Many clinical systems maintain current lists of some kind of resources for a patient. Some of the commonly maintained lists include:

• Current Problem List: a list of the problems that are of concern for care of the patient

• Current Medication List: a list of the medications that a patient is known to be on at the current time

Because of the way that resources are used, there is no simple way to determine, from examination of a resource, whether it is 'current' or not. Take, as an example, the Condition resource. In a typical EHR, condition resources might be published on the RESTful interface for the following reasons:

• to represent an item in a patient's curated problem list

• to represent a complaint or a diagnosis from an encounter record

• to represent a problem for investigation provided by a diagnostic system as part of a ServiceRequest/DiagnosticReport pair

• the resources were received from another system as part of a referral package, and were current for that system when they were received

There is no element on the Condition resource that can convey the difference between these usages. In particular, there can be no way to differentiate between current and past resources without having to retrospectively alter resources, which is problematic with regard to integrity and digital signatures.

One consequence of this is that searching the condition resource for a given patient will return more than just the patient's current problems. Though this is somewhat counter-intuitive to some implementers, restricting searches on Condition to only include the patient's current curated problem list excludes all the other - important - uses of the Condition resource.

Determining whether a Condition is an entry on a patient's current problem list is done by checking with the Condition resource is referenced from the correct list.

On the RESTful API, this is done using the list search mechanism:

```
GET [base]/AllergyIntolerance?patient=42&_list=$current-allergies
```

This is a request to fetch all the allergies in the patient 42's "Current Problem List". Note that the server is not required to actually make a resource representation of the current allergy list available, though doing so assists clients in their audit/integrity tasks. See List Operation "Find" for further information.

In a document, current lists are determined by the code on a Composition section.

FHIR defines the following names for functional lists:

List | ResourceType | Description | Possible LOINC codes in documents / sections

$current-problems | Condition | The "Current Problem List" - A list of current and active diagnoses as well as past diagnoses relevant to the current care of the patient | 46105-3 (Problem conditions Set)

$current-medications | MedicationStatement/ MedicationRequest | A list of all medications that the patient is taking. The 'current medications list' sometimes may include a mix of prescribed and over-the-counter medications - or only some of them. The list may contain a mix of prescriptions and more general statements, or only one of the two. The list may also correspond to a formal reconciled medication administration schedule, but more often does not | 57828-6 (Prescription list), 10160-0 (History of medication)

$current-allergies | AllergyIntolerance | A list of known or suspected propensities to medications, foods, or environmental agents that is provided to help prevent reactions while care is occurring | 18716-1 (Allergy studies (set)), 52472-8 (Allergies and Adverse Drug Reactions), and 48765-2 (Allergies and adverse reactions Document)

$current-drug-allergies | AllergyIntolerance | A list of known or suspected propensities to medications that is provided to help prevent reactions while care is occurring. This list is a subset of the full allergies list | (same as above?)

2.2.7 Entered in Error Summary

The entered-in-error state indicates the resource was created accidentally and should be ignored. This state can apply to resources created by manual entry. It is usually not associated with the Clinical Workflow Process life cycle, but can be associated with the Request/Order and the Entity Availability life cycles.

This table summarizes what is expected to happen for each resource in the case that the data it contains is subsequently found to be an erroneous entry.

Resource | Status

Account | .status = entered-in-error

ActivityDefinition | .status = retired

AdverseEvent

AllergyIntolerance | .verificationStatus = entered-in-error

Appointment | .status=entered-in-error

AppointmentResponse | .participantStatus=entered-in-error

AuditEvent | Audit Events are recorded, there is no update or delete.

Basic | There are no fixed arrangements for Basic - profiles should describe how this works as arrangements will depend on the 'type' of Basic resource

Binary | n/a (This would be handled where the binary is linked from)

BiologicallyDerivedProduct | .verificationStatus = entered-in-error

BodyStructure | .active = false

Bundle | Depends on the type: document - see for Composition; message - see for MessageHeader; transaction / transaction-response / history / searchset - not expected to be stored; collection: just delete it if it's stored, and in error

CapabilityStatement | .status = retired

CarePlan | .status = entered-in-error

CareTeam | .status = entered-in-error

CatalogEntry | .status = retired

ChargeItem | .status = entered-in-error

ChargeItemDefinition | .status = retired

Claim | .status = entered-in-error

ClaimResponse | .status = entered-in-error

ClinicalImpression

CodeSystem | .status = retired

Communication | .status = entered-in-error

CommunicationRequest | .status = entered-in-error

CompartmentDefinition | .status = retired

Composition | .status = entered-in-error

ConceptMap | .status = retired

Condition | .verificationStatus = entered-in-error

Consent | .status = entered-in-error

Contract | .status = entered-in-error

Coverage | .status = entered-in-error

CoverageEligibilityRequest | .status = entered-in-error

CoverageEligibilityResponse | .status = entered-in-error

DetectedIssue | .status = entered-in-error

Device | .status = entered-in-error

DeviceDefinition | .operationalStatus = entered-in-error

DeviceMetric | .operationalStatus = entered-in-error

DeviceRequest | .status = entered-in-error

DeviceUseStatement | .status = entered-in-error

DiagnosticReport | .status = entered-in-error

DocumentManifest | .status = entered-in-error

DocumentReference | .status = entered-in-error
EffectEvidenceSynthesis | .status = retired
Encounter | .status=entered-in-error
Endpoint | .status=entered-in-error
EnrollmentRequest | .status = entered-in-error
EnrollmentResponse | .status = entered-in-error
EpisodeOfCare | .status=entered-in-error
EventDefinition | .status = retired
Evidence | Delete the report if it is
EvidenceVariable | Delete the report if it is
ExampleScenario | .status = retired
ExplanationOfBenefit | .status = entered-in-error
FamilyMemberHistory | .status = entered-in-error
Flag | .status = entered-in-error
Goal | .status = entered-in-error
GraphDefinition | .status = retired
Group | .active = false
GuidanceResponse | .status = entered-in-error
HealthcareService | .active = false
ImagingStudy | Imaging Study state is managed in PACS.
Immunization | .status = entered-in-error
ImmunizationEvaluation | .status = entered-in-error
ImmunizationRecommendation | Immunization recommendations are generated on "just in time" and typically are not retained by the system generating them so there isn't an expected workflow where a previous recommendation would be re-evaluated and found to be "in error".
ImplementationGuide | .status = retired
InsurancePlan | .status = retired
Invoice | .status = entered-in-error
Library | .status = retired
Linkage | .active = false
List | .status = entered-in-error
Location | .status = inactive
Measure | Delete the report if it is
MeasureReport | Delete the report if it is
Media | n/a - this would be handled whereever the media is linked from
Medication | Medication does not have a status
MedicationAdministration | .status = entered-in-error
MedicationDispense | .status = entered-in-error
MedicationKnowledge | .status = entered-in-error
MedicationRequest | .status = entered-in-error
MedicationStatement | .status = entered-in-error
MedicinalProduct
MedicinalProductAuthorization
MedicinalProductContraindication
MedicinalProductIndication
MedicinalProductIngredient
MedicinalProductInteraction
MedicinalProductManufactured
MedicinalProductPackaged
MedicinalProductPharmaceutical
MedicinalProductUndesirableEffect
MessageDefinition | mostly n/a, but in the cases where messages are stored in error, they would simply be deleted
MessageHeader | mostly n/a, but in the cases where messages are stored in error, they would simply be deleted
MolecularSequence | .status = entered-in-error
NamingSystem | .status = retired
NutritionOrder | .status = entered-in-error
Observation | .status = entered-in-error
ObservationDefinition
OperationDefinition | .status = retired
OperationOutcome | n/a - this resource is not expected to be stored
Organization | .active = false
OrganizationAffiliation | .active = false
Patient | .active = false
PaymentNotice | .status = entered-in-error
PaymentReconciliation | .status = entered-in-error
Person | .active = false
PlanDefinition | .status = retired
Practitioner | .active = false
PractitionerRole | .active = false
Procedure | .status = entered-in-error
Provenance | Provenance are recorded, there is no update or delete.
Questionnaire | .status = retired

QuestionnaireResponse | .status = entered-in-error
RelatedPerson | .active = false
RequestGroup | .status = entered-in-error
ResearchDefinition | Delete the report if it is
ResearchElementDefinition | Delete the report if it is
ResearchStudy | .status = entered-in-error
ResearchSubject | .status = entered-in-error
RiskAssessment | .status = entered-in-error
RiskEvidenceSynthesis | .status = retired
Schedule | .active = false
SearchParameter | .status = retired
ServiceRequest | .status = entered-in-error
Slot | .status = entered-in-error
Specimen | .status = entered-in-error
SpecimenDefinition
StructureDefinition | .status = retired
StructureMap | .status = retired
Subscription | .status = off (just turn it off, maybe update the error message)
Substance | .status = entered-in-error
SubstanceNucleicAcid
SubstancePolymer
SubstanceProtein
SubstanceReferenceInformation
SubstanceSourceMaterial
SubstanceSpecification
SupplyDelivery | .status = entered-in-error
SupplyRequest | .status = entered-in-error
Task | .status = entered-in-error
TerminologyCapabilities | .status = retired
TestReport | .status = entered-in-error
TestScript | .status = retired
ValueSet | .status = retired
VerificationResult
VisionPrescription | .status = entered-in-error

Note: Resources that are not listed in this table do not have any explicit documentation with regard to being entered in error.

2.3 Resource References

URL: https://hl7.org/fhir/R4/references.html

2.3.0 Resource References

FHIR Infrastructure Work Group | Maturity Level: Normative | Standards Status: Normative

| This page has been approved as part of an ANSI standard. See the Infrastructure Package for further details.

Many of the defined elements in a resource are references to other resources. Using these references, the resources combine to build a web of information about healthcare.

References are always defined and represented in one direction - from one resource (source) to another (target). The corresponding reverse relationship from the target to the source exists in a logical sense, but is not typically represented explicitly in the target resource. For external references, navigating these reverse relationships requires some external infrastructure to track the relationship between resources (the REST API provides one such infrastructure by providing the ability to search the reverse relationship by naming search parameters for the references and by providing support for reverse includes).

Because resources are processed independently, relationships are not considered to be transitive. For example, if a Procedure resource references a Patient as its subject, and has a Procedure resource as its reason, there is no automatic rule or implication that the condition has the same patient for its subject. Instead, the subject of the condition must be established directly in the Condition resource itself. Another way to state this is that the context of the subject is not "inherited", nor does it "conduct" along the relationship to condition. The only exception to this is the case of contained resources (see below). Note that in practice, the relationships need to describe a logical and coherent record, and in the case of the Condition and Procedure described here, they would usually be required to have the same patient for their subjects. Profiles and/or implementation guides may make rules about this (also see GraphDefinition).

This specification describes the use of references to other resources that have the same FHIR version as the source of the reference. Expected behavior for references across FHIR versions is presently undefined.

References are made to resource based on their identity; there are several different identities to refer to.

Resources contain two types of references to other resources:

• Resource references - general references between resources

• Canonical references - references to resources by their canonical URL (see below)

For canonical references, see below. The general references are handled by the Reference type. which contains at least one of a reference (literal reference), an identifier (logical reference), and a display (text description of target). In addition, it may contain a target type.

• Structure

• UML

• XML

• JSON

• Turtle

• R3 Diff

• All

Structure

Name | Flags | Card. | Type | Description & Constraints
Reference | ΣIN | | Element | A reference from one resource to another + Rule: SHALL have a contained resource if a local reference is provided Elements defined in Ancestors: id, extension
reference | ΣI | 0..1 | string | Literal reference, Relative, internal or absolute URL
type | Σ | 0..1 | uri | Type the reference refers to (e.g. "Patient") ResourceType (Extensible)
identifier | Σ | 0..1 | Identifier | Logical reference, when literal reference is not known
display | Σ | 0..1 | string | Text alternative for the resource
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <reference value="[string]"/><!--  0..1 Literal reference, Relative, internal or absolute URL -->
 <type value="[uri]"/><!-- 0..1 Type the reference refers to (e.g. "Patient") -->
 <identifier><!-- 0..1 Identifier Logical reference, when literal reference is not known --></identifier>
 <display value="[string]"/><!-- 0..1 Text alternative for the resource -->
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "reference" : "<string>", // C? Literal reference, Relative, internal or absolute URL
  "type" : "<uri>", // Type the reference refers to (e.g. "Patient")
  "identifier" : { Identifier }, // Logical reference, when literal reference is not known
  "display" : "<string>" // Text alternative for the resource
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:Reference.reference [ string ]; # 0..1 Literal reference, Relative, internal or absolute URL
  fhir:Reference.type [ uri ]; # 0..1 Type the reference refers to (e.g. "Patient")
  fhir:Reference.identifier [ Identifier ]; # 0..1 Logical reference, when literal reference is not known
  fhir:Reference.display [ string ]; # 0..1 Text alternative for the resource
]
```

Changes since Release 3

Reference
Reference.type | Added Element

See the Full Difference for further information

Structure

Name | Flags | Card. | Type | Description & Constraints
Reference | ΣIN | | Element | A reference from one resource to another + Rule: SHALL have a contained resource if a local reference is provided Elements defined in Ancestors: id, extension
reference | ΣI | 0..1 | string | Literal reference, Relative, internal or absolute URL
type | Σ | 0..1 | uri | Type the reference refers to (e.g. "Patient") ResourceType (Extensible)
identifier | Σ | 0..1 | Identifier | Logical reference, when literal reference is not known
display | Σ | 0..1 | string | Text alternative for the resource
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <reference value="[string]"/><!--  0..1 Literal reference, Relative, internal or absolute URL -->
 <type value="[uri]"/><!-- 0..1 Type the reference refers to (e.g. "Patient") -->
 <identifier><!-- 0..1 Identifier Logical reference, when literal reference is not known --></identifier>
 <display value="[string]"/><!-- 0..1 Text alternative for the resource -->
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "reference" : "<string>", // C? Literal reference, Relative, internal or absolute URL
  "type" : "<uri>", // Type the reference refers to (e.g. "Patient")
  "identifier" : { Identifier }, // Logical reference, when literal reference is not known
  "display" : "<string>" // Text alternative for the resource
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:Reference.reference [ string ]; # 0..1 Literal reference, Relative, internal or absolute URL
  fhir:Reference.type [ uri ]; # 0..1 Type the reference refers to (e.g. "Patient")
  fhir:Reference.identifier [ Identifier ]; # 0..1 Logical reference, when literal reference is not known
  fhir:Reference.display [ string ]; # 0..1 Text alternative for the resource
]
```

Changes since Release 3

Reference
Reference.type | Added Element

See the Full Difference for further information

Constraints

At least one of reference, identifier and display SHALL be present (unless an extension is provided).

id | Level | Location | Description | Expression
ref-1 | Rule | (base) | SHALL have a contained resource if a local reference is provided | reference.startsWith('#').not() or (reference.substring(1).trace('url') in %rootResource.contained.id.trace('ids'))

2.3.0.1 Target Type

In resources, a Reference always point to another resource, which has a fixed and known type. If appropriate, this type can be specified in the reference itself. In principle, the type of the target reference can be determined by resolving the reference (using the approaches described below), and examining the content returned to determine the type; for this reason, specifying the type is almost always duplication of information. However, resolving references may be a very slow operation, or impossible in practice due to various practical concerns. For this reason, the reference may indicate directly the target resource type:

```
"subject": {
  "reference" : "http://someserver/some-path",
  "type" : "Patient"
}
```

When the type is provided directly, it SHALL agree with the type determined by resolving the resource.

Note that in practice, it is often not necessary to know the type of the target resource is, unless it's going to be resolved anyway, so in many cases, specifying the type of the target resource is unnecessary.

In order to support Logical Models, the type element has the type of "uri". Whenever type appears in resources, the uri is represented relative to the base URI http://hl7.org/fhir/StructureDefinition/. For resources, then, the value is simply a code that is the type of resource - e.g. "Patient".

2.3.0.2 Literal References

The reference is the key element - resources are identified and addressed by their URL. It contains a URL that is either

• an absolute URL

• a relative URL, which is relative to the Service Base URL, or, if processing a resource from a bundle, which is relative to the base URL implied by the Bundle.entry.fullUrl (see Resolving References in Bundles)

• an internal fragment reference (see "Contained Resources" below)

Notes:

• Using absolute URLs provides a stable, scalable approach suitable for a cloud/web context, while using relative/logical references provides a flexible approach suitable for use when trading across closed ecosystem boundaries. (see "Managing Resource Identity" for further discussion)

FHIR RESTful server

```
  ((http|https):\/\/([A-Za-z0-9\-
   \\\.\:\%\$]*\/)+)?(Account|ActivityDefinition|AdverseEvent|AllergyIntolerance|Appointment|AppointmentResponse|AuditE
   vent|Basic|Binary|BiologicallyDerivedProduct|BodyStructure|Bundle|CapabilityStatement|CarePlan|CareTeam|CatalogEntry
   |ChargeItem|ChargeItemDefinition|Claim|ClaimResponse|ClinicalImpression|CodeSystem|Communication|CommunicationReques
   t|CompartmentDefinition|Composition|ConceptMap|Condition|Consent|Contract|Coverage|CoverageEligibilityRequest|Covera
   geEligibilityResponse|DetectedIssue|Device|DeviceDefinition|DeviceMetric|DeviceRequest|DeviceUseStatement|Diagnostic
   Report|DocumentManifest|DocumentReference|EffectEvidenceSynthesis|Encounter|Endpoint|EnrollmentRequest|EnrollmentRes
   ponse|EpisodeOfCare|EventDefinition|Evidence|EvidenceVariable|ExampleScenario|ExplanationOfBenefit|FamilyMemberHisto
   ry|Flag|Goal|GraphDefinition|Group|GuidanceResponse|HealthcareService|ImagingStudy|Immunization|ImmunizationEvaluati
   on|ImmunizationRecommendation|ImplementationGuide|InsurancePlan|Invoice|Library|Linkage|List|Location|Measure|Measur
   eReport|Media|Medication|MedicationAdministration|MedicationDispense|MedicationKnowledge|MedicationRequest|Medicatio
```

```
nStatement|MedicinalProduct|MedicinalProductAuthorization|MedicinalProductContraindication|MedicinalProductIndicatio
n|MedicinalProductIngredient|MedicinalProductInteraction|MedicinalProductManufactured|MedicinalProductPackaged|Medic
inalProductPharmaceutical|MedicinalProductUndesirableEffect|MessageDefinition|MessageHeader|MolecularSequence|Naming
System|NutritionOrder|Observation|ObservationDefinition|OperationDefinition|OperationOutcome|Organization|Organizati
onAffiliation|Patient|PaymentNotice|PaymentReconciliation|Person|PlanDefinition|Practitioner|PractitionerRole|Proced
ure|Provenance|Questionnaire|QuestionnaireResponse|RelatedPerson|RequestGroup|ResearchDefinition|ResearchElementDefi
nition|ResearchStudy|ResearchSubject|RiskAssessment|RiskEvidenceSynthesis|Schedule|SearchParameter|ServiceRequest|Sl
ot|Specimen|SpecimenDefinition|StructureDefinition|StructureMap|Subscription|Substance|SubstanceNucleicAcid|Substanc
ePolymer|SubstanceProtein|SubstanceReferenceInformation|SubstanceSourceMaterial|SubstanceSpecification|SupplyDeliver
y|SupplyRequest|Task|TerminologyCapabilities|TestReport|TestScript|ValueSet|VerificationResult|VisionPrescription)\/
[A-Za-z0-9\-\.]{1,64}(\/_history\/[A-Za-z0-9\-\.]{1,64})?
```

• URLs are always considered to be case-sensitive

• The URL may contain a reference to a canonical URL (see below) and applications can use the canonical URL resolution methods they support when resolving references, though the |[version] syntax part of the canonical reference is not supported

• References SHALL be a reference to an actual FHIR resource, and SHALL be resolvable (given that access control works, there is no temporary unavailability, etc.). Resolution can be either by retrieval from the URL, or, where applicable by resource type, by treating an absolute reference as a canonical URL (see below) and looking it up in a local registry/repository

A relative reference to the Patient "034AB16" in an element named subject on a FHIR RESTful server:

```
<subject>
  <reference value="Patient/034AB16" />
</subject>
```

An absolute reference to a Structure Definition in an element named profile:

```
{
  "profile" : {
    "reference" : "http://fhir.hl7.org/svc/StructureDefinition/c8973a22-2b5b-4e76-9c66-00639c99e61b"
  }
}
```

Note that in a bundle during a transaction, reference URLs may actually contain logical URIs (e.g. OIDs or UUIDSs) that resolve within the transaction. When processing the transaction, the server replaces the logical URL with what is the correct literal URL at the completion of the transaction.

References can be version specific - that is, a reference may point to a specific version of a resource. e.g.:

```
<target>
  <reference value="http://example.org/fhir/Observation/1x2/_history/2" />
</target>
```

This is usually associated with audit trail or provenance information where it is important to reference a specific version of a record, not the most current information.

2.3.0.3 Logical References

In many contexts where FHIR is used, applications building a resource may know an identifier for the target of the reference, but there is no way for the application to convert this to a literal reference that directly references an actual resource. This situation may arise for several reasons:

• There is no server exposing any such resource. This is often the case with national identifiers (e.g. US SSN or NPI), and such identifiers are widely used

• The server that exposes the resource is not available to the source application, so it has no way to resolve an identifier to a reference

• The application is not in a RESTful environment - it is creating a message or a document

For further discussion of the use of identifiers on resources, see Consistent Resource Identification. In these cases, the source application may provide the identifier as a logical reference to the entity that the target resource would describe.

A logical reference to the Patient with an SSN of 000111111:

```
<patient>
  <identifier>
    <system value="http://hl7.org/fhir/sid/us-ssn" />
    <value value="000111111" />
  </identifier>
</patient>
```

There is no requirement that a Reference.identifier point to something that is actually exposed or exists as a FHIR instance (except, of course, that the reference will need to be resolved to a target resource if any information from it is required in a FHIR context). The reference SHALL point to a business concept that would be expected to be exposed as a FHIR instance, and that instance would need to be of a FHIR resource type allowed by the reference For example, it would not be legitimate to send the identifier for a drug prescription if the type were Reference(Observation|DiagnosticReport). One of the use-cases for Reference.identifier is the situation where no FHIR representation exists (where the type is Reference (Any).

When processing a resource, an application may be able to use the identifier directly, on the grounds that all it needs is the identifier, or it may be able to resolve the identifier directly. Alternatively, it may be able to use a server to resolve the logical reference to a literal reference to a resource.

Irrespective of how the resolution occurs, any system processing a logical reference will only be able to resolve the identifier to a reference if it understands the business context in which the identifier is used. Sometimes this is global (e.g. a national identifier) but often it is not.

For this reason, none of the useful mechanisms described for working with references (e.g. chaining, includes) are possible, nor should servers be expected to be able to automatically resolve the reference. Servers may accept an identifier based reference untouched, resolve it, and/or reject it - see CapabilityStatement.rest.resource.referencePolicy.

When both logical and literal references are provided, the literal reference is preferred. Applications processing the resource are allowed - but not required - to check that the identifier matches the literal reference, if they understand how to resolve the logical reference.

Applications converting a logical reference to a literal reference may choose to leave the logical reference present; or may remove it.

2.3.0.4 Reference Description

Irrespective of whether a literal and/or logical reference is provided, or neither, the display element may be used to provide a very short description of the target resource.

```
<custodian>
  <reference value="Organization/123" />
  <display value="HL7, Inc" />
</custodian>
```

This text can be used by any application that cannot resolve the reference to fill out the text portion of a hyperlink referring to the target resource, for instance. It can also save time fetching a target resource and determining how to convert it to a very short textual description.

In general, the display, if populated, does not have identical content to the Resource.text of the referenced resource. The purpose is to identify what's being referenced, not to more fully describe it.

2.3.0.5 Canonical URLs

Many resource types have a defined element "url" which is the canonical URL that always identifies the resource across all contexts of use. Typically, terminology, conformance or knowledge resources have canonical URLs. The canonical URL is the preferred way to reference a resource instance for the resource types on which it is defined. See Canonical Resource Identity for further information.

References to canonical URLs use the canonical type to refer to the url element on the conformance/knowledge resources (which has the type uri):

```
<valueSet value="http://hl7.org/fhir/ValueSet/my-valueset"/>
```

References of type canonical may include a version, in order be precise about which version of the resource is being referred to. To do this, append the version to the reference with a '|' like this:

```
<valueSet value="http://hl7.org/fhir/ValueSet/my-valueset|0.8"/>
```

This is a version specific reference to a value set. Note that this refers to the ValueSet.version not the ValueSet.meta.versionId. Searching for this on a FHIR server would look like this:

```
GET fhir/ValueSet?url=http://hl7.org/fhir/ValueSet/my-valueset&version=0.8
```

Note that if a References to a canonical URL does not have a version, and the server finds multiple versions for the value set, the system using the reference should pick the latest version of the target resource and use that. Servers SHOULD support version specific searching for canonical URLs by automatically detecting the presence of a |[version] and performing the appropriate search. Additional notes about searching on versioned references to canonical URLs:

• Search only regards the latest version for each different logical id

• If there is no match (either because .version is empty or is different), the instance will not be matched, and will not appear in the result bundle

• This search only works for specific data elements of type of uri that act as canonical URL's (see list below)

Systems resolving references to canonical URLs SHOULD first try to resolve the reference using the canonical reference (e.g. search on a known registry of terminology, conformance, or knowledge resources as appropriate), and then fall back to direct resolution using the URL as a literal reference if a local version of the canonical resource cannot be found. This approach is safe because the approaches must refer to the same artifact, though implementations will need to make appropriate arrangements regarding the version and/or currency of their local copy of the artifact.

The following resources have canonical URLs and are allowed to be the target of a references to a canonical URLs:

• ActivityDefinition

• CapabilityStatement

• ChargeItemDefinition

• CodeSystem

- CompartmentDefinition
- ConceptMap
- EffectEvidenceSynthesis
- EventDefinition
- Evidence
- EvidenceVariable
- ExampleScenario
- GraphDefinition
- ImplementationGuide
- Library
- Measure
- MessageDefinition
- NamingSystem
- OperationDefinition
- PlanDefinition
- Questionnaire
- ResearchDefinition
- ResearchElementDefinition
- RiskEvidenceSynthesis
- SearchParameter
- StructureDefinition
- StructureMap
- TerminologyCapabilities
- TestScript
- ValueSet

A few elements that have references to canonical URLs have a targetProfile of http://hl7.org/fhir/StructureDefinition/Resource, which is shown as canonical(Any) in this specification. Such references SHALL only reference one of these types of resources.

2.3.0.6 Canonical Reference and Fragments

Canonical references may include a fragment when the reference is to a contained resource. When the target of a canonical reference is a contained resource, the canonical reference will have a fragment as part of the URL:

```
  <valueSet value="http://fhir.acme.com/Questionnaire/example|1.0#vs1"/>
```

This is a reference to the value set with id "vs1" in version 1.0 of the identified questionnaire. The common case for this is internal references:

```
  <answerValueSet value="#vs1"/>
```

Which is the form for references to the contained value set inside the above example questionnaire.

2.3.0.7 Rendering References in Resource Narratives

References between resources create a challenge when rendering resource narratives: the narrative includes information from the target resources. As an example, consider an Observation with a patient reference:

```
{
  "resourceType" : "Observation",
  "subject" : {
    "reference" : "Patient/example"
  }
}
```

When represented in XHTML, this reference will generally become something like:

```
  ...
  <p>Patient: <a href="Patient/example">Peter James CHALMERS (12345)</a></p>  ...
```

This implies that the system generating the resource narrative will need to resolve the reference and generate a summary, or that the reference will contain the information (generated by an application that is able to resolve the reference):

```
{
  "resourceType" : "Observation",
  "subject" : {
    "reference" : "Patient/example",
    "display" : "Peter James CHALMERS (12345)"
  }
}
```

Narrative that contains information derived from other resources like this is still regarded as 'generated' (for Narrative.status).

It's not mandatory to generate narrative this way. Narrative may be generated like this:

```
  ...
  <p>Patient: <a href="Patient/example">(link)</a></p>  ...
```

However, users generally prefer a more informative narrative, so this is not always acceptable.

Applications (and networks of applications) will need some kind of systematic approach for resolving references and/or maintaining the currency of narratives as the resources they reference are changed. The exact details of this are out of scope for the FHIR standard.

Applications are encouraged to use the Reference.display element to store a user presentable representation of the resource for when the resource cannot be resolved (e.g. due to network errors). The canonical type does not have a display element because applications are generally expected to carry cached copies of the resources that are the target of the canonical references. If this functionality is still needed or desired anyway, the Rendered Value extension can be used.

2.3.0.8 Contained Resources

In some circumstances, the content referred to in the resource reference does not have an independent existence apart from the resource that contains it - it cannot be identified independently, and nor can it have its own independent transaction scope. Typically, such circumstances arise where resources are being assembled by a secondary user of the source data, such as a middleware engine. If the data available when the resource is constructed does not include record keys or absolute identification information, then a properly identified resource cannot be assembled, and even if an arbitrary identification was associated with it, the resource could never be the subject of a transaction outside the context of the resource that refers to it.

For example, consider a situation where an interface engine is creating a Condition record on a patient from an HL7 v2 message, and the only information about the primary surgeon is her first name and last name (REL-7.2 & REL-7.3). In the absence of a controlled practitioner directory, this is not enough information to create an identified Practitioner resource since more than one practitioner might have the same name.

In these circumstances, the resource is placed directly in-line in the resource. This SHOULD NOT be done when the content can be identified properly, as once the identification is lost, it is extremely difficult (and context dependent) to restore it again. The FHIR version of a contained resource SHALL always be the same as the resource that contains it.

An example of a contained resource:

```
 <Condition xmlns="http://hl7.org/fhir">
  <contained>
    <Practitioner>
      <id value="p1"/>
      <name>
        <family value="Person"/>
        <given value="Patricia"/>
      </name>
    </Practitioner>
  </contained>
  <!-- other attributes -->
  <asserter>
    <reference value="#p1" />
  </asserter>
  <!-- other attributes -->
 </Condition>
```

The same example in JSON:

```
{
  "resourceType" : "Condition",
  "contained": [
    {
      "resourceType" : "Practitioner",
      "id" : "p1",
      "name" : [{
        "family" : "Person",
        "given" : ["Patricia"]
      }]
    }],
   "asserter" : {
     "reference" : "#p1"
  }
}
```

Implementation Note: Contained resources are still a reference rather than being inlined directly into the element that is the reference (e.g. "custodian" above) to ensure that a single approach to resolving resource references can be used. Though direct containment would seem simpler, it would still be necessary to support internal references where the same contained resource is referenced more than once. In the end, all that it would achieve is creating additional options in the syntax. For users using XPath to process the resource, the following XPath fragment resolves the internal reference:

```
ancestor::f:*[not(parent::f:*)]/f:contained/*[@id=substring-after(current()/f:reference/@value, '#')]
```

Some notes about use and interpretation of contained resources:

• The contained element SHALL NOT have extensions on it (though contained resources can still contain extensions).

• The contained resource can be put in any resource that inherits from DomainResource. The contained element is then located at the beginning of the resource after any text narrative and before any extension.

• Contained resources share the same internal id resolution space as the parent resource (for id attributes, see Narrative references).

• When resolving references, references are resolved by looking through the 'container' resource - the one that contains the other resources. Since there are no nested contained resources, there is only one container resource.

• References to contained resources are never resolved outside the container resource. Specifically, resolution stops at the elements Bundle.entry.resource and Parameters.parameter.resource, but not at DomainResource.contained.

• Both Reference and canonical types may refer to contained resources

• Contained resources SHALL NOT contain additional contained resources.

• Contained resources SHALL NOT contain meta.versionId, meta.lastUpdated, or meta.security.

• Contained resources MAY contain meta.tag, though there are many tags that do not make sense on contained resources.

• A contained resource SHALL only be included in a resource if something in that resource (potentially another contained resource) has a reference to it.

Like other resources, contained resources can contain narrative. However, when rendering the containing resource, the narrative of the contained resources is ignored, so relevant information about contained resources SHALL appear in the narrative of the containing resource.

Resources that are contained inline do not "inherit" context from their parent resource. For instance, if the parent resource contains a "subject", and the contained resource also has a "subject" element defined, there is no implication that the contained resource has the same subject as the parent resource.

Resources can only be contained in other resources if there is a reference from the resource to the contained resource, or if the contained resource references the container resource. This is intended to ensure that the meaning of the contained resource is clear, and that there is no confusion as to its significance.

For a resource that references the container, the reference is "#", like this:

```
<Patient xmlns="http://hl7.org/fhir">
  <id value="something"/>
  <contained>
    <Provenance>
      <!-- no id necessary (though still allowed) -->
      <target>
        <reference value="#"/>
      </target>
    </Provenance>
  </contained>
  <!-- other attributes -->
</Patient>
```

There is no way to search for contained resources that reference their container. Is this a problem?

Feedback is welcome here .

2.3.0.9 Circular Resource References

Some references are circular - that is, the reference points to another resource of the same type. There are several reasons why a resource may refer to other resources of the same type:

• part-of: the resources describe a complex hierarchy composed of parts e.g. an organization structure

• builds-on: resources may add additional content to another resource by deriving from it e.g. extending type declarations

• replaces: one resource may be replaced by another due to changes over time e.g. replacing a prescription or to support patient merging

• uses/re-uses: a resource may use another resource in a modular arrangement e.g. library references or value sets including other value sets

For parameters where the relationship is a strict hierarchy (i.e. where it would be wrong to have circular references, even transitively), there is additional search support for traversing the hierarchy using :above and :below modifiers on the search parameters.

Hierarchical references with search parameters (can have :above and :below modifiers on search parameters):

• CarePlan.basedOn (based-on)

- CarePlan.replaces (replaces)
- CarePlan.partOf (part-of)
- Encounter.partOf (part-of)
- Location.partOf (partof)
- Organization.partOf (partof)
- StructureDefinition.baseDefinition (base)

References that may refer back to the source instance:

- Patient.link.other (link)

References for which the hierarchical behavior is not specified:

- Account.partOf
- BiologicallyDerivedProduct.parent
- CapabilityStatement.instantiates
- CapabilityStatement.imports
- CareTeam.participant.member (participant)
- CatalogEntry.relatedEntry.item
- ChargeItem.partOf
- ChargeItemDefinition.partOf
- ChargeItemDefinition.replaces
- Claim.related.claim
- ClaimResponse.insurance.claimResponse
- ClinicalImpression.previous (previous)
- CodeSystem.supplements (supplements)
- Communication.inResponseTo
- CommunicationRequest.replaces (replaces)
- Composition.relatesTo.target[x]
- ConceptMap.group.unmapped.url (other)
- Consent.source[x] (source-reference)
- Contract.instantiatesCanonical
- Contract.legallyBinding[x]
- Device.parent
- DeviceDefinition.parentDevice (parent)
- DocumentReference.relatesTo.target (relatesto)
- ExampleScenario.workflow
- Group.member.entity (member)
- ImplementationGuide.dependsOn.uri (depends-on)
- Medication.ingredient.item[x]
- MedicationAdministration.partOf
- MedicationKnowledge.relatedMedicationKnowledge.reference
- MedicationRequest.basedOn
- MedicationRequest.priorPrescription
- MedicationStatement.partOf (part-of)
- MessageDefinition.replaces
- MessageDefinition.base
- MessageDefinition.allowedResponse.message
- MolecularSequence.referenceSeq.referenceSeqPointer
- MolecularSequence.pointer
- Observation.hasMember (has-member)
- Observation.derivedFrom (derived-from)
- OperationDefinition.base (base)
- Person.link.target (link,patient,practitioner,relatedperson)
- PlanDefinition.action.definition[x] (definition)
- Procedure.partOf (part-of)

- Procedure.reasonReference (reason-reference)
- Questionnaire.derivedFrom
- ResearchStudy.partOf (partof)
- SearchParameter.derivedFrom (derived-from)
- SearchParameter.component.definition (component)
- ServiceRequest.basedOn (based-on)
- ServiceRequest.replaces (replaces)
- Specimen.parent (parent)
- StructureMap.import
- Substance.ingredient.substance[x]
- SubstanceSpecification.property.definingSubstance[x]
- SubstanceSpecification.relationship.substance[x]
- SupplyDelivery.partOf
- Task.partOf (part-of)
- ValueSet.compose.include.valueSet

2.4 Narrative

URL: https://hl7.org/fhir/R4/narrative.html

2.4.0 Narrative

FHIR Infrastructure Work Group | Maturity Level: Normative | Standards Status: Normative

| This page has been approved as part of an ANSI standard. See the Infrastructure Package for further details.

Any resource that is a DomainResource (all resources except Bundle, Parameters and Binary) may include a human-readable narrative that contains a summary of the resource and may be used to represent the content of the resource to a human.

If narrative is present with a status other than 'empty', it SHALL reflect all content needed for a human to understand the essential clinical and business information for the resource. It SHALL be safe to render only the narrative of the resource without displaying any of the resource's discrete/encoded information. Resource definitions and/or profiles on resources MAY define what content should be represented in the narrative to ensure clinical safety.

The narrative for a resource MAY contain additional information that is not in the structured data, including human-edited content. Such additional information SHALL be in the scope of the definition of the resource, though it is common for the narrative to include additional descriptive information extracted from other referenced resources when describing references. Narrative for a resource SHOULD include summary information about any referenced resources that would be required for a consumer of the resource to be able to understand the key, essential information about a resource without retrieving any additional resources. If the Narrative.status = extensions, the narrative SHALL reflect the impact of all modifier extensions that extend elements that are themselves described by the narrative. Narrative MAY include generated content from other resources and still be considered generated.

For example, the narrative for a MedicationRequest might include brief summary information about the referenced patient, prescriber and medication. Some resources (e.g. List) can provide specific rules about what content must (or must not) be included in the resource narrative. Consideration would be given to the fact that referenced resources may be updated without updating referencing resources, so the proportion of content of a referenced resource included in a referencing resource should be limited.

Systems MAY choose how narrative is generated, including how much de-referencing to perform, but SHALL NOT assume that the resource is rendered in any particular context when generating narrative, since resources will be used in multiple contexts.

Resource instances that permit narrative SHOULD always contain narrative to support human-consumption as a fallback. Structured data SHOULD NOT generally contain information of importance to human readers that is omitted from the narrative. Creators of FHIR resources should not assume that systems will render (or that humans will see) data that is not in the narrative. However, in strictly managed trading systems where all systems share a common data model and additional text is unnecessary or even a clinical safety risk, the narrative may be omitted. Implementers should give careful consideration before doing this, as it will mean that such resources can only be understood in the limited trading environment. Closed trading partner environments are very likely to open up during the lifetime of the resources they define. Also, many workflow steps involving finding and aggregating resources are much more difficult or tedious if the resources involved do not have their own text.

A resource MAY only have text with little or no additional discrete data (as long as all minOccurs=1 elements are satisfied). This can be necessary for data from legacy systems where information is captured as a "text

blob" or where text is additionally entered raw or narrated and encoded information is added later.

The narrative is an XHTML fragment with a flag to indicate its relationship to the data:

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints

Narrative | N | | Element | Human-readable summary of the resource (essential clinical and business information) Elements defined in Ancestors: id, extension

status | | 1..1 | code | generated | extensions | additional | empty NarrativeStatus (Required)

div | I | 1..1 | xhtml | Limited xhtml content + Rule: The narrative SHALL contain only the basic html formatting elements and attributes described in chapters 7-11 (except section 4 of chapter 9) and 15 of the HTML 4.0 standard, <a> elements (either name or href), images and internally contained style attributes + Rule: The narrative SHALL have some non-whitespace content

Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <status value="[code]"/><!-- 1..1 generated | extensions | additional | empty -->
 <div xmlns="http://www.w3.org/1999/xhtml"> <!-- Limited xhtml content< --> </div>
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "status" : "<code>", // R!  generated | extensions | additional | empty
  "div" : "(Escaped XHTML)" // R!  Limited xhtml content
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:Narrative.status [ code ]; # 1..1 generated | extensions | additional | empty
  fhir:Narrative.div [ fhir:value "[escaped xhtml]"^^xsd:string ]; # 1..1 Limited xhtml content
]
```

Changes since Release 3

Narrative

Narrative.status | Change value set from http://hl7.org/fhir/ValueSet/narrative-status to http://hl7.org/fhir/ValueSet/narrative-status|4.0.1

See the Full Difference for further information

Structure

Name | Flags | Card. | Type | Description & Constraints

Narrative | N | | Element | Human-readable summary of the resource (essential clinical and business information) Elements defined in Ancestors: id, extension

status | | 1..1 | code | generated | extensions | additional | empty NarrativeStatus (Required)

div | I | 1..1 | xhtml | Limited xhtml content + Rule: The narrative SHALL contain only the basic html formatting elements and attributes described in chapters 7-11 (except section 4 of chapter 9) and 15 of the HTML 4.0 standard, <a> elements (either name or href), images and internally contained style attributes + Rule: The narrative SHALL have some non-whitespace content

Documentation for this format

UML Diagram (Legend)

XML Template


```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <status value="[code]"/><!-- 1..1 generated | extensions | additional | empty -->
 <div xmlns="http://www.w3.org/1999/xhtml"> <!-- Limited xhtml content< --> </div>
</[name]>
```


JSON Template


```
{
  // from Element: extension
  "status" : "<code>", // R!  generated | extensions | additional | empty
  "div" : "(Escaped XHTML)" // R!  Limited xhtml content
}
```


Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:Narrative.status [ code ]; # 1..1 generated | extensions | additional | empty
  fhir:Narrative.div [ fhir:value "[escaped xhtml]"^^xsd:string ]; # 1..1 Limited xhtml content
]
```

Changes since Release 3

Narrative	
Narrative.status	Change value set from http://hl7.org/fhir/ValueSet/narrative-status to http://hl7.org/fhir/ValueSet/narrative-status|4.0.1

See the Full Difference for further information

The contents of the div element are an XHTML fragment that SHALL contain only the basic HTML formatting elements described in chapters 7-11 (except section 4 of chapter 9) and 15 of the HTML 4.0 standard, <a> elements (either name or href), images and internally contained style attributes. The XHTML content SHALL NOT contain a head, a body element, external stylesheet references, deprecated elements, scripts, forms, base/link/xlink, frames, iframes, objects or event related attributes (e.g. onClick). This is to ensure that the content of the narrative is contained within the resource and that there is no active content. Such content would introduce security issues and potentially safety issues with regard to extracting text from the XHTML. Note that even with these restrictions, there are still several important security risks associated with displaying the narrative.

The div element SHALL have some non-whitespace content (text or an image).

```
  <narrative>
    <status value="additional"/>
    <div xmlns="http://www.w3.org/1999/xhtml">This is a simple
          example with only plain text</div>
  </narrative>

  <narrative>
    <status value="additional"/>
   <div xmlns="http://www.w3.org/1999/xhtml">
     <p>
       This is an <i>example</i> with some <b>xhtml</b> formatting.
     </p>
   </div>
  </narrative>
```

The inner portion of the div content is often used for the innerHTML property in a browser. In order to simplify this kind of processing, when the narrative is represented in JSON, it SHALL be encoded so that the characters between the first '>' and the last '<' delimiters is the content of the element; e.g.

```
  "div": "<div xmlns=\"http://www.w3.org/1999/xhtml\">text</div>"
```

is legal, but this is not:

```
  "div": "<?xml ...><div>text</div>"
```

Note that the XHTML is contained in general XML so there is no support for HTML entities like   or © etc. Unicode characters SHALL be used instead. Unicode   substitutes for .

The narrative content SHOULD be in the language of the resource, but there is no reason to expect that HTML enabled tooling would understand the resource language element. For this reason, a lang attribute on the SHOULD also be used if language is declared on the resource (see the note in the HTML 5 specification about use of language).

2.4.0.1 Multi-language support for Narratives

A narrative may contain content from multiple languages. This is always possible in that the narrative may contain ad-hoc content in any language, often with inline translation provided - this is not unusual in clinical systems when dealing with patients/care providers speaking multiple languages. Some resources - specially standing documentation such as preparation notes - will contain the same documentation in several different languages.

Resources containing the same information in multiple different languages should use a div element with an XHTML lang attribute in the root div. By default, all languages will be displayed to a user, but multi-language aware applications can filter the content by language and only display the language that ia relevant to or chosend by the patient.

See the W3C documentation on Language declarations for further information.

2.4.0.2 Image References

Image source data (the src attribute) may refer to an image found in the resource (as a contained Media or Binary resource) by its id:

```
<Patient xmlns="http://hl7.org/fhir">
  <text>
    <status value="generated"/>
    <div xmlns="http://www.w3.org/1999/xhtml">
      <p>... <img src="#pic1"/>. ....</p>
    </div>
```

```
  </text>
  <contained>
    <Binary><id value="pic1"/><contentType value="image/gif"/><data value="MEKH....SD/Z"/></Binary>
  </contained>
</Patient>
```

2.4.0.3 Internal Id References

References between the narrative and the resource data (in either direction) are mediated by the XML id/idref attributes. in JSON, the property "id" is used which is equivalent to the XML attribute "id".

The id attribute SHALL have a unique value within the resource with regard to any other id attributes: the uniqueness and resolution scope of these id references is within the resource that contains them. Contained resources are included in the id uniqueness scope of the resource that contains them.

If multiple resources are combined into a single combined document, such as a Bundle, duplicate values of the id attribute may occur between resources. This SHALL be managed by applications reading the resources.

Since images that are not contained in the resource cannot be guaranteed to be available when the resource is presented to a user, the source for any images that are an essential part of the narrative SHOULD always be embedded as a data: url , in an attachment or a contained resource

2.4.0.4 Styling the XHTML

The XHTML fragment in the narrative may be styled using cascading stylesheets with either external or internal styles. External styles are applied using the class and id attributes on the XHTML elements and internal styles are applied using a style attribute on the XHTML elements directly.

In order to minimize manageability and security issues, authoring systems cannot specify the CSS stylesheet to use directly, unless the stylesheet is included in a FHIR Document. Instead, the application that displays the resource provides the stylesheets. This means that the rendering system chooses what styles can be used, but the authoring system must use them in advance. Authoring systems can use these classes, which SHALL be supported by all rendering systems:

```
bold | Bold Text | { font-weight: bold }
italics | Italics Text | { font-style: italic }
underline | Underlined Text | { text-decoration: underline }
strikethrough | Strikethrough Text | { text-decoration: line-through }
left | Left Aligned | { text-align : left }
right | Right Aligned | { text-align : right }
center | Center Aligned | { text-align : center }
justify | Justified | { text-align : justify }
border-left | Border on the left | { border-left: 1px solid grey }
border-right | Border on the right | { border-right: 1px solid grey }
border-top | Border on the top | { border-top: 1px solid grey }
border-bottom | Border on the bottom | { border-bottom: 1px solid grey }
arabic | List is ordered using Arabic numerals: 1, 2, 3 | { list-style-type: decimal }
little-roman | List is ordered using little Roman numerals: i, ii, iii | { list-style-type: lower-roman }
big-roman | List is ordered using big Roman numerals: I, II, III | { list-style-type: upper-roman }
little-alpha | List is ordered using little alpha characters: a, b, c | { list-style-type: lower-alpha }
big-alpha | List is ordered using big alpha characters: A, B, C | { list-style-type: upper-alpha }
disc | List bullets are simple solid discs | { list-style-type: disc }
circle | List bullets are hollow discs | { list-style-type : circle }
square | List bullets are solid squares | { list-style-type: square }
unlist | List with no bullets | { list-style-type: none }
```

Note: for testing purposes, there is an example resource that includes all of these styles. It is also available as XHTML and a standard stylesheet that includes all of these styles. Use of styles not on this list will require an arrangement between producing and consuming systems.

Authoring systems may refer to additional classes, but cannot rely on the fact that they will be supported. If the additional classes are critical for safe rendering, trading partner agreements will be required.

Authoring systems may also use internal styles using the style attribute. This has the advantage of not depending on external interpretation, but also has the side effect of making content more difficult to manage when rendering, so applications should use this approach with care.

Authoring systems may fix the following styling aspects of the content:

- bold, italic, underline, strikethrough
- font color, family and size
- background color, text alignment
- whitespace interpretation
- ordered list number format (since it may be referred to in text)

These style properties are specified in-line using the style attribute. Rendering systems SHOULD respect any of these rendering styles when they are specified in the style attribute, although appropriate interpretation is allowed in certain contexts (e.g. a low-contrast display for dark rooms or a high-contrast display for the

visually impaired may adjust colors accordingly).

Note that rendering systems can ignore or override any of the internal or external styles described above, but SHOULD be careful to ensure that this is only done in the context of well-maintained trading partner agreements, as altering the presentation of the text may create clinical safety issues.

Authors MAY specify additional styles and style properties as specified in the CSS specification, but these are extensions to this specification and renderers are not required to heed them. It SHOULD be safe to view the narrative without these additional styling features available.

Note that there are additional rules around styling for documents presentation.

2.4.0.5 Linking between Data and Narrative

In some contexts, it is useful to link between the two representations of the same content: structured data, and human readable narrative. This can be used to assert that the text is a representation of the data, or specifically that the data is derived from some particular text. This specification defines the extensions narrativeLink and originalText to establish these links. Here's an example of using originalText:

```
{
  "resourceType" : "Condition",
  "text" : {
    "div": "<div xmlns=\"http://www.w3.org/1999/xhtml\">There is a history of <span id=\"a1\">Asthma</span></div>"
  },
  "code" : {
    "coding" : {
      "system" : "http://snomed.info/sct",
      "code" : "195967001",
      "display" : "Asthma (disorder)"
    },
    "extension" : [{
      "url" : "http://hl7.org/fhir/StructureDefinition/originalText",
      "valueUrl" : "#a1"
    }]
  }
}
```

This indicates that the word "Asthma" in the narrative is the original text from which the SNOMED CT code "asthma" was derived by some text processing method. Typically, this method is associated with resources built from CDA documents. The same method can be used to reference across resource boundaries, e.g. between a resource and the composition that represents it:

```
<Bundle xmlns="http://hl7.org/fhir">
  <entry>
    <fullUrl value="http://example.org/fhir/Composition/abcdefghij"/>
    <Composition>
      <id value="c1"/>
      <text>
        <div xmlns="http://www.w3.org/1999/xhtml">
          ...
          <p>There is a history of <span id="a1">Asthma</span></p>
          ...
        </div>
      </text>
    </Composition>
  </entry>
  <entry>
    <Condition>
      <code>
        <extension url="http://hl7.org/fhir/StructureDefinition/originalText">
          <valueUrl value="http://example.org/fhir/Composition/abcdefghij#a1"/>
        </extension>
        <coding>
          <system value="http://snomed.info/sct"/>
          <code value="195967001"/>
          <display value="Asthma (disorder)"/>
        </coding>
      </code>
    </Condition>
  </entry>
</Bundle>
```

2.4.0.6 Clinical Safety Concerns

Health care records are often associated with legislative and business requirements for very long retention times (up to a century) and extreme risk aversion with regards to inconsistent display across a variety of devices, rendering engines, and display constraints. Although the narrative is allowed to use the standard XHTML and CSS features as described above, implementations are encouraged to show restraint when using the features available. Even when trading partner arrangements limit the current requirements made on a system, experience shows that these trading arrangements will likely broaden over time.

In particular:

• complex layered layouts requiring careful testing of the match between the XHTML div and span elements and styles, and those that include nested tables (possibly with images) as well, make rendering consistency difficult, and implementations SHOULD avoid these

• The use of styles e.g. bolding, italics and color; SHOULD NOT be used as the sole way to convey meaning or semantics, but should be used in conjunction with other data elements to ensure consistent, long term interoperability.

2.4.1 Example Narrative

URL: https://hl7.org/fhir/R4/narrative-example.html
FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative

2.4.1.1 Plain HTML, No Styles

2.4.1.1.1 Heading 3

2.4.1.1.1.1 Heading 4

2.4.1.1.1.1.1 Heading 5

2.4.1.1.1.1.1.1 Heading 6

Paragraph. span. Link. Bold, br: em, Italics, strong, small, big Teletype Text, small, Definition term, q, var. All provided by HL7, for FHIR (cite).

Paragraph in a blockquote, with an hr after it:

Paragraph in a div (Link Target)

• Unordered List Item

• Ordered List Item

DT Item

DD Item

```
Some Pre Text
  with a line break
```

Table:

Head Cell 1 | Head Cell 2 | Head Cell 3
Foot Cell 1 | Foot Cell 2 | Foot Cell 3
Body Cell 1 | Body Cell 2 | Body Cell 3

Code Block

Sample Block

2.4.1.2 External Styles

2.4.1.2.1 Text:

Example Text: bold, italics, underline and strikethrough

This paragraph is left aligned. The content should be laid out aligned at the left of the screen. The content should be laid out aligned at the left of the screen. The content should be laid out aligned at the left of the screen. The content should be laid out aligned at the left of the screen. The content should be laid out aligned at the left of the screen.

This paragraph is right aligned. The content should be laid out aligned at the right of the screen. The content should be laid out aligned at the right of the screen. The content should be laid out aligned at the right of the screen. The content should be laid out aligned at the right of the screen. The content should be laid out aligned at the right of the screen.

This paragraph is center aligned. The content should be laid out aligned at the center of the screen. The content should be laid out aligned at the center of the screen. The content should be laid out aligned at the center of the screen. The content should be laid out aligned at the center of the screen. The content should be laid out aligned at the center of the screen.

This paragraph is justified. The content should be laid out aligned at both the left and right of the screen. The content should be laid out aligned at both the left and right of the screen. The content should be laid out aligned at both the left and right of the screen. The content should be laid out aligned at both the left and right of the screen. The content should be laid out aligned at both the left and right of the screen.

2.4.1.2.2 Table:

Border Left | Border Right | Border Top | Border Bottom

2.4.1.2.3 List:

• arabic (Item 1)

• Item Two

• little-roman (Item 1)

- Item Two
- big-roman (Item 1)
- Item Two
- little-alpha (Item 1)
- Item Two
- big-alpha (Item 1)
- Item Two
- disc (Item 1)
- Item Two
- unlist(Item 1)
- Item Two
- circle (Item 1)
- Item Two
- square (Item 1)
- Item Two

2.4.1.3 Internal Styles

Example Text: bold, italics, underline and strikethrough. Font-Family Serif and Sans Serif, Font-size 50% 80% 150%, Font-Color Navy Maroon Brown, Background-color Aqua Silver Pink.

Whitespace Control:

Normal Whitespace Test, long long long

- decimal-leading-zero (Item 1)
- Item Two
- georgian (Item 1)
- Item Two
- hebrew (Item 1)
- Item Two
- hiragana (Item 1)
- Item Two
- hiragana-iroha (Item 1)
- Item Two
- inherit (Item 1)
- Item Two
- katakana (Item 1)
- Item Two
- katakana-iroha (Item 1)
- Item Two
- lower-alpha (Item 1)
- Item Two
- lower-greek (Item 1)
- Item Two
- lower-latin (Item 1)
- Item Two
- lower-roman (Item 1)
- Item Two
- none (Item 1)
- Item Two
- upper-alpha (Item 1)
- Item Two
- upper-latin (Item 1)
- Item Two
- upper-roman (Item 1)
- Item Two
- upper-roman (Item 1)
- Item Two

2.5 Extensibility

URL: https://hl7.org/fhir/R4/extensibility.html
FHIR Infrastructure Work Group | Maturity Level: Normative | Standards Status: Normative

| This page has been approved as part of an ANSI standard. See the Infrastructure Package for further details.

This exchange specification is based on generally agreed common requirements across healthcare - covering many jurisdictions, domains, and different functional approaches. It is common for specific implementations to have valid requirements that are not part of these agreed common requirements. Incorporating all valid requirements would make this specification very cumbersome and difficult to implement. Instead, this specification expects that additional valid requirements will be implemented as extensions.

As such, extensibility is a fundamental part of the design of this specification. Every element in a resource can have extension child elements to represent additional information that is not part of the basic definition of the resource. Applications should not reject resources merely because they contain extensions, though they may need to reject resources because of the specific contents of the extensions.

Note that, unlike in many other specifications, there can be no stigma associated with the use of extensions by any application, project, or standard - regardless of the institution or jurisdiction that uses or defines the extensions. The use of extensions is what allows the FHIR specification to retain a core simplicity for everyone.

To make the use of extensions safe and manageable, there is strict governance applied to the definition and use of extensions. Although any implementer can define and use extensions, there is a set of requirements that must be met as part of their use and definition.

2.5.0.1 Extension Element

Every element in a resource or data type includes an optional "extension" child element that may be present any number of times. This is the content model of the extension as it appears in each resource:

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
Extension | IN | | Element | Optional Extensions Element + Rule: Must have either extensions or value[x], not both Elements defined in Ancestors: id, extension
url | | 1..1 | uri | identifies the meaning of the extension
value[x] | | 0..1 | * | Value of extension
Documentation for this format

UML Diagram (Legend)

XML Template

```
<extension|modifierExtension xmlns="http://hl7.org/fhir" url="identifies the meaning of the extension (uri)">
 <!-- from Element: extension -->
 <value[x]><!-- 0..1 * Value of extension --></value[x]>
</extension|modifierExtension>
```

JSON Template

```
{
  // from Element: extension
  "url" : "<uri>", // R!  identifies the meaning of the extension
  // value[x]: Value of extension. One of these 50:
  "valueBase64Binary" : "<base64Binary>"
  "valueBoolean" : <boolean>
  "valueCanonical" : "<canonical>"
  "valueCode" : "<code>"
  "valueDate" : "<date>"
  "valueDateTime" : "<dateTime>"
  "valueDecimal" : <decimal>
  "valueId" : "<id>"
  "valueInstant" : "<instant>"
  "valueInteger" : <integer>
  "valueMarkdown" : "<markdown>"
  "valueOid" : "<oid>"
  "valuePositiveInt" : "<positiveInt>"
  "valueString" : "<string>"
  "valueTime" : "<time>"
  "valueUnsignedInt" : "<unsignedInt>"
  "valueUri" : "<uri>"
  "valueUrl" : "<url>"
  "valueUuid" : "<uuid>"
  "valueAddress" : { Address }
  "valueAge" : { Age }
  "valueAnnotation" : { Annotation }
  "valueAttachment" : { Attachment }
  "valueCodeableConcept" : { CodeableConcept }
  "valueCoding" : { Coding }
  "valueContactPoint" : { ContactPoint }
  "valueCount" : { Count }
  "valueDistance" : { Distance }
  "valueDuration" : { Duration }
  "valueHumanName" : { HumanName }
  "valueIdentifier" : { Identifier }
  "valueMoney" : { Money }
  "valuePeriod" : { Period }
  "valueQuantity" : { Quantity }
  "valueRange" : { Range }
  "valueRatio" : { Ratio }
  "valueReference" : { Reference }
  "valueSampledData" : { SampledData }
  "valueSignature" : { Signature }
  "valueTiming" : { Timing }
  "valueContactDetail" : { ContactDetail }
  "valueContributor" : { Contributor }
  "valueDataRequirement" : { DataRequirement }
  "valueExpression" : { Expression }
  "valueParameterDefinition" : { ParameterDefinition }
  "valueRelatedArtifact" : { RelatedArtifact }
  "valueTriggerDefinition" : { TriggerDefinition }
  "valueUsageContext" : { UsageContext }
  "valueDosage" : { Dosage }
```

```
  "valueMeta" : { Meta }
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:Extension.url [ uri ]; # 1..1 identifies the meaning of the extension
  # Extension.value[x] : 0..1 Value of extension. One of these 50
    fhir:Extension.valueBase64Binary [ base64Binary ]
    fhir:Extension.valueBoolean [ boolean ]
    fhir:Extension.valueCanonical [ canonical ]
    fhir:Extension.valueCode [ code ]
    fhir:Extension.valueDate [ date ]
    fhir:Extension.valueDateTime [ dateTime ]
    fhir:Extension.valueDecimal [ decimal ]
    fhir:Extension.valueId [ id ]
    fhir:Extension.valueInstant [ instant ]
    fhir:Extension.valueInteger [ integer ]
    fhir:Extension.valueMarkdown [ markdown ]
    fhir:Extension.valueOid [ oid ]
    fhir:Extension.valuePositiveInt [ positiveInt ]
    fhir:Extension.valueString [ string ]
    fhir:Extension.valueTime [ time ]
    fhir:Extension.valueUnsignedInt [ unsignedInt ]
    fhir:Extension.valueUri [ uri ]
    fhir:Extension.valueUrl [ url ]
    fhir:Extension.valueUuid [ uuid ]
    fhir:Extension.valueAddress [ Address ]
    fhir:Extension.valueAge [ Age ]
    fhir:Extension.valueAnnotation [ Annotation ]
    fhir:Extension.valueAttachment [ Attachment ]
    fhir:Extension.valueCodeableConcept [ CodeableConcept ]
    fhir:Extension.valueCoding [ Coding ]
    fhir:Extension.valueContactPoint [ ContactPoint ]
    fhir:Extension.valueCount [ Count ]
    fhir:Extension.valueDistance [ Distance ]
    fhir:Extension.valueDuration [ Duration ]
    fhir:Extension.valueHumanName [ HumanName ]
    fhir:Extension.valueIdentifier [ Identifier ]
    fhir:Extension.valueMoney [ Money ]
    fhir:Extension.valuePeriod [ Period ]
    fhir:Extension.valueQuantity [ Quantity ]
    fhir:Extension.valueRange [ Range ]
    fhir:Extension.valueRatio [ Ratio ]
    fhir:Extension.valueReference [ Reference ]
    fhir:Extension.valueSampledData [ SampledData ]
    fhir:Extension.valueSignature [ Signature ]
    fhir:Extension.valueTiming [ Timing ]
    fhir:Extension.valueContactDetail [ ContactDetail ]
    fhir:Extension.valueContributor [ Contributor ]
    fhir:Extension.valueDataRequirement [ DataRequirement ]
    fhir:Extension.valueExpression [ Expression ]
    fhir:Extension.valueParameterDefinition [ ParameterDefinition ]
    fhir:Extension.valueRelatedArtifact [ RelatedArtifact ]
    fhir:Extension.valueTriggerDefinition [ TriggerDefinition ]
    fhir:Extension.valueUsageContext [ UsageContext ]
    fhir:Extension.valueDosage [ Dosage ]
    fhir:Extension.valueMeta [ Meta ]
]
```

Changes since Release 3

Extension

Extension.value[x] | Add Types canonical, url, uuid, ContactDetail, Contributor, DataRequirement, Expression, ParameterDefinition, RelatedArtifact, TriggerDefinition, UsageContext, Dosage

See the Full Difference for further information

Structure

Name | Flags | Card. | Type | Description & Constraints

Extension | IN | | Element | Optional Extensions Element + Rule: Must have either extensions or value[x], not both Elements defined in Ancestors: id, extension

url | | 1..1 | uri | identifies the meaning of the extension

value[x] | | 0..1 | * | Value of extension

Documentation for this format

UML Diagram (Legend)

XML Template

```
<extension|modifierExtension xmlns="http://hl7.org/fhir" url="identifies the meaning of the extension (uri)">
 <!-- from Element: extension -->
 <value[x]><!-- 0..1 * Value of extension --></value[x]>
</extension|modifierExtension>
```

JSON Template

```
{
```

```
  // from Element: extension
  "url" : "<uri>", // R!  identifies the meaning of the extension
  // value[x]: Value of extension. One of these 50:
  "valueBase64Binary" : "<base64Binary>"
  "valueBoolean" : <boolean>
  "valueCanonical" : "<canonical>"
  "valueCode" : "<code>"
  "valueDate" : "<date>"
  "valueDateTime" : "<dateTime>"
  "valueDecimal" : <decimal>
  "valueId" : "<id>"
  "valueInstant" : "<instant>"
  "valueInteger" : <integer>
  "valueMarkdown" : "<markdown>"
  "valueOid" : "<oid>"
  "valuePositiveInt" : "<positiveInt>"
  "valueString" : "<string>"
  "valueTime" : "<time>"
  "valueUnsignedInt" : "<unsignedInt>"
  "valueUri" : "<uri>"
  "valueUrl" : "<url>"
  "valueUuid" : "<uuid>"
  "valueAddress" : { Address }
  "valueAge" : { Age }
  "valueAnnotation" : { Annotation }
  "valueAttachment" : { Attachment }
  "valueCodeableConcept" : { CodeableConcept }
  "valueCoding" : { Coding }
  "valueContactPoint" : { ContactPoint }
  "valueCount" : { Count }
  "valueDistance" : { Distance }
  "valueDuration" : { Duration }
  "valueHumanName" : { HumanName }
  "valueIdentifier" : { Identifier }
  "valueMoney" : { Money }
  "valuePeriod" : { Period }
  "valueQuantity" : { Quantity }
  "valueRange" : { Range }
  "valueRatio" : { Ratio }
  "valueReference" : { Reference }
  "valueSampledData" : { SampledData }
  "valueSignature" : { Signature }
  "valueTiming" : { Timing }
  "valueContactDetail" : { ContactDetail }
  "valueContributor" : { Contributor }
  "valueDataRequirement" : { DataRequirement }
  "valueExpression" : { Expression }
  "valueParameterDefinition" : { ParameterDefinition }
  "valueRelatedArtifact" : { RelatedArtifact }
  "valueTriggerDefinition" : { TriggerDefinition }
  "valueUsageContext" : { UsageContext }
  "valueDosage" : { Dosage }
  "valueMeta" : { Meta }
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:Extension.url [ uri ]; # 1..1 identifies the meaning of the extension
  # Extension.value[x] : 0..1 Value of extension. One of these 50
    fhir:Extension.valueBase64Binary [ base64Binary ]
    fhir:Extension.valueBoolean [ boolean ]
    fhir:Extension.valueCanonical [ canonical ]
    fhir:Extension.valueCode [ code ]
    fhir:Extension.valueDate [ date ]
    fhir:Extension.valueDateTime [ dateTime ]
    fhir:Extension.valueDecimal [ decimal ]
    fhir:Extension.valueId [ id ]
    fhir:Extension.valueInstant [ instant ]
    fhir:Extension.valueInteger [ integer ]
    fhir:Extension.valueMarkdown [ markdown ]
    fhir:Extension.valueOid [ oid ]
    fhir:Extension.valuePositiveInt [ positiveInt ]
    fhir:Extension.valueString [ string ]
    fhir:Extension.valueTime [ time ]
    fhir:Extension.valueUnsignedInt [ unsignedInt ]
    fhir:Extension.valueUri [ uri ]
    fhir:Extension.valueUrl [ url ]
    fhir:Extension.valueUuid [ uuid ]
    fhir:Extension.valueAddress [ Address ]
    fhir:Extension.valueAge [ Age ]
    fhir:Extension.valueAnnotation [ Annotation ]
    fhir:Extension.valueAttachment [ Attachment ]
    fhir:Extension.valueCodeableConcept [ CodeableConcept ]
    fhir:Extension.valueCoding [ Coding ]
    fhir:Extension.valueContactPoint [ ContactPoint ]
    fhir:Extension.valueCount [ Count ]
```

```
    fhir:Extension.valueDistance [ Distance ]
    fhir:Extension.valueDuration [ Duration ]
    fhir:Extension.valueHumanName [ HumanName ]
    fhir:Extension.valueIdentifier [ Identifier ]
    fhir:Extension.valueMoney [ Money ]
    fhir:Extension.valuePeriod [ Period ]
    fhir:Extension.valueQuantity [ Quantity ]
    fhir:Extension.valueRange [ Range ]
    fhir:Extension.valueRatio [ Ratio ]
    fhir:Extension.valueReference [ Reference ]
    fhir:Extension.valueSampledData [ SampledData ]
    fhir:Extension.valueSignature [ Signature ]
    fhir:Extension.valueTiming [ Timing ]
    fhir:Extension.valueContactDetail [ ContactDetail ]
    fhir:Extension.valueContributor [ Contributor ]
    fhir:Extension.valueDataRequirement [ DataRequirement ]
    fhir:Extension.valueExpression [ Expression ]
    fhir:Extension.valueParameterDefinition [ ParameterDefinition ]
    fhir:Extension.valueRelatedArtifact [ RelatedArtifact ]
    fhir:Extension.valueTriggerDefinition [ TriggerDefinition ]
    fhir:Extension.valueUsageContext [ UsageContext ]
    fhir:Extension.valueDosage [ Dosage ]
    fhir:Extension.valueMeta [ Meta ]
]
```

Changes since Release 3

Extension
Extension.value[x] | Add Types canonical, url, uuid, ContactDetail, Contributor, DataRequirement, Expression, ParameterDefinition, RelatedArtifact, TriggerDefinition, UsageContext, Dosage

See the Full Difference for further information

Notes:

• The url is a mandatory attribute / property and identifies a retrievable extension definition that defines the content and meaning of the extension

• The url SHALL be a URL, not a URN (e.g. not an OID or a UUID), and it SHALL be the canonical URL of a StructureDefinition that defines the extension. Except for child extensions defined within complex extensions, the URL SHALL be an absolute URL.

• The structure definitions for the extension SHOULD be available to consumers of an instance. The preferred mechanisms for achieving this availability are the direct resolvability of the extension's canonical URL and/or the publishing of the extension definition on a registry that is available and known to those systems that will be consuming the instances containing the extension

• An extension SHALL have either a value (i.e. a value[x] element) or sub-extensions, but not both. If present, the value[x] element SHALL have content (value attribute or other elements)

• If it is not safe for an application processing the content of the resource to ignore the extension it SHALL be represented using a Modifier Extension rather than an extension element

• There are a few resources (ones that do not specialize DomainResource) where extensions are not allowed on the root element, but extensions can appear elsewhere through the resource

• The value[x] element has an actual name of "value" and then the TitleCased name of one of these defined types, and its contents are those as defined for that type: valueBase64Binary: base64Binary valueBoolean: boolean valueCanonical: canonical valueCode: code (only if the extension definition provides a fixed binding to a suitable set of codes) valueDate: date valueDateTime: dateTime valueDecimal: decimal valueId: id valueInstant: instant valueInteger: integer valueMarkdown: markdown valueOid: oid valuePositiveInt: positiveInt valueString: string valueTime: time valueUnsignedInt: unsignedInt valueUri: uri valueUrl: url valueUuid: uuid valueAddress: Address valueAge: Age valueAnnotation: Annotation valueAttachment: Attachment valueCodeableConcept: CodeableConcept valueCoding: Coding valueContactPoint: ContactPoint valueCount: Count valueDistance: Distance valueDuration: Duration valueHumanName: HumanName valueIdentifier: Identifier valueMoney: Money valuePeriod: Period valueQuantity: Quantity valueRange: Range valueRatio: Ratio valueReference: Reference - a reference to another resource valueSampledData: SampledData valueSignature: Signature valueTiming: Timing valueContactDetail: ContactDetail valueContributor: Contributor valueDataRequirement: DataRequirement valueExpression: Expression valueParameterDefinition: ParameterDefinition valueRelatedArtifact: RelatedArtifact valueTriggerDefinition: TriggerDefinition valueUsageContext: UsageContext valueDosage: Dosage valueMeta: Meta

Here is an example of an extension in XML (see definition):

```
<name>
  <extension url="http://hl7.org/fhir/StructureDefinition/iso-21090-EN-use" >
    <valueCode value="I" />
  </extension>
  <text value="Chief Red Cloud"/>
</name>
```

In this example the name with text = "Chief Red Cloud" is extended to have a name use code of "Indigenous" (defined in ISO 21090, but very rarely used in practice).

In JSON, extensions are represented similarly:

```
{
  "extension" : [{
    "url" : "http://hl7.org/fhir/StructureDefinition/iso-21090-EN-use",
    "valueCode" : "I"
  }],
  "text" : "Chief Red Cloud"
}
```

Making the types explicit in the representation means that all systems can read and write (and therefore store and/or exchange) extensions correctly without needing to access the definition of the extension.

Note that the JSON representation for extensions on primitive data types is handled differently. See Representing primitive types in JSON for further information.

Extensions can also contain extensions, either because the extension definition itself defines complex content - that is, a nested tree of values in the extension - or because the extension is extended with an additional extension defined separately.

In the case where an extension defines complex content, the identity of the parts of the extension are local/relative to the reference to the extension definition.

As an example, consider extending a patient with information about citizenship (see definition) containing 2 fields: code and period. In XML:

```
<Patient>
  <extension url="http://hl7.org/fhir/StructureDefinition/patient-citizenship" >
    <extension url="code" >
      <valueCodeableConcept>
        <coding>
          <system value="urn:iso:std:iso:3166" />
          <code value="DE" />
        </coding>
      </valueCodeableConcept>
    </extension>
    <extension url="period" >
      <valuePeriod>
        <start value="2009-03-14" />
      </valuePeriod>
    </extension>
  </extension>
  <!-- other data for patient -->
</Patient>
```

Or in JSON:

```
{
  "resourceType" : "Patient",
  "extension" : [{
    "url" : "http://hl7.org/fhir/StructureDefinition/patient-citizenship",
    "extension" : [{
      "url" : "code",
      "valueCodeableConcept" : {
          "coding" : [{
             "system" : "urn:iso:std:iso:3166",
             "code" : "DE"
          }]
       }
    }, {
      "url" : "period",
      "valuePeriod" : {
         "start" : "2009-03-14"
      }
    }]
  }]
}
```

This extension can be extended again, by adding a "passport-number" extension:

The passport number is defined as a separate extension (e.g. by an implementing organization, not in this specification) rather than part of the official citizenship extension. The URL of the extension is thus different. In XML:

```
<Patient>
  <extension url="http://hl7.org/fhir/StructureDefinition/patient-citizenship" >
    <extension url="code" >
      <valueCodeableConcept>
        <coding>
          <system value="urn:iso:std:iso:3166" />
          <code value="DE" />
        </coding>
      </valueCodeableConcept>
    </extension>
    <extension url="period" >
      <valuePeriod>
        <start value="2009-03-14" />
      </valuePeriod>
    </extension>
    <extension url="http://acme.org/fhir/StructureDefinition/passport-number" >
        <valueString value="12345ABC" />
      </extension>
```

```
    <!-- other data for patient -->
  </extension>
</Patient>
```

or in JSON:

```
{
  "resourceType" : "Patient",
  "extension" : [{
    "url" : "http://hl7.org/fhir/StructureDefinition/patient-citizenship",
    "extension" : [{
      "url" : "code",
      "valueCodeableConcept" : {
        "coding" : [{
          "system" : "urn:iso:std:iso:3166",
          "code" : "DE"
        }]
      }
    },{
      "url" : "period",
      "valuePeriod" : {
        "start" : "2009-03-14"
      }
    },
    {
      "url": "http://acme.org/fhir/StructureDefinition/passport-number",
      "valueString": "12345ABC"
    }]
  }]
}
```

Note that this passport number extension is shown here as an example defined by a fake organization, and not appropriate for re-use by implementers.

2.5.0.2 Modifier Extensions

There are some cases where the information provided in an extension modifies the meaning of the element that contains it. Typically, this means information that qualifies or negates the primary meaning of the element that contains it. Some examples:

• A flag on a [Patient.contact] indicating they are not to be contacted - i.e. a next of kin for record-keeping purposes only

• Using the Condition resource to record an assertion that a patient has a family history of the condition rather than the condition itself

• Asserting that a performer was not actually involved in a Procedure

• Asserting an additional subsumption relationship on a concept in a value set

Such extensions are known as "modifier extensions". For further information, see the definition of what makes an element - or an extension - a modifier. If modifier extensions are present, an application is restricted in its ability to safely process the resource unless it knows what the extension means for its own use of the data.

Implementers SHOULD avoid the use of modifier extensions where possible. Any use should be carefully considered against its possible downstream consequences. Inclusion of modifier extensions in an instance would be expected to significantly limit the ability of other systems to process the instance. However, implementers are often forced into these situations by the business arrangements around the use of resources, so this specification creates a framework for handling such cases. Implementers who are introducing an extension and are uncertain whether the extension should be marked as a modifier are encouraged to raise the question on chat.fhir.org .

This specification allows for such modifier elements to be included at the base of a resource or in any elements that do not have a data type (e.g. the elements that correspond to classes in the resource UML diagrams), and on a few specially selected data types. Other data types and elements inside data types SHALL NOT have modifier extensions, and extensions SHALL NOT have modifier extensions internally (except for the reusable structures allowed to appear in extensions, listed above).

Note that complex extensions are allowed to have elements in the complex extension that are marked Is-Modifier = true, which means that these elements modify the extension value itself. Internal extensions like this marked "Is-Modifier" are still represented using the extension element, not modifierExtension because the impact of the modifier element is expected to be known by applications that understand the containing extension.

Modifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource (including cannot change the meaning of modifierExtension itself). Is-modifier elements not defined as part of an extension's definition cannot be conveyed.

In XML, these modifier elements are represented using an element named modifierExtension, which has same content as the extension element documented above:

Example: There's no element on MedicationRequest to write an "anti-prescription" - an instruction not to take a medication for a particular period. Classical clinical recording systems do not record this as a prescription - but one particular system does, and these "anti-prescription" records need to be shared within the institution

where this happens as they are an important part of the workflow. Hence, applications are allowed to extend a resource with data like this:

```
<MedicationRequest>
  <modifierExtension url="http://example.org/fhir/StructureDefinition/anti-prescription">
    <valueBoolean value="true"/>
  </modifierExtension>
  <!-- ... other content ... -->
</MedicationRequest>
```

Or in JSON:

```
{
  "resourceType" : "MedicationRequest",
  "modifierExtension" : [{
    "url" : "http://example.org/fhir/StructureDefinition/anti-prescription",
    "valueBoolean" : true
  }],
  .. other content ...
}
```

Implementations processing the data in resources SHALL check for modifiers anywhere they could appear, and if a modifier extension is present on a data element that the application 'processes', SHALL do one of these things:

• recognize the modifier extension, and understand the impact of the extension when using the data

• reject instances that contain the modifier extension

• treat such instances as "for rendering purposes only" - i.e. check that the narrative status = extensions or generated, display the narrative and don't process the discrete data

• Ask for a human to check the modifier extension before proceeding to process the data

• carry a warning concerning the data along with any action or output that results from processing the data to inform users that it has not fully understood the source information

Processing the data of a resource typically means copying or filtering data out of a resource for use in another context (display to a human, decision support, exchange in another format where not all information is included, or storing it for this kind of use). Servers and background processes that simply move whole resources around unchanged are not "processing the data of the resource", and therefore these applications are not required to check for unknown modifier extensions.

#1: When an application understands this extension, it means that some developer has provided appropriate instructions for what to do with the data contained in it because of the existence of the modifier extension or has determined that the modifier does not impact on the system's computational functions. Note that this assessment needs to be repeated each time a system's computational behavior changes

#2: This means that implementations are not inherently required to "support" a modifier extension in any meaningful way - they can achieve this understanding by rejecting instances that contain this extension (a server, for instance, could return a HTTP 422 status code with an OperationOutcome if a client PUTs or POSTs a modifier extension it does not know). Applications might also be able to ignore a modifier extension if they know that it is safe to do so in their own context, though this would not usually be the case.

Implementations SHALL ensure that they do not process data containing unrecognized modifier extensions. Note that implementations might be able to be sure, due to their implementation environment (e.g. specific trading partner agreement), that modifier extensions will never occur, and can therefore meet the requirement to check for modifiers at the design stage. However, since integration and deployment options often change, applications SHOULD always check for modifier extensions when processing resources.

#3: One way to warn the user is to download the extension definition from the given URL, and then use the defined display name to present the extension to the user. An error message could look something like this:

Note that the narrative of the resource SHALL contain modifying information, so it is safe to show this to the user as an expression of the resource's content. A warning dialog box could be extended to offer the user the choice to see the original narrative.

Here is the prescription example from above with narrative:

```
<MedicationRequest xmlns="http://hl7.org/fhir">
  <text>
    <status value="generated"/>
    <div xmlns="http://www.w3.org/1999/xhtml">
      <p><b>Note: This prescription is an instruction NOT to take a medication</b></p>
      <!-- snip actual narrative -->
    </div>
  </text>
  <!-- ...data... -->
  <modifierExtension url="http://example.org/fhir/StructureDefinition/anti-prescription">
    <valueBoolean value="true"/>
  </modifierExtension>
  <!-- ...data... -->
</MedicationRequest>
```

An application only needs to concern itself with modifier extensions on elements that it processes. Take, for example, a case where a procedure resource has a modifier extension on one of the performer elements

indicating that they did not participate in the procedure. If an application is not using the performer details at all, the fact that one of the performers has a modifier extension is irrelevant and the application is free to ignore it. If the application does process the performers, and it sees the modifier extension, it must act in one of the ways outlined above.

Implementation Guides might place limitations on the appearance of modifier extensions within instances that comply with the implementation guide.

2.5.0.2.1 Summary: Conformance Rules for Modifier Extensions

• A Modifier Extension SHALL only modify the element which contains it and/or that element's children

• It SHALL always be safe to show the narrative to humans; any modifier extension SHALL be represented in the narrative

• Applications SHALL always ensure unrecognized modifier extensions are not present when processing the data from any element that might have carry modifier extensions

• If a Modifier Extension that an application does not understand is present, the application SHALL refuse to process the resource or affected element, or SHALL provide an appropriate warning to its users

2.5.0.2.2 Special Case: Missing data

In some cases, implementers might find that they do not have appropriate data for an element with minimum cardinality = 1. In this case, the element must be present, but unless the resource or a profile on it has made the actual value of the primitive data type mandatory, it is possible to provide an extension that explains why the primitive value is not present:

```
<uri>
  <extension url="http://hl7.org/fhir/StructureDefinition/data-absent-reason">
    <valueCode="unknown"/>
  </extension>
</uri>
```

In this example, instead of a value, a data missing code is provided. Note that it is not required that this particular extension be used. This extension is not a modifier extension, because the primitive data type has no value.

It is not valid to create a fictional piece of data for the primitive value, and then add an extension indicating that the data has been constructed to meet the data rules. This would be both a bad idea as well as a modifier extension, which is not allowed on simple data types.

2.5.0.3 Exchanging Extensions

Note: This section describes the use of "non-modifier extensions", except where "modifier extensions" are explicitly mentioned (see Modifier Extensions above for details).

Extensions are a way of allowing local requirements to be reflected in a resource using a common information based approach so that all systems can confidently process them using the same tools. However, when it comes to processing the information, applications will be constrained in their ability to handle extensions by the degree to which they are informed about them.

While the structured definition of an extension should always be available (see below for details), the mere availability of a definition does not automatically mean that applications know how to handle them correctly - generally, human decisions are required to determine how the data in extensions should be handled, along with the implicit obligations that surround the information.

For this reason, local requirements that manifest as extensions are an obstacle to integration and interoperability. The more the requirements are shared (i.e. regional or national scale), the less of an obstacle the extensions (and the requirements they represent) will represent. The consistent representation, definition and registration of extensions that this specification defines cannot resolve that problem - it only provides a framework within which such local variations can be handled more easily.

When it comes to deploying applications that support local requirements, situations will very likely arise where different applications exchanging information with each other are supporting different sets of extensions. This specification defines some basic rules that are intended to make management of these situations easier, but it cannot resolve them.

• When exchanging resources, systems SHOULD retain unknown extensions when they are capable of doing so (just as they SHOULD retain core elements when they are capable of doing so)

• If a system modifies a resource it SHOULD remove any extensions that it does not understand from the modified element and its descendants, because it cannot know whether the modifications it has made might invalidate the value of the unknown extension

• Systems that drop existing elements are considered to be "processing the containing element"

• A system SHALL NOT modify a resource or element that contains "modifier" extensions it doesn't understand

• Applications SHOULD ignore extensions that they do not recognize if they are not "modifier" extensions

The degree to which a system can retain unknown extensions is a function of the type of system it is: a general purpose FHIR server, or a middleware engine would be expected to retain all extensions, while an application that manages patient registration through a user interface can only retain extensions to the degree that the information in them is part of the set managed by the user. Other applications will fall somewhere between these two extremes.

2.5.0.3.1 Summary: Handling extensions

Use the following rules as a guideline for handling resources:

• When writing extensions, make sure they are defined and published

• When reading, navigating through or searching on elements that can have modifier extensions, check whether there are any modifier extensions present

• When reading elements, read and process the extensions you know and use, and ignore other extensions except for modifier extensions

• Retain extensions whenever you can

2.5.1 Defining Extensions

URL: https://hl7.org/fhir/R4/defining-extensions.html

2.5.1 Defining Extensions

FHIR Infrastructure Work Group | Maturity Level: 3 | Standards Status: Normative

| This page has been approved as part of an ANSI standard. See the Infrastructure Package for further details.

All extensions used in resources require a formal published definition that can be used by application developers or the applications themselves, to help integrate extensions into the healthcare process they support.

Every extension in a resource refers directly to its definition, which is made available as a StructureDefinition. A resource can be profiled to specify where particular extensions are required or expected.

Whenever resources containing extensions are exchanged, the definitions of the extensions SHALL be available to all the parties that share the resources. Each extension contains a URI that references the source of the definitions as a StructureDefinition. The source SHOULD be a literal reference, such as an http: URL that refers to an end-point that responds with the contents of the definitions - preferably a FHIR RESTful server supporting the StructureDefinition, or a logical reference (e.g. using a urn:) - for instance, to a national published standard. Extensions may be defined by any project or jurisdiction, up to and including international standards organizations such as HL7 itself.

Before defining a new extension, attempt to reuse existing extensions defined in one of the shared registries described below. Also consider that some concepts may be appropriate to add as part of the core specification.

See also Best Practice Guidance for defining extensions .

2.5.1.1 Core Elements

Elements are included as part of FHIR resources and data types principally on the basis of current world-wide usage patterns. Policy is that if a significant majority of systems throughout the world that would use a resource or data type would use an element, then that element will be included as part of the resource/data type. If not, it will be left to an extension. This holds even if the element is very common or even mandatory in one or two specific jurisdictions.

Proposals suggesting a new core element can be raised by anyone. (Free registration is required.) However, given the timelines for new FHIR releases as well as the uncertainties associated with vetting the specification through a ballot process, it may still be necessary to define extensions even for elements that are likely to be supported as part of the core specification in a future release.

2.5.1.2 Context

Extensions are always defined against some particular context - which elements they can extend. There are three different ways to specify which elements an extension can be found on:

Code | Context type | Context format | Examples
fhirpath | A FHIRPath expression that selects the set of elements on which the extension can appear | The FHIRPath statement always starts from the root of the resource that might contain the element | Condition (Condition | Observation).code
element | Formal Element Id for the element | Element Ids are unique within the base specification, and within a structure definition. The full path for the element is [url]#[elementid]. If there is no #, the Element id is one defined in the base specification | Address.part.value http://hl7.org/fhir/StructureDefinition/resprate#Observation.category:vscat.coding
extension | Another extension | The canonical URL of the extension, optionally followed by #code for extension that appear within a complex extension | http://hl7.org/fhir/StructureDefinition/device-din

Extensions SHALL only be used on a target that appears in their context list.

2.5.1.3 Cardinality

The cardinality constraints asserted by the extension definition itself apply to any contexts where the extension is used.

Minimum Cardinality

If the Extension minimum cardinality is 0, then the extension is, by default, optional anywhere it appears. A profile that defines the use of an extension may make the minimum cardinality any number up to the maximum cardinality of the extension itself. Example: Patient birthplace.

If the Extension minimum cardinality is > 1, then the extension must have a minimum cardinality of at least the minimum cardinality in any profile that defines the use of the extension. The minimum cardinality may be any number up to the maximum cardinality of the extension. Even with a minimum cardinality > 0, the extension is only required to be present in instances if the instances explicitly or implicitly conform to a profile that defines the use of the extension. Example: CapabilityStatement Expectation.

Maximum Cardinality

If the Extension maximum cardinality is 1, then the extension is only allowed once on any element on which it appears. A profile that defines the use of an extension can only make the maximum cardinality 1 (or zero if the minimum cardinality is 0, and the profile constrains another profile that allows the extension). Example: Mother's Maiden Name.

If the Extension maximum cardinality is >1, then the extension is allowed up to the specified number of times on any element on which it appears. A profile that defines the use of an extension may make the maximum cardinality any value up to the specified maximum. Example: Patient Disability.

Context Invariants

Extensions are also able to define context invariants, which is a rule that is executed on the element that contains the extension when it is present to check that local requirements around cardinality and other business rules are met.

For example, if you use the extension for house number, the Address.line needs to be filled as well:

```
<contextInvariant value="line.exists()"/>
```

Note that since the context invariant is only evaluated when the extension is present, there is an implicit "line.extension('http://hl7.org/fhir/StructureDefinition/iso21090-ADXP-streetName').exists() implies" at the start of the expression. A context invariant that enforces that the extension is present under certain circumstances cannot be defined in the extension itself - it must be defined in a profile that makes use of the extension.

2.5.1.4 Use of ElementDefinition in Extension Definitions

An extension is a wrapper for an identifying url and either a value or other extensions. As such, some of the properties of the extension are defined on the extension itself, while others are defined on the Extension.value. This list provides guidance for the correct usage:

• Extension root element: Cardinality Short, Definition, Comments IsModifier MustSupport (is used on invocation of the extension) Conditions & Constraints. These SHOULD never be on url/value[x] Mappings. these SHALL never be on url/value[x]

• Extension.url: Cardinality = 1...1 (fixed) value = canonical URL (fixed)

• Extension.value[x]: Type Cardinality for Simple extensions (not nested): 1..1. Use 0..0 if nested. Note that the actual extension cardinality is defined by the root element Binding MaxLength, DefaultValue, Pattern, Example, MinValue, MaxValue

• Extension.extension: Cardinality for Complex extensions (nested): 0..*, though at least one nested extension must be present. Use 0..0 if a simple (non-nested extension) .extension is automatically sliced by url nested extensions can be defined in-line (as children elements of Extension.extension), or by reference (using .type.profile)

Note: Extensions are always sliced by their url property. Re-slicing extensions by additional properties is allowed (see Profiling/Slicing).

2.5.1.5 Publishing Extension Definitions

As well as defining the base element structure for resources, HL7 also publishes extensions, including as part of this specification. HL7 publishes such data definitions as extensions rather than as part of the base resource structure to keep the base resource structure simple and concise, and to allow implementers freedom from needing to engage with an entire world's worth of functionality up front.

Before extensions can be used in instances, their definition SHALL be published. HL7 maintains two extension registries:

• HL7 approved extensions, approved by an appropriate part of the HL7 community following a review process, and that have formal standing

• Provided as a service to the community, where anyone can register an extension

Users are encouraged to register their extensions in the second registry, though this is not required. All that is required is that the extension is published in a context that is available for users of the extension. So, for

example, if a particular extension is only used within a single institution, the definition of the extension can be placed on the institution's intranet. However since, by their nature, resources tend to travel well, it's always better to publish it on a public facing web site. Using the FHIR registry or other publicly accessible extension registries not only publishes the extension, but also makes it easy to find and encourages reuse and therefore consistency across implementations.

The HL7 FHIR registry can be found at http://hl7.org/fhir/registry .

HL7 extension definitions may be balloted alongside resource content as part of the FHIR specification or may be published as part of separate specifications. When HL7 publishes extension definitions as part of the FHIR specification, these extensions SHALL be used for this data whenever the data is represented in instances. Applications SHOULD use other HL7-defined extensions published to represent equivalent data in the interest of maximum interoperability.

To minimize complexity for implementers, HL7 will not elevate widely adopted extensions (defined by HL7 or other organizations) to be content defined in a core resource in future versions of the resource unless there is widespread endorsement of such a migration from the implementer community. This policy ensures that widespread adoption of an extension does not result in a forced migration to a core element. Extensions labeled as draft may be moved in either direction, but after extensions are finalized as normative they won't be moved.

In some cases, an HL7 work group or other body may publish a profile whose sole purpose is to define extensions expected to be needed by implementers in a particular context; e.g. extensions needed to map a particular set of HL7 v2 segments or a HL7 v3 model.

Implementations are encouraged to share their extensions with HL7 and register them with the HL7 extension registry. The domain committees will work to elevate the extensions into HL7 published extensions or, if adopted by a broad enough portion of the implementer community, into the base resource structure itself.

To avoid interoperability issues, extensions SHALL NOT change their definition once published. (Small clarifications to descriptions that do not affect interoperability are permitted.) Rather than modifying an existing extension, a new extension should be introduced. Revisions to an extension may extend the set of contexts in which the extension apply but SHALL NOT remove or constrain any context previously listed

2.5.2 Examples

URL: https://hl7.org/fhir/R4/extensibility-examples.html

2.5.2 Extensibility Examples

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative

In order to use an extension, there is a three-step process:

• Define the extension

• Register the extension

• Use it in the instance

This page contains examples of how this process executes.

2.5.2.1 Patient Consent for Record Sharing

The basic patient resource contains no information relating to patient consent, and/or the policy under which the patient consents to their registration details. A social web provider of personal healthcare record (PHR) services might be obliged to keep track of the particular policy under which a patient has created their relationship with the PHR provider, and share this with their participants via their FHIR API. If they wish, they can extend the patient resource to represent the patient's participation agreement. Note that other approaches to this problem are possible and preferred, but this example suffices to demonstrate the extension process.

For the purposes of this example, we assume that the patient agrees to a participation policy as part of their sign up, and that as the provider has to change their policy, they ask patients to agree to new participation details. Each participation agreement has a URI by which it is identified, and the patient resource will carry this URI for each policy agreement that the patient has agreed to.

2.5.2.1.1 Define the Extension

For each extension, the first thing to do is to fill out the definitional properties of the extension:

Code | "participation-agreement"
Context | This extension is used in the patient resource
Short Defn | Agreed agreement/policy
Definition | A URI that identifies a participation agreement/policy to which the patient has agreed
Comment | URI is a literal reference to agreement text (html)
Cardinality | 1..* (patient cannot participate without at least one agreement)
Type | uri
Invariants | No invariants

is Modifier | No (the participation agreements do not affect that interpretation of the elements of the patient, though they will likely influence how the system interacts with the patient)
Binding | (no binding - not a coded value)

2.5.2.1.2 Write the Definition of the Extension

From this table, we can build a formal extension definition. In this case, it looks like this:

```
<StructureDefinition xmlns="http://hl7.org/fhir">
  <url value="http://example.org/fhir/StructureDefinition/participation-agreement"/>
  <name value="Example Extension Definition"/>
  <!-- snip other metadata -->
  <kind value="complex-type"/>
  <context>
    <type value="element"/>
    <expression value="Patient"/>
  </context>
  <type value="Extension"/>
  <baseDefinition value="http://hl7.org/fhir/StructureDefinition/Extension"/>
  <derivation value="constraint"/>

  <differential>
    <element>
      <path value="Extension"/>
      <short value="Agreed agreement/policy"/>
      <definition value="A URI that identifies a participation agreement/policy
        to which the patient has agreed"/>
      <comment value="URI is a literal reference to agreement text (html).
        Systems SHALL conform to the policies as indicated.
        For further information, see the partnership agreement..."/>
      <mustSupport value="true"/>
      <isModifier value="false"/>
    </element>
    <element>
      <path value="Extension.url"/>
      <fixedUri value="http://example.org/fhir/StructureDefinition/participation-agreement"/>
    </element>
    <element>
      <path value="Extension.valueUri"/>
      <short value="The URI value"/>
      <min value="1"/>
      <max value="*"/>
      <type>
         <code value="uri"/>
      </type>
      <mustSupport value="true"/>
    </element>
  </differential>
</StructureDefinition>
```

Note that usually one would build the actual profile using a tool. This example was built by hand for this example.

2.5.2.1.3 Register the Extension

This means the profile shown above that defines the extension is placed on the web somewhere. By preference, it will be hosted at a FHIR Profile endpoint, and the best location of all is the HL7 FHIR Registry .

For this example, we assume that it has been uploaded to the PHR provider's own website at http://example.org/phr/documents/fhir/extensions.

2.5.2.1.4 Use it in the instance

To use the extension in an instance, the extension is placed in the root of the resource, because that was the declared context for the extension. Note that the URL of the extension refers to the registered location.

```
<Patient xmlns="http://hl7.org/fhir">
  <extension url="http://example.org/phr/documents/fhir/extensions/participation-agreement" >
    <valueUri value="http://example.org/phr/documents/patient/general/v1"/>
  </extension>
  <!-- ... -->
</Patient>
```

2.5.2.1.5 Using it in a Patient Profile

The extension definition above simply defines the extension "participation-agreement" and says that it is used with patient. But the profile above doesn't say that the server actually uses it. For the PHR provider to indicate that all Patient resources will use this resource, a StructureDefinition on the patient resource is used:

```
<StructureDefinition xmlns="http://hl7.org/fhir">
  <id value="patient-profile"/>
  <!-- snip other metadata, narrative -->
  <differential>

    <!-- first, the patient root element
      - can be copy/paste from the base patient profile -->
    <element>
      <path value="Patient"/>
      <!-- snip definition -->
```

```
    </element>

    <!-- now, the general definition for extensions
      - can be copied/pasted from the base patient profile,
      with changes for slicing  -->
    <element>
      <path value="Patient.extension"/>
      <!-- we're going to slice the extension element, and
        one of the extensions is one we have defined -->
      <slicing>
        <!-- extension is always sliced on url -->
        <discriminator>
          <type value="value"/>
          <path value="url"/>
        </discriminator>
        <!-- we don't care what the order of any extensions is -->
        <ordered value="false"/>
        <!-- Other extensions are allowed in addition to this one -->
        <rules value="open"/>
      </slicing>
      <!-- snip definition -->
    </element>

    <!-- now, the slice that contains our extension -->
    <element>
      <path value="Patient.extension"/>
      <sliceName value="agreement"/>
      <!-- clone information from the extension definition.
        duplicative, but this duplication makes it simpler overall -->
      <short value="Agreed agreement/policy"/>
      <definition value="A URI that identifies a participation agreement/policy
      to which the patient has agreed"/>
      <!--  min has to be 1, since the extension itself has min = 1 -->
      <min value="1"/>
      <max value="*"/>
      <type>
        <!-- obviously it has to be an extension -->
        <code value="Extension"/>
        <!-- and here is the link to the extension definition:
          this extension has to conform to the rules laid down in its definition -->
        <profile value="http://example.org/phr/documents/fhir/StructureDefinition/participation-agreement"/>
      </type>
      <isModifier value="false"/>
    </element>

  <!-- snip the rest of the profile -->

  </differential>
</StructureDefinition>
```

Note - this step is optional.

2.5.2.2 Patient Name Parts

ISO 21090 (Healthcare Data Types) defines a concept called a "name part qualifier" that contains extra information about how a particular name part should be used or interpreted. In practice, this field is used rarely except in particular cultural contexts, where certain part qualifiers are used as a matter of practice. Following the FHIR design policy, such a field is not included in the overall definition of the core name data type, instead is it added as an extension.

In practice, for cases such as these in ISO 21090, HL7 defines common extensions, and these are defined either in this specification, or in the FHIR registry .

2.5.2.2.1 Define the Extension

For each extension, the first thing to do is to fill out the definitional properties of the extension:

Code	"name-qualifier"
Context	This extension can be used anywhere a HumanName.part appears
Short Defn	(one of the codes) LS | AC | NB | PR | HON | BR | AD | SP | MID | CL | IN
Definition	A set of codes each of which specifies a certain subcategory of the name part in addition to the main name part type
Comment	Used to indicate additional information about the name part and how it should be used
Cardinality	0..* (this is always optional, but more than one can be used if required)
Type	code
Invariants	N/A
Is Modifier	No (Qualifiers do not change the fact that the part is a given or family name)
RIM Mapping	ENXP.qualifier
v2 Mapping	N/A
Binding	Bound to a subset of the codes specified for EntityNamePartQualifierR2 in ISO 21090

Not all the codes of the EntityNamePartQualifierR2 are required in this context, because prefix and suffix are explicitly part of the name types. Rather than simply refer to the OID for EntityNamePartQualifierR2 (2.16.840.1.113883.5.1122), in this case we enumerate the available codes, and set the type of the extension to code. The type of "code" is only allowed if the profile itself defines the codes that can be used. Here is a table of the codes (see the EntityNamePartQualifierR2 reference for the full definitions):

LS | Legal status | For organizations, a suffix...
AC | Academic | Indicates that a prefix like "D...
NB | Nobility | In Europe and Asia, there are s...
PR | Professional | Primarily in the British Im...
HON | Honorific | An honorific such as 'The Rig...
BR | Birth | A name that a person was given at ...
AD | Acquired | A name part a person acquired. ...
SP | Spouse | The name assumed from the partner...
MID | Middle Name | Indicates that the name par...
CL | Call me | Call me is used to indicate which...
IN | Initial | Indicates that a name part is ju...

This is all then represented formally in a profile. Such profiles do not need to include resource constraint statements; instead, they include just extension declarations and their associated bindings. In this case, the definition looks like this:

```
<StructureDefinition xmlns="http://hl7.org/fhir">
  <url value="http://hl7.org/fhir/StructureDefinition/iso21090-EN-qualifier"/>
  <name value="Iso21090NameQualifier"/>
  <title value="iso-21090 Name Qualifier"/>
  <!-- snip other metadata, including definition of RIM Mapping -->
  <kind value="complex-type"/>
  <context>
    <type value="element"/>
    <expression value="HumanName.given"/>
  </context>
  <context>
    <type value="element"/>
    <expression value="HumanName.prefix"/>
  </context>
  <context>
    <type value="element"/>
    <expression value="HumanName.family"/>
  </context>
  <context>
    <type value="element"/>
    <expression value="HumanName.suffix"/>
  </context>
  <type value="Extension"/>
  <baseDefinition value="http://hl7.org/fhir/StructureDefinition/Extension"/>
  <derivation value="constraint"/>
  <differential>
    <element>
      <path value="Extension"/>
      <short value="LS | AC | NB | PR | HON | BR | AD | SP | MID | CL | IN"/>
      <definition value="A set of codes each of which specifies a certain subcategory
          of the name part in addition to the main name part type"/>
      <comment value="Used to indicate additional information about the
            name part and how it should be used"/>
      <mustSupport value="false"/>
      <isModifier value="false"/>
    </element>
    <element>
      <path value="Extension.url"/>
      <fixedUri value="http://hl7.org/fhir/StructureDefinition/iso21090-EN-qualifier"/>
    </element>
    <element>
      <path value="Extension.value"/>
      <short value="LS | AC | NB | PR | HON | BR | AD | SP | MID | CL | IN"/>
      <min value="0"/>
      <max value="*"/>
      <type>
         <code value="code"/>
      </type>
      <binding>
        <strength value="required"/>
        <description value="A set of codes each of which specifies a certain subcategory
            of the name part in addition to the main name part type"/>
        <valueSet value="http://hl7.org/fhir/ValueSet/name-part-qualifier"/>
      </binding>
      <mapping>
         <identity value="RIM"/>
         <map value="ENXP.qualifier"/>
      </mapping>
    </element>
  </differential>
</StructureDefinition>
```

Note that usually one would build the actual profile using a tool. This example was built from a spreadsheet definition by the FHIR build tooling.

2.5.2.2.2 Register the Extension

For this example, it is registered at http://hl7.org/fhir/StructureDefinition/iso21090-EN-qualifier . This is the URL that will appear in the definition element when the extension is used.

2.5.2.2.3 Use it in the instance

To use the extension in an instance, the extension is nested within the attribute that is extended. Note that the URL of the extension refers to the registered location.

```
<name>
  <use value="official"/>
  <given value="Östlund">
     <extension url="http://hl7.org/fhir/StructureDefinition/iso21090-EN-qualifier" >
        <valueCode value="MID"/>
     </extension>
  </given>
</name>
```

This example is a Scandinavian mellannamn. See Datatypes examples for additional examples.

2.5.2.3 Complex Extension: Patient Clinical Trial

Defining complex extensions is a little different. They have the same metadata - context, etc. - but differ in the internal structure. As an example, consider enrolling a patient in a clinical trial. There are 3 data items to collect:

Code | Name | Cardinality | Type | Content
NCT | Clinical Trial number | 1..1 | string | The format for the US ClinicalTrials.gov registry number is "NCT" followed by an 8-digit number, e.g.: NCT00000419
period | trialPeriod | 0..1 | Period | The start and end times of the participation of this patient in the clinical trial
reason | reason enrolled | 0..1 | CodableConcept | Indication or reason that the patient is part of this trial

NOTE: This extension is included for example purposes only. In most cases, participation in clinical trials would be handled using the ResearchStudy resource, possibly referenced via the workflow-researchstudy extension.

Like simple extensions, the first thing to do is to assign a URI to the extension, e.g.: http://example.org/fhir/StructureDefinition/patient-clinicalTrial.

Internally, in the extension, the maximum cardinality of the value[x] element is set to 0 since it will not (and cannot) be used. The Extension.extension element is sliced by URL, and 3 slices are defined, each with a fixed relative URI which is the code from the table above. The relevant parts of the definition of this extension are as follows:

```
<StructureDefinition xmlns="http://hl7.org/fhir">
  <!-- metadata - setting up the base definition -->
  <url value="http://example.org/fhir/StructureDefinition/patient-clinicalTrial"/>
  <name value="ClinicalTrialParticipation"/>
  <title value="The patient's participation in clinical trials"/>
  <context>
    <type value="element"/>
    <expression value="Patient"/>
  </context>
  <type value="Extension"/>
  <baseDefinition value="http://hl7.org/fhir/StructureDefinition/Extension"/>
  <derivation value="constraint"/>
  <snapshot>
    <element>
      <path value="Extension"/>
      <!-- etc.-->
    </element>

    <!-- set up the slicing -->
    <element>
      <path value="Extension.extension"/>
      <slicing>
        <discriminator>
          <type value="value"/>
          <path value="url"/>
        </discriminator>
        <ordered value="true"/>
        <rules value="openAtEnd"/>
      </slicing>
    </element>

    <!-- first slice, NCT -->
    <element>
      <path value="Extension.extension"/>
      <sliceName value="NCT"/>
      <short value="National Clinical Trial number"/>
      <min value="1"/>
      <max value="1"/>
    </element>
    <element>
      <path value="Extension.extension.extension"/>
      <min value="0"/>
      <max value="0"/> <!-- not allowed to be used -->
    </element>
    <element>
      <path value="Extension.extension.url"/>
      <min value="1"/>
      <max value="1"/>
```

```
      <fixedUri value="NCT"/>
    </element>
    <element>
      <path value="Extension.extension.valueString"/>
      <min value="1"/>
      <max value="1"/>
      <type>
        <code value="string"/>
      </type>
    </element>

    <!-- second slice, period -->
    <element>
      <path value="Extension.extension"/>
      <sliceName value="period"/>
      <short value="The period of participation in the clinical trial"/>
      <min value="0"/>
      <max value="1"/>
    </element>
    </element>
    <element>
      <path value="Extension.extension.extension"/>
      <min value="0"/>
      <max value="0"/>
    </element>
    <element>
      <path value="Extension.extension.url"/>
      <min value="1"/>
      <fixedUri value="period"/>
    </element>
    <element>
      <path value="Extension.extension.valuePeriod"/>
      <type>
        <code value="Period"/>
      </type>
    </element>

    <!-- third slice, reason -->
    <element>
      <path value="Extension.extension"/>
      <sliceName value="reason"/>
      <short value="The reason for participation in the clinical trial"/>
      <min value="0"/>
      <max value="1"/>
    </element>
    <element>
      <path value="Extension.extension.extension"/>
      <min value="0"/>
      <max value="0"/>
    </element>
    <element>
      <path value="Extension.extension.url"/>
      <min value="1"/>
      <fixedUri value="reason"/>
    </element>
    <element>
      <path value="Extension.extension.valueCodeableConcept"/>
      <type>
        <code value="CodeableConcept"/>
      </type>
    </element>

    <!-- last (for order reasons): the fixed URI -->
    <element>
      <path value="Extension.url"/>
      <fixedUri value="http://example.org/fhir/StructureDefinition/patient-clinicalTrial"/>
    </element>
    <!-- and no value in the root -->
    <element>
      <path value="Extension.value[x]"/>
      <min value="0"/>
      <max value="0"/>
    </element>
  </snapshot>
</StructureDefinition>
```

2.5.2.3.1 Use it in the instance

With complex extensions, only the first URL is an absolute URL:

```
<Patient xmlns="http://hl7.org/fhir">
  <extension url="http://example.org/fhir/StructureDefinition/patient-clinicalTrial" >
    <extension url="NCT" >
      <valueString value="NCT00000419"/>
    </extension>
    <extension url="period" >
      <valuePeriod>
        <start value="200140105"/>
        <end value="20120105"/>
```

```
      </valuePeriod>
    </extension>
    <extension url="reason" >
      <valueCodeableConcept>
        <text value="healthy-volunteer"/>
      </valueCodeableConcept>
    </extension>
  </extension>
  <!-- ... -->
</Patient>
```

2.5.2.3.2 Using it in a Patient Profile

This is the same as for a simple extension:

```
<StructureDefinition xmlns="http://hl7.org/fhir">
  <id value="patient-profile"/>
  <!-- snip other metadata, narrative -->
  <differential>

    <!-- first, the patient root element
      - can be copy/paste from the base patient profile -->
    <element>
      <path value="Patient"/>
      <!-- snip definition -->
    </element>

    <!-- now, the general definition for extensions
      - can be copy/paste from the base patient profile,
      with changes for slicing  -->
    <element>
      <path value="Patient.extension"/>
      <!-- we're going to slice the extension element, and
        one of the extensions is one we have defined -->
      <slicing>
        <!-- extension is always sliced on url -->
        <discriminator>
          <type value="value"/>
          <path value="url"/>
        </discriminator>
        <!-- we don't care what the order of any extensions is -->
        <ordered value="false"/>
        <!-- Other extensions are allowed in addition to this one -->
        <rules value="open"/>
      </slicing>
      <!-- snip definition -->
    </element>

    <!-- now, the slice that contains our extension -->
    <element>
      <max value="1"/>
      <type>
        <!-- obviously it has to be an extension -->
        <code value="Extension"/>
        <!-- and here is the link to the extension definition:
          this extension has to conform to the rules laid down in its definition -->
        <profile value="http://example.org/fhir/StructureDefinition/patient-clinicalTrial"/>
      </type>
    </element>

  <!-- snip the rest of the profile -->

  </differential>
</StructureDefinition>
```

2.5.3 Extension - Detailed Descriptions

URL: https://hl7.org/fhir/R4/extensibility-definitions.html

2.5.3 Extension - Detailed Descriptions

FHIR Infrastructure Work Group | Maturity Level: Normative | Standards Status: Normative

| This page has been approved as part of an ANSI standard. See the Infrastructure Package for further details.

Extension
Standards Status | This element has a standards status of "Normative"
Definition | Optional Extension Element - found in all resources.
Cardinality | 0..*
Type | Element
Requirements | The ability to add extensions in a structured way is what keeps FHIR resources simple.
Invariants | Defined on this element ext-1 Rule Must have either extensions or value[x], not both extension.exists() != value.exists()
Defined on this element
ext-1 | Rule | Must have either extensions or value[x], not both | extension.exists() != value.exists()
Extension.url
Definition | Source of the definition for the extension code - a logical name or a URL.

Cardinality | 1..1
Type | uri
Comments | The definition may point directly to a computable or human-readable definition of the extensibility codes, or it may be a logical URI as declared in some other specification. The definition SHALL be a URI for the Structure Definition defining the extension.
Extension.value[x]
Definition | Value of extension - must be one of a constrained set of the data types (see Extensibility for a list).
Cardinality | 0..1
Type | *
[x] Note | See Choice of Data Types for further information about how to use [x]

2.6 Resource Formats

URL: https://hl7.org/fhir/R4/formats.html
FHIR Infrastructure Work Group | Maturity Level: Normative | Standards Status: Normative

| This page has been approved as part of an ANSI standard. See the Infrastructure Package for further details.

This page documents how the content of the resources are described. In actual exchange, resources can be represented in the following formats: XML, JSON and Turtle. Additional Bulk Data Formats are also undergoing exploration. Other representations are allowed, but are not described by this specification (though see link below .

2.6.0.1 Resource Definition

The resources are described in several different ways:

• a hierarchical table that presents a logical view of the content

• a UML diagram that summarizes the content graphically

• a pseudo-XML syntax that provides a visual sense of what the end resource instances will look like in XML

• a pseudo-JSON syntax that provides a visual sense of what the end resource instances will look like in JSON

• a pseudo-Turtle syntax that provides a visual sense of what the end resource instances will look like in Turtle

In addition to this descriptive syntax, other definitional forms are available, including W3C schema, Schematron, JSON Schema, and the StructureDefinition syntax defined internally.

2.6.0.1.1 Logical table

The Logical View shows the resources as a tree structure with the following columns:

Column | Content
Name | The name of the element in the resource (manifests as XML element name or JSON or RDF property name). Some names finish with [x] - the meaning of this is discussed below. In addition, this column contains an icon that denotes the underlying type of the content. The icons are described below
Flags | A set of information about the element that impacts how implementers handle them. The flags are described below
Card. | Cardinality: the lower and upper bounds on how many times this element is allowed to appear in the resource
Type | The type of the element (hyperlinked to the definition of the type). Note that the type of the element has one of two meanings, depending on whether the element has defined children. If the element has children, then the element has an anonymous type that specializes the given type. If the element has no children, then the element has properties and children as specified by the nominated type
Description & Constraints | A description of the element, and details about constraints that are applied to it. Particularly, for coded elements, information about which codes can be used. The description comes from ElementDefinition.short

Here's an example:

Name | Flags | Card. | Type | Description & Constraints
Resource Name | | | Base Type | Definition
nameA | Σ | 1..1 | TypeA | description of content
nameB[x] | ?! Σ | 0..1 | | description SHALL at least have a value
nameBType1 | | 0..1 | TypeB
nameBType2 | I | 0..1 | typeC
nameC | | 1..* | BackboneElement | Definition
nameD | | 1..1 | TypeD | Relevant Records

Key to Type Icons

• : The base element for a resource (see Resources)

• : An element that is part of the resource and has elements within it defined in the same resource or profile

• : An element which can have one of several different types (see below)

• : An element of a data type which describes an element that has a value attribute/property. These are also known as primitive types. All primitive type names start with a lower case letter

• : An element of a data type which describes an element that has other elements. These are known as complex types. All complex type names defined in this specification start with an uppwer case letter

• : An element that contains a reference to another resource (see references)

• : This element has the same content as another element defined within this resource or profile

• : Introduction of a set of slices (see Slicing)

• : A complex extension - one with nested extensions (see Extensibility)

• : An extension that has a value and no nested extensions (see Extensibility)

• : A complex modifier extension - one with nested extensions (see Extensibility)

• : A modifier extension that has a value and no nested extensions (see Extensibility)

• : The root of a logical profile

Key to Flags

• ?!: This element is a modifying element - see Modifier Elements

• S: This element is an element that must be supported - see MustSupport Elements

• Σ: This element is an element that is part of the summary set - see Summary Searches

• I: This element defines or is affected by constraints - see Constraints

• NE: This element cannot have extensions (some infrastructural elements only)

• TU: This element has a standards status of Trial Use (for discussion about mixing standards status in a resource, see Mixed Normative content)

• N: This element has a standards status of Normative

• D: This element has a standards status of Draft

Notes:

• Resource and Element names are case-sensitive (though duplicates that differ only in case are never defined)

• Any elements that have a primitive type will have a value attribute/property to contain the actual value of the element

• This value attribute/property can never be empty. Either it is absent, or it is present with at least one character of non-whitespace content

• Elements are assigned a cardinality that specifies how many times the element may or must appear.

• Unless elements have children defined directly (as nameC does above) they are assigned one or more types (see the data types). All the type names are hyperlinked to the source definition

• Element reuse: Some data types that have children have the same set of children as some other element defined in the resource. In that case, the type of that element has a "see [name]" where [name] is the name of the element that has the defined children

• Each element name is also a hyperlink to the formal definition of the element in the data dictionary that underlies the exchange formats.

• Any of the elements may have an id attribute to serve as the target of an internal reference. The id attribute is not shown in this format. Extensions are not always shown, but may appear except where the flag NE appears

• FHIR elements can never be empty. If an element is present in the resource, it SHALL have either a value, child elements as defined for its type, or 1 or more extensions

• Infrastructural elements that are common to all resources are not shown in the logical representation. These are described in the common base classes Resource, and DomainResource

The data type for a particular element is typically expressed as the name of the specified type with a hyperlink to its definition. However, there are two exceptions:

• If the element supports multiple types (name ends with [x]), then the type will be a list of data type options, each separated by "|"

• If one of the types is Reference or canonical, the data type might be followed by a list of allowed targets the reference is allowed to be. These might be resource names, data type names, or profile URLs, depending on the context. As well, the following symbols may appear that represent expectations of where the referenced resource is located: b: Resource must appear within the same Bundle; c: Resource must be sent as a contained resource; r: Resource is a non-contained reference - i.e. to a resource within the same Bundle or to an external resource

In profiles, references to types may be profiled - i.e. Instances of the element must comply with a specified profile or one of a list of profiles. The canonical URLs of any applicable profiles are listed inside {}.

Where an element can have a choice of data types, or is a Reference these are represented by showing the common type (Reference or Type), and then showing the applicable data type names or resource types in a stereotype, separated by the | character. Type is not formally otherwise defined by this specification, but is a super type of all the data types.

2.6.0.1.2 Choice of Data Types

A few elements have a choice of more than one data type for their content. All such elements have a name that takes the form nnn[x]. The "nnn" part of the name is constant, and the "[x]" is replaced with the title-cased name of the type that is actually used. The table view shows each of these names explicitly.

Elements that have a choice of data type cannot repeat - they must have a maximum cardinality of 1. When constructing an instance of an element with a choice of types, the authoring system must create a single element with a data type chosen from among the list of permitted data types.

Note: In object-orientated implementations, this is naturally represented as a polymorphic property. However this is not necessary and the correct implementation varies according to the particular features of the language. In XML schema, these become an xs:choice of element. To help with code generation, a list of choice elements is published.

2.6.0.1.3 UML

The UML diagrams represent the same content as a series of classes that represent the elements of a resource.

The elements and the data types are hyperlinks to the formal definitions of the parts. The UML diagrams also show the vocabulary bindings. These are hyperlinks to the value set details.

The actual order of the elements in XML cannot be determined from the diagram, nor whether a UML property becomes an element or an attribute in the XML representation.

Bindings to value sets are indicated by a stereotype on the element. The stereotype has 2 parts: the value set name, and a symbol that denotes the strength of the binding:

• ??: Example Binding

• ?: Preferred Binding

• +: Extensible Binding

• !: Required Binding

2.6.0.1.4 Serialization Format Representations

This specification defines the following ways to represent resources when they are exchanged:

• JSON

• XML

• RDF (Turtle)

Systems SHALL declare which format(s) they support in their Capability Statement. If a server receives a request for its Capability Statement in a format it does not otherwise support, it SHALL return a 406 Not Acceptable. Note: 406 is the appropriate response when the Accept header requests a format that the server does not support, and 415 Unsupported Media Type when the client posts a format that is not supported to the server.

Clients and servers can choose what syntax(s) to implement. In the interests of interoperability, servers SHOULD support both the XML and JSON formats, which have the same functionality, for different technical stacks. The RDF format has quite different benefits - primarily around data analysis rather than exchange.

2.6.0.1.4.1 Bulk Data Formats

Unlike this rest of this page, the bulk use formats are draft until further experience is gained with their use. Their status will be reviewed in a future version of FHIR.

The XML and JSON formats are designed to support typical system process-based data exchange uses. FHIR is also used to exchange large amounts of data- 1000s of records, or more (up to billions). The formats above can be used for this, but more suitable formats exist. This specification documents (or is exploring documenting) the following formats:

• ND-Json (New line delimited JSON)

• Google Protobuf (under consideration)

• Apache Parquet/Avro (bulk data formats under consideration)

2.6.1 XML Representation of Resources

URL: https://hl7.org/fhir/R4/xml.html

2.6.1 XML Representation of Resources

Implementable Technology Specifications Work Group | Maturity Level: Normative | Standards Status: Normative

| This page has been approved as part of an ANSI standard. See the Infrastructure Package for further details.

The XML representation for a resource is described using this format:

```
<name xmlns="http://hl7.org/fhir" (attrA="value")>
  <!-- from Resource: id, meta, implicitRules, and language -->
  <nameA><!--  1..1 type description of content  --><nameA>
  <nameB[x]><!-- 0..1 type1|type1 description  --></nameB[x]>
```

```
   <nameC> <!--  1..* -->
     <nameD><!-- 1..1 type>Relevant elements  --></nameD>
   </nameC>
 <name>
```

Using this format:

• To build a valid XML instance of a resource, simply replace the contents of the elements and attributes with valid content as described by the cardinality, type rules and content description found in the comment in each element

• Resource and Element names are case-sensitive (though duplicates that differ only in case are never defined)

• Elements must always appear in the order documented

• When an element is allowed to repeat, the elements are ordered, and the technical infrastructure needs to be able to access the items in the right order (see also Cardinality Rules for a further description of elements with cardinality > 1)

• A few properties are represented as attributes: values of primitive types in a value attribute, extension URLs in the url attribute on an extension, and the id property on elements (but not on resources, where the resource id is an element)

• Any of the XML elements may have an id attribute to serve as the target of an internal reference. The id attribute is not shown in this format

• FHIR elements are always in the namespace http://hl7.org/fhir . This is usually specified as the default namespace on the root element. The only other namespace that occurs in FHIR resources is the XHTML namespace - XHTML is found in most resources

• Infrastructural elements must appear prior to any other defined child elements in the following order: First, the elements from the base resource, in order Second, the elements from the domain resource, in order

• FHIR elements are never empty. If an element is present in the resource, it SHALL have either a value attribute, child elements as defined for its type, or 1 or more extensions

• Attributes cannot be empty. Either they are absent, or they are present with at least one character of non-whitespace content

• Implementers SHOULD trim leading and trailing whitespace before writing and SHOULD trim leading and trailing whitespace when reading attribute values (for XML schema conformance)

• The lock icon () denotes that an element defines or is affected by additional rules that control its presence and/or content

• XML comments, processing instructions and formatting are not part of the contents of a resource

• There SHALL be no DTD references in FHIR resources (because of the XXE security exploit)

• XML resources SHALL be exchanged using UTF-8 encoding. Specifying the character encoding using an XML declaration (<?xml encoding="UTF-8" ?>) is optional but recommended

• Other processing instructions SHOULD not be included and SHALL NOT be required to properly understand and/or present the data or narrative of the resource. Applications MAY preserve processing instructions when handling resources, but are not required to do so

• The MIME-type for this format is application/fhir+xml.

2.6.1.1 XML Schema and Schematron

This specification provides schema definitions for all the resource and data type content models it describes.

The base schema is called "fhir-base.xsd" and defines all the datatypes and base infrastructure types. In addition, there is a schema for each resource and a common schema fhir-all.xsd that includes all the resource schemas. For schema processors that do not like circular includes, there is a single schema that contains everything.

In addition to the w3c schema files, this specification also provides Schematron files that enforce most of the constraints defined for the datatypes and resources, though some are only expressible and validatable using FHIRPath. These are packaged as files for each resource.

XML that is exchanged SHALL be valid against the w3c schema and Schematron, though being valid against the schema and Schematron is not sufficient to be a conformant instance: this specification makes several rules that cannot be checked by either mechanism. Operational systems may choose to use schema tools to check validation, but are not required to do so. Exchanged content SHALL NOT specify the schema or even contain the schema instance namespace in the resource itself.

Given the way extensions work, applications reading XML resources will never encounter unknown elements. However, once an application starts trading with other applications that conform to later versions of this specification, unknown XML elements may be encountered. Applications MAY choose to ignore unknown elements to foster forwards compatibility in this regard, but may also choose not to - which would be the normal behavior for schema generated applications.

2.6.1.2 Code Generation Schema

In addition to the validation schema, this specification provides a set of schemas suitable for code generation. These schemas describe the same XML syntax, but apply less validation to create schemas that work better with code generation tooling.

Specifically, these schemas are generated without any xsd:choice elements, for code generators that don't deal with choices well. Implementers that use these schemas will need to enforce the correct usage of the choice elements without schema support.

Implementers making use of schema-driven code generation tooling need to consider how to handle the decimal data type. The decimal data type is defined to be precision aware - that is, that implementers need to preserve the difference between "2.0" and "2.00" - this is ubiquitously considered important in handling observed data in healthcare. Both schemas map this data type to a union of xsd:decimal and xsd:double, but the base W3C schema decimal type is specified not to be precision aware. Schema driven implementations vary as to how precision is handled. Implementers will need to determine how their generated code handles decimals/doubles and consider changing the type for decimal in their schema from xsd:decimal/double to xsd:string. Specifically, implementers may wish to change:

```
<xs:simpleType name="decimal-primitive">
  <xs:union memberTypes="xs:decimal xs:double"/>
</xs:simpleType>
```

to this:

```
<xs:simpleType name="decimal-primitive">
  <xs:restriction base="xs:string">
    <xs:pattern value="-?(0|[1-9][0-9]*)(\.[0-9]+)?([eE][+-]?[0-9]+)?"/>
  </xs:restriction>
</xs:simpleType>
```

Alternatively, if supported, implementers may wish to use the precisionDecimal from the XSD 1.1 framework.

Note that most code generation frameworks ignore the pattern restriction.

2.6.1.3 Canonical XML

Resources and/or Bundles may be digitally signed (see Bundle and Provenance).

This specification defines the following method for canonicalizing FHIR resources, when represented as XML:

- No whitespace other than single spaces in attribute values and in the XHTML in the Narrative
- Use default namespaces for the FHIR and XHTML namespaces
- Omit all comments
- Always use the Unicode character representation for any XML entities (e.g. ' instead of ")
- Include the XML processing instruction <?xml version="1.0" encoding="UTF-8"?>
- Using the XML canonical method Canonical XML 1.1 (http://www.w3.org/2006/12/xml-c14n11)

This canonicalization method is identified by the URI http://hl7.org/fhir/canonicalization/xml. The following additional canonicalization URIs are also defined:

http://hl7.org/fhir/canonicalization/xml#data | The narrative (Resource.text) is omitted prior to signing (note the deletion is at Resource.text, not Resource.text.div)

http://hl7.org/fhir/canonicalization/xml#static | In addition to narrative (Resource.text), the Resource.meta element is removed. This makes the signature robust as the content is moved from server to server, or workflow and access tags are added or removed. Note that workflow and security tags may contain information important to the handling of the resource, so meta elements should be protected from tampering by other means if unsigned.

http://hl7.org/fhir/canonicalization/xml#narrative | The method only retains the Resource.id and Narrative (Resource.text

http://hl7.org/fhir/canonicalization/xml#document | The signs everything in a Bundle, except for the Bundle.id and Bundle.metadata on the root Bundle (allows for a document to be copied from server to server)

These canonicalization methods allow systems the flexibility to sign the various portions of the resource that matter for the workflow the signature serves. These canonicalization algorithms do not work for enveloped signatures. This will be researched and addressed in a future release. This specification may define additional canonicalizations in the future, and other specifications might also define additional canonicalization methods.

Implementation Note: One consequence of signing the document is that URLs, identifiers and internal references are frozen and cannot be changed. This might be a desired feature, but it may also cripple interoperability between closed ecosystems where re-identification frequently occurs. For this reason, it is recommended that systems consider carefully the impact of any signature processes. The impact of signatures on Document bundles and their related processes is the most well understood use of digital signatures.

Note that following common normalization procedures in XML causes consecutive whitespace within attributes to be normalized. As a result, whitespace within markdown can be changed without breaking a digital signature. For elements with a type of markdown, this means that, in some cases, whitespace manipulation that results in significantly different visual rendering (e.g. changing indentation levels, causing content to appear in separate paragraphs rather than part of a single paragraph, etc.) might not be detected as a signature-breaking change. If this would present an unacceptable risk, systems should use the JSON signature approach as it does not normalize whitespace.

2.6.2 JSON Representation of Resources

URL: https://hl7.org/fhir/R4/json.html

2.6.2 JSON Representation of Resources

Implementable Technology Specifications Work Group | Maturity Level: Normative | Standards Status: Normative

| This page has been approved as part of an ANSI standard. See the Infrastructure Package for further details.

The JSON representation for a resource is based on the JSON format described in STD 90 (RFC 8259) , and is described using this format:

```
{
  "resourceType" : "[Resource Type]",
  // from Source: property0
  "property1" : "<[primitive]>", // short description
  "property2" : { [Data Type] }, // short description
  "property3" : { // Short Description
    "propertyA" : { CodeableConcept }, // Short Description (Example)
  },
  "property4" : [{ // Short Description
    "propertyB" : { Reference(ResourceType) } // R!  Short Description
  }]
}
```

Using this format:

• To build a valid JSON instance of a resource, replace the contents of the property values with valid content as described by the type rules and content description found in the property value for each element

• In this example: property1 has a primitive data type; the value of the property will be as described for the stated type property2 has a complex data type; the value of the property is an object that has the content as described for the stated type property3 is an object property that contains additional properties (e.g. propertyA; the allowable properties are listed (but also include extensions as appropriate) property4 is an array property that contains items which are objects themselves. The items may have any of the types already encountered in points 1-3 propertyA is an example of an object property that has a binding to a value set - the Short description is a link to the value set. In addition, the binding strength is shown propertyB is an example of an object property that has a reference to a particular kind of resource

• Property names are case-sensitive (though duplicates that differ only in case are never defined)

• Property names SHALL be unique. Note: this is not explicitly stated in the original JSON specification,so stated for clarity here

• Properties can appear in any order

• XHTML is represented as an escaped string

• Objects are never empty. If an element is present in the resource, it SHALL have properties as defined for its type, or 1 or more extensions

• String property values can never be empty. Either the property is absent, or it is present with at least one character of content

• The R! denotes that an element is mandatory - it must be present (or in an array, at least one item must be present)

• In this format, // is used for comments. While // is legal in Javascript, it is not legal in JSON, and comments SHALL not be in JSON instances irrespective of whether partiular applications ignore them

• The character encoding is always UTF-8

• The MIME-type for this format is application/fhir+json.

Given the way extensions work, applications reading JSON resources will never encounter unknown properties. However, once an application starts trading with other applications that conform to later versions of this specification, unknown properties may be encountered. Applications MAY choose to ignore unknown properties in order to foster forwards compatibility in this regard, but may also choose not to.

2.6.2.1 Comparison with XML

The JSON format is similar to the XML format:

• The names for the JSON object members are the same as the names of the elements and attributes in XML, including elements that may repeat. Property names are case sensitive

• Just as in XML, JSON objects and arrays are never empty, and properties never have null values (except for a special case documented below). Omit a property if it is empty

• JSON whitespace is not part of the contents of a resource. Applications MAY preserve the whitespace when handling resources, but are not required to do so. Note that digital signatures may depend on the whitespace

There are differences from XML:

• There are no namespaces in the JSON representation

• The type of the resource is represented differently in JSON - instead of being the name of the base object (there is none in JSON), it is carried as the property resourceType

• The order of properties of an object is not significant in the JSON representation, though order within an array SHALL be maintained

• JSON does not have a notion of attributes versus elements, so attributes (e.g. id, value) are handled differently (see below)

• JSON has the array notation, which is used to represent repeating elements. Note that arrays are used when the item might repeat, even if it does not repeat in a specific instance

• The XHTML element in the Narrative datatype is represented as a single escaped string of XHTML. This is to avoid problems in JSON with mixed content, etc. The XHTML SHALL still conform to the rules described for the Narrative

The JSON format for the resources follows the standard XML format closely to make interconversion easy, and so that XPath queries can easily be mapped to query the JSON structures. However, the differences - particularly the repeating element one, which cannot be avoided - mean that generic XML --> JSON converters are not able to perform correctly. The reference platforms provide XML <--> JSON conversion functionality that accommodates these FHIR-specific characteristics.

2.6.2.2 JSON Representation for repeating elements

An element that has a maximum cardinality of >1 (e.g. x..* in the definitions) may occur more than once in the instance. In XML, this is simply done by repeating the XML element multiple times. In JSON, this is done by using an array type. Note that:

• The name of the array is singular - the same as the XML element

• An item that may repeat is represented as an array even in the case that it doesn't repeat so that the process of parsing the resource is the same either way

```
 <code>
   <coding>
     <system value="http://snomed.info/sct"/>
     <code value="104934005"/>
   </coding>
   <coding>
     <system value="http://loinc.org"/>
     <code value="2947-0"/>
   </coding>
 </code>
```

is represented in JSON like this:

```
{
 "coding": [
   {
     "system" : "http://snomed.info/sct",
     "code" : "104934005"
   },
   {
     "system" : "http://loinc.org",
     "code" : "2947-0"
   }
 ]
}
```

2.6.2.3 JSON representation of primitive elements

FHIR elements with primitive data types are represented in two parts:

• A JSON property with the name of the element, which has a JSON type of number, boolean, or string

• a JSON property with _ prepended to the name of the element, which, if present, contains the value's id and/or extensions

The FHIR types integer and decimal are represented as a JSON number, the FHIR type boolean as a JSON boolean, and all other types are represented as a JSON string which has the same content as that specified for the relevant data type. Whitespace is always significant (i.e. no leading and trailing spaces for non-strings).

```
 <code value="abc"/> <!-- code -->
 <date value="1972-11-30"/> <!-- dateTime -->
 <deceased value="false" /> <!-- boolean -->
 <count value="23" />  <!-- integer -->
```

is represented in JSON as

```
 "code" : "abc",
 "date" : "1972-11-30",
 "deceased" : false,
 "count" : 23
```

When using a JavaScript JSON.parse() implementation, note that JavaScript natively supports only one numeric datatype, which is a floating point number. This can cause loss of precision for FHIR numbers. In particular, trailing 0s after a decimal point will be lost e.g. 2.00 will be converted to 2. The FHIR decimal data type is defined such that precision, including trailing zeros, is preserved for presentation purposes, and this is

widely regarded as critical for correct presentation of clinical measurements. Implementations should consider using a custom parser and big number library (e.g. https://github.com/jtobey/javascript-bignum) to meet these requirements.

If the value has an id attribute, or extensions, then this is represented as follows:

```
<birthDate id="314159" value="1970-03-30" >
  <extension url="http://example.org/fhir/StructureDefinition/text">
    <valueString value="Easter 1970"/>
  </extension>
</birthDate>
```

is represented in JSON as:

```
"birthDate": "1970-03-30",
"_birthDate": {
  "id": "314159",
  "extension" : [ {
     "url" : "http://example.org/fhir/StructureDefinition/text",
     "valueString" : "Easter 1970"
  }]
}
```

Note: If the primitive has an id attribute or extension, but no value, only the property with the _ is rendered.

In the case where the primitive element may repeat, it is represented in two arrays. JSON null values are used to fill out both arrays so that the id and/or extension are aligned with the matching value in the first array, as demonstrated in this example:

```
<code value="au"/>
<code value="nz">
  <extension url="http://hl7.org/fhir/StructureDefinition/display">
    <valueString value="New Zealand a.k.a Kiwiland"/>
  </extension>
</code>
```

is represented in JSON as:

```
"code": [ "au", "nz" ],
"_code": [
  null,
  {
    "extension" : [ {
       "url" : "http://hl7.org/fhir/StructureDefinition/display",
       "valueString" : "New Zealand a.k.a Kiwiland"
    }]
  }
 ]
```

Note: when one of the repeating elements has no value, it is represented in the first array using a null. When an element has a value but no extension/id, the second array will have a null at the position of that element.

Implementation Note: The representation of primitive data types has been split into two parts like this in order to simplify the representation of simple primitive values without id or extensions. This does have the cost of making the representation of the id attribute and extensions more ungainly, but these are both rarely used with primitive data types.

2.6.2.4 JSON representation of Elements and Complex Data types

Elements, and complex datatypes (types that contain named elements of other types) are represented using a JSON object, containing a member for each element in the datatype. Composites can have id attributes, which are converted to JSON member values, in the same manner as described for primitives. For example:

```
<Patient>
  <text>
    <status value="generated" />
    <div xmlns="http://www.w3.org/1999/xhtml"><p>...</p></div>
  </text>
  <name id="f2">
    <use value="official" />
    <given value="Karen" />
    <family id="a2" value="Van" />
  </name>
</Patient>
```

is represented in JSON as:

```
{
  "name" : [{
    "id" : "f2",
    "use" : "official" ,
    "given" : [ "Karen" ],
    "family" :  "Van",
    "_family" : {"id" : "a2"}
   }],
  "text" : {
    "status" : "generated" ,
    "div" : "<div xmlns=\"http://www.w3.org/1999/xhtml\"><p>...</p></div>"
  }
}
```

Things to note here are:

• Both given is a repeating XML elements, so it is serialized as an Array, whether or not it repeats in this instance

• In the family part of name, the id is added represented in _family as described above, while the id on the name itself is represented as just another property

• The XHTML content in the div element which is in the Narrative element text is represented as an escaped string in the value property in JSON. The xhtml root element needs to be a in the xhtml namespace

2.6.2.5 JSON representation of Resources

A resource is a JSON object with a property resourceType which informs the parser which resource type this is:

```
{
  "resourceType" : "Patient",
  "text" : {
    "status" : "generated" ,
    "div" : "<div xmlns=\"http://www.w3.org/1999/xhtml\"><p>...</p></div>"
  }
  // etc...
}
```

Note that parsers cannot assume that the resourceType property will come first.

Implementation Note: This is a problem for several JSON -> Object serializers that assume that the resourceType property does come first, including Json.NET . However, some JSON generators do not give the authoring application control of the order of the property values, and so these implementations cannot interoperate with implementations that make assumptions about order. Given that JSON says that the property values are an unordered map of name/value pairs, this specification cannot require that properties come in any particular order, though implementers may choose to fix the property order if they are able (and the reference platforms provided with this specification do so).

There is a sample file with many edge cases to help test JSON parsers.

2.6.2.6 Canonical JSON

Resources and/or Bundles may be digitally signed (see Bundle and Provenance).

This specification defines the following method for canonicalizing FHIR resources, when represented as JSON:

• No whitespace other than single spaces in property values and in the xhtml in the Narrative

• Order properties alphabetically

This canonicalization method is identified by the URI http://hl7.org/fhir/canonicalization/json. The following additional canonicalization URIs are also defined:

http://hl7.org/fhir/canonicalization/json#data	The narrative (Resource.text) is omitted prior to signing (note the deletion is at Resource.text, not Resource.text.div)
http://hl7.org/fhir/canonicalization/json#static	In addition to narrative (Resource.text), the Resource.meta element is removed. This makes the signature robust as the content is moved from server to server, or workflow and access tags are added or removed. Note that workflow and security tags may contain information important to the handling of the resource, so meta elements should be protected from tampering by other means if unsigned.
http://hl7.org/fhir/canonicalization/json#narrative	This method only retains the Resource.id and Narrative elements
http://hl7.org/fhir/canonicalization/json#document	The signs everything in a Bundle, except for the Bundle.id and Bundle.metadata on the root Bundle (allows for a document to be copied from server to server)

These canonicalization methods allow system the flexibility to sign the various portions of the resource that matter for the workflow the signature serves. These canonicalization algorithms do not work for enveloped signatures. This will be researched and addressed in a future release. This specification may define additional canonicalizations in the future, and other specifications might also define additional canonicalization methods.

Implementation Note: One consequence of signing the document is that URLs, identifiers and internal references are frozen and cannot be changed. This might be a desired feature, but it may also cripple interoperability between closed ecosystems where re-identification frequently occurs. For this reason, it is recommended that systems consider carefully the impact of any signature processes. The impact of signatures on Document bundles and their related processes is the most well understood use of digital signatures.

2.6.3 ND-JSON Representation of Resources

URL: https://hl7.org/fhir/R4/nd-json.html

2.6.3 ND-JSON Representation of Resources

Implementable Technology Specifications Work Group | Maturity Level: 2 | Standards Status: Draft

ns-json (New line delimited JSON) is a variant of the JSON format that is supported for bulk data transfer. In principle, nd-json is a simple variation on the JSON format, but where resources are serialized with no whitespace, and separated by a newline pair (characters 13 and 10).

```
{
  { "resourceType" : "[type]", .... }
```

```
  { "resourceType" : "[type]", .... }
}
```

The MIME-type for this format is application/fhir+ndjson.

In order to simplify nd-json processing, each nd-json document contains only resources of a single type - every line contains a resource of a particular type. (though note that resources may still contain contained resources of various types).

On the RESTful API, the nd-json format can only be retrieved using the Asynchronous Pattern.

2.6.4 Resource Description Framework (RDF) Representation

URL: https://hl7.org/fhir/R4/rdf.html

2.6.4 Resource Description Framework (RDF) Representation

FHIR Infrastructure Work Group | Maturity Level: 2 | Standards Status: Trial Use

| This page and the RDF forms are jointly maintained by the HL7 FHIR project and the W3C Semantic Web Health Care and Life Sciences Interest Group .

FHIR resources can be represented as an RDF graph serialized in the Turtle format . The RDF format is defined to assist the process of bridging between operational data exchange and formal knowledge processing systems. While the RDF form offers a fully functional representation of FHIR resources, it has different operational characteristics from the JSON and XML representations, and would be implemented for different reasons. Systems focused on operational exchange of data would not generally choose to use RDF.

This page describes:

• The relationship between the RDF, FHIR, and ontologies

• Representing FHIR resources in Turtle

• The FHIR definitions as Turtle

• How to exchange RDF using the RESTful API

• Implementation Advice for using RDF

Most of the major pieces of FHIR/RDF are now available in a complete enough form for developers to start trying them and providing feedback. Work still needed includes:

• Improved documentation on using the RDF format

• Reconciliation with the FHIR Ontology draft on W3C github

• Tutorial materials on using FHIR/RDF with other ontologies and ShExMap to translate other RDF data (non-FHIR) to/from FHIR/RDF

2.6.4.1 Relationship between the RDF, FHIR, and ontologies

This page documents an RDF format that can be used to exchange FHIR data, on the basis that RDF is a universal information representation. Using RDF enables FHIR data to be used with RDF-aware applications to support inference, shared semantics across multiple standards and data formats, data integration, semantic data validation, compliance enforcement, SPARQL queries and other uses. Implementers using FHIR in this fashion should be aware of the relationship between the RDF format the wider use of ontologies.

The RDF format is based on the same abstract information model as the XML and JSON formats and carries the same information content. Resources are losslessly round-trippable between XML, JSON and RDF formats and data expressed in the RDF format corresponds closely to the XML and JSON formats in its look and feel. However, there are a few additional terms that may appear in the RDF format, which are explained below: fhir:nodeRole, fhir:treeRoot, fhir:value, fhir:index and fhir:concept. ShEx schemas are available that describe the RDF format.

In addition to defining the RDF format, this specification provides an associated ontology that provides formal definitions for the relationships that appear in the RDF format. Ontologies that were designed independently almost always have some impedance mismatch when attempting to use them together. Many of the ontologies in the medical and life sciences domain are designed to capture facts about the world for research, such as the fact that the mitral valve is a kind of heart valve. But FHIR was designed to support the day-to-day operations of healthcare providers exchanging electronic health records (EHRs), and in this context, the orientation has historically been different. When using FHIR/RDF with other ontologies, impedance differences are likely to show up in two main ways:

• Records versus facts. FHIR is oriented toward recording who did what ("Dr. Jones diagnosed patient x with viral pneumonia") rather than stating absolute medical facts ("patient x has viral pneumonia").

• Non-monotonicity. RDF was designed to be monotonic, whereas FHIR has a few design aspects that would be non-monotonic if they were interpreted directly in RDF. (Monotonicity means that new data cannot invalidate previous conclusions; non-monotonicity means that previous conclusions can be invalidated by new data.) For example, a modifier extension indicates that the surrounding element's meaning will likely be misunderstood if the modifier extension is not understood.

For both of these reasons, to maintain monotonicity in RDF, FHIR/RDF should not be directly interpreted as stating facts; transformations are required to remove or isolate non-monotonic elements and reconcile the records across time and perspective.

Application developers should also be aware that some FHIR data attributes have a major impact on the interpretation of the enclosing data element: the meaning of the enclosing element cannot be determined in isolation. For example, a status of 'entered-in-error' means that the resource was created accidentally and should be ignored for most purposes.

2.6.4.2 Turtle Format for Resources

A FHIR resource is represented by a series of RDF triples. The Turtle representation for a resource is described using this format:

Turtle Template

```
[ a fhir:Observation; fhir:nodeRole fhir:treeRoot;
  # from Resource: id; meta; implicitRules; and language
  # from DomainResource: text; contained; extension; and modifierExtension
  fhir:Obervation.identifier [ Identifier ]; # 0..* Unique Id for this particular observation
  fhir:Obervation.status [ fhir:value "<code>" ]; # R!  registered | preliminary | final | amended +
  fhir:Obervation.code [ CodeableConcept ]; # 1..1 R!  Type of observation (code / type)
  fhir:Obervation.subject [ fhir:reference [ Patient|Group|Device|Location ] ]; # 0..1 Who and/or what this is about
  fhir:Obervation.encounter [ fhir:reference [ Encounter ] ]; # 0..1 Healthcare event during which this observation is
    made
  # effective[x]: 0..1 Clinically relevant time/time-period for observation. One of these 2:
    fhir:Obervation.effectiveDateTime [ fhir:value "<dateTime>" ];
    fhir:Obervation.effectivePeriod [ Period ];
]
```

Using this format:

• To build a valid Turtle instance of a resource, replace the contents of the property values with valid content as described by the type rules and content description found in the property value for each element

• Relationship names are case-sensitive (though duplicates that differ only in case are never defined)

• Relationships can appear in any order

• Content within a resource is always represented with anonymous nodes - only resources can be identified

• Nodes are never empty. If an element is present in the resource, it SHALL have properties as defined for its type, or 1 or more extensions

• The R! denotes that a relationship is mandatory - it must be present (or in an array, at least one item must be present)

• Note that this specification produces Turtle that is nicely formatted and well laid out, but this is not required or expected

• The MIME-type for this format is application/fhir+turtle. Other MIME types: text/shex for the SHEX (RDF schemas)

2.6.4.2.1 Representing resources

Each resource is represented as a set of RDF triples represented using the Turtle syntax. When a resource has a persistent identity (e.g. it can be found at a particular URL - usually a FHIR RESTful server), then that URL is its identity. Resources with no persistent identity (e.g. bundles from search results) have the identity of the root document - "<>" in Turtle syntax.

Some resources can contain other resources. Given that the relationships can appear in any order, it cannot be assumed that the first encountered element represents the resource of interest that is being represented by the set of Turtle statements. The focal resource - where to start when parsing - is the resource with the relationship fhir:nodeRole to fhir:treeRoot. If there is more than one node labeled as a 'treeRoot' in a set of Turtle statements, it cannot be determined how to parse them as a single resource.

2.6.4.2.2 Representing Repeating Elements

Elements that can repeat are represented with a relationship

```
  fhir:index [n]
```

where [n] is a zero-based integer offset (i.e. the first element has an index of 0). Lists are never sparse; it is an error if there are missing items in the repeat sequence.

Note: this means that the rdf:list structure is not used.

2.6.4.2.3 Representing Primitive Elements

Primitive elements - elements with a primitive type - are represented as regular nodes so that the elements extensions can be represented. The actual value of the primitive type is represented using the fhir:value predicate:

```
  fhir:value "[value]"^^xs:type
```

The value has two parts: a literal string that contains the value, and, if applicable, one of the following schema types:

• boolean
• integer
• decimal
• base64Binary
• dateTime
• date
• gYear
• gYearMonth
• time

The choice is made based on the types as specified for the primitive type. Note that the correct schema type for a date/dateTime must be determined by inspecting the value of the date for precision.

The fhir:value property can never be empty. Either the relationship is absent, or it is present with at least one character of content. XHTML is represented as an escaped xs:string.

2.6.4.2.4 Representing References

A Reference element is represented using the same rules as above:

```
fhir:Observation.subjectReference [
    fhir:Reference.reference [ fhir:value "Patient/example" ];
    fhir:Reference.display [ fhir:value "Example Patient" ];
  ];
```

This allows faithful round tripping of the resource between the Turtle format and the JSON and XML formats. However, it's very useful for an RDF processor if the RDF graph links to the target of the reference directly. This can be represented using the fhir:link property:

```
fhir:Observation.subjectReference [
    fhir:link <http://hl7.org/fhir/Patient/example>;
    fhir:Reference.reference [ fhir:value "Patient/example" ]
  ];
```

The correct value for the fhir:link relationship must be determined by resolving the rules for resolving references for the various reference types to a literal URL that refers to the correct content in the local RDF context.

The fhir:link relationship can be added automatically as part of generating the resource representation, or it can be injected by a post-processor that knows how to convert the raw references into RDF-suitable references.

2.6.4.2.5 Representing Inline Resources

Inline resources - when a resource is contained directly in another element occur in the following places:

• Bundle.entry.resource
• DomainResource.contained
• Parameters.parameter.resource

Inline resources are represented directly as anonymous nodes. This is an example for a contained Medication resource:

```
  fhir:DomainResource.contained [
     a fhir:Medication;
     fhir:index 0;
     # triples for the Medication
  ]
```

Note that DomainResource.contained has a cardinality of 0..*, so fhir:index is used to order the array.

2.6.4.2.6 Representing Concepts

The same logic applies to the Coding data type. These are represented directly in Turtle by serializing their properties as specified above:

```
  fhir:Observation.code [
     fhir:CodeableConcept.coding [
       fhir:index 0;
       fhir:Coding.system [ fhir:value "http://loinc.org" ];
       fhir:Coding.code [ fhir:value "29463-7" ];
       fhir:Coding.display [ fhir:value "Body Weight" ]
    ];
    fhir:CodeableConcept.coding [
       fhir:index 1;
       fhir:Coding.system [ fhir:value "http://snomed.info/sct" ];
       fhir:Coding.code [ fhir:value "27113001" ];
       fhir:Coding.display [ fhir:value "Body weight" ]
    ]
  ];
```

For reasoners using the RDF graph, it's very useful to make the implicit concept references in these Codings explicit using a rdf:type assertion ("a" in Turtle):

```
  fhir:Observation.code [
     fhir:CodeableConcept.coding [
       fhir:index 0;
       a loinc:29463-7;
       fhir:Coding.system [ fhir:value "http://loinc.org" ];
       fhir:Coding.code [ fhir:value "29463-7" ];
       fhir:Coding.display [ fhir:value "Body Weight" ]
     ];
     fhir:CodeableConcept.coding [
       fhir:index 2;
       a sct:27113001;
       fhir:Coding.system [ fhir:value "http://snomed.info/sct" ];
       fhir:Coding.code [ fhir:value "27113001" ];
       fhir:Coding.display [ fhir:value "Body weight" ]
    ]
  ];
```

These rdf:type assertions can be made by any agent that knows how to convert from the code system to the correct ontological representation on the RDF context. Note that a few code systems have standard ontological representations, but many don't. Again, these assertions can be made by the serializer, or injected by a post-processor.

2.6.4.2.7 Schema

FHIR uses ShEx for representing the Turtle schema. See fhir.shex for definitions.

2.6.4.3 RDF Representation of FHIR

In addition to the basic representation of FHIR resources in Turtle format, a Turtle representation of the FHIR infrastructure and definitions is also published, for the following purposes:

• Providing the class definitions to support RDF based representation of resource instances

• Supporting knowledge-based analysis of the FHIR specification itself

• Providing knowledge of use at run-time for converting between FHIR and other content models

• Supporting reasoning across the information/terminology model boundary

The RDF definitions are published as a series of Turtle files: HL7 v3 RIM (Reference Information Model) and FHIR.

2.6.4.4 Using RDF with the REST API

TODO

2.6.4.5 Using FHIR/RDF

Application developers wishing to use FHIR/RDF will often need to perform the following rough steps, though exact steps will depend on your application:

• Convert FHIR/XML or FHIR/JSON data to/from FHIR/RDF. Open source implementations that can perform this conversion include: HAPI java Reference library (Same code as used in the build) Reference Server (based on Delphi reference implementation) fhir-net-api , a .NET implementation. (should work on Mono) Check the HL7 Open Source FHIR Implementations page, in case new implementations have become available

• Download the FHIR ontology.

• Download other ontologies that you wish to use with your application, such as the SNOMED-CT ontology. (Because of license restrictions, the process of obtaining the SNOMED-CT ontology is more complex than a simple download. Some guidance is provided here .)

• Create or obtain a bridge ontology that relates terms in the FHIR ontology to terms in your other ontologies.

• Load your FHIR/RDF instance data and your ontologies into a reasoner and/or a triplestore, and perform SPARQL queries and/or reasoning.

• If you are creating FHIR/RDF then you might also wish to validate it, using ShEx . If so, you can download the ShEx schema for FHIR/RDF

Slides by Eric Prud'hommeaux illustrate how inference can be used on FHIR/RDF data to perform a query for FHIR Observations of rheumatoid arthritis, using the SNOMED-CT ontology . Caveat: those slides show some terms in the fhir: namespace that do not actually exist in the fhir: namespace, such as fhir:Observation-of-Rheumatoid_arthritis-disorder and fhir:Coding-of-Rheumatoid_arthritis-disorder. A different namespace should have been used in the example.

2.7 Version Management Policy

URL: https://hl7.org/fhir/R4/versions.html

The following rules apply once an artifact in the FHIR core specification or in an HL7-international published implementation guide has become normative

FHIR Infrastructure Work Group | Maturity Level: Normative | Standards Status: Normative

| This page has been approved as part of an ANSI standard. See the Infrastructure Package for further details.

2.7.0 Version Management Policy

This page documents the way version change is handled in FHIR. FHIR is a standard, so the way version change is handled is a bit different from an application API. This page describes:

- The standards development process
- The FHIR Maturity Model
- How FHIR versions work
- The rules for inter-version change once an artifact is Normative
- Managing Change during the development process
- Plans for future releases

See also Managing FHIR Versions for additional implementer advice about dealing with versions.

2.7.0.1 The Standards Development Process

FHIR is a standard. In order to be useful, standards need to evolve. At the same time, the evolution of standards needs to be predictable and manageable for the implementation community. This section describes how HL7 develops a standard so that implementers know what to expect as the standard evolves.

HL7 has five descriptive terms that describe the level of stability and implementation readiness associated with different aspects of the specification. They are as follows:

Standard Level	Description
Draft	This portion of the specification is not considered to be complete enough or sufficiently reviewed to be safe for implementation. It may have known issues or still be in the "in development" stage. It is included in the publication as a place-holder, to solicit feedback from the implementation community and/or to give implementers some insight as to functionality likely to be included in future versions of the specification. Content at this level should only be implemented by the brave or desperate and is very much "use at your own risk". The content that is Draft that will usually be elevated to Trial Use once review and correction is complete after it has been subjected to ballot
Trial Use	This content has been well reviewed and is considered by the authors to be ready for use in production systems. It has been subjected to ballot and approved as an official standard. However, it has not yet seen widespread use in production across the full spectrum of environments it is intended to be used in. In some cases, there may be documented known issues that require implementation experience to determine appropriate resolutions for. Future versions of FHIR may make significant changes to Trial Use content that are not compatible with previously published content.
Normative	This content has been subject to review and production implementation in a wide variety of environments. The content is considered to be stable and has been 'locked', subjecting it to FHIR Inter-version Compatibility Rules. While changes are possible, they are expected to be infrequent and are tightly constrained.
Informative	This portion of the specification is provided for implementer assistance and does not make rules that implementers are required to follow. Typical examples of this content in the FHIR specification are tables of contents, registries, examples, and implementer advice
Deprecated	This portion of the specification is outdated and may be withdrawn in a future version. Implementers who already support it should continue to do so for backward compatibility. Implementers should avoid adding new uses of this portion of the specification. The specification should include guidance on what implementers should use instead of the deprecated portion

2.7.0.1.1 Mixed Normative content

Some Normative artifacts contain a few parts labeled as 'Trial Use' even though the artifact itself is labeled 'Normative':

- Some normative resources contain elements labeled as 'trial-use'
- Some normative pages contain sections labeled as 'trial-use'

While HL7 prefers to avoid this outcome, there are several resources where the overall functionality of the artifact is clearly ready to be labeled as 'normative' while some very specific parts are known not to have the requisite level of implementation experience as the rest of the resource. E.g. Bundle.signature.

Where a Normative resource contains elements marked as trial-use, these elements are clearly marked in the resource definitions. Implementers should be aware that future versions of the FHIR specification may change these parts of the resources (in addition to the other changes allowed under the inter-version compatibility rules. While HL7 will carefully consider the consequences of breaking change to these elements, implementers should be aware that reading/using these elements has the potential to cause breaking change to their applications later.

Note that this same status will arise as a matter of process when new elements are introduced into normative resources in future versions - they will undergo a period of trial use as appropriate.

Note: it is also possible that some resources in the future will be labeled as 'trial use', but contain some elements labeled as 'normative'. There is no resource like this in this specification, though all Trial Use resources contain normative content from Resource and DomainResource, and the Data types.

Notes:

- The above statuses can apply to both the standard overall as well as to individual components of the FHIR specification
- Between FHIR release 2 and 3, HL7 changed from using "DSTU" (Draft Standard for Trial Use) to just simply "STU" (Standard for Trial Use) to reflect the maturity of the FHIR specification: Release 2 and particularly Release 3 were far beyond "draft" specifications and have been and will be widely implemented.

• Most pages in the specification identify their status explicitly. The few pages that don't are Table of contents pages, and are all Informative

• Some content is labeled with the status "External", which means that the content is maintained in another standard, and the status must be found by consulting that other standard. In this case, the Maturity Model does not apply

2.7.0.2 Maturity Levels

The content of this release has been subject to significant review through ballot and other HL7 processes and many aspects of it have been implemented and subjected to interoperability testing through Connectathons and early adoption. However, the degree of testing has varied. Some resources have been well tested in a variety of environments. Others have received relatively little real-world exercise. In general, the infrastructure should be considered to be more stable than the resources themselves. In some cases, there are issues on which input is specifically requested during the Trial Use period (see the Outstanding Issue List, and known issues will arise after publication (refer to the FHIR Change Request tracker for details.) Guidance from early implementation will help address these areas.

All artifacts in this specification are assigned a "Maturity Level", known as FMM (after the well-known CMM grades). The FMM level can be used by implementers to judge how advanced - and therefore stable - an artifact is. The following FMM levels are defined:

Draft (0) | the resource or profile (artifact) has been published on the current build. This level is synonymous with Draft

FMM 1 | PLUS the artifact produces no warnings during the build process and the responsible WG has indicated that they consider the artifact substantially complete and ready for implementation. For resources, profiles and implementation guides, the FHIR Management Group has approved the underlying resource/profile/IG proposal

FMM 2 | PLUS the artifact has been tested and successfully supports interoperability among at least three independently developed systems leveraging most of the scope (e.g. at least 80% of the core data elements) using semi-realistic data and scenarios based on at least one of the declared scopes of the artifact (e.g. at a connectathon). These interoperability results must have been reported to and accepted by the FMG

FMM 3 | PLUS + the artifact has been verified by the work group as meeting the Conformance Resource Quality Guidelines ; has been subject to a round of formal balloting; has at least 10 distinct implementer comments recorded in the tracker drawn from at least 3 organizations resulting in at least one substantive change

FMM 4 | PLUS the artifact has been tested across its scope (see below), published in a formal publication (e.g. Trial-Use), and implemented in multiple prototype projects. As well, the responsible work group agrees the artifact is sufficiently stable to require implementer consultation for subsequent non-backward compatible changes

FMM 5 | the artifact has been published in two formal publication release cycles at FMM1+ (i.e. Trial-Use level) and has been implemented in at least 5 independent production systems in more than one country

Normative | the artifact is now considered stable

Tested across scope means:

• The FMG has signed off on the list of "example contexts" defined for the artifact

• For each example context, the artifact has either been: reviewed and approved by a domain expert for that scope area, mapped to an existing implemented scope-area-specific standard or tested in an implementation

The Maturity level is strongly related to stability; the higher the maturity level, the more controls are enforced to restrict breaking changes to the resource. For further information, and discussion, see the FHIR Confluence page .

The maturity model is significantly influenced by the degree and type of implementation activity using an artifact. For this reason, we encourage implementers to register their implementations . A detailed analysis of the basis for the maturity metrics for FHIR artifacts can be found here .

2.7.0.3 FHIR Releases and Versioning

New versions of FHIR will be published on a release cycle of approximately 18-24 months. This frequency is based on the timelines necessary to consult with implementers, to develop and review new content, as well as to undertake the formal balloting and reconciliation processes required for ANSI-approved standards. This release cycle also ensures an opportunity to incorporate implementer feedback from earlier versions of the specification into subsequent versions. Limited-scope releases on a shorter timeline may occur occasionally where necessary to meet implementer needs.

Each new release is assigned a unique version number. The FHIR version policy is based on Semantic versioning , but with some differences due to the fact that FHIR is a specification, not a software API.

There is a single development version of FHIR. This undergoes cycles of development as managed by HL7. Each major cycle of development is concluded by a formal ballot (or more than one), and then a new specification is published. In version control terms, each published specification is a branch off the development trunk, which may then itself undergo further change as HL7 maintains the published specification (though such changes are usually minimal, limited to necessary technical corrections or security alerts).

Each FHIR version is identified by a string composed from 4 parts: publication.major.minor.revision.

publication | Incremented when HL7 publishes FHIR as an updated specification, e.g. a Trial Use or Normative version of FHIR The first Trial Use was version 0 FHIR Release 2 (DSTU) was version 1 FHIR Release 3 (STU) is version 3 (skipped '2' to align the major numbers at implementer request)

major | Increments every time a breaking change is made (see below) When a new publication is made, this is reset to 0 in the publication, and 1 in the development branch Since HL7 does not make breaking changes as technical corrections to a published specification, these versions of FHIR always have a version number X.0.n.r Because the development version is the subject of ongoing analysis, debate, ballot and repeated alterations, breaking changes are to be expected in STU content

minor | Increments every time an official snapshot release is generated that contains one or more substantive changes Resets to 0 any time the major version changes Snapshot releases are produced approximately 6 weeks in advance of the 3 annual HL7 working group meetings (and their associated connectathons), though they can also be produced for other major connectathons or to meet implementer requirements.

revision | The hash for the GIT version from which the specification was built, for tracing publication / tooling issues For the current build, changes are made numerous times a day, generally driven by change requests submitted by the implementation community The publication revision number only changes when technical corrections are made

Additional notes:

• Changes to a formally published specification (except for minor publishing corrections, such as correcting broken external links) are only made via announced technical corrections

• The reference implementations have 2 versions - the version of the specification that they implement and their own version. Consult the reference implementation documentation for policy regarding this version number

• The current build - published by the continuous integration service (http://build.fhir.org/) - does not conform to this version policy, in that the version is not updated as changes are made. To indicate this, the revision is always "cb" e.g. 3.1.cb immediately after the publication of Release 3

• The first DSTU was published prior to these rules being agreed as v0.80-2286. This has been updated to 0.0.81.2382 as a technical correction to align with this policy on 9-May 2014

2.7.0.4 Version identification

The FHIR version is usually known implicitly, but can be specified/determined by one of three methods:

• The fhirVersion element in the applicable CapabilityStatement, StructureDefinition, or ImplementationGuide

• The fhirVersion parameter on the MIME-type that applies to the resource

• Specifying a version specific profile on the resource itself in Resource.meta

For further information, see Managing Multiple FHIR Versions.

2.7.0.5 Rules for Inter-version change

The intent of these rules is to ensure that applications that are conformant to an existing specification are also conformant to subsequent versions. In practice, there are many subtle issues around inter-version change, and the exact rules are subject to further clarification based on feedback from implementers.

The following kinds of changes may be made to the specification:

• Breaking changes are changes that mean that previously conformant applications are no longer conformant to the updated specification

• Substantive changes are changes that introduce new functionality - changes to the specification that create new capabilities - but would not render unchanged existing applications non-conformant

• Non-substantive changes should not cause changes in any conformant application. For example, section renumbering, correcting broken links, changing styles, fixing typos, and providing clarifications that do not change the meaning of the specification. In addition, this covers corrections that are judged not to create any expectation of change to a conformant application

NOTE: The examples provided as part of this specification are never substantive. While every effort is made to ensure that FHIR examples are correct, changes to the examples in the specification are not considered substantive.

Content with a status of Draft or Trial Use can change - including Breaking Changes - from version to version, subject to the rules described by the Maturity Process. There are no rules for maintaining any sort of compatibility between versions for content with these statuses, though of we will only make breaking changes based on feedback from the community.

Once an artifact achieves Normative status, specific rules come into play around inter-version compatibility. These rules have implication for both forward and backward compatibility and are intended to allow implementations to exercise FHIR interfaces and process the content of FHIR resources safely while exchanging data between systems using different versions of FHIR.

Forward compatibility means that content that is conformant in an old release will remain conformant with future versions. Once normative, FHIR's rules try to enforce forward compatibility. However, that doesn't guarantee that all old systems will interoperate with future systems.

Backward compatibility means that instances created against future versions of the specification will interoperate with older versions of the specification. This is not guaranteed by FHIR, though there are strategies systems can adhere to that will increase their chances of such interoperability. Specifically, when dealing with content from a system supporting an unknown normative version and wishing to maximize backwards compatibility, applications SHOULD:

• Ignore elements that are unexpected (new elements will never be modifier elements)

• Ignore references to resources that are not recognized

• Ignore unrecognized codes in required and extensible bindings unless the element they appear on is a modifier (in which case, treat the element as an unrecognized modifier extension)

• Ignore unrecognized search criteria - see Handling Search Errors for further information.

• Respond to HTTP commands on unexpected URLs with an appropriate error code.

However, in a healthcare context, many implementers are unwilling to consider some of these steps because of concerns about clinical risk or technical limitations in their software (e.g. schema-based processing).

2.7.0.5.1 Forward and backward compatible rules

Category | Allowed changes

Resources | New artifacts resources may be introduced. Existing resources will not have their names changed

Artifacts (resources, profiles, code systems, etc.) | New artifacts including new resources and data types may be introduced. Existing artifacts will not have any computable identifiers (e.g. resource names) changed. Artifacts may be deprecated

Elements | New optional elements and/or content (e.g. XML attributes, etc.) may be introduced at any location in resource and data type structures provided they do not constitute "isModifier" elements. However, the names, path and meaning of previously existing data elements will not be changed. This means there will be no change to resource names and no changes to names assigned to slices and other elements within profiles.

Cardinality | Minimum element cardinalities will not be changed. Upper cardinality may change from 1 to * only in circumstances where all elements except for the first repetition can be safely ignored. Note that this may change the path to the element in some syntaxes (e.g. JSON). This may mean that an order is assigned to the repeating items or that there is no preference as to which element is retained. Systems should follow the rules above for unexpected elements.

Descriptions | Descriptive information about a resource - short labels, definitions, usage notes, aliases, examples, rationale, mappings, etc. may be updated or revised to provide additional clarity or guidance, but not in such a manner as to invalidate a reasonable interpretation of the previously documented use of an element. (This does not preclude fixing obvious errors.)

Value Sets and Code Systems | The definition of any value set that is marked as immutable will never change. The expansions for immutable value sets may still change if no "stable date" is declared and the value set does not restrict code system and/or value set references to specific versions and the codes in the referenced code system(s) or value set(s) change. For non-immutable value sets: Value sets with an enumerated list of codes and having a 'fixed' binding may have additional codes introduced but will never have codes removed, though they may be deprecated. Value sets making use of filters may have filters loosened or tightened to accommodate changes to underlying code systems. StableDates and referenced code system and value set versions may be adjusted to point to newer versions. Definitions and display values for codes may change, but only in a manner that would not change the reasonable interpretation of data captured using the previous definitions or names. Abstract codes may be made concrete. Concrete codes will not be made abstract. For both immutable and non-immutable value sets, additional designations may be declared. Normative CodeSystems whose content is generated from a mix of normative and non-normative contents may break these rules. For example, the code system containing the list of all resources may have codes removed or renamed as non-normative resources are removed or renamed. These expectations only apply to Value Sets and Code Systems maintained as part of the FHIR specification. HL7 cannot enforce these rules on terminology artifacts maintained by other authorities - e.g. UCUM unit codes, ISO language codes, etc.

Terminology Bindings | Required bindings will remain required and will continue to point to the same value set. If the reference is version-specific, it will not change Extensible bindings will remain extensible and will continue to point to the same value set. If the reference is version-specific, it will not change. Example bindings and preferred bindings may change to point to different value sets. Example bindings may be replaced with preferred bindings.

Data Types | Except as described in the preceding paragraph, Data types will not be removed or changed except as allowed above for elements. New data types may be introduced. Types declared on existing elements will not be removed or changed, except for the special case that string may be changed to markdown. Additional data types may be added to elements which are already expressed as a choice of data types only if those elements are optional (minimum cardinality = 0).

Value Constraints | The allowed list of Data types will not be added, removed or changed. Invariants, regular expressions, fixed values and patterns will not be added, removed or changed.

Flags | The Is Modifier and Is Summary flags will not be changed.

Slicing | Slicing rules and aggregation characteristics will not be changed.

Search Criteria | Search criteria may be added but not removed or renamed. Existing criteria will not have their type or path changed or have their description altered in any way that would invalidate the reasonable behavior of existing systems (except for correcting obvious errors).

Operations | New operations may be defined but operations will not be removed or renamed. Existing parameters will not be removed or renamed, nor may their type or lower cardinality be changed. Upper cardinality may be changed from 1 to *. (Systems should ignore unexpected repetitions.) Additional optional parameters may be introduced. Changes to operations that would violate the preceding constraints will be handled by defining new operations and, potentially, deprecating the old operations.

Restful interface | Existing endpoints will not be renamed or removed, nor have their expected behavior changed in a manner that would cause reasonable systems designed against prior versions to be non-interoperable. Additional endpoints and interactions may be introduced.

Profiles and extension definitions | Profile structure, extension definitions and search criteria definitions will not be removed or have their URIs changed. New profile structures, extension definitions and search criteria definitions may be introduced. Profiles may have their statuses changed to "retired". Profiles referenced by data elements for structures or data types may be replaced with a reference to a distinct profile that is "compatible" with the previously referenced profile according to these forward and backward compatibility rules.

Capability Statements | Within the CapabilityStatements for defined FHIR Services or 'core' implementation guides, additional operations may be added. These additions might be optional (MAY/SHOULD) or mandatory (SHALL). Note that the introduction of mandatory operations would break forwards compatibility and will only occur with community consultation.

Implementation Guides | Additional artifacts can be added, and artifacts can be changed. The list of global profiles will not change

References | Where one conformance resource points to another (e.g. CapabilityStatement to profile, profile to profiles, profile to value set, etc.), the reference may change to point to a newer version of the conformance resource or to a distinct conformance resource so long as the content of the newly referenced resource adheres to the compatibility rules with respect to the previously referenced version.

NOTE: In rare circumstances, HL7 may approve changes that technically break one of the above rules in situations where there is a high level of confidence that the change will not impact existing implementers. Such deviations from these declared rules will involve broad notification, extensive community consultation and reviews by multiple levels of HL7 governance processes.

2.7.0.5.2 Deprecation / Withdrawal of Normative Content

Once content is normative, there is a process for removing it from the standard by marking it as deprecated or withdrawn (from the HTML 4.0 Standard):

> Deprecated | Systems should continue to support the artifact/feature/concept, but are discouraged from making use of it
> Withdrawn | Documented for historical purposes, no longer supported

The specification will provide guidance with deprecated materials showing how to avoid using them. Deprecated materials are eligible to be balloted to be withdrawn two years after their deprecated status is published.

The computable artifact labels (e.g. codes, element names, urls, etc.) associated with withdrawn materials SHALL not be used in future versions of HL7 specifications. Materials marked "deprecated" may have that marking removed as part of a subsequent ballot at a later moment, while withdrawn materials SHALL NOT.

The following artifacts are deprecated in this version of FHIR:

• OperationOutcome.issue.location

Additional discussion on inter-versioning issues can be found here:
https://confluence.hl7.org/display/FHIR/Interversion+Compatibility .

2.7.0.6 Managing Change during the development process

Regardless of the degree of prior implementation, all aspects of the FHIR specification are potentially subject to change while an artifact has a status of Draft or Trial Use. These changes may be minor (clarifications of definitions, etc.) or major (refactoring of resources, changes to serialization rules, eliminating or adding data types, etc.) There is no commitment to backward or forward compatibility during the trial use process until content is normative. Changes will not be made without cause, however the interests of long-term implementability will generally trump the impact on early adopters when determining what changes should be made. This balance will shift more towards early adopters as maturity levels increase. I.e. Impact on existing implementations will be weighted more highly for an FMM-level 5 artifact than they would for an FMM-level 1 artifact.

Implementers who are willing to accept the risk of change (perhaps for the benefit of early implementation experience, first mover advantage and the ability to leverage FHIR's intrinsic benefits) are encouraged to implement those parts of FHIR that are early in the maturity cycle in real-world systems. However, those implementers should be aware that local adaptations may be necessary to meet real-world requirements. Furthermore, such implementers should architect their solutions to be tolerant of changes to the specification and, where necessary, to manage interoperability with systems that may be using different versions of the specification or different local adaptations.

During the Trial Use period, requests for change may be submitted using the HL7 issue tracker which can be found here . Where possible, updates to the "development" version of the specification will be made in a timely fashion. Implementers should be aware that the changes are not considered "official" until such time as they are balloted and approved as part of a subsequent Trial Use or Normative publication. Change requests might be fixes to allow implementation, clarifications or enhancements. In addition, HL7 will be developing and introducing additional resources and profiles as part of the FHIR specification.

SDOs and regulatory bodies that are interested in making use of the FHIR specification should feel free to do so, but should consider and plan for the possibility that the specification will evolve and change prior to becoming Normative.

A key aspect of the FHIR specification development process is gaining feedback from implementers making use of the specification. As well, the process is conditional on real world implementation in order to move through the maturity cycle. For this reason, all FHIR implementers are encouraged to register their usage here , which captures contact and other information that will allow HL7 to perform appropriate monitoring of FHIR usage. Survey information is confidential and reported in aggregate only.

2.7.0.7 Extensions for converting between versions

Many implementations need to convert resources from one FHIR version to another. Once resources become normative (once sufficiently mature and stable), converting resources forwards from past versions is not needed. Converting back to older versions presents a challenge, however, in that the newer version may add additional elements that are not present in the older version. In some cases, the elements are simply irrelevant since the requirements they represent are not in scope for older applications, but in other cases, it is necessary to represent the data in order to cater for round-tripping.

A more complex problem arises with resources that are not yet stable (early in the maturity process). If applications have implemented less stable resources, not only do they have the problem of new elements for new requirements, the specification may change in either compatible or incompatible ways, and it may be

necessary to carry data elements from past versions forward in order to allow seamless round-tripping.

In order to help implementers with this problem, any element defined in any version of FHIR is automatically assigned an extension URL that uniquely identifies the element and can be used in the relevant FHIR version. The extension URL for an element can automatically be derived:

```
http://hl7.org/fhir/[version]/StructureDefinition/extension-[Path]
```

where [version] is taken from this list:

This technique can be used with all versions of FHIR, including R2 and R3:

FHIR DSTU2 | 1.0
FHIR R3 (STU3, or just R3) | 3.0
FHIR R4 (this version) | 4.0

Note that this extension framework only applies back to DSTU2. The [Path] is actually the ElementDefinition.id from the relevant StructureDefinition for the element. This leads to URLs like the following:

http://hl7.org/fhir/4.0/StructureDefinition/extension-Bundle.signature | R4 Signature Element on Bundle
http://hl7.org/fhir/3.0/StructureDefinition/extension-Patient.animal.species | STU3 Species Element on Patient
http://hl7.org/fhir/1.0/StructureDefinition/extension-ValueSet.extensible | DSTU2 ValueSet.extensible

Implementers should be aware of the following issues when using these extensions:

• Implementations should always use the correct element for data when one exists. The version differences and maps (and their equivalent for other versions) can help

• There is no way to represent the few special elements that have the type "Resource" in extensions, so these are not defined (e.g. Bundle.entry.resource, DomainResource.contained, etc)

• Where complex data types have no equivalent in an earlier version, use a complex extension, containing extensions also following this pattern. Follow the same pattern for any elements not found in data types in earlier versions

• This table shows the data type mapping across versions. The mapping table SHALL be used.

This table shows the mapping between primitive data types across versions:

R4 | R3 | DSTU2
base64Binary | base64Binary | base64Binary
boolean | boolean | boolean
canonical | (uri) | (uri)
code | code | code
date | date | date
dateTime | dateTime | dateTime
decimal | decimal | decimal
id | id | id
instant | instant | instant
integer | integer | integer
markdown | markdown | markdown
oid | oid | oid
positiveInt | positiveInt | positiveInt
string | string | string
time | time | time
unsignedInt | unsignedInt | unsignedInt
uri | uri | uri
url | (uri) | (uri)
uuid | uuid | (id)

Formal Definitions for extensions:

Format: | R2 Package | R3 Package | R4 Package
R2 Extensions: | n/a | hl7.fhir.extensions.r2:3.0.1 | hl7.fhir.extensions.r2:4.0.1
R3 Extensions: | hl7.fhir.extensions.r3:1.0.2 | n/a` | hl7.fhir.extensions.r3:4.0.1
R4 Extensions: | hl7.fhir.extensions.r4:1.0.2 | hl7.fhir.extensions.r4:3.0.1 | n/a

Note for balloters: these packages will be created when R4 is finalized. Until then, these are broken links.

2.7.0.8 Plans for future releases

While implementation of this mixed Normative and Trial Use release is occurring, development will be progressing on the next release. This next release will include additional clarifications, resources, profiles and quality enhancements over the current release based on implementation experience and ongoing development work. It will also incorporate fixes for issues raised with the FHIR issue tracker . It may be useful for implementers of the STU to review the candidate current release (at http://build.fhir.org) to get a sense of what changes are likely coming and perhaps to find more robust definitions and guidance than are available in the this release. Some implementers who are dependent on content that exists in a draft release may choose to implement based on a particular snapshot of the development release, though in doing so, they will limit their potential communication partners.

The next major publication of FHIR will be Release 5. It is our hope release 5 will include more Normative content, including more of the core patient summary clinical content. Other content, including most if not all

clinical knowledge/reasoning, care planning, and financial resources, are likely to remain at the Trial Use level as they are not expected to meet the criteria for Normative.

More information on plans for Release 5 can be found on the HL7 product director's blog . (Subscribing to this blog is a good way to keep current on significant events in FHIR development, including ballot and publication timelines.)

There will be additional releases of FHIR with a frequency of somewhere between 18 and 24 months for the foreseeable future. These releases will include maintenance on Normative content, new content (e.g. in the public health, financial or clinical research spaces), revisions to trial use content reflecting implementer feedback and increasing maturity on Trial Use artifacts and the migration of additional content to normative status. As well, HL7 will continue to focus to providing additional guidance through the publication of implementation guides and profiles where consensus can be found at the international level.

2.8 Managing Multiple FHIR Versions

URL: https://hl7.org/fhir/R4/versioning.html

2.8 Managing Multiple FHIR Versions

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative

This FHIR specification is one in a series of publications of the FHIR specification . There have been previous releases, and there will be subsequent releases. Each release (or "version") introduces new features, and changes from the previous releases.

Implementers should be familiar with the kind of changes that can be made, and the change processes that apply - see the FHIR Version Management Policy for further information

Ongoing change to the specification means that implementers must be able to determine which version is in use in their context and implement accordingly.

There are 3 ways to determine the version of FHIR in use:

• The fhirVersion element in the applicable CapabilityStatement

• The fhirVersion parameter on the MIME-type that applies to the resource

• Specifying a version specific profile on the resource itself

2.8.1 CapabilityStatement.fhirVersion

Whenever resources are exchanged, the applicable FHIR version applies to the entire interaction, not just the resources. The semantics of the mime types, the RESTful URLs, the search parameters, and the overall interaction are bound to a particular FHIR version. The resource versions must be consistent with the overall interaction. It's not possible for client/server (or, for messaging, sender/receiver) to have a coherent conversation without first agreeing on the version. The version of the interaction is specified in the CapabilityStatement resource. The sequence works this way:

• The server decides what version it is implementing, and states this in it's CapabilityStatement

• The client knows what FHIR version it uses

• The first time the client connects to the server, it retrieves that CapabilityStatement and checks that the server version is the same

• All interactions between the client and server use the same version

Clients are advised to check the server's capability statement before first using the server, but how often is at the discretion of the implementer, and this varies wildly. Some implementers only check the statement during the development cycle, while others check on the first connection for each process. Which is appropriate depends on a variety of circumstances, but implementers should note that the CapabilityStatement can change over time - even changing FHIR versions as the system is subject to ongoing development.

Note that there are related places where the FHIR version in use can also be stated, and these may be consulted at different points in the development cycle in place (or as well as) CapabilityStatement.fhirVersion:

• ImplementationGuide.fhirVersion: specifies which FHIR versions(s) apply for a particular solution. This may imply a particular version for a server, or an exchange

• StructureDefinition.fhirVersion: may specify which version a resource conforms to - or, at least, what version it should be expected to conform to

2.8.2 Mime Type Parameter

The most common strategy for handling change between versions of FHIR is to use different end-points for different versions. e.g. http://acme.com/fhir/r2 and http://acme.com/fhir/r3. However, this can be problematic because it means that the same record has a different identity depending on the version of FHIR in use - though this can also be useful; it depends on the context.

A server can support multiple versions on the same end-point, if the client specifies which version to support using the fhirVersion mime type parameter:

```
GET [base]/metadata
Accept: application/fhir+json; fhirVersion=3.0
```

This is a request to return the CapabilityStatement for the server's support of FHIR Release 3. The client will know that the server doesn't support R3 if it gets an error in return, or if the capability statement that comes back has a different fhirVersion (which would mean that the server does not support the fhirVersion parameter).

Notes:

• The client must include the fhirVersion parameter on all following interactions.

```
--MIME_boundary
Content-Type: application/fhir+xml; fhirVersion=1.0
```

Clients can determine what versions a server supports by using the $versions operation.

Storage

The same technique can be used in a database, or any other kind of storage - just store the mime type (or just the version) alongside the resource. This means that when reading resources, the application knows what version the resource has, and therefore what parser to use, before reading the resource.

2.8.3 Version Specific Profile

There are some circumstances where applications handle 'naked' resources - that is, resources where there is no wrapping envelope to carry a mimetype version parameter, nor any suitable parallel location to store the version. The most obvious circumstance is when resources are stored in files, though this may arise in other circumstances due to (legacy) application or protocol design.

When there's no other alternative, the version has to be in the resources itself, and applications reading the resource have to scan the resource to determine the version and then read the resource as the correct version.

In these cases where the version must be placed in the resource itself, this is done by indicating that the resource conforms to specific version in meta.profile:

```
{
  "resourceType" : "Patient",
  "meta" : {
    "profile" : ["http://hl7.org/fhir/3.0/StructureDefinition/Patient"]
  }
}
```

This technique can be used with all versions of FHIR, including R2 and R3:

FHIR R1 (DSTU 1)	0.0
FHIR R2 (DSTU 2)	1.0
FHIR R3 (STU3, or just R3)	3.0
FHIR R4 (this version)	4.0 (once published)

2.8.4 Managing Version Identification

This means that there are 3 different methods for determining version, all of which may be applicable when exchanging data. Implementers are advised to adopt their use in the order documented:

• All exchange of FHIR resources will have a known version from the metadata - this always applies, and implementers are encouraged to prefer this method whenever possible.

• When this is not enough, or when resources are not exchanged under the rules of a CapabilityStatement, a version parameter should be stored alongside the resource, preferably using the mime type parameter documented above

• When there's no other choice, use the meta.profile to indicate which version(s) a resource conforms to

When determining the version, there's no order of preference between these methods: if the version of a resource or interaction is specified more than one way, the version statements must all agree with each other, or it is an error, and applications can process a request/resource how they wish - usually, with some kind of error.

2.8.5 Strategies for handling Changes Between Versions

All parts of the FHIR specification are subject to ongoing change. The types of changes allowed, and the change control process vary, as detailed in the Version Management Policy. Implementers need to be aware of what kind of changes are made and have a strategy for dealing with them. Where possible, implementers should design their solutions to be able to manage changes to the specification and, where necessary, to manage interoperability with systems that may be using different versions of the specification.

A key aspect of the FHIR specification development process is gaining feedback from implementers making use of the specification. As well, the process is conditional on real world implementation in order to move through the maturity cycle. For this reason, all FHIR implementers are encouraged to register their usage here , which captures contact and other information that will allow HL7 to perform appropriate monitoring of FHIR usage. Survey information is confidential and reported in aggregate only. (Note that implementers can advertise their use of FHIR here .

2.9 FHIRPath

URL: https://hl7.org/fhir/R4/fhirpath.html

2.9 FHIRPath

FHIR Infrastructure Work Group | Maturity Level: Normative | Standards Status: Normative

The FHIR Specification uses FHIRPath (release 2) for path-based navigation and extraction. FHIRPath is a separate specification published at http://hl7.org/fhirpath in order to support wider re-use across multiple specifications.

FHIRPath is used in several places in the FHIR and related specifications:

• invariants in ElementDefinition - used to apply co-occurrence and other rules to the contents (e.g. value.empty() or code!=component.code)

• slicing discriminator - used to indicate what element(s) define uniqueness (e.g. Observation.category)

• search parameter paths - used to define what contents the parameter refers to (e.g. Observation.dataAbsentReason)

• error message locations in OperationOutcome

• FHIRPath-based Patch

• Invariants in the TestScript resource

In addition, FHIRPath is used in pre-fetch templates in Smart on FHIR's CDS-Hooks .

2.9.1 Using FHIRPath with Resources

In FHIRPath, like XPath, operations are expressed in terms of the logical content of hierarchical data models, and support traversal, selection and filtering of data.

FHIRPath uses a tree model that abstracts away the actual underlying data model of the data being queried. For FHIR, this means that the contents of the resources and data types as described in the Logical views (or the UML diagrams) are used as the model, rather than the JSON and XML formats, so specific xml or json features are not visible to the FHIRPath language (such as comments and the split representation of primitives).

More specifically:

• A FHIRPath may optionally start with a full resource name

• Elements of datatypes and resources are used as the name of the nodes which can be navigated over, except for choice elements (ending with '[x]'), see below.

• The contained element node does not have the name of the Resource as its first and only child (instead it directly contains the contained resource’s children)

• There is no difference between an attribute and an element

• Repeating elements turn into multiple nodes with the same name

2.9.1.1 Polymorphism in FHIR

For choice elements, where elements can be one of multiple types, e.g. Patient.deceased[x]. In actual instances these will be present as either Patient.deceasedBoolean or Patient.deceasedDateTime. In FHIRPath, choice elements are labeled according to the name without the '[x]' suffix, and children can be explicitly treated as a specific type using the as operation:

```
(Observation.value as Quantity).unit
```

FHIRPath statements can start with a full resource name:

```
Patient.name.given
```

The name can also include super types such as DomainResource:

```
DomainResource.contained(id = 23).exists()
```

These statements apply to any resource that specializes DomainResource.

2.9.1.2 Using FHIR types in expressions

The namespace for the types defined in FHIR (primitive data types, data types, resources) is FHIR. So, for example:

```
Patient.is(FHIR.Patient)
```

The first element - the type name - is not namespaced, but the parameter to the is() operation is.

Understanding the primitive types is critical: FHIR.string is a different type to System.String. The FHIR.string type specializes FHIR.Element, and has the properties id, extension, and also the implicit value property that is actually of type of System.String.

The evaluation engine will automatically convert the value of FHIR types representing primitives to FHIRPath types when they are used in expressions according to the following mapping:

FHIR primitive type | FHIRPath type

FHIR.boolean | System.Boolean
FHIR.string, FHIR.uri, FHIR.code, FHIR.oid, FHIR.id, FHIR.uuid, FHIR.sid, FHIR.markdown, FHIR.base64Binary | System.String
FHIR.integer, FHIR.unsignedInt, FHIR.positiveInt | System.Integer
FHIR.decimal | System.Decimal
FHIR.date, FHIR.dateTime, FHIR.instant | System.DateTime
FHIR.time | System.Time
FHIR.Quantity | System.Quantity (see below)

Since FHIR primitives may contain extensions, so that the following expressions are not mutually exclusive:

```
Patient.name.given = 'Ewout'         // value of Patient.name.given as a string
Patient.name.given.extension.first().value = true   // extension of the primitive value
```

The automatic conversion means that in most respects, a FHIR primitive can generally be treated as if it was the equivalent FHIRPath system type. The primary exception is the is() operation, where the difference is explicit:

```
Patient.name.given.is(FHIR.string);
Patient.name.given.is(System.string).not();
Patient.name.given.getValue().is(System.string);
```

As shown, all FHIR primitives have the operation getValue() defined (see below) for the few edge cases where the automatic conversion isn't appropriate. Note that as() does not have such restrictions - both of the following are valid:

```
Patient.name.given.as(FHIR.string);
Patient.name.given.as(System.string);
```

2.9.1.3 Use of FHIR Quantity

The Mapping from FHIR Quantity to FHIRPath System.Quantity can only be applied if the FHIR Quantity has a UCUM code - i.e. a system of http://unitsofmeasure.org, and a code is present.

As part of the mapping, time-valued UCUM units are mapped to the calendar duration units defined in FHIRPath, according to the following map:

a | year
mo | month
d | day
h | hour
min | minute
s | second

e.g. the FHIR Quantity 1 'a' would be implicitly converted to the FHIRPath System.Quantity 1 'year'.

2.9.1.4 FHIR Specific Variables

FHIR defines two specific variables that are always in scope when FHIRPath is used in any of the contexts above:

```
%resource // the resource that contains the original node that is in %context
%rootResource // the container resource for the resource identified by %resource
```

The resource is very often the context, such that %resource = %context. When a DomainResource contains another resource, and that contained resource is the focus (%resource) then %rootResource refers to the container resource. Note that in most cases, the resource is not contained by another resource, and then %rootResource is the same as %resource.

2.9.1.5 Additional functions

FHIR adds (compatible) functionality to the common set of functions:

extension(url : string) : collection

Will filter the input collection for items named "extension" with the given url. This is a syntactical shortcut for .extension.where(url = string), but is simpler to write. Will return an empty collection if the input collection is empty or the url is empty.

hasValue() : Boolean

Returns true if the input collection contains a single value which is a FHIR primitive, and it has a primitive value (e.g. as opposed to not having a value and just having extensions).

Note to implementers: The FHIR conceptual model talks about "primitives" as subclasses of the type Element that also have id and extensions. What this actually means is that a FHIR primitive is not a primitive in an implementation language. The introduction (section 2 above) describes the navigation tree as if the FHIR model applies - primitives are both primitives and elements with children.

In FHIRPath, this means that FHIR primitives have a value child, but, as described above, they are automatically cast to FHIRPath primitives when comparisons are made, and that the primitive value will be included in the set returned by children() or descendants().

getValue() : System.[type]

Return the underlying system value for the FHIR primitive (see discussion above).

trace(name : string; selector : expression) : collection

When FHIRPath statements are used in an invariant, the log contents should be added to the error message constructed when the invariant is violated. For example:

```
"SHALL have a local reference if the resource is provided inline (url: height; ids: length,weight)"

  from

"reference.startsWith('#').not() or
  (%context.reference.substring(1).trace('url') in %resource.contained.id.trace('ids'))"
```

The FHIRPath specification adds an additional parameter to the trace() function, a selection expression that can be used to shape what is logged for the collection that is traced. E.g.

```
  contained.where(criteria).trace('ids', type().name+"/"+id).process...
```

This will log the type/name for each contained resource that meets the criteria.

Note: the selector parameter is planned to be added to a future version of the base FHIRPath specification.

resolve() : collection

For each item in the collection, if it is a string that is a uri (or canonical or url), locate the target of the reference, and add it to the resulting collection. If the item does not resolve to a resource, the item is ignored and nothing is added to the output collection.

The items in the collection may also represent a Reference, in which case the Reference.reference is resolved.

ofType(type : identifier) : collection

In FHIR, only concrete core types are allowed as an argument. All primitives are considered to be independent types (so markdown is not a subclass of string). Profiled types are not allowed, so to select SimpleQuantity one would pass Quantity as an argument.

elementDefinition() : collection

Returns the FHIR element definition information for each element in the input collection.

slice(structure : string, name : string) : collection

Returns the given slice as defined in the given structure definition. The structure argument is a uri that resolves to the structure definition, and the name must be the name of a slice within that structure definition. If the structure cannot be resolved, or the name of the slice within the resolved structure is not present, an error is thrown.

For every element in the input collection, if the resolved slice is present on the element, it will be returned. If the slice does not match any element in the input collection, or if the input collection is empty, the result is an empty collection ({ }).

checkModifiers([{modifier : string}]) : collection

For each element in the input collection, verifies that there are no modifying extensions defined other than the ones given by the modifier argument. If the check passes, the input collection is returned. Otherwise, an error is thrown.

conformsTo(structure : string) : Boolean

Returns true if the single input element conforms to the profile specified by the structure argument, and false otherwise. If the structure cannot be resolved to a valid profile, an error is thrown. If the input contains more than one element, an error is thrown. If the input is empty, the result is empty.

memberOf(valueset : string) : Boolean

When invoked on a code-valued element, returns true if the code is a member of the given valueset. When invoked on a concept-valued element, returns true if any code in the concept is a member of the given valueset. When invoked on a string, returns true if the string is equal to a code in the valueset, so long as the valueset only contains one codesystem. If the valueset in this case contains more than one codesystem, an error is thrown.

If the valueset cannot be resolved as a uri to a value set, an error is thrown.

Note that implementations are encouraged to make use of a terminology service to provide this functionality.

For example:

```
Observation.component.where(code.memberOf('http://hl7.org/fhir/ValueSet/observation-vitalsignresult'))
```

This expression returns components that have a code that is a member of the observation-vitalsignresult valueset.

subsumes(code : Coding | CodeableConcept) : Boolean

When invoked on a Coding-valued element and the given code is Coding-valued, returns true if the source code is equivalent to the given code, or if the source code subsumes the given code (i.e. the source code is an ancestor of the given code in a subsumption hierarchy), and false otherwise.

If the Codings are from different code systems, the relationships between the code systems must be well-defined or a run-time error is thrown.

When the source or given elements are CodeableConcepts, returns true if any Coding in the source or given elements is equivalent to or subsumes the given code.

Note that implementations are encouraged to make use of a terminology service to provide this functionality.

subsumedBy(code: Coding | CodeableConcept) : Boolean

When invoked on a Coding-valued element and the given code is Coding-valued, returns true if the source code is equivalent to the given code, or if the source code is subsumed by the given code (i.e. the given code is an ancestor of the source code in a subsumption hierarchy), and false otherwise.

If the Codings are from different code systems, the relationships between the code systems must be well-defined or a run-time error is thrown.

When the source or given elements are CodeableConcepts, returns true if any Coding in the source or given elements is equivalent to or subsumed by the given code.

Note that implementations are encouraged to make use of a terminology service to provide this functionality.

htmlChecks : Boolean

When invoked on an xhtml element returns true if the rules around HTML usage are met, and false if they are not. The return value is undefined (null) on any other kind of element.

2.9.1.6 Changes to operators

~ (Equivalence)

Equivalence works in exactly the same manner, but with the addition that for complex types, equality requires all child properties to be equal, except for "id" elements.

In addition, for Coding values, equivalence is defined based on the code and system elements only. The version, display, and userSelected elements are ignored for the purposes of determining Coding equivalence.

For CodeableConcept values, equivalence is defined as a non-empty intersection of Coding elements, using equivalence. In other words, two CodeableConcepts are considered equivalent if any Coding in one is equivalent to any Coding in the other.

2.9.1.7 Environment variables

The FHIR specification adds support for additional environment variables:

The following environmental values are set for all contexts:

```
%sct        // (string) url for snomed ct
%loinc      // (string) url for loinc
%"vs-[name]" // (string) full url for the provided HL7 value set with id [name]
%"ext-[name]" // (string) full url for the provided HL7 extension with id [name]
%resource // The original resource current context is part of. When evaluating a datatype, this would be the resource
    the element is part of. Do not go past a root resource into a bundle, if it is contained in a bundle.

// Note that the names of the `vs-` and `ext-` constants are quoted (just like paths) to allow "-" in the name.
```

For example:

```
Observation.component.where(code.memberOf(%"vs-observation-vitalsignresult"))
```

This expression returns components that have a code that is a member of the observation-vitalsignresult valueset.

Implementation Note: Implementation Guides are allowed to define their own externals, and implementers should provide some appropriate configuration framework to allow these constants to be provided to the evaluation engine at run-time. E.g.:

```
%"us-zip" = '[0-9]{5}(-[0-9]{4}){0,1}'
```

Authors of Implementation Guides should be aware that adding specific environment variables restricts the use of the FHIRPath to their particular context.

Note that these tokens are not restricted to simple types, and they may have fixed values that are not known before evaluation at run-time, though there is no way to define these kinds of values in implementation guides.

2.9.2 Restricted Subset ("Simple")

This page documents a restricted subset of the FHIRPath language that is used in a few contexts in this specification. When the restricted FHIRPath language subset is in use, the following rules apply:

• All statements SHALL start with the name of the context element (e.g. on a Patient resource, Patient.contact.name.), or SHALL be simply "$this" to refer to the element that has focus

• Operators SHALL NOT be used

• Only the following functions may be used: .resolve() .extension("url") .ofType(type) All other functions SHALL NOT be used

These rules exist to keep processing the path simple to support use of the path by processors that are not backed by a full FHIRPath implementation.

The following locations use this restricted FHIRPath language:

• ElementDefinition.slicing.discriminator.path

• DataRequirement.dateFilter.path

• OperationOutcome.issue.expression

2.9.3 Terminology Service API

Unlike this rest of this page, the FHIR Terminology service API is only draft (Maturity = 0). It will be advanced to a more mature status following the usual Maturity Model for FHIR.

In order to support terminological reasoning in FHIRPath statements, FHIR defines a general %terminologies object that FHIRPath implementations should make available. Calls to this object are passed through a standard FHIR terminology service.

Summary:

```
%terminologes.expand(valueSet, params) : ValueSet
%terminologies.lookup(coded, params) : Parameters
%terminologies.validateVS(valueSet, coded, params) : Parameters
%terminologies.validateCS(codeSystem, coded, params) : Parameters
%terminologies.subsumes(system, coded1, coded2, params) : code
%terminologies.translate(conceptMap, code, params) : Parameters
```

expand

```
%terminologes.expand(valueSet, params) : ValueSet
```

This calls the Terminology Service $expand operation (formal definition).

Parameters:

• valueSet: either an actual ValueSet, or a canonical URL reference to a value set.

• params: a URL encoded string with other parameters for the expand operation (e.g. 'displayLanguage=en&activeOnly=true')

Return Value: a ValueSet with an expansion, or null if an error occurs.

lookup

```
%terminologies.lookup(coded, params) : Parameters
```

This calls the Terminology Service $lookup operation (formal definition).

Parameters:

• coded: either a Coding, a CodeableConcept, or a resource element that is a code

• params: a URL encoded string with other parameters for the lookup operation (e.g. 'date=2011-03-04&displayLanguage=en')

Return Value:

validateVS

```
%terminologies.validateVS(valueSet, coded, params) : Parameters
```

This calls the Terminology Service $validate-code operation on a value set (formal definition).

Parameters:

• valueSet: either an actual ValueSet, or a canonical URL reference to a value set.

• coded: either a Coding, a CodeableConcept, or a resource element that is a code

• params: a URL encoded string with other parameters for the validate-code operation (e.g. 'date=2011-03-04&displayLanguage=en')

Return Value: A Parameters resource with the results of the validation operation.

validateCS

```
%terminologies.validateCS(codeSystem, coded, params) : Parameters
```

This calls the Terminology Service $validate-code operation on a code system (formal definition).

Parameters:

• codeSystem: either an actual CodeSystem, or a canonical URL reference to a code system.

• coded: either a Coding, a CodeableConcept, or a resource element that is a code

• params: a URL encoded string with other parameters for the validate-code operation (e.g. 'date=2011-03-04&displayLanguage=en')

Return Value: A Parameters resource with the results of the validation operation.

subsumes

```
%terminologies.subsumes(system, coded1, coded2, params) : code
```

This calls the Terminology Service $subsumes operation (formal definition).

Parameters:

• system: the URI of a code system within which the subsumption testing occurs

• coded1: A Coding or a resource element that is a code

• coded2: A Coding or a resource element that is a code

• params: a URL encoded string with other parameters for the validate-code operation (e.g. 'version=2014-05-06')

Return Value: a code as specified for the subsumes operation.

translate

```
%terminologies.translate(conceptMap, coded, params) : Parameters
```

This calls the Terminology Service $translate operation (formal definition).

Parameters:

• conceptMap: either an actual ConceptMap, or a canonical URL reference to a value set.

• coded: The source to translate: a Coding or a resource element that is a code

• params: a URL encoded string with other parameters for the validate-code operation (e.g. 'source=http://acme.org/valueset/23&target=http://acme.org/valueset/23')

Return Value: A Parameters resource with the results of the translation operation.

2.10 Version History since DSTU #1

URL: https://hl7.org/fhir/R4/history.html

2.10 Version History since DSTU #1

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative

For background to this version history, see FHIR Version Management Policy.

Major Milestones:

Dec 27, 2018 | 4.0.0 | Release 4 (1st Normative Content + Trial Use Developments)
Feb 21, 2017 | 3.0.0 | Release 3 (STU - Standard for Trial Use)
Oct 24, 2015 | 1.0.0 | DSTU2 (Second Draft Standard for Trial Use)
Sept 30, 2014 | 0.0.82 | DSTU1 (First Draft Standard for Trial Use)

This is the 4th milestone release of FHIR. The only changes tracked here are the changes made after the publication of the first DSTU release. For earlier changes, see the DSTU #1 Version History . Note that a full archive history of everything is available through the HL7 GitHub archives .

Version | Changes

4.0.1 | Technical Corrections to R4: Oct-30, 2019. Corrections to invariants & generated conformance resources, and add ANSI Normative Status Notes The FHIR Management Group has reviewed and approved all the changes, and confirmed that there is no substantive impact on the ANSI normative content of FHIR R4. Note, however, that the first set of changes should be reviewed for impact on non-normative aspects of implementations. Corrections to the specification - resolving internal conflicts, or correcting definitional errors: Add Meta as a Special Type for Open Type Element list, allowing it to be used as a Parameter in Parameters.parameter, and in other places. The $meta operations required their use, but the design did not allow for this. [Task] Change all code system OIDs to use the root 2.16.840.1.113883.4.642.4. instead of 2.16.840.1.113883.4.642.1. due to reassigning OID leaf values without updating the root in the past. This was causing problems for OID based terminology systems [Source] Add new FHIRPath context variable %rootResource, and use it in ref-1 constraint - FHIRPath engines will need to support the new context variable [Task] Fix genomics canonical URLS: http://www.ncbi.nlm.nih.gov/refseq/ -> http://www.ncbi.nlm.nih.gov/refseq, http://www.ncbi.nlm.nih.gov/clinvar/ -> http://www.ncbi.nlm.nih.gov/clinvar, and http://varnomen.hgvs.org/ -> http://varnomen.hgvs.org - these are now corrected back to what they had said in R3 [Task] Fix Binding for FamilyMemberHistory.sex - it had a reference to the administrative gender value set, but it wrongly referred to the one on http://build.fhir.org; corrected to reference the one in the specification [Task] Changes to Constraints and generated definitions (fixing problems in validation, no impact on implementations, except for code generators): Fix expressions for Search parameters SubstanceSpecification-code and StructureDefinition-ext-context [Task] Fix the definition of the Mother's Maiden name search parameter from extension(http://hl7.org/fhir/SearchParameter/patient-extensions-Patient-mothersMaidenName) to extension('http://hl7.org/fhir/StructureDefinition/patient-mothersMaidenName') [Task] Fix the extension context for iso21090-nullFlavor and iso21090-preferred to any Element, not any Resource [Task] Remove MultipleAnd/MultipleOr/Modifiers from generated Search Parameters [Task] Add "ge" as a comparator to generated search parameters [Task] Add extension operationdefinition-allowed-type, and use it to capture the definition of the property.value parameter is captured in the OperationDefinition that describes it [Task] Fix how the type of Element.id and Extension.url is defined in the generated StructureDefinitions [Task] Fix up the definition of ElementDefinition.type.code for special elements (Element.id, Extension.url, [Primitive].value) [Task] Regenerate StructureDefinition snapshots to fix various snapshot and id issues [Task , Task] Documentation Corrections and Clarifications (no implementer impact): Add ANSI notes Fix FMM levels for Device (0 to 2), DeviceRequest (0 to 1), ResearchStudy (0 to 1), and ResearchSubject (0 to 1) [Task] Fix up search parameter table [Task , Task] Fix definition of IssueType.exception [Task] Remove Erroneous Normative Marker on message-events value set [Task] Fix various broken internal links & R3 Comparison links + JSON Schema link in downloads [Task , Task , Task] Add note about markdown and XML signatures (to both XML and Signature pages) [Task] Update language about specification status in summary page [Task] Clarify copyright statement for NUBC derived codes (here and here) Remove spurious paragraph about R3 version status [Task] Add note about FHIRPath Quantity usage [Task] Fix profiling examples [Task] Fix the FHIRPath expressions for the following invariants: sdf-8, sdf-8a, sdf-9, sdf-15, sdf-17, sdf-16, con-3, eld-4, eld-18, ait-1, ait-2, obs-7, bnd-10 and vs-2 + fix the description for eld-8 [Task , Task , Task , Task , Task , Task , Task , Task , Task , Task] Fix the XPath invariant for dom-3 [Source] Fix the definition for entered-in-error for Event, Request and SupplyDelivery [Source] Fix the short descriptions for the following code elements: ActivityDefinition.intent, AppointmentResponse.participantStatus, CarePlan.status, CarePlan.activity.detail.kind, ClinicalImpression.status, CodeSystem.filter.operator, Communication.status, Communication.priority, CommunicationRequest.status, CommunicationRequest.priority, Contract.status, Contract.contentDefinition.publicationStatus, DeviceRequest.status, DeviceRequest.intent, DeviceRequest.priority, DocumentReference.docStatus, GraphDefinition.link.target.compartment.code, Media.status, MedicationDispense.status, MedicationRequest.intent, NutritionOrder.status, NutritionOrder.intent, Procedure.status, RequestGroup.status, RequestGroup.intent, ServiceRequest.status and ServiceRequest.intent - they had been listing the wrong codes [Source] Fix Implicit SNOMED CT Value sets in example concept maps [Task] This technical correction also includes earlier changes made without releasing a new version: Fix external links for the Implementation Guide Registry and the FHIR Registry [Task]

4.0.0 | R4: Dec 27, 2018. First Normative content, with many significant changes Note that between R3 and R4, nearly 3000 change proposals were applied to the specification, including >1000 substantive changes, of which 339 were labeled 'non-compatible'. As such, providing a comprehensive list of changes between specifications is not appropriate. To help see and understand the differences between the specification releases, implementers can: consult the R3/R4 difference analysis review (or use) the R3/R4 transforms Use the link at the foot of every page To help implementers, the more significant changes are listed here: Some content marked Normative using N - see Documentation and Resource indexes. The meaning of normative is described at length in the Rules for Inter-version change. New Resources: BiologicallyDerivedProduct CatalogEntry ChargeItemDefinition DeviceDefinition EffectEvidenceSynthesis EventDefinition Evidence EvidenceVariable ExampleScenario ImmunizationEvaluation InsurancePlan Invoice MedicationKnowledge MedicinalProduct MedicinalProductAuthorization MedicinalProductContraindication MedicinalProductIndication MedicinalProductIngredient MedicinalProductInteraction MedicinalProductManufactured MedicinalProductPackaged MedicinalProductPharmaceutical MedicinalProductUndesirableEffect ObservationDefinition OrganizationAffiliation ResearchDefinition ResearchElementDefinition RiskEvidenceSynthesis SpecimenDefinition SubstancePolymer SubstanceReferenceInformation SubstanceSpecification TerminologyCapabilities VerificationResult Renamed Resources: BodySite -> BodyStructure EligibilityRequest -> CoverageEligibilityRequest EligibilityResponse -> CoverageEligibilityResponse Sequence -> MolecularSequence ReferralRequest & ProcedureRequest -> ServiceRequest Deleted Resources: DeviceComponent ImagingManifest Clarification of the meaning and use of IsModifier Considerable development of the Version Management Policy, and support for multiple FHIR versions (specifically, the FHIR Version parameter) Datatypes: Rework Money (no longer a constraint on Quantity), and add Expression

3.6.0 | Current build after 3.5.0 release

3.5.0 | Release R4 Ballot #1: Sept 20, 2018. R4 ballot #2 Major changes from R4 Ballot 1 (3.3.0): Description Committee + Tasks Pages All Ballots Change the canonical URL for all v2 and v3 CodeSystems and ValueSets (and some FHIR ones too), to http://terminology.hl7.org (from the Unified Terminology Process) Breaking change! (no task: Vocab committee decision) todo Normative / Infrastructure Added reference.type FHIR-I: GF#13543 References between Resources ΔB Clarify definition of is-modifier + impacts on modifier extensions. Note: this lead to breaking changes on a few extensions (changed from modifierExtension to normal extension) FHIR-I: GF#16188 Conformance Rules ΔB, Extensibility ΔB Add mode parameter to /metadata FHIR-I: GF#14444 RESTful API ΔB Enhance/extend rules around changes between versions FHIR-I: GF#13089 Version Management Policy ΔB Add fhirVersion parameter to application/fhir mime type FHIR-I: GF#16165 RESTful API ΔB Add the $versions operation FHIR-I: GF#17009 Capability Statement Operations ΔB Allow exponential form for decimals (with corresponding consequences for precision) FHIR-I: GF#16874 Data Types ΔB, XML ΔB Describe use of exponential form when searching numbers (+ clarifications for precision) FHIR-I: GF#16369 Search ΔB Remove support for operations on historical resources Breaking change! FHIR-I: GF#17258 Operations ΔB Change Money Type to make it simpler Breaking change! FHIR-I: GF#16297 Data Types ΔB Change ElementDefinition.binding.valueSet to only be of type canonical Breaking change! FHIR-I: GF#16055 Element Definition ΔB Remove restriction on Bundle containing multiple versions of the same resource Breaking change! FHIR-I: GF#17085 Bundle ΔB Rename Binary.content to Binary.data and exclude it from summary (which makes it optional) Breaking change! FHIR-I: GF#16998 , GF#16898 Binary Resource ΔB Normative / Conformance + Terminology Remove ValueSet.$expand profile parameter, and add parameters from ExpansionProfile Breaking change! Vocab: GF#16337 & GF#16490 ValueSet.$expand ΔB Remove ValueSet.$expand.limitedExpansion parameter, and document how to use count instead Breaking change! Vocab: GF#16449 ValueSet $expand operation ΔB Move Valueset.extensible to an extension Vocab: GF#16427 ValueSet ΔB Add CapabilityStatement.implementation.custodian FHIR-I: GF#16342 CapabilityStatement ΔB Add CapabilityStatement.imports FHIR-I: GF#14299 CapabilityStatement ΔB Normative / Observation Update definition of subject and add note safety page the focus element OO: GF#16136 Observation ΔB Changed cardinality of Observation.interpretation and Observation.component.interpretation from 0..1 to 0..* OO: GF#16231 Observation ΔB Changed Observation.context to Observation.encounter with type Reference(Encounter). Breaking change! OO: GF#17661 Observation ΔB This is only the major changes. More comprehensive lists can be found in the ballot introduction (later removed). Major changes in the other parts of the specification: New Resources: MedicationKnowledge, DeviceDefinition, ChargeItemDefinition Renamed Resources: OrgnizationRole to OrganizationAffiliation, EligibilityRequest to CoverageEligibilityRequest, EligibilityResponse to CoverageEligibilityResponse, and ProductPlan to InsurancePlan Significant rework around devices - in the Device resource, remove DeviceComponent + new resource DeviceDefinition Update maps/diffs to R3 (see note above) New page listing best practices A sweeping change to code system and value set URLs to align with new terminology infrastructure Add information and operations related to supporting multiple versions of FHIR Rewrite Asynchronous Pattern Operation New security notes about suggested best practices, and new safety notes about deleting records Lots of changes to Financial Resources (especially Contract)

Description | Committee + Tasks | Pages

All Ballots

Change the canonical URL for all v2 and v3 CodeSystems and ValueSets (and some FHIR ones too), to http://terminology.hl7.org (from the Unified Terminology Process) Breaking change! | (no task: Vocab committee decision) | todo

Normative / Infrastructure

Added reference.type | FHIR-I: GF#13543 | References between Resources ΔB

Clarify definition of is-modifier + impacts on modifier extensions. Note: this lead to breaking changes on a few extensions (changed from modifierExtension to normal extension) | FHIR-I: GF#16188 | Conformance Rules ΔB, Extensibility ΔB

Add mode parameter to /metadata | FHIR-I: GF#14444 | RESTful API ΔB

Enhance/extend rules around changes between versions | FHIR-I: GF#13089 | Version Management Policy ΔB

Add fhirVersion parameter to application/fhir mime type | FHIR-I: GF#16165 | RESTful API ΔB

Add the $versions operation | FHIR-I: GF#17009 | Capability Statement Operations ΔB

Allow exponential form for decimals (with corresponding consequences for precision) | FHIR-I: GF#16874 | Data Types ΔB, XML ΔB

Describe use of exponential form when searching numbers (+ clarifications for precision) | FHIR-I: GF#16369 | Search ΔB

Remove support for operations on historical resources Breaking change! | FHIR-I: GF#17258 | Operations ΔB

Change Money Type to make it simpler Breaking change! | FHIR-I: GF#16297 | Data Types ΔB

Change ElementDefinition.binding.valueSet to only be of type canonical Breaking change! | FHIR-I: GF#16055 | Element Definition ΔB

Remove restriction on Bundle containing multiple versions of the same resource Breaking change! | FHIR-I: GF#17085 | Bundle ΔB

Rename Binary.content to Binary.data and exclude it from summary (which makes it optional) Breaking change! | FHIR-I: GF#16998 , GF#16898 | Binary Resource ΔB

Normative / Conformance + Terminology

Remove ValueSet.$expand profile parameter, and add parameters from ExpansionProfile Breaking change! | Vocab: GF#16337 & GF#16490 | ValueSet.$expand ΔB

Remove ValueSet.$expand.limitedExpansion parameter, and document how to use count instead Breaking change! | Vocab: GF#16449 | ValueSet $expand operation ΔB

Move Valueset.extensible to an extension | Vocab: GF#16427 | ValueSet ΔB

Add CapabilityStatement.implementation.custodian | FHIR-I: GF#16342 | CapabilityStatement ΔB

Add CapabilityStatement.imports | FHIR-I: GF#14299 | CapabilityStatement ΔB

Normative / Observation

Update definition of subject and add note safety page the focus element | OO: GF#16136 | Observation ΔB

Changed cardinality of Observation.interpretation and Observation.component.interpretation from 0..1 to 0..* | OO: GF#16231 | Observation ΔB

Changed Observation.context to Observation.encounter with type Reference(Encounter). Breaking change! | OO: GF#17661 | Observation ΔB

3.4.0 | Current build after 3.3.0 release

3.3.0 | Release R4 Ballot #1: Apr 3, 2018. R4 ballot #1. Major changes from R3: Mark all content with a standards status - Normative, Trial use, Draft, or external Many minor & major changes to data types, resources, operations New Resources: BiologicallyDerivedProduct, CatalogEntry, EventDefinition, ExampleScenario, ImmunizationEvaluation, Invoice, ItemInstance, ObservationDefinition, OccupationalData, OrganizationAffiliation, SpecimenDefinition, TerminologyCapabilities, UserSession and VerificationResult, and a new set of resources for medication knowledge management: MedicinalProduct, MedicinalProductAuthorization, MedicinalProductClinicals , MedicinalProductDeviceSpec , MedicinalProductIngredient, MedicinalProductPackaged, MedicinalProductPharmaceutical , InsurancePlan, SubstancePolymer , SubstanceReferenceInformation and SubstanceSpecification. Renamed Resources: ProcedureRequest to ServiceRequest & BodySite to BodyStructure Removed Resources: DataElement, ImagingManifest and ServiceDefinition New Pages: Asynchronous Use, Persistence/Data bases, FiveWs Pattern and Multi-language support

3.2.0 | Release R4 Draft: Dec 21, 2017. R4 draft ballot. Major changes: Ensure that all pages have a standards status, and enforce standards status throughout. Add dependency analysis Many minor changes to data types, resources, operations New Resources: CatalogEntry, SpecimenDefinition, ObservationDefinition, ExampleScenario, TerminologyCapabilities, UserSession, Invoice, InsurancePlan, OccupationalData, OrganizationAffiliation, VerificationResult, MedicinalProduct, MedicinalProductAuthorization MedicinalProductClinicals , MedicinalProductIngredient, MedicinalProductPackaged, MedicinalProductPharmaceutical MedicinalProductDeviceSpec , SubstanceSpecification, SubstancePolymer , SubstanceReferenceInformation, BiologicallyDerivedProduct, ItemInstance Add draft support for bulk data (Asynchronous API + nd-json format)

3.1.0 | R4 development begins

3.0.0 | Release 3: Mar 21, 2017. STU3 This is the full FHIR Release 3 (R3). R3 is a complete overhaul of the specification from R2, with over 2400 change proposals process, and applied in R3. Of those changes, 380+ are labeled as a breaking change. Given this level of change, there is no useful way to present a single change list. Users can use the R2/R3 difference comparison, the R2 <-> R3 transforms, or the "Compare to R2" link at the foot of every page to help visualise the differences between R2 and R3. Many of the changes made in this release (since 1.8.0) are in response to Quality Assurance processes in HL7, with a goal of readying the specification for Normative processes in R4: Reorder many resources and improve their definitions and bindings (Quality Assurance work) Add draft resources AdverseEvent and GraphDefinition Change to the way discriminators work in profiles (for profile/tooling authors) Finalization of the Turtle (RDF) format, along with JSON-LD (later removed) Consolidation of Vresion management issues into a single Version Policy page, and update to current policy Add rules about how missing data in search Roll up: Of all the many changes (thousands) between R2 and R3, these are the most significant: Change the FHIR mime type to application/fhir+xml|json instead of application/xml|json+fhir (breaking change, for conformance to W3C+IETF rules) Rename the "Conformance" resource to "CapabilityStatement" and Deprecate use of the OPTIONS command to retrieve the CapabilityStatement Search Changes around errors in search parameters, missing data, sorting Change HumanName.family from 0..* to 0..1 CodeSystem now separated out from ValueSet

1.9.0 | FHIR Rolling Build, 2016 Dec onwards

1.8.0 | FHIR QA Release, Dec 6, 2016 STU 3 Candidate, and also the basis for several implementation guide ballots and the San Antonio Jan 2017 Connectathon Breaking Changes Summary: Change HumanName.family from 0..* to 0..1 Rename MedicationOrder and DiagnosticOrder to MedicationRequest and DiagnosticRequest and make a series of changes to align with the Request Pattern Add cross resource search + rework composite parameter expressions Harmonise resource metadata across the Terminology, Conformance, and Knowledge resources Change the way versioned references to canonical URLs are handled REST - servers must return a location header on a create Condition status elements reworked These are only the more significant changes, there were many 100s of changes made in response to ballot comments and ongoing implementation experience. These include breaking changes to may resources. Structural changes are summarized here, and Transforms between DSTU 2 and STU 3 are provided for many resources.

1.6.0 | FHIR STU3 ballot, Aug 11, 2016 STU 3 ballot version, and also the basis for the Baltimore Sept 2016 Connectathon Publication Changes: Move the US Realm implementation guides out of the specification (see http://www.fhir.org/guides/registry) Rework the home page, and introduce modules as a way to navigate the specification + add significant new implementer advice (e.g. "Getting Started") Change the Navigation menu to focus on indexes to the content in the specification Move FHIRPath expression language to its long term home Technical Changes: RESTful API Change the FHIR mime type to application/fhir+xml|json instead of application/xml|json+fhir (breaking change, for conformance to W3C+IETF rules) Add new uses for the Prefer header (return OperationOutcome, and manage behavior related to unknown / unsupported search parameters Deprecate use of the OPTIONS command to retrieve the CapabilityStatement Add support for conditional references to the transaction interaction Add reverse chaining Formats: No change to XML + JSON formats. Add Turtle Format, and related validation tools For Data Types and Resources, there is formal difference analysis from DSTU2 (also found throughout the specification), and transforms between DSTU 2 and STU 3 are provided for many resources

1.5.0 | FHIR Current Build Update, July 8 2016 Update current version to 1.5 to prevent confusion with implementations still running the May 2016 version. There are many changes in this version from 1.4.0, but no formal list is provided. A formal list will be provided for 1.6 (as a diff to 1.4)

1.4.0 | FHIR Connectathon 12 Snapshot, Mar 30 2016 Frozen base for Connectathon 12 & For Comment ballots: FHIR API and Serialization Format Enhancement The _sort parameter has been reworked New or extended search parameters for read, search or history: _type, _at, _summary, _elements Change Turtle representation - now ready for trial Conformance resources with significant breaking changes and behavior: StructureDefinition - Type Handling: Changes to how the structure definition types work: baseType replaces constrainedType (now type) & baseDefinition replaces base CodeSystem now separated out from ValueSet: CodeSystem is now a first class resource type in order to support use throughout the FHIR eco-system, such as to support value set expansion and validation. It is intended to be used for distributing the smaller ad-hoc code systems that are ubiquitously encountered throughout the healthcare process. Other resources with significant breaking changes and behavior: AllergyIntolerance Appointment AuditEvent Claim ClaimResponse Device ConceptMap Conformance (now CapabilityStatement) Contract CoverageEligibilityRequest CoverageEligibilityResponse Device DiagnosticRequest ExplanationOfBenefit Library Measure Medication MedicationAdministration MedicationStatement Observation: the introduction of the Vital Signs Profile (proposed to be mandatory) OperationDefinition Patient PaymentNotice PaymentReconciliation Practitioner Provenance ReferralRequest Schedule

MolecularSequence Slot StructureDefinition TestScript ValueSet Add new draft resources: CareTeam CodeSystem CompartmentDefinition Linkage MeasureReport PractitionerRole Protocol StructureMap Task

1.2.0 | FHIR Connectathon 11 Snapshot, Dec 11 2015 Frozen base for Connectathon 11: Remove GuidanceRequest Add new draft resources: MolecularSequence, ExpansionProfile Modifications to Financial Resource & TestScript resource Note: this version is temporary, and was removed after Connectathon 11 was complete

1.1.0 | GAO Ballot + technical corrections, Dec 2 2015 A ballot publication for the GAO Ballot that also includes: Various technical corrections to the generated snapshots Populate FHIRPath expressions in extensions for all search parameters and invariants Add new draft resources: GuidanceRequest, GuidanceResponse, ModuleMetadata, ModuleDefinition, Library, DecisionSupportServiceModule, DecisionSupportRule, OrderSet, Measure Major restructure of Questionnaire

1.0.2 | Technical Correction 1, Oct 24 2015 A series of technical corrections to the specification following extensive review: Corrections to Extension cardinalities in implementation guides Corrections in the conformance resources that support the specifications Correct several erroneous invariants Various typos, broken links, and fixes in examples For a comprehensive list of corrections, see the Task list for FHIR DSTU2 Technical Correction 1

1.0.1 | DSTU 2, Sept 23 2015 Changes of significance during the QA process: Remove the Clinical Quality Improvement Framework (CQIF) from this published version made fixes to generated schematrons updated generated comformance resources (StructureDefinitions and SearchParameters) so they were consistent with the specification Many spelling / grammar / broken link fixes

1.0.0 | DSTU 2 QA Preview, Aug 31 2015 This version had extensive change as a result of the May 2015 DSTU ballot, ongoing testing, and the open change proposals (1317 tasks). The extent of the changes is best illustrated by the number of the list of changes labeled 'breaking change' - 158 changes of 1317 total tasks. Below is a list of the most important changes: General: introduced the maturity framework RESTful API: add batch, several clarifications around versioning & transactional integrity, changed Bundle URL resolution rules Search: changed the way <> etc.works, added _list parameter, changed rules around contained and included resources Formats: added a note about whitespace in XML, added code generation schemas Data Types: New data types: markdown, Annotation, changed data types: Coding, Quantity, Signature, Timing, Address, ContactPoint changes to ElementDefinition: add base, make type.profile repeat, remove invariant.name and replace with invariant.requirements, remove binding.name, add min/max value Resources: New resources: Account, ImplementationGuide, TestScript renamed: Contraindication -> DetectedIssue, MedicationPrescription -> MedicationRequest, QuestionnaireAnswers -> QuestionnaireResponse removed: Supply changed: almost all resources - too many to list (1317 tasks worth of changes) - add, remove elements, change types, references, definitions & value sets, re-order elements, provide much more documentation and new examples Implementation Guide: Move Argonaut content out

0.5.0 | DSTU Ballot, May 2015 This version had extensive change as a result of the January 2015 Draft ballot, ongoing testing, and the open change proposals (over 800 tasks). The list below is a summary of the major changes to resource content. It shows only a limited number of the overall changes. Enumerations All spaces removed Extensive content changes not noted here New Data Types unsignedInt positiveInt Signature Meta Changed Data Types Coding - remove valueSet property Attachment - add creation Identifier - replace label with type Timing - major rework of content ElementDefinition - add label, code, rename 'formal' to definition, rename synonym to alias, add language to mapping, remove conformance and isExtensible and replace with strength New Resources BodySite Claim Removed Resources CarePlan2 -> collapsed into CarePlan FamilyHistory -> broken up into FamilyMemberHistory InstitutionalClaim, OralHealthClaim, PharmacyClaim, ProfessionalClaim, VisionClaim -> collapsed into Claim Other - use Basic instead PendedRequest,Readjudicate, Reversal, StatusRequest, StatusResponse - use ProcessRequest/Response instead SupportingDocumentation - use DocumentManifest instead ProcessRequest, ProcessResponse - use Task instead Renamed Resources Alert -> Flag: 'alert' made people think it was an action like an alarm SecurityEvent -> AuditEvent: it wasn't just for security purposes ClinicalAssessment -> ClinicalImpression: people got confused with 'assessment' tools like APGAR score Profile -> StructureDefinition: 'Profile' is the process, a package of statements Changes Inside Resources Parameters - allow parameter.part to contain a resource AllergyIntolerance - rename subject to patient Appointment - remove lastModifiedBy/lastModified, add location AppointmentResponse - remove lastModifiedBy/lastModified, add rename individual to actor AuditEvent - add .event.purposeOfEvent, participant.location, .policy, and .purposeOfUse Bundle - major reorganization CarePlan - pull goal out + other reorganization ClinicalImpression - add status, replace careplan & referral with trigger, rename diagnosis to finding, make plan 0..*, Composition - change .section.content to refer to List only, not any ConceptMap - change identifier to url, add useContext, change telecom to contact, Condition - rename subject to patient, rename status to clinicalStatus, change to bodySite = code or Reference(BodySite), rename .codeableConcept to .code Conformance (now CapabilityStatement) - change identifier to url, add useContext, change telecom to contact, add requirements and copyright, add support for conditional operations, Contract - extensive rewrite Coverage - add bin, subscriberId DataElement (now StructureDefinition) - total rewrite to use ElementDefinition Device - add status, manufactureDate DeviceMetric - rename operationalState to operationalStatus, add measurementMode, rename calibrationInfo to calibration, change color to an enumerations DeviceRequest/DeviceUseStatement - change to bodySite = code or Reference(BodySite) DiagnosticRequest- change to bodySite = code or Reference(BodySite) DiagnosticReport - add encounter DocumentManifest - add options for how content is referred to DocumentReference - add format, remove policyManager, make content : Attachment, and remove several related attributes, remove service reference and add context.practiceSetting, sourcePatientInfo, and related Encounter - add incomingReferralRequest, allow reason to repeat, rename diet to dietPreference EpisodeOfCare - rename currentStatus to status, allow referralRequest to repeat, Flag - rename subject to patient, change from note to code Goal - add targetDate, statusDate, author, priority HealthcareService - extensive rewrite ImagingObjectSelection - remove retrieveAETitle, rename retrieveUrl to url, add frames ImagingStudy - add laterality, change url to attachment Immunization - add encounter, rename subject to patient, rename refusedIndicator to wasNotGiven, rename refusalReason to reasonNotGiven ImmunizationRecommendation - rename subject to patient List - add title, status, change ordered to orderedBy, add note Location - remove status Media - remove created (-> Attachment) Medication - add batch MedicationAdministration - add reasonGiven, note, text. remove timing & maxDosePerPeriod MedicationDispense - collapse to a single dispense, add daysSupply, note and substitution, change quantity to allow range MedicationRequest - add note, change quantity to allow range, MedicationStatement - add informationSource, status, dateAsserted, replace whenGiven with effective[x], remove device, add dosage.text NamingSystem - add date, publisher, NutritionOrder - extensive rewrite Observation - change name to code, allow more types of value[x], change type of dataAbsentReason, change to bodySite = code or Reference(BodySite), allow identifier to repeat, add device, OperationDefinition - change identifier to url, add useContext, change telecom to contact, change name to title, add reuqirements, idempotent, OperationOutcome - change type of .issue.type OrderResponse - rename code to orderStatus Organization - remove location and contact.gender Patient - communication to allow 'preferred' Person - rename other to target Practitioner - change type of birthDate, allow multiple roles per practitioner Procedure - add status and category, change to bodySite = code or Reference(BodySite), allow date to be period too, add location, change followUp to code 0..*, add device tracking ServiceRequest - change to bodySite = code or Reference(BodySite) Provenance - change integritySignature to signature & make it a type, allow reference by Reference as well as URI Questionnaire - add telecom Schedule - move lastModified SearchParameter - change telecom to contact, add status, experimental, date, Slot - move lastModified Specimen - change source to parent, change to bodySite = code or Reference(BodySite) StructureDefinition - complete rewrite Subscription - change type of tag, reanme url to

endPoint, ValueSet - change identifier to url, add useContext, change telecom to contact, replace purpose with useContext, add requirements, rename stableDate to lockedDate, change type of expansion.identifier, add expansion parameters

0.4.0 | Draft For Comment, January 2015 Ballot Breaking Changes (full list): Replace atom and taglist with a native Bundle format (3728 , 3558 , 2889) (and also Binary) JSON: change how extensions are represented (3471) RESTful API: change how version specific upgrades work (3451) DataTypes: Rename Schedule to Timing (3536 , 3236) Rename Contact to ContactPoint (3533) and swap order of elements (3108)) Address - change zip to postCode (2888) Quantity: Correct schema spelling for "QuantityCompararator" (3531) Change allowable values for the id type to include capital letters, and allow up to 64 chars (3750) Restructure Profile - only one structure, and pull ExtensionDefinition out of Profile (3647, 3498), and pull SearchParameter out (3626) Profile: allow 0..* discriminator (3131), and change the way discriminators work across resource boundaries (3124) + generate multiple types properly (2856) remove _validate interaction, and replace with $validate operation (3686) Patient: separate birth time from birthDate (3731), Change Administrative Gender from a CodableConcept to a Code. Also fixed the values as male|female|other|unknown with mappings to v2 and v3 (3070) DocumentReference: change encoding of Hash to Base64 (3291) Group: rename header to title (3126) Condition: split relatedItem into two (3111) Questionnaire: drop questionnaire.group.question.remarks (3255) and move omitReason from extension to base resource (3260) QuestionnaireResponse: allow multiple answers (3146) ValueSet: replace ValueSet.compose.include.code with ValueSet.compose.include.concept (3258), added new rules about expansion content (3138) Media: Rename element 'dateTime' to 'created' (3174) and length to duration (2866) Remove DeviceObservationReport and Query Collapse AdverseReaction into AllergyIntolerance Appointment changes - individual field renamed to actor, and added mappings to HL7 v2 and HL7 v3 FamilyMemberHistory combined with List replaces FamilyHistory (with corresponding updates to related profiles) Flag replaces Alert including improved clarification of how it is used and replacement of "note" with "code" CarePlan significantly refactored including splitting Goal out as a distinct resource, moving elements between activity and detail, introduction of several new elements and supported relationship types New Resources: Appointment AppointmentResponse Basic ClaimResponse ClinicalImpression Communication CommunicationRequest Contract Contraindication Coverage CoverageEligibilityRequest CoverageEligibilityResponse DataElement (now StructureDefinition) DeviceMetric DeviceRequest DeviceUseStatement EnrollmentRequest EnrollmentResponse EpisodeOfCare ExplanationOfBenefit StructureDefinition Goal HealthcareService ImagingObjectSelection InstitutionalClaim NamingSystem NutritionOrder OperationDefinition OralHealthClaim PaymentNotice PaymentReconciliation Person PharmacyClaim ServiceRequest ProfessionalClaim QuestionnaireResponse ReferralRequest RiskAssessment SearchParameter Schedule Slot Subscription SupportingDocumentation VisionClaim VisionPrescription New Implementation Guides (see discussion of status) Argonaut Project Data Access Framework Structured Data Capture & Structured Data Capture - Data Element Exchange US Laboratory Order, Report & Report to Public Health EHRS Functional model - Record Lifecycle Events

0.3.0 | Renamed Namespace to NamingSystem Split QuestionnaireResponse from Questionnaire and significantly revamped the Questionnaire in response to feedback from the Connectathon. Added DataElement (now StructureDefinition) resource (has been previously discussed as ObservationDefinition) Defined Subscription resource for evaluation Add time data type Define RPC-type operations on the RESTful interface, and add Operation Definition Defined operations on several resources including ConceptMap, Questionnaire and ValueSet Conformance (now CapabilityStatement): Change Conformance.rest.operation to Conformance.rest.interaction and Conformance.rest.resource.operation to Conformance.rest.resource.interaction, and add Conformance.rest.operation to point to Operation Definition Profile: add Profile.url, Profile.structure.snapshot and Profile.structure.differential, and remove query definition (now in OperationDefinition) Add pages for LOINC, RxNorm, and SNOMED CT Significant rework for ConceptMap so it can be used to map between structures as well as value sets Add Contraindication and Risk Assessment and added examples to other resources in support of these Add Referral Request Add supportingInformation to DiagnosticRequest Add fulfills to Encounter Add date and age[x] to FamilyHistory Change cardinality of Location.identifier to 0..* Change cardinality of Practitioner.address to 0..* Add Observation.encounter to 0..* Temporarily added content for the Structured Data Capture implementation guide , including profiles on several resources, conformance instances, etc. This content will be moved to an external specification once the necessary tooling is in place to support that form of publication.

0.2.1 | Minor new optional elements on value set for metadata, new extensions for all the rest of the VSD project metadata, formal profile to express basic minimum metadata for value set

0.2.0 | Namespace: adjustments based on Grahame's feedback

0.1.0 | Add Appointment, Appointment Response, Schedule and Slot Add Namespace

Note: a useful tool for displaying the differences between pages is the W3C HTML Diff engine .

2.11 Differences from Release 3

URL: https://hl7.org/fhir/R4/diff.html

2.11 Differences from Release 3

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative

This is an automatically generated list of all the structural differences between FHIR DSTU3 and this version. The following possible changes are checked when preparing this list (and others throughout the specification):

• Addition or deletion of elements, or name change to elements

• Changes to minimum and maximum cardinality

• Changes to allowed types

• Changes to bindings for Extensible/Required bindings, and changes to codes if it's a required binding on a code element

• Changes to the is-modifier status of an element

• Changes to the default value for an element (default values were removed after Release 3, but changes are noted for Release 3 to Release 4)

This analysis only considers changes to data types and resources. Comparison of extensions, profiles and value sets will be added in a future version.

See also the version transforms and the note about version specific extensions.

This analysis is available as XML or JSON.

Types
Address
Address.use | Change value set from http://hl7.org/fhir/ValueSet/address-use to http://hl7.org/fhir/ValueSet/address-use|4.0.1
Address.type | Change value set from http://hl7.org/fhir/ValueSet/address-type to http://hl7.org/fhir/ValueSet/address-type|4.0.1
Age
Age | Add Binding http://hl7.org/fhir/ValueSet/age-units (extensible), max =http://hl7.org/fhir/ValueSet/all-time-units
Annotation
Annotation.text | Type changed from string to markdown
Attachment
Attachment.contentType | Change value set from http://hl7.org/fhir/ValueSet/mimetypes to http://hl7.org/fhir/ValueSet/mimetypes|4.0.1
Attachment.language | Change binding strength from extensible to preferred
Attachment.url | Type changed from uri to url
BackboneElement | No Changes
CodeableConcept | No Changes
Coding | No Changes
ContactDetail | No Changes
ContactPoint
ContactPoint.system | Change value set from http://hl7.org/fhir/ValueSet/contact-point-system to http://hl7.org/fhir/ValueSet/contact-point-system|4.0.1
ContactPoint.use | Change value set from http://hl7.org/fhir/ValueSet/contact-point-use to http://hl7.org/fhir/ValueSet/contact-point-use|4.0.1
Contributor
Contributor.type | Change value set from http://hl7.org/fhir/ValueSet/contributor-type to http://hl7.org/fhir/ValueSet/contributor-type|4.0.1
Count | No Changes
DataRequirement
DataRequirement.type | Change value set from http://hl7.org/fhir/ValueSet/all-types to http://hl7.org/fhir/ValueSet/all-types|4.0.1
DataRequirement.profile | Type changed from uri to canonical(StructureDefinition)
DataRequirement.subject[x] | Added Element
DataRequirement.codeFilter.path | Min Cardinality changed from 1 to 0
DataRequirement.codeFilter.searchParam | Added Element
DataRequirement.codeFilter.valueSet | Renamed from valueSet[x] to valueSet Add Type canonical(ValueSet) Remove Types string, Reference(ValueSet)
DataRequirement.codeFilter.code | Added Element
DataRequirement.dateFilter.path | Min Cardinality changed from 1 to 0
DataRequirement.dateFilter.searchParam | Added Element
DataRequirement.limit | Added Element
DataRequirement.sort | Added Element
DataRequirement.sort.path | Added Mandatory Element
DataRequirement.sort.direction | Added Mandatory Element
DataRequirement.codeFilter.valueCode | deleted
DataRequirement.codeFilter.valueCoding | deleted
DataRequirement.codeFilter.valueCodeableConcept | deleted
Distance
Distance | Add Binding http://hl7.org/fhir/ValueSet/distance-units (extensible), max =http://hl7.org/fhir/ValueSet/all-distance-units
Dosage
Dosage.doseAndRate | Added Element
Dosage.doseAndRate.type | Added Element
Dosage.doseAndRate.dose[x] | Added Element
Dosage.doseAndRate.rate[x] | Added Element
Dosage.dose[x] | deleted
Dosage.rate[x] | deleted
Duration
Duration | Add Binding http://hl7.org/fhir/ValueSet/duration-units (extensible), max =http://hl7.org/fhir/ValueSet/all-time-units
Element | No Changes
ElementDefinition
ElementDefinition.representation | Change value set from http://hl7.org/fhir/ValueSet/property-representation to http://hl7.org/fhir/ValueSet/property-representation|4.0.1
ElementDefinition.sliceIsConstraining | Added Element
ElementDefinition.slicing.discriminator.type | Change value set from http://hl7.org/fhir/ValueSet/discriminator-type to http://hl7.org/fhir/ValueSet/discriminator-type|4.0.1
ElementDefinition.slicing.ordered | Default Value "false" removed
ElementDefinition.slicing.rules | Change value set from http://hl7.org/fhir/ValueSet/resource-slicing-rules to http://hl7.org/fhir/ValueSet/resource-slicing-rules|4.0.1
ElementDefinition.type.profile | Max Cardinality changed from 1 to * Type changed from uri to canonical(StructureDefinition | ImplementationGuide)
ElementDefinition.type.targetProfile | Max Cardinality changed from 1 to * Type changed from uri to canonical(StructureDefinition | ImplementationGuide)
ElementDefinition.type.aggregation | Change value set from http://hl7.org/fhir/ValueSet/resource-aggregation-mode to http://hl7.org/fhir/ValueSet/resource-aggregation-mode|4.0.1
ElementDefinition.type.versioning | Change value set from http://hl7.org/fhir/ValueSet/reference-version-rules to http://hl7.org/fhir/ValueSet/reference-version-rules|4.0.1 Default Value "either" removed

ElementDefinition.defaultValue[x] | Add Types canonical, url, uuid, ContactDetail, Contributor, DataRequirement, Expression, ParameterDefinition, RelatedArtifact, TriggerDefinition, UsageContext, Dosage
ElementDefinition.fixed[x] | Add Types canonical, url, uuid, ContactDetail, Contributor, DataRequirement, Expression, ParameterDefinition, RelatedArtifact, TriggerDefinition, UsageContext, Dosage
ElementDefinition.pattern[x] | Add Types canonical, url, uuid, ContactDetail, Contributor, DataRequirement, Expression, ParameterDefinition, RelatedArtifact, TriggerDefinition, UsageContext, Dosage
ElementDefinition.example.value[x] | Add Types canonical, url, uuid, ContactDetail, Contributor, DataRequirement, Expression, ParameterDefinition, RelatedArtifact, TriggerDefinition, UsageContext, Dosage
ElementDefinition.constraint.severity | Change value set from http://hl7.org/fhir/ValueSet/constraint-severity to http://hl7.org/fhir/ValueSet/constraint-severity|4.0.1
ElementDefinition.constraint.expression | Min Cardinality changed from 1 to 0
ElementDefinition.constraint.source | Type changed from uri to canonical(StructureDefinition)
ElementDefinition.mustSupport | Default Value "false" removed
ElementDefinition.isModifier | Default Value "false" removed
ElementDefinition.isModifierReason | Added Element
ElementDefinition.isSummary | Default Value "false" removed
ElementDefinition.binding.strength | Change value set from http://hl7.org/fhir/ValueSet/binding-strength to http://hl7.org/fhir/ValueSet/binding-strength|4.0.1
ElementDefinition.binding.valueSet | Renamed from valueSet[x] to valueSet Add Type canonical(ValueSet) Remove Types uri, Reference(ValueSet)
ElementDefinition.mapping.language | Change value set from http://hl7.org/fhir/ValueSet/mimetypes to http://hl7.org/fhir/ValueSet/mimetypes|4.0.1
Expression | Added Type
Extension
Extension.value[x] | Add Types canonical, url, uuid, ContactDetail, Contributor, DataRequirement, Expression, ParameterDefinition, RelatedArtifact, TriggerDefinition, UsageContext, Dosage
HumanName
HumanName.use | Change value set from http://hl7.org/fhir/ValueSet/name-use to http://hl7.org/fhir/ValueSet/name-use|4.0.1
Identifier
Identifier.use | Change value set from http://hl7.org/fhir/ValueSet/identifier-use to http://hl7.org/fhir/ValueSet/identifier-use|4.0.1
MarketingStatus | Added Type
Meta
Meta.source | Added Element
Meta.profile | Type changed from uri to canonical(StructureDefinition)
Money
Money.value | Added Element
Money.currency | Added Element
MoneyQuantity | Added Type
Narrative
Narrative.status | Change value set from http://hl7.org/fhir/ValueSet/narrative-status to http://hl7.org/fhir/ValueSet/narrative-status|4.0.1
ParameterDefinition
ParameterDefinition.use | Change value set from http://hl7.org/fhir/ValueSet/operation-parameter-use to http://hl7.org/fhir/ValueSet/operation-parameter-use|4.0.1
ParameterDefinition.type | Change value set from http://hl7.org/fhir/ValueSet/all-types to http://hl7.org/fhir/ValueSet/all-types|4.0.1
ParameterDefinition.profile | Type changed from Reference(StructureDefinition) to canonical(StructureDefinition)
Period | No Changes
Population | Added Type
ProdCharacteristic | Added Type
ProductShelfLife | Added Type
Quantity
Quantity.comparator | Change value set from http://hl7.org/fhir/ValueSet/quantity-comparator to http://hl7.org/fhir/ValueSet/quantity-comparator|4.0.1
Range | No Changes
Ratio | No Changes
Reference
Reference.type | Added Element
RelatedArtifact
RelatedArtifact.type | Change value set from http://hl7.org/fhir/ValueSet/related-artifact-type to http://hl7.org/fhir/ValueSet/related-artifact-type|4.0.1
RelatedArtifact.label | Added Element
RelatedArtifact.citation | Type changed from string to markdown
RelatedArtifact.url | Type changed from uri to url
RelatedArtifact.resource | Type changed from Reference(Resource) to canonical(Resource)
SampledData
SampledData.factor | Default Value "1" removed
SampledData.data | Min Cardinality changed from 1 to 0
Signature
Signature.who | Renamed from who[x] to who Remove Type uri
Signature.onBehalfOf | Renamed from onBehalfOf[x] to onBehalfOf Remove Type uri
Signature.targetFormat | Added Element
Signature.sigFormat | Added Element

Signature.data | Renamed from blob to data
Signature.contentType | deleted
SimpleQuantity | No Changes
SubstanceAmount | Added Type
Timing
Timing.repeat.count | Type changed from integer to positiveInt
Timing.repeat.countMax | Type changed from integer to positiveInt
Timing.repeat.durationUnit | Change value set from http://hl7.org/fhir/ValueSet/units-of-time to http://hl7.org/fhir/ValueSet/units-of-time|4.0.1
Timing.repeat.frequency | Type changed from integer to positiveInt Default Value "1" removed
Timing.repeat.frequencyMax | Type changed from integer to positiveInt
Timing.repeat.periodUnit | Change value set from http://hl7.org/fhir/ValueSet/units-of-time to http://hl7.org/fhir/ValueSet/units-of-time|4.0.1
Timing.repeat.dayOfWeek | Change value set from http://hl7.org/fhir/ValueSet/days-of-week to http://hl7.org/fhir/ValueSet/days-of-week|4.0.1
Timing.repeat.when | Change value set from http://hl7.org/fhir/ValueSet/event-timing to http://hl7.org/fhir/ValueSet/event-timing|4.0.1
TriggerDefinition
TriggerDefinition.type | Change value set from http://hl7.org/fhir/ValueSet/trigger-type to http://hl7.org/fhir/ValueSet/trigger-type|4.0.1
TriggerDefinition.name | Renamed from eventName to name
TriggerDefinition.timing[x] | Renamed from eventTiming[x] to timing[x]
TriggerDefinition.data | Renamed from eventData to data Max Cardinality changed from 1 to *
TriggerDefinition.condition | Added Element
UsageContext
UsageContext.code | Change code system for extensibly bound codes from "http://hl7.org/fhir/usage-context-type" to "http://terminology.hl7.org/CodeSystem/usage-context-type"
UsageContext.value[x] | Add Type Reference(PlanDefinition | ResearchStudy | InsurancePlan | HealthcareService | Group | Location | Organization)
base64Binary | (No Changes)
boolean | (No Changes)
canonical | Added Type
code | (No Changes)
date | (No Changes)
dateTime | (No Changes)
decimal | (No Changes)
id | (No Changes)
instant | (No Changes)
integer | (No Changes)
markdown | (No Changes)
oid | (No Changes)
positiveInt | (No Changes)
string | (No Changes)
time | (No Changes)
unsignedInt | (No Changes)
uri | (No Changes)
url | Added Type
uuid | (No Changes)
xhtml | (No Changes)
Resources
Account
Account.status | Min Cardinality changed from 0 to 1 Change value set from http://hl7.org/fhir/ValueSet/account-status to http://hl7.org/fhir/ValueSet/account-status|4.0.1
Account.subject | Max Cardinality changed from 1 to * Type Reference: Added Target Type PractitionerRole
Account.servicePeriod | Renamed from period to servicePeriod
Account.partOf | Added Element
Account.active | deleted
Account.balance | deleted
ActivityDefinition
ActivityDefinition.subtitle | Added Element
ActivityDefinition.status | Change value set from http://hl7.org/fhir/ValueSet/publication-status to http://hl7.org/fhir/ValueSet/publication-status|4.0.1
ActivityDefinition.experimental | No longer marked as Modifier
ActivityDefinition.subject[x] | Added Element
ActivityDefinition.author | Added Element
ActivityDefinition.editor | Added Element
ActivityDefinition.reviewer | Added Element
ActivityDefinition.endorser | Added Element
ActivityDefinition.library | Type changed from Reference(Library) to canonical(Library)
ActivityDefinition.kind | Change value set from http://hl7.org/fhir/ValueSet/resource-types to http://hl7.org/fhir/ValueSet/request-resource-types|4.0.1
ActivityDefinition.profile | Added Element
ActivityDefinition.intent | Added Element

ActivityDefinition.priority | Added Element
ActivityDefinition.doNotPerform | Added Element
ActivityDefinition.timing[x] | Add Types Age, Duration
ActivityDefinition.participant.type | Change value set from http://hl7.org/fhir/ValueSet/action-participant-type to http://hl7.org/fhir/ValueSet/action-participant-type|4.0.1
ActivityDefinition.specimenRequirement | Added Element
ActivityDefinition.observationRequirement | Added Element
ActivityDefinition.observationResultRequirement | Added Element
ActivityDefinition.transform | Type changed from Reference(StructureMap) to canonical(StructureMap)
ActivityDefinition.dynamicValue.path | Min Cardinality changed from 0 to 1
ActivityDefinition.dynamicValue.expression | Min Cardinality changed from 0 to 1 Type changed from string to Expression
ActivityDefinition.contributor | deleted
ActivityDefinition.dynamicValue.description | deleted
ActivityDefinition.dynamicValue.language | deleted
AdverseEvent
AdverseEvent.actuality | Added Mandatory Element
AdverseEvent.category | Max Cardinality changed from 1 to * Type changed from code to CodeableConcept Change binding strength from required to extensible
AdverseEvent.event | Added Element
AdverseEvent.subject | Min Cardinality changed from 0 to 1 Type Reference: Added Target Types Group, Practitioner, RelatedPerson Type Reference: Removed Target Types ResearchSubject, Medication, Device
AdverseEvent.encounter | Added Element
AdverseEvent.detected | Added Element
AdverseEvent.recordedDate | Added Element
AdverseEvent.resultingCondition | Added Element
AdverseEvent.severity | Added Element
AdverseEvent.outcome | Change value set from http://hl7.org/fhir/ValueSet/adverse-event-outcome to http://hl7.org/fhir/ValueSet/adverse-event-outcome|4.0.1
AdverseEvent.recorder | Type Reference: Added Target Type PractitionerRole
AdverseEvent.contributor | Added Element
AdverseEvent.suspectEntity.instance | Type Reference: Added Target Types Immunization, Procedure
AdverseEvent.suspectEntity.causality | Max Cardinality changed from 1 to * Type changed from code to BackboneElement Remove Binding http://hl7.org/fhir/ValueSet/adverse-event-causality (required)
AdverseEvent.suspectEntity.causality.assessment | Added Element
AdverseEvent.suspectEntity.causality.productRelatedness | Added Element
AdverseEvent.suspectEntity.causality.author | Added Element
AdverseEvent.suspectEntity.causality.method | Added Element
AdverseEvent.subjectMedicalHistory | Type Reference: Added Target Types Media, DocumentReference
AdverseEvent.type | deleted
AdverseEvent.reaction | deleted
AdverseEvent.eventParticipant | deleted
AdverseEvent.description | deleted
AdverseEvent.suspectEntity.causalityAssessment | deleted
AdverseEvent.suspectEntity.causalityProductRelatedness | deleted
AdverseEvent.suspectEntity.causalityMethod | deleted
AdverseEvent.suspectEntity.causalityAuthor | deleted
AdverseEvent.suspectEntity.causalityResult | deleted
AllergyIntolerance
AllergyIntolerance.clinicalStatus | Type changed from code to CodeableConcept Change value set from http://hl7.org/fhir/ValueSet/allergy-clinical-status to http://hl7.org/fhir/ValueSet/allergyintolerance-clinical|4.0.1
AllergyIntolerance.verificationStatus | Min Cardinality changed from 1 to 0 Type changed from code to CodeableConcept Change value set from http://hl7.org/fhir/ValueSet/allergy-verification-status to http://hl7.org/fhir/ValueSet/allergyintolerance-verification|4.0.1
AllergyIntolerance.type | Change value set from http://hl7.org/fhir/ValueSet/allergy-intolerance-type to http://hl7.org/fhir/ValueSet/allergy-intolerance-type|4.0.1
AllergyIntolerance.category | Change value set from http://hl7.org/fhir/ValueSet/allergy-intolerance-category to http://hl7.org/fhir/ValueSet/allergy-intolerance-category|4.0.1
AllergyIntolerance.criticality | Change value set from http://hl7.org/fhir/ValueSet/allergy-intolerance-criticality to http://hl7.org/fhir/ValueSet/allergy-intolerance-criticality|4.0.1
AllergyIntolerance.encounter | Added Element
AllergyIntolerance.recordedDate | Renamed from assertedDate to recordedDate
AllergyIntolerance.recorder | Type Reference: Added Target Types PractitionerRole, RelatedPerson
AllergyIntolerance.asserter | Type Reference: Added Target Type PractitionerRole
AllergyIntolerance.reaction.severity | Change value set from http://hl7.org/fhir/ValueSet/reaction-event-severity to http://hl7.org/fhir/ValueSet/reaction-event-severity|4.0.1
Appointment
Appointment.status | Change value set from http://hl7.org/fhir/ValueSet/appointmentstatus to http://hl7.org/fhir/ValueSet/appointmentstatus|4.0.1
Appointment.cancelationReason | Added Element
Appointment.serviceCategory | Max Cardinality changed from 1 to *
Appointment.reasonCode | Added Element
Appointment.reasonReference | Added Element
Appointment.patientInstruction | Added Element

Appointment.basedOn | Added Element
Appointment.participant.actor | Type Reference: Added Target Type PractitionerRole
Appointment.participant.required | Change value set from http://hl7.org/fhir/ValueSet/participantrequired to http://hl7.org/fhir/ValueSet/participantrequired|4.0.1
Appointment.participant.status | Change value set from http://hl7.org/fhir/ValueSet/participationstatus to http://hl7.org/fhir/ValueSet/participationstatus|4.0.1
Appointment.participant.period | Added Element
Appointment.reason | deleted
Appointment.indication | deleted
Appointment.incomingReferral | deleted
AppointmentResponse
AppointmentResponse.actor | Type Reference: Added Target Type PractitionerRole
AppointmentResponse.participantStatus | Change value set from http://hl7.org/fhir/ValueSet/participationstatus to http://hl7.org/fhir/ValueSet/participationstatus|4.0.1
AuditEvent
AuditEvent.action | Change value set from http://hl7.org/fhir/ValueSet/audit-event-action to http://hl7.org/fhir/ValueSet/audit-event-action|4.0.1
AuditEvent.period | Added Element
AuditEvent.outcome | Change value set from http://hl7.org/fhir/ValueSet/audit-event-outcome to http://hl7.org/fhir/ValueSet/audit-event-outcome|4.0.1
AuditEvent.agent.type | Added Element
AuditEvent.agent.role | Remove Binding http://hl7.org/fhir/ValueSet/security-role-type (extensible)
AuditEvent.agent.who | Added Element
AuditEvent.agent.network.type | Change value set from http://hl7.org/fhir/ValueSet/network-type to http://hl7.org/fhir/ValueSet/network-type|4.0.1
AuditEvent.source.observer | Added Mandatory Element
AuditEvent.source.type | Change code system for extensibly bound codes from "http://hl7.org/fhir/security-source-type" to "http://terminology.hl7.org/CodeSystem/security-source-type"
AuditEvent.entity.what | Added Element
AuditEvent.entity.role | Change code system for extensibly bound codes from "http://hl7.org/fhir/object-role" to "http://terminology.hl7.org/CodeSystem/object-role"
AuditEvent.entity.detail.value[x] | Renamed from value to value[x] Add Type string
AuditEvent.agent.reference | deleted
AuditEvent.agent.userId | deleted
AuditEvent.source.identifier | deleted
AuditEvent.entity.identifier | deleted
AuditEvent.entity.reference | deleted
Basic
Basic.author | Type Reference: Added Target Types PractitionerRole, Organization
Binary
Binary.contentType | Change value set from http://hl7.org/fhir/ValueSet/mimetypes to http://hl7.org/fhir/ValueSet/mimetypes|4.0.1
Binary.data | Renamed from content to data Min Cardinality changed from 1 to 0
BiologicallyDerivedProduct | Added Resource
BodyStructure | Added Resource
Bundle
Bundle.type | Change value set from http://hl7.org/fhir/ValueSet/bundle-type to http://hl7.org/fhir/ValueSet/bundle-type|4.0.1
Bundle.timestamp | Added Element
Bundle.entry.search.mode | Change value set from http://hl7.org/fhir/ValueSet/search-entry-mode to http://hl7.org/fhir/ValueSet/search-entry-mode|4.0.1
Bundle.entry.request.method | Change value set from http://hl7.org/fhir/ValueSet/http-verb to http://hl7.org/fhir/ValueSet/http-verb|4.0.1
CapabilityStatement
CapabilityStatement | Min Cardinality changed from 1 to 0 Max Cardinality changed from 1 to *
CapabilityStatement.status | Change value set from http://hl7.org/fhir/ValueSet/publication-status to http://hl7.org/fhir/ValueSet/publication-status|4.0.1
CapabilityStatement.experimental | No longer marked as Modifier
CapabilityStatement.kind | Change value set from http://hl7.org/fhir/ValueSet/capability-statement-kind to http://hl7.org/fhir/ValueSet/capability-statement-kind|4.0.1
CapabilityStatement.instantiates | Type changed from uri to canonical(CapabilityStatement)
CapabilityStatement.imports | Added Element
CapabilityStatement.implementation.url | Type changed from uri to url
CapabilityStatement.implementation.custodian | Added Element
CapabilityStatement.fhirVersion | Type changed from id to code Add Binding http://hl7.org/fhir/ValueSet/FHIR-version|4.0.1 (required)
CapabilityStatement.format | Change value set from http://hl7.org/fhir/ValueSet/mimetypes to http://hl7.org/fhir/ValueSet/mimetypes|4.0.1
CapabilityStatement.patchFormat | Change value set from http://hl7.org/fhir/ValueSet/mimetypes to http://hl7.org/fhir/ValueSet/mimetypes|4.0.1
CapabilityStatement.implementationGuide | Type changed from uri to canonical(ImplementationGuide)
CapabilityStatement.rest.mode | Change value set from http://hl7.org/fhir/ValueSet/restful-capability-mode to http://hl7.org/fhir/ValueSet/restful-capability-mode|4.0.1
CapabilityStatement.rest.documentation | Type changed from string to markdown
CapabilityStatement.rest.security.service | Change code system for extensibly bound codes from "http://hl7.org/fhir/restful-security-service" to "http://terminology.hl7.org/CodeSystem/restful-security-service"

CapabilityStatement.rest.security.description | Type changed from string to markdown
CapabilityStatement.rest.resource.type | Change value set from http://hl7.org/fhir/ValueSet/resource-types to http://hl7.org/fhir/ValueSet/resource-types|4.0.1
CapabilityStatement.rest.resource.profile | Type changed from Reference(StructureDefinition) to canonical(StructureDefinition)
CapabilityStatement.rest.resource.supportedProfile | Moved from CapabilityStatement.profile to supportedProfile Type changed from Reference(StructureDefinition) to canonical(StructureDefinition)
CapabilityStatement.rest.resource.interaction | Min Cardinality changed from 1 to 0
CapabilityStatement.rest.resource.interaction.code | Change value set from http://hl7.org/fhir/ValueSet/type-restful-interaction to http://hl7.org/fhir/ValueSet/type-restful-interaction|4.0.1
CapabilityStatement.rest.resource.interaction.documentation | Type changed from string to markdown
CapabilityStatement.rest.resource.versioning | Change value set from http://hl7.org/fhir/ValueSet/versioning-policy to http://hl7.org/fhir/ValueSet/versioning-policy|4.0.1
CapabilityStatement.rest.resource.conditionalRead | Change value set from http://hl7.org/fhir/ValueSet/conditional-read-status to http://hl7.org/fhir/ValueSet/conditional-read-status|4.0.1
CapabilityStatement.rest.resource.conditionalDelete | Change value set from http://hl7.org/fhir/ValueSet/conditional-delete-status to http://hl7.org/fhir/ValueSet/conditional-delete-status|4.0.1
CapabilityStatement.rest.resource.referencePolicy | Change value set from http://hl7.org/fhir/ValueSet/reference-handling-policy to http://hl7.org/fhir/ValueSet/reference-handling-policy|4.0.1
CapabilityStatement.rest.resource.searchParam.definition | Type changed from uri to canonical(SearchParameter)
CapabilityStatement.rest.resource.searchParam.type | Change value set from http://hl7.org/fhir/ValueSet/search-param-type to http://hl7.org/fhir/ValueSet/search-param-type|4.0.1
CapabilityStatement.rest.resource.searchParam.documentation | Type changed from string to markdown
CapabilityStatement.rest.resource.operation | Added Element
CapabilityStatement.rest.resource.operation.name | Added Mandatory Element
CapabilityStatement.rest.resource.operation.definition | Added Mandatory Element
CapabilityStatement.rest.resource.operation.documentation | Added Element
CapabilityStatement.rest.interaction.code | Change value set from http://hl7.org/fhir/ValueSet/system-restful-interaction to http://hl7.org/fhir/ValueSet/system-restful-interaction|4.0.1
CapabilityStatement.rest.interaction.documentation | Type changed from string to markdown
CapabilityStatement.rest.operation | Remove Type BackboneElement
CapabilityStatement.rest.compartment | Type changed from uri to canonical(CompartmentDefinition)
CapabilityStatement.messaging.endpoint.protocol | Change code system for extensibly bound codes from "http://hl7.org/fhir/message-transport" to "http://terminology.hl7.org/CodeSystem/message-transport"
CapabilityStatement.messaging.endpoint.address | Type changed from uri to url
CapabilityStatement.messaging.documentation | Type changed from string to markdown
CapabilityStatement.messaging.supportedMessage.mode | Change value set from http://hl7.org/fhir/ValueSet/event-capability-mode to http://hl7.org/fhir/ValueSet/event-capability-mode|4.0.1
CapabilityStatement.messaging.supportedMessage.definition | Type changed from Reference(MessageDefinition) to canonical(MessageDefinition)
CapabilityStatement.document.mode | Change value set from http://hl7.org/fhir/ValueSet/document-mode to http://hl7.org/fhir/ValueSet/document-mode|4.0.1
CapabilityStatement.document.documentation | Type changed from string to markdown
CapabilityStatement.document.profile | Type changed from Reference(StructureDefinition) to canonical(StructureDefinition)
CapabilityStatement.acceptUnknown | deleted
CapabilityStatement.rest.security.certificate | deleted
CapabilityStatement.rest.operation.name | deleted
CapabilityStatement.rest.operation.definition | deleted
CapabilityStatement.messaging.event | deleted
CarePlan
CarePlan.instantiatesCanonical | Added Element
CarePlan.instantiatesUri | Added Element
CarePlan.status | Change value set from http://hl7.org/fhir/ValueSet/care-plan-status to http://hl7.org/fhir/ValueSet/request-status|4.0.1
CarePlan.intent | Change value set from http://hl7.org/fhir/ValueSet/care-plan-intent to http://hl7.org/fhir/ValueSet/care-plan-intent|4.0.1
CarePlan.encounter | Added Element
CarePlan.created | Added Element
CarePlan.author | Max Cardinality changed from * to 1 Type Reference: Added Target Types PractitionerRole, Device
CarePlan.contributor | Added Element
CarePlan.activity.reference | Type Reference: Added Target Type ServiceRequest Type Reference: Removed Target Types ProcedureRequest, ReferralRequest
CarePlan.activity.detail.kind | Renamed from category to kind Type changed from CodeableConcept to code Add Binding http://hl7.org/fhir/ValueSet/care-plan-activity-kind|4.0.1 (required)
CarePlan.activity.detail.instantiatesCanonical | Added Element
CarePlan.activity.detail.instantiatesUri | Added Element
CarePlan.activity.detail.reasonReference | Type Reference: Added Target Types Observation, DiagnosticReport, DocumentReference
CarePlan.activity.detail.status | Change value set from http://hl7.org/fhir/ValueSet/care-plan-activity-status to http://hl7.org/fhir/ValueSet/care-plan-activity-status|4.0.1
CarePlan.activity.detail.statusReason | Type changed from string to CodeableConcept
CarePlan.activity.detail.doNotPerform | Renamed from prohibited to doNotPerform Default Value "false" removed
CarePlan.activity.detail.performer | Type Reference: Added Target Types PractitionerRole, HealthcareService, Device
CarePlan.definition | deleted
CarePlan.context | deleted

CarePlan.activity.detail.definition | deleted
CareTeam
CareTeam.status | Change value set from http://hl7.org/fhir/ValueSet/care-team-status to http://hl7.org/fhir/ValueSet/care-team-status|4.0.1
CareTeam.encounter | Added Element
CareTeam.participant.role | Max Cardinality changed from 1 to *
CareTeam.participant.member | Type Reference: Added Target Type PractitionerRole
CareTeam.telecom | Added Element
CareTeam.context | deleted
CatalogEntry | Added Resource
ChargeItem
ChargeItem.identifier | Max Cardinality changed from 1 to *
ChargeItem.definitionUri | Added Element
ChargeItem.definitionCanonical | Added Element
ChargeItem.status | Change value set from http://hl7.org/fhir/ValueSet/chargeitem-status to http://hl7.org/fhir/ValueSet/chargeitem-status|4.0.1
ChargeItem.performer | Added Element
ChargeItem.performer.function | Added Element
ChargeItem.performer.actor | Added Mandatory Element
ChargeItem.costCenter | Added Element
ChargeItem.enterer | Type Reference: Added Target Type PractitionerRole
ChargeItem.product[x] | Added Element
ChargeItem.definition | deleted
ChargeItem.participant | deleted
ChargeItemDefinition | Added Resource
Claim
Claim.status | Min Cardinality changed from 0 to 1 Change value set from http://hl7.org/fhir/ValueSet/fm-status to http://hl7.org/fhir/ValueSet/fm-status|4.0.1
Claim.type | Min Cardinality changed from 0 to 1 Change binding strength from required to extensible
Claim.subType | Max Cardinality changed from * to 1
Claim.use | Min Cardinality changed from 0 to 1 Change value set from http://hl7.org/fhir/ValueSet/claim-use to http://hl7.org/fhir/ValueSet/claim-use|4.0.1
Claim.patient | Min Cardinality changed from 0 to 1
Claim.created | Min Cardinality changed from 0 to 1
Claim.enterer | Type Reference: Added Target Type PractitionerRole
Claim.provider | Min Cardinality changed from 0 to 1 Type Reference: Added Target Types PractitionerRole, Organization
Claim.priority | Min Cardinality changed from 0 to 1
Claim.prescription | Type Reference: Added Target Type DeviceRequest
Claim.originalPrescription | Type Reference: Added Target Types DeviceRequest, VisionPrescription
Claim.payee.party | Type Reference: Added Target Type PractitionerRole
Claim.referral | Type Reference: Added Target Type ServiceRequest Type Reference: Removed Target Type ReferralRequest
Claim.careTeam.provider | Type Reference: Added Target Type PractitionerRole
Claim.supportingInfo | Renamed from information to supportingInfo
Claim.supportingInfo.sequence | Moved from Claim.information to Claim.supportingInfo
Claim.supportingInfo.category | Moved from Claim.information to Claim.supportingInfo
Claim.supportingInfo.code | Moved from Claim.information to Claim.supportingInfo
Claim.supportingInfo.timing[x] | Moved from Claim.information to Claim.supportingInfo
Claim.supportingInfo.value[x] | Moved from Claim.information to Claim.supportingInfo Add Type boolean
Claim.supportingInfo.reason | Moved from Claim.information to Claim.supportingInfo
Claim.diagnosis.onAdmission | Added Element
Claim.procedure.type | Added Element
Claim.procedure.udi | Added Element
Claim.insurance | Min Cardinality changed from 0 to 1
Claim.insurance.identifier | Added Element
Claim.accident.type | Change binding strength from required to extensible
Claim.item.careTeamSequence | Renamed from careTeamLinkId to careTeamSequence
Claim.item.diagnosisSequence | Renamed from diagnosisLinkId to diagnosisSequence
Claim.item.procedureSequence | Renamed from procedureLinkId to procedureSequence
Claim.item.informationSequence | Renamed from informationLinkId to informationSequence
Claim.item.productOrService | Renamed from service to productOrService Min Cardinality changed from 0 to 1
Claim.item.detail.productOrService | Renamed from service to productOrService Min Cardinality changed from 0 to 1
Claim.item.detail.subDetail.productOrService | Renamed from service to productOrService Min Cardinality changed from 0 to 1
Claim.organization | deleted
Claim.payee.resourceType | deleted
Claim.employmentImpacted | deleted
Claim.hospitalization | deleted
ClaimResponse
ClaimResponse.status | Min Cardinality changed from 0 to 1 Change value set from http://hl7.org/fhir/ValueSet/fm-status to http://hl7.org/fhir/ValueSet/fm-status|4.0.1
ClaimResponse.type | Added Mandatory Element
ClaimResponse.subType | Added Element
ClaimResponse.use | Added Mandatory Element
ClaimResponse.patient | Min Cardinality changed from 0 to 1

ClaimResponse.created | Min Cardinality changed from 0 to 1
ClaimResponse.insurer | Min Cardinality changed from 0 to 1
ClaimResponse.requestor | Renamed from requestProvider to requestor Type Reference: Added Target Types PractitionerRole, Organization
ClaimResponse.outcome | Min Cardinality changed from 0 to 1 Type changed from CodeableConcept to code Add Binding http://hl7.org/fhir/ValueSet/remittance-outcome|4.0.1 (required)
ClaimResponse.preAuthRef | Moved from ClaimResponse.insurance to ClaimResponse Max Cardinality changed from * to 1
ClaimResponse.preAuthPeriod | Added Element
ClaimResponse.item.itemSequence | Renamed from sequenceLinkId to itemSequence
ClaimResponse.item.adjudication | Min Cardinality changed from 0 to 1
ClaimResponse.item.detail.detailSequence | Renamed from sequenceLinkId to detailSequence
ClaimResponse.item.detail.adjudication | Min Cardinality changed from 0 to 1
ClaimResponse.item.detail.subDetail.subDetailSequence | Renamed from sequenceLinkId to subDetailSequence
ClaimResponse.addItem.itemSequence | Renamed from sequenceLinkId to itemSequence
ClaimResponse.addItem.detailSequence | Added Element
ClaimResponse.addItem.subdetailSequence | Added Element
ClaimResponse.addItem.provider | Added Element
ClaimResponse.addItem.productOrService | Renamed from service to productOrService Min Cardinality changed from 0 to 1
ClaimResponse.addItem.programCode | Added Element
ClaimResponse.addItem.serviced[x] | Added Element
ClaimResponse.addItem.location[x] | Added Element
ClaimResponse.addItem.quantity | Added Element
ClaimResponse.addItem.unitPrice | Added Element
ClaimResponse.addItem.factor | Added Element
ClaimResponse.addItem.net | Added Element
ClaimResponse.addItem.bodySite | Added Element
ClaimResponse.addItem.subSite | Added Element
ClaimResponse.addItem.adjudication | Min Cardinality changed from 0 to 1
ClaimResponse.addItem.detail.productOrService | Renamed from service to productOrService Min Cardinality changed from 0 to 1
ClaimResponse.addItem.detail.quantity | Added Element
ClaimResponse.addItem.detail.unitPrice | Added Element
ClaimResponse.addItem.detail.factor | Added Element
ClaimResponse.addItem.detail.net | Added Element
ClaimResponse.addItem.detail.adjudication | Min Cardinality changed from 0 to 1
ClaimResponse.addItem.detail.subDetail | Added Element
ClaimResponse.addItem.detail.subDetail.productOrService | Added Mandatory Element
ClaimResponse.addItem.detail.subDetail.modifier | Added Element
ClaimResponse.addItem.detail.subDetail.quantity | Added Element
ClaimResponse.addItem.detail.subDetail.unitPrice | Added Element
ClaimResponse.addItem.detail.subDetail.factor | Added Element
ClaimResponse.addItem.detail.subDetail.net | Added Element
ClaimResponse.addItem.detail.subDetail.noteNumber | Added Element
ClaimResponse.addItem.detail.subDetail.adjudication | Added Mandatory Element
ClaimResponse.adjudication | Added Element
ClaimResponse.total | Added Element
ClaimResponse.total.category | Added Mandatory Element
ClaimResponse.total.amount | Added Mandatory Element
ClaimResponse.payment.type | Min Cardinality changed from 0 to 1
ClaimResponse.payment.amount | Min Cardinality changed from 0 to 1
ClaimResponse.fundsReserve | Renamed from reserved to fundsReserve Type changed from Coding to CodeableConcept
ClaimResponse.formCode | Added Element
ClaimResponse.form | Type changed from CodeableConcept to Attachment
ClaimResponse.processNote.type | Type changed from CodeableConcept to code Change value set from http://hl7.org/fhir/ValueSet/note-type to http://hl7.org/fhir/ValueSet/note-type|4.0.1
ClaimResponse.processNote.text | Min Cardinality changed from 0 to 1
ClaimResponse.processNote.language | Change binding strength from extensible to preferred
ClaimResponse.error.itemSequence | Renamed from sequenceLinkId to itemSequence
ClaimResponse.error.detailSequence | Renamed from detailSequenceLinkId to detailSequence
ClaimResponse.error.subDetailSequence | Renamed from subdetailSequenceLinkId to subDetailSequence
ClaimResponse.requestOrganization | deleted
ClaimResponse.addItem.revenue | deleted
ClaimResponse.addItem.category | deleted
ClaimResponse.addItem.fee | deleted
ClaimResponse.addItem.detail.revenue | deleted
ClaimResponse.addItem.detail.category | deleted
ClaimResponse.addItem.detail.fee | deleted
ClaimResponse.totalCost | deleted
ClaimResponse.unallocDeductable | deleted
ClaimResponse.totalBenefit | deleted
ClinicalImpression
ClinicalImpression.status | Change value set from http://hl7.org/fhir/ValueSet/clinical-impression-status to http://hl7.org/fhir/ValueSet/clinicalimpression-status|4.0.1
ClinicalImpression.statusReason | Added Element

ClinicalImpression.encounter | Added Element
ClinicalImpression.assessor | Type Reference: Added Target Type PractitionerRole
ClinicalImpression.investigation.item | Type Reference: Added Target Type Media
ClinicalImpression.finding.itemCodeableConcept | Added Element
ClinicalImpression.finding.itemReference | Added Element
ClinicalImpression.supportingInfo | Added Element
ClinicalImpression.context | deleted
ClinicalImpression.finding.item[x] | deleted
ClinicalImpression.action | deleted
CodeSystem
CodeSystem.identifier | Max Cardinality changed from 1 to *
CodeSystem.status | Change value set from http://hl7.org/fhir/ValueSet/publication-status to http://hl7.org/fhir/ValueSet/publication-status|4.0.1
CodeSystem.experimental | No longer marked as Modifier
CodeSystem.valueSet | Type changed from uri to canonical(ValueSet)
CodeSystem.hierarchyMeaning | Change value set from http://hl7.org/fhir/ValueSet/codesystem-hierarchy-meaning to http://hl7.org/fhir/ValueSet/codesystem-hierarchy-meaning|4.0.1
CodeSystem.content | Change value set from http://hl7.org/fhir/ValueSet/codesystem-content-mode to http://hl7.org/fhir/ValueSet/codesystem-content-mode|4.0.1
CodeSystem.supplements | Added Element
CodeSystem.filter.operator | Change value set from http://hl7.org/fhir/ValueSet/filter-operator to http://hl7.org/fhir/ValueSet/filter-operator|4.0.1
CodeSystem.property.type | Change value set from http://hl7.org/fhir/ValueSet/concept-property-type to http://hl7.org/fhir/ValueSet/concept-property-type|4.0.1
CodeSystem.concept.designation.language | Change binding strength from extensible to preferred
CodeSystem.concept.property.value[x] | Add Type decimal
Communication
Communication.instantiatesCanonical | Added Element
Communication.instantiatesUri | Added Element
Communication.inResponseTo | Added Element
Communication.status | Change value set from http://hl7.org/fhir/ValueSet/event-status to http://hl7.org/fhir/ValueSet/event-status|4.0.1
Communication.statusReason | Added Element
Communication.priority | Added Element
Communication.topic | Max Cardinality changed from * to 1 Type changed from Reference(Resource) to CodeableConcept
Communication.about | Added Element
Communication.encounter | Added Element
Communication.recipient | Type Reference: Added Target Types PractitionerRole, CareTeam, HealthcareService
Communication.sender | Type Reference: Added Target Types PractitionerRole, HealthcareService
Communication.reasonReference | Type Reference: Added Target Types DiagnosticReport, DocumentReference
Communication.definition | deleted
Communication.notDone | deleted
Communication.notDoneReason | deleted
Communication.context | deleted
CommunicationRequest
CommunicationRequest.status | Change value set from http://hl7.org/fhir/ValueSet/request-status to http://hl7.org/fhir/ValueSet/request-status|4.0.1
CommunicationRequest.statusReason | Added Element
CommunicationRequest.priority | Change value set from http://hl7.org/fhir/ValueSet/request-priority to http://hl7.org/fhir/ValueSet/request-priority|4.0.1
CommunicationRequest.doNotPerform | Added Element
CommunicationRequest.about | Added Element
CommunicationRequest.encounter | Added Element
CommunicationRequest.requester | Type changed from BackboneElement to Reference(Practitioner | PractitionerRole | Organization | Patient | RelatedPerson | Device)
CommunicationRequest.recipient | Type Reference: Added Target Types PractitionerRole, HealthcareService
CommunicationRequest.sender | Type Reference: Added Target Types PractitionerRole, HealthcareService
CommunicationRequest.reasonReference | Type Reference: Added Target Types DiagnosticReport, DocumentReference
CommunicationRequest.topic | deleted
CommunicationRequest.context | deleted
CommunicationRequest.requester.agent | deleted
CommunicationRequest.requester.onBehalfOf | deleted
CompartmentDefinition
CompartmentDefinition | Min Cardinality changed from 1 to 0 Max Cardinality changed from 1 to *
CompartmentDefinition.version | Added Element
CompartmentDefinition.status | Change value set from http://hl7.org/fhir/ValueSet/publication-status to http://hl7.org/fhir/ValueSet/publication-status|4.0.1
CompartmentDefinition.experimental | No longer marked as Modifier
CompartmentDefinition.code | Change value set from http://hl7.org/fhir/ValueSet/compartment-type to http://hl7.org/fhir/ValueSet/compartment-type|4.0.1
CompartmentDefinition.resource.code | Change value set from http://hl7.org/fhir/ValueSet/resource-types to http://hl7.org/fhir/ValueSet/resource-types|4.0.1
CompartmentDefinition.title | deleted

CompartmentDefinition.jurisdiction | deleted
Composition
Composition.status | Change value set from http://hl7.org/fhir/ValueSet/composition-status to http://hl7.org/fhir/ValueSet/composition-status|4.0.1
Composition.category | Renamed from class to category Max Cardinality changed from 1 to *
Composition.subject | Min Cardinality changed from 1 to 0
Composition.author | Type Reference: Added Target Types PractitionerRole, Organization
Composition.confidentiality | Change value set from http://terminology.hl7.org/ValueSet/v3-ConfidentialityClassification to http://terminology.hl7.org/ValueSet/v3-ConfidentialityClassification|2014-03-26 No longer marked as Modifier
Composition.attester.mode | Max Cardinality changed from * to 1 Change value set from http://hl7.org/fhir/ValueSet/composition-attestation-mode to http://hl7.org/fhir/ValueSet/composition-attestation-mode|4.0.1
Composition.attester.party | Type Reference: Added Target Types RelatedPerson, PractitionerRole
Composition.relatesTo.code | Change value set from http://hl7.org/fhir/ValueSet/document-relationship-type to http://hl7.org/fhir/ValueSet/document-relationship-type|4.0.1
Composition.section.author | Added Element
Composition.section.focus | Added Element
Composition.section.mode | Change value set from http://hl7.org/fhir/ValueSet/list-mode to http://hl7.org/fhir/ValueSet/list-mode|4.0.1 No longer marked as Modifier
ConceptMap
ConceptMap.status | Change value set from http://hl7.org/fhir/ValueSet/publication-status to http://hl7.org/fhir/ValueSet/publication-status|4.0.1
ConceptMap.experimental | No longer marked as Modifier
ConceptMap.source[x] | Add Type canonical(ValueSet) Remove Type Reference(ValueSet)
ConceptMap.target[x] | Add Type canonical(ValueSet) Remove Type Reference(ValueSet)
ConceptMap.group.element.target.equivalence | Min Cardinality changed from 0 to 1 Change value set from http://hl7.org/fhir/ValueSet/concept-map-equivalence to http://hl7.org/fhir/ValueSet/concept-map-equivalence|4.0.1 Default Value "equivalent" removed
ConceptMap.group.element.target.dependsOn.system | Type changed from uri to canonical(CodeSystem)
ConceptMap.group.element.target.dependsOn.value | Renamed from code to value
ConceptMap.group.unmapped.mode | Change value set from http://hl7.org/fhir/ValueSet/conceptmap-unmapped-mode to http://hl7.org/fhir/ValueSet/conceptmap-unmapped-mode|4.0.1
ConceptMap.group.unmapped.url | Type changed from uri to canonical(ConceptMap)
Condition
Condition.clinicalStatus | Type changed from code to CodeableConcept Change value set from http://hl7.org/fhir/ValueSet/condition-clinical to http://hl7.org/fhir/ValueSet/condition-clinical|4.0.1
Condition.verificationStatus | Type changed from code to CodeableConcept Change value set from http://hl7.org/fhir/ValueSet/condition-ver-status to http://hl7.org/fhir/ValueSet/condition-ver-status|4.0.1 Default Value "unknown" removed
Condition.category | Add Binding http://hl7.org/fhir/ValueSet/condition-category (extensible)
Condition.encounter | Added Element
Condition.abatement[x] | Remove Type boolean
Condition.recordedDate | Renamed from assertedDate to recordedDate
Condition.recorder | Added Element
Condition.asserter | Type Reference: Added Target Type PractitionerRole
Condition.stage | Max Cardinality changed from 1 to *
Condition.stage.type | Added Element
Condition.context | deleted
Consent
Consent.identifier | Max Cardinality changed from 1 to *
Consent.status | Change value set from http://hl7.org/fhir/ValueSet/consent-state-codes to http://hl7.org/fhir/ValueSet/consent-state-codes|4.0.1
Consent.scope | Added Mandatory Element
Consent.category | Min Cardinality changed from 0 to 1 Add Binding http://hl7.org/fhir/ValueSet/consent-category (extensible)
Consent.patient | Min Cardinality changed from 1 to 0
Consent.performer | Added Element
Consent.source[x] | Remove Type Identifier
Consent.policyRule | Type changed from uri to CodeableConcept Add Binding http://hl7.org/fhir/ValueSet/consent-policy (extensible)
Consent.verification | Added Element
Consent.verification.verified | Added Mandatory Element
Consent.verification.verifiedWith | Added Element
Consent.verification.verificationDate | Added Element
Consent.provision | Added Element
Consent.provision.type | Added Element
Consent.provision.period | Added Element
Consent.provision.actor | Added Element
Consent.provision.actor.role | Added Mandatory Element
Consent.provision.actor.reference | Added Mandatory Element
Consent.provision.action | Added Element
Consent.provision.securityLabel | Added Element
Consent.provision.purpose | Added Element
Consent.provision.class | Added Element
Consent.provision.code | Added Element

Consent.provision.dataPeriod | Added Element
Consent.provision.data | Added Element
Consent.provision.data.meaning | Added Mandatory Element
Consent.provision.data.reference | Added Mandatory Element
Consent.provision.provision | Added Element
Consent.period | deleted
Consent.consentingParty | deleted
Consent.actor | deleted
Consent.action | deleted
Consent.securityLabel | deleted
Consent.purpose | deleted
Consent.dataPeriod | deleted
Consent.data | deleted
Consent.except | deleted
Contract
Contract.identifier | Max Cardinality changed from 1 to *
Contract.url | Added Element
Contract.version | Added Element
Contract.status | Change value set from http://hl7.org/fhir/ValueSet/contract-status to http://hl7.org/fhir/ValueSet/contract-status|4.0.1
Contract.legalState | Added Element
Contract.instantiatesCanonical | Added Element
Contract.instantiatesUri | Added Element
Contract.expirationType | Added Element
Contract.site | Added Element
Contract.name | Added Element
Contract.title | Added Element
Contract.subtitle | Added Element
Contract.alias | Added Element
Contract.author | Added Element
Contract.scope | Added Element
Contract.topic[x] | Renamed from topic to topic[x] Max Cardinality changed from * to 1 Add Type CodeableConcept
Contract.contentDefinition | Added Element
Contract.contentDefinition.type | Added Mandatory Element
Contract.contentDefinition.subType | Added Element
Contract.contentDefinition.publisher | Added Element
Contract.contentDefinition.publicationDate | Added Element
Contract.contentDefinition.publicationStatus | Added Mandatory Element
Contract.contentDefinition.copyright | Added Element
Contract.term.topic[x] | Renamed from topic to topic[x] Max Cardinality changed from * to 1 Add Type CodeableConcept
Contract.term.securityLabel | Type changed from Coding to BackboneElement Remove Binding http://hl7.org/fhir/ValueSet/security-labels (extensible)
Contract.term.securityLabel.number | Added Element
Contract.term.securityLabel.classification | Added Mandatory Element
Contract.term.securityLabel.category | Added Element
Contract.term.securityLabel.control | Added Element
Contract.term.offer | Added Mandatory Element
Contract.term.offer.identifier | Added Element
Contract.term.offer.party | Added Element
Contract.term.offer.party.reference | Added Mandatory Element
Contract.term.offer.party.role | Added Mandatory Element
Contract.term.offer.topic | Added Element
Contract.term.offer.type | Added Element
Contract.term.offer.decision | Added Element
Contract.term.offer.decisionMode | Added Element
Contract.term.offer.answer | Added Element
Contract.term.offer.answer.value[x] | Added Mandatory Element
Contract.term.offer.text | Added Element
Contract.term.offer.linkId | Added Element
Contract.term.offer.securityLabelNumber | Added Element
Contract.term.asset | Added Element
Contract.term.asset.scope | Added Element
Contract.term.asset.type | Added Element
Contract.term.asset.typeReference | Added Element
Contract.term.asset.subtype | Added Element
Contract.term.asset.relationship | Added Element
Contract.term.asset.context | Added Element
Contract.term.asset.context.reference | Added Element
Contract.term.asset.context.code | Added Element
Contract.term.asset.context.text | Added Element
Contract.term.asset.condition | Added Element
Contract.term.asset.periodType | Added Element

Contract.term.asset.period | Added Element
Contract.term.asset.usePeriod | Added Element
Contract.term.asset.text | Added Element
Contract.term.asset.linkId | Added Element
Contract.term.asset.answer | Added Element
Contract.term.asset.securityLabelNumber | Added Element
Contract.term.asset.valuedItem | Added Element
Contract.term.asset.valuedItem.entity[x] | Added Element
Contract.term.asset.valuedItem.identifier | Added Element
Contract.term.asset.valuedItem.effectiveTime | Added Element
Contract.term.asset.valuedItem.quantity | Added Element
Contract.term.asset.valuedItem.unitPrice | Added Element
Contract.term.asset.valuedItem.factor | Added Element
Contract.term.asset.valuedItem.points | Added Element
Contract.term.asset.valuedItem.net | Added Element
Contract.term.asset.valuedItem.payment | Added Element
Contract.term.asset.valuedItem.paymentDate | Added Element
Contract.term.asset.valuedItem.responsible | Added Element
Contract.term.asset.valuedItem.recipient | Added Element
Contract.term.asset.valuedItem.linkId | Added Element
Contract.term.asset.valuedItem.securityLabelNumber | Added Element
Contract.term.action | Type changed from CodeableConcept to BackboneElement
Contract.term.action.doNotPerform | Added Element
Contract.term.action.type | Added Mandatory Element
Contract.term.action.subject | Added Element
Contract.term.action.subject.reference | Added Mandatory Element
Contract.term.action.subject.role | Added Element
Contract.term.action.intent | Added Mandatory Element
Contract.term.action.linkId | Added Element
Contract.term.action.status | Added Mandatory Element
Contract.term.action.context | Added Element
Contract.term.action.contextLinkId | Added Element
Contract.term.action.occurrence[x] | Added Element
Contract.term.action.requester | Added Element
Contract.term.action.requesterLinkId | Added Element
Contract.term.action.performerType | Added Element
Contract.term.action.performerRole | Added Element
Contract.term.action.performer | Added Element
Contract.term.action.performerLinkId | Added Element
Contract.term.action.reasonCode | Added Element
Contract.term.action.reasonReference | Added Element
Contract.term.action.reason | Added Element
Contract.term.action.reasonLinkId | Added Element
Contract.term.action.note | Added Element
Contract.term.action.securityLabelNumber | Added Element
Contract.supportingInfo | Added Element
Contract.relevantHistory | Added Element
Contract.signer.party | Type Reference: Added Target Type PractitionerRole
Contract.legallyBinding[x] | Added Element
Contract.action | deleted
Contract.actionReason | deleted
Contract.decisionType | deleted
Contract.securityLabel | deleted
Contract.agent | deleted
Contract.valuedItem | deleted
Contract.term.actionReason | deleted
Contract.term.agent | deleted
Contract.term.valuedItem | deleted
Contract.binding[x] | deleted
Coverage
Coverage.status | Min Cardinality changed from 0 to 1 Change value set from http://hl7.org/fhir/ValueSet/fm-status to http://hl7.org/fhir/ValueSet/fm-status|4.0.1
Coverage.beneficiary | Min Cardinality changed from 0 to 1
Coverage.relationship | Add Binding http://hl7.org/fhir/ValueSet/subscriber-relationship (extensible)
Coverage.payor | Min Cardinality changed from 0 to 1
Coverage.class | Renamed from grouping to class Max Cardinality changed from 1 to *
Coverage.class.type | Added Mandatory Element
Coverage.class.value | Added Mandatory Element
Coverage.class.name | Added Element
Coverage.costToBeneficiary | Added Element
Coverage.costToBeneficiary.type | Added Element
Coverage.costToBeneficiary.value[x] | Added Mandatory Element

Coverage.costToBeneficiary.exception | Added Element
Coverage.costToBeneficiary.exception.type | Added Mandatory Element
Coverage.costToBeneficiary.exception.period | Added Element
Coverage.subrogation | Added Element
Coverage.grouping.group | deleted
Coverage.grouping.groupDisplay | deleted
Coverage.grouping.subGroup | deleted
Coverage.grouping.subGroupDisplay | deleted
Coverage.grouping.plan | deleted
Coverage.grouping.planDisplay | deleted
Coverage.grouping.subPlan | deleted
Coverage.grouping.subPlanDisplay | deleted
Coverage.grouping.class | deleted
Coverage.grouping.classDisplay | deleted
Coverage.grouping.subClass | deleted
Coverage.grouping.subClassDisplay | deleted
Coverage.sequence | deleted
CoverageEligibilityRequest | Added Resource
CoverageEligibilityResponse | Added Resource
DetectedIssue
DetectedIssue.identifier | Max Cardinality changed from 1 to *
DetectedIssue.status | Change value set from http://hl7.org/fhir/ValueSet/observation-status to http://hl7.org/fhir/ValueSet/observation-status|4.0.1
DetectedIssue.code | Renamed from category to code
DetectedIssue.severity | Change value set from http://hl7.org/fhir/ValueSet/detectedissue-severity to http://hl7.org/fhir/ValueSet/detectedissue-severity|4.0.1
DetectedIssue.identified[x] | Added Element
DetectedIssue.author | Type Reference: Added Target Type PractitionerRole
DetectedIssue.evidence | Added Element
DetectedIssue.evidence.code | Added Element
DetectedIssue.evidence.detail | Added Element
DetectedIssue.mitigation.author | Type Reference: Added Target Type PractitionerRole
DetectedIssue.date | deleted
Device
Device.definition | Added Element
Device.udiCarrier | Added Element
Device.udiCarrier.deviceIdentifier | Added Element
Device.udiCarrier.issuer | Added Element
Device.udiCarrier.jurisdiction | Added Element
Device.udiCarrier.carrierAIDC | Added Element
Device.udiCarrier.carrierHRF | Added Element
Device.udiCarrier.entryType | Added Element
Device.status | Change value set from http://hl7.org/fhir/ValueSet/device-status to http://hl7.org/fhir/ValueSet/device-status|4.0.1
Device.statusReason | Added Element
Device.distinctIdentifier | Added Element
Device.serialNumber | Added Element
Device.deviceName | Added Element
Device.deviceName.name | Added Mandatory Element
Device.deviceName.type | Added Mandatory Element
Device.modelNumber | Added Element
Device.partNumber | Added Element
Device.specialization | Added Element
Device.specialization.systemType | Added Mandatory Element
Device.specialization.version | Added Element
Device.version | Max Cardinality changed from 1 to * Type changed from string to BackboneElement
Device.version.type | Added Element
Device.version.component | Added Element
Device.version.value | Added Mandatory Element
Device.property | Added Element
Device.property.type | Added Mandatory Element
Device.property.valueQuantity | Added Element
Device.property.valueCode | Added Element
Device.parent | Added Element
Device.udi | deleted
Device.model | deleted
DeviceDefinition | Added Resource
DeviceMetric
DeviceMetric.identifier | Min Cardinality changed from 1 to 0 Max Cardinality changed from 1 to *
DeviceMetric.parent | Type Reference: Added Target Type Device Type Reference: Removed Target Type DeviceComponent
DeviceMetric.operationalStatus | Change value set from http://hl7.org/fhir/ValueSet/metric-operational-status to http://hl7.org/fhir/ValueSet/metric-operational-status|4.0.1
DeviceMetric.color | Change value set from http://hl7.org/fhir/ValueSet/metric-color to http://hl7.org/fhir/ValueSet/metric-color|4.0.1

DeviceMetric.category | Change value set from http://hl7.org/fhir/ValueSet/metric-category to http://hl7.org/fhir/ValueSet/metric-category|4.0.1
DeviceMetric.calibration.type | Change value set from http://hl7.org/fhir/ValueSet/metric-calibration-type to http://hl7.org/fhir/ValueSet/metric-calibration-type|4.0.1
DeviceMetric.calibration.state | Change value set from http://hl7.org/fhir/ValueSet/metric-calibration-state to http://hl7.org/fhir/ValueSet/metric-calibration-state|4.0.1
DeviceRequest
DeviceRequest.instantiatesCanonical | Added Element
DeviceRequest.instantiatesUri | Added Element
DeviceRequest.status | Change value set from http://hl7.org/fhir/ValueSet/request-status to http://hl7.org/fhir/ValueSet/request-status|4.0.1
DeviceRequest.intent | Type changed from CodeableConcept to code Change value set from http://build.fhir.org/valueset-request-intent.html to http://hl7.org/fhir/ValueSet/request-intent|4.0.1
DeviceRequest.priority | Change value set from http://hl7.org/fhir/ValueSet/request-priority to http://hl7.org/fhir/ValueSet/request-priority|4.0.1 Default Value "If missing, normal priority" removed
DeviceRequest.parameter | Added Element
DeviceRequest.parameter.code | Added Element
DeviceRequest.parameter.value[x] | Added Element
DeviceRequest.encounter | Added Element
DeviceRequest.requester | Type changed from BackboneElement to Reference(Device | Practitioner | PractitionerRole | Organization)
DeviceRequest.performer | Type Reference: Added Target Types PractitionerRole, CareTeam
DeviceRequest.reasonReference | Type Reference: Added Target Types Condition, Observation, DiagnosticReport, DocumentReference Type Reference: Removed Target Type Resource
DeviceRequest.insurance | Added Element
DeviceRequest.definition | deleted
DeviceRequest.context | deleted
DeviceRequest.requester.agent | deleted
DeviceRequest.requester.onBehalfOf | deleted
DeviceUseStatement
DeviceUseStatement.basedOn | Added Element
DeviceUseStatement.status | Change value set from http://hl7.org/fhir/ValueSet/device-statement-status to http://hl7.org/fhir/ValueSet/device-statement-status|4.0.1
DeviceUseStatement.derivedFrom | Added Element
DeviceUseStatement.source | Type Reference: Added Target Type PractitionerRole
DeviceUseStatement.reasonCode | Added Element
DeviceUseStatement.reasonReference | Added Element
DeviceUseStatement.whenUsed | deleted
DeviceUseStatement.indication | deleted
DiagnosticReport
DiagnosticReport.basedOn | Type Reference: Added Target Type ServiceRequest Type Reference: Removed Target Types ProcedureRequest, ReferralRequest
DiagnosticReport.status | Change value set from http://hl7.org/fhir/ValueSet/diagnostic-report-status to http://hl7.org/fhir/ValueSet/diagnostic-report-status|4.0.1
DiagnosticReport.category | Max Cardinality changed from 1 to *
DiagnosticReport.encounter | Added Element
DiagnosticReport.performer | Type changed from BackboneElement to Reference(Practitioner | PractitionerRole | Organization | CareTeam)
DiagnosticReport.resultsInterpreter | Added Element
DiagnosticReport.imagingStudy | Type Reference: Removed Target Type ImagingManifest
DiagnosticReport.media | Renamed from image to media
DiagnosticReport.media.comment | Moved from DiagnosticReport.image to DiagnosticReport.media
DiagnosticReport.media.link | Moved from DiagnosticReport.image to DiagnosticReport.media
DiagnosticReport.conclusionCode | Renamed from codedDiagnosis to conclusionCode
DiagnosticReport.context | deleted
DiagnosticReport.performer.role | deleted
DiagnosticReport.performer.actor | deleted
DocumentManifest
DocumentManifest.status | Change value set from http://hl7.org/fhir/ValueSet/document-reference-status to http://hl7.org/fhir/ValueSet/document-reference-status|4.0.1
DocumentManifest.author | Type Reference: Added Target Type PractitionerRole
DocumentManifest.recipient | Type Reference: Added Target Type PractitionerRole
DocumentManifest.content | Type changed from BackboneElement to Reference(Resource)
DocumentManifest.content.p[x] | deleted
DocumentReference
DocumentReference.status | Change value set from http://hl7.org/fhir/ValueSet/document-reference-status to http://hl7.org/fhir/ValueSet/document-reference-status|4.0.1
DocumentReference.docStatus | Change value set from http://hl7.org/fhir/ValueSet/composition-status to http://hl7.org/fhir/ValueSet/composition-status|4.0.1
DocumentReference.type | Min Cardinality changed from 1 to 0
DocumentReference.category | Renamed from class to category Max Cardinality changed from 1 to *
DocumentReference.date | Added Element
DocumentReference.author | Type Reference: Added Target Type PractitionerRole
DocumentReference.authenticator | Type Reference: Added Target Type PractitionerRole

DocumentReference.relatesTo | No longer marked as Modifier
DocumentReference.relatesTo.code | Change value set from http://hl7.org/fhir/ValueSet/document-relationship-type to http://hl7.org/fhir/ValueSet/document-relationship-type|4.0.1
DocumentReference.context.encounter | Max Cardinality changed from 1 to * Type Reference: Added Target Type EpisodeOfCare
DocumentReference.context.related | Type changed from BackboneElement to Reference(Resource)
DocumentReference.created | deleted
DocumentReference.indexed | deleted
DocumentReference.context.related.identifier | deleted
DocumentReference.context.related.ref | deleted
DomainResource | No Changes
EffectEvidenceSynthesis | Added Resource
Encounter
Encounter.status | Change value set from http://hl7.org/fhir/ValueSet/encounter-status to http://hl7.org/fhir/ValueSet/encounter-status|4.0.1
Encounter.statusHistory.status | Change value set from http://hl7.org/fhir/ValueSet/encounter-status to http://hl7.org/fhir/ValueSet/encounter-status|4.0.1
Encounter.class | Min Cardinality changed from 0 to 1
Encounter.serviceType | Added Element
Encounter.basedOn | Renamed from incomingReferral to basedOn Type Reference: Added Target Type ServiceRequest Type Reference: Removed Target Type ReferralRequest
Encounter.participant.individual | Type Reference: Added Target Type PractitionerRole
Encounter.appointment | Max Cardinality changed from 1 to *
Encounter.reasonCode | Added Element
Encounter.reasonReference | Added Element
Encounter.diagnosis.use | Added Element
Encounter.hospitalization.origin | Type Reference: Added Target Type Organization
Encounter.hospitalization.destination | Type Reference: Added Target Type Organization
Encounter.location.status | Change value set from http://hl7.org/fhir/ValueSet/encounter-location-status to http://hl7.org/fhir/ValueSet/encounter-location-status|4.0.1
Encounter.location.physicalType | Added Element
Encounter.reason | deleted
Encounter.diagnosis.role | deleted
Endpoint
Endpoint.status | Change value set from http://hl7.org/fhir/ValueSet/endpoint-status to http://hl7.org/fhir/ValueSet/endpoint-status|4.0.1
Endpoint.connectionType | Change code system for extensibly bound codes from "http://hl7.org/fhir/endpoint-connection-type" to "http://terminology.hl7.org/CodeSystem/endpoint-connection-type"
Endpoint.payloadMimeType | Change value set from http://hl7.org/fhir/ValueSet/mimetypes to http://hl7.org/fhir/ValueSet/mimetypes|4.0.1
Endpoint.address | Type changed from uri to url
EnrollmentRequest
EnrollmentRequest.status | Change value set from http://hl7.org/fhir/ValueSet/fm-status to http://hl7.org/fhir/ValueSet/fm-status|4.0.1
EnrollmentRequest.provider | Type Reference: Added Target Types PractitionerRole, Organization
EnrollmentRequest.candidate | Renamed from subject to candidate
EnrollmentRequest.organization | deleted
EnrollmentResponse
EnrollmentResponse.status | Change value set from http://hl7.org/fhir/ValueSet/fm-status to http://hl7.org/fhir/ValueSet/fm-status|4.0.1
EnrollmentResponse.outcome | Type changed from CodeableConcept to code Change value set from http://hl7.org/fhir/ValueSet/remittance-outcome to http://hl7.org/fhir/ValueSet/remittance-outcome|4.0.1
EnrollmentResponse.requestProvider | Type Reference: Added Target Types PractitionerRole, Organization
EnrollmentResponse.requestOrganization | deleted
EpisodeOfCare
EpisodeOfCare.status | Change value set from http://hl7.org/fhir/ValueSet/episode-of-care-status to http://hl7.org/fhir/ValueSet/episode-of-care-status|4.0.1
EpisodeOfCare.statusHistory.status | Change value set from http://hl7.org/fhir/ValueSet/episode-of-care-status to http://hl7.org/fhir/ValueSet/episode-of-care-status|4.0.1
EpisodeOfCare.referralRequest | Type Reference: Added Target Type ServiceRequest Type Reference: Removed Target Type ReferralRequest
EpisodeOfCare.careManager | Type Reference: Added Target Type PractitionerRole
EventDefinition | Added Resource
Evidence | Added Resource
EvidenceVariable | Added Resource
ExampleScenario | Added Resource
ExplanationOfBenefit
ExplanationOfBenefit.status | Min Cardinality changed from 0 to 1 Change value set from http://hl7.org/fhir/ValueSet/explanationofbenefit-status to http://hl7.org/fhir/ValueSet/explanationofbenefit-status|4.0.1
ExplanationOfBenefit.type | Min Cardinality changed from 0 to 1 Change binding strength from required to extensible
ExplanationOfBenefit.subType | Max Cardinality changed from * to 1
ExplanationOfBenefit.use | Added Mandatory Element
ExplanationOfBenefit.patient | Min Cardinality changed from 0 to 1
ExplanationOfBenefit.created | Min Cardinality changed from 0 to 1
ExplanationOfBenefit.enterer | Type Reference: Added Target Type PractitionerRole

ExplanationOfBenefit.insurer | Min Cardinality changed from 0 to 1
ExplanationOfBenefit.provider | Min Cardinality changed from 0 to 1 Type Reference: Added Target Types PractitionerRole, Organization
ExplanationOfBenefit.priority | Added Element
ExplanationOfBenefit.fundsReserveRequested | Added Element
ExplanationOfBenefit.fundsReserve | Added Element
ExplanationOfBenefit.payee.party | Type Reference: Added Target Type PractitionerRole
ExplanationOfBenefit.referral | Type Reference: Added Target Type ServiceRequest Type Reference: Removed Target Type ReferralRequest
ExplanationOfBenefit.outcome | Min Cardinality changed from 0 to 1 Type changed from CodeableConcept to code Add Binding http://hl7.org/fhir/ValueSet/remittance-outcome|4.0.1 (required)
ExplanationOfBenefit.preAuthRef | Added Element
ExplanationOfBenefit.preAuthRefPeriod | Added Element
ExplanationOfBenefit.careTeam.provider | Type Reference: Added Target Type PractitionerRole
ExplanationOfBenefit.supportingInfo | Renamed from information to supportingInfo
ExplanationOfBenefit.supportingInfo.sequence | Moved from ExplanationOfBenefit.information to ExplanationOfBenefit.supportingInfo
ExplanationOfBenefit.supportingInfo.category | Moved from ExplanationOfBenefit.information to ExplanationOfBenefit.supportingInfo
ExplanationOfBenefit.supportingInfo.code | Moved from ExplanationOfBenefit.information to ExplanationOfBenefit.supportingInfo
ExplanationOfBenefit.supportingInfo.timing[x] | Moved from ExplanationOfBenefit.information to ExplanationOfBenefit.supportingInfo
ExplanationOfBenefit.supportingInfo.value[x] | Moved from ExplanationOfBenefit.information to ExplanationOfBenefit.supportingInfo Add Type boolean
ExplanationOfBenefit.supportingInfo.reason | Moved from ExplanationOfBenefit.information to ExplanationOfBenefit.supportingInfo
ExplanationOfBenefit.diagnosis.onAdmission | Added Element
ExplanationOfBenefit.procedure.type | Added Element
ExplanationOfBenefit.procedure.udi | Added Element
ExplanationOfBenefit.insurance | Min Cardinality changed from 0 to 1 Max Cardinality changed from 1 to *
ExplanationOfBenefit.insurance.focal | Added Mandatory Element
ExplanationOfBenefit.insurance.coverage | Min Cardinality changed from 0 to 1
ExplanationOfBenefit.accident.type | Change binding strength from required to extensible
ExplanationOfBenefit.item.careTeamSequence | Renamed from careTeamLinkId to careTeamSequence
ExplanationOfBenefit.item.diagnosisSequence | Renamed from diagnosisLinkId to diagnosisSequence
ExplanationOfBenefit.item.procedureSequence | Renamed from procedureLinkId to procedureSequence
ExplanationOfBenefit.item.informationSequence | Renamed from informationLinkId to informationSequence
ExplanationOfBenefit.item.productOrService | Renamed from service to productOrService Min Cardinality changed from 0 to 1
ExplanationOfBenefit.item.detail.productOrService | Renamed from service to productOrService Min Cardinality changed from 0 to 1
ExplanationOfBenefit.item.detail.subDetail.productOrService | Renamed from service to productOrService Min Cardinality changed from 0 to 1
ExplanationOfBenefit.addItem.itemSequence | Renamed from sequenceLinkId to itemSequence
ExplanationOfBenefit.addItem.detailSequence | Added Element
ExplanationOfBenefit.addItem.subDetailSequence | Added Element
ExplanationOfBenefit.addItem.provider | Added Element
ExplanationOfBenefit.addItem.productOrService | Renamed from service to productOrService Min Cardinality changed from 0 to 1
ExplanationOfBenefit.addItem.programCode | Added Element
ExplanationOfBenefit.addItem.serviced[x] | Added Element
ExplanationOfBenefit.addItem.location[x] | Added Element
ExplanationOfBenefit.addItem.quantity | Added Element
ExplanationOfBenefit.addItem.unitPrice | Added Element
ExplanationOfBenefit.addItem.factor | Added Element
ExplanationOfBenefit.addItem.net | Renamed from fee to net
ExplanationOfBenefit.addItem.bodySite | Added Element
ExplanationOfBenefit.addItem.subSite | Added Element
ExplanationOfBenefit.addItem.detail.productOrService | Renamed from service to productOrService Min Cardinality changed from 0 to 1
ExplanationOfBenefit.addItem.detail.quantity | Added Element
ExplanationOfBenefit.addItem.detail.unitPrice | Added Element
ExplanationOfBenefit.addItem.detail.factor | Added Element
ExplanationOfBenefit.addItem.detail.net | Renamed from fee to net
ExplanationOfBenefit.addItem.detail.subDetail | Added Element
ExplanationOfBenefit.addItem.detail.subDetail.productOrService | Added Mandatory Element
ExplanationOfBenefit.addItem.detail.subDetail.modifier | Added Element
ExplanationOfBenefit.addItem.detail.subDetail.quantity | Added Element
ExplanationOfBenefit.addItem.detail.subDetail.unitPrice | Added Element
ExplanationOfBenefit.addItem.detail.subDetail.factor | Added Element
ExplanationOfBenefit.addItem.detail.subDetail.net | Added Element
ExplanationOfBenefit.addItem.detail.subDetail.noteNumber | Added Element
ExplanationOfBenefit.addItem.detail.subDetail.adjudication | Added Element
ExplanationOfBenefit.adjudication | Added Element
ExplanationOfBenefit.total | Added Element

ExplanationOfBenefit.total.category | Added Mandatory Element
ExplanationOfBenefit.total.amount | Added Mandatory Element
ExplanationOfBenefit.formCode | Added Element
ExplanationOfBenefit.form | Type changed from CodeableConcept to Attachment
ExplanationOfBenefit.processNote.type | Type changed from CodeableConcept to code Change value set from http://hl7.org/fhir/ValueSet/note-type to http://hl7.org/fhir/ValueSet/note-type|4.0.1
ExplanationOfBenefit.processNote.language | Change binding strength from extensible to preferred
ExplanationOfBenefit.benefitPeriod | Added Element
ExplanationOfBenefit.organization | deleted
ExplanationOfBenefit.payee.resourceType | deleted
ExplanationOfBenefit.employmentImpacted | deleted
ExplanationOfBenefit.hospitalization | deleted
ExplanationOfBenefit.item.detail.type | deleted
ExplanationOfBenefit.item.detail.subDetail.type | deleted
ExplanationOfBenefit.addItem.revenue | deleted
ExplanationOfBenefit.addItem.category | deleted
ExplanationOfBenefit.addItem.detail.revenue | deleted
ExplanationOfBenefit.addItem.detail.category | deleted
ExplanationOfBenefit.totalCost | deleted
ExplanationOfBenefit.unallocDeductable | deleted
ExplanationOfBenefit.totalBenefit | deleted
ExplanationOfBenefit.benefitBalance.subCategory | deleted
FamilyMemberHistory
FamilyMemberHistory.instantiatesCanonical | Added Element
FamilyMemberHistory.instantiatesUri | Added Element
FamilyMemberHistory.status | Change value set from http://hl7.org/fhir/ValueSet/history-status to http://hl7.org/fhir/ValueSet/history-status|4.0.1
FamilyMemberHistory.dataAbsentReason | Added Element
FamilyMemberHistory.sex | Added Element
FamilyMemberHistory.estimatedAge | No longer marked as Modifier
FamilyMemberHistory.reasonReference | Type Reference: Added Target Types DiagnosticReport, DocumentReference
FamilyMemberHistory.condition.contributedToDeath | Added Element
FamilyMemberHistory.definition | deleted
FamilyMemberHistory.notDone | deleted
FamilyMemberHistory.notDoneReason | deleted
FamilyMemberHistory.gender | deleted
Flag
Flag.status | Change value set from http://hl7.org/fhir/ValueSet/flag-status to http://hl7.org/fhir/ValueSet/flag-status|4.0.1
Flag.category | Max Cardinality changed from 1 to *
Flag.author | Type Reference: Added Target Type PractitionerRole
Goal
Goal.lifecycleStatus | Added Mandatory Element
Goal.achievementStatus | Added Element
Goal.subject | Min Cardinality changed from 0 to 1
Goal.target | Max Cardinality changed from 1 to *
Goal.target.detail[x] | Add Types string, boolean, integer, Ratio
Goal.expressedBy | Type Reference: Added Target Type PractitionerRole
Goal.addresses | Type Reference: Added Target Type ServiceRequest Type Reference: Removed Target Type ProcedureRequest
Goal.status | deleted
GraphDefinition
GraphDefinition.status | Change value set from http://hl7.org/fhir/ValueSet/publication-status to http://hl7.org/fhir/ValueSet/publication-status|4.0.1
GraphDefinition.experimental | No longer marked as Modifier
GraphDefinition.start | Change value set from http://hl7.org/fhir/ValueSet/resource-types to http://hl7.org/fhir/ValueSet/resource-types|4.0.1
GraphDefinition.profile | Type changed from uri to canonical(StructureDefinition)
GraphDefinition.link.path | Min Cardinality changed from 1 to 0
GraphDefinition.link.target | Min Cardinality changed from 1 to 0
GraphDefinition.link.target.type | Change value set from http://hl7.org/fhir/ValueSet/resource-types to http://hl7.org/fhir/ValueSet/resource-types|4.0.1
GraphDefinition.link.target.params | Added Element
GraphDefinition.link.target.profile | Type changed from uri to canonical(StructureDefinition)
GraphDefinition.link.target.compartment.use | Added Mandatory Element
GraphDefinition.link.target.compartment.code | Change value set from http://hl7.org/fhir/ValueSet/compartment-type to http://hl7.org/fhir/ValueSet/compartment-type|4.0.1
GraphDefinition.link.target.compartment.rule | Change value set from http://hl7.org/fhir/ValueSet/graph-compartment-rule to http://hl7.org/fhir/ValueSet/graph-compartment-rule|4.0.1
Group
Group.active | Default Value "true" removed
Group.type | Change value set from http://hl7.org/fhir/ValueSet/group-type to http://hl7.org/fhir/ValueSet/group-type|4.0.1
Group.managingEntity | Added Element
Group.characteristic.value[x] | Add Type Reference
Group.characteristic.exclude | No longer marked as Modifier

Group.member.entity | Type Reference: Added Target Types PractitionerRole, Group
Group.member.inactive | Default Value "false" removed
GuidanceResponse
GuidanceResponse.requestIdentifier | Renamed from requestId to requestIdentifier Type changed from id to Identifier
GuidanceResponse.identifier | Max Cardinality changed from 1 to *
GuidanceResponse.module[x] | Renamed from module to module[x] Add Types uri, canonical, CodeableConcept Remove Type Reference(ServiceDefinition)
GuidanceResponse.status | Change value set from http://hl7.org/fhir/ValueSet/guidance-response-status to http://hl7.org/fhir/ValueSet/guidance-response-status|4.0.1
GuidanceResponse.encounter | Added Element
GuidanceResponse.reasonCode | Added Element
GuidanceResponse.reasonReference | Added Element
GuidanceResponse.context | deleted
GuidanceResponse.reason[x] | deleted
HealthcareService
HealthcareService.active | Default Value "true" removed
HealthcareService.category | Max Cardinality changed from 1 to *
HealthcareService.extraDetails | Type changed from string to markdown
HealthcareService.eligibility | Max Cardinality changed from 1 to * Type changed from CodeableConcept to BackboneElement
HealthcareService.eligibility.code | Added Element
HealthcareService.eligibility.comment | Added Element
HealthcareService.program | Added Element
HealthcareService.communication | Added Element
HealthcareService.availableTime.daysOfWeek | Change value set from http://hl7.org/fhir/ValueSet/days-of-week to http://hl7.org/fhir/ValueSet/days-of-week|4.0.1
HealthcareService.eligibilityNote | deleted
HealthcareService.programName | deleted
ImagingStudy
ImagingStudy.status | Added Mandatory Element
ImagingStudy.modality | Renamed from modalityList to modality Change value set from http://hl7.org/fhir/ValueSet/dicom-cid29 to http://dicom.nema.org/medical/dicom/current/output/chtml/part16/sect_CID_29.html
ImagingStudy.subject | Renamed from patient to subject Type Reference: Added Target Types Device, Group
ImagingStudy.encounter | Added Element
ImagingStudy.basedOn | Type Reference: Added Target Types ServiceRequest, Appointment, AppointmentResponse, Task Type Reference: Removed Target Types ReferralRequest, ProcedureRequest
ImagingStudy.referrer | Type Reference: Added Target Type PractitionerRole
ImagingStudy.interpreter | Type Reference: Added Target Type PractitionerRole
ImagingStudy.procedureReference | Max Cardinality changed from * to 1
ImagingStudy.procedureCode | Add Binding http://www.rsna.org/RadLex_Playbook.aspx (extensible)
ImagingStudy.location | Added Element
ImagingStudy.reasonCode | Added Element
ImagingStudy.reasonReference | Added Element
ImagingStudy.note | Added Element
ImagingStudy.series.uid | Type changed from oid to id
ImagingStudy.series.modality | Change value set from http://hl7.org/fhir/ValueSet/dicom-cid29 to http://dicom.nema.org/medical/dicom/current/output/chtml/part16/sect_CID_29.html
ImagingStudy.series.specimen | Added Element
ImagingStudy.series.performer | Type changed from Reference(Practitioner) to BackboneElement
ImagingStudy.series.performer.function | Added Element
ImagingStudy.series.performer.actor | Added Mandatory Element
ImagingStudy.series.instance.uid | Type changed from oid to id
ImagingStudy.series.instance.sopClass | Type changed from oid to Coding Add Binding http://dicom.nema.org/medical/dicom/current/output/chtml/part04/sect_B.5.html#table_B.5-1 (extensible)
ImagingStudy.uid | deleted
ImagingStudy.accession | deleted
ImagingStudy.availability | deleted
ImagingStudy.context | deleted
ImagingStudy.reason | deleted
ImagingStudy.series.availability | deleted
Immunization
Immunization.status | Change value set from http://hl7.org/fhir/ValueSet/immunization-status to http://hl7.org/fhir/ValueSet/immunization-status|4.0.1
Immunization.statusReason | Renamed from notGiven to statusReason Min Cardinality changed from 1 to 0 Type changed from boolean to CodeableConcept No longer marked as Modifier
Immunization.occurrence[x] | Renamed from date to occurrence[x] Min Cardinality changed from 0 to 1 Add Type string
Immunization.recorded | Added Element
Immunization.primarySource | Min Cardinality changed from 1 to 0
Immunization.performer | Renamed from practitioner to performer
Immunization.performer.function | Moved from Immunization.practitioner.role to function Change value set from http://hl7.org/fhir/ValueSet/immunization-role to http://hl7.org/fhir/ValueSet/immunization-function
Immunization.performer.actor | Moved from Immunization.practitioner to Immunization.performer Type Reference: Added Target Types PractitionerRole, Organization
Immunization.reasonCode | Moved from Immunization.explanation.reason to reasonCode

Immunization.reasonReference | Added Element
Immunization.isSubpotent | Added Element
Immunization.subpotentReason | Added Element
Immunization.education | Added Element
Immunization.education.documentType | Added Element
Immunization.education.reference | Added Element
Immunization.education.publicationDate | Added Element
Immunization.education.presentationDate | Added Element
Immunization.programEligibility | Added Element
Immunization.fundingSource | Added Element
Immunization.protocolApplied | Renamed from vaccinationProtocol to protocolApplied
Immunization.protocolApplied.series | Moved from Immunization.vaccinationProtocol to Immunization.protocolApplied
Immunization.protocolApplied.authority | Moved from Immunization.vaccinationProtocol to Immunization.protocolApplied
Immunization.protocolApplied.targetDisease | Moved from Immunization.vaccinationProtocol to Immunization.protocolApplied Min Cardinality changed from 1 to 0
Immunization.protocolApplied.doseNumber[x] | Moved from Immunization.vaccinationProtocol.doseSequence to doseNumber[x] Min Cardinality changed from 0 to 1 Add Type string
Immunization.protocolApplied.seriesDoses[x] | Moved from Immunization.vaccinationProtocol.seriesDoses to seriesDoses[x] Add Type string
Immunization.explanation | deleted
Immunization.vaccinationProtocol.description | deleted
Immunization.vaccinationProtocol.doseStatus | deleted
Immunization.vaccinationProtocol.doseStatusReason | deleted
ImmunizationEvaluation | Added Resource
ImmunizationRecommendation
ImmunizationRecommendation.date | Moved from ImmunizationRecommendation.recommendation to ImmunizationRecommendation
ImmunizationRecommendation.authority | Moved from ImmunizationRecommendation.recommendation.protocol to ImmunizationRecommendation
ImmunizationRecommendation.recommendation.vaccineCode | Max Cardinality changed from 1 to *
ImmunizationRecommendation.recommendation.contraindicatedVaccineCode | Added Element
ImmunizationRecommendation.recommendation.forecastStatus | Now marked as Modifier
ImmunizationRecommendation.recommendation.forecastReason | Added Element
ImmunizationRecommendation.recommendation.description | Moved from ImmunizationRecommendation.recommendation.protocol to ImmunizationRecommendation.recommendation
ImmunizationRecommendation.recommendation.series | Moved from ImmunizationRecommendation.recommendation.protocol to ImmunizationRecommendation.recommendation
ImmunizationRecommendation.recommendation.doseNumber[x] | Renamed from doseNumber to doseNumber[x] Add Type string
ImmunizationRecommendation.recommendation.seriesDoses[x] | Moved from ImmunizationRecommendation.recommendation.protocol.doseSequence to seriesDoses[x] Add Type string
ImmunizationRecommendation.recommendation.supportingImmunization | Type Reference: Added Target Type ImmunizationEvaluation
ImmunizationRecommendation.recommendation.supportingPatientInformation | Type Reference: Added Target Type Resource Type Reference: Removed Target Types Observation, AllergyIntolerance
ImmunizationRecommendation.recommendation.protocol | deleted
ImplementationGuide
ImplementationGuide | Min Cardinality changed from 1 to 0 Max Cardinality changed from 1 to *
ImplementationGuide.title | Added Element
ImplementationGuide.status | Change value set from http://hl7.org/fhir/ValueSet/publication-status to http://hl7.org/fhir/ValueSet/publication-status|4.0.1
ImplementationGuide.experimental | No longer marked as Modifier
ImplementationGuide.packageId | Added Mandatory Element
ImplementationGuide.license | Added Element
ImplementationGuide.fhirVersion | Min Cardinality changed from 0 to 1 Max Cardinality changed from 1 to * Type changed from id to code Add Binding http://hl7.org/fhir/ValueSet/FHIR-version|4.0.1 (required)
ImplementationGuide.dependsOn | Renamed from dependency to dependsOn
ImplementationGuide.dependsOn.uri | Moved from ImplementationGuide.dependency to ImplementationGuide.dependsOn Type changed from uri to canonical(ImplementationGuide)
ImplementationGuide.dependsOn.packageId | Added Element
ImplementationGuide.dependsOn.version | Added Element
ImplementationGuide.global.type | Change value set from http://hl7.org/fhir/ValueSet/resource-types to http://hl7.org/fhir/ValueSet/resource-types|4.0.1
ImplementationGuide.global.profile | Type changed from Reference(StructureDefinition) to canonical(StructureDefinition)
ImplementationGuide.definition | Added Element
ImplementationGuide.definition.grouping | Added Element
ImplementationGuide.definition.grouping.name | Added Mandatory Element
ImplementationGuide.definition.grouping.description | Added Element
ImplementationGuide.definition.resource | Moved from ImplementationGuide.package to ImplementationGuide.definition
ImplementationGuide.definition.resource.reference | Added Mandatory Element
ImplementationGuide.definition.resource.fhirVersion | Added Element
ImplementationGuide.definition.resource.example[x] | Moved from ImplementationGuide.package.resource.example to example[x] Min Cardinality changed from 1 to 0 Add Type canonical(StructureDefinition)
ImplementationGuide.definition.resource.groupingId | Added Element
ImplementationGuide.definition.page | Moved from ImplementationGuide to ImplementationGuide.definition

ImplementationGuide.definition.page.name[x] | Added Mandatory Element
ImplementationGuide.definition.page.generation | Added Mandatory Element
ImplementationGuide.definition.parameter | Added Element
ImplementationGuide.definition.parameter.code | Added Mandatory Element
ImplementationGuide.definition.parameter.value | Added Mandatory Element
ImplementationGuide.definition.template | Added Element
ImplementationGuide.definition.template.code | Added Mandatory Element
ImplementationGuide.definition.template.source | Added Mandatory Element
ImplementationGuide.definition.template.scope | Added Element
ImplementationGuide.manifest | Added Element
ImplementationGuide.manifest.rendering | Added Element
ImplementationGuide.manifest.resource | Added Mandatory Element
ImplementationGuide.manifest.resource.reference | Added Mandatory Element
ImplementationGuide.manifest.resource.example[x] | Added Element
ImplementationGuide.manifest.resource.relativePath | Added Element
ImplementationGuide.manifest.page | Added Element
ImplementationGuide.manifest.page.name | Added Mandatory Element
ImplementationGuide.manifest.page.title | Added Element
ImplementationGuide.manifest.page.anchor | Added Element
ImplementationGuide.manifest.image | Added Element
ImplementationGuide.manifest.other | Added Element
ImplementationGuide.dependency.type | deleted
ImplementationGuide.package | deleted
ImplementationGuide.binary | deleted
ImplementationGuide.page.source | deleted
ImplementationGuide.page.kind | deleted
ImplementationGuide.page.type | deleted
ImplementationGuide.page.package | deleted
ImplementationGuide.page.format | deleted
InsurancePlan | Added Resource
Invoice | Added Resource
Library
Library.subtitle | Added Element
Library.status | Change value set from http://hl7.org/fhir/ValueSet/publication-status to http://hl7.org/fhir/ValueSet/publication-status|4.0.1
Library.experimental | No longer marked as Modifier
Library.type | Change code system for extensibly bound codes from "http://hl7.org/fhir/library-type" to "http://terminology.hl7.org/CodeSystem/library-type"
Library.subject[x] | Added Element
Library.author | Added Element
Library.editor | Added Element
Library.reviewer | Added Element
Library.endorser | Added Element
Library.contributor | deleted
Linkage
Linkage.active | Default Value "true" removed
Linkage.author | Type Reference: Added Target Type PractitionerRole
Linkage.item.type | Change value set from http://hl7.org/fhir/ValueSet/linkage-type to http://hl7.org/fhir/ValueSet/linkage-type|4.0.1
Linkage.item.resource | Type changed from Reference to Reference(Resource)
List
List.status | Change value set from http://hl7.org/fhir/ValueSet/list-status to http://hl7.org/fhir/ValueSet/list-status|4.0.1
List.mode | Change value set from http://hl7.org/fhir/ValueSet/list-mode to http://hl7.org/fhir/ValueSet/list-mode|4.0.1
List.source | Type Reference: Added Target Type PractitionerRole
List.entry.deleted | Default Value "false" removed
Location
Location.status | Change value set from http://hl7.org/fhir/ValueSet/location-status to http://hl7.org/fhir/ValueSet/location-status|4.0.1
Location.mode | Change value set from http://hl7.org/fhir/ValueSet/location-mode to http://hl7.org/fhir/ValueSet/location-mode|4.0.1 No longer marked as Modifier
Location.type | Max Cardinality changed from 1 to *
Location.hoursOfOperation | Added Element
Location.hoursOfOperation.daysOfWeek | Added Element
Location.hoursOfOperation.allDay | Added Element
Location.hoursOfOperation.openingTime | Added Element
Location.hoursOfOperation.closingTime | Added Element
Location.availabilityExceptions | Added Element
Measure
Measure.subtitle | Added Element
Measure.status | Change value set from http://hl7.org/fhir/ValueSet/publication-status to http://hl7.org/fhir/ValueSet/publication-status|4.0.1
Measure.experimental | No longer marked as Modifier
Measure.subject[x] | Added Element

Measure.author | Added Element
Measure.editor | Added Element
Measure.reviewer | Added Element
Measure.endorser | Added Element
Measure.library | Type changed from Reference(Library) to canonical(Library)
Measure.scoring | Change code system for extensibly bound codes from "http://hl7.org/fhir/measure-scoring" to "http://terminology.hl7.org/CodeSystem/measure-scoring"
Measure.compositeScoring | Change code system for extensibly bound codes from "http://hl7.org/fhir/composite-measure-scoring" to "http://terminology.hl7.org/CodeSystem/composite-measure-scoring"
Measure.type | Change code system for extensibly bound codes from "http://hl7.org/fhir/measure-type" to "http://terminology.hl7.org/CodeSystem/measure-type"
Measure.improvementNotation | Type changed from string to CodeableConcept Add Binding http://hl7.org/fhir/ValueSet/measure-improvement-notation|4.0.1 (required)
Measure.group.code | Added Element
Measure.group.population.code | Change code system for extensibly bound codes from "http://hl7.org/fhir/measure-population" to "http://terminology.hl7.org/CodeSystem/measure-population"
Measure.group.population.criteria | Type changed from string to Expression
Measure.group.stratifier.code | Added Element
Measure.group.stratifier.description | Added Element
Measure.group.stratifier.criteria | Type changed from string to Expression
Measure.group.stratifier.component | Added Element
Measure.group.stratifier.component.code | Added Element
Measure.group.stratifier.component.description | Added Element
Measure.group.stratifier.component.criteria | Added Mandatory Element
Measure.supplementalData.code | Added Element
Measure.supplementalData.usage | Change code system for extensibly bound codes from "http://hl7.org/fhir/measure-data-usage" to "http://terminology.hl7.org/CodeSystem/measure-data-usage"
Measure.supplementalData.description | Added Element
Measure.supplementalData.criteria | Min Cardinality changed from 0 to 1 Type changed from string to Expression
Measure.contributor | deleted
Measure.set | deleted
Measure.group.identifier | deleted
Measure.group.name | deleted
Measure.group.population.identifier | deleted
Measure.group.population.name | deleted
Measure.group.stratifier.identifier | deleted
Measure.group.stratifier.path | deleted
Measure.supplementalData.identifier | deleted
Measure.supplementalData.path | deleted
MeasureReport
MeasureReport.identifier | Max Cardinality changed from 1 to *
MeasureReport.status | Change value set from http://hl7.org/fhir/ValueSet/measure-report-status to http://hl7.org/fhir/ValueSet/measure-report-status|4.0.1
MeasureReport.type | Change value set from http://hl7.org/fhir/ValueSet/measure-report-type to http://hl7.org/fhir/ValueSet/measure-report-type|4.0.1
MeasureReport.measure | Type changed from Reference(Measure) to canonical(Measure)
MeasureReport.subject | Added Element
MeasureReport.reporter | Added Element
MeasureReport.improvementNotation | Added Element
MeasureReport.group.code | Added Element
MeasureReport.group.population.code | Change code system for extensibly bound codes from "http://hl7.org/fhir/measure-population" to "http://terminology.hl7.org/CodeSystem/measure-population"
MeasureReport.group.population.subjectResults | Added Element
MeasureReport.group.measureScore | Type changed from decimal to Quantity
MeasureReport.group.stratifier.code | Added Element
MeasureReport.group.stratifier.stratum.value | Min Cardinality changed from 1 to 0 Type changed from string to CodeableConcept
MeasureReport.group.stratifier.stratum.component | Added Element
MeasureReport.group.stratifier.stratum.component.code | Added Mandatory Element
MeasureReport.group.stratifier.stratum.component.value | Added Mandatory Element
MeasureReport.group.stratifier.stratum.population.code | Change code system for extensibly bound codes from "http://hl7.org/fhir/measure-population" to "http://terminology.hl7.org/CodeSystem/measure-population"
MeasureReport.group.stratifier.stratum.population.subjectResults | Added Element
MeasureReport.group.stratifier.stratum.measureScore | Type changed from decimal to Quantity
MeasureReport.evaluatedResource | Added Element
MeasureReport.patient | deleted
MeasureReport.reportingOrganization | deleted
MeasureReport.group.identifier | deleted
MeasureReport.group.population.identifier | deleted
MeasureReport.group.population.patients | deleted
MeasureReport.group.stratifier.identifier | deleted
MeasureReport.group.stratifier.stratum.population.identifier | deleted
MeasureReport.group.stratifier.stratum.population.patients | deleted
MeasureReport.evaluatedResources | deleted

Media
Media.basedOn | Type Reference: Added Target Types ServiceRequest, CarePlan Type Reference: Removed Target Type ProcedureRequest
Media.partOf | Added Element
Media.status | Added Mandatory Element
Media.type | Min Cardinality changed from 1 to 0 Type changed from code to CodeableConcept Change binding strength from required to extensible Change value set from http://hl7.org/fhir/ValueSet/digital-media-type to http://hl7.org/fhir/ValueSet/media-type
Media.modality | Renamed from subtype to modality
Media.subject | Type Reference: Added Target Types PractitionerRole, Location
Media.encounter | Added Element
Media.created[x] | Renamed from occurrence[x] to created[x]
Media.issued | Added Element
Media.operator | Type Reference: Added Target Types PractitionerRole, Organization, CareTeam, Patient, Device, RelatedPerson
Media.deviceName | Added Element
Media.frames | Default Value "1" removed
Media.duration | Type changed from unsignedInt to decimal
Media.context | deleted
Medication
Medication.identifier | Added Element
Medication.status | Change value set from http://hl7.org/fhir/ValueSet/medication-status to http://hl7.org/fhir/ValueSet/medication-status|4.0.1 Now marked as Modifier
Medication.amount | Added Element
Medication.ingredient.strength | Added Element
Medication.batch | Added Element
Medication.batch.lotNumber | Added Element
Medication.batch.expirationDate | Added Element
Medication.isBrand | deleted
Medication.isOverTheCounter | deleted
Medication.ingredient.amount | deleted
Medication.package | deleted
Medication.image | deleted
MedicationAdministration
MedicationAdministration.instantiates | Added Element
MedicationAdministration.status | Change value set from http://hl7.org/fhir/ValueSet/medication-admin-status to http://hl7.org/fhir/ValueSet/medication-admin-status|4.0.1
MedicationAdministration.statusReason | Added Element
MedicationAdministration.performer.function | Added Element
MedicationAdministration.performer.actor | Type Reference: Added Target Type PractitionerRole
MedicationAdministration.reasonReference | Type Reference: Added Target Type DiagnosticReport
MedicationAdministration.request | Added Element
MedicationAdministration.dosage.rate[x] | Add Type Quantity() Remove Type Quantity()
MedicationAdministration.definition | deleted
MedicationAdministration.performer.onBehalfOf | deleted
MedicationAdministration.notGiven | deleted
MedicationAdministration.reasonNotGiven | deleted
MedicationAdministration.prescription | deleted
MedicationDispense
MedicationDispense.status | Min Cardinality changed from 0 to 1 Change value set from http://hl7.org/fhir/ValueSet/medication-dispense-status to http://hl7.org/fhir/ValueSet/medicationdispense-status|4.0.1
MedicationDispense.statusReason[x] | Added Element
MedicationDispense.performer.function | Added Element
MedicationDispense.performer.actor | Type Reference: Added Target Type PractitionerRole
MedicationDispense.location | Added Element
MedicationDispense.substitution.responsibleParty | Type Reference: Added Target Type PractitionerRole
MedicationDispense.performer.onBehalfOf | deleted
MedicationDispense.notDone | deleted
MedicationDispense.notDoneReason[x] | deleted
MedicationKnowledge | Added Resource
MedicationRequest
MedicationRequest.status | Min Cardinality changed from 0 to 1 Change value set from http://hl7.org/fhir/ValueSet/medication-request-status to http://hl7.org/fhir/ValueSet/medicationrequest-status|4.0.1
MedicationRequest.statusReason | Added Element
MedicationRequest.intent | Change value set from http://hl7.org/fhir/ValueSet/medication-request-intent to http://hl7.org/fhir/ValueSet/medicationrequest-intent|4.0.1
MedicationRequest.category | Max Cardinality changed from 1 to *
MedicationRequest.priority | Change value set from http://hl7.org/fhir/ValueSet/medication-request-priority to http://hl7.org/fhir/ValueSet/request-priority|4.0.1
MedicationRequest.doNotPerform | Added Element
MedicationRequest.reported[x] | Added Element
MedicationRequest.encounter | Added Element
MedicationRequest.requester | Type changed from BackboneElement to Reference(Practitioner | PractitionerRole | Organization | Patient | RelatedPerson | Device)

MedicationRequest.performer | Added Element
MedicationRequest.performerType | Added Element
MedicationRequest.recorder | Type Reference: Added Target Type PractitionerRole
MedicationRequest.instantiatesCanonical | Added Element
MedicationRequest.instantiatesUri | Added Element
MedicationRequest.basedOn | Type Reference: Added Target Types ServiceRequest, ImmunizationRecommendation Type Reference: Removed Target Types ProcedureRequest, ReferralRequest
MedicationRequest.courseOfTherapyType | Added Element
MedicationRequest.insurance | Added Element
MedicationRequest.dispenseRequest.initialFill | Added Element
MedicationRequest.dispenseRequest.initialFill.quantity | Added Element
MedicationRequest.dispenseRequest.initialFill.duration | Added Element
MedicationRequest.dispenseRequest.dispenseInterval | Added Element
MedicationRequest.dispenseRequest.numberOfRepeatsAllowed | Type changed from positiveInt to unsignedInt
MedicationRequest.substitution.allowed[x] | Renamed from allowed to allowed[x] Add Type CodeableConcept No longer marked as Modifier
MedicationRequest.definition | deleted
MedicationRequest.context | deleted
MedicationRequest.requester.agent | deleted
MedicationRequest.requester.onBehalfOf | deleted
MedicationStatement
MedicationStatement.basedOn | Type Reference: Added Target Type ServiceRequest Type Reference: Removed Target Types ProcedureRequest, ReferralRequest
MedicationStatement.status | Change value set from http://hl7.org/fhir/ValueSet/medication-statement-status to http://hl7.org/fhir/ValueSet/medication-statement-status|4.0.1
MedicationStatement.statusReason | Added Element
MedicationStatement.informationSource | Type Reference: Added Target Type PractitionerRole
MedicationStatement.reasonReference | Type Reference: Added Target Type DiagnosticReport
MedicationStatement.taken | deleted
MedicationStatement.reasonNotTaken | deleted
MedicinalProduct | Added Resource
MedicinalProductAuthorization | Added Resource
MedicinalProductContraindication | Added Resource
MedicinalProductIndication | Added Resource
MedicinalProductIngredient | Added Resource
MedicinalProductInteraction | Added Resource
MedicinalProductManufactured | Added Resource
MedicinalProductPackaged | Added Resource
MedicinalProductPharmaceutical | Added Resource
MedicinalProductUndesirableEffect | Added Resource
MessageDefinition
MessageDefinition.identifier | Max Cardinality changed from 1 to *
MessageDefinition.replaces | Type changed from Reference(MessageDefinition) to canonical(MessageDefinition)
MessageDefinition.status | Change value set from http://hl7.org/fhir/ValueSet/publication-status to http://hl7.org/fhir/ValueSet/publication-status|4.0.1
MessageDefinition.experimental | No longer marked as Modifier
MessageDefinition.base | Type changed from Reference(MessageDefinition) to canonical(MessageDefinition)
MessageDefinition.parent | Type changed from Reference(ActivityDefinition | PlanDefinition) to canonical(ActivityDefinition | PlanDefinition)
MessageDefinition.event[x] | Renamed from event to event[x] Add Type uri
MessageDefinition.category | Change value set from http://hl7.org/fhir/ValueSet/message-significance-category to http://hl7.org/fhir/ValueSet/message-significance-category|4.0.1
MessageDefinition.focus.code | Change value set from http://hl7.org/fhir/ValueSet/resource-types to http://hl7.org/fhir/ValueSet/resource-types|4.0.1
MessageDefinition.focus.profile | Type changed from Reference(StructureDefinition) to canonical(StructureDefinition)
MessageDefinition.focus.min | Min Cardinality changed from 0 to 1 Default Value "0" removed
MessageDefinition.responseRequired | Type changed from boolean to code Add Binding http://hl7.org/fhir/ValueSet/messageheader-response-request|4.0.1 (required) Default Value "false" removed
MessageDefinition.allowedResponse.message | Type changed from Reference(MessageDefinition) to canonical(MessageDefinition)
MessageDefinition.graph | Added Element
MessageHeader
MessageHeader.event[x] | Renamed from event to event[x] Add Type uri
MessageHeader.destination.endpoint | Type changed from uri to url
MessageHeader.destination.receiver | Moved from MessageHeader to MessageHeader.destination Type Reference: Added Target Type PractitionerRole
MessageHeader.sender | Type Reference: Added Target Type PractitionerRole
MessageHeader.enterer | Type Reference: Added Target Type PractitionerRole
MessageHeader.author | Type Reference: Added Target Type PractitionerRole
MessageHeader.source.endpoint | Type changed from uri to url
MessageHeader.responsible | Type Reference: Added Target Type PractitionerRole
MessageHeader.response.code | Change value set from http://hl7.org/fhir/ValueSet/response-code to http://hl7.org/fhir/ValueSet/response-code|4.0.1
MessageHeader.definition | Added Element

MessageHeader.timestamp | deleted
MetadataResource
MetadataResource.status | Change value set from http://hl7.org/fhir/ValueSet/publication-status to http://hl7.org/fhir/ValueSet/publication-status|4.0.1
MetadataResource.experimental | No longer marked as Modifier
MolecularSequence | Added Resource
NamingSystem
NamingSystem.status | Change value set from http://hl7.org/fhir/ValueSet/publication-status to http://hl7.org/fhir/ValueSet/publication-status|4.0.1
NamingSystem.kind | Change value set from http://hl7.org/fhir/ValueSet/namingsystem-type to http://hl7.org/fhir/ValueSet/namingsystem-type|4.0.1
NamingSystem.uniqueId.type | Change value set from http://hl7.org/fhir/ValueSet/namingsystem-identifier-type to http://hl7.org/fhir/ValueSet/namingsystem-identifier-type|4.0.1
NamingSystem.replacedBy | deleted
NutritionOrder
NutritionOrder.instantiatesCanonical | Added Element
NutritionOrder.instantiatesUri | Added Element
NutritionOrder.instantiates | Added Element
NutritionOrder.status | Min Cardinality changed from 0 to 1 Change value set from http://hl7.org/fhir/ValueSet/nutrition-request-status to http://hl7.org/fhir/ValueSet/request-status|4.0.1
NutritionOrder.intent | Added Mandatory Element
NutritionOrder.orderer | Type Reference: Added Target Type PractitionerRole
NutritionOrder.enteralFormula.administration.rate[x] | Add Type Quantity() Remove Type Quantity()
NutritionOrder.note | Added Element
Observation
Observation.basedOn | Type Reference: Added Target Type ServiceRequest Type Reference: Removed Target Types ProcedureRequest, ReferralRequest
Observation.partOf | Added Element
Observation.status | Change value set from http://hl7.org/fhir/ValueSet/observation-status to http://hl7.org/fhir/ValueSet/observation-status|4.0.1
Observation.focus | Added Element
Observation.encounter | Renamed from context to encounter Type Reference: Removed Target Type EpisodeOfCare
Observation.effective[x] | Add Types Timing, instant
Observation.performer | Type Reference: Added Target Types PractitionerRole, CareTeam
Observation.value[x] | Add Type integer Remove Type Attachment
Observation.dataAbsentReason | Change value set from http://hl7.org/fhir/ValueSet/observation-valueabsentreason to http://hl7.org/fhir/ValueSet/data-absent-reason Change code system for extensibly bound codes from "http://hl7.org/fhir/data-absent-reason" to "http://terminology.hl7.org/CodeSystem/data-absent-reason"
Observation.interpretation | Max Cardinality changed from 1 to * Change code system for extensibly bound codes from "http://terminology.hl7.org/CodeSystem/v2-0078" to "http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation"
Observation.note | Renamed from comment to note Max Cardinality changed from 1 to * Type changed from string to Annotation
Observation.referenceRange.type | Remove Binding http://hl7.org/fhir/ValueSet/referencerange-meaning (extensible)
Observation.hasMember | Added Element
Observation.derivedFrom | Added Element
Observation.component.value[x] | Add Types boolean, integer Remove Type Attachment
Observation.component.dataAbsentReason | Change value set from http://hl7.org/fhir/ValueSet/observation-valueabsentreason to http://hl7.org/fhir/ValueSet/data-absent-reason Change code system for extensibly bound codes from "http://hl7.org/fhir/data-absent-reason" to "http://terminology.hl7.org/CodeSystem/data-absent-reason"
Observation.component.interpretation | Max Cardinality changed from 1 to * Change code system for extensibly bound codes from "http://terminology.hl7.org/CodeSystem/v2-0078" to "http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation"
Observation.related | deleted
ObservationDefinition | Added Resource
OperationDefinition
OperationDefinition.title | Added Element
OperationDefinition.status | Change value set from http://hl7.org/fhir/ValueSet/publication-status to http://hl7.org/fhir/ValueSet/publication-status|4.0.1
OperationDefinition.kind | Change value set from http://hl7.org/fhir/ValueSet/operation-kind to http://hl7.org/fhir/ValueSet/operation-kind|4.0.1
OperationDefinition.experimental | No longer marked as Modifier
OperationDefinition.affectsState | Renamed from idempotent to affectsState
OperationDefinition.comment | Type changed from string to markdown
OperationDefinition.base | Type changed from Reference(OperationDefinition) to canonical(OperationDefinition)
OperationDefinition.resource | Change value set from http://hl7.org/fhir/ValueSet/resource-types to http://hl7.org/fhir/ValueSet/resource-types|4.0.1
OperationDefinition.inputProfile | Added Element
OperationDefinition.outputProfile | Added Element
OperationDefinition.parameter.use | Change value set from http://hl7.org/fhir/ValueSet/operation-parameter-use to http://hl7.org/fhir/ValueSet/operation-parameter-use|4.0.1
OperationDefinition.parameter.type | Change value set from http://hl7.org/fhir/ValueSet/all-types to http://hl7.org/fhir/ValueSet/all-types|4.0.1
OperationDefinition.parameter.targetProfile | Renamed from profile to targetProfile Max Cardinality changed from 1 to * Type changed from Reference(StructureDefinition) to canonical(StructureDefinition)
OperationDefinition.parameter.searchType | Change value set from http://hl7.org/fhir/ValueSet/search-param-type to http://hl7.org/fhir/ValueSet/search-param-type|4.0.1

OperationDefinition.parameter.binding.strength | Change value set from http://hl7.org/fhir/ValueSet/binding-strength to http://hl7.org/fhir/ValueSet/binding-strength|4.0.1

OperationDefinition.parameter.binding.valueSet | Renamed from valueSet[x] to valueSet Add Type canonical(ValueSet) Remove Types uri, Reference(ValueSet)

OperationDefinition.parameter.referencedFrom | Added Element

OperationDefinition.parameter.referencedFrom.source | Added Mandatory Element

OperationDefinition.parameter.referencedFrom.sourceId | Added Element

OperationOutcome

OperationOutcome.issue.severity | Change value set from http://hl7.org/fhir/ValueSet/issue-severity to http://hl7.org/fhir/ValueSet/issue-severity|4.0.1 No longer marked as Modifier

OperationOutcome.issue.code | Change value set from http://hl7.org/fhir/ValueSet/issue-type to http://hl7.org/fhir/ValueSet/issue-type|4.0.1

Organization

Organization.active | Default Value "true" removed

Organization.contact.purpose | Change code system for extensibly bound codes from "http://hl7.org/fhir/contactentity-type" to "http://terminology.hl7.org/CodeSystem/contactentity-type"

OrganizationAffiliation | Added Resource

Parameters

Parameters.parameter.value[x] | Add Types canonical, url, uuid, ContactDetail, Contributor, DataRequirement, Expression, ParameterDefinition, RelatedArtifact, TriggerDefinition, UsageContext, Dosage

Patient

Patient.active | Default Value "true" removed

Patient.gender | Change value set from http://hl7.org/fhir/ValueSet/administrative-gender to http://hl7.org/fhir/ValueSet/administrative-gender|4.0.1

Patient.contact.relationship | Change value set from http://terminology.hl7.org/ValueSet/v2-0131 to http://hl7.org/fhir/ValueSet/patient-contactrelationship

Patient.contact.gender | Change value set from http://hl7.org/fhir/ValueSet/administrative-gender to http://hl7.org/fhir/ValueSet/administrative-gender|4.0.1

Patient.communication.language | Change binding strength from extensible to preferred

Patient.generalPractitioner | Type Reference: Added Target Type PractitionerRole

Patient.link.type | Change value set from http://hl7.org/fhir/ValueSet/link-type to http://hl7.org/fhir/ValueSet/link-type|4.0.1

Patient.animal | deleted

PaymentNotice

PaymentNotice.status | Min Cardinality changed from 0 to 1 Change value set from http://hl7.org/fhir/ValueSet/fm-status to http://hl7.org/fhir/ValueSet/fm-status|4.0.1

PaymentNotice.created | Min Cardinality changed from 0 to 1

PaymentNotice.provider | Renamed from organization to provider Type Reference: Added Target Types Practitioner, PractitionerRole

PaymentNotice.payment | Added Mandatory Element

PaymentNotice.paymentDate | Renamed from statusDate to paymentDate

PaymentNotice.payee | Added Element

PaymentNotice.recipient | Renamed from target to recipient Min Cardinality changed from 0 to 1

PaymentNotice.amount | Added Mandatory Element

PaymentNotice.provider | deleted

PaymentReconciliation

PaymentReconciliation.status | Min Cardinality changed from 0 to 1 Change value set from http://hl7.org/fhir/ValueSet/fm-status to http://hl7.org/fhir/ValueSet/fm-status|4.0.1

PaymentReconciliation.created | Min Cardinality changed from 0 to 1

PaymentReconciliation.paymentIssuer | Renamed from organization to paymentIssuer

PaymentReconciliation.request | Type Reference: Added Target Type Task Type Reference: Removed Target Type ProcessRequest

PaymentReconciliation.requestor | Renamed from requestOrganization to requestor Type Reference: Added Target Types Practitioner, PractitionerRole

PaymentReconciliation.outcome | Type changed from CodeableConcept to code Change value set from http://hl7.org/fhir/ValueSet/remittance-outcome to http://hl7.org/fhir/ValueSet/remittance-outcome|4.0.1

PaymentReconciliation.paymentDate | Added Mandatory Element

PaymentReconciliation.paymentAmount | Renamed from total to paymentAmount Min Cardinality changed from 0 to 1

PaymentReconciliation.paymentIdentifier | Added Element

PaymentReconciliation.detail.identifier | Added Element

PaymentReconciliation.detail.predecessor | Added Element

PaymentReconciliation.detail.submitter | Type Reference: Added Target Types Practitioner, PractitionerRole

PaymentReconciliation.detail.responsible | Added Element

PaymentReconciliation.detail.payee | Type Reference: Added Target Types Practitioner, PractitionerRole

PaymentReconciliation.formCode | Renamed from form to formCode

PaymentReconciliation.processNote.type | Type changed from CodeableConcept to code Change value set from http://hl7.org/fhir/ValueSet/note-type to http://hl7.org/fhir/ValueSet/note-type|4.0.1

PaymentReconciliation.requestProvider | deleted

Person

Person.gender | Change value set from http://hl7.org/fhir/ValueSet/administrative-gender to http://hl7.org/fhir/ValueSet/administrative-gender|4.0.1

Person.link.assurance | Change value set from http://hl7.org/fhir/ValueSet/identity-assuranceLevel to http://hl7.org/fhir/ValueSet/identity-assuranceLevel|4.0.1

PlanDefinition

PlanDefinition.subtitle | Added Element

PlanDefinition.type | Change code system for extensibly bound codes from "http://hl7.org/fhir/plan-definition-type" to "http://terminology.hl7.org/CodeSystem/plan-definition-type"
PlanDefinition.status | Change value set from http://hl7.org/fhir/ValueSet/publication-status to http://hl7.org/fhir/ValueSet/publication-status|4.0.1
PlanDefinition.experimental | No longer marked as Modifier
PlanDefinition.subject[x] | Added Element
PlanDefinition.author | Added Element
PlanDefinition.editor | Added Element
PlanDefinition.reviewer | Added Element
PlanDefinition.endorser | Added Element
PlanDefinition.library | Type changed from Reference(Library) to canonical(Library)
PlanDefinition.action.prefix | Added Element
PlanDefinition.action.priority | Added Element
PlanDefinition.action.subject[x] | Added Element
PlanDefinition.action.trigger | Added Element
PlanDefinition.action.condition.kind | Change value set from http://hl7.org/fhir/ValueSet/action-condition-kind to http://hl7.org/fhir/ValueSet/action-condition-kind|4.0.1
PlanDefinition.action.condition.expression | Type changed from string to Expression
PlanDefinition.action.relatedAction.relationship | Change value set from http://hl7.org/fhir/ValueSet/action-relationship-type to http://hl7.org/fhir/ValueSet/action-relationship-type|4.0.1
PlanDefinition.action.timing[x] | Add Type Age
PlanDefinition.action.participant.type | Change value set from http://hl7.org/fhir/ValueSet/action-participant-type to http://hl7.org/fhir/ValueSet/action-participant-type|4.0.1
PlanDefinition.action.type | Type changed from Coding to CodeableConcept Change binding strength from required to extensible
PlanDefinition.action.groupingBehavior | Change value set from http://hl7.org/fhir/ValueSet/action-grouping-behavior to http://hl7.org/fhir/ValueSet/action-grouping-behavior|4.0.1
PlanDefinition.action.selectionBehavior | Change value set from http://hl7.org/fhir/ValueSet/action-selection-behavior to http://hl7.org/fhir/ValueSet/action-selection-behavior|4.0.1
PlanDefinition.action.requiredBehavior | Change value set from http://hl7.org/fhir/ValueSet/action-required-behavior to http://hl7.org/fhir/ValueSet/action-required-behavior|4.0.1
PlanDefinition.action.precheckBehavior | Change value set from http://hl7.org/fhir/ValueSet/action-precheck-behavior to http://hl7.org/fhir/ValueSet/action-precheck-behavior|4.0.1
PlanDefinition.action.cardinalityBehavior | Change value set from http://hl7.org/fhir/ValueSet/action-cardinality-behavior to http://hl7.org/fhir/ValueSet/action-cardinality-behavior|4.0.1
PlanDefinition.action.definition[x] | Renamed from definition to definition[x] Add Types canonical(ActivityDefinition | PlanDefinition | Questionnaire), uri Remove Type Reference(ActivityDefinition | PlanDefinition)
PlanDefinition.action.transform | Type changed from Reference(StructureMap) to canonical(StructureMap)
PlanDefinition.action.dynamicValue.expression | Type changed from string to Expression
PlanDefinition.contributor | deleted
PlanDefinition.action.label | deleted
PlanDefinition.action.triggerDefinition | deleted
PlanDefinition.action.condition.description | deleted
PlanDefinition.action.condition.language | deleted
PlanDefinition.action.dynamicValue.description | deleted
PlanDefinition.action.dynamicValue.language | deleted
Practitioner
Practitioner.active | Default Value "true" removed
Practitioner.gender | Change value set from http://hl7.org/fhir/ValueSet/administrative-gender to http://hl7.org/fhir/ValueSet/administrative-gender|4.0.1
Practitioner.communication | Change binding strength from extensible to preferred
PractitionerRole
PractitionerRole.active | Default Value "true" removed
PractitionerRole.availableTime.daysOfWeek | Change value set from http://hl7.org/fhir/ValueSet/days-of-week to http://hl7.org/fhir/ValueSet/days-of-week|4.0.1
Procedure
Procedure.instantiatesCanonical | Added Element
Procedure.instantiatesUri | Added Element
Procedure.basedOn | Type Reference: Added Target Type ServiceRequest Type Reference: Removed Target Types ProcedureRequest, ReferralRequest
Procedure.status | Change value set from http://hl7.org/fhir/ValueSet/event-status to http://hl7.org/fhir/ValueSet/event-status|4.0.1
Procedure.statusReason | Added Element
Procedure.encounter | Added Element
Procedure.performed[x] | Add Types string, Age, Range
Procedure.recorder | Added Element
Procedure.asserter | Added Element
Procedure.performer.function | Added Element
Procedure.performer.actor | Type Reference: Added Target Type PractitionerRole
Procedure.reasonReference | Type Reference: Added Target Types Procedure, DiagnosticReport, DocumentReference
Procedure.report | Type Reference: Added Target Types DocumentReference, Composition
Procedure.definition | deleted
Procedure.notDone | deleted
Procedure.notDoneReason | deleted
Procedure.context | deleted

Procedure.performer.role | deleted
Provenance
Provenance.occurred[x] | Renamed from period to occurred[x] Add Type dateTime
Provenance.reason | Type changed from Coding to CodeableConcept
Provenance.activity | Type changed from Coding to CodeableConcept
Provenance.agent.type | Added Element
Provenance.agent.role | Remove Binding http://hl7.org/fhir/ValueSet/security-role-type (extensible)
Provenance.agent.who | Renamed from who[x] to who Remove Type uri
Provenance.agent.onBehalfOf | Renamed from onBehalfOf[x] to onBehalfOf Remove Type uri
Provenance.entity.role | Change value set from http://hl7.org/fhir/ValueSet/provenance-entity-role to http://hl7.org/fhir/ValueSet/provenance-entity-role|4.0.1
Provenance.entity.what | Renamed from what[x] to what Remove Types uri, Identifier
Provenance.agent.relatedAgentType | deleted
Questionnaire
Questionnaire.derivedFrom | Added Element
Questionnaire.status | Change value set from http://hl7.org/fhir/ValueSet/publication-status to http://hl7.org/fhir/ValueSet/publication-status|4.0.1
Questionnaire.experimental | No longer marked as Modifier
Questionnaire.subjectType | Change value set from http://hl7.org/fhir/ValueSet/resource-types to http://hl7.org/fhir/ValueSet/resource-types|4.0.1
Questionnaire.item.type | Change value set from http://hl7.org/fhir/ValueSet/item-type to http://hl7.org/fhir/ValueSet/item-type|4.0.1
Questionnaire.item.enableWhen.operator | Added Mandatory Element
Questionnaire.item.enableWhen.answer[x] | Min Cardinality changed from 0 to 1 Remove Types uri, Attachment
Questionnaire.item.enableBehavior | Added Element
Questionnaire.item.required | Default Value "false" removed
Questionnaire.item.repeats | Default Value "false" removed
Questionnaire.item.answerValueSet | Renamed from options to answerValueSet Type changed from Reference(ValueSet) to canonical(ValueSet)
Questionnaire.item.answerOption | Renamed from option to answerOption
Questionnaire.item.answerOption.value[x] | Moved from Questionnaire.item.option to Questionnaire.item.answerOption Add Type Reference(Resource)
Questionnaire.item.answerOption.initialSelected | Added Element
Questionnaire.item.initial | Renamed from initial[x] to initial Max Cardinality changed from 1 to * Remove Types boolean, decimal, integer, date, dateTime, time, string, uri, Attachment, Coding, Quantity, Reference(Resource)
Questionnaire.item.initial.value[x] | Added Mandatory Element
Questionnaire.item.enableWhen.hasAnswer | deleted
QuestionnaireResponse
QuestionnaireResponse.basedOn | Type Reference: Added Target Type ServiceRequest Type Reference: Removed Target Types ReferralRequest, ProcedureRequest
QuestionnaireResponse.partOf | Renamed from parent to partOf
QuestionnaireResponse.questionnaire | Type changed from Reference(Questionnaire) to canonical(Questionnaire)
QuestionnaireResponse.status | Change value set from http://hl7.org/fhir/ValueSet/questionnaire-answers-status to http://hl7.org/fhir/ValueSet/questionnaire-answers-status|4.0.1
QuestionnaireResponse.encounter | Added Element
QuestionnaireResponse.author | Type Reference: Added Target Types PractitionerRole, Organization
QuestionnaireResponse.source | Type Reference: Added Target Type PractitionerRole
QuestionnaireResponse.context | deleted
QuestionnaireResponse.item.subject | deleted
RelatedPerson
RelatedPerson.active | Default Value "true" removed
RelatedPerson.relationship | Max Cardinality changed from 1 to *
RelatedPerson.gender | Change value set from http://hl7.org/fhir/ValueSet/administrative-gender to http://hl7.org/fhir/ValueSet/administrative-gender|4.0.1
RelatedPerson.communication | Added Element
RelatedPerson.communication.language | Added Mandatory Element
RelatedPerson.communication.preferred | Added Element
RequestGroup
RequestGroup.instantiatesCanonical | Added Element
RequestGroup.instantiatesUri | Added Element
RequestGroup.status | Change value set from http://hl7.org/fhir/ValueSet/request-status to http://hl7.org/fhir/ValueSet/request-status|4.0.1
RequestGroup.intent | Change value set from http://hl7.org/fhir/ValueSet/request-intent to http://hl7.org/fhir/ValueSet/request-intent|4.0.1
RequestGroup.priority | Change value set from http://hl7.org/fhir/ValueSet/request-priority to http://hl7.org/fhir/ValueSet/request-priority|4.0.1
RequestGroup.code | Added Element
RequestGroup.encounter | Added Element
RequestGroup.author | Type Reference: Added Target Type PractitionerRole
RequestGroup.reasonCode | Added Element
RequestGroup.reasonReference | Added Element
RequestGroup.action.prefix | Added Element
RequestGroup.action.priority | Added Element

RequestGroup.action.condition.kind | Change value set from http://hl7.org/fhir/ValueSet/action-condition-kind to http://hl7.org/fhir/ValueSet/action-condition-kind|4.0.1
RequestGroup.action.condition.expression | Type changed from string to Expression
RequestGroup.action.relatedAction.relationship | Change value set from http://hl7.org/fhir/ValueSet/action-relationship-type to http://hl7.org/fhir/ValueSet/action-relationship-type|4.0.1
RequestGroup.action.timing[x] | Add Type Age
RequestGroup.action.participant | Type Reference: Added Target Types PractitionerRole, Device Type Reference: Removed Target Type Person
RequestGroup.action.type | Type changed from Coding to CodeableConcept Change code system for extensibly bound codes from "http://hl7.org/fhir/action-type" to "http://terminology.hl7.org/CodeSystem/action-type"
RequestGroup.action.groupingBehavior | Change value set from http://hl7.org/fhir/ValueSet/action-grouping-behavior to http://hl7.org/fhir/ValueSet/action-grouping-behavior|4.0.1
RequestGroup.action.selectionBehavior | Change value set from http://hl7.org/fhir/ValueSet/action-selection-behavior to http://hl7.org/fhir/ValueSet/action-selection-behavior|4.0.1
RequestGroup.action.requiredBehavior | Change value set from http://hl7.org/fhir/ValueSet/action-required-behavior to http://hl7.org/fhir/ValueSet/action-required-behavior|4.0.1
RequestGroup.action.precheckBehavior | Change value set from http://hl7.org/fhir/ValueSet/action-precheck-behavior to http://hl7.org/fhir/ValueSet/action-precheck-behavior|4.0.1
RequestGroup.action.cardinalityBehavior | Change value set from http://hl7.org/fhir/ValueSet/action-cardinality-behavior to http://hl7.org/fhir/ValueSet/action-cardinality-behavior|4.0.1
RequestGroup.definition | deleted
RequestGroup.context | deleted
RequestGroup.reason[x] | deleted
RequestGroup.action.label | deleted
RequestGroup.action.condition.description | deleted
RequestGroup.action.condition.language | deleted
ResearchDefinition | Added Resource
ResearchElementDefinition | Added Resource
ResearchStudy
ResearchStudy.status | Change value set from http://hl7.org/fhir/ValueSet/research-study-status to http://hl7.org/fhir/ValueSet/research-study-status|4.0.1
ResearchStudy.primaryPurposeType | Added Element
ResearchStudy.phase | Added Element
ResearchStudy.condition | Added Element
ResearchStudy.location | Added Element
ResearchStudy.principalInvestigator | Type Reference: Added Target Type PractitionerRole
ResearchStudy.arm.type | Added Element
ResearchStudy.objective | Added Element
ResearchStudy.objective.name | Added Element
ResearchStudy.objective.type | Added Element
ResearchStudy.jurisdiction | deleted
ResearchStudy.arm.code | deleted
ResearchSubject
ResearchSubject.identifier | Max Cardinality changed from 1 to *
ResearchSubject.status | Change value set from http://hl7.org/fhir/ValueSet/research-subject-status to http://hl7.org/fhir/ValueSet/research-subject-status|4.0.1
Resource
Resource.id | Type changed from id to string
Resource.language | Change binding strength from extensible to preferred
RiskAssessment
RiskAssessment.identifier | Max Cardinality changed from 1 to *
RiskAssessment.status | Change value set from http://hl7.org/fhir/ValueSet/observation-status to http://hl7.org/fhir/ValueSet/observation-status|4.0.1
RiskAssessment.subject | Min Cardinality changed from 0 to 1
RiskAssessment.encounter | Added Element
RiskAssessment.performer | Type Reference: Added Target Type PractitionerRole
RiskAssessment.reasonCode | Added Element
RiskAssessment.reasonReference | Added Element
RiskAssessment.prediction.outcome | Min Cardinality changed from 1 to 0
RiskAssessment.note | Renamed from comment to note Max Cardinality changed from 1 to * Type changed from string to Annotation
RiskAssessment.context | deleted
RiskAssessment.reason[x] | deleted
RiskEvidenceSynthesis | Added Resource
Schedule
Schedule.active | Default Value "true" removed
Schedule.serviceCategory | Max Cardinality changed from 1 to *
SearchParameter
SearchParameter | Min Cardinality changed from 1 to 0 Max Cardinality changed from 1 to *
SearchParameter.derivedFrom | Type changed from uri to canonical(SearchParameter)
SearchParameter.status | Change value set from http://hl7.org/fhir/ValueSet/publication-status to http://hl7.org/fhir/ValueSet/publication-status|4.0.1
SearchParameter.experimental | No longer marked as Modifier

SearchParameter.base | Change value set from http://hl7.org/fhir/ValueSet/resource-types to http://hl7.org/fhir/ValueSet/resource-types|4.0.1
SearchParameter.type | Change value set from http://hl7.org/fhir/ValueSet/search-param-type to http://hl7.org/fhir/ValueSet/search-param-type|4.0.1
SearchParameter.xpathUsage | Change value set from http://hl7.org/fhir/ValueSet/search-xpath-usage to http://hl7.org/fhir/ValueSet/search-xpath-usage|4.0.1
SearchParameter.target | Change value set from http://hl7.org/fhir/ValueSet/resource-types to http://hl7.org/fhir/ValueSet/resource-types|4.0.1
SearchParameter.multipleOr | Added Element
SearchParameter.multipleAnd | Added Element
SearchParameter.comparator | Change value set from http://hl7.org/fhir/ValueSet/search-comparator to http://hl7.org/fhir/ValueSet/search-comparator|4.0.1
SearchParameter.modifier | Change value set from http://hl7.org/fhir/ValueSet/search-modifier-code to http://hl7.org/fhir/ValueSet/search-modifier-code|4.0.1
SearchParameter.component.definition | Type changed from Reference(SearchParameter) to canonical(SearchParameter)
ServiceRequest | Added Resource
Slot
Slot.serviceCategory | Max Cardinality changed from 1 to *
Slot.status | Change value set from http://hl7.org/fhir/ValueSet/slotstatus to http://hl7.org/fhir/ValueSet/slotstatus|4.0.1
Specimen
Specimen.status | Change value set from http://hl7.org/fhir/ValueSet/specimen-status to http://hl7.org/fhir/ValueSet/specimen-status|4.0.1
Specimen.subject | Min Cardinality changed from 1 to 0 Type Reference: Added Target Type Location
Specimen.request | Type Reference: Added Target Type ServiceRequest Type Reference: Removed Target Type ProcedureRequest
Specimen.collection.collector | Type Reference: Added Target Type PractitionerRole
Specimen.collection.duration | Added Element
Specimen.collection.fastingStatus[x] | Added Element
Specimen.condition | Added Element
SpecimenDefinition | Added Resource
StructureDefinition
StructureDefinition.status | Change value set from http://hl7.org/fhir/ValueSet/publication-status to http://hl7.org/fhir/ValueSet/publication-status|4.0.1
StructureDefinition.experimental | No longer marked as Modifier
StructureDefinition.keyword | Add Binding http://hl7.org/fhir/ValueSet/definition-use (extensible)
StructureDefinition.fhirVersion | Type changed from id to code Add Binding http://hl7.org/fhir/ValueSet/FHIR-version|4.0.1 (required)
StructureDefinition.kind | Change value set from http://hl7.org/fhir/ValueSet/structure-definition-kind to http://hl7.org/fhir/ValueSet/structure-definition-kind|4.0.1
StructureDefinition.context | Type changed from string to BackboneElement
StructureDefinition.context.type | Added Mandatory Element
StructureDefinition.context.expression | Added Mandatory Element
StructureDefinition.type | Type changed from code to uri
StructureDefinition.baseDefinition | Type changed from uri to canonical(StructureDefinition)
StructureDefinition.derivation | Change value set from http://hl7.org/fhir/ValueSet/type-derivation-rule to http://hl7.org/fhir/ValueSet/type-derivation-rule|4.0.1
StructureDefinition.contextType | deleted
StructureMap
StructureMap | Min Cardinality changed from 1 to 0 Max Cardinality changed from 1 to *
StructureMap.status | Change value set from http://hl7.org/fhir/ValueSet/publication-status to http://hl7.org/fhir/ValueSet/publication-status|4.0.1
StructureMap.experimental | No longer marked as Modifier
StructureMap.structure.url | Type changed from uri to canonical(StructureDefinition)
StructureMap.structure.mode | Change value set from http://hl7.org/fhir/ValueSet/map-model-mode to http://hl7.org/fhir/ValueSet/map-model-mode|4.0.1
StructureMap.import | Type changed from uri to canonical(StructureMap)
StructureMap.group.typeMode | Change value set from http://hl7.org/fhir/ValueSet/map-group-type-mode to http://hl7.org/fhir/ValueSet/map-group-type-mode|4.0.1
StructureMap.group.input.mode | Change value set from http://hl7.org/fhir/ValueSet/map-input-mode to http://hl7.org/fhir/ValueSet/map-input-mode|4.0.1
StructureMap.group.rule.source.defaultValue[x] | Add Types canonical, url, uuid, ContactDetail, Contributor, DataRequirement, Expression, ParameterDefinition, RelatedArtifact, TriggerDefinition, UsageContext, Dosage
StructureMap.group.rule.source.listMode | Change value set from http://hl7.org/fhir/ValueSet/map-source-list-mode to http://hl7.org/fhir/ValueSet/map-source-list-mode|4.0.1
StructureMap.group.rule.source.logMessage | Added Element
StructureMap.group.rule.target.contextType | Change value set from http://hl7.org/fhir/ValueSet/map-context-type to http://hl7.org/fhir/ValueSet/map-context-type|4.0.1
StructureMap.group.rule.target.listMode | Change value set from http://hl7.org/fhir/ValueSet/map-target-list-mode to http://hl7.org/fhir/ValueSet/map-target-list-mode|4.0.1
StructureMap.group.rule.target.transform | Change value set from http://hl7.org/fhir/ValueSet/map-transform to http://hl7.org/fhir/ValueSet/map-transform|4.0.1
Subscription
Subscription.status | Change value set from http://hl7.org/fhir/ValueSet/subscription-status to http://hl7.org/fhir/ValueSet/subscription-status|4.0.1

Subscription.channel.type | Change value set from http://hl7.org/fhir/ValueSet/subscription-channel-type to http://hl7.org/fhir/ValueSet/subscription-channel-type|4.0.1
Subscription.channel.endpoint | Type changed from uri to url
Subscription.channel.payload | Type changed from string to code Add Binding http://hl7.org/fhir/ValueSet/mimetypes|4.0.1 (required)
Subscription.tag | deleted
Substance
Substance.status | Change value set from http://hl7.org/fhir/ValueSet/substance-status to http://hl7.org/fhir/ValueSet/substance-status|4.0.1 Now marked as Modifier
Substance.category | Change code system for extensibly bound codes from "http://hl7.org/fhir/substance-category" to "http://terminology.hl7.org/CodeSystem/substance-category"
SubstanceNucleicAcid | Added Resource
SubstancePolymer | Added Resource
SubstanceProtein | Added Resource
SubstanceReferenceInformation | Added Resource
SubstanceSourceMaterial | Added Resource
SubstanceSpecification | Added Resource
SupplyDelivery
SupplyDelivery.identifier | Max Cardinality changed from 1 to *
SupplyDelivery.status | Change value set from http://hl7.org/fhir/ValueSet/supplydelivery-status to http://hl7.org/fhir/ValueSet/supplydelivery-status|4.0.1
SupplyDelivery.type | Change value set from http://hl7.org/fhir/ValueSet/supplydelivery-type to http://hl7.org/fhir/ValueSet/supplydelivery-type|4.0.1
SupplyDelivery.supplier | Type Reference: Added Target Type PractitionerRole
SupplyDelivery.receiver | Type Reference: Added Target Type PractitionerRole
SupplyRequest
SupplyRequest.identifier | Max Cardinality changed from 1 to *
SupplyRequest.status | Change value set from http://hl7.org/fhir/ValueSet/supplyrequest-status to http://hl7.org/fhir/ValueSet/supplyrequest-status|4.0.1
SupplyRequest.priority | Change value set from http://hl7.org/fhir/ValueSet/request-priority to http://hl7.org/fhir/ValueSet/request-priority|4.0.1
SupplyRequest.item[x] | Added Mandatory Element
SupplyRequest.quantity | Added Mandatory Element
SupplyRequest.parameter | Added Element
SupplyRequest.parameter.code | Added Element
SupplyRequest.parameter.value[x] | Added Element
SupplyRequest.requester | Type changed from BackboneElement to Reference(Practitioner | PractitionerRole | Organization | Patient | RelatedPerson | Device)
SupplyRequest.supplier | Type Reference: Added Target Type HealthcareService
SupplyRequest.reasonCode | Added Element
SupplyRequest.reasonReference | Added Element
SupplyRequest.orderedItem | deleted
SupplyRequest.requester.agent | deleted
SupplyRequest.requester.onBehalfOf | deleted
SupplyRequest.reason[x] | deleted
Task
Task.instantiatesCanonical | Added Element
Task.instantiatesUri | Added Element
Task.status | Change value set from http://hl7.org/fhir/ValueSet/task-status to http://hl7.org/fhir/ValueSet/task-status|4.0.1 Now marked as Modifier
Task.intent | Change value set from http://hl7.org/fhir/ValueSet/request-intent to http://hl7.org/fhir/ValueSet/task-intent|4.0.1
Task.priority | Change value set from http://hl7.org/fhir/ValueSet/request-priority to http://hl7.org/fhir/ValueSet/request-priority|4.0.1
Task.encounter | Added Element
Task.requester | Type changed from BackboneElement to Reference(Device | Organization | Patient | Practitioner | PractitionerRole | RelatedPerson)
Task.owner | Type Reference: Added Target Types PractitionerRole, CareTeam, HealthcareService
Task.location | Added Element
Task.reasonCode | Added Element
Task.reasonReference | Added Element
Task.insurance | Added Element
Task.restriction.recipient | Type Reference: Added Target Type PractitionerRole
Task.input.value[x] | Add Types canonical, url, uuid, ContactDetail, Contributor, DataRequirement, Expression, ParameterDefinition, RelatedArtifact, TriggerDefinition, UsageContext, Dosage
Task.output.value[x] | Add Types canonical, url, uuid, ContactDetail, Contributor, DataRequirement, Expression, ParameterDefinition, RelatedArtifact, TriggerDefinition, UsageContext, Dosage
Task.definition[x] | deleted
Task.context | deleted
Task.requester.agent | deleted
Task.requester.onBehalfOf | deleted
Task.reason | deleted
TerminologyCapabilities | Added Resource
TestReport

TestReport.status | Change value set from http://hl7.org/fhir/ValueSet/report-status-codes to http://hl7.org/fhir/ValueSet/report-status-codes|4.0.1
TestReport.result | Change value set from http://hl7.org/fhir/ValueSet/report-result-codes to http://hl7.org/fhir/ValueSet/report-result-codes|4.0.1
TestReport.participant.type | Change value set from http://hl7.org/fhir/ValueSet/report-participant-type to http://hl7.org/fhir/ValueSet/report-participant-type|4.0.1
TestReport.setup.action.operation.result | Change value set from http://hl7.org/fhir/ValueSet/report-action-result-codes to http://hl7.org/fhir/ValueSet/report-action-result-codes|4.0.1
TestReport.setup.action.assert.result | Change value set from http://hl7.org/fhir/ValueSet/report-action-result-codes to http://hl7.org/fhir/ValueSet/report-action-result-codes|4.0.1
TestScript
TestScript.status | Change value set from http://hl7.org/fhir/ValueSet/publication-status to http://hl7.org/fhir/ValueSet/publication-status|4.0.1
TestScript.experimental | No longer marked as Modifier
TestScript.origin.profile | Change code system for extensibly bound codes from "http://hl7.org/fhir/testscript-profile-origin-types" to "http://terminology.hl7.org/CodeSystem/testscript-profile-origin-types"
TestScript.destination.profile | Change code system for extensibly bound codes from "http://hl7.org/fhir/testscript-profile-destination-types" to "http://terminology.hl7.org/CodeSystem/testscript-profile-destination-types"
TestScript.metadata.capability.required | Min Cardinality changed from 0 to 1 Default Value "false" removed
TestScript.metadata.capability.validated | Min Cardinality changed from 0 to 1 Default Value "false" removed
TestScript.metadata.capability.capabilities | Type changed from Reference(CapabilityStatement) to canonical(CapabilityStatement)
TestScript.fixture.autocreate | Min Cardinality changed from 0 to 1 Default Value "false" removed
TestScript.fixture.autodelete | Min Cardinality changed from 0 to 1 Default Value "false" removed
TestScript.setup.action.operation.type | Change code system for extensibly bound codes from "http://hl7.org/fhir/testscript-operation-codes" to "http://terminology.hl7.org/CodeSystem/testscript-operation-codes"
TestScript.setup.action.operation.resource | Change value set from http://hl7.org/fhir/ValueSet/defined-types to http://hl7.org/fhir/ValueSet/defined-types|4.0.1
TestScript.setup.action.operation.accept | Change value set from http://hl7.org/fhir/ValueSet/content-type to http://hl7.org/fhir/ValueSet/mimetypes|4.0.1
TestScript.setup.action.operation.contentType | Change value set from http://hl7.org/fhir/ValueSet/content-type to http://hl7.org/fhir/ValueSet/mimetypes|4.0.1
TestScript.setup.action.operation.encodeRequestUrl | Min Cardinality changed from 0 to 1 Default Value "true" removed
TestScript.setup.action.operation.method | Added Element
TestScript.setup.action.assert.direction | Change value set from http://hl7.org/fhir/ValueSet/assert-direction-codes to http://hl7.org/fhir/ValueSet/assert-direction-codes|4.0.1
TestScript.setup.action.assert.contentType | Change value set from http://hl7.org/fhir/ValueSet/content-type to http://hl7.org/fhir/ValueSet/mimetypes|4.0.1
TestScript.setup.action.assert.operator | Change value set from http://hl7.org/fhir/ValueSet/assert-operator-codes to http://hl7.org/fhir/ValueSet/assert-operator-codes|4.0.1
TestScript.setup.action.assert.requestMethod | Change value set from http://hl7.org/fhir/ValueSet/http-operations to http://hl7.org/fhir/ValueSet/http-operations|4.0.1
TestScript.setup.action.assert.resource | Change value set from http://hl7.org/fhir/ValueSet/defined-types to http://hl7.org/fhir/ValueSet/defined-types|4.0.1
TestScript.setup.action.assert.response | Change value set from http://hl7.org/fhir/ValueSet/assert-response-code-types to http://hl7.org/fhir/ValueSet/assert-response-code-types|4.0.1
TestScript.setup.action.assert.warningOnly | Min Cardinality changed from 0 to 1 Default Value "false" removed
TestScript.rule | deleted
TestScript.ruleset | deleted
TestScript.setup.action.assert.rule | deleted
TestScript.setup.action.assert.ruleset | deleted
ValueSet
ValueSet.status | Change value set from http://hl7.org/fhir/ValueSet/publication-status to http://hl7.org/fhir/ValueSet/publication-status|4.0.1
ValueSet.experimental | No longer marked as Modifier
ValueSet.compose.include.concept.designation.language | Change binding strength from extensible to preferred
ValueSet.compose.include.filter | No longer marked as Modifier
ValueSet.compose.include.filter.op | Change value set from http://hl7.org/fhir/ValueSet/filter-operator to http://hl7.org/fhir/ValueSet/filter-operator|4.0.1
ValueSet.compose.include.filter.value | Type changed from code to string
ValueSet.compose.include.valueSet | Type changed from uri to canonical(ValueSet)
ValueSet.expansion.identifier | Min Cardinality changed from 1 to 0
ValueSet.expansion.parameter.value[x] | Add Type dateTime
ValueSet.expansion.contains.abstract | Default Value "false" removed
ValueSet.expansion.contains.inactive | Default Value "false" removed
ValueSet.extensible | deleted
VerificationResult | Added Resource
VisionPrescription
VisionPrescription.status | Min Cardinality changed from 0 to 1 Change value set from http://hl7.org/fhir/ValueSet/fm-status to http://hl7.org/fhir/ValueSet/fm-status|4.0.1
VisionPrescription.created | Added Mandatory Element
VisionPrescription.patient | Min Cardinality changed from 0 to 1
VisionPrescription.dateWritten | Min Cardinality changed from 0 to 1

VisionPrescription.prescriber | Min Cardinality changed from 0 to 1 Type Reference: Added Target Type PractitionerRole
VisionPrescription.lensSpecification | Added Mandatory Element
VisionPrescription.lensSpecification.product | Added Mandatory Element
VisionPrescription.lensSpecification.eye | Added Mandatory Element
VisionPrescription.lensSpecification.sphere | Added Element
VisionPrescription.lensSpecification.cylinder | Added Element
VisionPrescription.lensSpecification.axis | Added Element
VisionPrescription.lensSpecification.prism | Added Element
VisionPrescription.lensSpecification.prism.amount | Added Mandatory Element
VisionPrescription.lensSpecification.prism.base | Added Mandatory Element
VisionPrescription.lensSpecification.add | Added Element
VisionPrescription.lensSpecification.power | Added Element
VisionPrescription.lensSpecification.backCurve | Added Element
VisionPrescription.lensSpecification.diameter | Added Element
VisionPrescription.lensSpecification.duration | Added Element
VisionPrescription.lensSpecification.color | Added Element
VisionPrescription.lensSpecification.brand | Added Element
VisionPrescription.lensSpecification.note | Added Element
VisionPrescription.reason[x] | deleted
VisionPrescription.dispense | deleted
BodySite | deleted
DataElement | deleted
DeviceComponent | deleted
EligibilityRequest | deleted
EligibilityResponse | deleted
ExpansionProfile | deleted
ImagingManifest | deleted
ProcedureRequest | deleted
ProcessRequest | deleted
ProcessResponse | deleted
ReferralRequest | deleted
Sequence | deleted
ServiceDefinition | deleted

2.12 Transforms between DSTU 3 and STU 4

URL: https://hl7.org/fhir/R4/r3maps.html
FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative

2.12 Transforms between DSTU 3 and STU 4

In addition to a list of all differences between DSTU 3 and STU 4, this specification also provides transforms between R3 and R4 for most resources that were in R3. These transforms are expressed using the FHIR Mapping Language, and both serve as formal documentation of the relationship between the resource structures in R3 and R4, and can also be used to convert between the formats automatically.

The following table summarizes the state of the R3:R4 maps for each resource type.

Note the following:

• Transforms are tested using the following method: For each example resource in the R3 release of the FHIR specification that has a transform to R4 apply the transform to R4 validate against the R4 specification apply the reverse transform back to R3 compare the outcome with the original resource - ideally, it should be identical

• All transforms should execute - that is, they should be able to take any R3 resource, represent the content in R4, and then generate an R3 representation for the same content. Note, though, that in a few cases, the differences between R3 and R4 designs are so significant that not all examples can be successfully transformed at all.

• For some resources, the differences in design between R3 and R4 are sufficiently great that the R3 examples cannot reproduce the same output after conversion to R4 and then back to R3. For example, the R4 resource may have a different set of status codes with imperfect maps

• For some resources, the information available in the R3 examples is not enough to generate valid R4 resources, and so the resources generate R4 validation errors. (e.g. R3 resources missing an element that is required in R4)

• The transform scripts are only tested to the degree that R3 provided example data

• For all these reasons, the maps provided in the specification are not authoritative; they are provided to help implementers who need to convert between the R2 and R3 format, but additional revision may be required in an implementation specific context

• The maps are subject to ongoing maintenance using the FHIR NPM Package "fhir.versions.r3r4" which is maintained on GitHub . See also FHIR Wiki for other ongoing work to support R3/R4 transformations

Resource | # Tests | % Execute OK | % RoundTrip Ok | % R4 Valid | R4 Error Count
Account | 2 | 100 | 0 | 100
ActivityDefinition | 9 | 100 | 77 | 0
AdverseEvent | 1 | 100 | 0 | 0
AllergyIntolerance | 3 | 100 | 100 | 100
Appointment | 3 | 100 | 100 | 100
AppointmentResponse | 2 | 100 | 100 | 100
AuditEvent | 8 | 100 | 100 | 100
Basic | 3 | 100 | 100 | 100
Binary | 2 | 100 | 100 | 100
BiologicallyDerivedProduct | No r2:r3 maps available
BodyStructure | 3 | 100 | 100 | 100
Bundle | 28 | 89 | 88 | 36
CapabilityStatement | 9 | 100 | 88 | 0
CarePlan | 11 | 100 | 100 | 9
CareTeam | 1 | 100 | 100 | 0
CatalogEntry | No r2:r3 maps available
ChargeItem | 1 | 100 | 100 | 100
ChargeItemDefinition | No r2:r3 maps available
Claim | 16 | 81 | 0 | 76
ClaimResponse | 1 | 100 | 0 | 0
ClinicalImpression | 1 | 100 | 100 | 100
CodeSystem | 4 | 100 | 100 | 25
Communication | 3 | 100 | 100 | 33
CommunicationRequest | 2 | 100 | 100 | 50
CompartmentDefinition | 6 | 100 | 83 | 16
Composition | 1 | 100 | 100 | 100
ConceptMap | 4 | 100 | 100 | 0
Condition | 12 | 100 | 100 | 91
Consent | 12 | 100 | 100 | 0
Contract | 7 | 28 | 50 | -150
Coverage | 4 | 100 | 75 | 100
CoverageEligibilityRequest | No r2:r3 maps available
CoverageEligibilityResponse | No r2:r3 maps available
DetectedIssue | 4 | 100 | 100 | 100
Device | 9 | 100 | 100 | 88
DeviceDefinition | No r2:r3 maps available
DeviceMetric | 1 | 100 | 100 | 100
DeviceRequest | 2 | 100 | 0 | 50
DeviceUseStatement | 1 | 100 | 100 | 100
DiagnosticReport | 14 | 100 | 100 | 35
DocumentManifest | 2 | 0 | 0 | 0
DocumentReference | 1 | 0 | 0 | 0
EffectEvidenceSynthesis | No r2:r3 maps available
Encounter | 10 | 100 | 100 | 70
Endpoint | 3 | 100 | 100 | 100
EnrollmentRequest | 1 | 100 | 0 | 100
EnrollmentResponse | 1 | 100 | 100 | 100
EpisodeOfCare | 1 | 100 | 100 | 100
EventDefinition | No r2:r3 maps available
Evidence | No r2:r3 maps available
EvidenceVariable | No r2:r3 maps available
ExampleScenario | No r2:r3 maps available
ExplanationOfBenefit | 1 | 100 | 0 | 0
FamilyMemberHistory | 2 | 100 | 100 | 50
Flag | 2 | 100 | 100 | 100
Goal | 2 | 100 | 100 | 100
GraphDefinition | 1 | 100 | 100 | 0
Group | 2 | 100 | 100 | 100
GuidanceResponse | 1 | 100 | 100 | 100
HealthcareService | 1 | 100 | 100 | 0
ImagingStudy | 2 | 100 | 50 | 0
Immunization | 3 | 100 | 100 | 0
ImmunizationEvaluation | No r2:r3 maps available
ImmunizationRecommendation | 1 | 100 | 100 | 100
ImplementationGuide | 1 | 100 | 100 | 0
InsurancePlan | No r2:r3 maps available
Invoice | No r2:r3 maps available
Library | 11 | 100 | 100 | 0
Linkage | 1 | 100 | 100 | 100
List | 9 | 100 | 100 | 44
Location | 6 | 100 | 100 | 83

Measure | 6 | 100 | 100 | 0
MeasureReport | 3 | 100 | 100 | 0
Media | 4 | 100 | 100 | 0
Medication | 23 | 100 | 4 | 4
MedicationAdministration | 14 | 100 | 71 | 0
MedicationDispense | 31 | 100 | 80 | 0
MedicationKnowledge | No r2:r3 maps available
MedicationRequest | 36 | 100 | 77 | 0
MedicationStatement | 7 | 100 | 57 | 0
MedicinalProduct | No r2:r3 maps available
MedicinalProductAuthorization | No r2:r3 maps available
MedicinalProductContraindication | No r2:r3 maps available
MedicinalProductIndication | No r2:r3 maps available
MedicinalProductIngredient | No r2:r3 maps available
MedicinalProductInteraction | No r2:r3 maps available
MedicinalProductManufactured | No r2:r3 maps available
MedicinalProductPackaged | No r2:r3 maps available
MedicinalProductPharmaceutical | No r2:r3 maps available
MedicinalProductUndesirableEffect | No r2:r3 maps available
MessageDefinition | 3 | 100 | 100 | 33
MessageHeader | 1 | 100 | 100 | 0
MolecularSequence | 14 | 100 | 100 | 100
NamingSystem | 3 | 100 | 66 | 0
NutritionOrder | 13 | 100 | 100 | 0
Observation | 48 | 100 | 100 | 52
ObservationDefinition | No r2:r3 maps available
OperationDefinition | 37 | 100 | 100 | 8
OperationOutcome | 6 | 100 | 100 | 100
Organization | 11 | 100 | 100 | 90
OrganizationAffiliation | No r2:r3 maps available
Patient | 16 | 100 | 100 | 100
PaymentNotice | 1 | 100 | 0 | 0
PaymentReconciliation | 1 | 100 | 0 | 0
Person | 5 | 100 | 100 | 100
PlanDefinition | 11 | 45 | 100 | 0
Practitioner | 14 | 100 | 100 | 100
PractitionerRole | 1 | 100 | 100 | 100
Procedure | 15 | 100 | 80 | 93
Provenance | 5 | 100 | 100 | 60
Questionnaire | 6 | 100 | 100 | 16
QuestionnaireResponse | 5 | 100 | 80 | 0
RelatedPerson | 4 | 100 | 100 | 100
RequestGroup | 2 | 100 | 100 | 0
ResearchDefinition | No r2:r3 maps available
ResearchElementDefinition | No r2:r3 maps available
ResearchStudy | 1 | 100 | 100 | 100
ResearchSubject | 1 | 100 | 100 | 100
RiskAssessment | 4 | 100 | 100 | 25
RiskEvidenceSynthesis | No r2:r3 maps available
Schedule | 3 | 100 | 100 | 100
SearchParameter | 3 | 100 | 100 | 0
ServiceRequest | No r2:r3 maps available
Slot | 4 | 100 | 100 | 100
Specimen | 4 | 100 | 100 | 50
SpecimenDefinition | No r2:r3 maps available
StructureDefinition | 206 | 100 | 98 | 0
StructureMap | 2 | 100 | 100 | 50
Subscription | 2 | 100 | 0 | 100
Substance | 6 | 100 | 100 | 83
SubstanceNucleicAcid | No r2:r3 maps available
SubstancePolymer | No r2:r3 maps available
SubstanceProtein | No r2:r3 maps available
SubstanceReferenceInformation | No r2:r3 maps available
SubstanceSourceMaterial | No r2:r3 maps available
SubstanceSpecification | No r2:r3 maps available
SupplyDelivery | 2 | 100 | 100 | 100
SupplyRequest | 1 | 100 | 100 | 0
Task | 6 | 100 | 50 | 83
TerminologyCapabilities | No r2:r3 maps available
TestReport | 1 | 100 | 100 | 100
TestScript | 7 | 100 | 85 | 0
ValueSet | 8 | 100 | 100 | 0

VerificationResult | No r2:r3 maps available
VisionPrescription | 2 | 100 | 50 | 0

2.13 FHIR Overview

URL: https://hl7.org/fhir/R4/overview.html

2.13 FHIR Overview

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative

Welcome to the FHIR (Fast Healthcare Interoperability Resources) Specification, which is a standard for exchanging healthcare information electronically. This page provides an overview of the standard, and serves as a road map for first-time readers of the specification to help find your way around FHIR quickly.

2.13.1 Background

Healthcare records are increasingly becoming digitized. As patients move around the healthcare ecosystem, their electronic health records must be available, discoverable, and understandable. Further, to support automated clinical decision support and other machine-based processing, the data must also be structured and standardized. (See Coming digital challenges in healthcare)

HL7 has been addressing these challenges by producing healthcare data exchange and information modeling standards for over 20 years. FHIR is a new specification based on emerging industry approaches, but informed by years of lessons around requirements, successes and challenges gained through defining and implementing HL7 v2 , HL7 v3 and the RIM, and CDA . FHIR can be used as a stand-alone data exchange standard, but can and will also be used in partnership with existing widely used standards. (See Comparing FHIR to other HL7 standards)

FHIR aims to simplify implementation without sacrificing information integrity. It leverages existing logical and theoretical models to provide a consistent, easy to implement, and rigorous mechanism for exchanging data between healthcare applications. FHIR has built-in mechanisms for traceability to the HL7 RIM and other important content models. This ensures alignment to HL7's previously defined patterns and best practices without requiring the implementer to have intimate knowledge of the RIM or any HL7 v3 derivations. (See Comparing FHIR to other HL7 standards)

2.13.2 Components

The basic building block in FHIR is a Resource. All exchangeable content is defined as a resource. Resources all share the following set of characteristics:

• A common way to define and represent them, building them from data types that define common reusable patterns of elements

• A common set of metadata

• A human readable part

2.13.3 Approach

2.13.3.1 Approach to information modeling

The philosophy behind FHIR is to build a base set of resources that, either by themselves or when combined, satisfy the majority of common use cases. FHIR resources aim to define the information contents and structure for the core information set that is shared by most implementations. There is a built-in extension mechanism to cover the remaining content as needed.

FHIR modeling uses a composition approach. In comparison, HL7 v3 modeling is based on "model by constraint" (see Comparing FHIR to other HL7 standards). With FHIR, specific use cases are usually implemented by combining resources together through the use of resource references. Although a single resource might be useful by itself for a given use case, it is more common that resources will be combined and tailored to meet use case specific requirements. Two special kinds of resources are used to describe how resources are defined and used:

• Capability Statement - describes the interfaces that an implementation exposes for exchange of data

• StructureDefinition - provide additional rules that serve to constrain the optionality, cardinality, terminology bindings, data types and extensions defined in the resources used by the implementation

2.13.4 The Specification

Broadly, the FHIR specification is broken up into a set of modules:

• Foundation: The basic definitional infrastructure on which the rest of the specification is built

• Implementer Support: Services to help implementers make use of the specification

• Security & Privacy: Documentation and services to create and maintain security, integrity and privacy

• Conformance: How to test conformance to the specification, and define implementation guides

• Terminology: Use and support of terminologies and related artifacts

• Linked Data: Defined methods of exchange for resources

• Administration: Basic resources for tracking patients, practitioners, organizations, devices, substances, etc.

• Clinical: Core clinical content such as problems, allergies, and the care process (care plans, referrals) + more

• Medications: Medication management and immunization tracking

• Diagnostics: Observations, Diagnostic reports and requests + related content

• Workflow: Managing the process of care, and technical artifacts to do with obligation management

• Financial: Billing and Claiming support

• Clinical Reasoning: Clinical Decision Support and Quality Measures

Resources have a wide range of uses, from pure clinical content such as care plans and diagnostic reports to pure infrastructure such as Message Header and Capability Statements. They all share common technical characteristics (see below for a more formal definition), but they are used in totally different fashions. Note that you do not have to use REST to make use of resources.

2.13.5 Where to Start

The best place to start is to quickly read the Resources list to get a sense of what resources exist and then look at the Patient resource definition to see what resource definitions look like, and then read these background pages:

• Resource Definitions - basic background to how resources are defined

• About Resources: the Narrative they all contain, and how Resources refer to each other

• Formats: XML and JSON

• About Extensibility: a key way that the specification is kept simple

• If you are coming to FHIR with a background in another HL7 standard (HL7 v2 , HL7 v3 or CDA), The Relationship between FHIR and other HL7 Standards may also be useful

2.13.5.1 Header Tabs

These header tabs found through-out the specification are important, and many readers miss them:

Resources and the data types that they use are presented in a concise easy to read XML-like format, but they also have detailed descriptions of their contents. In addition, most resources are mapped to several different formats, including HL7 v2 , the HL7 v3 RIM, CDA , DICOM, and others. Also, all resources come with at least one example (sometimes many more) and, where appropriate, with profiles that describe their use in specific circumstances. Finally, some resources include notes that help implementers understand the design rationale underlying them.

2.13.6 Finding Additional Information and Providing Feedback

While intended to be consumable by a variety of audiences, the FHIR specification is targeted to the implementation community - those who will actually write the software that uses the specification. To help meet the needs of the implementation community, the editors have strived to keep the specification concise to reduce the amount of reading that must be done before writing useful code. For this reason, information that is not essential to the implementation process, such as considered alternatives, points of contention, future plans, etc. have been excluded from this specification. As well, it is likely that from time-to-time, implementers will encounter situations where the specification is unclear or incomplete. Finally, there may be circumstances where the specification is broken or where a change could allow it to better meet implementer needs.

HL7 has therefore provided a number of mechanisms through which additional information about FHIR can be sought and maintained and through which support and requests for change can be made.

2.13.6.1 The FHIR Confluence Pages

The FHIR project team also maintains a set of Confluence pages where development processes, methodology and design decisions are documented. Implementers and others can also contribute to Confluence to provide additional guidance and supplemental information not found in the specification. Note that Confluence content is not authoritative and is not relevant for determining conformance to the FHIR specification. As well, some Confluence content might not be up to date with the most recent version of the FHIR specification.

Pages defined include FHIR methodology , use of the FHIR design tools , etc. To explore the FHIR Confluence pages, you can start at the root page

2.13.6.2 Formal Change requests

Formal requests for change can be submitted here . (There's a link at the bottom of each page as well.) These will be reviewed by the appropriate work group and a decision made on their incorporation into the specification, including which release (if any) they will be part of.

2.13.6.3 Additional Information sources/Engagement Mechanisms

In addition to the above mechanisms, HL7 provides a Stack Overflow tag, list servers and online chat system to provide various levels of implementer support and engagement. Instructions for accessing these other mechanisms (and instructions for how best to make use of them) can be found at the Support Links .

2.14 FHIR Overview - Developers

URL: https://hl7.org/fhir/R4/overview-dev.html

2.14 FHIR Overview - Developers

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative

FHIR (Fast Healthcare Interoperability Resources) is designed to enable information exchange to support the provision of healthcare in a wide variety of settings. The specification builds on and adapts modern, widely used RESTful practices to enable the provision of integrated healthcare across a wide range of teams and organizations.

The intended scope of FHIR is broad, covering human and veterinary, clinical care, public health, clinical trials, administration and financial aspects. The standard is intended for global use and in a wide variety of architectures and scenarios.

2.14.1 Framework

FHIR is based on "Resources" which are the common building blocks for all exchanges. Resources are an instance-level representation of some kind of healthcare entity. All resources have the following features in common:

• A URL that identifies the resource

• Common metadata

• A human-readable XHTML summary

• A set of defined data elements - a different set for each type of resource

• An extensibility framework to support variation in healthcare

Resource instances are represented as either XML, JSON or RDF and there are currently 145 different resource types defined in the FHIR specification.

This specification describes a set of resources - that is, a set of resource types that describe the set of resource instances that can actually be exchanged. The term 'Resource' is sometimes used without clarifying whether it specifically refers to types or instances - the context of use makes this clear.

2.14.2 Example Resource Instance

This is an example of how a patient is represented as a FHIR object in JSON. An XML encoding is also defined in the specification.

```
{
  "resourceType": "Patient",
  "id" : "23434",
  "meta" : {
    "versionId" : "12",
    "lastUpdated" : "2014-08-18T15:43:30Z"
  }
  "text": {
    "status": "generated",
    "div": "<!-- Snipped for Brevity -->"
  },
  "extension": [
    {
      "url": "http://example.org/consent#trials",
      "valueCode": "renal"
    }
  ],
  "identifier": [
    {
      "use": "usual",
      "label": "MRN",
      "system": "http://www.goodhealth.org/identifiers/mrn",
      "value": "123456"
    }
  ],
  "name": [
    {
      "family": [
        "Levin"
      ],
      "given": [
        "Henry"
      ],
      "suffix": [
        "The 7th"
      ]
    }
```

```
  ],
  "gender": {
    "text": "Male"
  },
  "birthDate": "1932-09-24",
  "active": true
}
```

Each instance of a resource consists of:

• resourceType (line 2) - Required: FHIR defines many different types of resources. See the full index

• id (line 3) - The id of this resource. Always present when a resource is exchanged, except during the create operation (below)

• meta (lines 4 - 7) - Usually Present: Common use/context data to all resources and managed by the infrastructure. Missing if there is no metadata

• text (lines 8 - 11) - Recommended: XHTML that provides a human readable representation for the resource

• extension (lines 12 - 17) - Optional: Extensions defined by the extensibility framework

• data (lines 18 - 43) - Optional: data elements - a different set, defined for each type of resource

Note that although this specification always shows the JSON properties in the order that they are defined, many JSON libraries order properties by other criteria.

2.14.3 URLs and Identities

All resources may have a URL that identifies the resource and specifies where it was/can be accessed from. This URL is not represented inside the resource; the value arises in a context use, and changes as copies of the resource are made, or following other deployment/security related changes. If the resource is accessed via the FHIR RESTful API (see immediately below) then the URL for the resource is [base]/[resourceType]/[id] where the resourceType and id come from the resource (see above).

Some resources - those typically associated with formal publication cycles, rather than operational healthcare - have an explicit URL in them, which is normally the URL of master copy of the resource. This URL remains constant as the resource is copied across systems. See Canonical URLs for further information.

2.14.4 Interactions

For manipulation of resources, FHIR provides a REST API with a rich but simple set of interactions:

• Create = POST https://example.com/path/{resourceType}

• Read = GET https://example.com/path/{resourceType}/{id}

• Update = PUT https://example.com/path/{resourceType}/{id}

• Delete = DELETE https://example.com/path/{resourceType}/{id}

• Search = GET https://example.com/path/{resourceType}?search parameters...

• History = GET https://example.com/path/{resourceType}/{id}/_history

• Transaction = POST https://example.com/path/ (POST a transaction bundle to the system)

• Operation = GET https://example.com/path/{resourceType}/{id}/${opname}

The FHIR specification describes other kinds of exchanges beyond this simple RESTful API, including exchange of groups of resources as Documents, as Messages, and by using various types of Services.

2.14.5 Managing Variability

There is a wide variation between different geo-political jurisdictions and segments of the healthcare industry, and no central authority to impose common business practices. Because of this, the FHIR specification defines an extension framework and defines a framework for managing variability.

Another key aspect of the variability encountered in healthcare is that the same information may be represented differently and with different levels of detail, granularity and nesting by various parties across the system. For example, in some cases a blood pressure measurement may be just a simple observation, a vital sign measure, while in other cases can be a rich set of highly defined data that includes things like controlled vocabularies for posture, exercise, etc. The resource types defined in this specification focus on the general, common use cases. Richer and more specific content can be supported and standardized by defining "profiles" on the base resource types.

2.14.6 Managing Versions

Versions, in the context of FHIR, means one of three different things:

• FHIR Version: Which FHIR version is in use?

• Record Version: for tracking changes to resources, and preventing changes overwriting each other

• Business Version: So humans know which version of the content they are dealing with (for some kinds of resources)

FHIR Version

Usually, the FHIR version is fixed by the context - the CapabilityStatement that a client can use to find out about the server, but there are other ways of managing multiple FHIR versions.

Record Version

FHIR Servers do not have to support versioning, though they are strongly encouraged to do so. There are three different levels of versioning support for FHIR servers:

• Versioning and .meta.version are not supported (and usually, .meta.lastUpdated is not supported either)

• Versioning and the VersionId meta-property are supported, but a history of old versions is not kept

• Versioning and the VersionId meta-property are supported, and a history of old versions is available

In addition, servers may require that version aware updates are used, to prevent over-writing changes, but this is not described on this page.

Business Version

Some resources - typically those that represent content that goes through a formal publishing cycle - carry a version element that explicitly states what version of the content the resource represents. This is changed explicitly by a human, or by some automated process in accordance with applicable business rules.

2.14.7 Creating a resource

To create a resource, send an HTTP POST request to the resource type's respective end point.

```
  POST https://example.com/path/{resourceType}
```

In the example below we see the creation of a Patient.

```
POST {some base path}/Patient HTTP/1.1
Authorization: Bearer 37CC0B0E-C15B-4578-9AC1-D83DCED2B2F9
Accept: application/fhir+json
Content-Type: application/fhir+json
Content-Length: 1198

{
  "resourceType": "Patient",
  ...(properties)
}
```

Submit a new patient to the server, and ask it to store the patient with an id of its own choice.

Notes:

• /Patient (line 1) - the manager for all patients - use the name of the type of resource

• Authorization (line 2) - see Security for FHIR

• Accept, Content-Type (lines 3-4) - the content type for all FHIR resources as represented in JSON (or application/fhir+xml for the XML version). FHIR resources are always represented in UTF-8

• id - The client does not need to provide an id for a resource that is being created - the server will assign one. If the client assigns one, the server will overwrite it

• Resource Content, lines 8+ - There's no meta property at this point. The rest of the resource is the same content as shown above

2.14.8 Create Response

A response will contain an HTTP code 201 to indicate that the Resource has been created successfully. A location header indicates where the resource can be fetched in subsequent requests. The server may choose to return an OperationOutcome resource, but is not required to do so.

```
HTTP/1.1 201 Created
Content-Length: 161
Content-Type: application/fhir+json
Date: Mon, 18 Aug 2014 01:43:30 GMT
ETag: W/"1"
Location: http://example.com/Patient/347

{
  "resourceType": "OperationOutcome",
  "text": {
    "status": "generated",
    "div": "<div xmlns=\"http://www.w3.org/1999/xhtml\">The operation was successful</div>"
  }
}
```

Notes:

• HTTP/1.1 201 (line 1) - the operation was successful. Note that use of HTTP v 1.1 is strongly recommended but not required

• ETag (line 5) - used in the version aware update pattern (if the server supports versioning)

• Location (line 6) - the id the server assigned to the resource. The id in the URL must match the id in the resource when the resource is subsequently returned

• OperationOutcome (line 9) - OperationOutcome resources in this context have no id or meta element (they have no managed identity)

2.14.8.1 Error response

For a variety of reasons, servers may need to return an error. Clients should be alert to authentication related responses, but FHIR content related errors should be returned using an appropriate HTTP status code, with an OperationOutcome resource to provide additional information. Here is an example of a server rejecting a resource because of server defined business rules:

```
HTTP/1.1 422 Unprocessable Entity
Content-Length: 161
Content-Type: application/fhir+json
Date: Mon, 18 Aug 2014 01:43:30 GMT

{
  "resourceType": "OperationOutcome",
  "text": {
    "status": "generated",
    "div": "<div xmlns=\"http://www.w3.org/1999/xhtml\">MRN conflict
   - the MRN 123456 is already assigned to a different patient</div>"
  },
}
```

Notes:

• The server can return additional structured information using the details of the OperationOutcome

2.14.9 Read Request

Reading a resource is done by sending HTTP GET requests to the desired Resource Type end-point.

```
  GET https://example.com/path/{resourceType}/{id}
```

Here's an example.

```
GET /Patient/347?_format=xml HTTP/1.1
Host: example.com
Accept: application/fhir+xml
Cache-Control: no-cache
```

Notes:

• 347 (line 1) - The id of the resource that is being fetched

• _format=xml (line 1) - this is another method for clients to indicate the desired response format, in addition to using the accept header, and is useful for clients that don't have access to the HTTP Headers (e.g. XSLT transforms) (see Mime Types)

• cache control (line 4) - Cache control is important, though FHIR itself says nothing about it - see http://www.w3.org/Protocols/rfc2616/rfc2616-sec13.html or https://www.mnot.net/cache_docs/

2.14.10 Read Response

The response to a GET contains the Resource.

```
HTTP/1.1 200 OK
Content-Length: 729
Content-Type: application/fhir+xml
Last-Modified: Sun, 17 Aug 2014 15:43:30 GMT
ETag: W/"1"

<?xml version="1.0" encoding="UTF-8"?>
<Patient xmlns="http://hl7.org/fhir">
  <id value="347"/>
  <meta>
    <versionId value="1"/>
    <lastUpdated value="2014-08-17T15:43:30Z"/>
  </meta>
  <!-- content as shown above for patient -->
</Patient>
```

Notes:

• id (line 9) - The id of the resource. This must match the id in the read request

• versionId (line 11) - The current version id of the resource (if the server supports versioning). Best practice is that this value matches the ETag (see version aware update), but clients must never assume this

• Note that servers are not required to support versioning, but are strongly encouraged to do so

• lastUpdated (line 12) - if present, this must match the value in the HTTP header

2.14.11 Search Request

In addition to getting single known resources it's possible to find a collection of resources by searching the resource type end-point with a set of criteria describing the set of resources that should be retrieved, and their order. The general pattern is:

```
  GET https://example.com/path/{resourceType}?criteria
```

The criteria is a set of HTTP parameters that specify which resources to return. The search operation

```
https://example.com/base/MedicationRequest?patient=347
```

returns all the prescriptions for the patient created above.

2.14.12 Search Response

The response to a search request is a Bundle: a list of matching resources with some metadata:

```
HTTP/1.1 200 OK
Content-Length: 14523
Content-Type: application/fhir+xml
Last-Modified: Sun, 17 Aug 2014 15:49:30 GMT

{
  "resourceType": "Bundle",
  "type": "searchset",
  "id" : "eceb4882-5c7e-4ca4-af62-995dfb8cef01"
  "timestamp": "2014-08-19T15:49:30Z",
  "total": "3",
  "link": [
    {
      "relation" : "next",
      "url" :
    "https://example.com/base/MedicationRequest?patient=347&searchId=ff15fd40-ff71-4b48-b366-09c706bed9d0&page=2"
    }, {
      "relation" : "self",
      "url" : "https://example.com/base/MedicationRequest?patient=347"
    }
  ],
  "entry": [
    {
      "resource" : {
        "resourceType": "MedicationRequest",
        "id" : "3123",
        "meta" : {
          "versionId" : "1",
          "lastUpdated" : "2014-08-16T05:31:17Z"
        },
        ... content of resource ...
      },
    },
    ... 2 additional resources ....
  ]
}
```

Notes:

• resourceType/type (line 7/8) - the result of a search is always a bundle of type "searchset"

• id (line 9) - An identifier assigned to this particular bundle. The server should assign a unique id to this bundle that it will not be re-used

• timestamp (line 11) - (if the server supports versioning) This should match the HTTP header, and should be the date the search was executed, or more recent, depending on how the server handles ongoing updates. The timestamp must be the same or more recent than the most recent resource in the results

• total (line 13) - The total number of matches in the search results. Not the number of matches in this particular bundle, which may be a paged view into the results

• link (line 14) - A set of named links that give related contexts to this bundle. Names defined in this specification: first, prev, next, last, self

• entry (line 23) - Actual resources in this set of results

• entry.resource.id (line 27) - Note that in some bundles, the entry.resource.id must be unique in the bundle

• The search operation is also able to return additional related resources as well

2.14.13 Update Request

The client sends the server a new version of the resource to replace the existing version - it PUTs it to the location of the existing resource:

```
  PUT https://example.com/path/{resourceType}/{id}
```

Note that there does not need to be a resource already existing at {id} - the server may elect to automatically create the resource at the specified address. Here is an example of updating a patient:

```
PUT /Patient/347 HTTP/1.1
Host: example.com
Content-Type: application/fhir+json
Content-Length: 1435
Accept: application/fhir+json
If-Match: 1

{
  "resourceType": "Patient",
  "id" : "347",
  "meta" : {
    "versionId" : "1",
    "lastUpdated" : "2014-08-18T15:43:30Z"
  },
  ...
}
```

Notes:

• resourceType (line 1) - "Patient" in the URL must match the resource type in the resource (line 9)

• resource id (line 1, "347") - This must match the id in the resource (line 10)

• If-Match (line 6) - if this is provided, it must match the value in meta.versionId (line 12), and the server must check the version integrity, or return 412 if it doesn't support versions

• meta.lastUpdated (line 13) - This value is ignored, and will be updated by the server (mostly, but not always, if the server does not support versioning)

• resource content (line 14) - Not shown here, the same as Patient above

2.14.14 Update Response

The response to an update request has metadata / status, and optionally an OperationOutcome:

```
HTTP/1.1 200 OK
Content-Length: 161
Content-Type: application/fhir+json
Date: Mon, 18 Aug 2014 01:43:30 GMT
ETag: W/"2"
Location: https://example.com/Patient/347/_history/2

{
  "resourceType": "OperationOutcome",
  "text": {
    "status": "generated",
    "div": "<div xmlns=\"http://www.w3.org/1999/xhtml\">The operation was successful</div>"
  }
}
```

Notes:

• ETag (line 5) - This is the versionId of the new version, and is also found in the location header (line 6) (if the server supports versioning)

2.14.15 Base Resource Content

Here is an example that shows all the information found in all resources, fully populated:

```
{
  "resourceType" : "X",
  "id" : "12",
  "meta" : {
    "versionId" : "12",
    "lastUpdated" : "2014-08-18T15:43:30Z",
    "profile" : ["http://example-consortium.org/fhir/profile/patient"],
    "security" : [{
      "system" : "http://terminology.hl7.org/CodeSystem/v3-ActCode",
      "code" : "EMP"
    }],
    "tag" : [{
      "system" : "http://example.com/codes/workflow",
      "code" : "needs-review"
    }]
  },
  "implicitRules" : "http://example-consortium.org/fhir/ehr-plugins",
  "language" : "X"
}
```

Implementers notes:

• resourceType (line 2) - always found in every resource. In XML, this is the name of the root element for the resource

• id (line 3) - defined when the resource is created, and never changed. Only missing when the resource is first created

• meta.versionId (line 5) - changes each time any resource contents change (except for the last 3 elements in meta - profile, security and tag)

• meta.lastUpdated (line 6) - Changes when the versionId changes. Systems that don't support versions usually don't track lastUpdated either

• meta.profile (line 7) - An assertion that the content conforms to a profile. See Extending and Restricting Resources for further discussion. Can be changed as profiles and value sets change or the system rechecks conformance

• meta.security (lines 8 - 11) - Security labels applied to this resource. These tags connect resources in specific ways to the overall security policy and infrastructure. Security tags can be updated when the resource changes, or whenever the security sub-system chooses to

• meta.tag (lines 12 - 16) - Tags applied to this resource. Tags are used to relate resources to process and workflow. Applications are not required to consider the tags when interpreting the meaning of a resource

• implicitRules (line 17) - indicates that there is a custom agreement about how the resources are used that must be understood in order to safely process the resource. Use of this is discouraged because it restricts sharing, but sometimes necessary

• language (line 18) - The base language of the resource. The resource is allowed to have content from other languages; this is just the base, but should be the main language of the resource

The base properties of all resources are defined on the resource types Resource and DomainResource.

Want more information?

• Getting Started
• Resource Index
• Resource Guide
• Documentation Index
• Support Links

2.15 FHIR Overview - Clinicians

URL: https://hl7.org/fhir/R4/overview-clinical.html

2.15 FHIR Overview - Clinicians

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative

FHIR (Fast Healthcare Interoperability Resources) is designed to enable the exchange of healthcare-related information. This includes clinical data as well as healthcare-related administrative, public health and research data. It covers both human and veterinary medicine and is intended to be usable world-wide in a wide variety of contexts, including in-patient, ambulatory care, acute care, long-term care, community care, allied health, etc.

The FHIR specification is targeted to individuals and organizations developing software and architecting interoperable solutions that will be using FHIR. The FHIR specification does not attempt to define good or best clinical practices, nor does it provide guidance on user interfaces or workflows. Guidance in these areas may be useful, but it is outside of FHIR's scope.

Because of FHIR's focus on implementation, many aspects of the specification deal with the technical underpinnings of the exchange of clinical information between electronic systems. This section provides an introduction to what FHIR provides, and tries to highlight those portions of the specification that are likely to be of most interest to the clinical community while skipping over some of the technical minutiae of interoperability. However, clinical readers are welcome to explore some of the more technical areas if they find them of interest.

2.15.1 Resources

From a clinical perspective, the most important parts of the FHIR specification to understand are the Resources. Think of Resources as paper "forms" reflecting different types of clinical and administrative information that can be captured and shared. The FHIR specification defines a generic "form template" for each type of clinical information - so one for allergies, one for prescriptions, one for referrals, etc.

FHIR data consists of repositories containing completed "forms" (resource instances). The resource instances describe patient-related information (such as demographics, health conditions and procedures) as well as administrative information (such as practitioners, organizations and locations). Some resources are infrastructure components used to support the technical exchange of information by describing what systems are able to do, defining allowed sets of codes, etc. FHIR repositories might be electronic health record (EHR) systems, pharmacy systems, hospital information systems (HIS), etc. Some systems, such as clinical decision support engines, may expose FHIR interfaces even though they don't actually store any patient or administrative information themselves.

Each Resource defines a small amount of highly-focused data. A single resource doesn't say very much, but a collection of Resources taken together creates a useful clinical record. Information systems map the actions that a user takes (look up patient records, make a note in their history, etc.) to operations on the relevant resources.

2.15.2 Extensibility and Profiling

The paper forms (Resources) in FHIR are somewhat generic. They have to be usable in different countries and by different types of clinicians in different contexts (human care, veterinary care, public health, research, etc.). Recognizing that a one size fits all approach is not appropriate in the healthcare space, FHIR provides the ability to adjust the forms (Resources) to be able to handle the needs of different implementation spaces by defining "extensions" as well as enforcing constraints. For example, a "prescription" form might have extension elements added to support tracking of restricted medications while also constraining the codes that can be used to communicate types of drugs to a particular national standard. Forms are designed in such a way that these changes can be made without changing how systems pass forms around, enabling any system to consume completed forms even if they have additional elements added, whether or not those additional elements are used by the receiving system.

To keep the base forms that everyone uses from being overly complex, FHIR has a rule that, in most cases, a resource will only include data elements if there's an expectation that most implementations will use that

particular data element. That doesn't mean the data must always exist. For example, most systems in the world are capable of tracking "deceased date" for a patient, even though that element will be blank for many patient records. On the other hand, not as many systems track hair color, so hair color would be omitted from the base form and those systems that need it (perhaps in a specialty clinical or research setting) can use a FHIR extension to capture it if needed.

To keep the number of resources reasonable, some of them are fairly broad. For example, the Observation resource is used for vital signs, lab results, psychological assessments and a variety of other things. To support setting rules for more narrow areas (e.g. "What should I send if I want to share a blood pressure?"), FHIR allows the creation of Profiles. There will be a great deal of clinical work involved in forming consensus around how different types of detailed clinical information should be captured and shared in particular settings. Tooling to support the creation of profiles directly by clinicians is part of the plan for FHIR, but is still in the very early stages.

2.15.3 Narrative

FHIR is intended to support sharing data in a computable manner. I.e. The information shared should be usable for computer-mediated processes such as decision support, rules triggering, trend analysis, etc. However, not every system is the same and not all systems are able to recognize all discrete data. Also, there is still considerable value in data exchange in circumstances where not all or even none of that data is captured in a discrete manner. For this reason, FHIR resources support sharing not only discrete data for computation, but also a human-readable view so that the humans on each end of a healthcare information exchange can still get a full picture of what's going on.

Narrative is expected to exist for most resource instances, although it can be omitted in a few limited circumstances. In some cases, the narrative will be generated from discrete information. For example, the narrative for a patient might look like this:

> Peter James Chalmers (OFFICIAL), Jim identifier: MRN = 12345 (USUAL) telecom: ph: (03) 5555 6473(WORK) gender: MALE birthDate: Dec 25, 1974 deceased: false address: 534 Erewhon St PleasantVille Vic 3999 (HOME)

In other cases, the narrative might be free-form text commentary entered directly by a practitioner, such as referral letters, pathology reports, etc. Certain parts of the narrative content could also later be exposed as discrete data.

2.15.4 Interfaces

In addition to defining the "forms" for data exchange (Resources), FHIR also defines a set of interfaces by which systems actually share that information. There are four primary mechanisms or "paradigms" of exchange supported by FHIR: via a REST interface, by exchanging Documents, by sending and receiving Messages and by exposing and invoking Services.

2.15.4.1 REST

REST is the simplest exchange mechanism. Continuing with the "form" metaphor, a RESTful server can be thought of as a room full of filing cabinets. Within the room is a cabinet for each "type" of form (or Resource) it supports. The cabinet contains folders where each folder has a unique number and represents one particular real-world thing: one Patient, one Encounter, one Medication, etc. Each folder (which represents a single Resource instance) contains multiple pieces of paper, with each piece of paper representing a specific "version" of that instance. Every time someone updates a record, a new piece of paper is added to the top of the file folder. To see the history of a resource, you simply have to flip through the pieces of paper in the folder.

Note that a typical medical record is generally a big "folder-of-folders" with many different types of 'forms' or 'reports' gathered together. This is convenient for someone who wants to review the whole record, but inconvenient for someone updating bits of it. There's always contention for access to it to update the right part. In the computer application the record will be decomposed to its smallest components for management purposes, and a computer will (or should) assemble the correct bits as required, by following references that exist from one piece of information to the next.

Now picture a clerk at the front door of that room. You can pass the clerk a requisition to have them do something with the information in those file cabinets. The "clerk" and the set of requisition forms on the clerk's counter make up the FHIR restful API. With that API, you can do the following:

• search: Have the clerk search through the folders for one(s) that meet a set of search criteria and give you a copy of the top piece of paper in each relevant folder

• read: Get a copy of the top piece of paper (which is the most current version) from a specific folder in one of the cabinets

• create: Add a new folder to the appropriate cabinet (with a new number)

• update: Add a new page (version) to the contents of a specific folder

• delete: Remove a folder from the cabinet (or more accurately, virtually remove it by putting a sticker on it saying "do not open")

• history: Look at all the pages in a single folder, in a particular file cabinet or possibly in the whole room. Such a broad request would be used primarily for administrative purposes rather than clinical purposes

• transaction: Give the clerk multiple pieces of paper to place in distinct folders at the same time

• operation: Ask the clerk to perform an action or procedure on papers from one or more of the folders - for example, averaging numbers across patients, producing a summary record, or perform a complex search just by ticking a box on a requisition saying "do that one"

EHRs and other systems may present a more sophisticated interface to their end users, but behind the scenes they're all making these same types of requests to the file clerk.

2.15.4.2 Documents

Documents are a familiar mechanism for sharing information in the healthcare space. They are useful whenever there's a desire to guide how a consumer of information will navigate it and there's a need to have a "frozen" view of information that can be reliably retrieved even years in the future. Examples of document-like things in healthcare include discharge summaries and lab reports.

In FHIR, there's a special resource called Composition that acts as the "cover page" for a document. It identifies the title, author date, relevant patient and the table of contents. A FHIR document can be thought of as a set of sheets (Resource instances) stacked together with a title page on top that's stapled together. That stapled collection can then be stored or passed around, conveying a complete set of information at once.

2.15.4.3 Messaging

Much healthcare information exchange happens using a messaging paradigm. In messaging, a set of information is sent from one system to another, typically triggered by an event in the sender system. For example, a patient being admitted, a lab test being ordered, a drug being administered, the clock striking 12:00 or someone pressing a button. The message serves to notify the receiver that the event occurred as well as provide details about any existing data that was modified or new data that was created. Typically receiving a message means there's an expectation that the receiving system will "do something" in response.

A message might request that a lab order be fulfilled or notify a system that two patient records have been merged or that a patient has been transferred from one bed to another. A message is similar to a document in that it collects resources together, however for a message, the "cover page" is a MessageHeader that acts as a requisition. And rather than using a staple, the resources are joined together with a paper-clip and there's no expectation that the receiving system will store the contents of the message exactly as received, if at all.

2.15.4.4 Services

Services can be thought of as a light-weight way of doing messaging. Rather than a full cover page, a small sticky note is attached to the front of a resource. And sometimes rather than sending a full piece of paper, the relevant pieces are cut out and sent as fragments. The response to a requisition is a similarly paper-clipped bundle of resource instances. Services are likely to be used for things like decision support. E.g. "Is there a problem with prescribing medication X for patient Y?" or "What's the recommended care plan for a patient with conditions A, B and C?"

2.15.5 Approaching the specification

A FHIR-based system's capabilities are defined by what the Resources can say and from a clinical perspective, these things define the clinical record:

• the kinds of Resources that are defined

• their data contents, and rules about the data such as what terminology codes are supported and/or required

• how resources reference to each other

• how you can search for information

This information can all be found in the resource definition pages. The resources most likely to be of interest can be found in the following modules:

• Clinical & Care provision

• Diagnostics

• Medications

• Administrative

Instructions on how to interpret the information found on the resource pages can be found here. The Logical table or the UML views are likely to be easiest to understand. Don't forget to look at the examples tab for an idea of what kind of information can be expressed. Seeing how elements are used to convey real data is often more useful than just looking at definitions. Also, look at the Profiles tab to see examples of how different resources can be constrained for use in particular contexts.

Clinician and other domain expertise and feedback is always welcome as we continue refining the FHIR specification. At the top of each resource page is a link to the home page for the work group responsible for that particular resource. If you have feedback on resource design, consider getting involved.

2.16 FHIR Overview - Architects

URL: https://hl7.org/fhir/R4/overview-arch.html

2.16 FHIR Overview - Architects

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative

At its core, FHIR contains two primary components:

• Resources - a collection of information models that define the data elements, constraints and relationships for the “business objects” most relevant to healthcare. From a model-driven architecture perspective, FHIR resources are notionally equivalent to a physical model implemented in XML or JSON. See the formal definition.

• APIs – a collection of well-defined interfaces for interoperating between two applications. Although not required, the FHIR specification targets RESTful interfaces for API implementation. See details on FHIR RESTful interfaces.

In the healthcare domain, the set of “business objects” is not universally defined, but there is a notional and ongoing evolutionary, consensus-based process for standardizing on a core set of common business objects including things like “a patient”, “a procedure”, “an observation”, “an order”, etc. (see a list of defined resources). The FHIR specification provides a framework for defining these healthcare business objects (“resources”), for relating them together in a compositional manner, for implementing them in a computable form, and for sharing them across well-defined interfaces. The framework contains a verifiable and testable syntax, a set of rules and constraints, methods and interface signatures for “FHIR-aware” APIs, and specifications for the implementation of a server capable of requesting and delivering FHIR business objects.

From an operational perspective, HL7’s internal standards development and governance processes determine what constitutes a resource and which resources exist. In addition, the FHIR specification also provides a mechanism for contextualizing resources for specific needs within specific bounds (see Profiling Resources).

2.16.1 Architecture Frameworks and FHIR Alignment

FHIR resources fit firmly within the information architecture domain and the FHIR APIs for data exchange address aspects of application architecture.

From a TOGAF perspective , FHIR addresses aspects of architecture views related to information model definition and data exchange, which are described in the Information Systems Architectures portion of the TOGAF Architecture Development Method.

With regards to the Zachman Framework , FHIR fits within the What and the How dimensions of the Architect, Engineer and Technician Perspectives

When considering the HL7 Services Aware Interoperability Framework (SAIF) , FHIR resources and RESTful APIs represent the “Physical Models” and “Interface Implementations” within the Platform Specific Specifications layer of the Information Models and the Behavioral Models Viewpoints, respectively.

2.16.2 FHIR and Architectural Principles

FHIR’s primary purpose is to address interoperability with well-structured, expressive data models and simple, efficient data exchange mechanisms. In addition, FHIR aligns to the following architectural principles:

• Reuse and Composability – FHIR resources are designed with the 80/20 rule in mind – focus on the 20% of requirements that satisfy 80% of the interoperability needs. To this end, resources are designed to meet the general or common data requirements of many use cases to avoid the proliferation of numerous, overlapping and redundant resources. Extension and customizations exist (see FHIR Profiles) to allow common, somewhat generic resources to be adopted and adapted as needed for specific use case requirements. In addition, FHIR resources are highly composable in that resources commonly refer to other resources. This further promotes reuse and allows for complex structures to be built from more atomic resources.

• Scalability – Aligning FHIR APIs to the REST architectural style ensure that all transactions are stateless which reduces memory usage, eliminates the needs for “sticky” sessions within a server farm and therefore supports horizontal scalability.

• Performance – FHIR resources are lean and suitable for exchange across the network. Highly optimized formats are available, which has the potential to improve performance in complex transactions across multiple systems connected via a shared and finite network, though most implementers find the standard JSON / XML formats adequate.

• Usability – FHIR resources are understood by technical experts and non-technical people alike. Even if the details of XML or JSON syntax are not understood, non-technical people can view these in any browser or text reader and understand the contents within them.

• Data Fidelity – FHIR is strongly typed and has mechanisms built in for clinical terminology linkage and validation. In addition, XML and JSON documents can be validated syntactically as well as against a defined set of business rules. This promotes high data fidelity and goes a long way towards using FHIR to achieve semantic interoperability.

• Implementability – One of the driving forces for FHIR is the need to create a standard with high adoption across disparate developer communities. FHIR is easily understood and readily implemented using industry standards and common mark-up and data exchange technologies.

There are additional architecture principles related to consistency, granularity, referential integrity, and others that are not as well established or proven. See the section below on Outstanding Issues for details.

2.16.3 FHIR Decomposition

As discussed, FHIR's principal components are resources and RESTful APIs. However, there is more to the FHIR specification including the components depicted below.

NOTE: The term "component" is used loosely to mean a part of something and does not intend to carry the specific meanings for this term provided by rigorous ontologies, modeling frameworks, or other architectural and organizational constructs. Diagrammatically, the components below are depicted below as UML classes. This is done purely to take advantage of the semantics afforded using this notation. FHIR is neither objected oriented in its modeling approach nor are the components that make up the FHIR specification UML classes or objects in the formal sense. Likewise, the UML packages shown below are notional and used for organizational purposes only.

As shown in the diagram below, it is convenient to think of the FHIR specification as having components that address the following:

• Information Model – the components of FHIR related to the creation of FHIR resources

• Constraints – the components of FHIR addressing constraints and validity

• Terminology – the components of FHIR related to clinical terminologies and ontologies

• Usage – the component of FHIR addressing the use of FHIR in a run-time capacity

Component definitions:

• FHIR Information Model Base Classes: Element and Resource Definitions for Base Classes: ElementDefinition and StructureDefinition Data Types

• FHIR Conformance Model Capability Statement Profile

• FHIR Terminology Code System Value Set

• FHIR Usage REST API

2.16.4 Creating Resources – Architectural Considerations

The following list provides general guidelines that apply when FHIR resources are defined. Most of these items are not enforced programmatically requiring human due diligence and governance to ensure adherence.

• Resources should have a clear boundary; one that matches one or more logical transaction scopes

• Resources should differ from each other in meaning, not just in usage (e.g., each different way to use a lab report should not result in a different resource)

• Resources need to have a natural identity

• Most resources should be very common and used in many different business transactions

• Resources should not be specific or detailed enough to preclude support for a wide range of business transactions

• Resources should be mutually exclusive [this is a very important consideration that helps to reduce redundancy and ambiguity]

• Resources should use other resources, but they should be more than just compositions of other resources; each resource should introduce novel content

• Resources should be organized into a logical framework based on the commonality of the resource and what it links to (see resource framework below)

• Resources should be large enough to provide meaningful context; resources that contain only a few attributes are likely too small to provide meaningful business value

• Resources should reflect general usage: if most systems treat something as a single concept, that suggests a single resource; if most systems treat something as distinct concepts, then that suggests multiple resources if two different uses of a "resource" would result in wildly different interpretations of what constitutes "core" then that suggests two resources might be appropriate.

• There is a bias towards fewer resources rather than more

2.16.5 Organizing FHIR Resources

It is impractical to model the entirety of health data in a single information model. Every modeling initiative in healthcare from HL7 version 2 message specifications to FHIR resources decomposes the healthcare domain into smaller, more manageable sub-domains or information model snippets. With FHIR, each resource is essentially a snippet of the larger healthcare information domain.

When breaking down the healthcare information model into smaller chunks (or resources for FHIR), it is important to have a framework and set of guidelines to promote consistency and integrity within the resource structures and in the way resources reference each other. The framework shown below includes health information model sub-categories organized into layers based on their degree of commonness. The layers and categories are useful for identifying which parts of healthcare information are the most common and

therefore need to be the most consistently defined and tightly governed. The categories at the top layers are the most common and contain the FHIR resources that support the largest number of common healthcare transactions.

Descriptions of the layers in the framework:

• Foundation Resources: Foundation resources are the most rudimentary, foundational resources. They are often used for infrastructural tasks. Although not prohibited, they are not always referenced by other resources.

• Base Resources: Layer two consists of base resources. These are often the leaf nodes of a resource graph. In other words, they are often referenced by other resources, but don't typically reference other resources themselves. These resources are typically the most commonly used, and therefore require the highest degree of consistency and architectural rigor. Governance is greatest for resources in layers one and two.

• Clinical Resources: Layer 3 includes the resources that are clinical in nature but are also very common across many use cases. This includes resources for clinical observations, clinical treatment, care provision, and medications. These resources can be used by themselves, but typically build on the resources in layer two. For example, an observation resource will reference the patient resource from layer two. These resources are also frequently contextualized when they are referenced by resources in layers three, four and five.

• Financial Resources: Layer four is dedicated to financial resources. Logically, financial resources build on clinical and base resources. For example, a billing resource will reference clinical events and activities as well as base resources like a patient.

• Specialized Resources: In layer five, we find more specialized resources for less common use cases. These resources almost always reference resources in lower layers. Given that FHIR places priority on satisfying the most common use cases, there are fewer resources in this layer.

• Resource Contextualization: Layer 6 does not contain resources. However, it does extend the composition framework made up by the first five layers of resources. Layer 6 includes profiles and graphs. Profiles are used to extend, constrain, or otherwise contextualize resources for a given purpose. Graphs are compositions of resources, or webs of resource, that contain attributes of their own.

The complete set of FHIR resources organized against this framework is found on the Resources page.

The framework serves three primary purposes:

• Organize resource for navigation and identification

• Classify resources into categories based on common sense groupings or patterns describing expected structures and/or behaviors amongst resources in the same category

• Disseminate resources across layers to stratify relative common-ness with the most common resources in the top layers

Purposes 2 and 3 set the foundation for future architectural rigor and resource governance to optimize consistency, integrity and predictability of new or refined resources in the future. The actual rules and patterns will be defined and refined in future FHIR releases. However, one general guideline to state now is that resources generally reference resources in the same layer or higher. In other words, a layer 4 resource will typically only reference resources in layers 4, 3, 2 or 1. There is nothing prohibiting a layer 4 resource from referencing a layer 5 resource, but this is not as common. Given this guideline, it is possible to identify the resources that are likely to be most common across use cases and therefore demand the highest degree of consistency and governance. Further, the framework helps identify the areas where creating new FHIR resources is the highest priority. It is generally a higher priority to create FHIR resources in the higher layers (layers 1, 2 and 3) than it is to create FHIR resources in the lower layers (layers 4 and 5) because the higher layer resources will provide the greatest value across the largest number of use cases and stakeholders. This is not to say that the business transactions needed for the higher layers are not important, it’s just that they are not as common across the whole healthcare space.

The 6th layer of the framework are not actually resources. Profiles and Graphs are extensions of resources or resource compositions that continue the progression through the FHIR Composition Framework. They provide additional contextualization required to satisfy certain use cases.

There are several benefits expected from aligning the creation of FHIR resources to this framework, including:

• Organization and manageability of health domains - the framework provides a basis for decomposition and modularity

• Identifying commonality - the framework teases out the common areas from the less common areas

• FHIR resources prioritization - the framework provides a structure for determining priorities and delegating work

• Tiered governance levels - the framework separates the areas needing the most stringent and universal governance from those that require more context-specific governance

The framework is further elaborated in the FHIR Resource Considerations page .

Another useful tool for visualizing how FHIR resources are organized relative to each other can be found using the Resource Reference Visualization tool on clinFHIR .

2.16.6 FHIR Implementation – High-Level Design Considerations

2.16.6.1 FHIR Servers and the FHIR APIs

A FHIR REST server is any software that implements the FHIR APIs and uses FHIR resources to exchange data. The diagram below describes the FHIR interface definitions. The methods are classified as:

• iServeInstance – methods that perform Get, Put or Delete operations on a resource

• iServeType – methods that get type information or metadata about resources

• iServeSystem – methods that expose or enable system behaviors.

Additional details on the FHIR APIs can be found at the FHIR RESTful API and the Operations Framework.

2.16.6.2 FHIR Transactions

As mentioned, FHIR resources are optimized for stateless transactions with RESTful APIs. Although this is not the only way FHIR resources can be used, these types of transactions are the only ones with defined interfaces and behaviors in the FHIR specification.

FHIR transactions follow a simple request and response transaction pattern. The request and response can be for a single payload or can operate as batch. The payload or a request and response consist of a header and the content of interest. See diagram below for details.

2.16.6.3 Security

(section to be filled out) (but see Security in the meantime).

Example Use Cases Using FHIR

For illustrative purposes, the following diagram depicts a simple use case of a patient accessing their personal health record (portal) enabled by an underlying electronic medical record (EMR) system. The EMR plays the role of the FHIR server in this example.

The pre-conditions for this use case are:

• the EMR implements the necessary FHIR APIs

• the EMR implements the necessary authentication and authorization mechanisms

• the patient is successfully authenticated and authorized to access FHIR resources

The basic flow of the use case is that the patient registers (if required), logs in, enters search criteria to identify a patient or patients of interest (the patient is most like themselves in this use case), retrieves clinical documents for the patient and retrieves clinical resources for the patient. The use cases utilize the GET methods on the iServeInstance interface and works with the following types of FHIR resources:

• The Patient resource

• One or more document resource(s)

• One or more clinical resource(s)

Although this example use case is very simple, more complex transactions using a combination of GETs, PUTs and DELETEs against resources and metadata can be envisioned. However, the exact details of these use cases including which methods are used, the orchestration of methods and the specific resources involved are outside the scope of the FHIR specification.

2.16.7 Outstanding Issues

• Resource Consistency and Granularity – there is nothing intrinsically prohibiting one resource from duplicating the same information as another resource. Further, there is nothing prohibiting resources with the same information from defining and modeling the data elements differently. HL7 has a number of processes to ensure that resources are consistently designed, but the question is when to be consistent within the specification, and when to be consistent with the real world practices of healthcare - these are sometimes in conflict with each other. Resource granularity is a related potential problem as there are variations in the size, complexity and comprehensiveness of the existing resources. Further, the degree to which the FHIR specification can impose consistency is limited to how much agreement can be gained across various communities. While the Implementers Safety Check List and the Considerations for FHIR Resource Considerations provide guidance and promote consistency, rules for achieving complete consistency of both content and granularity amongst resources are neither completely defined nor completely enforced. Considering that FHIR is still a new and emerging standard, an over-abundance of constraint and rigor has been avoided to maximize initial adoption. Further, there is a natural tension between consistency and an architectural virtue and the practicalities of supporting the real practice of health care. Considering that FHIR ultimately is a reflection of the health business processes it supports, FHIR will always carry forward some of the data discrepancies, inconsistencies and gaps that are present in the practice of healthcare across different organizations and practitioners. Nonetheless, the issues of resource consistency and granularity is a topic that gets considerable ongoing discussion, and may change as FHIR approaches a final normative

standard and as FHIR adoption approaches a level where more control is warranted, or more information/process consistency emerges in the existing healthcare systems.

• Resource References – there are currently a lack of strict rules for what resources should be referenced by other resources and under what circumstance. There is potential for ambiguity, duplication, inaccurate and/or conflicting information communicated by a resource graph (a collection of linked resources). Imagine the scenario where Resource Type A (e.g., procedure) references Resource Type B (e.g., encounter) and Resource Type C (e.g., patient), and Resource Type B (e.g., encounter) also references Resource Type C (e.g., patient). In this scenario, is a reference to Resource A to Resource C meant to provide the same information as the reference from Resource B to Resource C? If so, is this duplication of information problematic? Note that this is not unique to FHIR - it is an innate property of information systems. If an actual instance of A, and the B that it references, reference different instances of Resource C (e.g. the procedure references patient X and an encounter for patient Y), how does the system know that the references are intentionally different versus an error or data anomaly? The problem is that there is limited ability to describe the intent of the reference which leads to the possibility of ambiguity and error. The Linkage resource can be used to help with this problem, but additional capabilities may be considered in the future to allow systems to address referential integrity.

• Conditional Semantics – Currently, the constraints for element definitions including things like data types, value sets, optionality and cardinality are defined at design time with limited consideration for variable run-time semantics. Imagine the scenario where the value of Data Element Y (e.g., “intolerance type”) is constrained differently depending on the value of Data Element X (e.g., “causative agent”) in a given instance of a resource. For example, if the instance of an Intolerance Resource has the “intolerance type” data element populated with “food intolerance”, then “causative agent” should be constrained to only valid values for this value set (e.g., valid foods instead of medications or environmental agents). Tools for addressing deep semantic consistency in this regard are only gradually developing.

• Business Rule Enforcement and Validation – As governance increases and more resource rules are defined, it may be advantageous to have a resource validation tool that checks for things like resource consistency, duplication, referential integrity, circular or nonsensical references, and other defined and approved validation rules. Once rules are agreed to, this level of automation can help address the other issues outlined above. These kinds of facilities are planned for the future.

2.17 Introducing HL7 FHIR

URL: https://hl7.org/fhir/R4/summary.html
FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative

FHIR® – Fast Healthcare Interoperability Resources (hl7.org/fhir) – is a next generation standards framework created by HL7. FHIR combines the best features of HL7's v2 , HL7 v3 and CDA product lines while leveraging the latest web standards and applying a tight focus on implementability.

FHIR solutions are built from a set of modular components called "Resources". These resources can easily be assembled into working systems that solve real world clinical and administrative problems at a fraction of the price of existing alternatives. FHIR is suitable for use in a wide variety of contexts – mobile phone apps, cloud communications, EHR-based data sharing, server communication in large institutional healthcare providers, and much more.

2.17.1 Why FHIR is better

FHIR offers many improvements over existing standards:

• A strong focus on implementation: fast and easy to implement (multiple developers have had simple interfaces working in a single day)

• Multiple implementation libraries, many examples available to kick-start development

• Specification is free for use with no restrictions

• Interoperability out-of-the-box: base resources can be used as is, but can also be adapted as needed - which happens a lot - for local requirements using Profiles, Extensions, Terminologies and more

• Evolutionary development path from HL7 Version 2 and CDA: standards can co-exist and leverage each other

• Strong foundation in Web standards: XML, JSON, HTTP, OAuth, etc.

• Support for RESTful architectures, seamless exchange of information using messages or documents, and service-based architectures

• Concise and easily understood specifications

• A human-readable serialization format for ease of use by developers

• Ontology-based analysis with formal mapping for correctness (under development)

2.17.2 Flexibility / Adaptation

A central challenge for healthcare standards is how to handle the wide variability caused by diverse healthcare processes. Over time, more fields and optionality are added to the specification, gradually adding

cost and complexity to the resulting implementations. The alternative is relying on custom extensions, but these create many implementation problems too.

FHIR solves this challenge by defining a simple framework for extending the existing resources and describing their use with Profiles. All systems can read all resources, but applications can add more control and meaning using profiles. Many healthcare contexts require extensive local agreements.

In addition, each resource carries a human-readable text representation using html as a fall-back display option for clinical safety. This is particularly important for complex clinical information where many systems take a simple textual/document based approach.

2.17.3 Example Resource: Patient

This simple example shows the important parts of a resource: a local extension, the human readable HTML presentation, and the standard defined data content.

FHIR has resources for administrative concepts such as patient, provider, organization and device as well as a wide variety of clinical concepts covering problems, medications, diagnostics, care plans, financial concerns and more.

2.17.4 The FHIR development process

FHIR is published as a mixed standard with Normative portions, which are kept stable for implementers, and also parts still undergoing Trial Use. HL7 actively monitors implementations in order to continue to improve the specification to be responsive to their needs. Due to the many advantages FHIR offers, FHIR is widely used around the world, and the implementation community is growing quickly.

http://hl7.org/fhir . Follow us on Twitter using #FHIR

2.18 Glossary

URL: https://hl7.org/fhir/R4/help.html

2.18 Glossary

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative

This page will be the FHIR Glossary, but it has not, as yet, been created.

Looking for volunteers to take this on...

2.19 Language Glossary

URL: https://hl7.org/fhir/R4/glossary.html
FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative

This page lists a set of words and acronyms that this specification uses in a particular way that might not be inherently obvious from their english definition. These words often present a challenge when describing/documenting FHIR in a language other than english. This pages contains suggested alternatives for these words in some languages where implementers have asked for such support. To add additional languages to this page, ask on http://chat.fhir.org

English | Portuguese
FHIR = Fast Healthcare Interoperability Resources
Accept | Aceitar
Authentication | Autenticação
Authorization | Autorização
Client | Cliente
Conformance | Conformidade
Connectathon | Connectathon
Content-Type | Tipo de Conteúdo
Create | Criar
Delete | Remover
Documents | Documentos
Extensibility | Extensibilidade
History | Histórico
Implementers | Implementadores
Interface | Interface
Interoperability | Interoperabilidade
JSON | JSON
Messages | Mensagens
Metadata | Metadados
Ontology | Ontologia
Operation | Operação
Profile | Perfil
RESTful API | RESTful API
Read | Ler

Request | Pedido
Resource | Recurso
Resource Instance | Instância do recurso
Resource Type | Tipo de Recurso
Response | Resposta
Search | Pesquisa
Server | Servidor
Specification | Especificação
Standard | Standard
Transaction | Transação
URL | URL
Update | Actualizar
Version | Versão
XHTML | XHTML
XML | XML

2.20 License and Legal Terms

URL: https://hl7.org/fhir/R4/license.html

2.20 License and Legal Terms

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative

2.20.1 Disclaimer and Warning of Use

FHIR Resource definitions developed by HL7 are derived from the considerable collective experience of the HL7 membership and wide community feedback from the development and application of a spectrum of health care interoperability solutions. However, Resource definitions are generalized to support multiple contexts of use. It is the responsibility of the persons or organizations using these Resources to ensure their use is fit for the particular purpose in which they are used, including validation for clinical and operational use.

See also the specific warnings associated with use of the STU.

2.20.2 FHIR License

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2.20.3 Third-party artifacts and terminologies

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International Classification of Diseases (ICD) codes	Consult the World Health Organization (WHO)
Current Procedures Terminology (CPT) code set	American Medical Association (AMA) CPT copyright 2014 American Medical Association. All rights reserved.

2.21 Community and Credits

URL: https://hl7.org/fhir/R4/credits.html

2.21 Community and Credits

Maturity Level: N/A | Standards Status: Informative

2.21.1 Community

The FHIR community meets as part of the wider HL7 community and draws on its extensive human resources, institutional memory, previous standards and corporate support. HL7 itself owns FHIR and makes it freely available and the community relies on HL7-provided infrastructure.

The FHIR community gathers, meets, or communicates using the following infrastructure, provided by HL7 directly, or by the HL7 FHIR Foundation :

• http://chat.fhir.org - IM based FHIR support forum

• the FHIR issue tracker for task management

• http://community.fhir.org - general FHIR support forum

• the FHIR email list for general FHIR standards business

• HL7 Confluence : for recording informal documentation in support of the community

• and GitHub for source control

In addition, the community holds regular face to face connectathons and meetings as part of the HL7 Working Group meetings . The formal governance arrangements that manage FHIR development are documented on HL7's Confluence site .

2.21.2 Credits

FHIR is a specification produced by the HL7 Community. Many individuals and organizations (1000s) contribute to the FHIR specification through many roles:

• Balloters / QA reviewers

• Testers / Implementers who give feedback or help others

• Editors / Committee Co-chairs / Minute keepers

• Secretariat: Process support / ANSI compliance

• Regulators / Advocates (e.g. Social Media) / Question answerers

Of particular note as contributers:

• Contributors and Editors: James Agnew (Smile; formerly University Health Network) David Booth (DBooth) Yishen Chen (University of Illinois) Chris Courville (Epic) Bo Dagnall (HP) Richard Ettema (Aegis) Julie Evans (Samvit Solutions) Rick Geimer (Lantana) Grahame Grieve (Health Intersections) Eric Haas (Health eData) Rob Hausam (Hausam Consulting) David Hay (Orion Health) Simone Heckmann (Gefyra) Alexander Henket (NICTIZ) Tony Julian (Mayo Clinic) Paul Knapp (Knapp Consulting) Ewout Kramer (Furore) Brett Marquard (River Rock Associates) Lloyd McKenzie (Gevity Consulting) Josh Mandel (Microsoft; formerly Verily) John Moehrke (By Light) Michelle M Miller (Cerner) Craig Newman (Northrop Grumman) Viet Nguyen (Leidos) Vadim Peretokin (Firely) Melva Peters (Jenaker Consulting; formerly Gevity Consulting) Brian Postlethwaite (Telstra Health) Kevin Power (Cerner) Bryn Rhodes (Database Consulting Group) Chris Schuler (Database Consulting Group) Harold Solbrig (Mayo Clinic) Rik Smithies (NProgram) Andy Stechishin (CANA Software & Services Ltd) Cooper Thompson (Epic) Jose Costa Tiexiera (Equip-med) Jason Walonoski (Mitre) Patrick Werner (MOLIT Institut)

• In addition, the editors and contributors thank the following individuals for key past contributions: Lorraine Constable, Jean-Henri Duteau, Hugh Glover, Patrick Loyd, Joginder Madra, Claude Nanjo, Chris Nickerson

• The Management group is Hans Buitendijk, Grahame Grieve, David Hay, Paul Knapp, Josh Mandel, Lloyd McKenzie, John Moehrke, Brian Pech, Brian Postlethwaite and Wayne Kubick. (Former members: Woody Beeler, Lorraine Constable, Jean Duteau, Hugh Glover, Ron Parker and John Quinn)

• The FHIR specification includes content developed by or reviewed by the following HL7 Work Groups : Application Implementation and Design , Biomedical Research and Regulation , Community Based Collaborative Care , Clinical Decision Support , Clinical Genomics , Clinical Quality Information , Health Care Devices , Electronic Health Records , FHIR Infrastructure , Financial Management , Health Standards Integration , Imaging Integration , Infrastructure And Messaging , Implementable Technology Specifications , Modeling and Methodology , Orders and Observations , Patient Administration , Patient Care , Public Health and Emergency Response , Pharmacy , Structured Documents , Security , US Realm Taskforce , Vocabulary

• In addition, some resources were developed in coordination with the following organizations: DICOM , IHE , LOINC , SNOMED Intl ,

• Innumerable others have assisted through participation in formal QA, submission of ballot comments, discussions on the old Implementer's Skype chat, http://chat.fhir.org, the FHIR list server, participation at HL7 Working Group Meetings and through submission of change requests

• The basic HTML design done by studiojoyo.com - thanks

• THANK YOU ALL! - without the support and effort of this community, FHIR would not exist.

2.22 Outstanding Issues

URL: https://hl7.org/fhir/R4/todo.html

2.22 Outstanding Issues

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative

This specification is currently in its third round of trial use. While some parts of the specification are mature and stable, and HL7 is focusing on moving these to a formal standard, much work remains to be done in other areas. The following general areas of functionality have been deferred to a future version, or are incompletely covered in this version:

• An alarm resource to represent current issues with the patient (e.g. device created)

• Concern Tracking

• Aggregated Data Reporting including Public Health Reporting

• One or more resources for Advance Care Directive / Power of Attorney

• A full server side query framework

For some of these, some draft content is included in the specification for implementer consideration.

In addition, there are a number of specific notes in the specification requesting feedback from implementers:

• AllergyIntolerance: new codes needed for certainty?

• AllergyIntolerance: How should "No Known Allergies" be represented?

• Appointment: Values for Appointment.priority: how interoperable are they

• ClinicalImpression: General Questions about use

• Clinical Reasoning: Multiple questions about implementation and usage

• Composition: Is title different to type? and open questions about document signatures

• Condition: How should "No Known Problems" be represented?

• DeviceRequest: Should this be combined with SupplyRequest?

• Exchange: Should support for ProtocolBuffers be added to the specification?

• Observation: What is the best way to group Observations?

- Patient: Should linking/merging affect the RESTful API?
- Profiling Resources: Need feedback on using system profiles
- References between Resources: Do we need to allow contained resources that reference the container?
- Safety: Request for comments about further development
- Security: Feedback about signatures on RESTful interfaces sought
- Subscription: messaging details still to be resolved
- SupplyRequest: Should this be combined with DeviceRequest?
- Persistent Store/Database: Is a standard extension required for resolved links?
- Task: Question about pre-requisites
- Workflow: several implementation questions

2.23 Appendix: The Role of Informatics in the Shift from Reactive to Proactive Healthcare

URL: https://hl7.org/fhir/R4/change.html

2.23 Appendix: The Role of Informatics in the Shift from Reactive to Proactive Healthcare

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative

FHIR offers more than a standard solving existing problems in interoperability, it provides a platform for the future. Interoperability is "all about the people"; in order to get past the peak of inflated expectations to the plateau of productivity on the hype cycle it is imperative that there is a shared vision of what is trying to be achieved. Healthcare is in the midst of a transformation driven by the convergence of biological and information revolutions, by economic imperative and social change. FHIR's strength lies in it being grounded in the real world which is changing rapidly. This appendix offers a high-level view of the disruption underway and helps place FHIR into a wider context.

Just as John Snow used a branch of informatics and geospatial analysis to identify the source of cholera in London and so prove the germ theory and revolutionize medicine in 1854, so too will the application of informatics and associated technology be instrumental in the next big change in healthcare.

In countries with advanced economies, health systems are facing increased demand from aging populations and increased incidence of chronic disease. In many of these countries, there is also a shortage of skilled workers. Mounting evidence points to avoidable errors causing serious harm to patients. Indeed, optimal care only occurs about half the time in even the best performing health systems (1). Small incremental improvements will not be enough to address the looming crisis in sustainable healthcare.

Convergence of trends in health care, biology, informatics and technology together with the associated social changes and economic imperative is driving a paradigm shift (2) that may be the answer. Informatics has a role in most aspects of this. The figure below provides a summary of the convergence of these trends and the likely results of disruption that will move us from reactive to proactive healthcare.

In terms of our understanding of physiological pathways, informatics is now the major tool of molecular biology. For example, the time spent computing exceeds chemical analysis for gene sequencing by 3 to 1. Informatics is being used to map neural networks and to build the models of systems biology (3) with ever increasing levels of precision and complexity that can't be done without the help of machines. Our understanding has changed so much in recent times that the American Academy of Science is now arguing that it is time for a new taxonomy (4).

Personalization is occurring both because of social change and increased biological knowledge and is being facilitated by cheap mobile computing, sensors and devices. These social forces and the enabling technologies are allowing greater participation by 'non-experts' in decision making, treatment, discovery and knowledge management.

Greater knowledge about how we think (5), advances in the information sciences and the increased availability of computing power means that our capacity to acquire knowledge and use it to predict the course of pathology has increased enormously, and that is fortuitous, because the explosion of information is impossible to deal with otherwise. These new capabilities can be used to provide better advice and to better prevent disease through discovery, monitoring and treatment.

The health system itself can also benefit from what looks like a second phase of utilization of information technology through on-line care provision, real integrated measurement of quality and integration of knowledge in work-flow. With openness and transparency there is also the prospect of co-opetition (cooperative competition) and with integrated outcome measures that pay-for-success contracting can be used as an economic lever.

Large scale change in the way healthcare is done is both essential and inevitable. It is likely that this will derive from the merging of the knowledge and machines of the biological and information revolutions

facilitating a shift from reactive treatment to proactive personalized medicine. Only by such significant changes could the needed quantum improvement in the effectiveness and efficiency of healthcare be made.

The development of digital technology has disrupted other sectors, notably media, retail and manufacturing, and the health sector is unlikely to remain immune (6). Digitization of biology and health will allow machines to help, lead to a demystification of disease, the democratization of healthcare, and a move from the treatment of disease to the promotion and maintenance of wellness.

References

• CareTrack: assessing the appropriateness of health care delivery in Australia. Runciman WB, Hunt TD, Hannaford NA, Hibbert PD, Westbrook JI, Coiera EW, Day RO, Hindmarsh DM, McGlynn EA, Braithwaite J. 2, 16 July 2012, Med J Aust. 2012 Jul 16;197(2):100-5. PubMed PMID: 227, Vol. 197(2), pp. 100-5

• Kuhn, Thomas S. The Structure of Scientific Revolutions. Chicago : University of Chicago Press, 1962

• Hood, Leroy. Institute of Systems Biology. [Online] [Cited: 25 May 2013.] https://www.systemsbiology.org/leroy-hood

• Board on Life Sciences, Committee on A Framework for Developing a New Taxonomy of Disease. Toward Precision Medicine; Building a Knowledge Network for Biomedical Research and a New Taxonomy of Disease. Washington, DC : The National Academic Press, 2011

• Kahneman, Daniel. Thinking, Fast and Slow. New York : Farrar, Straus and Giroux, 2011

• Topol, Eric. The Creative Destruction of Medicine - How the Digital Revolution Will Create Better Health Care. New York : Basic Books, 2012.

This page maintained by Michael Legg

2.24 Data Types

URL: https://hl7.org/fhir/R4/datatypes.html
FHIR Infrastructure Work Group | Maturity Level: Normative | Standards Status: Partially Normative

| This page has been approved as part of an ANSI standard. See the Infrastructure Package for further details.

The FHIR specification defines a set of data types that are used for the resource elements. There are four categories of data types:

• Simple / primitive types, which are single elements with a primitive value (below)

• General-purpose complex types, which are re-usable clusters of elements (below)

• Metadata types: A set of types for use with metadata resources

• Special purpose data types - defined elsewhere in the specification for specific usages

This page describes the general-purpose data types (categories 1 and 2).

Data Types Summary.

Legend: see Standards Status Colors

Primitive Types

General-Purpose Data types

Metadata Types

Special Purpose Data types

A limited set of these data types may appear in extensions. All data types (including primitives) may have extensions, but only the following data types may include Modifier Extensions:

• Timing

• Dosage

• ElementDefinition

2.24.0.1 Primitive Types

The following table describes the primitive types that are used in this specification. Primitive types are those with only a value, and no additional elements as children (though, like all types, they have extensions). See also the Examples.

Primitive Types
FHIR Name | Value Domain | XML Representation | JSON representation
boolean | true | false | xs:boolean, except that 0 and 1 are not valid values | JSON boolean (true or false)
| Regex: true|false
integer | A signed integer in the range −2,147,483,648..2,147,483,647 (32-bit; for larger values, use decimal) | xs:int, except that leading 0 digits are not allowed | JSON number (with no decimal point)
| Regex: [0]|[-+]?[1-9][0-9]*
string | A sequence of Unicode characters | xs:string | JSON String
| Note that strings SHALL NOT exceed 1MB (1024*1024 characters) in size. Strings SHOULD not contain Unicode character points below 32, except for u0009 (horizontal tab), u0010 (carriage return) and u0013 (line feed). Leading and Trailing whitespace is allowed, but SHOULD be removed when using the XML format. Note: This means that a string that consists only of whitespace

could be trimmed to nothing, which would be treated as an invalid element value. Therefore strings SHOULD always contain non-whitespace content
| This data type can be bound to a ValueSet
| Regex: [\r\n\t\S]+ (see notes below)
decimal | Rational numbers that have a decimal representation. See below about the precision of the number | union of xs:decimal and xs:double (see below for limitations) | A JSON number (see below for limitations)
| Regex: -?(0|[1-9][0-9]*)(\.[0-9]+)?([eE][+-]?[0-9]+)?
uri | A Uniform Resource Identifier Reference (RFC 3986). Note: URIs are case sensitive. For UUID (urn:uuid:53fefa32-fcbb-4ff8-8a92-55ee120877b7) use all lowercase | xs:anyURI | A JSON string - a URI
| Regex: \S* (This regex is very permissive, but URIs must be valid. Implementers are welcome to use more specific regex statements for a URI in specific contexts)
| URIs can be absolute or relative, and may have an optional fragment identifier This data type can be bound to a ValueSet
url | A Uniform Resource Locator (RFC 1738). Note URLs are accessed directly using the specified protocol. Common URL protocols are http{s}:, ftp:, mailto: and mllp:, though many others are defined | xs:anyURI | A JSON string - a URL
canonical | A URI that refers to a resource by its canonical URL (resources with a url property). The canonical type differs from a uri in that it has special meaning in this specification, and in that it may have a version appended, separated by a vertical bar (|). Note that the type canonical is not used for the actual canonical URLs that are the target of these references, but for the URIs that refer to them, and may have the version suffix in them. Like other URIs, elements of type canonical may also have #fragment references | xs:anyURI | A JSON string - a canonical URL
base64Binary | A stream of bytes, base64 encoded (RFC 4648) | xs:base64Binary | A JSON string - base64 content
| Regex: (\s*([0-9a-zA-Z\+\=]){4}\s*)+
| There is no specified upper limit to the size of a binary, but systems will have to impose some implementation based limit to the size they support. This should be clearly documented, though there is no computable for this at this time
instant | An instant in time in the format YYYY-MM-DDThh:mm:ss.sss+zz:zz (e.g. 2015-02-07T13:28:17.239+02:00 or 2017-01-01T00:00:00Z). The time SHALL specified at least to the second and SHALL include a time zone. Note: This is intended for when precisely observed times are required (typically system logs etc.), and not human-reported times - for those, use date or dateTime (which can be as precise as instant, but is not required to be). instant is a more constrained dateTime | xs:dateTime | A JSON string - an xs:dateTime
| Note: This type is for system times, not human times (see date and dateTime below).
| Regex: ([0-9]([0-9]([0-9][1-9]|[1-9]0)|[1-9]00)|[1-9]000)-(0[1-9]|1[0-2])-(0[1-9]|[1-2][0-9]|3[0-1])T([01][0-9]|2[0-3]):[0-5][0-9]:([0-5][0-9]|60)(\.[0-9]+)?(Z|(\+|-)((0[0-9]|1[0-3]):[0-5][0-9]|14:00))
date | A date, or partial date (e.g. just year or year + month) as used in human communication. The format is YYYY, YYYY-MM, or YYYY-MM-DD, e.g. 2018, 1973-06, or 1905-08-23. There SHALL be no time zone. Dates SHALL be valid dates | union of xs:date, xs:gYearMonth, xs:gYear | A JSON string - a union of xs:date, xs:gYearMonth, xs:gYear
| Regex: ([0-9]([0-9]([0-9][1-9]|[1-9]0)|[1-9]00)|[1-9]000)(-(0[1-9]|1[0-2])(-(0[1-9]|[1-2][0-9]|3[0-1]))?)?
dateTime | A date, date-time or partial date (e.g. just year or year + month) as used in human communication. The format is YYYY, YYYY-MM, YYYY-MM-DD or YYYY-MM-DDThh:mm:ss+zz:zz, e.g. 2018, 1973-06, 1905-08-23, 2015-02-07T13:28:17-05:00 or 2017-01-01T00:00:00.000Z. If hours and minutes are specified, a time zone SHALL be populated. Seconds must be provided due to schema type constraints but may be zero-filled and may be ignored at receiver discretion. Dates SHALL be valid dates. The time "24:00" is not allowed. Leap Seconds are allowed - see below | union of xs:dateTime, xs:date, xs:gYearMonth, xs:gYear | A JSON string - a union of xs:dateTime, xs:date, xs:gYearMonth, xs:gYear
| Regex: ([0-9]([0-9]([0-9][1-9]|[1-9]0)|[1-9]00)|[1-9]000)(-(0[1-9]|1[0-2])(-(0[1-9]|[1-2][0-9]|3[0-1])(T([01][0-9]|2[0-3]):[0-5][0-9]:([0-5][0-9]|60)(\.[0-9]+)?(Z|(\+|-)((0[0-9]|1[0-3]):[0-5][0-9]|14:00)))?)?)?
time | A time during the day, in the format hh:mm:ss. There is no date specified. Seconds must be provided due to schema type constraints but may be zero-filled and may be ignored at receiver discretion. The time "24:00" SHALL NOT be used. A time zone SHALL NOT be present. Times can be converted to a Duration since midnight. | xs:time | A JSON string - an xs:time
| Regex: ([01][0-9]|2[0-3]):[0-5][0-9]:([0-5][0-9]|60)(\.[0-9]+)?
code | Indicates that the value is taken from a set of controlled strings defined elsewhere (see Using codes for further discussion). Technically, a code is restricted to a string which has at least one character and no leading or trailing whitespace, and where there is no whitespace other than single spaces in the contents | xs:token | JSON string
| Regex: [^\s]+(\s[^\s]+)* This data type can be bound to a ValueSet
oid | An OID represented as a URI (RFC 3001); e.g. urn:oid:1.2.3.4.5 | xs:anyURI | JSON string - uri
| Regex: urn:oid:[0-2](\.(0|[1-9][0-9]*))+
id | Any combination of upper- or lower-case ASCII letters ('A'..'Z', and 'a'..'z', numerals ('0'..'9'), '-' and '.', with a length limit of 64 characters. (This might be an integer, an un-prefixed OID, UUID or any other identifier pattern that meets these constraints.) | xs:string | JSON string
| Regex: [A-Za-z0-9\-\.]{1,64}
markdown | A FHIR string (see above) that may contain markdown syntax for optional processing by a markdown presentation engine, in the GFM extension of CommonMark format (see below) | xs:string | JSON string
| Regex: \s*(\S|\s)* (can't put size limit in the regex - too large)
unsignedInt | Any non-negative integer in the range 0..2,147,483,647 | xs:nonNegativeInteger | JSON number
| Regex: [0]|([1-9][0-9]*)
positiveInt | Any positive integer in the range 1..2,147,483,647 | xs:positiveInteger | JSON number
| Regex: +?[1-9][0-9]*
uuid | A UUID (aka GUID) represented as a URI (RFC 4122); e.g. urn:uuid:c757873d-ec9a-4326-a141-556f43239520 | xs:anyURI | JSON string - uri

Notes:

• For all the types, the XML, JSON and Turtle representations of the primitive values are the same except for different escaping in XML and JSON

• For decimal values, the XML special values INF, -INF and NaN are not allowed, and JSON is restricted to the precision limits documented in XML schema for xs:double and xs:decimal

• The precision of the decimal value has significance: e.g. 0.010 is regarded as different to 0.01, and the original precision should be preserved Implementations SHALL handle decimal values in ways that preserve

and respect the precision of the value as represented for presentation purposes Implementations are not required to perform calculations with these numbers differently, though they may choose to do so (i.e. preserve significance) See implementation comments for XML, JSON and RDF In object code, implementations that might meet this constraint are GMP implementations or equivalents to Java BigDecimal that implement arbitrary precision, or a combination of a (64 bit) floating point value with a precision field Note that large and/or highly precise values are extremely rare in medicine. One element where highly precise decimals may be encountered is the Location coordinates. Irrespective of this, the limits documented in XML Schema apply

• Boolean values can also be represented using coded values (such as HL7 v2 Table 0136). See Observation for one such use

• Issues with the specified regexes: The regexes are provided to assist with tooling, but are informative, not normative. There are several issues with the regexes The string regex has problems with unicode - specifically, it might or might not allow unicode whitespace to some degree depending on unicode support in the regex engine being used. The regexes [\r\n\t\x{0020}-\x{FFFF}]* or [\r\n\t\u0020-\uFFFF]* are better expressions of the constraints on string, but poorly supported (see Regex Tutorial for details). The string regex also applies to markdown as well. The regex does not enforce the length limit The unicode issues also apply to the regex for code The regexes should be qualified with start of string and end of string anchors based on the regex implementation used (e.g. caret '^' and dollar-sign '$' for JavaScript, POSIX, XML and XPath; '\A' and '\Z' for .NET, Java, Python and others; please verify these definitions with the regex implementation used). The regexes may allow a broader set of values than are actually valid (e.g. leap years) so additional validation is always needed

• Leap second are allowed in the datetime, instant and time types. Note, though, that many systems and libraries do not support leap seconds. Applications reading times SHOULD accept and handle leap seconds gracefully, and applications producing them MAY choose to avoid encoding leap seconds

• About the id datatype: Ids are case sensitive. UUIDs SHALL be sent using lowercase letters The ID type includes identifiers consistent with ISO 18232 , but also includes other identifier formats as well, and is not case insensitive like ISO 18232. In a typical FHIR URL, like http://example.com/fhir/Patient/1234, the last part "1234" (highlighted in red) is the part that is an id datatype A full UUID is a uri, not an id. UUIDs in URIs SHALL also be represented in lowercase (urn:uuid:59bf0ef4-e89c-4628-9b51-12ae3fdbe22b)

• About the uri, url and canonical datatypes: They all contain URIs, but differ in how applications resolve the reference Although the url and canonical are specializations of uri, they are never substituted for each other They are all case sensitive for comparison purposes. Applications SHOULD not create URIs that only differ by case A general URI may be either a URL or a canonical URL or some other kind of URI

• About the markdown datatype: This specification requires and uses the GFM (Github Flavored Markdown) extensions on CommonMark format Note that GFM prohibits Raw HTML Systems are not required to have markdown support, so the content of a string should be readable without markdown processing, per markdown philosophy Markdown content SHALL NOT contain Unicode character points below 32, except for u0009 (horizontal tab), u0010 (carriage return) and u0013 (line feed) Markdown is a string, and subject to the same rules (e.g. length limit) Converting an element that has the type string to markdown in a later version of this FHIR specification is not considered a breaking change (neither is adding markdown as a choice to an optional element that already has a choice of data types)

2.24.0.1.1 Representations in XML, JSON, and Turtle

All elements using these primitive types may have one or more of a value as described above, an internal identity (e.g. xml:id), and extensions. For an example, take an element of name "count" and type "integer".

XML

The value is represented in XML as an attribute named "value":

```
<count value="2"/>
```

The full representation, with id, extensions and value:

```
<count id="a1" value="2">
  <extension url="...">
    <valueXX.../>
  </extension>
</count>
```

JSON

In JSON, for convenience, the value is represented as the property itself:

```
"count" : 2
```

The full representation, with id, extensions and value, showing the id and extensions in the sibling property:

```
"count" : 2
"_count" : {
  "id" : "a1",
  "extension" : [{
    "url" : "...",
    "valueXXX" : "...."
  }]
}
```

RDF

The value is represented in RDF as a relationship with the URI "http://h;7.org/fhir/value". Using the normal prefix, this becomes:

```
fhir:Type.count [ fhir:value "2"^^xsd:integer ]
```

For the types date and DateTime, the type must be specified explicitly. For all other types, it is optional. The full representation, with id, extensions and value:

```
fhir:Type.count [
  Element.id "a1";
  fhir:value "2"^^xsd:integer;
  Element.extension [
    fhir:Extension.url "..";
    fhir:Extension.valueXX...
  ]
]
```

For additional details, see the XML, JSON and Turtle format definitions. When the value is missing, and there are no extensions, the element is not represented at all. This means that in xml, attributes are never present with a length of 0 (value=""), and properties are never a 0 length string or null in JSON ("name" : "" is not valid). (note: there is one specific use of the null in the JSON representation).

According to XML schema, leading and trailing whitespace in the value attribute is ignored for the types boolean, integer, decimal, base64Binary, instant, uri, date, dateTime, oid, and uri. Note that this means that the schema aware XML libraries give different attribute values to non-schema aware libraries when reading the XML instances. For this reason, the value attribute for these types SHOULD not have leading and trailing spaces. String values should only have leading and trailing spaces if they are part of the content of the value. In JSON and Turtle whitespace in string values is always significant. Primitive types other than string SHALL NOT have leading or trailing whitespace.

2.24.0.2 Complex Types

In XML, these types are represented as XML Elements with child elements with the name of the defined elements of the type. The name of the element is defined where the type is used. In JSON, the data type is represented by an object with properties named the same as the XML elements. Since the JSON representation is almost exactly the same, only the first example has an additional explicit JSON representation.

Complex data types may be "profiled". A Structure Definition or type "constraint" makes a set of rules about which elements SHALL have values and what the possible values are.

UML Diagrams of the Data types

2.24.0.3 Attachment

See also Examples, Detailed Descriptions, Mappings, Profiles & Extensions and R2 Conversions.

This type is for containing or referencing attachments - additional data content defined in other formats. The most common use of this type is to include images or reports in some report format such as PDF. However, it can be used for any data that has a MIME type.

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name	Flags	Card.	Type	Description & Constraints
Attachment	IN		Element	Content in a format defined elsewhere + Rule: If the Attachment has data, it SHALL have a contentType Elements defined in Ancestors: id, extension
contentType	Σ	0..1	code	Mime type of the content, with charset etc. MimeType (Required)
language	Σ	0..1	code	Human language of the content (BCP-47) Common Languages (Preferred but limited to AllLanguages)
data		0..1	base64Binary	Data inline, base64ed
url	Σ	0..1	url	Uri where the data can be found
size	Σ	0..1	unsignedInt	Number of bytes of content (if url provided)
hash	Σ	0..1	base64Binary	Hash of the data (sha-1, base64ed)
title	Σ	0..1	string	Label to display in place of the data
creation	Σ	0..1	dateTime	Date attachment was first created
Documentation for this format				

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <contentType value="[code]"/><!-- 0..1 Mime type of the content, with charset etc. -->
 <language value="[code]"/><!-- 0..1 Human language of the content (BCP-47) -->
 <data value="[base64Binary]"/><!-- 0..1 Data inline, base64ed -->
 <url value="[url]"/><!-- 0..1 Uri where the data can be found -->
 <size value="[unsignedInt]"/><!-- 0..1 Number of bytes of content (if url provided) -->
 <hash value="[base64Binary]"/><!-- 0..1 Hash of the data (sha-1, base64ed) -->
 <title value="[string]"/><!-- 0..1 Label to display in place of the data -->
 <creation value="[dateTime]"/><!-- 0..1 Date attachment was first created -->
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "contentType" : "<code>", // Mime type of the content, with charset etc.
  "language" : "<code>", // Human language of the content (BCP-47)
  "data" : "<base64Binary>", // Data inline, base64ed
  "url" : "<url>", // Uri where the data can be found
  "size" : "<unsignedInt>", // Number of bytes of content (if url provided)
  "hash" : "<base64Binary>", // Hash of the data (sha-1, base64ed)
  "title" : "<string>", // Label to display in place of the data
  "creation" : "<dateTime>" // Date attachment was first created
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:Attachment.contentType [ code ]; # 0..1 Mime type of the content, with charset etc.
  fhir:Attachment.language [ code ]; # 0..1 Human language of the content (BCP-47)
  fhir:Attachment.data [ base64Binary ]; # 0..1 Data inline, base64ed
  fhir:Attachment.url [ url ]; # 0..1 Uri where the data can be found
  fhir:Attachment.size [ unsignedInt ]; # 0..1 Number of bytes of content (if url provided)
  fhir:Attachment.hash [ base64Binary ]; # 0..1 Hash of the data (sha-1, base64ed)
  fhir:Attachment.title [ string ]; # 0..1 Label to display in place of the data
  fhir:Attachment.creation [ dateTime ]; # 0..1 Date attachment was first created
]
```

Changes since Release 3

Attachment

Attachment.contentType | Change value set from http://hl7.org/fhir/ValueSet/mimetypes to http://hl7.org/fhir/ValueSet/mimetypes|4.0.1

Attachment.language | Change binding strength from extensible to preferred

Attachment.url | Type changed from uri to url

See the Full Difference for further information

Structure

Name | Flags | Card. | Type | Description & Constraints

Attachment | IN | | Element | Content in a format defined elsewhere + Rule: If the Attachment has data, it SHALL have a contentType Elements defined in Ancestors: id, extension

contentType | Σ | 0..1 | code | Mime type of the content, with charset etc. MimeType (Required)

language | Σ | 0..1 | code | Human language of the content (BCP-47) Common Languages (Preferred but limited to AllLanguages)

data | | 0..1 | base64Binary | Data inline, base64ed

url | Σ | 0..1 | url | Uri where the data can be found

size | Σ | 0..1 | unsignedInt | Number of bytes of content (if url provided)

hash | Σ | 0..1 | base64Binary | Hash of the data (sha-1, base64ed)

title | Σ | 0..1 | string | Label to display in place of the data

creation | Σ | 0..1 | dateTime | Date attachment was first created

Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <contentType value="[code]"/><!-- 0..1 Mime type of the content, with charset etc. -->
 <language value="[code]"/><!-- 0..1 Human language of the content (BCP-47) -->
 <data value="[base64Binary]"/><!-- 0..1 Data inline, base64ed -->
 <url value="[url]"/><!-- 0..1 Uri where the data can be found -->
 <size value="[unsignedInt]"/><!-- 0..1 Number of bytes of content (if url provided) -->
 <hash value="[base64Binary]"/><!-- 0..1 Hash of the data (sha-1, base64ed) -->
 <title value="[string]"/><!-- 0..1 Label to display in place of the data -->
 <creation value="[dateTime]"/><!-- 0..1 Date attachment was first created -->
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "contentType" : "<code>", // Mime type of the content, with charset etc.
  "language" : "<code>", // Human language of the content (BCP-47)
  "data" : "<base64Binary>", // Data inline, base64ed
  "url" : "<url>", // Uri where the data can be found
```

```
  "size" : "<unsignedInt>", // Number of bytes of content (if url provided)
  "hash" : "<base64Binary>", // Hash of the data (sha-1, base64ed)
  "title" : "<string>", // Label to display in place of the data
  "creation" : "<dateTime>" // Date attachment was first created
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:Attachment.contentType [ code ]; # 0..1 Mime type of the content, with charset etc.
  fhir:Attachment.language [ code ]; # 0..1 Human language of the content (BCP-47)
  fhir:Attachment.data [ base64Binary ]; # 0..1 Data inline, base64ed
  fhir:Attachment.url [ url ]; # 0..1 Uri where the data can be found
  fhir:Attachment.size [ unsignedInt ]; # 0..1 Number of bytes of content (if url provided)
  fhir:Attachment.hash [ base64Binary ]; # 0..1 Hash of the data (sha-1, base64ed)
  fhir:Attachment.title [ string ]; # 0..1 Label to display in place of the data
  fhir:Attachment.creation [ dateTime ]; # 0..1 Date attachment was first created
]
```

Changes since Release 3

Attachment

Attachment.contentType | Change value set from http://hl7.org/fhir/ValueSet/mimetypes to http://hl7.org/fhir/ValueSet/mimetypes|4.0.1

Attachment.language | Change binding strength from extensible to preferred

Attachment.url | Type changed from uri to url

See the Full Difference for further information

The actual content of an Attachment can be conveyed directly using the data element or a URL reference can be provided. If both are provided, the reference SHALL point to the same content as found in the data. The reference can never be reused to point to some different data (i.e. the reference is version specific). The URL reference SHALL point to a location that resolves to actual data; some URIs such as cid: meet this requirement. If the URL is a relative reference, it is interpreted in the same way as a resource reference.

The contentType element SHALL always be populated when an Attachment contains data, and MAY be populated when there is a url. It can include charset information and other mime type extensions as appropriate. If there is no character set in the contentType then the correct course of action is undefined, though some media types may define a default character set and/or the correct character set may be able to be determined by inspection of the content.

The hash is included so that applications can verify that the content returned by the URL has not changed. The hash and size relate to the data before it is represented in base64 form. The hash is not intended to support digital signatures. Where protection against malicious threats a digital signature should be considered, see Provenance.signature for mechanism to protect a resource with a digital signature.

Attachment data are not constrained, and therefore can be of any content type and encoding. Therefore extra care needs to be taken to validate the content against malicious or malformed content. For more details see Security of Narrative.

In many cases where Attachment is used, the cardinality is >1. A valid use of repeats is to convey the same content in different mime types and languages. Guidance on the meaning of repeating elements SHALL be provided in the definition of the repeating resource element or extension that references this type. The language element describes the language of the attachment using the codes defined in BCP 47 .

Constraints

id | Level | Location | Description | Expression

att-1 | Rule | (base) | If the Attachment has data, it SHALL have a contentType | data.empty() or contentType.exists()

If neither data nor a URL is provided, the value should be understood as an assertion that no content for the specified mimeType and/or language is available for the combination of language and contentType.

The context of use may frequently make rules about the kind of attachment (and therefore, the kind of mime types) that can be used.

Terminology Bindings

Path | Definition | Type | Reference

Attachment.contentType | The mime type of an attachment. Any valid mime type is allowed. | Required | Mime Types

Attachment.language | A human language. | Preferred, but limited to AllLanguages | CommonLanguages

Attachment is used in the following places: RelatedArtifact, ProdCharacteristic, BodyStructure, Claim, ClaimResponse, Communication, CommunicationRequest, Consent, Contract, DiagnosticReport, DocumentReference, ExplanationOfBenefit, HealthcareService, Library, Media, Patient, Person, Practitioner, Questionnaire, QuestionnaireResponse, RelatedPerson, SubstanceNucleicAcid, SubstancePolymer, SubstanceProtein and SubstanceSpecification

2.24.0.4 Coding

See also Examples, Detailed Descriptions, Mappings, Profiles & Extensions and R2 Conversions.

A Coding is a representation of a defined concept using a symbol from a defined "code system" - see Using Codes in resources for more details.

This data type can be bound to a ValueSet.

• Structure
• UML
• XML
• JSON
• Turtle
• R3 Diff
• All

Structure

Name | Flags | Card. | Type | Description & Constraints
Coding | ΣN | | Element | A reference to a code defined by a terminology system Elements defined in Ancestors: id, extension
system | Σ | 0..1 | uri | Identity of the terminology system
version | Σ | 0..1 | string | Version of the system - if relevant
code | Σ | 0..1 | code | Symbol in syntax defined by the system
display | Σ | 0..1 | string | Representation defined by the system
userSelected | Σ | 0..1 | boolean | If this coding was chosen directly by the user
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <system value="[uri]"/><!-- 0..1 Identity of the terminology system -->
 <version value="[string]"/><!-- 0..1 Version of the system - if relevant -->
 <code value="[code]"/><!-- 0..1 Symbol in syntax defined by the system -->
 <display value="[string]"/><!-- 0..1 Representation defined by the system -->
 <userSelected value="[boolean]"/><!-- 0..1 If this coding was chosen directly by the user -->
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "system" : "<uri>", // Identity of the terminology system
  "version" : "<string>", // Version of the system - if relevant
  "code" : "<code>", // Symbol in syntax defined by the system
  "display" : "<string>", // Representation defined by the system
  "userSelected" : <boolean> // If this coding was chosen directly by the user
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:Coding.system [ uri ]; # 0..1 Identity of the terminology system
  fhir:Coding.version [ string ]; # 0..1 Version of the system - if relevant
  fhir:Coding.code [ code ]; # 0..1 Symbol in syntax defined by the system
  fhir:Coding.display [ string ]; # 0..1 Representation defined by the system
  fhir:Coding.userSelected [ boolean ]; # 0..1 If this coding was chosen directly by the user
]
```

Changes since Release 3

Coding | No Changes

See the Full Difference for further information

Structure

Name | Flags | Card. | Type | Description & Constraints
Coding | ΣN | | Element | A reference to a code defined by a terminology system Elements defined in Ancestors: id, extension
system | Σ | 0..1 | uri | Identity of the terminology system
version | Σ | 0..1 | string | Version of the system - if relevant
code | Σ | 0..1 | code | Symbol in syntax defined by the system
display | Σ | 0..1 | string | Representation defined by the system
userSelected | Σ | 0..1 | boolean | If this coding was chosen directly by the user
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <system value="[uri]"/><!-- 0..1 Identity of the terminology system -->
 <version value="[string]"/><!-- 0..1 Version of the system - if relevant -->
 <code value="[code]"/><!-- 0..1 Symbol in syntax defined by the system -->
 <display value="[string]"/><!-- 0..1 Representation defined by the system -->
```

```
 <userSelected value="[boolean]"/><!-- 0..1 If this coding was chosen directly by the user -->
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "system" : "<uri>", // Identity of the terminology system
  "version" : "<string>", // Version of the system - if relevant
  "code" : "<code>", // Symbol in syntax defined by the system
  "display" : "<string>", // Representation defined by the system
  "userSelected" : <boolean> // If this coding was chosen directly by the user
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:Coding.system [ uri ]; # 0..1 Identity of the terminology system
  fhir:Coding.version [ string ]; # 0..1 Version of the system - if relevant
  fhir:Coding.code [ code ]; # 0..1 Symbol in syntax defined by the system
  fhir:Coding.display [ string ]; # 0..1 Representation defined by the system
  fhir:Coding.userSelected [ boolean ]; # 0..1 If this coding was chosen directly by the user
]
```

Changes since Release 3

Coding | No Changes

See the Full Difference for further information

The meaning of the Coding is defined by the code. The system provides the source of the definition of the code, along with an optional version reference. The display is a human display for the text defined by the system - it is not intended for computation.

The system is a URI that identifies the code system that defines the code. Choosing the correct system is important; for more information about the code system URI, read Managing Terminology System URIs. If the code is taken from a CodeSystem resource, CodeSystem.url is the correct value for the system element. Resolvable URLs are generally preferred by implementers over non-resolvable URNs, particularly opaque URNs such as OIDs (urn:oid:) or UUIDs (urn:uuid:). The system URI SHALL NOT contain a reference to a value set (e.g. ValueSet.url), since value sets just define the set of codes which are intended for use in a specific context, not the meaning of the codes themselves.

A code system version may also be supplied. If the meaning of codes within the code system is consistent across releases, this is not required. The version SHOULD be exchanged when the system does not maintain consistent definitions across versions. Note that the following systems SHOULD always have a version specified:

• National releases of SNOMED CT (consistency of definitions varies amongst jurisdictions, and some jurisdictions may make their own rules on this)

• Various versions of ICD (note: the major releases are labeled as different code systems altogether, but there is variation within versions)

More generally, any classification (e.g. a code system that includes concepts with relative definitions such as "not otherwise coded" will require a version. See the discussion of code system versions in the Code System resource for further discussion on versioning.

If present, the code SHALL be a syntactically correct symbol as defined by the system. In some code systems such as SNOMED CT, the symbol may be an expression composed of other predefined symbol (e.g. post-coordination). Note that codes are case sensitive unless specified otherwise by the code system. The display is a text representation of the code defined by the system and is used to display the meaning of the code by an application that is not aware of the system.

If the 'display' element is populated, the string used in display SHALL be one of the display strings defined for that code by the code system (code systems may define multiple display strings for a single code). If one of the available display strings is labeled as preferred, it SHOULD be used. If the code system does not define a text representation for display (e.g. SNOMED CT Expressions) then the 'display' element cannot be populated, and the meaning of the code won't be accessible to systems that don't understand the code expression.

In some cases, the system might not be known - only the code is known. In this case, no useful processing of the code may be performed unless the system can be safely inferred by the context. This practice should be avoided where possible, as information sharing in a wider context is very likely to arise eventually, and codes cannot be used in the absence of a known system.

If the system is present, and there is no code, then this is understood to mean that there is no suitable code in the system in which to represent the code.

If two codings have the same system, version and code then they have the same meaning. If the version information is missing, or the system, version or the code elements differ, then how the codes are related can only be determined by consulting the definitions of the system(s) and any mappings available.

A coding may be marked as a "userSelected" if a user selected the particular coded value in a user interface (e.g. the user selects an item in a pick-list). If a user selected coding exists, it is the preferred choice for performing translations etc.

Constraints

The context of use (as defined in the resource or applicable profile) usually makes rules about what codes and systems are allowed or required in a particular context by binding the element to a value set.

Coding is used in the following places: Meta, DataRequirement, UsageContext, Signature, CodeableConcept, AuditEvent, CapabilityStatement, CodeSystem, Consent, Contract, DocumentReference, Encounter, Endpoint, ExplanationOfBenefit, ImagingStudy, Location, MedicinalProduct, MessageDefinition, MessageHeader, Questionnaire, QuestionnaireResponse, StructureDefinition, TestScript and ValueSet

Implementation Note: This specification defines two types for representing coded values:

• Coding: a simple direct reference to a code defined by a code system

• CodeableConcept: a text description and/or a list of Codings (i.e. a list of references to codes defined by code systems)

The Coding data type corresponds to the simple case of selecting a single code from a code list. However, this type is rarely used in the FHIR specifications; long experience with exchanging coded values in HL7 shows that in the general case, systems need to able to exchange multiple translation codes, and/or an original text.

The Coding data type is used directly when there is certainty that the value must be selected directly from one of the available codes, and the list of possible codes is agreed to by all participants. This is not usually the case in the context of FHIR - general interoperability - so Coding is mostly used in extensions, which are usually intended to be defined for a well-controlled context of use.

2.24.0.5 CodeableConcept

See also Examples, Detailed Descriptions, Mappings, Profiles & Extensions and R2 Conversions.

A CodeableConcept represents a value that is usually supplied by providing a reference to one or more terminologies or ontologies but may also be defined by the provision of text. This is a common pattern in healthcare data.

This data type can be bound to a ValueSet.

• Structure

• UML

• XML

• JSON

• Turtle

• R3 Diff

• All

Structure

Name | Flags | Card. | Type | Description & Constraints
CodeableConcept | ΣN | | Element | Concept - reference to a terminology or just text Elements defined in Ancestors: id, extension
coding | Σ | 0..* | Coding | Code defined by a terminology system
text | Σ | 0..1 | string | Plain text representation of the concept
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <coding><!-- 0..* Coding Code defined by a terminology system --></coding>
 <text value="[string]"/><!-- 0..1 Plain text representation of the concept -->
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "coding" : [{ Coding }], // Code defined by a terminology system
  "text" : "<string>" // Plain text representation of the concept
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:CodeableConcept.coding [ Coding ], ... ; # 0..* Code defined by a terminology system
  fhir:CodeableConcept.text [ string ]; # 0..1 Plain text representation of the concept
]
```

Changes since Release 3

CodeableConcept | No Changes

See the Full Difference for further information

Structure

Name | Flags | Card. | Type | Description & Constraints
CodeableConcept | ΣN | | Element | Concept - reference to a terminology or just text Elements defined in Ancestors: id, extension
coding | Σ | 0..* | Coding | Code defined by a terminology system
text | Σ | 0..1 | string | Plain text representation of the concept
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <coding><!-- 0..* Coding Code defined by a terminology system --></coding>
 <text value="[string]"/><!-- 0..1 Plain text representation of the concept -->
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "coding" : [{ Coding }], // Code defined by a terminology system
  "text" : "<string>" // Plain text representation of the concept
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:CodeableConcept.coding [ Coding ], ... ; # 0..* Code defined by a terminology system
  fhir:CodeableConcept.text [ string ]; # 0..1 Plain text representation of the concept
]
```

Changes since Release 3

CodeableConcept | No Changes

See the Full Difference for further information

Additional Codes

More than one code may be used in CodeableConcept. The concept may be coded multiple times in different code systems (or even multiple times in the same code systems, where multiple forms are possible, such as with SNOMED CT). Each coding (also referred to as a 'translation') is a representation of the concept as described above and may have slightly different granularity due to the differences in the definitions of the underlying codes. There is no meaning associated with the ordering of coding within a CodeableConcept. A typical use of CodeableConcept is to send the local code that the concept was coded with, and also one or more translations to publicly defined code systems such as LOINC or SNOMED CT. Sending local codes is useful and important for the purposes of debugging and integrity auditing.

For example, many qualitative laboratory test results values are typically represented with coded presence/absence concepts. Using the coded value for 'negative' with a standard SNOMED CT code translation, valueCodeableConcept would be:

```
  "valueCodeableConcept": {
    "coding": [
      {
        "system": "http://snomed.info/sct",
        "code": "260385009",
        "display": "Negative"
      }, {
        "system": "https://acme.lab/resultcodes",
        "code": "NEG",
        "display": "Negative"
      }
    ],
    "text": "Negative for Chlamydia Trachomatis rRNA"
  }
```

Note that these concepts may be cross mapped using the ConceptMap resource instead of or in addition to being represented as translations directly in the in CodeableConcept.

Using Text in CodeableConcept

The text is the representation of the concept as entered or chosen by the user, and which most closely represents the intended meaning of the user or concept. Very often the text is the same as a display of one of the codings. One or more of the codings may be flagged as the user selected code - the code or concept that the user actually selected directly. Note that in all but a few cases, only one of the codings may be flagged as the coding.userSelected = true - the code or concept that the user actually selected directly. If more than one code is marked as user selected, this means the user explicitly chose multiple codes. When none of the coding elements is marked as user selected, the text (if present) is the preferred source of meaning.

A free text only representation of the concept without any coding elements is permitted if there is no appropriate code and only free text is available (and not prohibited by the implementation). For example, using text only, the Observation.valueCodeableConcept element would be:

```
  "valueCodeableConcept": {
    "text": "uncoded free text result"
  }
```

Constraints

The context of use usually makes rules about what codes and systems are allowed or required in a particular context by binding the element to a value set.

CodeableConcept is used in the following places: DataRequirement, Dosage, MarketingStatus, Identifier, SubstanceAmount, Population, ProductShelfLife, UsageContext, Timing, ProdCharacteristic, Account, ActivityDefinition, AdverseEvent, AllergyIntolerance, Appointment, AppointmentResponse, AuditEvent, Basic, BiologicallyDerivedProduct, BodyStructure, CapabilityStatement, CarePlan, CareTeam, CatalogEntry, ChargeItem, ChargeItemDefinition, Claim, ClaimResponse, ClinicalImpression, CodeSystem, Communication, CommunicationRequest, Composition, ConceptMap, Condition, Consent, Contract, Coverage, CoverageEligibilityRequest, CoverageEligibilityResponse, DetectedIssue, Device, DeviceDefinition, DeviceMetric, DeviceRequest, DeviceUseStatement, DiagnosticReport, DocumentManifest, DocumentReference, EffectEvidenceSynthesis, Encounter, Endpoint, EpisodeOfCare, EventDefinition, Evidence, EvidenceVariable, ExampleScenario, ExplanationOfBenefit, FamilyMemberHistory, Flag, Goal, GraphDefinition, Group, GuidanceResponse, HealthcareService, ImagingStudy, Immunization, ImmunizationEvaluation, ImmunizationRecommendation, ImplementationGuide, InsurancePlan, Invoice, Library, List, Location, Measure, MeasureReport, Media, Medication, MedicationAdministration, MedicationDispense, MedicationKnowledge, MedicationRequest, MedicationStatement, MedicinalProduct, MedicinalProductAuthorization, MedicinalProductContraindication, MedicinalProductIndication, MedicinalProductIngredient, MedicinalProductInteraction, MedicinalProductManufactured, MedicinalProductPackaged, MedicinalProductPharmaceutical, MedicinalProductUndesirableEffect, MessageDefinition, MessageHeader, MolecularSequence, NamingSystem, NutritionOrder, Observation, ObservationDefinition, OperationDefinition, OperationOutcome, Organization, OrganizationAffiliation, Patient, PaymentNotice, PaymentReconciliation, PlanDefinition, Practitioner, PractitionerRole, Procedure, Provenance, Questionnaire, RelatedPerson, RequestGroup, ResearchDefinition, ResearchElementDefinition, ResearchStudy, RiskAssessment, RiskEvidenceSynthesis, Schedule, SearchParameter, ServiceRequest, Slot, Specimen, SpecimenDefinition, StructureDefinition, StructureMap, Substance, SubstanceNucleicAcid, SubstancePolymer, SubstanceProtein, SubstanceReferenceInformation, SubstanceSourceMaterial, SubstanceSpecification, SupplyDelivery, SupplyRequest, Task, TerminologyCapabilities, TestScript, ValueSet, VerificationResult and VisionPrescription

2.24.0.6 Quantity

See also Examples, Detailed Descriptions and Mappings, Profiles & Extensions and R2 Conversions.

A measured amount (or an amount that can potentially be measured).

This data type can be bound to a ValueSet.

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name	Flags	Card.	Type	Description & Constraints
Quantity	ΣIN		Element	A measured or measurable amount + Rule: If a code for the unit is present, the system SHALL also be present Elements defined in Ancestors: id, extension
value	Σ	0..1	decimal	Numerical value (with implicit precision)
comparator	?!Σ	0..1	code	< | <= | >= | > - how to understand the value QuantityComparator (Required)
unit	Σ	0..1	string	Unit representation
system	ΣI	0..1	uri	System that defines coded unit form
code	Σ	0..1	code	Coded form of the unit
Documentation for this format				

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <value value="[decimal]"/><!-- 0..1 Numerical value (with implicit precision) -->
 <comparator value="[code]"/><!-- 0..1 < | <= | >= | > - how to understand the value -->
 <unit value="[string]"/><!-- 0..1 Unit representation -->
```

```
 <system value="[uri]"/><!--  0..1 System that defines coded unit form -->
 <code value="[code]"/><!-- 0..1 Coded form of the unit -->
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "value" : <decimal>, // Numerical value (with implicit precision)
  "comparator" : "<code>", // < | <= | >= | > - how to understand the value
  "unit" : "<string>", // Unit representation
  "system" : "<uri>", // C? System that defines coded unit form
  "code" : "<code>" // Coded form of the unit
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:Quantity.value [ decimal ]; # 0..1 Numerical value (with implicit precision)
  fhir:Quantity.comparator [ code ]; # 0..1 < | <= | >= | > - how to understand the value
  fhir:Quantity.unit [ string ]; # 0..1 Unit representation
  fhir:Quantity.system [ uri ]; # 0..1 System that defines coded unit form
  fhir:Quantity.code [ code ]; # 0..1 Coded form of the unit
]
```

Changes since Release 3

Quantity

Quantity.comparator | Change value set from http://hl7.org/fhir/ValueSet/quantity-comparator to http://hl7.org/fhir/ValueSet/quantity-comparator|4.0.1

See the Full Difference for further information

Structure

Name | Flags | Card. | Type | Description & Constraints

Quantity | ΣIN | | Element | A measured or measurable amount + Rule: If a code for the unit is present, the system SHALL also be present Elements defined in Ancestors: id, extension

value | Σ | 0..1 | decimal | Numerical value (with implicit precision)

comparator | ?!Σ | 0..1 | code | < | <= | >= | > - how to understand the value QuantityComparator (Required)

unit | Σ | 0..1 | string | Unit representation

system | ΣI | 0..1 | uri | System that defines coded unit form

code | Σ | 0..1 | code | Coded form of the unit

Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <value value="[decimal]"/><!-- 0..1 Numerical value (with implicit precision) -->
 <comparator value="[code]"/><!-- 0..1 < | <= | >= | > - how to understand the value -->
 <unit value="[string]"/><!-- 0..1 Unit representation -->
 <system value="[uri]"/><!--  0..1 System that defines coded unit form -->
 <code value="[code]"/><!-- 0..1 Coded form of the unit -->
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "value" : <decimal>, // Numerical value (with implicit precision)
  "comparator" : "<code>", // < | <= | >= | > - how to understand the value
  "unit" : "<string>", // Unit representation
  "system" : "<uri>", // C? System that defines coded unit form
  "code" : "<code>" // Coded form of the unit
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:Quantity.value [ decimal ]; # 0..1 Numerical value (with implicit precision)
  fhir:Quantity.comparator [ code ]; # 0..1 < | <= | >= | > - how to understand the value
  fhir:Quantity.unit [ string ]; # 0..1 Unit representation
  fhir:Quantity.system [ uri ]; # 0..1 System that defines coded unit form
  fhir:Quantity.code [ code ]; # 0..1 Coded form of the unit
]
```

Changes since Release 3

Quantity

Quantity.comparator | Change value set from http://hl7.org/fhir/ValueSet/quantity-comparator to http://hl7.org/fhir/ValueSet/quantity-comparator|4.0.1

See the Full Difference for further information

The value contains the numerical value of the quantity, including an implicit precision. If no comparator is specified, the value is a point value (i.e. '='). The comparator element can never be ignored.

The unit element contains a displayable unit that defines what is measured. The unit may additionally be coded in some formal way using the code and the system (see Coding for further information about how to use the system element).

If the unit can be coded in UCUM and a code is provided, it SHOULD be a UCUM code. If a UCUM unit is provided in the code, then a canonical value can be generated for purposes of comparison between quantities. Note that the unit element will often contain text that is a valid UCUM unit, but it cannot be assumed that the unit actually contains a valid UCUM unit.

Constraints

id | Level | Location | Description | Expression
qty-3 | Rule | (base) | If a code for the unit is present, the system SHALL also be present | code.empty() or system.exists()

The context of use may frequently define what kind of measured quantity this is and therefore what kind of unit can be used. The context of use may additionally require a code from a particular system, or a value set - see Using Terminologies for information about binding a Quantity to a value set to constrain the unit codes. The context of use may also restrict the values for the value or comparator.

Terminology Bindings

Path | Definition | Type | Reference
Quantity.comparator | How the Quantity should be understood and represented. | Required | QuantityComparator

Quantity is used in the following places: Count, SubstanceAmount, Ratio, Distance, Age, Duration, ProductShelfLife, UsageContext, ProdCharacteristic, ChargeItem, Claim, Contract, Device, DeviceDefinition, DeviceRequest, ExplanationOfBenefit, Goal, Group, InsurancePlan, MeasureReport, MedicinalProductIndication, MedicinalProductManufactured, MedicinalProductPackaged, MedicinalProductPharmaceutical, MolecularSequence, Observation, PlanDefinition, Questionnaire, QuestionnaireResponse, ServiceRequest, SubstanceReferenceInformation, SubstanceSpecification and SupplyRequest

2.24.0.6.1 Defined Variations on Quantity

There are several additional data types that are specializations of Quantity that only introduce new restrictions on the existing elements defined as part of the Quantity data type.

The types Age, Distance and Count are marked as Trial Use because they are not used in this specification (though they may be used in extensions). These types may be converted back to a profile (see R2 definitions).

Type Name | Rules | Formal Definitions
Distance | id Level Location Description Expression dis-1 Rule (base) There SHALL be a code if there is a value and it SHALL be an expression of length. If system is present, it SHALL be UCUM. (code.exists() or value.empty()) and (system.empty() or system = %ucum) Terminology Bindings Path Definition Type Reference Distance Appropriate units for Distance. Extensible, but limited to AllUCUMExpressionForDistance CommonUCUMCodesForDistance | XML, JSON
id | Level | Location | Description | Expression
dis-1 | Rule | (base) | There SHALL be a code if there is a value and it SHALL be an expression of length. If system is present, it SHALL be UCUM. | (code.exists() or value.empty()) and (system.empty() or system = %ucum)
Path | Definition | Type | Reference
Distance | Appropriate units for Distance. | Extensible, but limited to AllUCUMExpressionForDistance | CommonUCUMCodesForDistance
| Usage: (not used as yet)
| Implementation Note: If the duration value is specified as a whole number (e.g. 1 month), then when the duration is added or subtracted to a given date(time), the outcome should be rounded to the nearest natural calendar division - e.g. Feb. 1 + 1 mo = March 1, not March 2 or 3 (since 1 month in is defined in UCUM as 30 days).
Age | id Level Location Description Expression age-1 Rule (base) There SHALL be a code if there is a value and it SHALL be an expression of time. If system is present, it SHALL be UCUM. If value is present, it SHALL be positive. (code.exists() or value.empty()) and (system.empty() or system = %ucum) and (value.empty() or value.hasValue().not() or value > 0) Terminology Bindings Path Definition Type Reference Age Appropriate units for Age. Extensible, but limited to AllUCUMExpressionForTime CommonUCUMCodesForAge | XML, JSON
id | Level | Location | Description | Expression
age-1 | Rule | (base) | There SHALL be a code if there is a value and it SHALL be an expression of time. If system is present, it SHALL be UCUM. If value is present, it SHALL be positive. | (code.exists() or value.empty()) and (system.empty() or system = %ucum) and (value.empty() or value.hasValue().not() or value > 0)
Path | Definition | Type | Reference
Age | Appropriate units for Age. | Extensible, but limited to AllUCUMExpressionForTime | CommonUCUMCodesForAge
| Usage: ActivityDefinition, AllergyIntolerance, Condition, FamilyMemberHistory, PlanDefinition, Procedure and RequestGroup
Count | id Level Location Description Expression cnt-3 Rule (base) There SHALL be a code with a value of "1" if there is a value. If system is present, it SHALL be UCUM. If present, the value SHALL be a whole number. (code.exists() or value.empty()) and (system.empty() or system = %ucum) and (code.empty() or code = '1') and (value.empty() or value.hasValue().not() or value.toString().contains('.').not()) | XML, JSON
id | Level | Location | Description | Expression
cnt-3 | Rule | (base) | There SHALL be a code with a value of "1" if there is a value. If system is present, it SHALL be UCUM. If present, the value SHALL be a whole number. | (code.exists() or value.empty()) and (system.empty() or system = %ucum) and (code.empty() or code = '1') and (value.empty() or value.hasValue().not() or value.toString().contains('.').not())

| Usage: (not used as yet)
Duration | id Level Location Description Expression drt-1 Rule (base) There SHALL be a code if there is a value and it SHALL be an expression of time. If system is present, it SHALL be UCUM. code.exists() implies ((system = %ucum) and value.exists())
Terminology Bindings Path Definition Type Reference Duration Appropriate units for Duration. Extensible, but limited to AllUCUMExpressionForTime CommonUCUMCodesForDuration | XML, JSON
id | Level | Location | Description | Expression
drt-1 | Rule | (base) | There SHALL be a code if there is a value and it SHALL be an expression of time. If system is present, it SHALL be UCUM. | code.exists() implies ((system = %ucum) and value.exists())
Path | Definition | Type | Reference
Duration | Appropriate units for Duration. | Extensible, but limited to AllUCUMExpressionForTime | CommonUCUMCodesForDuration
| Usage: DataRequirement, Timing, ActivityDefinition, Encounter, EvidenceVariable, Goal, MedicationKnowledge, MedicationRequest, MedicinalProductPharmaceutical, PlanDefinition, RequestGroup, ResearchElementDefinition, Specimen and SpecimenDefinition
In addition to the specializations, there is one constraint on Quantity used in several resources:
Profile Name | Rules | Formal Definitions
Simple Quantity | id Level Location Description Expression sqty-1 Rule (base) The comparator is not used on a SimpleQuantity comparator.empty()) | XML, JSON
id | Level | Location | Description | Expression
sqty-1 | Rule | (base) | The comparator is not used on a SimpleQuantity | comparator.empty())
| Usage: Dosage, SampledData, Range, ActivityDefinition, CarePlan, Claim, ClaimResponse, Contract, Coverage, CoverageEligibilityRequest, ExplanationOfBenefit, Immunization, MedicationAdministration, MedicationDispense, MedicationKnowledge, MedicationRequest, NutritionOrder, Observation, Specimen, SpecimenDefinition, Substance, SupplyDelivery and VisionPrescription

Note that the constraint is different from the other specializations of Quantity because it is not a type, just rules applied where the Quantity type is used. There's another constraint - see Money immediately below.

2.24.0.7 Money

See also Examples, Detailed Descriptions and Mappings, Profiles & Extensions and R2 Conversions.

An amount of currency.

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
Money | ΣN | | Element | An amount of economic utility in some recognized currency Elements defined in Ancestors: id, extension
value | Σ | 0..1 | decimal | Numerical value (with implicit precision)
currency | Σ | 0..1 | code | ISO 4217 Currency Code CurrencyCode (Required)
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <value value="[decimal]"/><!-- 0..1 Numerical value (with implicit precision) -->
 <currency value="[code]"/><!-- 0..1 ISO 4217 Currency Code -->
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "value" : <decimal>, // Numerical value (with implicit precision)
  "currency" : "<code>" // ISO 4217 Currency Code
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:Money.value [ decimal ]; # 0..1 Numerical value (with implicit precision)
  fhir:Money.currency [ code ]; # 0..1 ISO 4217 Currency Code
]
```

Changes since Release 3

Money
Money.value | Added Element
Money.currency | Added Element

See the Full Difference for further information

Structure

Name | Flags | Card. | Type | Description & Constraints
Money | ΣN | | Element | An amount of economic utility in some recognized currency Elements defined in Ancestors: id, extension
value | Σ | 0..1 | decimal | Numerical value (with implicit precision)
currency | Σ | 0..1 | code | ISO 4217 Currency Code CurrencyCode (Required)
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <value value="[decimal]"/><!-- 0..1 Numerical value (with implicit precision) -->
 <currency value="[code]"/><!-- 0..1 ISO 4217 Currency Code -->
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "value" : <decimal>, // Numerical value (with implicit precision)
  "currency" : "<code>" // ISO 4217 Currency Code
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:Money.value [ decimal ]; # 0..1 Numerical value (with implicit precision)
  fhir:Money.currency [ code ]; # 0..1 ISO 4217 Currency Code
]
```

Changes since Release 3

Money
Money.value | Added Element
Money.currency | Added Element

See the Full Difference for further information

The value contains the amount of the currency, including an implicit precision. Precision is always important for financial amounts. The currency element contains an ISO 4217 code for the currency.

Money is used in the following places: ChargeItem, ChargeItemDefinition, Claim, ClaimResponse, Contract, Coverage, CoverageEligibilityRequest, CoverageEligibilityResponse, ExplanationOfBenefit, InsurancePlan, Invoice, MedicationKnowledge, PaymentNotice and PaymentReconciliation

2.24.0.7.1 Alternate Representation

There are also circumstances where a financial amount must be represented as the numerator or denominator in a Ratio, where the type is currency. In this context, the Money amount is represented as a Quantity, using the MoneyQuantity constraint:

Profile Name | Rules | Formal Definitions
Money Quantity | id Level Location Description Expression mqty-1 Rule (base) There SHALL be a code if there is a value and it SHALL be an expression of currency. If system is present, it SHALL be ISO 4217 (system = "urn:iso:std:iso:4217" - currency). (code.exists() or value.empty()) and (system.empty() or system = 'urn:iso:std:iso:4217')) | XML, JSON
id | Level | Location | Description | Expression
mqty-1 | Rule | (base) | There SHALL be a code if there is a value and it SHALL be an expression of currency. If system is present, it SHALL be ISO 4217 (system = "urn:iso:std:iso:4217" - currency). | (code.exists() or value.empty()) and (system.empty() or system = 'urn:iso:std:iso:4217'))
| Usage: (not used as yet)

Note that the profile is different from the other specializations because it is not a type, just rules applied where the Quantity type is used to represent Money amounts.

2.24.0.8 Range

See also Examples, Detailed Descriptions, Mappings, Profiles & Extensions and R2 Conversions.

A set of ordered Quantity values defined by a low and high limit.

A Range specifies a set of possible values; usually, one value from the range applies (e.g. "give the patient between 2 and 4 tablets"). Ranges are typically used in instructions.

- Structure
- UML
- XML
- JSON
- Turtle

- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
Range | ΣIN | | Element | Set of values bounded by low and high + Rule: If present, low SHALL have a lower value than high
Elements defined in Ancestors: id, extension
low | Σ | 0..1 | SimpleQuantity | Low limit
high | Σ | 0..1 | SimpleQuantity | High limit
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <low><!-- 0..1 Quantity(SimpleQuantity) Low limit --></low>
 <high><!-- 0..1 Quantity(SimpleQuantity) High limit --></high>
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "low" : { Quantity(SimpleQuantity) }, // Low limit
  "high" : { Quantity(SimpleQuantity) } // High limit
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:Range.low [ Quantity(SimpleQuantity) ]; # 0..1 Low limit
  fhir:Range.high [ Quantity(SimpleQuantity) ]; # 0..1 High limit
]
```

Changes since Release 3

Range | No Changes

See the Full Difference for further information

Structure

Name | Flags | Card. | Type | Description & Constraints
Range | ΣIN | | Element | Set of values bounded by low and high + Rule: If present, low SHALL have a lower value than high
Elements defined in Ancestors: id, extension
low | Σ | 0..1 | SimpleQuantity | Low limit
high | Σ | 0..1 | SimpleQuantity | High limit
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <low><!-- 0..1 Quantity(SimpleQuantity) Low limit --></low>
 <high><!-- 0..1 Quantity(SimpleQuantity) High limit --></high>
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "low" : { Quantity(SimpleQuantity) }, // Low limit
  "high" : { Quantity(SimpleQuantity) } // High limit
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:Range.low [ Quantity(SimpleQuantity) ]; # 0..1 Low limit
  fhir:Range.high [ Quantity(SimpleQuantity) ]; # 0..1 High limit
]
```

Changes since Release 3

Range | No Changes

See the Full Difference for further information

The unit and code/system elements of the low or high elements SHALL match. If the low or high elements are missing, the meaning is that the low or high boundaries are not known and therefore neither is the complete range.

The comparator flag on the low or high elements cannot be present. Note that the Range type should not be used to represent out of range measurements: A quantity type with the comparator element should be used instead.

The low and the high values are inclusive and are assumed to have arbitrarily high precision; e.g. the range 1.5 to 2.5 includes 1.50, and 2.50 but not 1.49 or 2.51.

Constraints

id | Level | Location | Description | Expression
rng-2 | Rule | (base) | If present, low SHALL have a lower value than high | low.empty() or high.empty() or (low <= high)

Range is used in the following places: Dosage, SubstanceAmount, Population, UsageContext, Timing, ActivityDefinition, AllergyIntolerance, Condition, DeviceRequest, FamilyMemberHistory, Goal, Group, Observation, ObservationDefinition, PlanDefinition, Procedure, RequestGroup, RiskAssessment, ServiceRequest, SpecimenDefinition, SubstanceReferenceInformation, SubstanceSpecification and SupplyRequest

2.24.0.9 Ratio

See also Examples, Detailed Descriptions, Mappings, Profiles & Extensions and R2 Conversions.

A relationship between two Quantity values expressed as a numerator and a denominator.

The Ratio datatype should only be used to express a relationship of two numbers if the relationship cannot be suitably expressed using a Quantity and a common unit. Where the denominator value is known to be fixed to "1", Quantity should be used instead of Ratio.

• Structure
• UML
• XML
• JSON
• Turtle
• R3 Diff
• All

Structure

Name | Flags | Card. | Type | Description & Constraints
Ratio | ΣIN | | Element | A ratio of two Quantity values - a numerator and a denominator + Rule: Numerator and denominator SHALL both be present, or both are absent. If both are absent, there SHALL be some extension present Elements defined in Ancestors: id, extension
numerator | Σ | 0..1 | Quantity | Numerator value
denominator | Σ | 0..1 | Quantity | Denominator value
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <numerator><!-- 0..1 Quantity Numerator value --></numerator>
 <denominator><!-- 0..1 Quantity Denominator value --></denominator>
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "numerator" : { Quantity }, // Numerator value
  "denominator" : { Quantity } // Denominator value
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:Ratio.numerator [ Quantity ]; # 0..1 Numerator value
  fhir:Ratio.denominator [ Quantity ]; # 0..1 Denominator value
]
```

Changes since Release 3

Ratio | No Changes

See the Full Difference for further information

Structure

Name | Flags | Card. | Type | Description & Constraints
Ratio | ΣIN | | Element | A ratio of two Quantity values - a numerator and a denominator + Rule: Numerator and denominator SHALL both be present, or both are absent. If both are absent, there SHALL be some extension present Elements defined in Ancestors: id, extension

numerator | Σ | 0..1 | Quantity | Numerator value
denominator | Σ | 0..1 | Quantity | Denominator value
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <numerator><!-- 0..1 Quantity Numerator value --></numerator>
 <denominator><!-- 0..1 Quantity Denominator value --></denominator>
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "numerator" : { Quantity }, // Numerator value
  "denominator" : { Quantity } // Denominator value
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:Ratio.numerator [ Quantity ]; # 0..1 Numerator value
  fhir:Ratio.denominator [ Quantity ]; # 0..1 Denominator value
]
```

Changes since Release 3

Ratio | No Changes

See the Full Difference for further information

Examples where a Quantity is typically used are rates, densities, concentrations. Examples where a Ratio is used are: titers (e.g. 1:128); concentration ratios where the denominator is significant (e.g. 5mg/10mL); observed frequencies (e.g. 2 repetitions/8 hr), and where the numerator or denominator is an amount of a currency (no UCUM code for $ etc.).

Common factors in the numerator and denominator are not automatically cancelled out. Ratios are not simply "structured numbers" - for example, blood pressure measurements (e.g. "120/60") are not ratios.

A proper ratio has both a numerator and a denominator; however, these are not mandatory in order to allow an invalid ratio with an extension with further information.

Constraints

id | Level | Location | Description | Expression
rat-1 | Rule | (base) | Numerator and denominator SHALL both be present, or both are absent. If both are absent, there SHALL be some extension present | (numerator.empty() xor denominator.exists()) and (numerator.exists() or extension.exists())

The context of use may require particular types of Quantity for the numerator or denominator.

Ratio is used in the following places: Dosage, Goal, Medication, MedicationAdministration, MedicationKnowledge, MedicinalProductIngredient, MedicinalProductPharmaceutical, NutritionOrder, Observation, ServiceRequest, Substance and SubstanceSpecification

2.24.0.10 Period

See also Examples, Detailed Descriptions, Mappings, Profiles & Extensions and R2 Conversions.

A time period defined by a start and end date/time.

A period specifies a range of times. The context of use will specify whether the entire range applies (e.g. "the patient was an inpatient of the hospital for this time range") or one value from the period applies (e.g. "give to the patient between 2 and 4 pm on 24-Jun 2013").

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
Period | ΣIN | | Element | Time range defined by start and end date/time + Rule: If present, start SHALL have a lower value than end Elements defined in Ancestors: id, extension
start | ΣI | 0..1 | dateTime | Starting time with inclusive boundary
end | ΣI | 0..1 | dateTime | End time with inclusive boundary, if not ongoing
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <start value="[dateTime]"/><!--  0..1 Starting time with inclusive boundary -->
 <end value="[dateTime]"/><!--  0..1 End time with inclusive boundary, if not ongoing -->
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "start" : "<dateTime>", // C? Starting time with inclusive boundary
  "end" : "<dateTime>" // C? End time with inclusive boundary, if not ongoing
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:Period.start [ dateTime ]; # 0..1 Starting time with inclusive boundary
  fhir:Period.end [ dateTime ]; # 0..1 End time with inclusive boundary, if not ongoing
]
```

Changes since Release 3

Period | No Changes

See the Full Difference for further information

Structure

Name | Flags | Card. | Type | Description & Constraints

Period | ΣIN | | Element | Time range defined by start and end date/time + Rule: If present, start SHALL have a lower value than end Elements defined in Ancestors: id, extension

start | ΣI | 0..1 | dateTime | Starting time with inclusive boundary

end | ΣI | 0..1 | dateTime | End time with inclusive boundary, if not ongoing

Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <start value="[dateTime]"/><!--  0..1 Starting time with inclusive boundary -->
 <end value="[dateTime]"/><!--  0..1 End time with inclusive boundary, if not ongoing -->
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "start" : "<dateTime>", // C? Starting time with inclusive boundary
  "end" : "<dateTime>" // C? End time with inclusive boundary, if not ongoing
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:Period.start [ dateTime ]; # 0..1 Starting time with inclusive boundary
  fhir:Period.end [ dateTime ]; # 0..1 End time with inclusive boundary, if not ongoing
]
```

Changes since Release 3

Period | No Changes

See the Full Difference for further information

If the start element is missing, the start of the period is not known. If the end element is missing, it means that the period is ongoing, or the start may be in the past, and the end date in the future, which means that period is expected/planned to end at the specified time

The end value includes any matching date/time. For example, the period 2011-05-23 to 2011-05-27 includes all the times from the start of the 23rd May through to the end of the 27th of May.

Period is used in the following places: Address, DataRequirement, HumanName, ContactPoint, MarketingStatus, Identifier, Timing, Account, ActivityDefinition, AllergyIntolerance, Appointment, AuditEvent, BiologicallyDerivedProduct, CarePlan, CareTeam, CatalogEntry, ChargeItem, ChargeItemDefinition, Claim, ClaimResponse, ClinicalImpression, CommunicationRequest, Composition, Condition, Consent, Contract, Coverage, CoverageEligibilityRequest, CoverageEligibilityResponse, DetectedIssue, DeviceRequest, DeviceUseStatement, DiagnosticReport, DocumentReference, EffectEvidenceSynthesis, Encounter, Endpoint, EpisodeOfCare, EventDefinition, Evidence, EvidenceVariable, ExplanationOfBenefit, FamilyMemberHistory, Flag, Group, HealthcareService, InsurancePlan, Library, Measure, MeasureReport, Media,

MedicationAdministration, MedicationRequest, MedicationStatement, MedicinalProductAuthorization, NamingSystem, Observation, OrganizationAffiliation, Patient, PaymentReconciliation, PlanDefinition, Practitioner, PractitionerRole, Procedure, Provenance, Questionnaire, RelatedPerson, RequestGroup, ResearchDefinition, ResearchElementDefinition, ResearchStudy, ResearchSubject, RiskAssessment, RiskEvidenceSynthesis, Schedule, ServiceRequest, Specimen, SupplyDelivery, SupplyRequest and Task

2.24.0.11 SampledData

Normative Candidate Note: This DataType is not normative - it is still undergoing Trial Use while more experience is gathered.

See also Examples, Detailed Descriptions, Mappings, Profiles & Extensions and R2 Conversions.

Data that comes from a series of measurements taken by a device, which may have upper and lower limits. The data type also supports more than one dimension in the data.

A SampledData provides a concise way to handle the data produced by devices that sample a particular physical state at a high frequency. A typical use for this is for the output of an ECG or EKG device. The data type includes a series of raw decimal values (which are mostly simple integers), along with adjustments for scale and factor. These are interpreted such that

```
original measured value[i] = SampledData.data[i] * SampledData.scaleFactor + SampledData.origin.value
```

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name	Flags	Card.	Type	Description & Constraints
SampledData	TU		Element	A series of measurements taken by a device Elements defined in Ancestors: id, extension
origin	Σ	1..1	SimpleQuantity	Zero value and units
period	Σ	1..1	decimal	Number of milliseconds between samples
factor	Σ	0..1	decimal	Multiply data by this before adding to origin
lowerLimit	Σ	0..1	decimal	Lower limit of detection
upperLimit	Σ	0..1	decimal	Upper limit of detection
dimensions	Σ	1..1	positiveInt	Number of sample points at each time point
data		0..1	string	Decimal values with spaces, or "E" | "U" | "L"
Documentation for this format				

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <origin><!-- 1..1 Quantity(SimpleQuantity) Zero value and units --></origin>
 <period value="[decimal]"/><!-- 1..1 Number of milliseconds between samples -->
 <factor value="[decimal]"/><!-- 0..1 Multiply data by this before adding to origin -->
 <lowerLimit value="[decimal]"/><!-- 0..1 Lower limit of detection -->
 <upperLimit value="[decimal]"/><!-- 0..1 Upper limit of detection -->
 <dimensions value="[positiveInt]"/><!-- 1..1 Number of sample points at each time point -->
 <data value="[string]"/><!-- 0..1 Decimal values with spaces, or "E" | "U" | "L" -->
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "origin" : { Quantity(SimpleQuantity) }, // R!  Zero value and units
  "period" : <decimal>, // R!  Number of milliseconds between samples
  "factor" : <decimal>, // Multiply data by this before adding to origin
  "lowerLimit" : <decimal>, // Lower limit of detection
  "upperLimit" : <decimal>, // Upper limit of detection
  "dimensions" : "<positiveInt>", // R!  Number of sample points at each time point
  "data" : "<string>" // Decimal values with spaces, or "E" | "U" | "L"
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:SampledData.origin [ Quantity(SimpleQuantity) ]; # 1..1 Zero value and units
  fhir:SampledData.period [ decimal ]; # 1..1 Number of milliseconds between samples
  fhir:SampledData.factor [ decimal ]; # 0..1 Multiply data by this before adding to origin
  fhir:SampledData.lowerLimit [ decimal ]; # 0..1 Lower limit of detection
  fhir:SampledData.upperLimit [ decimal ]; # 0..1 Upper limit of detection
  fhir:SampledData.dimensions [ positiveInt ]; # 1..1 Number of sample points at each time point
```

```
  fhir:SampledData.data [ string ]; # 0..1 Decimal values with spaces, or "E" | "U" | "L"
]
```

Changes since Release 3

SampledData
SampledData.factor | Default Value "1" removed
SampledData.data | Min Cardinality changed from 1 to 0

See the Full Difference for further information

Structure

Name	Flags	Card.	Type	Description & Constraints
SampledData	TU		Element	A series of measurements taken by a device Elements defined in Ancestors: id, extension
origin	Σ	1..1	SimpleQuantity	Zero value and units
period	Σ	1..1	decimal	Number of milliseconds between samples
factor	Σ	0..1	decimal	Multiply data by this before adding to origin
lowerLimit	Σ	0..1	decimal	Lower limit of detection
upperLimit	Σ	0..1	decimal	Upper limit of detection
dimensions	Σ	1..1	positiveInt	Number of sample points at each time point
data		0..1	string	Decimal values with spaces, or "E" | "U" | "L"
Documentation for this format				

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <origin><!-- 1..1 Quantity(SimpleQuantity) Zero value and units --></origin>
 <period value="[decimal]"/><!-- 1..1 Number of milliseconds between samples -->
 <factor value="[decimal]"/><!-- 0..1 Multiply data by this before adding to origin -->
 <lowerLimit value="[decimal]"/><!-- 0..1 Lower limit of detection -->
 <upperLimit value="[decimal]"/><!-- 0..1 Upper limit of detection -->
 <dimensions value="[positiveInt]"/><!-- 1..1 Number of sample points at each time point -->
 <data value="[string]"/><!-- 0..1 Decimal values with spaces, or "E" | "U" | "L" -->
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "origin" : { Quantity(SimpleQuantity) }, // R!  Zero value and units
  "period" : <decimal>, // R!  Number of milliseconds between samples
  "factor" : <decimal>, // Multiply data by this before adding to origin
  "lowerLimit" : <decimal>, // Lower limit of detection
  "upperLimit" : <decimal>, // Upper limit of detection
  "dimensions" : "<positiveInt>", // R!  Number of sample points at each time point
  "data" : "<string>" // Decimal values with spaces, or "E" | "U" | "L"
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:SampledData.origin [ Quantity(SimpleQuantity) ]; # 1..1 Zero value and units
  fhir:SampledData.period [ decimal ]; # 1..1 Number of milliseconds between samples
  fhir:SampledData.factor [ decimal ]; # 0..1 Multiply data by this before adding to origin
  fhir:SampledData.lowerLimit [ decimal ]; # 0..1 Lower limit of detection
  fhir:SampledData.upperLimit [ decimal ]; # 0..1 Upper limit of detection
  fhir:SampledData.dimensions [ positiveInt ]; # 1..1 Number of sample points at each time point
  fhir:SampledData.data [ string ]; # 0..1 Decimal values with spaces, or "E" | "U" | "L"
]
```

Changes since Release 3

SampledData
SampledData.factor | Default Value "1" removed
SampledData.data | Min Cardinality changed from 1 to 0

See the Full Difference for further information

The digits are a set of decimal values separated by a single space (Unicode character u20). In addition to decimal values, the special values "E" (error), "L" (below detection limit) and "U" (above detection limit) can also be used. If there is more than one dimension, the different dimensions are interlaced - all the data points for a particular time are represented together.

SampledData is used in the following places: Observation

2.24.0.12 Identifier

See also Examples, Detailed Descriptions, Mappings, Profiles & Extensions and R2 Conversions.

A numeric or alphanumeric string that is associated with a single object or entity within a given system. Typically, identifiers are used to connect content in resources to external content available in other frameworks or protocols. Identifiers are associated with objects and may be changed or retired due to human or system process and errors.

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
Identifier | ΣN | | Element | An identifier intended for computation Elements defined in Ancestors: id, extension
use | ?!Σ | 0..1 | code | usual | official | temp | secondary | old (If known) IdentifierUse (Required)
type | Σ | 0..1 | CodeableConcept | Description of identifier IdentifierType (Extensible)
system | Σ | 0..1 | uri | The namespace for the identifier value
value | Σ | 0..1 | string | The value that is unique
period | Σ | 0..1 | Period | Time period when id is/was valid for use
assigner | Σ | 0..1 | Reference(Organization) | Organization that issued id (may be just text)
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <use value="[code]"/><!-- 0..1 usual | official | temp | secondary | old (If known) -->
 <type><!-- 0..1 CodeableConcept Description of identifier --></type>
 <system value="[uri]"/><!-- 0..1 The namespace for the identifier value -->
 <value value="[string]"/><!-- 0..1 The value that is unique -->
 <period><!-- 0..1 Period Time period when id is/was valid for use --></period>
 <assigner><!-- 0..1 Reference(Organization) Organization that issued id (may be just text) --></assigner>
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "use" : "<code>", // usual | official | temp | secondary | old (If known)
  "type" : { CodeableConcept }, // Description of identifier
  "system" : "<uri>", // The namespace for the identifier value
  "value" : "<string>", // The value that is unique
  "period" : { Period }, // Time period when id is/was valid for use
  "assigner" : { Reference(Organization) } // Organization that issued id (may be just text)
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:Identifier.use [ code ]; # 0..1 usual | official | temp | secondary | old (If known)
  fhir:Identifier.type [ CodeableConcept ]; # 0..1 Description of identifier
  fhir:Identifier.system [ uri ]; # 0..1 The namespace for the identifier value
  fhir:Identifier.value [ string ]; # 0..1 The value that is unique
  fhir:Identifier.period [ Period ]; # 0..1 Time period when id is/was valid for use
  fhir:Identifier.assigner [ Reference(Organization) ]; # 0..1 Organization that issued id (may be just text)
]
```

Changes since Release 3

Identifier
Identifier.use | Change value set from http://hl7.org/fhir/ValueSet/identifier-use to http://hl7.org/fhir/ValueSet/identifier-use|4.0.1

See the Full Difference for further information

Structure

Name | Flags | Card. | Type | Description & Constraints
Identifier | ΣN | | Element | An identifier intended for computation Elements defined in Ancestors: id, extension
use | ?!Σ | 0..1 | code | usual | official | temp | secondary | old (If known) IdentifierUse (Required)
type | Σ | 0..1 | CodeableConcept | Description of identifier IdentifierType (Extensible)
system | Σ | 0..1 | uri | The namespace for the identifier value
value | Σ | 0..1 | string | The value that is unique
period | Σ | 0..1 | Period | Time period when id is/was valid for use
assigner | Σ | 0..1 | Reference(Organization) | Organization that issued id (may be just text)
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <use value="[code]"/><!-- 0..1 usual | official | temp | secondary | old (If known) -->
 <type><!-- 0..1 CodeableConcept Description of identifier --></type>
```

```
 <system value="[uri]"/><!-- 0..1 The namespace for the identifier value -->
 <value value="[string]"/><!-- 0..1 The value that is unique -->
 <period><!-- 0..1 Period Time period when id is/was valid for use --></period>
 <assigner><!-- 0..1 Reference(Organization) Organization that issued id (may be just text) --></assigner>
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "use" : "<code>", // usual | official | temp | secondary | old (If known)
  "type" : { CodeableConcept }, // Description of identifier
  "system" : "<uri>", // The namespace for the identifier value
  "value" : "<string>", // The value that is unique
  "period" : { Period }, // Time period when id is/was valid for use
  "assigner" : { Reference(Organization) } // Organization that issued id (may be just text)
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:Identifier.use [ code ]; # 0..1 usual | official | temp | secondary | old (If known)
  fhir:Identifier.type [ CodeableConcept ]; # 0..1 Description of identifier
  fhir:Identifier.system [ uri ]; # 0..1 The namespace for the identifier value
  fhir:Identifier.value [ string ]; # 0..1 The value that is unique
  fhir:Identifier.period [ Period ]; # 0..1 Time period when id is/was valid for use
  fhir:Identifier.assigner [ Reference(Organization) ]; # 0..1 Organization that issued id (may be just text)
]
```

Changes since Release 3

Identifier

Identifier.use | Change value set from http://hl7.org/fhir/ValueSet/identifier-use to http://hl7.org/fhir/ValueSet/identifier-use|4.0.1

See the Full Difference for further information

The value SHALL be unique within the defined system and have a consistent meaning wherever it appears. Identifier.system is always case sensitive. Identifier.value is to be treated as case sensitive unless knowledge of the Identifier.system allows the processer to be confident that non-case-sensitive processing is safe.

The system is a URI that defines a set of identifiers (i.e. how the value is made unique). It might be a specific application or a recognized standard/specification for a set of identifiers or a way of making identifiers unique. FHIR defines some useful or important system URIs directly. Here are some example identifier namespaces:

• http://hl7.org/fhir/sid/us-ssn for United States Social Security Number (SSN) values

• http://ns.electronichealth.net.au/id/hi/ihi/1.0 for Australian Individual Healthcare Identifier (IHI) numbers

• urn:ietf:rfc:3986 for when the value of the identifier is itself a globally unique URI

If the system is a URL, it SHOULD resolve. Resolution might be to a web page that describes the identifier system and/or supports look-up of identifiers. Alternatively, it could be to a NamingSystem resource instance. Resolvable URLs are generally preferred by implementers over non-resolvable URNs, particularly opaque URNs such as OIDs (urn:oid:) or UUIDs (urn:uuid:). If used, OIDs and UUIDs may be registered in the HL7 OID registry and SHOULD be registered if the content is shared or exchanged across institutional boundaries.

It is up to the implementer organization to determine an appropriate URL or URN structure that will avoid collisions and to manage that space (and the resolvability of URLs) over time.

Note that the scope of a given identifier system may extend beyond identifiers that might be captured by a single resource. For example, some systems might draw all "order" identifiers from a single namespace, though some might be used on MedicationRequest while others would appear on ServiceRequest.

If the identifier value itself is naturally a globally unique URI (e.g. an OID, a UUID, or a URI with no trailing local part), then the system SHALL be "urn:ietf:rfc:3986", and the URI is in the value (OIDs and UUIDs using urn:oid: and urn:uuid: - see note on the V3 mapping and the examples). Naturally globally unique identifiers are those for which no system has been assigned and where the value of the identifier is reasonably expected to not be re-used. Typically, these are absolute URIs of some kind.

In some cases, the system might not be known - only the value is known (e.g. a simple device that scans a barcode), or the system is known implicitly (simple exchange in a limited context, often driven by barcode readers). In this case, no useful matching may be performed using the value unless the system can be safely inferred by the context. Applications should provide a system wherever possible, as information sharing in a wider context is very likely to arise eventually, and values without a system are inherently limited in use.

In addition to the system (which provides a uniqueness scope) and the value, identifiers may also have a type, which may be useful when a system encounters identifiers with unknown system values. Note, however, that the type of an identifier is not a well-controlled vocabulary with wide variations in practice. The type deals only with general categories of identifiers and SHOULD not be used for codes that correspond 1..1 with the Identifier.system. Some identifiers may fall into multiple categories due to variations in common usage.

The assigner is used to indicate what registry/state/facility/etc. assigned the identifier. As a Reference, the assigner can include just a text description in the display.

Constraints

Terminology Bindings

Path	Definition	Type	Reference
Identifier.use	Identifies the purpose for this identifier, if known .	Required	IdentifierUse
Identifier.type	A coded type for an identifier that can be used to determine which identifier to use for a specific purpose.	Extensible	Identifier Type Codes

Identifier is used in the following places: Reference, ProductShelfLife, Account, ActivityDefinition, AdverseEvent, AllergyIntolerance, Appointment, AppointmentResponse, Basic, BiologicallyDerivedProduct, BodyStructure, Bundle, CarePlan, CareTeam, CatalogEntry, ChargeItem, ChargeItemDefinition, Claim, ClaimResponse, ClinicalImpression, CodeSystem, Communication, CommunicationRequest, Composition, ConceptMap, Condition, Consent, Contract, Coverage, CoverageEligibilityRequest, CoverageEligibilityResponse, DetectedIssue, Device, DeviceDefinition, DeviceMetric, DeviceRequest, DeviceUseStatement, DiagnosticReport, DocumentManifest, DocumentReference, EffectEvidenceSynthesis, Encounter, Endpoint, EnrollmentRequest, EnrollmentResponse, EpisodeOfCare, EventDefinition, Evidence, EvidenceVariable, ExampleScenario, ExplanationOfBenefit, FamilyMemberHistory, Flag, Goal, Group, GuidanceResponse, HealthcareService, ImagingStudy, Immunization, ImmunizationEvaluation, ImmunizationRecommendation, InsurancePlan, Invoice, Library, List, Location, Measure, MeasureReport, Media, Medication, MedicationAdministration, MedicationDispense, MedicationRequest, MedicationStatement, MedicinalProduct, MedicinalProductAuthorization, MedicinalProductIngredient, MedicinalProductPackaged, MedicinalProductPharmaceutical, MessageDefinition, MolecularSequence, NutritionOrder, Observation, ObservationDefinition, Organization, OrganizationAffiliation, Patient, PaymentNotice, PaymentReconciliation, Person, PlanDefinition, Practitioner, PractitionerRole, Procedure, Questionnaire, QuestionnaireResponse, RelatedPerson, RequestGroup, ResearchDefinition, ResearchElementDefinition, ResearchStudy, ResearchSubject, RiskAssessment, RiskEvidenceSynthesis, Schedule, ServiceRequest, Slot, Specimen, SpecimenDefinition, StructureDefinition, StructureMap, Substance, SubstanceNucleicAcid, SubstanceProtein, SubstanceReferenceInformation, SubstanceSourceMaterial, SubstanceSpecification, SupplyDelivery, SupplyRequest, Task, TestReport, TestScript, ValueSet and VisionPrescription

2.24.0.13 HumanName

See also Examples, Detailed Descriptions, Mappings, Profiles & Extensions and R2 Conversions.

A name of a human with text, parts and usage information.

Names may be changed or repudiated. People may have different names in different contexts. Names may be divided into parts of different type that have variable significance depending on context, though the division into parts is not always significant. With personal names, the different parts might or might not be imbued with some implicit meaning; various cultures associate different importance with the name parts and the degree to which systems SHALL care about name parts around the world varies widely.

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name	Flags	Card.	Type	Description & Constraints
HumanName	ΣN		Element	Name of a human - parts and usage Elements defined in Ancestors: id, extension
use	?!Σ	0..1	code	usual | official | temp | nickname | anonymous | old | maiden NameUse (Required)
text	Σ	0..1	string	Text representation of the full name
family	Σ	0..1	string	Family name (often called 'Surname')
given	Σ	0..*	string	Given names (not always 'first'). Includes middle names This repeating element order: Given Names appear in the correct order for presenting the name
prefix	Σ	0..*	string	Parts that come before the name This repeating element order: Prefixes appear in the correct order for presenting the name
suffix	Σ	0..*	string	Parts that come after the name This repeating element order: Suffixes appear in the correct order for presenting the name
period	Σ	0..1	Period	Time period when name was/is in use
Documentation for this format				

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <use value="[code]"/><!-- 0..1 usual | official | temp | nickname | anonymous | old | maiden -->
 <text value="[string]"/><!-- 0..1 Text representation of the full name -->
 <family value="[string]"/><!-- 0..1 Family name (often called 'Surname') -->
```

```
 <given value="[string]"/><!-- 0..* Given names (not always 'first'). Includes middle names -->
 <prefix value="[string]"/><!-- 0..* Parts that come before the name -->
 <suffix value="[string]"/><!-- 0..* Parts that come after the name -->
 <period><!-- 0..1 Period Time period when name was/is in use --></period>
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "use" : "<code>", // usual | official | temp | nickname | anonymous | old | maiden
  "text" : "<string>", // Text representation of the full name
  "family" : "<string>", // Family name (often called 'Surname')
  "given" : ["<string>"], // Given names (not always 'first'). Includes middle names
  "prefix" : ["<string>"], // Parts that come before the name
  "suffix" : ["<string>"], // Parts that come after the name
  "period" : { Period } // Time period when name was/is in use
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:HumanName.use [ code ]; # 0..1 usual | official | temp | nickname | anonymous | old | maiden
  fhir:HumanName.text [ string ]; # 0..1 Text representation of the full name
  fhir:HumanName.family [ string ]; # 0..1 Family name (often called 'Surname')
  fhir:HumanName.given [ string ], ... ; # 0..* Given names (not always 'first'). Includes middle names
  fhir:HumanName.prefix [ string ], ... ; # 0..* Parts that come before the name
  fhir:HumanName.suffix [ string ], ... ; # 0..* Parts that come after the name
  fhir:HumanName.period [ Period ]; # 0..1 Time period when name was/is in use
]
```

Changes since Release 3

HumanName

HumanName.use | Change value set from http://hl7.org/fhir/ValueSet/name-use to http://hl7.org/fhir/ValueSet/name-use|4.0.1

See the Full Difference for further information

Structure

Name | Flags | Card. | Type | Description & Constraints

HumanName | ΣN | | Element | Name of a human - parts and usage Elements defined in Ancestors: id, extension

use | ?!Σ | 0..1 | code | usual | official | temp | nickname | anonymous | old | maiden NameUse (Required)

text | Σ | 0..1 | string | Text representation of the full name

family | Σ | 0..1 | string | Family name (often called 'Surname')

given | Σ | 0..* | string | Given names (not always 'first'). Includes middle names This repeating element order: Given Names appear in the correct order for presenting the name

prefix | Σ | 0..* | string | Parts that come before the name This repeating element order: Prefixes appear in the correct order for presenting the name

suffix | Σ | 0..* | string | Parts that come after the name This repeating element order: Suffixes appear in the correct order for presenting the name

period | Σ | 0..1 | Period | Time period when name was/is in use

Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <use value="[code]"/><!-- 0..1 usual | official | temp | nickname | anonymous | old | maiden -->
 <text value="[string]"/><!-- 0..1 Text representation of the full name -->
 <family value="[string]"/><!-- 0..1 Family name (often called 'Surname') -->
 <given value="[string]"/><!-- 0..* Given names (not always 'first'). Includes middle names -->
 <prefix value="[string]"/><!-- 0..* Parts that come before the name -->
 <suffix value="[string]"/><!-- 0..* Parts that come after the name -->
 <period><!-- 0..1 Period Time period when name was/is in use --></period>
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "use" : "<code>", // usual | official | temp | nickname | anonymous | old | maiden
  "text" : "<string>", // Text representation of the full name
  "family" : "<string>", // Family name (often called 'Surname')
  "given" : ["<string>"], // Given names (not always 'first'). Includes middle names
  "prefix" : ["<string>"], // Parts that come before the name
  "suffix" : ["<string>"], // Parts that come after the name
  "period" : { Period } // Time period when name was/is in use
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:HumanName.use [ code ]; # 0..1 usual | official | temp | nickname | anonymous | old | maiden
  fhir:HumanName.text [ string ]; # 0..1 Text representation of the full name
```

```
  fhir:HumanName.family [ string ]; # 0..1 Family name (often called 'Surname')
  fhir:HumanName.given [ string ], ... ; # 0..* Given names (not always 'first'). Includes middle names
  fhir:HumanName.prefix [ string ], ... ; # 0..* Parts that come before the name
  fhir:HumanName.suffix [ string ], ... ; # 0..* Parts that come after the name
  fhir:HumanName.period [ Period ]; # 0..1 Time period when name was/is in use
]
```

Changes since Release 3

HumanName
HumanName.use | Change value set from http://hl7.org/fhir/ValueSet/name-use to http://hl7.org/fhir/ValueSet/name-use|4.0.1

See the Full Difference for further information

This table summarizes where common parts of a person's name are found.

Name | Example | Destination / Comments
Surname | Smith | Family Name
First name | John | Given Name
Title | Mr. | Prefix
Middle Name | Samuel | Subsequent Given Names
Patronymic | bin Osman | Family Name
Multiple family names | Carreño Quiñones | Family Name. See note below about decomposition of family name
Initials | Q. | Given Name as initial ("." recommended)
Nick Name | Jock | Given name, with Use = common
Qualifications | PhD | Suffix
Honorifics | Senior | Suffix
Voorvoegsel / Nobility | van Beethoven | Family Name. See note below about decomposition of family name

For further information, including all W3C International Examples , consult the examples. Note: Implementers should read the name examples for a full understanding of how name works.

The multiple given parts and family name combine to form a single name. Where a person has alternate names that may be used in place of each other (e.g. Nicknames, Aliases), these are different instances of HumanName.

The text element specifies the entire name as it should be displayed e.g. in an application UI. This may be provided instead of or as well as the specific parts. Applications updating a name SHALL ensure that when both text and parts are present, no content is included in the text that isn't found in a part. The correct order of assembly of the parts is culture dependent: the order of the parts within a given part type has significance and SHALL be observed. The appropriate order between family name and given names depends on culture and context of use. Note that there is an extension for the few times name assembly order is not fixed by the culture.

The given name parts may contain whitespace, though generally they don't. Initials may be used in place of the full name if that is all that is recorded. Systems that operate across cultures should generally rely on the text form for presentation and use the parts for index/search functionality. For this reason, applications SHOULD populate the text element for future robustness.

In some cultures (e.g. German, Dutch, Spanish, Portuguese), family names are complex and composed of various parts that may need to be managed separately, e.g. they have differing significance for searching. In these cases, the full family name is populated in family, and a decomposition of the name can be provided using the family extensions own-name, own-prefix, partner-name, partner-prefix, fathers-family and mothers-family.

For robust search, servers should search the parts of a family name independently. E.g. Searching either Carreno or Quinones should match a family name of "Carreno Quinones". HL7 affiliates, and others producing implementation guides, may make more specific recommendations about how search should work in specific cultures or environments.

Constraints

Terminology Bindings

Path | Definition | Type | Reference
HumanName.use | The use of a human name. | Required | NameUse

HumanName is used in the following places: InsurancePlan, Organization, Patient, Person, Practitioner and RelatedPerson

2.24.0.14 Address

See also Examples, Detailed Descriptions, Mappings, Profiles & Extensions and R2 Conversions.

An address expressed using postal conventions (as opposed to GPS or other location definition formats). This data type may be used to convey addresses for use in delivering mail as well as for visiting locations which might not be valid for mail delivery. There are a variety of postal address formats defined around the world.

- Structure
- UML
- XML

- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
Address | ΣN | | Element | An address expressed using postal conventions (as opposed to GPS or other location definition formats)
Elements defined in Ancestors: id, extension
use | ?!Σ | 0..1 | code | home | work | temp | old | billing - purpose of this address AddressUse (Required)
type | Σ | 0..1 | code | postal | physical | both AddressType (Required)
text | Σ | 0..1 | string | Text representation of the address
line | Σ | 0..* | string | Street name, number, direction & P.O. Box etc. This repeating element order: The order in which lines should appear in an address label
city | Σ | 0..1 | string | Name of city, town etc.
district | Σ | 0..1 | string | District name (aka county)
state | Σ | 0..1 | string | Sub-unit of country (abbreviations ok)
postalCode | Σ | 0..1 | string | Postal code for area
country | Σ | 0..1 | string | Country (e.g. can be ISO 3166 2 or 3 letter code)
period | Σ | 0..1 | Period | Time period when address was/is in use
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <use value="[code]"/><!-- 0..1 home | work | temp | old | billing - purpose of this address -->
 <type value="[code]"/><!-- 0..1 postal | physical | both -->
 <text value="[string]"/><!-- 0..1 Text representation of the address -->
 <line value="[string]"/><!-- 0..* Street name, number, direction & P.O. Box etc. -->
 <city value="[string]"/><!-- 0..1 Name of city, town etc. -->
 <district value="[string]"/><!-- 0..1 District name (aka county) -->
 <state value="[string]"/><!-- 0..1 Sub-unit of country (abbreviations ok) -->
 <postalCode value="[string]"/><!-- 0..1 Postal code for area -->
 <country value="[string]"/><!-- 0..1 Country (e.g. can be ISO 3166 2 or 3 letter code) -->
 <period><!-- 0..1 Period Time period when address was/is in use --></period>
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "use" : "<code>", // home | work | temp | old | billing - purpose of this address
  "type" : "<code>", // postal | physical | both
  "text" : "<string>", // Text representation of the address
  "line" : ["<string>"], // Street name, number, direction & P.O. Box etc.
  "city" : "<string>", // Name of city, town etc.
  "district" : "<string>", // District name (aka county)
  "state" : "<string>", // Sub-unit of country (abbreviations ok)
  "postalCode" : "<string>", // Postal code for area
  "country" : "<string>", // Country (e.g. can be ISO 3166 2 or 3 letter code)
  "period" : { Period } // Time period when address was/is in use
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:Address.use [ code ]; # 0..1 home | work | temp | old | billing - purpose of this address
  fhir:Address.type [ code ]; # 0..1 postal | physical | both
  fhir:Address.text [ string ]; # 0..1 Text representation of the address
  fhir:Address.line [ string ], ... ; # 0..* Street name, number, direction & P.O. Box etc.
  fhir:Address.city [ string ]; # 0..1 Name of city, town etc.
  fhir:Address.district [ string ]; # 0..1 District name (aka county)
  fhir:Address.state [ string ]; # 0..1 Sub-unit of country (abbreviations ok)
  fhir:Address.postalCode [ string ]; # 0..1 Postal code for area
  fhir:Address.country [ string ]; # 0..1 Country (e.g. can be ISO 3166 2 or 3 letter code)
  fhir:Address.period [ Period ]; # 0..1 Time period when address was/is in use
]
```

Changes since Release 3

Address
Address.use | Change value set from http://hl7.org/fhir/ValueSet/address-use to http://hl7.org/fhir/ValueSet/address-use|4.0.1
Address.type | Change value set from http://hl7.org/fhir/ValueSet/address-type to http://hl7.org/fhir/ValueSet/address-type|4.0.1

See the Full Difference for further information

Structure

Name | Flags | Card. | Type | Description & Constraints
Address | ΣN | | Element | An address expressed using postal conventions (as opposed to GPS or other location definition formats)
Elements defined in Ancestors: id, extension

use | ?!Σ | 0..1 | code | home | work | temp | old | billing - purpose of this address AddressUse (Required)
type | Σ | 0..1 | code | postal | physical | both AddressType (Required)
text | Σ | 0..1 | string | Text representation of the address
line | Σ | 0..* | string | Street name, number, direction & P.O. Box etc. This repeating element order: The order in which lines should appear in an address label
city | Σ | 0..1 | string | Name of city, town etc.
district | Σ | 0..1 | string | District name (aka county)
state | Σ | 0..1 | string | Sub-unit of country (abbreviations ok)
postalCode | Σ | 0..1 | string | Postal code for area
country | Σ | 0..1 | string | Country (e.g. can be ISO 3166 2 or 3 letter code)
period | Σ | 0..1 | Period | Time period when address was/is in use
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <use value="[code]"/><!-- 0..1 home | work | temp | old | billing - purpose of this address -->
 <type value="[code]"/><!-- 0..1 postal | physical | both -->
 <text value="[string]"/><!-- 0..1 Text representation of the address -->
 <line value="[string]"/><!-- 0..* Street name, number, direction & P.O. Box etc. -->
 <city value="[string]"/><!-- 0..1 Name of city, town etc. -->
 <district value="[string]"/><!-- 0..1 District name (aka county) -->
 <state value="[string]"/><!-- 0..1 Sub-unit of country (abbreviations ok) -->
 <postalCode value="[string]"/><!-- 0..1 Postal code for area -->
 <country value="[string]"/><!-- 0..1 Country (e.g. can be ISO 3166 2 or 3 letter code) -->
 <period><!-- 0..1 Period Time period when address was/is in use --></period>
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "use" : "<code>", // home | work | temp | old | billing - purpose of this address
  "type" : "<code>", // postal | physical | both
  "text" : "<string>", // Text representation of the address
  "line" : ["<string>"], // Street name, number, direction & P.O. Box etc.
  "city" : "<string>", // Name of city, town etc.
  "district" : "<string>", // District name (aka county)
  "state" : "<string>", // Sub-unit of country (abbreviations ok)
  "postalCode" : "<string>", // Postal code for area
  "country" : "<string>", // Country (e.g. can be ISO 3166 2 or 3 letter code)
  "period" : { Period } // Time period when address was/is in use
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:Address.use [ code ]; # 0..1 home | work | temp | old | billing - purpose of this address
  fhir:Address.type [ code ]; # 0..1 postal | physical | both
  fhir:Address.text [ string ]; # 0..1 Text representation of the address
  fhir:Address.line [ string ], ... ; # 0..* Street name, number, direction & P.O. Box etc.
  fhir:Address.city [ string ]; # 0..1 Name of city, town etc.
  fhir:Address.district [ string ]; # 0..1 District name (aka county)
  fhir:Address.state [ string ]; # 0..1 Sub-unit of country (abbreviations ok)
  fhir:Address.postalCode [ string ]; # 0..1 Postal code for area
  fhir:Address.country [ string ]; # 0..1 Country (e.g. can be ISO 3166 2 or 3 letter code)
  fhir:Address.period [ Period ]; # 0..1 Time period when address was/is in use
]
```

Changes since Release 3

Address
Address.use | Change value set from http://hl7.org/fhir/ValueSet/address-use to http://hl7.org/fhir/ValueSet/address-use|4.0.1
Address.type | Change value set from http://hl7.org/fhir/ValueSet/address-type to http://hl7.org/fhir/ValueSet/address-type|4.0.1

See the Full Difference for further information

The text element specifies the entire address as it should be displayed e.g. on a postal label. This may be provided instead of or as well as the specific parts. Applications updating an address SHALL ensure that when both text and parts are present, no content is included in the text that isn't found in a part.

Constraints

Terminology Bindings

Path | Definition | Type | Reference
Address.use | The use of an address. | Required | AddressUse
Address.type | The type of an address (physical / postal). | Required | AddressType

Address is used in the following places: Claim, ClaimResponse, ExplanationOfBenefit, InsurancePlan, Location, Organization, Patient, Person, Practitioner and RelatedPerson

2.24.0.15 ContactPoint

See also Examples, Detailed Descriptions, Mappings, Profiles & Extensions and R2 Conversions.

Details for all kinds of technology-mediated contact points for a person or organization, including telephone, email, etc.

• Structure
• UML
• XML
• JSON
• Turtle
• R3 Diff
• All

Structure

Name | Flags | Card. | Type | Description & Constraints
ContactPoint | ΣIN | | Element | Details of a Technology mediated contact point (phone, fax, email, etc.) + Rule: A system is required if a value is provided. Elements defined in Ancestors: id, extension
system | ΣI | 0..1 | code | phone | fax | email | pager | url | sms | other ContactPointSystem (Required)
value | Σ | 0..1 | string | The actual contact point details
use | ?!Σ | 0..1 | code | home | work | temp | old | mobile - purpose of this contact point ContactPointUse (Required)
rank | Σ | 0..1 | positiveInt | Specify preferred order of use (1 = highest)
period | Σ | 0..1 | Period | Time period when the contact point was/is in use
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <system value="[code]"/><!--  0..1 phone | fax | email | pager | url | sms | other -->
 <value value="[string]"/><!-- 0..1 The actual contact point details -->
 <use value="[code]"/><!-- 0..1 home | work | temp | old | mobile - purpose of this contact point -->
 <rank value="[positiveInt]"/><!-- 0..1 Specify preferred order of use (1 = highest) -->
 <period><!-- 0..1 Period Time period when the contact point was/is in use --></period>
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "system" : "<code>", // C? phone | fax | email | pager | url | sms | other
  "value" : "<string>", // The actual contact point details
  "use" : "<code>", // home | work | temp | old | mobile - purpose of this contact point
  "rank" : "<positiveInt>", // Specify preferred order of use (1 = highest)
  "period" : { Period } // Time period when the contact point was/is in use
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:ContactPoint.system [ code ]; # 0..1 phone | fax | email | pager | url | sms | other
  fhir:ContactPoint.value [ string ]; # 0..1 The actual contact point details
  fhir:ContactPoint.use [ code ]; # 0..1 home | work | temp | old | mobile - purpose of this contact point
  fhir:ContactPoint.rank [ positiveInt ]; # 0..1 Specify preferred order of use (1 = highest)
  fhir:ContactPoint.period [ Period ]; # 0..1 Time period when the contact point was/is in use
]
```

Changes since Release 3

ContactPoint
ContactPoint.system | Change value set from http://hl7.org/fhir/ValueSet/contact-point-system to http://hl7.org/fhir/ValueSet/contact-point-system|4.0.1
ContactPoint.use | Change value set from http://hl7.org/fhir/ValueSet/contact-point-use to http://hl7.org/fhir/ValueSet/contact-point-use|4.0.1

See the Full Difference for further information

Structure

Name | Flags | Card. | Type | Description & Constraints
ContactPoint | ΣIN | | Element | Details of a Technology mediated contact point (phone, fax, email, etc.) + Rule: A system is required if a value is provided. Elements defined in Ancestors: id, extension
system | ΣI | 0..1 | code | phone | fax | email | pager | url | sms | other ContactPointSystem (Required)
value | Σ | 0..1 | string | The actual contact point details
use | ?!Σ | 0..1 | code | home | work | temp | old | mobile - purpose of this contact point ContactPointUse (Required)
rank | Σ | 0..1 | positiveInt | Specify preferred order of use (1 = highest)
period | Σ | 0..1 | Period | Time period when the contact point was/is in use
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <system value="[code]"/><!--  0..1 phone | fax | email | pager | url | sms | other -->
 <value value="[string]"/><!-- 0..1 The actual contact point details -->
 <use value="[code]"/><!-- 0..1 home | work | temp | old | mobile - purpose of this contact point -->
 <rank value="[positiveInt]"/><!-- 0..1 Specify preferred order of use (1 = highest) -->
 <period><!-- 0..1 Period Time period when the contact point was/is in use --></period>
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "system" : "<code>", // C? phone | fax | email | pager | url | sms | other
  "value" : "<string>", // The actual contact point details
  "use" : "<code>", // home | work | temp | old | mobile - purpose of this contact point
  "rank" : "<positiveInt>", // Specify preferred order of use (1 = highest)
  "period" : { Period } // Time period when the contact point was/is in use
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:ContactPoint.system [ code ]; # 0..1 phone | fax | email | pager | url | sms | other
  fhir:ContactPoint.value [ string ]; # 0..1 The actual contact point details
  fhir:ContactPoint.use [ code ]; # 0..1 home | work | temp | old | mobile - purpose of this contact point
  fhir:ContactPoint.rank [ positiveInt ]; # 0..1 Specify preferred order of use (1 = highest)
  fhir:ContactPoint.period [ Period ]; # 0..1 Time period when the contact point was/is in use
]
```

Changes since Release 3

ContactPoint
ContactPoint.system | Change value set from http://hl7.org/fhir/ValueSet/contact-point-system to http://hl7.org/fhir/ValueSet/contact-point-system|4.0.1
ContactPoint.use | Change value set from http://hl7.org/fhir/ValueSet/contact-point-use to http://hl7.org/fhir/ValueSet/contact-point-use|4.0.1

See the Full Difference for further information

If capturing a phone, fax or similar contact point, the value should be a properly formatted telephone number according to ITU-T E.123 . However, this is frequently not possible due to legacy data and/or clerical practices when recording contact details. For this reason, phone, fax, pager, and email addresses are not handled as formal URLs. For other kinds of contact points, the system is "other" and the value SHOULD be a URL so that its use can be determined automatically. Typical URL schemes used in the value are http{s}: for web addresses, and URL schemes for various kinds of messaging systems. If the value is not a URL, then human interpretation will be required.

The rank element can be used to specify a preference for the order in which a set of contacts is used. ContactPoints with lower rank values are more preferred than those with higher rank values. Note that rank does not necessarily follow the order in which the contacts are represented in the instance.

Constraints

id | Level | Location | Description | Expression
cpt-2 | Rule | (base) | A system is required if a value is provided. | value.empty() or system.exists()

Terminology Bindings

Path | Definition | Type | Reference
ContactPoint.system | Telecommunications form for contact point. | Required | ContactPointSystem
ContactPoint.use | Use of contact point. | Required | ContactPointUse

ContactPoint is used in the following places: ContactDetail, CareTeam, Device, DeviceDefinition, Endpoint, HealthcareService, InsurancePlan, Location, MessageHeader, Organization, OrganizationAffiliation, Patient, Person, Practitioner, PractitionerRole, RelatedPerson and Subscription

2.24.0.16 Timing

See also Examples, Detailed Descriptions, Mappings, Profiles & Extensions and R2 Conversions.

Describes the occurrence of an event that may occur multiple times. Timing schedules are used for specifying when events are expected or requested to occur and may also be used to represent the summary of a past or ongoing event. For simplicity, the definitions of Timing components are expressed as 'future' events, but such components can also be used to describe historic or ongoing events.

A Timing schedule can be a list of events and/or criteria for when the event happens, which can be expressed in a structured form and/or as a code. When both event and a repeating specification are provided, the list of events should be understood as an interpretation of the information in the repeat structure.

Note: The Timing data type allows modifier extensions.

• Structure

- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
Timing | ΣN | | BackBoneElement | A timing schedule that specifies an event that may occur multiple times Elements defined in Ancestors: id, extension, modifierExtension
event | Σ | 0..* | dateTime | When the event occurs
repeat | ΣI | 0..1 | Element | When the event is to occur + Rule: if there's a duration, there needs to be duration units + Rule: if there's a period, there needs to be period units + Rule: duration SHALL be a non-negative value + Rule: period SHALL be a non-negative value + Rule: If there's a periodMax, there must be a period + Rule: If there's a durationMax, there must be a duration + Rule: If there's a countMax, there must be a count + Rule: If there's an offset, there must be a when (and not C, CM, CD, CV) + Rule: If there's a timeOfDay, there cannot be a when, or vice versa
bounds[x] | Σ | 0..1 | | Length/Range of lengths, or (Start and/or end) limits
boundsDuration | | | Duration
boundsRange | | | Range
boundsPeriod | | | Period
count | Σ | 0..1 | positiveInt | Number of times to repeat
countMax | Σ | 0..1 | positiveInt | Maximum number of times to repeat
duration | Σ | 0..1 | decimal | How long when it happens
durationMax | Σ | 0..1 | decimal | How long when it happens (Max)
durationUnit | Σ | 0..1 | code | s | min | h | d | wk | mo | a - unit of time (UCUM) UnitsOfTime (Required)
frequency | Σ | 0..1 | positiveInt | Event occurs frequency times per period
frequencyMax | Σ | 0..1 | positiveInt | Event occurs up to frequencyMax times per period
period | Σ | 0..1 | decimal | Event occurs frequency times per period
periodMax | Σ | 0..1 | decimal | Upper limit of period (3-4 hours)
periodUnit | Σ | 0..1 | code | s | min | h | d | wk | mo | a - unit of time (UCUM) UnitsOfTime (Required)
dayOfWeek | Σ | 0..* | code | mon | tue | wed | thu | fri | sat | sun DaysOfWeek (Required)
timeOfDay | Σ | 0..* | time | Time of day for action
when | Σ | 0..* | code | Code for time period of occurrence EventTiming (Required)
offset | Σ | 0..1 | unsignedInt | Minutes from event (before or after)
code | Σ | 0..1 | CodeableConcept | BID | TID | QID | AM | PM | QD | QOD | + TimingAbbreviation (Preferred)
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Timing xmlns="http://hl7.org/fhir">
 <!-- from BackboneElement: extension, modifierExtension -->
 <event value="[dateTime]"/><!-- 0..* When the event occurs -->
 <repeat>  <!-- 0..1 When the event is to occur -->
  <bounds[x]><!-- 0..1 Duration|Range|Period Length/Range of lengths, or (Start and/or end) limits --></bounds[x]>
  <count value="[positiveInt]"/><!-- 0..1 Number of times to repeat -->
  <countMax value="[positiveInt]"/><!-- 0..1 Maximum number of times to repeat -->
  <duration value="[decimal]"/><!-- 0..1 How long when it happens -->
  <durationMax value="[decimal]"/><!-- 0..1 How long when it happens (Max) -->
  <durationUnit value="[code]"/><!-- 0..1 s | min | h | d | wk | mo | a - unit of time (UCUM) -->
  <frequency value="[positiveInt]"/><!-- 0..1 Event occurs frequency times per period -->
  <frequencyMax value="[positiveInt]"/><!-- 0..1 Event occurs up to frequencyMax times per period -->
  <period value="[decimal]"/><!-- 0..1 Event occurs frequency times per period -->
  <periodMax value="[decimal]"/><!-- 0..1 Upper limit of period (3-4 hours) -->
  <periodUnit value="[code]"/><!-- 0..1 s | min | h | d | wk | mo | a - unit of time (UCUM) -->
  <dayOfWeek value="[code]"/><!-- 0..* mon | tue | wed | thu | fri | sat | sun -->
  <timeOfDay value="[time]"/><!-- 0..* Time of day for action -->
  <when value="[code]"/><!-- 0..* Code for time period of occurrence -->
  <offset value="[unsignedInt]"/><!-- 0..1 Minutes from event (before or after) -->
 </repeat>
 <code><!-- 0..1 CodeableConcept BID | TID | QID | AM | PM | QD | QOD | + --></code>
</Timing>
```

JSON Template

```
{
  // from BackboneElement: extension, modifierExtension
  "event" : ["<dateTime>"], // When the event occurs
  "repeat" : { // When the event is to occur
    // bounds[x]: Length/Range of lengths, or (Start and/or end) limits. One of these 3:
    "boundsDuration" : { Duration },
    "boundsRange" : { Range },
    "boundsPeriod" : { Period },
    "count" : "<positiveInt>", // Number of times to repeat
    "countMax" : "<positiveInt>", // Maximum number of times to repeat
    "duration" : <decimal>, // How long when it happens
    "durationMax" : <decimal>, // How long when it happens (Max)
    "durationUnit" : "<code>", // s | min | h | d | wk | mo | a - unit of time (UCUM)
    "frequency" : "<positiveInt>", // Event occurs frequency times per period
```

```
    "frequencyMax" : "<positiveInt>", // Event occurs up to frequencyMax times per period
    "period" : <decimal>, // Event occurs frequency times per period
    "periodMax" : <decimal>, // Upper limit of period (3-4 hours)
    "periodUnit" : "<code>", // s | min | h | d | wk | mo | a - unit of time (UCUM)
    "dayOfWeek" : ["<code>"], // mon | tue | wed | thu | fri | sat | sun
    "timeOfDay" : ["<time>"], // Time of day for action
    "when" : ["<code>"], // Code for time period of occurrence
    "offset" : "<unsignedInt>" // Minutes from event (before or after)
  },
  "code" : { CodeableConcept } // BID | TID | QID | AM | PM | QD | QOD | +
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from BackboneElement: Element.extension, BackboneElement.modifierextension
  fhir:Timing.event [ dateTime ], ... ; # 0..* When the event occurs
  fhir:Timing.repeat [ # 0..1 When the event is to occur
    # Timing.repeat.bounds[x] : 0..1 Length/Range of lengths, or (Start and/or end) limits. One of these 3
      fhir:Timing.repeat.boundsDuration [ Duration ]
      fhir:Timing.repeat.boundsRange [ Range ]
      fhir:Timing.repeat.boundsPeriod [ Period ]
    fhir:Timing.repeat.count [ positiveInt ]; # 0..1 Number of times to repeat
    fhir:Timing.repeat.countMax [ positiveInt ]; # 0..1 Maximum number of times to repeat
    fhir:Timing.repeat.duration [ decimal ]; # 0..1 How long when it happens
    fhir:Timing.repeat.durationMax [ decimal ]; # 0..1 How long when it happens (Max)
    fhir:Timing.repeat.durationUnit [ code ]; # 0..1 s | min | h | d | wk | mo | a - unit of time (UCUM)
    fhir:Timing.repeat.frequency [ positiveInt ]; # 0..1 Event occurs frequency times per period
    fhir:Timing.repeat.frequencyMax [ positiveInt ]; # 0..1 Event occurs up to frequencyMax times per period
    fhir:Timing.repeat.period [ decimal ]; # 0..1 Event occurs frequency times per period
    fhir:Timing.repeat.periodMax [ decimal ]; # 0..1 Upper limit of period (3-4 hours)
    fhir:Timing.repeat.periodUnit [ code ]; # 0..1 s | min | h | d | wk | mo | a - unit of time (UCUM)
    fhir:Timing.repeat.dayOfWeek [ code ], ... ; # 0..* mon | tue | wed | thu | fri | sat | sun
    fhir:Timing.repeat.timeOfDay [ time ], ... ; # 0..* Time of day for action
    fhir:Timing.repeat.when [ code ], ... ; # 0..* Code for time period of occurrence
    fhir:Timing.repeat.offset [ unsignedInt ]; # 0..1 Minutes from event (before or after)
  ];
  fhir:Timing.code [ CodeableConcept ]; # 0..1 BID | TID | QID | AM | PM | QD | QOD | +
]
```

Changes since Release 3

Timing
Timing.repeat.count | Type changed from integer to positiveInt
Timing.repeat.countMax | Type changed from integer to positiveInt
Timing.repeat.durationUnit | Change value set from http://hl7.org/fhir/ValueSet/units-of-time to http://hl7.org/fhir/ValueSet/units-of-time|4.0.1
Timing.repeat.frequency | Type changed from integer to positiveInt Default Value "1" removed
Timing.repeat.frequencyMax | Type changed from integer to positiveInt
Timing.repeat.periodUnit | Change value set from http://hl7.org/fhir/ValueSet/units-of-time to http://hl7.org/fhir/ValueSet/units-of-time|4.0.1
Timing.repeat.dayOfWeek | Change value set from http://hl7.org/fhir/ValueSet/days-of-week to http://hl7.org/fhir/ValueSet/days-of-week|4.0.1
Timing.repeat.when | Change value set from http://hl7.org/fhir/ValueSet/event-timing to http://hl7.org/fhir/ValueSet/event-timing|4.0.1

See the Full Difference for further information

Structure

Name | Flags | Card. | Type | Description & Constraints
Timing | ΣN | | BackBoneElement | A timing schedule that specifies an event that may occur multiple times Elements defined in Ancestors: id, extension, modifierExtension
event | Σ | 0..* | dateTime | When the event occurs
repeat | ΣI | 0..1 | Element | When the event is to occur + Rule: if there's a duration, there needs to be duration units + Rule: if there's a period, there needs to be period units + Rule: duration SHALL be a non-negative value + Rule: period SHALL be a non-negative value + Rule: If there's a periodMax, there must be a period + Rule: If there's a durationMax, there must be a duration + Rule: If there's a countMax, there must be a count + Rule: If there's an offset, there must be a when (and not C, CM, CD, CV) + Rule: If there's a timeOfDay, there cannot be a when, or vice versa
bounds[x] | Σ | 0..1 | | Length/Range of lengths, or (Start and/or end) limits
boundsDuration | | | Duration
boundsRange | | | Range
boundsPeriod | | | Period
count | Σ | 0..1 | positiveInt | Number of times to repeat
countMax | Σ | 0..1 | positiveInt | Maximum number of times to repeat
duration | Σ | 0..1 | decimal | How long when it happens
durationMax | Σ | 0..1 | decimal | How long when it happens (Max)
durationUnit | Σ | 0..1 | code | s | min | h | d | wk | mo | a - unit of time (UCUM) UnitsOfTime (Required)
frequency | Σ | 0..1 | positiveInt | Event occurs frequency times per period
frequencyMax | Σ | 0..1 | positiveInt | Event occurs up to frequencyMax times per period
period | Σ | 0..1 | decimal | Event occurs frequency times per period
periodMax | Σ | 0..1 | decimal | Upper limit of period (3-4 hours)
periodUnit | Σ | 0..1 | code | s | min | h | d | wk | mo | a - unit of time (UCUM) UnitsOfTime (Required)

dayOfWeek | Σ | 0..* | code | mon | tue | wed | thu | fri | sat | sun DaysOfWeek (Required)
timeOfDay | Σ | 0..* | time | Time of day for action
when | Σ | 0..* | code | Code for time period of occurrence EventTiming (Required)
offset | Σ | 0..1 | unsignedInt | Minutes from event (before or after)
code | Σ | 0..1 | CodeableConcept | BID | TID | QID | AM | PM | QD | QOD | + TimingAbbreviation (Preferred)
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Timing xmlns="http://hl7.org/fhir">
 <!-- from BackboneElement: extension, modifierExtension -->
 <event value="[dateTime]"/><!-- 0..* When the event occurs -->
 <repeat>  <!-- 0..1 When the event is to occur -->
  <bounds[x]><!-- 0..1 Duration|Range|Period Length/Range of lengths, or (Start and/or end) limits --></bounds[x]>
  <count value="[positiveInt]"/><!-- 0..1 Number of times to repeat -->
  <countMax value="[positiveInt]"/><!-- 0..1 Maximum number of times to repeat -->
  <duration value="[decimal]"/><!-- 0..1 How long when it happens -->
  <durationMax value="[decimal]"/><!-- 0..1 How long when it happens (Max) -->
  <durationUnit value="[code]"/><!-- 0..1 s | min | h | d | wk | mo | a - unit of time (UCUM) -->
  <frequency value="[positiveInt]"/><!-- 0..1 Event occurs frequency times per period -->
  <frequencyMax value="[positiveInt]"/><!-- 0..1 Event occurs up to frequencyMax times per period -->
  <period value="[decimal]"/><!-- 0..1 Event occurs frequency times per period -->
  <periodMax value="[decimal]"/><!-- 0..1 Upper limit of period (3-4 hours) -->
  <periodUnit value="[code]"/><!-- 0..1 s | min | h | d | wk | mo | a - unit of time (UCUM) -->
  <dayOfWeek value="[code]"/><!-- 0..* mon | tue | wed | thu | fri | sat | sun -->
  <timeOfDay value="[time]"/><!-- 0..* Time of day for action -->
  <when value="[code]"/><!-- 0..* Code for time period of occurrence -->
  <offset value="[unsignedInt]"/><!-- 0..1 Minutes from event (before or after) -->
 </repeat>
 <code><!-- 0..1 CodeableConcept BID | TID | QID | AM | PM | QD | QOD | + --></code>
</Timing>
```

JSON Template

```
{
  // from BackboneElement: extension, modifierExtension
  "event" : ["<dateTime>"], // When the event occurs
  "repeat" : { // When the event is to occur
    // bounds[x]: Length/Range of lengths, or (Start and/or end) limits. One of these 3:
    "boundsDuration" : { Duration },
    "boundsRange" : { Range },
    "boundsPeriod" : { Period },
    "count" : "<positiveInt>", // Number of times to repeat
    "countMax" : "<positiveInt>", // Maximum number of times to repeat
    "duration" : <decimal>, // How long when it happens
    "durationMax" : <decimal>, // How long when it happens (Max)
    "durationUnit" : "<code>", // s | min | h | d | wk | mo | a - unit of time (UCUM)
    "frequency" : "<positiveInt>", // Event occurs frequency times per period
    "frequencyMax" : "<positiveInt>", // Event occurs up to frequencyMax times per period
    "period" : <decimal>, // Event occurs frequency times per period
    "periodMax" : <decimal>, // Upper limit of period (3-4 hours)
    "periodUnit" : "<code>", // s | min | h | d | wk | mo | a - unit of time (UCUM)
    "dayOfWeek" : ["<code>"], // mon | tue | wed | thu | fri | sat | sun
    "timeOfDay" : ["<time>"], // Time of day for action
    "when" : ["<code>"], // Code for time period of occurrence
    "offset" : "<unsignedInt>" // Minutes from event (before or after)
  },
  "code" : { CodeableConcept } // BID | TID | QID | AM | PM | QD | QOD | +
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from BackboneElement: Element.extension, BackboneElement.modifierextension
  fhir:Timing.event [ dateTime ], ... ; # 0..* When the event occurs
  fhir:Timing.repeat [ # 0..1 When the event is to occur
    # Timing.repeat.bounds[x] : 0..1 Length/Range of lengths, or (Start and/or end) limits. One of these 3
      fhir:Timing.repeat.boundsDuration [ Duration ]
      fhir:Timing.repeat.boundsRange [ Range ]
      fhir:Timing.repeat.boundsPeriod [ Period ]
    fhir:Timing.repeat.count [ positiveInt ]; # 0..1 Number of times to repeat
    fhir:Timing.repeat.countMax [ positiveInt ]; # 0..1 Maximum number of times to repeat
    fhir:Timing.repeat.duration [ decimal ]; # 0..1 How long when it happens
    fhir:Timing.repeat.durationMax [ decimal ]; # 0..1 How long when it happens (Max)
    fhir:Timing.repeat.durationUnit [ code ]; # 0..1 s | min | h | d | wk | mo | a - unit of time (UCUM)
    fhir:Timing.repeat.frequency [ positiveInt ]; # 0..1 Event occurs frequency times per period
    fhir:Timing.repeat.frequencyMax [ positiveInt ]; # 0..1 Event occurs up to frequencyMax times per period
    fhir:Timing.repeat.period [ decimal ]; # 0..1 Event occurs frequency times per period
    fhir:Timing.repeat.periodMax [ decimal ]; # 0..1 Upper limit of period (3-4 hours)
    fhir:Timing.repeat.periodUnit [ code ]; # 0..1 s | min | h | d | wk | mo | a - unit of time (UCUM)
    fhir:Timing.repeat.dayOfWeek [ code ], ... ; # 0..* mon | tue | wed | thu | fri | sat | sun
    fhir:Timing.repeat.timeOfDay [ time ], ... ; # 0..* Time of day for action
    fhir:Timing.repeat.when [ code ], ... ; # 0..* Code for time period of occurrence
    fhir:Timing.repeat.offset [ unsignedInt ]; # 0..1 Minutes from event (before or after)
  ];
  fhir:Timing.code [ CodeableConcept ]; # 0..1 BID | TID | QID | AM | PM | QD | QOD | +
]
```

Changes since Release 3

Timing
Timing.repeat.count | Type changed from integer to positiveInt
Timing.repeat.countMax | Type changed from integer to positiveInt
Timing.repeat.durationUnit | Change value set from http://hl7.org/fhir/ValueSet/units-of-time to http://hl7.org/fhir/ValueSet/units-of-time|4.0.1
Timing.repeat.frequency | Type changed from integer to positiveInt Default Value "1" removed
Timing.repeat.frequencyMax | Type changed from integer to positiveInt
Timing.repeat.periodUnit | Change value set from http://hl7.org/fhir/ValueSet/units-of-time to http://hl7.org/fhir/ValueSet/units-of-time|4.0.1
Timing.repeat.dayOfWeek | Change value set from http://hl7.org/fhir/ValueSet/days-of-week to http://hl7.org/fhir/ValueSet/days-of-week|4.0.1
Timing.repeat.when | Change value set from http://hl7.org/fhir/ValueSet/event-timing to http://hl7.org/fhir/ValueSet/event-timing|4.0.1

See the Full Difference for further information

If the timing schedule has repeating criteria, the repeat can occur a given number of times per the specified duration or in relation to some repeating real-world event. If no end condition is specified, the schedule will terminate on some criteria that are expressed elsewhere.

This table summarizes some common uses of the Timing Data Type criteria.

description | duration | durationUnit | frequency | frequencyMax | period | periodUnit | periodMax | Day of Week | Time Of Day | when | offset | bounds[x] | count
Every 8 hours | | | 1 | | 8 | h
Every 7 days | | | 1 | | 7 | d
3 times a day | | | 3 | | 1 | d
3-4 times a day | | | 3 | 4 | 1 | d
Every 4-6 hours | | | 1 | | 4 | h | 6
Every 21 days for 1 hour | 1 | hr | 1 | | 21 | d
Three times a week for ½ hour | 0.5 | hr | 3 | | 1 | wk
With breakfast | | | | | | | | | CM
For 5 minutes, 10 minutes before meals | 5 | min | | | | | | | AC | 10
1 tablet 3 times daily, 30 minutes before meals | | | 3 | | 1 | d | | | | AC | 30
BID, 30 mins before meal, for next 10 days | | | 2 | | 1 | d | | | | AC | 30 | Duration = 10 days
TID, for 14 days | | | 3 | | 1 | d | | | | | | Duration = 14 days
BID, start on 7/1/2015 at 1:00 PM | | | 2 | | 1 | d | | | | | | Period.start = 2015-07-01T13:00:00
Mon, Wed, Fri Morning | | | 1 | | 1 | d | | mon | wed | fri | | MORN
Every day at 10am | | | 1 | | | 1 | d | | | 10:00
Take once, at any time | | | | | | | | | | | | | | 1
Take every second day, in the morning, until 20 have been taken | | | 1 | | 2 | d | | | | MORN | | | 20

Many systems avoid the complexity of the Timing structure by using a text field for timing instructions. This maps to Timing.code.text. For example, the text instruction "take medication in the morning on weekends and days off work' would be represented as:

```
"timing": {
  "code" : {
    "text" : "Take medication in the morning on weekends and days off work"
  }
}
```

Note, though, that some systems include timing details in something like 'Dosage instructions' which is wider than just Timing; those systems do not use the Timing data type. Other systems use a set of 'common' codes - including, but usually not limited to, widely understood acronyms such as "BID". If a Timing.code is provided, the code is understood to be a complete statement of whatever is specified in the structured timing data (except for Timing.repeat.bounds, which applies to the code), and either the code or the data may be used to interpret the Timing. A structured timing specification SHOULD be provided whenever possible, unless the code is BID, TID, QID, AM or PM, which have a ubiquitous meaning.

This table shows the relationship between the codes provided as part of the base specification, and the structured data portions of the Timing type:

description | duration | durationUnit | frequency | frequencyMax | period | periodUnit | periodMax | when | bounds[x]
QOD | | | 1 | | 2 | d
QD | | | 1 | | 1 | d
BID | | | 2 | | 1 | d
TID | | | 3 | | 1 | d
QID | | | 4 | | 1 | d
Q4H | | | 1 | | 4 | h
Q6H | | | 1 | | 6 | h
AM | | | 1 | | 1 | d | | MORN
PM | | | 1 | | 1 | d | | AFT or EVE

These codes SHALL be understood as having the formal meanings documented in this table. Note that BID, etc. are defined as 'at institutionally specified times'. For example, an institution may choose that BID is "always at 7am and 6pm". If it is inappropriate for this choice to be made, the code BID should not be used.

Instead, a distinct organization-specific code should be used in place of the HL7-defined BID code and/or a structured representation should be used (in this case, timeOfDay).

Constraints

id | Level | Location | Description | Expression
tim-1 | Rule | Timing.repeat | if there's a duration, there needs to be duration units | duration.empty() or durationUnit.exists()
tim-2 | Rule | Timing.repeat | if there's a period, there needs to be period units | period.empty() or periodUnit.exists()
tim-4 | Rule | Timing.repeat | duration SHALL be a non-negative value | duration.exists() implies duration >= 0
tim-5 | Rule | Timing.repeat | period SHALL be a non-negative value | period.exists() implies period >= 0
tim-6 | Rule | Timing.repeat | If there's a periodMax, there must be a period | periodMax.empty() or period.exists()
tim-7 | Rule | Timing.repeat | If there's a durationMax, there must be a duration | durationMax.empty() or duration.exists()
tim-8 | Rule | Timing.repeat | If there's a countMax, there must be a count | countMax.empty() or count.exists()
tim-9 | Rule | Timing.repeat | If there's an offset, there must be a when (and not C, CM, CD, CV) | offset.empty() or (when.exists() and ((when in ('C' | 'CM' | 'CD' | 'CV')).not()))
tim-10 | Rule | Timing.repeat | If there's a timeOfDay, there cannot be a when, or vice versa | timeOfDay.empty() or when.empty()

Note that these constraints still allow for nonsensical timing specifications such as "Once per day at 2:00 and 4:00" or "every 3 days on Friday". Implementers must take care to ensure that their configuration and data collection designs do not lead to these non-interpretable timing specifications. The elements dayOfWeek, timeOfDay, and when are particularly likely to be at issue here.

Terminology Bindings

Path | Definition | Type | Reference
Timing.repeat.durationUnit Timing.repeat.periodUnit | A unit of time (units from UCUM). | Required | UnitsOfTime
Timing.repeat.dayOfWeek | | Required | DaysOfWeek
Timing.repeat.when | Real world event relating to the schedule. | Required | EventTiming
Timing.code | Code for a known / defined timing pattern. | Preferred | TimingAbbreviation

Timing is used in the following places: Dosage, TriggerDefinition, ActivityDefinition, CarePlan, ChargeItem, Contract, DeviceMetric, DeviceRequest, DeviceUseStatement, EvidenceVariable, NutritionOrder, Observation, PlanDefinition, RequestGroup, ResearchElementDefinition, ServiceRequest, SupplyDelivery, SupplyRequest and VerificationResult

2.24.0.17 Signature

Normative Candidate Note: This DataType is not normative - it is still undergoing Trial Use while more experience is gathered.

See also Examples, Detailed Descriptions, Mappings, Profiles & Extensions and R2 Conversions.

A Signature holds an electronic representation of a signature and its supporting context in a FHIR accessible form. The signature may either be a cryptographic type (XML DigSig or a JWS), which is able to provide non-repudiation proof, or it may be a graphical image that represents a signature or a signature process.

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
Signature | TU | | Element | A Signature - XML DigSig, JWS, Graphical image of signature, etc. Elements defined in Ancestors: id, extension
type | Σ | 1..* | Coding | Indication of the reason the entity signed the object(s) Signature Type Codes (Preferred)
when | Σ | 1..1 | instant | When the signature was created
who | Σ | 1..1 | Reference(Practitioner | PractitionerRole | RelatedPerson | Patient | Device | Organization) | Who signed
onBehalfOf | Σ | 0..1 | Reference(Practitioner | PractitionerRole | RelatedPerson | Patient | Device | Organization) | The party represented
targetFormat | | 0..1 | code | The technical format of the signed resources MimeType (Required)
sigFormat | | 0..1 | code | The technical format of the signature MimeType (Required)
data | | 0..1 | base64Binary | The actual signature content (XML DigSig. JWS, picture, etc.)
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <type><!-- 1..* Coding Indication of the reason the entity signed the object(s) --></type>
 <when value="[instant]"/><!-- 1..1 When the signature was created -->
 <who><!-- 1..1 Reference(Practitioner|PractitionerRole|RelatedPerson|Patient|
   Device|Organization) Who signed --></who>
 <onBehalfOf><!-- 0..1 Reference(Practitioner|PractitionerRole|RelatedPerson|
```

```
   Patient|Device|Organization) The party represented --></onBehalfOf>
 <targetFormat value="[code]"/><!-- 0..1 The technical format of the signed resources -->
 <sigFormat value="[code]"/><!-- 0..1 The technical format of the signature -->
 <data value="[base64Binary]"/><!-- 0..1 The actual signature content (XML DigSig. JWS, picture, etc.) -->
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "type" : [{ Coding }], // R!  Indication of the reason the entity signed the object(s)
  "when" : "<instant>", // R!  When the signature was created
  "who" : { Reference(Practitioner|PractitionerRole|RelatedPerson|Patient|
   Device|Organization) }, // R!  Who signed
  "onBehalfOf" : { Reference(Practitioner|PractitionerRole|RelatedPerson|
   Patient|Device|Organization) }, // The party represented
  "targetFormat" : "<code>", // The technical format of the signed resources
  "sigFormat" : "<code>", // The technical format of the signature
  "data" : "<base64Binary>" // The actual signature content (XML DigSig. JWS, picture, etc.)
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:Signature.type [ Coding ], ... ; # 1..* Indication of the reason the entity signed the object(s)
  fhir:Signature.when [ instant ]; # 1..1 When the signature was created
  fhir:Signature.who [ Reference(Practitioner|PractitionerRole|RelatedPerson|Patient|Device|Organization) ]; # 1..1 Who
    signed
  fhir:Signature.onBehalfOf [ Reference(Practitioner|PractitionerRole|RelatedPerson|Patient|Device|Organization) ]; #
    0..1 The party represented
  fhir:Signature.targetFormat [ code ]; # 0..1 The technical format of the signed resources
  fhir:Signature.sigFormat [ code ]; # 0..1 The technical format of the signature
  fhir:Signature.data [ base64Binary ]; # 0..1 The actual signature content (XML DigSig. JWS, picture, etc.)
]
```

Changes since Release 3

Signature
Signature.who | Renamed from who[x] to who Remove Type uri
Signature.onBehalfOf | Renamed from onBehalfOf[x] to onBehalfOf Remove Type uri
Signature.targetFormat | Added Element
Signature.sigFormat | Added Element
Signature.data | Renamed from blob to data
Signature.contentType | deleted

See the Full Difference for further information

Structure

Name | Flags | Card. | Type | Description & Constraints
Signature | TU | | Element | A Signature - XML DigSig, JWS, Graphical image of signature, etc. Elements defined in Ancestors: id, extension
type | Σ | 1..* | Coding | Indication of the reason the entity signed the object(s) Signature Type Codes (Preferred)
when | Σ | 1..1 | instant | When the signature was created
who | Σ | 1..1 | Reference(Practitioner | PractitionerRole | RelatedPerson | Patient | Device | Organization) | Who signed
onBehalfOf | Σ | 0..1 | Reference(Practitioner | PractitionerRole | RelatedPerson | Patient | Device | Organization) | The party represented
targetFormat | | 0..1 | code | The technical format of the signed resources MimeType (Required)
sigFormat | | 0..1 | code | The technical format of the signature MimeType (Required)
data | | 0..1 | base64Binary | The actual signature content (XML DigSig. JWS, picture, etc.)
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <type><!-- 1..* Coding Indication of the reason the entity signed the object(s) --></type>
 <when value="[instant]"/><!-- 1..1 When the signature was created -->
 <who><!-- 1..1 Reference(Practitioner|PractitionerRole|RelatedPerson|Patient|
   Device|Organization) Who signed --></who>
 <onBehalfOf><!-- 0..1 Reference(Practitioner|PractitionerRole|RelatedPerson|
   Patient|Device|Organization) The party represented --></onBehalfOf>
 <targetFormat value="[code]"/><!-- 0..1 The technical format of the signed resources -->
 <sigFormat value="[code]"/><!-- 0..1 The technical format of the signature -->
 <data value="[base64Binary]"/><!-- 0..1 The actual signature content (XML DigSig. JWS, picture, etc.) -->
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "type" : [{ Coding }], // R!  Indication of the reason the entity signed the object(s)
  "when" : "<instant>", // R!  When the signature was created
  "who" : { Reference(Practitioner|PractitionerRole|RelatedPerson|Patient|
   Device|Organization) }, // R!  Who signed
  "onBehalfOf" : { Reference(Practitioner|PractitionerRole|RelatedPerson|
```

```
   Patient|Device|Organization) }, // The party represented
  "targetFormat" : "<code>", // The technical format of the signed resources
  "sigFormat" : "<code>", // The technical format of the signature
  "data" : "<base64Binary>" // The actual signature content (XML DigSig. JWS, picture, etc.)
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:Signature.type [ Coding ], ... ; # 1..* Indication of the reason the entity signed the object(s)
  fhir:Signature.when [ instant ]; # 1..1 When the signature was created
  fhir:Signature.who [ Reference(Practitioner|PractitionerRole|RelatedPerson|Patient|Device|Organization) ]; # 1..1 Who
    signed
  fhir:Signature.onBehalfOf [ Reference(Practitioner|PractitionerRole|RelatedPerson|Patient|Device|Organization) ]; #
    0..1 The party represented
  fhir:Signature.targetFormat [ code ]; # 0..1 The technical format of the signed resources
  fhir:Signature.sigFormat [ code ]; # 0..1 The technical format of the signature
  fhir:Signature.data [ base64Binary ]; # 0..1 The actual signature content (XML DigSig. JWS, picture, etc.)
]
```

Changes since Release 3

Signature
Signature.who | Renamed from who[x] to who Remove Type uri
Signature.onBehalfOf | Renamed from onBehalfOf[x] to onBehalfOf Remove Type uri
Signature.targetFormat | Added Element
Signature.sigFormat | Added Element
Signature.data | Renamed from blob to data
Signature.contentType | deleted

See the Full Difference for further information

Constraints

Notes:

• One consequence of signing the document is that URLs, identifiers and internal references are frozen and cannot be changed. This might be a desired feature, but it may also cripple interoperability between closed ecosystems where re-identification frequently occurs. For this reason, it is recommended that systems consider carefully the impact of any signature processes. The impact of signatures on Document bundles and their related processes is the most well understood use of digital signatures.

• Note that following common normalization procedures in XML causes consecutive whitespace within attributes to be normalized. As a result, whitespace within markdown can be changed without breaking a digital signature. For elements with a type of markdown, this means that, in some cases, whitespace manipulation that results in significantly different visual rendering (e.g. changing indentation levels, causing content to appear in separate paragraphs rather than part of a single paragraph, etc.) might not be detected as a signature-breaking change. If this would present an unacceptable risk, systems should use the JSON signature approach as it does not normalize whitespace.

2.24.0.17.1 XML Signature rules

When the signature is an XML Digital Signature (contentType = application/signature+xml), the following rules apply:

• The Signature.data is base64 encoded XML-Signature

• The XML-Signature is a Detached Signature (where the content that is signed is separate from the signature itself)

• The Signature SHOULD conform to XAdES-X-L for support of Long Term signatures. The XAdES-X-L specification adds the timestamp of the signing, inclusion of the signing certificate, and statement of revocation

• When FHIR Resources are signed, the signature is across the Canonical XML form of the resource(s)

• The Signature SHOULD use the hashing algorithm SHA-256. Signature validation policy will apply to the signature and determine acceptability

• The Signature SHALL include a "CommitmentTypeIndication" element for the Purpose(s) of Signature. The Purpose can be the action being attested to, or the role associated with the signature. The value shall come from ASTM E1762-95(2013). The Signature.type shall contain the same values as the CommitmentTypeIndication element.

There are three levels of signature verification:

• Verifying that the Digital Signature block itself has integrity through verifying the signature across the XML-Signature.

• Confirming that the signer was authentic, not revoked, and appropriate to the signature purpose.

• Confirming that the signed content of interest is unmodified using the hash algorithm.

Deviations from these guidelines would need to be expressed in site policy and would be enumerated in the XML-Signature block. For example, some environments may choose a different XAdES profile, hashing algorithm, policy identifier, or signature purpose vocabulary.

2.24.0.17.2 JSON Signature rules

When the signature is an JSON Digital Signature (contentType = application/jose), the following rules apply:

• The Signature.data is base64 encoded JWS-Signature RFC 7515: JSON Web Signature (JWS)

• The signature is a Detached Signature (where the content that is signed is separate from the signature itself)

• When FHIR Resources are signed, the signature is across the Canonical JSON form of the resource(s)

• The Signature SHOULD use the hashing algorithm SHA256. Signature validation policy will apply to the signature and determine acceptability

• The Signature SHALL include a "CommitmentTypeIndication" element for the Purpose(s) of Signature. The Purpose can be the action being attested to, or the role associated with the signature. The value shall come from ASTM E1762-95(2013). The Signature.type shall contain the same values as the CommitmentTypeIndication element.

There are three levels of signature verification:

• Verifying that the Digital Signature block itself has integrity through verifying the signature across the JWS-Signature.

• Confirming that the signer was authentic, not revoked, and appropriate to the signature purpose.

• Confirming that the signed content of interest is unmodified using the hash algorithm.

Deviations from these guidelines would need to be expressed in site policy and would be enumerated in the JWS-Signature block. For example, some environments may choose a different hashing algorithm, policy identifier, or signature purpose vocabulary.

Signature is used in the following places: Bundle, Contract, Provenance and VerificationResult

2.24.0.18 Annotation

See also Examples, Detailed Descriptions, Mappings, Profiles & Extensions and R2 Conversions.

A text note which also contains information about who made the statement and when.

• Structure

• UML

• XML

• JSON

• Turtle

• R3 Diff

• All

Structure

Name	Flags	Card.	Type	Description & Constraints
Annotation	N		Element	Text node with attribution Elements defined in Ancestors: id, extension
author[x]	Σ	0..1		Individual responsible for the annotation
authorReference			Reference(Practitioner | Patient | RelatedPerson | Organization)	
authorString			string	
time	Σ	0..1	dateTime	When the annotation was made
text	Σ	1..1	markdown	The annotation - text content (as markdown)

Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <author[x]><!-- 0..1 Reference(Practitioner|Patient|RelatedPerson|Organization)|
   string Individual responsible for the annotation --></author[x]>
 <time value="[dateTime]"/><!-- 0..1 When the annotation was made -->
 <text value="[markdown]"/><!-- 1..1 The annotation  - text content (as markdown) -->
</[name]>
```

JSON Template

```
{
  // from Element: extension
  // author[x]: Individual responsible for the annotation. One of these 2:
  "authorReference" : { Reference(Practitioner|Patient|RelatedPerson|Organization) },
  "authorString" : "<string>",
  "time" : "<dateTime>", // When the annotation was made
  "text" : "<markdown>" // R!  The annotation  - text content (as markdown)
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  # Annotation.author[x] : 0..1 Individual responsible for the annotation. One of these 2
    fhir:Annotation.authorReference [ Reference(Practitioner|Patient|RelatedPerson|Organization) ]
    fhir:Annotation.authorString [ string ]
  fhir:Annotation.time [ dateTime ]; # 0..1 When the annotation was made
  fhir:Annotation.text [ markdown ]; # 1..1 The annotation  - text content (as markdown)
]
```

Changes since Release 3

Annotation

Annotation.text | Type changed from string to markdown

See the Full Difference for further information

Structure

Name	Flags	Card.	Type	Description & Constraints
Annotation	N		Element	Text node with attribution Elements defined in Ancestors: id, extension
author[x]	Σ	0..1		Individual responsible for the annotation
authorReference			Reference(Practitioner | Patient | RelatedPerson | Organization)	
authorString			string	
time	Σ	0..1	dateTime	When the annotation was made
text	Σ	1..1	markdown	The annotation - text content (as markdown)

Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <author[x]><!-- 0..1 Reference(Practitioner|Patient|RelatedPerson|Organization)|
   string Individual responsible for the annotation --></author[x]>
 <time value="[dateTime]"/><!-- 0..1 When the annotation was made -->
 <text value="[markdown]"/><!-- 1..1 The annotation  - text content (as markdown) -->
</[name]>
```

JSON Template

```
{
  // from Element: extension
  // author[x]: Individual responsible for the annotation. One of these 2:
  "authorReference" : { Reference(Practitioner|Patient|RelatedPerson|Organization) },
  "authorString" : "<string>",
  "time" : "<dateTime>", // When the annotation was made
  "text" : "<markdown>" // R!  The annotation  - text content (as markdown)
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  # Annotation.author[x] : 0..1 Individual responsible for the annotation. One of these 2
    fhir:Annotation.authorReference [ Reference(Practitioner|Patient|RelatedPerson|Organization) ]
    fhir:Annotation.authorString [ string ]
  fhir:Annotation.time [ dateTime ]; # 0..1 When the annotation was made
  fhir:Annotation.text [ markdown ]; # 1..1 The annotation  - text content (as markdown)
]
```

Changes since Release 3

Annotation

Annotation.text | Type changed from string to markdown

See the Full Difference for further information

Systems that do not have structured annotations simply communicate a single annotation with no author or time.

This element may need to be included in narrative because of the potential for modifying information.

Annotations SHOULD NOT be used to communicate "modifying" information that could be computable (this is a SHOULD because enforcing user behavior is nearly impossible).

Annotation is used in the following places: AllergyIntolerance, CarePlan, CareTeam, ChargeItem, ClinicalImpression, Communication, CommunicationRequest, Condition, Contract, Device, DeviceDefinition, DeviceRequest, DeviceUseStatement, EffectEvidenceSynthesis, Evidence, EvidenceVariable, FamilyMemberHistory, Goal, GuidanceResponse, ImagingStudy, Immunization, Invoice, List, Media, MedicationAdministration, MedicationDispense, MedicationRequest, MedicationStatement, NutritionOrder, Observation, Procedure, RequestGroup, ResearchStudy, RiskAssessment, RiskEvidenceSynthesis, ServiceRequest, Specimen, Task and VisionPrescription

2.24.0.19 Open Type Element

Some elements do not have a specified type. The type is represented by the wildcard symbol "*". In these cases, the element type may be one of the following:

Primitive Types

- base64Binary
- boolean
- canonical
- code
- date
- dateTime
- decimal
- id
- instant
- integer
- markdown
- oid
- positiveInt
- string
- time
- unsignedInt
- uri
- url
- uuid

Data Types

- Address
- Age
- Annotation
- Attachment
- CodeableConcept
- Coding
- ContactPoint
- Count
- Distance
- Duration
- HumanName
- Identifier
- Money
- Period
- Quantity
- Range
- Ratio
- Reference
- SampledData
- Signature
- Timing

MetaDataTypes

- ContactDetail
- Contributor
- DataRequirement
- Expression
- ParameterDefinition
- RelatedArtifact

• TriggerDefinition
• UsageContext
Special Types
• Dosage
• Meta
The element name ends with "[x]", and this is replaced with the Title cased name of the data type.
Open references are used in the following places: Parameters, StructureMap and Task

2.24.0.20 Other Types

The following types are defined as part of the data types, but are documented elsewhere in the specification:
• Resource - the conceptual base class for all resources
• Reference - for references from one resource to another
• Extension - used to convey additional data in a resource
• Narrative - conveys a human-readable representation of the content of a resource

2.24.1 Data Type Examples

URL: https://hl7.org/fhir/R4/datatypes-examples.html

2.24.1 Data Type Examples

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative

This page includes additional examples of the data types, based on common usages and questions
Table of Contents
Primitive Types
Complex Types

2.24.1.1 Primitive Types

A boolean true value:

```
<active value="true" />
```

A negative integer value:

```
<score value="-14" />
```

A high-precision decimal value:

```
<pi value="3.14159265358979323846264338327950288419716939937510" />
```

A stream of bytes, base64 encoded:

```
<data value="/9j/4...KAP//Z" /> <!-- covers many lines -->
```

A Unicode string:

```
<caption value="Noodles are called ?? in Chinese" />
```

A URI that points to a website:

```
<reference value="http://hl7.org/fhir" />
```

A URI that is a urn:

```
<id value="urn:isbn:0451450523" />
```

A date of birth:

```
<date value="1951-06-04" />
```

An approximate date of birth:

```
<date value="1951-06" />
```

The instant a document was created, including time zone:

```
<instant value="2013-06-08T10:57:34+01:00" />
```

The instant a document was created, expressed in UTC, with milliseconds:

```
<instant value="2013-06-08T09:57:34.2112Z" />
```

2:35pm in the afternoon:

```
<time value="14:35" />
```

2.24.1.2 String Patterns

A URI that is the root oid of HL7:

```
<root value="urn:oid:2.16.840.1.113883" />
```

A URI that is a uuid:

```
<id value="urn:uuid:a5afddf4-e880-459b-876e-e4591b0acc11" />
```

A code:

```
<code value="acq4+acq5" />
```

A code with single internal space:

```
<code value="Question 4b" />
```

A numeric id:

```
<id value="314" />
```

An alphanumeric id:

```
<id value="alpha-gamma-14" />
```

2.24.1.3 Attachment

See also Base Definition, Detailed Descriptions, Mappings, Profiles & Extensions and R2 Conversions.

A PDF document:

```
<document>
  <contentType value="application/pdf" />
  <language value="en" />
  <data value="/9j/4...KAP//Z" /> <!-- covers many lines -->
  <title value="Definition of Procedure" />
</document>
```

```
document : {
  contentType :  { value : "application/pdf" },
  language : { value : "en" },
  data :  { value : "/9j/4...KAP//Z"},
  title :  { value : "Definition of Procedure" }
}
```

Since the JSON examples have the same structure as the XML, only XML is shown for the rest of the examples.

A reference to a DICOM image via WADO:

```
<image>
  <contentType value="application/dicom" />
  <url value="http://10.1.2.3:1000/wado?requestType=WADO&amp;wado_details..." />
  <hash value="EQH/..AgME" />
</image>
```

2.24.1.4 Identifier

See also Base Definition, Detailed Descriptions, Mappings, Profiles & Extensions and R2 Conversions.

Examples

A primary key from an application table (an OID in the space allocated by HL7 to some organization to further sub-allocate):

```
<identifier>
  <use value="official" />
  <system value="urn:oid:2.16.840.1.113883.16.4.3.2.5" />
  <value value="123" />
</identifier>
```

A patient identifier defined by a hospital:

```
<identifier>
  <use value="official" />
  <system value="http://www.acmehosp.com/patients" />
  <value value="44552" />
  <period>
    <start value="2003-05-03" />
  </period>
</identifier>
```

In this case, the period is used to track when the identifier was first assigned to the patient.

An identifier that refers to a patient FHIR resource on a particular system:

```
<identifier>
 <system value="urn:ietf:rfc:3986" />
 <value value="http://pas-server/xxx/Patient/443556" />
</identifier>
```

This is not a resource reference - it's a logical reference by the patient identifier.

A UUID:

```
<identifier>
  <use value="temp" />
  <system value="urn:ietf:rfc:3986" />
  <value value="urn:uuid:a76d9bbf-f293-4fb7-ad4c-2851cac77162" />
</identifier>
```

UUIDs are often used for temporary identifiers, though this is not necessary.

A DICOM OID:

```
<identifier>
  <system value="urn:dicom:uid" />
  <value value="urn:oid:2.16.124.113543.6003.189642796.63084.16748.2599092901" />
</identifier>
```

The system value urn:dicom:uid can be used for any OID issued under the DICOM arrangements for OIDs. Such OIDs are typically found in DICOM messages, but this is not a requirement for using urn:dicom:uid

A US SSN:

```
<identifier>
  <use value="usual" />
  <type>
    <coding>
      <system value="http://terminology.hl7.org/CodeSystem/v2-0203"/>
      <code value="SSN"/>
    </coding>
  </type>
  <system value="http://hl7.org/fhir/sid/us-ssn" />
  <value value="000111111" />
</identifier>
```

Notes:

• US SSNs are often presented like this: 000-11-1111, the dashes are for presentation and should be removed, as specified in the definition of ssn-us

• The use of "usual" means that this institution prefers to use SSN when identifying the patient

A medical record number assigned on 5-July 2009:

```
<identifier>
  <use value="usual" />
  <type>
    <coding>
      <system value="http://terminology.hl7.org/CodeSystem/v2-0203"/>
      <code value="MRN"/>
    </coding>
  </type>
  <system value="urn:oid:0.1.2.3.4.5.6.7" />
  <value value="2356" />
  <period>
    <start value="2009-07-05" />
  </period>
</identifier>
```

2.24.1.4.1 V3 Mapping Examples for Identifier

A root only:

```
<v3:id root="0.1.2.3.4.5"/>
```

becomes:

```
<identifier>
  <system value="urn:ietf:rfc:3986" />
  <value value="urn:oid:0.1.2.3.4.5" />
</identifier>
```

A root with an extension:

```
<v3:id root="0.1.2.3.4.5" extension="13412"/>
```

becomes:

```
<identifier>
  <system value="urn:oid:0.1.2.3.4.5" />
  <value value="13412" />
</identifier>
```

See v3 Mapping Notes

2.24.1.5 Coding

See also Base Definition, Detailed Descriptions, Mappings, Profiles & Extensions and R2 Conversions.

Examples

A simple code for headache, in ICD-10:

```
<code>
  <system value="http://hl7.org/fhir/sid/icd-10" />
  <code value="G44.1" />
</code>
```

A SNOMED CT expression:

```
<problem>
  <system value="http://snomed.info/sct" />
  <code value="128045006:{363698007=56459004}" />
</problem>
```

2.24.1.6 CodeableConcept

See also Base Definition, Detailed Descriptions, Mappings, Profiles & Extensions and R2 Conversions.

Examples

A simple code for headache initially coded in SNOMED CT (by picking the SNOMED CT code from a pick-list), and then translated to ICD-10:

```
<concept>
```

```
  <coding>
    <system value="http://hl7.org/fhir/sid/icd-10" />
    <code value="R51" />
  </coding>
  <coding>
    <system value="http://snomed.info/sct" />
    <code value="25064002" />
    <display value="Headache" />
    <userSelected value="true" />
  </coding>
  <text value="general headache" />
</concept>
```

A concept represented in an institution's local coding systems for unit for which no UCUM equivalent exists:

```
<unit>
  <coding>
    <system value="urn:oid:2.16.840.1.113883.19.5.2" />
    <code value="tab" />
    <display value="Tablet" />
  </coding>
  <coding>
    <system value="http://unitsofmeasure.org" />
  </coding>
</unit>
```

A SNOMED CT expression:

```
<diagnosis>
  <coding>
    <system value="http://snomed.info/sct" />
    <code value="128045006:{363698007=56459004}" />
  </coding>
  <text value="Cellulitis of the foot" />
</diagnosis>
```

In this case, there is no display element, because no display is defined for SNOMED CT expressions.

Using the value set:

The results on a urinalysis strip:

```
<valueCoding>
  <system value="http://example.org/codes/simple-grades" />
  <code value="+" />
</valueCoding>
```

And where the value set would be something like this:

```
<ValueSet xmlns="http://hl7.org/fhir">
  <text>
    <status value="generated"/>
    <div xmlns="http://www.w3.org/1999/xhtml">
      <p>Possible Clinistix codes: neg, trace, +, ++, and +++</p>
    </div>
  </text>
  <url value="http://hl7.org/fhir/ValueSet/clinistix"/>
  <name value="Codes for Clinistix"/>
  <publisher value="HL7"/>
  <contact>
      <name value="FHIR project team"/>
      <telecom>
        <system value="url"/>
        <value value="http://hl7.org/fhir"/>
      </telecom>
  </contact>
  <description value="Clinistix Codes"/>
  <status value="draft"/>
  <experimental value="true"/>
  <date value="2013-10-01"/>
  <compose>
    <include>
      <system value="http://example.org/codes/simple-grades"/>
      <concept>
        <code value="neg"/>
      </concept>
      <concept>
        <code value="trace"/>
      </concept>
      <concept>
        <code value="+"/>
      </concept>
      <concept>
        <code value="+"/>
      </concept>
      <concept>
        <code value="++"/>
      </concept>
      <concept>
        <code value="+++"/>
      </concept>
    </include>
  </compose>
```

```
</ValueSet>
```

2.24.1.7 Quantity

See also Base Definition, Detailed Descriptions, Mappings, Profiles & Extensions and R2 Conversions.

Examples

A duration:

```
<time>
  <value value="25" />
  <unit value="sec" />
  <system value="http://unitsofmeasure.org" />
  <code value="s" />
</time>
```

A concentration where the value was out of range:

```
<result>
  <value value="40000" />
  <comparator value="&gt;" />
  <unit value="mcg/L" />
  <system value="http://unitsofmeasure.org" />
  <code value="ug" />
</result>
```

An amount of prescribed medication:

```
<dose>
  <value value="3" />
  <unit value="capsules" />
  <system value="http://snomed.info/sct" />
  <code value="385049006" />
</dose>
```

2.24.1.8 Money

See also Base Definition, Detailed Descriptions, Mappings, Profiles & Extensions and R2 Conversions.

Examples

US Dollars:

```
<time>
  <value value="25.00" />
  <currency value="USD" />
</time>
```

Vietnamese Dong:

```
<result>
  <value value="410000" />
  <currency value="VND" />
</result>
```

A quantity that represents a currency amount - used in Ratio where Money is not used:

```
<numerator>
  <value value="25.45" />
  <unit value="US$" />
  <system value="urn:iso:std:iso:4217" />
  <code value="USD" />
</numerator>
```

(See full example on Ratio)

2.24.1.9 Range

See also Base Definition, Detailed Descriptions, Mappings, Profiles & Extensions and R2 Conversions.

Examples

Range of Quantity (distance):

```
<estimate>
 <low>
   <value value="1.6" />
   <unit value="m" />
 </low>
 <high>
   <value value="1.9" />
   <unit value="m" />
 </high>
</estimate>
```

2.24.1.10 Ratio

See also Base Definition, Detailed Descriptions, Mappings, Profiles & Extensions and R2 Conversions.

Examples

Titer (Ratio of integer:integer)

```
<result>
 <numerator>
   <value value="1" />
 </numerator>
```

```
   <denominator>
     <value value="128" />
   </denominator>
  </result>
```

Unit cost (Ratio of Money(as Quantity):Quantity):

```
  <charge>
   <numerator>
     <value value="103.50" />
     <unit value="US$" />
     <code value="USD" />
     <system value="urn:iso:std:iso:4217" />
   </numerator>
   <denominator>
     <value value="1" />
     <unit value="day" />
     <code value="day" />
     <system value="http://unitsofmeasure.org" />
   </denominator>
  </charge>
```

2.24.1.11 Period

See also Base Definition, Detailed Descriptions, Mappings, Profiles & Extensions and R2 Conversions.

Examples

23rd May 2011 to 27th May, including 27th May:

```
  <coverage>
   <start value="2011-05-23" />
   <end value="2011-05-27" />
  </coverage>
```

2.24.1.12 SampledData

See also Base Definition, Detailed Descriptions, Mappings, Profiles & Extensions and R2 Conversions.

Example

The output from an EKG device:

```
 <sampledData>
  <origin>
   <value value="0"/>
   <unit value="µV"/>
   <system value="http://unitsofmeasure.org"/>
   <code value="uV"/>
  </origin>
  <period value="2"/>
  <factor value="2.5"/>
  <dimensions value="1"/>
  <data value="-4 -13 -18 -18 -18 -17 -16 -16 -16 -16 -16 -17 -18 -18 -18 ...."/>
 </sampledData>
```

2.24.1.13 HumanName

See also Base Definition, Detailed Descriptions, Mappings, Profiles & Extensions and R2 Conversions.

A Simple example

```
<name>
  <family value="Everyman" />
  <given value="Adam" />
  <given value="A." />
</name>
```

Composite names

```
<name>
  <family value="Contrata" />
  <given value="Mary Jane" />
</name>
```

These cases can be quite ambiguous - is "Mary Jane" one name, or two? Different systems, and data enterers may treat this differently, and the person themselves might not know. Parts are allowed to contain spaces, but systems should consider how to treat these cases. Composite names separated by "-" should be treated as a single name part.

A common pattern: a person is called by a name other than that expected from their official name (first given name in most cultures).

```
<name>
  <use value="official" />
  <family value="Chalmers" />
  <given value="Peter" />
  <given value="James" />
</name>
<name>
  <use value="usual" />
  <given value="Jim" />
</name>
```

This same pattern is often encountered with immigrants, who retain their real name for official use, but adopt a localized name for everyday use:

```
<name>
  <use value="official" />
  <family value="Sczypinski" />
  <given value="Piotr" />
  <given value="Andre" />
</name>
<name>
  <use value="usual" />
  <family value="Skipper" />
  <given value="Jim" />
</name>
```

Some people may also have a nickname that needs to be recorded:

```
<name>
  <use value="official" />
  <family value="Brown" />
  <given value="Steven" />
</name>
<name>
  <use value="nickname" />
  <given value="Junior" />
</name>
```

Note that there is some ambiguity in real life as to whether a non-official name is the patient's usual name, or whether it's their nickname. In principle, nick names are used occasionally and informally, whereas the usual name is used consistently, and in formal contexts. However, there is no formal criteria for the differentiating these usages, and even within a culture, people may vary in their cases.

Karen van Hentenryck is of Dutch origin, and the "van" is a voorvoegsel.

```
<name>
  <use value="official" />
  <family value="van Hentenryck" >
    <extension url="http://hl7.org/fhir/StructureDefinition/humanname-own-prefix" >
     <valueString value="van" />
    </extension>
    <extension url="http://hl7.org/fhir/StructureDefinition/humanname-own-name">
      <valueString value="Hentenryck" />
    </extension>
  </family>
  <given value="Karen" />
</name>
```

See the Extensibility Example for more information about the use of extensions.

Complex example from Germany: Dr.phil. Regina Johanna Maria von Hochheim-Weilenfels, NCFSA. This example shows extensive use of multiple given names, prefixes, suffixes, for academic degrees, nobility titles, and professional designations.

```
<name>
  <use value="official" />
  <family value="von Hochheim-Weilenfels">
    <extension url="http://hl7.org/fhir/StructureDefinition/humanname-own-prefix" >
      <valueString value="von" />
    </extension>
    <extension url="http://hl7.org/fhir/StructureDefinition/humanname-own-name">
      <valueString value="Hochheim-Weilenfels" />
    </extension>
  </family>
  <given value="Regina" />
  <given value="Johanna" />
  <given value="Maria" />
  <prefix value="Dr. phil.">
    <extension url="http://hl7.org/fhir/StructureDefinition/iso21090-EN-qualifier" >
      <valueCode value="AC" />
    </extension>
  </prefix>
  <suffix value="NCFSA" />
</name>
<name>
  <use value="maiden" />
  <family value="Hochheim" />
</name>
```

This example makes use of the ISO 21090 extensions to carry the rarely used ISO 21090 qualifier attribute "AC".

Japanese example in the three forms: ideographic (Kanji), syllabic (Hiragana) and alphabetic (Romaji).

```
<name>
  <extension url="http://hl7.org/fhir/StructureDefinition/iso21090-EN-representation">
      <valueCode value="IDE" />
  </extension>
  <family value="□□" />
  <given value="□□" />
  </name>
<name>
```

```
  <extension url="http://hl7.org/fhir/StructureDefinition/iso21090-EN-representation">
      <valueCode value="SYL" />
  </extension>
 <family value="□□□" />
 <given value="□□□" />
</name>
<name>
  <extension url="http://hl7.org/fhir/StructureDefinition/iso21090-EN-representation">
      <valueCode value="ABC" />
  </extension>
  <family value="KIMURA" />
  <given value="MICHIO" />
</name>
```

The three forms may be differentiated by the character subset each contains, but some systems require the differentiation to be made manually, which can be done using the ED representation extension.

Russian example in the two forms: Cyrillic and Latin:

```
<name>
  <family value="ЕМЕЛИН" />
  <given value="ИВАН" />
  <given value="ВЛАДИМИРОВИЧ" />
</name>
<name>
  <family value="EMELIN" />
  <given value="IVAN" />
  <given value="VLADIMIROVICH" />
</name>
```

In Russian usage, these names are known as the domestic and foreign names respectively. The two forms are differentiated by the character subset each contains.

Scandinavian example: Erikson is the family name. Jan Erik are the given names, and Östlund the family name of the mother, which is taken as a Mellannamn.

```
<name>
  <use value="official" />
  <family value="Erikson" />
  <given value="Jan" />
  <given value="Erik" />
  <given value="Östlund">
    <extension url="http://hl7.org/fhir/StructureDefinitioniso-20190#name-qualifier" >
      <valueCoding>
         <code value="MID" />
         <system value="http://terminology.hl7.org/CodeSystem/v3-EntityNamePartQualifier2" />
      </valueCoding>
    </extension>
  </given>
</name>
```

This example makes use of the ISO 21090 extension to carry the culture specific ISO 21090 qualifier attribute "MID" for the Mellannamn.

Then Jan Erikson has a daughter, Karin, with his wife Margrete Hansen. The first communication of the new born name is "Margrete Jente" (Margrete's Girl) and the mother's family name, not the given name (Karin). The father's Family name is not used at all. This is a known temporary name assigned directly after the birth of the child.

```
<name>
  <use value="temp" />
  <!-- use could be OR+OLD, depends how record keeping is done -->
  <family value="Hansen" />
  <given value="Margrete Jente" />
</name>
```

The baby's name is subsequently changed to the fathers' family name, and to use the mother's name as mellomnamn.

```
<name>
  <use value="official" />
  <family value="Erikson" />
  <given value="Karin" />
  <given value="Hansen">
    <extension url="http://hl7.org/fhir/StructureDefinitioniso-20190#name-qualifier" >
      <valueCoding>
         <code value="MID" />
         <system value="http://terminology.hl7.org/CodeSystem/v3-EntityNamePartQualifier2" />
      </valueCoding>
    </extension>
  </given>
</name>
```

Later, Karin gets married to Per Berg, and decides to adopt Berg as her family name, and also decides to use Erikson as the mellom navn. (Note: Karin could have chosen to use another mellom navn, e.g. the family name of her mother, her father or other family names as specified by naming laws of the country in question).

```
<name>
  <use value="old" />
  <family value="Erikson" />
  <given value="Karin" />
```

```
  <given value="Hansen">
    <extension url="http://hl7.org/fhir/StructureDefinitioniso-20190#name-qualifier" >
      <valueCoding>
        <code value="MID" />
        <system value="http://terminology.hl7.org/CodeSystem/v3-EntityNamePartQualifier2" />
      </valueCoding>
    </extension>
  </given>
</name>
<name>
  <use value="official" />
  <family value="Berg" />
  <given value="Karin" />
  <given value="Erikson">
    <extension url="http://hl7.org/fhir/StructureDefinitioniso-20190#name-qualifier" >
      <valueCoding>
        <code value="MID" />
        <system value="http://terminology.hl7.org/CodeSystem/v3-EntityNamePartQualifier2" />
      </valueCoding>
    </extension>
  </given>
</name>
<name>
  <use value="usual" />
  <family value="Berg" />
  <given value="Karin" />
</name>
```

2.24.1.13.1 W3C International Examples

These examples are taken from the W3C International Examples , which should be consulted for further information.

A patronymic is "The part of a name that links to the genealogy":

```
<name>
  <text value="Björk Guðmundsdóttir"/>
  <family value="Guðmundsdóttir"/>
  <given value="Björk"/>
</name>
```

A patronymic with a "son/daughter of" appellation:

```
<name>
  <text value="Isa bin Osman"/>
  <family value="bin Osman"/>
  <given value="Isa"/>
</name>
```

A Chinese name with a generational name:

```
<name>
  <text value="□□□"/> <!-- left to right -->
  <family value="□"/>
  <given value="□□"/>
</name>
<name>
  <text value="Mao Ze Dong"/> <!-- left to right -->
  <family value="Mao"/>
  <given value="Ze Dong"/>
</name>
```

Note that many systems in China do not store family and given names separately, and just use text.

Additional Western name (see also example above):

```
<name>
  <use value="official" />
  <family value="Yao" />
  <given value="Ming" />
</name>
<name>
  <use value="usual" />
  <given value="Fred" />
</name>
```

Composite Family name:

```
<name>
  <family value="Carreño Quiñones" >
    <extension url="http://hl7.org/fhir/StructureDefinition/humanname-fathers">
        <valueString value="Carreño" />
    </extension>
    <extension url="http://hl7.org/fhir/StructureDefinition/humanname-mothers">
        <valueString value="Quiñones" />
    </extension>
  </family>
  <given value="María-Jose" />
</name>
```

Note that it is optional whether to break down the family name to mother's and father's parts, and not always of value, as in this Brazilian Example:

```
<name>
```

```
  <family value="Santos Tavares Melo Silva" />
  <given value="José" />
  <given value="Eduardo" />
</name>
```

Note that this naming pattern can become quite extreme ("Pedro de Alcântara João Carlos Leopoldo Salvador Bibiano Francisco Xavier de Paula Leocádio Miguel Gabriel Rafael Gonzaga"). A more practical example is

```
<name>
  <family value="Costa Teixeira" >
    <extension url="http://hl7.org/fhir/StructureDefinition/humanname-fathers">
        <valueString value="Costa" />
    </extension>
    <extension url="http://hl7.org/fhir/StructureDefinition/humanname-mothers">
        <valueString value="Teixeira" />
    </extension>
  </family>
  <given value="Manuel" />
</name>
```

After marriage, this may change to

```
<name>
  <family value="Costa Teixeira Sanches" >
    <extension url="http://hl7.org/fhir/StructureDefinition/humanname-fathers">
        <valueString value="Costa" />
    </extension>
    <extension url="http://hl7.org/fhir/StructureDefinition/humanname-mothers">
        <valueString value="Teixeira" />
    </extension>
    <extension url="http://hl7.org/fhir/StructureDefinition/humanname-partners-name">
        <valueString value="Sanches" />
    </extension>
  </family>
  <given value="Manuel" />
</name>
```

Russian Examples (using Cyrillic):

```
<name>
  <family value="Ельцин" />
  <given value="Борис" />
  <given value="Николаевич" />
</name>

<name>
  <family value="Ельцина" />
  <given value="Наина" />
  <given value="Иосифовна" />
</name>
```

Example with Initial:

```
<name>
  <family value="Public" />
  <given value="John" />
  <given value="Q." />
</name>
```

Other Examples:

```
<name>
  <text value="Velikkakathu Sankaran Achuthanandan"/>
  <family value="Velikkakathu" />
  <given value="Sankaran" />
  <given value="Achuthanandan" />
</name>
<name>
  <text value="Kogaddu Birappa Timappa Nair"/>
  <family value="Nair" />
  <given value="Birappa" />
  <given value="Timappa" />
  <prefix value="Kogaddu" />
</name>
<name>
  <text value="Aditya Pratap Singh Chauhan"/>
  <family value="Singh" />
  <given value="Aditya" />
  <given value="Pratap" />
  <suffix value="Chauhan" />
</name>
<name>
  <text value="Madurai Mani Iyer"/>
  <given value="Mani" />
  <prefix value="Madurai" />
  <suffix value="Iyer" />
</name>
<name>
  <text value="Abu Karim Muhammad al-Jamil ibn Nidal ibn Abdulaziz al-Filistini"/>
  <family value="ibn Nidal ibn Abdulaziz" />
  <given value="Muhammad" />
  <given value="al-Jamil" />
  <prefix value="Abu Karim" />
```

```
  <suffix value="al-Filistini" />
</name>
```

Todo: need to discuss this with Indian / Arabic implementers. Note that collecting and storing the text element makes the primary purpose of the structured parts for index/searching, and fidelity of the name parts is not critical.

2.24.1.14 Address

See also Base Definition, Detailed Descriptions, Mappings, Profiles & Extensions and R2 Conversions.

Note about these examples: the most important element in an address is the 'text' element - this defines what is printed on the envelope, the actual postal address. The other data elements are provided to support either data analysis based on a patient's address, or for the many systems that exchange fully structured addresses specific to a particular culture. For this reason, these examples focus on the structured data elements of the address.

Example

HL7 office's address.

```
  <address>
   <use value="work" />
   <text value="1050 W Wishard Blvd
RG
         5th floor
Indianapolis, IN 46240" />
   <line value="1050 W Wishard Blvd" />
   <line value="RG 5th floor" />
   <city value="Indianapolis" />
   <state value="IN" />
   <postalCode value="46240" />
  </address>
```

A UK example address, with the county 'HUDDERSFIELD'.

```
  <address>
    <text value="1 Back Lane&#13;&#10;Holmfirth&#13;&#10;HUDDERSFIELD&#13;&#10;HD7 1HQ"/>
    <line value="1 Back Lane"/>
    <city value="Holmfirth"/>
    <district value="HUDDERSFIELD"/>
    <postalCode value="HD7 1HQ"/>
  </address>
```

A temporary postal address - i.e. an address that it doesn't make sense to try and visit.

```
  <address>
    <use value="temp"/>
    <type value="postal"/>
    <line value="PO Box 31445"/>
    <city value="Erewhon"/>
    <postalCode value="0001"/>
  </address>
```

The next set of examples are taken from the official international postal union examples.

Rue Lougoraïa 12, app. 10 | thoroughfare type, name and number, apartment
Korolevo | locality
223016 NOVY DVOR | postcode + post office name
Minsk | District name of district
Minsk | Region name of region
BELARUS | Country

```
 <address>
   <line value="Rue Lougoraïa 12, app. 10"/>
   <city value="Korolevo"/>
   <district value="Minsk"/>
   <state value="Minsk" />
   <country value="BELARUS" />
 </address>
```

Protea Apt 12 | apartment number
22 Ally Hassan Mwinyi | premise + street
14111 MSASANI | postcode + locality
DAR ES SALAM | province
TANZANIA | country

```
 <address>
   <line value="Protea Apt 12"/>
   <line value="22 Ally Hassan Mwinyi"/>
   <city value="MSASANI"/>
   <state value="DAR ES SALAM" />
   <postalCode value="14111"/>
   <country value="TANZANIA" />
 </address>
```

15 Shiri | premise + sub-locality 2
Kimashuku Village | sub-locality
25204 MACHAME KUSINI | postcode + locality
HAI | sub-province

KILIMANJARO | province
TANZANIA | country

```
<address>
  <extension url="http://hl7.org/fhir/StructureDefinition/iso21090-ADXP-precinct">
    <valueCode value="Kimashuku Village"/>
  </extension>
  <line value="15 Shiri"/>
  <city value="MACHAME KUSINI"/>
  <district value="HAI"/>
  <state value="KILIMANJARO" />
  <postalCode value="25204"/>
  <country value="TANZANIA" />
</address>
```

705-1104 | building number + door
56, Dalgubeol-daero 323beon-gil, Suseong-gu | street
Daegu | city
Rep. OF KOREA | country
706-907 | postcode

```
<address>
  <line value="705-1104"/>
  <line value="56, Dalgubeol-daero 323beon-gil, Suseong-gu"/>
  <city value="Daegu"/>
  <country value="Rep. OF KOREA" />
  <postalCode value="706-907"/>
</address>
```

3F | building floor
42, Toegye-ro 77beon-gil | street
Chuncheon-si | city
Gangwon-do | province
Rep. OF KOREA | country
200-066 | postcode

```
<address>
  <line value="3F"/>
  <line value="42, Toegye-ro 77beon-gil"/>
  <city value="Chuncheon-si"/>
  <state value="Gangwon-do" />
  <country value="Rep. OF KOREA" />
  <postalCode value="200-066"/>
</address>
```

Ga-B101 | building number + door
136, Sesil-ro | street
Busan | city
Rep. OF KOREA | country
612-837 | postcode

```
<address>
  <line value="Ga-B101"/>
  <line value="136, Sesil-ro"/>
  <city value="Busan"/>
  <country value="Rep. OF KOREA" />
  <postalCode value="612-837"/>
</address>
```

2.24.1.15 ContactPoint

See also Base Definition, Detailed Descriptions, Mappings, Profiles & Extensions and R2 Conversions.

Example

Home phone number:

```
<telecom>
 <system value="phone" />
 <value value="(555) 675 5745" />
 <use value="home" />
</telecom>
```

In ISO 21090, CDA and other HL7 v3 specifications, the TEL data type requires that the telephone number be represented as a formal URI in RFC 3166 syntax. Many CDA documents violate this rule, since the telephone numbers are not collected from the user in conformance with the underlying rules (particularly around extensions, notes about use, etc.), so the base FHIR type does not require the phone number to be conformant. An ISO 21090 extension can be used to convey this if desired:

Example

Formally correct URI:

```
<telecom>
  <extension url="http://hl7.org/fhir/StructureDefinition/iso21090-TEL-address" >
    <valueUri value="tel:+15556755745" />
  </extension>
  <system value="phone" />
  <value value="(555) 675 5745" />
  <use value="home" />
```

```
</telecom>
```

2.24.1.16 Timing

See also Base Definition, Detailed Descriptions, Mappings, Profiles & Extensions and R2 Conversions.

Example

A series of appointments for radiotherapy:

```
<schedule>
  <event value="2012-01-07T09:00:00+10:00" />
  <event value="2012-01-14T09:00:00+10:00" />
  <event value="2012-01-22T11:00:00+10:00" />
</schedule>
```

BID (twice a day) (no start or end specified):

```
<schedule>
 <repeat>
   <frequency value="2" />
   <period value="1" />
   <periodUnit value="d" />
 </repeat>
</schedule>
```

1/2 an hour before breakfast for 10 days from 23-Dec 2011:

```
<schedule>
  <repeat>
    <boundsPeriod>
      <start value="2011-12-23" />
      <end value="2012-01-02" />
    </boundsPeriod>
    <when value="ACM" />
    <offset value="30" />
  </repeat>
</schedule>
```

Note that the end date is inclusive like the end date of a Period.

TID, for 14 days:

```
<schedule>
  <repeat>
    <boundsDuration>
      <value value="14" />
      <unit value="d" />
      <system value="http://unitsofmeasure.org" />
      <code value="d" />
    </boundsDuration>
    <frequency value="3" />
    <period value="1" />
    <periodUnit value="d" />
  </repeat>
</schedule>
```

BID, start on 7/1/2015 at 1:00 PM:

```
<schedule>
  <repeat>
    <boundsPeriod>
      <start value="2015-07-01T13:00:00" />
    </boundsPeriod>
    <frequency value="2" />
    <period value="1" />
    <periodUnit value="d" />
  </repeat>
</schedule>
```

Take just once, with no specified time

```
<schedule>
  <repeat>
    <count value="1" />
  </repeat>
</schedule>
```

2.24.1.17 Signature

See also Base Definition, Detailed Descriptions, Mappings, Profiles & Extensions and R2 Conversions.

Example

todo

```
<signature>
  <!-- todo -->
</signature>
```

2.24.1.18 Annotation

See also Base Definition, Detailed Descriptions, Mappings, Profiles & Extensions and R2 Conversions.

Example

todo

```
<Annotation>
  <!-- todo -->
</Annotation>
```

2.24.2 Data Type Detailed Descriptions

URL: https://hl7.org/fhir/R4/datatypes-definitions.html

2.24.2 Data Type Detailed Descriptions

FHIR Infrastructure Work Group | Maturity Level: Normative | Standards Status: Normative

| This page has been approved as part of an ANSI standard. See the Infrastructure Package for further details.

This page provides the detailed descriptions for the data types

Table of Contents

Primitive Types ElementinstanttimedatedateTimedecimalbooleanintegerstringuribase64Binarycodeidoidunsignedintpositiveintmarkdownurlcanonicaluuid | Complex Types ElementIdentifierHumanNameAddressContactPointTimingQuantitySimpleQuantityAttachmentRangePeriodRatioCodeableConceptCodingSampledDataAgeDistanceDurationCountMoneyMoneyQuantityAnnotationSignatureBackboneElement

2.24.2.1 Primitive Types

See also Base Definition, Examples, Profiles & Extensions, Mappings and R2 Conversions.

todo

2.24.2.2 Attachment

See also Base Definition, Examples, Profiles & Extensions, Mappings and R2 Conversions.

Attachment
Standards Status | This element has a standards status of "Normative"
Element Id | Attachment
Definition | For referring to data content defined in other formats.
Cardinality | 0..*
Type | Element
Requirements | Many models need to include data defined in other specifications that is complex and opaque to the healthcare model. This includes documents, media recordings, structured data, etc.
Comments | When providing a summary view (for example with Observation.value[x]) Attachment should be represented with a brief display text such as "Signed Procedure Consent".
Invariants | Defined on this element att-1 Rule If the Attachment has data, it SHALL have a contentType data.empty() or contentType.exists()
Defined on this element
att-1 | Rule | If the Attachment has data, it SHALL have a contentType | data.empty() or contentType.exists()
Attachment.contentType
Element Id | Attachment.contentType
Definition | Identifies the type of the data in the attachment and allows a method to be chosen to interpret or render the data. Includes mime type parameters such as charset where appropriate.
Cardinality | 0..1
Terminology Binding | MimeType (Required)
Type | code
Requirements | Processors of the data need to be able to know how to interpret the data.
Summary | true
Attachment.language
Element Id | Attachment.language
Definition | The human language of the content. The value can be any valid value according to BCP 47.
Cardinality | 0..1
Terminology Binding | Common Languages (Preferred but limited to All Languages)
Type | code
Requirements | Users need to be able to choose between the languages in a set of attachments.
Summary | true
Attachment.data
Element Id | Attachment.data
Definition | The actual data of the attachment - a sequence of bytes, base64 encoded.
Cardinality | 0..1
Type | base64Binary
Requirements | The data needs to able to be transmitted inline.
Comments | The base64-encoded data SHALL be expressed in the same character set as the base resource XML or JSON.
Attachment.url
Element Id | Attachment.url
Definition | A location where the data can be accessed.
Cardinality | 0..1
Type | url
Requirements | The data needs to be transmitted by reference.
Summary | true

Comments | If both data and url are provided, the url SHALL point to the same content as the data contains. Urls may be relative references or may reference transient locations such as a wrapping envelope using cid: though this has ramifications for using signatures. Relative URLs are interpreted relative to the service url, like a resource reference, rather than relative to the resource itself. If a URL is provided, it SHALL resolve to actual data.
Attachment.size
Element Id | Attachment.size
Definition | The number of bytes of data that make up this attachment (before base64 encoding, if that is done).
Cardinality | 0..1
Type | unsignedInt
Requirements | Representing the size allows applications to determine whether they should fetch the content automatically in advance, or refuse to fetch it at all.
Summary | true
Comments | The number of bytes is redundant if the data is provided as a base64binary, but is useful if the data is provided as a url reference.
Attachment.hash
Element Id | Attachment.hash
Definition | The calculated hash of the data using SHA-1. Represented using base64.
Cardinality | 0..1
Type | base64Binary
Requirements | Included so that applications can verify that the contents of a location have not changed due to technical failures (e.g., storage rot, transport glitch, incorrect version).
Summary | true
Comments | The hash is calculated on the data prior to base64 encoding, if the data is based64 encoded. The hash is not intended to support digital signatures. Where protection against malicious threats a digital signature should be considered, see Provenance.signature for mechanism to protect a resource with a digital signature.
Attachment.title
Element Id | Attachment.title
Definition | A label or set of text to display in place of the data.
Cardinality | 0..1
Type | string
Requirements | Applications need a label to display to a human user in place of the actual data if the data cannot be rendered or perceived by the viewer.
Summary | true
Attachment.creation
Element Id | Attachment.creation
Definition | The date that the attachment was first created.
Cardinality | 0..1
Type | dateTime
Requirements | This is often tracked as an integrity issue for use of the attachment.
Summary | true

2.24.2.3 Identifier

See also Base Definition, Examples, Profiles & Extensions, Mappings and R2 Conversions.

Identifier
Standards Status | This element has a standards status of "Normative"
Element Id | Identifier
Definition | An identifier - identifies some entity uniquely and unambiguously. Typically this is used for business identifiers.
Cardinality | 0..*
Type | Element
Requirements | Need to be able to identify things with confidence and be sure that the identification is not subject to misinterpretation.
Summary | true
Identifier.use
Element Id | Identifier.use
Definition | The purpose of this identifier.
Cardinality | 0..1
Terminology Binding | IdentifierUse (Required)
Type | code
Is Modifier | true (Reason: This is labeled as "Is Modifier" because applications should not mistake a temporary id for a permanent one.)
Requirements | Allows the appropriate identifier for a particular context of use to be selected from among a set of identifiers.
Summary | true
Comments | Applications can assume that an identifier is permanent unless it explicitly says that it is temporary.
Identifier.type
Element Id | Identifier.type
Definition | A coded type for the identifier that can be used to determine which identifier to use for a specific purpose.
Cardinality | 0..1
Terminology Binding | IdentifierType (Extensible)
Type | CodeableConcept
Requirements | Allows users to make use of identifiers when the identifier system is not known.
Summary | true
Comments | This element deals only with general categories of identifiers. It SHOULD not be used for codes that correspond 1..1 with the Identifier.system. Some identifiers may fall into multiple categories due to common usage. Where the system is known, a

type is unnecessary because the type is always part of the system definition. However systems often need to handle identifiers where the system is not known. There is not a 1:1 relationship between type and system, since many different systems have the same type.
Identifier.system
Element Id | Identifier.system
Definition | Establishes the namespace for the value - that is, a URL that describes a set values that are unique.
Cardinality | 0..1
Type | uri
Requirements | There are many sets of identifiers. To perform matching of two identifiers, we need to know what set we're dealing with. The system identifies a particular set of unique identifiers.
Summary | true
Comments | Identifier.system is always case sensitive.
Identifier.value
Element Id | Identifier.value
Definition | The portion of the identifier typically relevant to the user and which is unique within the context of the system.
Cardinality | 0..1
Type | string
Summary | true
Comments | If the value is a full URI, then the system SHALL be urn:ietf:rfc:3986. The value's primary purpose is computational mapping. As a result, it may be normalized for comparison purposes (e.g. removing non-significant whitespace, dashes, etc.) A value formatted for human display can be conveyed using the Rendered Value extension. Identifier.value is to be treated as case sensitive unless knowledge of the Identifier.system allows the processer to be confident that non-case-sensitive processing is safe.
Identifier.period
Element Id | Identifier.period
Definition | Time period during which identifier is/was valid for use.
Cardinality | 0..1
Type | Period
Summary | true
Identifier.assigner
Element Id | Identifier.assigner
Definition | Organization that issued/manages the identifier.
Cardinality | 0..1
Type | Reference(Organization)
Summary | true
Comments | The Identifier.assigner may omit the .reference element and only contain a .display element reflecting the name or other textual information about the assigning organization.

2.24.2.4 Coding

See also Base Definition, Examples, Profiles & Extensions, Mappings and R2 Conversions.

Coding
Standards Status | This element has a standards status of "Normative"
Element Id | Coding
Definition | A reference to a code defined by a terminology system.
Cardinality | 0..*
Type | Element
Requirements | References to codes are very common in healthcare models.
Summary | true
Comments | Codes may be defined very casually in enumerations or code lists, up to very formal definitions such as SNOMED CT - see the HL7 v3 Core Principles for more information.
Coding.system
Element Id | Coding.system
Definition | The identification of the code system that defines the meaning of the symbol in the code.
Cardinality | 0..1
Type | uri
Requirements | Need to be unambiguous about the source of the definition of the symbol.
Summary | true
Comments | The URI may be an OID (urn:oid:...) or a UUID (urn:uuid:...). OIDs and UUIDs SHALL be references to the HL7 OID registry. Otherwise, the URI should come from HL7's list of FHIR defined special URIs or it should reference to some definition that establishes the system clearly and unambiguously.
Coding.version
Element Id | Coding.version
Definition | The version of the code system which was used when choosing this code. Note that a well-maintained code system does not need the version reported, because the meaning of codes is consistent across versions. However this cannot consistently be assured, and when the meaning is not guaranteed to be consistent, the version SHOULD be exchanged.
Note | This is a business versionId, not a resource version id (see discussion)
Cardinality | 0..1
Type | string
Summary | true
Comments | Where the terminology does not clearly define what string should be used to identify code system versions, the recommendation is to use the date (expressed in FHIR date format) on which that version was officially published as the version date.
Coding.code
Element Id | Coding.code

Definition | A symbol in syntax defined by the system. The symbol may be a predefined code or an expression in a syntax defined by the coding system (e.g. post-coordination).
Cardinality | 0..1
Type | code
Requirements | Need to refer to a particular code in the system.
Summary | true
Coding.display
Element Id | Coding.display
Definition | A representation of the meaning of the code in the system, following the rules of the system.
Cardinality | 0..1
Type | string
Requirements | Need to be able to carry a human-readable meaning of the code for readers that do not know the system.
Summary | true
Coding.userSelected
Element Id | Coding.userSelected
Definition | Indicates that this coding was chosen by a user directly - e.g. off a pick list of available items (codes or displays).
Cardinality | 0..1
Type | boolean
Requirements | This has been identified as a clinical safety criterium - that this exact system/code pair was chosen explicitly, rather than inferred by the system based on some rules or language processing.
Summary | true
Comments | Amongst a set of alternatives, a directly chosen code is the most appropriate starting point for new translations. There is some ambiguity about what exactly 'directly chosen' implies, and trading partner agreement may be needed to clarify the use of this element and its consequences more completely.

2.24.2.5 CodeableConcept

See also Base Definition, Examples, Profiles & Extensions, Mappings and R2 Conversions.

CodeableConcept
Standards Status | This element has a standards status of "Normative"
Element Id | CodeableConcept
Definition | A concept that may be defined by a formal reference to a terminology or ontology or may be provided by text.
Cardinality | 0..*
Type | Element
Requirements | This is a common pattern in healthcare - a concept that may be defined by one or more codes from formal definitions including LOINC and SNOMED CT, and/or defined by the provision of text that captures a human sense of the concept.
Summary | true
Comments | Not all terminology uses fit this general pattern. In some cases, models should not use CodeableConcept and use Coding directly and provide their own structure for managing text, codings, translations and the relationship between elements and pre- and post-coordination.
CodeableConcept.coding
Element Id | CodeableConcept.coding
Definition | A reference to a code defined by a terminology system.
Cardinality | 0..*
Type | Coding
Requirements | Allows for alternative encodings within a code system, and translations to other code systems.
Summary | true
Comments | Codes may be defined very casually in enumerations, or code lists, up to very formal definitions such as SNOMED CT - see the HL7 v3 Core Principles for more information. Ordering of codings is undefined and SHALL NOT be used to infer meaning. Generally, at most only one of the coding values will be labeled as UserSelected = true.
CodeableConcept.text
Element Id | CodeableConcept.text
Definition | A human language representation of the concept as seen/selected/uttered by the user who entered the data and/or which represents the intended meaning of the user.
Cardinality | 0..1
Type | string
Requirements | The codes from the terminologies do not always capture the correct meaning with all the nuances of the human using them, or sometimes there is no appropriate code at all. In these cases, the text is used to capture the full meaning of the source.
Summary | true
Comments | Very often the text is the same as a displayName of one of the codings.

2.24.2.6 Quantity

See also Base Definition, Examples, Profiles & Extensions, Mappings and R2 Conversions.

Quantity
Standards Status | This element has a standards status of "Normative"
Element Id | Quantity
Definition | A measured amount (or an amount that can potentially be measured). Note that measured amounts include amounts that are not precisely quantified, including amounts involving arbitrary units and floating currencies.
Cardinality | 0..*
Type | Element
Requirements | Need to able to capture all sorts of measured values, even if the measured value are not precisely quantified. Values include exact measures such as 3.51g, customary units such as 3 tablets, and currencies such as $100.32USD.

Summary | true
Comments | The context of use may frequently define what kind of quantity this is and therefore what kind of units can be used. The context of use may also restrict the values for the comparator.
Invariants | Defined on this element qty-3 Rule If a code for the unit is present, the system SHALL also be present code.empty() or system.exists()
Defined on this element
qty-3 | Rule | If a code for the unit is present, the system SHALL also be present | code.empty() or system.exists()
Quantity.value
Element Id | Quantity.value
Definition | The value of the measured amount. The value includes an implicit precision in the presentation of the value.
Cardinality | 0..1
Type | decimal
Requirements | Precision is handled implicitly in almost all cases of measurement.
Summary | true
Comments | The implicit precision in the value should always be honored. Monetary values have their own rules for handling precision (refer to standard accounting text books).
Quantity.comparator
Element Id | Quantity.comparator
Definition | How the value should be understood and represented - whether the actual value is greater or less than the stated value due to measurement issues; e.g. if the comparator is "<" , then the real value is < stated value.
Cardinality | 0..1
Terminology Binding | QuantityComparator (Required)
Type | code
Is Modifier | true (Reason: This is labeled as "Is Modifier" because the comparator modifies the interpretation of the value significantly. If there is no comparator, then there is no modification of the value)
Meaning if Missing | If there is no comparator, then there is no modification of the value
Requirements | Need a framework for handling measures where the value is <5ug/L or >400mg/L due to the limitations of measuring methodology.
Summary | true
Quantity.unit
Element Id | Quantity.unit
Definition | A human-readable form of the unit.
Cardinality | 0..1
Type | string
Requirements | There are many representations for units of measure and in many contexts, particular representations are fixed and required. I.e. mcg for micrograms.
Summary | true
Quantity.system
Element Id | Quantity.system
Definition | The identification of the system that provides the coded form of the unit.
Cardinality | 0..1
Type | uri
Requirements | Need to know the system that defines the coded form of the unit.
Summary | true
Invariants | Affect this element qty-3 Rule If a code for the unit is present, the system SHALL also be present code.empty() or system.exists()
Affect this element
qty-3 | Rule | If a code for the unit is present, the system SHALL also be present | code.empty() or system.exists()
Quantity.code
Element Id | Quantity.code
Definition | A computer processable form of the unit in some unit representation system.
Cardinality | 0..1
Type | code
Requirements | Need a computable form of the unit that is fixed across all forms. UCUM provides this for quantities, but SNOMED CT provides many units of interest.
Summary | true
Comments | The preferred system is UCUM, but SNOMED CT can also be used (for customary units) or ISO 4217 for currency. The context of use may additionally require a code from a particular system.

2.24.2.7 Money

See also Base Definition, Examples, Profiles & Extensions, Mappings and R2 Conversions.

Money
Standards Status | This element has a standards status of "Normative"
Element Id | Money
Definition | An amount of economic utility in some recognized currency.
Cardinality | 0..*
Type | Element
Summary | true
Money.value
Element Id | Money.value
Definition | Numerical value (with implicit precision).
Cardinality | 0..1

Type | decimal
Requirements | The amount of the currency. The value includes an implicit precision in the presentation of the value.
Summary | true
Comments | Monetary values have their own rules for handling precision (refer to standard accounting text books).
Money.currency
Element Id | Money.currency
Definition | ISO 4217 Currency Code.
Cardinality | 0..1
Terminology Binding | CurrencyCode (Required)
Type | code
Requirements | A code indicating the currency, taken from ISO 4217.
Summary | true

2.24.2.8 Range

See also Base Definition, Examples, Profiles & Extensions, Mappings and R2 Conversions.

Range
Standards Status | This element has a standards status of "Normative"
Element Id | Range
Definition | A set of ordered Quantities defined by a low and high limit.
Cardinality | 0..*
Type | Element
Requirements | Need to be able to specify ranges of values.
Summary | true
Comments | The stated low and high value are assumed to have arbitrarily high precision when it comes to determining which values are in the range. I.e. 1.99 is not in the range 2 -> 3.
Invariants | Defined on this element rng-2 Rule If present, low SHALL have a lower value than high low.empty() or high.empty() or (low <= high)
Defined on this element
rng-2 | Rule | If present, low SHALL have a lower value than high | low.empty() or high.empty() or (low <= high)
Range.low
Element Id | Range.low
Definition | The low limit. The boundary is inclusive.
Cardinality | 0..1
Type | SimpleQuantity
Summary | true
Comments | If the low element is missing, the low boundary is not known.
Range.high
Element Id | Range.high
Definition | The high limit. The boundary is inclusive.
Cardinality | 0..1
Type | SimpleQuantity
Summary | true
Comments | If the high element is missing, the high boundary is not known.

2.24.2.9 Ratio

See also Base Definition, Examples, Profiles & Extensions, Mappings and R2 Conversions.

Ratio
Standards Status | This element has a standards status of "Normative"
Element Id | Ratio
Definition | A relationship of two Quantity values - expressed as a numerator and a denominator.
Cardinality | 0..*
Type | Element
Requirements | Need to able to capture ratios for some measurements (titers) and some rates (costs).
Summary | true
Comments | The Ratio datatype should only be used to express a relationship of two numbers if the relationship cannot be suitably expressed using a Quantity and a common unit. Where the denominator value is known to be fixed to "1", Quantity should be used instead of Ratio.
Invariants | Defined on this element rat-1 Rule Numerator and denominator SHALL both be present, or both are absent. If both are absent, there SHALL be some extension present (numerator.empty() xor denominator.exists()) and (numerator.exists() or extension.exists())
Defined on this element
rat-1 | Rule | Numerator and denominator SHALL both be present, or both are absent. If both are absent, there SHALL be some extension present | (numerator.empty() xor denominator.exists()) and (numerator.exists() or extension.exists())
Ratio.numerator
Element Id | Ratio.numerator
Definition | The value of the numerator.
Cardinality | 0..1
Type | Quantity
Summary | true
Ratio.denominator
Element Id | Ratio.denominator

Definition | The value of the denominator.
Cardinality | 0..1
Type | Quantity
Summary | true

2.24.2.10 Period

See also Base Definition, Examples, Profiles & Extensions, Mappings and R2 Conversions.

Period
Standards Status | This element has a standards status of "Normative"
Element Id | Period
Definition | A time period defined by a start and end date and optionally time.
Cardinality | 0..*
Type | Element
Summary | true
Comments | A Period specifies a range of time; the context of use will specify whether the entire range applies (e.g. "the patient was an inpatient of the hospital for this time range") or one value from the range applies (e.g. "give to the patient between these two times"). Period is not used for a duration (a measure of elapsed time). See Duration.
Invariants | Defined on this element per-1 Rule If present, start SHALL have a lower value than end start.hasValue().not() or end.hasValue().not() or (start <= end)
Defined on this element
per-1 | Rule | If present, start SHALL have a lower value than end | start.hasValue().not() or end.hasValue().not() or (start <= end)
Period.start
Element Id | Period.start
Definition | The start of the period. The boundary is inclusive.
Cardinality | 0..1
Type | dateTime
Summary | true
Comments | If the low element is missing, the meaning is that the low boundary is not known.
Invariants | Affect this element per-1 Rule If present, start SHALL have a lower value than end start.hasValue().not() or end.hasValue().not() or (start <= end)
Affect this element
per-1 | Rule | If present, start SHALL have a lower value than end | start.hasValue().not() or end.hasValue().not() or (start <= end)
Period.end
Element Id | Period.end
Definition | The end of the period. If the end of the period is missing, it means no end was known or planned at the time the instance was created. The start may be in the past, and the end date in the future, which means that period is expected/planned to end at that time.
Cardinality | 0..1
Type | dateTime
Meaning if Missing | If the end of the period is missing, it means that the period is ongoing
Summary | true
Comments | The high value includes any matching date/time. i.e. 2012-02-03T10:00:00 is in a period that has an end value of 2012-02-03.
Invariants | Affect this element per-1 Rule If present, start SHALL have a lower value than end start.hasValue().not() or end.hasValue().not() or (start <= end)
Affect this element
per-1 | Rule | If present, start SHALL have a lower value than end | start.hasValue().not() or end.hasValue().not() or (start <= end)

2.24.2.11 SampledData

Normative Candidate Note: This DataType is not normative - it is still undergoing Trial Use while more experience is gathered.

See also Base Definition, Examples, Profiles & Extensions, Mappings and R2 Conversions.

SampledData
Standards Status | This element has a standards status of "Trial Use"
Element Id | SampledData
Definition | A series of measurements taken by a device, with upper and lower limits. There may be more than one dimension in the data.
Cardinality | 0..*
Type | Element
Requirements | There is a need for a concise way to handle the data produced by devices that sample a physical state at a high frequency.
Comments | The data is not interpretable without at least origin, period, and dimensions, but these are optional to allow a separation between the template of measurement and the actual measurement, such as between DeviceCapabilities and DeviceLog. When providing a summary view (for example with Observation.value[x]) SampledData should be represented with a brief display text such as "Sampled Data".
SampledData.origin
Element Id | SampledData.origin
Definition | The base quantity that a measured value of zero represents. In addition, this provides the units of the entire measurement series.
Cardinality | 1..1
Type | SimpleQuantity
Summary | true

SampledData.period
Element Id | SampledData.period
Definition | The length of time between sampling times, measured in milliseconds.
Cardinality | 1..1
Type | decimal
Summary | true
Comments | This is usually a whole number.
SampledData.factor
Element Id | SampledData.factor
Definition | A correction factor that is applied to the sampled data points before they are added to the origin.
Cardinality | 0..1
Type | decimal
Meaning if Missing | If no factor is assigned, the data is not adjusted before adding to the origin
Summary | true
SampledData.lowerLimit
Element Id | SampledData.lowerLimit
Definition | The lower limit of detection of the measured points. This is needed if any of the data points have the value "L" (lower than detection limit).
Cardinality | 0..1
Type | decimal
Summary | true
SampledData.upperLimit
Element Id | SampledData.upperLimit
Definition | The upper limit of detection of the measured points. This is needed if any of the data points have the value "U" (higher than detection limit).
Cardinality | 0..1
Type | decimal
Summary | true
SampledData.dimensions
Element Id | SampledData.dimensions
Definition | The number of sample points at each time point. If this value is greater than one, then the dimensions will be interlaced - all the sample points for a point in time will be recorded at once.
Cardinality | 1..1
Type | positiveInt
Summary | true
Comments | If there is more than one dimension, the code for the type of data will define the meaning of the dimensions (typically ECG data).
SampledData.data
Element Id | SampledData.data
Definition | A series of data points which are decimal values separated by a single space (character u20). The special values "E" (error), "L" (below detection limit) and "U" (above detection limit) can also be used in place of a decimal value.
Cardinality | 0..1
Type | string
Comments | Data may be missing if it is omitted for summarization purposes. In general, data is required for any actual use of a SampledData.

2.24.2.12 HumanName

See also Base Definition, Examples, Profiles & Extensions, Mappings and R2 Conversions.

HumanName
Standards Status | This element has a standards status of "Normative"
Element Id | HumanName
Definition | A human's name with the ability to identify parts and usage.
Cardinality | 0..*
Type | Element
Requirements | Need to be able to record names, along with notes about their use.
Summary | true
Comments | Names may be changed, or repudiated, or people may have different names in different contexts. Names may be divided into parts of different type that have variable significance depending on context, though the division into parts does not always matter. With personal names, the different parts might or might not be imbued with some implicit meaning; various cultures associate different importance with the name parts and the degree to which systems must care about name parts around the world varies widely.
HumanName.use
Element Id | HumanName.use
Definition | Identifies the purpose for this name.
Cardinality | 0..1
Terminology Binding | NameUse (Required)
Type | code
Is Modifier | true (Reason: This is labeled as "Is Modifier" because applications should not mistake a temporary or old name etc.for a current/permanent one)
Requirements | Allows the appropriate name for a particular context of use to be selected from among a set of names.
Summary | true
Comments | Applications can assume that a name is current unless it explicitly says that it is temporary or old.

HumanName.text
Element Id | HumanName.text
Definition | Specifies the entire name as it should be displayed e.g. on an application UI. This may be provided instead of or as well as the specific parts.
Cardinality | 0..1
Type | string
Requirements | A renderable, unencoded form.
Summary | true
Comments | Can provide both a text representation and parts. Applications updating a name SHALL ensure that when both text and parts are present, no content is included in the text that isn't found in a part.
HumanName.family
Element Id | HumanName.family
Definition | The part of a name that links to the genealogy. In some cultures (e.g. Eritrea) the family name of a son is the first name of his father.
Cardinality | 0..1
Type | string
Alternate Names | surname
Summary | true
Comments | Family Name may be decomposed into specific parts using extensions (de, nl, es related cultures).
HumanName.given
Element Id | HumanName.given
Definition | Given name.
Cardinality | 0..*
Type | string
Element Order Meaning | Given Names appear in the correct order for presenting the name
Alternate Names | first name; middle name
Summary | true
Comments | If only initials are recorded, they may be used in place of the full name parts. Initials may be separated into multiple given names but often aren't due to paractical limitations. This element is not called "first name" since given names do not always come first.
HumanName.prefix
Element Id | HumanName.prefix
Definition | Part of the name that is acquired as a title due to academic, legal, employment or nobility status, etc. and that appears at the start of the name.
Cardinality | 0..*
Type | string
Element Order Meaning | Prefixes appear in the correct order for presenting the name
Summary | true
HumanName.suffix
Element Id | HumanName.suffix
Definition | Part of the name that is acquired as a title due to academic, legal, employment or nobility status, etc. and that appears at the end of the name.
Cardinality | 0..*
Type | string
Element Order Meaning | Suffixes appear in the correct order for presenting the name
Summary | true
HumanName.period
Element Id | HumanName.period
Definition | Indicates the period of time when this name was valid for the named person.
Cardinality | 0..1
Type | Period
Requirements | Allows names to be placed in historical context.
Summary | true

2.24.2.13 Address

See also Base Definition, Examples, Profiles & Extensions, Mappings and R2 Conversions.

Address
Standards Status | This element has a standards status of "Normative"
Element Id | Address
Definition | An address expressed using postal conventions (as opposed to GPS or other location definition formats). This data type may be used to convey addresses for use in delivering mail as well as for visiting locations which might not be valid for mail delivery. There are a variety of postal address formats defined around the world.
Cardinality | 0..*
Type | Element
Requirements | Need to be able to record postal addresses, along with notes about their use.
Summary | true
Comments | Note: address is intended to describe postal addresses for administrative purposes, not to describe absolute geographical coordinates. Postal addresses are often used as proxies for physical locations (also see the Location resource).
Address.use
Element Id | Address.use
Definition | The purpose of this address.
Cardinality | 0..1

Terminology Binding | AddressUse (Required)
Type | code
Is Modifier | true (Reason: This is labeled as "Is Modifier" because applications should not mistake a temporary or old address etc.for a current/permanent one)
Requirements | Allows an appropriate address to be chosen from a list of many.
Summary | true
Comments | Applications can assume that an address is current unless it explicitly says that it is temporary or old.
Address.type
Element Id | Address.type
Definition | Distinguishes between physical addresses (those you can visit) and mailing addresses (e.g. PO Boxes and care-of addresses). Most addresses are both.
Cardinality | 0..1
Terminology Binding | AddressType (Required)
Type | code
Summary | true
Comments | The definition of Address states that "address is intended to describe postal addresses, not physical locations". However, many applications track whether an address has a dual purpose of being a location that can be visited as well as being a valid delivery destination, and Postal addresses are often used as proxies for physical locations (also see the Location resource).
Address.text
Element Id | Address.text
Definition | Specifies the entire address as it should be displayed e.g. on a postal label. This may be provided instead of or as well as the specific parts.
Cardinality | 0..1
Type | string
Requirements | A renderable, unencoded form.
Summary | true
Comments | Can provide both a text representation and parts. Applications updating an address SHALL ensure that when both text and parts are present, no content is included in the text that isn't found in a part.
Address.line
Element Id | Address.line
Definition | This component contains the house number, apartment number, street name, street direction, P.O. Box number, delivery hints, and similar address information.
Cardinality | 0..*
Type | string
Element Order Meaning | The order in which lines should appear in an address label
Summary | true
Address.city
Element Id | Address.city
Definition | The name of the city, town, suburb, village or other community or delivery center.
Cardinality | 0..1
Type | string
Alternate Names | Municpality
Summary | true
Address.district
Element Id | Address.district
Definition | The name of the administrative area (county).
Cardinality | 0..1
Type | string
Alternate Names | County
Summary | true
Comments | District is sometimes known as county, but in some regions 'county' is used in place of city (municipality), so county name should be conveyed in city instead.
Address.state
Element Id | Address.state
Definition | Sub-unit of a country with limited sovereignty in a federally organized country. A code may be used if codes are in common use (e.g. US 2 letter state codes).
Cardinality | 0..1
Type | string
Alternate Names | Province; Territory
Summary | true
Address.postalCode
Element Id | Address.postalCode
Definition | A postal code designating a region defined by the postal service.
Cardinality | 0..1
Type | string
Alternate Names | Zip
Summary | true
Address.country
Element Id | Address.country
Definition | Country - a nation as commonly understood or generally accepted.
Cardinality | 0..1
Type | string
Summary | true

Comments | ISO 3166 3 letter codes can be used in place of a human readable country name.
Address.period
Element Id | Address.period
Definition | Time period when address was/is in use.
Cardinality | 0..1
Type | Period
Requirements | Allows addresses to be placed in historical context.
Summary | true

2.24.2.14 ContactPoint

See also Base Definition, Examples, Profiles & Extensions, Mappings and R2 Conversions.

ContactPoint
Standards Status | This element has a standards status of "Normative"
Element Id | ContactPoint
Definition | Details for all kinds of technology mediated contact points for a person or organization, including telephone, email, etc.
Cardinality | 0..*
Type | Element
Requirements | Need to track phone, fax, mobile, sms numbers, email addresses, twitter tags, etc.
Summary | true
Invariants | Defined on this element cpt-2 Rule A system is required if a value is provided. value.empty() or system.exists()
Defined on this element
cpt-2 | Rule | A system is required if a value is provided. | value.empty() or system.exists()
ContactPoint.system
Element Id | ContactPoint.system
Definition | Telecommunications form for contact point - what communications system is required to make use of the contact.
Cardinality | 0..1
Terminology Binding | ContactPointSystem (Required)
Type | code
Summary | true
Invariants | Affect this element cpt-2 Rule A system is required if a value is provided. value.empty() or system.exists()
Affect this element
cpt-2 | Rule | A system is required if a value is provided. | value.empty() or system.exists()
ContactPoint.value
Element Id | ContactPoint.value
Definition | The actual contact point details, in a form that is meaningful to the designated communication system (i.e. phone number or email address).
Cardinality | 0..1
Type | string
Requirements | Need to support legacy numbers that are not in a tightly controlled format.
Summary | true
Comments | Additional text data such as phone extension numbers, or notes about use of the contact are sometimes included in the value.
ContactPoint.use
Element Id | ContactPoint.use
Definition | Identifies the purpose for the contact point.
Cardinality | 0..1
Terminology Binding | ContactPointUse (Required)
Type | code
Is Modifier | true (Reason: This is labeled as "Is Modifier" because applications should not mistake a temporary or old contact etc.for a current/permanent one)
Requirements | Need to track the way a person uses this contact, so a user can choose which is appropriate for their purpose.
Summary | true
Comments | Applications can assume that a contact is current unless it explicitly says that it is temporary or old.
ContactPoint.rank
Element Id | ContactPoint.rank
Definition | Specifies a preferred order in which to use a set of contacts. ContactPoints with lower rank values are more preferred than those with higher rank values.
Cardinality | 0..1
Type | positiveInt
Summary | true
Comments | Note that rank does not necessarily follow the order in which the contacts are represented in the instance.
ContactPoint.period
Element Id | ContactPoint.period
Definition | Time period when the contact point was/is in use.
Cardinality | 0..1
Type | Period
Summary | true

2.24.2.15 Timing

See also Base Definition, Examples, Profiles & Extensions, Mappings and R2 Conversions.

Timing

Standards Status | This element has a standards status of "Normative"
Element Id | Timing
Definition | Specifies an event that may occur multiple times. Timing schedules are used to record when things are planned, expected or requested to occur. The most common usage is in dosage instructions for medications. They are also used when planning care of various kinds, and may be used for reporting the schedule to which past regular activities were carried out.
Cardinality | 0..*
Type | BackboneElement
Requirements | Need to able to track proposed timing schedules. There are several different ways to do this: one or more specified times, a simple rules like three times a day, or before/after meals.
Summary | true
Comments | Describes the occurrence of an event that may occur multiple times. Timing schedules are used for specifying when events are expected or requested to occur, and may also be used to represent the summary of a past or ongoing event. For simplicity, the definitions of Timing components are expressed as 'future' events, but such components can also be used to describe historic or ongoing events. A Timing schedule can be a list of events and/or criteria for when the event happens, which can be expressed in a structured form and/or as a code. When both event and a repeating specification are provided, the list of events should be understood as an interpretation of the information in the repeat structure.
Timing.event
Element Id | Timing.event
Definition | Identifies specific times when the event occurs.
Cardinality | 0..*
Type | dateTime
Requirements | In a Medication Administration Record, for instance, you need to take a general specification, and turn it into a precise specification.
Summary | true
Timing.repeat
Element Id | Timing.repeat
Definition | A set of rules that describe when the event is scheduled.
Cardinality | 0..1
Requirements | Many timing schedules are determined by regular repetitions.
Summary | true
Invariants | Defined on this element tim-1 Rule if there's a duration, there needs to be duration units duration.empty() or durationUnit.exists() tim-2 Rule if there's a period, there needs to be period units period.empty() or periodUnit.exists() tim-4 Rule duration SHALL be a non-negative value duration.exists() implies duration >= 0 tim-5 Rule period SHALL be a non-negative value period.exists() implies period >= 0 tim-6 Rule If there's a periodMax, there must be a period periodMax.empty() or period.exists() tim-7 Rule If there's a durationMax, there must be a duration durationMax.empty() or duration.exists() tim-8 Rule If there's a countMax, there must be a count countMax.empty() or count.exists() tim-9 Rule If there's an offset, there must be a when (and not C, CM, CD, CV) offset.empty() or (when.exists() and ((when in ('C' | 'CM' | 'CD' | 'CV')).not())) tim-10 Rule If there's a timeOfDay, there cannot be a when, or vice versa timeOfDay.empty() or when.empty()
Defined on this element
tim-1 | Rule | if there's a duration, there needs to be duration units | duration.empty() or durationUnit.exists()
tim-2 | Rule | if there's a period, there needs to be period units | period.empty() or periodUnit.exists()
tim-4 | Rule | duration SHALL be a non-negative value | duration.exists() implies duration >= 0
tim-5 | Rule | period SHALL be a non-negative value | period.exists() implies period >= 0
tim-6 | Rule | If there's a periodMax, there must be a period | periodMax.empty() or period.exists()
tim-7 | Rule | If there's a durationMax, there must be a duration | durationMax.empty() or duration.exists()
tim-8 | Rule | If there's a countMax, there must be a count | countMax.empty() or count.exists()
tim-9 | Rule | If there's an offset, there must be a when (and not C, CM, CD, CV) | offset.empty() or (when.exists() and ((when in ('C' | 'CM' | 'CD' | 'CV')).not()))
tim-10 | Rule | If there's a timeOfDay, there cannot be a when, or vice versa | timeOfDay.empty() or when.empty()
Timing.repeat.bounds[x]
Element Id | Timing.repeat.bounds[x]
Definition | Either a duration for the length of the timing schedule, a range of possible length, or outer bounds for start and/or end limits of the timing schedule.
Cardinality | 0..1
Type | Duration|Range|Period
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
Timing.repeat.count
Element Id | Timing.repeat.count
Definition | A total count of the desired number of repetitions across the duration of the entire timing specification. If countMax is present, this element indicates the lower bound of the allowed range of count values.
Cardinality | 0..1
Type | positiveInt
Requirements | Repetitions may be limited by end time or total occurrences.
Summary | true
Comments | If you have both bounds and count, then this should be understood as within the bounds period, until count times happens.
Timing.repeat.countMax
Element Id | Timing.repeat.countMax
Definition | If present, indicates that the count is a range - so to perform the action between [count] and [countMax] times.
Cardinality | 0..1
Type | positiveInt
Summary | true
Timing.repeat.duration

Element Id | Timing.repeat.duration
Definition | How long this thing happens for when it happens. If durationMax is present, this element indicates the lower bound of the allowed range of the duration.
Cardinality | 0..1
Type | decimal
Requirements | Some activities are not instantaneous and need to be maintained for a period of time.
Summary | true
Comments | For some events the duration is part of the definition of the event (e.g. IV infusions, where the duration is implicit in the specified quantity and rate). For others, it's part of the timing specification (e.g. exercise).
Timing.repeat.durationMax
Element Id | Timing.repeat.durationMax
Definition | If present, indicates that the duration is a range - so to perform the action between [duration] and [durationMax] time length.
Cardinality | 0..1
Type | decimal
Requirements | Some activities are not instantaneous and need to be maintained for a period of time.
Summary | true
Comments | For some events the duration is part of the definition of the event (e.g. IV infusions, where the duration is implicit in the specified quantity and rate). For others, it's part of the timing specification (e.g. exercise).
Timing.repeat.durationUnit
Element Id | Timing.repeat.durationUnit
Definition | The units of time for the duration, in UCUM units.
Cardinality | 0..1
Terminology Binding | UnitsOfTime (Required)
Type | code
Summary | true
Timing.repeat.frequency
Element Id | Timing.repeat.frequency
Definition | The number of times to repeat the action within the specified period. If frequencyMax is present, this element indicates the lower bound of the allowed range of the frequency.
Cardinality | 0..1
Type | positiveInt
Meaning if Missing | If no frequency is stated, the assumption is that the event occurs once per period, but systems SHOULD always be specific about this
Summary | true
Timing.repeat.frequencyMax
Element Id | Timing.repeat.frequencyMax
Definition | If present, indicates that the frequency is a range - so to repeat between [frequency] and [frequencyMax] times within the period or period range.
Cardinality | 0..1
Type | positiveInt
Summary | true
Timing.repeat.period
Element Id | Timing.repeat.period
Definition | Indicates the duration of time over which repetitions are to occur; e.g. to express "3 times per day", 3 would be the frequency and "1 day" would be the period. If periodMax is present, this element indicates the lower bound of the allowed range of the period length.
Cardinality | 0..1
Type | decimal
Summary | true
Timing.repeat.periodMax
Element Id | Timing.repeat.periodMax
Definition | If present, indicates that the period is a range from [period] to [periodMax], allowing expressing concepts such as "do this once every 3-5 days.
Cardinality | 0..1
Type | decimal
Summary | true
Timing.repeat.periodUnit
Element Id | Timing.repeat.periodUnit
Definition | The units of time for the period in UCUM units.
Cardinality | 0..1
Terminology Binding | UnitsOfTime (Required)
Type | code
Summary | true
Timing.repeat.dayOfWeek
Element Id | Timing.repeat.dayOfWeek
Definition | If one or more days of week is provided, then the action happens only on the specified day(s).
Cardinality | 0..*
Terminology Binding | DaysOfWeek (Required)
Type | code
Summary | true
Comments | If no days are specified, the action is assumed to happen every day as otherwise specified. The elements frequency and period cannot be used as well as dayOfWeek.

Timing.repeat.timeOfDay
Element Id | Timing.repeat.timeOfDay
Definition | Specified time of day for action to take place.
Cardinality | 0..*
Type | time
Summary | true
Comments | When time of day is specified, it is inferred that the action happens every day (as filtered by dayofWeek) on the specified times. The elements when, frequency and period cannot be used as well as timeOfDay.
Timing.repeat.when
Element Id | Timing.repeat.when
Definition | An approximate time period during the day, potentially linked to an event of daily living that indicates when the action should occur.
Cardinality | 0..*
Terminology Binding | EventTiming (Required)
Type | code
Requirements | Timings are frequently determined by occurrences such as waking, eating and sleep.
Summary | true
Comments | When more than one event is listed, the event is tied to the union of the specified events.
Timing.repeat.offset
Element Id | Timing.repeat.offset
Definition | The number of minutes from the event. If the event code does not indicate whether the minutes is before or after the event, then the offset is assumed to be after the event.
Cardinality | 0..1
Type | unsignedInt
Summary | true
Timing.code
Element Id | Timing.code
Definition | A code for the timing schedule (or just text in code.text). Some codes such as BID are ubiquitous, but many institutions define their own additional codes. If a code is provided, the code is understood to be a complete statement of whatever is specified in the structured timing data, and either the code or the data may be used to interpret the Timing, with the exception that .repeat.bounds still applies over the code (and is not contained in the code).
Cardinality | 0..1
Terminology Binding | TimingAbbreviation (Preferred)
Type | CodeableConcept
Summary | true
Comments | BID etc. are defined as 'at institutionally specified times'. For example, an institution may choose that BID is "always at 7am and 6pm". If it is inappropriate for this choice to be made, the code BID should not be used. Instead, a distinct organization-specific code should be used in place of the HL7-defined BID code and/or a structured representation should be used (in this case, specifying the two event times).

2.24.2.16 Signature

Normative Candidate Note: This DataType is not normative - it is still undergoing Trial Use while more experience is gathered.

See also Base Definition, Examples, Profiles & Extensions, Mappings and R2 Conversions.

Signature
Standards Status | This element has a standards status of "Trial Use"
Element Id | Signature
Definition | A signature along with supporting context. The signature may be a digital signature that is cryptographic in nature, or some other signature acceptable to the domain. This other signature may be as simple as a graphical image representing a hand-written signature, or a signature ceremony Different signature approaches have different utilities.
Cardinality | 0..*
Type | Element
Requirements | There are a number of places where content must be signed in healthcare.
Comments | The elements of the Signature Resource are for ease of access of these elements. For digital signatures (Xml DigSig, JWS), the non-repudiation proof comes from the Signature validation, which includes validation of the referenced objects (e.g. Resources) (a.k.a., Content) in the XML-Signature Detached form.
Signature.type
Element Id | Signature.type
Definition | An indication of the reason that the entity signed this document. This may be explicitly included as part of the signature information and can be used when determining accountability for various actions concerning the document.
Cardinality | 1..*
Terminology Binding | Signature Type Codes (Preferred)
Type | Coding
Summary | true
Comments | Examples include attesting to: authorship, correct transcription, and witness of specific event. Also known as a "Commitment Type Indication".
Signature.when
Element Id | Signature.when
Definition | When the digital signature was signed.
Cardinality | 1..1
Type | instant
Summary | true

Comments | This should agree with the information in the signature.
Signature.who
Element Id | Signature.who
Definition | A reference to an application-usable description of the identity that signed (e.g. the signature used their private key).
Cardinality | 1..1
Type | Reference(Practitioner | PractitionerRole | RelatedPerson | Patient | Device | Organization)
Summary | true
Comments | This should agree with the information in the signature.
Signature.onBehalfOf
Element Id | Signature.onBehalfOf
Definition | A reference to an application-usable description of the identity that is represented by the signature.
Cardinality | 0..1
Type | Reference(Practitioner | PractitionerRole | RelatedPerson | Patient | Device | Organization)
Requirements | used when the signature is on behalf of a non-signer.
Summary | true
Comments | The party that can't sign. For example a child.
Signature.targetFormat
Element Id | Signature.targetFormat
Definition | A mime type that indicates the technical format of the target resources signed by the signature.
Cardinality | 0..1
Terminology Binding | MimeType (Required)
Type | code
Comments | "xml", "json" and "ttl" are allowed, which describe the simple encodings described in the specification (and imply appropriate bundle support). Otherwise, mime types are legal here.
Signature.sigFormat
Element Id | Signature.sigFormat
Definition | A mime type that indicates the technical format of the signature. Important mime types are application/signature+xml for X ML DigSig, application/jose for JWS, and image/* for a graphical image of a signature, etc.
Cardinality | 0..1
Terminology Binding | MimeType (Required)
Type | code
Signature.data
Element Id | Signature.data
Definition | The base64 encoding of the Signature content. When signature is not recorded electronically this element would be empty.
Cardinality | 0..1
Type | base64Binary
Comments | Where the signature type is an XML DigSig, the signed content is a FHIR Resource(s), the signature is of the XML form of the Resource(s) using XML-Signature (XMLDIG) "Detached Signature" form.

2.24.2.17 Annotation

See also Base Definition, Examples, Profiles & Extensions, Mappings and R2 Conversions.

Annotation
Standards Status | This element has a standards status of "Normative"
Element Id | Annotation
Definition | A text note which also contains information about who made the statement and when.
Cardinality | 0..*
Type | Element
Comments | For systems that do not have structured annotations, they can simply communicate a single annotation with no author or time. This element may need to be included in narrative because of the potential for modifying information. Annotations SHOULD NOT be used to communicate "modifying" information that could be computable. (This is a SHOULD because enforcing user behavior is nearly impossible).
Annotation.author[x]
Element Id | Annotation.author[x]
Definition | The individual responsible for making the annotation.
Cardinality | 0..1
Type | Reference(Practitioner | Patient | RelatedPerson | Organization)|string
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
Comments | Organization is used when there's no need for specific attribution as to who made the comment.
Annotation.time
Element Id | Annotation.time
Definition | Indicates when this particular annotation was made.
Cardinality | 0..1
Type | dateTime
Summary | true
Annotation.text
Element Id | Annotation.text
Definition | The text of the annotation in markdown format.
Cardinality | 1..1
Type | markdown
Summary | true

2.24.3 Data Type Mappings

URL: https://hl7.org/fhir/R4/datatypes-mappings.html

2.24.3 Data Type Mappings

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative

This page provides mappings for the data types. There are mappings to HL7 v2 , HL7 v3 , and (where appropriate) vCard (see Mappings to Other Standards for further information & status).

Table of Contents

Primitive Types ElementinstanttimedatedateTimedecimalbooleanintegerstringuribase64Binarycodeidoidunsignedintpositiveintmarkdownurlcanonicaluuid | Complex Types ElementIdentifierHumanNameAddressContactPointTimingQuantitySimpleQuantityAttachmentRangePeriodRatioCodeableConceptCodingSampledDataAgeDistanceDurationCountMoneyMoneyQuantityAnnotationSignatureBackboneElement

2.24.3.1 Primitive Types

See also Base Definition, Examples, Profiles & Extensions, Detailed Descriptions and R2 Conversions.

Data Type	V2	RIM
base64Binary	~ED	BIN
boolean	n/a	BL
date	DT	TS
dateTime	DTM	TS
decimal	~NM	REAL
instant	DTM	TS
integer	~NM/~SI	INT
string	ST	ST
time	TM	n/a
uri	n/a	URL

2.24.3.2 Attachment

See also Base Definition, Examples, Profiles & Extensions, Detailed Descriptions and R2 Conversions.

2.24.3.3 HL7 v2 Mapping (http://hl7.org/v2)

Attachment	ED/RP
contentType	ED.2+ED.3/RP.2+RP.3. Note conversion may be needed if old style values are being used
language	
data	ED.5
url	RP.1+RP.2 - if they refer to a URL (see v2.6)
size	
hash	
title	
creation	

2.24.3.4 RIM Mapping (http://hl7.org/v3)

Attachment	ED
contentType	./mediaType, ./charset
language	./language
data	./data
url	./reference/literal
size	N/A (needs data type R3 proposal)
hash	.integrityCheck[parent::ED/integrityCheckAlgorithm="SHA-1"]
title	./title/data
creation	N/A (needs data type R3 proposal)

2.24.3.5 Identifier

See also Base Definition, Examples, Profiles & Extensions, Detailed Descriptions and R2 Conversions.

2.24.3.6 HL7 v2 Mapping (http://hl7.org/v2)

Identifier	CX / EI (occasionally, more often EI maps to a resource id or a URL)
use	N/A
type	CX.5
system	CX.4 / EI-2-4
value	CX.1 / EI.1
period	CX.7 + CX.8
assigner	CX.4 / (CX.4,CX.9,CX.10)

2.24.3.7 RIM Mapping (http://hl7.org/v3)

Identifier	II - The Identifier class is a little looser than the v3 type II because it allows URIs as well as registered OIDs or GUIDs. Also maps to Role[classCode=IDENT]

use | Role.code or implied by context
type | Role.code or implied by context
system | II.root or Role.id.root
value | II.extension or II.root if system indicates OID or GUID (Or Role.id.extension or root)
period | Role.effectiveTime or implied by context
assigner | II.assigningAuthorityName but note that this is an improper use by the definition of the field. Also Role.scoper

2.24.3.8 ServD (http://www.omg.org/spec/ServD/1.0/)

Identifier | Identifier
use
type
system | ./IdentifierType
value | ./Value
period | ./StartDate and ./EndDate
assigner | ./IdentifierIssuingAuthority

2.24.3.8.1 V3 Mapping Issues for Identifier

When mapping an HL7 v3 II data type to a FHIR Identifier, there are two possibilities:

• If the II has only a root and no extension, the FHIR Identifier will have a system of 'urn:ietf:rfc:3986' and the II.root will appear in Identifier.value

• If the II has both a root and an extension, the II.root will be mapped to a URI - either a human-friendly URL or URN or (less preferred) an OID expressed as a urn, and the II.extension will be sent in Identifier.value

To search on a CDA II.root - which may appear in either Identifier.system or Identifier.value, use the syntax identifier=|[root],[root].

See v3 Mapping examples

2.24.3.9 Coding

See also Base Definition, Examples, Profiles & Extensions, Detailed Descriptions and R2 Conversions.

2.24.3.10 Ontological RIM Mapping (http://hl7.org/orim)

Coding | fhir:Coding rdfs:subClassOf dt:CDCoding
system | fhir:Coding.system rdfs:subPropertyOf dt:CDCoding.codeSystem
version | fhir:Coding.version rdfs:subPropertyOf dt:CDCoding.codeSystemVersion
code | fhir:Coding.code rdfs:subPropertyOf dt:CDCoding.code
display | fhir:Coding.display rdfs:subPropertyOf dt:CDCoding.displayName
userSelected | fhir:Coding.userSelected fhir:mapsTo dt:CDCoding.codingRationale. fhir:Coding.userSelected fhir:hasMap fhir:Coding.userSelected.map. fhir:Coding.userSelected.map a fhir:Map; fhir:target dt:CDCoding.codingRationale. fhir:Coding.userSelected\#true a [fhir:source "true"; fhir:target dt:CDCoding.codingRationale\#O]

2.24.3.11 HL7 v2 Mapping (http://hl7.org/v2)

Coding | CE/CNE/CWE subset one of the sets of component 1-3 or 4-6
system | C*E.3
version | C*E.7
code | C*E.1
display | C*E.2 - but note this is not well followed
userSelected | Sometimes implied by being first

2.24.3.12 RIM Mapping (http://hl7.org/v3)

Coding | CV
system | ./codeSystem
version | ./codeSystemVersion
code | ./code
display | CV.displayName
userSelected | CD.codingRationale

2.24.3.13 CodeableConcept

See also Base Definition, Examples, Profiles & Extensions, Detailed Descriptions and R2 Conversions.

2.24.3.14 Ontological RIM Mapping (http://hl7.org/orim)

CodeableConcept | fhir:CodeableConcept rdfs:subClassOf dt:CD
coding | fhir:CodeableConcept.coding rdfs:subPropertyOf dt:CD.coding
text | fhir:CodeableConcept.text rdfs:subPropertyOf dt:CD.originalText

2.24.3.15 HL7 v2 Mapping (http://hl7.org/v2)

CodeableConcept | CE/CNE/CWE
coding | C*E.1-8, C*E.10-22
text | C*E.9. But note many systems use C*E.2 for this

2.24.3.16 RIM Mapping (http://hl7.org/v3)

CodeableConcept | CD
coding | union(., ./translation)
text | ./originalText[mediaType/code="text/plain"]/data

2.24.3.17 Quantity

See also Base Definition, Examples, Profiles & Extensions, Detailed Descriptions and R2 Conversions.

2.24.3.18 HL7 v2 Mapping (http://hl7.org/v2)

Quantity | SN (see also Range) or CQ
value | SN.2 / CQ - N/A
comparator | SN.1 / CQ.1
unit | (see OBX.6 etc.) / CQ.2
system | (see OBX.6 etc.) / CQ.2
code | (see OBX.6 etc.) / CQ.2

2.24.3.19 RIM Mapping (http://hl7.org/v3)

Quantity | PQ, IVL<PQ>, MO, CO, depending on the values
value | PQ.value, CO.value, MO.value, IVL.high or IVL.low depending on the value
comparator | IVL properties
unit | PQ.unit
system | CO.codeSystem, PQ.translation.codeSystem
code | PQ.code, MO.currency, PQ.translation.code

2.24.3.20 Money

See also Base Definition, Examples, Profiles & Extensions, Detailed Descriptions and R2 Conversions.

2.24.3.21 HL7 v2 Mapping (http://hl7.org/v2)

Money
value | MO.Value
currency | MO.Currency

2.24.3.22 RIM Mapping (http://hl7.org/v3)

Money | MO

2.24.3.23 Range

See also Base Definition, Examples, Profiles & Extensions, Detailed Descriptions and R2 Conversions.

2.24.3.24 HL7 v2 Mapping (http://hl7.org/v2)

Range | NR and also possibly SN (but see also quantity)
low | NR.1
high | NR.2

2.24.3.25 RIM Mapping (http://hl7.org/v3)

Range | IVL<QTY[not(type="TS")]> [lowClosed="true" and highClosed="true"]or URG<QTY[not(type="TS")]>
low | ./low
high | ./high

2.24.3.26 Ratio

See also Base Definition, Examples, Profiles & Extensions, Detailed Descriptions and R2 Conversions.

2.24.3.27 HL7 v2 Mapping (http://hl7.org/v2)

Ratio | N/A

2.24.3.28 RIM Mapping (http://hl7.org/v3)

Ratio | RTO
numerator | .numerator
denominator | .denominator

2.24.3.29 Period

See also Base Definition, Examples, Profiles & Extensions, Detailed Descriptions and R2 Conversions.

2.24.3.30 HL7 v2 Mapping (http://hl7.org/v2)

Period | DR
start | DR.1
end | DR.2

2.24.3.31 RIM Mapping (http://hl7.org/v3)

Period | IVL<TS>[lowClosed="true" and highClosed="true"] or URG<TS>[lowClosed="true" and highClosed="true"]
start | ./low
end | ./high

2.24.3.32 SampledData

See also Base Definition, Examples, Profiles & Extensions, Detailed Descriptions and R2 Conversions.

2.24.3.33 HL7 v2 Mapping (http://hl7.org/v2)

SampledData | N/A (but see section 7.14 in v2.7)
origin
period
factor
lowerLimit
upperLimit
dimensions
data

2.24.3.34 RIM Mapping (http://hl7.org/v3)

SampledData | SLIST + GLIST
origin | SLIST.origin
period | GLIST.increment
factor | SLIST.factor
lowerLimit | N/A (use SLIST.digits.nullFlavor)
upperLimit | N/A (use SLIST.digits.nullFlavor)
dimensions | N/A (fixed to 1)
data | SLIST.digits

2.24.3.35 HumanName

See also Base Definition, Examples, Profiles & Extensions, Detailed Descriptions and R2 Conversions.

2.24.3.36 HL7 v2 Mapping (http://hl7.org/v2)

HumanName | XPN
use | XPN.7, but often indicated by which field contains the name
text | implied by XPN.11
family | XPN.1/FN.1
given | XPN.2 + XPN.3
prefix | XPN.5
suffix | XPN/4
period | XPN.13 + XPN.14

2.24.3.37 RIM Mapping (http://hl7.org/v3)

HumanName | EN (actually, PN)
use | unique(./use)
text | ./formatted
family | ./part[partType = FAM]
given | ./part[partType = GIV]
prefix | ./part[partType = PFX]
suffix | ./part[partType = SFX]
period | ./usablePeriod[type="IVL<TS>"]

2.24.3.38 ServD (http://www.omg.org/spec/ServD/1.0/)

HumanName | ProviderName
use | ./NamePurpose
text
family | ./FamilyName
given | ./GivenNames
prefix | ./TitleCode
suffix
period | ./StartDate and ./EndDate

vCard Mappings

• HumanName.text = vCard "FN" field

• HumanName.use = use of the vCard "TYPE" parameter

• HumanName.family, .given, .prefix, .suffix = parts of vCard "N" field. Note that there is no FHIR equivalent for the poorly defined "additional" name field. In FHIR, given names go in "middle" names

• The vCard nickname corresponds to a name with the use "nickname"

2.24.3.39 Address

See also Base Definition, Examples, Profiles & Extensions, Detailed Descriptions and R2 Conversions.

2.24.3.40 HL7 v2 Mapping (http://hl7.org/v2)

Address | XAD
use | XAD.7
type | XAD.18
text | XAD.1 + XAD.2 + XAD.3 + XAD.4 + XAD.5 + XAD.6
line | XAD.1 + XAD.2 (note: XAD.1 and XAD.2 have different meanings for a company address than for a person address)
city | XAD.3
district | XAD.9
state | XAD.4
postalCode | XAD.5
country | XAD.6
period | XAD.12 / XAD.13 + XAD.14

2.24.3.41 RIM Mapping (http://hl7.org/v3)

Address | AD
use | unique(./use)
type | unique(./use)
text | ./formatted
line | AD.part[parttype = AL]
city | AD.part[parttype = CTY]
district | AD.part[parttype = CNT | CPA]
state | AD.part[parttype = STA]
postalCode | AD.part[parttype = ZIP]
country | AD.part[parttype = CNT]
period | ./usablePeriod[type="IVL<TS>"]

2.24.3.42 vCard Mapping (http://w3.org/vcard)

Address
use
type | address type parameter
text | address label parameter
line | street
city | locality
district
state | region
postalCode | code
country | country
period

2.24.3.43 ServD (http://www.omg.org/spec/ServD/1.0/)

Address | Address
use | ./AddressPurpose
type
text
line | ./StreetAddress (newline delimitted)
city | ./Jurisdiction
district
state | ./Region
postalCode | ./PostalIdentificationCode
country | ./Country
period | ./StartDate and ./EndDate

2.24.3.44 ContactPoint

See also Base Definition, Examples, Profiles & Extensions, Detailed Descriptions and R2 Conversions.

2.24.3.45 HL7 v2 Mapping (http://hl7.org/v2)

ContactPoint | XTN
system | XTN.3
value | XTN.1 (or XTN.12)
use | XTN.2 - but often indicated by field
rank | n/a
period | N/A

2.24.3.46 RIM Mapping (http://hl7.org/v3)

ContactPoint | TEL
system | ./scheme
value | ./url
use | unique(./use)
rank | n/a
period | ./usablePeriod[type="IVL<TS>"]

2.24.3.47 ServD (http://www.omg.org/spec/ServD/1.0/)

ContactPoint | ContactPoint
system | ./ContactPointType
value | ./Value
use | ./ContactPointPurpose
rank
period | ./StartDate and ./EndDate

2.24.3.48 Timing

See also Base Definition, Examples, Profiles & Extensions, Detailed Descriptions and R2 Conversions.

2.24.3.49 HL7 v2 Mapping (http://hl7.org/v2)

Timing | N/A

2.24.3.50 RIM Mapping (http://hl7.org/v3)

Timing | QSET<TS> (GTS)
event | QLIST<TS>
repeat | Implies PIVL or EIVL
bounds[x] | IVL(TS) used in a QSI
count | PIVL.count
countMax | PIVL.count
duration | PIVL.phase
durationMax | PIVL.phase
durationUnit | PIVL.phase.unit
frequency | PIVL.phase
frequencyMax | PIVL.phase
period | PIVL.phase
periodMax | PIVL.phase
periodUnit | PIVL.phase.unit
dayOfWeek | n/a
timeOfDay | n/a
when | EIVL.event
offset | EIVL.offset
code | QSC.code

2.24.3.51 Signature

See also Base Definition, Examples, Profiles & Extensions, Detailed Descriptions and R2 Conversions.

2.24.3.52 RIM Mapping (http://hl7.org/v3)

Signature | N/A

2.24.3.53 Annotation

See also Base Definition, Examples, Profiles & Extensions, Detailed Descriptions and R2 Conversions.

2.24.3.54 HL7 v2 Mapping (http://hl7.org/v2)

Annotation | N/A

2.24.3.55 RIM Mapping (http://hl7.org/v3)

Annotation | Act
author[x] | Act.participant[typeCode=AUT].role
time | Act.effectiveTime
text | Act.text

2.24.4 Data Type Profiles and Extensions

URL: https://hl7.org/fhir/R4/datatypes-extras.html

2.24.4 Data Type Profiles and Extensions

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative

This page lists profiles and extensions defined in this specification itself for the data types. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

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Primitive Types ElementinstanttimedatedateTimedecimalbooleanintegerstringuribase64Binarycodeidoidunsignedintpositiveintmarkdownurlcanonicaluuid | Complex Types ElementIdentifierHumanNameAddressContactPointTimingQuantitySimpleQuantityAttachmentRangePeriodRatioCodeableConceptCodingSampledDataAgeDistanceDurationCountMoneyMoneyQuantityAnnotationSignature BackboneElement

2.24.4.1 Primitive Types

See also Base Definition, Examples, Detailed Descriptions, Mappings and R2 Conversions.

Extensions (+ see extensions on all Elements)
display | Display Name | for General Extensions for use by FHIR Implementers
iso21090-SC-coding | SC-coding | for ISO 21090 Data Type Extensions
quantity-precision | precision | for Common extensions for Quantity data type
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
rendered-value | Rendered Value | for General Extensions for use by FHIR Implementers
rendering-markdown | markdown | for Extensions for rendering data elements
rendering-xhtml | xhtml | for Extensions for rendering data elements
translation | Translation | for General Extensions for use by FHIR Implementers
tz-code | Timezone Code | for General Extensions for use by FHIR Implementers
tz-offset | Timezone Offset | for General Extensions for use by FHIR Implementers

2.24.4.2 Attachment

See also Base Definition, Examples, Detailed Descriptions, Mappings and R2 Conversions.

Extensions (+ see extensions on all Elements)
cqf-citation | citation | for Clinical Reasoning Extensions
cqf-qualityOfEvidence | qualityOfEvidence | for Clinical Reasoning Extensions
cqf-strengthOfRecommendation | strengthOfRecommendation | for Clinical Reasoning Extensions

2.24.4.3 Identifier

See also Base Definition, Examples, Detailed Descriptions, Mappings and R2 Conversions.

Extensions (+ see extensions on all Elements)
identifier-validDate | validDate | for Common extensions for Identifier data type
rendered-value | Rendered Value | for General Extensions for use by FHIR Implementers

2.24.4.4 Coding

See also Base Definition, Examples, Detailed Descriptions, Mappings and R2 Conversions.

Extensions (+ see extensions on all Elements)
coding-sctdescid | sctdescid | for Common extensions for Coding data type
ordinalValue | Ordinal Value | for General Extensions for use by FHIR Implementers
valueset-reference | reference | for ValueSet HL7 Extensions

2.24.4.5 CodeableConcept

See also Base Definition, Examples, Detailed Descriptions, Mappings and R2 Conversions.

Extensions (+ see extensions on all Elements)
No Extensions defined for this type (though extensions on all Elements)

2.24.4.6 Quantity

See also Base Definition, Examples, Detailed Descriptions, Mappings and R2 Conversions.

Extensions (+ see extensions on all Elements)
iso21090-PQ-translation | PQ-translation | for ISO 21090 Data Type Extensions
iso21090-uncertainty | uncertainty | for ISO 21090 Data Type Extensions
iso21090-uncertaintyType | uncertaintyType | for ISO 21090 Data Type Extensions

Profile Name | Rules | Formal Definitions
Simple Quantity | id Level Location Description Expression sqty-1 Rule (base) The comparator is not used on a SimpleQuantity comparator.empty()) | XML, JSON
id | Level | Location | Description | Expression
sqty-1 | Rule | (base) | The comparator is not used on a SimpleQuantity | comparator.empty())
| Usage: Dosage, SampledData, Range, ActivityDefinition, CarePlan, Claim, ClaimResponse, Contract, Coverage, CoverageEligibilityRequest, ExplanationOfBenefit, Immunization, MedicationAdministration, MedicationDispense, MedicationKnowledge, MedicationRequest, NutritionOrder, Observation, Specimen, SpecimenDefinition, Substance, SupplyDelivery and VisionPrescription

2.24.4.7 Money

See also Base Definition, Examples, Detailed Descriptions, Mappings and R2 Conversions.

Extensions (+ see extensions on all Elements)
No Extensions defined for this type (though extensions on all Elements)

2.24.4.8 Range

See also Base Definition, Examples, Detailed Descriptions, Mappings and R2 Conversions.

Extensions (+ see extensions on all Elements)
No Extensions defined for this type (though extensions on all Elements)

2.24.4.9 Ratio

See also Base Definition, Examples, Detailed Descriptions, Mappings and R2 Conversions.

Extensions (+ see extensions on all Elements)
No Extensions defined for this type (though extensions on all Elements)

2.24.4.10 Period

See also Base Definition, Examples, Detailed Descriptions, Mappings and R2 Conversions.

Extensions (+ see extensions on all Elements)
No Extensions defined for this type (though extensions on all Elements)

2.24.4.11 SampledData

See also Base Definition, Examples, Detailed Descriptions, Mappings and R2 Conversions.

Extensions (+ see extensions on all Elements)
No Extensions defined for this type (though extensions on all Elements)

2.24.4.12 HumanName

See also Base Definition, Examples, Detailed Descriptions, Mappings and R2 Conversions.

Extensions (+ see extensions on all Elements)
humanname-assembly-order | assembly-order | for Common extensions for HumanName data type
humanname-fathers-family | fathers-family | for Common extensions for HumanName data type
humanname-mothers-family | mothers-family | for Common extensions for HumanName data type
humanname-own-name | own-name | for Common extensions for HumanName data type
humanname-own-prefix | own-prefix | for Common extensions for HumanName data type
humanname-partner-name | partner-name | for Common extensions for HumanName data type
humanname-partner-prefix | partner-prefix | for Common extensions for HumanName data type
iso21090-EN-qualifier | EN-qualifier | for ISO 21090 Data Type Extensions
iso21090-EN-representation | EN-representation | for ISO 21090 Data Type Extensions
iso21090-EN-use | EN-use | for ISO 21090 Data Type Extensions
language | Human Language | for General Extensions for use by FHIR Implementers

2.24.4.13 Address

See also Base Definition, Examples, Detailed Descriptions, Mappings and R2 Conversions.

Extensions (+ see extensions on all Elements)
geolocation | Geolocation | for General Extensions for use by FHIR Implementers
iso21090-AD-use | AD-use | for ISO 21090 Data Type Extensions
iso21090-ADXP-additionalLocator | ADXP-additionalLocator | for ISO 21090 Data Type Extensions
iso21090-ADXP-buildingNumberSuffix | ADXP-buildingNumberSuffix | for ISO 21090 Data Type Extensions
iso21090-ADXP-careOf | ADXP-careOf | for ISO 21090 Data Type Extensions
iso21090-ADXP-censusTract | ADXP-censusTract | for ISO 21090 Data Type Extensions
iso21090-ADXP-delimiter | ADXP-delimiter | for ISO 21090 Data Type Extensions
iso21090-ADXP-deliveryAddressLine | ADXP-deliveryAddressLine | for ISO 21090 Data Type Extensions
iso21090-ADXP-deliveryInstallationArea | ADXP-deliveryInstallationArea | for ISO 21090 Data Type Extensions
iso21090-ADXP-deliveryInstallationQualifier | ADXP-deliveryInstallationQualifier | for ISO 21090 Data Type Extensions
iso21090-ADXP-deliveryInstallationType | ADXP-deliveryInstallationType | for ISO 21090 Data Type Extensions
iso21090-ADXP-deliveryMode | ADXP-deliveryMode | for ISO 21090 Data Type Extensions
iso21090-ADXP-deliveryModeIdentifier | ADXP-deliveryModeIdentifier | for ISO 21090 Data Type Extensions
iso21090-ADXP-direction | ADXP-direction | for ISO 21090 Data Type Extensions
iso21090-ADXP-houseNumber | ADXP-houseNumber | for ISO 21090 Data Type Extensions
iso21090-ADXP-houseNumberNumeric | ADXP-houseNumberNumeric | for ISO 21090 Data Type Extensions
iso21090-ADXP-postBox | ADXP-postBox | for ISO 21090 Data Type Extensions
iso21090-ADXP-precinct | ADXP-precinct | for ISO 21090 Data Type Extensions
iso21090-ADXP-streetAddressLine | ADXP-streetAddressLine | for ISO 21090 Data Type Extensions
iso21090-ADXP-streetName | ADXP-streetName | for ISO 21090 Data Type Extensions
iso21090-ADXP-streetNameBase | ADXP-streetNameBase | for ISO 21090 Data Type Extensions
iso21090-ADXP-streetNameType | ADXP-streetNameType | for ISO 21090 Data Type Extensions
iso21090-ADXP-unitID | ADXP-unitID | for ISO 21090 Data Type Extensions
iso21090-ADXP-unitType | ADXP-unitType | for ISO 21090 Data Type Extensions
language | Human Language | for General Extensions for use by FHIR Implementers

2.24.4.14 ContactPoint

See also Base Definition, Examples, Detailed Descriptions, Mappings and R2 Conversions.

Extensions (+ see extensions on all Elements)
contactpoint-area | area | for Common extensions for ContactPoint data type
contactpoint-country | country | for Common extensions for ContactPoint data type
contactpoint-extension | extension | for Common extensions for ContactPoint data type
contactpoint-local | local | for Common extensions for ContactPoint data type
iso21090-TEL-address | TEL-address | for ISO 21090 Data Type Extensions

2.24.4.15 Timing

See also Base Definition, Examples, Detailed Descriptions, Mappings and R2 Conversions.

Extensions (+ see extensions on all Elements)
timing-dayOfMonth | dayOfMonth | for Common extensions for Timing data type
timing-exact | exact | for Common extensions for Timing data type

2.24.4.16 Signature

See also Base Definition, Examples, Detailed Descriptions, Mappings and R2 Conversions.

Extensions (+ see extensions on all Elements)
No Extensions defined for this type (though extensions on all Elements)

2.24.4.17 Annotation

See also Base Definition, Examples, Detailed Descriptions, Mappings and R2 Conversions.

Extensions (+ see extensions on all Elements)
language | Human Language | for General Extensions for use by FHIR Implementers

2.24.5 Data Type Detailed Descriptions

URL: https://hl7.org/fhir/R4/datatypes-version-maps.html

2.24.5 Data Type Detailed Descriptions

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative

This page provides the detailed descriptions for the data types

Table of Contents

Primitive Types ElementinstanttimedatedateTimedecimalbooleanintegerstringuribase64Binarycodeidoidunsignedintpositiveintmarkdownurlcanonicaluuid | Complex Types ElementIdentifierHumanNameAddressContactPointTimingQuantitySimpleQuantityAttachmentRangePeriodRatioCodeableConceptCodingSampledDataAgeDistanceDurationCountMoneyMoneyQuantityAnnotationSignature BackboneElement

2.24.5.1 Primitive Types

See also Base Definition, Detailed Descriptions, Examples, Profiles & Extensions and Mappings.

Functional status for this map: Not Mapped (based on R2 -> R3 -> R2 round tripping)

2.24.5.1.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/primitives3to4" = "R3 to R4 Primitive Conversions"

uses "http://hl7.org/fhir/3.0/StructureDefinition/base64Binary" alias base64BinaryR3 as source
uses "http://hl7.org/fhir/StructureDefinition/base64Binary" as target
uses "http://hl7.org/fhir/3.0/StructureDefinition/boolean" alias booleanR3 as source
uses "http://hl7.org/fhir/StructureDefinition/boolean" as target
uses "http://hl7.org/fhir/3.0/StructureDefinition/canonical" alias canonicalR3 as source
uses "http://hl7.org/fhir/StructureDefinition/canonical" as target
uses "http://hl7.org/fhir/3.0/StructureDefinition/code" alias codeR3 as source
uses "http://hl7.org/fhir/StructureDefinition/code" as target
uses "http://hl7.org/fhir/3.0/StructureDefinition/date" alias dateR3 as source
uses "http://hl7.org/fhir/StructureDefinition/date" as target
uses "http://hl7.org/fhir/3.0/StructureDefinition/dateTime" alias dateTimeR3 as source
uses "http://hl7.org/fhir/StructureDefinition/dateTime" as target
uses "http://hl7.org/fhir/3.0/StructureDefinition/decimal" alias decimalR3 as source
uses "http://hl7.org/fhir/StructureDefinition/decimal" as target
uses "http://hl7.org/fhir/3.0/StructureDefinition/id" alias idR3 as source
uses "http://hl7.org/fhir/StructureDefinition/id" as target
uses "http://hl7.org/fhir/3.0/StructureDefinition/instant" alias instantR3 as source
uses "http://hl7.org/fhir/StructureDefinition/instant" as target
uses "http://hl7.org/fhir/3.0/StructureDefinition/integer" alias integerR3 as source
uses "http://hl7.org/fhir/StructureDefinition/integer" as target
uses "http://hl7.org/fhir/3.0/StructureDefinition/markdown" alias markdownR3 as source
uses "http://hl7.org/fhir/StructureDefinition/markdown" as target
uses "http://hl7.org/fhir/3.0/StructureDefinition/oid" alias oidR3 as source
uses "http://hl7.org/fhir/StructureDefinition/oid" as target
uses "http://hl7.org/fhir/3.0/StructureDefinition/positiveInt" alias positiveIntR3 as source
uses "http://hl7.org/fhir/StructureDefinition/positiveInt" as target
uses "http://hl7.org/fhir/3.0/StructureDefinition/string" alias stringR3 as source
uses "http://hl7.org/fhir/StructureDefinition/string" as target
uses "http://hl7.org/fhir/3.0/StructureDefinition/time" alias timeR3 as source
uses "http://hl7.org/fhir/StructureDefinition/time" as target
uses "http://hl7.org/fhir/3.0/StructureDefinition/unsignedInt" alias unsignedIntR3 as source
uses "http://hl7.org/fhir/StructureDefinition/unsignedInt" as target
uses "http://hl7.org/fhir/3.0/StructureDefinition/uri" alias uriR3 as source
uses "http://hl7.org/fhir/StructureDefinition/uri" as target
uses "http://hl7.org/fhir/3.0/StructureDefinition/url" alias urlR3 as source
uses "http://hl7.org/fhir/StructureDefinition/url" as target
uses "http://hl7.org/fhir/3.0/StructureDefinition/uuid" alias uuidR3 as source
uses "http://hl7.org/fhir/StructureDefinition/uuid" as target
uses "http://hl7.org/fhir/3.0/StructureDefinition/xhtml" alias xhtmlR3 as source
uses "http://hl7.org/fhir/StructureDefinition/xhtml" as target

imports "http://hl7.org/fhir/StructureMap/Element3to4"
```

```
group base64Binary(source src : base64BinaryR3, target tgt : base64Binary) extends Element <<type+>> {
  src.value as v -> tgt.value = v "base64Binary-value";
}

group boolean(source src : booleanR3, target tgt : boolean) extends Element <<type+>> {
  src.value as v -> tgt.value = v "boolean-value";
}

group canonical(source src : canonicalR3, target tgt : canonical) extends Element <<type+>> {
  src.value as v -> tgt.value = v "canonical-value";
}

group code(source src : codeR3, target tgt : code) extends Element <<type+>> {
  src.value as v -> tgt.value = v "code-value";
}

group date(source src : dateR3, target tgt : date) extends Element <<type+>> {
  src.value as v -> tgt.value = v "date-value";
}

group dateTime(source src : dateTimeR3, target tgt : dateTime) extends Element <<type+>> {
  src.value as v -> tgt.value = v "dateTime-value";
}

group decimal(source src : decimalR3, target tgt : decimal) extends Element <<type+>> {
  src.value as v -> tgt.value = v "decimal-value";
}

group id(source src : idR3, target tgt : id) extends Element <<type+>> {
  src.value as v -> tgt.value = v "id-value";
}

group instant(source src : instantR3, target tgt : instant) extends Element <<type+>> {
  src.value as v -> tgt.value = v "instant-value";
}

group integer(source src : integerR3, target tgt : integer) extends Element <<type+>> {
  src.value as v -> tgt.value = v "integer-value";
}

group markdown(source src : markdownR3, target tgt : markdown) extends Element <<type+>> {
  src.value as v -> tgt.value = v "markdown-value";
}

group oid(source src : oidR3, target tgt : oid) extends Element <<type+>> {
  src.value as v -> tgt.value = v "oid-value";
}

group positiveInt(source src : positiveIntR3, target tgt : positiveInt) extends Element <<type+>> {
  src.value as v -> tgt.value = v "positiveInt-value";
}

group string(source src : stringR3, target tgt : string) extends Element <<type+>> {
  src.value as v -> tgt.value = v "string-value";
}

group time(source src : timeR3, target tgt : time) extends Element <<type+>> {
  src.value as v -> tgt.value = v "time-value";
}

group unsignedInt(source src : unsignedIntR3, target tgt : unsignedInt) extends Element <<type+>> {
  src.value as v -> tgt.value = v "unsignedInt-value";
}

group uri(source src : uriR3, target tgt : uri) extends Element <<type+>> {
  src.value as v -> tgt.value = v "uri-value";
}

group uri2Canonical(source src : uriR3, target tgt : canonical) extends Element <<types>> {
  src.value as v -> tgt.value = v "uri-value";
}

group url(source src : urlR3, target tgt : url) extends Element <<type+>> {
  src.value as v -> tgt.value = v "url-value";
}
group uuid(source src : uuidR3, target tgt : uuid) extends Element <<type+>> {
  src.value as v -> tgt.value = v "uuid-value";
}

group xhtml(source src : xhtmlR3, target tgt : xhtml) extends Element <<type+>> {
  src.value as v -> tgt.value = v "xhtml-value";
}

group stringToMarkdown(source src : stringR3, target tgt : markdown) extends Element <<types>> {
  src.value as v -> tgt.value = v "stringToMarkdown-value";
}

group id2code(source src : idR3, target tgt : code) extends Element <<types>> {
  src.value as v -> tgt.value = v "id2code-value";
```

```
}

group code2uri(source src : codeR3, target tgt : uri) extends Element <<types>> {
  src.value as v -> tgt.value = v "code2uri-value";
}

group code2string(source src : codeR3, target tgt : string) extends Element <<types>> {
  src.value as v -> tgt.value = v "code2uri-value";
}

group string2code(source src : stringR3, target tgt : code) extends Element <<types>> {
  src.value as v -> tgt.value = v "code2uri-value";
}

group uri2url(source src : uriR3, target tgt : url) extends Element <<types>> {
  src.value as v -> tgt.value = v "code2uri-value";
}

group id2string(source src : idR3, target tgt : string) extends Element <<type+>> {
  src.value as v -> tgt.value = v;
}

group uri2string(source src : uriR3, target tgt : string) extends Element <<types>> {
  src.value as v -> tgt.value = v "code2uri-value";
}

group unsignedInt2decimal(source src : unsignedIntR3, target tgt : decimal) extends Element <<types>> {
  src.value as v -> tgt.value = v "unsignedInt2decimal-value";
}

group oid2string(source src : oidR3, target tgt : string) extends Element <<type+>> {
  src.value as v -> tgt.value = v;
}

group oid2id(source src : oidR3, target tgt : id) extends Element <<type+>> {
  src.value as v -> tgt.value = v;
}

group positiveInt2integer(source src : positiveIntR3, target tgt : integer) extends Element <<types>> {
  src.value as v -> tgt.value = v "unsignedInt2decimal-value";
}

group integer2positiveInt(source src : integerR3, target tgt : positiveInt) extends Element <<types>> {
  src.value as v -> tgt.value = v "unsignedInt2decimal-value";
}

group positiveInt2unsignedInt(source src : positiveIntR3, target tgt : unsignedInt) extends Element <<types>> {
  src.value as v -> tgt.value = v;
}
```

2.24.5.1.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/primitives4to3" = "R4 to R3 Primitive Conversions"

uses "http://hl7.org/fhir/StructureDefinition/base64Binary" as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/base64Binary" alias base64BinaryR3 as target
uses "http://hl7.org/fhir/StructureDefinition/boolean" as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/boolean" alias booleanR3 as target
uses "http://hl7.org/fhir/StructureDefinition/canonical" as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/canonical" alias canonicalR3 as target
uses "http://hl7.org/fhir/StructureDefinition/code" as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/code" alias codeR3 as target
uses "http://hl7.org/fhir/StructureDefinition/date" as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/date" alias dateR3 as target
uses "http://hl7.org/fhir/StructureDefinition/dateTime" as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/dateTime" alias dateTimeR3 as target
uses "http://hl7.org/fhir/StructureDefinition/decimal" as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/decimal" alias decimalR3 as target
uses "http://hl7.org/fhir/StructureDefinition/id" as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/id" alias idR3 as target
uses "http://hl7.org/fhir/StructureDefinition/instant" as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/instant" alias instantR3 as target
uses "http://hl7.org/fhir/StructureDefinition/integer" as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/integer" alias integerR3 as target
uses "http://hl7.org/fhir/StructureDefinition/markdown" as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/markdown" alias markdownR3 as target
uses "http://hl7.org/fhir/StructureDefinition/oid" as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/oid" alias oidR3 as target
uses "http://hl7.org/fhir/StructureDefinition/positiveInt" as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/positiveInt" alias positiveIntR3 as target
uses "http://hl7.org/fhir/StructureDefinition/string" as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/string" alias stringR3 as target
uses "http://hl7.org/fhir/StructureDefinition/time" as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/time" alias timeR3 as target
uses "http://hl7.org/fhir/StructureDefinition/unsignedInt" as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/unsignedInt" alias unsignedIntR3 as target
uses "http://hl7.org/fhir/StructureDefinition/uri" as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/uri" alias uriR3 as target
uses "http://hl7.org/fhir/StructureDefinition/url" as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/url" alias urlR3 as target
```

```
uses "http://hl7.org/fhir/StructureDefinition/uuid" as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/uuid" alias uuidR3 as target
uses "http://hl7.org/fhir/StructureDefinition/xhtml" as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/xhtml" alias xhtmlR3 as target

imports "http://hl7.org/fhir/StructureMap/Element4to3"

group base64Binary(source src : base64Binary, target tgt : base64BinaryR3) extends Element <<type+>> {
  src.value as v -> tgt.value = v "base64Binary-value";
}

group boolean(source src : boolean, target tgt : booleanR3) extends Element <<type+>> {
  src.value as v -> tgt.value = v "boolean-value";
}

group canonical(source src : canonical, target tgt : canonicalR3) extends Element <<type+>> {
  src.value as v -> tgt.value = v "canonical-value";
}

group code(source src : code, target tgt : codeR3) extends Element <<type+>> {
  src.value as v -> tgt.value = v "code-value";
}

group date(source src : date, target tgt : dateR3) extends Element <<type+>> {
  src.value as v -> tgt.value = v "date-value";
}

group dateTime(source src : dateTime, target tgt : dateTimeR3) extends Element <<type+>> {
  src.value as v -> tgt.value = v "dateTime-value";
}

group decimal(source src : decimal, target tgt : decimalR3) extends Element <<type+>> {
  src.value as v -> tgt.value = v "decimal-value";
}

group id(source src : id, target tgt : idR3) extends Element <<type+>> {
  src.value as v -> tgt.value = v "id-value";
}

group instant(source src : instant, target tgt : instantR3) extends Element <<type+>> {
  src.value as v -> tgt.value = v "instant-value";
}

group integer(source src : integer, target tgt : integerR3) extends Element <<type+>> {
  src.value as v -> tgt.value = v "integer-value";
}

group markdown(source src : markdown, target tgt : markdownR3) extends Element <<type+>> {
  src.value as v -> tgt.value = v "markdown-value";
}

group oid(source src : oid, target tgt : oidR3) extends Element <<type+>> {
  src.value as v -> tgt.value = v "oid-value";
}

group positiveInt(source src : positiveInt, target tgt : positiveIntR3) extends Element <<type+>> {
  src.value as v -> tgt.value = v "positiveInt-value";
}

group string(source src : string, target tgt : stringR3) extends Element <<type+>> {
  src.value as v -> tgt.value = v "string-value";
}

group time(source src : time, target tgt : timeR3) extends Element <<type+>> {
  src.value as v -> tgt.value = v "time-value";
}

group unsignedInt(source src : unsignedInt, target tgt : unsignedIntR3) extends Element <<type+>> {
  src.value as v -> tgt.value = v "unsignedInt-value";
}

group uri(source src : uri, target tgt : uriR3) extends Element <<type+>> {
  src.value as v -> tgt.value = v "uri-value";
}

group url(source src : url, target tgt : urlR3) extends Element <<type+>> {
  src.value as v -> tgt.value = v "url-value";
}

group uuid(source src : uuid, target tgt : uuidR3) extends Element <<type+>> {
  src.value as v -> tgt.value = v "uuid-value";
}
group xhtml(source src : xhtml, target tgt : xhtmlR3) extends Element <<type+>> {
  src.value as v -> tgt.value = v "xhtml-value";
}

group markdownToString(source src : markdown, target tgt : stringR3) extends Element <<types>> {
  src.value as v -> tgt.value = v "markdownToString-value";
}
```

```
group code2id(source src : code, target tgt : idR3) extends Element <<types>> {
  src.value as v -> tgt.value = v "code2id-value";
}

group uri2code(source src : uriR3, target tgt : code) extends Element <<types>> {
  src.value as v -> tgt.value = v "uri2code-value";
}

group string2code(source src : stringR3, target tgt : code) extends Element <<types>> {
  src.value as v -> tgt.value = v "uri2code-value";
}

group code2string(source src : codeR3, target tgt : string) extends Element <<types>> {
  src.value as v -> tgt.value = v "uri2code-value";
}

group canonical2Uri(source src : canonicalR3, target tgt : uri) extends Element <<types>> {
  src.value as v -> tgt.value = v "canonical2uri-value";
}

group url2uri(source src : url, target tgt : uriR3) extends Element <<types>> {
  src.value as v -> tgt.value = v "code2uri-value";
}

group string2id(source src : string, target tgt : idR3) extends Element <<types>> {
  src.value as v -> tgt.value = v "uri2code-value";
}

group string2uri(source src : string, target tgt : uriR3) extends Element <<types>> {
  src.value as v -> tgt.value = v "code2uri-value";
}

group decimal2unsignedInt(source src : decimal, target tgt : unsignedIntR3) extends Element <<types>> {
  src.value as v -> tgt.value = v "unsignedInt2decimal-value";
}

group canonical2string(source src : canonical, target tgt : stringR3) extends Element <<types>> {
  src.value as v -> tgt.value = v "code2uri-value";
}

group oid2string(source src : string, target tgt : oidR3) extends Element <<types>> {
  src.value as v -> tgt.value = v;
}

group id2oid(source src : id, target tgt : oidR3) extends Element <<types>> {
  src.value as v -> tgt.value = v;
}

group id2string(source src : id, target tgt : stringR3) extends Element <<type+>> {
  src.value as v -> tgt.value = v;
}

group code2oid(source src : code, target tgt : oidR3) extends Element <<type+>> {
  src.value as v -> tgt.value = v;
}

group integer2positiveInt(source src : integer, target tgt : positiveIntR3) extends Element <<types>> {
  src.value as v -> tgt.value = v "unsignedInt2decimal-value";
}

group positiveInt2integer(source src : positiveInt, target tgt : integerR3) extends Element <<types>> {
  src.value as v -> tgt.value = v "unsignedInt2decimal-value";
}

group unsignedInt2positiveInt(source src : unsignedInt, target tgt : positiveIntR3) extends Element <<types>> {
  src.value as v -> tgt.value = v;
}

group code2uri(source src : code, target tgt : uriR3) extends Element <<types>> {
  src.value as v -> tgt.value = v;
}
```

2.24.5.2 Attachment

See also Base Definition, Detailed Descriptions, Examples, Profiles & Extensions and Mappings.

Functional status for this map: Not Mapped (based on R2 -> R3 -> R2 round tripping)

2.24.5.2.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/Attachment3to4" = "R3 to R4 Conversions for Attachment"

uses "http://hl7.org/fhir/3.0/StructureDefinition/Attachment" alias AttachmentR3 as source
uses "http://hl7.org/fhir/StructureDefinition/Attachment" alias Attachment as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group Attachment(source src : AttachmentR3, target tgt : Attachment) extends Element <<type+>> {
  src.contentType -> tgt.contentType;
  src.language -> tgt.language;
```

```
  src.data -> tgt.data;
  src.url -> tgt.url;
  src.size -> tgt.size;
  src.hash -> tgt.hash;
  src.title -> tgt.title;
  src.creation -> tgt.creation;
}
```

2.24.5.2.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/Attachment4to3" = "R4 to R3 Conversion for Attachment"

uses "http://hl7.org/fhir/StructureDefinition/Attachment" alias Attachment as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/Attachment" alias AttachmentR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group Attachment(source src : Attachment, target tgt : AttachmentR3) extends Element <<type+>> {
  src.contentType -> tgt.contentType;
  src.language -> tgt.language;
  src.data -> tgt.data;
  src.url -> tgt.url;
  src.size -> tgt.size;
  src.hash -> tgt.hash;
  src.title -> tgt.title;
  src.creation -> tgt.creation;
}
```

2.24.5.3 Identifier

See also Base Definition, Detailed Descriptions, Examples, Profiles & Extensions and Mappings.

Functional status for this map: Not Mapped (based on R2 -> R3 -> R2 round tripping)

2.24.5.3.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/Identifier3to4" = "R3 to R4 Conversions for Identifier"

uses "http://hl7.org/fhir/3.0/StructureDefinition/Identifier" alias IdentifierR3 as source
uses "http://hl7.org/fhir/StructureDefinition/Identifier" alias Identifier as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group Identifier(source src : IdentifierR3, target tgt : Identifier) extends Element <<type+>> {
  src.use -> tgt.use;
  src.type -> tgt.type;
  src.system -> tgt.system;
  src.value -> tgt.value;
  src.period -> tgt.period;
  src.assigner -> tgt.assigner;
}

group Identifier2Codeable(source src, target tgt) extends Element {
  src -> tgt.coding as c then {
    src.system -> c.system;
    src.value -> c.code;
  } "coding";
}
```

2.24.5.3.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/Identifier4to3" = "R4 to R3 Conversion for Identifier"

uses "http://hl7.org/fhir/StructureDefinition/Identifier" alias Identifier as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/Identifier" alias IdentifierR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group Identifier(source src : Identifier, target tgt : IdentifierR3) extends Element <<type+>> {
  src.use -> tgt.use;
  src.type -> tgt.type;
  src.system -> tgt.system;
  src.value -> tgt.value;
  src.period -> tgt.period;
  src.assigner -> tgt.assigner;
}

group Codeable2Identifier(source src, target tgt) extends Element {
  src.coding as c then {
    c.system -> tgt.system;
    c.code -> tgt.value;
  };
}
```

2.24.5.4 Coding

See also Base Definition, Detailed Descriptions, Examples, Profiles & Extensions and Mappings.

Functional status for this map: Not Mapped (based on R2 -> R3 -> R2 round tripping)

2.24.5.4.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/Coding3to4" = "R3 to R4 Conversions for Coding"

uses "http://hl7.org/fhir/3.0/StructureDefinition/Coding" alias CodingR3 as source
uses "http://hl7.org/fhir/StructureDefinition/Coding" alias Coding as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group Coding(source src : CodingR3, target tgt : Coding) extends Element <<type+>> {
  src.system -> tgt.system;
  src.version -> tgt.version;
  src.code -> tgt.code;
  src.display -> tgt.display;
  src.userSelected -> tgt.userSelected;
}
```

2.24.5.4.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/Coding4to3" = "R4 to R3 Conversion for Coding"

uses "http://hl7.org/fhir/StructureDefinition/Coding" alias Coding as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/Coding" alias CodingR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group Coding(source src : Coding, target tgt : CodingR3) extends Element <<type+>> {
  src.system -> tgt.system;
  src.version -> tgt.version;
  src.code -> tgt.code;
  src.display -> tgt.display;
  src.userSelected -> tgt.userSelected;
}

group Coding2code(source src : Coding, target tgt : codeR3) extends Element <<types>> {
  src.code as vs0 then code(vs0, tgt);
}
```

2.24.5.5 CodeableConcept

See also Base Definition, Detailed Descriptions, Examples, Profiles & Extensions and Mappings.

Functional status for this map: Not Mapped (based on R2 -> R3 -> R2 round tripping)

2.24.5.5.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/CodeableConcept3to4" = "R3 to R4 Conversions for CodeableConcept"

uses "http://hl7.org/fhir/3.0/StructureDefinition/CodeableConcept" alias CodeableConceptR3 as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/Coding" alias CodingR3 as source
uses "http://hl7.org/fhir/StructureDefinition/CodeableConcept" alias CodeableConcept as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group CodeableConcept(source src : CodeableConceptR3, target tgt : CodeableConcept) extends Element <<type+>> {
  src.coding -> tgt.coding;
  src.text -> tgt.text;
}

group Coding2CodeableConcept(source src : CodingR3, target tgt : CodeableConcept) extends Element <<types>> {
  src -> tgt.coding as vt0 then Coding(src, vt0) "coding";
}
```

2.24.5.5.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/CodeableConcept4to3" = "R4 to R3 Conversion for CodeableConcept"

uses "http://hl7.org/fhir/StructureDefinition/CodeableConcept" alias CodeableConcept as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/CodeableConcept" alias CodeableConceptR3 as target
uses "http://hl7.org/fhir/3.0/StructureDefinition/Coding" alias CodingR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group CodeableConcept(source src : CodeableConcept, target tgt : CodeableConceptR3) extends Element <<type+>> {
  src.coding -> tgt.coding;
  src.text -> tgt.text;
}

group CodeableConceptNoExtensions(source src : CodeableConcept, target tgt : CodeableConceptR3) {
  src.coding -> tgt.coding;
  src.text -> tgt.text;
}

group CodeableConcept2Coding(source src : CodeableConcept, target tgt : CodingR3) extends Element <<types>> {
  src.coding first as vs0 then Coding(vs0, tgt);
}

group CodeableConcept2code(source src : CodeableConcept, target tgt : codeR3) extends Element <<types>> {
  src.coding first as vs0 -> tgt.code = create('code') as vt then Coding2code(vs0, vt) "CodeableConcept2code-coding";
}
```

2.24.5.6 Quantity

See also Base Definition, Detailed Descriptions, Examples, Profiles & Extensions and Mappings.

Functional status for this map: Not Mapped (based on R2 -> R3 -> R2 round tripping)

2.24.5.6.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/Quantity3to4" = "R3 to R4 Conversions for Quantity"

uses "http://hl7.org/fhir/3.0/StructureDefinition/Quantity" alias QuantityR3 as source
uses "http://hl7.org/fhir/StructureDefinition/Quantity" alias Quantity as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group Quantity(source src : QuantityR3, target tgt : Quantity) extends Element <<type+>> {
  src.value -> tgt.value;
  src.comparator -> tgt.comparator;
  src.unit -> tgt.unit;
  src.system -> tgt.system;
  src.code -> tgt.code;
}
```

2.24.5.6.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/Quantity4to3" = "R4 to R3 Conversion for Quantity"

uses "http://hl7.org/fhir/StructureDefinition/Quantity" alias Quantity as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/Quantity" alias QuantityR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group Quantity(source src : Quantity, target tgt : QuantityR3) extends Element <<type+>> {
  src.value -> tgt.value;
  src.comparator -> tgt.comparator;
  src.unit -> tgt.unit;
  src.system -> tgt.system;
  src.code -> tgt.code;
}
```

2.24.5.7 Money

See also Base Definition, Detailed Descriptions, Examples, Profiles & Extensions and Mappings.

Functional status for this map: Not Mapped (based on R2 -> R3 -> R2 round tripping)

2.24.5.7.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/Money3to4" = "R3 to R4 Conversions for Money"

uses "http://hl7.org/fhir/3.0/StructureDefinition/Money" alias MoneyR3 as source
uses "http://hl7.org/fhir/StructureDefinition/Money" alias Money as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group Money(source src : MoneyR3, target tgt : Money) extends Element <<type+>> {
  src.value -> tgt.value;
  src.code -> tgt.currency;
}
```

2.24.5.7.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/Money4to3" = "R4 to R3 Conversion for Money"

uses "http://hl7.org/fhir/StructureDefinition/Money" alias Money as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/Money" alias MoneyR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group Money(source src : Money, target tgt : MoneyR3) extends Element <<type+>> {
  src.value -> tgt.value;
  src.currency -> tgt.code "code";
  src.currency as s ->  tgt.unit as t,  t.value = s;
  src ->  tgt.system as vt,  vt.value = 'urn:iso:std:iso:4217' "code";
}
```

2.24.5.8 Range

See also Base Definition, Detailed Descriptions, Examples, Profiles & Extensions and Mappings.

Functional status for this map: Not Mapped (based on R2 -> R3 -> R2 round tripping)

2.24.5.8.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/Range3to4" = "R3 to R4 Conversions for Range"

uses "http://hl7.org/fhir/3.0/StructureDefinition/Range" alias RangeR3 as source
uses "http://hl7.org/fhir/StructureDefinition/Range" alias Range as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group Range(source src : RangeR3, target tgt : Range) extends Element <<type+>> {
  src.low -> tgt.low;
  src.high -> tgt.high;
```

```
}
```

2.24.5.8.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/Range4to3" = "R4 to R3 Conversion for Range"

uses "http://hl7.org/fhir/StructureDefinition/Range" alias Range as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/Range" alias RangeR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group Range(source src : Range, target tgt : RangeR3) extends Element <<type+>> {
  src.low -> tgt.low;
  src.high -> tgt.high;
}
```

2.24.5.9 Ratio

See also Base Definition, Detailed Descriptions, Examples, Profiles & Extensions and Mappings.

Functional status for this map: Not Mapped (based on R2 -> R3 -> R2 round tripping)

2.24.5.9.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/Ratio3to4" = "R3 to R4 Conversions for Ratio"

uses "http://hl7.org/fhir/3.0/StructureDefinition/Ratio" alias RatioR3 as source
uses "http://hl7.org/fhir/StructureDefinition/Ratio" alias Ratio as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group Ratio(source src : RatioR3, target tgt : Ratio) extends Element <<type+>> {
  src.numerator -> tgt.numerator;
  src.denominator -> tgt.denominator;
}
```

2.24.5.9.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/Ratio4to3" = "R4 to R3 Conversion for Ratio"

uses "http://hl7.org/fhir/StructureDefinition/Ratio" alias Ratio as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/Ratio" alias RatioR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group Ratio(source src : Ratio, target tgt : RatioR3) extends Element <<type+>> {
  src.numerator -> tgt.numerator;
  src.denominator -> tgt.denominator;
}
```

2.24.5.10 Period

See also Base Definition, Detailed Descriptions, Examples, Profiles & Extensions and Mappings.

Functional status for this map: Not Mapped (based on R2 -> R3 -> R2 round tripping)

2.24.5.10.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/Period3to4" = "R3 to R4 Conversions for Period"

uses "http://hl7.org/fhir/3.0/StructureDefinition/Period" alias PeriodR3 as source
uses "http://hl7.org/fhir/StructureDefinition/Period" alias Period as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group Period(source src : PeriodR3, target tgt : Period) extends Element <<type+>> {
  src.start -> tgt.start;
  src.end -> tgt.end;
}
```

2.24.5.10.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/Period4to3" = "R4 to R3 Conversion for Period"

uses "http://hl7.org/fhir/StructureDefinition/Period" alias Period as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/Period" alias PeriodR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group Period(source src : Period, target tgt : PeriodR3) extends Element <<type+>> {
  src.start -> tgt.start;
  src.end -> tgt.end;
}
```

2.24.5.11 SampledData

See also Base Definition, Detailed Descriptions, Examples, Profiles & Extensions and Mappings.

Functional status for this map: Not Mapped (based on R2 -> R3 -> R2 round tripping)

2.24.5.11.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/SampledData3to4" = "R3 to R4 Conversions for SampledData"
```

```
uses "http://hl7.org/fhir/3.0/StructureDefinition/SampledData" alias SampledDataR3 as source
uses "http://hl7.org/fhir/StructureDefinition/SampledData" alias SampledData as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group SampledData(source src : SampledDataR3, target tgt : SampledData) extends Element <<type+>> {
  src.origin -> tgt.origin;
  src.period -> tgt.period;
  src.factor -> tgt.factor;
  src.lowerLimit -> tgt.lowerLimit;
  src.upperLimit -> tgt.upperLimit;
  src.dimensions -> tgt.dimensions;
  src.data -> tgt.data;
}
```

2.24.5.11.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/SampledData4to3" = "R4 to R3 Conversion for SampledData"

uses "http://hl7.org/fhir/StructureDefinition/SampledData" alias SampledData as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/SampledData" alias SampledDataR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group SampledData(source src : SampledData, target tgt : SampledDataR3) extends Element <<type+>> {
  src.origin -> tgt.origin;
  src.period -> tgt.period;
  src.factor -> tgt.factor;
  src.lowerLimit -> tgt.lowerLimit;
  src.upperLimit -> tgt.upperLimit;
  src.dimensions -> tgt.dimensions;
  src.data -> tgt.data;
}
```

2.24.5.12 HumanName

See also Base Definition, Detailed Descriptions, Examples, Profiles & Extensions and Mappings.

Functional status for this map: Not Mapped (based on R2 -> R3 -> R2 round tripping)

2.24.5.12.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/HumanName3to4" = "R3 to R4 Conversions for HumanName"

uses "http://hl7.org/fhir/3.0/StructureDefinition/HumanName" alias HumanNameR3 as source
uses "http://hl7.org/fhir/StructureDefinition/HumanName" alias HumanName as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group HumanName(source src : HumanNameR3, target tgt : HumanName) extends Element <<type+>> {
  src.use -> tgt.use;
  src.text -> tgt.text;
  src.family -> tgt.family;
  src.given -> tgt.given;
  src.prefix -> tgt.prefix;
  src.suffix -> tgt.suffix;
  src.period -> tgt.period;
}
```

2.24.5.12.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/HumanName4to3" = "R4 to R3 Conversion for HumanName"

uses "http://hl7.org/fhir/StructureDefinition/HumanName" alias HumanName as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/HumanName" alias HumanNameR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group HumanName(source src : HumanName, target tgt : HumanNameR3) extends Element <<type+>> {
  src.use -> tgt.use;
  src.text -> tgt.text;
  src.family -> tgt.family;
  src.given -> tgt.given;
  src.prefix -> tgt.prefix;
  src.suffix -> tgt.suffix;
  src.period -> tgt.period;
}
```

2.24.5.13 Address

See also Base Definition, Detailed Descriptions, Examples, Profiles & Extensions and Mappings.

Functional status for this map: Not Mapped (based on R2 -> R3 -> R2 round tripping)

2.24.5.13.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/Address3to4" = "R3 to R4 Conversions for Address"

uses "http://hl7.org/fhir/3.0/StructureDefinition/Address" alias AddressR3 as source
uses "http://hl7.org/fhir/StructureDefinition/Address" alias Address as target

imports "http://hl7.org/fhir/StructureMap/*3to4"
```

```
group Address(source src : AddressR3, target tgt : Address) extends Element <<type+>> {
  src.use -> tgt.use;
  src.type -> tgt.type;
  src.text -> tgt.text;
  src.line -> tgt.line;
  src.city -> tgt.city;
  src.district -> tgt.district;
  src.state -> tgt.state;
  src.postalCode -> tgt.postalCode;
  src.country -> tgt.country;
  src.period -> tgt.period;
}
```

2.24.5.13.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/Address4to3" = "R4 to R3 Conversion for Address"

uses "http://hl7.org/fhir/StructureDefinition/Address" alias Address as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/Address" alias AddressR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group Address(source src : Address, target tgt : AddressR3) extends Element <<type+>> {
  src.use -> tgt.use;
  src.type -> tgt.type;
  src.text -> tgt.text;
  src.line -> tgt.line;
  src.city -> tgt.city;
  src.district -> tgt.district;
  src.state -> tgt.state;
  src.postalCode -> tgt.postalCode;
  src.country -> tgt.country;
  src.period -> tgt.period;
}
```

2.24.5.14 ContactPoint

See also Base Definition, Detailed Descriptions, Examples, Profiles & Extensions and Mappings.

Functional status for this map: Not Mapped (based on R2 -> R3 -> R2 round tripping)

2.24.5.14.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/ContactPoint3to4" = "R3 to R4 Conversions for ContactPoint"

uses "http://hl7.org/fhir/3.0/StructureDefinition/ContactPoint" alias ContactPointR3 as source
uses "http://hl7.org/fhir/StructureDefinition/ContactPoint" alias ContactPoint as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group ContactPoint(source src : ContactPointR3, target tgt : ContactPoint) extends Element <<type+>> {
  src.system -> tgt.system;
  src.value -> tgt.value;
  src.use -> tgt.use;
  src.rank -> tgt.rank;
  src.period -> tgt.period;
}
```

2.24.5.14.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/ContactPoint4to3" = "R4 to R3 Conversion for ContactPoint"

uses "http://hl7.org/fhir/StructureDefinition/ContactPoint" alias ContactPoint as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/ContactPoint" alias ContactPointR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group ContactPoint(source src : ContactPoint, target tgt : ContactPointR3) extends Element <<type+>> {
  src.system -> tgt.system;
  src.value -> tgt.value;
  src.use -> tgt.use;
  src.rank -> tgt.rank;
  src.period -> tgt.period;
}
```

2.24.5.15 Timing

See also Base Definition, Detailed Descriptions, Examples, Profiles & Extensions and Mappings.

Functional status for this map: Not Mapped (based on R2 -> R3 -> R2 round tripping)

2.24.5.15.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/Timing3to4" = "R3 to R4 Conversions for Timing"

uses "http://hl7.org/fhir/3.0/StructureDefinition/Timing" alias TimingR3 as source
uses "http://hl7.org/fhir/StructureDefinition/Timing" alias Timing as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group Timing(source src : TimingR3, target tgt : Timing) extends BackboneElement <<type+>> {
  src.event -> tgt.event;
```

```
  src.repeat as vs0 -> tgt.repeat as vt0 then repeat(vs0, vt0);
  src.code -> tgt.code;
}

group repeat(source src, target tgt) extends Element {
  src.bounds : Duration as vs -> tgt.bounds = create('Duration') as vt then Duration(vs, vt);
  src.bounds : Range as vs -> tgt.bounds = create('Range') as vt then Range(vs, vt);
  src.bounds : Period as vs -> tgt.bounds = create('Period') as vt then Period(vs, vt);
  src.count -> tgt.count;
  src.countMax -> tgt.countMax;
  src.duration -> tgt.duration;
  src.durationMax -> tgt.durationMax;
  src.durationUnit -> tgt.durationUnit;
  src.frequency -> tgt.frequency;
  src.frequencyMax -> tgt.frequencyMax;
  src.period -> tgt.period;
  src.periodMax -> tgt.periodMax;
  src.periodUnit -> tgt.periodUnit;
  src.dayOfWeek -> tgt.dayOfWeek;
  src.timeOfDay -> tgt.timeOfDay;
  src.when -> tgt.when;
  src.offset -> tgt.offset;
}
```

2.24.5.15.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/Timing4to3" = "R4 to R3 Conversion for Timing"

uses "http://hl7.org/fhir/StructureDefinition/Timing" alias Timing as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/Timing" alias TimingR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group Timing(source src : Timing, target tgt : TimingR3) extends BackboneElement <<type+>> {
  src.event -> tgt.event;
  src.repeat as vs0 -> tgt.repeat as vt0 then repeat(vs0, vt0);
  src.code -> tgt.code;
}

group repeat(source src, target tgt) extends Element {
  src.bounds : Duration as vs -> tgt.bounds = create('Duration') as vt then Duration(vs, vt);
  src.bounds : Range as vs -> tgt.bounds = create('Range') as vt then Range(vs, vt);
  src.bounds : Period as vs -> tgt.bounds = create('Period') as vt then Period(vs, vt);
  src.count -> tgt.count;
  src.countMax -> tgt.countMax;
  src.duration -> tgt.duration;
  src.durationMax -> tgt.durationMax;
  src.durationUnit -> tgt.durationUnit;
  src.frequency -> tgt.frequency;
  src.frequencyMax -> tgt.frequencyMax;
  src.period -> tgt.period;
  src.periodMax -> tgt.periodMax;
  src.periodUnit -> tgt.periodUnit;
  src.dayOfWeek -> tgt.dayOfWeek;
  src.timeOfDay -> tgt.timeOfDay;
  src.when -> tgt.when;
  src.offset -> tgt.offset;
}
```

2.24.5.16 Signature

See also Base Definition, Detailed Descriptions, Examples, Profiles & Extensions and Mappings.

Functional status for this map: Not Mapped (based on R2 -> R3 -> R2 round tripping)

2.24.5.16.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/Signature3to4" = "R3 to R4 Conversions for Signature"

uses "http://hl7.org/fhir/3.0/StructureDefinition/Signature" alias SignatureR3 as source
uses "http://hl7.org/fhir/StructureDefinition/Signature" alias Signature as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group Signature(source src : SignatureR3, target tgt : Signature) extends Element <<type+>> {
  src.type -> tgt.type;
  src.when -> tgt.when;
  src.who : uri as vs ->  tgt.who as ref,  ref.reference = vs,  ref.extension as ext,  ext.url =
    'http://hl7.org/fhir/3.0/StructureDefinition/BaseType',  ext.value = 'uri';
  src.who : Reference as vs0 -> tgt.who as vt0 then Reference(vs0, vt0);
  src.onBehalfOf : uri as vs ->  tgt.onBehalfOf as ref,  ref.reference = vs,  ref.extension as ext,  ext.url =
    'http://hl7.org/fhir/3.0/StructureDefinition/BaseType',  ext.value = 'uri';
  src.onBehalfOf : Reference as vs0 -> tgt.onBehalfOf as vt0 then Reference(vs0, vt0);
  src.contentType -> tgt.sigFormat;
  src.blob -> tgt.data;
}
```

2.24.5.16.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/Signature4to3" = "R4 to R3 Conversion for Signature"
```

```
uses "http://hl7.org/fhir/StructureDefinition/Signature" alias Signature as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/Signature" alias SignatureR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group Signature(source src : SignatureR3, target tgt : Signature) extends Element <<type+>> {
  src.type -> tgt.type;
  src.when -> tgt.when;
  src.who as vs0 where extension('http://hl7.org/fhir/3.0/StructureDefinition/BaseType').empty() -> tgt.who =
    create('Reference') as vt0 then Reference(vs0, vt0);
  src.who as vs0 where extension('http://hl7.org/fhir/3.0/StructureDefinition/BaseType').exists() -> tgt.who =
    create('uri') as vt0 then {
    vs0.reference as vs1 then uri(vs1, vt0);
  };
  src.onBehalfOf as vs0 where extension('http://hl7.org/fhir/3.0/StructureDefinition/BaseType').empty() ->
    tgt.onBehalfOf = create('Reference') as vt0 then Reference(vs0, vt0);
  src.onBehalfOf as vs0 where extension('http://hl7.org/fhir/3.0/StructureDefinition/BaseType').exists() ->
    tgt.onBehalfOf = create('uri') as vt0 then {
    vs0.reference as vs1 then uri(vs1, vt0);
  };
  src.sigFormat -> tgt.contentType;
  src.data -> tgt.blob;
}
```

2.24.5.17 Annotation

See also Base Definition, Detailed Descriptions, Examples, Profiles & Extensions and Mappings.

Functional status for this map: Not Mapped (based on R2 -> R3 -> R2 round tripping)

2.24.5.17.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/Annotation3to4" = "R3 to R4 Conversions for Annotation"

uses "http://hl7.org/fhir/3.0/StructureDefinition/Annotation" alias AnnotationR3 as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/string" alias StringR3 as source
uses "http://hl7.org/fhir/StructureDefinition/Annotation" alias Annotation as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group Annotation(source src : AnnotationR3, target tgt : Annotation) extends Element <<type+>> {
  src.author : Reference as vs -> tgt.author = create('Reference') as vt then Reference(vs, vt);
  src.author : string as vs -> tgt.author = create('string') as vt then string(vs, vt);
  src.time -> tgt.time;
  src.text -> tgt.text;
}

group string2Annotation(source src : stringR3, target tgt : Annotation) extends Element <<types>> {
  src -> tgt.text as vt0 then string(src, vt0) "text";
}
```

2.24.5.17.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/Annotation4to3" = "R4 to R3 Conversion for Annotation"

uses "http://hl7.org/fhir/StructureDefinition/Annotation" alias Annotation as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/Annotation" alias AnnotationR3 as target
uses "http://hl7.org/fhir/3.0/StructureDefinition/string" alias StringR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group Annotation(source src : Annotation, target tgt : AnnotationR3) extends Element <<type+>> {
  src.author : Reference as vs -> tgt.author = create('Reference') as vt then Reference(vs, vt);
  src.author : string as vs -> tgt.author = create('string') as vt then string(vs, vt);
  src.time -> tgt.time;
  src.text -> tgt.text;
}

group Annotation2string(source src : Annotation, target tgt : stringR3) extends Element <<types>> {
  src.text as vs0 then string(vs0, tgt);
}
```

2.25 MetaData Types

URL: https://hl7.org/fhir/R4/metadatatypes.html

FHIR Infrastructure Work Group | Maturity Level: 4 | Standards Status: Partially Normative

| This page has been approved as part of an ANSI standard. See the Infrastructure Package for further details.

This page describes a set of types used for conveying metadata about knowledge resources (mainly for Terminology, Conformance and Clinical Definition resources).

Table of Contents

ContactDetail | Contributor | DataRequirement | ParameterDefinition
RelatedArtifact | TriggerDefinition | Expression | UsageContext

For an index of all data types, see the Data types page.

2.25.0.0.1 ContactDetail

| This DataType has been approved as part of an ANSI standard. See the Infrastructure Package for further details.

See also Examples, Detailed Descriptions and Mappings.

The ContactDetail structure defines general contact details.

• Structure
• UML
• XML
• JSON
• Turtle
• R3 Diff
• All

Structure

Name | Flags | Card. | Type | Description & Constraints
ContactDetail | ΣN | | Element | Contact information Elements defined in Ancestors: id, extension
name | Σ | 0..1 | string | Name of an individual to contact
telecom | Σ | 0..* | ContactPoint | Contact details for individual or organization
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <name value="[string]"/><!-- 0..1 Name of an individual to contact -->
 <telecom><!-- 0..* ContactPoint Contact details for individual or organization --></telecom>
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "name" : "<string>", // Name of an individual to contact
  "telecom" : [{ ContactPoint }] // Contact details for individual or organization
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:ContactDetail.name [ string ]; # 0..1 Name of an individual to contact
  fhir:ContactDetail.telecom [ ContactPoint ], ... ; # 0..* Contact details for individual or organization
]
```

Changes since Release 3

ContactDetail | No Changes

See the Full Difference for further information

Structure

Name | Flags | Card. | Type | Description & Constraints
ContactDetail | ΣN | | Element | Contact information Elements defined in Ancestors: id, extension
name | Σ | 0..1 | string | Name of an individual to contact
telecom | Σ | 0..* | ContactPoint | Contact details for individual or organization
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <name value="[string]"/><!-- 0..1 Name of an individual to contact -->
 <telecom><!-- 0..* ContactPoint Contact details for individual or organization --></telecom>
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "name" : "<string>", // Name of an individual to contact
  "telecom" : [{ ContactPoint }] // Contact details for individual or organization
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:ContactDetail.name [ string ]; # 0..1 Name of an individual to contact
  fhir:ContactDetail.telecom [ ContactPoint ], ... ; # 0..* Contact details for individual or organization
]
```

Changes since Release 3

ContactDetail | No Changes

See the Full Difference for further information

Constraints

ContactDetail is used in the following places: Contributor, ActivityDefinition, CapabilityStatement, ChargeItemDefinition, CodeSystem, CompartmentDefinition, ConceptMap, EffectEvidenceSynthesis, EventDefinition, Evidence, EvidenceVariable, ExampleScenario, GraphDefinition, ImplementationGuide, Library, Measure, MessageDefinition, NamingSystem, OperationDefinition, PlanDefinition, Questionnaire, ResearchDefinition, ResearchElementDefinition, ResearchStudy, RiskEvidenceSynthesis, SearchParameter, StructureDefinition, StructureMap, TerminologyCapabilities, TestScript and ValueSet

2.25.0.0.1.1 Interpretation of ContactDetail in different contexts

2.25.0.0.2 Contributor

Normative Candidate Note: This DataType (and the rest on this page) is not normative - it is still undergoing Trial Use while more experience is gathered.

See also Examples, Detailed Descriptions and Mappings.

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
Contributor | ΣTU | | Element | Contributor information Elements defined in Ancestors: id, extension
type | Σ | 1..1 | code | author | editor | reviewer | endorser ContributorType (Required)
name | Σ | 1..1 | string | Who contributed the content
contact | Σ | 0..* | ContactDetail | Contact details of the contributor
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <type value="[code]"/><!-- 1..1 author | editor | reviewer | endorser -->
 <name value="[string]"/><!-- 1..1 Who contributed the content -->
 <contact><!-- 0..* ContactDetail Contact details of the contributor --></contact>
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "type" : "<code>", // R!  author | editor | reviewer | endorser
  "name" : "<string>", // R!  Who contributed the content
  "contact" : [{ ContactDetail }] // Contact details of the contributor
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:Contributor.type [ code ]; # 1..1 author | editor | reviewer | endorser
  fhir:Contributor.name [ string ]; # 1..1 Who contributed the content
  fhir:Contributor.contact [ ContactDetail ], ... ; # 0..* Contact details of the contributor
]
```

Changes since Release 3

Contributor
Contributor.type | Change value set from http://hl7.org/fhir/ValueSet/contributor-type to http://hl7.org/fhir/ValueSet/contributor-type|4.0.1

See the Full Difference for further information

Structure

Name | Flags | Card. | Type | Description & Constraints
Contributor | ΣTU | | Element | Contributor information Elements defined in Ancestors: id, extension
type | Σ | 1..1 | code | author | editor | reviewer | endorser ContributorType (Required)
name | Σ | 1..1 | string | Who contributed the content
contact | Σ | 0..* | ContactDetail | Contact details of the contributor

Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <type value="[code]"/><!-- 1..1 author | editor | reviewer | endorser -->
 <name value="[string]"/><!-- 1..1 Who contributed the content -->
 <contact><!-- 0..* ContactDetail Contact details of the contributor --></contact>
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "type" : "<code>", // R!  author | editor | reviewer | endorser
  "name" : "<string>", // R!  Who contributed the content
  "contact" : [{ ContactDetail }] // Contact details of the contributor
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:Contributor.type [ code ]; # 1..1 author | editor | reviewer | endorser
  fhir:Contributor.name [ string ]; # 1..1 Who contributed the content
  fhir:Contributor.contact [ ContactDetail ], ... ; # 0..* Contact details of the contributor
]
```

Changes since Release 3

Contributor

Contributor.type | Change value set from http://hl7.org/fhir/ValueSet/contributor-type to http://hl7.org/fhir/ValueSet/contributor-type|4.0.1

See the Full Difference for further information

Constraints

Terminology Bindings

Path | Definition | Type | Reference

Contributor.type | The type of contributor. | Required | ContributorType

Contributor is used in the following places: (not used as yet)

2.25.0.0.2.1 Interpretation of Contributor in different contexts

2.25.0.0.3 DataRequirement

See also Examples, Detailed Descriptions and Mappings.

The DataRequirement structure defines a general data requirement for a knowledge asset such as a decision support rule or quality measure.

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints

DataRequirement | ΣTU | | Element | Describes a required data item Elements defined in Ancestors: id, extension

type | Σ | 1..1 | code | The type of the required data FHIRAllTypes (Required)

profile | Σ | 0..* | canonical(StructureDefinition) | The profile of the required data

subject[x] | Σ | 0..1 | | E.g. Patient, Practitioner, RelatedPerson, Organization, Location, Device Subject type (Extensible)

subjectCodeableConcept | | | CodeableConcept

subjectReference | | | Reference(Group)

mustSupport | Σ | 0..* | string | Indicates specific structure elements that are referenced by the knowledge module

codeFilter | ΣI | 0..* | Element | What codes are expected + Rule: Either a path or a searchParam must be provided, but not both

path | Σ | 0..1 | string | A code-valued attribute to filter on

searchParam | Σ | 0..1 | string | A coded (token) parameter to search on

valueSet | Σ | 0..1 | canonical(ValueSet) | Valueset for the filter

code | Σ | 0..* | Coding | What code is expected

dateFilter | ΣI | 0..* | Element | What dates/date ranges are expected + Rule: Either a path or a searchParam must be provided, but not both

path | Σ | 0..1 | string | A date-valued attribute to filter on

searchParam | Σ | 0..1 | string | A date valued parameter to search on

value[x] | Σ | 0..1 | | The value of the filter, as a Period, DateTime, or Duration value
valueDateTime | | | dateTime
valuePeriod | | | Period
valueDuration | | | Duration
limit | Σ | 0..1 | positiveInt | Number of results
sort | Σ | 0..* | Element | Order of the results
path | Σ | 1..1 | string | The name of the attribute to perform the sort
direction | Σ | 1..1 | code | ascending | descending SortDirection (Required)
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <type value="[code]"/><!-- 1..1 The type of the required data -->
 <profile><!-- 0..* canonical(StructureDefinition) The profile of the required data --></profile>
 <subject[x]><!-- 0..1 CodeableConcept|Reference(Group) E.g. Patient, Practitioner, RelatedPerson, Organization,
    Location, Device --></subject[x]>
 <mustSupport value="[string]"/><!-- 0..* Indicates specific structure elements that are referenced by the knowledge
    module -->
 <codeFilter>  <!-- 0..* What codes are expected -->
  <path value="[string]"/><!-- 0..1 A code-valued attribute to filter on -->
  <searchParam value="[string]"/><!-- 0..1 A coded (token) parameter to search on -->
  <valueSet><!-- 0..1 canonical(ValueSet) Valueset for the filter --></valueSet>
  <code><!-- 0..* Coding What code is expected --></code>
 </codeFilter>
 <dateFilter>  <!-- 0..* What dates/date ranges are expected -->
  <path value="[string]"/><!-- 0..1 A date-valued attribute to filter on -->
  <searchParam value="[string]"/><!-- 0..1 A date valued parameter to search on -->
  <value[x]><!-- 0..1 dateTime|Period|Duration The value of the filter, as a Period, DateTime, or Duration value
    --></value[x]>
 </dateFilter>
 <limit value="[positiveInt]"/><!-- 0..1 Number of results -->
 <sort>  <!-- 0..* Order of the results -->
  <path value="[string]"/><!-- 1..1 The name of the attribute to perform the sort -->
  <direction value="[code]"/><!-- 1..1 ascending | descending -->
 </sort>
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "type" : "<code>", // R!  The type of the required data
  "profile" : [{ canonical(StructureDefinition) }], // The profile of the required data
  // subject[x]: E.g. Patient, Practitioner, RelatedPerson, Organization, Location, Device. One of these 2:
  "subjectCodeableConcept" : { CodeableConcept },
  "subjectReference" : { Reference(Group) },
  "mustSupport" : ["<string>"], // Indicates specific structure elements that are referenced by the knowledge module
  "codeFilter" : [{ // What codes are expected
    "path" : "<string>", // A code-valued attribute to filter on
    "searchParam" : "<string>", // A coded (token) parameter to search on
    "valueSet" : { canonical(ValueSet) }, // Valueset for the filter
    "code" : [{ Coding }] // What code is expected
  }],
  "dateFilter" : [{ // What dates/date ranges are expected
    "path" : "<string>", // A date-valued attribute to filter on
    "searchParam" : "<string>", // A date valued parameter to search on
    // value[x]: The value of the filter, as a Period, DateTime, or Duration value. One of these 3:
    "valueDateTime" : "<dateTime>"
    "valuePeriod" : { Period }
    "valueDuration" : { Duration }
  }],
  "limit" : "<positiveInt>", // Number of results
  "sort" : [{ // Order of the results
    "path" : "<string>", // R!  The name of the attribute to perform the sort
    "direction" : "<code>" // R!  ascending | descending
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:DataRequirement.type [ code ]; # 1..1 The type of the required data
  fhir:DataRequirement.profile [ canonical(StructureDefinition) ], ... ; # 0..* The profile of the required data
  # DataRequirement.subject[x] : 0..1 E.g. Patient, Practitioner, RelatedPerson, Organization, Location, Device. One of
    these 2
    fhir:DataRequirement.subjectCodeableConcept [ CodeableConcept ]
    fhir:DataRequirement.subjectReference [ Reference(Group) ]
  fhir:DataRequirement.mustSupport [ string ], ... ; # 0..* Indicates specific structure elements that are referenced by
     the knowledge module
  fhir:DataRequirement.codeFilter [ # 0..* What codes are expected
    fhir:DataRequirement.codeFilter.path [ string ]; # 0..1 A code-valued attribute to filter on
    fhir:DataRequirement.codeFilter.searchParam [ string ]; # 0..1 A coded (token) parameter to search on
    fhir:DataRequirement.codeFilter.valueSet [ canonical(ValueSet) ]; # 0..1 Valueset for the filter
```

```
    fhir:DataRequirement.codeFilter.code [ Coding ], ... ; # 0..* What code is expected
  ], ...;
  fhir:DataRequirement.dateFilter [ # 0..* What dates/date ranges are expected
    fhir:DataRequirement.dateFilter.path [ string ]; # 0..1 A date-valued attribute to filter on
    fhir:DataRequirement.dateFilter.searchParam [ string ]; # 0..1 A date valued parameter to search on
    # DataRequirement.dateFilter.value[x] : 0..1 The value of the filter, as a Period, DateTime, or Duration value. One
    of these 3
      fhir:DataRequirement.dateFilter.valueDateTime [ dateTime ]
      fhir:DataRequirement.dateFilter.valuePeriod [ Period ]
      fhir:DataRequirement.dateFilter.valueDuration [ Duration ]
  ], ...;
  fhir:DataRequirement.limit [ positiveInt ]; # 0..1 Number of results
  fhir:DataRequirement.sort [ # 0..* Order of the results
    fhir:DataRequirement.sort.path [ string ]; # 1..1 The name of the attribute to perform the sort
    fhir:DataRequirement.sort.direction [ code ]; # 1..1 ascending | descending
  ], ...;
]
```

Changes since Release 3

DataRequirement
DataRequirement.type | Change value set from http://hl7.org/fhir/ValueSet/all-types to http://hl7.org/fhir/ValueSet/all-types|4.0.1
DataRequirement.profile | Type changed from uri to canonical(StructureDefinition)
DataRequirement.subject[x] | Added Element
DataRequirement.codeFilter.path | Min Cardinality changed from 1 to 0
DataRequirement.codeFilter.searchParam | Added Element
DataRequirement.codeFilter.valueSet | Renamed from valueSet[x] to valueSet Add Type canonical(ValueSet) Remove Types string, Reference(ValueSet)
DataRequirement.codeFilter.code | Added Element
DataRequirement.dateFilter.path | Min Cardinality changed from 1 to 0
DataRequirement.dateFilter.searchParam | Added Element
DataRequirement.limit | Added Element
DataRequirement.sort | Added Element
DataRequirement.sort.path | Added Mandatory Element
DataRequirement.sort.direction | Added Mandatory Element
DataRequirement.codeFilter.valueCode | deleted
DataRequirement.codeFilter.valueCoding | deleted
DataRequirement.codeFilter.valueCodeableConcept | deleted

See the Full Difference for further information

Structure

Name | Flags | Card. | Type | Description & Constraints
DataRequirement | ΣTU | | Element | Describes a required data item Elements defined in Ancestors: id, extension
type | Σ | 1..1 | code | The type of the required data FHIRAllTypes (Required)
profile | Σ | 0..* | canonical(StructureDefinition) | The profile of the required data
subject[x] | Σ | 0..1 | | E.g. Patient, Practitioner, RelatedPerson, Organization, Location, Device Subject type (Extensible)
subjectCodeableConcept | | | CodeableConcept
subjectReference | | | Reference(Group)
mustSupport | Σ | 0..* | string | Indicates specific structure elements that are referenced by the knowledge module
codeFilter | ΣI | 0..* | Element | What codes are expected + Rule: Either a path or a searchParam must be provided, but not both
path | Σ | 0..1 | string | A code-valued attribute to filter on
searchParam | Σ | 0..1 | string | A coded (token) parameter to search on
valueSet | Σ | 0..1 | canonical(ValueSet) | Valueset for the filter
code | Σ | 0..* | Coding | What code is expected
dateFilter | ΣI | 0..* | Element | What dates/date ranges are expected + Rule: Either a path or a searchParam must be provided, but not both
path | Σ | 0..1 | string | A date-valued attribute to filter on
searchParam | Σ | 0..1 | string | A date valued parameter to search on
value[x] | Σ | 0..1 | | The value of the filter, as a Period, DateTime, or Duration value
valueDateTime | | | dateTime
valuePeriod | | | Period
valueDuration | | | Duration
limit | Σ | 0..1 | positiveInt | Number of results
sort | Σ | 0..* | Element | Order of the results
path | Σ | 1..1 | string | The name of the attribute to perform the sort
direction | Σ | 1..1 | code | ascending | descending SortDirection (Required)
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <type value="[code]"/><!-- 1..1 The type of the required data -->
 <profile><!-- 0..* canonical(StructureDefinition) The profile of the required data --></profile>
 <subject[x]><!-- 0..1 CodeableConcept|Reference(Group) E.g. Patient, Practitioner, RelatedPerson, Organization,
    Location, Device --></subject[x]>
 <mustSupport value="[string]"/><!-- 0..* Indicates specific structure elements that are referenced by the knowledge
    module -->
```

```
 <codeFilter>  <!-- 0..* What codes are expected -->
  <path value="[string]"/><!-- 0..1 A code-valued attribute to filter on -->
  <searchParam value="[string]"/><!-- 0..1 A coded (token) parameter to search on -->
  <valueSet><!-- 0..1 canonical(ValueSet) Valueset for the filter --></valueSet>
  <code><!-- 0..* Coding What code is expected --></code>
 </codeFilter>
 <dateFilter>  <!-- 0..* What dates/date ranges are expected -->
  <path value="[string]"/><!-- 0..1 A date-valued attribute to filter on -->
  <searchParam value="[string]"/><!-- 0..1 A date valued parameter to search on -->
  <value[x]><!-- 0..1 dateTime|Period|Duration The value of the filter, as a Period, DateTime, or Duration value
    --></value[x]>
 </dateFilter>
 <limit value="[positiveInt]"/><!-- 0..1 Number of results -->
 <sort>  <!-- 0..* Order of the results -->
  <path value="[string]"/><!-- 1..1 The name of the attribute to perform the sort -->
  <direction value="[code]"/><!-- 1..1 ascending | descending -->
 </sort>
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "type" : "<code>", // R!  The type of the required data
  "profile" : [{ canonical(StructureDefinition) }], // The profile of the required data
  // subject[x]: E.g. Patient, Practitioner, RelatedPerson, Organization, Location, Device. One of these 2:
  "subjectCodeableConcept" : { CodeableConcept },
  "subjectReference" : { Reference(Group) },
  "mustSupport" : ["<string>"], // Indicates specific structure elements that are referenced by the knowledge module
  "codeFilter" : [{ // What codes are expected
    "path" : "<string>", // A code-valued attribute to filter on
    "searchParam" : "<string>", // A coded (token) parameter to search on
    "valueSet" : { canonical(ValueSet) }, // Valueset for the filter
    "code" : [{ Coding }] // What code is expected
  }],
  "dateFilter" : [{ // What dates/date ranges are expected
    "path" : "<string>", // A date-valued attribute to filter on
    "searchParam" : "<string>", // A date valued parameter to search on
    // value[x]: The value of the filter, as a Period, DateTime, or Duration value. One of these 3:
    "valueDateTime" : "<dateTime>"
    "valuePeriod" : { Period }
    "valueDuration" : { Duration }
  }],
  "limit" : "<positiveInt>", // Number of results
  "sort" : [{ // Order of the results
    "path" : "<string>", // R!  The name of the attribute to perform the sort
    "direction" : "<code>" // R!  ascending | descending
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:DataRequirement.type [ code ]; # 1..1 The type of the required data
  fhir:DataRequirement.profile [ canonical(StructureDefinition) ], ... ; # 0..* The profile of the required data
  # DataRequirement.subject[x] : 0..1 E.g. Patient, Practitioner, RelatedPerson, Organization, Location, Device. One of
    these 2
    fhir:DataRequirement.subjectCodeableConcept [ CodeableConcept ]
    fhir:DataRequirement.subjectReference [ Reference(Group) ]
  fhir:DataRequirement.mustSupport [ string ], ... ; # 0..* Indicates specific structure elements that are referenced by
     the knowledge module
  fhir:DataRequirement.codeFilter [ # 0..* What codes are expected
    fhir:DataRequirement.codeFilter.path [ string ]; # 0..1 A code-valued attribute to filter on
    fhir:DataRequirement.codeFilter.searchParam [ string ]; # 0..1 A coded (token) parameter to search on
    fhir:DataRequirement.codeFilter.valueSet [ canonical(ValueSet) ]; # 0..1 Valueset for the filter
    fhir:DataRequirement.codeFilter.code [ Coding ], ... ; # 0..* What code is expected
  ], ...;
  fhir:DataRequirement.dateFilter [ # 0..* What dates/date ranges are expected
    fhir:DataRequirement.dateFilter.path [ string ]; # 0..1 A date-valued attribute to filter on
    fhir:DataRequirement.dateFilter.searchParam [ string ]; # 0..1 A date valued parameter to search on
    # DataRequirement.dateFilter.value[x] : 0..1 The value of the filter, as a Period, DateTime, or Duration value. One
    of these 3
      fhir:DataRequirement.dateFilter.valueDateTime [ dateTime ]
      fhir:DataRequirement.dateFilter.valuePeriod [ Period ]
      fhir:DataRequirement.dateFilter.valueDuration [ Duration ]
  ], ...;
  fhir:DataRequirement.limit [ positiveInt ]; # 0..1 Number of results
  fhir:DataRequirement.sort [ # 0..* Order of the results
    fhir:DataRequirement.sort.path [ string ]; # 1..1 The name of the attribute to perform the sort
    fhir:DataRequirement.sort.direction [ code ]; # 1..1 ascending | descending
  ], ...;
]
```

Changes since Release 3

DataRequirement

DataRequirement.type | Change value set from http://hl7.org/fhir/ValueSet/all-types to http://hl7.org/fhir/ValueSet/all-types|4.0.1

DataRequirement.profile | Type changed from uri to canonical(StructureDefinition)

DataRequirement.subject[x] | Added Element
DataRequirement.codeFilter.path | Min Cardinality changed from 1 to 0
DataRequirement.codeFilter.searchParam | Added Element
DataRequirement.codeFilter.valueSet | Renamed from valueSet[x] to valueSet Add Type canonical(ValueSet) Remove Types string, Reference(ValueSet)
DataRequirement.codeFilter.code | Added Element
DataRequirement.dateFilter.path | Min Cardinality changed from 1 to 0
DataRequirement.dateFilter.searchParam | Added Element
DataRequirement.limit | Added Element
DataRequirement.sort | Added Element
DataRequirement.sort.path | Added Mandatory Element
DataRequirement.sort.direction | Added Mandatory Element
DataRequirement.codeFilter.valueCode | deleted
DataRequirement.codeFilter.valueCoding | deleted
DataRequirement.codeFilter.valueCodeableConcept | deleted

See the Full Difference for further information

Constraints

id | Level | Location | Description | Expression
drq-1 | Rule | DataRequirement.codeFilter | Either a path or a searchParam must be provided, but not both | path.exists() xor searchParam.exists()
drq-2 | Rule | DataRequirement.dateFilter | Either a path or a searchParam must be provided, but not both | path.exists() xor searchParam.exists()

Terminology Bindings

Path | Definition | Type | Reference
DataRequirement.type | A list of all the concrete types defined in this version of the FHIR specification - Abstract Types, Data Types and Resource Types. | Required | All Types
DataRequirement.subject[x] | The possible types of subjects for a data requirement (E.g., Patient, Practitioner, Organization, Location, etc.). | Extensible | SubjectType
DataRequirement.sort.direction | The possible sort directions, ascending or descending. | Required | SortDirection

A DataRequirement differs from a parameter in that it specifies the data to be provided in terms of the type of data, and specific filters on code-valued and/or date-valued attributes. Data requirements are not named because they are referenced by type within the evaluation context.

DataRequirements are used by knowledge modules to communicate the set of required data to a consumer in a way that is computable (as opposed to a set of named parameters which must be integrated by hand based on the meaning of the parameter as communicated through the documentation).

DataRequirements are typically used to communicate patient-dependent information such as MedicationStatements and Encounters, whereas Parameters are typically used to communicate patient-independent information such as configuration values.

For example, consider the following CQL expression:

```
define "Total Colectomy Procedures":
  [Procedure: "Total Colectomy Value Set"] P
    where P.performedPeriod during "Measurement Period"
```

The criteria is looking for procedures matching the "Total Colectomy Value Set" that were performed during the "Measurement Period". In this case, "Measurement Period" is a parameter, referenced by name, whereas the reference to Procedure uses the name of the resource type, and so constitutes a data requirement of the criteria:

```
<dataRequirement>
  <type value="Procedure"/>
  <codeFilter>
    <path value="code"/>
    <valueSetString value="Total Colectomy Value Set"/>
  </codeFilter>
  <dateFilter>
    <path value="performedPeriod"/>
    <valuePeriod>
      <start value="2016-01-01"/>
      <end value="2016-12-31"/>
    </valuePeriod>
  </dateFilter>
</dataRequirement>
```

If a resource type has multiple date attributes, then it may be necessary to include multiple date criteria. For instance, in the example above the Procedure resource is using the performedPeriod date attribute. However, the Procedure resource also provides a performedDateTime (specific date and/or time when procedure was performed) attribute. Therefore, if the date criteria for the Procedure in the example was recorded using the performedDateTime attribute, the date criteria would be incomplete. To account for this additional date attribute, the example could be expanded as follows:

```
define "Total Colectomy Procedures":
  [Procedure: "Total Colectomy Value Set"] P
    where exists (P.performedPeriod during "Measurement Period")
    or exists (P.performedDateTime during "Measurement Period")
```

The data requirement for the expanded criteria:

```
<dataRequirement>
  <type value="Procedure"/>
  <codeFilter>
    <path value="code"/>
    <valueSetString value="Total Colectomy Value Set"/>
  </codeFilter>
  <dateFilter>
    <path value="performedPeriod"/>
    <path value="performedDateTime"/>
  </dateFilter>
</dataRequirement>
```

DataRequirement is used in the following places: TriggerDefinition, EvidenceVariable, GuidanceResponse, Library, PlanDefinition and ResearchElementDefinition

2.25.0.0.4 ParameterDefinition

See also Examples, Detailed Descriptions and Mappings.

The ParameterDefinition structure defines a parameter to a knowledge asset such as a decision support rule or quality measure.

Parameters are typically used to communicate patient-independent information such as configuration values, whereas DataRequirements are typically used to communicate patient-dependent information such as MedicationStatements and Encounters.

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name	Flags	Card.	Type	Description & Constraints
ParameterDefinition	ΣTU		Element	Definition of a parameter to a module Elements defined in Ancestors: id, extension
name	Σ	0..1	code	Name used to access the parameter value
use	Σ	1..1	code	in | out OperationParameterUse (Required)
min	Σ	0..1	integer	Minimum cardinality
max	Σ	0..1	string	Maximum cardinality (a number of *)
documentation	Σ	0..1	string	A brief description of the parameter
type	Σ	1..1	code	What type of value FHIRAllTypes (Required)
profile	Σ	0..1	canonical(StructureDefinition)	What profile the value is expected to be

Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <name value="[code]"/><!-- 0..1 Name used to access the parameter value -->
 <use value="[code]"/><!-- 1..1 in | out -->
 <min value="[integer]"/><!-- 0..1 Minimum cardinality -->
 <max value="[string]"/><!-- 0..1 Maximum cardinality (a number of *) -->
 <documentation value="[string]"/><!-- 0..1 A brief description of the parameter -->
 <type value="[code]"/><!-- 1..1 What type of value -->
 <profile><!-- 0..1 canonical(StructureDefinition) What profile the value is expected to be --></profile>
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "name" : "<code>", // Name used to access the parameter value
  "use" : "<code>", // R!  in | out
  "min" : <integer>, // Minimum cardinality
  "max" : "<string>", // Maximum cardinality (a number of *)
  "documentation" : "<string>", // A brief description of the parameter
  "type" : "<code>", // R!  What type of value
  "profile" : { canonical(StructureDefinition) } // What profile the value is expected to be
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:ParameterDefinition.name [ code ]; # 0..1 Name used to access the parameter value
  fhir:ParameterDefinition.use [ code ]; # 1..1 in | out
  fhir:ParameterDefinition.min [ integer ]; # 0..1 Minimum cardinality
```

```
  fhir:ParameterDefinition.max [ string ]; # 0..1 Maximum cardinality (a number of *)
  fhir:ParameterDefinition.documentation [ string ]; # 0..1 A brief description of the parameter
  fhir:ParameterDefinition.type [ code ]; # 1..1 What type of value
  fhir:ParameterDefinition.profile [ canonical(StructureDefinition) ]; # 0..1 What profile the value is expected to be
]
```

Changes since Release 3

ParameterDefinition

ParameterDefinition.use | Change value set from http://hl7.org/fhir/ValueSet/operation-parameter-use to http://hl7.org/fhir/ValueSet/operation-parameter-use|4.0.1

ParameterDefinition.type | Change value set from http://hl7.org/fhir/ValueSet/all-types to http://hl7.org/fhir/ValueSet/all-types|4.0.1

ParameterDefinition.profile | Type changed from Reference(StructureDefinition) to canonical(StructureDefinition)

See the Full Difference for further information

Structure

Name | Flags | Card. | Type | Description & Constraints

ParameterDefinition | ΣTU | | Element | Definition of a parameter to a module Elements defined in Ancestors: id, extension

name | Σ | 0..1 | code | Name used to access the parameter value

use | Σ | 1..1 | code | in | out OperationParameterUse (Required)

min | Σ | 0..1 | integer | Minimum cardinality

max | Σ | 0..1 | string | Maximum cardinality (a number of *)

documentation | Σ | 0..1 | string | A brief description of the parameter

type | Σ | 1..1 | code | What type of value FHIRAllTypes (Required)

profile | Σ | 0..1 | canonical(StructureDefinition) | What profile the value is expected to be

Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <name value="[code]"/><!-- 0..1 Name used to access the parameter value -->
 <use value="[code]"/><!-- 1..1 in | out -->
 <min value="[integer]"/><!-- 0..1 Minimum cardinality -->
 <max value="[string]"/><!-- 0..1 Maximum cardinality (a number of *) -->
 <documentation value="[string]"/><!-- 0..1 A brief description of the parameter -->
 <type value="[code]"/><!-- 1..1 What type of value -->
 <profile><!-- 0..1 canonical(StructureDefinition) What profile the value is expected to be --></profile>
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "name" : "<code>", // Name used to access the parameter value
  "use" : "<code>", // R!  in | out
  "min" : <integer>, // Minimum cardinality
  "max" : "<string>", // Maximum cardinality (a number of *)
  "documentation" : "<string>", // A brief description of the parameter
  "type" : "<code>", // R!  What type of value
  "profile" : { canonical(StructureDefinition) } // What profile the value is expected to be
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:ParameterDefinition.name [ code ]; # 0..1 Name used to access the parameter value
  fhir:ParameterDefinition.use [ code ]; # 1..1 in | out
  fhir:ParameterDefinition.min [ integer ]; # 0..1 Minimum cardinality
  fhir:ParameterDefinition.max [ string ]; # 0..1 Maximum cardinality (a number of *)
  fhir:ParameterDefinition.documentation [ string ]; # 0..1 A brief description of the parameter
  fhir:ParameterDefinition.type [ code ]; # 1..1 What type of value
  fhir:ParameterDefinition.profile [ canonical(StructureDefinition) ]; # 0..1 What profile the value is expected to be
]
```

Changes since Release 3

ParameterDefinition

ParameterDefinition.use | Change value set from http://hl7.org/fhir/ValueSet/operation-parameter-use to http://hl7.org/fhir/ValueSet/operation-parameter-use|4.0.1

ParameterDefinition.type | Change value set from http://hl7.org/fhir/ValueSet/all-types to http://hl7.org/fhir/ValueSet/all-types|4.0.1

ParameterDefinition.profile | Type changed from Reference(StructureDefinition) to canonical(StructureDefinition)

See the Full Difference for further information

Constraints

Terminology Bindings

Path | Definition | Type | Reference

ParameterDefinition.use | Whether the parameter is input or output. | Required | OperationParameterUse

ParameterDefinition.type | A list of all the concrete types defined in this version of the FHIR specification - Abstract Types, Data Types and Resource Types. | Required | All Types

2.25.0.0.5 RelatedArtifact

See also Examples, Detailed Descriptions and Mappings.

The RelatedArtifact structure defines resources related to a module such as previous and next versions of documents, documentation, citations, etc. Note that the name resource here is being used in a more general sense than the FHIR-specific Resource. The related resource may be a FHIR resource, or it may be another type of resource, represented using the Attachment data type.

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
RelatedArtifact | ΣTU | | Element | Related artifacts for a knowledge resource Elements defined in Ancestors: id, extension
type | Σ | 1..1 | code | documentation | justification | citation | predecessor | successor | derived-from | depends-on | composed-of RelatedArtifactType (Required)
label | Σ | 0..1 | string | Short label
display | Σ | 0..1 | string | Brief description of the related artifact
citation | Σ | 0..1 | markdown | Bibliographic citation for the artifact
url | Σ | 0..1 | url | Where the artifact can be accessed
document | Σ | 0..1 | Attachment | What document is being referenced
resource | Σ | 0..1 | canonical(Any) | What resource is being referenced
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <type value="[code]"/><!-- 1..1 documentation | justification | citation | predecessor | successor | derived-from |
    depends-on | composed-of -->
 <label value="[string]"/><!-- 0..1 Short label -->
 <display value="[string]"/><!-- 0..1 Brief description of the related artifact -->
 <citation value="[markdown]"/><!-- 0..1 Bibliographic citation for the artifact -->
 <url value="[url]"/><!-- 0..1 Where the artifact can be accessed -->
 <document><!-- 0..1 Attachment What document is being referenced --></document>
 <resource><!-- 0..1 canonical(Any) What resource is being referenced --></resource>
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "type" : "<code>", // R!  documentation | justification | citation | predecessor | successor | derived-from | depends-
    on | composed-of
  "label" : "<string>", // Short label
  "display" : "<string>", // Brief description of the related artifact
  "citation" : "<markdown>", // Bibliographic citation for the artifact
  "url" : "<url>", // Where the artifact can be accessed
  "document" : { Attachment }, // What document is being referenced
  "resource" : { canonical(Any) } // What resource is being referenced
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:RelatedArtifact.type [ code ]; # 1..1 documentation | justification | citation | predecessor | successor |
    derived-from | depends-on | composed-of
  fhir:RelatedArtifact.label [ string ]; # 0..1 Short label
  fhir:RelatedArtifact.display [ string ]; # 0..1 Brief description of the related artifact
  fhir:RelatedArtifact.citation [ markdown ]; # 0..1 Bibliographic citation for the artifact
  fhir:RelatedArtifact.url [ url ]; # 0..1 Where the artifact can be accessed
  fhir:RelatedArtifact.document [ Attachment ]; # 0..1 What document is being referenced
  fhir:RelatedArtifact.resource [ canonical(Any) ]; # 0..1 What resource is being referenced
]
```

Changes since Release 3

RelatedArtifact
RelatedArtifact.type | Change value set from http://hl7.org/fhir/ValueSet/related-artifact-type to http://hl7.org/fhir/ValueSet/related-artifact-type|4.0.1

RelatedArtifact.label | Added Element
RelatedArtifact.citation | Type changed from string to markdown
RelatedArtifact.url | Type changed from uri to url
RelatedArtifact.resource | Type changed from Reference(Resource) to canonical(Resource)

See the Full Difference for further information

Structure

Name | Flags | Card. | Type | Description & Constraints
RelatedArtifact | ΣTU | | Element | Related artifacts for a knowledge resource Elements defined in Ancestors: id, extension
type | Σ | 1..1 | code | documentation | justification | citation | predecessor | successor | derived-from | depends-on | composed-of
RelatedArtifactType (Required)
label | Σ | 0..1 | string | Short label
display | Σ | 0..1 | string | Brief description of the related artifact
citation | Σ | 0..1 | markdown | Bibliographic citation for the artifact
url | Σ | 0..1 | url | Where the artifact can be accessed
document | Σ | 0..1 | Attachment | What document is being referenced
resource | Σ | 0..1 | canonical(Any) | What resource is being referenced
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <type value="[code]"/><!-- 1..1 documentation | justification | citation | predecessor | successor | derived-from |
    depends-on | composed-of -->
 <label value="[string]"/><!-- 0..1 Short label -->
 <display value="[string]"/><!-- 0..1 Brief description of the related artifact -->
 <citation value="[markdown]"/><!-- 0..1 Bibliographic citation for the artifact -->
 <url value="[url]"/><!-- 0..1 Where the artifact can be accessed -->
 <document><!-- 0..1 Attachment What document is being referenced --></document>
 <resource><!-- 0..1 canonical(Any) What resource is being referenced --></resource>
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "type" : "<code>", // R!  documentation | justification | citation | predecessor | successor | derived-from | depends-
    on | composed-of
  "label" : "<string>", // Short label
  "display" : "<string>", // Brief description of the related artifact
  "citation" : "<markdown>", // Bibliographic citation for the artifact
  "url" : "<url>", // Where the artifact can be accessed
  "document" : { Attachment }, // What document is being referenced
  "resource" : { canonical(Any) } // What resource is being referenced
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:RelatedArtifact.type [ code ]; # 1..1 documentation | justification | citation | predecessor | successor |
    derived-from | depends-on | composed-of
  fhir:RelatedArtifact.label [ string ]; # 0..1 Short label
  fhir:RelatedArtifact.display [ string ]; # 0..1 Brief description of the related artifact
  fhir:RelatedArtifact.citation [ markdown ]; # 0..1 Bibliographic citation for the artifact
  fhir:RelatedArtifact.url [ url ]; # 0..1 Where the artifact can be accessed
  fhir:RelatedArtifact.document [ Attachment ]; # 0..1 What document is being referenced
  fhir:RelatedArtifact.resource [ canonical(Any) ]; # 0..1 What resource is being referenced
]
```

Changes since Release 3

RelatedArtifact
RelatedArtifact.type | Change value set from http://hl7.org/fhir/ValueSet/related-artifact-type to http://hl7.org/fhir/ValueSet/related-artifact-type|4.0.1
RelatedArtifact.label | Added Element
RelatedArtifact.citation | Type changed from string to markdown
RelatedArtifact.url | Type changed from uri to url
RelatedArtifact.resource | Type changed from Reference(Resource) to canonical(Resource)

See the Full Difference for further information

Constraints

Terminology Bindings

Path | Definition | Type | Reference
RelatedArtifact.type | The type of relationship to the related artifact. | Required | RelatedArtifactType

The following examples illustrate the use of relatedArtifact elements to provide citations:

```
<relatedArtifact>
    <label value="1"/>
    <type value="citation"/>
```

```
    <citation value="Linder, J.A., D.W. Bates, G.M. Lee, J.A. Finkelstein. 2005. _Antibiotic treatment of children with
    sore throat_ JAMA 294(18):2315-2322. "/>
  </relatedArtifact>
  <relatedArtifact>
    <type value="citation"/>
    <label value="2"/>
    <citation value="Infectious Diseases Society of America. 2012. _Clinical Practice Guideline for the Diagnosis and
    Management of Group A Streptococcal Pharyngitis: 2012 Update._ "/>
  </relatedArtifact>
  <relatedArtifact>
    <type value="citation"/>
    <label value="3"/>
    <citation value="Roberts, R.R., B. Hota, I. Ahmad, et al. _Hospital and Societal Costs of Antimicrobial-Resistant
    Infections in a Chicago Teaching Hospital: Implications for Antibiotic Stewardship._ Clin Infect Dis. Oct 2009;
    49(8):1175-84."/>
  </relatedArtifact>
```

RelatedArtifact is used in the following places: ActivityDefinition, EffectEvidenceSynthesis, EventDefinition, Evidence, EvidenceVariable, Library, Measure, PlanDefinition, RequestGroup, ResearchDefinition, ResearchElementDefinition, ResearchStudy and RiskEvidenceSynthesis

2.25.0.0.5.1 Interpretation of RelatedArtifact in different contexts

2.25.0.0.6 TriggerDefinition

See also Examples, Detailed Descriptions and Mappings.

The TriggerDefinition structure defines when a knowledge artifact is expected to be evaluated. The structure can represent three main kinds of triggering events, depending on the value of type:

- Named Event
- Scheduled Event
- Data Event

A named event is an event identified by the implementation environment. This allows any event generated within the implementation environment to be used as a trigger, but it requires pre-coordination of the names involved with the consuming environments. HL7 v2 events are assigned the URI http://terminology.hl7.org/CodeSystem/v2-0003#[code] e.g. http://terminology.hl7.org/CodeSystem/v2-0003/A01, and reference any data change that would trigger the sending of the matching HL7 v2 version, if the application providing the FHIR API supports v2 events internally.

A scheduled event occurs on a fixed or periodic schedule.

And finally, a data event occurs in response to some data-related event in the integrated environment such as a record being added or updated. The data-of-interest for a data event is described using a DataRequirement. This allows for systems to automatically invoke based on data activity occurring within the system. A condition may also be specified to further refine the trigger

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
TriggerDefinition | ΣITU | | Element | Defines an expected trigger for a module + Rule: Either timing, or a data requirement, but not both + Rule: A condition only if there is a data requirement + Rule: A named event requires a name, a periodic event requires timing, and a data event requires data Elements defined in Ancestors: id, extension
type | Σ | 1..1 | code | named-event | periodic | data-changed | data-added | data-modified | data-removed | data-accessed | data-access-ended TriggerType (Required)
name | Σ | 0..1 | string | Name or URI that identifies the event
timing[x] | Σ | 0..1 | | Timing of the event
timingTiming | | | Timing
timingReference | | | Reference(Schedule)
timingDate | | | date
timingDateTime | | | dateTime
data | Σ | 0..* | DataRequirement | Triggering data of the event (multiple = 'and')
condition | Σ | 0..1 | Expression | Whether the event triggers (boolean expression)
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
```

```
 <!-- from Element: extension -->
 <type value="[code]"/><!-- 1..1 named-event | periodic | data-changed | data-added | data-modified | data-removed |
    data-accessed | data-access-ended -->
 <name value="[string]"/><!-- 0..1 Name or URI that identifies the event -->
 <timing[x]><!-- 0..1 Timing|Reference(Schedule)|date|dateTime Timing of the event --></timing[x]>
 <data><!-- 0..* DataRequirement Triggering data of the event (multiple = 'and') --></data>
 <condition><!-- 0..1 Expression Whether the event triggers (boolean expression) --></condition>
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "type" : "<code>", // R!  named-event | periodic | data-changed | data-added | data-modified | data-removed | data-
    accessed | data-access-ended
  "name" : "<string>", // Name or URI that identifies the event
  // timing[x]: Timing of the event. One of these 4:
  "timingTiming" : { Timing },
  "timingReference" : { Reference(Schedule) },
  "timingDate" : "<date>",
  "timingDateTime" : "<dateTime>",
  "data" : [{ DataRequirement }], // Triggering data of the event (multiple = 'and')
  "condition" : { Expression } // Whether the event triggers (boolean expression)
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:TriggerDefinition.type [ code ]; # 1..1 named-event | periodic | data-changed | data-added | data-modified |
    data-removed | data-accessed | data-access-ended
  fhir:TriggerDefinition.name [ string ]; # 0..1 Name or URI that identifies the event
  # TriggerDefinition.timing[x] : 0..1 Timing of the event. One of these 4
    fhir:TriggerDefinition.timingTiming [ Timing ]
    fhir:TriggerDefinition.timingReference [ Reference(Schedule) ]
    fhir:TriggerDefinition.timingDate [ date ]
    fhir:TriggerDefinition.timingDateTime [ dateTime ]
  fhir:TriggerDefinition.data [ DataRequirement ], ... ; # 0..* Triggering data of the event (multiple = 'and')
  fhir:TriggerDefinition.condition [ Expression ]; # 0..1 Whether the event triggers (boolean expression)
]
```

Changes since Release 3

TriggerDefinition

TriggerDefinition.type | Change value set from http://hl7.org/fhir/ValueSet/trigger-type to http://hl7.org/fhir/ValueSet/trigger-type|4.0.1

TriggerDefinition.name | Renamed from eventName to name

TriggerDefinition.timing[x] | Renamed from eventTiming[x] to timing[x]

TriggerDefinition.data | Renamed from eventData to data Max Cardinality changed from 1 to *

TriggerDefinition.condition | Added Element

See the Full Difference for further information

Structure

Name | Flags | Card. | Type | Description & Constraints

TriggerDefinition | ΣITU | | Element | Defines an expected trigger for a module + Rule: Either timing, or a data requirement, but not both + Rule: A condition only if there is a data requirement + Rule: A named event requires a name, a periodic event requires timing, and a data event requires data Elements defined in Ancestors: id, extension

type | Σ | 1..1 | code | named-event | periodic | data-changed | data-added | data-modified | data-removed | data-accessed | data-access-ended TriggerType (Required)

name | Σ | 0..1 | string | Name or URI that identifies the event

timing[x] | Σ | 0..1 | | Timing of the event

timingTiming | | | Timing

timingReference | | | Reference(Schedule)

timingDate | | | date

timingDateTime | | | dateTime

data | Σ | 0..* | DataRequirement | Triggering data of the event (multiple = 'and')

condition | Σ | 0..1 | Expression | Whether the event triggers (boolean expression)

Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <type value="[code]"/><!-- 1..1 named-event | periodic | data-changed | data-added | data-modified | data-removed |
    data-accessed | data-access-ended -->
 <name value="[string]"/><!-- 0..1 Name or URI that identifies the event -->
 <timing[x]><!-- 0..1 Timing|Reference(Schedule)|date|dateTime Timing of the event --></timing[x]>
 <data><!-- 0..* DataRequirement Triggering data of the event (multiple = 'and') --></data>
 <condition><!-- 0..1 Expression Whether the event triggers (boolean expression) --></condition>
</[name]>
```

JSON Template

```
{
```

```
  // from Element: extension
  "type" : "<code>", // R!  named-event | periodic | data-changed | data-added | data-modified | data-removed | data-
    accessed | data-access-ended
  "name" : "<string>", // Name or URI that identifies the event
  // timing[x]: Timing of the event. One of these 4:
  "timingTiming" : { Timing },
  "timingReference" : { Reference(Schedule) },
  "timingDate" : "<date>",
  "timingDateTime" : "<dateTime>",
  "data" : [{ DataRequirement }], // Triggering data of the event (multiple = 'and')
  "condition" : { Expression } // Whether the event triggers (boolean expression)
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:TriggerDefinition.type [ code ]; # 1..1 named-event | periodic | data-changed | data-added | data-modified |
    data-removed | data-accessed | data-access-ended
  fhir:TriggerDefinition.name [ string ]; # 0..1 Name or URI that identifies the event
  # TriggerDefinition.timing[x] : 0..1 Timing of the event. One of these 4
    fhir:TriggerDefinition.timingTiming [ Timing ]
    fhir:TriggerDefinition.timingReference [ Reference(Schedule) ]
    fhir:TriggerDefinition.timingDate [ date ]
    fhir:TriggerDefinition.timingDateTime [ dateTime ]
  fhir:TriggerDefinition.data [ DataRequirement ], ... ; # 0..* Triggering data of the event (multiple = 'and')
  fhir:TriggerDefinition.condition [ Expression ]; # 0..1 Whether the event triggers (boolean expression)
]
```

Changes since Release 3

TriggerDefinition

TriggerDefinition.type | Change value set from http://hl7.org/fhir/ValueSet/trigger-type to http://hl7.org/fhir/ValueSet/trigger-type|4.0.1

TriggerDefinition.name | Renamed from eventName to name

TriggerDefinition.timing[x] | Renamed from eventTiming[x] to timing[x]

TriggerDefinition.data | Renamed from eventData to data Max Cardinality changed from 1 to *

TriggerDefinition.condition | Added Element

See the Full Difference for further information

Constraints

id | Level | Location | Description | Expression

trd-1 | Rule | (base) | Either timing, or a data requirement, but not both | data.empty() or timing.empty()

trd-2 | Rule | (base) | A condition only if there is a data requirement | condition.exists() implies data.exists()

trd-3 | Rule | (base) | A named event requires a name, a periodic event requires timing, and a data event requires data | (type = 'named-event' implies name.exists()) and (type = 'periodic' implies timing.exists()) and (type.startsWith('data-') implies data.exists())

Terminology Bindings

Path | Definition | Type | Reference

TriggerDefinition.type | The type of trigger. | Required | TriggerType

TriggerDefinition is used in the following places: EventDefinition, EvidenceVariable and PlanDefinition

2.25.0.0.7 Expression

See also Examples, Detailed Descriptions and Mappings.

The Expression structure defines an expression that generates a value. The expression is provided in a specifed language (by mime type)

The context of use of the expression must specify the context in which the expression is evaluated, and how the result of the expression is used.

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints

Expression | ΣITU | | Element | An expression that can be used to generate a value + Rule: An expression or a reference must be provided Elements defined in Ancestors: id, extension

description | Σ | 0..1 | string | Natural language description of the condition

name | Σ | 0..1 | id | Short name assigned to expression for reuse

language | Σ | 1..1 | code | text/cql | text/fhirpath | application/x-fhir-query | etc. ExpressionLanguage (Extensible but limited to http://www.rfc-editor.org/bcp/bcp13.txt)

expression | Σ | 0..1 | string | Expression in specified language
reference | Σ | 0..1 | uri | Where the expression is found
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Expression xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <description value="[string]"/><!-- 0..1 Natural language description of the condition -->
 <name value="[id]"/><!-- 0..1 Short name assigned to expression for reuse -->
 <language value="[code]"/><!-- 1..1 text/cql | text/fhirpath | application/x-fhir-query | etc. -->
 <expression value="[string]"/><!-- 0..1 Expression in specified language -->
 <reference value="[uri]"/><!-- 0..1 Where the expression is found -->
</Expression>
```

JSON Template

```
{
  // from Element: extension
  "description" : "<string>", // Natural language description of the condition
  "name" : "<id>", // Short name assigned to expression for reuse
  "language" : "<code>", // R!  text/cql | text/fhirpath | application/x-fhir-query | etc.
  "expression" : "<string>", // Expression in specified language
  "reference" : "<uri>" // Where the expression is found
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:Expression.description [ string ]; # 0..1 Natural language description of the condition
  fhir:Expression.name [ id ]; # 0..1 Short name assigned to expression for reuse
  fhir:Expression.language [ code ]; # 1..1 text/cql | text/fhirpath | application/x-fhir-query | etc.
  fhir:Expression.expression [ string ]; # 0..1 Expression in specified language
  fhir:Expression.reference [ uri ]; # 0..1 Where the expression is found
]
```

Changes since Release 3

This complex-type did not exist in Release 2

Structure

Name | Flags | Card. | Type | Description & Constraints
Expression | ΣITU | | Element | An expression that can be used to generate a value + Rule: An expression or a reference must be provided Elements defined in Ancestors: id, extension
description | Σ | 0..1 | string | Natural language description of the condition
name | Σ | 0..1 | id | Short name assigned to expression for reuse
language | Σ | 1..1 | code | text/cql | text/fhirpath | application/x-fhir-query | etc. ExpressionLanguage (Extensible but limited to http://www.rfc-editor.org/bcp/bcp13.txt)
expression | Σ | 0..1 | string | Expression in specified language
reference | Σ | 0..1 | uri | Where the expression is found
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Expression xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <description value="[string]"/><!-- 0..1 Natural language description of the condition -->
 <name value="[id]"/><!-- 0..1 Short name assigned to expression for reuse -->
 <language value="[code]"/><!-- 1..1 text/cql | text/fhirpath | application/x-fhir-query | etc. -->
 <expression value="[string]"/><!-- 0..1 Expression in specified language -->
 <reference value="[uri]"/><!-- 0..1 Where the expression is found -->
</Expression>
```

JSON Template

```
{
  // from Element: extension
  "description" : "<string>", // Natural language description of the condition
  "name" : "<id>", // Short name assigned to expression for reuse
  "language" : "<code>", // R!  text/cql | text/fhirpath | application/x-fhir-query | etc.
  "expression" : "<string>", // Expression in specified language
  "reference" : "<uri>" // Where the expression is found
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:Expression.description [ string ]; # 0..1 Natural language description of the condition
  fhir:Expression.name [ id ]; # 0..1 Short name assigned to expression for reuse
  fhir:Expression.language [ code ]; # 1..1 text/cql | text/fhirpath | application/x-fhir-query | etc.
  fhir:Expression.expression [ string ]; # 0..1 Expression in specified language
  fhir:Expression.reference [ uri ]; # 0..1 Where the expression is found
```

]

Changes since Release 3

This complex-type did not exist in Release 2

Constraints

id | Level | Location | Description | Expression
exp-1 | Rule | (base) | An expression or a reference must be provided | expression.exists() or reference.exists()

Terminology Bindings

Path | Definition | Type | Reference
Expression.language | The media type of the expression language. | Extensible, but limited to http://www.rfc-editor.org/bcp/bcp13.txt | ExpressionLanguage

Expression is used in the following places: TriggerDefinition, ActivityDefinition, EvidenceVariable, Measure, PlanDefinition, RequestGroup and ResearchElementDefinition

2.25.0.0.8 UsageContext

See also Examples, Detailed Descriptions and Mappings.

The UsageContext structure defines the context of use for a module.

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
UsageContext | ΣTU | | Element | Describes the context of use for a conformance or knowledge resource Elements defined in Ancestors: id, extension
code | Σ | 1..1 | Coding | Type of context being specified UsageContextType (Extensible)
value[x] | Σ | 1..1 | | Value that defines the context ConformanceUseContext (Example)
valueCodeableConcept | | | CodeableConcept
valueQuantity | | | Quantity
valueRange | | | Range
valueReference | | | Reference(PlanDefinition | ResearchStudy | InsurancePlan | HealthcareService | Group | Location | Organization)
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <code><!-- 1..1 Coding Type of context being specified --></code>
 <value[x]><!-- 1..1 CodeableConcept|Quantity|Range|Reference(PlanDefinition|
   ResearchStudy|InsurancePlan|HealthcareService|Group|Location|Organization) Value that defines the context
    --></value[x]>
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "code" : { Coding }, // R!  Type of context being specified
  // value[x]: Value that defines the context. One of these 4:
  "valueCodeableConcept" : { CodeableConcept }
  "valueQuantity" : { Quantity }
  "valueRange" : { Range }
  "valueReference" : { Reference(PlanDefinition|ResearchStudy|InsurancePlan|
   HealthcareService|Group|Location|Organization) }
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:UsageContext.code [ Coding ]; # 1..1 Type of context being specified
  # UsageContext.value[x] : 1..1 Value that defines the context. One of these 4
    fhir:UsageContext.valueCodeableConcept [ CodeableConcept ]
    fhir:UsageContext.valueQuantity [ Quantity ]
    fhir:UsageContext.valueRange [ Range ]
    fhir:UsageContext.valueReference [
    Reference(PlanDefinition|ResearchStudy|InsurancePlan|HealthcareService|Group|Location|
  Organization) ]
]
```

Changes since Release 3

UsageContext

UsageContext.code | Change code system for extensibly bound codes from "http://hl7.org/fhir/usage-context-type" to "http://terminology.hl7.org/CodeSystem/usage-context-type"

UsageContext.value[x] | Add Type Reference(PlanDefinition | ResearchStudy | InsurancePlan | HealthcareService | Group | Location | Organization)

See the Full Difference for further information

Structure

Name | Flags | Card. | Type | Description & Constraints

UsageContext | ΣTU | | Element | Describes the context of use for a conformance or knowledge resource Elements defined in Ancestors: id, extension

code | Σ | 1..1 | Coding | Type of context being specified UsageContextType (Extensible)

value[x] | Σ | 1..1 | | Value that defines the context ConformanceUseContext (Example)

valueCodeableConcept | | | CodeableConcept

valueQuantity | | | Quantity

valueRange | | | Range

valueReference | | | Reference(PlanDefinition | ResearchStudy | InsurancePlan | HealthcareService | Group | Location | Organization)

Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <code><!-- 1..1 Coding Type of context being specified --></code>
 <value[x]><!-- 1..1 CodeableConcept|Quantity|Range|Reference(PlanDefinition|
   ResearchStudy|InsurancePlan|HealthcareService|Group|Location|Organization) Value that defines the context
    --></value[x]>
</[name]>
```

JSON Template

```
{
  // from Element: extension
  "code" : { Coding }, // R!  Type of context being specified
  // value[x]: Value that defines the context. One of these 4:
  "valueCodeableConcept" : { CodeableConcept }
  "valueQuantity" : { Quantity }
  "valueRange" : { Range }
  "valueReference" : { Reference(PlanDefinition|ResearchStudy|InsurancePlan|
   HealthcareService|Group|Location|Organization) }
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:UsageContext.code [ Coding ]; # 1..1 Type of context being specified
  # UsageContext.value[x] : 1..1 Value that defines the context. One of these 4
    fhir:UsageContext.valueCodeableConcept [ CodeableConcept ]
    fhir:UsageContext.valueQuantity [ Quantity ]
    fhir:UsageContext.valueRange [ Range ]
    fhir:UsageContext.valueReference [
    Reference(PlanDefinition|ResearchStudy|InsurancePlan|HealthcareService|Group|Location|
  Organization) ]
]
```

Changes since Release 3

UsageContext

UsageContext.code | Change code system for extensibly bound codes from "http://hl7.org/fhir/usage-context-type" to "http://terminology.hl7.org/CodeSystem/usage-context-type"

UsageContext.value[x] | Add Type Reference(PlanDefinition | ResearchStudy | InsurancePlan | HealthcareService | Group | Location | Organization)

See the Full Difference for further information

Constraints

Terminology Bindings

Path | Definition | Type | Reference

UsageContext.code | A code that specifies a type of context being specified by a usage context. | Extensible | UsageContextType

UsageContext.value[x] | A code that defines the specific value for the context being specified. | Example | Context of Use ValueSet

UsageContext is used in the following places: ActivityDefinition, CapabilityStatement, ChargeItemDefinition, CodeSystem, CompartmentDefinition, ConceptMap, EffectEvidenceSynthesis, EventDefinition, Evidence, EvidenceVariable, ExampleScenario, GraphDefinition, ImplementationGuide, Library, Measure, MessageDefinition, NamingSystem, OperationDefinition, PlanDefinition, Questionnaire, ResearchDefinition, ResearchElementDefinition, RiskEvidenceSynthesis, SearchParameter, StructureDefinition, StructureMap, TerminologyCapabilities, TestScript and ValueSet

2.25.0.1 Interpretation of UsageContext in different contexts

2.25.1 Examples

URL: https://hl7.org/fhir/R4/metadatatypes-examples.html

2.25.1 MetaData Types - Examples

FHIR Infrastructure Work Group | Maturity Level: 4 | Standards Status: Informative

Table of Contents

ContactDetail | Contributor | DataRequirement | ParameterDefinition
RelatedArtifact | TriggerDefinition | Expression | UsageContext

2.25.1.1 ContactDetail

See also Base Definition, Detailed Descriptions and Mappings.

For ContactDetail examples, see the examples in Library, Measure, and PlanDefinition.

2.25.1.2 Contributor

See also Base Definition, Detailed Descriptions and Mappings.

For Contributor examples, see the examples in Library, Measure, and PlanDefinition.

2.25.1.3 DataRequirement

See also Base Definition, Detailed Descriptions and Mappings.

For DataRequirement examples, see the examples in Library, Measure, and PlanDefinition.

2.25.1.4 ParameterDefinition

See also Base Definition, Detailed Descriptions and Mappings.

For ParameterDefinition examples, see the examples in Library, Measure, and PlanDefinition.

2.25.1.5 RelatedArtifact

See also Base Definition, Detailed Descriptions and Mappings.

For RelatedArtifact examples, see the examples in Library, Measure, and PlanDefinition.

2.25.1.6 TriggerDefinition

See also Base Definition, Detailed Descriptions and Mappings.

For TriggerDefinition examples, see the examples in PlanDefinition.

2.25.1.7 Expression

See also Base Definition, Detailed Descriptions and Mappings.

For Expression examples, see the examples in PlanDefinition.

2.25.1.8 UsageContext

See also Base Definition, Detailed Descriptions and Mappings.

For UsageContext examples, see the examples in Library, Measure, and PlanDefinition.

2.25.2 Metadata Types - Detailed Descriptions

URL: https://hl7.org/fhir/R4/metadatatypes-definitions.html

2.25.2 Metadata Types - Detailed Descriptions

FHIR Infrastructure Work Group | Maturity Level: 4 | Standards Status: Partially Normative

| This page has been approved as part of an ANSI standard. See the Infrastructure Package for further details.

Table of Contents

ContactDetail | Contributor | DataRequirement | ParameterDefinition
RelatedArtifact | TriggerDefinition | Expression | UsageContext

2.25.2.1 ContactDetail

See also Base Definition, Examples and Mappings.

| This DataType has been approved as part of an ANSI standard. See the Infrastructure Package for further details.

ContactDetail
Standards Status | This element has a standards status of "Normative"
Element Id | ContactDetail
Definition | Specifies contact information for a person or organization.
Cardinality | 0..*
Type | Element

Requirements | Need to track contact information in the same way across multiple resources.
Summary | true
ContactDetail.name
Element Id | ContactDetail.name
Definition | The name of an individual to contact.
Cardinality | 0..1
Type | string
Summary | true
Comments | If there is no named individual, the telecom information is for the organization as a whole.
ContactDetail.telecom
Element Id | ContactDetail.telecom
Definition | The contact details for the individual (if a name was provided) or the organization.
Cardinality | 0..*
Type | ContactPoint
Summary | true

2.25.2.2 Contributor

See also Base Definition, Examples and Mappings.

Normative Candidate Note: This DataType (and the rest on this page) is not normative - it is still undergoing Trial Use while more experience is gathered.

Contributor
Standards Status | This element has a standards status of "Trial Use"
Element Id | Contributor
Definition | A contributor to the content of a knowledge asset, including authors, editors, reviewers, and endorsers.
Cardinality | 0..*
Type | Element
Requirements | Need to track contributor information in the same way across multiple resources.
Summary | true
Contributor.type
Element Id | Contributor.type
Definition | The type of contributor.
Cardinality | 1..1
Terminology Binding | ContributorType (Required)
Type | code
Summary | true
Contributor.name
Element Id | Contributor.name
Definition | The name of the individual or organization responsible for the contribution.
Cardinality | 1..1
Type | string
Summary | true
Contributor.contact
Element Id | Contributor.contact
Definition | Contact details to assist a user in finding and communicating with the contributor.
Cardinality | 0..*
Type | ContactDetail
Summary | true

2.25.2.3 DataRequirement

See also Base Definition, Examples and Mappings.

DataRequirement
Standards Status | This element has a standards status of "Trial Use"
Element Id | DataRequirement
Definition | Describes a required data item for evaluation in terms of the type of data, and optional code or date-based filters of the data.
Cardinality | 0..*
Type | Element
Summary | true
DataRequirement.type
Element Id | DataRequirement.type
Definition | The type of the required data, specified as the type name of a resource. For profiles, this value is set to the type of the base resource of the profile.
Cardinality | 1..1
Terminology Binding | Any defined Resource or Data Type name (including "Any" and "Type")
Type | code
Summary | true
DataRequirement.profile
Element Id | DataRequirement.profile
Definition | The profile of the required data, specified as the uri of the profile definition.
Cardinality | 0..*

Type | canonical(StructureDefinition)
Summary | true
DataRequirement.subject[x]
Element Id | DataRequirement.subject[x]
Definition | The intended subjects of the data requirement. If this element is not provided, a Patient subject is assumed.
Cardinality | 0..1
Terminology Binding | Subject type (Extensible)
Type | CodeableConcept|Reference(Group)
[x] Note | See Choice of Data Types for further information about how to use [x]
Meaning if Missing | Patient
Summary | true
Comments | The subject of a data requirement is critical, as the data being specified is determined with respect to a particular subject. This corresponds roughly to the notion of a Compartment in that it limits what data is available based on its relationship to the subject. In CQL, this corresponds to the context declaration.
DataRequirement.mustSupport
Element Id | DataRequirement.mustSupport
Definition | Indicates that specific elements of the type are referenced by the knowledge module and must be supported by the consumer in order to obtain an effective evaluation. This does not mean that a value is required for this element, only that the consuming system must understand the element and be able to provide values for it if they are available. The value of mustSupport SHALL be a FHIRPath resolveable on the type of the DataRequirement. The path SHALL consist only of identifiers, constant indexers, and .resolve() (see the Simple FHIRPath Profile for full details).
Cardinality | 0..*
Type | string
Summary | true
DataRequirement.codeFilter
Element Id | DataRequirement.codeFilter
Definition | Code filters specify additional constraints on the data, specifying the value set of interest for a particular element of the data. Each code filter defines an additional constraint on the data, i.e. code filters are AND'ed, not OR'ed.
Cardinality | 0..*
Summary | true
Invariants | Defined on this element drq-1 Rule Either a path or a searchParam must be provided, but not both path.exists() xor searchParam.exists()
Defined on this element
drq-1 | Rule | Either a path or a searchParam must be provided, but not both | path.exists() xor searchParam.exists()
DataRequirement.codeFilter.path
Element Id | DataRequirement.codeFilter.path
Definition | The code-valued attribute of the filter. The specified path SHALL be a FHIRPath resolveable on the specified type of the DataRequirement, and SHALL consist only of identifiers, constant indexers, and .resolve(). The path is allowed to contain qualifiers (.) to traverse sub-elements, as well as indexers ([x]) to traverse multiple-cardinality sub-elements (see the Simple FHIRPath Profile for full details). Note that the index must be an integer constant. The path must resolve to an element of type code, Coding, or CodeableConcept.
Cardinality | 0..1
Type | string
Summary | true
Comments | The path attribute contains a Simple FHIRPath Subset that allows path traversal, but not calculation.
DataRequirement.codeFilter.searchParam
Element Id | DataRequirement.codeFilter.searchParam
Definition | A token parameter that refers to a search parameter defined on the specified type of the DataRequirement, and which searches on elements of type code, Coding, or CodeableConcept.
Cardinality | 0..1
Type | string
Summary | true
DataRequirement.codeFilter.valueSet
Element Id | DataRequirement.codeFilter.valueSet
Definition | The valueset for the code filter. The valueSet and code elements are additive. If valueSet is specified, the filter will return only those data items for which the value of the code-valued element specified in the path is a member of the specified valueset.
Cardinality | 0..1
Type | canonical(ValueSet)
Summary | true
DataRequirement.codeFilter.code
Element Id | DataRequirement.codeFilter.code
Definition | The codes for the code filter. If values are given, the filter will return only those data items for which the code-valued attribute specified by the path has a value that is one of the specified codes. If codes are specified in addition to a value set, the filter returns items matching a code in the value set or one of the specified codes.
Cardinality | 0..*
Type | Coding
Summary | true
DataRequirement.dateFilter
Element Id | DataRequirement.dateFilter
Definition | Date filters specify additional constraints on the data in terms of the applicable date range for specific elements. Each date filter specifies an additional constraint on the data, i.e. date filters are AND'ed, not OR'ed.
Cardinality | 0..*
Summary | true

Invariants | Defined on this element drq-2 Rule Either a path or a searchParam must be provided, but not both path.exists() xor searchParam.exists()
Defined on this element
drq-2 | Rule | Either a path or a searchParam must be provided, but not both | path.exists() xor searchParam.exists()
DataRequirement.dateFilter.path
Element Id | DataRequirement.dateFilter.path
Definition | The date-valued attribute of the filter. The specified path SHALL be a FHIRPath resolveable on the specified type of the DataRequirement, and SHALL consist only of identifiers, constant indexers, and .resolve(). The path is allowed to contain qualifiers (.) to traverse sub-elements, as well as indexers ([x]) to traverse multiple-cardinality sub-elements (see the Simple FHIRPath Profile for full details). Note that the index must be an integer constant. The path must resolve to an element of type date, dateTime, Period, Schedule, or Timing.
Cardinality | 0..1
Type | string
Summary | true
Comments | The path attribute contains a Simple FHIR Subset that allows path traversal, but not calculation.
DataRequirement.dateFilter.searchParam
Element Id | DataRequirement.dateFilter.searchParam
Definition | A date parameter that refers to a search parameter defined on the specified type of the DataRequirement, and which searches on elements of type date, dateTime, Period, Schedule, or Timing.
Cardinality | 0..1
Type | string
Summary | true
DataRequirement.dateFilter.value[x]
Element Id | DataRequirement.dateFilter.value[x]
Definition | The value of the filter. If period is specified, the filter will return only those data items that fall within the bounds determined by the Period, inclusive of the period boundaries. If dateTime is specified, the filter will return only those data items that are equal to the specified dateTime. If a Duration is specified, the filter will return only those data items that fall within Duration before now.
Cardinality | 0..1
Type | dateTime|Period|Duration
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
DataRequirement.limit
Element Id | DataRequirement.limit
Definition | Specifies a maximum number of results that are required (uses the _count search parameter).
Cardinality | 0..1
Type | positiveInt
Requirements | Enables the requirement "most recent 5 results" to be expressed.
Summary | true
Comments | This element can be used in combination with the sort element to specify quota requirements such as "the most recent 5" or "the highest 5".
DataRequirement.sort
Element Id | DataRequirement.sort
Definition | Specifies the order of the results to be returned.
Cardinality | 0..*
Requirements | Enables the requirement "most recent 5 results" to be expressed.
Summary | true
Comments | This element can be used in combination with the sort element to specify quota requirements such as "the most recent 5" or "the highest 5". When multiple sorts are specified, they are applied in the order they appear in the resource.
DataRequirement.sort.path
Element Id | DataRequirement.sort.path
Definition | The attribute of the sort. The specified path must be resolvable from the type of the required data. The path is allowed to contain qualifiers (.) to traverse sub-elements, as well as indexers ([x]) to traverse multiple-cardinality sub-elements. Note that the index must be an integer constant.
Cardinality | 1..1
Type | string
Summary | true
DataRequirement.sort.direction
Element Id | DataRequirement.sort.direction
Definition | The direction of the sort, ascending or descending.
Cardinality | 1..1
Terminology Binding | SortDirection (Required)
Type | code
Summary | true

2.25.2.4 ParameterDefinition

See also Base Definition, Examples and Mappings.

ParameterDefinition
Standards Status | This element has a standards status of "Trial Use"
Element Id | ParameterDefinition
Definition | The parameters to the module. This collection specifies both the input and output parameters. Input parameters are provided by the caller as part of the $evaluate operation. Output parameters are included in the GuidanceResponse.
Cardinality | 0..*

Type | Element
Summary | true
ParameterDefinition.name
Element Id | ParameterDefinition.name
Definition | The name of the parameter used to allow access to the value of the parameter in evaluation contexts.
Cardinality | 0..1
Type | code
Summary | true
ParameterDefinition.use
Element Id | ParameterDefinition.use
Definition | Whether the parameter is input or output for the module.
Cardinality | 1..1
Terminology Binding | OperationParameterUse (Required)
Type | code
Summary | true
ParameterDefinition.min
Element Id | ParameterDefinition.min
Definition | The minimum number of times this parameter SHALL appear in the request or response.
Cardinality | 0..1
Type | integer
Summary | true
ParameterDefinition.max
Element Id | ParameterDefinition.max
Definition | The maximum number of times this element is permitted to appear in the request or response.
Cardinality | 0..1
Type | string
Summary | true
ParameterDefinition.documentation
Element Id | ParameterDefinition.documentation
Definition | A brief discussion of what the parameter is for and how it is used by the module.
Cardinality | 0..1
Type | string
Summary | true
ParameterDefinition.type
Element Id | ParameterDefinition.type
Definition | The type of the parameter.
Cardinality | 1..1
Terminology Binding | Any defined Resource or Data Type name (including "Any" and "Type")
Type | code
Summary | true
ParameterDefinition.profile
Element Id | ParameterDefinition.profile
Definition | If specified, this indicates a profile that the input data must conform to, or that the output data will conform to.
Cardinality | 0..1
Type | canonical(StructureDefinition)
Summary | true

2.25.2.5 RelatedArtifact

See also Base Definition, Examples and Mappings.

RelatedArtifact
Standards Status | This element has a standards status of "Trial Use"
Element Id | RelatedArtifact
Definition | Related artifacts such as additional documentation, justification, or bibliographic references.
Cardinality | 0..*
Type | Element
Requirements | Knowledge resources must be able to provide enough information for consumers of the content (and/or interventions or results produced by the content) to be able to determine and understand the justification for and evidence in support of the content.
Summary | true
Comments | Each related artifact is either an attachment, or a reference to another knowledge resource, but not both.
RelatedArtifact.type
Element Id | RelatedArtifact.type
Definition | The type of relationship to the related artifact.
Cardinality | 1..1
Terminology Binding | RelatedArtifactType (Required)
Type | code
Summary | true
RelatedArtifact.label
Element Id | RelatedArtifact.label
Definition | A short label that can be used to reference the citation from elsewhere in the containing artifact, such as a footnote index.
Cardinality | 0..1

Type | string
Summary | true
RelatedArtifact.display
Element Id | RelatedArtifact.display
Definition | A brief description of the document or knowledge resource being referenced, suitable for display to a consumer.
Cardinality | 0..1
Type | string
Summary | true
RelatedArtifact.citation
Element Id | RelatedArtifact.citation
Definition | A bibliographic citation for the related artifact. This text SHOULD be formatted according to an accepted citation format.
Cardinality | 0..1
Type | markdown
Summary | true
Comments | Additional structured information about citations should be captured as extensions.
RelatedArtifact.url
Element Id | RelatedArtifact.url
Definition | A url for the artifact that can be followed to access the actual content.
Cardinality | 0..1
Type | url
Summary | true
Comments | If a document or resource element is present, this element SHALL NOT be provided (use the url or reference in the Attachment or resource reference).
RelatedArtifact.document
Element Id | RelatedArtifact.document
Definition | The document being referenced, represented as an attachment. This is exclusive with the resource element.
Cardinality | 0..1
Type | Attachment
Summary | true
RelatedArtifact.resource
Element Id | RelatedArtifact.resource
Definition | The related resource, such as a library, value set, profile, or other knowledge resource.
Cardinality | 0..1
Type | canonical(Any)
Summary | true
Comments | If the type is predecessor, this is a reference to the succeeding knowledge resource. If the type is successor, this is a reference to the prior knowledge resource.

2.25.2.6 TriggerDefinition

See also Base Definition, Examples and Mappings.

TriggerDefinition
Standards Status | This element has a standards status of "Trial Use"
Element Id | TriggerDefinition
Definition | A description of a triggering event. Triggering events can be named events, data events, or periodic, as determined by the type element.
Cardinality | 0..*
Type | Element
Summary | true
Comments | If an event is a named-event, it means the event is completely pre-coordinated, and no other information can be specified for the event. If the event is one of the data- events, the data and condition elements specify the triggering criteria. The data element specifies the structured component, and the condition element provides additional optional refinement of that structured component. If the event is periodic, the timing element defines when the event is triggered. For both data- and periodic events, a name can be provided as a shorthand for the formal semantics provided by the other elements.
Invariants | Defined on this element trd-1 Rule Either timing, or a data requirement, but not both data.empty() or timing.empty() trd-2 Rule A condition only if there is a data requirement condition.exists() implies data.exists() trd-3 Rule A named event requires a name, a periodic event requires timing, and a data event requires data (type = 'named-event' implies name.exists()) and (type = 'periodic' implies timing.exists()) and (type.startsWith('data-') implies data.exists())
Defined on this element
trd-1 | Rule | Either timing, or a data requirement, but not both | data.empty() or timing.empty()
trd-2 | Rule | A condition only if there is a data requirement | condition.exists() implies data.exists()
trd-3 | Rule | A named event requires a name, a periodic event requires timing, and a data event requires data | (type = 'named-event' implies name.exists()) and (type = 'periodic' implies timing.exists()) and (type.startsWith('data-') implies data.exists())
TriggerDefinition.type
Element Id | TriggerDefinition.type
Definition | The type of triggering event.
Cardinality | 1..1
Terminology Binding | TriggerType (Required)
Type | code
Summary | true
TriggerDefinition.name
Element Id | TriggerDefinition.name

Definition | A formal name for the event. This may be an absolute URI that identifies the event formally (e.g. from a trigger registry), or a simple relative URI that identifies the event in a local context.
Cardinality | 0..1
Type | string
Summary | true
Comments | An event name can be provided for all event types, but is required for named events. If a name is provided for a type other than named events, it is considered to be a shorthand for the semantics described by the formal description of the event.
TriggerDefinition.timing[x]
Element Id | TriggerDefinition.timing[x]
Definition | The timing of the event (if this is a periodic trigger).
Cardinality | 0..1
Type | Timing|Reference(Schedule)|date|dateTime
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
TriggerDefinition.data
Element Id | TriggerDefinition.data
Definition | The triggering data of the event (if this is a data trigger). If more than one data is requirement is specified, then all the data requirements must be true.
Cardinality | 0..*
Type | DataRequirement
Summary | true
Comments | This element shall be present for any data type trigger.
TriggerDefinition.condition
Element Id | TriggerDefinition.condition
Definition | A boolean-valued expression that is evaluated in the context of the container of the trigger definition and returns whether or not the trigger fires.
Cardinality | 0..1
Type | Expression
Requirements | Need to be able to formally describe the triggering criteria.
Summary | true
Comments | This element can be only be specified for data type triggers and provides additional semantics for the trigger. The context available within the condition is based on the type of data event. For all events, the current resource will be available as context. In addition, for modification events, the previous resource will also be available. The expression may be inlined, or may be a simple absolute URI, which is a reference to a named expression within a logic library referenced by a library element or extension within the containing resource. If the expression is a FHIR Path expression, it evaluates in the context of a resource of one of the type identified in the data requirement, and may also refer to the variable %previous for delta comparisons on events of type data-changed, data-modified, and data-deleted which will always have the same type.

2.25.2.7 Expression

See also Base Definition, Examples and Mappings.

Expression
Standards Status | This element has a standards status of "Trial Use"
Element Id | Expression
Definition | A expression that is evaluated in a specified context and returns a value. The context of use of the expression must specify the context in which the expression is evaluated, and how the result of the expression is used.
Cardinality | 0..*
Type | Element
Summary | true
Invariants | Defined on this element exp-1 Rule An expression or a reference must be provided expression.exists() or reference.exists()
Defined on this element
exp-1 | Rule | An expression or a reference must be provided | expression.exists() or reference.exists()
Expression.description
Element Id | Expression.description
Definition | A brief, natural language description of the condition that effectively communicates the intended semantics.
Cardinality | 0..1
Type | string
Summary | true
Expression.name
Element Id | Expression.name
Definition | A short name assigned to the expression to allow for multiple reuse of the expression in the context where it is defined.
Cardinality | 0..1
Type | id
Summary | true
Expression.language
Element Id | Expression.language
Definition | The media type of the language for the expression.
Cardinality | 1..1
Terminology Binding | ExpressionLanguage (Extensible but limited to ??)
Type | code
Summary | true
Expression.expression
Element Id | Expression.expression

Definition | An expression in the specified language that returns a value.
Cardinality | 0..1
Type | string
Summary | true
Expression.reference
Element Id | Expression.reference
Definition | A URI that defines where the expression is found.
Cardinality | 0..1
Type | uri
Summary | true
Comments | If both a reference and an expression is found, the reference SHALL point to the same expression.

2.25.2.8 UsageContext

See also Base Definition, Examples and Mappings.

UsageContext
Standards Status | This element has a standards status of "Trial Use"
Element Id | UsageContext
Definition | Specifies clinical/business/etc. metadata that can be used to retrieve, index and/or categorize an artifact. This metadata can either be specific to the applicable population (e.g., age category, DRG) or the specific context of care (e.g., venue, care setting, provider of care).
Cardinality | 0..*
Type | Element
Requirements | Consumers of the resource must be able to determine the intended applicability for the resource. Ideally, this information would be used programmatically to determine when and how it should be incorporated or exposed.
Summary | true
UsageContext.code
Element Id | UsageContext.code
Definition | A code that identifies the type of context being specified by this usage context.
Cardinality | 1..1
Terminology Binding | UsageContextType (Extensible)
Type | Coding
Summary | true
UsageContext.value[x]
Element Id | UsageContext.value[x]
Definition | A value that defines the context specified in this context of use. The interpretation of the value is defined by the code.
Cardinality | 1..1
Terminology Binding | ConformanceUseContext (Example)
Type | CodeableConcept|Quantity|Range|Reference(PlanDefinition | ResearchStudy | InsurancePlan | HealthcareService | Group | Location | Organization)
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true

2.25.3 Metadata Types - Mappings

URL: https://hl7.org/fhir/R4/metadatatypes-mappings.html

2.25.3 Metadata Types - Mappings

FHIR Infrastructure Work Group | Maturity Level: 4 | Standards Status: Informative

This page provides mappings for the data types. There are mappings to HL7 v2 , HL7 v3 , and (where appropriate) vCard (see Mappings to Other Standards for further information & status).

Table of Contents

ContactDetail | Contributor | DataRequirement | ParameterDefinition
RelatedArtifact | TriggerDefinition | Expression | UsageContext

2.25.3.1 ContactDetail

See also Base Definition, Examples and Detailed Descriptions.

2.25.3.2 Contributor

See also Base Definition, Examples and Detailed Descriptions.

2.25.3.3 DataRequirement

See also Base Definition, Examples and Detailed Descriptions.

2.25.3.4 ParameterDefinition

See also Base Definition, Examples and Detailed Descriptions.

2.25.3.5 RelatedArtifact

See also Base Definition, Examples and Detailed Descriptions.

2.25.3.6 TriggerDefinition

See also Base Definition, Examples and Detailed Descriptions.

2.25.3.7 Expression

See also Base Definition, Examples and Detailed Descriptions.

2.25.3.8 UsageContext

See also Base Definition, Examples and Detailed Descriptions.

2.26 Base Resource Definitions

URL: https://hl7.org/fhir/R4/resource.html
http://www.hl7.org/Special/committees/fiwg/index.cfm | Maturity Level: N | Normative (from v4.0.0) | Compartments: Not linked to any defined compartments

| This page has been approved as part of an ANSI standard. See the Infrastructure Package for further details.

2.26.1 Scope and Usage

This specification defines a series of different types of resource that can be used to exchange and/or store data in order to solve a wide range of healthcare related problems, both clinical and administrative. In addition, this specification defines several different ways of exchanging the resources.

A resource is an entity that:

• has a known identity (a URL) by which it can be addressed
• identifies itself as one of the types of resource defined in this specification
• contains a set of structured data items as described by the definition of the resource type
• has an identified version that changes if the contents of the resource change

Resources have multiple representations.

2.26.2 Boundaries and Relationships

The following optional elements and properties are defined for all resources:

• An identity
• Meta data
• A base language
• A reference to "Implicit Rules"

Most resources are derived from DomainResource - so they also can contain text, contained resources, extensions, and data elements specific to the particular domain of the resource. There is a special type of resource called Bundle for collections of resources.

Note: there is documentation for the Structure, UML, XML, and JSON representations of the resource structure.

2.26.3 Resource Content

• Structure
• UML
• XML
• JSON
• Turtle
• All

Structure

Name | Flags | Card. | Type | Description & Constraints
Resource | N | | n/a | Base Resource
id | Σ | 0..1 | id | Logical id of this artifact
meta | Σ | 0..1 | Meta | Metadata about the resource
implicitRules | ?!Σ | 0..1 | uri | A set of rules under which this content was created
language | | 0..1 | code | Language of the resource content Common Languages (Preferred but limited to AllLanguages)
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <id value="[id]"/><!-- 0..1 Logical id of this artifact -->
 <meta><!-- 0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!-- 0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!-- 0..1 Language of the resource content -->
</[name]>
```

JSON Template

```
{
  "resourceType" : "[name]",
  "id" : "<id>", // Logical id of this artifact
  "meta" : { Meta }, // Metadata about the resource
  "implicitRules" : "<uri>", // A set of rules under which this content was created
  "language" : "<code>" // Language of the resource content
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:[name];
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

 # from Element: Element.extension
  fhir:Resource.id [ id ]; # 0..1 Logical id of this artifact
  fhir:Resource.meta [ Meta ]; # 0..1 Metadata about the resource
  fhir:Resource.implicitRules [ uri ]; # 0..1 A set of rules under which this content was created
  fhir:Resource.language [ code ]; # 0..1 Language of the resource content
]
```

Structure

Name | Flags | Card. | Type | Description & Constraints
Resource | N | | n/a | Base Resource
id | Σ | 0..1 | id | Logical id of this artifact
meta | Σ | 0..1 | Meta | Metadata about the resource
implicitRules | ?!Σ | 0..1 | uri | A set of rules under which this content was created
language | | 0..1 | code | Language of the resource content Common Languages (Preferred but limited to AllLanguages)
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <id value="[id]"/><!-- 0..1 Logical id of this artifact -->
 <meta><!-- 0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!-- 0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!-- 0..1 Language of the resource content -->
</[name]>
```

JSON Template

```
{
  "resourceType" : "[name]",
  "id" : "<id>", // Logical id of this artifact
  "meta" : { Meta }, // Metadata about the resource
  "implicitRules" : "<uri>", // A set of rules under which this content was created
  "language" : "<code>" // Language of the resource content
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:[name];
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

 # from Element: Element.extension
  fhir:Resource.id [ id ]; # 0..1 Logical id of this artifact
  fhir:Resource.meta [ Meta ]; # 0..1 Metadata about the resource
  fhir:Resource.implicitRules [ uri ]; # 0..1 A set of rules under which this content was created
  fhir:Resource.language [ code ]; # 0..1 Language of the resource content
]
```

2.26.3.1 Terminology Bindings

Path | Definition | Type | Reference
Resource.language | A human language. | Preferred, but limited to AllLanguages | CommonLanguages

2.26.3.2 Resource Identity

There are two different ways to identify a resource:

• By a "Location" URL that identifies where is can be accessed (based on the "Logical ID"). This location will be change as it is copied/moved around

• By some inherent identifier ("Business Identifier" or "Canonical URL") that is part of the resource and remains fixed as it is copied/moved around

2.26.3.3 Logical ID

Each resource has an id element which contains the "logical id" of the resource assigned by the server responsible for storing it. Resources always have a known logical id except for a few special cases (e.g. when

a new resource is being sent to a server to assign a logical id in the create interaction). The logical id is unique within the space of all resources of the same type on the same server. Once assigned by the server, the id is never changed.

The location of a resource instance is an absolute URI constructed from the server base address at which the instance is found, the resource type and the Logical ID, such as http://test.fhir.org/rest/Patient/123 (where 123 is the Logical Id of a Patient resource). When the location is an HTTP address, this address can generally be used to retrieve or manipulate the resource. Note that implementations SHOULD NOT assume that the location of a resource is always resolvable to an accessible server - it may be temporarily unavailable, or not available by policy (e.g. firewalls) or in some cases, it might not actually exist (e.g. use of resource outside a RESTful environment). Resources reference each other by their location. These references are allowed to be absolute or relative (see Resource References for further discussion).

When a resource is copied from one server to another server, the copy might or might not keep the same logical id on the new server. This depends on replication and server policy. For further details, see Managing Resource Identity (including "Consistent Resource Identification").

Logical ids (and therefore locations) are case sensitive. Logical Ids are always opaque, and external systems need not and should not attempt to determine their internal structure. A logical id SHALL always be represented in the same way in resource references and URLs. Ids can be up to 64 characters long, and contain any combination of upper and lowercase ASCII letters, numerals, "-" and ".".

In some contexts, resources are not associated with location on a RESTful server, either because they are only created transiently for transfer between systems, or the systems are not using RESTful servers. In these cases, resources may be assigned some kind of location anyway, for purposes of consistency, or they might not have an assigned logical id, and they are identified based on other kinds of identifiers. See Resolving references in Bundles for one method of using resources not associated with RESTful servers.

2.26.3.3.1 "Business" Identifiers

Although the logical id of a resource changes as it moves from server to server, all copies of the resource refer to the same underlying concept, and this concept may also be represented in other formats (variously, HL7 v2 , CDA , XDS, and many more). Each representation carries common identifiers that identify the underlying concept (also called "real world entity") consistently across all contexts of use. This is known as the business identifier, and is found in the identifier element, which has the type Identifier.

All resources that have an identifier element support searching by the identifier, so that records can be located by that method. So if an HL7 v2 message has the following OBR:

```
OBR|1|845439^GHH OE|1045813^GHH LAB|1554-5^GLUCOSE^LN|||200202150730|...
```

Then the DiagnosticReport it represents can be located using the following query:

```
 GET [base]/DiagnosticReport?identifier=1045813
```

Most resources have a logical master server, the system on which the record first is created, and which manages the record directly. In the master system, the Logical ID might be the same value found in the business identifier since the master server can enforce business practices that make for stable identifiers. This is usually not a safe practice on secondary systems which are not in a position to enforce their required identifier consistency.

2.26.3.3.2 Canonical URLs

Some resource types have a defined element url which is the 'canonical URL' that always identifies the resource. This is a special kind of Business Identifier. Note that the element actually contains a URI, but is named url for legacy reasons.

The canonical URL serves as a stable logical identifier for the resource, and is the preferred way to reference a conformance or knowledge resource. The canonical URL SHOULD also resolve to the location where the master copy of the artifact is found, though it is not always possible to arrange this. The canonical URL SHALL NOT refer to some other resource (though it may resolve to a different version of the same resource).

See Canonical references for further discussion of canonical references and the canonical data type. The following resources have canonical URLs:

- ActivityDefinition
- CapabilityStatement
- ChargeItemDefinition
- CodeSystem
- CompartmentDefinition
- ConceptMap
- EffectEvidenceSynthesis
- EventDefinition
- Evidence
- EvidenceVariable

• ExampleScenario
• GraphDefinition
• ImplementationGuide
• Library
• Measure
• MessageDefinition
• NamingSystem
• OperationDefinition
• PlanDefinition
• Questionnaire
• ResearchDefinition
• ResearchElementDefinition
• RiskEvidenceSynthesis
• SearchParameter
• StructureDefinition
• StructureMap
• TerminologyCapabilities
• TestScript
• ValueSet

2.26.3.4 Consistent Resource Identification

Business Identifiers are the preferred basis to recognize the same content on different systems. For example:

```
GET http://a.company.example.com/Patient/23

<Patient xmlns="http://hl7.org/fhir">
  <id value="23"/>
  <identifier>
    <system value="http://a.particular.system/identifier"/>
    <value value="123456"/>
  </identifier>
</Patient>

GET http://other.company.example.com/fhir/Patient/5860200e-0ee3-42f5-8095-506e18dc9ca2

<Patient xmlns="http://hl7.org/fhir">
  <id value="5860200e-0ee3-42f5-8095-506e18dc9ca2"/>
  <identifier>
    <system value="http://a.particular.system/identifier"/>
    <value value="123456"/>
  </identifier>
</Patient>
```

Given that the identifier is the same, the patient resources are understood to mean that they describe the same patient. For this reason, systems SHOULD:

• Only assign identifiers to resources when they uniquely identify the real world entity that the resources match (e.g. do not use account numbers as patient identifiers when multiple different patients share the same account number)

• Preserve identifiers as much as possible and not throw them away. Even if the identifiers are not useful to the system itself, they are likely to be useful to downstream consumers

• When serving resources, use consistent identifiers from the master persistent store where ever possible (e.g. when a resource is created from a primary record, populate the identifier, and store the identifier somewhere so the same identifier is used next time)

Following these practices will make for consistent identification of concepts and real-world entities across healthcare eco-systems. For example, medication orders can be represented as MedicationRequest and MedicationStatement. Some systems - primary sources - will provide the medication order as a MedicationRequest while other secondary systems will provide them as a MedicationStatement. Any system processing the resources will need to reconcile these, and the presence of consistent identifiers will considerably simplify the reconciliation process.

Note that this specification provides two other related methods by which common source derivation can be indicated:

• Provenance - resources that have provenance statements where Provenance.entity.what[x] refers to the same source, and the .entity.role is derivation

• Resource.meta.source - two resources that have the same source

The granularity of Provenance.entity.what[x] / Resource.meta.source is not fixed to a single resource - multiple resources can be derived from a single source entity (e.g. a single CDA document), so applications need to be careful assuming that matching entity source information means that the resources are identifying the same real world concept.

2.26.3.5 Implicit Rules

A reference to a custom agreement that describes how the resource is being used (e.g. an implementation guide) that was followed when the resource was constructed, where the implementation guide must be known and understood in order to safely processing the content.

Asserting this rule set restricts the content to be only understood by a limited set of trading partners. This inherently limits the usefulness of the data in the long term, and should be avoided where possible. However, the existing health eco-system is highly fractured, and not yet ready to define, collect, and exchange data in a generally exchangeable sense.

Note that resources are almost always constructed following some custom agreement. Best practice - which is recommended throughout the conformance resources - is that such agreements make all knowledge about the content of the resource explicit in the content of the resource; if custom agreements follow this advice, and declare their extensions as required, then it is not necessary to understand the agreement in order to safely process the resource content. For this reason, use of implicitRules is rare.

2.26.3.6 Language

Each resource may have a language element that specifies the base language of the content using a code defined in BCP 47 . The language element is provided to support indexing and accessibility (e.g. text-to-speech use the language tag).

If a language is specified, it should also be specified on the Narrative Text. The html language tag in the narrative is used when processing the narrative. The language tag on the resource is provided so that applications processing the data in the resource can specify the language of any alternate presentations generated from the data.

There is no default language, though one may be inferred from the context of use. Not all of the content of the resource has to be in the specified language. Multiple languages can be supported.

Any valid language code as defined in BCP 47 is valid. The language code system is complex, with parts for Primary Language, Script, Region, and other variants and extensions. In order to simplify implementations, the language element has a preferred binding to an enumerated list of common language codes encountered in applications, with translations, and also a maximum binding of any language; this allows most applications to simply use a look up table, without preventing the use of any language code.

2.26.3.7 Resource Metadata

Each resource contains an element "meta", of type "Meta", which is a set of metadata that provides technical and workflow context to the resource. The metadata items are all optional, though some or all of them may be required in particular implementations or contexts of use.

Metadata Item | Type | Usage

versionId (0..1) | id | Changes each time the content of the resource changes. Can be referenced in a resource reference. Can be used to ensure that updates are based on the latest version of the resource. The version can be globally unique, or scoped by the Logical Id of the resource. Version identifiers are generally either a serially incrementing id scoped by the logical id, or a uuid, though neither of these approaches is required. There is no fixed order for version ids - clients cannot assume that a versionId that comes after another one either numerically or alphabetically represents a later version. The same versionId can never be used for more than one version of the same resource. On the RESTful API: On receiving a write operation, the server SHALL update this item to the current value, or remove it. Note that servers SHOULD support versions, but some are unable to

lastUpdated (0..1) | instant | If populated, this value changes each time the content of the resource changes. It can be used by a system or a human to judge the currency of the resource content. Note that version aware updates do not depend on this element. Note that a timezone code extension may be present on Meta.lastUpdated. If present, the timezone code applies to the server copy of the resource, and not necessarily to other time related elements within the resource even if the timezone offsets are the same. On the RESTful API: On receiving a write operation, the server SHALL update this item to the current time on the server

source (0..1) | uri | A uri that identifies the source system of the resource. This provides a minimal amount of Provenance information that can be used to track or differentiate the source of information in the resource. The source may identify another FHIR server, document, message, database, etc. In the provenance resource, this corresponds to Provenance.entity.what[x]. The exact use of the source (and the implied Provenance.entity.role) is left to implementer discretion. Only one nominated source is allowed; for additional provenance details, a full Provenance resource should be used. On the RESTful API: On receiving a write operation, the server SHOULD generally leave this unchanged, unless applicable business rules, along with available provenance, dictate otherwise

profile (0..*) | canonical | An assertion that the content conforms to a resource profile (a StructureDefinition). See Extending and Restricting Resources for further discussion. Can be changed as profiles and value sets change or the system rechecks conformance. The profile can be used to indicate which version(s) of FHIR a resource conforms to. On the RESTful API: On receiving a write operation, the server MAY elect to remove invalid claims, SHOULD retain claims that are correct or untested, and MAY add additional claims it believes are valid

security (0..*) | Coding | Security labels applied to this resource. These tags connect resources in specific ways to the overall security policy and infrastructure. Security tags can be updated when the resource changes, or whenever the security sub-system chooses to. On the RESTful API: On receiving a write operation, the server SHOULD preserve the labels unless applicable business rules dictate otherwise

tag (0..*) | Coding | Tags applied to this resource. Tags are used to relate resources to process and workflow. Applications are not required to consider the tags when interpreting the meaning of a resource. On the RESTful API: On receiving a write operation, the server SHOULD preserve the labels unless applicable business rules dictate otherwise

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
Meta | ΣN | | Element | Metadata about a resource Elements defined in Ancestors: id, extension
versionId | Σ | 0..1 | id | Version specific identifier
lastUpdated | Σ | 0..1 | instant | When the resource version last changed
source | Σ | 0..1 | uri | Identifies where the resource comes from
profile | Σ | 0..* | canonical(StructureDefinition) | Profiles this resource claims to conform to
security | Σ | 0..* | Coding | Security Labels applied to this resource SecurityLabels (Extensible)
tag | Σ | 0..* | Coding | Tags applied to this resource Common Tags (Example)
Documentation for this format

UML Diagram (Legend)

XML Template

```
<meta xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <versionId value="[id]"/><!-- 0..1 Version specific identifier -->
 <lastUpdated value="[instant]"/><!-- 0..1 When the resource version last changed -->
 <source value="[uri]"/><!-- 0..1 Identifies where the resource comes from -->
 <profile><!-- 0..* canonical(StructureDefinition) Profiles this resource claims to conform to --></profile>
 <security><!-- 0..* Coding Security Labels applied to this resource --></security>
 <tag><!-- 0..* Coding Tags applied to this resource --></tag>
</meta>
```

JSON Template

```
{
  // from Element: extension
  "versionId" : "<id>", // Version specific identifier
  "lastUpdated" : "<instant>", // When the resource version last changed
  "source" : "<uri>", // Identifies where the resource comes from
  "profile" : [{ canonical(StructureDefinition) }], // Profiles this resource claims to conform to
  "security" : [{ Coding }], // Security Labels applied to this resource
  "tag" : [{ Coding }] // Tags applied to this resource
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:Meta.versionId [ id ]; # 0..1 Version specific identifier
  fhir:Meta.lastUpdated [ instant ]; # 0..1 When the resource version last changed
  fhir:Meta.source [ uri ]; # 0..1 Identifies where the resource comes from
  fhir:Meta.profile [ canonical(StructureDefinition) ], ... ; # 0..* Profiles this resource claims to conform to
  fhir:Meta.security [ Coding ], ... ; # 0..* Security Labels applied to this resource
  fhir:Meta.tag [ Coding ], ... ; # 0..* Tags applied to this resource
]
```

Changes since Release 3

Meta
Meta.source | Added Element
Meta.profile | Type changed from uri to canonical(StructureDefinition)

See the Full Difference for further information

Structure

Name | Flags | Card. | Type | Description & Constraints
Meta | ΣN | | Element | Metadata about a resource Elements defined in Ancestors: id, extension
versionId | Σ | 0..1 | id | Version specific identifier
lastUpdated | Σ | 0..1 | instant | When the resource version last changed
source | Σ | 0..1 | uri | Identifies where the resource comes from
profile | Σ | 0..* | canonical(StructureDefinition) | Profiles this resource claims to conform to
security | Σ | 0..* | Coding | Security Labels applied to this resource SecurityLabels (Extensible)
tag | Σ | 0..* | Coding | Tags applied to this resource Common Tags (Example)
Documentation for this format

UML Diagram (Legend)

XML Template

```
<meta xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <versionId value="[id]"/><!-- 0..1 Version specific identifier -->
 <lastUpdated value="[instant]"/><!-- 0..1 When the resource version last changed -->
 <source value="[uri]"/><!-- 0..1 Identifies where the resource comes from -->
 <profile><!-- 0..* canonical(StructureDefinition) Profiles this resource claims to conform to --></profile>
 <security><!-- 0..* Coding Security Labels applied to this resource --></security>
 <tag><!-- 0..* Coding Tags applied to this resource --></tag>
</meta>
```

JSON Template

```
{
  // from Element: extension
  "versionId" : "<id>", // Version specific identifier
  "lastUpdated" : "<instant>", // When the resource version last changed
  "source" : "<uri>", // Identifies where the resource comes from
  "profile" : [{ canonical(StructureDefinition) }], // Profiles this resource claims to conform to
  "security" : [{ Coding }], // Security Labels applied to this resource
  "tag" : [{ Coding }] // Tags applied to this resource
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:Meta.versionId [ id ]; # 0..1 Version specific identifier
  fhir:Meta.lastUpdated [ instant ]; # 0..1 When the resource version last changed
  fhir:Meta.source [ uri ]; # 0..1 Identifies where the resource comes from
  fhir:Meta.profile [ canonical(StructureDefinition) ], ... ; # 0..* Profiles this resource claims to conform to
  fhir:Meta.security [ Coding ], ... ; # 0..* Security Labels applied to this resource
  fhir:Meta.tag [ Coding ], ... ; # 0..* Tags applied to this resource
]
```

Changes since Release 3

Meta
Meta.source | Added Element
Meta.profile | Type changed from uri to canonical(StructureDefinition)

See the Full Difference for further information

Note that the RESTful API defines some Operations that provide direct read and write access to the meta element.

2.26.3.7.1 Record Versions vs Business Versions vs FHIR Versions

There are 3 different versions of interest for a resource:

• The Record Version: changes each time the resource changes (usually managed by a server)

• The Business version: changes each time the content in the resources changes (managed by a human author or by business policy)

• The FHIR Version: the version of FHIR in which the resource is represented (controlled by the author of the resource)

Record (Resource) Version

All resources are conceptually versioned, and each resource sits at the head of a linear list of past versions. The past versions are superseded by the current version, and only available for audit/integrity purposes. The current version is e.g. http://acme.org/fhir/ResourceType/id123, and a past version would be http://acme.org/fhir/ResourceType/id123/_history/v2.

Note that there's no requirement for servers to keep a history. The history interaction is provided for where this is an appropriate service to provide. However, whether a server keeps them or not, past versions are dead and gone. The current version of the resource is in the Resource.meta.versionId. For a value set this would be:

```
<ValueSet>
  <meta>
    <versionId value="v2"/>
  </meta>
</ValueSet>
```

The versionId changes every time the server updates the resource and writes a new version over the top of an existing one.

Business Version

Some resources have another version marker in them. For instance, ValueSet has another version in it:

```
<ValueSet>
  <url value="http://acme.com/fhir/ValueSet/example"/>
  <version value="2.0"/>
</ValueSet>
```

This says that this is version 2.0 of the 'example' value set. This is the business version of the value set, the one that humans get involved with. These 2 versions elements have quite different lifecycles. To illustrate, take these cases:

• A value set is posted to a server (POST [base]/ValueSet) with ValueSet.url = "http://acme.com/valuesets/example". This is identified as the 1st version of the value set (ValueSet.version = 1). When the server gets it, it assigns an identity e.g. ValueSet.id = x1, and ValueSet.meta.versionId = 1. Later, another user creates a revised version of the value set, and this is called version 2. It is committed to the server as an update (PUT [base]/ValueSet/x1). Now, ValueSet.url = http://acme.com/valuesets/example, ValueSet.id = x1, ValueSet.version = 2 and ValueSet.meta.versionId = 2

• A value set is posted to a server (POST [base]/ValueSet) with ValueSet.url = "http://acme.com/valuesets/example". This is identified as the 1st version of the value set (ValueSet.version = 1). When the server gets it, it assigns an identity e.g. ValueSet.id = x1, and ValueSet.meta.versionId = 1. Then a typo is found in the definition, so this is fixed, but it's still v1 of the value set. This is PUT to [base]/ValueSet/x1. Now, ValueSet.url = http://acme.com/valuesets/example, ValueSet.id = x1, ValueSet.version = 1 and ValueSet.meta.versionId = 2. Later, another user creates a revised version of the value set, and this is called version 2. It is commited to the server as an update (PUT [base]/ValueSet/x1). Now, ValueSet.url = http://acme.com/valuesets/example, ValueSet.id = x1, ValueSet.version = 2 and ValueSet.meta.versionId = 3

• A value set is posted to a server (POST [base]/ValueSet) with ValueSet.url = "http://acme.com/valuesets/example". This is identified as the 1st version of the value set (ValueSet.version = 1). When the server gets it, it assigns an identity e.g. ValueSet.id = x1, and ValueSet.meta.versionId = 1. Later, another user creates a revised version of the value set, and this is called version 2. This time, as well as supporting this new version 2, there are production systems still using version 1, and both need to be valid on the value set server. So a new value set is created on the server (POST [base]/ValueSet) and is assigned the identiity 'x2'. Now, there are two different value sets, both with URL "http://acme.com/valuesets/example". One has ValueSet.id = x1, ValueSet.version = 1 and ValueSet.meta.versionId = 1 and the other has ValueSet.id = x2, ValueSet.version = 2 and ValueSet.meta.versionId = 1.

The version of resource SHOULD change every time some non-significant change is made. Exactly what change constitutes 'significant' depends on context, and is not specified here. Note, though, that it is possible to change elements such as status and date without changing the version, such as when marking a past version as retired. In general, any other changes should be associated with a change in version.

The Semantic Versioning specification recommend that versions be represented in three parts using the form MAJOR.MINOR.PATCH where, for the kind of artefacts described in this specification:

• MAJOR version increments when incompatible changes are made to the artifact

• MINOR version increments functionality is added to an artefact in a backwards-compatible manner

• PATCH version increments when minor changes are made that don't impact on implementations of the artefact (e.g. documentation clarifications)

For the kind of resources that have canonical URLs, this specification recommends that implementers follow the Semantic Versioning principles, but does not require this. Implementers should consider, however, making the MAJOR version part of the canonical URL rather than putting it in [Resource].version. So instead of this:

```
{
  "resourceType" : "ValueSet",
  "url" : "http://example.org/fhir/ValueSet/blah",
  "version" : "2.1.1"
 }
```

implementers should instead consider:

```
{
  "resourceType" : "ValueSet",
  "url" : "http://example.org/fhir/ValueSet/blah2",
  "version" : "1.1"
 }
```

The principle advantage is that other artefacts that reference the first artefact can do so by major version:

```
  "binding" : {
     "valueSet" : "http://example.org/fhir/ValueSet/blah2"
  }
```

The alternative - not having the major version in the URL - means that version dependencies must be managed by some other method (e.g. managing a set of resources with specific versioned references via an Implementation Guide).

There is no single best practice here; the correct choice depends on many factors outside the scope of this specification.

FHIR Version

All resources always have a FHIR version against which they were written. Note, however, that resources may be valid against multiple different FHIR versions, and this becomes progressively more likely as the maturity of the resource progresses.

The version is often not stated explicitly in the resource; it is controlled by the context. See Managing Multiple FHIR versions for further information.

2.26.3.8 Tags, Profiles, and Security Labels

These 3 metadata attributes are part of the resource, but are never used to keep information that needs to be understood when interpreting the content of a resource; their function is limited to finding and controlling access to the resource, and connecting resources to technical or clinical workflow.

2.26.3.8.1 Tags

Tags are used to associate additional operational information with the Resources, such as workflow management. A typical use of tagging would be to maintain an informal list of resources that need review.

In a general tag, the coded concept may be a reference to a healthcare vocabulary, including ones defined in this specification, or vocabularies such as those defined by HL7 for other purposes (e.g. HL7 v2 and HL7 v3 /CDA), LOINC, or SNOMED CT. Alternatively, the concept may be one defined by the implementation in the local context.

The list of tags on a resource is a set, where each tag is unique based on the system+code combination.

2.26.3.8.2 Profile Tags

A profile assertion represents a claim that the resource conforms to the identified StructureDefinition, which makes rules about what content is allowed to be in a resource. In a profile tag, the term is a URL that references an identified StructureDefinition resource.

It's always possible to determine whether a resource conforms to any profile simply by testing it against the profile (the validation tools provide the functionality to perform this test in a variety of contexts). However there are several circumstances where simply examining whether a resource conforms to a particular profile as needed is impractical:

- A server searching a set of resources for ones that conform to a particular profile
- A receiver that has many profiles to choose when validating resource

Profile Tags serve these use cases - a client/creator of a resource can tag the resource with an assertion that the resource conforms to a particular structure definition. The server/receiver of the resource can choose to take this assertion at face value, or to assist in locating the correct StructureDefinition against which to validate the resource.

Note: resources can conform to multiple profiles at once. A resource can conform to a profile without ever being labeled that it does, or a resource may falsely claim to conform to a profile. For this reason, applications processing resources SHOULD always depend on the contents of the resource when processing them, and/or check resources against the StructureDefinitions directly rather than relying the existence of profile tags for meaning. Profile Tags are provided as a method of finding resources that conform to a particular StructureDefinition, not statements of meaning about the resource.

Many trading partner agreements will make rules about what claims can be made and when they must be tested, which will make the profile assertion more reliable.

The list of profiles on a resource is a set, where each profile is unique based on the value of the URI.

2.26.3.8.3 Security Labels

A security label is attached to a resource to provide specific security metadata about the information in the resource. For further information, see Security Labels.

The list of security labels on a resource is a set, where each tag is unique based on the system+code combination.

2.26.3.9 Updates to Tags, Profiles, and Security Labels

When a resource is updated (e.g. on the RESTful interface), servers generally follow this pattern:

- Merge existing and new tags
- Replace existing profile tags with new profile tags
- Merge existing and new security labels

However, in some cases, an update may invalidate existing tags. Servers may update or remove previously existing recognized tags if this is known to be appropriate.

2.26.3.10 Further Information

- Conformance Rules
- Resource Definitions
- References between Resources
- Narratives
- Formats: XML, JSON
- Extensibility (Examples)

• Detailed Descriptions
• Inter-version Compatibility

2.26.3.11 Maturity Levels

The Maturity level documentation has moved.

2.26.4 Search Parameters

Common search parameters defined by this resource. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Paths
_content | string | Search on the entire content of the resource
_id | token | Logical id of this artifact | Resource.id
_lastUpdated | date | When the resource version last changed | Resource.meta.lastUpdated
_profile | uri | Profiles this resource claims to conform to | Resource.meta.profile
_query | token | A custom search profile that describes a specific defined query operation
_security | token | Security Labels applied to this resource | Resource.meta.security
_source | uri | Identifies where the resource comes from | Resource.meta.source
_tag | token | Tags applied to this resource | Resource.meta.tag

2.26.5 Resource Resource - Detailed Descriptions

URL: https://hl7.org/fhir/R4/resource-definitions.html

2.26.5 Resource Resource - Detailed Descriptions

http://www.hl7.org/Special/committees/fiwg/index.cfm | Maturity Level: N | Normative (from v4.0.0) | Compartments: Not linked to any defined compartments

| This page has been approved as part of an ANSI standard. See the Infrastructure Package for further details.

Detailed Descriptions for the elements in the Base Resource Definitions.

Resource
Definition | This is the base resource type for everything.
Cardinality | 0..*
Resource.id
Definition | The logical id of the resource, as used in the URL for the resource. Once assigned, this value never changes.
Cardinality | 0..1
Type | id
Summary | true
Comments | The only time that a resource does not have an id is when it is being submitted to the server using a create operation.
Resource.meta
Definition | The metadata about the resource. This is content that is maintained by the infrastructure. Changes to the content might not always be associated with version changes to the resource.
Cardinality | 0..1
Type | Meta
Summary | true
Resource.implicitRules
Definition | A reference to a set of rules that were followed when the resource was constructed, and which must be understood when processing the content. Often, this is a reference to an implementation guide that defines the special rules along with other profiles etc.
Cardinality | 0..1
Type | uri
Is Modifier | true (Reason: This element is labeled as a modifier because the implicit rules may provide additional knowledge about the resource that modifies it's meaning or interpretation)
Summary | true
Comments | Asserting this rule set restricts the content to be only understood by a limited set of trading partners. This inherently limits the usefulness of the data in the long term. However, the existing health eco-system is highly fractured, and not yet ready to define, collect, and exchange data in a generally computable sense. Wherever possible, implementers and/or specification writers should avoid using this element. Often, when used, the URL is a reference to an implementation guide that defines these special rules as part of it's narrative along with other profiles, value sets, etc.
Resource.language
Definition | The base language in which the resource is written.
Cardinality | 0..1
Terminology Binding | Common Languages (Preferred but limited to All Languages)
Type | code
Comments | Language is provided to support indexing and accessibility (typically, services such as text to speech use the language tag). The html language tag in the narrative applies to the narrative. The language tag on the resource may be used to specify the language of other presentations generated from the data in the resource. Not all the content has to be in the base language. The Resource.language should not be assumed to apply to the narrative automatically. If a language is specified, it should it also be specified on the div element in the html (see rules in HTML5 for information about the relationship between xml:lang and the html lang attribute).

Meta
Standards Status | This element has a standards status of "Normative"

Element Id | Meta
Definition | The metadata about a resource. This is content in the resource that is maintained by the infrastructure. Changes to the content might not always be associated with version changes to the resource.
Cardinality | 0..*
Type | Element
Summary | true
Meta.versionId
Element Id | Meta.versionId
Definition | The version specific identifier, as it appears in the version portion of the URL. This value changes when the resource is created, updated, or deleted.
Cardinality | 0..1
Type | id
Summary | true
Comments | The server assigns this value, and ignores what the client specifies, except in the case that the server is imposing version integrity on updates/deletes.
Meta.lastUpdated
Element Id | Meta.lastUpdated
Definition | When the resource last changed - e.g. when the version changed.
Cardinality | 0..1
Type | instant
Summary | true
Comments | This value is always populated except when the resource is first being created. The server / resource manager sets this value; what a client provides is irrelevant. This is equivalent to the HTTP Last-Modified and SHOULD have the same value on a read interaction.
Meta.source
Element Id | Meta.source
Definition | A uri that identifies the source system of the resource. This provides a minimal amount of Provenance information that can be used to track or differentiate the source of information in the resource. The source may identify another FHIR server, document, message, database, etc.
Cardinality | 0..1
Type | uri
Summary | true
Comments | In the provenance resource, this corresponds to Provenance.entity.what[x]. The exact use of the source (and the implied Provenance.entity.role) is left to implementer discretion. Only one nominated source is allowed; for additional provenance details, a full Provenance resource should be used. This element can be used to indicate where the current master source of a resource that has a canonical URL if the resource is no longer hosted at the canonical URL.
Meta.profile
Element Id | Meta.profile
Definition | A list of profiles (references to StructureDefinition resources) that this resource claims to conform to. The URL is a reference to StructureDefinition.url.
Cardinality | 0..*
Type | canonical(StructureDefinition)
Summary | true
Comments | It is up to the server and/or other infrastructure of policy to determine whether/how these claims are verified and/or updated over time. The list of profile URLs is a set.
Meta.security
Element Id | Meta.security
Definition | Security labels applied to this resource. These tags connect specific resources to the overall security policy and infrastructure.
Cardinality | 0..*
Terminology Binding | SecurityLabels (Extensible)
Type | Coding
Summary | true
Comments | The security labels can be updated without changing the stated version of the resource. The list of security labels is a set. Uniqueness is based the system/code, and version and display are ignored.
Meta.tag
Element Id | Meta.tag
Definition | Tags applied to this resource. Tags are intended to be used to identify and relate resources to process and workflow, and applications are not required to consider the tags when interpreting the meaning of a resource.
Cardinality | 0..*
Terminology Binding | Common Tags (Example)
Type | Coding
Summary | true
Comments | The tags can be updated without changing the stated version of the resource. The list of tags is a set. Uniqueness is based the system/code, and version and display are ignored.

2.26.6 Resource Resource - Operations

URL: https://hl7.org/fhir/R4/resource-operations.html

2.26.6 Resource Resource - Operations

FHIR Infrastructure Work Group | Maturity Level: N | Normative (from v4.0.0) | | Compartments: Not linked to any defined compartments

This resource has 7 operations operations associated with it:

$validate | Validate a resource | Normative
$meta | Access a list of profiles, tags, and security labels | Trial Use
$meta-add | Add profiles, tags, and security labels to a resource | Trial Use
$meta-delete | Delete profiles, tags, and security labels for a resource | Trial Use
$convert | Convert from one form to another | Draft
$graphql | Execute a graphql statement | Trial Use
$graph | Return a graph of resources | Trial Use

For more information about operations, including how they are invoked, see Operations.

2.26.6.1 Operation $validate on Resource

URL: https://hl7.org/fhir/R4/resource-operation-validate.html

2.26.6.1 Operation $validate on Resource

FHIR Infrastructure Work Group | Maturity Level: N | Normative (from v4.0.1) | Compartments: N/A

| This page has been approved as part of an ANSI standard. See the Infrastructure Package for further details.

The validate operation checks whether the attached content would be acceptable either generally, as a create, an update or as a delete to an existing resource. The action the server takes depends on the mode parameter:

• [mode not provided]: The server checks the content of the resource against any schema, constraint rules, and other general terminology rules

• create: The server checks the content, and then checks that the content would be acceptable as a create (e.g. that the content would not violate any uniqueness constraints)

• update: The server checks the content, and then checks that it would accept it as an update against the nominated specific resource (e.g. that there are no changes to immutable fields the server does not allow to change, and checking version integrity if appropriate)

• delete: The server ignores the content, and checks that the nominated resource is allowed to be deleted (e.g. checking referential integrity rules)

Modes update and delete can only be used when the operation is invoked at the resource instance level. The return from this operation is an OperationOutcome

Note that this operation is not the only way to validate resources - see Validating Resources for further information.

The official URL for this operation definition is

```
http://hl7.org/fhir/OperationDefinition/Resource-validate
```

Formal Definition (as a OperationDefinition).

URL: [base]/[Resource]/$validate

URL: [base]/[Resource]/[id]/$validate

This is an idempotent operation

In Parameters:
Name | Cardinality | Type | Binding | Profile | Documentation
resource | 0..1 | Resource | | | Must be present unless the mode is "delete"
mode | 0..1 | code | ResourceValidationMode (Required) | | Default is 'no action'; (e.g. general validation)
profile | 0..1 | uri | | | If this is nominated, then the resource is validated against this specific profile. If a profile is nominated, and the server cannot validate against the nominated profile, it SHALL return an error
Out Parameters:
Name | Cardinality | Type | Binding | Profile | Documentation
return | 1..1 | OperationOutcome | | | If the operation outcome does not list any errors, and a mode was specified, then this is an indication that the operation would be expected to succeed (excepting for transactional integrity issues, see below) Note: as this is the only out parameter, it is a resource, and it has the name 'return', the result of this operation is returned directly as a resource

This operation may be used during design and development to validate application design. It can also be used at run-time. One possible use might be that a client asks the server whether a proposed update is valid as the user is editing a dialog and displays an updated error to the user. The operation can be used as part of a light-weight two phase commit protocol but there is no expectation that the server will hold the content of the resource after this operation is used, or that the server guarantees to successfully perform an actual create, update or delete after the validation operation completes.

This operation returns a 200 OK whether or not the resource is valid. A 4xx or 5xx error means that the validation itself could not be performed, and it is unknown whether the resource is valid or not.

Note: the correct behaviour of validation with regard to language (especially for Coding.display) is currently undefined, and further development and testing may lead to specific requirements or recommendations in subsequent releases

Future versions of this specifcation may add additional validation parameters. A candidate list is maintained with the FHIR Validator Documentation

2.26.6.1.1 Examples

Request: Validate a Patient against the DAF patient profile using POST directly.

```
POST /open/Patient/$validate?profile=http://hl7.org/fhir/StructureDefinition/daf-patient
[other headers]
```

```
<?xml version="1.0" encoding="UTF-8"?>
<Patient xmlns="http://hl7.org/fhir">
  <id value="us01"/>
  <text>
    <status value="generated"/>
    <div xmlns="http://www.w3.org/1999/xhtml">
      <p>
        <b>Generated Narrative with Details</b>
      </p>
      <p>
        <b>id</b>: us01</p>
      <p>
        <b>active</b>: true</p>
        <p>
        <b>birthsex</b>: Male</p>
      <p>
        <b>name</b>: Peter James Chalmers (OFFICIAL)</p>
      <p>
        <b>telecom</b>: ph: (03) 5555 6473(WORK), person@example.org(WORK)</p>
      <p>
        <b>address</b>: 534 Erewhon St PleasantVille Vic 3999 (HOME)</p>
      <p>
        <b>race</b>: White, American Indian or Alaska Native, Asian, Shoshone, Filipino</p>
      <p>
        <b>ethnicity</b>: Hispanic or Latino, Dominican, Mexican</p>
    </div>
  </text>
  <extension url="http://hl7.org/fhir/StructureDefinition/us-core-race">
    <extension url="ombCategory">
      <valueCoding>
        <system value="http://terminology.hl7.org/CodeSystem/v3-Race"/>
        <code value="2106-3"/>
        <display value="White"/>
      </valueCoding>
    </extension>
    <extension url="ombCategory">
      <valueCoding>
        <system value="http://terminology.hl7.org/CodeSystem/v3-Race"/>
        <code value="1002-5"/>
        <display value="American Indian or Alaska Native"/>
      </valueCoding>
    </extension>
    <extension url="ombCategory">
      <valueCoding>
        <system value="http://terminology.hl7.org/CodeSystem/v3-Race"/>
        <code value="2028-9"/>
        <display value="Asian"/>
      </valueCoding>
    </extension>
    <extension url="detailed">
      <valueCoding>
        <system value="http://terminology.hl7.org/CodeSystem/v3-Race"/>
        <code value="1586-7"/>
        <display value="Shoshone"/>
      </valueCoding>
    </extension>
    <extension url="detailed">
      <valueCoding>
        <system value="http://terminology.hl7.org/CodeSystem/v3-Race"/>
        <code value="2036-2"/>
        <display value="Filipino"/>
      </valueCoding>
    </extension>
  </extension>
  <extension url="http://hl7.org/fhir/us/core/StructureDefinition/us-core-ethnicity">
    <extension url="ombCategory">
      <valueCoding>
        <system value="http://terminology.hl7.org/CodeSystem/v3-Ethnicity"/>
        <code value="2135-2"/>
        <display value="Hispanic or Latino"/>
      </valueCoding>
    </extension>
    <extension url="detailed">
      <valueCoding>
        <system value="http://terminology.hl7.org/CodeSystem/v3-Ethnicity"/>
        <code value="2184-0"/>
        <display value="Dominican"/>
      </valueCoding>
    </extension>
    <extension url="detailed">
      <valueCoding>
        <system value="http://terminology.hl7.org/CodeSystem/v3-Ethnicity"/>
        <code value="2148-5"/>
```

```
        <display value="Mexican"/>
      </valueCoding>
    </extension>
  </extension>
  <extension url="http://hl7.org/fhir/StructureDefinition/us-core-birthsex">
    <valueCode value="M"/>
  </extension>
  <active value="true"/>
  <!--  Peter James Chalmers, but called "Jim"  -->
  <name>
    <use value="official"/>
    <family value="Chalmers"/>
    <given value="Peter"/>
    <given value="James"/>
  </name>
  <telecom>
    <system value="phone"/>
    <value value="(03) 5555 6473"/>
    <use value="work"/>
  </telecom>
  <telecom>
    <extension url="http://hl7.org/fhir/StructureDefinition/us-core-direct">
      <valueBoolean value="true"/>
    </extension>
    <system value="email"/>
    <value value="person@example.org"/>
    <use value="work"/>
  </telecom>
  <address>
    <use value="home"/>
    <line value="534 Erewhon St"/>
    <city value="PleasantVille"/>
    <district value="Orange County"/>
    <state value="Vic"/>
    <postalCode value="3999"/>
  </address>
</Patient>
```

Request: Validate a proposed patient update - and for this example, using the Parameters syntax

```
POST /open/Patient/example/$validate
[other headers]

<Parameters xmlns="http://hl7.org/fhir">
  <parameter>
    <name value="mode"/>
    <valueCode value="update"/>
  </parameter>
  <parameter>
    <name value="resource"/>
    <resource>
      <?xml version="1.0" encoding="UTF-8"?>
      <Patient xmlns="http://hl7.org/fhir">
        <id value="example"/>
        <text>
          <status value="generated"/>
          <div xmlns="http://www.w3.org/1999/xhtml">
            <table>
              <tbody>
                <tr>
                  <td>Name</td>
                  <td>Peter James
                    <b>Chalmers</b> (&quot;Jim&quot;)
                  </td>
                </tr>
                <tr>
                  <td>Address</td>
                  <td>534 Erewhon, Pleasantville, Vic, 3999</td>
                </tr>
                <tr>
                  <td>Contacts</td>
                  <td>Home: unknown. Work: (03) 5555 6473</td>
                </tr>
                <tr>
                  <td>Id</td>
                  <td>MRN: 12345 (Acme Healthcare)</td>
                </tr>
              </tbody>
            </table>
          </div>
        </text>
        <!--  MRN assigned by ACME healthcare on 6-May 2001  -->
        <identifier>
          <use value="usual"/>
          <type>
            <coding>
              <system value="http://terminology.hl7.org/CodeSystem/v2-0203"/>
              <code value="MR"/>
            </coding>
```

```
    </type>
    <system value="urn:oid:1.2.36.146.595.217.0.1"/>
    <value value="12345"/>
    <period>
      <start value="2001-05-06"/>
    </period>
    <assigner>
      <display value="Acme Healthcare"/>
    </assigner>
  </identifier>
  <active value="true"/>
  <!--  Peter James Chalmers, but called "Jim"  -->
  <name>
    <use value="official"/>
    <family value="Chalmers"/>
    <given value="Peter"/>
    <given value="James"/>
  </name>
  <name>
    <use value="usual"/>
    <given value="Jim"/>
  </name>
  <name>
    <!-- Maiden names apply for anyone whose name changes as a result of marriage - irrespective of gender -->
    <use value="maiden"/>
    <family value="Windsor"/>
    <given value="Peter"/>
    <given value="James"/>
    <period>
      <end value="2002"/>
    </period>
  </name>
  <telecom>
    <use value="home"/>
    <!--  home communication details aren't known  -->
  </telecom>
  <telecom>
    <system value="phone"/>
    <value value="(03) 5555 6473"/>
    <use value="work"/>
    <rank value="1"/>
  </telecom>
  <telecom>
    <system value="phone"/>
    <value value="(03) 3410 5613"/>
    <use value="mobile"/>
    <rank value="2"/>
  </telecom>
  <telecom>
    <system value="phone"/>
    <value value="(03) 5555 8834"/>
    <use value="old"/>
    <period>
      <end value="2014"/>
    </period>
  </telecom>
  <!--  use FHIR code system for male / female  -->
  <gender value="male"/>
  <birthDate value="1974-12-25">
    <extension url="http://hl7.org/fhir/StructureDefinition/patient-birthTime">
      <valueDateTime value="1974-12-25T14:35:45-05:00"/>
    </extension>
  </birthDate>
  <deceasedBoolean value="false"/>
  <address>
    <use value="home"/>
    <type value="both"/>
    <text value="534 Erewhon St
PeasantVille, Rainbow, Vic  3999"/>
    <line value="534 Erewhon St"/>
    <city value="PleasantVille"/>
    <district value="Rainbow"/>
    <state value="Vic"/>
    <postalCode value="3999"/>
    <period>
      <start value="1974-12-25"/>
    </period>
  </address>
  <contact>
    <relationship>
      <coding>
        <system value="http://terminology.hl7.org/CodeSystem/v2-0131"/>
        <code value="N"/>
      </coding>
    </relationship>
    <name>
      <family value="du Marché">
        <!--  the "du" part is a family name prefix (VV in iso 21090)  -->
        <extension url="http://hl7.org/fhir/StructureDefinition/humanname-own-prefix">
```

```
              <valueString value="VV"/>
            </extension>
          </family>
          <given value="Bénédicte"/>
        </name>
        <telecom>
          <system value="phone"/>
          <value value="+33 (237) 998327"/>
        </telecom>
        <address>
          <use value="home"/>
          <type value="both"/>
          <line value="534 Erewhon St"/>
          <city value="PleasantVille"/>
          <district value="Rainbow"/>
          <state value="Vic"/>
          <postalCode value="3999"/>
          <period>
            <start value="1974-12-25"/>
          </period>
        </address>
        <gender value="female"/>
        <period>
          <!--  The contact relationship started in 2012  -->
          <start value="2012"/>
        </period>
      </contact>
      <managingOrganization>
        <reference value="Organization/1"/>
      </managingOrganization>
    </Patient>
  </resource>
 </parameter>
</Parameters>
```

Response: Returned when the resource is valid

```
HTTP/1.1 200 OK
[other headers]

<?xml version="1.0" encoding="UTF-8"?>
<OperationOutcome xmlns="http://hl7.org/fhir">
  <id value="allok"/>
  <text>
    <status value="generated"/>
    <div xmlns="http://www.w3.org/1999/xhtml">
      <p>All OK</p>
    </div>
  </text>
  <issue>
    <severity value="information"/>
    <code value="informational"/>
    <details>
      <text value="All OK"/>
    </details>
 </issue>
</OperationOutcome>
```

Response: Returned when the resource is not valid

```
HTTP/1.1 200 OK
[other headers]

<?xml version="1.0" encoding="UTF-8"?>
<OperationOutcome xmlns="http://hl7.org/fhir">
  <id value="validationfail"/>
  <text>
    <status value="generated"/>
    <div xmlns="http://www.w3.org/1999/xhtml">
      <p>Error parsing resource XML (Unknown Content &quot;label&quot; @ /Patient/identifier/label) at
    Patient.identifier</p>
    </div>
  </text>
  <issue>
    <severity value="error"/>
    <code value="structure"/>
    <details>
      <text value="Error parsing resource XML (Unknown Content &quot;label&quot;"/>
    </details>
    <location value="/f:Patient/f:identifier"/>
    <expression value="Patient.identifier"/>
  </issue>
</OperationOutcome>
```

For more information about operations, including how they are invoked, see Operations.

2.26.6.2 Operation $meta on Resource

URL: https://hl7.org/fhir/R4/resource-operation-meta.html

2.26.6.2 Operation $meta on Resource

FHIR Infrastructure Work Group | Maturity Level: 3 | Trial Use | Compartments: N/A

This operation retrieves a summary of the profiles, tags, and security labels for the given scope; e.g. for each scope:

• system-wide: a list of all profiles, tags and security labels in use by the system

• resource-type level: A list of all profiles, tags, and security labels for the resource type

• individual resource level: A list of all profiles, tags, and security labels for the current version of the resource. Also, as a special case, this operation (and other meta operations) can be performed on a historical version of a resource).

The official URL for this operation definition is

`http://hl7.org/fhir/OperationDefinition/Resource-meta`

Formal Definition (as a OperationDefinition).

URL: [base]/$meta

URL: [base]/[Resource]/$meta

URL: [base]/[Resource]/[id]/$meta

This is an idempotent operation

Out Parameters:

Name	Cardinality	Type	Binding	Profile	Documentation
return	1..1	Meta			The meta returned by the operation

At the system and type levels, the $meta operation is used to get a summary of all the labels that are in use across the system. The principal use for this operation is to support search e.g. what tags can be searched for. At these levels, the meta will not contain versionId, lastUpdated etc. Systems are not obligated to implement the operation at this level (and should return a 4xx error if they don't). At the resource and historical entry level, the $meta operation returns the same meta as would be returned by accessing the resource directly. This can be used to allow a system to get access to the meta-information for the resource without accessing the resource itself, e.g. for security reasons

2.26.6.2.1 Examples

Request: Get all the tags, profiles, and security labels used on patient resources

```
GET /fhir/Patient/$meta
```

Response: All the tags, profiles, and security labels used on patient resources

```
HTTP/1.1 200 OK
[other headers]

<Parameters xmlns="http://hl7.org/fhir">
  <parameter>
    <name value="return"/>
    <valueMeta>
      <profile value="http://hl7.org/fhir/StructureDefinition/daf-patient"/>
      <profile value="http://hl7.org/fhir/StructureDefinition/uslab-patient"/>
      <security>
        <system value="http://hl7.org/fhir/v3/ActCode"/>
        <code value="EMP"/>
        <display value="employee information sensitivity"/>
      </security>
      <tag>
        <system value="http://example.org/codes/tags"/>
        <code value="current"/>
        <display value="Current Inpatient"/>
      </tag>
    </valueMeta>
  </parameter>
</Parameters>
```

For more information about operations, including how they are invoked, see Operations.

2.26.6.3 Operation $meta-add on Resource

URL: https://hl7.org/fhir/R4/resource-operation-meta-add.html

2.26.6.3 Operation $meta-add on Resource

FHIR Infrastructure Work Group | Maturity Level: 3 | Trial Use | Compartments: N/A

This operation takes a meta, and adds the profiles, tags, and security labels found in it to the nominated resource.

The official URL for this operation definition is

`http://hl7.org/fhir/OperationDefinition/Resource-meta-add`

Formal Definition (as a OperationDefinition).

URL: [base]/[Resource]/[id]/$meta-add

This is not an idempotent operation

In Parameters:

Name | Cardinality | Type | Binding | Profile | Documentation

meta | 1..1 | Meta | | | Profiles, tags, and security labels to add to the existing resource. Note that profiles, tags, and security labels are sets, and duplicates are not created. The identity of a tag or security label is the system+code. When matching existing tags during adding, version and display are ignored. For profiles, matching is based on the full URL

Out Parameters:

Name | Cardinality | Type | Binding | Profile | Documentation

return | 1..1 | Meta | | | Resulting meta for the resource

2.26.6.3.1 Examples

Request: Add a 'paper record lost' tag to a resource

```
POST /fhir/Patient/example/$meta-add
[Other headers]

<Parameters xmlns="http://hl7.org/fhir">
  <parameter>
    <name value="meta"/>
    <valueMeta>
      <tag>
        <system value="http://example.org/codes/tags"/>
        <code value="record-lost"/>
        <display value="Patient File Lost"/>
      </tag>
    </valueMeta>
  </parameter>
</Parameters>
```

Response: All the tags, profiles, and security labels used on patient resources

```
HTTP/1.1 200 OK
[other headers]

<Parameters xmlns="http://hl7.org/fhir">
  <parameter>
    <name value="return"/>
    <valueMeta>
      <profile value="http://hl7.org/fhir/StructureDefinition/daf-patient"/>
      <tag>
        <system value="http://example.org/codes/tags"/>
        <code value="current"/>
        <display value="Current Inpatient"/>
      </tag>
      <tag>
        <system value="http://example.org/codes/tags"/>
        <code value="record-lost"/>
        <display value="Patient File Lost"/>
      </tag>
    </valueMeta>
  </parameter>
</Parameters>
```

For more information about operations, including how they are invoked, see Operations.

2.26.6.4 Operation $meta-delete on Resource

URL: https://hl7.org/fhir/R4/resource-operation-meta-delete.html

2.26.6.4 Operation $meta-delete on Resource

FHIR Infrastructure Work Group | Maturity Level: 3 | Trial Use | Compartments: N/A

This operation takes a meta, and deletes the profiles, tags, and security labels found in it from the nominated resource.

The official URL for this operation definition is

```
http://hl7.org/fhir/OperationDefinition/Resource-meta-delete
```

Formal Definition (as a OperationDefinition).

URL: [base]/[Resource]/[id]/$meta-delete

This is not an idempotent operation

In Parameters:

Name | Cardinality | Type | Binding | Profile | Documentation

meta | 1..1 | Meta | | | Profiles, tags, and security labels to delete from the existing resource. It is not an error if these tags, profiles, and labels do not exist. The identity of a tag or security label is the system+code. When matching existing tags during deletion, version and display are ignored. For profiles, matching is based on the full URL

Out Parameters:

Name | Cardinality | Type | Binding | Profile | Documentation

return | 1..1 | Meta | | | Resulting meta for the resource

2.26.6.4.1 Examples

Request: Remove the 'current patient' tag from a resource

```
POST /fhir/Patient/example/$meta-delete
[Other headers]

<Parameters xmlns="http://hl7.org/fhir">
  <parameter>
    <name value="meta"/>
    <valueMeta>
      <tag>
        <system value="http://example.org/codes/tags"/>
        <code value="current"/>
        <display value="Current Inpatient"/>
      </tag>
    </valueMeta>
  </parameter>
</Parameters>
```

Response: All the tags, profiles, and security labels used on patient resources

```
HTTP/1.1 200 OK
[other headers]

<Parameters xmlns="http://hl7.org/fhir">
  <parameter>
    <name value="return"/>
    <valueMeta>
      <profile value="http://hl7.org/fhir/StructureDefinition/daf-patient"/>
      <tag>
        <system value="http://example.org/codes/tags"/>
        <code value="record-lost"/>
        <display value="Patient File Lost"/>
      </tag>
    </valueMeta>
  </parameter>
</Parameters>
```

For more information about operations, including how they are invoked, see Operations.

2.26.6.5 Operation $convert on Resource

URL: https://hl7.org/fhir/R4/resource-operation-convert.html

2.26.6.5 Operation $convert on Resource

FHIR Infrastructure Work Group | Maturity Level: 1 | Draft | Compartments: N/A

This operation takes a resource in one form, and returns to in another form. Both input and output are a single resource. The primary use of this operation is to convert between formats (e.g. (XML -> JSON or vice versa).

The official URL for this operation definition is

```
http://hl7.org/fhir/OperationDefinition/Resource-convert
```

Formal Definition (as a OperationDefinition).

URL: [base]/$convert

This is an idempotent operation

In Parameters:

Name	Cardinality	Type	Binding	Profile	Documentation
input	1..1	Resource			The resource that is to be converted

Out Parameters:

Name	Cardinality	Type	Binding	Profile	Documentation
output	1..1	Resource			The resource after conversion

While the primary use of this operation is simple - converting a resource from one format to another, there are many potential uses including:

• converting resources from one version to another

• restructuring information in a resource (e.g. moving method into/out of Observation.code)

• extracting data from a questionnaire

• converting CDA documents or v2 messages (as a binary resource) to a bundle (or vice versa) (or even openEHR or openMHealth).

These variants would all be associated with parameters that define and control these kind of conversions, though such parameters are not defined at this time. In the absence of any parameters, simple format conversion is all that will occur.

For this reason, implementers should be aware that:

• the output resource type may be different from the input resource (particularly, it might be a bundle)

• binary resources may be represented directly using some other content-type (i.e. just post the content directly)

Implementers are encouraged to provide feedback to HL7 about their use of this operation

2.26.6.5.1 Examples

Request: Convert to a different version of FHIR (in this case, from 3 to 4

```
POST /base/$convert
Accept: application/fhir+json
Content-Type: application/fhir+xml
[other headers]

<?xml version="1.0" encoding="UTF-8"?>
<Patient xmlns="http://hl7.org/fhir">
  <id value="example"/>
  <text>
    <status value="generated"/>
    <div xmlns="http://www.w3.org/1999/xhtml">
      <table>
        <tbody>
          <tr>
            <td>Name</td>
            <td>Peter James
              <b>Chalmers</b> (&quot;Jim&quot;)
            </td>
          </tr>
          <tr>
            <td>Address</td>
            <td>534 Erewhon, Pleasantville, Vic, 3999</td>
          </tr>
          <tr>
            <td>Contacts</td>
            <td>Home: unknown. Work: (03) 5555 6473</td>
          </tr>
          <tr>
            <td>Id</td>
            <td>MRN: 12345 (Acme Healthcare)</td>
          </tr>
        </tbody>
      </table>
    </div>
  </text>
  <!--  MRN assigned by ACME healthcare on 6-May 2001  -->
  <identifier>
    <use value="usual"/>
    <type>
      <coding>
        <system value="http://terminology.hl7.org/CodeSystem/v2-0203"/>
        <code value="MR"/>
      </coding>
    </type>
    <system value="urn:oid:1.2.36.146.595.217.0.1"/>
    <value value="12345"/>
    <period>
      <start value="2001-05-06"/>
    </period>
    <assigner>
      <display value="Acme Healthcare"/>
    </assigner>
  </identifier>
  <active value="true"/>
  <!--  Peter James Chalmers, but called "Jim"  -->
  <name>
    <use value="official"/>
    <family value="Chalmers"/>
    <given value="Peter"/>
    <given value="James"/>
  </name>
  <name>
    <use value="usual"/>
    <given value="Jim"/>
  </name>
  <name>
    <!-- Maiden names apply for anyone whose name changes as a result of marriage - irrespective of gender -->
    <use value="maiden"/>
    <family value="Windsor"/>
    <given value="Peter"/>
    <given value="James"/>
    <period>
      <end value="2002"/>
    </period>
  </name>
  <telecom>
    <use value="home"/>
    <!--  home communication details aren't known  -->
  </telecom>
  <telecom>
    <system value="phone"/>
    <value value="(03) 5555 6473"/>
    <use value="work"/>
    <rank value="1"/>
  </telecom>
  <telecom>
```

```
    <system value="phone"/>
    <value value="(03) 3410 5613"/>
    <use value="mobile"/>
    <rank value="2"/>
  </telecom>
  <telecom>
    <system value="phone"/>
    <value value="(03) 5555 8834"/>
    <use value="old"/>
    <period>
      <end value="2014"/>
    </period>
  </telecom>
  <!--  use FHIR code system for male / female  -->
  <gender value="male"/>
  <birthDate value="1974-12-25">
    <extension url="http://hl7.org/fhir/StructureDefinition/patient-birthTime">
      <valueDateTime value="1974-12-25T14:35:45-05:00"/>
    </extension>
  </birthDate>
  <deceasedBoolean value="false"/>
  <address>
    <use value="home"/>
    <type value="both"/>
    <text value="534 Erewhon St
PeasantVille, Rainbow, Vic  3999"/>
    <line value="534 Erewhon St"/>
    <city value="PleasantVille"/>
    <district value="Rainbow"/>
    <state value="Vic"/>
    <postalCode value="3999"/>
    <period>
      <start value="1974-12-25"/>
    </period>
  </address>
  <contact>
    <relationship>
      <coding>
        <system value="http://terminology.hl7.org/CodeSystem/v2-0131"/>
        <code value="N"/>
      </coding>
    </relationship>
    <name>
      <family value="du Marché">
        <!--  the "du" part is a family name prefix (VV in iso 21090)  -->
        <extension url="http://hl7.org/fhir/StructureDefinition/humanname-own-prefix">
          <valueString value="VV"/>
        </extension>
      </family>
      <given value="Bénédicte"/>
    </name>
    <telecom>
      <system value="phone"/>
      <value value="+33 (237) 998327"/>
    </telecom>
    <address>
      <use value="home"/>
      <type value="both"/>
      <line value="534 Erewhon St"/>
      <city value="PleasantVille"/>
      <district value="Rainbow"/>
      <state value="Vic"/>
      <postalCode value="3999"/>
      <period>
        <start value="1974-12-25"/>
      </period>
    </address>
    <gender value="female"/>
    <period>
      <!--  The contact relationship started in 2012  -->
      <start value="2012"/>
    </period>
  </contact>
  <managingOrganization>
    <reference value="Organization/1"/>
  </managingOrganization>
</Patient>

-------------------------------------------------

POST /base/$convert
Accept: application/fhir+json; fhirVersion=3.0
Content-Type: application/fhir+json; fhirVersion=4.0
[other headers]

-- snip - R3 version of a resource --
```

Response: Return the JSON format

```
HTTP/1.1 200 OK
Content-Type: application/fhir+json
[other headers]
```


```
{
  "resourceType" : "Patient",
  "id" : "example",
  "text" : {
    "status" : "generated",
    "div" : "<div xmlns=\"http://www.w3.org/1999/xhtml\">\n\t\t\t<table>\n\t\t\t\t<tbody>\n\t\t\t\t\t<tr>\n\t\t\t\t\t\t<
    td>Name</td>\n\t\t\t\t\t\t<td>Peter James \n              <b>Chalmers</b> (&quot;Jim&quot;)\n
    </td>\n\t\t\t\t\t</tr>\n\t\t\t\t\t<tr>\n\t\t\t\t\t\t<td>Address</td>\n\t\t\t\t\t\t<td>534 Erewhon, Pleasantville,
    Vic, 3999</td>\n\t\t\t\t</tr>\n\t\t\t\t<tr>\n\t\t\t\t\t<td>Contacts</td>\n\t\t\t\t\t<td>Home: unknown. Work:
     (03) 5555 6473</td>\n\t\t\t\t</tr>\n\t\t\t\t<tr>\n\t\t\t\t\t<td>Id</td>\n\t\t\t\t\t<td>MRN: 12345 (Acme
    Healthcare)</td>\n\t\t\t\t</tr>\n\t\t\t</tbody>\n\t\t</table>\n\t</div>"
  },
  "identifier" : [{
    "use" : "usual",
    "type" : {
      "coding" : [{
        "system" : "http://terminology.hl7.org/CodeSystem/v2-0203",
        "code" : "MR"
      }]
    },
    "system" : "urn:oid:1.2.36.146.595.217.0.1",
    "value" : "12345",
    "period" : {
      "start" : "2001-05-06"
    },
    "assigner" : {
      "display" : "Acme Healthcare"
    }
  }],
  "active" : true,
  "name" : [{
    "use" : "official",
    "family" : "Chalmers",
    "given" : ["Peter",
    "James"]
  },
  {
    "use" : "usual",
    "given" : ["Jim"]
  },
  {
    "use" : "maiden",
    "family" : "Windsor",
    "given" : ["Peter",
    "James"],
    "period" : {
      "end" : "2002"
    }
  }],
  "telecom" : [{
    "use" : "home"
  },
  {
    "system" : "phone",
    "value" : "(03) 5555 6473",
    "use" : "work",
    "rank" : 1
  },
  {
    "system" : "phone",
    "value" : "(03) 3410 5613",
    "use" : "mobile",
    "rank" : 2
  },
  {
    "system" : "phone",
    "value" : "(03) 5555 8834",
    "use" : "old",
    "period" : {
      "end" : "2014"
    }
  }],
  "gender" : "male",
  "birthDate" : "1974-12-25",
  "_birthDate" : {
    "extension" : [{
      "url" : "http://hl7.org/fhir/StructureDefinition/patient-birthTime",
      "valueDateTime" : "1974-12-25T14:35:45-05:00"
    }]
  },
  "deceasedBoolean" : false,
  "address" : [{
    "use" : "home",
    "type" : "both",
    "text" : "534 Erewhon St PeasantVille, Rainbow, Vic  3999",
```

```
    "line" : ["534 Erewhon St"],
    "city" : "PleasantVille",
    "district" : "Rainbow",
    "state" : "Vic",
    "postalCode" : "3999",
    "period" : {
      "start" : "1974-12-25"
    }
  }],
  "contact" : [{
    "relationship" : [{
      "coding" : [{
        "system" : "http://terminology.hl7.org/CodeSystem/v2-0131",
        "code" : "N"
      }]
    }],
    "name" : {
      "family" : "du Marché",
      "_family" : {
        "extension" : [{
          "url" : "http://hl7.org/fhir/StructureDefinition/humanname-own-prefix",
          "valueString" : "VV"
        }]
      },
      "given" : ["Bénédicte"]
    },
    "telecom" : [{
      "system" : "phone",
      "value" : "+33 (237) 998327"
    }],
    "address" : {
      "use" : "home",
      "type" : "both",
      "line" : ["534 Erewhon St"],
      "city" : "PleasantVille",
      "district" : "Rainbow",
      "state" : "Vic",
      "postalCode" : "3999",
      "period" : {
        "start" : "1974-12-25"
      }
    },
    "gender" : "female",
    "period" : {
      "start" : "2012"
    }
  }],
  "managingOrganization" : {
    "reference" : "Organization/1"
  }
}
```


For more information about operations, including how they are invoked, see Operations.

2.26.6.6 Operation $graphql on Resource

URL: https://hl7.org/fhir/R4/resource-operation-graphql.html

2.26.6.6 Operation $graphql on Resource

FHIR Infrastructure Work Group | Maturity Level: 1 | Trial Use | Compartments: N/A

Execute a graphql statement on a since resource or against the entire system. See the Using GraphQL with FHIR page for further details.

For the purposes of graphQL compatibility, this operation can also be invoked using a POST with the graphQL as the body, or a JSON body (see graphQL spec for details).

The official URL for this operation definition is

`http://hl7.org/fhir/OperationDefinition/Resource-graphql`

Formal Definition (as a OperationDefinition).

URL: [base]/$graphql

URL: [base]/[Resource]/[id]/$graphql

This is an idempotent operation

In Parameters:

Name	Cardinality	Type	Binding	Profile	Documentation
query	1..1	string			

Out Parameters:

Name	Cardinality	Type	Binding	Profile	Documentation
result	1..1	Binary			The content is always returned as application/json; this SHOULD be specified in the Accept header

For more information about operations, including how they are invoked, see Operations.

2.26.6.7 Operation $graph on Resource

URL: https://hl7.org/fhir/R4/resource-operation-graph.html

2.26.6.7 Operation $graph on Resource

FHIR Infrastructure Work Group | Maturity Level: 1 | Trial Use | Compartments: N/A

Return an entire graph of resources based on a GraphDefinition. The operation is invoked on a specific instance of a resource, and the graph definition tells the server what other resources to return in the same packaage.

The official URL for this operation definition is

```
http://hl7.org/fhir/OperationDefinition/Resource-graph
```

Formal Definition (as a OperationDefinition).

URL: [base]/[Resource]/[id]/$graph

This is an idempotent operation

In Parameters:

Name | Cardinality | Type | Binding | Profile | Documentation

graph | 1..1 | uri | | | Servers MAY choose to allow any graph definition to be specified, but MAY require that the client choose a graph definition from a specific list of known supported definitions. The server is not required to support a formal definition of the graph on the end point

Out Parameters:

Name | Cardinality | Type | Binding | Profile | Documentation

result | 1..1 | Bundle | | | The set of resources that were in the graph based on the provided definition

For more information about operations, including how they are invoked, see Operations.

2.26.7 Resource Resource - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/resource-version-maps.html

n/a

2.26.7 Resource Resource - R3/R4 Conversions

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | | Compartments: Not linked to any defined compartments

R3 : R4 Conversion maps for Resource.

Functional status for this map: Not Mapped (see documentation)

2.26.7.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/Resource3to4" = "R2 to R3 Conversions for Resource"

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Resource" alias ResourceR2 as source
uses "http://hl7.org/fhir/StructureDefinition/Resource" alias Resource as target

imports "http://hl7.org/fhir/StructureMap/primitives3to4"
imports "http://hl7.org/fhir/StructureMap/Meta3to4"

group Resource(source src : ResourceR2, target tgt : Resource) <<type+>> {
  src.id -> tgt.id "Resource-id";
  src.meta -> tgt.meta "Resource-meta";
  src.implicitRules -> tgt.implicitRules "Resource-implicitRules";
  src.language -> tgt.language "Resource-language";
}
```

2.26.7.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/Resource4to3" = "R3 to R2 Conversion for Resource"

uses "http://hl7.org/fhir/StructureDefinition/Resource" alias Resource as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Resource" alias ResourceR2 as target

imports "http://hl7.org/fhir/StructureMap/primitives4to3"
imports "http://hl7.org/fhir/StructureMap/Meta4to3"

group Resource(source src : Resource, target tgt : ResourceR2) <<type+>> {
  src.id -> tgt.id "Resource-id";
  src.meta -> tgt.meta "Resource-meta";
  src.implicitRules -> tgt.implicitRules "Resource-implicitRules";
  src.language -> tgt.language "Resource-language";
}
```

2.26.7.3 R4 Validation Errors

2.26.8 Resource HL7 Extensions

URL: https://hl7.org/fhir/R4/resource-extensions.html

Defines common extensions used across multiple different resources

2.26.8 Resource HL7 Extensions

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative

2.26.8.1 Content

Extensions:

resource-approvalDate | approvalDate : The date on which the asset content was approved by the publisher. Approval happens once when the content is officially approved for usage.

resource-lastReviewDate | lastReviewDate : The date on which the asset content was last reviewed. Review happens periodically after that, but doesn't change the original approval date.

resource-effectivePeriod | effectivePeriod : The period during which the resource content was or is planned to be effective.

resource-pertainsToGoal | pertainsToGoal : Indicates that the resource is related to either the measurement, achievement or progress towards the referenced goal. For example, a Procedure to exercise pertainsToGoal of losing weight.

2.26.9 JSON Schema for Resource

URL: https://hl7.org/fhir/R4/resource.schema.json.html

2.26.9 JSON Schema for Resource

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is a representation of the json schema for Resource, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/Resource",
  "$ref": "#/definitions/Resource",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "Resource": {
      "description": "This is the base resource type for everything.",
      "properties": {
        "resourceType": {
          "description": "This is a Resource resource",
          "const": "Resource"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        }
      },
      "additionalProperties": false,
      "required": [
        "resourceType"
      ]
    }
  }
}
```


2.26.10 Resource Resource - Maturity Dependency View

URL: https://hl7.org/fhir/R4/resource-dependencies.html

2.26.10 Resource Resource - Maturity Dependency View

FHIR Infrastructure Work Group | Maturity Level: N | Normative (from v4.0.0) | Compartments: Not linked to any defined compartments

| Some of the content on this page (marked clearly) has been approved as part of an ANSI standard. See the Infrastructure Package for further details.

Dependency Graph for Resource FMM level 5

Name | Card. | Type | Dependency Analysis
Resource | | n/a
id | 0..1 | id
meta | 0..1 | Meta
implicitRules | 0..1 | uri
language | 0..1 | code

2.27 DomainResource Resource

URL: https://hl7.org/fhir/R4/domainresource.html
http://www.hl7.org/Special/committees/fiwg/index.cfm | Maturity Level: N | Normative (from v4.0.0) | Compartments: Not linked to any defined compartments

| This page has been approved as part of an ANSI standard. See the Infrastructure Package for further details.

2.27.1 Scope and Usage

A domain resource is a resource that:

• has a human-readable XHTML representation of the content of the resource (see Human Narrative in resources)

• can contain additional related resources inside the resource (see Contained Resources)

• can have additional extensions and modifierExtensions as well as the defined data (See Extensibility)

As an abstract resource, this resource is never created directly; instead, one of its descendant resources (see List of Resources) is created.

2.27.2 Boundaries and Relationships

This resource extends the base Resource. All of the listed Resources except Bundle, Parameters and Binary extend this resource.

2.27.3 Resource Content

• Structure

• UML

• XML

• JSON

• Turtle

• All

Structure

Name | Flags | Card. | Type | Description & Constraints
DomainResource | IN | | Resource | A resource with narrative, extensions, and contained resources + Rule: If the resource is contained in another resource, it SHALL NOT contain nested Resources + Rule: If the resource is contained in another resource, it SHALL be referred to from elsewhere in the resource or SHALL refer to the containing resource + Rule: If a resource is contained in another resource, it SHALL NOT have a meta.versionId or a meta.lastUpdated + Rule: If a resource is contained in another resource, it SHALL NOT have a security label + Guideline: A resource should have narrative for robust management Elements defined in Ancestors: id, meta, implicitRules, language
text | | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?! | 0..* | Extension | Extensions that cannot be ignored
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <text><!-- 0..1 Narrative Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <extension><!-- 0..* Extension Additional content defined by implementations --></extension>
 <modifierExtension><!-- 0..* Extension Extensions that cannot be ignored --></modifierExtension>
</[name]>
```

JSON Template

```
{
  "resourceType" : "[name]",
  // from Resource: id, meta, implicitRules, and language
  "text" : { Narrative }, // Text summary of the resource, for human interpretation
  "contained" : [{ Resource }], // Contained, inline Resources
  (Extensions - see JSON page)
```

```
  (Modifier Extensions - see JSON page)
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:[name];
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

 # from Element: Element.extension
  fhir:DomainResource.text [ Narrative ]; # 0..1 Text summary of the resource, for human interpretation
  fhir:DomainResource.contained [ Resource ], ... ; # 0..* Contained, inline Resources
  fhir:DomainResource.extension [ Extension ], ... ; # 0..* Additional content defined by implementations
  fhir:DomainResource.modifierExtension [ Extension ], ... ; # 0..* Extensions that cannot be ignored
]
```

Structure

Name | Flags | Card. | Type | Description & Constraints

DomainResource | IN | | Resource | A resource with narrative, extensions, and contained resources + Rule: If the resource is contained in another resource, it SHALL NOT contain nested Resources + Rule: If the resource is contained in another resource, it SHALL be referred to from elsewhere in the resource or SHALL refer to the containing resource + Rule: If a resource is contained in another resource, it SHALL NOT have a meta.versionId or a meta.lastUpdated + Rule: If a resource is contained in another resource, it SHALL NOT have a security label + Guideline: A resource should have narrative for robust management Elements defined in Ancestors: id, meta, implicitRules, language

text | | 0..1 | Narrative | Text summary of the resource, for human interpretation

contained | | 0..* | Resource | Contained, inline Resources

extension | | 0..* | Extension | Additional content defined by implementations

modifierExtension | ?! | 0..* | Extension | Extensions that cannot be ignored

Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <text><!-- 0..1 Narrative Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <extension><!-- 0..* Extension Additional content defined by implementations --></extension>
 <modifierExtension><!-- 0..* Extension Extensions that cannot be ignored --></modifierExtension>
</[name]>
```

JSON Template

```
{
  "resourceType" : "[name]",
  // from Resource: id, meta, implicitRules, and language
  "text" : { Narrative }, // Text summary of the resource, for human interpretation
  "contained" : [{ Resource }], // Contained, inline Resources
  (Extensions - see JSON page)
  (Modifier Extensions - see JSON page)
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:[name];
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

 # from Element: Element.extension
  fhir:DomainResource.text [ Narrative ]; # 0..1 Text summary of the resource, for human interpretation
  fhir:DomainResource.contained [ Resource ], ... ; # 0..* Contained, inline Resources
  fhir:DomainResource.extension [ Extension ], ... ; # 0..* Additional content defined by implementations
  fhir:DomainResource.modifierExtension [ Extension ], ... ; # 0..* Extensions that cannot be ignored
]
```

2.27.3.1 Constraints

id | Level | Location | Description | Expression

dom-2 | Rule | (base) | If the resource is contained in another resource, it SHALL NOT contain nested Resources | contained.contained.empty()

dom-3 | Rule | (base) | If the resource is contained in another resource, it SHALL be referred to from elsewhere in the resource or SHALL refer to the containing resource | contained.where((('#'+id in (%resource.descendants().reference | %resource.descendants().as(canonical) | %resource.descendants().as(uri) | %resource.descendants().as(url))) or descendants().where(reference = '#').exists() or descendants().where(as(canonical) = '#').exists() or descendants().where(as(canonical) = '#').exists()).not()).trace('unmatched', id).empty()

dom-4 | Rule | (base) | If a resource is contained in another resource, it SHALL NOT have a meta.versionId or a meta.lastUpdated | contained.meta.versionId.empty() and contained.meta.lastUpdated.empty()

dom-5 | Rule | (base) | If a resource is contained in another resource, it SHALL NOT have a security label | contained.meta.security.empty()

dom-6 | Guideline | (base) | A resource should have narrative for robust management | text.`div`.exists() This is (only) a best practice guideline because: When a resource has no narrative, only systems that fully understand the data can display the resource to a human safely. Including a human readable representation in the resource makes for a much more robust eco-system and cheaper handling of resources by intermediary systems. Some ecosystems restrict distribution of resources to only those

systems that do fully understand the resources, and as a consequence implementers may believe that the narrative is superfluous. However experience shows that such eco-systems often open up to new participants over time.

2.27.4 Searching Extensions

To search for extensions, define a SearchParameter for the extension. All other search parameters are named aliases for existing content in the resource. In some cases, though not all, the search parameter name is the same as the element that it searches, but this is not required. Searching for extensions is the same - define a name that identifies the value extension by its URL, and then searches can filter based on the value of the extension.

2.27.5 Search Parameters

Common search parameters defined by this resource. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Paths
_text | string | Search on the narrative of the resource

2.27.6 Resource DomainResource - Detailed Descriptions

URL: https://hl7.org/fhir/R4/domainresource-definitions.html

2.27.6 Resource DomainResource - Detailed Descriptions

http://www.hl7.org/Special/committees/fiwg/index.cfm | Maturity Level: N | Normative (from v4.0.0) | Compartments: Not linked to any defined compartments

| This page has been approved as part of an ANSI standard. See the Infrastructure Package for further details.

Detailed Descriptions for the elements in the DomainResource Resource.

DomainResource
Definition | A resource that includes narrative, extensions, and contained resources.
Cardinality | 0..*
Type | Resource
Invariants | Defined on this element dom-2 Rule If the resource is contained in another resource, it SHALL NOT contain nested Resources contained.contained.empty() dom-3 Rule If the resource is contained in another resource, it SHALL be referred to from elsewhere in the resource or SHALL refer to the containing resource contained.where((('#'+id in (%resource.descendants().reference | %resource.descendants().as(canonical) | %resource.descendants().as(uri) | %resource.descendants().as(url))) or descendants().where(reference = '#').exists() or descendants().where(as(canonical) = '#').exists() or descendants().where(as(canonical) = '#').exists()).not()).trace('unmatched', id).empty() dom-4 Rule If a resource is contained in another resource, it SHALL NOT have a meta.versionId or a meta.lastUpdated contained.meta.versionId.empty() and contained.meta.lastUpdated.empty() dom-5 Rule If a resource is contained in another resource, it SHALL NOT have a security label contained.meta.security.empty() dom-6 Guideline A resource should have narrative for robust management text.`div`.exists() This is (only) a best practice guideline because: When a resource has no narrative, only systems that fully understand the data can display the resource to a human safely. Including a human readable representation in the resource makes for a much more robust eco-system and cheaper handling of resources by intermediary systems. Some ecosystems restrict distribution of resources to only those systems that do fully understand the resources, and as a consequence implementers may believe that the narrative is superfluous. However experience shows that such eco-systems often open up to new participants over time.
Defined on this element
dom-2 | Rule | If the resource is contained in another resource, it SHALL NOT contain nested Resources | contained.contained.empty()
dom-3 | Rule | If the resource is contained in another resource, it SHALL be referred to from elsewhere in the resource or SHALL refer to the containing resource | contained.where((('#'+id in (%resource.descendants().reference | %resource.descendants().as(canonical) | %resource.descendants().as(uri) | %resource.descendants().as(url))) or descendants().where(reference = '#').exists() or descendants().where(as(canonical) = '#').exists() or descendants().where(as(canonical) = '#').exists()).not()).trace('unmatched', id).empty()
dom-4 | Rule | If a resource is contained in another resource, it SHALL NOT have a meta.versionId or a meta.lastUpdated | contained.meta.versionId.empty() and contained.meta.lastUpdated.empty()
dom-5 | Rule | If a resource is contained in another resource, it SHALL NOT have a security label | contained.meta.security.empty()
dom-6 | Guideline | A resource should have narrative for robust management | text.`div`.exists() This is (only) a best practice guideline because: When a resource has no narrative, only systems that fully understand the data can display the resource to a human safely. Including a human readable representation in the resource makes for a much more robust eco-system and cheaper handling of resources by intermediary systems. Some ecosystems restrict distribution of resources to only those systems that do fully understand the resources, and as a consequence implementers may believe that the narrative is superfluous. However experience shows that such eco-systems often open up to new participants over time.
DomainResource.text
Definition | A human-readable narrative that contains a summary of the resource and can be used to represent the content of the resource to a human. The narrative need not encode all the structured data, but is required to contain sufficient detail to make it "clinically safe" for a human to just read the narrative. Resource definitions may define what content should be represented in the narrative to ensure clinical safety.
Cardinality | 0..1
Type | Narrative
Alternate Names | narrative; html; xhtml; display
Comments | Contained resources do not have narrative. Resources that are not contained SHOULD have a narrative. In some cases, a resource may only have text with little or no additional discrete data (as long as all minOccurs=1 elements are satisfied). This may be necessary for data from legacy systems where information is captured as a "text blob" or where text is additionally

entered raw or narrated and encoded information is added later.

DomainResource.contained

Definition | These resources do not have an independent existence apart from the resource that contains them - they cannot be identified independently, and nor can they have their own independent transaction scope.

Cardinality | 0..*

Type | Resource

Alternate Names | inline resources; anonymous resources; contained resources

Comments | This should never be done when the content can be identified properly, as once identification is lost, it is extremely difficult (and context dependent) to restore it again. Contained resources may have profiles and tags In their meta elements, but SHALL NOT have security labels.

DomainResource.extension

Definition | May be used to represent additional information that is not part of the basic definition of the resource. To make the use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension.

Cardinality | 0..*

Type | Extension

Alternate Names | extensions; user content

Comments | There can be no stigma associated with the use of extensions by any application, project, or standard - regardless of the institution or jurisdiction that uses or defines the extensions. The use of extensions is what allows the FHIR specification to retain a core level of simplicity for everyone.

DomainResource.modifierExtension

Definition | May be used to represent additional information that is not part of the basic definition of the resource and that modifies the understanding of the element that contains it and/or the understanding of the containing element's descendants. Usually modifier elements provide negation or qualification. To make the use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension. Applications processing a resource are required to check for modifier extensions. Modifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource (including cannot change the meaning of modifierExtension itself).

Cardinality | 0..*

Type | Extension

Is Modifier | true (Reason: Modifier extensions are expected to modify the meaning or interpretation of the resource that contains them)

Requirements | Modifier extensions allow for extensions that cannot be safely ignored to be clearly distinguished from the vast majority of extensions which can be safely ignored. This promotes interoperability by eliminating the need for implementers to prohibit the presence of extensions. For further information, see the definition of modifier extensions.

Alternate Names | extensions; user content

Comments | There can be no stigma associated with the use of extensions by any application, project, or standard - regardless of the institution or jurisdiction that uses or defines the extensions. The use of extensions is what allows the FHIR specification to retain a core level of simplicity for everyone.

2.27.7 Resource DomainResource - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/domainresource-version-maps.html

n/a

2.27.7 Resource DomainResource - R3/R4 Conversions

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | | Compartments: Not linked to any defined compartments

R3 : R4 Conversion maps for DomainResource.

Functional status for this map: Not Mapped (see documentation)

2.27.7.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/DomainResource3to4" = "R3 to R4 Conversions for DomainResource"

uses "http://hl7.org/fhir/3.0/StructureDefinition/DomainResource" alias DomainResourceR3 as source
uses "http://hl7.org/fhir/StructureDefinition/DomainResource" alias DomainResource as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group DomainResource(source src : DomainResourceR3, target tgt : DomainResource) extends Resource <<type+>> {
  src.text -> tgt.text;
  src.contained -> tgt.contained;
  src.extension -> tgt.extension;
  src.modifierExtension -> tgt.modifierExtension;
}
```

2.27.7.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/DomainResource4to3" = "R4 to R3 Conversion for DomainResource"

uses "http://hl7.org/fhir/StructureDefinition/DomainResource" alias DomainResource as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/DomainResource" alias DomainResourceR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group DomainResource(source src : DomainResourceR3, target tgt : DomainResource) extends Resource <<type+>> {
  src.text -> tgt.text;
```

```
  src.contained -> tgt.contained;
  src.extension where url.startsWith('http://hl7.org/fhir/3.0/StructureDefinition').not() and (url !=
    'http://hl7.org/fhir/StructureDefinition/patient-animal') -> tgt.extension;
  src.modifierExtension -> tgt.modifierExtension;
}
```

2.27.7.3 R4 Validation Errors

2.27.8 JSON Schema for DomainResource

URL: https://hl7.org/fhir/R4/domainresource.schema.json.html

2.27.8 JSON Schema for DomainResource

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is a representation of the json schema for DomainResource, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/DomainResource",
  "$ref": "#/definitions/DomainResource",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "DomainResource": {
      "description": "A resource that includes narrative, extensions, and contained resources.",
      "properties": {
        "resourceType": {
          "description": "This is a DomainResource resource",
          "const": "DomainResource"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
          "description": "A human-readable narrative that contains a summary of the resource and can be used to
    represent the content of the resource to a human. The narrative need not encode all the structured data, but is
    required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
    Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
          "$ref": "Narrative.schema.json#/definitions/Narrative"
        },
        "contained": {
          "description": "These resources do not have an independent existence apart from the resource that contains
    them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
          "items": {
            "$ref": "ResourceList.schema.json#/definitions/ResourceList"
          },
          "type": "array"
        },
        "extension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
     definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "modifierExtension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource and that modifies the understanding of the element that contains it and/or the understanding of the
```

```
    containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
    of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
    met as part of the definition of the extension. Applications processing a resource are required to check for
    modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
    DomainResource (including cannot change the meaning of modifierExtension itself).",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        }
      },
      "additionalProperties": false,
      "required": [
        "resourceType"
      ]
    }
  }
}
```


2.27.9 Resource DomainResource - Maturity Dependency View

URL: https://hl7.org/fhir/R4/domainresource-dependencies.html

2.27.9 Resource DomainResource - Maturity Dependency View

FHIR Infrastructure Work Group | Maturity Level: N | Normative (from v4.0.0) | Compartments: Not linked to any defined compartments

| This page has been approved as part of an ANSI standard. See the Infrastructure Package for further details.

Dependency Graph for DomainResource FMM level 5

Name | Card. | Type | Dependency Analysis
DomainResource | | Resource
text | 0..1 | Narrative
contained | 0..* | Resource
extension | 0..* | Extension
modifierExtension | 0..* | Extension

2.28 Element

URL: https://hl7.org/fhir/R4/element.html
FHIR Infrastructure Work Group | Maturity Level: Normative | Standards Status: Normative

| This page has been approved as part of an ANSI standard. See the Infrastructure Package for further details.

The base definition for all elements contained inside a resource. All elements, whether defined as a Data Type (including primitives) or as part of a resource structure, have this base content:

• Extensions

• An internal id

There are 3 kinds of descendant types that specialize Element:

• Primitive data types, that add a primitive value property of the specified type

• Complex data types, that add their own children (all of which are also elements)

• BackboneElement, A specialization that adds modifierExtension, which is the super-type of all the element types defined in resource definitions (e.g. Patient.contact)

Note that resources themselves all specialize the base type Resource.

2.28.0.1 Content

• Structure

• UML

• XML

• JSON

• All

Structure

Name | Flags | Card. | Type | Description & Constraints
Element | I | | n/a | Base for all elements All FHIR elements must have a @value or children
id | | 0..1 | string | Unique id for inter-element referencing
extension | | 0..* | Extension | Additional content defined by implementations
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir" id="Internal id (e.g. like xml:id) (string)">
```

```
 <extension><!-- 0..* Extension Additional content defined by implementations --></extension>
</[name]>
```

JSON Template

```
// complex types:

{
  "id" : "<string>", // Internal Id
  "extension" : [{//  Additional content defined by implementations
    // from Element: extension (recursive)
    "url" : "<uri>", // R!  identifies the meaning of the extension
    "value[x]" : <*> // identifies the meaning of the extension
  }]
}
// primitive types:

{
  "{name}" : "value",  // The primitive value
  "{name}_" : { // Special syntax for the id and extensions
    "id" : "<string>", // Internal Id
    "extension" : [{//  Additional content defined by implementations
      // from Element: extension (recursive)
      "url" : "<uri>", // R!  identifies the meaning of the extension
      "value[x]" : <*> // identifies the meaning of the extension
    }]
  }
}
```

Structure

Name	Flags	Card.	Type	Description & Constraints
Element	I		Element	Base for all elements All FHIR elements must have a @value or children
id		0..1	string	Unique id for inter-element referencing
extension		0..*	Extension	Additional content defined by implementations

Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir" id="Internal id (e.g. like xml:id) (string)">
 <extension><!-- 0..* Extension Additional content defined by implementations --></extension>
</[name]>
```

JSON Template

```
// complex types:

{
  "id" : "<string>", // Internal Id
  "extension" : [{//  Additional content defined by implementations
    // from Element: extension (recursive)
    "url" : "<uri>", // R!  identifies the meaning of the extension
    "value[x]" : <*> // identifies the meaning of the extension
  }]
}
// primitive types:

{
  "{name}" : "value",  // The primitive value
  "{name}_" : { // Special syntax for the id and extensions
    "id" : "<string>", // Internal Id
    "extension" : [{//  Additional content defined by implementations
      // from Element: extension (recursive)
      "url" : "<uri>", // R!  identifies the meaning of the extension
      "value[x]" : <*> // identifies the meaning of the extension
    }]
  }
}
```

Constraints

id	Level	Location	Description	Expression
ele-1	Rule	(base)	All FHIR elements must have a @value or children	hasValue() or (children().count() > id.count())

This constraint exists to reduce syntactical variation in resource contents. If an element has no children, then it is always omitted from the resource, as opposed to optionally present without any content.

2.28.0.2 Representation of Element

As the base type for all elements included in a resource, Element is an important structural element of FHIR. Even the primitive types inherit the base features and representation rules that apply to the Element type.

2.28.0.3 XML Representation

Elements are represented by an XML element. The name of the element comes from the context in which it is used, not from the type. The internal id is represented as an attribute (similar to xml:id, but see below about scope). Extensions are represented as XML elements. Here is the representation for an element 'use' of type Coding:

```
<use id="[internal id]">
  <extension url="..."/>
    ... if there are any extensions
  <extension>
  .. elements of Coding type...
</use>
```

Primitive types have the same representation; the actual primitive value appears as an XML attribute named value on the XML element. For example, a code property named "status" will be represented like this:

```
<status id="[internal id]" value="[value of code]"">
  <extension url="..."/>
    ... if there are any extensions
  <extension>
</status>
```

2.28.0.4 JSON Representation

Elements (except for primitive types, see below) are represented by a JSON object property. The name of the property comes from the context in which it is used, not from the type. The internal id is represented as a JSON string property named "id". Extensions are represented in a JSON array of objects named "extension". Here is the representation for a property 'use' of type Coding:

```
{
  "use" : {
    "id" : "[internal id]",
    "extension" : [
      ..extensions, if present...
    ],
    .. properties of Coding type...
  }
}
```

Primitive types are represented differently; the actual primitive value appears as a JSON string or number property. If an internal id or extensions are present, they appear in a JSON object with the name of the primitive value property with "_" prepended. For example, a code property named "status" will be represented like this:

```
{
  "status" : "[value of code]",
  "_status" : {
    "id" : "[internal id]",
    "extension" : [
      ..extensions, if present...
    ]
  }
}
```

The exact use of this pattern is described here.

2.28.0.5 Internal Id Scope

The id property of the element is defined to allow implementers to build implementation functionality that makes use of internal references inside the resource. This specification does not define any general use for the internal id, though some resources (e.g. StructureDefinition) and extensions (e.g. originalText, narrativeLink) make use of it.

The internal id is unique within the scope of the resource that contains it. Specifically, this means:

• The id SHALL be unique within a given resource

• The uniqueness boundary extends into contained resources. i.e. a contained resource cannot have the same id as any element in the resource that contains it or any other contained resource

• The uniqueness boundary is broken at Bundle.entry.resource and Parameters.parameter.resource, since these are elements that aggregate different resources

• The id element does not have extensions itself

These rules ensure that there is no need to change internal identifiers while exchanging resources.

2.28.1 Detailed Descriptions

URL: https://hl7.org/fhir/R4/element-definitions.html

2.28.1 Element - Detailed Descriptions

FHIR Infrastructure Work Group | Maturity Level: Normative | Standards Status: Normative

| This page has been approved as part of an ANSI standard. See the Infrastructure Package for further details.

2.28.1.1 Element

Element
Standards Status | This element has a standards status of "Normative"
Definition | Base definition for all elements in a resource.
Cardinality | 0..*

Invariants | Defined on this element ele-1 Rule All FHIR elements must have a @value or children hasValue() or (children().count() > id.count()) Affect this element
Defined on this element
ele-1 | Rule | All FHIR elements must have a @value or children | hasValue() or (children().count() > id.count())
Affect this element
Element.id
Definition | Unique id for the element within a resource (for internal references). This may be any string value that does not contain spaces.
Cardinality | 0..1
Type | string
Element.extension
Definition | May be used to represent additional information that is not part of the basic definition of the element. To make the use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension.
Cardinality | 0..*
Type | Extension
Alternate Names | extensions; user content
Comments | There can be no stigma associated with the use of extensions by any application, project, or standard - regardless of the institution or jurisdiction that uses or defines the extensions. The use of extensions is what allows the FHIR specification to retain a core level of simplicity for everyone.

2.28.2 Extensions (can be used anywhere)

URL: https://hl7.org/fhir/R4/element-extras.html

2.28.2 Element Extensions (can be used anywhere)

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative

The extensions defined here can be used on any element. For background information, see Extensibility. Additional extensions may be found in published Implementation Guides , or in the Conformance resource registry .

Extensions (+ see extensions on all Elements)
bodySite | BodyStructure Reference | for General Extensions for use by FHIR Implementers
cqf-calculatedValue | calculatedValue | for Clinical Reasoning Extensions
cqf-expression | expression | for Clinical Reasoning Extensions
cqf-initialValue | initialValue | for Clinical Reasoning Extensions
cqf-library | library | for Clinical Reasoning Extensions
cqf-measureInfo | measureInfo | for Clinical Reasoning Extensions
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
data-absent-reason | Data Absent Reason | for General Extensions for use by FHIR Implementers
iso21090-nullFlavor | nullFlavor | for ISO 21090 Data Type Extensions
iso21090-preferred | preferred | for ISO 21090 Data Type Extensions
narrativeLink | Narrative Link | for General Extensions for use by FHIR Implementers
originalText | Original Text | for General Extensions for use by FHIR Implementers
rendering-style | style | for Extensions for rendering data elements
rendering-styleSensitive | styleSensitive | for Extensions for rendering data elements
structuredefinition-fmm | fmm | for Standard Structure Definition Extensions
structuredefinition-standards-status | standards-status | for Standard Structure Definition Extensions
structuredefinition-wg | wg | for Standard Structure Definition Extensions
Profiles
No Profiles defined for for this type

2.29 BackboneElement

URL: https://hl7.org/fhir/R4/backboneelement.html
FHIR Infrastructure Work Group | Maturity Level: Normative | Standards Status: Normative

| This page has been approved as part of an ANSI standard. See the Infrastructure Package for further details.

The base definition for complex elements defined as part of a resource definition - that is, elements that have children that are defined in the resource. Data Type elements do not use this type, though a few data types specialize it (Timing, Dosage, ElementDefinition). For instance, Patient.contact is an element that is defined as part of the patient resource, so it automatically has the type BackboneElement.

Note that the descendant types of BackboneElement are all declared implicitly as part of the definitions of the resources.

2.29.0.1 Content

- Structure
- UML
- XML

• JSON
• Turtle
• R3 Diff
• All

Structure

Name | Flags | Card. | Type | Description & Constraints
BackboneElement | N | | Element | Base for elements defined inside a resource Elements defined in Ancestors: id, extension
modifierExtension | ?!Σ | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <modifierExtension><!-- 0..* Extension Extensions that cannot be ignored even if unrecognized --></modifierExtension>
</[name]>
```

JSON Template

```
{
  // from Element: extension
  (Modifier Extensions - see JSON page)
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:BackboneElement.modifierExtension [ Extension ], ... ; # 0..* Extensions that cannot be ignored even if
    unrecognized
]
```

Changes since Release 3

BackboneElement | No Changes

See the Full Difference for further information

Structure

Name | Flags | Card. | Type | Description & Constraints
BackboneElement | N | | Element | Base for elements defined inside a resource Elements defined in Ancestors: id, extension
modifierExtension | ?!Σ | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <modifierExtension><!-- 0..* Extension Extensions that cannot be ignored even if unrecognized --></modifierExtension>
</[name]>
```

JSON Template

```
{
  // from Element: extension
  (Modifier Extensions - see JSON page)
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:BackboneElement.modifierExtension [ Extension ], ... ; # 0..* Extensions that cannot be ignored even if
    unrecognized
]
```

Changes since Release 3

BackboneElement | No Changes

See the Full Difference for further information

2.29.1 Detailed Descriptions

URL: https://hl7.org/fhir/R4/backboneelement-definitions.html

2.29.1 BackboneElement - Detailed Descriptions

FHIR Infrastructure Work Group | Maturity Level: Normative | Standards Status: Normative

| This page has been approved as part of an ANSI standard. See the Infrastructure Package for further details.

2.29.1.1 BackboneElement

BackboneElement
Standards Status | This element has a standards status of "Normative"
Definition | Base definition for all elements that are defined inside a resource - but not those in a data type.
Cardinality | 0..*
Type | Element
BackboneElement.modifierExtension
Definition | May be used to represent additional information that is not part of the basic definition of the element and that modifies the understanding of the element in which it is contained and/or the understanding of the containing element's descendants. Usually modifier elements provide negation or qualification. To make the use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension. Applications processing a resource are required to check for modifier extensions. Modifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource (including cannot change the meaning of modifierExtension itself).
Cardinality | 0..*
Type | Extension
Is Modifier | true (Reason: Modifier extensions are expected to modify the meaning or interpretation of the element that contains them)
Requirements | Modifier extensions allow for extensions that cannot be safely ignored to be clearly distinguished from the vast majority of extensions which can be safely ignored. This promotes interoperability by eliminating the need for implementers to prohibit the presence of extensions. For further information, see the definition of modifier extensions.
Alternate Names | extensions; user content; modifiers
Summary | true
Comments | There can be no stigma associated with the use of extensions by any application, project, or standard - regardless of the institution or jurisdiction that uses or defines the extensions. The use of extensions is what allows the FHIR specification to retain a core level of simplicity for everyone.

2.30 Element Definition

URL: https://hl7.org/fhir/R4/elementdefinition.html
FHIR Infrastructure Work Group | Maturity Level: Normative | Standards Status: Normative

| This page has been approved as part of an ANSI standard. See the Infrastructure Package for further details.

The definition of an element in a resource or an extension. The definition includes:

- Path (name), Cardinality, and data type
- Definitions, usage notes, and requirements
- Default or fixed values
- Constraints, Length limits, and other usage rules
- Terminology Binding
- Mappings to other specifications
- Structural Usage Information such as Slicing

The ElementDefinition type is the core of the FHIR metadata layer, and is closely (conceptually) aligned to ISO 11179. All the data elements defined in this specification are published as a collection of data elements (XML or JSON).

ElementDefinition is used in StructureDefinition

2.30.0.1 Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
ElementDefinition | ΣIN | | BackBoneElement | Definition of an element in a resource or extension + Rule: Min <= Max + Rule: if the element definition has a contentReference, it cannot have type, defaultValue, fixed, pattern, example, minValue, maxValue, maxLength, or binding + Rule: Fixed value may only be specified if there is one type + Rule: Pattern may only be specified if there is one type + Rule: Pattern and fixed are mutually exclusive + Rule: Binding can only be present for coded elements, string, and uri + Rule: Types must be unique by code + Rule: Constraints must be unique by key + Rule: default value and meaningWhenMissing are mutually exclusive + Rule: sliceName must be composed of proper tokens separated by "/" + Rule: Must have a modifier reason if isModifier = true + Rule: Element names cannot include some special characters + Warning: Element names should be simple alphanumerics with a max of 64 characters, or code generation tools may be broken + Rule: sliceIsConstraining can only appear if slicename is present Elements defined in Ancestors: id, extension, modifierExtension
path | Σ | 1..1 | string | Path of the element in the hierarchy of elements

representation | Σ | 0..* | code | xmlAttr | xmlText | typeAttr | cdaText | xhtml PropertyRepresentation (Required)
sliceName | Σ | 0..1 | string | Name for this particular element (in a set of slices)
sliceIsConstraining | ΣTU | 0..1 | boolean | If this slice definition constrains an inherited slice definition (or not)
label | Σ | 0..1 | string | Name for element to display with or prompt for element
code | Σ | 0..* | Coding | Corresponding codes in terminologies LOINC Codes (Example)
slicing | ΣI | 0..1 | Element | This element is sliced - slices follow + Rule: If there are no discriminators, there must be a definition
discriminator | Σ | 0..* | Element | Element values that are used to distinguish the slices
type | Σ | 1..1 | code | value | exists | pattern | type | profile DiscriminatorType (Required)
path | Σ | 1..1 | string | Path to element value
description | ΣI | 0..1 | string | Text description of how slicing works (or not)
ordered | Σ | 0..1 | boolean | If elements must be in same order as slices
rules | Σ | 1..1 | code | closed | open | openAtEnd SlicingRules (Required)
short | Σ | 0..1 | string | Concise definition for space-constrained presentation
definition | Σ | 0..1 | markdown | Full formal definition as narrative text
comment | Σ | 0..1 | markdown | Comments about the use of this element
requirements | Σ | 0..1 | markdown | Why this resource has been created
alias | Σ | 0..* | string | Other names
min | ΣI | 0..1 | unsignedInt | Minimum Cardinality
max | ΣI | 0..1 | string | Maximum Cardinality (a number or *) + Rule: Max SHALL be a number or "*"
base | Σ | 0..1 | Element | Base definition information for tools
path | Σ | 1..1 | string | Path that identifies the base element
min | Σ | 1..1 | unsignedInt | Min cardinality of the base element
max | Σ | 1..1 | string | Max cardinality of the base element
contentReference | ΣI | 0..1 | uri | Reference to definition of content for the element
type | ΣI | 0..* | Element | Data type and Profile for this element + Rule: Aggregation may only be specified if one of the allowed types for the element is a reference + Rule: targetProfile is only allowed if the type is Reference or canonical
code | Σ | 1..1 | uri | Data type or Resource (reference to definition) FHIRDefinedType (Extensible)
profile | Σ | 0..* | canonical(StructureDefinition | ImplementationGuide) | Profiles (StructureDefinition or IG) - one must apply
targetProfile | Σ | 0..* | canonical(StructureDefinition | ImplementationGuide) | Profile (StructureDefinition or IG) on the Reference/canonical target - one must apply
aggregation | ΣI | 0..* | code | contained | referenced | bundled - how aggregated AggregationMode (Required)
versioning | Σ | 0..1 | code | either | independent | specific ReferenceVersionRules (Required)
defaultValue[x] | ΣI | 0..1 | * | Specified value if missing from instance
meaningWhenMissing | ΣI | 0..1 | markdown | Implicit meaning when this element is missing
orderMeaning | Σ | 0..1 | string | What the order of the elements means
fixed[x] | ΣI | 0..1 | * | Value must be exactly this
pattern[x] | ΣI | 0..1 | * | Value must have at least these property values
example | Σ | 0..* | Element | Example value (as defined for type)
label | Σ | 1..1 | string | Describes the purpose of this example
value[x] | Σ | 1..1 | * | Value of Example (one of allowed types)
minValue[x] | Σ | 0..1 | | Minimum Allowed Value (for some types)
minValueDate | | | date
minValueDateTime | | | dateTime
minValueInstant | | | instant
minValueTime | | | time
minValueDecimal | | | decimal
minValueInteger | | | integer
minValuePositiveInt | | | positiveInt
minValueUnsignedInt | | | unsignedInt
minValueQuantity | | | Quantity
maxValue[x] | Σ | 0..1 | | Maximum Allowed Value (for some types)
maxValueDate | | | date
maxValueDateTime | | | dateTime
maxValueInstant | | | instant
maxValueTime | | | time
maxValueDecimal | | | decimal
maxValueInteger | | | integer
maxValuePositiveInt | | | positiveInt
maxValueUnsignedInt | | | unsignedInt
maxValueQuantity | | | Quantity
maxLength | Σ | 0..1 | integer | Max length for strings
condition | Σ | 0..* | id | Reference to invariant about presence
constraint | ΣI | 0..* | Element | Condition that must evaluate to true + Warning: Constraints should have an expression or else validators will not be able to enforce them
key | ΣI | 1..1 | id | Target of 'condition' reference above
requirements | Σ | 0..1 | string | Why this constraint is necessary or appropriate
severity | Σ | 1..1 | code | error | warning ConstraintSeverity (Required)
human | Σ | 1..1 | string | Human description of constraint
expression | Σ | 0..1 | string | FHIRPath expression of constraint
xpath | ΣTU | 0..1 | string | XPath expression of constraint
source | Σ | 0..1 | canonical(StructureDefinition) | Reference to original source of constraint
mustSupport | Σ | 0..1 | boolean | If the element must be supported

isModifier | Σ | 0..1 | boolean | If this modifies the meaning of other elements
isModifierReason | Σ | 0..1 | string | Reason that this element is marked as a modifier
isSummary | Σ | 0..1 | boolean | Include when _summary = true?
binding | ΣI | 0..1 | Element | ValueSet details if this is coded + Rule: ValueSet SHALL start with http:// or https:// or urn:
strength | Σ | 1..1 | code | required | extensible | preferred | example BindingStrength (Required)
description | Σ | 0..1 | string | Human explanation of the value set
valueSet | ΣI | 0..1 | canonical(ValueSet) | Source of value set
mapping | Σ | 0..* | Element | Map element to another set of definitions
identity | Σ | 1..1 | id | Reference to mapping declaration
language | Σ | 0..1 | code | Computable language of mapping MimeType (Required)
map | Σ | 1..1 | string | Details of the mapping
comment | Σ | 0..1 | string | Comments about the mapping or its use
Documentation for this format

UML Diagram (Legend)

XML Template

```
<ElementDefinition xmlns="http://hl7.org/fhir">
 <!-- from BackboneElement: extension, modifierExtension -->
 <path value="[string]"/><!-- 1..1 Path of the element in the hierarchy of elements -->
 <representation value="[code]"/><!-- 0..* xmlAttr | xmlText | typeAttr | cdaText | xhtml -->
 <sliceName value="[string]"/><!-- 0..1 Name for this particular element (in a set of slices) -->
 <sliceIsConstraining value="[boolean]"/><!-- 0..1 If this slice definition constrains an inherited slice definition (or
     not) -->
 <label value="[string]"/><!-- 0..1 Name for element to display with or prompt for element -->
 <code><!-- 0..* Coding Corresponding codes in terminologies --></code>
 <slicing>  <!-- 0..1 This element is sliced - slices follow -->
  <discriminator>  <!-- 0..* Element values that are used to distinguish the slices -->
   <type value="[code]"/><!-- 1..1 value | exists | pattern | type | profile -->
   <path value="[string]"/><!-- 1..1 Path to element value -->
  </discriminator>
  <description value="[string]"/><!--  0..1 Text description of how slicing works (or not) -->
  <ordered value="[boolean]"/><!-- 0..1 If elements must be in same order as slices -->
  <rules value="[code]"/><!-- 1..1 closed | open | openAtEnd -->
 </slicing>
 <short value="[string]"/><!-- 0..1 Concise definition for space-constrained presentation -->
 <definition value="[markdown]"/><!-- 0..1 Full formal definition as narrative text -->
 <comment value="[markdown]"/><!-- 0..1 Comments about the use of this element -->
 <requirements value="[markdown]"/><!-- 0..1 Why this resource has been created -->
 <alias value="[string]"/><!-- 0..* Other names -->
 <min value="[unsignedInt]"/><!--  0..1 Minimum Cardinality -->
 <max value="[string]"/><!--  0..1 Maximum Cardinality (a number or *) -->
 <base>  <!-- 0..1 Base definition information for tools -->
  <path value="[string]"/><!-- 1..1 Path that identifies the base element -->
  <min value="[unsignedInt]"/><!-- 1..1 Min cardinality of the base element -->
  <max value="[string]"/><!-- 1..1 Max cardinality of the base element -->
 </base>
 <contentReference value="[uri]"/><!--  0..1 Reference to definition of content for the element -->
 <type>  <!--  0..* Data type and Profile for this element -->
  <code value="[uri]"/><!-- 1..1 Data type or Resource (reference to definition) -->
  <profile><!-- 0..* canonical(StructureDefinition|ImplementationGuide) Profiles (StructureDefinition or IG) - one must
    apply --></profile>
  <targetProfile><!-- 0..* canonical(StructureDefinition|ImplementationGuide) Profile (StructureDefinition or IG) on the
     Reference/canonical target - one must apply --></targetProfile>
  <aggregation value="[code]"/><!--  0..* contained | referenced | bundled - how aggregated -->
  <versioning value="[code]"/><!-- 0..1 either | independent | specific -->
 </type>
 <defaultValue[x]><!--  0..1 * Specified value if missing from instance --></defaultValue[x]>
 <meaningWhenMissing value="[markdown]"/><!--  0..1 Implicit meaning when this element is missing -->
 <orderMeaning value="[string]"/><!-- 0..1 What the order of the elements means -->
 <fixed[x]><!--  0..1 * Value must be exactly this --></fixed[x]>
 <pattern[x]><!--  0..1 * Value must have at least these property values --></pattern[x]>
 <example>  <!-- 0..* Example value (as defined for type) -->
  <label value="[string]"/><!-- 1..1 Describes the purpose of this example -->
  <value[x]><!-- 1..1 * Value of Example (one of allowed types) --></value[x]>
 </example>
 <minValue[x]><!-- 0..1 date|dateTime|instant|time|decimal|integer|positiveInt|
   unsignedInt|Quantity Minimum Allowed Value (for some types) --></minValue[x]>
 <maxValue[x]><!-- 0..1 date|dateTime|instant|time|decimal|integer|positiveInt|
   unsignedInt|Quantity Maximum Allowed Value (for some types) --></maxValue[x]>
 <maxLength value="[integer]"/><!-- 0..1 Max length for strings -->
 <condition value="[id]"/><!-- 0..* Reference to invariant about presence -->
 <constraint>  <!-- 0..* Condition that must evaluate to true -->
  <key value="[id]"/><!--  1..1 Target of 'condition' reference above -->
  <requirements value="[string]"/><!-- 0..1 Why this constraint is necessary or appropriate -->
  <severity value="[code]"/><!-- 1..1 error | warning -->
  <human value="[string]"/><!-- 1..1 Human description of constraint -->
  <expression value="[string]"/><!-- 0..1 FHIRPath expression of constraint -->
  <xpath value="[string]"/><!-- 0..1 XPath expression of constraint -->
  <source><!-- 0..1 canonical(StructureDefinition) Reference to original source of constraint --></source>
 </constraint>
 <mustSupport value="[boolean]"/><!-- 0..1 If the element must be supported -->
 <isModifier value="[boolean]"/><!-- 0..1 If this modifies the meaning of other elements -->
 <isModifierReason value="[string]"/><!-- 0..1 Reason that this element is marked as a modifier -->
 <isSummary value="[boolean]"/><!-- 0..1 Include when _summary = true? -->
 <binding>  <!--  0..1 ValueSet details if this is coded -->
```

```
  <strength value="[code]"/><!-- 1..1 required | extensible | preferred | example -->
  <description value="[string]"/><!-- 0..1 Human explanation of the value set -->
  <valueSet><!--  0..1 canonical(ValueSet) Source of value set --></valueSet>
 </binding>
 <mapping>  <!-- 0..* Map element to another set of definitions -->
  <identity value="[id]"/><!-- 1..1 Reference to mapping declaration -->
  <language value="[code]"/><!-- 0..1 Computable language of mapping -->
  <map value="[string]"/><!-- 1..1 Details of the mapping -->
  <comment value="[string]"/><!-- 0..1 Comments about the mapping or its use -->
 </mapping>
</ElementDefinition>
```

JSON Template

```
{
  // from BackboneElement: extension, modifierExtension
  "path" : "<string>", // R!  Path of the element in the hierarchy of elements
  "representation" : ["<code>"], // xmlAttr | xmlText | typeAttr | cdaText | xhtml
  "sliceName" : "<string>", // Name for this particular element (in a set of slices)
  "sliceIsConstraining" : <boolean>, // If this slice definition constrains an inherited slice definition (or not)
  "label" : "<string>", // Name for element to display with or prompt for element
  "code" : [{ Coding }], // Corresponding codes in terminologies
  "slicing" : { // This element is sliced - slices follow
    "discriminator" : [{ // Element values that are used to distinguish the slices
      "type" : "<code>", // R!  value | exists | pattern | type | profile
      "path" : "<string>" // R!  Path to element value
    }],
    "description" : "<string>", // C? Text description of how slicing works (or not)
    "ordered" : <boolean>, // If elements must be in same order as slices
    "rules" : "<code>" // R!  closed | open | openAtEnd
  },
  "short" : "<string>", // Concise definition for space-constrained presentation
  "definition" : "<markdown>", // Full formal definition as narrative text
  "comment" : "<markdown>", // Comments about the use of this element
  "requirements" : "<markdown>", // Why this resource has been created
  "alias" : ["<string>"], // Other names
  "min" : "<unsignedInt>", // C? Minimum Cardinality
  "max" : "<string>", // C? Maximum Cardinality (a number or *)
  "base" : { // Base definition information for tools
    "path" : "<string>", // R!  Path that identifies the base element
    "min" : "<unsignedInt>", // R!  Min cardinality of the base element
    "max" : "<string>" // R!  Max cardinality of the base element
  },
  "contentReference" : "<uri>", // C? Reference to definition of content for the element
  "type" : [{ // C? Data type and Profile for this element
    "code" : "<uri>", // R!  Data type or Resource (reference to definition)
    "profile" : [{ canonical(StructureDefinition|ImplementationGuide) }], // Profiles (StructureDefinition or IG) - one
    must apply
    "targetProfile" : [{ canonical(StructureDefinition|ImplementationGuide) }], // Profile (StructureDefinition or IG)
    on the Reference/canonical target - one must apply
    "aggregation" : ["<code>"], // C? contained | referenced | bundled - how aggregated
    "versioning" : "<code>" // either | independent | specific
  }],
  // defaultValue[x]: Specified value if missing from instance. One of these 50:
  "defaultValueBase64Binary" : "<base64Binary>",
  "defaultValueBoolean" : <boolean>,
  "defaultValueCanonical" : "<canonical>",
  "defaultValueCode" : "<code>",
  "defaultValueDate" : "<date>",
  "defaultValueDateTime" : "<dateTime>",
  "defaultValueDecimal" : <decimal>,
  "defaultValueId" : "<id>",
  "defaultValueInstant" : "<instant>",
  "defaultValueInteger" : <integer>,
  "defaultValueMarkdown" : "<markdown>",
  "defaultValueOid" : "<oid>",
  "defaultValuePositiveInt" : "<positiveInt>",
  "defaultValueString" : "<string>",
  "defaultValueTime" : "<time>",
  "defaultValueUnsignedInt" : "<unsignedInt>",
  "defaultValueUri" : "<uri>",
  "defaultValueUrl" : "<url>",
  "defaultValueUuid" : "<uuid>",
  "defaultValueAddress" : { Address },
  "defaultValueAge" : { Age },
  "defaultValueAnnotation" : { Annotation },
  "defaultValueAttachment" : { Attachment },
  "defaultValueCodeableConcept" : { CodeableConcept },
  "defaultValueCoding" : { Coding },
  "defaultValueContactPoint" : { ContactPoint },
  "defaultValueCount" : { Count },
  "defaultValueDistance" : { Distance },
  "defaultValueDuration" : { Duration },
  "defaultValueHumanName" : { HumanName },
  "defaultValueIdentifier" : { Identifier },
  "defaultValueMoney" : { Money },
  "defaultValuePeriod" : { Period },
  "defaultValueQuantity" : { Quantity },
  "defaultValueRange" : { Range },
  "defaultValueRatio" : { Ratio },
```

```
"defaultValueReference" : { Reference },
"defaultValueSampledData" : { SampledData },
"defaultValueSignature" : { Signature },
"defaultValueTiming" : { Timing },
"defaultValueContactDetail" : { ContactDetail },
"defaultValueContributor" : { Contributor },
"defaultValueDataRequirement" : { DataRequirement },
"defaultValueExpression" : { Expression },
"defaultValueParameterDefinition" : { ParameterDefinition },
"defaultValueRelatedArtifact" : { RelatedArtifact },
"defaultValueTriggerDefinition" : { TriggerDefinition },
"defaultValueUsageContext" : { UsageContext },
"defaultValueDosage" : { Dosage },
"defaultValueMeta" : { Meta },
"meaningWhenMissing" : "<markdown>", // C? Implicit meaning when this element is missing
"orderMeaning" : "<string>", // What the order of the elements means
// fixed[x]: Value must be exactly this. One of these 50:
"fixedBase64Binary" : "<base64Binary>",
"fixedBoolean" : <boolean>,
"fixedCanonical" : "<canonical>",
"fixedCode" : "<code>",
"fixedDate" : "<date>",
"fixedDateTime" : "<dateTime>",
"fixedDecimal" : <decimal>,
"fixedId" : "<id>",
"fixedInstant" : "<instant>",
"fixedInteger" : <integer>,
"fixedMarkdown" : "<markdown>",
"fixedOid" : "<oid>",
"fixedPositiveInt" : "<positiveInt>",
"fixedString" : "<string>",
"fixedTime" : "<time>",
"fixedUnsignedInt" : "<unsignedInt>",
"fixedUri" : "<uri>",
"fixedUrl" : "<url>",
"fixedUuid" : "<uuid>",
"fixedAddress" : { Address },
"fixedAge" : { Age },
"fixedAnnotation" : { Annotation },
"fixedAttachment" : { Attachment },
"fixedCodeableConcept" : { CodeableConcept },
"fixedCoding" : { Coding },
"fixedContactPoint" : { ContactPoint },
"fixedCount" : { Count },
"fixedDistance" : { Distance },
"fixedDuration" : { Duration },
"fixedHumanName" : { HumanName },
"fixedIdentifier" : { Identifier },
"fixedMoney" : { Money },
"fixedPeriod" : { Period },
"fixedQuantity" : { Quantity },
"fixedRange" : { Range },
"fixedRatio" : { Ratio },
"fixedReference" : { Reference },
"fixedSampledData" : { SampledData },
"fixedSignature" : { Signature },
"fixedTiming" : { Timing },
"fixedContactDetail" : { ContactDetail },
"fixedContributor" : { Contributor },
"fixedDataRequirement" : { DataRequirement },
"fixedExpression" : { Expression },
"fixedParameterDefinition" : { ParameterDefinition },
"fixedRelatedArtifact" : { RelatedArtifact },
"fixedTriggerDefinition" : { TriggerDefinition },
"fixedUsageContext" : { UsageContext },
"fixedDosage" : { Dosage },
"fixedMeta" : { Meta },
// pattern[x]: Value must have at least these property values. One of these 50:
"patternBase64Binary" : "<base64Binary>",
"patternBoolean" : <boolean>,
"patternCanonical" : "<canonical>",
"patternCode" : "<code>",
"patternDate" : "<date>",
"patternDateTime" : "<dateTime>",
"patternDecimal" : <decimal>,
"patternId" : "<id>",
"patternInstant" : "<instant>",
"patternInteger" : <integer>,
"patternMarkdown" : "<markdown>",
"patternOid" : "<oid>",
"patternPositiveInt" : "<positiveInt>",
"patternString" : "<string>",
"patternTime" : "<time>",
"patternUnsignedInt" : "<unsignedInt>",
"patternUri" : "<uri>",
"patternUrl" : "<url>",
"patternUuid" : "<uuid>",
"patternAddress" : { Address },
"patternAge" : { Age },
```

```
"patternAnnotation" : { Annotation },
"patternAttachment" : { Attachment },
"patternCodeableConcept" : { CodeableConcept },
"patternCoding" : { Coding },
"patternContactPoint" : { ContactPoint },
"patternCount" : { Count },
"patternDistance" : { Distance },
"patternDuration" : { Duration },
"patternHumanName" : { HumanName },
"patternIdentifier" : { Identifier },
"patternMoney" : { Money },
"patternPeriod" : { Period },
"patternQuantity" : { Quantity },
"patternRange" : { Range },
"patternRatio" : { Ratio },
"patternReference" : { Reference },
"patternSampledData" : { SampledData },
"patternSignature" : { Signature },
"patternTiming" : { Timing },
"patternContactDetail" : { ContactDetail },
"patternContributor" : { Contributor },
"patternDataRequirement" : { DataRequirement },
"patternExpression" : { Expression },
"patternParameterDefinition" : { ParameterDefinition },
"patternRelatedArtifact" : { RelatedArtifact },
"patternTriggerDefinition" : { TriggerDefinition },
"patternUsageContext" : { UsageContext },
"patternDosage" : { Dosage },
"patternMeta" : { Meta },
"example" : [{ // Example value (as defined for type)
  "label" : "<string>", // R!  Describes the purpose of this example
  // value[x]: Value of Example (one of allowed types). One of these 50:
  "valueBase64Binary" : "<base64Binary>"
  "valueBoolean" : <boolean>
  "valueCanonical" : "<canonical>"
  "valueCode" : "<code>"
  "valueDate" : "<date>"
  "valueDateTime" : "<dateTime>"
  "valueDecimal" : <decimal>
  "valueId" : "<id>"
  "valueInstant" : "<instant>"
  "valueInteger" : <integer>
  "valueMarkdown" : "<markdown>"
  "valueOid" : "<oid>"
  "valuePositiveInt" : "<positiveInt>"
  "valueString" : "<string>"
  "valueTime" : "<time>"
  "valueUnsignedInt" : "<unsignedInt>"
  "valueUri" : "<uri>"
  "valueUrl" : "<url>"
  "valueUuid" : "<uuid>"
  "valueAddress" : { Address }
  "valueAge" : { Age }
  "valueAnnotation" : { Annotation }
  "valueAttachment" : { Attachment }
  "valueCodeableConcept" : { CodeableConcept }
  "valueCoding" : { Coding }
  "valueContactPoint" : { ContactPoint }
  "valueCount" : { Count }
  "valueDistance" : { Distance }
  "valueDuration" : { Duration }
  "valueHumanName" : { HumanName }
  "valueIdentifier" : { Identifier }
  "valueMoney" : { Money }
  "valuePeriod" : { Period }
  "valueQuantity" : { Quantity }
  "valueRange" : { Range }
  "valueRatio" : { Ratio }
  "valueReference" : { Reference }
  "valueSampledData" : { SampledData }
  "valueSignature" : { Signature }
  "valueTiming" : { Timing }
  "valueContactDetail" : { ContactDetail }
  "valueContributor" : { Contributor }
  "valueDataRequirement" : { DataRequirement }
  "valueExpression" : { Expression }
  "valueParameterDefinition" : { ParameterDefinition }
  "valueRelatedArtifact" : { RelatedArtifact }
  "valueTriggerDefinition" : { TriggerDefinition }
  "valueUsageContext" : { UsageContext }
  "valueDosage" : { Dosage }
  "valueMeta" : { Meta }
}],
// minValue[x]: Minimum Allowed Value (for some types). One of these 9:
"minValueDate" : "<date>",
"minValueDateTime" : "<dateTime>",
"minValueInstant" : "<instant>",
"minValueTime" : "<time>",
"minValueDecimal" : <decimal>,
```

```
  "minValueInteger" : <integer>,
  "minValuePositiveInt" : "<positiveInt>",
  "minValueUnsignedInt" : "<unsignedInt>",
  "minValueQuantity" : { Quantity },
  // maxValue[x]: Maximum Allowed Value (for some types). One of these 9:
  "maxValueDate" : "<date>",
  "maxValueDateTime" : "<dateTime>",
  "maxValueInstant" : "<instant>",
  "maxValueTime" : "<time>",
  "maxValueDecimal" : <decimal>,
  "maxValueInteger" : <integer>,
  "maxValuePositiveInt" : "<positiveInt>",
  "maxValueUnsignedInt" : "<unsignedInt>",
  "maxValueQuantity" : { Quantity },
  "maxLength" : <integer>, // Max length for strings
  "condition" : ["<id>"], // Reference to invariant about presence
  "constraint" : [{ // Condition that must evaluate to true
    "key" : "<id>", // C? R!  Target of 'condition' reference above
    "requirements" : "<string>", // Why this constraint is necessary or appropriate
    "severity" : "<code>", // R!  error | warning
    "human" : "<string>", // R!  Human description of constraint
    "expression" : "<string>", // FHIRPath expression of constraint
    "xpath" : "<string>", // XPath expression of constraint
    "source" : { canonical(StructureDefinition) } // Reference to original source of constraint
  }],
  "mustSupport" : <boolean>, // If the element must be supported
  "isModifier" : <boolean>, // If this modifies the meaning of other elements
  "isModifierReason" : "<string>", // Reason that this element is marked as a modifier
  "isSummary" : <boolean>, // Include when _summary = true?
  "binding" : { // C? ValueSet details if this is coded
    "strength" : "<code>", // R!  required | extensible | preferred | example
    "description" : "<string>", // Human explanation of the value set
    "valueSet" : { canonical(ValueSet) } // C? Source of value set
  },
  "mapping" : [{ // Map element to another set of definitions
    "identity" : "<id>", // R!  Reference to mapping declaration
    "language" : "<code>", // Computable language of mapping
    "map" : "<string>", // R!  Details of the mapping
    "comment" : "<string>" // Comments about the mapping or its use
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from BackboneElement: Element.extension, BackboneElement.modifierextension
  fhir:ElementDefinition.path [ string ]; # 1..1 Path of the element in the hierarchy of elements
  fhir:ElementDefinition.representation [ code ], ... ; # 0..* xmlAttr | xmlText | typeAttr | cdaText | xhtml
  fhir:ElementDefinition.sliceName [ string ]; # 0..1 Name for this particular element (in a set of slices)
  fhir:ElementDefinition.sliceIsConstraining [ boolean ]; # 0..1 If this slice definition constrains an inherited slice
    definition (or not)
  fhir:ElementDefinition.label [ string ]; # 0..1 Name for element to display with or prompt for element
  fhir:ElementDefinition.code [ Coding ], ... ; # 0..* Corresponding codes in terminologies
  fhir:ElementDefinition.slicing [ # 0..1 This element is sliced - slices follow
    fhir:ElementDefinition.slicing.discriminator [ # 0..* Element values that are used to distinguish the slices
      fhir:ElementDefinition.slicing.discriminator.type [ code ]; # 1..1 value | exists | pattern | type | profile
      fhir:ElementDefinition.slicing.discriminator.path [ string ]; # 1..1 Path to element value
    ], ...;
    fhir:ElementDefinition.slicing.description [ string ]; # 0..1 Text description of how slicing works (or not)
    fhir:ElementDefinition.slicing.ordered [ boolean ]; # 0..1 If elements must be in same order as slices
    fhir:ElementDefinition.slicing.rules [ code ]; # 1..1 closed | open | openAtEnd
  ];
  fhir:ElementDefinition.short [ string ]; # 0..1 Concise definition for space-constrained presentation
  fhir:ElementDefinition.definition [ markdown ]; # 0..1 Full formal definition as narrative text
  fhir:ElementDefinition.comment [ markdown ]; # 0..1 Comments about the use of this element
  fhir:ElementDefinition.requirements [ markdown ]; # 0..1 Why this resource has been created
  fhir:ElementDefinition.alias [ string ], ... ; # 0..* Other names
  fhir:ElementDefinition.min [ unsignedInt ]; # 0..1 Minimum Cardinality
  fhir:ElementDefinition.max [ string ]; # 0..1 Maximum Cardinality (a number or *)
  fhir:ElementDefinition.base [ # 0..1 Base definition information for tools
    fhir:ElementDefinition.base.path [ string ]; # 1..1 Path that identifies the base element
    fhir:ElementDefinition.base.min [ unsignedInt ]; # 1..1 Min cardinality of the base element
    fhir:ElementDefinition.base.max [ string ]; # 1..1 Max cardinality of the base element
  ];
  fhir:ElementDefinition.contentReference [ uri ]; # 0..1 Reference to definition of content for the element
  fhir:ElementDefinition.type [ # 0..* Data type and Profile for this element
    fhir:ElementDefinition.type.code [ uri ]; # 1..1 Data type or Resource (reference to definition)
    fhir:ElementDefinition.type.profile [ canonical(StructureDefinition|ImplementationGuide) ], ... ; # 0..* Profiles
    (StructureDefinition or IG) - one must apply
    fhir:ElementDefinition.type.targetProfile [ canonical(StructureDefinition|ImplementationGuide) ], ... ; # 0..*
    Profile (StructureDefinition or IG) on the Reference/canonical target - one must apply
    fhir:ElementDefinition.type.aggregation [ code ], ... ; # 0..* contained | referenced | bundled - how aggregated
    fhir:ElementDefinition.type.versioning [ code ]; # 0..1 either | independent | specific
  ], ...;
  # ElementDefinition.defaultValue[x] : 0..1 Specified value if missing from instance. One of these 50
    fhir:ElementDefinition.defaultValueBase64Binary [ base64Binary ]
    fhir:ElementDefinition.defaultValueBoolean [ boolean ]
    fhir:ElementDefinition.defaultValueCanonical [ canonical ]
```

```
  fhir:ElementDefinition.defaultValueCode [ code ]
  fhir:ElementDefinition.defaultValueDate [ date ]
  fhir:ElementDefinition.defaultValueDateTime [ dateTime ]
  fhir:ElementDefinition.defaultValueDecimal [ decimal ]
  fhir:ElementDefinition.defaultValueId [ id ]
  fhir:ElementDefinition.defaultValueInstant [ instant ]
  fhir:ElementDefinition.defaultValueInteger [ integer ]
  fhir:ElementDefinition.defaultValueMarkdown [ markdown ]
  fhir:ElementDefinition.defaultValueOid [ oid ]
  fhir:ElementDefinition.defaultValuePositiveInt [ positiveInt ]
  fhir:ElementDefinition.defaultValueString [ string ]
  fhir:ElementDefinition.defaultValueTime [ time ]
  fhir:ElementDefinition.defaultValueUnsignedInt [ unsignedInt ]
  fhir:ElementDefinition.defaultValueUri [ uri ]
  fhir:ElementDefinition.defaultValueUrl [ url ]
  fhir:ElementDefinition.defaultValueUuid [ uuid ]
  fhir:ElementDefinition.defaultValueAddress [ Address ]
  fhir:ElementDefinition.defaultValueAge [ Age ]
  fhir:ElementDefinition.defaultValueAnnotation [ Annotation ]
  fhir:ElementDefinition.defaultValueAttachment [ Attachment ]
  fhir:ElementDefinition.defaultValueCodeableConcept [ CodeableConcept ]
  fhir:ElementDefinition.defaultValueCoding [ Coding ]
  fhir:ElementDefinition.defaultValueContactPoint [ ContactPoint ]
  fhir:ElementDefinition.defaultValueCount [ Count ]
  fhir:ElementDefinition.defaultValueDistance [ Distance ]
  fhir:ElementDefinition.defaultValueDuration [ Duration ]
  fhir:ElementDefinition.defaultValueHumanName [ HumanName ]
  fhir:ElementDefinition.defaultValueIdentifier [ Identifier ]
  fhir:ElementDefinition.defaultValueMoney [ Money ]
  fhir:ElementDefinition.defaultValuePeriod [ Period ]
  fhir:ElementDefinition.defaultValueQuantity [ Quantity ]
  fhir:ElementDefinition.defaultValueRange [ Range ]
  fhir:ElementDefinition.defaultValueRatio [ Ratio ]
  fhir:ElementDefinition.defaultValueReference [ Reference ]
  fhir:ElementDefinition.defaultValueSampledData [ SampledData ]
  fhir:ElementDefinition.defaultValueSignature [ Signature ]
  fhir:ElementDefinition.defaultValueTiming [ Timing ]
  fhir:ElementDefinition.defaultValueContactDetail [ ContactDetail ]
  fhir:ElementDefinition.defaultValueContributor [ Contributor ]
  fhir:ElementDefinition.defaultValueDataRequirement [ DataRequirement ]
  fhir:ElementDefinition.defaultValueExpression [ Expression ]
  fhir:ElementDefinition.defaultValueParameterDefinition [ ParameterDefinition ]
  fhir:ElementDefinition.defaultValueRelatedArtifact [ RelatedArtifact ]
  fhir:ElementDefinition.defaultValueTriggerDefinition [ TriggerDefinition ]
  fhir:ElementDefinition.defaultValueUsageContext [ UsageContext ]
  fhir:ElementDefinition.defaultValueDosage [ Dosage ]
  fhir:ElementDefinition.defaultValueMeta [ Meta ]
fhir:ElementDefinition.meaningWhenMissing [ markdown ]; # 0..1 Implicit meaning when this element is missing
fhir:ElementDefinition.orderMeaning [ string ]; # 0..1 What the order of the elements means
# ElementDefinition.fixed[x] : 0..1 Value must be exactly this. One of these 50
  fhir:ElementDefinition.fixedBase64Binary [ base64Binary ]
  fhir:ElementDefinition.fixedBoolean [ boolean ]
  fhir:ElementDefinition.fixedCanonical [ canonical ]
  fhir:ElementDefinition.fixedCode [ code ]
  fhir:ElementDefinition.fixedDate [ date ]
  fhir:ElementDefinition.fixedDateTime [ dateTime ]
  fhir:ElementDefinition.fixedDecimal [ decimal ]
  fhir:ElementDefinition.fixedId [ id ]
  fhir:ElementDefinition.fixedInstant [ instant ]
  fhir:ElementDefinition.fixedInteger [ integer ]
  fhir:ElementDefinition.fixedMarkdown [ markdown ]
  fhir:ElementDefinition.fixedOid [ oid ]
  fhir:ElementDefinition.fixedPositiveInt [ positiveInt ]
  fhir:ElementDefinition.fixedString [ string ]
  fhir:ElementDefinition.fixedTime [ time ]
  fhir:ElementDefinition.fixedUnsignedInt [ unsignedInt ]
  fhir:ElementDefinition.fixedUri [ uri ]
  fhir:ElementDefinition.fixedUrl [ url ]
  fhir:ElementDefinition.fixedUuid [ uuid ]
  fhir:ElementDefinition.fixedAddress [ Address ]
  fhir:ElementDefinition.fixedAge [ Age ]
  fhir:ElementDefinition.fixedAnnotation [ Annotation ]
  fhir:ElementDefinition.fixedAttachment [ Attachment ]
  fhir:ElementDefinition.fixedCodeableConcept [ CodeableConcept ]
  fhir:ElementDefinition.fixedCoding [ Coding ]
  fhir:ElementDefinition.fixedContactPoint [ ContactPoint ]
  fhir:ElementDefinition.fixedCount [ Count ]
  fhir:ElementDefinition.fixedDistance [ Distance ]
  fhir:ElementDefinition.fixedDuration [ Duration ]
  fhir:ElementDefinition.fixedHumanName [ HumanName ]
  fhir:ElementDefinition.fixedIdentifier [ Identifier ]
  fhir:ElementDefinition.fixedMoney [ Money ]
  fhir:ElementDefinition.fixedPeriod [ Period ]
  fhir:ElementDefinition.fixedQuantity [ Quantity ]
  fhir:ElementDefinition.fixedRange [ Range ]
  fhir:ElementDefinition.fixedRatio [ Ratio ]
  fhir:ElementDefinition.fixedReference [ Reference ]
  fhir:ElementDefinition.fixedSampledData [ SampledData ]
  fhir:ElementDefinition.fixedSignature [ Signature ]
```

```
    fhir:ElementDefinition.fixedTiming [ Timing ]
    fhir:ElementDefinition.fixedContactDetail [ ContactDetail ]
    fhir:ElementDefinition.fixedContributor [ Contributor ]
    fhir:ElementDefinition.fixedDataRequirement [ DataRequirement ]
    fhir:ElementDefinition.fixedExpression [ Expression ]
    fhir:ElementDefinition.fixedParameterDefinition [ ParameterDefinition ]
    fhir:ElementDefinition.fixedRelatedArtifact [ RelatedArtifact ]
    fhir:ElementDefinition.fixedTriggerDefinition [ TriggerDefinition ]
    fhir:ElementDefinition.fixedUsageContext [ UsageContext ]
    fhir:ElementDefinition.fixedDosage [ Dosage ]
    fhir:ElementDefinition.fixedMeta [ Meta ]
  # ElementDefinition.pattern[x] : 0..1 Value must have at least these property values. One of these 50
    fhir:ElementDefinition.patternBase64Binary [ base64Binary ]
    fhir:ElementDefinition.patternBoolean [ boolean ]
    fhir:ElementDefinition.patternCanonical [ canonical ]
    fhir:ElementDefinition.patternCode [ code ]
    fhir:ElementDefinition.patternDate [ date ]
    fhir:ElementDefinition.patternDateTime [ dateTime ]
    fhir:ElementDefinition.patternDecimal [ decimal ]
    fhir:ElementDefinition.patternId [ id ]
    fhir:ElementDefinition.patternInstant [ instant ]
    fhir:ElementDefinition.patternInteger [ integer ]
    fhir:ElementDefinition.patternMarkdown [ markdown ]
    fhir:ElementDefinition.patternOid [ oid ]
    fhir:ElementDefinition.patternPositiveInt [ positiveInt ]
    fhir:ElementDefinition.patternString [ string ]
    fhir:ElementDefinition.patternTime [ time ]
    fhir:ElementDefinition.patternUnsignedInt [ unsignedInt ]
    fhir:ElementDefinition.patternUri [ uri ]
    fhir:ElementDefinition.patternUrl [ url ]
    fhir:ElementDefinition.patternUuid [ uuid ]
    fhir:ElementDefinition.patternAddress [ Address ]
    fhir:ElementDefinition.patternAge [ Age ]
    fhir:ElementDefinition.patternAnnotation [ Annotation ]
    fhir:ElementDefinition.patternAttachment [ Attachment ]
    fhir:ElementDefinition.patternCodeableConcept [ CodeableConcept ]
    fhir:ElementDefinition.patternCoding [ Coding ]
    fhir:ElementDefinition.patternContactPoint [ ContactPoint ]
    fhir:ElementDefinition.patternCount [ Count ]
    fhir:ElementDefinition.patternDistance [ Distance ]
    fhir:ElementDefinition.patternDuration [ Duration ]
    fhir:ElementDefinition.patternHumanName [ HumanName ]
    fhir:ElementDefinition.patternIdentifier [ Identifier ]
    fhir:ElementDefinition.patternMoney [ Money ]
    fhir:ElementDefinition.patternPeriod [ Period ]
    fhir:ElementDefinition.patternQuantity [ Quantity ]
    fhir:ElementDefinition.patternRange [ Range ]
    fhir:ElementDefinition.patternRatio [ Ratio ]
    fhir:ElementDefinition.patternReference [ Reference ]
    fhir:ElementDefinition.patternSampledData [ SampledData ]
    fhir:ElementDefinition.patternSignature [ Signature ]
    fhir:ElementDefinition.patternTiming [ Timing ]
    fhir:ElementDefinition.patternContactDetail [ ContactDetail ]
    fhir:ElementDefinition.patternContributor [ Contributor ]
    fhir:ElementDefinition.patternDataRequirement [ DataRequirement ]
    fhir:ElementDefinition.patternExpression [ Expression ]
    fhir:ElementDefinition.patternParameterDefinition [ ParameterDefinition ]
    fhir:ElementDefinition.patternRelatedArtifact [ RelatedArtifact ]
    fhir:ElementDefinition.patternTriggerDefinition [ TriggerDefinition ]
    fhir:ElementDefinition.patternUsageContext [ UsageContext ]
    fhir:ElementDefinition.patternDosage [ Dosage ]
    fhir:ElementDefinition.patternMeta [ Meta ]
  fhir:ElementDefinition.example [ # 0..* Example value (as defined for type)
    fhir:ElementDefinition.example.label [ string ]; # 1..1 Describes the purpose of this example
    # ElementDefinition.example.value[x] : 1..1 Value of Example (one of allowed types). One of these 50
      fhir:ElementDefinition.example.valueBase64Binary [ base64Binary ]
      fhir:ElementDefinition.example.valueBoolean [ boolean ]
      fhir:ElementDefinition.example.valueCanonical [ canonical ]
      fhir:ElementDefinition.example.valueCode [ code ]
      fhir:ElementDefinition.example.valueDate [ date ]
      fhir:ElementDefinition.example.valueDateTime [ dateTime ]
      fhir:ElementDefinition.example.valueDecimal [ decimal ]
      fhir:ElementDefinition.example.valueId [ id ]
      fhir:ElementDefinition.example.valueInstant [ instant ]
      fhir:ElementDefinition.example.valueInteger [ integer ]
      fhir:ElementDefinition.example.valueMarkdown [ markdown ]
      fhir:ElementDefinition.example.valueOid [ oid ]
      fhir:ElementDefinition.example.valuePositiveInt [ positiveInt ]
      fhir:ElementDefinition.example.valueString [ string ]
      fhir:ElementDefinition.example.valueTime [ time ]
      fhir:ElementDefinition.example.valueUnsignedInt [ unsignedInt ]
      fhir:ElementDefinition.example.valueUri [ uri ]
      fhir:ElementDefinition.example.valueUrl [ url ]
      fhir:ElementDefinition.example.valueUuid [ uuid ]
      fhir:ElementDefinition.example.valueAddress [ Address ]
      fhir:ElementDefinition.example.valueAge [ Age ]
      fhir:ElementDefinition.example.valueAnnotation [ Annotation ]
      fhir:ElementDefinition.example.valueAttachment [ Attachment ]
      fhir:ElementDefinition.example.valueCodeableConcept [ CodeableConcept ]
```

```
      fhir:ElementDefinition.example.valueCoding [ Coding ]
      fhir:ElementDefinition.example.valueContactPoint [ ContactPoint ]
      fhir:ElementDefinition.example.valueCount [ Count ]
      fhir:ElementDefinition.example.valueDistance [ Distance ]
      fhir:ElementDefinition.example.valueDuration [ Duration ]
      fhir:ElementDefinition.example.valueHumanName [ HumanName ]
      fhir:ElementDefinition.example.valueIdentifier [ Identifier ]
      fhir:ElementDefinition.example.valueMoney [ Money ]
      fhir:ElementDefinition.example.valuePeriod [ Period ]
      fhir:ElementDefinition.example.valueQuantity [ Quantity ]
      fhir:ElementDefinition.example.valueRange [ Range ]
      fhir:ElementDefinition.example.valueRatio [ Ratio ]
      fhir:ElementDefinition.example.valueReference [ Reference ]
      fhir:ElementDefinition.example.valueSampledData [ SampledData ]
      fhir:ElementDefinition.example.valueSignature [ Signature ]
      fhir:ElementDefinition.example.valueTiming [ Timing ]
      fhir:ElementDefinition.example.valueContactDetail [ ContactDetail ]
      fhir:ElementDefinition.example.valueContributor [ Contributor ]
      fhir:ElementDefinition.example.valueDataRequirement [ DataRequirement ]
      fhir:ElementDefinition.example.valueExpression [ Expression ]
      fhir:ElementDefinition.example.valueParameterDefinition [ ParameterDefinition ]
      fhir:ElementDefinition.example.valueRelatedArtifact [ RelatedArtifact ]
      fhir:ElementDefinition.example.valueTriggerDefinition [ TriggerDefinition ]
      fhir:ElementDefinition.example.valueUsageContext [ UsageContext ]
      fhir:ElementDefinition.example.valueDosage [ Dosage ]
      fhir:ElementDefinition.example.valueMeta [ Meta ]
  ], ...;
  # ElementDefinition.minValue[x] : 0..1 Minimum Allowed Value (for some types). One of these 9
    fhir:ElementDefinition.minValueDate [ date ]
    fhir:ElementDefinition.minValueDateTime [ dateTime ]
    fhir:ElementDefinition.minValueInstant [ instant ]
    fhir:ElementDefinition.minValueTime [ time ]
    fhir:ElementDefinition.minValueDecimal [ decimal ]
    fhir:ElementDefinition.minValueInteger [ integer ]
    fhir:ElementDefinition.minValuePositiveInt [ positiveInt ]
    fhir:ElementDefinition.minValueUnsignedInt [ unsignedInt ]
    fhir:ElementDefinition.minValueQuantity [ Quantity ]
  # ElementDefinition.maxValue[x] : 0..1 Maximum Allowed Value (for some types). One of these 9
    fhir:ElementDefinition.maxValueDate [ date ]
    fhir:ElementDefinition.maxValueDateTime [ dateTime ]
    fhir:ElementDefinition.maxValueInstant [ instant ]
    fhir:ElementDefinition.maxValueTime [ time ]
    fhir:ElementDefinition.maxValueDecimal [ decimal ]
    fhir:ElementDefinition.maxValueInteger [ integer ]
    fhir:ElementDefinition.maxValuePositiveInt [ positiveInt ]
    fhir:ElementDefinition.maxValueUnsignedInt [ unsignedInt ]
    fhir:ElementDefinition.maxValueQuantity [ Quantity ]
  fhir:ElementDefinition.maxLength [ integer ]; # 0..1 Max length for strings
  fhir:ElementDefinition.condition [ id ], ... ; # 0..* Reference to invariant about presence
  fhir:ElementDefinition.constraint [ # 0..* Condition that must evaluate to true
    fhir:ElementDefinition.constraint.key [ id ]; # 1..1 Target of 'condition' reference above
    fhir:ElementDefinition.constraint.requirements [ string ]; # 0..1 Why this constraint is necessary or appropriate
    fhir:ElementDefinition.constraint.severity [ code ]; # 1..1 error | warning
    fhir:ElementDefinition.constraint.human [ string ]; # 1..1 Human description of constraint
    fhir:ElementDefinition.constraint.expression [ string ]; # 0..1 FHIRPath expression of constraint
    fhir:ElementDefinition.constraint.xpath [ string ]; # 0..1 XPath expression of constraint
    fhir:ElementDefinition.constraint.source [ canonical(StructureDefinition) ]; # 0..1 Reference to original source of
    constraint
  ], ...;
  fhir:ElementDefinition.mustSupport [ boolean ]; # 0..1 If the element must be supported
  fhir:ElementDefinition.isModifier [ boolean ]; # 0..1 If this modifies the meaning of other elements
  fhir:ElementDefinition.isModifierReason [ string ]; # 0..1 Reason that this element is marked as a modifier
  fhir:ElementDefinition.isSummary [ boolean ]; # 0..1 Include when _summary = true?
  fhir:ElementDefinition.binding [ # 0..1 ValueSet details if this is coded
    fhir:ElementDefinition.binding.strength [ code ]; # 1..1 required | extensible | preferred | example
    fhir:ElementDefinition.binding.description [ string ]; # 0..1 Human explanation of the value set
    fhir:ElementDefinition.binding.valueSet [ canonical(ValueSet) ]; # 0..1 Source of value set
  ];
  fhir:ElementDefinition.mapping [ # 0..* Map element to another set of definitions
    fhir:ElementDefinition.mapping.identity [ id ]; # 1..1 Reference to mapping declaration
    fhir:ElementDefinition.mapping.language [ code ]; # 0..1 Computable language of mapping
    fhir:ElementDefinition.mapping.map [ string ]; # 1..1 Details of the mapping
    fhir:ElementDefinition.mapping.comment [ string ]; # 0..1 Comments about the mapping or its use
  ], ...;
]
```

Changes since Release 3

ElementDefinition

ElementDefinition.representation | Change value set from http://hl7.org/fhir/ValueSet/property-representation to http://hl7.org/fhir/ValueSet/property-representation|4.0.1

ElementDefinition.sliceIsConstraining | Added Element

ElementDefinition.slicing.discriminator.type | Change value set from http://hl7.org/fhir/ValueSet/discriminator-type to http://hl7.org/fhir/ValueSet/discriminator-type|4.0.1

ElementDefinition.slicing.ordered | Default Value "false" removed

ElementDefinition.slicing.rules | Change value set from http://hl7.org/fhir/ValueSet/resource-slicing-rules to http://hl7.org/fhir/ValueSet/resource-slicing-rules|4.0.1

ElementDefinition.type.profile | Max Cardinality changed from 1 to * Type changed from uri to canonical(StructureDefinition | ImplementationGuide)
ElementDefinition.type.targetProfile | Max Cardinality changed from 1 to * Type changed from uri to canonical(StructureDefinition | ImplementationGuide)
ElementDefinition.type.aggregation | Change value set from http://hl7.org/fhir/ValueSet/resource-aggregation-mode to http://hl7.org/fhir/ValueSet/resource-aggregation-mode|4.0.1
ElementDefinition.type.versioning | Change value set from http://hl7.org/fhir/ValueSet/reference-version-rules to http://hl7.org/fhir/ValueSet/reference-version-rules|4.0.1 Default Value "either" removed
ElementDefinition.defaultValue[x] | Add Types canonical, url, uuid, ContactDetail, Contributor, DataRequirement, Expression, ParameterDefinition, RelatedArtifact, TriggerDefinition, UsageContext, Dosage
ElementDefinition.fixed[x] | Add Types canonical, url, uuid, ContactDetail, Contributor, DataRequirement, Expression, ParameterDefinition, RelatedArtifact, TriggerDefinition, UsageContext, Dosage
ElementDefinition.pattern[x] | Add Types canonical, url, uuid, ContactDetail, Contributor, DataRequirement, Expression, ParameterDefinition, RelatedArtifact, TriggerDefinition, UsageContext, Dosage
ElementDefinition.example.value[x] | Add Types canonical, url, uuid, ContactDetail, Contributor, DataRequirement, Expression, ParameterDefinition, RelatedArtifact, TriggerDefinition, UsageContext, Dosage
ElementDefinition.constraint.severity | Change value set from http://hl7.org/fhir/ValueSet/constraint-severity to http://hl7.org/fhir/ValueSet/constraint-severity|4.0.1
ElementDefinition.constraint.expression | Min Cardinality changed from 1 to 0
ElementDefinition.constraint.source | Type changed from uri to canonical(StructureDefinition)
ElementDefinition.mustSupport | Default Value "false" removed
ElementDefinition.isModifier | Default Value "false" removed
ElementDefinition.isModifierReason | Added Element
ElementDefinition.isSummary | Default Value "false" removed
ElementDefinition.binding.strength | Change value set from http://hl7.org/fhir/ValueSet/binding-strength to http://hl7.org/fhir/ValueSet/binding-strength|4.0.1
ElementDefinition.binding.valueSet | Renamed from valueSet[x] to valueSet Add Type canonical(ValueSet) Remove Types uri, Reference(ValueSet)
ElementDefinition.mapping.language | Change value set from http://hl7.org/fhir/ValueSet/mimetypes to http://hl7.org/fhir/ValueSet/mimetypes|4.0.1

See the Full Difference for further information

Structure

Name | Flags | Card. | Type | Description & Constraints
ElementDefinition | ΣIN | | BackBoneElement | Definition of an element in a resource or extension + Rule: Min <= Max + Rule: if the element definition has a contentReference, it cannot have type, defaultValue, fixed, pattern, example, minValue, maxValue, maxLength, or binding + Rule: Fixed value may only be specified if there is one type + Rule: Pattern may only be specified if there is one type + Rule: Pattern and fixed are mutually exclusive + Rule: Binding can only be present for coded elements, string, and uri + Rule: Types must be unique by code + Rule: Constraints must be unique by key + Rule: default value and meaningWhenMissing are mutually exclusive + Rule: sliceName must be composed of proper tokens separated by "/" + Rule: Must have a modifier reason if isModifier = true + Rule: Element names cannot include some special characters + Warning: Element names should be simple alphanumerics with a max of 64 characters, or code generation tools may be broken + Rule: sliceIsConstraining can only appear if slicename is present Elements defined in Ancestors: id, extension, modifierExtension
path | Σ | 1..1 | string | Path of the element in the hierarchy of elements
representation | Σ | 0..* | code | xmlAttr | xmlText | typeAttr | cdaText | xhtml PropertyRepresentation (Required)
sliceName | Σ | 0..1 | string | Name for this particular element (in a set of slices)
sliceIsConstraining | ΣTU | 0..1 | boolean | If this slice definition constrains an inherited slice definition (or not)
label | Σ | 0..1 | string | Name for element to display with or prompt for element
code | Σ | 0..* | Coding | Corresponding codes in terminologies LOINC Codes (Example)
slicing | ΣI | 0..1 | Element | This element is sliced - slices follow + Rule: If there are no discriminators, there must be a definition
discriminator | Σ | 0..* | Element | Element values that are used to distinguish the slices
type | Σ | 1..1 | code | value | exists | pattern | type | profile DiscriminatorType (Required)
path | Σ | 1..1 | string | Path to element value
description | ΣI | 0..1 | string | Text description of how slicing works (or not)
ordered | Σ | 0..1 | boolean | If elements must be in same order as slices
rules | Σ | 1..1 | code | closed | open | openAtEnd SlicingRules (Required)
short | Σ | 0..1 | string | Concise definition for space-constrained presentation
definition | Σ | 0..1 | markdown | Full formal definition as narrative text
comment | Σ | 0..1 | markdown | Comments about the use of this element
requirements | Σ | 0..1 | markdown | Why this resource has been created
alias | Σ | 0..* | string | Other names
min | ΣI | 0..1 | unsignedInt | Minimum Cardinality
max | ΣI | 0..1 | string | Maximum Cardinality (a number or *) + Rule: Max SHALL be a number or "*"
base | Σ | 0..1 | Element | Base definition information for tools
path | Σ | 1..1 | string | Path that identifies the base element
min | Σ | 1..1 | unsignedInt | Min cardinality of the base element
max | Σ | 1..1 | string | Max cardinality of the base element
contentReference | ΣI | 0..1 | uri | Reference to definition of content for the element
type | ΣI | 0..* | Element | Data type and Profile for this element + Rule: Aggregation may only be specified if one of the allowed types for the element is a reference + Rule: targetProfile is only allowed if the type is Reference or canonical
code | Σ | 1..1 | uri | Data type or Resource (reference to definition) FHIRDefinedType (Extensible)
profile | Σ | 0..* | canonical(StructureDefinition | ImplementationGuide) | Profiles (StructureDefinition or IG) - one must apply
targetProfile | Σ | 0..* | canonical(StructureDefinition | ImplementationGuide) | Profile (StructureDefinition or IG) on the Reference/canonical target - one must apply

aggregation | ΣI | 0..* | code | contained | referenced | bundled - how aggregated AggregationMode (Required)
versioning | Σ | 0..1 | code | either | independent | specific ReferenceVersionRules (Required)
defaultValue[x] | ΣI | 0..1 | * | Specified value if missing from instance
meaningWhenMissing | ΣI | 0..1 | markdown | Implicit meaning when this element is missing
orderMeaning | Σ | 0..1 | string | What the order of the elements means
fixed[x] | ΣI | 0..1 | * | Value must be exactly this
pattern[x] | ΣI | 0..1 | * | Value must have at least these property values
example | Σ | 0..* | Element | Example value (as defined for type)
label | Σ | 1..1 | string | Describes the purpose of this example
value[x] | Σ | 1..1 | * | Value of Example (one of allowed types)
minValue[x] | Σ | 0..1 | | Minimum Allowed Value (for some types)
minValueDate | | | date
minValueDateTime | | | dateTime
minValueInstant | | | instant
minValueTime | | | time
minValueDecimal | | | decimal
minValueInteger | | | integer
minValuePositiveInt | | | positiveInt
minValueUnsignedInt | | | unsignedInt
minValueQuantity | | | Quantity
maxValue[x] | Σ | 0..1 | | Maximum Allowed Value (for some types)
maxValueDate | | | date
maxValueDateTime | | | dateTime
maxValueInstant | | | instant
maxValueTime | | | time
maxValueDecimal | | | decimal
maxValueInteger | | | integer
maxValuePositiveInt | | | positiveInt
maxValueUnsignedInt | | | unsignedInt
maxValueQuantity | | | Quantity
maxLength | Σ | 0..1 | integer | Max length for strings
condition | Σ | 0..* | id | Reference to invariant about presence
constraint | ΣI | 0..* | Element | Condition that must evaluate to true + Warning: Constraints should have an expression or else validators will not be able to enforce them
key | ΣI | 1..1 | id | Target of 'condition' reference above
requirements | Σ | 0..1 | string | Why this constraint is necessary or appropriate
severity | Σ | 1..1 | code | error | warning ConstraintSeverity (Required)
human | Σ | 1..1 | string | Human description of constraint
expression | Σ | 0..1 | string | FHIRPath expression of constraint
xpath | ΣTU | 0..1 | string | XPath expression of constraint
source | Σ | 0..1 | canonical(StructureDefinition) | Reference to original source of constraint
mustSupport | Σ | 0..1 | boolean | If the element must be supported
isModifier | Σ | 0..1 | boolean | If this modifies the meaning of other elements
isModifierReason | Σ | 0..1 | string | Reason that this element is marked as a modifier
isSummary | Σ | 0..1 | boolean | Include when _summary = true?
binding | ΣI | 0..1 | Element | ValueSet details if this is coded + Rule: ValueSet SHALL start with http:// or https:// or urn:
strength | Σ | 1..1 | code | required | extensible | preferred | example BindingStrength (Required)
description | Σ | 0..1 | string | Human explanation of the value set
valueSet | ΣI | 0..1 | canonical(ValueSet) | Source of value set
mapping | Σ | 0..* | Element | Map element to another set of definitions
identity | Σ | 1..1 | id | Reference to mapping declaration
language | Σ | 0..1 | code | Computable language of mapping MimeType (Required)
map | Σ | 1..1 | string | Details of the mapping
comment | Σ | 0..1 | string | Comments about the mapping or its use
Documentation for this format

UML Diagram (Legend)

XML Template

```
<ElementDefinition xmlns="http://hl7.org/fhir">
 <!-- from BackboneElement: extension, modifierExtension -->
 <path value="[string]"/><!-- 1..1 Path of the element in the hierarchy of elements -->
 <representation value="[code]"/><!-- 0..* xmlAttr | xmlText | typeAttr | cdaText | xhtml -->
 <sliceName value="[string]"/><!-- 0..1 Name for this particular element (in a set of slices) -->
 <sliceIsConstraining value="[boolean]"/><!-- 0..1 If this slice definition constrains an inherited slice definition (or
     not) -->
 <label value="[string]"/><!-- 0..1 Name for element to display with or prompt for element -->
 <code><!-- 0..* Coding Corresponding codes in terminologies --></code>
 <slicing>  <!-- 0..1 This element is sliced - slices follow -->
  <discriminator>  <!-- 0..* Element values that are used to distinguish the slices -->
   <type value="[code]"/><!-- 1..1 value | exists | pattern | type | profile -->
   <path value="[string]"/><!-- 1..1 Path to element value -->
  </discriminator>
  <description value="[string]"/><!--  0..1 Text description of how slicing works (or not) -->
  <ordered value="[boolean]"/><!-- 0..1 If elements must be in same order as slices -->
  <rules value="[code]"/><!-- 1..1 closed | open | openAtEnd -->
 </slicing>
```

```
 <short value="[string]"/><!-- 0..1 Concise definition for space-constrained presentation -->
 <definition value="[markdown]"/><!-- 0..1 Full formal definition as narrative text -->
 <comment value="[markdown]"/><!-- 0..1 Comments about the use of this element -->
 <requirements value="[markdown]"/><!-- 0..1 Why this resource has been created -->
 <alias value="[string]"/><!-- 0..* Other names -->
 <min value="[unsignedInt]"/><!--  0..1 Minimum Cardinality -->
 <max value="[string]"/><!--  0..1 Maximum Cardinality (a number or *) -->
 <base>  <!-- 0..1 Base definition information for tools -->
  <path value="[string]"/><!-- 1..1 Path that identifies the base element -->
  <min value="[unsignedInt]"/><!-- 1..1 Min cardinality of the base element -->
  <max value="[string]"/><!-- 1..1 Max cardinality of the base element -->
 </base>
 <contentReference value="[uri]"/><!--  0..1 Reference to definition of content for the element -->
 <type>  <!--  0..* Data type and Profile for this element -->
  <code value="[uri]"/><!-- 1..1 Data type or Resource (reference to definition) -->
  <profile><!-- 0..* canonical(StructureDefinition|ImplementationGuide) Profiles (StructureDefinition or IG) - one must
    apply --></profile>
  <targetProfile><!-- 0..* canonical(StructureDefinition|ImplementationGuide) Profile (StructureDefinition or IG) on the
     Reference/canonical target - one must apply --></targetProfile>
  <aggregation value="[code]"/><!--  0..* contained | referenced | bundled - how aggregated -->
  <versioning value="[code]"/><!-- 0..1 either | independent | specific -->
 </type>
 <defaultValue[x]><!--  0..1 * Specified value if missing from instance --></defaultValue[x]>
 <meaningWhenMissing value="[markdown]"/><!--  0..1 Implicit meaning when this element is missing -->
 <orderMeaning value="[string]"/><!-- 0..1 What the order of the elements means -->
 <fixed[x]><!--  0..1 * Value must be exactly this --></fixed[x]>
 <pattern[x]><!--  0..1 * Value must have at least these property values --></pattern[x]>
 <example>  <!-- 0..* Example value (as defined for type) -->
  <label value="[string]"/><!-- 1..1 Describes the purpose of this example -->
  <value[x]><!-- 1..1 * Value of Example (one of allowed types) --></value[x]>
 </example>
 <minValue[x]><!-- 0..1 date|dateTime|instant|time|decimal|integer|positiveInt|
   unsignedInt|Quantity Minimum Allowed Value (for some types) --></minValue[x]>
 <maxValue[x]><!-- 0..1 date|dateTime|instant|time|decimal|integer|positiveInt|
   unsignedInt|Quantity Maximum Allowed Value (for some types) --></maxValue[x]>
 <maxLength value="[integer]"/><!-- 0..1 Max length for strings -->
 <condition value="[id]"/><!-- 0..* Reference to invariant about presence -->
 <constraint>  <!-- 0..* Condition that must evaluate to true -->
  <key value="[id]"/><!--  1..1 Target of 'condition' reference above -->
  <requirements value="[string]"/><!-- 0..1 Why this constraint is necessary or appropriate -->
  <severity value="[code]"/><!-- 1..1 error | warning -->
  <human value="[string]"/><!-- 1..1 Human description of constraint -->
  <expression value="[string]"/><!-- 0..1 FHIRPath expression of constraint -->
  <xpath value="[string]"/><!-- 0..1 XPath expression of constraint -->
  <source><!-- 0..1 canonical(StructureDefinition) Reference to original source of constraint --></source>
 </constraint>
 <mustSupport value="[boolean]"/><!-- 0..1 If the element must be supported -->
 <isModifier value="[boolean]"/><!-- 0..1 If this modifies the meaning of other elements -->
 <isModifierReason value="[string]"/><!-- 0..1 Reason that this element is marked as a modifier -->
 <isSummary value="[boolean]"/><!-- 0..1 Include when _summary = true? -->
 <binding>  <!--  0..1 ValueSet details if this is coded -->
  <strength value="[code]"/><!-- 1..1 required | extensible | preferred | example -->
  <description value="[string]"/><!-- 0..1 Human explanation of the value set -->
  <valueSet><!--  0..1 canonical(ValueSet) Source of value set --></valueSet>
 </binding>
 <mapping>  <!-- 0..* Map element to another set of definitions -->
  <identity value="[id]"/><!-- 1..1 Reference to mapping declaration -->
  <language value="[code]"/><!-- 0..1 Computable language of mapping -->
  <map value="[string]"/><!-- 1..1 Details of the mapping -->
  <comment value="[string]"/><!-- 0..1 Comments about the mapping or its use -->
 </mapping>
</ElementDefinition>
```

JSON Template

```
{
  // from BackboneElement: extension, modifierExtension
  "path" : "<string>", // R!  Path of the element in the hierarchy of elements
  "representation" : ["<code>"], // xmlAttr | xmlText | typeAttr | cdaText | xhtml
  "sliceName" : "<string>", // Name for this particular element (in a set of slices)
  "sliceIsConstraining" : <boolean>, // If this slice definition constrains an inherited slice definition (or not)
  "label" : "<string>", // Name for element to display with or prompt for element
  "code" : [{ Coding }], // Corresponding codes in terminologies
  "slicing" : { // This element is sliced - slices follow
    "discriminator" : [{ // Element values that are used to distinguish the slices
      "type" : "<code>", // R!  value | exists | pattern | type | profile
      "path" : "<string>" // R!  Path to element value
    }],
    "description" : "<string>", // C? Text description of how slicing works (or not)
    "ordered" : <boolean>, // If elements must be in same order as slices
    "rules" : "<code>" // R!  closed | open | openAtEnd
  },
  "short" : "<string>", // Concise definition for space-constrained presentation
  "definition" : "<markdown>", // Full formal definition as narrative text
  "comment" : "<markdown>", // Comments about the use of this element
  "requirements" : "<markdown>", // Why this resource has been created
  "alias" : ["<string>"], // Other names
  "min" : "<unsignedInt>", // C? Minimum Cardinality
  "max" : "<string>", // C? Maximum Cardinality (a number or *)
  "base" : { // Base definition information for tools
```

```
  "path" : "<string>", // R!  Path that identifies the base element
  "min" : "<unsignedInt>", // R!  Min cardinality of the base element
  "max" : "<string>" // R!  Max cardinality of the base element
},
"contentReference" : "<uri>", // C? Reference to definition of content for the element
"type" : [{ // C? Data type and Profile for this element
  "code" : "<uri>", // R!  Data type or Resource (reference to definition)
  "profile" : [{ canonical(StructureDefinition|ImplementationGuide) }], // Profiles (StructureDefinition or IG) - one
  must apply
  "targetProfile" : [{ canonical(StructureDefinition|ImplementationGuide) }], // Profile (StructureDefinition or IG)
  on the Reference/canonical target - one must apply
  "aggregation" : ["<code>"], // C? contained | referenced | bundled - how aggregated
  "versioning" : "<code>" // either | independent | specific
}],
// defaultValue[x]: Specified value if missing from instance. One of these 50:
"defaultValueBase64Binary" : "<base64Binary>",
"defaultValueBoolean" : <boolean>,
"defaultValueCanonical" : "<canonical>",
"defaultValueCode" : "<code>",
"defaultValueDate" : "<date>",
"defaultValueDateTime" : "<dateTime>",
"defaultValueDecimal" : <decimal>,
"defaultValueId" : "<id>",
"defaultValueInstant" : "<instant>",
"defaultValueInteger" : <integer>,
"defaultValueMarkdown" : "<markdown>",
"defaultValueOid" : "<oid>",
"defaultValuePositiveInt" : "<positiveInt>",
"defaultValueString" : "<string>",
"defaultValueTime" : "<time>",
"defaultValueUnsignedInt" : "<unsignedInt>",
"defaultValueUri" : "<uri>",
"defaultValueUrl" : "<url>",
"defaultValueUuid" : "<uuid>",
"defaultValueAddress" : { Address },
"defaultValueAge" : { Age },
"defaultValueAnnotation" : { Annotation },
"defaultValueAttachment" : { Attachment },
"defaultValueCodeableConcept" : { CodeableConcept },
"defaultValueCoding" : { Coding },
"defaultValueContactPoint" : { ContactPoint },
"defaultValueCount" : { Count },
"defaultValueDistance" : { Distance },
"defaultValueDuration" : { Duration },
"defaultValueHumanName" : { HumanName },
"defaultValueIdentifier" : { Identifier },
"defaultValueMoney" : { Money },
"defaultValuePeriod" : { Period },
"defaultValueQuantity" : { Quantity },
"defaultValueRange" : { Range },
"defaultValueRatio" : { Ratio },
"defaultValueReference" : { Reference },
"defaultValueSampledData" : { SampledData },
"defaultValueSignature" : { Signature },
"defaultValueTiming" : { Timing },
"defaultValueContactDetail" : { ContactDetail },
"defaultValueContributor" : { Contributor },
"defaultValueDataRequirement" : { DataRequirement },
"defaultValueExpression" : { Expression },
"defaultValueParameterDefinition" : { ParameterDefinition },
"defaultValueRelatedArtifact" : { RelatedArtifact },
"defaultValueTriggerDefinition" : { TriggerDefinition },
"defaultValueUsageContext" : { UsageContext },
"defaultValueDosage" : { Dosage },
"defaultValueMeta" : { Meta },
"meaningWhenMissing" : "<markdown>", // C? Implicit meaning when this element is missing
"orderMeaning" : "<string>", // What the order of the elements means
// fixed[x]: Value must be exactly this. One of these 50:
"fixedBase64Binary" : "<base64Binary>",
"fixedBoolean" : <boolean>,
"fixedCanonical" : "<canonical>",
"fixedCode" : "<code>",
"fixedDate" : "<date>",
"fixedDateTime" : "<dateTime>",
"fixedDecimal" : <decimal>,
"fixedId" : "<id>",
"fixedInstant" : "<instant>",
"fixedInteger" : <integer>,
"fixedMarkdown" : "<markdown>",
"fixedOid" : "<oid>",
"fixedPositiveInt" : "<positiveInt>",
"fixedString" : "<string>",
"fixedTime" : "<time>",
"fixedUnsignedInt" : "<unsignedInt>",
"fixedUri" : "<uri>",
"fixedUrl" : "<url>",
"fixedUuid" : "<uuid>",
"fixedAddress" : { Address },
"fixedAge" : { Age },
```

```
"fixedAnnotation" : { Annotation },
"fixedAttachment" : { Attachment },
"fixedCodeableConcept" : { CodeableConcept },
"fixedCoding" : { Coding },
"fixedContactPoint" : { ContactPoint },
"fixedCount" : { Count },
"fixedDistance" : { Distance },
"fixedDuration" : { Duration },
"fixedHumanName" : { HumanName },
"fixedIdentifier" : { Identifier },
"fixedMoney" : { Money },
"fixedPeriod" : { Period },
"fixedQuantity" : { Quantity },
"fixedRange" : { Range },
"fixedRatio" : { Ratio },
"fixedReference" : { Reference },
"fixedSampledData" : { SampledData },
"fixedSignature" : { Signature },
"fixedTiming" : { Timing },
"fixedContactDetail" : { ContactDetail },
"fixedContributor" : { Contributor },
"fixedDataRequirement" : { DataRequirement },
"fixedExpression" : { Expression },
"fixedParameterDefinition" : { ParameterDefinition },
"fixedRelatedArtifact" : { RelatedArtifact },
"fixedTriggerDefinition" : { TriggerDefinition },
"fixedUsageContext" : { UsageContext },
"fixedDosage" : { Dosage },
"fixedMeta" : { Meta },
// pattern[x]: Value must have at least these property values. One of these 50:
"patternBase64Binary" : "<base64Binary>",
"patternBoolean" : <boolean>,
"patternCanonical" : "<canonical>",
"patternCode" : "<code>",
"patternDate" : "<date>",
"patternDateTime" : "<dateTime>",
"patternDecimal" : <decimal>,
"patternId" : "<id>",
"patternInstant" : "<instant>",
"patternInteger" : <integer>,
"patternMarkdown" : "<markdown>",
"patternOid" : "<oid>",
"patternPositiveInt" : "<positiveInt>",
"patternString" : "<string>",
"patternTime" : "<time>",
"patternUnsignedInt" : "<unsignedInt>",
"patternUri" : "<uri>",
"patternUrl" : "<url>",
"patternUuid" : "<uuid>",
"patternAddress" : { Address },
"patternAge" : { Age },
"patternAnnotation" : { Annotation },
"patternAttachment" : { Attachment },
"patternCodeableConcept" : { CodeableConcept },
"patternCoding" : { Coding },
"patternContactPoint" : { ContactPoint },
"patternCount" : { Count },
"patternDistance" : { Distance },
"patternDuration" : { Duration },
"patternHumanName" : { HumanName },
"patternIdentifier" : { Identifier },
"patternMoney" : { Money },
"patternPeriod" : { Period },
"patternQuantity" : { Quantity },
"patternRange" : { Range },
"patternRatio" : { Ratio },
"patternReference" : { Reference },
"patternSampledData" : { SampledData },
"patternSignature" : { Signature },
"patternTiming" : { Timing },
"patternContactDetail" : { ContactDetail },
"patternContributor" : { Contributor },
"patternDataRequirement" : { DataRequirement },
"patternExpression" : { Expression },
"patternParameterDefinition" : { ParameterDefinition },
"patternRelatedArtifact" : { RelatedArtifact },
"patternTriggerDefinition" : { TriggerDefinition },
"patternUsageContext" : { UsageContext },
"patternDosage" : { Dosage },
"patternMeta" : { Meta },
"example" : [{ // Example value (as defined for type)
  "label" : "<string>", // R!  Describes the purpose of this example
  // value[x]: Value of Example (one of allowed types). One of these 50:
  "valueBase64Binary" : "<base64Binary>"
  "valueBoolean" : <boolean>
  "valueCanonical" : "<canonical>"
  "valueCode" : "<code>"
  "valueDate" : "<date>"
  "valueDateTime" : "<dateTime>"
```

```
  "valueDecimal" : <decimal>
  "valueId" : "<id>"
  "valueInstant" : "<instant>"
  "valueInteger" : <integer>
  "valueMarkdown" : "<markdown>"
  "valueOid" : "<oid>"
  "valuePositiveInt" : "<positiveInt>"
  "valueString" : "<string>"
  "valueTime" : "<time>"
  "valueUnsignedInt" : "<unsignedInt>"
  "valueUri" : "<uri>"
  "valueUrl" : "<url>"
  "valueUuid" : "<uuid>"
  "valueAddress" : { Address }
  "valueAge" : { Age }
  "valueAnnotation" : { Annotation }
  "valueAttachment" : { Attachment }
  "valueCodeableConcept" : { CodeableConcept }
  "valueCoding" : { Coding }
  "valueContactPoint" : { ContactPoint }
  "valueCount" : { Count }
  "valueDistance" : { Distance }
  "valueDuration" : { Duration }
  "valueHumanName" : { HumanName }
  "valueIdentifier" : { Identifier }
  "valueMoney" : { Money }
  "valuePeriod" : { Period }
  "valueQuantity" : { Quantity }
  "valueRange" : { Range }
  "valueRatio" : { Ratio }
  "valueReference" : { Reference }
  "valueSampledData" : { SampledData }
  "valueSignature" : { Signature }
  "valueTiming" : { Timing }
  "valueContactDetail" : { ContactDetail }
  "valueContributor" : { Contributor }
  "valueDataRequirement" : { DataRequirement }
  "valueExpression" : { Expression }
  "valueParameterDefinition" : { ParameterDefinition }
  "valueRelatedArtifact" : { RelatedArtifact }
  "valueTriggerDefinition" : { TriggerDefinition }
  "valueUsageContext" : { UsageContext }
  "valueDosage" : { Dosage }
  "valueMeta" : { Meta }
}],
// minValue[x]: Minimum Allowed Value (for some types). One of these 9:
"minValueDate" : "<date>",
"minValueDateTime" : "<dateTime>",
"minValueInstant" : "<instant>",
"minValueTime" : "<time>",
"minValueDecimal" : <decimal>,
"minValueInteger" : <integer>,
"minValuePositiveInt" : "<positiveInt>",
"minValueUnsignedInt" : "<unsignedInt>",
"minValueQuantity" : { Quantity },
// maxValue[x]: Maximum Allowed Value (for some types). One of these 9:
"maxValueDate" : "<date>",
"maxValueDateTime" : "<dateTime>",
"maxValueInstant" : "<instant>",
"maxValueTime" : "<time>",
"maxValueDecimal" : <decimal>,
"maxValueInteger" : <integer>,
"maxValuePositiveInt" : "<positiveInt>",
"maxValueUnsignedInt" : "<unsignedInt>",
"maxValueQuantity" : { Quantity },
"maxLength" : <integer>, // Max length for strings
"condition" : ["<id>"], // Reference to invariant about presence
"constraint" : [{ // Condition that must evaluate to true
  "key" : "<id>", // C? R!  Target of 'condition' reference above
  "requirements" : "<string>", // Why this constraint is necessary or appropriate
  "severity" : "<code>", // R!  error | warning
  "human" : "<string>", // R!  Human description of constraint
  "expression" : "<string>", // FHIRPath expression of constraint
  "xpath" : "<string>", // XPath expression of constraint
  "source" : { canonical(StructureDefinition) } // Reference to original source of constraint
}],
"mustSupport" : <boolean>, // If the element must be supported
"isModifier" : <boolean>, // If this modifies the meaning of other elements
"isModifierReason" : "<string>", // Reason that this element is marked as a modifier
"isSummary" : <boolean>, // Include when _summary = true?
"binding" : { // C? ValueSet details if this is coded
  "strength" : "<code>", // R!  required | extensible | preferred | example
  "description" : "<string>", // Human explanation of the value set
  "valueSet" : { canonical(ValueSet) } // C? Source of value set
},
"mapping" : [{ // Map element to another set of definitions
  "identity" : "<id>", // R!  Reference to mapping declaration
  "language" : "<code>", // Computable language of mapping
  "map" : "<string>", // R!  Details of the mapping
```

```
    "comment" : "<string>" // Comments about the mapping or its use
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from BackboneElement: Element.extension, BackboneElement.modifierextension
  fhir:ElementDefinition.path [ string ]; # 1..1 Path of the element in the hierarchy of elements
  fhir:ElementDefinition.representation [ code ], ... ; # 0..* xmlAttr | xmlText | typeAttr | cdaText | xhtml
  fhir:ElementDefinition.sliceName [ string ]; # 0..1 Name for this particular element (in a set of slices)
  fhir:ElementDefinition.sliceIsConstraining [ boolean ]; # 0..1 If this slice definition constrains an inherited slice
    definition (or not)
  fhir:ElementDefinition.label [ string ]; # 0..1 Name for element to display with or prompt for element
  fhir:ElementDefinition.code [ Coding ], ... ; # 0..* Corresponding codes in terminologies
  fhir:ElementDefinition.slicing [ # 0..1 This element is sliced - slices follow
    fhir:ElementDefinition.slicing.discriminator [ # 0..* Element values that are used to distinguish the slices
      fhir:ElementDefinition.slicing.discriminator.type [ code ]; # 1..1 value | exists | pattern | type | profile
      fhir:ElementDefinition.slicing.discriminator.path [ string ]; # 1..1 Path to element value
    ], ...;
    fhir:ElementDefinition.slicing.description [ string ]; # 0..1 Text description of how slicing works (or not)
    fhir:ElementDefinition.slicing.ordered [ boolean ]; # 0..1 If elements must be in same order as slices
    fhir:ElementDefinition.slicing.rules [ code ]; # 1..1 closed | open | openAtEnd
  ];
  fhir:ElementDefinition.short [ string ]; # 0..1 Concise definition for space-constrained presentation
  fhir:ElementDefinition.definition [ markdown ]; # 0..1 Full formal definition as narrative text
  fhir:ElementDefinition.comment [ markdown ]; # 0..1 Comments about the use of this element
  fhir:ElementDefinition.requirements [ markdown ]; # 0..1 Why this resource has been created
  fhir:ElementDefinition.alias [ string ], ... ; # 0..* Other names
  fhir:ElementDefinition.min [ unsignedInt ]; # 0..1 Minimum Cardinality
  fhir:ElementDefinition.max [ string ]; # 0..1 Maximum Cardinality (a number or *)
  fhir:ElementDefinition.base [ # 0..1 Base definition information for tools
    fhir:ElementDefinition.base.path [ string ]; # 1..1 Path that identifies the base element
    fhir:ElementDefinition.base.min [ unsignedInt ]; # 1..1 Min cardinality of the base element
    fhir:ElementDefinition.base.max [ string ]; # 1..1 Max cardinality of the base element
  ];
  fhir:ElementDefinition.contentReference [ uri ]; # 0..1 Reference to definition of content for the element
  fhir:ElementDefinition.type [ # 0..* Data type and Profile for this element
    fhir:ElementDefinition.type.code [ uri ]; # 1..1 Data type or Resource (reference to definition)
    fhir:ElementDefinition.type.profile [ canonical(StructureDefinition|ImplementationGuide) ], ... ; # 0..* Profiles
    (StructureDefinition or IG) - one must apply
    fhir:ElementDefinition.type.targetProfile [ canonical(StructureDefinition|ImplementationGuide) ], ... ; # 0..*
    Profile (StructureDefinition or IG) on the Reference/canonical target - one must apply
    fhir:ElementDefinition.type.aggregation [ code ], ... ; # 0..* contained | referenced | bundled - how aggregated
    fhir:ElementDefinition.type.versioning [ code ]; # 0..1 either | independent | specific
  ], ...;
  # ElementDefinition.defaultValue[x] : 0..1 Specified value if missing from instance. One of these 50
    fhir:ElementDefinition.defaultValueBase64Binary [ base64Binary ]
    fhir:ElementDefinition.defaultValueBoolean [ boolean ]
    fhir:ElementDefinition.defaultValueCanonical [ canonical ]
    fhir:ElementDefinition.defaultValueCode [ code ]
    fhir:ElementDefinition.defaultValueDate [ date ]
    fhir:ElementDefinition.defaultValueDateTime [ dateTime ]
    fhir:ElementDefinition.defaultValueDecimal [ decimal ]
    fhir:ElementDefinition.defaultValueId [ id ]
    fhir:ElementDefinition.defaultValueInstant [ instant ]
    fhir:ElementDefinition.defaultValueInteger [ integer ]
    fhir:ElementDefinition.defaultValueMarkdown [ markdown ]
    fhir:ElementDefinition.defaultValueOid [ oid ]
    fhir:ElementDefinition.defaultValuePositiveInt [ positiveInt ]
    fhir:ElementDefinition.defaultValueString [ string ]
    fhir:ElementDefinition.defaultValueTime [ time ]
    fhir:ElementDefinition.defaultValueUnsignedInt [ unsignedInt ]
    fhir:ElementDefinition.defaultValueUri [ uri ]
    fhir:ElementDefinition.defaultValueUrl [ url ]
    fhir:ElementDefinition.defaultValueUuid [ uuid ]
    fhir:ElementDefinition.defaultValueAddress [ Address ]
    fhir:ElementDefinition.defaultValueAge [ Age ]
    fhir:ElementDefinition.defaultValueAnnotation [ Annotation ]
    fhir:ElementDefinition.defaultValueAttachment [ Attachment ]
    fhir:ElementDefinition.defaultValueCodeableConcept [ CodeableConcept ]
    fhir:ElementDefinition.defaultValueCoding [ Coding ]
    fhir:ElementDefinition.defaultValueContactPoint [ ContactPoint ]
    fhir:ElementDefinition.defaultValueCount [ Count ]
    fhir:ElementDefinition.defaultValueDistance [ Distance ]
    fhir:ElementDefinition.defaultValueDuration [ Duration ]
    fhir:ElementDefinition.defaultValueHumanName [ HumanName ]
    fhir:ElementDefinition.defaultValueIdentifier [ Identifier ]
    fhir:ElementDefinition.defaultValueMoney [ Money ]
    fhir:ElementDefinition.defaultValuePeriod [ Period ]
    fhir:ElementDefinition.defaultValueQuantity [ Quantity ]
    fhir:ElementDefinition.defaultValueRange [ Range ]
    fhir:ElementDefinition.defaultValueRatio [ Ratio ]
    fhir:ElementDefinition.defaultValueReference [ Reference ]
    fhir:ElementDefinition.defaultValueSampledData [ SampledData ]
    fhir:ElementDefinition.defaultValueSignature [ Signature ]
    fhir:ElementDefinition.defaultValueTiming [ Timing ]
    fhir:ElementDefinition.defaultValueContactDetail [ ContactDetail ]
```

```
  fhir:ElementDefinition.defaultValueContributor [ Contributor ]
  fhir:ElementDefinition.defaultValueDataRequirement [ DataRequirement ]
  fhir:ElementDefinition.defaultValueExpression [ Expression ]
  fhir:ElementDefinition.defaultValueParameterDefinition [ ParameterDefinition ]
  fhir:ElementDefinition.defaultValueRelatedArtifact [ RelatedArtifact ]
  fhir:ElementDefinition.defaultValueTriggerDefinition [ TriggerDefinition ]
  fhir:ElementDefinition.defaultValueUsageContext [ UsageContext ]
  fhir:ElementDefinition.defaultValueDosage [ Dosage ]
  fhir:ElementDefinition.defaultValueMeta [ Meta ]
fhir:ElementDefinition.meaningWhenMissing [ markdown ]; # 0..1 Implicit meaning when this element is missing
fhir:ElementDefinition.orderMeaning [ string ]; # 0..1 What the order of the elements means
# ElementDefinition.fixed[x] : 0..1 Value must be exactly this. One of these 50
  fhir:ElementDefinition.fixedBase64Binary [ base64Binary ]
  fhir:ElementDefinition.fixedBoolean [ boolean ]
  fhir:ElementDefinition.fixedCanonical [ canonical ]
  fhir:ElementDefinition.fixedCode [ code ]
  fhir:ElementDefinition.fixedDate [ date ]
  fhir:ElementDefinition.fixedDateTime [ dateTime ]
  fhir:ElementDefinition.fixedDecimal [ decimal ]
  fhir:ElementDefinition.fixedId [ id ]
  fhir:ElementDefinition.fixedInstant [ instant ]
  fhir:ElementDefinition.fixedInteger [ integer ]
  fhir:ElementDefinition.fixedMarkdown [ markdown ]
  fhir:ElementDefinition.fixedOid [ oid ]
  fhir:ElementDefinition.fixedPositiveInt [ positiveInt ]
  fhir:ElementDefinition.fixedString [ string ]
  fhir:ElementDefinition.fixedTime [ time ]
  fhir:ElementDefinition.fixedUnsignedInt [ unsignedInt ]
  fhir:ElementDefinition.fixedUri [ uri ]
  fhir:ElementDefinition.fixedUrl [ url ]
  fhir:ElementDefinition.fixedUuid [ uuid ]
  fhir:ElementDefinition.fixedAddress [ Address ]
  fhir:ElementDefinition.fixedAge [ Age ]
  fhir:ElementDefinition.fixedAnnotation [ Annotation ]
  fhir:ElementDefinition.fixedAttachment [ Attachment ]
  fhir:ElementDefinition.fixedCodeableConcept [ CodeableConcept ]
  fhir:ElementDefinition.fixedCoding [ Coding ]
  fhir:ElementDefinition.fixedContactPoint [ ContactPoint ]
  fhir:ElementDefinition.fixedCount [ Count ]
  fhir:ElementDefinition.fixedDistance [ Distance ]
  fhir:ElementDefinition.fixedDuration [ Duration ]
  fhir:ElementDefinition.fixedHumanName [ HumanName ]
  fhir:ElementDefinition.fixedIdentifier [ Identifier ]
  fhir:ElementDefinition.fixedMoney [ Money ]
  fhir:ElementDefinition.fixedPeriod [ Period ]
  fhir:ElementDefinition.fixedQuantity [ Quantity ]
  fhir:ElementDefinition.fixedRange [ Range ]
  fhir:ElementDefinition.fixedRatio [ Ratio ]
  fhir:ElementDefinition.fixedReference [ Reference ]
  fhir:ElementDefinition.fixedSampledData [ SampledData ]
  fhir:ElementDefinition.fixedSignature [ Signature ]
  fhir:ElementDefinition.fixedTiming [ Timing ]
  fhir:ElementDefinition.fixedContactDetail [ ContactDetail ]
  fhir:ElementDefinition.fixedContributor [ Contributor ]
  fhir:ElementDefinition.fixedDataRequirement [ DataRequirement ]
  fhir:ElementDefinition.fixedExpression [ Expression ]
  fhir:ElementDefinition.fixedParameterDefinition [ ParameterDefinition ]
  fhir:ElementDefinition.fixedRelatedArtifact [ RelatedArtifact ]
  fhir:ElementDefinition.fixedTriggerDefinition [ TriggerDefinition ]
  fhir:ElementDefinition.fixedUsageContext [ UsageContext ]
  fhir:ElementDefinition.fixedDosage [ Dosage ]
  fhir:ElementDefinition.fixedMeta [ Meta ]
# ElementDefinition.pattern[x] : 0..1 Value must have at least these property values. One of these 50
  fhir:ElementDefinition.patternBase64Binary [ base64Binary ]
  fhir:ElementDefinition.patternBoolean [ boolean ]
  fhir:ElementDefinition.patternCanonical [ canonical ]
  fhir:ElementDefinition.patternCode [ code ]
  fhir:ElementDefinition.patternDate [ date ]
  fhir:ElementDefinition.patternDateTime [ dateTime ]
  fhir:ElementDefinition.patternDecimal [ decimal ]
  fhir:ElementDefinition.patternId [ id ]
  fhir:ElementDefinition.patternInstant [ instant ]
  fhir:ElementDefinition.patternInteger [ integer ]
  fhir:ElementDefinition.patternMarkdown [ markdown ]
  fhir:ElementDefinition.patternOid [ oid ]
  fhir:ElementDefinition.patternPositiveInt [ positiveInt ]
  fhir:ElementDefinition.patternString [ string ]
  fhir:ElementDefinition.patternTime [ time ]
  fhir:ElementDefinition.patternUnsignedInt [ unsignedInt ]
  fhir:ElementDefinition.patternUri [ uri ]
  fhir:ElementDefinition.patternUrl [ url ]
  fhir:ElementDefinition.patternUuid [ uuid ]
  fhir:ElementDefinition.patternAddress [ Address ]
  fhir:ElementDefinition.patternAge [ Age ]
  fhir:ElementDefinition.patternAnnotation [ Annotation ]
  fhir:ElementDefinition.patternAttachment [ Attachment ]
  fhir:ElementDefinition.patternCodeableConcept [ CodeableConcept ]
  fhir:ElementDefinition.patternCoding [ Coding ]
  fhir:ElementDefinition.patternContactPoint [ ContactPoint ]
```

```
  fhir:ElementDefinition.patternCount [ Count ]
  fhir:ElementDefinition.patternDistance [ Distance ]
  fhir:ElementDefinition.patternDuration [ Duration ]
  fhir:ElementDefinition.patternHumanName [ HumanName ]
  fhir:ElementDefinition.patternIdentifier [ Identifier ]
  fhir:ElementDefinition.patternMoney [ Money ]
  fhir:ElementDefinition.patternPeriod [ Period ]
  fhir:ElementDefinition.patternQuantity [ Quantity ]
  fhir:ElementDefinition.patternRange [ Range ]
  fhir:ElementDefinition.patternRatio [ Ratio ]
  fhir:ElementDefinition.patternReference [ Reference ]
  fhir:ElementDefinition.patternSampledData [ SampledData ]
  fhir:ElementDefinition.patternSignature [ Signature ]
  fhir:ElementDefinition.patternTiming [ Timing ]
  fhir:ElementDefinition.patternContactDetail [ ContactDetail ]
  fhir:ElementDefinition.patternContributor [ Contributor ]
  fhir:ElementDefinition.patternDataRequirement [ DataRequirement ]
  fhir:ElementDefinition.patternExpression [ Expression ]
  fhir:ElementDefinition.patternParameterDefinition [ ParameterDefinition ]
  fhir:ElementDefinition.patternRelatedArtifact [ RelatedArtifact ]
  fhir:ElementDefinition.patternTriggerDefinition [ TriggerDefinition ]
  fhir:ElementDefinition.patternUsageContext [ UsageContext ]
  fhir:ElementDefinition.patternDosage [ Dosage ]
  fhir:ElementDefinition.patternMeta [ Meta ]
fhir:ElementDefinition.example [ # 0..* Example value (as defined for type)
  fhir:ElementDefinition.example.label [ string ]; # 1..1 Describes the purpose of this example
  # ElementDefinition.example.value[x] : 1..1 Value of Example (one of allowed types). One of these 50
    fhir:ElementDefinition.example.valueBase64Binary [ base64Binary ]
    fhir:ElementDefinition.example.valueBoolean [ boolean ]
    fhir:ElementDefinition.example.valueCanonical [ canonical ]
    fhir:ElementDefinition.example.valueCode [ code ]
    fhir:ElementDefinition.example.valueDate [ date ]
    fhir:ElementDefinition.example.valueDateTime [ dateTime ]
    fhir:ElementDefinition.example.valueDecimal [ decimal ]
    fhir:ElementDefinition.example.valueId [ id ]
    fhir:ElementDefinition.example.valueInstant [ instant ]
    fhir:ElementDefinition.example.valueInteger [ integer ]
    fhir:ElementDefinition.example.valueMarkdown [ markdown ]
    fhir:ElementDefinition.example.valueOid [ oid ]
    fhir:ElementDefinition.example.valuePositiveInt [ positiveInt ]
    fhir:ElementDefinition.example.valueString [ string ]
    fhir:ElementDefinition.example.valueTime [ time ]
    fhir:ElementDefinition.example.valueUnsignedInt [ unsignedInt ]
    fhir:ElementDefinition.example.valueUri [ uri ]
    fhir:ElementDefinition.example.valueUrl [ url ]
    fhir:ElementDefinition.example.valueUuid [ uuid ]
    fhir:ElementDefinition.example.valueAddress [ Address ]
    fhir:ElementDefinition.example.valueAge [ Age ]
    fhir:ElementDefinition.example.valueAnnotation [ Annotation ]
    fhir:ElementDefinition.example.valueAttachment [ Attachment ]
    fhir:ElementDefinition.example.valueCodeableConcept [ CodeableConcept ]
    fhir:ElementDefinition.example.valueCoding [ Coding ]
    fhir:ElementDefinition.example.valueContactPoint [ ContactPoint ]
    fhir:ElementDefinition.example.valueCount [ Count ]
    fhir:ElementDefinition.example.valueDistance [ Distance ]
    fhir:ElementDefinition.example.valueDuration [ Duration ]
    fhir:ElementDefinition.example.valueHumanName [ HumanName ]
    fhir:ElementDefinition.example.valueIdentifier [ Identifier ]
    fhir:ElementDefinition.example.valueMoney [ Money ]
    fhir:ElementDefinition.example.valuePeriod [ Period ]
    fhir:ElementDefinition.example.valueQuantity [ Quantity ]
    fhir:ElementDefinition.example.valueRange [ Range ]
    fhir:ElementDefinition.example.valueRatio [ Ratio ]
    fhir:ElementDefinition.example.valueReference [ Reference ]
    fhir:ElementDefinition.example.valueSampledData [ SampledData ]
    fhir:ElementDefinition.example.valueSignature [ Signature ]
    fhir:ElementDefinition.example.valueTiming [ Timing ]
    fhir:ElementDefinition.example.valueContactDetail [ ContactDetail ]
    fhir:ElementDefinition.example.valueContributor [ Contributor ]
    fhir:ElementDefinition.example.valueDataRequirement [ DataRequirement ]
    fhir:ElementDefinition.example.valueExpression [ Expression ]
    fhir:ElementDefinition.example.valueParameterDefinition [ ParameterDefinition ]
    fhir:ElementDefinition.example.valueRelatedArtifact [ RelatedArtifact ]
    fhir:ElementDefinition.example.valueTriggerDefinition [ TriggerDefinition ]
    fhir:ElementDefinition.example.valueUsageContext [ UsageContext ]
    fhir:ElementDefinition.example.valueDosage [ Dosage ]
    fhir:ElementDefinition.example.valueMeta [ Meta ]
], ...;
# ElementDefinition.minValue[x] : 0..1 Minimum Allowed Value (for some types). One of these 9
  fhir:ElementDefinition.minValueDate [ date ]
  fhir:ElementDefinition.minValueDateTime [ dateTime ]
  fhir:ElementDefinition.minValueInstant [ instant ]
  fhir:ElementDefinition.minValueTime [ time ]
  fhir:ElementDefinition.minValueDecimal [ decimal ]
  fhir:ElementDefinition.minValueInteger [ integer ]
  fhir:ElementDefinition.minValuePositiveInt [ positiveInt ]
  fhir:ElementDefinition.minValueUnsignedInt [ unsignedInt ]
  fhir:ElementDefinition.minValueQuantity [ Quantity ]
# ElementDefinition.maxValue[x] : 0..1 Maximum Allowed Value (for some types). One of these 9
```

```
    fhir:ElementDefinition.maxValueDate [ date ]
    fhir:ElementDefinition.maxValueDateTime [ dateTime ]
    fhir:ElementDefinition.maxValueInstant [ instant ]
    fhir:ElementDefinition.maxValueTime [ time ]
    fhir:ElementDefinition.maxValueDecimal [ decimal ]
    fhir:ElementDefinition.maxValueInteger [ integer ]
    fhir:ElementDefinition.maxValuePositiveInt [ positiveInt ]
    fhir:ElementDefinition.maxValueUnsignedInt [ unsignedInt ]
    fhir:ElementDefinition.maxValueQuantity [ Quantity ]
  fhir:ElementDefinition.maxLength [ integer ]; # 0..1 Max length for strings
  fhir:ElementDefinition.condition [ id ], ... ; # 0..* Reference to invariant about presence
  fhir:ElementDefinition.constraint [ # 0..* Condition that must evaluate to true
    fhir:ElementDefinition.constraint.key [ id ]; # 1..1 Target of 'condition' reference above
    fhir:ElementDefinition.constraint.requirements [ string ]; # 0..1 Why this constraint is necessary or appropriate
    fhir:ElementDefinition.constraint.severity [ code ]; # 1..1 error | warning
    fhir:ElementDefinition.constraint.human [ string ]; # 1..1 Human description of constraint
    fhir:ElementDefinition.constraint.expression [ string ]; # 0..1 FHIRPath expression of constraint
    fhir:ElementDefinition.constraint.xpath [ string ]; # 0..1 XPath expression of constraint
    fhir:ElementDefinition.constraint.source [ canonical(StructureDefinition) ]; # 0..1 Reference to original source of
    constraint
  ], ...;
  fhir:ElementDefinition.mustSupport [ boolean ]; # 0..1 If the element must be supported
  fhir:ElementDefinition.isModifier [ boolean ]; # 0..1 If this modifies the meaning of other elements
  fhir:ElementDefinition.isModifierReason [ string ]; # 0..1 Reason that this element is marked as a modifier
  fhir:ElementDefinition.isSummary [ boolean ]; # 0..1 Include when _summary = true?
  fhir:ElementDefinition.binding [ # 0..1 ValueSet details if this is coded
    fhir:ElementDefinition.binding.strength [ code ]; # 1..1 required | extensible | preferred | example
    fhir:ElementDefinition.binding.description [ string ]; # 0..1 Human explanation of the value set
    fhir:ElementDefinition.binding.valueSet [ canonical(ValueSet) ]; # 0..1 Source of value set
  ];
  fhir:ElementDefinition.mapping [ # 0..* Map element to another set of definitions
    fhir:ElementDefinition.mapping.identity [ id ]; # 1..1 Reference to mapping declaration
    fhir:ElementDefinition.mapping.language [ code ]; # 0..1 Computable language of mapping
    fhir:ElementDefinition.mapping.map [ string ]; # 1..1 Details of the mapping
    fhir:ElementDefinition.mapping.comment [ string ]; # 0..1 Comments about the mapping or its use
  ], ...;
]
```

Changes since Release 3

ElementDefinition

ElementDefinition.representation | Change value set from http://hl7.org/fhir/ValueSet/property-representation to http://hl7.org/fhir/ValueSet/property-representation|4.0.1

ElementDefinition.sliceIsConstraining | Added Element

ElementDefinition.slicing.discriminator.type | Change value set from http://hl7.org/fhir/ValueSet/discriminator-type to http://hl7.org/fhir/ValueSet/discriminator-type|4.0.1

ElementDefinition.slicing.ordered | Default Value "false" removed

ElementDefinition.slicing.rules | Change value set from http://hl7.org/fhir/ValueSet/resource-slicing-rules to http://hl7.org/fhir/ValueSet/resource-slicing-rules|4.0.1

ElementDefinition.type.profile | Max Cardinality changed from 1 to * Type changed from uri to canonical(StructureDefinition | ImplementationGuide)

ElementDefinition.type.targetProfile | Max Cardinality changed from 1 to * Type changed from uri to canonical(StructureDefinition | ImplementationGuide)

ElementDefinition.type.aggregation | Change value set from http://hl7.org/fhir/ValueSet/resource-aggregation-mode to http://hl7.org/fhir/ValueSet/resource-aggregation-mode|4.0.1

ElementDefinition.type.versioning | Change value set from http://hl7.org/fhir/ValueSet/reference-version-rules to http://hl7.org/fhir/ValueSet/reference-version-rules|4.0.1 Default Value "either" removed

ElementDefinition.defaultValue[x] | Add Types canonical, url, uuid, ContactDetail, Contributor, DataRequirement, Expression, ParameterDefinition, RelatedArtifact, TriggerDefinition, UsageContext, Dosage

ElementDefinition.fixed[x] | Add Types canonical, url, uuid, ContactDetail, Contributor, DataRequirement, Expression, ParameterDefinition, RelatedArtifact, TriggerDefinition, UsageContext, Dosage

ElementDefinition.pattern[x] | Add Types canonical, url, uuid, ContactDetail, Contributor, DataRequirement, Expression, ParameterDefinition, RelatedArtifact, TriggerDefinition, UsageContext, Dosage

ElementDefinition.example.value[x] | Add Types canonical, url, uuid, ContactDetail, Contributor, DataRequirement, Expression, ParameterDefinition, RelatedArtifact, TriggerDefinition, UsageContext, Dosage

ElementDefinition.constraint.severity | Change value set from http://hl7.org/fhir/ValueSet/constraint-severity to http://hl7.org/fhir/ValueSet/constraint-severity|4.0.1

ElementDefinition.constraint.expression | Min Cardinality changed from 1 to 0

ElementDefinition.constraint.source | Type changed from uri to canonical(StructureDefinition)

ElementDefinition.mustSupport | Default Value "false" removed

ElementDefinition.isModifier | Default Value "false" removed

ElementDefinition.isModifierReason | Added Element

ElementDefinition.isSummary | Default Value "false" removed

ElementDefinition.binding.strength | Change value set from http://hl7.org/fhir/ValueSet/binding-strength to http://hl7.org/fhir/ValueSet/binding-strength|4.0.1

ElementDefinition.binding.valueSet | Renamed from valueSet[x] to valueSet Add Type canonical(ValueSet) Remove Types uri, Reference(ValueSet)

ElementDefinition.mapping.language | Change value set from http://hl7.org/fhir/ValueSet/mimetypes to http://hl7.org/fhir/ValueSet/mimetypes|4.0.1

See the Full Difference for further information

Constraints

id | Level | Location | Description | Expression
eld-1 | Rule | ElementDefinition.slicing | If there are no discriminators, there must be a definition | discriminator.exists() or description.exists()
eld-2 | Rule | (base) | Min <= Max | min.empty() or max.empty() or (max = '*') or iif(max != '*', min <= max.toInteger())
eld-3 | Rule | ElementDefinition.max | Max SHALL be a number or "*" | empty() or ($this = '*') or (toInteger() >= 0)
eld-4 | Rule | ElementDefinition.type | Aggregation may only be specified if one of the allowed types for the element is a reference | aggregation.empty() or (code = 'Reference') or (code = 'canonical')
eld-5 | Rule | (base) | if the element definition has a contentReference, it cannot have type, defaultValue, fixed, pattern, example, minValue, maxValue, maxLength, or binding | contentReference.empty() or (type.empty() and defaultValue.empty() and fixed.empty() and pattern.empty() and example.empty() and minValue.empty() and maxValue.empty() and maxLength.empty() and binding.empty())
eld-6 | Rule | (base) | Fixed value may only be specified if there is one type | fixed.empty() or (type.count() <= 1)
eld-7 | Rule | (base) | Pattern may only be specified if there is one type | pattern.empty() or (type.count() <= 1)
eld-8 | Rule | (base) | Pattern and fixed are mutually exclusive | pattern.empty() or fixed.empty()
eld-11 | Rule | (base) | Binding can only be present for coded elements, string, and uri | binding.empty() or type.code.empty() or type.select((code = 'code') or (code = 'Coding') or (code='CodeableConcept') or (code = 'Quantity') or (code = 'string') or (code = 'uri')).exists()
eld-12 | Rule | ElementDefinition.binding | ValueSet SHALL start with http:// or https:// or urn: | valueSet.exists() implies (valueSet.startsWith('http:') or valueSet.startsWith('https') or valueSet.startsWith('urn:'))
eld-13 | Rule | (base) | Types must be unique by code | type.select(code).isDistinct()
eld-14 | Rule | (base) | Constraints must be unique by key | constraint.select(key).isDistinct()
eld-15 | Rule | (base) | default value and meaningWhenMissing are mutually exclusive | defaultValue.empty() or meaningWhenMissing.empty()
eld-16 | Rule | (base) | sliceName must be composed of proper tokens separated by "/" | sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\/\\-_\\[\\]\\@]+$')
eld-17 | Rule | ElementDefinition.type | targetProfile is only allowed if the type is Reference or canonical | (code='Reference' or code = 'canonical') or targetProfile.empty()
eld-18 | Rule | (base) | Must have a modifier reason if isModifier = true | (isModifier.exists() and isModifier) implies isModifierReason.exists()
eld-19 | Rule | (base) | Element names cannot include some special characters | path.matches('[^\\s\\.,:;\\\'"\\/|?!@#$%&*()\\[\\]{}]{1,64}(\\.[^\\s\\.,:;\\\'"\\/|?!@#$%&*()\\[\\]{}]{1,64}(\\[x\\])?(\\:[^\\s\\.]+)?)*')
eld-20 | Warning | (base) | Element names should be simple alphanumerics with a max of 64 characters, or code generation tools may be broken | path.matches('[A-Za-z][A-Za-z0-9]*(\\.[a-z][A-Za-z0-9]*(\\[x])?)*')
eld-21 | Warning | ElementDefinition.constraint | Constraints should have an expression or else validators will not be able to enforce them | expression.exists()
eld-22 | Rule | (base) | sliceIsConstraining can only appear if slicename is present | sliceIsConstraining.exists() implies sliceName.exists()

2.30.0.2 Use of ElementDefinition.path

The path element is the most important property of the element definition. It both names the element, and locates the element within a hierarchy defined within a particular context. Within the FHIR specification, there is only one original definition for each path. This is the master definition to which all the other definitions with the same path must conform.

All elements defined within the FHIR specification itself are defined within a StructureDefinition that defines a resource, or a datatype. This defines the identity of the element and provides the context in which the meaning of the element is understood. When Elements are defined, the following rules apply:

• Element names (the parts of a path delineated by the '.' character) SHALL NOT contain whitespace (i.e. Unicode characters marked as whitespace)

• Element names SHALL NOT contain the characters ,:;'"/|?!@#$%^&*()[]{}

• Element names SHOULD not contain non-ASCII characters

• Element names SHALL NOT exceed 64 characters in length

• Element paths cannot imply elements that are not explicitly defined i.e. a.b.c.d cannot be defined unless a.b.c is explicitly defined

• By convention, each path starts with an uppercase letter (type) but all the element names that follow this are lowercase (not type names). All resources and data types (except for Primitive data types) follow this convention, but logical models are not required to do so

Elements may be defined in:

• StructureDefinitions of kind = resource, complex-type or primitive-type, where derivation = specialization. These are either Resources or Data Types defined in the specification

• StructureDefinitions of kind = logical

• Data Elements

StructureDefinitions with derivation = constraint (i.e. Resource and Data Type profiles) are not allowed to define or include ElementDefinitions with a path not defined within the base type definition from which they derive (e.g. in the FHIR specification).

2.30.0.3 ElementDefinition.id

In addition to the path, every ElementDefinition SHALL have a populated id, and the id SHALL have a unique value populated by following this algorithm:

• The id will be constructed as a dot separated string, each part corresponding to a token in the path

• For each token in the path, use the syntax pathpart:slicename/reslicename

• For type choice elements, the id reflects the type slice. e.g. For path = Patient.deceasedBoolean, the id is Patient.deceased[x]:deceasedBoolean

Note that in a profile with no slices, this id will match the path exactly and entirely. id values constructed in this fashion are unique, and persistent, and may be used as the target of external references into the definition, where necessary.

2.30.0.4 Interpretation of ElementDefinition in different contexts

The data type ElementDefinition is used in StructureDefinition. The way its elements are to be used and interpreted depends on the context:

ElementDefinition field	Type definition, first element	Type definition, following elements	Constraint Definition, first element	Constraint Definition, following elements
sliceName	prohibited	prohibited	prohibited	required for slices, else prohibited
label	optional	optional	recommended	recommended
code	optional	optional	optional	optional
slicing	prohibited	prohibited	prohibited	optional
short/definition	required	required	required‡	required‡
requirements/ comments/alias	prohibited	optional	prohibited‡	optional‡
base	snapshot: required differential: optional	snapshot: required differential: optional	required	required
type	required	required	optional	optional
nameReference	prohibited	optional	prohibited	optional
min/max	optional (irrelevant)	required	optional	optional†
defaultValue[x]	prohibited	optional	prohibited	optional†
meaningWhenMissing	prohibited	optional	prohibited	optional†
fixed[x]	prohibited	prohibited	prohibited	optional
pattern[x]	prohibited	prohibited	prohibited	optional
example[x]	prohibited	optional	prohibited	optional
minValue[x]	prohibited	prohibited	prohibited	optional
maxValue[x]	prohibited	prohibited	prohibited	optional
maxLength	prohibited	prohibited	prohibited	optional
mustSupport	prohibited	prohibited	optional	optional
isModifier	prohibited	optional	prohibited	optional†
isSummary	prohibited	optional	prohibited	optional†
binding	prohibited	optional	prohibited	optional
constraint	optional	optional	optionalΔ	optionalΔ
condition	prohibited	optional	prohibited	optionalΔ
mapping	optional	optional	optionalΔ	optionalΔ

Notes:

• Type definition: A StructureDefinition without a baseDefinition element, or where the derivation type is 'specialization'

• Constraint definition: A StructureDefinition with a baseDefinition element and a derivation of 'constraint' - e.g. a definition of a structure that constrains another base structure, referring to the differential portion

• †: The element's presence, and value, must match the definition in the base definition

• ‡: The element content must be consistent with that matching element in the base definition

• Δ: Additional constraints and mappings can be defined, but they do not replace the ones in the base definition

The use of Path and type depends more deeply on the context where the ElementDefinition is used:

Context	path (1st element)	path (following elements)	type (1st element)
Base definition of a data type (example: Quantity - XML, JSON)	Name of the type	Path inside the datatype	Element
A constrained data type (example: Money - XML, JSON)	Name of the base type	Path inside the datatype	Name of the base type
Base definition of a resource (example: Patient - XML, JSON)	The name of the resource	Path inside the resource	DomainResource or sometimes Resource
Constraint on a resource (example: DAF Patient - XML, JSON)	The name of the resource	Path inside the resource (including into the data types)	The name of the resource
Base Extension (which is a standard data type) (example: Extension - XML, JSON)	Extension	Extension.value[x] or Extension.extension	Extension
A defined Extension (example: Extension - XML, JSON)	Extension	Extension.value[x] or Extension.extension (for complex extensions)	Extension

There are additional notes about the use of ElementDefinition when defining Extensions on the Defining Extensions page.

2.30.0.4.1 Rules about Slicing

For a description of slicing, see Slicing

• Slicing is only allowed when constraining an existing structure

• slicing can only be used on the first repetition of an element. This first element that declares slicing is considered to be the slicing entry

• All elements following the first repeat that containing a slicing SHALL have a sliceName

• The special slice name @default applies to all entries that are not in any other slice

• The first entry (the one having the slicing information) is understood to be the set of constraints that apply to all slices and entries, whether they have a defined slice or not It's use follows the "normal case", except: slicing must be present min governs the number of total occurrences of the sliced element including the number of occurrences in the open portion of the slice (individual slices may have a different min value).

2.30.0.4.2 Constraining elements with a choice of Type

Elements that allow a choice of multiple types can be constrained. In principle, there are two different types of constraints to apply:

• A constraint that applies to the element as a whole - e.g. as restricting the cardinality, or limiting the choice of types

• A constraint that applies to the use of a particular type - e.g. value set binding

When constraining elements with multiple types, the following rules apply:

• Constraints limiting the acceptable list of types must be applied to the original "[x]" element as this is where the list of acceptable types is defined

• The inclusion of a type specific path (such as "Patient.deceasedBoolean") SHALL NOT be interpreted as constraining allowed types, but instead, it constrains the use of a particular type

• the original element SHALL always be represented in a snapshot; the type specific variants are only represented when needed

2.30.0.4.3 Rules about min and max

• If there is no StructureDefinition.baseDefinition: min and max are always required

• Otherwise, in StructureDefinition.differential: min and max are always optional; if they are not present, they default to the min and max from the base definition

• In StructureDefinition.snapshot: min and max are always required

2.30.0.4.4 Rules about Aggregation

• If an aggregationMode is present in the definition, the 'reference' element SHALL be present and have a value and the target of the reference SHALL be aggregated as defined

• If type.versioning is present in the definition, the 'reference' element SHALL be present and have a value and the reference SHALL be populated as the versioning constraint dictates.

2.30.0.4.5 Missing Elements

Most elements have a minimum cardinality of 0, which means that they may be missing from a resource when it is exchanged between systems. Generally, when an element is missing, all that an application processing the resource can say about the element is that the value is unknown - it may have a correct value, but it has not been provided for security or workflow reasons. On the other hand, it might not have a value at all. All the application can say is that the value is unknown.

This also applies when the element is present, but has no value or child elements, and only has extensions instead.

However, for some elements, this specification makes specific rules about what it means if the element is missing. Constraints on other structures cannot change the missing meaning of an element. Here is a list of all elements with a default value or a missing meaning:

• DataRequirement.subject[x]: Patient

• Period.end: If the end of the period is missing, it means that the period is ongoing

• Quantity.comparator: If there is no comparator, then there is no modification of the value

• SampledData.factor: If no factor is assigned, the data is not adjusted before adding to the origin

• Timing.repeat.frequency: If no frequency is stated, the assumption is that the event occurs once per period, but systems SHOULD always be specific about this

• ElementDefinition.slicing.ordered: Order is not required unless specified

• ElementDefinition.mustSupport: In a base type, where the element is being defined for the first time, the element is assumed to be mustSupport = false. In a profile (a constraint on an existing type), if mustSupport is not specific in either the differentil or the snapshot, the mustSupport value is not changed fro the base definition. Not, though, that the snapshot SHOULD always populate the mustSuppot value

• ElementDefinition.isModifier: An element is not a modifier element unless it is explicitly specified to be one

• ElementDefinition.isSummary: An element is not included in the summary unless it is explicitly specified to be so

• ActivityDefinition.subject[x]: Patient

• BodyStructure.active: This resource is generally assumed to be active if no value is provided for the active element

• CarePlan.activity.detail.doNotPerform: If missing indicates that the described activity is one that should be engaged in when following the plan.

• Communication.priority: If missing, this communication should be treated with normal priority

• CommunicationRequest.priority: If missing, this task should be performed with normal priority

• CommunicationRequest.doNotPerform: If do not perform is not specified, the request is a positive request e.g. "do perform"

• ConceptMap.group.element.target.dependsOn.system: The code is in the source (dependsOn) or target (product) system

• Consent.provision.actor: There is no specific actor associated with the exception

• Consent.provision.action: all actions

• Consent.provision.data: all data

• DeviceRequest.priority: If missing, normal priority

• EventDefinition.subject[x]: Patient

• EvidenceVariable.characteristic.exclude: False

• FamilyMemberHistory.estimatedAge: It is unknown whether the age is an estimate or not

• Group.active: This resource is generally assumed to be active if no value is provided for the active element

• Group.member.period: The member is in the group at this time

• Group.member.inactive: Members are considered active unless explicitly specified otherwise

• HealthcareService.active: This resource is generally assumed to be active if no value is provided for the active element

• Immunization.isSubpotent: By default, a dose should be considered to be potent.

• InsurancePlan.status: Default interpretation is active.

• Library.subject[x]: Patient

• Linkage.active: This resource is generally assumed to be active if no value is provided for the active element

• List.entry.deleted: List items are generally only treated as deleted when this element explicitly carries a value of true. Systems SHOULD always populate this value when mode is 'changes'

• Measure.subject[x]: Patient

• NamingSystem.uniqueId.preferred: If there are multiple ids, and one is labeled "preferred", then the assumption is that the others are not preferred. In the absence of any id marked as preferred, no inference can be drawn

• Organization.active: This resource is generally assumed to be active if no value is provided for the active element

• OrganizationAffiliation.active: This resource is generally assumed to be active if no value is provided for the active element

• Patient.active: This resource is generally assumed to be active if no value is provided for the active element

• PlanDefinition.subject[x]: Patient

• PlanDefinition.action.subject[x]: Patient

• Practitioner.active: This resource is generally assumed to be active if no value is provided for the active element

• PractitionerRole.active: This resource is generally assumed to be active if no value is provided for the active element

• Questionnaire.item.required: Items are generally assumed not to be required unless explicitly specified. Systems SHOULD always populate this value

• Questionnaire.item.repeats: Items are generally assumed not to repeat unless explicitly specified. Systems SHOULD always populate this value

• Questionnaire.item.answerOption.initialSelected: Only selected items explicitly marked to be selected

• RelatedPerson.active: This resource is generally assumed to be active if no value is provided for the active element

• RequestGroup.priority: If missing, this request should be performed with normal priority

• ResearchDefinition.subject[x]: Patient

• ResearchElementDefinition.subject[x]: Patient

• ResearchElementDefinition.characteristic.exclude: False

• Schedule.active: This resource is generally assumed to be active if no value is provided for the active element

• ServiceRequest.priority: If missing, this task should be performed with normal priority

• ServiceRequest.doNotPerform: If missing, the request is a positive request e.g. "do perform"

• Slot.overbooked: If overbooked is missing, systems may assume that there are still appointments available

• Substance.instance: If this element is not present, then the substance resource describes a kind of substance

• SupplyRequest.priority: If missing, this task should be performed with normal priority

• Task.priority: If missing, this task should be performed with normal priority

• TestScript.origin.profile: FHIR-Client

• TestScript.destination.profile: FHIR-Server

• ValueSet.expansion.offset: Paging is not being used

• ValueSet.expansion.contains.abstract: If this is not populated, systems can assume that the concept it not abstract. This SHALL always be populated for abstract concepts

This specification does not define any default values for resources or data types because:

• The value must be known by all implementations

• When an element has a default value, it can never be unknown - e.g. it is implicitly mandatory

• The default value can never be changed

• The presence of a default value interacts with minimum cardinality and the :missing search token in ways that create confusion for implementations

Note that default values can be defined in Logical Models.

2.30.1 Examples

URL: https://hl7.org/fhir/R4/elementdefinition-examples.html

2.30.1 Element Definition Examples

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative

For ElementDefinition examples, see the extensive StructureDefinition example list.

2.30.2 Detailed Descriptions

URL: https://hl7.org/fhir/R4/elementdefinition-definitions.html

2.30.2 Element Definition Detailed Descriptions

FHIR Infrastructure Work Group | Maturity Level: Normative | Standards Status: Normative

| This page has been approved as part of an ANSI standard. See the Infrastructure Package for further details.

2.30.2.1 ElementDefinition

ElementDefinition
Standards Status | This element has a standards status of "Normative"
Element Id | ElementDefinition
Definition | Captures constraints on each element within the resource, profile, or extension.
Cardinality | 0..*
Type | BackboneElement
Summary | true
Invariants | Defined on this element eld-2 Rule Min <= Max min.empty() or max.empty() or (max = '*') or iif(max != '*', min <= max.toInteger()) eld-5 Rule if the element definition has a contentReference, it cannot have type, defaultValue, fixed, pattern, example, minValue, maxValue, maxLength, or binding contentReference.empty() or (type.empty() and defaultValue.empty() and fixed.empty() and pattern.empty() and example.empty() and minValue.empty() and maxValue.empty() and maxLength.empty() and binding.empty()) eld-6 Rule Fixed value may only be specified if there is one type fixed.empty() or (type.count() <= 1) eld-7 Rule Pattern may only be specified if there is one type pattern.empty() or (type.count() <= 1) eld-8 Rule Pattern and fixed are mutually exclusive pattern.empty() or fixed.empty() eld-11 Rule Binding can only be present for coded elements, string, and uri binding.empty() or type.code.empty() or type.select((code = 'code') or (code = 'Coding') or (code='CodeableConcept') or (code = 'Quantity') or (code = 'string') or (code = 'uri')).exists() eld-13 Rule Types must be unique by code type.select(code).isDistinct() eld-14 Rule Constraints must be unique by key constraint.select(key).isDistinct() eld-15 Rule default value and meaningWhenMissing are mutually exclusive defaultValue.empty() or meaningWhenMissing.empty() eld-16 Rule sliceName must be composed of proper tokens separated by "/" sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\/\\-_\\[\\]\\@]+$') eld-18 Rule Must have a modifier reason if isModifier = true (isModifier.exists() and isModifier) implies isModifierReason.exists() eld-19 Rule Element names cannot include some special characters path.matches('[^\\s\\.,:;\\\'"\\/|?!@#$%&*()\\[\\]{}]{1,64}(\\.[^\\s\\.,:;\\\'"\\/|?!@#$%&*()\\[\\]{}]{1,64}(\\[x\\])?(\\:[^\\s\\.]+)?)*') eld-20 Warning Element names should be simple alphanumerics with a max of 64 characters, or code generation tools may be broken path.matches('[A-Za-z][A-Za-z0-9]*(\\.[a-z][A-Za-z0-9]*(\\[x])?)*') eld-22 Rule sliceIsConstraining can only appear if slicename is present sliceIsConstraining.exists() implies sliceName.exists()

Defined on this element
eld-2 | Rule | Min <= Max | min.empty() or max.empty() or (max = '*') or iif(max != '*', min <= max.toInteger())
eld-5 | Rule | if the element definition has a contentReference, it cannot have type, defaultValue, fixed, pattern, example, minValue, maxValue, maxLength, or binding | contentReference.empty() or (type.empty() and defaultValue.empty() and fixed.empty() and pattern.empty() and example.empty() and minValue.empty() and maxValue.empty() and maxLength.empty() and binding.empty())
eld-6 | Rule | Fixed value may only be specified if there is one type | fixed.empty() or (type.count() <= 1)
eld-7 | Rule | Pattern may only be specified if there is one type | pattern.empty() or (type.count() <= 1)
eld-8 | Rule | Pattern and fixed are mutually exclusive | pattern.empty() or fixed.empty()
eld-11 | Rule | Binding can only be present for coded elements, string, and uri | binding.empty() or type.code.empty() or type.select((code = 'code') or (code = 'Coding') or (code='CodeableConcept') or (code = 'Quantity') or (code = 'string') or (code = 'uri')).exists()
eld-13 | Rule | Types must be unique by code | type.select(code).isDistinct()
eld-14 | Rule | Constraints must be unique by key | constraint.select(key).isDistinct()
eld-15 | Rule | default value and meaningWhenMissing are mutually exclusive | defaultValue.empty() or meaningWhenMissing.empty()
eld-16 | Rule | sliceName must be composed of proper tokens separated by "/" | sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\/\\-_\\[\\]\\@]+$')
eld-18 | Rule | Must have a modifier reason if isModifier = true | (isModifier.exists() and isModifier) implies isModifierReason.exists()
eld-19 | Rule | Element names cannot include some special characters | path.matches('[^\\s\\.,:;\\\'"\\/|?!@#$%&*()\\[\\]{}]{1,64}(\\.[^\\s\\.,:;\\\'"\\/|?!@#$%&*()\\[\\]{}]{1,64}(\\[x\\])?(\\:[^\\s\\.]+)?)*')
eld-20 | Warning | Element names should be simple alphanumerics with a max of 64 characters, or code generation tools may be broken | path.matches('[A-Za-z][A-Za-z0-9]*(\\.[a-z][A-Za-z0-9]*(\\[x])?)*')
eld-22 | Rule | sliceIsConstraining can only appear if slicename is present | sliceIsConstraining.exists() implies sliceName.exists()
ElementDefinition.path
Element Id | ElementDefinition.path
Definition | The path identifies the element and is expressed as a "."-separated list of ancestor elements, beginning with the name of the resource or extension.
Cardinality | 1..1
Type | string
Summary | true
ElementDefinition.representation
Element Id | ElementDefinition.representation
Definition | Codes that define how this element is represented in instances, when the deviation varies from the normal case.
Cardinality | 0..*
Terminology Binding | PropertyRepresentation (Required)
Type | code
Summary | true
Comments | In resources, this is rarely used except for special cases where the representation deviates from the normal, and can only be done in the base standard (and profiles must reproduce what the base standard does). This element is used quite commonly in Logical models when the logical models represent a specific serialization format (e.g. CDA, v2 etc.).
ElementDefinition.sliceName
Element Id | ElementDefinition.sliceName
Definition | The name of this element definition slice, when slicing is working. The name must be a token with no dots or spaces. This is a unique name referring to a specific set of constraints applied to this element, used to provide a name to different slices of the same element.
Cardinality | 0..1
Type | string
Requirements | May also be used for code generation purposes.
Summary | true
Comments | The name SHALL be unique within the structure within the context of the constrained resource element. (Though to avoid confusion, uniqueness across all elements is recommended.).
ElementDefinition.sliceIsConstraining
Element Id | ElementDefinition.sliceIsConstraining
Definition | If true, indicates that this slice definition is constraining a slice definition with the same name in an inherited profile. If false, the slice is not overriding any slice in an inherited profile. If missing, the slice might or might not be overriding a slice in an inherited profile, depending on the sliceName.
Cardinality | 0..1
Type | boolean
Requirements | Allows detection of a situation where an ancestor profile adds or removes slicing with the same name where that might impact the child profile.
Summary | true
Comments | If set to true, an ancestor profile SHALL have a slicing definition with this name. If set to false, no ancestor profile is permitted to have a slicing definition with this name.
ElementDefinition.label
Element Id | ElementDefinition.label
Definition | A single preferred label which is the text to display beside the element indicating its meaning or to use to prompt for the element in a user display or form.
Cardinality | 0..1
Type | string
Summary | true
Comments | See also the extension (http://hl7.org/fhir/StructureDefinition/elementdefinition-question)[extension-elementdefinition-question.html].

ElementDefinition.code
Element Id | ElementDefinition.code
Definition | A code that has the same meaning as the element in a particular terminology.
Cardinality | 0..*
Terminology Binding | LOINC Codes (Example)
Type | Coding
Requirements | Links the meaning of an element to an external terminology, and is very useful for searching and indexing.
Summary | true
Comments | The concept SHALL be properly aligned with the data element definition and other constraints, as defined in the code system, including relationships, of any code listed here. Where multiple codes exist in a terminology that could correspond to the data element, the most granular code(s) should be selected, so long as they are not more restrictive than the data element itself. The mappings may be used to provide more or less granular or structured equivalences in the code system.
LOINC Code | LOINC_NUM
ElementDefinition.slicing
Element Id | ElementDefinition.slicing
Definition | Indicates that the element is sliced into a set of alternative definitions (i.e. in a structure definition, there are multiple different constraints on a single element in the base resource). Slicing can be used in any resource that has cardinality ..* on the base resource, or any resource with a choice of types. The set of slices is any elements that come after this in the element sequence that have the same path, until a shorter path occurs (the shorter path terminates the set).
Cardinality | 0..1
Summary | true
Comments | The first element in the sequence, the one that carries the slicing, is the definition that applies to all the slices. This is based on the unconstrained element, but can apply any constraints as appropriate. This may include the common constraints on the children of the element.
Invariants | Defined on this element eld-1 Rule If there are no discriminators, there must be a definition discriminator.exists() or description.exists()
Defined on this element
eld-1 | Rule | If there are no discriminators, there must be a definition | discriminator.exists() or description.exists()
ElementDefinition.slicing.discriminator
Element Id | ElementDefinition.slicing.discriminator
Definition | Designates which child elements are used to discriminate between the slices when processing an instance. If one or more discriminators are provided, the value of the child elements in the instance data SHALL completely distinguish which slice the element in the resource matches based on the allowed values for those elements in each of the slices.
Cardinality | 0..*
Summary | true
Comments | If there is no discriminator, the content is hard to process, so this should be avoided.
ElementDefinition.slicing.discriminator.type
Element Id | ElementDefinition.slicing.discriminator.type
Definition | How the element value is interpreted when discrimination is evaluated.
Cardinality | 1..1
Terminology Binding | DiscriminatorType (Required)
Type | code
Summary | true
ElementDefinition.slicing.discriminator.path
Element Id | ElementDefinition.slicing.discriminator.path
Definition | A FHIRPath expression, using the simple subset of FHIRPath, that is used to identify the element on which discrimination is based.
Cardinality | 1..1
Type | string
Summary | true
Comments | The only FHIRPath functions that are allowed are as(type), resolve(), and extension(url).
ElementDefinition.slicing.description
Element Id | ElementDefinition.slicing.description
Definition | A human-readable text description of how the slicing works. If there is no discriminator, this is required to be present to provide whatever information is possible about how the slices can be differentiated.
Cardinality | 0..1
Type | string
Summary | true
Comments | If it's really not possible to differentiate them, the design should be re-evaluated to make the content usable.
Invariants | Affect this element eld-1 Rule If there are no discriminators, there must be a definition discriminator.exists() or description.exists()
Affect this element
eld-1 | Rule | If there are no discriminators, there must be a definition | discriminator.exists() or description.exists()
ElementDefinition.slicing.ordered
Element Id | ElementDefinition.slicing.ordered
Definition | If the matching elements have to occur in the same order as defined in the profile.
Cardinality | 0..1
Type | boolean
Meaning if Missing | Order is not required unless specified
Summary | true
Comments | Order should only be required when it is a pressing concern for presentation. Profile authors should consider making the order a feature of the rules about the narrative, not the rules about the data - requiring ordered data makes the profile much less re-usable.
ElementDefinition.slicing.rules

Element Id | ElementDefinition.slicing.rules
Definition | Whether additional slices are allowed or not. When the slices are ordered, profile authors can also say that additional slices are only allowed at the end.
Cardinality | 1..1
Terminology Binding | SlicingRules (Required)
Type | code
Summary | true
Comments | Allowing additional elements makes for a much for flexible template - it's open for use in wider contexts, but also means that the content of the resource is not closed, and applications have to decide how to handle content not described by the profile.
ElementDefinition.short
Element Id | ElementDefinition.short
Definition | A concise description of what this element means (e.g. for use in autogenerated summaries).
Cardinality | 0..1
Type | string
Summary | true
Comments | It is easy for a different short definition to change the meaning of an element and this can have nasty downstream consequences. Please be careful when providing short definitions in a profile.
ElementDefinition.definition
Element Id | ElementDefinition.definition
Definition | Provides a complete explanation of the meaning of the data element for human readability. For the case of elements derived from existing elements (e.g. constraints), the definition SHALL be consistent with the base definition, but convey the meaning of the element in the particular context of use of the resource. (Note: The text you are reading is specified in ElementDefinition.definition).
Cardinality | 0..1
Type | markdown
Requirements | To allow a user to state the usage of an element in a particular context.
Alternate Names | Description
Summary | true
Comments | It is easy for a different definition to change the meaning of an element and this can have nasty downstream consequences. Please be careful when providing definitions in a profile.
LOINC Code | TERM DEFINITION/DESCRIPTION(S)
ElementDefinition.comment
Element Id | ElementDefinition.comment
Definition | Explanatory notes and implementation guidance about the data element, including notes about how to use the data properly, exceptions to proper use, etc. (Note: The text you are reading is specified in ElementDefinition.comment).
Cardinality | 0..1
Type | markdown
Summary | true
Comments | If it is possible to capture usage rules using constraints, that mechanism should be used in preference to this element.
LOINC Code | COMMENTS
ElementDefinition.requirements
Element Id | ElementDefinition.requirements
Definition | This element is for traceability of why the element was created and why the constraints exist as they do. This may be used to point to source materials or specifications that drove the structure of this element.
Cardinality | 0..1
Type | markdown
Summary | true
Comments | This element does not describe the usage of the element (that's done in comments), rather it's for traceability of why the element is either needed or why the constraints exist as they do. This may be used to point to source materials or specifications that drove the structure of this data element.
ElementDefinition.alias
Element Id | ElementDefinition.alias
Definition | Identifies additional names by which this element might also be known.
Cardinality | 0..*
Type | string
Requirements | Allows for better easier recognition of the element by multiple communities, including international communities.
Alternate Names | synonym; other name
Summary | true
LOINC Code | RELATED NAMES (only some of these apply)
ElementDefinition.min
Element Id | ElementDefinition.min
Definition | The minimum number of times this element SHALL appear in the instance.
Cardinality | 0..1
Type | unsignedInt
Summary | true
Invariants | Affect this element eld-2 Rule Min <= Max min.empty() or max.empty() or (max = '*') or iif(max != '*', min <= max.toInteger())
Affect this element
eld-2 | Rule | Min <= Max | min.empty() or max.empty() or (max = '*') or iif(max != '*', min <= max.toInteger())
ElementDefinition.max
Element Id | ElementDefinition.max

Definition | The maximum number of times this element is permitted to appear in the instance.
Cardinality | 0..1
Type | string
Summary | true
Invariants | Defined on this element eld-3 Rule Max SHALL be a number or "*" empty() or ($this = '*') or (toInteger() >= 0) Affect this element eld-2 Rule Min <= Max min.empty() or max.empty() or (max = '*') or iif(max != '*', min <= max.toInteger())
Defined on this element
eld-3 | Rule | Max SHALL be a number or "*" | empty() or ($this = '*') or (toInteger() >= 0)
Affect this element
eld-2 | Rule | Min <= Max | min.empty() or max.empty() or (max = '*') or iif(max != '*', min <= max.toInteger())
ElementDefinition.base
Element Id | ElementDefinition.base
Definition | Information about the base definition of the element, provided to make it unnecessary for tools to trace the deviation of the element through the derived and related profiles. When the element definition is not the original definition of an element - i.g. either in a constraint on another type, or for elements from a super type in a snap shot - then the information in provided in the element definition may be different to the base definition. On the original definition of the element, it will be same.
Cardinality | 0..1
Summary | true
Comments | The base information does not carry any information that could not be determined from the path and related profiles, but making this determination requires both that the related profiles are available, and that the algorithm to determine them be available. For tooling simplicity, the base information must always be populated in element definitions in snap shots, even if it is the same.
ElementDefinition.base.path
Element Id | ElementDefinition.base.path
Definition | The Path that identifies the base element - this matches the ElementDefinition.path for that element. Across FHIR, there is only one base definition of any element - that is, an element definition on a StructureDefinition without a StructureDefinition.base.
Cardinality | 1..1
Type | string
Summary | true
ElementDefinition.base.min
Element Id | ElementDefinition.base.min
Definition | Minimum cardinality of the base element identified by the path.
Cardinality | 1..1
Type | unsignedInt
Summary | true
Comments | This is provided for consistency with max, and may affect code generation of mandatory elements of the base resource are generated differently (some reference implementations have done this).
ElementDefinition.base.max
Element Id | ElementDefinition.base.max
Definition | Maximum cardinality of the base element identified by the path.
Cardinality | 1..1
Type | string
Summary | true
Comments | This is provided to code generation, since the serialization representation in JSON differs depending on whether the base element has max > 1. Also, some forms of code generation may differ.
ElementDefinition.contentReference
Element Id | ElementDefinition.contentReference
Definition | Identifies an element defined elsewhere in the definition whose content rules should be applied to the current element. ContentReferences bring across all the rules that are in the ElementDefinition for the element, including definitions, cardinality constraints, bindings, invariants etc.
Cardinality | 0..1
Type | uri
Summary | true
Comments | ContentReferences can only be defined in specializations, not constrained types, and they cannot be changed and always reference the non-constrained definition.
Invariants | Affect this element eld-5 Rule if the element definition has a contentReference, it cannot have type, defaultValue, fixed, pattern, example, minValue, maxValue, maxLength, or binding contentReference.empty() or (type.empty() and defaultValue.empty() and fixed.empty() and pattern.empty() and example.empty() and minValue.empty() and maxValue.empty() and maxLength.empty() and binding.empty())
Affect this element
eld-5 | Rule | if the element definition has a contentReference, it cannot have type, defaultValue, fixed, pattern, example, minValue, maxValue, maxLength, or binding | contentReference.empty() or (type.empty() and defaultValue.empty() and fixed.empty() and pattern.empty() and example.empty() and minValue.empty() and maxValue.empty() and maxLength.empty() and binding.empty())
ElementDefinition.type
Element Id | ElementDefinition.type
Definition | The data type or resource that the value of this element is permitted to be.
Cardinality | 0..*
Summary | true
Comments | The Type of the element can be left blank in a differential constraint, in which case the type is inherited from the resource. Abstract types are not permitted to appear as a type when multiple types are listed. (I.e. Abstract types cannot be part of a choice).

Invariants | Defined on this element eld-4 Rule Aggregation may only be specified if one of the allowed types for the element is a reference aggregation.empty() or (code = 'Reference') or (code = 'canonical') eld-17 Rule targetProfile is only allowed if the type is Reference or canonical (code='Reference' or code = 'canonical') or targetProfile.empty() Affect this element eld-13 Rule Types must be unique by code type.select(code).isDistinct()
Defined on this element
eld-4 | Rule | Aggregation may only be specified if one of the allowed types for the element is a reference | aggregation.empty() or (code = 'Reference') or (code = 'canonical')
eld-17 | Rule | targetProfile is only allowed if the type is Reference or canonical | (code='Reference' or code = 'canonical') or targetProfile.empty()
Affect this element
eld-13 | Rule | Types must be unique by code | type.select(code).isDistinct()
LOINC Code | HL7_V2_DATATYPE (translation required), HL7_V3_DATATYPE (translation required)
ElementDefinition.type.code
Element Id | ElementDefinition.type.code
Definition | URL of Data type or Resource that is a(or the) type used for this element. References are URLs that are relative to http://hl7.org/fhir/StructureDefinition e.g. "string" is a reference to http://hl7.org/fhir/StructureDefinition/string. Absolute URLs are only allowed in logical models.
Cardinality | 1..1
Terminology Binding | FHIRDefinedType (Extensible)
Type | uri
Summary | true
Comments | If the element is a reference to another resource, this element contains "Reference", and the targetProfile element defines what resources can be referenced. The targetProfile may be a reference to the general definition of a resource (e.g. http://hl7.org/fhir/StructureDefinition/Patient).
LOINC Code | HL7_V2_DATATYPE (translation required), HL7_V3_DATATYPE (translation required)
ElementDefinition.type.profile
Element Id | ElementDefinition.type.profile
Definition | Identifies a profile structure or implementation Guide that applies to the datatype this element refers to. If any profiles are specified, then the content must conform to at least one of them. The URL can be a local reference - to a contained StructureDefinition, or a reference to another StructureDefinition or Implementation Guide by a canonical URL. When an implementation guide is specified, the type SHALL conform to at least one profile defined in the implementation guide.
Cardinality | 0..*
Type | canonical(StructureDefinition | ImplementationGuide)
Summary | true
Comments | It is possible to profile backbone element (e.g. part of a resource), using the profile-element extension.
ElementDefinition.type.targetProfile
Element Id | ElementDefinition.type.targetProfile
Definition | Used when the type is "Reference" or "canonical", and identifies a profile structure or implementation Guide that applies to the target of the reference this element refers to. If any profiles are specified, then the content must conform to at least one of them. The URL can be a local reference - to a contained StructureDefinition, or a reference to another StructureDefinition or Implementation Guide by a canonical URL. When an implementation guide is specified, the target resource SHALL conform to at least one profile defined in the implementation guide.
Cardinality | 0..*
Type | canonical(StructureDefinition | ImplementationGuide)
Summary | true
ElementDefinition.type.aggregation
Element Id | ElementDefinition.type.aggregation
Definition | If the type is a reference to another resource, how the resource is or can be aggregated - is it a contained resource, or a reference, and if the context is a bundle, is it included in the bundle.
Cardinality | 0..*
Terminology Binding | AggregationMode (Required)
Type | code
Summary | true
Comments | See Aggregation Rules for further clarification.
Invariants | Affect this element eld-4 Rule Aggregation may only be specified if one of the allowed types for the element is a reference aggregation.empty() or (code = 'Reference') or (code = 'canonical')
Affect this element
eld-4 | Rule | Aggregation may only be specified if one of the allowed types for the element is a reference | aggregation.empty() or (code = 'Reference') or (code = 'canonical')
ElementDefinition.type.versioning
Element Id | ElementDefinition.type.versioning
Definition | Whether this reference needs to be version specific or version independent, or whether either can be used.
Cardinality | 0..1
Terminology Binding | ReferenceVersionRules (Required)
Type | code
Summary | true
Comments | The base specification never makes a rule as to which form is allowed, but implementation guides may do this. See Aggregation Rules for further clarification.
ElementDefinition.defaultValue[x]
Element Id | ElementDefinition.defaultValue[x]
Definition | The value that should be used if there is no value stated in the instance (e.g. 'if not otherwise specified, the abstract is false').
Cardinality | 0..1
Type | *

[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
Comments | Specifying a default value means that the property can never been unknown - it must always have a value. Further, the default value can never be changed, or changed in constraints on content models. Defining default values creates many difficulties in implementation (e.g. when is a value missing?). For these reasons, default values are (and should be) used extremely sparingly. No default values are ever defined in the FHIR specification, nor can they be defined in constraints ("profiles") on data types or resources. This element only exists so that default values may be defined in logical models.
Invariants | Affect this element eld-15 Rule default value and meaningWhenMissing are mutually exclusive defaultValue.empty() or meaningWhenMissing.empty()
Affect this element
eld-15 | Rule | default value and meaningWhenMissing are mutually exclusive | defaultValue.empty() or meaningWhenMissing.empty()
ElementDefinition.meaningWhenMissing
Element Id | ElementDefinition.meaningWhenMissing
Definition | The Implicit meaning that is to be understood when this element is missing (e.g. 'when this element is missing, the period is ongoing').
Cardinality | 0..1
Type | markdown
Summary | true
Comments | Implicit meanings for missing values can only be specified on a resource, data type, or extension definition, and never in a profile that applies to one of these. An implicit meaning for a missing value can never be changed, and specifying one has the consequence that constraining its use in profiles eliminates use cases as possibilities, not merely moving them out of scope.
Invariants | Affect this element eld-15 Rule default value and meaningWhenMissing are mutually exclusive defaultValue.empty() or meaningWhenMissing.empty()
Affect this element
eld-15 | Rule | default value and meaningWhenMissing are mutually exclusive | defaultValue.empty() or meaningWhenMissing.empty()
ElementDefinition.orderMeaning
Element Id | ElementDefinition.orderMeaning
Definition | If present, indicates that the order of the repeating element has meaning and describes what that meaning is. If absent, it means that the order of the element has no meaning.
Cardinality | 0..1
Type | string
Summary | true
Comments | This element can only be asserted on repeating elements and can only be introduced when defining resources or data types. It can be further refined profiled elements but if absent in the base type, a profile cannot assert meaning.
ElementDefinition.fixed[x]
Element Id | ElementDefinition.fixed[x]
Definition | Specifies a value that SHALL be exactly the value for this element in the instance. For purposes of comparison, non-significant whitespace is ignored, and all values must be an exact match (case and accent sensitive). Missing elements/attributes must also be missing.
Cardinality | 0..1
Type | *
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
Comments | This is not recommended for Coding and CodeableConcept since these often have highly contextual properties such as version or display.
Invariants | Affect this element eld-5 Rule if the element definition has a contentReference, it cannot have type, defaultValue, fixed, pattern, example, minValue, maxValue, maxLength, or binding contentReference.empty() or (type.empty() and defaultValue.empty() and fixed.empty() and pattern.empty() and example.empty() and minValue.empty() and maxValue.empty() and maxLength.empty() and binding.empty()) eld-6 Rule Fixed value may only be specified if there is one type fixed.empty() or (type.count() <= 1) eld-8 Rule Pattern and fixed are mutually exclusive pattern.empty() or fixed.empty()
Affect this element
eld-5 | Rule | if the element definition has a contentReference, it cannot have type, defaultValue, fixed, pattern, example, minValue, maxValue, maxLength, or binding | contentReference.empty() or (type.empty() and defaultValue.empty() and fixed.empty() and pattern.empty() and example.empty() and minValue.empty() and maxValue.empty() and maxLength.empty() and binding.empty())
eld-6 | Rule | Fixed value may only be specified if there is one type | fixed.empty() or (type.count() <= 1)
eld-8 | Rule | Pattern and fixed are mutually exclusive | pattern.empty() or fixed.empty()
ElementDefinition.pattern[x]
Element Id | ElementDefinition.pattern[x]
Definition | Specifies a value that the value in the instance SHALL follow - that is, any value in the pattern must be found in the instance. Other additional values may be found too. This is effectively constraint by example. When pattern[x] is used to constrain a primitive, it means that the value provided in the pattern[x] must match the instance value exactly. When pattern[x] is used to constrain an array, it means that each element provided in the pattern[x] array must (recursively) match at least one element from the instance array. When pattern[x] is used to constrain a complex object, it means that each property in the pattern must be present in the complex object, and its value must recursively match -- i.e., If primitive: it must match exactly the pattern value If a complex object: it must match (recursively) the pattern value If an array: it must match (recursively) the pattern value.
Cardinality | 0..1
Type | *
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
Comments | Mostly used for fixing values of CodeableConcept. In general, pattern[x] is not intended for use with primitive types, where is has the same meaning as fixed[x].

Invariants | Affect this element eld-5 Rule if the element definition has a contentReference, it cannot have type, defaultValue, fixed, pattern, example, minValue, maxValue, maxLength, or binding contentReference.empty() or (type.empty() and defaultValue.empty() and fixed.empty() and pattern.empty() and example.empty() and minValue.empty() and maxValue.empty() and maxLength.empty() and binding.empty()) eld-7 Rule Pattern may only be specified if there is one type pattern.empty() or (type.count() <= 1) eld-8 Rule Pattern and fixed are mutually exclusive pattern.empty() or fixed.empty()
Affect this element
eld-5 | Rule | if the element definition has a contentReference, it cannot have type, defaultValue, fixed, pattern, example, minValue, maxValue, maxLength, or binding | contentReference.empty() or (type.empty() and defaultValue.empty() and fixed.empty() and pattern.empty() and example.empty() and minValue.empty() and maxValue.empty() and maxLength.empty() and binding.empty())
eld-7 | Rule | Pattern may only be specified if there is one type | pattern.empty() or (type.count() <= 1)
eld-8 | Rule | Pattern and fixed are mutually exclusive | pattern.empty() or fixed.empty()
ElementDefinition.example
Element Id | ElementDefinition.example
Definition | A sample value for this element demonstrating the type of information that would typically be found in the element.
Cardinality | 0..*
Summary | true
Comments | Examples will most commonly be present for data where it's not implicitly obvious from either the data type or value set what the values might be. (I.e. Example values for dates or quantities would generally be unnecessary.) If the example value is fully populated, the publication tool can generate an instance automatically.
ElementDefinition.example.label
Element Id | ElementDefinition.example.label
Definition | Describes the purpose of this example amoung the set of examples.
Cardinality | 1..1
Type | string
Summary | true
ElementDefinition.example.value[x]
Element Id | ElementDefinition.example.value[x]
Definition | The actual value for the element, which must be one of the types allowed for this element.
Cardinality | 1..1
Type | *
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
LOINC Code | EXMPL_ANSWERS
ElementDefinition.minValue[x]
Element Id | ElementDefinition.minValue[x]
Definition | The minimum allowed value for the element. The value is inclusive. This is allowed for the types date, dateTime, instant, time, decimal, integer, and Quantity.
Cardinality | 0..1
Type | date|dateTime|instant|time|decimal|integer|positiveInt|unsignedInt|Quantity
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
Comments | Except for date/date/instant, the type of the minValue[x] SHALL be the same as the specified type of the element. For the date/dateTime/instant values, the type of minValue[x] SHALL be either the same, or a Duration which specifies a relative time limit to the current time. The duration value is positive, and is subtracted from the current clock to determine the minimum allowable value. A minimum value for a Quantity is interpreted as an canonical minimum - e.g. you cannot provide 100mg if the minimum value is 10g.
ElementDefinition.maxValue[x]
Element Id | ElementDefinition.maxValue[x]
Definition | The maximum allowed value for the element. The value is inclusive. This is allowed for the types date, dateTime, instant, time, decimal, integer, and Quantity.
Cardinality | 0..1
Type | date|dateTime|instant|time|decimal|integer|positiveInt|unsignedInt|Quantity
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
Comments | Except for date/date/instant, the type of the maxValue[x] SHALL be the same as the specified type of the element. For the date/dateTime/instant values, the type of maxValue[x] SHALL be either the same, or a Duration which specifies a relative time limit to the current time. The duration value is positive, and is added to the current clock to determine the maximum allowable value. A maximum value for a Quantity is interpreted as an canonical maximum - e.g. you cannot provide 10g if the maximum value is 50mg.
ElementDefinition.maxLength
Element Id | ElementDefinition.maxLength
Definition | Indicates the maximum length in characters that is permitted to be present in conformant instances and which is expected to be supported by conformant consumers that support the element.
Cardinality | 0..1
Type | integer
Summary | true
Comments | Receivers are not required to reject instances that exceed the maximum length. The full length could be stored. In some cases, data might be truncated, though truncation should be undertaken with care and an understanding of the consequences of doing so. If not specified, there is no conformance expectation for length support.
ElementDefinition.condition
Element Id | ElementDefinition.condition
Definition | A reference to an invariant that may make additional statements about the cardinality or value in the instance.
Cardinality | 0..*

Type | id
Summary | true
ElementDefinition.constraint
Element Id | ElementDefinition.constraint
Definition | Formal constraints such as co-occurrence and other constraints that can be computationally evaluated within the context of the instance.
Cardinality | 0..*
Summary | true
Comments | Constraints should be declared on the "context" element - the lowest element in the hierarchy that is common to all nodes referenced by the constraint.
Invariants | Defined on this element eld-21 Warning Constraints should have an expression or else validators will not be able to enforce them expression.exists()
Defined on this element
eld-21 | Warning | Constraints should have an expression or else validators will not be able to enforce them | expression.exists()
ElementDefinition.constraint.key
Element Id | ElementDefinition.constraint.key
Definition | Allows identification of which elements have their cardinalities impacted by the constraint. Will not be referenced for constraints that do not affect cardinality.
Cardinality | 1..1
Type | id
Summary | true
Invariants | Affect this element eld-14 Rule Constraints must be unique by key constraint.select(key).isDistinct()
Affect this element
eld-14 | Rule | Constraints must be unique by key | constraint.select(key).isDistinct()
ElementDefinition.constraint.requirements
Element Id | ElementDefinition.constraint.requirements
Definition | Description of why this constraint is necessary or appropriate.
Cardinality | 0..1
Type | string
Summary | true
Comments | To be used if the reason for the constraint might not be intuitive to all implementers.
ElementDefinition.constraint.severity
Element Id | ElementDefinition.constraint.severity
Definition | Identifies the impact constraint violation has on the conformance of the instance.
Cardinality | 1..1
Terminology Binding | ConstraintSeverity (Required)
Type | code
Summary | true
Comments | This allows constraints to be asserted as "shall" (error) and "should" (warning).
ElementDefinition.constraint.human
Element Id | ElementDefinition.constraint.human
Definition | Text that can be used to describe the constraint in messages identifying that the constraint has been violated.
Cardinality | 1..1
Type | string
Summary | true
Comments | Should be expressed in business terms as much as possible.
ElementDefinition.constraint.expression
Element Id | ElementDefinition.constraint.expression
Definition | A FHIRPath expression of constraint that can be executed to see if this constraint is met.
Cardinality | 0..1
Type | string
Requirements | Used by validation tooling tests of the validity of the resource.
Summary | true
Comments | In the absense of an expression, the expression is likely not enforceable by validators, and might be missed by many systems.
ElementDefinition.constraint.xpath
Element Id | ElementDefinition.constraint.xpath
Definition | An XPath expression of constraint that can be executed to see if this constraint is met.
Cardinality | 0..1
Type | string
Requirements | Used in Schematron tests of the validity of the resource.
Summary | true
Comments | Elements SHALL use "f" as the namespace prefix for the FHIR namespace, and "x" for the xhtml namespace, and SHALL NOT use any other prefixes. Note: XPath is generally considered not useful because it does not apply to JSON and other formats and because of XSLT implementation issues, and may be removed in the future.
ElementDefinition.constraint.source
Element Id | ElementDefinition.constraint.source
Definition | A reference to the original source of the constraint, for traceability purposes.
Cardinality | 0..1
Type | canonical(StructureDefinition)
Summary | true
Comments | This is used when, e.g. rendering, where it is not useful to present inherited constraints when rendering the snapshot.
ElementDefinition.mustSupport

Element Id | ElementDefinition.mustSupport
Definition | If true, implementations that produce or consume resources SHALL provide "support" for the element in some meaningful way. If false, the element may be ignored and not supported. If false, whether to populate or use the data element in any way is at the discretion of the implementation.
Cardinality | 0..1
Type | boolean
Meaning if Missing | In a base type, where the element is being defined for the first time, the element is assumed to be mustSupport = false. In a profile (a constraint on an existing type), if mustSupport is not specific in either the differentil or the snapshot, the mustSupport value is not changed fro the base definition. Not, though, that the snapshot SHOULD always populate the mustSuppot value
Requirements | Allows a profile to set expectations for system capabilities beyond merely respecting cardinality constraints.
Summary | true
Comments | "Something useful" is context dependent and impossible to describe in the base FHIR specification. For this reason, tue mustSupport flag is never set to true by the FHIR specification itself - it is only set to true in profiles. A profile on a type can always make musSupport = true if it is false in the base type but cannot make mustSupport = false if it is true in the base type. This is done in Resource Profiles, where the profile labels an element as mustSupport=true. When a profile does this, it SHALL also make clear exactly what kind of "support" is required, as this can mean many things. Note that an element that has the property IsModifier is not necessarily a "key" element (e.g. one of the important elements to make use of the resource), nor is it automatically mustSupport - however both of these things are more likely to be true for IsModifier elements than for other elements.
ElementDefinition.isModifier
Element Id | ElementDefinition.isModifier
Definition | If true, the value of this element affects the interpretation of the element or resource that contains it, and the value of the element cannot be ignored. Typically, this is used for status, negation and qualification codes. The effect of this is that the element cannot be ignored by systems: they SHALL either recognize the element and process it, and/or a pre-determination has been made that it is not relevant to their particular system.
Cardinality | 0..1
Type | boolean
Meaning if Missing | An element is not a modifier element unless it is explicitly specified to be one
Requirements | Allows elements to be introduced into a specification that can't safely be ignored by applications that don't recognize them.
Summary | true
Comments | Only the definition of an element can set IsModifier true - either the specification itself or where an extension is originally defined. Once set, it cannot be changed in derived profiles. An element/extension that has isModifier=true SHOULD also have a minimum cardinality of 1, so that there is no lack of clarity about what to do if it is missing. If it can be missing, the definition SHALL make the meaning of a missing element clear.
ElementDefinition.isModifierReason
Element Id | ElementDefinition.isModifierReason
Definition | Explains how that element affects the interpretation of the resource or element that contains it.
Cardinality | 0..1
Type | string
Summary | true
ElementDefinition.isSummary
Element Id | ElementDefinition.isSummary
Definition | Whether the element should be included if a client requests a search with the parameter _summary=true.
Cardinality | 0..1
Type | boolean
Meaning if Missing | An element is not included in the summary unless it is explicitly specified to be so
Requirements | Allow clients to search through large resources quickly.
Summary | true
Comments | Some resources include a set of simple metadata, and some very large data. This element is used to reduce the quantity of data returned in searches. Note that servers may pre-cache summarized resources for optimal performance, so servers might not support per-profile use of the isSummary flag. When a request is made with _summary=true, serailisers only include elements marked as 'isSummary = true'. Other than Attachment.data, all data type properties are included in the summary form. In resource and data type definitions, if an element is at the root or has a parent that is 'mustSupport' and the minimum cardinality is 1 or the element is a modifier, it must be marked as isSummary=true.
ElementDefinition.binding
Element Id | ElementDefinition.binding
Definition | Binds to a value set if this element is coded (code, Coding, CodeableConcept, Quantity), or the data types (string, uri).
Cardinality | 0..1
Summary | true
Comments | For a CodeableConcept, when no codes are allowed - only text, use a binding of strength "required" with a description explaining that no coded values are allowed and what sort of information to put in the "text" element.
Invariants | Defined on this element eld-12 Rule ValueSet SHALL start with http:// or https:// or urn: valueSet.exists() implies (valueSet.startsWith('http:') or valueSet.startsWith('https') or valueSet.startsWith('urn:')) Affect this element eld-11 Rule Binding can only be present for coded elements, string, and uri binding.empty() or type.code.empty() or type.select((code = 'code') or (code = 'Coding') or (code='CodeableConcept') or (code = 'Quantity') or (code = 'string') or (code = 'uri')).exists()
Defined on this element
eld-12 | Rule | ValueSet SHALL start with http:// or https:// or urn: | valueSet.exists() implies (valueSet.startsWith('http:') or valueSet.startsWith('https') or valueSet.startsWith('urn:'))
Affect this element
eld-11 | Rule | Binding can only be present for coded elements, string, and uri | binding.empty() or type.code.empty() or type.select((code = 'code') or (code = 'Coding') or (code='CodeableConcept') or (code = 'Quantity') or (code = 'string') or (code = 'uri')).exists()
LOINC Code | ANSWER LIST, NORMATIVE ANSWER LIST

ElementDefinition.binding.strength
Element Id | ElementDefinition.binding.strength
Definition | Indicates the degree of conformance expectations associated with this binding - that is, the degree to which the provided value set must be adhered to in the instances.
Cardinality | 1..1
Terminology Binding | BindingStrength (Required)
Type | code
Alternate Names | conformance; extensibility
Summary | true
Comments | For further discussion, see Using Terminologies.
LOINC Code | N/A
ElementDefinition.binding.description
Element Id | ElementDefinition.binding.description
Definition | Describes the intended use of this particular set of codes.
Cardinality | 0..1
Type | string
Summary | true
ElementDefinition.binding.valueSet
Element Id | ElementDefinition.binding.valueSet
Definition | Refers to the value set that identifies the set of codes the binding refers to.
Cardinality | 0..1
Type | canonical(ValueSet)
Summary | true
Comments | The reference may be version-specific or not (e.g. have a |[version] at the end of the canonical URL).
Invariants | Affect this element eld-12 Rule ValueSet SHALL start with http:// or https:// or urn: valueSet.exists() implies (valueSet.startsWith('http:') or valueSet.startsWith('https') or valueSet.startsWith('urn:'))
Affect this element
eld-12 | Rule | ValueSet SHALL start with http:// or https:// or urn: | valueSet.exists() implies (valueSet.startsWith('http:') or valueSet.startsWith('https') or valueSet.startsWith('urn:'))
LOINC Code | CODE_TABLE
ElementDefinition.mapping
Element Id | ElementDefinition.mapping
Definition | Identifies a concept from an external specification that roughly corresponds to this element.
Cardinality | 0..*
Requirements | Provides guidance to implementers familiar with or converting content from other specifications.
Summary | true
Comments | Mappings are not necessarily specific enough for safe translation.
ElementDefinition.mapping.identity
Element Id | ElementDefinition.mapping.identity
Definition | An internal reference to the definition of a mapping.
Cardinality | 1..1
Type | id
Summary | true
ElementDefinition.mapping.language
Element Id | ElementDefinition.mapping.language
Definition | Identifies the computable language in which mapping.map is expressed.
Cardinality | 0..1
Terminology Binding | MimeType (Required)
Type | code
Summary | true
Comments | If omitted, then there can be no expectation of computational interpretation of the mapping.
ElementDefinition.mapping.map
Element Id | ElementDefinition.mapping.map
Definition | Expresses what part of the target specification corresponds to this element.
Cardinality | 1..1
Type | string
Summary | true
Comments | For most mappings, the syntax is undefined. Syntax will be provided for mappings to the RIM. Multiple mappings may be possible and may include constraints on other resource elements that identify when a particular mapping applies.
ElementDefinition.mapping.comment
Element Id | ElementDefinition.mapping.comment
Definition | Comments that provide information about the mapping or its use.
Cardinality | 0..1
Type | string
Summary | true

2.30.3 Mappings

URL: https://hl7.org/fhir/R4/elementdefinition-mappings.html

2.30.3 Element Definition Mappings

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative

This page provides mappings for Element Definition (see Mappings to Other Standards for further information & status).

Many other standard frameworks define frameworks for defining "data elements". These overlap to some degree with the ElementDefinition type. This page provides general mappings between ElementDefinition and the other frameworks. One important consideration with regard to other Element definition frameworks is the scope of the definition. Many of these definition frameworks use the term "Data Element" to refer to a data item that is collected and stored with provenance, such as when it was collected, who recorded it, etc. In this specification, that's an Observation, and the definition of this kind of data element is an ObservationDefinition, or a StructureDefinition that contains several different atomic data items. An ElementDefinition in this specification is a narrower notion, solely around the characteristics of a single element, it's definition, and it's value domain . Other frameworks - especially ISO 11179 - are used in both ways, sometimes with no formal differentiation between them. For this reason, the mappings on this page are very provisional, and are provided with the intent of helping implementers understand the possible relationships.

On this page:

v2 | Mappings to the V2 segment OM1 which is used to define observations (i.e. also related to ObservationDefinition)
RIM | Mappings to the RIM class Observation in definition mood which is used to define observations (i.e. also related to ObservationDefinition)
IHE DEX | Mappings IHE Data Element Exchange, which is closer to defining atomic data elements - still very incomplete
LOINC | Mappings to the LOINC master table which defines observations (i.e. also related to ObservationDefinition)
ISO 11179 | Mappings to ISO 11179 which details how the ElementDefinition class relates to the ISO 11179 framework. Note that the principle differences are that FHIR does not differentiate between a Data Element and a Data Element Value, and the FHIR specification is heavily type dependent. Also, the FHIR specification includes constraints and other concerns that are outside the scope of ISO 11179

2.30.3.1 HL7 v2 Mapping (http://hl7.org/v2)

ElementDefinition | OM1
path | N/A
representation | N/A
sliceName | N/A
sliceIsConstraining | N/A
label
code | OM1.7
slicing
discriminator
type
path
description
ordered
rules
short
definition | OM1.6, OM1.41
comment | NTE, OM1.32, OM1.33, OM1.39
requirements
alias | OM1.8, OM1.52
min
max | OMC.10
base
path
min
max
contentReference
type | OM1.3, OMC.9
code | OM1.3
profile
targetProfile
aggregation
versioning
defaultValue[x]
meaningWhenMissing
orderMeaning
fixed[x]
pattern[x]
example
label
value[x]
minValue[x]
maxValue[x]
maxLength | OMC.12
condition

constraint
key
requirements
severity
human
expression
xpath
source
mustSupport
isModifier
isModifierReason
isSummary
binding | OM3, OMC.11
strength | N/A
description
valueSet
mapping
identity
language
map
comment

2.30.3.2 RIM Mapping (http://hl7.org/v3)

ElementDefinition | Observation[classCode=OBS, moodCode=DEFN]
path | N/A
representation | N/A
sliceName | N/A
sliceIsConstraining | N/A
label | ./code/display
code | ./code (root or translation)
slicing | N/A
discriminator | N/A
type | N/A
path | N/A
description | N/A
ordered | N/A
rules | N/A
short | N/A
definition | ./text
comment | ./inboundRelationship[typeCode=SUBJ]/source[classCode=LIST, moodCode=EVN]/code
requirements | ./inboundRelationship[typeCode=META]/source[classCode=ACT, moodCode=EVN, isCriterionInd=true]/reasonCode/originalText
alias | Not currently expressible
min | N/A (MIF territory)
max | N/A (MIF territory)
base | N/A (MIF territory)
path | N/A (MIF territory)
min | N/A (MIF territory)
max | N/A (MIF territory)
contentReference | N/A
type | N/A (MIF territory)
code | N/A (MIF territory)
profile | N/A (MIF territory)
targetProfile | N/A (MIF territory)
aggregation | N/A (MIF territory)
versioning | N/A (MIF territory)
defaultValue[x] | N/A (MIF territory)
meaningWhenMissing | N/A (MIF territory)
orderMeaning | N/A (MIF territory)
fixed[x] | N/A (MIF territory)
pattern[x] | N/A (MIF territory)
example | N/A (MIF territory)
label | N/A (MIF territory)
value[x] | N/A (MIF territory)
minValue[x] | N/A (MIF territory)
maxValue[x] | N/A (MIF territory)
maxLength | N/A (MIF territory)
condition | N/A (MIF territory)
constraint | N/A (MIF territory)
key | N/A (MIF territory)
requirements | N/A (MIF territory)
severity | N/A (MIF territory)

human | N/A (MIF territory)
expression | N/A (MIF territory)
xpath | N/A (MIF territory)
source | N/A (MIF territory)
mustSupport | N/A (MIF territory)
isModifier | N/A (MIF territory)
isModifierReason | N/A (MIF territory)
isSummary | N/A (MIF territory)
binding | N/A (MIF territory)
strength | N/A (MIF territory)
description | N/A (MIF territory)
valueSet | N/A (MIF territory)
mapping | N/A (MIF territory)
identity | N/A (MIF territory)
language | N/A (MIF territory)
map | N/A (MIF territory)
comment | N/A (MIF territory)

2.30.3.3 IHE Data Element Exchange (DEX) (http://ihe.net/data-element-exchange)

ElementDefinition | RetrieveMetadata response
path
representation
sliceName
sliceIsConstraining
label
code
slicing
discriminator
type
path
description
ordered
rules
short
definition
comment
requirements
alias
min
max
base
path
min
max
contentReference
type | valueDomain.dataType
code
profile
targetProfile
aggregation
versioning
defaultValue[x]
meaningWhenMissing
orderMeaning
fixed[x]
pattern[x]
example
label
value[x]
minValue[x]
maxValue[x]
maxLength
condition
constraint
key
requirements
severity
human
expression
xpath
source
mustSupport

isModifier
isModifierReason
isSummary
binding
strength
description
valueSet | valueDomain.valueSet
mapping | mappingSpecification
identity
language
map
comment

2.30.3.4 LOINC code for the element (http://loinc.org)

ElementDefinition
path
representation
sliceName
sliceIsConstraining
label
code | LOINC_NUM
slicing
discriminator
type
path
description
ordered
rules
short
definition | TERM DEFINITION/DESCRIPTION(S)
comment | COMMENTS
requirements
alias | RELATED NAMES (only some of these apply)
min
max
base
path
min
max
contentReference
type | HL7_V2_DATATYPE (translation required), HL7_V3_DATATYPE (translation required)
code | HL7_V2_DATATYPE (translation required), HL7_V3_DATATYPE (translation required)
profile
targetProfile
aggregation
versioning
defaultValue[x]
meaningWhenMissing
orderMeaning
fixed[x]
pattern[x]
example
label
value[x] | EXMPL_ANSWERS
minValue[x]
maxValue[x]
maxLength
condition
constraint
key
requirements
severity
human
expression
xpath
source
mustSupport
isModifier
isModifierReason
isSummary
binding | ANSWER LIST, NORMATIVE ANSWER LIST
strength | N/A

description
valueSet | CODE_TABLE
mapping
identity
language
map
comment

2.30.3.5 ISO 11179 (http://metadata-standards.org/11179/)

ElementDefinition | Data_Element
path | Name, Identifier & Context
representation | n/a
sliceName | n/a
sliceIsConstraining | n/a
label | n/a
code | (Data_Element).data_element_concept.identifier
slicing | n/a
discriminator | n/a
type
path | n/a
description
ordered | n/a
rules | n/a
short | N/A
definition | (Designatable_Item).definition.text acceptability=preferred in default context
comment | (Administered_Item).explanatory_comment
requirements | (Registered_item).document_reference[document_type=requirements].notation
alias | (Designatable_Item).designation.sign acceptability!=preferred or context is other than default
min | Minimum size of data element values?
max | Maximum size of data element values?
base | n/a
path | n/a
min | n/a
max | n/a
contentReference
type | .domain.data+Q14type
code | .domain.data+Q14type
profile | n/a
targetProfile | n/a
aggregation | n/a
versioning
defaultValue[x]
meaningWhenMissing
orderMeaning
fixed[x] | N/A (only relevant when constraining, which 11179 doesn't do)
pattern[x]
example
label
value[x] | .example
minValue[x]
maxValue[x]
maxLength | .domain.maximum_character_quantity
condition
constraint | ??
key
requirements
severity
human
expression
xpath
source
mustSupport | ??
isModifier | ??
isModifierReason
isSummary | ??
binding | .domain
strength
description | .domain.description
valueSet | points to explicit list or expression that evaluates to list of (Enumerated_Value_Domain).member
mapping | Registered_item).document_reference[document_type=mapping] Also, .meaning linkage to Data_Element_Concept is done as a mapping to a reference model. (Data_Element_Concepts are all defined in some sort of reference model, be that Object_Class and Property or some other mechanism)

identity
language
map | ObjectClass, Property (this is one possible data model that can be mapped to - the uri would identify the data model
mappingSpecification.mappingScript
comment

2.30.4 ElementDefinition Extensions

URL: https://hl7.org/fhir/R4/elementdefinition-extras.html

2.30.4 ElementDefinition Extensions

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative

This extensions defined here can be used on any ElementDefinition. For background information, see Extensibility. Additional extensions may be found in published Implementation Guides , or in the Conformance resource registry .

Extensions (+ see extensions on all Elements)
11179-objectClass | objectClass | for ISO 11179 Element Definition Profile
11179-objectClassProperty | objectClassProperty | for ISO 11179 Element Definition Profile
designNote | Design Note | for General Extensions for use by FHIR Implementers
elementdefinition-allowedUnits | allowedUnits | for Common extensions for ElementDefinition data type
elementdefinition-bestpractice | bestpractice | for Common extensions for ElementDefinition data type
elementdefinition-bestpractice-explanation | bestpractice-explanation | for Common extensions for ElementDefinition data type
elementdefinition-bindingName | bindingName | for Common extensions for ElementDefinition data type
elementdefinition-equivalence | equivalence | for Common extensions for ElementDefinition data type
elementdefinition-identifier | identifier | for Common extensions for ElementDefinition data type
elementdefinition-inheritedExtensibleValueSet | inheritedExtensibleValueSet | for Common extensions for ElementDefinition data type
elementdefinition-isCommonBinding | isCommonBinding | for Common extensions for ElementDefinition data type
elementdefinition-maxValueSet | maxValueSet | for Common extensions for ElementDefinition data type
elementdefinition-minValueSet | minValueSet | for Common extensions for ElementDefinition data type
elementdefinition-namespace | namespace | for Common extensions for ElementDefinition data type
elementdefinition-profile-element | profile-element | for Common extensions for ElementDefinition data type
elementdefinition-question | question | for Common extensions for ElementDefinition data type
elementdefinition-selector | selector | for Common extensions for ElementDefinition data type
elementdefinition-translatable | translatable | for Common extensions for ElementDefinition data type
entryFormat | entryFormat | for Element-definition Extensions for use by FHIR Implementers
maxDecimalPlaces | maxDecimalPlaces | for Element-definition Extensions for use by FHIR Implementers
maxSize | maxSize | for Element-definition Extensions for use by FHIR Implementers
mimeType | mimeType | for Element-definition Extensions for use by FHIR Implementers
minLength | minLength | for Element-definition Extensions for use by FHIR Implementers
regex | regex | for Element-definition Extensions for use by FHIR Implementers
structuredefinition-display-hint | display-hint | for Standard Structure Definition Extensions
structuredefinition-explicit-type-name | explicit-type-name | for Standard Structure Definition Extensions
structuredefinition-fhir-type | fhir-type | for Standard Structure Definition Extensions
structuredefinition-hierarchy | hierarchy | for Standard Structure Definition Extensions
Profiles
elementdefinition-de | Identifies how the ElementDefinition data type is used when it appears within a data element | for DataElement constraint on ElementDefinition data type

2.31 Guide to Resources

URL: https://hl7.org/fhir/R4/resourceguide.html

2.31 Guide to Resources

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative

The FHIR specification defines a set of resources, and an infrastructure for handling resources. In order to use FHIR to create solutions for integration requirements, implementers must map their problems to resources and their content.

The Resource List provides views of the resources organized in several different ways, and is the primary entry point to the resources.

This page describes the resources and their functional intent in more detail to assist implementers to understand their purpose and scope, and their supporting classifications.

Where to find common concepts in this specification:

Concept | Example | Where to find
Clinical Findings
Laboratory Results | Blood panels such as CBC with Differential, Liver Panel, etc. | DiagnosticReport with Observations
Imaging Study Findings | CT Scans, MRI, Plain Radiographs, Ultrasounds) | DiagnosticReport (some with Observations)
Diagnostic Test Results | EKG, pulmonary function test, EEG | Observations (and maybe a DiagnosticReport)

Vital Signs | Temperature, Blood Pressure, Heart Rate, Respiratory Rate | Observation
Other Physical Exam Findings | Auscultation findings | Observation
Pulmonary Artery Catheter readings | Pulmonary artery pressure | Observation
Patient Problems, Allergies and Adverse Events
Allergy | Food or drug allergies | AllergyIntolerance
Clinical Diagnosis | Diabetes, Congestive Heart Failure | Condition
Adverse Event | Adverse reaction to an agent, falls, adverse surgical events, hospital infections | AdverseEvent
Patient History
Chief Complaint | Cough, Pain, Fever, Fatigue | Condition
Past Surgical History | Appendectomy, Hernia repair | Procedure
Past Medical History | Diabetes, Congestive heart failure | Condition
MAR (Medication Administration Record) | Warfarin 5mg PO administered on 12/10/2013 at 3pm | MedicationAdministration
Home Meds | Warfarin 5mg, 30 day supply, dispensed on 12/01/2013 | MedicationStatement
Social History | Sexual behavior, Smoking status, Alcohol intake, Illicit drug use | Observation
Family History | Mother has diabetes | FamilyMemberHistory
Signs & Symptoms | from a review of systems- Pain, Fever | Condition
Suggested Physician Orders
Proposal for a laboratory test | A blood panel, a stool analysis | ServiceRequest
Proposal for an imaging procedure | CT Scan, MRI, X-Rays | ServiceRequest
Proposed Diet Order | An oral diet order | NutritionOrder
Proposed respiratory order | Oxygen delivery | Not done yet
Proposed Medications | Aspirin, Lisinopril | MedicationRequest
Proposed Supply | Wheel Chair, Food Tray | SupplyRequest
Interdisciplinary Care Planning
Patient Goal | Reduce risk of falls, lose weight | Goal (as part of a CarePlan)
Intervention | Patient assessments | ServiceRequest

In addition, to the information on this page, see also Common Use Cases.

2.31.1 Resource Classification

The following frameworks are provided for helping to understand the set of resources provided in this specification:

- Modules
- Workflow Patterns: Event, Request, Definition
- RIM Mappings (todo)

2.33 FMG Summary Page

URL: https://hl7.org/fhir/R4/fmg.html

2.33 FMG Summary Page

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative

FMG Approvals

Not yet proposed

- EffectEvidenceSynthesis
- EventDefinition
- Evidence
- EvidenceVariable
- ExampleScenario
- Linkage
- MedicinalProductAuthorization
- MedicinalProductContraindication
- MedicinalProductIndication
- MedicinalProductIngredient
- MedicinalProductInteraction
- MedicinalProductManufactured
- MedicinalProductPackaged
- MedicinalProductPharmaceutical
- MedicinalProductUndesirableEffect
- OperationDefinition N
- Parameters N
- ResearchDefinition
- ResearchElementDefinition

- RiskEvidenceSynthesis
- SubstanceNucleicAcid
- SubstancePolymer
- SubstanceProtein
- SubstanceReferenceInformation
- SubstanceSourceMaterial
- TerminologyCapabilities

Pending

- AdverseEvent
- BiologicallyDerivedProduct
- CatalogEntry
- ChargeItemDefinition
- ClinicalImpression
- DeviceDefinition
- DeviceUseStatement
- InsurancePlan
- Invoice
- MedicationKnowledge
- MedicinalProduct
- MessageDefinition
- ResearchStudy
- ResearchSubject
- SubstanceSpecification
- TestReport
- VerificationResult

Approved

- Account
- ActivityDefinition
- AllergyIntolerance
- Appointment
- AppointmentResponse
- AuditEvent
- Basic
- Binary N
- BodyStructure
- Bundle N
- CapabilityStatement N
- CarePlan
- CareTeam
- ChargeItem
- Claim
- ClaimResponse
- CodeSystem N
- Communication
- CommunicationRequest
- CompartmentDefinition
- Composition
- ConceptMap
- Condition
- Consent
- Contract
- Coverage

- CoverageEligibilityRequest
- CoverageEligibilityResponse
- DetectedIssue
- Device
- DeviceMetric
- DeviceRequest
- DiagnosticReport
- DocumentManifest
- DocumentReference
- Encounter
- Endpoint
- EnrollmentRequest
- EnrollmentResponse
- EpisodeOfCare
- ExplanationOfBenefit
- FamilyMemberHistory
- Flag
- Goal
- GraphDefinition
- Group
- GuidanceResponse
- HealthcareService
- ImagingStudy
- Immunization
- ImmunizationEvaluation
- ImmunizationRecommendation
- ImplementationGuide
- Library
- List
- Location
- Measure
- MeasureReport
- Media
- Medication
- MedicationAdministration
- MedicationDispense
- MedicationRequest
- MedicationStatement
- MessageHeader
- MolecularSequence
- NamingSystem
- NutritionOrder
- Observation N
- ObservationDefinition
- OperationOutcome N
- Organization
- OrganizationAffiliation
- Patient N
- PaymentNotice
- PaymentReconciliation
- Person
- PlanDefinition

• Practitioner
• PractitionerRole
• Procedure
• Provenance
• Questionnaire
• QuestionnaireResponse
• RelatedPerson
• RequestGroup
• RiskAssessment
• Schedule
• SearchParameter
• ServiceRequest
• Slot
• Specimen
• SpecimenDefinition
• StructureDefinition N
• StructureMap
• Subscription
• Substance
• SupplyDelivery
• SupplyRequest
• Task
• TestScript
• ValueSet N
• VisionPrescription

2.34 Resource Basic - Content

URL: https://hl7.org/fhir/R4/basic.html
FHIR Infrastructure Work Group | Maturity Level: 1 | Trial Use | Security Category: Not Classified | Compartments: Patient, Practitioner, RelatedPerson

Basic is used for handling concepts not yet defined in FHIR, narrative-only resources that don't map to an existing resource, and custom resources not appropriate for inclusion in the FHIR specification.

2.34.1 Scope and Usage

Basic is a special type of resource. Unlike all other resources, it doesn't correspond to a specific pre-defined HL7 concept. Instead, it's a placeholder for any resource-like concept that isn't already defined in the HL7 specification.

The Basic resource is intended for use in three circumstances:

• When an implementer needs a resource concept that is likely to be defined by HL7 in the future but they have not yet done so (due to bandwidth issues, lack of sufficient requirements, lower prioritization, etc.)

• When there's a need to convey a narrative-only construct that doesn't neatly correspond to one of the other resources, either because it combines aspects of several resources (e.g. Assessment + Plan) or because the allowed content is flexible such that the system can't be totally sure what sort of content might have been included in the narrative text.

• Other than the circumstances above, this resource will see minimal use. To keep the FHIR specification manageable, it cannot incorporate every site-specific requirement that might be needed in some implementation somewhere. This set of resources likely won't ever be officially defined in HL7.

There's also a fourth circumstance: An implementer wishes to convey information that could/should be conveyed using a standard resource, however they want to represent the information in a custom format that isn't aligned with the official resource's elements. While this resource would be the preferred way of meeting that use-case because it will at least be wire-format compatible, such a use would not be conformant because making use of the Basic resource would prevent the healthcare-related information from being safely processed, queried and analyzed by other conformant systems.

Implementers don't need to be concerned with which of the three categories their desired resource fits within. If they need a resource and it clearly doesn't fit one of the ones currently defined, they should use Basic.

2.34.2 Background and Context

Basic defines only a minimal set of data elements - those necessary to identify what kind of resource it represents and those necessary to support resource compartmenting. All other data elements are represented using the extension mechanism. It's entirely possible to have a Basic resource instance with nothing other than narrative, a subject and code. And, in practice, that's all many systems will understand.

2.34.3 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
Basic | ΣTU | | DomainResource | Resource for non-supported content Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | Business identifier
code | ?!Σ | 1..1 | CodeableConcept | Kind of Resource Basic Resource Types (Example)
subject | Σ | 0..1 | Reference(Any) | Identifies the focus of this resource
created | Σ | 0..1 | date | When created
author | Σ | 0..1 | Reference(Practitioner | PractitionerRole | Patient | RelatedPerson | Organization) | Who created
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Basic xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business identifier --></identifier>
 <code><!-- 1..1 CodeableConcept Kind of Resource --></code>
 <subject><!-- 0..1 Reference(Any) Identifies the focus of this resource --></subject>
 <created value="[date]"/><!-- 0..1 When created -->
 <author><!-- 0..1 Reference(Practitioner|PractitionerRole|Patient|RelatedPerson|
   Organization) Who created --></author>
</Basic>
```

JSON Template

```
{
  "resourceType" : "Basic",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business identifier
  "code" : { CodeableConcept }, // R!  Kind of Resource
  "subject" : { Reference(Any) }, // Identifies the focus of this resource
  "created" : "<date>", // When created
  "author" : { Reference(Practitioner|PractitionerRole|Patient|RelatedPerson|
   Organization) } // Who created
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Basic;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Basic.identifier [ Identifier ], ... ; # 0..* Business identifier
  fhir:Basic.code [ CodeableConcept ]; # 1..1 Kind of Resource
  fhir:Basic.subject [ Reference(Any) ]; # 0..1 Identifies the focus of this resource
  fhir:Basic.created [ date ]; # 0..1 When created
  fhir:Basic.author [ Reference(Practitioner|PractitionerRole|Patient|RelatedPerson|Organization) ]; # 0..1 Who created
]
```

Changes since R3

Basic
Basic.author | Type Reference: Added Target Types PractitionerRole, Organization

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 3 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.)

Structure

Name | Flags | Card. | Type | Description & Constraints
Basic | ΣTU | | DomainResource | Resource for non-supported content Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | Business identifier
code | ?!Σ | 1..1 | CodeableConcept | Kind of Resource Basic Resource Types (Example)
subject | Σ | 0..1 | Reference(Any) | Identifies the focus of this resource
created | Σ | 0..1 | date | When created
author | Σ | 0..1 | Reference(Practitioner | PractitionerRole | Patient | RelatedPerson | Organization) | Who created
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Basic xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business identifier --></identifier>
 <code><!-- 1..1 CodeableConcept Kind of Resource --></code>
 <subject><!-- 0..1 Reference(Any) Identifies the focus of this resource --></subject>
 <created value="[date]"/><!-- 0..1 When created -->
 <author><!-- 0..1 Reference(Practitioner|PractitionerRole|Patient|RelatedPerson|
   Organization) Who created --></author>
</Basic>
```

JSON Template

```
{
  "resourceType" : "Basic",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business identifier
  "code" : { CodeableConcept }, // R!  Kind of Resource
  "subject" : { Reference(Any) }, // Identifies the focus of this resource
  "created" : "<date>", // When created
  "author" : { Reference(Practitioner|PractitionerRole|Patient|RelatedPerson|
   Organization) } // Who created
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Basic;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Basic.identifier [ Identifier ], ... ; # 0..* Business identifier
  fhir:Basic.code [ CodeableConcept ]; # 1..1 Kind of Resource
  fhir:Basic.subject [ Reference(Any) ]; # 0..1 Identifies the focus of this resource
  fhir:Basic.created [ date ]; # 0..1 When created
  fhir:Basic.author [ Reference(Practitioner|PractitionerRole|Patient|RelatedPerson|Organization) ]; # 0..1 Who created
]
```

Changes since Release 3

Basic
Basic.author | Type Reference: Added Target Types PractitionerRole, Organization

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 3 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

2.34.3.1 Terminology Bindings

Path | Definition | Type | Reference
Basic.code | Codes for identifying types of resources not yet defined by FHIR. | Example | BasicResourceTypes

2.34.4 Why not custom resources?

Technically, nothing prevents implementers from going off and defining their own resources containing whatever data elements they wish. However, doing so causes several issues:

• Custom resources don't have a discoverability mechanism in the same way custom codes and extensions do using the ValueSet and StructureDefinition resources. As a result, any implementer that receives a custom resource would have no way of looking up what the meaning of the resource or its elements were. While they could get some sense of meaning from XML or JSON tag names, this often isn't sufficient for safe healthcare interoperability.

• Custom resource names would not be present in the FHIR schemas as allowed elements within the FHIR Bundle schema, would not be present in the enumeration of resources in the Reference type, and would not

be supported by any of the autogenerated reference implementations and software interfaces. This would cause issues for any receiving system making use of the schemas directly or via code-generation.

• There is no means of preventing two implementers from independently coming up with the same name for a resource but defining it differently in terms of meaning as well as allowed elements. This would also cause interoperability issues.

All of these concerns are mitigated when there's an assumption that the custom resource will only be used within a narrow constrained environment where all participants will be aware of the semantics, will be using the same custom schemas and there's no chance of collisions. However, HL7's experience is that closed implementation environments rarely remain that way over the long term. Eventually data will need to be shared with others outside the closed environment and all of the above issues will again come into play.

Therefore, use of 'custom' resources is NOT considered to be conformant with FHIR. While the use of extensions may make the Basic resource slightly more complex and less visually appealing, it is the only safe and approved mechanism for sharing resource concepts not representable using standard HL7-defined resources.

It is expected that future versions of the interface tooling will be able to generate object interfaces on the basis of profiles. Where this occurs, the complexity of custom resource elements being expressed as extensions should be transparent to the internal code of systems that support that particular variant of the Basic resource. This should further reduce the cost of using 'Basic' as opposed to custom resources.

NOTE: This position is subject to change based on implementation experience. Alternative mechanisms for handling custom resource requirements in a safe manner may be explored. Ideas around alternative technical strategies for managing this issue are welcome.

2.34.5 Documents and narrative-only resources

Documents are constructed of sections, where a key part of each section is the narrative. The narratives are stitched together to form the overall text of the document. Many document sections will correspond neatly to resources that are already defined - List, DiagnosticReport, FamilyMemberHistory, etc. However, oddly enough, alignment with FHIR resources isn't always in mind when clinicians and others design documents, and some sections won't neatly align with the boundaries of resources. Sometimes there's simply a need for a place where a document author can say "stuff" without any particular constraints on what they may choose to talk about. Basic is intended to provide a mechanism to handle those circumstances.

Wherever possible, the "standard" FHIR resources should be used, even for narrative-only content. That's because subsequent revisions of the narrative-only content might choose to encode pieces or even all of the narrative content. Encoding can occur with "Basic" as well. Extensions can point to other resources (contained or stand-alone) that fully encode pieces of the free-form narrative found in the Basic resource. If no appropriate other resource exists for the meaning of the content, extensions can also be used.

2.34.6 Best practices for using 'Basic'

There are several good practices to follow when making use of the Basic resource:

• Before using Basic, post a description of the desired resource type on HL7's FHIR list-server or on Stack Overflow to see whether the use-case can be met by an existing resource. (Sometimes the intended scope of an existing resource won't be clear, even if the intent is to cover your space.) Using an existing resource is usually preferable to using Basic as it significantly increases the likelihood of interoperability.

• If an existing resource would normally be a good fit for your use-case but can't be used due to overly prescriptive constraints your implementation is unable to meet, again raise the problem on Stack Overflow so the problem with the specification can be addressed.

• If it is necessary to make use of the Basic resource, try to use one of the HL7-defined codes for resource type or submit your requirement for a new type for inclusion in the HL7 vocabulary (using the Propose a change link), as this will increase the likelihood of interoperability. Alternate code systems are conformant, but are less likely to be recognized or re-used across the healthcare implementation space.

• Architect your interface in a way that will make it less painful to swap your use of Basic with an 'official' resource in the event that a future release of FHIR formally defines a resource that encompasses your use-case.

• Use a StructureDefinition to define the extensions relevant to each type of other resource used. Profiles can also be used to define additional search criteria appropriate for the resource.

• When defining a profile on Basic, include mappings to the w5 categories to allow systems to easily manage AuditEvent and Provenance uses as well as other potential higher-level abstractions of the data.

• As well, profiles should consider how best to handle common notions such as "entered in error" status and alignment with common practices within similar resource families (other request resources, medication-related resources), etc.

2.34.7 Referencing Basic resources

None of the standard resources will have direct references to Basic, aside from those that allow linking to "Any" resource. As a result, most references to "Basic" will need to be performed using extensions.

2.34.8 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
author | reference | Who created | Basic.author (Practitioner, Organization, Patient, PractitionerRole, RelatedPerson)
code | token | Kind of Resource | Basic.code
created | date | When created | Basic.created
identifier | token | Business identifier | Basic.identifier
patient | reference | Identifies the focus of this resource | Basic.subject.where(resolve() is Patient) (Patient)
subject | reference | Identifies the focus of this resource | Basic.subject (Any)

2.34.9 Resource Basic - Examples

URL: https://hl7.org/fhir/R4/basic-examples.html

2.34.9 Resource Basic - Examples

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Not Classified | Compartments: Patient, Practitioner, RelatedPerson

Example Name | id | Format
How 'basic' can be used for future resources - in this case referral | referral | XML | JSON | Turtle
How 'Basic' can be used for essoteric resources - in this case, a UML class model | classModel | XML | JSON | Turtle
A resource that uses all narrative features for testing rendering applications | basic-example-narrative | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

2.34.10 Resource Basic - Detailed Descriptions

URL: https://hl7.org/fhir/R4/basic-definitions.html

2.34.10 Resource Basic - Detailed Descriptions

FHIR Infrastructure Work Group | Maturity Level: 1 | Trial Use | Security Category: Not Classified | Compartments: Patient, Practitioner, RelatedPerson

Detailed Descriptions for the elements in the Basic resource.

Basic
Element Id | Basic
Definition | Basic is used for handling concepts not yet defined in FHIR, narrative-only resources that don't map to an existing resource, and custom resources not appropriate for inclusion in the FHIR specification.
Cardinality | 0..*
Type | DomainResource
Requirements | Need some way to safely (without breaking interoperability) allow implementers to exchange content not supported by the initial set of declared resources.
Alternate Names | Z-resource; Extension-resource; Custom-resource
Summary | true
Basic.identifier
Element Id | Basic.identifier
Definition | Identifier assigned to the resource for business purposes, outside the context of FHIR.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Summary | true
Basic.code
Element Id | Basic.code
Definition | Identifies the 'type' of resource - equivalent to the resource name for other resources.
Cardinality | 1..1
Terminology Binding | Basic Resource Types (Example)
Type | CodeableConcept
Is Modifier | true (Reason: This element determines what kind of resource is being represented which drives the meaning of all of the other elements.)
Requirements | Must be able to distinguish different types of Basic resources.
Summary | true
Comments | Because resource references will only be able to indicate 'Basic', the type of reference will need to be specified in a Profile identified as part of the resource. Refer to the resource notes section for information on appropriate terminologies for this code. This element is labeled as a modifier because it defines the meaning of the resource and cannot be ignored.
Basic.subject
Element Id | Basic.subject
Definition | Identifies the patient, practitioner, device or any other resource that is the "focus" of this resource.

Cardinality | 0..1
Type | Reference(Any)
Requirements | Needed for partitioning the resource by Patient.
Summary | true
Comments | Optional as not all potential resources will have subjects. Resources associated with multiple subjects can handle this via extension.
Basic.created
Element Id | Basic.created
Definition | Identifies when the resource was first created.
Cardinality | 0..1
Type | date
Requirements | Allows ordering resource instances by time.
Summary | true
Basic.author
Element Id | Basic.author
Definition | Indicates who was responsible for creating the resource instance.
Cardinality | 0..1
Type | Reference(Practitioner | PractitionerRole | Patient | RelatedPerson | Organization)
Requirements | Needed for partitioning the resource.
Summary | true

2.34.11 Resource Basic - Mappings

URL: https://hl7.org/fhir/R4/basic-mappings.html

2.34.11 Resource Basic - Mappings

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Not Classified | Compartments: Patient, Practitioner, RelatedPerson

Mappings:

FiveWs Pattern Mapping

RIM Mapping

Mappings for the basic resource (see Mappings to Other Standards for further information & status).

2.34.11.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

Basic
identifier | FiveWs.identifier
code | FiveWs.what[x]
subject | FiveWs.subject[x]
created | FiveWs.recorded
author | FiveWs.author

2.34.11.2 RIM Mapping (http://hl7.org/v3)

Basic | Act, Entity or Role
identifier | ./identifier
code | ./code
subject | ./participation[typeCode='SBJ'] (possibly through a ControlAct and Role)
created | ./participation[typeCode='AUT']/time (possibly through a ControlAct and Role)
author | ./participation[typeCode='SUB'] (possibly through a ControlAct and Role)

2.34.12 Resource Basic - Extensions & Profiles

URL: https://hl7.org/fhir/R4/basic-profiles.html

2.34.12 Resource Basic - Extensions & Profiles

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Not Classified | Compartments: Patient, Practitioner, RelatedPerson

This table lists profiles and extensions for the Basic resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

2.34.12.1 Profiles

No Profiles defined for this resource

2.34.12.2 Extensions

cqf-encounterClass | Basic | encounterClass | for Clinical Reasoning Extensions
cqf-encounterType | Basic | encounterType | for Clinical Reasoning Extensions
cqf-initiatingOrganization | Basic | initiatingOrganization | for Clinical Reasoning Extensions
cqf-initiatingPerson | Basic | initiatingPerson | for Clinical Reasoning Extensions

cqf-receivingOrganization | Basic | receivingOrganization | for Clinical Reasoning Extensions
cqf-receivingPerson | Basic | receivingPerson | for Clinical Reasoning Extensions
cqf-recipientLanguage | Basic | recipientLanguage | for Clinical Reasoning Extensions
cqf-recipientType | Basic | recipientType | for Clinical Reasoning Extensions
cqf-systemUserLanguage | Basic | systemUserLanguage | for Clinical Reasoning Extensions
cqf-systemUserTaskContext | Basic | systemUserTaskContext | for Clinical Reasoning Extensions
cqf-systemUserType | Basic | systemUserType | for Clinical Reasoning Extensions
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

2.34.12.3 Search Extensions

No Search Extensions defined for this resource

2.34.13 Resource Basic - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/basic-version-maps.html

No validation errors - all conversions are clean

2.34.13 Resource Basic - R3/R4 Conversions

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Not Classified | Compartments: Patient, Practitioner, RelatedPerson

R3 : R4 Conversion maps for Basic.

Functional status for this map: 3 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid. (see documentation)

2.34.13.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/Basic3to4" = "R3 to R4 Conversions for Basic"

uses "http://hl7.org/fhir/3.0/StructureDefinition/Basic" alias BasicR3 as source
uses "http://hl7.org/fhir/StructureDefinition/Basic" alias Basic as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group Basic(source src : BasicR3, target tgt : Basic) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.code -> tgt.code;
  src.subject -> tgt.subject;
  src.created -> tgt.created;
  src.author -> tgt.author;
}
```

2.34.13.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/Basic4to3" = "R4 to R3 Conversion for Basic"

uses "http://hl7.org/fhir/StructureDefinition/Basic" alias Basic as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/Basic" alias BasicR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group Basic(source src : Basic, target tgt : BasicR3) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.code -> tgt.code;
  src.subject -> tgt.subject;
  src.created -> tgt.created;
  src.author -> tgt.author;
}
```

2.34.13.3 R4 Validation Errors

2.34.14 JSON Schema for Basic

URL: https://hl7.org/fhir/R4/basic.schema.json.html

2.34.14 JSON Schema for Basic

This is a representation of the json schema for Basic, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/Basic",
  "$ref": "#/definitions/Basic",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "Basic": {
      "description": "Basic is used for handling concepts not yet defined in FHIR, narrative-only resources that
    don\u0027t map to an existing resource, and custom resources not appropriate for inclusion in the FHIR
    specification.",
      "properties": {
        "resourceType": {
          "description": "This is a Basic resource",
          "const": "Basic"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
          "description": "A human-readable narrative that contains a summary of the resource and can be used to
    represent the content of the resource to a human. The narrative need not encode all the structured data, but is
    required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
    Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
          "$ref": "Narrative.schema.json#/definitions/Narrative"
        },
        "contained": {
          "description": "These resources do not have an independent existence apart from the resource that contains
    them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
          "items": {
            "$ref": "ResourceList.schema.json#/definitions/ResourceList"
          },
          "type": "array"
        },
        "extension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
     definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "modifierExtension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource and that modifies the understanding of the element that contains it and/or the understanding of the
    containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
    of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
    met as part of the definition of the extension. Applications processing a resource are required to check for
    modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
    DomainResource (including cannot change the meaning of modifierExtension itself).",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "identifier": {
          "description": "Identifier assigned to the resource for business purposes, outside the context of FHIR.",
          "items": {
```

```
            "$ref": "Identifier.schema.json#/definitions/Identifier"
          },
          "type": "array"
        },
        "code": {
          "description": "Identifies the \u0027type\u0027 of resource - equivalent to the resource name for other
    resources.",
          "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
        },
        "subject": {
          "description": "Identifies the patient, practitioner, device or any other resource that is the \"focus\" of
    this resource.",
          "$ref": "Reference.schema.json#/definitions/Reference"
        },
        "created": {
          "description": "Identifies when the resource was first created.",
          "$ref": "#/definitions/date"
        },
        "_created": {
          "description": "Extensions for created",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "author": {
          "description": "Indicates who was responsible for creating the resource instance.",
          "$ref": "Reference.schema.json#/definitions/Reference"
        }
      },
      "additionalProperties": false,
      "required": [
        "code",
        "resourceType"
      ]
    }
  }
}
```


2.34.15 Resource Basic - Maturity Dependency View

URL: https://hl7.org/fhir/R4/basic-dependencies.html

2.34.15 Resource Basic - Maturity Dependency View

FHIR Infrastructure Work Group | Maturity Level: 1 | Trial Use | Compartments: Patient, Practitioner, RelatedPerson

Dependency Graph for Basic FMM level 1

Name | Card. | Type | Dependency Analysis
Basic | | DomainResource
identifier | 0..* | Identifier
code | 1..1 | CodeableConcept
subject | 0..1 | Reference(Any)
created | 0..1 | date
author | 0..1 | Reference(Practitioner | PractitionerRole | Patient | RelatedPerson | Organization)

2.35 Resource Binary - Content

URL: https://hl7.org/fhir/R4/binary.html
FHIR Infrastructure Work Group | Maturity Level: N | Normative (from v4.0.0) | Security Category: Not Classified | Compartments: Not linked to any defined compartments

| This page has been approved as part of an ANSI standard. See the Infrastructure Package for further details.

A resource that represents the data of a single raw artifact as digital content accessible in its native format. A Binary resource can contain any content, whether text, image, pdf, zip archive, etc.

2.35.1 Scope and Usage

There are situations where it is useful or required to handle pure binary content using the same framework as other resources. Typically, this is when the binary content is referred to from other FHIR Resources. Using the same framework means that the existing servers, security arrangements, code libraries, etc. can handle additional related content. Typically, Binary resources are used for handling content such as:

• CDA Documents (i.e. with XDS)

• PDF Documents

• Images

A binary resource can contain any content, whether text, image, pdf, zip archive, etc. These resources are served in their native form on the rest interface, but can also be represented in XML, JSON, or other formats, such as when including these resources in a Bundle (used when it is convenient to include these in the response directly rather than leaving them by reference).

2.35.2 Boundaries and Relationships

When a FHIR server finds it convenient to manage the content within the same overall REST framework as the other resources, the Binary resource is generally used as the target in the Attachment data type to reference content via the .url element, such as in the DocumentReference and Media resources. Consequently, the Binary resource can be a target wherever the Attachment data type is used such as in the DocumentReference resource.

The DocumentReference and Media resources allow conveying binary content (via attachment) or pointing to one (as a Binary or non-FHIR URI) along with the metadata around that resource, and as such are searchable. Binary resources do not support 'search'.

While CDA and PDF documents are conveyed as Binary (because they cannot be expressed natively in FHIR), FHIR Documents do not need to be similarly encoded and can be sent natively in FHIR using Bundle. However, in some situations FHIR Documents may be sent as a Binary if there is a need to treat them the same as other types of documents or binary files.

The Binary resource does not convey context of the file. If the context (information such as author, procedure, technique, etc.) should be conveyed, Media or DocumentReference resources are appropriate. The Binary resource may be used to convey actual binary file content conveyed by those resources.

The Media resource is preferred for handling images, but this is not possible when the content is already binary (e.g. in some uses of IHE XDS).

This resource is referenced by CatalogEntry and ImplementationGuide

2.35.3 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints

Binary | N | | Resource | Pure binary content defined by a format other than FHIR Elements defined in Ancestors: id, meta, implicitRules, language

contentType | Σ | 1..1 | code | MimeType of the binary content MimeType (Required)

securityContext | Σ | 0..1 | Reference(Any) | Identifies another resource to use as proxy when enforcing access control

data | | 0..1 | base64Binary | The actual content

Documentation for this format

UML Diagram (Legend)

XML Template

```
<Binary xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <contentType value="[code]"/><!-- 1..1 MimeType of the binary content -->
 <securityContext><!-- 0..1 Reference(Any) Identifies another resource to use as proxy when enforcing access control
    --></securityContext>
 <data value="[base64Binary]"/><!-- 0..1 The actual content -->
</Binary>
```

JSON Template

```
{
  "resourceType" : "Binary",
  // from Resource: id, meta, implicitRules, and language
  "contentType" : "<code>", // R!  MimeType of the binary content
  "securityContext" : { Reference(Any) }, // Identifies another resource to use as proxy when enforcing access control
  "data" : "<base64Binary>" // The actual content
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Binary;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  fhir:Binary.contentType [ code ]; # 1..1 MimeType of the binary content
  fhir:Binary.securityContext [ Reference(Any) ]; # 0..1 Identifies another resource to use as proxy when enforcing
    access control
  fhir:Binary.data [ base64Binary ]; # 0..1 The actual content
]
```

Changes since R3

Binary

Binary.contentType | Change value set from http://hl7.org/fhir/ValueSet/mimetypes to http://hl7.org/fhir/ValueSet/mimetypes|4.0.1
Binary.data | Renamed from content to data Min Cardinality changed from 1 to 0

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 2 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.)

Structure

Name | Flags | Card. | Type | Description & Constraints
Binary | N | | Resource | Pure binary content defined by a format other than FHIR Elements defined in Ancestors: id, meta, implicitRules, language
contentType | Σ | 1..1 | code | MimeType of the binary content MimeType (Required)
securityContext | Σ | 0..1 | Reference(Any) | Identifies another resource to use as proxy when enforcing access control
data | | 0..1 | base64Binary | The actual content
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Binary xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <contentType value="[code]"/><!-- 1..1 MimeType of the binary content -->
 <securityContext><!-- 0..1 Reference(Any) Identifies another resource to use as proxy when enforcing access control
    --></securityContext>
 <data value="[base64Binary]"/><!-- 0..1 The actual content -->
</Binary>
```

JSON Template

```
{
  "resourceType" : "Binary",
  // from Resource: id, meta, implicitRules, and language
  "contentType" : "<code>", // R!  MimeType of the binary content
  "securityContext" : { Reference(Any) }, // Identifies another resource to use as proxy when enforcing access control
  "data" : "<base64Binary>" // The actual content
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Binary;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  fhir:Binary.contentType [ code ]; # 1..1 MimeType of the binary content
  fhir:Binary.securityContext [ Reference(Any) ]; # 0..1 Identifies another resource to use as proxy when enforcing
    access control
  fhir:Binary.data [ base64Binary ]; # 0..1 The actual content
]
```

Changes since Release 3

Binary
Binary.contentType | Change value set from http://hl7.org/fhir/ValueSet/mimetypes to http://hl7.org/fhir/ValueSet/mimetypes|4.0.1
Binary.data | Renamed from content to data Min Cardinality changed from 1 to 0

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 2 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) & the dependency analysis

2.35.3.1 Terminology Bindings

Path | Definition | Type | Reference
Binary.contentType | The mime type of an attachment. Any valid mime type is allowed. | Required | Mime Types

2.35.3.2 Content Optionality

Binary.data is optional because it is excluded when a summary view of the Binary resource is requested (this can be useful when the binary resource is part of a wider request e.g. a search _include).

Otherwise, the data is not optional; a binary resource without any specified content it not useful.

2.35.3.3 Serving Binary Resources using the RESTful API

Binary resources behave slightly differently from all other resources on the RESTful API:

• When a read request is made with a FHIR type in the Accept header (e.g. "application/fhir+xml" or "application/fhir+json") the Binary resource is returned in the requested FHIR format. This applies even when

the binary data itself has a FHIR mime type

• When a read request is made with a FHIR type in the Accept header (e.g. "application/fhir+xml" or "application/fhir+json") the binary resource is returned in the requested FHIR format. This applies even when the binary data itself has a FHIR mime type

• The Binary resource is always represented in the native FHIR format when wrapped in a Bundle

• The _format overrides the accept header and SHALL be interpreted as using the standard FHIR mime types, even if the more generic mime types are given as a value.

• When the read request has some other type in the Accept header, then the content should be returned with the content type stated in the resource in the Content-Type header. E.g. if the content type in the resource is "application/pdf", then the content should be returned as a PDF directly. The _summary parameter does not apply in this case.

• due to the way the web infrastructure works, it is not possible to make blanket rules about the relationship between the "Accept" field in the HTTP request, and the return type, which is why there is no hard rule about this. However, the intent is that unless specifically requested, the FHIR XML/JSON representation is not returned

• When binary data is written to the server (create/update - POST or PUT), the data is accepted as is and treated as the content of a Binary, including when the content type is "application/fhir+xml" or "application/fhir+json", except for the special case where the content is actually a Binary resource.

• Note that when client requests a Binary resource using a generic mime type (application/xml, text/xml, or application/json), the server SHOULD return the content directly if the content-type format matches the requested mime type (e.g. if the Accept header is application/json, and the contentType is vnd.xacml+json). However, servers might not always be able to interpret mime types correctly, and clients SHOULD be prepared to receive either format.

Note that in the native binary representation, the normal resource metadata is not available directly, though some of it is replicated in the HTTP response headers. Specifically, the following elements of the resource have matching HTTP Headers:

• Binary.meta.lastUpdated: Last-Modified

• Binary.meta.versionId: ETag header

• Binary.contentType: Content-Type

• Binary.securityContext: X-Security-Context - this is a FHIR specific extension, primarily intended to allow the security context for a Binary resource to be specified when it is posted in native form. The content of the header is Binary.securityContext.reference

2.35.3.4 Security Considerations

Binary resources are not constrained to any list of safe content types (content types without active elements such as scripting or executable code), and therefore can be of any content type and encoding. Therefore, extra care needs to be taken to validate the content of the Binary resource against malicious or malformed content. For more details see Security of Narrative, since the security issues are similar.

Very often, the content of a Binary resource is sensitive, and the server should apply appropriate access control to the content. When the server itself generates the content, it implicitly knows what access control to apply. When the client provides the binary to the server itself, it uses the securityContext element (or the matching X-Security-Context HTTP header) to inform the server that the Binary resource should be treated as if it was the other resource. Typically, the other resource is a DocumentReference or similar resource that refers directly to the Binary resource, but that is not mandatory.

2.35.3.5 Read

This specification makes no rules concerning advanced read functionality for the Binary resource, such as:

• retrieving byte ranges

• Finding the size of the content before retrieving it (hough this can be achieved using a HEAD request)

• Stream content from Binary?

• generating content dynamically on the fly

Servers are free to support these features, but they are not required.

2.35.3.6 Search

This specification does not support searching on Binary resources.

2.35.3.7 Patch

The semantics of a PATCH operation on a Binary resource are not currently specified.

2.35.4 Resource Binary - Examples

URL: https://hl7.org/fhir/R4/binary-examples.html

2.35.4 Resource Binary - Examples

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Not Classified | Compartments: Not linked to any defined compartments

Example Name | id | Format
Binary Example - a PDF document | example | XML | JSON | Turtle
Example Patient photo as a binary | f006 | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

2.35.5 Resource Binary - Detailed Descriptions

URL: https://hl7.org/fhir/R4/binary-definitions.html

2.35.5 Resource Binary - Detailed Descriptions

FHIR Infrastructure Work Group | Maturity Level: N | Normative (from v4.0.0) | Security Category: Not Classified | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the Binary resource.

Binary
Element Id | Binary
Definition | A resource that represents the data of a single raw artifact as digital content accessible in its native format. A Binary resource can contain any content, whether text, image, pdf, zip archive, etc.
Cardinality | 0..*
Type | Resource
Requirements | There are situations where it is useful or required to handle pure binary content using the same framework as other resources.
Comments | Typically, Binary resources are used for handling content such as: CDA Documents (i.e. with XDS) PDF Documents Images (the Media resource is preferred for handling images, but not possible when the content is already binary - e.g. XDS).
Binary.contentType
Element Id | Binary.contentType
Definition | MimeType of the binary content represented as a standard MimeType (BCP 13).
Cardinality | 1..1
Terminology Binding | MimeType (Required)
Type | code
Summary | true
Binary.securityContext
Element Id | Binary.securityContext
Definition | This element identifies another resource that can be used as a proxy of the security sensitivity to use when deciding and enforcing access control rules for the Binary resource. Given that the Binary resource contains very few elements that can be used to determine the sensitivity of the data and relationships to individuals, the referenced resource stands in as a proxy equivalent for this purpose. This referenced resource may be related to the Binary (e.g. Media, DocumentReference), or may be some non-related Resource purely as a security proxy. E.g. to identify that the binary resource relates to a patient, and access should only be granted to applications that have access to the patient.
Cardinality | 0..1
Type | Reference(Any)
Summary | true
Comments | Very often, a server will also know of a resource that references the binary, and can automatically apply the appropriate access rules based on that reference. However, there are some circumstances where this is not appropriate, e.g. the binary is uploaded directly to the server without any linking resource, the binary is referred to from multiple different resources, and/or the binary is content such as an application logo that has less protection than any of the resources that reference it.
Binary.data
Element Id | Binary.data
Definition | The actual content, base64 encoded.
Cardinality | 0..1
Type | base64Binary
Summary | false
Comments | If the content type is itself base64 encoding, then this will be base64 encoded twice - what is created by un-base64ing the content must be the specified content type.

2.35.6 Resource Binary - Mappings

URL: https://hl7.org/fhir/R4/binary-mappings.html

2.35.6 Resource Binary - Mappings

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Not Classified | Compartments: Not linked to any defined compartments

Mappings:

RIM Mapping

Mappings for the binary resource (see Mappings to Other Standards for further information & status).

2.35.6.1 RIM Mapping (http://hl7.org/v3)

Binary | ED
contentType | ED.mediaType
securityContext | n/a
data | ED.data

2.35.7 Resource Binary - Extensions & Profiles

URL: https://hl7.org/fhir/R4/binary-profiles.html

2.35.7 Resource Binary - Extensions & Profiles

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Not Classified | Compartments: Not linked to any defined compartments

This table lists profiles and extensions for the Binary resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

2.35.7.1 Profiles

Resources of type Binary do not have extensions at the root element, but extensions MAY be present on the elements in the resource

No Profiles defined for this resource

2.35.7.2 Extensions

No Extensions defined for this resource
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

2.35.7.3 Search Extensions

No Search Extensions defined for this resource

2.35.8 Resource Binary - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/binary-version-maps.html

No validation errors - all conversions are clean

2.35.8 Resource Binary - R3/R4 Conversions

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Not Classified | Compartments: Not linked to any defined compartments

R3 : R4 Conversion maps for Binary.

Functional status for this map: 2 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid. (see documentation)

2.35.8.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/Binary3to4" = "R3 to R4 Conversions for Binary"

uses "http://hl7.org/fhir/3.0/StructureDefinition/Binary" alias BinaryR3 as source
uses "http://hl7.org/fhir/StructureDefinition/Binary" alias Binary as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group Binary(source src : BinaryR3, target tgt : Binary) extends Resource <<type+>> {
  src.contentType -> tgt.contentType;
  src.securityContext -> tgt.securityContext;
  src.content -> tgt.data;
}
```

2.35.8.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/Binary4to3" = "R4 to R3 Conversion for Binary"
```

```
uses "http://hl7.org/fhir/StructureDefinition/Binary" alias Binary as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/Binary" alias BinaryR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group Binary(source src : Binary, target tgt : BinaryR3) extends Resource <<type+>> {
  src.contentType -> tgt.contentType;
  src.securityContext -> tgt.securityContext;
  src.data -> tgt.content "content";
}
```

2.35.8.3 R4 Validation Errors

2.35.9 JSON Schema for Binary

URL: https://hl7.org/fhir/R4/binary.schema.json.html

2.35.9 JSON Schema for Binary

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is a representation of the json schema for Binary, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/Binary",
  "$ref": "#/definitions/Binary",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "Binary": {
      "description": "A resource that represents the data of a single raw artifact as digital content accessible in its
    native format.  A Binary resource can contain any content, whether text, image, pdf, zip archive, etc.",
      "properties": {
        "resourceType": {
          "description": "This is a Binary resource",
          "const": "Binary"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "contentType": {
          "description": "MimeType of the binary content represented as a standard MimeType (BCP 13).",
          "$ref": "#/definitions/code"
        },
        "_contentType": {
          "description": "Extensions for contentType",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "securityContext": {
          "description": "This element identifies another resource that can be used as a proxy of the security
    sensitivity to use when deciding and enforcing access control rules for the Binary resource. Given that the Binary
    resource contains very few elements that can be used to determine the sensitivity of the data and relationships to
    individuals, the referenced resource stands in as a proxy equivalent for this purpose. This referenced resource may
    be related to the Binary (e.g. Media, DocumentReference), or may be some non-related Resource purely as a security
    proxy. E.g. to identify that the binary resource relates to a patient, and access should only be granted to
    applications that have access to the patient.",
          "$ref": "Reference.schema.json#/definitions/Reference"
        },
        "data": {
          "description": "The actual content, base64 encoded.",
          "$ref": "#/definitions/base64Binary"
        },
```

```
        "_data": {
          "description": "Extensions for data",
          "$ref": "Element.schema.json#/definitions/Element"
        }
      },
      "additionalProperties": false,
      "required": [
        "resourceType"
      ]
    }
  }
}
```

2.35.10 Resource Binary - Maturity Dependency View

URL: https://hl7.org/fhir/R4/binary-dependencies.html

2.35.10 Resource Binary - Maturity Dependency View

FHIR Infrastructure Work Group | Maturity Level: N | Normative (from v4.0.0) | Compartments: Not linked to any defined compartments

| This page has been approved as part of an ANSI standard. See the Infrastructure Package for further details.

Dependency Graph for Binary FMM level 5

Name | Card. | Type | Dependency Analysis
Binary | | Resource
contentType | 1..1 | code
securityContext | 0..1 | Reference(Any) | Type Warning: (Any = FMM1-Trial Use vs. Element = FMM5-Normative)
data | 0..1 | base64Binary

2.36 Resource Bundle - Content

URL: https://hl7.org/fhir/R4/bundle.html
FHIR Infrastructure Work Group | Maturity Level: N | Normative (from v4.0.0) | Security Category: Not Classified | Compartments: Not linked to any defined compartments

| This page has been approved as part of an ANSI standard. See the Infrastructure Package for further details.

A container for a collection of resources.

2.36.1 Scope and Usage

One common operation performed with resources is to gather a collection of resources into a single instance with containing context. In FHIR this is referred to as "bundling" the resources together. These resource bundles are useful for a variety of different reasons, including:

• Returning a set of resources that meet some criteria as part of a server operation (see RESTful Search)

• Returning a set of versions of resources as part of the history operation on a server (see History)

• Sending a set of resources as part of a message exchange (see Messaging)

• Grouping a self-contained set of resources to act as an exchangeable and persistable collection with clinical integrity - e.g. a clinical document (see Documents)

• Creating/updating/deleting a set of resources on a server as a single operation (including doing so as a single atomic transaction) (see Transactions)

• Storing a collection of resources

2.36.2 Boundaries and Relationships

There are two ways to collect resources together for transport and persistence purposes - contained resources, and bundles. There is an important difference between the two:

• Contained resources are "in" the container resource - they can only ever be interpreted and/or changed in the context of the container

• A bundle is a collection of resources that can have an independent existence - for example, they might also be accessed directly using the RESTful API

In addition to these two technical mechanisms, there are three administrative and infrastructure resources which also support grouping of content. These resources do not contain resources directly, but instead use [Reference] to point to the grouped resources:

• The List resource – Enumerates a flat collection of resources and provides features for managing the collection. While a particular List instance may represent a "snapshot", from a business process perspective the notion of "List" is dynamic – items are added and removed over time. The list resource references other resources. Lists may be curated and have specific business meaning.

• The Group resource – Defines a group of specific people, animals, devices, etc. by enumerating them, or by describing qualities that group members have. The group resource refers to other resources, possibly

implicitly. Groups are intended to be acted upon or observed as a whole; e.g. performing therapy on a group, calculating risk for a group, etc. This resource will commonly be used for public health (e.g. describing an at-risk population), clinical trials (e.g. defining a test subject pool) and similar purposes.

• The Composition resource – Defines a set of healthcare-related information that is assembled together into a single logical document that provides a single coherent statement of meaning, establishes its own context and that has clinical attestation with regard to who is making the statement. The composition resource provides the basic structure of a FHIR document. The full content of the document is expressed using a Bundle. Compositions will often reference Lists as the focus of particular sections.

These three resources represent meaningful groupings of the resources they refer to (e.g. a discharge medication List, a Group of participants for a clinical trial, a set of resources that form a signed document), while a Bundle is merely is a container for resources used for transfer and storage. This list is not exhaustive; other resources also provde grouping functionality.

2.36.3 Resource Content

• Structure
• UML
• XML
• JSON
• Turtle
• R3 Diff
• All

Structure

Name | Flags | Card. | Type | Description & Constraints
Bundle | ΣIN | | Resource | Contains a collection of resources + Rule: total only when a search or history + Rule: entry.search only when a search + Rule: entry.request mandatory for batch/transaction/history, otherwise prohibited + Rule: entry.response mandatory for batch-response/transaction-response/history, otherwise prohibited + Rule: FullUrl must be unique in a bundle, or else entries with the same fullUrl must have different meta.versionId (except in history bundles) + Rule: A document must have an identifier with a system and a value + Rule: A document must have a date + Rule: A document must have a Composition as the first resource + Rule: A message must have a MessageHeader as the first resource Elements defined in Ancestors: id, meta, implicitRules, language
identifier | Σ | 0..1 | Identifier | Persistent identifier for the bundle
type | Σ | 1..1 | code | document | message | transaction | transaction-response | batch | batch-response | history | searchset | collection BundleType (Required)
timestamp | Σ | 0..1 | instant | When the bundle was assembled
total | ΣI | 0..1 | unsignedInt | If search, the total number of matches
link | Σ | 0..* | BackboneElement | Links related to this Bundle
relation | Σ | 1..1 | string | See http://www.iana.org/assignments/link-relations/link-relations.xhtml#link-relations-1
url | Σ | 1..1 | uri | Reference details for the link
entry | ΣI | 0..* | BackboneElement | Entry in the bundle - will have a resource or information + Rule: must be a resource unless there's a request or response + Rule: fullUrl cannot be a version specific reference This repeating element order: For bundles of type 'document' and 'message', the first resource is special (must be Composition or MessageHeader respectively). For all bundles, the meaning of the order of entries depends on the bundle type
link | Σ | 0..* | see link | Links related to this entry
fullUrl | Σ | 0..1 | uri | URI for resource (Absolute URL server address or URI for UUID/OID)
resource | Σ | 0..1 | Resource | A resource in the bundle
search | ΣI | 0..1 | BackboneElement | Search related information
mode | Σ | 0..1 | code | match | include | outcome - why this is in the result set SearchEntryMode (Required)
score | Σ | 0..1 | decimal | Search ranking (between 0 and 1)
request | ΣI | 0..1 | BackboneElement | Additional execution information (transaction/batch/history)
method | Σ | 1..1 | code | GET | HEAD | POST | PUT | DELETE | PATCH HTTPVerb (Required)
url | Σ | 1..1 | uri | URL for HTTP equivalent of this entry
ifNoneMatch | Σ | 0..1 | string | For managing cache currency
ifModifiedSince | Σ | 0..1 | instant | For managing cache currency
ifMatch | Σ | 0..1 | string | For managing update contention
ifNoneExist | Σ | 0..1 | string | For conditional creates
response | ΣI | 0..1 | BackboneElement | Results of execution (transaction/batch/history)
status | Σ | 1..1 | string | Status response code (text optional)
location | Σ | 0..1 | uri | The location (if the operation returns a location)
etag | Σ | 0..1 | string | The Etag for the resource (if relevant)
lastModified | Σ | 0..1 | instant | Server's date time modified
outcome | Σ | 0..1 | Resource | OperationOutcome with hints and warnings (for batch/transaction)
signature | ΣTU | 0..1 | Signature | Digital Signature
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Bundle xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
```

```
 <identifier><!-- 0..1 Identifier Persistent identifier for the bundle --></identifier>
 <type value="[code]"/><!-- 1..1 document | message | transaction | transaction-response | batch | batch-response |
    history | searchset | collection -->
 <timestamp value="[instant]"/><!-- 0..1 When the bundle was assembled -->
 <total value="[unsignedInt]"/><!--  0..1 If search, the total number of matches -->
 <link>  <!-- 0..* Links related to this Bundle -->
  <relation value="[string]"/><!-- 1..1 See http://www.iana.org/assignments/link-relations/link-relations.xhtml#link-
    relations-1 -->
  <url value="[uri]"/><!-- 1..1 Reference details for the link -->
 </link>
 <entry>  <!-- 0..* Entry in the bundle - will have a resource or information -->
  <link><!-- 0..* Content as for Bundle.link Links related to this entry --></link>
  <fullUrl value="[uri]"/><!-- 0..1 URI for resource (Absolute URL server address or URI for UUID/OID) -->
  <resource><!-- 0..1 Resource A resource in the bundle --></resource>
  <search>  <!--  0..1 Search related information -->
   <mode value="[code]"/><!-- 0..1 match | include | outcome - why this is in the result set -->
   <score value="[decimal]"/><!-- 0..1 Search ranking (between 0 and 1) -->
  </search>
  <request>  <!--  0..1 Additional execution information (transaction/batch/history) -->
   <method value="[code]"/><!-- 1..1 GET | HEAD | POST | PUT | DELETE | PATCH -->
   <url value="[uri]"/><!-- 1..1 URL for HTTP equivalent of this entry -->
   <ifNoneMatch value="[string]"/><!-- 0..1 For managing cache currency -->
   <ifModifiedSince value="[instant]"/><!-- 0..1 For managing cache currency -->
   <ifMatch value="[string]"/><!-- 0..1 For managing update contention -->
   <ifNoneExist value="[string]"/><!-- 0..1 For conditional creates -->
  </request>
  <response>  <!--  0..1 Results of execution (transaction/batch/history) -->
   <status value="[string]"/><!-- 1..1 Status response code (text optional) -->
   <location value="[uri]"/><!-- 0..1 The location (if the operation returns a location) -->
   <etag value="[string]"/><!-- 0..1 The Etag for the resource (if relevant) -->
   <lastModified value="[instant]"/><!-- 0..1 Server's date time modified -->
   <outcome><!-- 0..1 Resource OperationOutcome with hints and warnings (for batch/transaction) --></outcome>
  </response>
 </entry>
 <signature><!-- 0..1 Signature Digital Signature --></signature>
</Bundle>
```

JSON Template

```
{
  "resourceType" : "Bundle",
  // from Resource: id, meta, implicitRules, and language
  "identifier" : { Identifier }, // Persistent identifier for the bundle
  "type" : "<code>", // R!  document | message | transaction | transaction-response | batch | batch-response | history |
     searchset | collection
  "timestamp" : "<instant>", // When the bundle was assembled
  "total" : "<unsignedInt>", // C? If search, the total number of matches
  "link" : [{ // Links related to this Bundle
    "relation" : "<string>", // R!  See http://www.iana.org/assignments/link-relations/link-relations.xhtml#link-
    relations-1
    "url" : "<uri>" // R!  Reference details for the link
  }],
  "entry" : [{ // Entry in the bundle - will have a resource or information
    "link" : [{ Content as for Bundle.link }], // Links related to this entry
    "fullUrl" : "<uri>", // URI for resource (Absolute URL server address or URI for UUID/OID)
    "resource" : { Resource }, // A resource in the bundle
    "search" : { // C? Search related information
      "mode" : "<code>", // match | include | outcome - why this is in the result set
      "score" : <decimal> // Search ranking (between 0 and 1)
    },
    "request" : { // C? Additional execution information (transaction/batch/history)
      "method" : "<code>", // R!  GET | HEAD | POST | PUT | DELETE | PATCH
      "url" : "<uri>", // R!  URL for HTTP equivalent of this entry
      "ifNoneMatch" : "<string>", // For managing cache currency
      "ifModifiedSince" : "<instant>", // For managing cache currency
      "ifMatch" : "<string>", // For managing update contention
      "ifNoneExist" : "<string>" // For conditional creates
    },
    "response" : { // C? Results of execution (transaction/batch/history)
      "status" : "<string>", // R!  Status response code (text optional)
      "location" : "<uri>", // The location (if the operation returns a location)
      "etag" : "<string>", // The Etag for the resource (if relevant)
      "lastModified" : "<instant>", // Server's date time modified
      "outcome" : { Resource } // OperationOutcome with hints and warnings (for batch/transaction)
    }
  }],
  "signature" : { Signature } // Digital Signature
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Bundle;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  fhir:Bundle.identifier [ Identifier ]; # 0..1 Persistent identifier for the bundle
  fhir:Bundle.type [ code ]; # 1..1 document | message | transaction | transaction-response | batch | batch-response |
    history | searchset | collection
```

```
  fhir:Bundle.timestamp [ instant ]; # 0..1 When the bundle was assembled
  fhir:Bundle.total [ unsignedInt ]; # 0..1 If search, the total number of matches
  fhir:Bundle.link [ # 0..* Links related to this Bundle
    fhir:Bundle.link.relation [ string ]; # 1..1 See http://www.iana.org/assignments/link-relations/link-
    relations.xhtml#link-relations-1
    fhir:Bundle.link.url [ uri ]; # 1..1 Reference details for the link
  ], ...;
  fhir:Bundle.entry [ # 0..* Entry in the bundle - will have a resource or information
    fhir:Bundle.entry.link [ See Bundle.link ], ... ; # 0..* Links related to this entry
    fhir:Bundle.entry.fullUrl [ uri ]; # 0..1 URI for resource (Absolute URL server address or URI for UUID/OID)
    fhir:Bundle.entry.resource [ Resource ]; # 0..1 A resource in the bundle
    fhir:Bundle.entry.search [ # 0..1 Search related information
      fhir:Bundle.entry.search.mode [ code ]; # 0..1 match | include | outcome - why this is in the result set
      fhir:Bundle.entry.search.score [ decimal ]; # 0..1 Search ranking (between 0 and 1)
    ];
    fhir:Bundle.entry.request [ # 0..1 Additional execution information (transaction/batch/history)
      fhir:Bundle.entry.request.method [ code ]; # 1..1 GET | HEAD | POST | PUT | DELETE | PATCH
      fhir:Bundle.entry.request.url [ uri ]; # 1..1 URL for HTTP equivalent of this entry
      fhir:Bundle.entry.request.ifNoneMatch [ string ]; # 0..1 For managing cache currency
      fhir:Bundle.entry.request.ifModifiedSince [ instant ]; # 0..1 For managing cache currency
      fhir:Bundle.entry.request.ifMatch [ string ]; # 0..1 For managing update contention
      fhir:Bundle.entry.request.ifNoneExist [ string ]; # 0..1 For conditional creates
    ];
    fhir:Bundle.entry.response [ # 0..1 Results of execution (transaction/batch/history)
      fhir:Bundle.entry.response.status [ string ]; # 1..1 Status response code (text optional)
      fhir:Bundle.entry.response.location [ uri ]; # 0..1 The location (if the operation returns a location)
      fhir:Bundle.entry.response.etag [ string ]; # 0..1 The Etag for the resource (if relevant)
      fhir:Bundle.entry.response.lastModified [ instant ]; # 0..1 Server's date time modified
      fhir:Bundle.entry.response.outcome [ Resource ]; # 0..1 OperationOutcome with hints and warnings (for
    batch/transaction)
    ];
  ], ...;
  fhir:Bundle.signature [ Signature ]; # 0..1 Digital Signature
]
```

Changes since R3

Bundle

Bundle.type | Change value set from http://hl7.org/fhir/ValueSet/bundle-type to http://hl7.org/fhir/ValueSet/bundle-type|4.0.1

Bundle.timestamp | Added Element

Bundle.entry.search.mode | Change value set from http://hl7.org/fhir/ValueSet/search-entry-mode to http://hl7.org/fhir/ValueSet/search-entry-mode|4.0.1

Bundle.entry.request.method | Change value set from http://hl7.org/fhir/ValueSet/http-verb to http://hl7.org/fhir/ValueSet/http-verb|4.0.1

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 28 tests of which 3 fail to execute. 3 fail round-trip testing and 16 r3 resources are invalid (0 errors).)

Structure

Name | Flags | Card. | Type | Description & Constraints

Bundle | ΣIN | | Resource | Contains a collection of resources + Rule: total only when a search or history + Rule: entry.search only when a search + Rule: entry.request mandatory for batch/transaction/history, otherwise prohibited + Rule: entry.response mandatory for batch-response/transaction-response/history, otherwise prohibited + Rule: FullUrl must be unique in a bundle, or else entries with the same fullUrl must have different meta.versionId (except in history bundles) + Rule: A document must have an identifier with a system and a value + Rule: A document must have a date + Rule: A document must have a Composition as the first resource + Rule: A message must have a MessageHeader as the first resource Elements defined in Ancestors: id, meta, implicitRules, language

identifier | Σ | 0..1 | Identifier | Persistent identifier for the bundle

type | Σ | 1..1 | code | document | message | transaction | transaction-response | batch | batch-response | history | searchset | collection BundleType (Required)

timestamp | Σ | 0..1 | instant | When the bundle was assembled

total | ΣI | 0..1 | unsignedInt | If search, the total number of matches

link | Σ | 0..* | BackboneElement | Links related to this Bundle

relation | Σ | 1..1 | string | See http://www.iana.org/assignments/link-relations/link-relations.xhtml#link-relations-1

url | Σ | 1..1 | uri | Reference details for the link

entry | ΣI | 0..* | BackboneElement | Entry in the bundle - will have a resource or information + Rule: must be a resource unless there's a request or response + Rule: fullUrl cannot be a version specific reference This repeating element order: For bundles of type 'document' and 'message', the first resource is special (must be Composition or MessageHeader respectively). For all bundles, the meaning of the order of entries depends on the bundle type

link | Σ | 0..* | see link | Links related to this entry

fullUrl | Σ | 0..1 | uri | URI for resource (Absolute URL server address or URI for UUID/OID)

resource | Σ | 0..1 | Resource | A resource in the bundle

search | ΣI | 0..1 | BackboneElement | Search related information

mode | Σ | 0..1 | code | match | include | outcome - why this is in the result set SearchEntryMode (Required)

score | Σ | 0..1 | decimal | Search ranking (between 0 and 1)

request | ΣI | 0..1 | BackboneElement | Additional execution information (transaction/batch/history)

method | Σ | 1..1 | code | GET | HEAD | POST | PUT | DELETE | PATCH HTTPVerb (Required)

url | Σ | 1..1 | uri | URL for HTTP equivalent of this entry

ifNoneMatch | Σ | 0..1 | string | For managing cache currency

ifModifiedSince | Σ | 0..1 | instant | For managing cache currency
ifMatch | Σ | 0..1 | string | For managing update contention
ifNoneExist | Σ | 0..1 | string | For conditional creates
response | ΣI | 0..1 | BackboneElement | Results of execution (transaction/batch/history)
status | Σ | 1..1 | string | Status response code (text optional)
location | Σ | 0..1 | uri | The location (if the operation returns a location)
etag | Σ | 0..1 | string | The Etag for the resource (if relevant)
lastModified | Σ | 0..1 | instant | Server's date time modified
outcome | Σ | 0..1 | Resource | OperationOutcome with hints and warnings (for batch/transaction)
signature | ΣTU | 0..1 | Signature | Digital Signature
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Bundle xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <identifier><!-- 0..1 Identifier Persistent identifier for the bundle --></identifier>
 <type value="[code]"/><!-- 1..1 document | message | transaction | transaction-response | batch | batch-response |
    history | searchset | collection -->
 <timestamp value="[instant]"/><!-- 0..1 When the bundle was assembled -->
 <total value="[unsignedInt]"/><!--  0..1 If search, the total number of matches -->
 <link>  <!-- 0..* Links related to this Bundle -->
  <relation value="[string]"/><!-- 1..1 See http://www.iana.org/assignments/link-relations/link-relations.xhtml#link-
    relations-1 -->
  <url value="[uri]"/><!-- 1..1 Reference details for the link -->
 </link>
 <entry>  <!-- 0..* Entry in the bundle - will have a resource or information -->
  <link><!-- 0..* Content as for Bundle.link Links related to this entry --></link>
  <fullUrl value="[uri]"/><!-- 0..1 URI for resource (Absolute URL server address or URI for UUID/OID) -->
  <resource><!-- 0..1 Resource A resource in the bundle --></resource>
  <search>  <!--  0..1 Search related information -->
   <mode value="[code]"/><!-- 0..1 match | include | outcome - why this is in the result set -->
   <score value="[decimal]"/><!-- 0..1 Search ranking (between 0 and 1) -->
  </search>
  <request>  <!--  0..1 Additional execution information (transaction/batch/history) -->
   <method value="[code]"/><!-- 1..1 GET | HEAD | POST | PUT | DELETE | PATCH -->
   <url value="[uri]"/><!-- 1..1 URL for HTTP equivalent of this entry -->
   <ifNoneMatch value="[string]"/><!-- 0..1 For managing cache currency -->
   <ifModifiedSince value="[instant]"/><!-- 0..1 For managing cache currency -->
   <ifMatch value="[string]"/><!-- 0..1 For managing update contention -->
   <ifNoneExist value="[string]"/><!-- 0..1 For conditional creates -->
  </request>
  <response>  <!--  0..1 Results of execution (transaction/batch/history) -->
   <status value="[string]"/><!-- 1..1 Status response code (text optional) -->
   <location value="[uri]"/><!-- 0..1 The location (if the operation returns a location) -->
   <etag value="[string]"/><!-- 0..1 The Etag for the resource (if relevant) -->
   <lastModified value="[instant]"/><!-- 0..1 Server's date time modified -->
   <outcome><!-- 0..1 Resource OperationOutcome with hints and warnings (for batch/transaction) --></outcome>
  </response>
 </entry>
 <signature><!-- 0..1 Signature Digital Signature --></signature>
</Bundle>
```

JSON Template

```
{
  "resourceType" : "Bundle",
  // from Resource: id, meta, implicitRules, and language
  "identifier" : { Identifier }, // Persistent identifier for the bundle
  "type" : "<code>", // R!  document | message | transaction | transaction-response | batch | batch-response | history |
     searchset | collection
  "timestamp" : "<instant>", // When the bundle was assembled
  "total" : "<unsignedInt>", // C? If search, the total number of matches
  "link" : [{ // Links related to this Bundle
    "relation" : "<string>", // R!  See http://www.iana.org/assignments/link-relations/link-relations.xhtml#link-
    relations-1
    "url" : "<uri>" // R!  Reference details for the link
  }],
  "entry" : [{ // Entry in the bundle - will have a resource or information
    "link" : [{ Content as for Bundle.link }], // Links related to this entry
    "fullUrl" : "<uri>", // URI for resource (Absolute URL server address or URI for UUID/OID)
    "resource" : { Resource }, // A resource in the bundle
    "search" : { // C? Search related information
      "mode" : "<code>", // match | include | outcome - why this is in the result set
      "score" : <decimal> // Search ranking (between 0 and 1)
    },
    "request" : { // C? Additional execution information (transaction/batch/history)
      "method" : "<code>", // R!  GET | HEAD | POST | PUT | DELETE | PATCH
      "url" : "<uri>", // R!  URL for HTTP equivalent of this entry
      "ifNoneMatch" : "<string>", // For managing cache currency
      "ifModifiedSince" : "<instant>", // For managing cache currency
      "ifMatch" : "<string>", // For managing update contention
      "ifNoneExist" : "<string>" // For conditional creates
    },
    "response" : { // C? Results of execution (transaction/batch/history)
      "status" : "<string>", // R!  Status response code (text optional)
      "location" : "<uri>", // The location (if the operation returns a location)
```

```
      "etag" : "<string>", // The Etag for the resource (if relevant)
      "lastModified" : "<instant>", // Server's date time modified
      "outcome" : { Resource } // OperationOutcome with hints and warnings (for batch/transaction)
    }
  }],
  "signature" : { Signature } // Digital Signature
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Bundle;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  fhir:Bundle.identifier [ Identifier ]; # 0..1 Persistent identifier for the bundle
  fhir:Bundle.type [ code ]; # 1..1 document | message | transaction | transaction-response | batch | batch-response |
    history | searchset | collection
  fhir:Bundle.timestamp [ instant ]; # 0..1 When the bundle was assembled
  fhir:Bundle.total [ unsignedInt ]; # 0..1 If search, the total number of matches
  fhir:Bundle.link [ # 0..* Links related to this Bundle
    fhir:Bundle.link.relation [ string ]; # 1..1 See http://www.iana.org/assignments/link-relations/link-
    relations.xhtml#link-relations-1
    fhir:Bundle.link.url [ uri ]; # 1..1 Reference details for the link
  ], ...;
  fhir:Bundle.entry [ # 0..* Entry in the bundle - will have a resource or information
    fhir:Bundle.entry.link [ See Bundle.link ], ... ; # 0..* Links related to this entry
    fhir:Bundle.entry.fullUrl [ uri ]; # 0..1 URI for resource (Absolute URL server address or URI for UUID/OID)
    fhir:Bundle.entry.resource [ Resource ]; # 0..1 A resource in the bundle
    fhir:Bundle.entry.search [ # 0..1 Search related information
      fhir:Bundle.entry.search.mode [ code ]; # 0..1 match | include | outcome - why this is in the result set
      fhir:Bundle.entry.search.score [ decimal ]; # 0..1 Search ranking (between 0 and 1)
    ];
    fhir:Bundle.entry.request [ # 0..1 Additional execution information (transaction/batch/history)
      fhir:Bundle.entry.request.method [ code ]; # 1..1 GET | HEAD | POST | PUT | DELETE | PATCH
      fhir:Bundle.entry.request.url [ uri ]; # 1..1 URL for HTTP equivalent of this entry
      fhir:Bundle.entry.request.ifNoneMatch [ string ]; # 0..1 For managing cache currency
      fhir:Bundle.entry.request.ifModifiedSince [ instant ]; # 0..1 For managing cache currency
      fhir:Bundle.entry.request.ifMatch [ string ]; # 0..1 For managing update contention
      fhir:Bundle.entry.request.ifNoneExist [ string ]; # 0..1 For conditional creates
    ];
    fhir:Bundle.entry.response [ # 0..1 Results of execution (transaction/batch/history)
      fhir:Bundle.entry.response.status [ string ]; # 1..1 Status response code (text optional)
      fhir:Bundle.entry.response.location [ uri ]; # 0..1 The location (if the operation returns a location)
      fhir:Bundle.entry.response.etag [ string ]; # 0..1 The Etag for the resource (if relevant)
      fhir:Bundle.entry.response.lastModified [ instant ]; # 0..1 Server's date time modified
      fhir:Bundle.entry.response.outcome [ Resource ]; # 0..1 OperationOutcome with hints and warnings (for
    batch/transaction)
    ];
  ], ...;
  fhir:Bundle.signature [ Signature ]; # 0..1 Digital Signature
]
```

Changes since Release 3

Bundle
Bundle.type | Change value set from http://hl7.org/fhir/ValueSet/bundle-type to http://hl7.org/fhir/ValueSet/bundle-type|4.0.1
Bundle.timestamp | Added Element
Bundle.entry.search.mode | Change value set from http://hl7.org/fhir/ValueSet/search-entry-mode to http://hl7.org/fhir/ValueSet/search-entry-mode|4.0.1
Bundle.entry.request.method | Change value set from http://hl7.org/fhir/ValueSet/http-verb to http://hl7.org/fhir/ValueSet/http-verb|4.0.1

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 28 tests of which 3 fail to execute. 3 fail round-trip testing and 16 r3 resources are invalid (0 errors).)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) & the dependency analysis

2.36.3.1 Terminology Bindings

Path | Definition | Type | Reference
Bundle.type | Indicates the purpose of a bundle - how it is intended to be used. | Required | BundleType
Bundle.entry.search.mode | Why an entry is in the result set - whether it's included as a match or because of an _include requirement, or to convey information or warning information about the search process. | Required | SearchEntryMode
Bundle.entry.request.method | HTTP verbs (in the HTTP command line). See [HTTP rfc](https://tools.ietf.org/html/rfc7231) for details. | Required | HTTPVerb

2.36.3.2 Constraints

id | Level | Location | Description | Expression
bdl-1 | Rule | (base) | total only when a search or history | total.empty() or (type = 'searchset') or (type = 'history')
bdl-2 | Rule | (base) | entry.search only when a search | entry.search.empty() or (type = 'searchset')

bdl-3 | Rule | (base) | entry.request mandatory for batch/transaction/history, otherwise prohibited | entry.all(request.exists() = (%resource.type = 'batch' or %resource.type = 'transaction' or %resource.type = 'history'))
bdl-4 | Rule | (base) | entry.response mandatory for batch-response/transaction-response/history, otherwise prohibited | entry.all(response.exists() = (%resource.type = 'batch-response' or %resource.type = 'transaction-response' or %resource.type = 'history'))
bdl-5 | Rule | Bundle.entry | must be a resource unless there's a request or response | resource.exists() or request.exists() or response.exists()
bdl-7 | Rule | (base) | FullUrl must be unique in a bundle, or else entries with the same fullUrl must have different meta.versionId (except in history bundles) | (type = 'history') or entry.where(fullUrl.exists()).select(fullUrl&resource.meta.versionId).isDistinct()
bdl-8 | Rule | Bundle.entry | fullUrl cannot be a version specific reference | fullUrl.contains('/_history/').not()
bdl-9 | Rule | (base) | A document must have an identifier with a system and a value | type = 'document' implies (identifier.system.exists() and identifier.value.exists())
bdl-10 | Rule | (base) | A document must have a date | type = 'document' implies (timestamp.hasValue())
bdl-11 | Rule | (base) | A document must have a Composition as the first resource | type = 'document' implies entry.first().resource.is(Composition)
bdl-12 | Rule | (base) | A message must have a MessageHeader as the first resource | type = 'message' implies entry.first().resource.is(MessageHeader)

2.36.3.3 Notes about Bundle

• Conceptually, a bundle is a list of resources with some context (named links, and status on the entries)

• Since a Bundle is itself a Resource it has the same common metadata as all resources, including profile assertions, tags, and security labels

• Although there are no extensions on the Bundle itself, link, entry, and search/request/response can all have extensions. See Patient and Location for examples on search

• Both Bundle.link and Bundle.entry.link are defined to support providing additional context when Bundles are used (e.g. HATEOAS). Bundle.entry.link corresponds to links found in the HTTP header if the resource in the entry was read directly. This specification defines some specific uses of Bundle.link for searching and paging, but no specific uses for Bundle.entry.link, and no defined function in a transaction - meaning is implementation specific

• Bundles have both .id and .identifier - see Resource Identities for further information

2.36.3.4 Using Bundles

The content and rules for using a Bundle depend on the type of the bundle. Note that all bundle types use resource identity resolution as described below.

2.36.3.4.1 Document

A document Bundle (type = "document") consists of a series of entries, the first of which is a Composition. Each entry element SHALL contain a resource. See Documents for further information.

Example

2.36.3.4.2 Message

A message Bundle (type = "message") consists of a series of entries, the first of which is a MessageHeader. Each entry element SHALL contain a resource. See Messaging for further information.

Example Request and Response

2.36.3.4.3 Search Results

A set of search results (type = "searchset") consists of a series of 0 or more entries. Each entry element SHALL contain a resource. See Search for further information.

In addition, Bundle.total may be used to return the total number of resources that match the search, and that may be returned by following the "next" link.

For each entry, a search set can also contain two specific pieces of search related information:

• search.mode: An indication of whether the resource is in the search set because it matched the search criteria or whether it is included because another resource refers to it (e.g. by the _include parameter)

• search.score: The server's search ranking score for the entry. Servers are not required to return a ranking score, but if they do, 1 is most relevant, and 0 is least relevant. Note: often, search results are sorted by score, but the client may specify a different sort order (see Search Relevance)

Example

2.36.3.4.4 History

A change history (type = "history") consists of a series of 0 or more entries. Each entry element SHALL contain a request element that describes the change that was made and, if the method is a POST or PUT, a resource that represents the state of the resource at the conclusion of the operation. A response element SHALL also be present so that consumers can access the location header. See History for further information.

In addition, Bundle.total may be used to return the total number of resources that are included in the change history, including those that may be returned by following the "next" link.

2.36.3.4.5 Transaction / Batch

A transaction (type = "transaction") or batch (type = "batch") consists of a series of 0 or more entries. Each entry element SHALL contain a request element has the details of an HTTP operation that informs the system processing the transaction what to do with the entry. If the entry method is a 'PUT' or 'POST', then the entry SHALL contain a resource that becomes the body of the HTTP operation. See Transactions for further information.

Example

2.36.3.4.6 Transaction/Batch Response

A transaction response (type = "transaction-response") or batch response (type="batch-response") consists of a series of 0 or more entries: 1 for each entry in the transaction or batch it is in response to. Each entry element SHALL contain a response element which indicates the outcome of the HTTP operation that the server performed for the entry.

Example

2.36.3.4.7 Collection

A collection (type = "collection") consists of a series of 0 or more entries. No particular use with respect to the FHIR specification is associated with this Bundle. Each entry element SHALL contain a resource.

Example

2.36.4 Resource URL & Uniqueness rules in a bundle

Except for transactions and batches, each entry in a Bundle must have a fullUrl which is the identity of the resource in the entry. Note that this is not a versioned reference to the resource, but its identity. Where a resource is not assigned a persistent identity that can be used in the Bundle, a UUID should be used (urn:uuid:...).

For transactions and batches, entries MAY not have fullURLs when the entry.request.method = POST, and the resource has no identity. Note that even in this case, there may still be a fullURL in a transaction on a POST so that relationships between resources can be represented (see Transactions).

A given version of a resource SHALL only appear once in each Bundle. There might, however, be multiple versions of a single resource present in a single bundle. This would be expected in Bundles of type history, and also might be necessitated by closely tracking Provenance.

Note that the meaning of an unversioned reference to a resource that appears multiple times is potentially ambiguous, though processors may have additional informaton to help resolve this (e.g. change order in a history bundle).

When processing batches and transactions, it is at server discretion how to behave if multiple versions of a single resource are present.

2.36.4.1 Resolving references in Bundles

The Bundle resource is a packaging construct that has one of more entries that are other kinds of resources. Those resources themselves have references to other resources - e.g. an Observation that refers to a Patient. The referenced resources may also be found in the Bundle. For example, the system that constructed the Bundle may have included both the Observation and the Patient. The content of the references between resources doesn't change because of the bundle.

This section documents a method that resolves references correctly within a bundle. Note that this method does not define any new semantics; resolution is based on the way resource identity and resource references work.

Applications reading a Bundle should always look for a resource by its identity in the bundle first before trying to access it by its URL externally.

How to resolve a reference in a Bundle:

• If the reference is not an absolute reference, convert it to an absolute URL: if the reference has the format [type]/[id], and if the fullUrl for the bundle entry containing the resource is a RESTful one (see the RESTful URL regex) extract the [root] from the fullUrl, and append the reference (type/id) to it then try to resolve within the bundle as for a RESTful URL reference. If no resolution is possible, then the reference has no defined meaning within this specification else no resolution is possible and the reference has no defined meaning within this specification

• else Look for an entry with a fullUrl that matches the URI in the reference if no match is found, and the URI is a URL that can be resolved (e.g. if an http: URL), try accessing it directly)

Note, in addition, that a reference may be by identifier, and if it is, and there is no URL, it may be resolved by scanning the ids in the bundle. Note also that transactions may contain conditional references that must be resolved by the server before processing the matches.

If the reference is version specific (either relative or absolute), then remove the version from the URL before matching fullUrl, and then match the version based on Resource.meta.versionId. Note that the rules for

resolving references in contained resources are the same as those for resolving resources in the resource that contains the contained resource.

If multiple matches are found, it is ambiguous which is correct. Applications MAY return an error or take some other action as they deem appropriate.

There is an example Bundle that demonstrates these rules.

2.36.4.2 Serving Bundles using the RESTful API

The Bundle resource type has an end-point like all most other resources. This end-point serves the usual interactions. Bundles are treated as static resources on the /Bundle end-point (i.e. when a batch, transaction, or message is POSTed to /Bundle, it is stored as is, and the content is not processed as batch, transaction, or message - instead, they are processed like normal resource, with indexing / auditing etc. Performing a GET /Bundle/[location] will return the same resource.

The Bundle end point does have two special search parameters - composition and message, which allow for chained search into the first (special) entries in document and message resources.

2.36.5 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
composition TU | reference | The first resource in the bundle, if the bundle type is "document" - this is a composition, and this parameter provides access to search its contents | Bundle.entry[0].resource (Composition)
identifier TU | token | Persistent identifier for the bundle | Bundle.identifier
message TU | reference | The first resource in the bundle, if the bundle type is "message" - this is a message header, and this parameter provides access to search its contents | Bundle.entry[0].resource (MessageHeader)
timestamp TU | date | When the bundle was assembled | Bundle.timestamp
type TU | token | document | message | transaction | transaction-response | batch | batch-response | history | searchset | collection | Bundle.type

2.36.6 Resource Bundle - Examples

URL: https://hl7.org/fhir/R4/bundle-examples.html

2.36.6 Resource Bundle - Examples

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Not Classified | Compartments: Not linked to any defined compartments

In addition to the examples below, there are other examples of Bundles through the specification:

- Document
- Message Request
- Message Response
- Collection

Example Name | id | Format
An example of a search response | bundle-example | XML | JSON | Turtle
An example transaction | bundle-transaction | XML | JSON | Turtle
Response for the example transaction | bundle-response | XML | JSON | Turtle
Retrieve a patients medications, allergies, problems and immunizations | bundle-request-medsallergies | XML | JSON | Turtle
Response for Retrieve a patients medications, allergies, problems and immunizations | bundle-response-medsallergies | XML | JSON | Turtle
Retrieve a patient’s conditions, medication statements and blood pressure readings since a given date | bundle-request-simplesummary | XML | JSON | Turtle
Response for Retrieve a patient’s conditions, medication statements and blood pressure readings since a given date | bundle-response-simplesummary | XML | JSON | Turtle
An example an empty search result with a warning | bundle-search-warning | XML | JSON | Turtle
An example bundle that demonstrates various reference resolution paths | bundle-references | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

2.36.7 Resource Bundle - Detailed Descriptions

URL: https://hl7.org/fhir/R4/bundle-definitions.html

2.36.7 Resource Bundle - Detailed Descriptions

FHIR Infrastructure Work Group | Maturity Level: N | Normative (from v4.0.0) | Security Category: Not Classified | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the Bundle resource.

Bundle

Element Id | Bundle
Definition | A container for a collection of resources.
Cardinality | 0..*
Type | Resource
Summary | true
Invariants | Defined on this element bdl-1 Rule total only when a search or history total.empty() or (type = 'searchset') or (type = 'history') bdl-2 Rule entry.search only when a search entry.search.empty() or (type = 'searchset') bdl-3 Rule entry.request mandatory for batch/transaction/history, otherwise prohibited entry.all(request.exists() = (%resource.type = 'batch' or %resource.type = 'transaction' or %resource.type = 'history')) bdl-4 Rule entry.response mandatory for batch-response/transaction-response/history, otherwise prohibited entry.all(response.exists() = (%resource.type = 'batch-response' or %resource.type = 'transaction-response' or %resource.type = 'history')) bdl-7 Rule FullUrl must be unique in a bundle, or else entries with the same fullUrl must have different meta.versionId (except in history bundles) (type = 'history') or entry.where(fullUrl.exists()).select(fullUrl&resource.meta.versionId).isDistinct() bdl-9 Rule A document must have an identifier with a system and a value type = 'document' implies (identifier.system.exists() and identifier.value.exists()) bdl-10 Rule A document must have a date type = 'document' implies (timestamp.hasValue()) bdl-11 Rule A document must have a Composition as the first resource type = 'document' implies entry.first().resource.is(Composition) bdl-12 Rule A message must have a MessageHeader as the first resource type = 'message' implies entry.first().resource.is(MessageHeader)
Defined on this element
bdl-1 | Rule | total only when a search or history | total.empty() or (type = 'searchset') or (type = 'history')
bdl-2 | Rule | entry.search only when a search | entry.search.empty() or (type = 'searchset')
bdl-3 | Rule | entry.request mandatory for batch/transaction/history, otherwise prohibited | entry.all(request.exists() = (%resource.type = 'batch' or %resource.type = 'transaction' or %resource.type = 'history'))
bdl-4 | Rule | entry.response mandatory for batch-response/transaction-response/history, otherwise prohibited | entry.all(response.exists() = (%resource.type = 'batch-response' or %resource.type = 'transaction-response' or %resource.type = 'history'))
bdl-7 | Rule | FullUrl must be unique in a bundle, or else entries with the same fullUrl must have different meta.versionId (except in history bundles) | (type = 'history') or entry.where(fullUrl.exists()).select(fullUrl&resource.meta.versionId).isDistinct()
bdl-9 | Rule | A document must have an identifier with a system and a value | type = 'document' implies (identifier.system.exists() and identifier.value.exists())
bdl-10 | Rule | A document must have a date | type = 'document' implies (timestamp.hasValue())
bdl-11 | Rule | A document must have a Composition as the first resource | type = 'document' implies entry.first().resource.is(Composition)
bdl-12 | Rule | A message must have a MessageHeader as the first resource | type = 'message' implies entry.first().resource.is(MessageHeader)
Bundle.identifier
Element Id | Bundle.identifier
Definition | A persistent identifier for the bundle that won't change as a bundle is copied from server to server.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..1
Type | Identifier
Summary | true
Comments | Persistent identity generally only matters for batches of type Document, Message, and Collection. It would not normally be populated for search and history results and servers ignore Bundle.identifier when processing batches and transactions. For Documents the .identifier SHALL be populated such that the .identifier is globally unique.
Bundle.type
Element Id | Bundle.type
Definition | Indicates the purpose of this bundle - how it is intended to be used.
Cardinality | 1..1
Terminology Binding | BundleType (Required)
Type | code
Summary | true
Comments | It's possible to use a bundle for other purposes (e.g. a document can be accepted as a transaction). This is primarily defined so that there can be specific rules for some of the bundle types.
Bundle.timestamp
Element Id | Bundle.timestamp
Definition | The date/time that the bundle was assembled - i.e. when the resources were placed in the bundle.
Cardinality | 0..1
Type | instant
Summary | true
Comments | For many bundles, the timestamp is equal to .meta.lastUpdated, because they are not stored (e.g. search results). When a bundle is placed in a persistent store, .meta.lastUpdated will be usually be changed by the server. When the bundle is a message, a middleware agent altering the message (even if not stored) SHOULD update .meta.lastUpdated. .timestamp is used to track the original time of the Bundle, and SHOULD be populated. Usage: document : the date the document was created. Note: the composition may predate the document, or be associated with multiple documents. The date of the composition - the authoring time - may be earlier than the document assembly time message : the date that the content of the message was assembled. This date is not changed by middleware engines unless they add additional data that changes the meaning of the time of the message history : the date that the history was assembled. This time would be used as the _since time to ask for subsequent updates searchset : the time that the search set was assembled. Note that different pages MAY have different timestamps but need not. Having different timestamps does not imply that subsequent pages will represent or include changes made since the initial query transaction | transaction-response | batch | batch-response | collection : no particular assigned meaning The timestamp value should be greater than the lastUpdated and other timestamps in the resources in the bundle, and it should be equal or earlier than the .meta.lastUpdated on the Bundle itself.
Bundle.total
Element Id | Bundle.total

Definition | If a set of search matches, this is the total number of entries of type 'match' across all pages in the search. It does not include search.mode = 'include' or 'outcome' entries and it does not provide a count of the number of entries in the Bundle.
Cardinality | 0..1
Type | unsignedInt
Summary | true
Comments | Only used if the bundle is a search result set. The total does not include resources such as OperationOutcome and included resources, only the total number of matching resources.
Invariants | Affect this element bdl-1 Rule total only when a search or history total.empty() or (type = 'searchset') or (type = 'history')
Affect this element
bdl-1 | Rule | total only when a search or history | total.empty() or (type = 'searchset') or (type = 'history')
Bundle.link
Element Id | Bundle.link
Definition | A series of links that provide context to this bundle.
Cardinality | 0..*
Summary | true
Comments | Both Bundle.link and Bundle.entry.link are defined to support providing additional context when Bundles are used (e.g. HATEOAS). Bundle.entry.link corresponds to links found in the HTTP header if the resource in the entry was read directly. This specification defines some specific uses of Bundle.link for searching and paging, but no specific uses for Bundle.entry.link, and no defined function in a transaction - the meaning is implementation specific.
Bundle.link.relation
Element Id | Bundle.link.relation
Definition | A name which details the functional use for this link - see http://www.iana.org/assignments/link-relations/link-relations.xhtml#link-relations-1 .
Cardinality | 1..1
Type | string
Summary | true
Bundle.link.url
Element Id | Bundle.link.url
Definition | The reference details for the link.
Cardinality | 1..1
Type | uri
Summary | true
Bundle.entry
Element Id | Bundle.entry
Definition | An entry in a bundle resource - will either contain a resource or information about a resource (transactions and history only).
Cardinality | 0..*
Element Order Meaning | For bundles of type 'document' and 'message', the first resource is special (must be Composition or MessageHeader respectively). For all bundles, the meaning of the order of entries depends on the bundle type
Summary | true
Invariants | Defined on this element bdl-5 Rule must be a resource unless there's a request or response resource.exists() or request.exists() or response.exists() bdl-8 Rule fullUrl cannot be a version specific reference fullUrl.contains('/_history/').not()
Defined on this element
bdl-5 | Rule | must be a resource unless there's a request or response | resource.exists() or request.exists() or response.exists()
bdl-8 | Rule | fullUrl cannot be a version specific reference | fullUrl.contains('/_history/').not()
Bundle.entry.link
Element Id | Bundle.entry.link
Definition | A series of links that provide context to this entry.
Cardinality | 0..*
Type | See Bundle.link
Summary | true
Bundle.entry.fullUrl
Element Id | Bundle.entry.fullUrl
Definition | The Absolute URL for the resource. The fullUrl SHALL NOT disagree with the id in the resource - i.e. if the fullUrl is not a urn:uuid, the URL shall be version-independent URL consistent with the Resource.id. The fullUrl is a version independent reference to the resource. The fullUrl element SHALL have a value except that: fullUrl can be empty on a POST (although it does not need to when specifying a temporary id for reference in the bundle) Results from operations might involve resources that are not identified.
Cardinality | 0..1
Type | uri
Summary | true
Comments | fullUrl might not be unique in the context of a resource. Note that since FHIR resources do not need to be served through the FHIR API, the fullURL might be a URN or an absolute URL that does not end with the logical id of the resource (Resource.id). However, but if the fullUrl does look like a RESTful server URL (e.g. meets the regex, then the 'id' portion of the fullUrl SHALL end with the Resource.id. Note that the fullUrl is not the same as the canonical URL - it's an absolute url for an endpoint serving the resource (these will happen to have the same value on the canonical server for the resource with the canonical URL).
Bundle.entry.resource
Element Id | Bundle.entry.resource
Definition | The Resource for the entry. The purpose/meaning of the resource is determined by the Bundle.type.
Cardinality | 0..1
Type | Resource
Summary | true

Bundle.entry.search
Element Id | Bundle.entry.search
Definition | Information about the search process that lead to the creation of this entry.
Cardinality | 0..1
Summary | true
Invariants | Affect this element bdl-2 Rule entry.search only when a search entry.search.empty() or (type = 'searchset')
Affect this element
bdl-2 | Rule | entry.search only when a search | entry.search.empty() or (type = 'searchset')
Bundle.entry.search.mode
Element Id | Bundle.entry.search.mode
Definition | Why this entry is in the result set - whether it's included as a match or because of an _include requirement, or to convey information or warning information about the search process.
Cardinality | 0..1
Terminology Binding | SearchEntryMode (Required)
Type | code
Summary | true
Comments | There is only one mode. In some corner cases, a resource may be included because it is both a match and an include. In these circumstances, 'match' takes precedence.
Bundle.entry.search.score
Element Id | Bundle.entry.search.score
Definition | When searching, the server's search ranking score for the entry.
Cardinality | 0..1
Type | decimal
Summary | true
Comments | Servers are not required to return a ranking score. 1 is most relevant, and 0 is least relevant. Often, search results are sorted by score, but the client may specify a different sort order. See Patient Match for the EMPI search which relates to this element.
Bundle.entry.request
Element Id | Bundle.entry.request
Definition | Additional information about how this entry should be processed as part of a transaction or batch. For history, it shows how the entry was processed to create the version contained in the entry.
Cardinality | 0..1
Summary | true
Invariants | Affect this element bdl-3 Rule entry.request mandatory for batch/transaction/history, otherwise prohibited entry.all(request.exists() = (%resource.type = 'batch' or %resource.type = 'transaction' or %resource.type = 'history'))
Affect this element
bdl-3 | Rule | entry.request mandatory for batch/transaction/history, otherwise prohibited | entry.all(request.exists() = (%resource.type = 'batch' or %resource.type = 'transaction' or %resource.type = 'history'))
Bundle.entry.request.method
Element Id | Bundle.entry.request.method
Definition | In a transaction or batch, this is the HTTP action to be executed for this entry. In a history bundle, this indicates the HTTP action that occurred.
Cardinality | 1..1
Terminology Binding | HTTPVerb (Required)
Type | code
Summary | true
Bundle.entry.request.url
Element Id | Bundle.entry.request.url
Definition | The URL for this entry, relative to the root (the address to which the request is posted).
Cardinality | 1..1
Type | uri
Summary | true
Comments | E.g. for a Patient Create, the method would be "POST" and the URL would be "Patient". For a Patient Update, the method would be PUT and the URL would be "Patient/[id]".
Bundle.entry.request.ifNoneMatch
Element Id | Bundle.entry.request.ifNoneMatch
Definition | If the ETag values match, return a 304 Not Modified status. See the API documentation for "Conditional Read".
Cardinality | 0..1
Type | string
Summary | true
Bundle.entry.request.ifModifiedSince
Element Id | Bundle.entry.request.ifModifiedSince
Definition | Only perform the operation if the last updated date matches. See the API documentation for "Conditional Read".
Cardinality | 0..1
Type | instant
Summary | true
Bundle.entry.request.ifMatch
Element Id | Bundle.entry.request.ifMatch
Definition | Only perform the operation if the Etag value matches. For more information, see the API section "Managing Resource Contention".
Cardinality | 0..1
Type | string
Summary | true

Bundle.entry.request.ifNoneExist
Element Id | Bundle.entry.request.ifNoneExist
Definition | Instruct the server not to perform the create if a specified resource already exists. For further information, see the API documentation for "Conditional Create". This is just the query portion of the URL - what follows the "?" (not including the "?").
Cardinality | 0..1
Type | string
Summary | true
Bundle.entry.response
Element Id | Bundle.entry.response
Definition | Indicates the results of processing the corresponding 'request' entry in the batch or transaction being responded to or what the results of an operation where when returning history.
Cardinality | 0..1
Summary | true
Invariants | Affect this element bdl-4 Rule entry.response mandatory for batch-response/transaction-response/history, otherwise prohibited entry.all(response.exists() = (%resource.type = 'batch-response' or %resource.type = 'transaction-response' or %resource.type = 'history'))
Affect this element
bdl-4 | Rule | entry.response mandatory for batch-response/transaction-response/history, otherwise prohibited | entry.all(response.exists() = (%resource.type = 'batch-response' or %resource.type = 'transaction-response' or %resource.type = 'history'))
Bundle.entry.response.status
Element Id | Bundle.entry.response.status
Definition | The status code returned by processing this entry. The status SHALL start with a 3 digit HTTP code (e.g. 404) and may contain the standard HTTP description associated with the status code.
Cardinality | 1..1
Type | string
Summary | true
Bundle.entry.response.location
Element Id | Bundle.entry.response.location
Definition | The location header created by processing this operation, populated if the operation returns a location.
Cardinality | 0..1
Type | uri
Summary | true
Bundle.entry.response.etag
Element Id | Bundle.entry.response.etag
Definition | The Etag for the resource, if the operation for the entry produced a versioned resource (see Resource Metadata and Versioning and Managing Resource Contention).
Cardinality | 0..1
Type | string
Summary | true
Comments | Etags match the Resource.meta.versionId. The ETag has to match the version id in the header if a resource is included.
Bundle.entry.response.lastModified
Element Id | Bundle.entry.response.lastModified
Definition | The date/time that the resource was modified on the server.
Cardinality | 0..1
Type | instant
Summary | true
Comments | This has to match the same time in the meta header (meta.lastUpdated) if a resource is included.
Bundle.entry.response.outcome
Element Id | Bundle.entry.response.outcome
Definition | An OperationOutcome containing hints and warnings produced as part of processing this entry in a batch or transaction.
Cardinality | 0..1
Type | Resource
Summary | true
Comments | For a POST/PUT operation, this is the equivalent outcome that would be returned for prefer = operationoutcome - except that the resource is always returned whether or not the outcome is returned. This outcome is not used for error responses in batch/transaction, only for hints and warnings. In a batch operation, the error will be in Bundle.entry.response, and for transaction, there will be a single OperationOutcome instead of a bundle in the case of an error.
Bundle.signature
Element Id | Bundle.signature
Definition | Digital Signature - base64 encoded. XML-DSig or a JWT.
Cardinality | 0..1
Type | Signature
Requirements | A Signature holds an electronic representation of a signature and its supporting context in a FHIR accessible form. The signature may either be a cryptographic type (XML DigSig or a JWS), which is able to provide non-repudiation proof, or it may be a graphical image that represents a signature or a signature process. This element allows capturing signatures on documents, messages, transactions or even search responses, to support content-authentication, non-repudiation or other business cases. This is primarily relevant where the bundle may travel through multiple hops or via other mechanisms where HTTPS non-repudiation is insufficient.
Summary | true
Comments | The signature could be created by the "author" of the bundle or by the originating device. Requirements around inclusion of a signature, verification of signatures and treatment of signed/non-signed bundles is implementation-environment

specific.

2.36.8 Resource Bundle - Mappings

URL: https://hl7.org/fhir/R4/bundle-mappings.html

2.36.8 Resource Bundle - Mappings

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Not Classified | Compartments: Not linked to any defined compartments

Mappings:

FiveWs Pattern Mapping

HL7 v2 Mapping

CDA (R2)

RIM Mapping

Mappings for the bundle resource (see Mappings to Other Standards for further information & status).

2.36.8.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

Bundle
identifier | FiveWs.identifier
type | FiveWs.class
timestamp | FiveWs.init

2.36.8.2 HL7 v2 Mapping (http://hl7.org/v2)

Bundle | N/A

2.36.8.3 CDA (R2) (http://hl7.org/v3/cda)

Bundle | ClinicalDocument, if bundle.type = document
identifier | ClinicalDocument.id
type
timestamp | ClinicalDocument.effectiveTime
total
link
relation
url
entry
link
fullUrl
resource
search
mode
score
request
method
url
ifNoneMatch
ifModifiedSince
ifMatch
ifNoneExist
response
status
location
etag
lastModified
outcome
signature

2.36.8.4 RIM Mapping (http://hl7.org/v3)

Bundle | N/A

2.36.9 Resource Bundle - Extensions & Profiles

URL: https://hl7.org/fhir/R4/bundle-profiles.html

2.36.9 Resource Bundle - Extensions & Profiles

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Not Classified | Compartments: Not linked to any defined compartments

This table lists profiles and extensions for the Bundle resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

2.36.9.1 Profiles

Resources of type Bundle do not have extensions at the root element, but extensions MAY be present on the elements in the resource

No Profiles defined for this resource

2.36.9.2 Extensions

http-response-header | Bundle.entry.response | http-response-header | for Common extensions for bundle-related types
location-distance | Bundle.entry.search | location-distance | for Common extensions for bundle-related types
match-grade | Bundle.entry.search | match-grade | for Common extensions for bundle-related types
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

2.36.9.3 Search Extensions

No Search Extensions defined for this resource

2.36.10 Resource Bundle - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/bundle-version-maps.html

No validation errors - all conversions are clean

2.36.10 Resource Bundle - R3/R4 Conversions

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Not Classified | Compartments: Not linked to any defined compartments

R3 : R4 Conversion maps for Bundle.

Functional status for this map: 28 tests of which 3 fail to execute. 3 fail round-trip testing and 16 r3 resources are invalid (0 errors). (see documentation)

2.36.10.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/Bundle3to4" = "R3 to R4 Conversions for Bundle"

uses "http://hl7.org/fhir/3.0/StructureDefinition/Bundle" alias BundleR3 as source
uses "http://hl7.org/fhir/StructureDefinition/Bundle" alias Bundle as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group Bundle(source src : BundleR3, target tgt : Bundle) extends Resource <<type+>> {
  src.identifier -> tgt.identifier;
  src.type -> tgt.type;
  src.total -> tgt.total;
  src.link as vs0 -> tgt.link as vt0 then link(vs0, vt0);
  src.entry as vs0 -> tgt.entry as vt0 then entry(vs0, vt0);
  src.signature -> tgt.signature;
}

group link(source src, target tgt) extends BackboneElement {
  src.relation -> tgt.relation;
  src.url -> tgt.url;
}

group entry(source src, target tgt) extends BackboneElement {
  src.fullUrl -> tgt.fullUrl;
  src.resource -> tgt.resource;
  src.search as vs0 -> tgt.search as vt0 then search(vs0, vt0);
  src.request as vs0 -> tgt.request as vt0 then request(vs0, vt0);
  src.response as vs0 -> tgt.response as vt0 then response(vs0, vt0);
}

group search(source src, target tgt) extends BackboneElement {
  src.mode -> tgt.mode;
```

```
  src.score -> tgt.score;
}

group request(source src, target tgt) extends BackboneElement {
  src.method -> tgt.method;
  src.url -> tgt.url;
  src.ifNoneMatch -> tgt.ifNoneMatch;
  src.ifModifiedSince -> tgt.ifModifiedSince;
  src.ifMatch -> tgt.ifMatch;
  src.ifNoneExist -> tgt.ifNoneExist;
}

group response(source src, target tgt) extends BackboneElement {
  src.status -> tgt.status;
  src.location -> tgt.location;
  src.etag -> tgt.etag;
  src.lastModified -> tgt.lastModified;
  src.outcome -> tgt.outcome;
}
```

2.36.10.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/Bundle4to3" = "R4 to R3 Conversion for Bundle"

uses "http://hl7.org/fhir/StructureDefinition/Bundle" alias Bundle as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/Bundle" alias BundleR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group Bundle(source src : Bundle, target tgt : BundleR3) extends Resource <<type+>> {
  src.identifier -> tgt.identifier;
  src.type -> tgt.type;
  src.total -> tgt.total;
  src.link as vs0 -> tgt.link as vt0 then link(vs0, vt0);
  src.entry as vs0 -> tgt.entry as vt0 then entry(vs0, vt0);
  src.signature -> tgt.signature;
}

group link(source src, target tgt) extends BackboneElement {
  src.relation -> tgt.relation;
  src.url -> tgt.url;
}

group entry(source src, target tgt) extends BackboneElement {
  src.fullUrl -> tgt.fullUrl;
  src.resource -> tgt.resource;
  src.search as vs0 -> tgt.search as vt0 then search(vs0, vt0);
  src.request as vs0 -> tgt.request as vt0 then request(vs0, vt0);
  src.response as vs0 -> tgt.response as vt0 then response(vs0, vt0);
}

group search(source src, target tgt) extends BackboneElement {
  src.mode -> tgt.mode;
  src.score -> tgt.score;
}

group request(source src, target tgt) extends BackboneElement {
  src.method -> tgt.method;
  src.url -> tgt.url;
  src.ifNoneMatch -> tgt.ifNoneMatch;
  src.ifModifiedSince -> tgt.ifModifiedSince;
  src.ifMatch -> tgt.ifMatch;
  src.ifNoneExist -> tgt.ifNoneExist;
}

group response(source src, target tgt) extends BackboneElement {
  src.status -> tgt.status;
  src.location -> tgt.location;
  src.etag -> tgt.etag;
  src.lastModified -> tgt.lastModified;
  src.outcome -> tgt.outcome;
}
```

2.36.10.3 R4 Validation Errors

2.36.11 Common extensions for bundle-related types

URL: https://hl7.org/fhir/R4/bundle-extensions.html

Defines "common" extensions for use with the Bundle resource and contained elements

2.36.11 Common extensions for bundle-related types

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative

2.36.11.1 Content

Extensions:

http-response-header | http-response-header : In a transaction, every single interaction can have multiple HTTP response headers returned as a result of the interaction.
match-grade | match-grade : Assessment of resource match outcome - how likely this resource is to be a match.
location-distance | location-distance : A calculated distance between the resource and a provided location.

2.36.12 JSON Schema for Bundle

URL: https://hl7.org/fhir/R4/bundle.schema.json.html

2.36.12 JSON Schema for Bundle

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is a representation of the json schema for Bundle, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/Bundle",
  "$ref": "#/definitions/Bundle",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "Bundle": {
      "description": "A container for a collection of resources.",
      "properties": {
        "resourceType": {
          "description": "This is a Bundle resource",
          "const": "Bundle"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "identifier": {
          "description": "A persistent identifier for the bundle that won\u0027t change as a bundle is copied from
    server to server.",
          "$ref": "Identifier.schema.json#/definitions/Identifier"
        },
        "type": {
          "description": "Indicates the purpose of this bundle - how it is intended to be used.",
          "enum": [
            "document",
            "message",
            "transaction",
            "transaction-response",
            "batch",
            "batch-response",
            "history",
            "searchset",
            "collection"
          ]
        },
        "_type": {
          "description": "Extensions for type",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "timestamp": {
          "description": "The date/time that the bundle was assembled - i.e. when the resources were placed in the
    bundle.",
          "$ref": "#/definitions/instant"
        },
        "_timestamp": {
          "description": "Extensions for timestamp",
```

```
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "total": {
      "description": "If a set of search matches, this is the total number of entries of type \u0027match\u0027
across all pages in the search.  It does not include search.mode \u003d \u0027include\u0027 or \u0027outcome\u0027
entries and it does not provide a count of the number of entries in the Bundle.",
      "$ref": "#/definitions/unsignedInt"
    },
    "_total": {
      "description": "Extensions for total",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "link": {
      "description": "A series of links that provide context to this bundle.",
      "items": {
        "$ref": "#/definitions/Bundle_Link"
      },
      "type": "array"
    },
    "entry": {
      "description": "An entry in a bundle resource - will either contain a resource or information about a resource
 (transactions and history only).",
      "items": {
        "$ref": "#/definitions/Bundle_Entry"
      },
      "type": "array"
    },
    "signature": {
      "description": "Digital Signature - base64 encoded. XML-DSig or a JWT.",
      "$ref": "Signature.schema.json#/definitions/Signature"
    }
  },
  "additionalProperties": false,
  "required": [
    "resourceType"
  ]
},
"Bundle_Link": {
  "description": "A container for a collection of resources.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "relation": {
      "description": "A name which details the functional use for this link - see
[http://www.iana.org/assignments/link-relations/link-relations.xhtml#link-
relations-1](http://www.iana.org/assignments/link-relations/link-relations.xhtml#link-relations-1).",
      "$ref": "#/definitions/string"
    },
    "_relation": {
      "description": "Extensions for relation",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "url": {
      "description": "The reference details for the link.",
      "$ref": "#/definitions/uri"
    },
    "_url": {
      "description": "Extensions for url",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
```

```
},
"Bundle_Entry": {
  "description": "A container for a collection of resources.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "link": {
      "description": "A series of links that provide context to this entry.",
      "items": {
        "$ref": "#/definitions/Bundle_Link"
      },
      "type": "array"
    },
    "fullUrl": {
      "description": "The Absolute URL for the resource.  The fullUrl SHALL NOT disagree with the id in the resource
 - i.e. if the fullUrl is not a urn:uuid, the URL shall be version-independent URL consistent with the Resource.id.
The fullUrl is a version independent reference to the resource. The fullUrl element SHALL have a value except that:
\n* fullUrl can be empty on a POST (although it does not need to when specifying a temporary id for reference in the
 bundle)\n* Results from operations might involve resources that are not identified.",
      "$ref": "#/definitions/uri"
    },
    "_fullUrl": {
      "description": "Extensions for fullUrl",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "resource": {
      "description": "The Resource for the entry. The purpose/meaning of the resource is determined by the
Bundle.type.",
      "$ref": "ResourceList.schema.json#/definitions/ResourceList"
    },
    "search": {
      "description": "Information about the search process that lead to the creation of this entry.",
      "$ref": "#/definitions/Bundle_Search"
    },
    "request": {
      "description": "Additional information about how this entry should be processed as part of a transaction or
batch.  For history, it shows how the entry was processed to create the version contained in the entry.",
      "$ref": "#/definitions/Bundle_Request"
    },
    "response": {
      "description": "Indicates the results of processing the corresponding \u0027request\u0027 entry in the batch
or transaction being responded to or what the results of an operation where when returning history.",
      "$ref": "#/definitions/Bundle_Response"
    }
  },
  "additionalProperties": false
},
"Bundle_Search": {
  "description": "A container for a collection of resources.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
```

```
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "mode": {
      "description": "Why this entry is in the result set - whether it\u0027s included as a match or because of an
_include requirement, or to convey information or warning information about the search process.",
      "enum": [
        "match",
        "include",
        "outcome"
      ]
    },
    "_mode": {
      "description": "Extensions for mode",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "score": {
      "description": "When searching, the server\u0027s search ranking score for the entry.",
      "$ref": "#/definitions/decimal"
    },
    "_score": {
      "description": "Extensions for score",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
},
"Bundle_Request": {
  "description": "A container for a collection of resources.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "method": {
      "description": "In a transaction or batch, this is the HTTP action to be executed for this entry. In a history
 bundle, this indicates the HTTP action that occurred.",
      "enum": [
        "GET",
        "HEAD",
        "POST",
        "PUT",
        "DELETE",
        "PATCH"
      ]
    },
    "_method": {
      "description": "Extensions for method",
      "$ref": "Element.schema.json#/definitions/Element"
    },
```

```
    "url": {
      "description": "The URL for this entry, relative to the root (the address to which the request is posted).",
      "$ref": "#/definitions/uri"
    },
    "_url": {
      "description": "Extensions for url",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "ifNoneMatch": {
      "description": "If the ETag values match, return a 304 Not Modified status. See the API documentation for
[\"Conditional Read\"](http.html#cread).",
      "$ref": "#/definitions/string"
    },
    "_ifNoneMatch": {
      "description": "Extensions for ifNoneMatch",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "ifModifiedSince": {
      "description": "Only perform the operation if the last updated date matches. See the API documentation for
[\"Conditional Read\"](http.html#cread).",
      "$ref": "#/definitions/instant"
    },
    "_ifModifiedSince": {
      "description": "Extensions for ifModifiedSince",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "ifMatch": {
      "description": "Only perform the operation if the Etag value matches. For more information, see the API
section [\"Managing Resource Contention\"](http.html#concurrency).",
      "$ref": "#/definitions/string"
    },
    "_ifMatch": {
      "description": "Extensions for ifMatch",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "ifNoneExist": {
      "description": "Instruct the server not to perform the create if a specified resource already exists. For
further information, see the API documentation for [\"Conditional Create\"](http.html#ccreate). This is just the
query portion of the URL - what follows the \"?\" (not including the \"?\").",
      "$ref": "#/definitions/string"
    },
    "_ifNoneExist": {
      "description": "Extensions for ifNoneExist",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
},
"Bundle_Response": {
  "description": "A container for a collection of resources.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "status": {
      "description": "The status code returned by processing this entry. The status SHALL start with a 3 digit HTTP
code (e.g. 404) and may contain the standard HTTP description associated with the status code.",
      "$ref": "#/definitions/string"
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
```

```
        "location": {
          "description": "The location header created by processing this operation, populated if the operation returns a
     location.",
          "$ref": "#/definitions/uri"
        },
        "_location": {
          "description": "Extensions for location",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "etag": {
          "description": "The Etag for the resource, if the operation for the entry produced a versioned resource (see
    [Resource Metadata and Versioning](http.html#versioning) and [Managing Resource
    Contention](http.html#concurrency)).",
          "$ref": "#/definitions/string"
        },
        "_etag": {
          "description": "Extensions for etag",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "lastModified": {
          "description": "The date/time that the resource was modified on the server.",
          "$ref": "#/definitions/instant"
        },
        "_lastModified": {
          "description": "Extensions for lastModified",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "outcome": {
          "description": "An OperationOutcome containing hints and warnings produced as part of processing this entry in
     a batch or transaction.",
          "$ref": "ResourceList.schema.json#/definitions/ResourceList"
        }
      },
      "additionalProperties": false
    }
  }
}
```


2.36.13 Resource Bundle - Maturity Dependency View

URL: https://hl7.org/fhir/R4/bundle-dependencies.html

2.36.13 Resource Bundle - Maturity Dependency View

FHIR Infrastructure Work Group | Maturity Level: N | Normative (from v4.0.0) | Compartments: Not linked to any defined compartments

| This page has been approved as part of an ANSI standard. See the Infrastructure Package for further details.

Dependency Graph for Bundle FMM level 5

Name | Card. | Type | Dependency Analysis
Bundle | | Resource
identifier | 0..1 | Identifier
type | 1..1 | code
timestamp | 0..1 | instant
total | 0..1 | unsignedInt
link | 0..* | BackboneElement
relation | 1..1 | string
url | 1..1 | uri
entry | 0..* | BackboneElement
link | 0..* | see link
fullUrl | 0..1 | uri
resource | 0..1 | Resource
search | 0..1 | BackboneElement
mode | 0..1 | code
score | 0..1 | decimal
request | 0..1 | BackboneElement
method | 1..1 | code
url | 1..1 | uri
ifNoneMatch | 0..1 | string
ifModifiedSince | 0..1 | instant
ifMatch | 0..1 | string
ifNoneExist | 0..1 | string
response | 0..1 | BackboneElement
status | 1..1 | string
location | 0..1 | uri
etag | 0..1 | string
lastModified | 0..1 | instant
outcome | 0..1 | Resource
signature TU | 0..1 | Signature

2.37 Resource Questionnaire - Content

URL: https://hl7.org/fhir/R4/questionnaire.html
FHIR Infrastructure Work Group | Maturity Level: 3 | Trial Use | Security Category: Business | Compartments: Not linked to any defined compartments

A structured set of questions intended to guide the collection of answers from end-users. Questionnaires provide detailed control over order, presentation, phraseology and grouping to allow coherent, consistent data collection.

2.37.1 Scope and Usage

A Questionnaire is an organized collection of questions intended to solicit information from patients, providers or other individuals involved in the healthcare domain. They may be simple flat lists of questions or can be hierarchically organized in groups and sub-groups, each containing questions. The Questionnaire defines the questions to be asked, how they are ordered and grouped, any intervening instructional text and what the constraints are on the allowed answers. The results of a Questionnaire can be communicated using the QuestionnaireResponse resource.

Questionnaires cover the need to communicate data originating from forms used in medical history examinations, research questionnaires and sometimes full clinical specialty records. In many systems this data is collected using user-defined screens and forms. Questionnaires define specifics about data capture - exactly what questions were asked, in what order, what choices for answers were, etc. Each of these questions is part of the Questionnaire, and as such the Questionnaire is a separately identifiable Resource, whereas the individual questions are not. (Questionnaire questions can be linked to shared data elements using the Questionnaire.item.definition element.)

Examples of Questionnaires include:

• Past medical history (PMH)

• Family diseases

• Social history

• Research questionnaires/Clinical research forms (CRFs)

• Quality and evaluation forms

• Patient intake form (e.g. clipboard)

• Forms to support insurance claims

This resource is limited in scope to support the characteristics of simple questionnaires. However, common extensions have been defined to allow more sophisticated behavior. This includes:

• Questionnaire core extensions which defines the additional descriptive characteristics for questionnaires and their groups and questions

• Element extensions which can describe additional constraints on allowed answers for questionnaires such as string length and date and numeric ranges

Additional implementation guides, in particular the Structured Data Capture Questionnaire IG provide guidance, extensions and operations that support additional capabilities and allow the definition of more sophisticated questionnaires and their corresponding responses.

2.37.2 Boundaries and Relationships

Questionnaires differ from Lists because Lists group existing resources, while Questionnaires group arbitrary questions. Questionnaires are distinct from Observations and DiagnosticReports in that both of these resources are intended to capture only certain types of information (lab, imaging, vitals, etc.) and should not be used to capture the full breadth of healthcare information (allergies, medications, care plans, etc.), while Questionnaires are able to capture any information at all. More importantly, Observation and DiagnosticReport focus on capturing the discrete information in a standardized form so that the information can be used consistently regardless of where or how it is captured. Questionnaire focuses on information capture. The same information can be captured using a wide variety of questionnaires with differently phrased questions organized in different manners. As such, the Questionnaire resource provides a means to standardize the information gathering process (how information is captured), but not how data is interoperably compared, analyzed or computed upon (typically managed using Observation, DiagnosticReport as well as other resources).

Questionnaire supports data-collection workflow to a limited extent, in that - once triggered - a Questionnaire can guide a user through a data collection process that ensures appropriate information is collected based on answers to particular questions. However, Questionnaire doesn't provide support for capturing sets of information at different times or highly interactive data capture. Broader workflow is typically managed using PlanDefinition and Task or using other mechanisms such as CDSHooks.

Questionnaires are similar to the notion of "logical models" supported by the StructureDefinition resource. Both support the representation of a collection of data points with labels, data types and hierarchy. It will be

common to find the two of them mapped together. The primary difference is that Questionnaire is focused on user-facing data collection. It describes specific questions and includes information such as what number/label should be displayed beside each question, conditions in which questions should be displayed (or not), what instructions should be provided to the user, etc. StructureDefinition, on the other hand, merely defines a data structure with no guidance on display or rules around capture mechanism, only what data should exist in the end. As well, logical models are not intended to capture data directly. Rather, they provide a basis for mapping between data capture structures.

2.37.3 Background and Context

Groups and questions that make up a Questionnaire can use the definition element to refer to data elements in external code systems and StructureDefinitions. In this way, it's possible to leverage standard definitions of questions and their associated data types and answer sets across questionnaires. This also allows extraction of the data from a questionnaire response into other data structures (e.g. populating a ServiceRequest or CarePlan from information in a completed questionnaire response. It can also support other sorts of post-processing of the data contained in a questionnaire response. However, use of the definition element is not required and Questionnaires may be quite loosely defined. The section Questionnaire versus Resources below discusses the issues of collecting data in such loosely defined Questionnaires versus collecting data as well-defined separate Resources.

This resource is referenced by CarePlan, Communication, Contract, FamilyMemberHistory, PlanDefinition, Procedure, itself and QuestionnaireResponse

2.37.4 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
Questionnaire | ITU | | DomainResource | A structured set of questions + Warning: Name should be usable as an identifier for the module by machine processing applications such as code generation + Rule: The link ids for groups and questions must be unique within the questionnaire Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
url | Σ | 0..1 | uri | Canonical identifier for this questionnaire, represented as a URI (globally unique)
identifier | Σ | 0..* | Identifier | Additional identifier for the questionnaire
version | Σ | 0..1 | string | Business version of the questionnaire
name | ΣI | 0..1 | string | Name for this questionnaire (computer friendly)
title | Σ | 0..1 | string | Name for this questionnaire (human friendly)
derivedFrom | | 0..* | canonical(Questionnaire) | Instantiates protocol or definition
status | ?!Σ | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)
experimental | Σ | 0..1 | boolean | For testing purposes, not real usage
subjectType | Σ | 0..* | code | Resource that can be subject of QuestionnaireResponse ResourceType (Required)
date | Σ | 0..1 | dateTime | Date last changed
publisher | Σ | 0..1 | string | Name of the publisher (organization or individual)
contact | Σ | 0..* | ContactDetail | Contact details for the publisher
description | | 0..1 | markdown | Natural language description of the questionnaire
useContext | Σ | 0..* | UsageContext | The context that the content is intended to support
jurisdiction | Σ | 0..* | CodeableConcept | Intended jurisdiction for questionnaire (if applicable) Jurisdiction (Extensible)
purpose | | 0..1 | markdown | Why this questionnaire is defined
copyright | | 0..1 | markdown | Use and/or publishing restrictions
approvalDate | | 0..1 | date | When the questionnaire was approved by publisher
lastReviewDate | | 0..1 | date | When the questionnaire was last reviewed
effectivePeriod | Σ | 0..1 | Period | When the questionnaire is expected to be used
code | Σ | 0..* | Coding | Concept that represents the overall questionnaire Questionnaire Question Codes (Example)
item | I | 0..* | BackboneElement | Questions and sections within the Questionnaire + Rule: Group items must have nested items, display items cannot have nested items + Rule: Display items cannot have a "code" asserted + Rule: A question cannot have both answerOption and answerValueSet + Rule: Only 'choice' and 'open-choice' items can have answerValueSet + Rule: Required and repeat aren't permitted for display items + Rule: Initial values can't be specified for groups or display items + Rule: Read-only can't be specified for "display" items + Rule: Maximum length can only be declared for simple question types + Rule: If one or more answerOption is present, initial[x] must be missing + Rule: If there are more than one enableWhen, enableBehavior must be specified + Rule: Can only have multiple initial values for repeating items
linkId | | 1..1 | string | Unique id for item in questionnaire
definition | | 0..1 | uri | ElementDefinition - details for the item
code | I | 0..* | Coding | Corresponding concept for this item in a terminology Questionnaire Question Codes (Example)
prefix | | 0..1 | string | E.g. "1(a)", "2.5.3"

text | | 0..1 | string | Primary text for the item
type | | 1..1 | code | group | display | boolean | decimal | integer | date | dateTime + QuestionnaireItemType (Required)
enableWhen | ?!I | 0..* | BackboneElement | Only allow data when + Rule: If the operator is 'exists', the value must be a boolean
question | | 1..1 | string | Question that determines whether item is enabled
operator | | 1..1 | code | exists | = | != | > | < | >= | <= QuestionnaireItemOperator (Required)
answer[x] | I | 1..1 | | Value for question comparison based on operator Questionnaire Answer Codes (Example)
answerBoolean | | | boolean
answerDecimal | | | decimal
answerInteger | | | integer
answerDate | | | date
answerDateTime | | | dateTime
answerTime | | | time
answerString | | | string
answerCoding | | | Coding
answerQuantity | | | Quantity
answerReference | | | Reference(Any)
enableBehavior | I | 0..1 | code | all | any EnableWhenBehavior (Required)
required | I | 0..1 | boolean | Whether the item must be included in data results
repeats | I | 0..1 | boolean | Whether the item may repeat
readOnly | I | 0..1 | boolean | Don't allow human editing
maxLength | I | 0..1 | integer | No more than this many characters
answerValueSet | I | 0..1 | canonical(ValueSet) | Valueset containing permitted answers
answerOption | I | 0..* | BackboneElement | Permitted answer
value[x] | | 1..1 | | Answer value Questionnaire Answer Codes (Example)
valueInteger | | | integer
valueDate | | | date
valueTime | | | time
valueString | | | string
valueCoding | | | Coding
valueReference | | | Reference(Any)
initialSelected | | 0..1 | boolean | Whether option is selected by default
initial | I | 0..* | BackboneElement | Initial value(s) when item is first rendered
value[x] | | 1..1 | | Actual value for initializing the question Questionnaire Answer Codes (Example)
valueBoolean | | | boolean
valueDecimal | | | decimal
valueInteger | | | integer
valueDate | | | date
valueDateTime | | | dateTime
valueTime | | | time
valueString | | | string
valueUri | | | uri
valueAttachment | | | Attachment
valueCoding | | | Coding
valueQuantity | | | Quantity
valueReference | | | Reference(Any)
item | I | 0..* | see item | Nested questionnaire items
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Questionnaire xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 0..1 Canonical identifier for this questionnaire, represented as a URI (globally unique) -->
 <identifier><!-- 0..* Identifier Additional identifier for the questionnaire --></identifier>
 <version value="[string]"/><!-- 0..1 Business version of the questionnaire -->
 <name value="[string]"/><!--  0..1 Name for this questionnaire (computer friendly) -->
 <title value="[string]"/><!-- 0..1 Name for this questionnaire (human friendly) -->
 <derivedFrom><!-- 0..* canonical(Questionnaire) Instantiates protocol or definition --></derivedFrom>
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
 <subjectType value="[code]"/><!-- 0..* Resource that can be subject of QuestionnaireResponse -->
 <date value="[dateTime]"/><!-- 0..1 Date last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!-- 0..1 Natural language description of the questionnaire -->
 <useContext><!-- 0..* UsageContext The context that the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for questionnaire (if applicable) --></jurisdiction>
 <purpose value="[markdown]"/><!-- 0..1 Why this questionnaire is defined -->
 <copyright value="[markdown]"/><!-- 0..1 Use and/or publishing restrictions -->
 <approvalDate value="[date]"/><!-- 0..1 When the questionnaire was approved by publisher -->
 <lastReviewDate value="[date]"/><!-- 0..1 When the questionnaire was last reviewed -->
 <effectivePeriod><!-- 0..1 Period When the questionnaire is expected to be used --></effectivePeriod>
 <code><!-- 0..* Coding Concept that represents the overall questionnaire --></code>
 <item>  <!--  0..* Questions and sections within the Questionnaire -->
  <linkId value="[string]"/><!-- 1..1 Unique id for item in questionnaire -->
  <definition value="[uri]"/><!-- 0..1 ElementDefinition - details for the item -->
```

```
  <code><!--  0..* Coding Corresponding concept for this item in a terminology --></code>
  <prefix value="[string]"/><!-- 0..1 E.g. "1(a)", "2.5.3" -->
  <text value="[string]"/><!-- 0..1 Primary text for the item -->
  <type value="[code]"/><!-- 1..1 group | display | boolean | decimal | integer | date | dateTime + -->
  <enableWhen>  <!-- 0..* Only allow data when -->
   <question value="[string]"/><!-- 1..1 Question that determines whether item is enabled -->
   <operator value="[code]"/><!-- 1..1 exists | = | != | > | < | >= | <= -->
   <answer[x]><!--  1..1 boolean|decimal|integer|date|dateTime|time|string|Coding|
     Quantity|Reference(Any) Value for question comparison based on operator --></answer[x]>
  </enableWhen>
  <enableBehavior value="[code]"/><!--  0..1 all | any -->
  <required value="[boolean]"/><!--  0..1 Whether the item must be included in data results -->
  <repeats value="[boolean]"/><!--  0..1 Whether the item may repeat -->
  <readOnly value="[boolean]"/><!--  0..1 Don't allow human editing -->
  <maxLength value="[integer]"/><!--  0..1 No more than this many characters -->
  <answerValueSet><!--  0..1 canonical(ValueSet) Valueset containing permitted answers --></answerValueSet>
  <answerOption>  <!--  0..* Permitted answer -->
   <value[x]><!-- 1..1 integer|date|time|string|Coding|Reference(Any) Answer value --></value[x]>
   <initialSelected value="[boolean]"/><!-- 0..1 Whether option is selected by default -->
  </answerOption>
  <initial>  <!--  0..* Initial value(s) when item is first rendered -->
   <value[x]><!-- 1..1 boolean|decimal|integer|date|dateTime|time|string|uri|
     Attachment|Coding|Quantity|Reference(Any) Actual value for initializing the question --></value[x]>
  </initial>
  <item><!--  0..* Content as for Questionnaire.item Nested questionnaire items --></item>
 </item>
</Questionnaire>
```

JSON Template

```
{
  "resourceType" : "Questionnaire",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // Canonical identifier for this questionnaire, represented as a URI (globally unique)
  "identifier" : [{ Identifier }], // Additional identifier for the questionnaire
  "version" : "<string>", // Business version of the questionnaire
  "name" : "<string>", // C? Name for this questionnaire (computer friendly)
  "title" : "<string>", // Name for this questionnaire (human friendly)
  "derivedFrom" : [{ canonical(Questionnaire) }], // Instantiates protocol or definition
  "status" : "<code>", // R!  draft | active | retired | unknown
  "experimental" : <boolean>, // For testing purposes, not real usage
  "subjectType" : ["<code>"], // Resource that can be subject of QuestionnaireResponse
  "date" : "<dateTime>", // Date last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "description" : "<markdown>", // Natural language description of the questionnaire
  "useContext" : [{ UsageContext }], // The context that the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for questionnaire (if applicable)
  "purpose" : "<markdown>", // Why this questionnaire is defined
  "copyright" : "<markdown>", // Use and/or publishing restrictions
  "approvalDate" : "<date>", // When the questionnaire was approved by publisher
  "lastReviewDate" : "<date>", // When the questionnaire was last reviewed
  "effectivePeriod" : { Period }, // When the questionnaire is expected to be used
  "code" : [{ Coding }], // Concept that represents the overall questionnaire
  "item" : [{ // C? Questions and sections within the Questionnaire
    "linkId" : "<string>", // R!  Unique id for item in questionnaire
    "definition" : "<uri>", // ElementDefinition - details for the item
    "code" : [{ Coding }], // C? Corresponding concept for this item in a terminology
    "prefix" : "<string>", // E.g. "1(a)", "2.5.3"
    "text" : "<string>", // Primary text for the item
    "type" : "<code>", // R!  group | display | boolean | decimal | integer | date | dateTime +
    "enableWhen" : [{ // Only allow data when
      "question" : "<string>", // R!  Question that determines whether item is enabled
      "operator" : "<code>", // R!  exists | = | != | > | < | >= | <=
      // answer[x]: Value for question comparison based on operator. One of these 10:
      "answerBoolean" : <boolean>
      "answerDecimal" : <decimal>
      "answerInteger" : <integer>
      "answerDate" : "<date>"
      "answerDateTime" : "<dateTime>"
      "answerTime" : "<time>"
      "answerString" : "<string>"
      "answerCoding" : { Coding }
      "answerQuantity" : { Quantity }
      "answerReference" : { Reference(Any) }
    }],
    "enableBehavior" : "<code>", // C? all | any
    "required" : <boolean>, // C? Whether the item must be included in data results
    "repeats" : <boolean>, // C? Whether the item may repeat
    "readOnly" : <boolean>, // C? Don't allow human editing
    "maxLength" : <integer>, // C? No more than this many characters
    "answerValueSet" : { canonical(ValueSet) }, // C? Valueset containing permitted answers
    "answerOption" : [{ // C? Permitted answer
      // value[x]: Answer value. One of these 6:
      "valueInteger" : <integer>,
      "valueDate" : "<date>",
      "valueTime" : "<time>",
      "valueString" : "<string>",
      "valueCoding" : { Coding },
```

```
      "valueReference" : { Reference(Any) },
      "initialSelected" : <boolean> // Whether option is selected by default
    }],
    "initial" : [{ // C? Initial value(s) when item is first rendered
      // value[x]: Actual value for initializing the question. One of these 12:
      "valueBoolean" : <boolean>
      "valueDecimal" : <decimal>
      "valueInteger" : <integer>
      "valueDate" : "<date>"
      "valueDateTime" : "<dateTime>"
      "valueTime" : "<time>"
      "valueString" : "<string>"
      "valueUri" : "<uri>"
      "valueAttachment" : { Attachment }
      "valueCoding" : { Coding }
      "valueQuantity" : { Quantity }
      "valueReference" : { Reference(Any) }
    }],
    "item" : [{ Content as for Questionnaire.item }] // C? Nested questionnaire items
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Questionnaire;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Questionnaire.url [ uri ]; # 0..1 Canonical identifier for this questionnaire, represented as a URI (globally
    unique)
  fhir:Questionnaire.identifier [ Identifier ], ... ; # 0..* Additional identifier for the questionnaire
  fhir:Questionnaire.version [ string ]; # 0..1 Business version of the questionnaire
  fhir:Questionnaire.name [ string ]; # 0..1 Name for this questionnaire (computer friendly)
  fhir:Questionnaire.title [ string ]; # 0..1 Name for this questionnaire (human friendly)
  fhir:Questionnaire.derivedFrom [ canonical(Questionnaire) ], ... ; # 0..* Instantiates protocol or definition
  fhir:Questionnaire.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:Questionnaire.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
  fhir:Questionnaire.subjectType [ code ], ... ; # 0..* Resource that can be subject of QuestionnaireResponse
  fhir:Questionnaire.date [ dateTime ]; # 0..1 Date last changed
  fhir:Questionnaire.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:Questionnaire.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:Questionnaire.description [ markdown ]; # 0..1 Natural language description of the questionnaire
  fhir:Questionnaire.useContext [ UsageContext ], ... ; # 0..* The context that the content is intended to support
  fhir:Questionnaire.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for questionnaire (if
    applicable)
  fhir:Questionnaire.purpose [ markdown ]; # 0..1 Why this questionnaire is defined
  fhir:Questionnaire.copyright [ markdown ]; # 0..1 Use and/or publishing restrictions
  fhir:Questionnaire.approvalDate [ date ]; # 0..1 When the questionnaire was approved by publisher
  fhir:Questionnaire.lastReviewDate [ date ]; # 0..1 When the questionnaire was last reviewed
  fhir:Questionnaire.effectivePeriod [ Period ]; # 0..1 When the questionnaire is expected to be used
  fhir:Questionnaire.code [ Coding ], ... ; # 0..* Concept that represents the overall questionnaire
  fhir:Questionnaire.item [ # 0..* Questions and sections within the Questionnaire
    fhir:Questionnaire.item.linkId [ string ]; # 1..1 Unique id for item in questionnaire
    fhir:Questionnaire.item.definition [ uri ]; # 0..1 ElementDefinition - details for the item
    fhir:Questionnaire.item.code [ Coding ], ... ; # 0..* Corresponding concept for this item in a terminology
    fhir:Questionnaire.item.prefix [ string ]; # 0..1 E.g. "1(a)", "2.5.3"
    fhir:Questionnaire.item.text [ string ]; # 0..1 Primary text for the item
    fhir:Questionnaire.item.type [ code ]; # 1..1 group | display | boolean | decimal | integer | date | dateTime +
    fhir:Questionnaire.item.enableWhen [ # 0..* Only allow data when
      fhir:Questionnaire.item.enableWhen.question [ string ]; # 1..1 Question that determines whether item is enabled
      fhir:Questionnaire.item.enableWhen.operator [ code ]; # 1..1 exists | = | != | > | < | >= | <=
      # Questionnaire.item.enableWhen.answer[x] : 1..1 Value for question comparison based on operator. One of these 10
        fhir:Questionnaire.item.enableWhen.answerBoolean [ boolean ]
        fhir:Questionnaire.item.enableWhen.answerDecimal [ decimal ]
        fhir:Questionnaire.item.enableWhen.answerInteger [ integer ]
        fhir:Questionnaire.item.enableWhen.answerDate [ date ]
        fhir:Questionnaire.item.enableWhen.answerDateTime [ dateTime ]
        fhir:Questionnaire.item.enableWhen.answerTime [ time ]
        fhir:Questionnaire.item.enableWhen.answerString [ string ]
        fhir:Questionnaire.item.enableWhen.answerCoding [ Coding ]
        fhir:Questionnaire.item.enableWhen.answerQuantity [ Quantity ]
        fhir:Questionnaire.item.enableWhen.answerReference [ Reference(Any) ]
    ], ...;
    fhir:Questionnaire.item.enableBehavior [ code ]; # 0..1 all | any
    fhir:Questionnaire.item.required [ boolean ]; # 0..1 Whether the item must be included in data results
    fhir:Questionnaire.item.repeats [ boolean ]; # 0..1 Whether the item may repeat
    fhir:Questionnaire.item.readOnly [ boolean ]; # 0..1 Don't allow human editing
    fhir:Questionnaire.item.maxLength [ integer ]; # 0..1 No more than this many characters
    fhir:Questionnaire.item.answerValueSet [ canonical(ValueSet) ]; # 0..1 Valueset containing permitted answers
    fhir:Questionnaire.item.answerOption [ # 0..* Permitted answer
      # Questionnaire.item.answerOption.value[x] : 1..1 Answer value. One of these 6
        fhir:Questionnaire.item.answerOption.valueInteger [ integer ]
        fhir:Questionnaire.item.answerOption.valueDate [ date ]
        fhir:Questionnaire.item.answerOption.valueTime [ time ]
        fhir:Questionnaire.item.answerOption.valueString [ string ]
        fhir:Questionnaire.item.answerOption.valueCoding [ Coding ]
```

```
        fhir:Questionnaire.item.answerOption.valueReference [ Reference(Any) ]
      fhir:Questionnaire.item.answerOption.initialSelected [ boolean ]; # 0..1 Whether option is selected by default
    ], ...;
    fhir:Questionnaire.item.initial [ # 0..* Initial value(s) when item is first rendered
      # Questionnaire.item.initial.value[x] : 1..1 Actual value for initializing the question. One of these 12
        fhir:Questionnaire.item.initial.valueBoolean [ boolean ]
        fhir:Questionnaire.item.initial.valueDecimal [ decimal ]
        fhir:Questionnaire.item.initial.valueInteger [ integer ]
        fhir:Questionnaire.item.initial.valueDate [ date ]
        fhir:Questionnaire.item.initial.valueDateTime [ dateTime ]
        fhir:Questionnaire.item.initial.valueTime [ time ]
        fhir:Questionnaire.item.initial.valueString [ string ]
        fhir:Questionnaire.item.initial.valueUri [ uri ]
        fhir:Questionnaire.item.initial.valueAttachment [ Attachment ]
        fhir:Questionnaire.item.initial.valueCoding [ Coding ]
        fhir:Questionnaire.item.initial.valueQuantity [ Quantity ]
        fhir:Questionnaire.item.initial.valueReference [ Reference(Any) ]
    ], ...;
    fhir:Questionnaire.item.item [ See Questionnaire.item ], ... ; # 0..* Nested questionnaire items
  ], ...;
]
```

Changes since R3

Questionnaire

Questionnaire.derivedFrom | Added Element

Questionnaire.status | Change value set from http://hl7.org/fhir/ValueSet/publication-status to http://hl7.org/fhir/ValueSet/publication-status|4.0.1

Questionnaire.experimental | No longer marked as Modifier

Questionnaire.subjectType | Change value set from http://hl7.org/fhir/ValueSet/resource-types to http://hl7.org/fhir/ValueSet/resource-types|4.0.1

Questionnaire.item.type | Change value set from http://hl7.org/fhir/ValueSet/item-type to http://hl7.org/fhir/ValueSet/item-type|4.0.1

Questionnaire.item.enableWhen.operator | Added Mandatory Element

Questionnaire.item.enableWhen.answer[x] | Min Cardinality changed from 0 to 1 Remove Types uri, Attachment

Questionnaire.item.enableBehavior | Added Element

Questionnaire.item.required | Default Value "false" removed

Questionnaire.item.repeats | Default Value "false" removed

Questionnaire.item.answerValueSet | Renamed from options to answerValueSet Type changed from Reference(ValueSet) to canonical(ValueSet)

Questionnaire.item.answerOption | Renamed from option to answerOption

Questionnaire.item.answerOption.value[x] | Moved from Questionnaire.item.option to Questionnaire.item.answerOption Add Type Reference(Resource)

Questionnaire.item.answerOption.initialSelected | Added Element

Questionnaire.item.initial | Renamed from initial[x] to initial Max Cardinality changed from 1 to * Remove Types boolean, decimal, integer, date, dateTime, time, string, uri, Attachment, Coding, Quantity, Reference(Resource)

Questionnaire.item.initial.value[x] | Added Mandatory Element

Questionnaire.item.enableWhen.hasAnswer | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 6 tests that all execute ok. All tests pass round-trip testing and 5 r3 resources are invalid (0 errors).)

Structure

Name | Flags | Card. | Type | Description & Constraints

Questionnaire | ITU | | DomainResource | A structured set of questions + Warning: Name should be usable as an identifier for the module by machine processing applications such as code generation + Rule: The link ids for groups and questions must be unique within the questionnaire Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

url | Σ | 0..1 | uri | Canonical identifier for this questionnaire, represented as a URI (globally unique)

identifier | Σ | 0..* | Identifier | Additional identifier for the questionnaire

version | Σ | 0..1 | string | Business version of the questionnaire

name | ΣI | 0..1 | string | Name for this questionnaire (computer friendly)

title | Σ | 0..1 | string | Name for this questionnaire (human friendly)

derivedFrom | | 0..* | canonical(Questionnaire) | Instantiates protocol or definition

status | ?!Σ | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)

experimental | Σ | 0..1 | boolean | For testing purposes, not real usage

subjectType | Σ | 0..* | code | Resource that can be subject of QuestionnaireResponse ResourceType (Required)

date | Σ | 0..1 | dateTime | Date last changed

publisher | Σ | 0..1 | string | Name of the publisher (organization or individual)

contact | Σ | 0..* | ContactDetail | Contact details for the publisher

description | | 0..1 | markdown | Natural language description of the questionnaire

useContext | Σ | 0..* | UsageContext | The context that the content is intended to support

jurisdiction | Σ | 0..* | CodeableConcept | Intended jurisdiction for questionnaire (if applicable) Jurisdiction (Extensible)

purpose | | 0..1 | markdown | Why this questionnaire is defined

copyright | | 0..1 | markdown | Use and/or publishing restrictions

approvalDate | | 0..1 | date | When the questionnaire was approved by publisher

lastReviewDate | | 0..1 | date | When the questionnaire was last reviewed
effectivePeriod | Σ | 0..1 | Period | When the questionnaire is expected to be used
code | Σ | 0..* | Coding | Concept that represents the overall questionnaire Questionnaire Question Codes (Example)
item | I | 0..* | BackboneElement | Questions and sections within the Questionnaire + Rule: Group items must have nested items, display items cannot have nested items + Rule: Display items cannot have a "code" asserted + Rule: A question cannot have both answerOption and answerValueSet + Rule: Only 'choice' and 'open-choice' items can have answerValueSet + Rule: Required and repeat aren't permitted for display items + Rule: Initial values can't be specified for groups or display items + Rule: Read-only can't be specified for "display" items + Rule: Maximum length can only be declared for simple question types + Rule: If one or more answerOption is present, initial[x] must be missing + Rule: If there are more than one enableWhen, enableBehavior must be specified + Rule: Can only have multiple initial values for repeating items
linkId | | 1..1 | string | Unique id for item in questionnaire
definition | | 0..1 | uri | ElementDefinition - details for the item
code | I | 0..* | Coding | Corresponding concept for this item in a terminology Questionnaire Question Codes (Example)
prefix | | 0..1 | string | E.g. "1(a)", "2.5.3"
text | | 0..1 | string | Primary text for the item
type | | 1..1 | code | group | display | boolean | decimal | integer | date | dateTime + QuestionnaireItemType (Required)
enableWhen | ?!I | 0..* | BackboneElement | Only allow data when + Rule: If the operator is 'exists', the value must be a boolean
question | | 1..1 | string | Question that determines whether item is enabled
operator | | 1..1 | code | exists | = | != | > | < | >= | <= QuestionnaireItemOperator (Required)
answer[x] | I | 1..1 | | Value for question comparison based on operator Questionnaire Answer Codes (Example)
answerBoolean | | | boolean
answerDecimal | | | decimal
answerInteger | | | integer
answerDate | | | date
answerDateTime | | | dateTime
answerTime | | | time
answerString | | | string
answerCoding | | | Coding
answerQuantity | | | Quantity
answerReference | | | Reference(Any)
enableBehavior | I | 0..1 | code | all | any EnableWhenBehavior (Required)
required | I | 0..1 | boolean | Whether the item must be included in data results
repeats | I | 0..1 | boolean | Whether the item may repeat
readOnly | I | 0..1 | boolean | Don't allow human editing
maxLength | I | 0..1 | integer | No more than this many characters
answerValueSet | I | 0..1 | canonical(ValueSet) | Valueset containing permitted answers
answerOption | I | 0..* | BackboneElement | Permitted answer
value[x] | | 1..1 | | Answer value Questionnaire Answer Codes (Example)
valueInteger | | | integer
valueDate | | | date
valueTime | | | time
valueString | | | string
valueCoding | | | Coding
valueReference | | | Reference(Any)
initialSelected | | 0..1 | boolean | Whether option is selected by default
initial | I | 0..* | BackboneElement | Initial value(s) when item is first rendered
value[x] | | 1..1 | | Actual value for initializing the question Questionnaire Answer Codes (Example)
valueBoolean | | | boolean
valueDecimal | | | decimal
valueInteger | | | integer
valueDate | | | date
valueDateTime | | | dateTime
valueTime | | | time
valueString | | | string
valueUri | | | uri
valueAttachment | | | Attachment
valueCoding | | | Coding
valueQuantity | | | Quantity
valueReference | | | Reference(Any)
item | I | 0..* | see item | Nested questionnaire items
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Questionnaire xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 0..1 Canonical identifier for this questionnaire, represented as a URI (globally unique) -->
 <identifier><!-- 0..* Identifier Additional identifier for the questionnaire --></identifier>
 <version value="[string]"/><!-- 0..1 Business version of the questionnaire -->
 <name value="[string]"/><!--  0..1 Name for this questionnaire (computer friendly) -->
 <title value="[string]"/><!-- 0..1 Name for this questionnaire (human friendly) -->
 <derivedFrom><!-- 0..* canonical(Questionnaire) Instantiates protocol or definition --></derivedFrom>
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
```

```
 <subjectType value="[code]"/><!-- 0..* Resource that can be subject of QuestionnaireResponse -->
 <date value="[dateTime]"/><!-- 0..1 Date last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!-- 0..1 Natural language description of the questionnaire -->
 <useContext><!-- 0..* UsageContext The context that the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for questionnaire (if applicable) --></jurisdiction>
 <purpose value="[markdown]"/><!-- 0..1 Why this questionnaire is defined -->
 <copyright value="[markdown]"/><!-- 0..1 Use and/or publishing restrictions -->
 <approvalDate value="[date]"/><!-- 0..1 When the questionnaire was approved by publisher -->
 <lastReviewDate value="[date]"/><!-- 0..1 When the questionnaire was last reviewed -->
 <effectivePeriod><!-- 0..1 Period When the questionnaire is expected to be used --></effectivePeriod>
 <code><!-- 0..* Coding Concept that represents the overall questionnaire --></code>
 <item>  <!--  0..* Questions and sections within the Questionnaire -->
  <linkId value="[string]"/><!-- 1..1 Unique id for item in questionnaire -->
  <definition value="[uri]"/><!-- 0..1 ElementDefinition - details for the item -->
  <code><!--  0..* Coding Corresponding concept for this item in a terminology --></code>
  <prefix value="[string]"/><!-- 0..1 E.g. "1(a)", "2.5.3" -->
  <text value="[string]"/><!-- 0..1 Primary text for the item -->
  <type value="[code]"/><!-- 1..1 group | display | boolean | decimal | integer | date | dateTime + -->
  <enableWhen>  <!-- 0..* Only allow data when -->
   <question value="[string]"/><!-- 1..1 Question that determines whether item is enabled -->
   <operator value="[code]"/><!-- 1..1 exists | = | != | > | < | >= | <= -->
   <answer[x]><!--  1..1 boolean|decimal|integer|date|dateTime|time|string|Coding|
     Quantity|Reference(Any) Value for question comparison based on operator --></answer[x]>
  </enableWhen>
  <enableBehavior value="[code]"/><!--  0..1 all | any -->
  <required value="[boolean]"/><!--  0..1 Whether the item must be included in data results -->
  <repeats value="[boolean]"/><!--  0..1 Whether the item may repeat -->
  <readOnly value="[boolean]"/><!--  0..1 Don't allow human editing -->
  <maxLength value="[integer]"/><!--  0..1 No more than this many characters -->
  <answerValueSet><!--  0..1 canonical(ValueSet) Valueset containing permitted answers --></answerValueSet>
  <answerOption>  <!--  0..* Permitted answer -->
   <value[x]><!-- 1..1 integer|date|time|string|Coding|Reference(Any) Answer value --></value[x]>
   <initialSelected value="[boolean]"/><!-- 0..1 Whether option is selected by default -->
  </answerOption>
  <initial>  <!--  0..* Initial value(s) when item is first rendered -->
   <value[x]><!-- 1..1 boolean|decimal|integer|date|dateTime|time|string|uri|
     Attachment|Coding|Quantity|Reference(Any) Actual value for initializing the question --></value[x]>
  </initial>
  <item><!--  0..* Content as for Questionnaire.item Nested questionnaire items --></item>
 </item>
</Questionnaire>
```

JSON Template

```
{
  "resourceType" : "Questionnaire",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // Canonical identifier for this questionnaire, represented as a URI (globally unique)
  "identifier" : [{ Identifier }], // Additional identifier for the questionnaire
  "version" : "<string>", // Business version of the questionnaire
  "name" : "<string>", // C? Name for this questionnaire (computer friendly)
  "title" : "<string>", // Name for this questionnaire (human friendly)
  "derivedFrom" : [{ canonical(Questionnaire) }], // Instantiates protocol or definition
  "status" : "<code>", // R!  draft | active | retired | unknown
  "experimental" : <boolean>, // For testing purposes, not real usage
  "subjectType" : ["<code>"], // Resource that can be subject of QuestionnaireResponse
  "date" : "<dateTime>", // Date last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "description" : "<markdown>", // Natural language description of the questionnaire
  "useContext" : [{ UsageContext }], // The context that the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for questionnaire (if applicable)
  "purpose" : "<markdown>", // Why this questionnaire is defined
  "copyright" : "<markdown>", // Use and/or publishing restrictions
  "approvalDate" : "<date>", // When the questionnaire was approved by publisher
  "lastReviewDate" : "<date>", // When the questionnaire was last reviewed
  "effectivePeriod" : { Period }, // When the questionnaire is expected to be used
  "code" : [{ Coding }], // Concept that represents the overall questionnaire
  "item" : [{ // C? Questions and sections within the Questionnaire
    "linkId" : "<string>", // R!  Unique id for item in questionnaire
    "definition" : "<uri>", // ElementDefinition - details for the item
    "code" : [{ Coding }], // C? Corresponding concept for this item in a terminology
    "prefix" : "<string>", // E.g. "1(a)", "2.5.3"
    "text" : "<string>", // Primary text for the item
    "type" : "<code>", // R!  group | display | boolean | decimal | integer | date | dateTime +
    "enableWhen" : [{ // Only allow data when
      "question" : "<string>", // R!  Question that determines whether item is enabled
      "operator" : "<code>", // R!  exists | = | != | > | < | >= | <=
      // answer[x]: Value for question comparison based on operator. One of these 10:
      "answerBoolean" : <boolean>
      "answerDecimal" : <decimal>
      "answerInteger" : <integer>
      "answerDate" : "<date>"
      "answerDateTime" : "<dateTime>"
      "answerTime" : "<time>"
      "answerString" : "<string>"
      "answerCoding" : { Coding }
```

```
      "answerQuantity" : { Quantity }
      "answerReference" : { Reference(Any) }
    }],
    "enableBehavior" : "<code>", // C? all | any
    "required" : <boolean>, // C? Whether the item must be included in data results
    "repeats" : <boolean>, // C? Whether the item may repeat
    "readOnly" : <boolean>, // C? Don't allow human editing
    "maxLength" : <integer>, // C? No more than this many characters
    "answerValueSet" : { canonical(ValueSet) }, // C? Valueset containing permitted answers
    "answerOption" : [{ // C? Permitted answer
      // value[x]: Answer value. One of these 6:
      "valueInteger" : <integer>,
      "valueDate" : "<date>",
      "valueTime" : "<time>",
      "valueString" : "<string>",
      "valueCoding" : { Coding },
      "valueReference" : { Reference(Any) },
      "initialSelected" : <boolean> // Whether option is selected by default
    }],
    "initial" : [{ // C? Initial value(s) when item is first rendered
      // value[x]: Actual value for initializing the question. One of these 12:
      "valueBoolean" : <boolean>
      "valueDecimal" : <decimal>
      "valueInteger" : <integer>
      "valueDate" : "<date>"
      "valueDateTime" : "<dateTime>"
      "valueTime" : "<time>"
      "valueString" : "<string>"
      "valueUri" : "<uri>"
      "valueAttachment" : { Attachment }
      "valueCoding" : { Coding }
      "valueQuantity" : { Quantity }
      "valueReference" : { Reference(Any) }
    }],
    "item" : [{ Content as for Questionnaire.item }] // C? Nested questionnaire items
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Questionnaire;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Questionnaire.url [ uri ]; # 0..1 Canonical identifier for this questionnaire, represented as a URI (globally
    unique)
  fhir:Questionnaire.identifier [ Identifier ], ... ; # 0..* Additional identifier for the questionnaire
  fhir:Questionnaire.version [ string ]; # 0..1 Business version of the questionnaire
  fhir:Questionnaire.name [ string ]; # 0..1 Name for this questionnaire (computer friendly)
  fhir:Questionnaire.title [ string ]; # 0..1 Name for this questionnaire (human friendly)
  fhir:Questionnaire.derivedFrom [ canonical(Questionnaire) ], ... ; # 0..* Instantiates protocol or definition
  fhir:Questionnaire.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:Questionnaire.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
  fhir:Questionnaire.subjectType [ code ], ... ; # 0..* Resource that can be subject of QuestionnaireResponse
  fhir:Questionnaire.date [ dateTime ]; # 0..1 Date last changed
  fhir:Questionnaire.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:Questionnaire.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:Questionnaire.description [ markdown ]; # 0..1 Natural language description of the questionnaire
  fhir:Questionnaire.useContext [ UsageContext ], ... ; # 0..* The context that the content is intended to support
  fhir:Questionnaire.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for questionnaire (if
    applicable)
  fhir:Questionnaire.purpose [ markdown ]; # 0..1 Why this questionnaire is defined
  fhir:Questionnaire.copyright [ markdown ]; # 0..1 Use and/or publishing restrictions
  fhir:Questionnaire.approvalDate [ date ]; # 0..1 When the questionnaire was approved by publisher
  fhir:Questionnaire.lastReviewDate [ date ]; # 0..1 When the questionnaire was last reviewed
  fhir:Questionnaire.effectivePeriod [ Period ]; # 0..1 When the questionnaire is expected to be used
  fhir:Questionnaire.code [ Coding ], ... ; # 0..* Concept that represents the overall questionnaire
  fhir:Questionnaire.item [ # 0..* Questions and sections within the Questionnaire
    fhir:Questionnaire.item.linkId [ string ]; # 1..1 Unique id for item in questionnaire
    fhir:Questionnaire.item.definition [ uri ]; # 0..1 ElementDefinition - details for the item
    fhir:Questionnaire.item.code [ Coding ], ... ; # 0..* Corresponding concept for this item in a terminology
    fhir:Questionnaire.item.prefix [ string ]; # 0..1 E.g. "1(a)", "2.5.3"
    fhir:Questionnaire.item.text [ string ]; # 0..1 Primary text for the item
    fhir:Questionnaire.item.type [ code ]; # 1..1 group | display | boolean | decimal | integer | date | dateTime +
    fhir:Questionnaire.item.enableWhen [ # 0..* Only allow data when
      fhir:Questionnaire.item.enableWhen.question [ string ]; # 1..1 Question that determines whether item is enabled
      fhir:Questionnaire.item.enableWhen.operator [ code ]; # 1..1 exists | = | != | > | < | >= | <=
      # Questionnaire.item.enableWhen.answer[x] : 1..1 Value for question comparison based on operator. One of these 10
        fhir:Questionnaire.item.enableWhen.answerBoolean [ boolean ]
        fhir:Questionnaire.item.enableWhen.answerDecimal [ decimal ]
        fhir:Questionnaire.item.enableWhen.answerInteger [ integer ]
        fhir:Questionnaire.item.enableWhen.answerDate [ date ]
        fhir:Questionnaire.item.enableWhen.answerDateTime [ dateTime ]
        fhir:Questionnaire.item.enableWhen.answerTime [ time ]
        fhir:Questionnaire.item.enableWhen.answerString [ string ]
        fhir:Questionnaire.item.enableWhen.answerCoding [ Coding ]
```

```
        fhir:Questionnaire.item.enableWhen.answerQuantity [ Quantity ]
        fhir:Questionnaire.item.enableWhen.answerReference [ Reference(Any) ]
    ], ...;
    fhir:Questionnaire.item.enableBehavior [ code ]; # 0..1 all | any
    fhir:Questionnaire.item.required [ boolean ]; # 0..1 Whether the item must be included in data results
    fhir:Questionnaire.item.repeats [ boolean ]; # 0..1 Whether the item may repeat
    fhir:Questionnaire.item.readOnly [ boolean ]; # 0..1 Don't allow human editing
    fhir:Questionnaire.item.maxLength [ integer ]; # 0..1 No more than this many characters
    fhir:Questionnaire.item.answerValueSet [ canonical(ValueSet) ]; # 0..1 Valueset containing permitted answers
    fhir:Questionnaire.item.answerOption [ # 0..* Permitted answer
      # Questionnaire.item.answerOption.value[x] : 1..1 Answer value. One of these 6
        fhir:Questionnaire.item.answerOption.valueInteger [ integer ]
        fhir:Questionnaire.item.answerOption.valueDate [ date ]
        fhir:Questionnaire.item.answerOption.valueTime [ time ]
        fhir:Questionnaire.item.answerOption.valueString [ string ]
        fhir:Questionnaire.item.answerOption.valueCoding [ Coding ]
        fhir:Questionnaire.item.answerOption.valueReference [ Reference(Any) ]
      fhir:Questionnaire.item.answerOption.initialSelected [ boolean ]; # 0..1 Whether option is selected by default
    ], ...;
    fhir:Questionnaire.item.initial [ # 0..* Initial value(s) when item is first rendered
      # Questionnaire.item.initial.value[x] : 1..1 Actual value for initializing the question. One of these 12
        fhir:Questionnaire.item.initial.valueBoolean [ boolean ]
        fhir:Questionnaire.item.initial.valueDecimal [ decimal ]
        fhir:Questionnaire.item.initial.valueInteger [ integer ]
        fhir:Questionnaire.item.initial.valueDate [ date ]
        fhir:Questionnaire.item.initial.valueDateTime [ dateTime ]
        fhir:Questionnaire.item.initial.valueTime [ time ]
        fhir:Questionnaire.item.initial.valueString [ string ]
        fhir:Questionnaire.item.initial.valueUri [ uri ]
        fhir:Questionnaire.item.initial.valueAttachment [ Attachment ]
        fhir:Questionnaire.item.initial.valueCoding [ Coding ]
        fhir:Questionnaire.item.initial.valueQuantity [ Quantity ]
        fhir:Questionnaire.item.initial.valueReference [ Reference(Any) ]
    ], ...;
    fhir:Questionnaire.item.item [ See Questionnaire.item ], ... ; # 0..* Nested questionnaire items
  ], ...;
]
```

Changes since Release 3

Questionnaire

Questionnaire.derivedFrom | Added Element

Questionnaire.status | Change value set from http://hl7.org/fhir/ValueSet/publication-status to http://hl7.org/fhir/ValueSet/publication-status|4.0.1

Questionnaire.experimental | No longer marked as Modifier

Questionnaire.subjectType | Change value set from http://hl7.org/fhir/ValueSet/resource-types to http://hl7.org/fhir/ValueSet/resource-types|4.0.1

Questionnaire.item.type | Change value set from http://hl7.org/fhir/ValueSet/item-type to http://hl7.org/fhir/ValueSet/item-type|4.0.1

Questionnaire.item.enableWhen.operator | Added Mandatory Element

Questionnaire.item.enableWhen.answer[x] | Min Cardinality changed from 0 to 1 Remove Types uri, Attachment

Questionnaire.item.enableBehavior | Added Element

Questionnaire.item.required | Default Value "false" removed

Questionnaire.item.repeats | Default Value "false" removed

Questionnaire.item.answerValueSet | Renamed from options to answerValueSet Type changed from Reference(ValueSet) to canonical(ValueSet)

Questionnaire.item.answerOption | Renamed from option to answerOption

Questionnaire.item.answerOption.value[x] | Moved from Questionnaire.item.option to Questionnaire.item.answerOption Add Type Reference(Resource)

Questionnaire.item.answerOption.initialSelected | Added Element

Questionnaire.item.initial | Renamed from initial[x] to initial Max Cardinality changed from 1 to * Remove Types boolean, decimal, integer, date, dateTime, time, string, uri, Attachment, Coding, Quantity, Reference(Resource)

Questionnaire.item.initial.value[x] | Added Mandatory Element

Questionnaire.item.enableWhen.hasAnswer | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 6 tests that all execute ok. All tests pass round-trip testing and 5 r3 resources are invalid (0 errors).)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

2.37.4.1 Terminology Bindings

Path | Definition | Type | Reference

Questionnaire.status | The lifecycle status of an artifact. | Required | PublicationStatus

Questionnaire.subjectType | One of the resource types defined as part of this version of FHIR. | Required | Resource Types

Questionnaire.jurisdiction | Countries and regions within which this artifact is targeted for use. | Extensible | Jurisdiction ValueSet

Questionnaire.code Questionnaire.item.code | Codes for questionnaires, groups and individual questions. | Example | QuestionnaireQuestionCodes

Questionnaire.item.type | Distinguishes groups from questions and display text and indicates data type for questions. | Required | QuestionnaireItemType
Questionnaire.item.enableWhen.operator | The criteria by which a question is enabled. | Required | QuestionnaireItemOperator
Questionnaire.item.enableWhen.answer[x] | Allowed values to answer questions. | Example | QuestionnaireAnswerCodes
Questionnaire.item.enableBehavior | Controls how multiple enableWhen values are interpreted - whether all or any must be true. | Required | EnableWhenBehavior
Questionnaire.item.answerOption.value[x] | Allowed values to answer questions. | Example | QuestionnaireAnswerCodes
Questionnaire.item.initial.value[x] | Allowed values to answer questions. | Example | QuestionnaireAnswerCodes

2.37.4.2 Constraints

id | Level | Location | Description | Expression
que-0 | Warning | (base) | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
que-1 | Rule | Questionnaire.item | Group items must have nested items, display items cannot have nested items | (type='group' implies item.empty().not()) and (type.trace('type')='display' implies item.trace('item').empty())
que-2 | Rule | (base) | The link ids for groups and questions must be unique within the questionnaire | descendants().linkId.isDistinct()
que-3 | Rule | Questionnaire.item | Display items cannot have a "code" asserted | type!='display' or code.empty()
que-4 | Rule | Questionnaire.item | A question cannot have both answerOption and answerValueSet | answerOption.empty() or answerValueSet.empty()
que-5 | Rule | Questionnaire.item | Only 'choice' and 'open-choice' items can have answerValueSet | (type ='choice' or type = 'open-choice' or type = 'decimal' or type = 'integer' or type = 'date' or type = 'dateTime' or type = 'time' or type = 'string' or type = 'quantity') or (answerValueSet.empty() and answerOption.empty())
que-6 | Rule | Questionnaire.item | Required and repeat aren't permitted for display items | type!='display' or (required.empty() and repeats.empty())
que-7 | Rule | Questionnaire.item.enableWhen | If the operator is 'exists', the value must be a boolean | operator = 'exists' implies (answer is Boolean)
que-8 | Rule | Questionnaire.item | Initial values can't be specified for groups or display items | (type!='group' and type!='display') or initial.empty()
que-9 | Rule | Questionnaire.item | Read-only can't be specified for "display" items | type!='display' or readOnly.empty()
que-10 | Rule | Questionnaire.item | Maximum length can only be declared for simple question types | (type in ('boolean' | 'decimal' | 'integer' | 'string' | 'text' | 'url' | 'open-choice')) or maxLength.empty()
que-11 | Rule | Questionnaire.item | If one or more answerOption is present, initial[x] must be missing | answerOption.empty() or initial.empty()
que-12 | Rule | Questionnaire.item | If there are more than one enableWhen, enableBehavior must be specified | enableWhen.count() > 2 implies enableBehavior.exists()
que-13 | Rule | Questionnaire.item | Can only have multiple initial values for repeating items | repeats=true or initial.count() <= 1

2.37.5 Notes:

• Questionnaires may be used to represent predefined forms or panels, referenced using items of type "group"

• Questions may be nested. The Questionnaire resource supports nested items beneath both items of type "group" (which doesn't capture an answer of its own) as well as beneath questions

• Groups and Questions may have linkIds allowing groups and question answers captured in a QuestionnaireResponse to be associated with their corresponding group or question.

• Questionnaire allows for flexible naming and structuring of its contents to reflect the flexible and varying nature of forms and questionnaires. It explicitly does not try to standardize or streamline exchange of its contents outside its context of use, although exchanging partners may further constrain its structure and flexibility using profiles to define standardized, reusable forms.

• Because of the lack of explicit support for Questionnaires in HL7 v3 , CDA Documents frequently used named sections with Observations to model Questionnaires. Such use cases should now use the Questionnaire Resource instead.

• The order of questions within groups, groups within groups and groups within questions is relevant and must be retained for display and capture.

• Display items allow the inclusion of instructions, background information and similar content within a questionnaire.

• In cases where it is necessary or appropriate to capture the reason why a response is missing (that is, why a particular question was not answered), there should be an explicit child item defined for the relevant question(s) in the Questionnaire; e.g. add a child question of "Reason not specified:" with an enableWhen operator "not exists" to the parent question.

2.37.5.1 Structured Data Capture

The base questionnaire resource is aimed at the capabilities of "most" healthcare systems that deal with forms and questionnaires. While the resource structure supports a reasonable degree of complexity, it doesn't provide the degree of control over rendering, question flow and data management needed by some portions of the healthcare space. Many of these more sophisticated requirements are addressed by a supplemental implementation guide called Structured Data Capture (SDC) . This implementation guide covers the following capabilities, plus several others:

• Rendering grids of questions, multi-column drop-downs, multi-language questionnaires, and other 'advanced' formatting capabilities

• Advanced dynamic questionnaires including calculated scores, enabling of questions and groups based on complex logic, drop-downs driven by answers to other questions

• Support for populating questionnaires from existing clinical data based on the current patient, encounter, etc., including the ability to render the form as HTML or to redirect to a website to complete unanswered items

• Support for extracting information from a completed QuestionnaireResponse to generate other FHIR resources such as Observations, MedicationStatements, etc.

• Support for completing 'adaptive' questionnaires (see below)

2.37.5.2 Adaptive Questionnaires

Some questionnaires might not provide a full listing of questions, instead relying on a service to interactively sending partially completed forms to a server which generates the next set of questions. As such, some questionnaires might be 'active' but have either no questions or an insufficient set of questions for their purpose. Further guidance on the use of such forms, can be found in the SDC implementation guide.

2.37.5.3 Using Questionnaires versus using Resources

There is considerable overlap between the information covered by Questionnaires and other Resources (especially FamilyMemberHistory, MedicationStatement, Observation, Procedure, etc.). Questionnaire's flexible structure can easily be misused to capture any data, even data that should be captured as separate Resources. The choice between using Questionnaires or separate Resources may be dictated by the procedure of collection and recording; e.g. if the data is captured as a physician-agreed (electronic) form, it might be impossible or undesirable to distill separate resources from it and capturing the data in a Questionnaire would be most appropriate.

However, data captured only in Questionnaires can be difficult to query after-the-fact. Queries against other Resources will not return data captured only in Questionnaires and querying against Questionnaires directly might not find all desired data, depending on how the questions may have been phrased or encoded over time or by different clinicians. Moreover, interoperability of such Questionnaires is limited, as interpretation of its contents is only known to the circle of parties that were involved in its definition: encoding data from such Questionnaires using more specific resources increases the availability of the information they contain.

It is entirely possible for data to exist in both QuestionnaireResponse and in other resources. For example, data may be captured in QuestionnaireResponse as part of an intake questionnaire. That data may then be propagated into the Patient resource (demographics), FamilyMemberHistory, AllergyIntolerance, MedicationStatement and Observation resources to allow the data to be queried and analyzed. The original QuestionnaireResponse instance can be retained for traceability purposes. If desired, a Provenance resource can be used to identify the Questionnaire as the data source for the other resources. For example, if a questionnaire question asks "what is your weight", an Observation with the appropriate Observation.code and the specified answer as the Observation.valueQuantity can be created.

2.37.5.4 Structure of the Questionnaire

A Questionnaire is built out of two components, Questionnaire and item:

• The main component is Questionnaire, which holds information about the Questionnaire, like the identifier, publisher, date authored, title, etc. The Questionnaire contains zero or more items which define the content of the questionnaire (zero items allows for the possibility of narrative-only questionnaires, though these provide little computable information).

• Items have one of three sub-types, distinguished by the type element: display items convey text to be rendered on the form that won't capture data and which won't contain nested items. This might include copyright or authorship information, instructions or other background information. group items organize content of the questionnaire into sections, sub-sections, etc. Groups don't have answers associated with them directly, but generally contain child 'question' items which do. question items ask a specific question to which an answer may be given. There isn't a single type for question items. Instead, the type conveys the data type of the answer for the question. If an item isn't a 'display' or a 'group', then it's a question which allows a particular type of answer to be gathered. Questions may also have nested content - groups and/or other questions. This way, one can model "panels" of questions which would only be relevant (and thus would only be displayed) depending on the answers of a parent question; e.g. a question "Did the patient receive treatment in the past six months?" would contain a nested group of questions asking for further details about the treatment. See an example from the Australian New South Wales blue book

Specific controls on dynamic display of groups, questions, etc. based on the answers to other questions is outside the scope of the base resource and are handled using extensions.

Group text is intended to provide a "label" for a collection of nested items. The text will often be rendered in a "heading" style. As such, it should be kept relatively short. Detailed instructions should be conveyed by a nested "display" item. It's not necessary to create groups in order to relate an instruction to a collection of questions - you can simply intermix display items and questions or groups, with the expectation that users will

see the display text before encountering the following questions.

2.37.5.5 Identifiers within Questionnaire

There are three different "identifying" elements within Questionnaire: identifier, id and linkId. Each serves very distinct purposes:

• The identifier element is used to reference the overall questionnaire in business terms. It is the number printed across the top of the form or listed beside the form when making a choice between alternate forms.

• The id attribute supported on each element is used for references within a resource, for example linking narrative to discrete elements. Ids are not generally displayed to end users

• The linkId element on questions and groups establishes a link between elements in a QuestionnaireResponse and their definition inside a Questionnaire. This is the only way to link between the groups and questions in a Questionnaire and QuestionnaireResponse. These are also not displayed to end users

2.37.5.6 Question Definitions

Questionnaires can be crafted using any questions the author can conceive and can be phrased in whatever manner is appropriate or convenient. However, standardization of questions and the constraints enforced on the accompanying answers increases the ability to compare data gathered across questionnaires. Items can be directly linked to an ElementDefinition using Questionnaire.item.definition which provides details for the item. If a definition is provided, then the following element values can be derived from the definition:

• concept (from ElementDefinition.code)

• type (from ElementDefinition.type)

• required (from ElementDefinition.min)

• repeats (from ElementDefinition.max)

• maxLength (from ElementDefinition.maxLength)

• answerValueSet (from ElementDefinition.binding.valueSet)

Any information provided in these elements overrides the information from the definition, and some are always required to be populated so that invariants can be enforced, etc. The url refers to an ElementDefinition in StructureDefinition, and always starts with the canonical URL for the target resource. When referring to a StructureDefinition, a fragment identifier is used to specify the element definition by its id (Element.id). E.g. http://hl7.org/fhir/StructureDefinition/Observation#Observation.value[x]. In the absence of a fragment identifier, the first/root element definition in the target is the matching element definition. Examples:

Value of definition	Explanation
http://hl7.org/fhir/StructureDefinition/Observation#Observation.comment	A reference to the standard Observation element comment
http://example.org/definitions/xxx	A reference to the logical model "xxx" that represents a data element
http://example.org/StructureDefinition/Oedema#Observation.category:XXX	A reference to slice XXX on the Observation category element on an Oedema Scale profile
http://example.org/StructureDefinition/Oedema|2.0#Observation.category:XXX	A reference to slice XXX on the Observation category element on an Oedema Scale profile, specifying version 2.0

The definitions may also be used to automatically pre-populate answers, or extract data from a corresponding pre-existing QuestionnaireResponse resource.

Note that LOINC codes implicitly define logical models whose elements can be considered question definitions.

NOTE: Even if a standard question definition is provided , information such as the question text, data type and value set SHOULD still be declared for the question. Systems processing the questionnaire might not have access to or support the ability to look up the question definitions from external sources. If the information is not included in-line within the questionnaire, other systems might not be able to render or use the Questionnaire.

If an external definition is included along with question information such as question text, data type or value set, it is expected that the content for the question and the referenced definition would be in alignment. However, FHIR does not dictate what degree of alignment is required; e.g. Does the question text need to match exactly? Is it legitimate for a question to constrain the allowed value set of possible answers? These rules will need to be established within implementation environments.

2.37.5.7 Question types

The codes for the data type for each question are slightly different than the data types allowed in the QuestionnaireResponse resource. Where the names are the same (ignoring case), the "answer" must use the specified type. Where the names are different, the mappings are as follows:

Questionnaire AnswerFormat code	QuestionnaireResponse data type
text	string
url	uri
choice	Coding

open-choice | Coding (if choosing from one of the listed choices) or string (if a different value is provided)

2.37.5.8 Permitted values for questions

Many questionnaires place constraints on the allowed list of values for a given question. FHIR supports this notion through the item.answerValueSet or the item.answerOption element.

The "answerOption" mechanism is simplest - all possible answers are listed in-line with the question using the item.answerOption element. Maintenance of the set of permitted question answers involves maintenance of the questionnaire itself.

The "answerValueSet" mechanism is more sophisticated. Rather than listing the possible answers directly, the item.answerValueSet element refers to a ValueSet resource. This approach adds complexity for questionnaires having a simple list of strings as choices for a question, but provides several benefits, including:

• Questionnaires that reference codes from externally defined code systems have a means of doing so

• Answer sets can be shared across questions (and there are many questionnaires where this capability is useful)

• The full capability of value sets can be brought into play, including the ability to use ConceptMap to link and translate from questionnaire-specific codes to other codes, the use of multiple display names (e.g. different languages), the ability to use coded ordinals, and the ability to allow choices from larger value sets (e.g. "all SNOMED CT procedure codes")

In many cases, the set of code choices will be specific to a given questionnaire and should be maintained as part of the questionnaire. In this case, the referenced ValueSet can be included as a contained resource. All contained ValueSets are listed together and then are referenced by the individual questions as necessary. Alternatively, the item.answerValueSet element can reference the value set in a version-specific manner. This ensures that the Questionnaire will not adopt a new version of the referenced value set without revising the Questionnaire to point to that new version - putting the author of the Questionnaire in control of any changes.

2.37.5.9 Useful Value Sets

Many questions have a common set of possible codes for their answer. These value sets may be useful to help, and when referenced by full URL, do not need to be provided as part of the questionnaire:

• Yes | No | Don't Know: http://hl7.org/fhir/ValueSet/yesnodontknow

2.37.5.10 Questions with multiple answer types

In some questionnaires, it may be possible to respond to a question with multiple types of answers. For example, capturing a coded answer plus free text, capturing a coded value plus a numeric priority, etc. In FHIR, this is handled through nested questions. Each question has a single data type. However, each question can have nested child questions. These nested child questions can have text (e.g. "Please specify") or can have no text at all if the appropriate guidance is provided by the parent question or group.

2.37.5.11 Questionnaires with Math

Many questionnaires involve computations on questionnaire answers, e.g. tabulating scores. To help with associating numeric values to coded answers, the ordinalValue extension can be used to define the numeric weighting of a code defined in a value set or in the questionnaire itself.

In addition, there are extensions for defining the dynamic value of an element using expressions. For more information, see the Using Expressions and Documentation Template topics in the Clinical Reasoning module.

2.37.5.12 More advanced questionnaires

The capabilities of the Questionnaire resource are fairly simple and try to adhere to what most systems that deal with questionnaires would be capable of. However, there are lots of systems that require more sophisticated capabilities, such as:

• Greater control over rendering, including tables, read-only values, hidden answers, multi-column drop-downs

• Greater control over question flow, including formula-based rules for conditional display of elements

• Ability to pre-populate forms with data found in the client system

• Ability to define how to convert a completed QuestionnaireResponse for a Questionnaire into a collection of Observations and/or other resources

Extensions have been defined for all of these purposes and many more. (Questionnaire has the most 'core' extensions of any resource to-date.) A list of these extensions can be found here. However, some of the extensions are complex and work together in particular ways. To support more advanced questionnaire implementation, the Structured Data Capture (SDC) implementation guide has been created to provide non-country-specific guidance on the use of most of these extensions and sets expectations for systems wishing to support more complex forms.

2.37.6 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
code | token | A code that corresponds to one of its items in the questionnaire | Questionnaire.item.code
context | token | A use context assigned to the questionnaire | (Questionnaire.useContext.value as CodeableConcept)
context-quantity | quantity | A quantity- or range-valued use context assigned to the questionnaire | (Questionnaire.useContext.value as Quantity) | (Questionnaire.useContext.value as Range)
context-type | token | A type of use context assigned to the questionnaire | Questionnaire.useContext.code
context-type-quantity | composite | A use context type and quantity- or range-based value assigned to the questionnaire | On Questionnaire.useContext: context-type: code context-quantity: value.as(Quantity) | value.as(Range)
context-type-value | composite | A use context type and value assigned to the questionnaire | On Questionnaire.useContext: context-type: code context: value.as(CodeableConcept)
date | date | The questionnaire publication date | Questionnaire.date
definition | uri | ElementDefinition - details for the item | Questionnaire.item.definition
description | string | The description of the questionnaire | Questionnaire.description
effective | date | The time during which the questionnaire is intended to be in use | Questionnaire.effectivePeriod
identifier | token | External identifier for the questionnaire | Questionnaire.identifier
jurisdiction | token | Intended jurisdiction for the questionnaire | Questionnaire.jurisdiction
name | string | Computationally friendly name of the questionnaire | Questionnaire.name
publisher | string | Name of the publisher of the questionnaire | Questionnaire.publisher
status | token | The current status of the questionnaire | Questionnaire.status
subject-type | token | Resource that can be subject of QuestionnaireResponse | Questionnaire.subjectType
title | string | The human-friendly name of the questionnaire | Questionnaire.title
url | uri | The uri that identifies the questionnaire | Questionnaire.url
version | token | The business version of the questionnaire | Questionnaire.version

2.37.7 Resource Questionnaire - Examples

URL: https://hl7.org/fhir/R4/questionnaire-examples.html

2.37.7 Resource Questionnaire - Examples

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

Example Name | id | Format
General questionnaire example | 3141 | XML | JSON | Turtle
Real-world lifelines questionnaire (fictively taken from the patient) | f201 | XML | JSON | Turtle
Neonate record from New South Wales, Australia "My Personal Health Record" example | bb | XML | JSON | Turtle
Glasgow Coma Score example form | gcs | XML | JSON | Turtle
Example instrument for assessing Zika virus exposure potential | zika-virus-exposure-assessment | XML | JSON | Turtle
LOINC US Surgeon General family history including value sets | ussg-fht | XML | JSON | Turtle
Questionnaire used to define the questions involved in the PHQ-9 health questionnaire | phq-9-questionnaire | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

2.37.8 Resource Questionnaire - Detailed Descriptions

URL: https://hl7.org/fhir/R4/questionnaire-definitions.html

2.37.8 Resource Questionnaire - Detailed Descriptions

FHIR Infrastructure Work Group | Maturity Level: 3 | Trial Use | Security Category: Business | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the Questionnaire resource.

Questionnaire
Element Id | Questionnaire
Definition | A structured set of questions intended to guide the collection of answers from end-users. Questionnaires provide detailed control over order, presentation, phraseology and grouping to allow coherent, consistent data collection.
Cardinality | 0..*
Type | DomainResource
Requirements | To support structured, hierarchical registration of data gathered using digital forms and other questionnaires. Questionnaires provide greater control over presentation and allow capture of data in a domain-independent way (i.e. capturing information that would otherwise require multiple distinct types of resources).
Alternate Names | Form; CRF; Survey
Invariants | Defined on this element que-0 Warning Name should be usable as an identifier for the module by machine processing applications such as code generation name.matches('[A-Z]([A-Za-z0-9_]){0,254}') que-2 Rule The link ids for groups and questions must be unique within the questionnaire descendants().linkId.isDistinct()
Defined on this element
que-0 | Warning | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
que-2 | Rule | The link ids for groups and questions must be unique within the questionnaire | descendants().linkId.isDistinct()
Questionnaire.url

Element Id | Questionnaire.url
Definition | An absolute URI that is used to identify this questionnaire when it is referenced in a specification, model, design or an instance; also called its canonical identifier. This SHOULD be globally unique and SHOULD be a literal address at which at which an authoritative instance of this questionnaire is (or will be) published. This URL can be the target of a canonical reference. It SHALL remain the same when the questionnaire is stored on different servers.
Cardinality | 0..1
Type | uri
Requirements | ... This is the id that will be used to link a QuestionnaireResponse to the Questionnaire the response is for.
Summary | true
Comments | The name of the referenced questionnaire can be conveyed using the http://hl7.org/fhir/StructureDefinition/display extension.
Questionnaire.identifier
Element Id | Questionnaire.identifier
Definition | A formal identifier that is used to identify this questionnaire when it is represented in other formats, or referenced in a specification, model, design or an instance.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Requirements | Allows externally provided and/or usable business identifiers to be easily associated with the module.
Summary | true
Comments | Typically, this is used for identifiers that can go in an HL7 V3 II (instance identifier) data type, and can then identify this questionnaire outside of FHIR, where it is not possible to use the logical URI.
Questionnaire.version
Element Id | Questionnaire.version
Definition | The identifier that is used to identify this version of the questionnaire when it is referenced in a specification, model, design or instance. This is an arbitrary value managed by the questionnaire author and is not expected to be globally unique. For example, it might be a timestamp (e.g. yyyymmdd) if a managed version is not available. There is also no expectation that versions can be placed in a lexicographical sequence.
Note | This is a business versionId, not a resource version id (see discussion)
Cardinality | 0..1
Type | string
Summary | true
Comments | There may be different questionnaire instances that have the same identifier but different versions. The version can be appended to the url in a reference to allow a reference to a particular business version of the questionnaire with the format [url]|[version].
Questionnaire.name
Element Id | Questionnaire.name
Definition | A natural language name identifying the questionnaire. This name should be usable as an identifier for the module by machine processing applications such as code generation.
Cardinality | 0..1
Type | string
Requirements | Support human navigation and code generation.
Summary | true
Comments | The name is not expected to be globally unique. The name should be a simple alphanumeric type name to ensure that it is machine-processing friendly.
Invariants | Affect this element inv-0 Warning Name should be usable as an identifier for the module by machine processing applications such as code generation name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
Affect this element
inv-0 | Warning | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
Questionnaire.title
Element Id | Questionnaire.title
Definition | A short, descriptive, user-friendly title for the questionnaire.
Cardinality | 0..1
Type | string
Summary | true
Comments | This name does not need to be machine-processing friendly and may contain punctuation, white-space, etc.
Questionnaire.derivedFrom
Element Id | Questionnaire.derivedFrom
Definition | The URL of a Questionnaire that this Questionnaire is based on.
Cardinality | 0..*
Type | canonical(Questionnaire)
Requirements | Allows a Questionnaire to be created based on another Questionnaire.
Questionnaire.status
Element Id | Questionnaire.status
Definition | The status of this questionnaire. Enables tracking the life-cycle of the content.
Cardinality | 1..1
Terminology Binding | PublicationStatus (Required)
Type | code
Is Modifier | true (Reason: This is labeled as "Is Modifier" because applications should not use a retired {{title}} without due consideration)
Summary | true
Comments | Allows filtering of questionnaires that are appropriate for use versus not.
Questionnaire.experimental

Element Id | Questionnaire.experimental
Definition | A Boolean value to indicate that this questionnaire is authored for testing purposes (or education/evaluation/marketing) and is not intended to be used for genuine usage.
Cardinality | 0..1
Type | boolean
Requirements | Enables experimental content to be developed following the same lifecycle that would be used for a production-level questionnaire.
Summary | true
Comments | Allows filtering of questionnaires that are appropriate for use versus not.
Questionnaire.subjectType
Element Id | Questionnaire.subjectType
Definition | The types of subjects that can be the subject of responses created for the questionnaire.
Cardinality | 0..*
Terminology Binding | Any defined Resource Type name
Type | code
Summary | true
Comments | If none are specified, then the subject is unlimited.
Questionnaire.date
Element Id | Questionnaire.date
Definition | The date (and optionally time) when the questionnaire was published. The date must change when the business version changes and it must change if the status code changes. In addition, it should change when the substantive content of the questionnaire changes.
Cardinality | 0..1
Type | dateTime
Alternate Names | Revision Date
Summary | true
Comments | Note that this is not the same as the resource last-modified-date, since the resource may be a secondary representation of the questionnaire. Additional specific dates may be added as extensions or be found by consulting Provenances associated with past versions of the resource.
Questionnaire.publisher
Element Id | Questionnaire.publisher
Definition | The name of the organization or individual that published the questionnaire.
Cardinality | 0..1
Type | string
Requirements | Helps establish the "authority/credibility" of the questionnaire. May also allow for contact.
Summary | true
Comments | Usually an organization but may be an individual. The publisher (or steward) of the questionnaire is the organization or individual primarily responsible for the maintenance and upkeep of the questionnaire. This is not necessarily the same individual or organization that developed and initially authored the content. The publisher is the primary point of contact for questions or issues with the questionnaire. This item SHOULD be populated unless the information is available from context.
Questionnaire.contact
Element Id | Questionnaire.contact
Definition | Contact details to assist a user in finding and communicating with the publisher.
Cardinality | 0..*
Type | ContactDetail
Summary | true
Comments | May be a web site, an email address, a telephone number, etc.
Questionnaire.description
Element Id | Questionnaire.description
Definition | A free text natural language description of the questionnaire from a consumer's perspective.
Cardinality | 0..1
Type | markdown
Comments | This description can be used to capture details such as why the questionnaire was built, comments about misuse, instructions for clinical use and interpretation, literature references, examples from the paper world, etc. It is not a rendering of the questionnaire as conveyed in the 'text' field of the resource itself. This item SHOULD be populated unless the information is available from context (e.g. the language of the questionnaire is presumed to be the predominant language in the place the questionnaire was created).
Questionnaire.useContext
Element Id | Questionnaire.useContext
Definition | The content was developed with a focus and intent of supporting the contexts that are listed. These contexts may be general categories (gender, age, ...) or may be references to specific programs (insurance plans, studies, ...) and may be used to assist with indexing and searching for appropriate questionnaire instances.
Cardinality | 0..*
Type | UsageContext
Requirements | Assist in searching for appropriate content.
Summary | true
Comments | When multiple useContexts are specified, there is no expectation that all or any of the contexts apply.
Questionnaire.jurisdiction
Element Id | Questionnaire.jurisdiction
Definition | A legal or geographic region in which the questionnaire is intended to be used.
Cardinality | 0..*
Terminology Binding | Jurisdiction (Extensible)
Type | CodeableConcept
Summary | true

Comments | It may be possible for the questionnaire to be used in jurisdictions other than those for which it was originally designed or intended.
Questionnaire.purpose
Element Id | Questionnaire.purpose
Definition | Explanation of why this questionnaire is needed and why it has been designed as it has.
Cardinality | 0..1
Type | markdown
Comments | This element does not describe the usage of the questionnaire. Instead, it provides traceability of ''why'' the resource is either needed or ''why'' it is defined as it is. This may be used to point to source materials or specifications that drove the structure of this questionnaire.
Questionnaire.copyright
Element Id | Questionnaire.copyright
Definition | A copyright statement relating to the questionnaire and/or its contents. Copyright statements are generally legal restrictions on the use and publishing of the questionnaire.
Cardinality | 0..1
Type | markdown
Requirements | Consumers must be able to determine any legal restrictions on the use of the questionnaire and/or its content.
Alternate Names | License; Restrictions
Questionnaire.approvalDate
Element Id | Questionnaire.approvalDate
Definition | The date on which the resource content was approved by the publisher. Approval happens once when the content is officially approved for usage.
Cardinality | 0..1
Type | date
Comments | The 'date' element may be more recent than the approval date because of minor changes or editorial corrections.
Questionnaire.lastReviewDate
Element Id | Questionnaire.lastReviewDate
Definition | The date on which the resource content was last reviewed. Review happens periodically after approval but does not change the original approval date.
Cardinality | 0..1
Type | date
Requirements | Gives a sense of how "current" the content is. Resources that have not been reviewed in a long time may have a risk of being less appropriate/relevant.
Comments | If specified, this date follows the original approval date.
Questionnaire.effectivePeriod
Element Id | Questionnaire.effectivePeriod
Definition | The period during which the questionnaire content was or is planned to be in active use.
Cardinality | 0..1
Type | Period
Requirements | Allows establishing a transition before a resource comes into effect and also allows for a sunsetting process when new versions of the questionnaire are or are expected to be used instead.
Summary | true
Comments | The effective period for a questionnaire determines when the content is applicable for usage and is independent of publication and review dates. For example, a measure intended to be used for the year 2016 might be published in 2015.
Questionnaire.code
Element Id | Questionnaire.code
Definition | An identifier for this question or group of questions in a particular terminology such as LOINC.
Cardinality | 0..*
Terminology Binding | Questionnaire Question Codes (Example)
Type | Coding
Requirements | Allows linking of the complete Questionnaire resources to formal terminologies. It's common for "panels" of questions to be identified by a code.
Summary | true
Questionnaire.item
Element Id | Questionnaire.item
Definition | A particular question, question grouping or display text that is part of the questionnaire.
Cardinality | 0..*
Comments | The content of the questionnaire is constructed from an ordered, hierarchical collection of items.
Invariants | Defined on this element que-1 Rule Group items must have nested items, display items cannot have nested items (type='group' implies item.empty().not()) and (type.trace('type')='display' implies item.trace('item').empty()) que-3 Rule Display items cannot have a "code" asserted type!='display' or code.empty() que-4 Rule A question cannot have both answerOption and answerValueSet answerOption.empty() or answerValueSet.empty() que-5 Rule Only 'choice' and 'open-choice' items can have answerValueSet (type ='choice' or type = 'open-choice' or type = 'decimal' or type = 'integer' or type = 'date' or type = 'dateTime' or type = 'time' or type = 'string' or type = 'quantity') or (answerValueSet.empty() and answerOption.empty()) que-6 Rule Required and repeat aren't permitted for display items type!='display' or (required.empty() and repeats.empty()) que-8 Rule Initial values can't be specified for groups or display items (type!='group' and type!='display') or initial.empty() que-9 Rule Read-only can't be specified for "display" items type!='display' or readOnly.empty() que-10 Rule Maximum length can only be declared for simple question types (type in ('boolean' | 'decimal' | 'integer' | 'string' | 'text' | 'url' | 'open-choice')) or maxLength.empty() que-11 Rule If one or more answerOption is present, initial[x] must be missing answerOption.empty() or initial.empty() que-12 Rule If there are more than one enableWhen, enableBehavior must be specified enableWhen.count() > 2 implies enableBehavior.exists() que-13 Rule Can only have multiple initial values for repeating items repeats=true or initial.count() <= 1 Affect this element
Defined on this element

que-1 | Rule | Group items must have nested items, display items cannot have nested items | (type='group' implies item.empty().not()) and (type.trace('type')='display' implies item.trace('item').empty())
que-3 | Rule | Display items cannot have a "code" asserted | type!='display' or code.empty()
que-4 | Rule | A question cannot have both answerOption and answerValueSet | answerOption.empty() or answerValueSet.empty()
que-5 | Rule | Only 'choice' and 'open-choice' items can have answerValueSet | (type ='choice' or type = 'open-choice' or type = 'decimal' or type = 'integer' or type = 'date' or type = 'dateTime' or type = 'time' or type = 'string' or type = 'quantity') or (answerValueSet.empty() and answerOption.empty())
que-6 | Rule | Required and repeat aren't permitted for display items | type!='display' or (required.empty() and repeats.empty())
que-8 | Rule | Initial values can't be specified for groups or display items | (type!='group' and type!='display') or initial.empty()
que-9 | Rule | Read-only can't be specified for "display" items | type!='display' or readOnly.empty()
que-10 | Rule | Maximum length can only be declared for simple question types | (type in ('boolean' | 'decimal' | 'integer' | 'string' | 'text' | 'url' | 'open-choice')) or maxLength.empty()
que-11 | Rule | If one or more answerOption is present, initial[x] must be missing | answerOption.empty() or initial.empty()
que-12 | Rule | If there are more than one enableWhen, enableBehavior must be specified | enableWhen.count() > 2 implies enableBehavior.exists()
que-13 | Rule | Can only have multiple initial values for repeating items | repeats=true or initial.count() <= 1
Affect this element
Questionnaire.item.linkId
Element Id | Questionnaire.item.linkId
Definition | An identifier that is unique within the Questionnaire allowing linkage to the equivalent item in a QuestionnaireResponse resource.
Cardinality | 1..1
Type | string
Requirements | QuestionnaireResponse does not require omitted items to be included and may have some items that repeat, so linkage based on position alone is not sufficient.
Comments | This ''can'' be a meaningful identifier (e.g. a LOINC code) but is not intended to have any meaning. GUIDs or sequential numbers are appropriate here.
Questionnaire.item.definition
Element Id | Questionnaire.item.definition
Definition | This element is a URI that refers to an ElementDefinition that provides information about this item, including information that might otherwise be included in the instance of the Questionnaire resource. A detailed description of the construction of the URI is shown in Comments, below. If this element is present then the following element values MAY be derived from the Element Definition if the corresponding elements of this Questionnaire resource instance have no value: code (ElementDefinition.code) type (ElementDefinition.type) required (ElementDefinition.min) repeats (ElementDefinition.max) maxLength (ElementDefinition.maxLength) answerValueSet (ElementDefinition.binding) options (ElementDefinition.binding).
Cardinality | 0..1
Type | uri
Requirements | A common pattern is to define a set of data elements and then build multiple questionnaires for different circumstances to gather the data. This element provides traceability to the common definition and allows the content for the question to come from the underlying definition.
Comments | The uri refers to an ElementDefinition in a StructureDefinition and always starts with the canonical URL for the target resource. When referring to a StructureDefinition, a fragment identifier is used to specify the element definition by its id Element.id. E.g. http://hl7.org/fhir/StructureDefinition/Observation#Observation.value[x]. In the absence of a fragment identifier, the first/root element definition in the target is the matching element definition.
Questionnaire.item.code
Element Id | Questionnaire.item.code
Definition | A terminology code that corresponds to this group or question (e.g. a code from LOINC, which defines many questions and answers).
Cardinality | 0..*
Terminology Binding | Questionnaire Question Codes (Example)
Type | Coding
Requirements | Allows linking of groups of questions to formal terminologies.
Comments | The value may come from the ElementDefinition referred to by .definition.
Invariants | Affect this element que-3 Rule Display items cannot have a "code" asserted type!='display' or code.empty()
Affect this element
que-3 | Rule | Display items cannot have a "code" asserted | type!='display' or code.empty()
Questionnaire.item.prefix
Element Id | Questionnaire.item.prefix
Definition | A short label for a particular group, question or set of display text within the questionnaire used for reference by the individual completing the questionnaire.
Cardinality | 0..1
Type | string
Requirements | Separating the label from the question text allows improved rendering. Also, instructions will often refer to specific prefixes, so there's a need for the questionnaire design to have control over what labels are used.
Alternate Names | label
Comments | These are generally unique within a questionnaire, though this is not guaranteed. Some questionnaires may have multiple questions with the same label with logic to control which gets exposed. Typically, these won't be used for "display" items, though such use is not prohibited. Systems SHOULD NOT generate their own prefixes if prefixes are defined for any items within a Questionnaire.
Questionnaire.item.text
Element Id | Questionnaire.item.text
Definition | The name of a section, the text of a question or text content for a display item.
Cardinality | 0..1
Type | string

Comments | When using this element to represent the name of a section, use group type item and also make sure to limit the text element to a short string suitable for display as a section heading. Group item instructions should be included as a display type item within the group.
Questionnaire.item.type
Element Id | Questionnaire.item.type
Definition | The type of questionnaire item this is - whether text for display, a grouping of other items or a particular type of data to be captured (string, integer, coded choice, etc.).
Cardinality | 1..1
Terminology Binding | QuestionnaireItemType (Required)
Type | code
Requirements | Defines the format in which the user is to be prompted for the answer.
Comments | Additional constraints on the type of answer can be conveyed by extensions. The value may come from the ElementDefinition referred to by .definition.
Questionnaire.item.enableWhen
Element Id | Questionnaire.item.enableWhen
Definition | A constraint indicating that this item should only be enabled (displayed/allow answers to be captured) when the specified condition is true.
Cardinality | 0..*
Is Modifier | true (Reason: If enableWhen is present and the condition evaluates to false, then the Questionnaire behaves as though the elements weren't actually present)
Requirements | Allows questionnaires to adapt based on answers to other questions. E.g. If physical gender is specified as a male, no need to capture pregnancy history. Also allows conditional display of instructions or groups of questions.
Comments | If multiple repetitions of this extension are present, the item should be enabled when the condition for any of the repetitions is true. I.e. treat "enableWhen"s as being joined by an "or" clause. This element is a modifier because if enableWhen is present for an item, "required" is ignored unless one of the enableWhen conditions is met. When an item is disabled, all of its descendants are disabled, regardless of what their own enableWhen logic might evaluate to.
Invariants | Defined on this element que-7 Rule If the operator is 'exists', the value must be a boolean operator = 'exists' implies (answer is Boolean)
Defined on this element
que-7 | Rule | If the operator is 'exists', the value must be a boolean | operator = 'exists' implies (answer is Boolean)
Questionnaire.item.enableWhen.question
Element Id | Questionnaire.item.enableWhen.question
Definition | The linkId for the question whose answer (or lack of answer) governs whether this item is enabled.
Cardinality | 1..1
Type | string
Comments | If multiple question occurrences are present for the same question (same linkId), then this refers to the nearest question occurrence reachable by tracing first the "ancestor" axis and then the "preceding" axis and then the "following" axis.
Questionnaire.item.enableWhen.operator
Element Id | Questionnaire.item.enableWhen.operator
Definition | Specifies the criteria by which the question is enabled.
Cardinality | 1..1
Terminology Binding | QuestionnaireItemOperator (Required)
Type | code
Questionnaire.item.enableWhen.answer[x]
Element Id | Questionnaire.item.enableWhen.answer[x]
Definition | A value that the referenced question is tested using the specified operator in order for the item to be enabled.
Cardinality | 1..1
Terminology Binding | Questionnaire Answer Codes (Example)
Type | boolean|decimal|integer|date|dateTime|time|string|Coding|Quantity|Reference(Any)
[x] Note | See Choice of Data Types for further information about how to use [x]
Invariants | Affect this element que-7 Rule If the operator is 'exists', the value must be a boolean operator = 'exists' implies (answer is Boolean)
Affect this element
que-7 | Rule | If the operator is 'exists', the value must be a boolean | operator = 'exists' implies (answer is Boolean)
Questionnaire.item.enableBehavior
Element Id | Questionnaire.item.enableBehavior
Definition | Controls how multiple enableWhen values are interpreted - whether all or any must be true.
Cardinality | 0..1
Terminology Binding | EnableWhenBehavior (Required)
Type | code
Comments | This element must be specified if more than one enableWhen value is provided.
Invariants | Affect this element que-12 Rule If there are more than one enableWhen, enableBehavior must be specified enableWhen.count() > 2 implies enableBehavior.exists()
Affect this element
que-12 | Rule | If there are more than one enableWhen, enableBehavior must be specified | enableWhen.count() > 2 implies enableBehavior.exists()
Questionnaire.item.required
Element Id | Questionnaire.item.required
Definition | An indication, if true, that the item must be present in a "completed" QuestionnaireResponse. If false, the item may be skipped when answering the questionnaire.
Cardinality | 0..1
Type | boolean
Meaning if Missing | Items are generally assumed not to be required unless explicitly specified. Systems SHOULD always populate this value

Comments | Questionnaire.item.required only has meaning for elements that are conditionally enabled with enableWhen if the condition evaluates to true. If an item that contains other items is marked as required, that does not automatically make the contained elements required (though required groups must contain at least one child element). The value may come from the ElementDefinition referred to by .definition.
Invariants | Affect this element que-6 Rule Required and repeat aren't permitted for display items type!='display' or (required.empty() and repeats.empty())
Affect this element
que-6 | Rule | Required and repeat aren't permitted for display items | type!='display' or (required.empty() and repeats.empty())
Questionnaire.item.repeats
Element Id | Questionnaire.item.repeats
Definition | An indication, if true, that the item may occur multiple times in the response, collecting multiple answers for questions or multiple sets of answers for groups.
Cardinality | 0..1
Type | boolean
Meaning if Missing | Items are generally assumed not to repeat unless explicitly specified. Systems SHOULD always populate this value
Requirements | Items may be used to create set of (related) questions that can be repeated to give multiple answers to such a set.
Comments | If a question is marked as repeats=true, then multiple answers can be provided for the question in the corresponding QuestionnaireResponse. When rendering the questionnaire, it is up to the rendering software whether to render the question text for each answer repetition (i.e. "repeat the question") or to simply allow entry/selection of multiple answers for the question (repeat the answers). Which is most appropriate visually may depend on the type of answer as well as whether there are nested items. The resulting QuestionnaireResponse will be populated the same way regardless of rendering - one 'question' item with multiple answer values. The value may come from the ElementDefinition referred to by .definition.
Invariants | Affect this element que-6 Rule Required and repeat aren't permitted for display items type!='display' or (required.empty() and repeats.empty())
Affect this element
que-6 | Rule | Required and repeat aren't permitted for display items | type!='display' or (required.empty() and repeats.empty())
Questionnaire.item.readOnly
Element Id | Questionnaire.item.readOnly
Definition | An indication, when true, that the value cannot be changed by a human respondent to the Questionnaire.
Cardinality | 0..1
Type | boolean
Requirements | Allows certain information to be phrased (and rendered) as a question and an answer, while keeping users from changing it. May also be useful for preventing changes to pre-populated portions of a questionnaire, for calculated values, etc.
Comments | The value of readOnly elements can be established by asserting extensions for defaultValues, linkages that support pre-population and/or extensions that support calculation based on other answers.
Invariants | Affect this element que-9 Rule Read-only can't be specified for "display" items type!='display' or readOnly.empty()
Affect this element
que-9 | Rule | Read-only can't be specified for "display" items | type!='display' or readOnly.empty()
Questionnaire.item.maxLength
Element Id | Questionnaire.item.maxLength
Definition | The maximum number of characters that are permitted in the answer to be considered a "valid" QuestionnaireResponse.
Cardinality | 0..1
Type | integer
Comments | For base64binary, reflects the number of characters representing the encoded data, not the number of bytes of the binary data. The value may come from the ElementDefinition referred to by .definition.
Invariants | Affect this element que-10 Rule Maximum length can only be declared for simple question types (type in ('boolean' | 'decimal' | 'integer' | 'string' | 'text' | 'url' | 'open-choice')) or maxLength.empty()
Affect this element
que-10 | Rule | Maximum length can only be declared for simple question types | (type in ('boolean' | 'decimal' | 'integer' | 'string' | 'text' | 'url' | 'open-choice')) or maxLength.empty()
Questionnaire.item.answerValueSet
Element Id | Questionnaire.item.answerValueSet
Definition | A reference to a value set containing a list of codes representing permitted answers for a "choice" or "open-choice" question.
Cardinality | 0..1
Type | canonical(ValueSet)
Comments | LOINC defines many useful value sets for questionnaire responses. See LOINC Answer Lists. The value may come from the ElementDefinition referred to by .definition.
Invariants | Affect this element que-4 Rule A question cannot have both answerOption and answerValueSet answerOption.empty() or answerValueSet.empty() que-5 Rule Only 'choice' and 'open-choice' items can have answerValueSet (type ='choice' or type = 'open-choice' or type = 'decimal' or type = 'integer' or type = 'date' or type = 'dateTime' or type = 'time' or type = 'string' or type = 'quantity') or (answerValueSet.empty() and answerOption.empty())
Affect this element
que-4 | Rule | A question cannot have both answerOption and answerValueSet | answerOption.empty() or answerValueSet.empty()
que-5 | Rule | Only 'choice' and 'open-choice' items can have answerValueSet | (type ='choice' or type = 'open-choice' or type = 'decimal' or type = 'integer' or type = 'date' or type = 'dateTime' or type = 'time' or type = 'string' or type = 'quantity') or (answerValueSet.empty() and answerOption.empty())
Questionnaire.item.answerOption
Element Id | Questionnaire.item.answerOption
Definition | One of the permitted answers for a "choice" or "open-choice" question.
Cardinality | 0..*
Comments | This element can be used when the value set machinery of answerValueSet is deemed too cumbersome or when there's a need to capture possible answers that are not codes.

Invariants | Affect this element que-4 Rule A question cannot have both answerOption and answerValueSet answerOption.empty() or answerValueSet.empty() que-5 Rule Only 'choice' and 'open-choice' items can have answerValueSet (type ='choice' or type = 'open-choice' or type = 'decimal' or type = 'integer' or type = 'date' or type = 'dateTime' or type = 'time' or type = 'string' or type = 'quantity') or (answerValueSet.empty() and answerOption.empty())
Affect this element
que-4 | Rule | A question cannot have both answerOption and answerValueSet | answerOption.empty() or answerValueSet.empty()
que-5 | Rule | Only 'choice' and 'open-choice' items can have answerValueSet | (type ='choice' or type = 'open-choice' or type = 'decimal' or type = 'integer' or type = 'date' or type = 'dateTime' or type = 'time' or type = 'string' or type = 'quantity') or (answerValueSet.empty() and answerOption.empty())
Questionnaire.item.answerOption.value[x]
Element Id | Questionnaire.item.answerOption.value[x]
Definition | A potential answer that's allowed as the answer to this question.
Cardinality | 1..1
Terminology Binding | Questionnaire Answer Codes (Example)
Type | integer|date|time|string|Coding|Reference(Any)
[x] Note | See Choice of Data Types for further information about how to use [x]
Comments | The data type of the value must agree with the item.type.
Questionnaire.item.answerOption.initialSelected
Element Id | Questionnaire.item.answerOption.initialSelected
Definition | Indicates whether the answer value is selected when the list of possible answers is initially shown.
Cardinality | 0..1
Type | boolean
Meaning if Missing | Only selected items explicitly marked to be selected
Comments | Use this instead of initial[v] if answerValueSet is present.
Questionnaire.item.initial
Element Id | Questionnaire.item.initial
Definition | One or more values that should be pre-populated in the answer when initially rendering the questionnaire for user input.
Cardinality | 0..*
Requirements | In some workflows, having defaults saves time.
Comments | The user is allowed to change the value and override the default (unless marked as read-only). If the user doesn't change the value, then this initial value will be persisted when the QuestionnaireResponse is initially created. Note that initial values can influence results. The data type of initial[x] must agree with the item.type, and only repeating items can have more then one initial value.
Invariants | Affect this element que-8 Rule Initial values can't be specified for groups or display items (type!='group' and type!='display') or initial.empty() que-13 Rule Can only have multiple initial values for repeating items repeats=true or initial.count() <= 1
Affect this element
que-8 | Rule | Initial values can't be specified for groups or display items | (type!='group' and type!='display') or initial.empty()
que-13 | Rule | Can only have multiple initial values for repeating items | repeats=true or initial.count() <= 1
Questionnaire.item.initial.value[x]
Element Id | Questionnaire.item.initial.value[x]
Definition | The actual value to for an initial answer.
Cardinality | 1..1
Terminology Binding | Questionnaire Answer Codes (Example)
Type | boolean|decimal|integer|date|dateTime|time|string|uri|Attachment|Coding|Quantity|Reference(Any)
[x] Note | See Choice of Data Types for further information about how to use [x]
Comments | The type of the initial value must be consistent with the type of the item.
Questionnaire.item.item
Element Id | Questionnaire.item.item
Definition | Text, questions and other groups to be nested beneath a question or group.
Cardinality | 0..*
Type | See Questionnaire.item
Requirements | Reports can consist of complex nested groups.
Comments | There is no specified limit to the depth of nesting. However, Questionnaire authors are encouraged to consider the impact on the user and user interface of overly deep nesting.
Invariants | Affect this element que-1 Rule Group items must have nested items, display items cannot have nested items (type='group' implies item.empty().not()) and (type.trace('type')='display' implies item.trace('item').empty())
Affect this element
que-1 | Rule | Group items must have nested items, display items cannot have nested items | (type='group' implies item.empty().not()) and (type.trace('type')='display' implies item.trace('item').empty())

2.37.9 Resource Questionnaire - Mappings

URL: https://hl7.org/fhir/R4/questionnaire-mappings.html

2.37.9 Resource Questionnaire - Mappings

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

HL7 v2 Mapping

RIM Mapping

Mappings for the questionnaire resource (see Mappings to Other Standards for further information & status).

2.37.9.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

Questionnaire
url | FiveWs.identifier
identifier | FiveWs.identifier
version | FiveWs.version
status | FiveWs.status
experimental | FiveWs.class
date | FiveWs.recorded
publisher | FiveWs.witness
purpose | FiveWs.why[x]

2.37.9.2 Workflow Pattern (http://hl7.org/fhir/workflow)

Questionnaire | Definition
url | Definition.url
identifier | Definition.identifier
version | Definition.version
title | Definition.title
derivedFrom | Definition.derivedFrom
status | Definition.status
experimental | Definition.experimental
subjectType | Definition.subject[x]
date | Definition.date
publisher | Definition.publisher
contact | Definition.contact
description | Definition.description
useContext | Definition.useContext
jurisdiction | Definition.jurisdiction
purpose | Definition.purpose
copyright | Definition.copyright
approvalDate | Definition.approvalDate
lastReviewDate | Definition.lastReviewDate
effectivePeriod | Definition.effectivePeriod

2.37.9.3 HL7 v2 Mapping (http://hl7.org/v2)

Questionnaire
url
identifier
version
name
title
derivedFrom
status
experimental
subjectType
date
publisher
contact
description
useContext
jurisdiction
purpose
copyright
approvalDate
lastReviewDate
effectivePeriod
code
item
linkId
definition
code
prefix
text
type
enableWhen
question
operator
answer[x]

enableBehavior
required
repeats
readOnly
maxLength | N/A
answerValueSet
answerOption
value[x]
initialSelected
initial
value[x]
item

2.37.9.4 RIM Mapping (http://hl7.org/v3)

Questionnaire | Observation[moodCode=DEF]
url | .identifier[scope=BUSN;reliability=ISS]
identifier | .identifier
version | N/A (to add?)
name | N/A
title | .title
derivedFrom | .outboundRelationship[typeCode=DRIV].target[classCode=OBS, moodCode=DEFN]
status | .status
experimental | N/A
subjectType | .outboundRelationship[typeCode=META].target[classCode=OBS, moodCode=DEFN, isCriterion=true].participation.role.classCode
date | .participation[typeCode=AUT].time
publisher | .participation[typeCode=AUT].role
contact | .participation[typeCode=CALLBCK].role
description | .text
useContext | N/A (to add?)
jurisdiction | N/A (to add?)
purpose | .reasonCode.text
copyright | N/A (to add?)
approvalDate | .outboundRelationship[typeCode="SUBJ"].act[classCode=CACT;moodCode=EVN;code="approval"].effectiveTime
lastReviewDate | .outboundRelationship[typeCode="SUBJ"; subsetCode="RECENT"].act[classCode=CACT;moodCode=EVN;code="review"].effectiveTime
effectivePeriod | N/A (to add?)
code | .code
item | .outboundRelationship[typeCode=COMP].target[classCode=OBS, moodCode=DEF]
linkId | .id
definition | .outboundRelationship[typeCode=INST].target[classCode=OBS, moodCode=DEF]
code | .code
prefix | Not supported
text | .text
type | N/A - MIF rather than RIM level
enableWhen | N/A - MIF rather than RIM level
question | N/A - MIF rather than RIM level
operator | N/A - MIF rather than RIM level
answer[x] | N/A - MIF rather than RIM level
enableBehavior | N/A - MIF rather than RIM level
required | N/A - MIF rather than RIM level
repeats | N/A - MIF rather than RIM level
readOnly | N/A - MIF rather than RIM level
maxLength | N/A - MIF rather than RIM level
answerValueSet | N/A - MIF rather than RIM level
answerOption | N/A - MIF rather than RIM level
value[x] | N/A - MIF rather than RIM level
initialSelected | N/A - MIF rather than RIM level
initial | N/A - MIF rather than RIM level
value[x] | N/A - MIF rather than RIM level
item | .outboundRelationship[typeCode=COMP].target

2.37.10 Resource Questionnaire - Extensions & Profiles

URL: https://hl7.org/fhir/R4/questionnaire-profiles.html

2.37.10 Resource Questionnaire - Extensions & Profiles

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

This table lists profiles and extensions for the Questionnaire resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published

Implementation Guides , or in the Conformance resource registry .

2.37.10.1 Profiles

Profile | Description | Context
CQF-Questionnaire | Adds the ability to define behavior as part of a questionnaire. | CQF-Questionnaire

2.37.10.2 Extensions

11179-permitted-value-conceptmap | Questionnaire.item.answerValueSet | permitted-value-conceptmap | for ISO 11179 Element Definition Profile
11179-permitted-value-valueset | Questionnaire.item.answerValueSet | permitted-value-valueset | for ISO 11179 Element Definition Profile
designNote | Questionnaire, Questionnaire.item, Questionnaire.item.item | Design Note | for General Extensions for use by FHIR Implementers
entryFormat | Questionnaire.item | entryFormat | for Element-definition Extensions for use by FHIR Implementers
maxDecimalPlaces | Questionnaire.item | maxDecimalPlaces | for Element-definition Extensions for use by FHIR Implementers
maxSize | Questionnaire.item | maxSize | for Element-definition Extensions for use by FHIR Implementers
maxValue | Questionnaire.item | maxValue | for Element-definition Extensions for use by FHIR Implementers
mimeType | Questionnaire.item | mimeType | for Element-definition Extensions for use by FHIR Implementers
minLength | Questionnaire.item | minLength | for Element-definition Extensions for use by FHIR Implementers
minValue | Questionnaire.item | minValue | for Element-definition Extensions for use by FHIR Implementers
ordinalValue | Questionnaire.item.answerOption | Ordinal Value | for General Extensions for use by FHIR Implementers
questionnaire-baseType | Questionnaire.item, Questionnaire.item.item | baseType | for Core extensions for Questionnaire
questionnaire-choiceOrientation | Questionnaire.item, Questionnaire.item.item | choiceOrientation | for Core extensions for Questionnaire
questionnaire-constraint | Questionnaire, Questionnaire.item, Questionnaire.item.item | constraint | for Core extensions for Questionnaire
questionnaire-displayCategory | Questionnaire.item, Questionnaire.item.item | displayCategory | for Core extensions for Questionnaire
questionnaire-fhirType | Questionnaire.item, Questionnaire.item.item | fhirType | for Core extensions for Questionnaire
questionnaire-hidden | Questionnaire.item, Questionnaire.item.item | hidden | for Core extensions for Questionnaire
questionnaire-itemControl | Questionnaire.item, Questionnaire.item.item | itemControl | for Core extensions for Questionnaire
questionnaire-maxOccurs | Questionnaire.item, Questionnaire.item.item | maxOccurs | for Core extensions for Questionnaire
questionnaire-minOccurs | Questionnaire.item, Questionnaire.item.item | minOccurs | for Core extensions for Questionnaire
questionnaire-optionExclusive | Questionnaire.item.answerOption | optionExclusive | for Core extensions for Questionnaire
questionnaire-optionPrefix | Questionnaire.item.answerOption | optionPrefix | for Core extensions for Questionnaire
questionnaire-referenceFilter | Questionnaire.item, Questionnaire.item.item | referenceFilter | for Core extensions for Questionnaire
questionnaire-referenceProfile | Questionnaire.item, Questionnaire.item.item | referenceProfile | for Core extensions for Questionnaire
questionnaire-referenceResource | Questionnaire.item, Questionnaire.item.item | referenceResource | for Core extensions for Questionnaire
questionnaire-signatureRequired | Questionnaire, Questionnaire.item, Questionnaire.item.item | signatureRequired | for Core extensions for Questionnaire
questionnaire-sliderStepValue | Questionnaire.item, Questionnaire.item.item | sliderStepValue | for Core extensions for Questionnaire
questionnaire-supportLink | Questionnaire.item, Questionnaire.item.item | supportLink | for Core extensions for Questionnaire
questionnaire-unit | Questionnaire.item, Questionnaire.item.item | unit | for Core extensions for Questionnaire
questionnaire-unitOption | Questionnaire.item, Questionnaire.item.item | unitOption | for Core extensions for Questionnaire
questionnaire-unitValueSet | Questionnaire.item, Questionnaire.item.item | unitValueSet | for Core extensions for Questionnaire
questionnaire-usageMode | Questionnaire.item, Questionnaire.item.item | usageMode | for Core extensions for Questionnaire
regex | Questionnaire.item | regex | for Element-definition Extensions for use by FHIR Implementers
replaces | Questionnaire | replaces | for General Extensions for use by FHIR Implementers
variable | Questionnaire, Questionnaire.item, Questionnaire.item.item | Variable | for General Extensions for use by FHIR Implementers
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

servicerequest-questionnaireRequest | questionnaireRequest | for Service Request HL7 Extensions
workflow-instantiatesCanonical | instantiatesCanonical | for Workflow Pattern HL7 Extensions
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

2.37.10.3 Search Extensions

No Search Extensions defined for this resource

2.37.11 Resource Questionnaire - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/questionnaire-version-maps.html

No validation errors - all conversions are clean

2.37.11 Resource Questionnaire - R3/R4 Conversions

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

R3 : R4 Conversion maps for Questionnaire.

Functional status for this map: 6 tests that all execute ok. All tests pass round-trip testing and 5 r3 resources are invalid (0 errors). (see documentation)

2.37.11.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/Questionnaire3to4" = "R3 to R4 Conversions for Questionnaire"

uses "http://hl7.org/fhir/3.0/StructureDefinition/Questionnaire" alias QuestionnaireR3 as source
uses "http://hl7.org/fhir/StructureDefinition/Questionnaire" alias Questionnaire as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group Questionnaire(source src : QuestionnaireR3, target tgt : Questionnaire) extends DomainResource <<type+>> {
  src.url -> tgt.url;
  src.identifier -> tgt.identifier;
  src.version -> tgt.version;
  src.name -> tgt.name;
  src.title -> tgt.title;
  src.status -> tgt.status;
  src.experimental -> tgt.experimental;
  src.date -> tgt.date;
  src.publisher -> tgt.publisher;
  src.description -> tgt.description;
  src.purpose -> tgt.purpose;
  src.approvalDate -> tgt.approvalDate;
  src.lastReviewDate -> tgt.lastReviewDate;
  src.effectivePeriod -> tgt.effectivePeriod;
  src.useContext -> tgt.useContext;
  src.jurisdiction -> tgt.jurisdiction;
  src.contact -> tgt.contact;
  src.copyright -> tgt.copyright;
  src.code -> tgt.code;
  src.subjectType -> tgt.subjectType;
  src.item as s -> tgt.item as t then item(s, t);
}

group item(source src, target tgt) extends BackboneElement {
  src.linkId -> tgt.linkId;
  src.definition -> tgt.definition;
  src.code -> tgt.code;
  src.prefix -> tgt.prefix;
  src.text -> tgt.text;
  src.type -> tgt.type;
  src.enableWhen as s -> tgt.enableWhen as t then enableWhen(s, t);
  src.required -> tgt.required;
  src.repeats -> tgt.repeats;
  src.readOnly -> tgt.readOnly;
  src.maxLength -> tgt.maxLength;
  src.options -> tgt.answerValueSet;
  src.option as s -> tgt.answerOption as t then option(s, t);
  src.initial -> tgt.initial as t then initial(src, t);
  src.item as s -> tgt.item as t then item(s, t);
}

group enableWhen(source src, target tgt) extends BackboneElement {
  src.question -> tgt.question;
  src.hasAnswer as v where src.answer.empty() ->  tgt.operator = 'exists',  tgt.answer = v;
  src.answer as v ->  tgt.answer = v,  tgt.operator = '=';
}

group option(source src, target tgt) extends BackboneElement {
  src.value -> tgt.value;
}

group initial(source src, target tgt) extends BackboneElement {
  src.initial -> tgt.value;
}
```

2.37.11.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/Questionnaire4to3" = "R4 to R3 Conversion for Questionnaire"

uses "http://hl7.org/fhir/StructureDefinition/Questionnaire" alias Questionnaire as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/Questionnaire" alias QuestionnaireR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"
```

```
group Questionnaire(source src : Questionnaire, target tgt : QuestionnaireR3) extends DomainResource <<type+>> {
  src.url -> tgt.url;
  src.identifier -> tgt.identifier;
  src.version -> tgt.version;
  src.name -> tgt.name;
  src.title -> tgt.title;
  src.status -> tgt.status;
  src.experimental -> tgt.experimental;
  src.date -> tgt.date;
  src.publisher -> tgt.publisher;
  src.description -> tgt.description;
  src.purpose -> tgt.purpose;
  src.approvalDate -> tgt.approvalDate;
  src.lastReviewDate -> tgt.lastReviewDate;
  src.effectivePeriod -> tgt.effectivePeriod;
  src.useContext -> tgt.useContext;
  src.jurisdiction -> tgt.jurisdiction;
  src.contact -> tgt.contact;
  src.copyright -> tgt.copyright;
  src.code -> tgt.code;
  src.subjectType -> tgt.subjectType;
  src.item as s -> tgt.item as t then item(s, t);
}

group item(source src, target tgt) extends BackboneElement {
  src.linkId -> tgt.linkId;
  src.definition -> tgt.definition;
  src.code -> tgt.code;
  src.prefix -> tgt.prefix;
  src.text -> tgt.text;
  src.type -> tgt.type;
  src.enableWhen as s -> tgt.enableWhen as t then enableWhen(s, t);
  src.required -> tgt.required;
  src.repeats -> tgt.repeats;
  src.readOnly -> tgt.readOnly;
  src.maxLength -> tgt.maxLength;
  src.answerValueSet -> tgt.options;
  src.answerOption as s -> tgt.option as t then option(s, t);
  src.initial as s then initial(s, tgt);
  src.item as s -> tgt.item as t then item(s, t);
}

group enableWhen(source src, target tgt) extends BackboneElement {
  src.question -> tgt.question;
  src.answer as v where src.operator = 'exists' -> tgt.hasAnswer = v;
  src.answer : boolean as vs where src.operator = '=' -> tgt.answer = create('boolean') as vt then boolean(vs, vt);
  src.answer : decimal as vs where src.operator = '=' -> tgt.answer = create('decimal') as vt then decimal(vs, vt);
  src.answer : integer as vs where src.operator = '=' -> tgt.answer = create('integer') as vt then integer(vs, vt);
  src.answer : date as vs where src.operator = '=' -> tgt.answer = create('date') as vt then date(vs, vt);
  src.answer : dateTime as vs where src.operator = '=' -> tgt.answer = create('dateTime') as vt then dateTime(vs, vt);
  src.answer : time as vs where src.operator = '=' -> tgt.answer = create('time') as vt then time(vs, vt);
  src.answer : string as vs where src.operator = '=' -> tgt.answer = create('string') as vt then string(vs, vt);
  src.answer : Coding as vs where src.operator = '=' -> tgt.answer = create('Coding') as vt then Coding(vs, vt);
  src.answer : Quantity as vs where src.operator = '=' -> tgt.answer = create('Quantity') as vt then Quantity(vs, vt);
  src.answer : Reference as vs where src.operator = '=' -> tgt.answer = create('Reference') as vt then Reference(vs,
    vt);
}

group option(source src, target tgt) extends BackboneElement {
  src.value -> tgt.value;
}

group initial(source src, target tgt) extends BackboneElement {
  src.value -> tgt.initial;
}
```

2.37.11.3 R4 Validation Errors

2.37.12 Core extensions for Questionnaire

URL: https://hl7.org/fhir/R4/questionnaire-extensions.html

2.37.12 Core extensions for Questionnaire

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative

2.37.12.1 Scope and Usage

The core extensions provide HL7 provided extensions to the Questionnaire resource.

2.37.12.2 Content

Extensions:

questionnaire-minOccurs | minOccurs : The minimum number of times the group must appear, or the minimum number of answers for a question - when greater than 1.

questionnaire-maxOccurs | maxOccurs : The maximum number of times the group must appear, or the maximum number of answers for a question - when greater than 1 and not unlimited.

questionnaire-hidden | hidden : If true, indicates that the extended item should not be displayed to the user.
questionnaire-itemControl | itemControl : The type of data entry control or structure that should be used to render the item.
questionnaire-choiceOrientation | choiceOrientation : Identifies the desired orientation when rendering a list of choices (typically radio-box or check-box lists).
questionnaire-referenceResource | referenceResource : Where the type for a question is "Reference", indicates a type of resource that is permitted.
questionnaire-referenceProfile | referenceProfile : Where the type for a question is "Reference", indicates a profile that the resource instances pointed to in answers to this question must be valid against.
questionnaire-referenceFilter | referenceFilter : Identifies a filter to apply when looking up candidate answers for the question.
questionnaire-displayCategory | displayCategory : Describes the intended purpose of the rendered text. For example - instructions, guidance on access control, maintenance information, etc.
questionnaire-fhirType | fhirType : For questionnaires generated from FHIR profiles, indicates the FHIR data type or resource type that corresponds to this node.
questionnaire-unit | unit : Provides a computable unit of measure associated with numeric questions to support subsequent computation on responses. This is for use on items of type integer and decimal, and it's purpose is to support converting the integer or decimal answer into a Quantity when extracting the data into a resource.
questionnaire-unitOption | unitOption : A unit that the user may choose when providing a quantity value.
questionnaire-unitValueSet | unitValueSet : A set of units that the user may choose when providing a quantity value.
questionnaire-optionPrefix | optionPrefix : The label to list in front of a code when presenting a list of possible values in a questionnaire-like fashion.
questionnaire-optionExclusive | optionExclusive : If true, indicates that if this answerOption is selected, no other possible answers may be selected, even if the item is a repeating question.
questionnaire-usageMode | usageMode : Identifies that the specified element should only appear in certain "modes" of operation.
questionnaire-supportLink | supportLink : A URL that resolves to additional supporting information or guidance related to the question.
questionnaire-baseType | baseType : This identifies the underlying type in a profile, when a questionnaire is generated from a profile.
questionnaire-signatureRequired | signatureRequired : Indicates that a signature (of the specified type) is needed when completing the QuestionnaireResponse.
questionnaire-constraint | constraint : An invariant that must be satisfied before responses to the questionnaire can be considered "complete".
questionnaire-sliderStepValue | sliderStepValue : For slider-based controls, indicates the step size to use when toggling the control up or down.

2.37.13 CQF-Questionnaire

URL: https://hl7.org/fhir/R4/cqf.html

Adds the ability to define behavior as part of a questionnaire.

2.37.13 CQF-Questionnaire

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative

2.37.13.1 Content

Profiles:
CQF-Questionnaire | A questionnaire with the ability to specify behavior associated with questions or groups of questions

2.37.13.2 Questionnaire Examples

• PHQ-9: Questionnaire used to define the questions involved in the PHQ-9 health questionnaire

2.37.13.3 StructureDefinition: CQF-Questionnaire

URL: https://hl7.org/fhir/R4/cqf-questionnaire.html

2.37.13.3 StructureDefinition: CQF-Questionnaire

FHIR Infrastructure Work Group | Maturity Level: 1 | Trial Use | Use Context: Any

The official URL for this profile is:

`http://hl7.org/fhir/StructureDefinition/cqf-questionnaire`

A questionnaire with the ability to specify behavior associated with questions or groups of questions

This profile was published on Sat, May 30, 2015 00:00+1000 as a draft by Health Level Seven, Inc. - CDS WG.

2.37.13.3.1 Formal Views of Profile Content

Description of Profiles, Differentials, Snapshots, and how the XML and JSON presentations work.

• Text Summary
• Differential Table
• Snapshot Table
• XML Template
• JSON Template
• All

This structure is derived from Questionnaire.

Summary

Extensions

This structure refers to these extensions:

• http://hl7.org/fhir/StructureDefinition/cqf-library

This structure is derived from Questionnaire.

Name | Flags | Card. | Type | Description & Constraints
Questionnaire | | 0..* | | A questionnaire with the ability to specify behavior associated with questions or groups of questions
cqf-library | | 0..* | canonical(Library) | A library containing logic referenced by the questionnaire URL: http://hl7.org/fhir/StructureDefinition/cqf-library

Documentation for this format

Name | Flags | Card. | Type | Description & Constraints
Questionnaire | I | 0..* | | A questionnaire with the ability to specify behavior associated with questions or groups of questions
id | Σ | 0..1 | string | Logical id of this artifact
meta | ΣI | 0..1 | Meta | Metadata about the resource
implicitRules | ?!ΣI | 0..1 | uri | A set of rules under which this content was created
language | I | 0..1 | code | Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | I | 0..* | Extension | Extension Slice: Unordered, Open by value:url
cqf-library | I | 0..* | canonical(Library) | A library containing logic referenced by the questionnaire URL: http://hl7.org/fhir/StructureDefinition/cqf-library
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
url | ΣI | 0..1 | uri | Canonical identifier for this questionnaire, represented as a URI (globally unique)
identifier | ΣI | 0..* | Identifier | Additional identifier for the questionnaire
version | ΣI | 0..1 | string | Business version of the questionnaire
name | ΣI | 0..1 | string | Name for this questionnaire (computer friendly)
title | ΣI | 0..1 | string | Name for this questionnaire (human friendly)
derivedFrom | I | 0..* | canonical(Questionnaire) | Instantiates protocol or definition
status | ?!ΣI | 1..1 | code | draft | active | retired | unknown Binding: PublicationStatus (required)
experimental | ΣI | 0..1 | boolean | For testing purposes, not real usage
subjectType | ΣI | 0..* | code | Resource that can be subject of QuestionnaireResponse Binding: ResourceType (required)
date | ΣI | 0..1 | dateTime | Date last changed
publisher | ΣI | 0..1 | string | Name of the publisher (organization or individual)
contact | ΣI | 0..* | ContactDetail | Contact details for the publisher
description | I | 0..1 | markdown | Natural language description of the questionnaire
useContext | ΣI | 0..* | UsageContext | The context that the content is intended to support
jurisdiction | ΣI | 0..* | CodeableConcept | Intended jurisdiction for questionnaire (if applicable) Binding: Jurisdiction (extensible)
purpose | I | 0..1 | markdown | Why this questionnaire is defined
copyright | I | 0..1 | markdown | Use and/or publishing restrictions
approvalDate | I | 0..1 | date | When the questionnaire was approved by publisher
lastReviewDate | I | 0..1 | date | When the questionnaire was last reviewed
effectivePeriod | ΣI | 0..1 | Period | When the questionnaire is expected to be used
code | ΣI | 0..* | Coding | Concept that represents the overall questionnaire Binding: Questionnaire Question Codes (example)
item | I | 0..* | BackboneElement | Questions and sections within the Questionnaire que-1: Group items must have nested items, display items cannot have nested items que-3: Display items cannot have a "code" asserted que-4: A question cannot have both answerOption and answerValueSet que-5: Only 'choice' and 'open-choice' items can have answerValueSet que-6: Required and repeat aren't permitted for display items que-8: Initial values can't be specified for groups or display items que-9: Read-only can't be specified for "display" items que-10: Maximum length can only be declared for simple question types que-11: If one or more answerOption is present, initial[x] must be missing que-12: If there are more than one enableWhen, enableBehavior must be specified que-13: Can only have multiple initial values for repeating items
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
linkId | I | 1..1 | string | Unique id for item in questionnaire
definition | I | 0..1 | uri | ElementDefinition - details for the item
code | I | 0..* | Coding | Corresponding concept for this item in a terminology Binding: Questionnaire Question Codes (example)
prefix | I | 0..1 | string | E.g. "1(a)", "2.5.3"
text | I | 0..1 | string | Primary text for the item
type | I | 1..1 | code | group | display | boolean | decimal | integer | date | dateTime + Binding: QuestionnaireItemType (required)
enableWhen | ?!I | 0..* | BackboneElement | Only allow data when que-7: If the operator is 'exists', the value must be a boolean
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
question | I | 1..1 | string | Question that determines whether item is enabled
operator | I | 1..1 | code | exists | = | != | > | < | >= | <= Binding: QuestionnaireItemOperator (required)
answer[x] | I | 1..1 | | Value for question comparison based on operator Binding: Questionnaire Answer Codes (example)
answerBoolean | | | boolean
answerDecimal | | | decimal
answerInteger | | | integer

answerDate | | | date
answerDateTime | | | dateTime
answerTime | | | time
answerString | | | string
answerCoding | | | Coding
answerQuantity | | | Quantity
answerReference | | | Reference(resource.html)
enableBehavior | I | 0..1 | code | all | any Binding: EnableWhenBehavior (required)
required | I | 0..1 | boolean | Whether the item must be included in data results
repeats | I | 0..1 | boolean | Whether the item may repeat
readOnly | I | 0..1 | boolean | Don't allow human editing
maxLength | I | 0..1 | integer | No more than this many characters
answerValueSet | I | 0..1 | canonical(ValueSet) | Valueset containing permitted answers
answerOption | I | 0..* | BackboneElement | Permitted answer
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
value[x] | I | 1..1 | | Answer value Binding: Questionnaire Answer Codes (example)
valueInteger | | | integer
valueDate | | | date
valueTime | | | time
valueString | | | string
valueCoding | | | Coding
valueReference | | | Reference(resource.html)
initialSelected | I | 0..1 | boolean | Whether option is selected by default
initial | I | 0..* | BackboneElement | Initial value(s) when item is first rendered
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
value[x] | I | 1..1 | | Actual value for initializing the question Binding: Questionnaire Answer Codes (example)
valueBoolean | | | boolean
valueDecimal | | | decimal
valueInteger | | | integer
valueDate | | | date
valueDateTime | | | dateTime
valueTime | | | time
valueString | | | string
valueUri | | | uri
valueAttachment | | | Attachment
valueCoding | | | Coding
valueQuantity | | | Quantity
valueReference | | | Reference(resource.html)
item | I | 0..* | | Unknown reference to #Questionnaire.item Nested questionnaire items
Documentation for this format

```
<!-- CQF-Questionnaire -->

<Questionnaire xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <url value="[uri]"/><!--  0..1
     Canonical identifier for this questionnaire, represented as a URI (globally unique) -->
 <identifier><!--  0..* Identifier
     Additional identifier for the questionnaire --></identifier>
 <version value="[string]"/><!--  0..1 Business version of the questionnaire -->
 <name value="[string]"/><!--  0..1 Name for this questionnaire (computer friendly) -->
 <title value="[string]"/><!--  0..1 Name for this questionnaire (human friendly) -->
 <derivedFrom value="[canonical]"/><!--  0..* Instantiates protocol or definition -->
 <status value="[code]"/><!--  1..1 draft | active | retired | unknown   -->
 <experimental value="[boolean]"/><!--  0..1 For testing purposes, not real usage -->
 <subjectType value="[code]"/><!--  0..* Resource that can be subject of QuestionnaireResponse   -->
 <date value="[dateTime]"/><!--  0..1 Date last changed -->
 <publisher value="[string]"/><!--  0..1 Name of the publisher (organization or individual) -->
 <contact><!--  0..* ContactDetail
     Contact details for the publisher --></contact>
 <description value="[markdown]"/><!--  0..1 Natural language description of the questionnaire -->
 <useContext><!--  0..* UsageContext
     The context that the content is intended to support --></useContext>
 <jurisdiction><!--  0..* CodeableConcept Intended jurisdiction for questionnaire (if applicable)   --></jurisdiction>
 <purpose value="[markdown]"/><!--  0..1 Why this questionnaire is defined -->
 <copyright value="[markdown]"/><!--  0..1 Use and/or publishing restrictions -->
```

```
 <approvalDate value="[date]"/><!--  0..1 When the questionnaire was approved by publisher -->
 <lastReviewDate value="[date]"/><!--  0..1 When the questionnaire was last reviewed -->
 <effectivePeriod><!--  0..1 Period
     When the questionnaire is expected to be used --></effectivePeriod>
 <code><!--  0..* Coding Concept that represents the overall questionnaire   --></code>
 <item>  0..* BackboneElement  <!--  0..* Questions and sections within the Questionnaire -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <linkId value="[string]"/><!--  1..1 Unique id for item in questionnaire -->
  <definition value="[uri]"/><!--  0..1 ElementDefinition - details for the item -->
  <code><!--  0..* Coding Corresponding concept for this item in a terminology   --></code>
  <prefix value="[string]"/><!--  0..1 E.g. "1(a)", "2.5.3" -->
  <text value="[string]"/><!--  0..1 Primary text for the item -->
  <type value="[code]"/><!--  1..1 group | display | boolean | decimal | integer | date | dateTime +   -->
  <enableWhen>  0..* BackboneElement  <!--  0..* Only allow data when -->
   <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
   <question value="[string]"/><!--  1..1 Question that determines whether item is enabled -->
   <operator value="[code]"/><!--  1..1 exists | = | != | > | < | >= | <=   -->
   <answer[x]><!--  1..1 boolean|decimal|integer|
     date|dateTime|time|string|Coding|Quantity|Reference(Resource) Value for question comparison based on operator
    --></answer[x]>
  </enableWhen>
  <enableBehavior value="[code]"/><!--  0..1 all | any   -->
  <required value="[boolean]"/><!--  0..1 Whether the item must be included in data results -->
  <repeats value="[boolean]"/><!--  0..1 Whether the item may repeat -->
  <readOnly value="[boolean]"/><!--  0..1 Don't allow human editing -->
  <maxLength value="[integer]"/><!--  0..1 No more than this many characters -->
  <answerValueSet value="[canonical]"/><!--  0..1 Valueset containing permitted answers -->
  <answerOption>  0..* BackboneElement  <!--  0..* Permitted answer -->
   <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
   <value[x]><!--  1..1 integer|date|time|string|
     Coding|Reference(Resource) Answer value   --></value[x]>
   <initialSelected value="[boolean]"/><!--  0..1 Whether option is selected by default -->
  </answerOption>
  <initial>  0..* BackboneElement  <!--  0..* Initial value(s) when item is first rendered -->
   <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
   <value[x]><!--  1..1 boolean|decimal|integer|date|
     dateTime|time|string|uri|Attachment|Coding|Quantity|Reference(Resource) Actual value for initializing the question
       --></value[x]>
  </initial>
  <item><!-- See #Questionnaire.item  Nested questionnaire items --></item>
 </item>
</Questionnaire>
```

```
{ // CQF-Questionnaire
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "extension" : [ //  sliced by value:url  in the specified order, Open
      { // A library containing logic referenced by the questionnaire // C?
        // from Element: extension
      }
    ],
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "url" : "<uri>", // C?Canonical identifier for this questionnaire, represented as a URI (globally unique)
    "identifier" : [{ Identifier }], // C?Additional identifier for the questionnaire
    "version" : "<string>", // C?Business version of the questionnaire
    "name" : "<string>", // C?Name for this questionnaire (computer friendly)
    "title" : "<string>", // C?Name for this questionnaire (human friendly)
    "derivedFrom" : ["<canonical>"], // C?Instantiates protocol or definition
    "status" : "<code>", // C? R! draft | active | retired | unknown
    "experimental" : <boolean>, // C?For testing purposes, not real usage
    "subjectType" : ["<code>"], // C?Resource that can be subject of QuestionnaireResponse
    "date" : "<dateTime>", // C?Date last changed
    "publisher" : "<string>", // C?Name of the publisher (organization or individual)
    "contact" : [{ ContactDetail }], // C?Contact details for the publisher
    "description" : "<markdown>", // C?Natural language description of the questionnaire
    "useContext" : [{ UsageContext }], // C?The context that the content is intended to support
    "jurisdiction" : [{ CodeableConcept }], // C?Intended jurisdiction for questionnaire (if applicable)
    "purpose" : "<markdown>", // C?Why this questionnaire is defined
    "copyright" : "<markdown>", // C?Use and/or publishing restrictions
    "approvalDate" : "<date>", // C?When the questionnaire was approved by publisher
    "lastReviewDate" : "<date>", // C?When the questionnaire was last reviewed
    "effectivePeriod" : { Period }, // C?When the questionnaire is expected to be used
    "code" : [{ Coding }], // C?Concept that represents the overall questionnaire
    "item" : [{ BackboneElement }], // C?Questions and sections within the Questionnaire
  }
```

This structure is derived from Questionnaire.

Summary

Extensions

This structure refers to these extensions:

• http://hl7.org/fhir/StructureDefinition/cqf-library

Differential View

This structure is derived from Questionnaire.

Name | Flags | Card. | Type | Description & Constraints
Questionnaire | | 0..* | | A questionnaire with the ability to specify behavior associated with questions or groups of questions
cqf-library | | 0..* | canonical(Library) | A library containing logic referenced by the questionnaire URL: http://hl7.org/fhir/StructureDefinition/cqf-library
Documentation for this format

Snapshot View

Name | Flags | Card. | Type | Description & Constraints
Questionnaire | I | 0..* | | A questionnaire with the ability to specify behavior associated with questions or groups of questions
id | Σ | 0..1 | string | Logical id of this artifact
meta | ΣI | 0..1 | Meta | Metadata about the resource
implicitRules | ?!ΣI | 0..1 | uri | A set of rules under which this content was created
language | I | 0..1 | code | Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | I | 0..* | Extension | Extension Slice: Unordered, Open by value:url
cqf-library | I | 0..* | canonical(Library) | A library containing logic referenced by the questionnaire URL: http://hl7.org/fhir/StructureDefinition/cqf-library
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
url | ΣI | 0..1 | uri | Canonical identifier for this questionnaire, represented as a URI (globally unique)
identifier | ΣI | 0..* | Identifier | Additional identifier for the questionnaire
version | ΣI | 0..1 | string | Business version of the questionnaire
name | ΣI | 0..1 | string | Name for this questionnaire (computer friendly)
title | ΣI | 0..1 | string | Name for this questionnaire (human friendly)
derivedFrom | I | 0..* | canonical(Questionnaire) | Instantiates protocol or definition
status | ?!ΣI | 1..1 | code | draft | active | retired | unknown Binding: PublicationStatus (required)
experimental | ΣI | 0..1 | boolean | For testing purposes, not real usage
subjectType | ΣI | 0..* | code | Resource that can be subject of QuestionnaireResponse Binding: ResourceType (required)
date | ΣI | 0..1 | dateTime | Date last changed
publisher | ΣI | 0..1 | string | Name of the publisher (organization or individual)
contact | ΣI | 0..* | ContactDetail | Contact details for the publisher
description | I | 0..1 | markdown | Natural language description of the questionnaire
useContext | ΣI | 0..* | UsageContext | The context that the content is intended to support
jurisdiction | ΣI | 0..* | CodeableConcept | Intended jurisdiction for questionnaire (if applicable) Binding: Jurisdiction (extensible)
purpose | I | 0..1 | markdown | Why this questionnaire is defined
copyright | I | 0..1 | markdown | Use and/or publishing restrictions
approvalDate | I | 0..1 | date | When the questionnaire was approved by publisher
lastReviewDate | I | 0..1 | date | When the questionnaire was last reviewed
effectivePeriod | ΣI | 0..1 | Period | When the questionnaire is expected to be used
code | ΣI | 0..* | Coding | Concept that represents the overall questionnaire Binding: Questionnaire Question Codes (example)
item | I | 0..* | BackboneElement | Questions and sections within the Questionnaire que-1: Group items must have nested items, display items cannot have nested items que-3: Display items cannot have a "code" asserted que-4: A question cannot have both answerOption and answerValueSet que-5: Only 'choice' and 'open-choice' items can have answerValueSet que-6: Required and repeat aren't permitted for display items que-8: Initial values can't be specified for groups or display items que-9: Read-only can't be specified for "display" items que-10: Maximum length can only be declared for simple question types que-11: If one or more answerOption is present, initial[x] must be missing que-12: If there are more than one enableWhen, enableBehavior must be specified que-13: Can only have multiple initial values for repeating items
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
linkId | I | 1..1 | string | Unique id for item in questionnaire
definition | I | 0..1 | uri | ElementDefinition - details for the item
code | I | 0..* | Coding | Corresponding concept for this item in a terminology Binding: Questionnaire Question Codes (example)
prefix | I | 0..1 | string | E.g. "1(a)", "2.5.3"
text | I | 0..1 | string | Primary text for the item
type | I | 1..1 | code | group | display | boolean | decimal | integer | date | dateTime + Binding: QuestionnaireItemType (required)
enableWhen | ?!I | 0..* | BackboneElement | Only allow data when que-7: If the operator is 'exists', the value must be a boolean
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
question | I | 1..1 | string | Question that determines whether item is enabled
operator | I | 1..1 | code | exists | = | != | > | < | >= | <= Binding: QuestionnaireItemOperator (required)
answer[x] | I | 1..1 | | Value for question comparison based on operator Binding: Questionnaire Answer Codes (example)
answerBoolean | | | boolean
answerDecimal | | | decimal
answerInteger | | | integer
answerDate | | | date
answerDateTime | | | dateTime
answerTime | | | time

answerString | | | string
answerCoding | | | Coding
answerQuantity | | | Quantity
answerReference | | | Reference(resource.html)
enableBehavior | I | 0..1 | code | all | any Binding: EnableWhenBehavior (required)
required | I | 0..1 | boolean | Whether the item must be included in data results
repeats | I | 0..1 | boolean | Whether the item may repeat
readOnly | I | 0..1 | boolean | Don't allow human editing
maxLength | I | 0..1 | integer | No more than this many characters
answerValueSet | I | 0..1 | canonical(ValueSet) | Valueset containing permitted answers
answerOption | I | 0..* | BackboneElement | Permitted answer
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
value[x] | I | 1..1 | | Answer value Binding: Questionnaire Answer Codes (example)
valueInteger | | | integer
valueDate | | | date
valueTime | | | time
valueString | | | string
valueCoding | | | Coding
valueReference | | | Reference(resource.html)
initialSelected | I | 0..1 | boolean | Whether option is selected by default
initial | I | 0..* | BackboneElement | Initial value(s) when item is first rendered
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
value[x] | I | 1..1 | | Actual value for initializing the question Binding: Questionnaire Answer Codes (example)
valueBoolean | | | boolean
valueDecimal | | | decimal
valueInteger | | | integer
valueDate | | | date
valueDateTime | | | dateTime
valueTime | | | time
valueString | | | string
valueUri | | | uri
valueAttachment | | | Attachment
valueCoding | | | Coding
valueQuantity | | | Quantity
valueReference | | | Reference(resource.html)
item | I | 0..* | | Unknown reference to #Questionnaire.item Nested questionnaire items
Documentation for this format

XML Template

```
<!-- CQF-Questionnaire -->

<Questionnaire xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <url value="[uri]"/><!--  0..1
     Canonical identifier for this questionnaire, represented as a URI (globally unique) -->
 <identifier><!--  0..* Identifier
     Additional identifier for the questionnaire --></identifier>
 <version value="[string]"/><!--  0..1 Business version of the questionnaire -->
 <name value="[string]"/><!--  0..1 Name for this questionnaire (computer friendly) -->
 <title value="[string]"/><!--  0..1 Name for this questionnaire (human friendly) -->
 <derivedFrom value="[canonical]"/><!--  0..* Instantiates protocol or definition -->
 <status value="[code]"/><!--  1..1 draft | active | retired | unknown   -->
 <experimental value="[boolean]"/><!--  0..1 For testing purposes, not real usage -->
 <subjectType value="[code]"/><!--  0..* Resource that can be subject of QuestionnaireResponse   -->
 <date value="[dateTime]"/><!--  0..1 Date last changed -->
 <publisher value="[string]"/><!--  0..1 Name of the publisher (organization or individual) -->
 <contact><!--  0..* ContactDetail
     Contact details for the publisher --></contact>
 <description value="[markdown]"/><!--  0..1 Natural language description of the questionnaire -->
 <useContext><!--  0..* UsageContext
     The context that the content is intended to support --></useContext>
 <jurisdiction><!--  0..* CodeableConcept Intended jurisdiction for questionnaire (if applicable)   --></jurisdiction>
 <purpose value="[markdown]"/><!--  0..1 Why this questionnaire is defined -->
 <copyright value="[markdown]"/><!--  0..1 Use and/or publishing restrictions -->
 <approvalDate value="[date]"/><!--  0..1 When the questionnaire was approved by publisher -->
 <lastReviewDate value="[date]"/><!--  0..1 When the questionnaire was last reviewed -->
```

```
 <effectivePeriod><!--  0..1 Period
     When the questionnaire is expected to be used --></effectivePeriod>
 <code><!--  0..* Coding Concept that represents the overall questionnaire   --></code>
 <item>  0..* BackboneElement  <!--  0..* Questions and sections within the Questionnaire -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <linkId value="[string]"/><!--  1..1 Unique id for item in questionnaire -->
  <definition value="[uri]"/><!--  0..1 ElementDefinition - details for the item -->
  <code><!--  0..* Coding Corresponding concept for this item in a terminology   --></code>
  <prefix value="[string]"/><!--  0..1 E.g. "1(a)", "2.5.3" -->
  <text value="[string]"/><!--  0..1 Primary text for the item -->
  <type value="[code]"/><!--  1..1 group | display | boolean | decimal | integer | date | dateTime +   -->
  <enableWhen>  0..* BackboneElement  <!--  0..* Only allow data when -->
   <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
   <question value="[string]"/><!--  1..1 Question that determines whether item is enabled -->
   <operator value="[code]"/><!--  1..1 exists | = | != | > | < | >= | <=   -->
   <answer[x]><!--  1..1 boolean|decimal|integer|
     date|dateTime|time|string|Coding|Quantity|Reference(Resource) Value for question comparison based on operator
    --></answer[x]>
  </enableWhen>
  <enableBehavior value="[code]"/><!--  0..1 all | any   -->
  <required value="[boolean]"/><!--  0..1 Whether the item must be included in data results -->
  <repeats value="[boolean]"/><!--  0..1 Whether the item may repeat -->
  <readOnly value="[boolean]"/><!--  0..1 Don't allow human editing -->
  <maxLength value="[integer]"/><!--  0..1 No more than this many characters -->
  <answerValueSet value="[canonical]"/><!--  0..1 Valueset containing permitted answers -->
  <answerOption>  0..* BackboneElement  <!--  0..* Permitted answer -->
   <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
   <value[x]><!--  1..1 integer|date|time|string|
     Coding|Reference(Resource) Answer value   --></value[x]>
   <initialSelected value="[boolean]"/><!--  0..1 Whether option is selected by default -->
  </answerOption>
  <initial>  0..* BackboneElement  <!--  0..* Initial value(s) when item is first rendered -->
   <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
   <value[x]><!--  1..1 boolean|decimal|integer|date|
     dateTime|time|string|uri|Attachment|Coding|Quantity|Reference(Resource) Actual value for initializing the question
       --></value[x]>
  </initial>
  <item><!-- See #Questionnaire.item  Nested questionnaire items --></item>
 </item>
</Questionnaire>
```

JSON Template

```
{ // CQF-Questionnaire
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "extension" : [ //  sliced by value:url  in the specified order, Open
      { // A library containing logic referenced by the questionnaire // C?
        // from Element: extension
      }
    ],
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "url" : "<uri>", // C?Canonical identifier for this questionnaire, represented as a URI (globally unique)
    "identifier" : [{ Identifier }], // C?Additional identifier for the questionnaire
    "version" : "<string>", // C?Business version of the questionnaire
    "name" : "<string>", // C?Name for this questionnaire (computer friendly)
    "title" : "<string>", // C?Name for this questionnaire (human friendly)
    "derivedFrom" : ["<canonical>"], // C?Instantiates protocol or definition
    "status" : "<code>", // C? R! draft | active | retired | unknown
    "experimental" : <boolean>, // C?For testing purposes, not real usage
    "subjectType" : ["<code>"], // C?Resource that can be subject of QuestionnaireResponse
    "date" : "<dateTime>", // C?Date last changed
    "publisher" : "<string>", // C?Name of the publisher (organization or individual)
    "contact" : [{ ContactDetail }], // C?Contact details for the publisher
    "description" : "<markdown>", // C?Natural language description of the questionnaire
    "useContext" : [{ UsageContext }], // C?The context that the content is intended to support
    "jurisdiction" : [{ CodeableConcept }], // C?Intended jurisdiction for questionnaire (if applicable)
    "purpose" : "<markdown>", // C?Why this questionnaire is defined
    "copyright" : "<markdown>", // C?Use and/or publishing restrictions
    "approvalDate" : "<date>", // C?When the questionnaire was approved by publisher
    "lastReviewDate" : "<date>", // C?When the questionnaire was last reviewed
    "effectivePeriod" : { Period }, // C?When the questionnaire is expected to be used
    "code" : [{ Coding }], // C?Concept that represents the overall questionnaire
    "item" : [{ BackboneElement }], // C?Questions and sections within the Questionnaire
  }
```

Alternate definitions: Master Definition (XML, JSON), Schematron

2.37.13.3.2 Terminology Bindings

Path	Name	Conformance	ValueSet
Questionnaire.language	Common Languages	preferred	Common Languages

Questionnaire.status | PublicationStatus | required | PublicationStatus
Questionnaire.subjectType | ResourceType | required | ResourceType
Questionnaire.jurisdiction | Jurisdiction | extensible | Jurisdiction
Questionnaire.code | Questionnaire Question Codes | example | Questionnaire Question Codes
Questionnaire.item.code | Questionnaire Question Codes | example | Questionnaire Question Codes
Questionnaire.item.type | QuestionnaireItemType | required | QuestionnaireItemType
Questionnaire.item.enableWhen.operator | QuestionnaireItemOperator | required | QuestionnaireItemOperator
Questionnaire.item.enableWhen.answer[x] | Questionnaire Answer Codes | example | Questionnaire Answer Codes
Questionnaire.item.enableBehavior | EnableWhenBehavior | required | EnableWhenBehavior
Questionnaire.item.answerOption.value[x] | Questionnaire Answer Codes | example | Questionnaire Answer Codes
Questionnaire.item.initial.value[x] | Questionnaire Answer Codes | example | Questionnaire Answer Codes

2.37.13.3.3 Constraints

Id | Path | Details | Requirements
que-1 | Questionnaire.item | Group items must have nested items, display items cannot have nested items Expression : (type='group' implies item.empty().not()) and (type.trace('type')='display' implies item.trace('item').empty())
que-3 | Questionnaire.item | Display items cannot have a "code" asserted Expression : type!='display' or code.empty()
que-4 | Questionnaire.item | A question cannot have both answerOption and answerValueSet Expression : answerOption.empty() or answerValueSet.empty()
que-5 | Questionnaire.item | Only 'choice' and 'open-choice' items can have answerValueSet Expression : (type ='choice' or type = 'open-choice' or type = 'decimal' or type = 'integer' or type = 'date' or type = 'dateTime' or type = 'time' or type = 'string' or type = 'quantity') or (answerValueSet.empty() and answerOption.empty())
que-6 | Questionnaire.item | Required and repeat aren't permitted for display items Expression : type!='display' or (required.empty() and repeats.empty())
que-8 | Questionnaire.item | Initial values can't be specified for groups or display items Expression : (type!='group' and type!='display') or initial.empty()
que-9 | Questionnaire.item | Read-only can't be specified for "display" items Expression : type!='display' or readOnly.empty()
que-10 | Questionnaire.item | Maximum length can only be declared for simple question types Expression : (type in ('boolean' | 'decimal' | 'integer' | 'string' | 'text' | 'url' | 'open-choice')) or maxLength.empty()
que-11 | Questionnaire.item | If one or more answerOption is present, initial[x] must be missing Expression : answerOption.empty() or initial.empty()
que-12 | Questionnaire.item | If there are more than one enableWhen, enableBehavior must be specified Expression : enableWhen.count() > 2 implies enableBehavior.exists()
que-13 | Questionnaire.item | Can only have multiple initial values for repeating items Expression : repeats=true or initial.count() <= 1
que-7 | Questionnaire.item.enableWhen | If the operator is 'exists', the value must be a boolean Expression : operator = 'exists' implies (answer is Boolean)

2.37.14 JSON Schema for Questionnaire

URL: https://hl7.org/fhir/R4/questionnaire.schema.json.html

2.37.14 JSON Schema for Questionnaire

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is a representation of the json schema for Questionnaire, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/Questionnaire",
  "$ref": "#/definitions/Questionnaire",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "Questionnaire": {
      "description": "A structured set of questions intended to guide the collection of answers from end-users.
    Questionnaires provide detailed control over order, presentation, phraseology and grouping to allow coherent,
    consistent data collection.",
      "properties": {
        "resourceType": {
          "description": "This is a Questionnaire resource",
          "const": "Questionnaire"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
```

```
    "_implicitRules": {
      "description": "Extensions for implicitRules",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "language": {
      "description": "The base language in which the resource is written.",
      "$ref": "#/definitions/code"
    },
    "_language": {
      "description": "Extensions for language",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "text": {
      "description": "A human-readable narrative that contains a summary of the resource and can be used to
represent the content of the resource to a human. The narrative need not encode all the structured data, but is
required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
      "$ref": "Narrative.schema.json#/definitions/Narrative"
    },
    "contained": {
      "description": "These resources do not have an independent existence apart from the resource that contains
them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
 definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "url": {
      "description": "An absolute URI that is used to identify this questionnaire when it is referenced in a
specification, model, design or an instance; also called its canonical identifier. This SHOULD be globally unique
and SHOULD be a literal address at which at which an authoritative instance of this questionnaire is (or will be)
published. This URL can be the target of a canonical reference. It SHALL remain the same when the questionnaire is
stored on different servers.",
      "$ref": "#/definitions/uri"
    },
    "_url": {
      "description": "Extensions for url",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "identifier": {
      "description": "A formal identifier that is used to identify this questionnaire when it is represented in
other formats, or referenced in a specification, model, design or an instance.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "version": {
      "description": "The identifier that is used to identify this version of the questionnaire when it is
referenced in a specification, model, design or instance. This is an arbitrary value managed by the questionnaire
author and is not expected to be globally unique. For example, it might be a timestamp (e.g. yyyymmdd) if a managed
version is not available. There is also no expectation that versions can be placed in a lexicographical sequence.",
      "$ref": "#/definitions/string"
    },
    "_version": {
      "description": "Extensions for version",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "name": {
      "description": "A natural language name identifying the questionnaire. This name should be usable as an
identifier for the module by machine processing applications such as code generation.",
      "$ref": "#/definitions/string"
    },
    "_name": {
      "description": "Extensions for name",
```

```
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "title": {
      "description": "A short, descriptive, user-friendly title for the questionnaire.",
      "$ref": "#/definitions/string"
    },
    "_title": {
      "description": "Extensions for title",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "derivedFrom": {
      "description": "The URL of a Questionnaire that this Questionnaire is based on.",
      "items": {
        "$ref": "canonical.schema.json#/definitions/canonical"
      },
      "type": "array"
    },
    "status": {
      "description": "The status of this questionnaire. Enables tracking the life-cycle of the content.",
      "enum": [
        "draft",
        "active",
        "retired",
        "unknown"
      ]
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "experimental": {
      "description": "A Boolean value to indicate that this questionnaire is authored for testing purposes (or
education/evaluation/marketing) and is not intended to be used for genuine usage.",
      "$ref": "#/definitions/boolean"
    },
    "_experimental": {
      "description": "Extensions for experimental",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "subjectType": {
      "description": "The types of subjects that can be the subject of responses created for the questionnaire.",
      "items": {
        "$ref": "#/definitions/code"
      },
      "type": "array"
    },
    "_subjectType": {
      "description": "Extensions for subjectType",
      "items": {
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "type": "array"
    },
    "date": {
      "description": "The date  (and optionally time) when the questionnaire was published. The date must change
when the business version changes and it must change if the status code changes. In addition, it should change when
the substantive content of the questionnaire changes.",
      "$ref": "#/definitions/dateTime"
    },
    "_date": {
      "description": "Extensions for date",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "publisher": {
      "description": "The name of the organization or individual that published the questionnaire.",
      "$ref": "#/definitions/string"
    },
    "_publisher": {
      "description": "Extensions for publisher",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "contact": {
      "description": "Contact details to assist a user in finding and communicating with the publisher.",
      "items": {
        "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
      },
      "type": "array"
    },
    "description": {
      "description": "A free text natural language description of the questionnaire from a consumer\u0027s
perspective.",
      "$ref": "#/definitions/markdown"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "useContext": {
      "description": "The content was developed with a focus and intent of supporting the contexts that are listed.
```

```
These contexts may be general categories (gender, age, ...) or may be references to specific programs (insurance
plans, studies, ...) and may be used to assist with indexing and searching for appropriate questionnaire
instances.",
      "items": {
        "$ref": "UsageContext.schema.json#/definitions/UsageContext"
      },
      "type": "array"
    },
    "jurisdiction": {
      "description": "A legal or geographic region in which the questionnaire is intended to be used.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "purpose": {
      "description": "Explanation of why this questionnaire is needed and why it has been designed as it has.",
      "$ref": "#/definitions/markdown"
    },
    "_purpose": {
      "description": "Extensions for purpose",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "copyright": {
      "description": "A copyright statement relating to the questionnaire and/or its contents. Copyright statements
are generally legal restrictions on the use and publishing of the questionnaire.",
      "$ref": "#/definitions/markdown"
    },
    "_copyright": {
      "description": "Extensions for copyright",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "approvalDate": {
      "description": "The date on which the resource content was approved by the publisher. Approval happens once
when the content is officially approved for usage.",
      "$ref": "#/definitions/date"
    },
    "_approvalDate": {
      "description": "Extensions for approvalDate",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "lastReviewDate": {
      "description": "The date on which the resource content was last reviewed. Review happens periodically after
approval but does not change the original approval date.",
      "$ref": "#/definitions/date"
    },
    "_lastReviewDate": {
      "description": "Extensions for lastReviewDate",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "effectivePeriod": {
      "description": "The period during which the questionnaire content was or is planned to be in active use.",
      "$ref": "Period.schema.json#/definitions/Period"
    },
    "code": {
      "description": "An identifier for this question or group of questions in a particular terminology such as
LOINC.",
      "items": {
        "$ref": "Coding.schema.json#/definitions/Coding"
      },
      "type": "array"
    },
    "item": {
      "description": "A particular question, question grouping or display text that is part of the questionnaire.",
      "items": {
        "$ref": "#/definitions/Questionnaire_Item"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "resourceType"
  ]
},
"Questionnaire_Item": {
  "description": "A structured set of questions intended to guide the collection of answers from end-users.
Questionnaires provide detailed control over order, presentation, phraseology and grouping to allow coherent,
consistent data collection.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
```

```
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "linkId": {
      "description": "An identifier that is unique within the Questionnaire allowing linkage to the equivalent item
in a QuestionnaireResponse resource.",
      "$ref": "#/definitions/string"
    },
    "_linkId": {
      "description": "Extensions for linkId",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "definition": {
      "description": "This element is a URI that refers to an [[[ElementDefinition]]] that provides information
about this item, including information that might otherwise be included in the instance of the Questionnaire
resource. A detailed description of the construction of the URI is shown in Comments, below. If this element is
present then the following element values MAY be derived from the Element Definition if the corresponding elements
of this Questionnaire resource instance have no value:\n\n* code (ElementDefinition.code) \n* type
(ElementDefinition.type) \n* required (ElementDefinition.min) \n* repeats (ElementDefinition.max) \n* maxLength
(ElementDefinition.maxLength) \n* answerValueSet (ElementDefinition.binding)\n* options
(ElementDefinition.binding).",
      "$ref": "#/definitions/uri"
    },
    "_definition": {
      "description": "Extensions for definition",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "code": {
      "description": "A terminology code that corresponds to this group or question (e.g. a code from LOINC, which
defines many questions and answers).",
      "items": {
        "$ref": "Coding.schema.json#/definitions/Coding"
      },
      "type": "array"
    },
    "prefix": {
      "description": "A short label for a particular group, question or set of display text within the questionnaire
 used for reference by the individual completing the questionnaire.",
      "$ref": "#/definitions/string"
    },
    "_prefix": {
      "description": "Extensions for prefix",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "text": {
      "description": "The name of a section, the text of a question or text content for a display item.",
      "$ref": "#/definitions/string"
    },
    "_text": {
      "description": "Extensions for text",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "type": {
      "description": "The type of questionnaire item this is - whether text for display, a grouping of other items
or a particular type of data to be captured (string, integer, coded choice, etc.).",
      "enum": [
        "group",
        "display",
        "boolean",
        "decimal",
        "integer",
        "date",
        "dateTime",
        "time",
        "string",
        "text",
        "url",
        "choice",
        "open-choice",
        "attachment",
        "reference",
        "quantity"
```

```
      ]
    },
    "_type": {
      "description": "Extensions for type",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "enableWhen": {
      "description": "A constraint indicating that this item should only be enabled (displayed/allow answers to be
captured) when the specified condition is true.",
      "items": {
        "$ref": "#/definitions/Questionnaire_EnableWhen"
      },
      "type": "array"
    },
    "enableBehavior": {
      "description": "Controls how multiple enableWhen values are interpreted -  whether all or any must be true.",
      "enum": [
        "all",
        "any"
      ]
    },
    "_enableBehavior": {
      "description": "Extensions for enableBehavior",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "required": {
      "description": "An indication, if true, that the item must be present in a \"completed\"
QuestionnaireResponse.  If false, the item may be skipped when answering the questionnaire.",
      "$ref": "#/definitions/boolean"
    },
    "_required": {
      "description": "Extensions for required",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "repeats": {
      "description": "An indication, if true, that the item may occur multiple times in the response, collecting
multiple answers for questions or multiple sets of answers for groups.",
      "$ref": "#/definitions/boolean"
    },
    "_repeats": {
      "description": "Extensions for repeats",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "readOnly": {
      "description": "An indication, when true, that the value cannot be changed by a human respondent to the
Questionnaire.",
      "$ref": "#/definitions/boolean"
    },
    "_readOnly": {
      "description": "Extensions for readOnly",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "maxLength": {
      "description": "The maximum number of characters that are permitted in the answer to be considered a \"valid\"
 QuestionnaireResponse.",
      "$ref": "#/definitions/integer"
    },
    "_maxLength": {
      "description": "Extensions for maxLength",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "answerValueSet": {
      "description": "A reference to a value set containing a list of codes representing permitted answers for a
\"choice\" or \"open-choice\" question.",
      "$ref": "canonical.schema.json#/definitions/canonical"
    },
    "answerOption": {
      "description": "One of the permitted answers for a \"choice\" or \"open-choice\" question.",
      "items": {
        "$ref": "#/definitions/Questionnaire_AnswerOption"
      },
      "type": "array"
    },
    "initial": {
      "description": "One or more values that should be pre-populated in the answer when initially rendering the
questionnaire for user input.",
      "items": {
        "$ref": "#/definitions/Questionnaire_Initial"
      },
      "type": "array"
    },
    "item": {
      "description": "Text, questions and other groups to be nested beneath a question or group.",
      "items": {
        "$ref": "#/definitions/Questionnaire_Item"
      },
      "type": "array"
    }
  },
```

```
  "additionalProperties": false
},
"Questionnaire_EnableWhen": {
  "description": "A structured set of questions intended to guide the collection of answers from end-users.
Questionnaires provide detailed control over order, presentation, phraseology and grouping to allow coherent,
consistent data collection.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "question": {
      "description": "The linkId for the question whose answer (or lack of answer) governs whether this item is
enabled.",
      "$ref": "#/definitions/string"
    },
    "_question": {
      "description": "Extensions for question",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "operator": {
      "description": "Specifies the criteria by which the question is enabled.",
      "enum": [
        "exists",
        "\u003d",
        "!\u003d",
        "\u003e",
        "\u003c",
        "\u003e\u003d",
        "\u003c\u003d"
      ]
    },
    "_operator": {
      "description": "Extensions for operator",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "answerBoolean": {
      "description": "A value that the referenced question is tested using the specified operator in order for the
item to be enabled.",
      "pattern": "^true|false$",
      "type": "boolean"
    },
    "_answerBoolean": {
      "description": "Extensions for answerBoolean",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "answerDecimal": {
      "description": "A value that the referenced question is tested using the specified operator in order for the
item to be enabled.",
      "pattern": "^-?(0|[1-9][0-9]*)(\\.[0-9]+)?([eE][+-]?[0-9]+)?$",
      "type": "number"
    },
    "_answerDecimal": {
      "description": "Extensions for answerDecimal",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "answerInteger": {
      "description": "A value that the referenced question is tested using the specified operator in order for the
item to be enabled.",
      "pattern": "^-?([0]|([1-9][0-9]*))$",
      "type": "number"
    },
    "_answerInteger": {
      "description": "Extensions for answerInteger",
```

```
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "answerDate": {
      "description": "A value that the referenced question is tested using the specified operator in order for the
item to be enabled.",
      "pattern":
"^([0-9]([0-9]([0-9][1-9]|[1-9]0)|[1-9]00)|[1-9]000)(-(0[1-9]|1[0-2])(-(0[1-9]|[1-2][0-9]|3[0-1]))?)?$",
      "type": "string"
    },
    "_answerDate": {
      "description": "Extensions for answerDate",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "answerDateTime": {
      "description": "A value that the referenced question is tested using the specified operator in order for the
item to be enabled.",
      "pattern": "^([0-9]([0-9]([0-9][1-9]|[1-9]0)|[1-9]00)|[1-9]000)(-(0[1-9]|1[0-2])(-(0[1-9]|[1-2][0-9]|3[0-
1])(T([01][0-9]|2[0-3]):[0-5][0-9]:([0-5][0-9]|60)(\\.[0-9]+)?(Z|(\\+|-)((0[0-9]|1[0-3]):[0-5][0-9]|14:00)))?)?)?$",
      "type": "string"
    },
    "_answerDateTime": {
      "description": "Extensions for answerDateTime",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "answerTime": {
      "description": "A value that the referenced question is tested using the specified operator in order for the
item to be enabled.",
      "pattern": "^([01][0-9]|2[0-3]):[0-5][0-9]:([0-5][0-9]|60)(\\.[0-9]+)?$",
      "type": "string"
    },
    "_answerTime": {
      "description": "Extensions for answerTime",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "answerString": {
      "description": "A value that the referenced question is tested using the specified operator in order for the
item to be enabled.",
      "pattern": "^[ \\r\\n\\t\\S]+$",
      "type": "string"
    },
    "_answerString": {
      "description": "Extensions for answerString",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "answerCoding": {
      "description": "A value that the referenced question is tested using the specified operator in order for the
item to be enabled.",
      "$ref": "Coding.schema.json#/definitions/Coding"
    },
    "answerQuantity": {
      "description": "A value that the referenced question is tested using the specified operator in order for the
item to be enabled.",
      "$ref": "Quantity.schema.json#/definitions/Quantity"
    },
    "answerReference": {
      "description": "A value that the referenced question is tested using the specified operator in order for the
item to be enabled.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    }
  },
  "additionalProperties": false
},
"Questionnaire_AnswerOption": {
  "description": "A structured set of questions intended to guide the collection of answers from end-users.
Questionnaires provide detailed control over order, presentation, phraseology and grouping to allow coherent,
consistent data collection.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
```

```
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "valueInteger": {
      "description": "A potential answer that\u0027s allowed as the answer to this question.",
      "pattern": "^-?([0]|([1-9][0-9]*))$",
      "type": "number"
    },
    "_valueInteger": {
      "description": "Extensions for valueInteger",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueDate": {
      "description": "A potential answer that\u0027s allowed as the answer to this question.",
      "pattern":
"^([0-9]([0-9]([0-9][1-9]|[1-9]0)|[1-9]00)|[1-9]000)(-(0[1-9]|1[0-2])(-(0[1-9]|[1-2][0-9]|3[0-1]))?)?$",
      "type": "string"
    },
    "_valueDate": {
      "description": "Extensions for valueDate",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueTime": {
      "description": "A potential answer that\u0027s allowed as the answer to this question.",
      "pattern": "^([01][0-9]|2[0-3]):[0-5][0-9]:([0-5][0-9]|60)(\\.[0-9]+)?$",
      "type": "string"
    },
    "_valueTime": {
      "description": "Extensions for valueTime",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueString": {
      "description": "A potential answer that\u0027s allowed as the answer to this question.",
      "pattern": "^[ \\r\\n\\t\\S]+$",
      "type": "string"
    },
    "_valueString": {
      "description": "Extensions for valueString",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueCoding": {
      "description": "A potential answer that\u0027s allowed as the answer to this question.",
      "$ref": "Coding.schema.json#/definitions/Coding"
    },
    "valueReference": {
      "description": "A potential answer that\u0027s allowed as the answer to this question.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "initialSelected": {
      "description": "Indicates whether the answer value is selected when the list of possible answers is initially
shown.",
      "$ref": "#/definitions/boolean"
    },
    "_initialSelected": {
      "description": "Extensions for initialSelected",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
},
"Questionnaire_Initial": {
  "description": "A structured set of questions intended to guide the collection of answers from end-users.

Questionnaires provide detailed control over order, presentation, phraseology and grouping to allow coherent,
consistent data collection.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
```

```
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "valueBoolean": {
      "description": "The actual value to for an initial answer.",
      "pattern": "^true|false$",
      "type": "boolean"
    },
    "_valueBoolean": {
      "description": "Extensions for valueBoolean",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueDecimal": {
      "description": "The actual value to for an initial answer.",
      "pattern": "^-?(0|[1-9][0-9]*)(\\.[0-9]+)?([eE][+-]?[0-9]+)?$",
      "type": "number"
    },
    "_valueDecimal": {
      "description": "Extensions for valueDecimal",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueInteger": {
      "description": "The actual value to for an initial answer.",
      "pattern": "^-?([0]|([1-9][0-9]*))$",
      "type": "number"
    },
    "_valueInteger": {
      "description": "Extensions for valueInteger",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueDate": {
      "description": "The actual value to for an initial answer.",
      "pattern":
"^([0-9]([0-9]([0-9][1-9]|[1-9]0)|[1-9]00)|[1-9]000)(-(0[1-9]|1[0-2])(-(0[1-9]|[1-2][0-9]|3[0-1]))?)?$",
      "type": "string"
    },
    "_valueDate": {
      "description": "Extensions for valueDate",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueDateTime": {
      "description": "The actual value to for an initial answer.",
      "pattern": "^([0-9]([0-9]([0-9][1-9]|[1-9]0)|[1-9]00)|[1-9]000)(-(0[1-9]|1[0-2])(-(0[1-9]|[1-2][0-9]|3[0-
1])(T([01][0-9]|2[0-3]):[0-5][0-9]:([0-5][0-9]|60)(\\.[0-9]+)?(Z|(\\+|-)((0[0-9]|1[0-3]):[0-5][0-9]|14:00)))?)?)?$",
      "type": "string"
    },
    "_valueDateTime": {
      "description": "Extensions for valueDateTime",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueTime": {
      "description": "The actual value to for an initial answer.",
      "pattern": "^([01][0-9]|2[0-3]):[0-5][0-9]:([0-5][0-9]|60)(\\.[0-9]+)?$",
      "type": "string"
    },
    "_valueTime": {
      "description": "Extensions for valueTime",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueString": {
      "description": "The actual value to for an initial answer.",
      "pattern": "^[ \\r\\n\\t\\S]+$",
      "type": "string"
    },
    "_valueString": {
      "description": "Extensions for valueString",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueUri": {
      "description": "The actual value to for an initial answer.",
      "pattern": "^\\S*$",
      "type": "string"
    },
    "_valueUri": {
      "description": "Extensions for valueUri",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueAttachment": {
      "description": "The actual value to for an initial answer.",
      "$ref": "Attachment.schema.json#/definitions/Attachment"
    },
    "valueCoding": {
      "description": "The actual value to for an initial answer.",
```

```
          "$ref": "Coding.schema.json#/definitions/Coding"
        },
        "valueQuantity": {
          "description": "The actual value to for an initial answer.",
          "$ref": "Quantity.schema.json#/definitions/Quantity"
        },
        "valueReference": {
          "description": "The actual value to for an initial answer.",
          "$ref": "Reference.schema.json#/definitions/Reference"
        }
      },
      "additionalProperties": false
    }
  }
}
```

2.37.15 Resource Questionnaire - Maturity Dependency View

URL: https://hl7.org/fhir/R4/questionnaire-dependencies.html

2.37.15 Resource Questionnaire - Maturity Dependency View

FHIR Infrastructure Work Group | Maturity Level: 3 | Trial Use | Compartments: Not linked to any defined compartments

Dependency Graph for Questionnaire FMM level 3

Name | Card. | Type | Dependency Analysis
Questionnaire | | DomainResource
url | 0..1 | uri
identifier | 0..* | Identifier
version | 0..1 | string
name | 0..1 | string
title | 0..1 | string
derivedFrom | 0..* | canonical(Questionnaire)
status | 1..1 | code
experimental | 0..1 | boolean
subjectType | 0..* | code
date | 0..1 | dateTime
publisher | 0..1 | string
contact | 0..* | ContactDetail
description | 0..1 | markdown
useContext | 0..* | UsageContext
jurisdiction | 0..* | CodeableConcept
purpose | 0..1 | markdown
copyright | 0..1 | markdown
approvalDate | 0..1 | date
lastReviewDate | 0..1 | date
effectivePeriod | 0..1 | Period
code | 0..* | Coding
item | 0..* | BackboneElement
linkId | 1..1 | string
definition | 0..1 | uri
code | 0..* | Coding
prefix | 0..1 | string
text | 0..1 | string
type | 1..1 | code
enableWhen | 0..* | BackboneElement
question | 1..1 | string
operator | 1..1 | code
answer[x] | 1..1
answerBoolean | | boolean
answerDecimal | | decimal
answerInteger | | integer
answerDate | | date
answerDateTime | | dateTime
answerTime | | time
answerString | | string
answerCoding | | Coding
answerQuantity | | Quantity
answerReference | | Reference(Any
enableBehavior | 0..1 | code
required | 0..1 | boolean
repeats | 0..1 | boolean
readOnly | 0..1 | boolean
maxLength | 0..1 | integer
answerValueSet | 0..1 | canonical(ValueSet)

answerOption | 0..* | BackboneElement
value[x] | 1..1
valueInteger | | integer
valueDate | | date
valueTime | | time
valueString | | string
valueCoding | | Coding
valueReference | | Reference(Any
initialSelected | 0..1 | boolean
initial | 0..* | BackboneElement
value[x] | 1..1
valueBoolean | | boolean
valueDecimal | | decimal
valueInteger | | integer
valueDate | | date
valueDateTime | | dateTime
valueTime | | time
valueString | | string
valueUri | | uri
valueAttachment | | Attachment
valueCoding | | Coding
valueQuantity | | Quantity
valueReference | | Reference(Any
item | 0..* | see item

2.38 Resource QuestionnaireResponse - Content

URL: https://hl7.org/fhir/R4/questionnaireresponse.html
FHIR Infrastructure Work Group | Maturity Level: 3 | Trial Use | Security Category: Patient | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

A structured set of questions and their answers. The questions are ordered and grouped into coherent subsets, corresponding to the structure of the grouping of the questionnaire being responded to.

2.38.1 Scope and Usage

QuestionnaireResponse provides a complete or partial list of answers to a set of questions filled when responding to a questionnaire. The questions may be included directly or by reference to a Questionnaire resource that defines the questions as well as the constraints on the allowed answers. In some cases, both formal rules for editing the questionnaire (via link to Questionnaire) as well as sufficient local information to allow rendering of the questionnaire may be provided.

Each time a questionnaire is completed for a different subject or at a different time, a distinct QuestionnaireResponse is generated, though it may be possible for a previously entered set of answers to be edited or updated.

Questionnaire responses cover the need to communicate data originating from forms used in medical history examinations, research questionnaires and sometimes full clinical specialty records. In many systems this data is collected using user-defined screens and forms. Questionnaire responses record specifics about data capture - exactly what questions were asked, in what order, what answers were given, etc. Each of these questions is part of the Questionnaire, and as such the Questionnaire is a separately identifiable Resource, whereas the individual questions are not.

Examples of Questionnaires include:

- Past medical history (PMH)
- Family diseases
- Social history
- Research questionnaires and Case report forms (CRFs)
- Quality and evaluation forms
- Patient intake form (e.g. clipboard)
- Insurance claim form

QuestionnaireResponse resources can be validated against their corresponding Questionnaire to verify that required groups and questions are answered and that answers fit constraints in terms of cardinality, data type, etc.

2.38.2 Boundaries and Relationships

The QuestionnaireResponse resource captures the responses to a questionnaire, while Questionnaire represents the definition of the questionnaire form, including what questions are asked, how they're organized and the constraints on the allowed answers.

While Observation, with its nested relatedObservation structure, can create complex hierarchies of questions and answers, the focus is different. First, Observation is used primarily for capturing data elements that are "true" observations - lab measurements, vital signs, social assessments, etc., while QuestionnaireResponse can be used to capture any types of data, including data that would typically map to other resources (Procedure, Patient, MedicationStatement, etc.). Second, the focus of QuestionnaireResponse includes the specific phrasing and organization of the questions. All data must be explicitly captured as a question. With Observation, the focus is only on the meaning of the answer, not what question was asked (assuming a question was even asked at all). Additional information such as normal ranges, interpretation, date, etc., may also be captured.

This resource is referenced by ClinicalImpression, Consent, Contract, DeviceUseStatement, FamilyMemberHistory and Observation

2.38.3 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
QuestionnaireResponse | TU | | DomainResource | A structured set of questions and their answers Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..1 | Identifier | Unique id for this set of answers
basedOn | Σ | 0..* | Reference(CarePlan | ServiceRequest) | Request fulfilled by this QuestionnaireResponse
partOf | Σ | 0..* | Reference(Observation | Procedure) | Part of this action
questionnaire | Σ | 0..1 | canonical(Questionnaire) | Form being answered
status | ?!Σ | 1..1 | code | in-progress | completed | amended | entered-in-error | stopped QuestionnaireResponseStatus (Required)
subject | Σ | 0..1 | Reference(Any) | The subject of the questions
encounter | Σ | 0..1 | Reference(Encounter) | Encounter created as part of
authored | Σ | 0..1 | dateTime | Date the answers were gathered
author | Σ | 0..1 | Reference(Device | Practitioner | PractitionerRole | Patient | RelatedPerson | Organization) | Person who received and recorded the answers
source | Σ | 0..1 | Reference(Patient | Practitioner | PractitionerRole | RelatedPerson) | The person who answered the questions
item | I | 0..* | BackboneElement | Groups and questions + Rule: Nested item can't be beneath both item and answer
linkId | | 1..1 | string | Pointer to specific item from Questionnaire
definition | | 0..1 | uri | ElementDefinition - details for the item
text | | 0..1 | string | Name for group or question text
answer | | 0..* | BackboneElement | The response(s) to the question
value[x] | | 0..1 | | Single-valued answer to the question Questionnaire Answer Codes (Example)
valueBoolean | | | boolean
valueDecimal | | | decimal
valueInteger | | | integer
valueDate | | | date
valueDateTime | | | dateTime
valueTime | | | time
valueString | | | string
valueUri | | | uri
valueAttachment | | | Attachment
valueCoding | | | Coding
valueQuantity | | | Quantity
valueReference | | | Reference(Any)
item | | 0..* | see item | Nested groups and questions
item | | 0..* | see item | Nested questionnaire response items
Documentation for this format

UML Diagram (Legend)

XML Template

```
<QuestionnaireResponse xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..1 Identifier Unique id for this set of answers --></identifier>
 <basedOn><!-- 0..* Reference(CarePlan|ServiceRequest) Request fulfilled by this QuestionnaireResponse --></basedOn>
 <partOf><!-- 0..* Reference(Observation|Procedure) Part of this action --></partOf>
 <questionnaire><!-- 0..1 canonical(Questionnaire) Form being answered --></questionnaire>
 <status value="[code]"/><!-- 1..1 in-progress | completed | amended | entered-in-error | stopped -->
 <subject><!-- 0..1 Reference(Any) The subject of the questions --></subject>
 <encounter><!-- 0..1 Reference(Encounter) Encounter created as part of --></encounter>
```

```
 <authored value="[dateTime]"/><!-- 0..1 Date the answers were gathered -->
 <author><!-- 0..1 Reference(Device|Practitioner|PractitionerRole|Patient|
   RelatedPerson|Organization) Person who received and recorded the answers --></author>
 <source><!-- 0..1 Reference(Patient|Practitioner|PractitionerRole|RelatedPerson) The person who answered the questions
    --></source>
 <item>  <!-- 0..* Groups and questions -->
  <linkId value="[string]"/><!-- 1..1 Pointer to specific item from Questionnaire -->
  <definition value="[uri]"/><!-- 0..1 ElementDefinition - details for the item -->
  <text value="[string]"/><!-- 0..1 Name for group or question text -->
  <answer>  <!-- 0..* The response(s) to the question -->
   <value[x]><!-- 0..1 boolean|decimal|integer|date|dateTime|time|string|uri|
     Attachment|Coding|Quantity|Reference(Any) Single-valued answer to the question --></value[x]>
   <item><!-- 0..* Content as for QuestionnaireResponse.item Nested groups and questions --></item>
  </answer>
  <item><!-- 0..* Content as for QuestionnaireResponse.item Nested questionnaire response items --></item>
 </item>
</QuestionnaireResponse>
```

JSON Template

```
{
  "resourceType" : "QuestionnaireResponse",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : { Identifier }, // Unique id for this set of answers
  "basedOn" : [{ Reference(CarePlan|ServiceRequest) }], // Request fulfilled by this QuestionnaireResponse
  "partOf" : [{ Reference(Observation|Procedure) }], // Part of this action
  "questionnaire" : { canonical(Questionnaire) }, // Form being answered
  "status" : "<code>", // R!  in-progress | completed | amended | entered-in-error | stopped
  "subject" : { Reference(Any) }, // The subject of the questions
  "encounter" : { Reference(Encounter) }, // Encounter created as part of
  "authored" : "<dateTime>", // Date the answers were gathered
  "author" : { Reference(Device|Practitioner|PractitionerRole|Patient|
   RelatedPerson|Organization) }, // Person who received and recorded the answers
  "source" : { Reference(Patient|Practitioner|PractitionerRole|RelatedPerson) }, // The person who answered the
    questions
  "item" : [{ // Groups and questions
    "linkId" : "<string>", // R!  Pointer to specific item from Questionnaire
    "definition" : "<uri>", // ElementDefinition - details for the item
    "text" : "<string>", // Name for group or question text
    "answer" : [{ // The response(s) to the question
      // value[x]: Single-valued answer to the question. One of these 12:
      "valueBoolean" : <boolean>,
      "valueDecimal" : <decimal>,
      "valueInteger" : <integer>,
      "valueDate" : "<date>",
      "valueDateTime" : "<dateTime>",
      "valueTime" : "<time>",
      "valueString" : "<string>",
      "valueUri" : "<uri>",
      "valueAttachment" : { Attachment },
      "valueCoding" : { Coding },
      "valueQuantity" : { Quantity },
      "valueReference" : { Reference(Any) },
      "item" : [{ Content as for QuestionnaireResponse.item }] // Nested groups and questions
    }],
    "item" : [{ Content as for QuestionnaireResponse.item }] // Nested questionnaire response items
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:QuestionnaireResponse;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:QuestionnaireResponse.identifier [ Identifier ]; # 0..1 Unique id for this set of answers
  fhir:QuestionnaireResponse.basedOn [ Reference(CarePlan|ServiceRequest) ], ... ; # 0..* Request fulfilled by this
    QuestionnaireResponse
  fhir:QuestionnaireResponse.partOf [ Reference(Observation|Procedure) ], ... ; # 0..* Part of this action
  fhir:QuestionnaireResponse.questionnaire [ canonical(Questionnaire) ]; # 0..1 Form being answered
  fhir:QuestionnaireResponse.status [ code ]; # 1..1 in-progress | completed | amended | entered-in-error | stopped
  fhir:QuestionnaireResponse.subject [ Reference(Any) ]; # 0..1 The subject of the questions
  fhir:QuestionnaireResponse.encounter [ Reference(Encounter) ]; # 0..1 Encounter created as part of
  fhir:QuestionnaireResponse.authored [ dateTime ]; # 0..1 Date the answers were gathered
  fhir:QuestionnaireResponse.author [ Reference(Device|Practitioner|PractitionerRole|Patient|RelatedPerson|Organization)
     ]; # 0..1 Person who received and recorded the answers
  fhir:QuestionnaireResponse.source [ Reference(Patient|Practitioner|PractitionerRole|RelatedPerson) ]; # 0..1 The
    person who answered the questions
  fhir:QuestionnaireResponse.item [ # 0..* Groups and questions
    fhir:QuestionnaireResponse.item.linkId [ string ]; # 1..1 Pointer to specific item from Questionnaire
    fhir:QuestionnaireResponse.item.definition [ uri ]; # 0..1 ElementDefinition - details for the item
    fhir:QuestionnaireResponse.item.text [ string ]; # 0..1 Name for group or question text
    fhir:QuestionnaireResponse.item.answer [ # 0..* The response(s) to the question
      # QuestionnaireResponse.item.answer.value[x] : 0..1 Single-valued answer to the question. One of these 12
        fhir:QuestionnaireResponse.item.answer.valueBoolean [ boolean ]
        fhir:QuestionnaireResponse.item.answer.valueDecimal [ decimal ]
```

```
        fhir:QuestionnaireResponse.item.answer.valueInteger [ integer ]
        fhir:QuestionnaireResponse.item.answer.valueDate [ date ]
        fhir:QuestionnaireResponse.item.answer.valueDateTime [ dateTime ]
        fhir:QuestionnaireResponse.item.answer.valueTime [ time ]
        fhir:QuestionnaireResponse.item.answer.valueString [ string ]
        fhir:QuestionnaireResponse.item.answer.valueUri [ uri ]
        fhir:QuestionnaireResponse.item.answer.valueAttachment [ Attachment ]
        fhir:QuestionnaireResponse.item.answer.valueCoding [ Coding ]
        fhir:QuestionnaireResponse.item.answer.valueQuantity [ Quantity ]
        fhir:QuestionnaireResponse.item.answer.valueReference [ Reference(Any) ]
      fhir:QuestionnaireResponse.item.answer.item [ See QuestionnaireResponse.item ], ... ; # 0..* Nested groups and
    questions
    ], ...;
    fhir:QuestionnaireResponse.item.item [ See QuestionnaireResponse.item ], ... ; # 0..* Nested questionnaire response
    items
  ], ...;
]
```

Changes since R3

QuestionnaireResponse

QuestionnaireResponse.basedOn | Type Reference: Added Target Type ServiceRequest Type Reference: Removed Target Types ReferralRequest, ProcedureRequest

QuestionnaireResponse.partOf | Renamed from parent to partOf

QuestionnaireResponse.questionnaire | Type changed from Reference(Questionnaire) to canonical(Questionnaire)

QuestionnaireResponse.status | Change value set from http://hl7.org/fhir/ValueSet/questionnaire-answers-status to http://hl7.org/fhir/ValueSet/questionnaire-answers-status|4.0.1

QuestionnaireResponse.encounter | Added Element

QuestionnaireResponse.author | Type Reference: Added Target Types PractitionerRole, Organization

QuestionnaireResponse.source | Type Reference: Added Target Type PractitionerRole

QuestionnaireResponse.context | deleted

QuestionnaireResponse.item.subject | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 5 tests that all execute ok. 1 fail round-trip testing and 5 r3 resources are invalid (0 errors).)

Structure

Name | Flags | Card. | Type | Description & Constraints

QuestionnaireResponse | TU | | DomainResource | A structured set of questions and their answers Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | Σ | 0..1 | Identifier | Unique id for this set of answers

basedOn | Σ | 0..* | Reference(CarePlan | ServiceRequest) | Request fulfilled by this QuestionnaireResponse

partOf | Σ | 0..* | Reference(Observation | Procedure) | Part of this action

questionnaire | Σ | 0..1 | canonical(Questionnaire) | Form being answered

status | ?!Σ | 1..1 | code | in-progress | completed | amended | entered-in-error | stopped QuestionnaireResponseStatus (Required)

subject | Σ | 0..1 | Reference(Any) | The subject of the questions

encounter | Σ | 0..1 | Reference(Encounter) | Encounter created as part of

authored | Σ | 0..1 | dateTime | Date the answers were gathered

author | Σ | 0..1 | Reference(Device | Practitioner | PractitionerRole | Patient | RelatedPerson | Organization) | Person who received and recorded the answers

source | Σ | 0..1 | Reference(Patient | Practitioner | PractitionerRole | RelatedPerson) | The person who answered the questions

item | I | 0..* | BackboneElement | Groups and questions + Rule: Nested item can't be beneath both item and answer

linkId | | 1..1 | string | Pointer to specific item from Questionnaire

definition | | 0..1 | uri | ElementDefinition - details for the item

text | | 0..1 | string | Name for group or question text

answer | | 0..* | BackboneElement | The response(s) to the question

value[x] | | 0..1 | | Single-valued answer to the question Questionnaire Answer Codes (Example)

valueBoolean | | | boolean

valueDecimal | | | decimal

valueInteger | | | integer

valueDate | | | date

valueDateTime | | | dateTime

valueTime | | | time

valueString | | | string

valueUri | | | uri

valueAttachment | | | Attachment

valueCoding | | | Coding

valueQuantity | | | Quantity

valueReference | | | Reference(Any)

item | | 0..* | see item | Nested groups and questions

item | | 0..* | see item | Nested questionnaire response items

Documentation for this format

UML Diagram (Legend)

XML Template

```
<QuestionnaireResponse xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..1 Identifier Unique id for this set of answers --></identifier>
 <basedOn><!-- 0..* Reference(CarePlan|ServiceRequest) Request fulfilled by this QuestionnaireResponse --></basedOn>
 <partOf><!-- 0..* Reference(Observation|Procedure) Part of this action --></partOf>
 <questionnaire><!-- 0..1 canonical(Questionnaire) Form being answered --></questionnaire>
 <status value="[code]"/><!-- 1..1 in-progress | completed | amended | entered-in-error | stopped -->
 <subject><!-- 0..1 Reference(Any) The subject of the questions --></subject>
 <encounter><!-- 0..1 Reference(Encounter) Encounter created as part of --></encounter>
 <authored value="[dateTime]"/><!-- 0..1 Date the answers were gathered -->
 <author><!-- 0..1 Reference(Device|Practitioner|PractitionerRole|Patient|
   RelatedPerson|Organization) Person who received and recorded the answers --></author>
 <source><!-- 0..1 Reference(Patient|Practitioner|PractitionerRole|RelatedPerson) The person who answered the questions
    --></source>
 <item>  <!-- 0..* Groups and questions -->
  <linkId value="[string]"/><!-- 1..1 Pointer to specific item from Questionnaire -->
  <definition value="[uri]"/><!-- 0..1 ElementDefinition - details for the item -->
  <text value="[string]"/><!-- 0..1 Name for group or question text -->
  <answer>  <!-- 0..* The response(s) to the question -->
   <value[x]><!-- 0..1 boolean|decimal|integer|date|dateTime|time|string|uri|
     Attachment|Coding|Quantity|Reference(Any) Single-valued answer to the question --></value[x]>
   <item><!-- 0..* Content as for QuestionnaireResponse.item Nested groups and questions --></item>
  </answer>
  <item><!-- 0..* Content as for QuestionnaireResponse.item Nested questionnaire response items --></item>
 </item>
</QuestionnaireResponse>
```

JSON Template

```
{
  "resourceType" : "QuestionnaireResponse",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : { Identifier }, // Unique id for this set of answers
  "basedOn" : [{ Reference(CarePlan|ServiceRequest) }], // Request fulfilled by this QuestionnaireResponse
  "partOf" : [{ Reference(Observation|Procedure) }], // Part of this action
  "questionnaire" : { canonical(Questionnaire) }, // Form being answered
  "status" : "<code>", // R!  in-progress | completed | amended | entered-in-error | stopped
  "subject" : { Reference(Any) }, // The subject of the questions
  "encounter" : { Reference(Encounter) }, // Encounter created as part of
  "authored" : "<dateTime>", // Date the answers were gathered
  "author" : { Reference(Device|Practitioner|PractitionerRole|Patient|
   RelatedPerson|Organization) }, // Person who received and recorded the answers
  "source" : { Reference(Patient|Practitioner|PractitionerRole|RelatedPerson) }, // The person who answered the
    questions
  "item" : [{ // Groups and questions
    "linkId" : "<string>", // R!  Pointer to specific item from Questionnaire
    "definition" : "<uri>", // ElementDefinition - details for the item
    "text" : "<string>", // Name for group or question text
    "answer" : [{ // The response(s) to the question
      // value[x]: Single-valued answer to the question. One of these 12:
      "valueBoolean" : <boolean>,
      "valueDecimal" : <decimal>,
      "valueInteger" : <integer>,
      "valueDate" : "<date>",
      "valueDateTime" : "<dateTime>",
      "valueTime" : "<time>",
      "valueString" : "<string>",
      "valueUri" : "<uri>",
      "valueAttachment" : { Attachment },
      "valueCoding" : { Coding },
      "valueQuantity" : { Quantity },
      "valueReference" : { Reference(Any) },
      "item" : [{ Content as for QuestionnaireResponse.item }] // Nested groups and questions
    }],
    "item" : [{ Content as for QuestionnaireResponse.item }] // Nested questionnaire response items
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:QuestionnaireResponse;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:QuestionnaireResponse.identifier [ Identifier ]; # 0..1 Unique id for this set of answers
  fhir:QuestionnaireResponse.basedOn [ Reference(CarePlan|ServiceRequest) ], ... ; # 0..* Request fulfilled by this
    QuestionnaireResponse
  fhir:QuestionnaireResponse.partOf [ Reference(Observation|Procedure) ], ... ; # 0..* Part of this action
  fhir:QuestionnaireResponse.questionnaire [ canonical(Questionnaire) ]; # 0..1 Form being answered
  fhir:QuestionnaireResponse.status [ code ]; # 1..1 in-progress | completed | amended | entered-in-error | stopped
  fhir:QuestionnaireResponse.subject [ Reference(Any) ]; # 0..1 The subject of the questions
  fhir:QuestionnaireResponse.encounter [ Reference(Encounter) ]; # 0..1 Encounter created as part of
  fhir:QuestionnaireResponse.authored [ dateTime ]; # 0..1 Date the answers were gathered
  fhir:QuestionnaireResponse.author [ Reference(Device|Practitioner|PractitionerRole|Patient|RelatedPerson|Organization)
     ]; # 0..1 Person who received and recorded the answers
```

```
  fhir:QuestionnaireResponse.source [ Reference(Patient|Practitioner|PractitionerRole|RelatedPerson) ]; # 0..1 The
    person who answered the questions
  fhir:QuestionnaireResponse.item [ # 0..* Groups and questions
    fhir:QuestionnaireResponse.item.linkId [ string ]; # 1..1 Pointer to specific item from Questionnaire
    fhir:QuestionnaireResponse.item.definition [ uri ]; # 0..1 ElementDefinition - details for the item
    fhir:QuestionnaireResponse.item.text [ string ]; # 0..1 Name for group or question text
    fhir:QuestionnaireResponse.item.answer [ # 0..* The response(s) to the question
      # QuestionnaireResponse.item.answer.value[x] : 0..1 Single-valued answer to the question. One of these 12
        fhir:QuestionnaireResponse.item.answer.valueBoolean [ boolean ]
        fhir:QuestionnaireResponse.item.answer.valueDecimal [ decimal ]
        fhir:QuestionnaireResponse.item.answer.valueInteger [ integer ]
        fhir:QuestionnaireResponse.item.answer.valueDate [ date ]
        fhir:QuestionnaireResponse.item.answer.valueDateTime [ dateTime ]
        fhir:QuestionnaireResponse.item.answer.valueTime [ time ]
        fhir:QuestionnaireResponse.item.answer.valueString [ string ]
        fhir:QuestionnaireResponse.item.answer.valueUri [ uri ]
        fhir:QuestionnaireResponse.item.answer.valueAttachment [ Attachment ]
        fhir:QuestionnaireResponse.item.answer.valueCoding [ Coding ]
        fhir:QuestionnaireResponse.item.answer.valueQuantity [ Quantity ]
        fhir:QuestionnaireResponse.item.answer.valueReference [ Reference(Any) ]
      fhir:QuestionnaireResponse.item.answer.item [ See QuestionnaireResponse.item ], ... ; # 0..* Nested groups and
    questions
    ], ...;
    fhir:QuestionnaireResponse.item.item [ See QuestionnaireResponse.item ], ... ; # 0..* Nested questionnaire response
    items
  ], ...;
]
```

Changes since Release 3

QuestionnaireResponse
QuestionnaireResponse.basedOn | Type Reference: Added Target Type ServiceRequest Type Reference: Removed Target Types ReferralRequest, ProcedureRequest
QuestionnaireResponse.partOf | Renamed from parent to partOf
QuestionnaireResponse.questionnaire | Type changed from Reference(Questionnaire) to canonical(Questionnaire)
QuestionnaireResponse.status | Change value set from http://hl7.org/fhir/ValueSet/questionnaire-answers-status to http://hl7.org/fhir/ValueSet/questionnaire-answers-status|4.0.1
QuestionnaireResponse.encounter | Added Element
QuestionnaireResponse.author | Type Reference: Added Target Types PractitionerRole, Organization
QuestionnaireResponse.source | Type Reference: Added Target Type PractitionerRole
QuestionnaireResponse.context | deleted
QuestionnaireResponse.item.subject | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 5 tests that all execute ok. 1 fail round-trip testing and 5 r3 resources are invalid (0 errors).)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

2.38.3.1 Terminology Bindings

Path | Definition | Type | Reference
QuestionnaireResponse.status | Lifecycle status of the questionnaire response. | Required | QuestionnaireResponseStatus
QuestionnaireResponse.item.answer.value[x] | Code indicating the response provided for a question. | Example | QuestionnaireAnswerCodes

2.38.3.2 Constraints

id | Level | Location | Description | Expression
qrs-1 | Rule | QuestionnaireResponse.item | Nested item can't be beneath both item and answer | (answer.exists() and item.exists()).not()

2.38.4 Notes

• QuestionnaireResponses can be authored by clinicians, patients, or patients' relatives (or even owners in the case of animals). Clinicians may author questionnaire responses, where the answers are provided by others on behalf of the patient. Additionally, information gathered on behalf of a patient may be about the patient's relatives (e.g. in family anamnesis). Therefore, QuestionnaireResponse makes a distinction between the author, the subject and the source of information.

• The QuestionnaireResponse's encounter element can be used to link to the encounter when the questionnaire response was authored. This can be relevant since the encounter gives context to the answers and can be used to relate information in the QuestionnaireResponse to orders and observations that were done during the same Encounter.

• The QuestionnaireResponse's language element can be used to indicte the language in which the patient completed the survey/questionnaire

• The order of questions within groups, groups within groups and groups within questions is relevant and must be retained for display and capture. The hierarchy items within the Questionnaire SHALL mirror the hierarchy

of the corresponding Questionnaire (if any).

• For use of QuestionnaireResponse.item.definition, see notes on Questionnaire

• Because of the lack of explicit support for Questionnaires in HL7 v3 , CDA Documents frequently used named sections with Observations to model questionnaires. Such use cases should now utilize the QuestionnaireResponse resource instead.

Refer to additional guidance provided in the Questionnaire resource dealing with the design of questionnaires.

2.38.4.1 Links to Questionnaire

A QuestionnaireResponse may be stand-alone or may point to the definition of the questions in Questionnaire. If the QuestionnaireResponse refers to a Questionnaire:

• The QuestionnaireResponse structure must be consistent with the Questionnaire (i.e. questions must be organized into the same groups, nested questions must still be nested, etc.

• The linkage between the questions and groups in the two resources is established using the linkId element

• When a QuestionnaireResponse references a Questionnaire, all the items in the questionnaire SHOULD be included in the QuestionnaireResponse if they are relevant to the interpretation of the answers, including items of type = display. This might also include items that were >not enabled when the Questionnaire was answered, though of course these will not have answers

2.38.4.2 Security

QuestionnaireResponse resources can have answers with values of type Attachment. These attachments will typically be selected by the user answering the questionnaire and this selection may be done in an uncontrolled environment. Systems should ensure that the attachment is of the desired type and should take precautions before rendering or executing any attached content.

2.38.4.3 Access Control

For most resources, the type of information that can be conveyed in the resource is determined by the resource, and the key attributes that determine the sensitivity level of the information are also known; e.g., drug, observation type, clinical trial randomization status, etc. However, for QuestionnaireResponse, the sensitivity of an instance is dependent on what type of Questionnaire it is associated with. And the data elements that determine that sensitivity could be the answers to any of the questions. This makes automatically enforcing access control rules more challenging. Designers should take these challenges into account and may need to place stricter access controls around QuestionnaireResponse to ensure that access to information is not granted improperly.

2.38.5 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
author | reference | The author of the questionnaire response | QuestionnaireResponse.author (Practitioner, Organization, Device, Patient, PractitionerRole, RelatedPerson)
authored | date | When the questionnaire response was last changed | QuestionnaireResponse.authored
based-on | reference | Plan/proposal/order fulfilled by this questionnaire response | QuestionnaireResponse.basedOn (CarePlan, ServiceRequest)
encounter | reference | Encounter associated with the questionnaire response | QuestionnaireResponse.encounter (Encounter)
identifier | token | The unique identifier for the questionnaire response | QuestionnaireResponse.identifier
part-of | reference | Procedure or observation this questionnaire response was performed as a part of | QuestionnaireResponse.partOf (Observation, Procedure)
patient | reference | The patient that is the subject of the questionnaire response | QuestionnaireResponse.subject.where(resolve() is Patient) (Patient)
questionnaire | reference | The questionnaire the answers are provided for | QuestionnaireResponse.questionnaire (Questionnaire)
source | reference | The individual providing the information reflected in the questionnaire respose | QuestionnaireResponse.source (Practitioner, Patient, PractitionerRole, RelatedPerson)
status | token | The status of the questionnaire response | QuestionnaireResponse.status
subject | reference | The subject of the questionnaire response | QuestionnaireResponse.subject (Any)

2.38.6 Resource QuestionnaireResponse - Examples

URL: https://hl7.org/fhir/R4/questionnaireresponse-examples.html

2.38.6 Resource QuestionnaireResponse - Examples

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

Example Name | id | Format
General questionnaire response example | 3141 | XML | JSON | Turtle
Real-world lifelines questionnaire response (fictively taken from the patient) | f201 | XML | JSON | Turtle
Real-world NSW My Personal Health Record example | bb | XML | JSON | Turtle

Glasgow Coma Score example answers | gcs | XML | JSON | Turtle
Example response to the SDC-LOINC USSG Family History questionnaire | ussg-fht-answers | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

2.38.7 Resource QuestionnaireResponse - Detailed Descriptions

URL: https://hl7.org/fhir/R4/questionnaireresponse-definitions.html

2.38.7 Resource QuestionnaireResponse - Detailed Descriptions

FHIR Infrastructure Work Group | Maturity Level: 3 | Trial Use | Security Category: Patient | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

Detailed Descriptions for the elements in the QuestionnaireResponse resource.

QuestionnaireResponse
Element Id | QuestionnaireResponse
Definition | A structured set of questions and their answers. The questions are ordered and grouped into coherent subsets, corresponding to the structure of the grouping of the questionnaire being responded to.
Cardinality | 0..*
Type | DomainResource
Requirements | To support structured, hierarchical reporting of data gathered using digital forms and other questionnaires.
Alternate Names | Form; QuestionnaireAnswers
Comments | The QuestionnaireResponse contains enough information about the questions asked and their organization that it can be interpreted somewhat independently from the Questionnaire it is based on. I.e. You don't need access to the Questionnaire in order to extract basic information from a QuestionnaireResponse.
QuestionnaireResponse.identifier
Element Id | QuestionnaireResponse.identifier
Definition | A business identifier assigned to a particular completed (or partially completed) questionnaire.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..1
Type | Identifier
Requirements | Used for tracking, registration and other business purposes.
Summary | true
QuestionnaireResponse.basedOn
Element Id | QuestionnaireResponse.basedOn
Definition | The order, proposal or plan that is fulfilled in whole or in part by this QuestionnaireResponse. For example, a ServiceRequest seeking an intake assessment or a decision support recommendation to assess for post-partum depression.
Cardinality | 0..*
Type | Reference(CarePlan | ServiceRequest)
Requirements | Supports traceability of responsibility for the action and allows linkage of an action to the recommendations acted upon.
Alternate Names | order
Summary | true
QuestionnaireResponse.partOf
Element Id | QuestionnaireResponse.partOf
Definition | A procedure or observation that this questionnaire was performed as part of the execution of. For example, the surgery a checklist was executed as part of.
Cardinality | 0..*
Type | Reference(Observation | Procedure)
Summary | true
Comments | Composition of questionnaire responses will be handled by the parent questionnaire having answers that reference the child questionnaire. For relationships to referrals, and other types of requests, use basedOn.
QuestionnaireResponse.questionnaire
Element Id | QuestionnaireResponse.questionnaire
Definition | The Questionnaire that defines and organizes the questions for which answers are being provided.
Cardinality | 0..1
Type | canonical(Questionnaire)
Requirements | Needed to allow editing of the questionnaire response in a manner that enforces the constraints of the original form.
Alternate Names | Form
Summary | true
Comments | If a QuestionnaireResponse references a Questionnaire, then the QuestionnaireResponse structure must be consistent with the Questionnaire (i.e. questions must be organized into the same groups, nested questions must still be nested, etc.).
QuestionnaireResponse.status
Element Id | QuestionnaireResponse.status
Definition | The position of the questionnaire response within its overall lifecycle.
Cardinality | 1..1
Terminology Binding | QuestionnaireResponseStatus (Required)
Type | code

Is Modifier | true (Reason: This element is labelled as a modifier because it is a status element that contains status entered-in-error which means that the resource should not be treated as valid)
Requirements | The information on Questionnaire resources may possibly be gathered during multiple sessions and altered after considered being finished.
Summary | true
Comments | This element is labeled as a modifier because the status contains codes that mark the resource as not currently valid.
QuestionnaireResponse.subject
Element Id | QuestionnaireResponse.subject
Definition | The subject of the questionnaire response. This could be a patient, organization, practitioner, device, etc. This is who/what the answers apply to, but is not necessarily the source of information.
Cardinality | 0..1
Type | Reference(Any)
Requirements | Allows linking the answers to the individual the answers describe. May also affect access control.
Alternate Names | Patient; Focus
Summary | true
Comments | If the Questionnaire declared a subjectType, the resource pointed to by this element must be an instance of one of the listed types.
QuestionnaireResponse.encounter
Element Id | QuestionnaireResponse.encounter
Definition | The Encounter during which this questionnaire response was created or to which the creation of this record is tightly associated.
Cardinality | 0..1
Type | Reference(Encounter)
Requirements | Provides context for the information that was captured. May also affect access control.
Summary | true
Comments | This will typically be the encounter the event occurred within, but some activities may be initiated prior to or after the official completion of an encounter but still be tied to the context of the encounter. A questionnaire that was initiated during an encounter but not fully completed during the encounter would still generally be associated with the encounter.
QuestionnaireResponse.authored
Element Id | QuestionnaireResponse.authored
Definition | The date and/or time that this set of answers were last changed.
Cardinality | 0..1
Type | dateTime
Requirements | Clinicians need to be able to check the date that the information in the questionnaire was collected, to derive the context of the answers.
Alternate Names | Date Created; Date published; Date Issued; Date updated
Summary | true
Comments | May be different from the lastUpdateTime of the resource itself, because that reflects when the data was known to the server, not when the data was captured. This element is optional to allow for systems that might not know the value, however it SHOULD be populated if possible.
QuestionnaireResponse.author
Element Id | QuestionnaireResponse.author
Definition | Person who received the answers to the questions in the QuestionnaireResponse and recorded them in the system.
Cardinality | 0..1
Type | Reference(Device | Practitioner | PractitionerRole | Patient | RelatedPerson | Organization)
Requirements | Need to know who interpreted the subject's answers to the questions in the questionnaire, and selected the appropriate options for answers.
Alternate Names | Laboratory; Service; Practitioner; Department; Company; Performer
Summary | true
Comments | Mapping a subject's answers to multiple choice options and determining what to put in the textual answer is a matter of interpretation. Authoring by device would indicate that some portion of the questionnaire had been auto-populated.
QuestionnaireResponse.source
Element Id | QuestionnaireResponse.source
Definition | The person who answered the questions about the subject.
Cardinality | 0..1
Type | Reference(Patient | Practitioner | PractitionerRole | RelatedPerson)
Requirements | When answering questions about a subject that is minor, incapable of answering or an animal, another human source may answer the questions.
Summary | true
Comments | If not specified, no inference can be made about who provided the data.
QuestionnaireResponse.item
Element Id | QuestionnaireResponse.item
Definition | A group or question item from the original questionnaire for which answers are provided.
Cardinality | 0..*
Comments | Groups cannot have answers and therefore must nest directly within item. When dealing with questions, nesting must occur within each answer because some questions may have multiple answers (and the nesting occurs for each answer).
Invariants | Defined on this element qrs-1 Rule Nested item can't be beneath both item and answer (answer.exists() and item.exists()).not()
Defined on this element
qrs-1 | Rule | Nested item can't be beneath both item and answer | (answer.exists() and item.exists()).not()
QuestionnaireResponse.item.linkId
Element Id | QuestionnaireResponse.item.linkId
Definition | The item from the Questionnaire that corresponds to this item in the QuestionnaireResponse resource.
Cardinality | 1..1

Type | string
Requirements | Items can repeat in the answers, so a direct 1..1 correspondence by position might not exist - requiring correspondence by identifier.
QuestionnaireResponse.item.definition
Element Id | QuestionnaireResponse.item.definition
Definition | A reference to an ElementDefinition that provides the details for the item.
Cardinality | 0..1
Type | uri
Requirements | A common pattern is to define a set of data elements, and then build multiple different questionnaires for different circumstances to gather the data. This element provides traceability to the common definition.
Comments | The ElementDefinition must be in a StructureDefinition, and must have a fragment identifier that identifies the specific data element by its id (Element.id). E.g. http://hl7.org/fhir/StructureDefinition/Observation#Observation.value[x]. There is no need for this element if the item pointed to by the linkId has a definition listed.
QuestionnaireResponse.item.text
Element Id | QuestionnaireResponse.item.text
Definition | Text that is displayed above the contents of the group or as the text of the question being answered.
Cardinality | 0..1
Type | string
Requirements | Allows the questionnaire response to be read without access to the questionnaire.
QuestionnaireResponse.item.answer
Element Id | QuestionnaireResponse.item.answer
Definition | The respondent's answer(s) to the question.
Cardinality | 0..*
Comments | The value is nested because we cannot have a repeating structure that has variable type.
QuestionnaireResponse.item.answer.value[x]
Element Id | QuestionnaireResponse.item.answer.value[x]
Definition | The answer (or one of the answers) provided by the respondent to the question.
Cardinality | 0..1
Terminology Binding | Questionnaire Answer Codes (Example)
Type | boolean|decimal|integer|date|dateTime|time|string|uri|Attachment|Coding|Quantity|Reference(Any)
[x] Note | See Choice of Data Types for further information about how to use [x]
Requirements | Ability to retain a single-valued answer to a question.
Comments | More complex structures (Attachment, Resource and Quantity) will typically be limited to electronic forms that can expose an appropriate user interface to capture the components and enforce the constraints of a complex data type. Additional complex types can be introduced through extensions. Must match the datatype specified by Questionnaire.item.type in the corresponding Questionnaire.
QuestionnaireResponse.item.answer.item
Element Id | QuestionnaireResponse.item.answer.item
Definition | Nested groups and/or questions found within this particular answer.
Cardinality | 0..*
Type | See QuestionnaireResponse.item
Requirements | It is useful to have "sub-questions", questions which normally appear when certain answers are given and which collect additional details.
QuestionnaireResponse.item.item
Element Id | QuestionnaireResponse.item.item
Definition | Questions or sub-groups nested beneath a question or group.
Cardinality | 0..*
Type | See QuestionnaireResponse.item
Requirements | Reports can consist of complex nested groups.

2.38.8 Resource QuestionnaireResponse - Mappings

URL: https://hl7.org/fhir/R4/questionnaireresponse-mappings.html

2.38.8 Resource QuestionnaireResponse - Mappings

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

RIM Mapping

Mappings for the questionnaireresponse resource (see Mappings to Other Standards for further information & status).

2.38.8.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

QuestionnaireResponse
identifier | FiveWs.identifier
status | FiveWs.status
subject | FiveWs.subject[x]
encounter | FiveWs.context

authored | FiveWs.recorded
author | FiveWs.author
source | FiveWs.source

2.38.8.2 Workflow Pattern (http://hl7.org/fhir/workflow)

QuestionnaireResponse | Event
identifier | Event.identifier
basedOn | Event.basedOn
partOf | Event.partOf
questionnaire | Event.instantiates
status | Event.status
subject | Event.subject
encounter | Event.encounter
authored | Event.occurrence[x]
author | Event.performer

2.38.8.3 RIM Mapping (http://hl7.org/v3)

QuestionnaireResponse | Observation[moodCode=EVN]
identifier | .id
basedOn | .outboundRelationship[typeCode=FLFS].target
partOf | .inboundRelationship[typeCode=COMP].source[moodCode=EVN]
questionnaire | ./outboundRelationship[typeCode=INST]/target[classCode=OBS, moodCode=DEFN]
status | .statusCode (also whether there's a revisionControlAct - and possibly mood to distinguish "in-progress" from "published)
subject | .participation[typeCode=SBJ].role
encounter | .inboundRelationship(typeCode=COMP].source[classCode<=PCPR, moodCode=EVN]
authored | .participation[typeCode=AUT].time
author | .participation[typeCode=AUT].role
source | .participation[typeCode=INF].role
item | .outboundRelationship[typeCode=COMP].target[classCode=OBS, moodCode=EVN]
linkId | .outboundRelationship[typeCode=DEFN].target[classCode=OBS, moodCode=DEFN].id
definition | .outboundRelationship[typeCode=DEFN].target[classCode=OBS, moodCode=DEFN].code
text | .text
answer | .value[type=LIST_ANY]
value[x] | .item
item | .outboundRelationship[typeCode=COMP].target[classCode=OBS, moodCode=EVN]
item | .outboundRelationship[typeCode=COMP].target[classCode=OBS, moodCode=EVN]

2.38.9 Resource QuestionnaireResponse - Extensions & Profiles

URL: https://hl7.org/fhir/R4/questionnaireresponse-profiles.html

2.38.9 Resource QuestionnaireResponse - Extensions & Profiles

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

This table lists profiles and extensions for the QuestionnaireResponse resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

2.38.9.1 Profiles

No Profiles defined for this resource

2.38.9.2 Extensions

questionnaireresponse-author | QuestionnaireResponse.item, QuestionnaireResponse.item.item | author | for Core extensions for QuestionnaireResponse
questionnaireresponse-completionMode | QuestionnaireResponse | completionMode | for Core extensions for QuestionnaireResponse
questionnaireresponse-reason | QuestionnaireResponse | reason | for Core extensions for QuestionnaireResponse
questionnaireresponse-reviewer | QuestionnaireResponse | reviewer | for Core extensions for QuestionnaireResponse
questionnaireresponse-signature | QuestionnaireResponse, QuestionnaireResponse.item, QuestionnaireResponse.item.item | signature | for Core extensions for QuestionnaireResponse
workflow-episodeOfCare | QuestionnaireResponse | episodeOfCare | for Workflow Pattern HL7 Extensions
workflow-researchStudy | QuestionnaireResponse | researchStudy | for Workflow Pattern HL7 Extensions
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions

event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

2.38.9.3 Search Extensions

item-subject | reference | Allows searching for QuestionnaireResponses by item value where the item has isSubject=true
Expression: QuestionnaireResponse.item.where(hasExtension('http://hl7.org/fhir/StructureDefinition/questionnaireresponse-isSubject')).answer.value.ofType(Reference)

2.38.10 Resource QuestionnaireResponse - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/questionnaireresponse-version-maps.html
No validation errors - all conversions are clean

2.38.10 Resource QuestionnaireResponse - R3/R4 Conversions

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

R3 : R4 Conversion maps for QuestionnaireResponse.

Functional status for this map: 5 tests that all execute ok. 1 fail round-trip testing and 5 r3 resources are invalid (0 errors). (see documentation)

2.38.10.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/QuestionnaireResponse3to4" = "R3 to R4 Conversions for QuestionnaireResponse"

uses "http://hl7.org/fhir/3.0/StructureDefinition/QuestionnaireResponse" alias QuestionnaireResponseR3 as source
uses "http://hl7.org/fhir/StructureDefinition/QuestionnaireResponse" alias QuestionnaireResponse as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group QuestionnaireResponse(source src : QuestionnaireResponseR3, target tgt : QuestionnaireResponse) extends
    DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.basedOn -> tgt.basedOn;
  src.parent -> tgt.partOf;
  src.questionnaire -> tgt.questionnaire;
  src.status -> tgt.status;
  src.subject -> tgt.subject;
  src.context -> tgt.encounter;
  src.authored -> tgt.authored;
  src.author -> tgt.author;
  src.source -> tgt.source;
  src.item as vs0 -> tgt.item as vt0 then item(vs0, vt0);
}

group item(source src, target tgt) extends BackboneElement {
  src.linkId -> tgt.linkId;
  src.definition -> tgt.definition;
  src.text -> tgt.text;
  src.subject as v ->  tgt.extension as vt,  vt.url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    QuestionnaireResponse.item.subject',  vt.value = v;
  src.answer as vs0 -> tgt.answer as vt0 then answer(vs0, vt0);
  src.item as vs0 -> tgt.item as vt0 then item(vs0, vt0);
}

group answer(source src, target tgt) extends BackboneElement {
  src.value -> tgt.value;
  src.item as vs0 -> tgt.item as vt0 then item(vs0, vt0);
}
```

2.38.10.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/QuestionnaireResponse4to3" = "R4 to R3 Conversion for QuestionnaireResponse"

uses "http://hl7.org/fhir/StructureDefinition/QuestionnaireResponse" alias QuestionnaireResponse as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/QuestionnaireResponse" alias QuestionnaireResponseR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group QuestionnaireResponse(source src : QuestionnaireResponse, target tgt : QuestionnaireResponseR3) extends
    DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.basedOn -> tgt.basedOn;
  src.partOf -> tgt.parent;
  src.questionnaire -> tgt.questionnaire;
  src.status -> tgt.status;
  src.subject -> tgt.subject;
  src.encounter -> tgt.context;
```

```
  src.authored -> tgt.authored;
  src.author -> tgt.author;
  src.source -> tgt.source;
  src.item as vs0 -> tgt.item as vt0 then item(vs0, vt0);
}

group item(source src, target tgt) extends BackboneElement {
  src.linkId -> tgt.linkId;
  src.definition -> tgt.definition;
  src.text -> tgt.text;
  src.extension as e where url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    QuestionnaireResponse.item.subject' then {
    e.value -> tgt.subject;
  };
  src.answer as vs0 -> tgt.answer as vt0 then answer(vs0, vt0);
  src.item as vs0 -> tgt.item as vt0 then item(vs0, vt0);
}

group answer(source src, target tgt) extends BackboneElement {
  src.value -> tgt.value;
  src.item as vs0 -> tgt.item as vt0 then item(vs0, vt0);
}
```

2.38.10.3 R4 Validation Errors

2.38.11 Core extensions for QuestionnaireResponse

URL: https://hl7.org/fhir/R4/questionnaireresponse-extensions.html

Contains standard extensions for QuestionnaireResponse, amongst others for validation

2.38.11 Core extensions for QuestionnaireResponse

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative

2.38.11.1 Content

Extensions:

questionnaireresponse-author | author : Allows capturing, on a specific question or group of questions, exactly who was responsible for providing the answer(s).

questionnaireresponse-reviewer | reviewer : Individual responsible for ensuring that the questionnaire response have been completed appropriately and signs off on the content.

questionnaireresponse-reason | reason : The factor(s) that caused the questionnaire to be answered.

questionnaireresponse-signature | signature : Represents a wet or electronic signature for either the form overall or for the question or item it's associated with.

questionnaireresponse-completionMode | completionMode : Indicates how the individual completing the QuestionnaireResponse provided their responses.

2.38.11.2 Search Parameters

Search parameters defined by this package. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Paths | Source

item-subject | reference | Allows searching for QuestionnaireResponses by item value where the item has isSubject=true | | XML / JSON

2.38.11.3 Questionnaireresponse-extensions-QuestionnaireResponse-item-subject

URL: https://hl7.org/fhir/R4/questionnaireresponse-extensions-QuestionnaireResponse-item-subject.html

2.38.11.3 Questionnaireresponse-extensions-QuestionnaireResponse-item-subject

Search Parameter item-subject (type: reference) defined as part of the questionnaireresponse-extensions Profile.

FHIR Infrastructure | Maturity Level: 0 | Informative

Allows searching for QuestionnaireResponses by item value where the item has isSubject=true

Other Details:

• Full URL : http://hl7.org/fhir/SearchParameter/questionnaireresponse-extensions-QuestionnaireResponse-item-subject

• Paths :

• Targets :

2.38.12 JSON Schema for QuestionnaireResponse

URL: https://hl7.org/fhir/R4/questionnaireresponse.schema.json.html

2.38.12 JSON Schema for QuestionnaireResponse

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

This is a representation of the json schema for QuestionnaireResponse, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/QuestionnaireResponse",
  "$ref": "#/definitions/QuestionnaireResponse",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "QuestionnaireResponse": {
      "description": "A structured set of questions and their answers. The questions are ordered and grouped into
    coherent subsets, corresponding to the structure of the grouping of the questionnaire being responded to.",
      "properties": {
        "resourceType": {
          "description": "This is a QuestionnaireResponse resource",
          "const": "QuestionnaireResponse"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
          "description": "A human-readable narrative that contains a summary of the resource and can be used to
    represent the content of the resource to a human. The narrative need not encode all the structured data, but is
    required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
    Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
          "$ref": "Narrative.schema.json#/definitions/Narrative"
        },
        "contained": {
          "description": "These resources do not have an independent existence apart from the resource that contains
    them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
          "items": {
            "$ref": "ResourceList.schema.json#/definitions/ResourceList"
          },
          "type": "array"
        },
        "extension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
     definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "modifierExtension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource and that modifies the understanding of the element that contains it and/or the understanding of the
    containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
    of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
    met as part of the definition of the extension. Applications processing a resource are required to check for
    modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
    DomainResource (including cannot change the meaning of modifierExtension itself).",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
```

```
    "identifier": {
      "description": "A business identifier assigned to a particular completed (or partially completed)
questionnaire.",
      "$ref": "Identifier.schema.json#/definitions/Identifier"
    },
    "basedOn": {
      "description": "The order, proposal or plan that is fulfilled in whole or in part by this
QuestionnaireResponse.  For example, a ServiceRequest seeking an intake assessment or a decision support
recommendation to assess for post-partum depression.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "partOf": {
      "description": "A procedure or observation that this questionnaire was performed as part of the execution of.
  For example, the surgery a checklist was executed as part of.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "questionnaire": {
      "description": "The Questionnaire that defines and organizes the questions for which answers are being
provided.",
      "$ref": "canonical.schema.json#/definitions/canonical"
    },
    "status": {
      "description": "The position of the questionnaire response within its overall lifecycle.",
      "enum": [
        "in-progress",
        "completed",
        "amended",
        "entered-in-error",
        "stopped"
      ]
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "subject": {
      "description": "The subject of the questionnaire response.  This could be a patient, organization,
practitioner, device, etc.  This is who/what the answers apply to, but is not necessarily the source of
information.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "encounter": {
      "description": "The Encounter during which this questionnaire response was created or to which the creation of
 this record is tightly associated.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "authored": {
      "description": "The date and/or time that this set of answers were last changed.",
      "$ref": "#/definitions/dateTime"
    },
    "_authored": {
      "description": "Extensions for authored",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "author": {
      "description": "Person who received the answers to the questions in the QuestionnaireResponse and recorded
them in the system.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "source": {
      "description": "The person who answered the questions about the subject.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "item": {
      "description": "A group or question item from the original questionnaire for which answers are provided.",
      "items": {
        "$ref": "#/definitions/QuestionnaireResponse_Item"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "resourceType"
  ]
},
"QuestionnaireResponse_Item": {
  "description": "A structured set of questions and their answers. The questions are ordered and grouped into
coherent subsets, corresponding to the structure of the grouping of the questionnaire being responded to.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
```

```
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "linkId": {
      "description": "The item from the Questionnaire that corresponds to this item in the QuestionnaireResponse
resource.",
      "$ref": "#/definitions/string"
    },
    "_linkId": {
      "description": "Extensions for linkId",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "definition": {
      "description": "A reference to an [[[ElementDefinition]]] that provides the details for the item.",
      "$ref": "#/definitions/uri"
    },
    "_definition": {
      "description": "Extensions for definition",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "text": {
      "description": "Text that is displayed above the contents of the group or as the text of the question being
answered.",
      "$ref": "#/definitions/string"
    },
    "_text": {
      "description": "Extensions for text",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "answer": {
      "description": "The respondent\u0027s answer(s) to the question.",
      "items": {
        "$ref": "#/definitions/QuestionnaireResponse_Answer"
      },
      "type": "array"
    },
    "item": {
      "description": "Questions or sub-groups nested beneath a question or group.",
      "items": {
        "$ref": "#/definitions/QuestionnaireResponse_Item"
      },
      "type": "array"
    }
  },
  "additionalProperties": false
},
"QuestionnaireResponse_Answer": {
  "description": "A structured set of questions and their answers. The questions are ordered and grouped into
coherent subsets, corresponding to the structure of the grouping of the questionnaire being responded to.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
```

```
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "valueBoolean": {
      "description": "The answer (or one of the answers) provided by the respondent to the question.",
      "pattern": "^true|false$",
      "type": "boolean"
    },
    "_valueBoolean": {
      "description": "Extensions for valueBoolean",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueDecimal": {
      "description": "The answer (or one of the answers) provided by the respondent to the question.",
      "pattern": "^-?(0|[1-9][0-9]*)(\\.[0-9]+)?([eE][+-]?[0-9]+)?$",
      "type": "number"
    },
    "_valueDecimal": {
      "description": "Extensions for valueDecimal",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueInteger": {
      "description": "The answer (or one of the answers) provided by the respondent to the question.",
      "pattern": "^-?([0]|([1-9][0-9]*))$",
      "type": "number"
    },
    "_valueInteger": {
      "description": "Extensions for valueInteger",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueDate": {
      "description": "The answer (or one of the answers) provided by the respondent to the question.",
      "pattern":
"^([0-9]([0-9]([0-9][1-9]|[1-9]0)|[1-9]00)|[1-9]000)(-(0[1-9]|1[0-2])(-(0[1-9]|[1-2][0-9]|3[0-1]))?)?$",
      "type": "string"
    },
    "_valueDate": {
      "description": "Extensions for valueDate",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueDateTime": {
      "description": "The answer (or one of the answers) provided by the respondent to the question.",
      "pattern": "^([0-9]([0-9]([0-9][1-9]|[1-9]0)|[1-9]00)|[1-9]000)(-(0[1-9]|1[0-2])(-(0[1-9]|[1-2][0-9]|3[0-
1])(T([01][0-9]|2[0-3]):[0-5][0-9]:([0-5][0-9]|60)(\\.[0-9]+)?(Z|(\\+|-)((0[0-9]|1[0-3]):[0-5][0-9]|14:00))?)?)?$",
      "type": "string"
    },
    "_valueDateTime": {
      "description": "Extensions for valueDateTime",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueTime": {
      "description": "The answer (or one of the answers) provided by the respondent to the question.",
      "pattern": "^([01][0-9]|2[0-3]):[0-5][0-9]:([0-5][0-9]|60)(\\.[0-9]+)?$",
      "type": "string"
    },
    "_valueTime": {
      "description": "Extensions for valueTime",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueString": {
      "description": "The answer (or one of the answers) provided by the respondent to the question.",
      "pattern": "^[ \\r\\n\\t\\S]+$",
      "type": "string"
    },
    "_valueString": {
      "description": "Extensions for valueString",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueUri": {
      "description": "The answer (or one of the answers) provided by the respondent to the question.",
      "pattern": "^\\S*$",
      "type": "string"
    },
    "_valueUri": {
      "description": "Extensions for valueUri",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueAttachment": {
      "description": "The answer (or one of the answers) provided by the respondent to the question.",
```

```
          "$ref": "Attachment.schema.json#/definitions/Attachment"
        },
        "valueCoding": {
          "description": "The answer (or one of the answers) provided by the respondent to the question.",
          "$ref": "Coding.schema.json#/definitions/Coding"
        },
        "valueQuantity": {
          "description": "The answer (or one of the answers) provided by the respondent to the question.",
          "$ref": "Quantity.schema.json#/definitions/Quantity"
        },
        "valueReference": {
          "description": "The answer (or one of the answers) provided by the respondent to the question.",
          "$ref": "Reference.schema.json#/definitions/Reference"
        },
        "item": {
          "description": "Nested groups and/or questions found within this particular answer.",
          "items": {
            "$ref": "#/definitions/QuestionnaireResponse_Item"
          },
          "type": "array"
        }
      },
      "additionalProperties": false
    }
  }
}
```


2.38.13 Resource QuestionnaireResponse - Maturity Dependency View

URL: https://hl7.org/fhir/R4/questionnaireresponse-dependencies.html

2.38.13 Resource QuestionnaireResponse - Maturity Dependency View

FHIR Infrastructure Work Group | Maturity Level: 3 | Trial Use | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

Dependency Graph for QuestionnaireResponse FMM level 3

Name | Card. | Type | Dependency Analysis
QuestionnaireResponse | | DomainResource
identifier | 0..1 | Identifier
basedOn | 0..* | Reference(CarePlan | ServiceRequest) | Type Warning: (CarePlan = FMM2-Trial Use vs. Element = FMM3-Trial Use) Type Warning: (ServiceRequest = FMM2-Trial Use vs. Element = FMM3-Trial Use)
partOf | 0..* | Reference(Observation | Procedure)
questionnaire | 0..1 | canonical(Questionnaire)
status | 1..1 | code
subject | 0..1 | Reference(Any) | Type Warning: (Any = FMM1-Trial Use vs. Element = FMM3-Trial Use)
encounter | 0..1 | Reference(Encounter) | Type Warning: (Encounter = FMM2-Trial Use vs. Element = FMM3-Trial Use)
authored | 0..1 | dateTime
author | 0..1 | Reference(Device | Practitioner | PractitionerRole | Patient | RelatedPerson | Organization) | Type Warning: (Device = FMM2-Trial Use vs. Element = FMM3-Trial Use) Type Warning: (PractitionerRole = FMM2-Trial Use vs. Element = FMM3-Trial Use) Type Warning: (RelatedPerson = FMM2-Trial Use vs. Element = FMM3-Trial Use)
source | 0..1 | Reference(Patient | Practitioner | PractitionerRole | RelatedPerson) | Type Warning: (PractitionerRole = FMM2-Trial Use vs. Element = FMM3-Trial Use) Type Warning: (RelatedPerson = FMM2-Trial Use vs. Element = FMM3-Trial Use)
item | 0..* | BackboneElement
linkId | 1..1 | string
definition | 0..1 | uri
text | 0..1 | string
answer | 0..* | BackboneElement
value[x] | 0..1
valueBoolean | | boolean
valueDecimal | | decimal
valueInteger | | integer
valueDate | | date
valueDateTime | | dateTime
valueTime | | time
valueString | | string
valueUri | | uri
valueAttachment | | Attachment
valueCoding | | Coding
valueQuantity | | Quantity
valueReference | | Reference(Any
item | 0..* | see item
item | 0..* | see item

2.38.14 Value Set Sc-valueset-questionnaire-answers-status - Status Code Mapping

URL: https://hl7.org/fhir/R4/sc-valueset-questionnaire-answers-status.html

2.38.14 Value Set Sc-valueset-questionnaire-answers-status - Status Code Mapping

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

This is the narrative for the resource. See also the XML, JSON or Turtle format.

2.38.15 QuestionnaireResponseStatusCanonicalMap (http://hl7.org/fhir/ConceptMap/sc-questionnaire-answers-status)

Mapping from http://hl7.org/fhir/ValueSet/questionnaire-answers-status to http://hl7.org/fhir/ValueSet/resource-status

DRAFT. Published on 01/11/2019 9:29:23 AM by HL7 (FHIR Project) (http://hl7.org/fhir , fhir@lists.hl7.org).

Canonical Mapping for "Lifecycle status of the questionnaire response."

Source Code | Equivalence | Destination Code
entered-in-error | equivalent | error
in-progress | equivalent | active
stopped | equivalent | failed
amended | equivalent | replaced
completed | equivalent | complete

2.39 Resource List - Content

URL: https://hl7.org/fhir/R4/list.html
FHIR Infrastructure Work Group | Maturity Level: 1 | Trial Use | Security Category: Not Classified | Compartments: Device, Patient, Practitioner

A list is a curated collection of resources.

2.39.1 Scope and Usage

The List resource is a flat, possibly ordered collection of records. List resources are used in many places, including allergies, medications, alerts, family history, medical history, etc. List resources can be used to support patient-specific clinical lists as well as lists that manage workflows such as tracking patients, managing teaching cases, etc. Resources supported by the List resource can be homogeneous – consisting of only one type of resource (e.g. allergy lists) as well as heterogeneous – containing a variety of resources (e.g. a problem list including Conditions, AllergyIntolerances, recent Procedures, etc.).

Lists will typically include references to the resources that make up the list, however in some cases the details of the content of the list might be expressed in narrative only; e.g. a text record of a family history. The List resource is only needed if there is a need to filter the set of resources by a mechanism that cannot be accomplished via a simple query; e.g. there is no need to have a list for all AllergyIntolerances that exist on a server for a given patient. However, List is an appropriate mechanism to provide a filtered list of the subset of AllergyIntolerances that are deemed to be "current". Lists are allowed to contain other Lists, to create a nested collection of Lists.

Querying a List of resources such as AllergyIntolerance, Condition or Medication-related resources is different than querying the resource-specific endpoint. For example, a List of AllergyIntolerance resources would represent a curated point-in-time snapshot of the patient's allergies and intolerances. On the other hand, querying the AllergyIntolerance endpoint would typically produce a larger set of records as it would both be non-curated (potentially containing duplicate or out-of-date records) and current - generated based on information as of "now" rather than the last time a human manually revised the List resource instance. Which mechanism is most appropriate for data retrieval will vary by use-case. In some cases, systems might not have an appropriate curated List to query.

Note that the presence of an item in a List resource SHALL NOT change the meaning of any information that would be understood by looking at the item outside the context of the List, because items may be accessed directly outside the List by RESTful means or after a document is processed. For example, a List with a code that means "refuted conditions" cannot have items that are Condition resources that do not have a verificationStatus of refuted.

2.39.2 Boundaries and Relationships

There are five mechanisms in FHIR for communicating collections of resources:

• This List resource - enumerates a flat collection of resources and provides features for managing the collection. While a particular List instance may represent a "snapshot", from a business process perspective

the notion of "List" is dynamic – items are added and removed over time. The List resource references other resources. Lists may be curated and have specific business meaning.

• The Group resource - defines a group of specific people, animals, devices, etc. by enumerating them, or by describing qualities that group members have. The group resource refers to other resources, possibly implicitly. Groups are intended to be acted upon or observed as a whole; e.g. performing therapy on a group, calculating risk for a group, etc. This resource will commonly be used for public health (e.g. describing an at-risk population), clinical trials (e.g. defining a test subject pool) and similar purposes.

• The Composition resource - defines a set of healthcare-related information that is assembled together into a single logical document that provides a single coherent statement of meaning, establishes its own context and that has clinical attestation with regard to who is making the statement. The Composition resource provides the basic structure of a FHIR document. The full content of the document is expressed using a bundle. Compositions will often reference Lists as the focus of particular sections.

• The Bundle resource - is an infrastructure container for a group of resources. It does not have a narrative and is used to group collections of resources for transmission, persistence or processing (e.g. messages, documents, transactions, query responses, etc.) The content of bundles is typically algorithmically determined for a particular exchange or persistence purpose.

• The DomainResource.contained element - allows multiple resources to be nested inside any DomainResource. This is a special type of grouping where the grouped resources lose independent existence - they no longer have their own identifiers, can't easily be queried independently, etc. Use of this grouping is a technical mechanism for managing the independence of resources and has no impact on meaning. Contained, bundled, and remotely referenced resources convey the same meaning.

This resource is referenced by MeasureReport

2.39.3 Resource Content

• Structure

• UML

• XML

• JSON

• Turtle

• R3 Diff

• All

Structure

Name | Flags | Card. | Type | Description & Constraints
List | ITU | | DomainResource | A list is a curated collection of resources + Rule: A list can only have an emptyReason if it is empty + Rule: The deleted flag can only be used if the mode of the list is "changes" + Rule: An entry date can only be used if the mode of the list is "working" Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | | 0..* | Identifier | Business identifier
status | ?!Σ | 1..1 | code | current | retired | entered-in-error ListStatus (Required)
mode | ?!Σ | 1..1 | code | working | snapshot | changes ListMode (Required)
title | Σ | 0..1 | string | Descriptive name for the list
code | Σ | 0..1 | CodeableConcept | What the purpose of this list is Example Use Codes for List (Example)
subject | Σ | 0..1 | Reference(Patient | Group | Device | Location) | If all resources have the same subject
encounter | | 0..1 | Reference(Encounter) | Context in which list created
date | Σ | 0..1 | dateTime | When the list was prepared
source | Σ | 0..1 | Reference(Practitioner | PractitionerRole | Patient | Device) | Who and/or what defined the list contents (aka Author)
orderedBy | | 0..1 | CodeableConcept | What order the list has List Order Codes (Preferred)
note | | 0..* | Annotation | Comments about the list
entry | I | 0..* | BackboneElement | Entries in the list
flag | | 0..1 | CodeableConcept | Status/Workflow information about this item Patient Medicine Change Types (Example)
deleted | ?!I | 0..1 | boolean | If this item is actually marked as deleted
date | | 0..1 | dateTime | When item added to list
item | | 1..1 | Reference(Any) | Actual entry
emptyReason | I | 0..1 | CodeableConcept | Why list is empty List Empty Reasons (Preferred)
Documentation for this format

UML Diagram (Legend)

XML Template

```
<List xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business identifier --></identifier>
 <status value="[code]"/><!-- 1..1 current | retired | entered-in-error -->
 <mode value="[code]"/><!-- 1..1 working | snapshot | changes -->
 <title value="[string]"/><!-- 0..1 Descriptive name for the list -->
 <code><!-- 0..1 CodeableConcept What the purpose of this list is --></code>
```

```
 <subject><!-- 0..1 Reference(Patient|Group|Device|Location) If all resources have the same subject --></subject>
 <encounter><!-- 0..1 Reference(Encounter) Context in which list created --></encounter>
 <date value="[dateTime]"/><!-- 0..1 When the list was prepared -->
 <source><!-- 0..1 Reference(Practitioner|PractitionerRole|Patient|Device) Who and/or what defined the list contents
    (aka Author) --></source>
 <orderedBy><!-- 0..1 CodeableConcept What order the list has --></orderedBy>
 <note><!-- 0..* Annotation Comments about the list --></note>
 <entry>  <!--  0..* Entries in the list -->
  <flag><!-- 0..1 CodeableConcept Status/Workflow information about this item --></flag>
  <deleted value="[boolean]"/><!--  0..1 If this item is actually marked as deleted -->
  <date value="[dateTime]"/><!-- 0..1 When item added to list -->
  <item><!-- 1..1 Reference(Any) Actual entry --></item>
 </entry>
 <emptyReason><!--  0..1 CodeableConcept Why list is empty --></emptyReason>
</List>
```

JSON Template

```
{
  "resourceType" : "List",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business identifier
  "status" : "<code>", // R!  current | retired | entered-in-error
  "mode" : "<code>", // R!  working | snapshot | changes
  "title" : "<string>", // Descriptive name for the list
  "code" : { CodeableConcept }, // What the purpose of this list is
  "subject" : { Reference(Patient|Group|Device|Location) }, // If all resources have the same subject
  "encounter" : { Reference(Encounter) }, // Context in which list created
  "date" : "<dateTime>", // When the list was prepared
  "source" : { Reference(Practitioner|PractitionerRole|Patient|Device) }, // Who and/or what defined the list contents
    (aka Author)
  "orderedBy" : { CodeableConcept }, // What order the list has
  "note" : [{ Annotation }], // Comments about the list
  "entry" : [{ // C? Entries in the list
    "flag" : { CodeableConcept }, // Status/Workflow information about this item
    "deleted" : <boolean>, // C? If this item is actually marked as deleted
    "date" : "<dateTime>", // When item added to list
    "item" : { Reference(Any) } // R!  Actual entry
  }],
  "emptyReason" : { CodeableConcept } // C? Why list is empty
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:List;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:List.identifier [ Identifier ], ... ; # 0..* Business identifier
  fhir:List.status [ code ]; # 1..1 current | retired | entered-in-error
  fhir:List.mode [ code ]; # 1..1 working | snapshot | changes
  fhir:List.title [ string ]; # 0..1 Descriptive name for the list
  fhir:List.code [ CodeableConcept ]; # 0..1 What the purpose of this list is
  fhir:List.subject [ Reference(Patient|Group|Device|Location) ]; # 0..1 If all resources have the same subject
  fhir:List.encounter [ Reference(Encounter) ]; # 0..1 Context in which list created
  fhir:List.date [ dateTime ]; # 0..1 When the list was prepared
  fhir:List.source [ Reference(Practitioner|PractitionerRole|Patient|Device) ]; # 0..1 Who and/or what defined the list
    contents (aka Author)
  fhir:List.orderedBy [ CodeableConcept ]; # 0..1 What order the list has
  fhir:List.note [ Annotation ], ... ; # 0..* Comments about the list
  fhir:List.entry [ # 0..* Entries in the list
    fhir:List.entry.flag [ CodeableConcept ]; # 0..1 Status/Workflow information about this item
    fhir:List.entry.deleted [ boolean ]; # 0..1 If this item is actually marked as deleted
    fhir:List.entry.date [ dateTime ]; # 0..1 When item added to list
    fhir:List.entry.item [ Reference(Any) ]; # 1..1 Actual entry
  ], ...;
  fhir:List.emptyReason [ CodeableConcept ]; # 0..1 Why list is empty
]
```

Changes since R3

List
List.status | Change value set from http://hl7.org/fhir/ValueSet/list-status to http://hl7.org/fhir/ValueSet/list-status|4.0.1
List.mode | Change value set from http://hl7.org/fhir/ValueSet/list-mode to http://hl7.org/fhir/ValueSet/list-mode|4.0.1
List.source | Type Reference: Added Target Type PractitionerRole
List.entry.deleted | Default Value "false" removed

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 9 tests that all execute ok. All tests pass round-trip testing and 5 r3 resources are invalid (0 errors).)

Structure

Name | Flags | Card. | Type | Description & Constraints

List | ITU | | DomainResource | A list is a curated collection of resources + Rule: A list can only have an emptyReason if it is empty + Rule: The deleted flag can only be used if the mode of the list is "changes" + Rule: An entry date can only be used if the mode of the list is "working" Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | | 0..* | Identifier | Business identifier
status | ?!Σ | 1..1 | code | current | retired | entered-in-error ListStatus (Required)
mode | ?!Σ | 1..1 | code | working | snapshot | changes ListMode (Required)
title | Σ | 0..1 | string | Descriptive name for the list
code | Σ | 0..1 | CodeableConcept | What the purpose of this list is Example Use Codes for List (Example)
subject | Σ | 0..1 | Reference(Patient | Group | Device | Location) | If all resources have the same subject
encounter | | 0..1 | Reference(Encounter) | Context in which list created
date | Σ | 0..1 | dateTime | When the list was prepared
source | Σ | 0..1 | Reference(Practitioner | PractitionerRole | Patient | Device) | Who and/or what defined the list contents (aka Author)
orderedBy | | 0..1 | CodeableConcept | What order the list has List Order Codes (Preferred)
note | | 0..* | Annotation | Comments about the list
entry | I | 0..* | BackboneElement | Entries in the list
flag | | 0..1 | CodeableConcept | Status/Workflow information about this item Patient Medicine Change Types (Example)
deleted | ?!I | 0..1 | boolean | If this item is actually marked as deleted
date | | 0..1 | dateTime | When item added to list
item | | 1..1 | Reference(Any) | Actual entry
emptyReason | I | 0..1 | CodeableConcept | Why list is empty List Empty Reasons (Preferred)
Documentation for this format

UML Diagram (Legend)

XML Template

```
<List xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business identifier --></identifier>
 <status value="[code]"/><!-- 1..1 current | retired | entered-in-error -->
 <mode value="[code]"/><!-- 1..1 working | snapshot | changes -->
 <title value="[string]"/><!-- 0..1 Descriptive name for the list -->
 <code><!-- 0..1 CodeableConcept What the purpose of this list is --></code>
 <subject><!-- 0..1 Reference(Patient|Group|Device|Location) If all resources have the same subject --></subject>
 <encounter><!-- 0..1 Reference(Encounter) Context in which list created --></encounter>
 <date value="[dateTime]"/><!-- 0..1 When the list was prepared -->
 <source><!-- 0..1 Reference(Practitioner|PractitionerRole|Patient|Device) Who and/or what defined the list contents
    (aka Author) --></source>
 <orderedBy><!-- 0..1 CodeableConcept What order the list has --></orderedBy>
 <note><!-- 0..* Annotation Comments about the list --></note>
 <entry>  <!--  0..* Entries in the list -->
  <flag><!-- 0..1 CodeableConcept Status/Workflow information about this item --></flag>
  <deleted value="[boolean]"/><!--  0..1 If this item is actually marked as deleted -->
  <date value="[dateTime]"/><!-- 0..1 When item added to list -->
  <item><!-- 1..1 Reference(Any) Actual entry --></item>
 </entry>
 <emptyReason><!--  0..1 CodeableConcept Why list is empty --></emptyReason>
</List>
```

JSON Template

```
{
  "resourceType" : "List",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business identifier
  "status" : "<code>", // R!  current | retired | entered-in-error
  "mode" : "<code>", // R!  working | snapshot | changes
  "title" : "<string>", // Descriptive name for the list
  "code" : { CodeableConcept }, // What the purpose of this list is
  "subject" : { Reference(Patient|Group|Device|Location) }, // If all resources have the same subject
  "encounter" : { Reference(Encounter) }, // Context in which list created
  "date" : "<dateTime>", // When the list was prepared
  "source" : { Reference(Practitioner|PractitionerRole|Patient|Device) }, // Who and/or what defined the list contents
    (aka Author)
  "orderedBy" : { CodeableConcept }, // What order the list has
  "note" : [{ Annotation }], // Comments about the list
  "entry" : [{ // C? Entries in the list
    "flag" : { CodeableConcept }, // Status/Workflow information about this item
    "deleted" : <boolean>, // C? If this item is actually marked as deleted
    "date" : "<dateTime>", // When item added to list
    "item" : { Reference(Any) } // R!  Actual entry
  }],
  "emptyReason" : { CodeableConcept } // C? Why list is empty
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:List;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
```

```
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:List.identifier [ Identifier ], ... ; # 0..* Business identifier
  fhir:List.status [ code ]; # 1..1 current | retired | entered-in-error
  fhir:List.mode [ code ]; # 1..1 working | snapshot | changes
  fhir:List.title [ string ]; # 0..1 Descriptive name for the list
  fhir:List.code [ CodeableConcept ]; # 0..1 What the purpose of this list is
  fhir:List.subject [ Reference(Patient|Group|Device|Location) ]; # 0..1 If all resources have the same subject
  fhir:List.encounter [ Reference(Encounter) ]; # 0..1 Context in which list created
  fhir:List.date [ dateTime ]; # 0..1 When the list was prepared
  fhir:List.source [ Reference(Practitioner|PractitionerRole|Patient|Device) ]; # 0..1 Who and/or what defined the list
    contents (aka Author)
  fhir:List.orderedBy [ CodeableConcept ]; # 0..1 What order the list has
  fhir:List.note [ Annotation ], ... ; # 0..* Comments about the list
  fhir:List.entry [ # 0..* Entries in the list
    fhir:List.entry.flag [ CodeableConcept ]; # 0..1 Status/Workflow information about this item
    fhir:List.entry.deleted [ boolean ]; # 0..1 If this item is actually marked as deleted
    fhir:List.entry.date [ dateTime ]; # 0..1 When item added to list
    fhir:List.entry.item [ Reference(Any) ]; # 1..1 Actual entry
  ], ...;
  fhir:List.emptyReason [ CodeableConcept ]; # 0..1 Why list is empty
]
```

Changes since Release 3

List

List.status | Change value set from http://hl7.org/fhir/ValueSet/list-status to http://hl7.org/fhir/ValueSet/list-status|4.0.1

List.mode | Change value set from http://hl7.org/fhir/ValueSet/list-mode to http://hl7.org/fhir/ValueSet/list-mode|4.0.1

List.source | Type Reference: Added Target Type PractitionerRole

List.entry.deleted | Default Value "false" removed

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 9 tests that all execute ok. All tests pass round-trip testing and 5 r3 resources are invalid (0 errors).)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

2.39.3.1 Terminology Bindings

Path | Definition | Type | Reference

List.status | The current state of the list. | Required | ListStatus

List.mode | The processing mode that applies to this list. | Required | ListMode

List.code | What the purpose of a list is. | Example | ExampleUseCodesForList

List.orderedBy | What order applies to the items in a list. | Preferred | ListOrderCodes

List.entry.flag | Codes that provide further information about the reason and meaning of the item in the list. | Example | PatientMedicineChangeTypes

List.emptyReason | If a list is empty, why it is empty. | Preferred | ListEmptyReasons

2.39.3.2 Constraints

id | Level | Location | Description | Expression

lst-1 | Rule | (base) | A list can only have an emptyReason if it is empty | emptyReason.empty() or entry.empty()

lst-2 | Rule | (base) | The deleted flag can only be used if the mode of the list is "changes" | mode = 'changes' or entry.deleted.empty()

lst-3 | Rule | (base) | An entry date can only be used if the mode of the list is "working" | mode = 'working' or entry.date.empty()

2.39.4 Querying Lists vs. the underlying resources

When a client system is interested in a patient's medications, allergies, problems, family history or other information typically handled via List, they have three options:

• They can query for List instances of the appropriate type; or

• They can query against the resource endpoint for the resources that make up the list (e.g. AllergyIntolerance, Condition MedicationStatement, etc.), possibly filtering by time, etc.

• They can query against the resource endpoint, requesting one of the Current resource lists

Querying directly against the clinical resource endpoints will provide an un-curated view of information. A server may contain records that were part of various clinical documents, referrals and other submission sources that might not necessarily be considered wholly accurate or current, but which must be retained "as is" to provide a record of what data was received or seen by a clinician at a particular point in time.

On the other hand, lists are almost always curated. They will include only those records deemed by the author of the list to be both current and accurate. This can be both an advantage and a disadvantage. Lists are less likely to include irrelevant content, but there's a risk that they won't be completely up-to-date (new content may have been added that's more recent than when a given list was last updated). It's also possible that the list author's notion of "current" or "relevant" may differ from the perspective of the user performing the query. Also, there's the challenge that multiple lists may exist, with different author and different perspectives

The Current resource lists get around the problem of multiple lists and make it clear which list is considered authoritative, but not all systems will necessarily support this capability, and there's still the possibility that the list won't be completely up-to-date or will exclude information that is of interest to the querying system.

There's no right and wrong way to retrieve allergy, medication and other such information. Sometimes retrieving the "current" medication list will provide the best results, sometimes querying the raw records will provide better results. Sometimes providing the contents of the list plus a filtered view of the raw records may give the most useful information. Determining the most appropriate approach will depend on the needs of the user as well as the types of data sources for information held on the server and patterns of list maintenance. Client systems may well need to adapt their behavior in different environments if they can't count on consistent server behavior.

2.39.5 List Order

All lists are considered ordered - the order in which items literally appear in the list may be an important part of the meaning of the list. Reordering the items in a list may change the meaning of the list.

While a list always contains an ordered set of items, the significance of the order may be unknown, or it may be insignificant. As an example, consider a list of patients for a practitioner to visit. The list may be in the order in which the patients are to be visited, or it may be an unsorted list of patients to be visited in any order.

The List resource has an orderedBy element that, if present, specifies the meaning of the item order. Note, however, that the meaning of the order may be known implicitly rather than specified in the orderedBy element.

Applications SHOULD NOT reorder the elements in a list unless they understand the impact of this on the meaning of the list.

2.39.6 List Mode & Item Deleted

There are several different kinds of uses for a List resource:

2.39.6.1 The processing mode that applies to this list.

working | This list is the master list, maintained in an ongoing fashion with regular updates as the real world list it is tracking changes.

snapshot | This list was prepared as a snapshot. It should not be assumed to be current.

changes | A point-in-time list that shows what changes have been made or recommended. E.g. a discharge medication list showing what was added and removed during an encounter.

The most common mode is "snapshot" - a list that is accurate within the context it is used in but not current or maintained after that; e.g., medications on discharge in a discharge summary. Note that these lists usually have a status of 'current' - they were current when they were prepared. Some kinds of lists may be explicitly retired (particularly if mode = working), but most will not be maintained after creation.

A change list may include deleted items. Some examples of change lists are a reconciled list of allergies, a discharge medication list and a list with new, updated and deleted items in it - though these might not be lists that include changes (this is an implementation decision). In order to ensure that the list is safe to process, any item where the flag implies that the item has actually been deleted SHALL have the deleted element set to true.

Note that there is no implication about the status of a resource that has been deleted. The only statement that is made is that the resource has been dropped from the list. However, applications should ensure that the implication of adding or deleting items from the list is consistent with the logical status of the resource and its contents.

A proper use of List.mode = "changes" with a deleted resource is in a medications list section of a discharge summary. See Example "med-list". An improper use would be if the list was a working list of patient medications in a clinical tracking system, and list item flags were used to implement version tracking history within the resource.

2.39.7 Efficiency and size issues

Some kinds of List resources may grow to a considerable size, and handling them may require more care than typical resources. Some approaches to consider include:

- Reading portions of the list resource using GraphQL
- Using the _list parameter rather than retrieving the list
- Changing the resource using the PATCH interaction rather than PUTting the whole resource

2.39.8 Narrative Content

The narrative portion of the List resource should contain a summary of the items in the list, their key information, along with a human-readable summary of their flags (if present). The narrative may be generated from the data content and/or narrative of the resources referred to in the list, or it may be a narrative written by a human, which is partially or completely matched by structured data in the linked resources. The human written narrative may be the only content if the list has no entries (which would equate

to a narrative only section in a document).

An HTML table is the recommended approach, though this is not required. Each List.item should appear in the narrative for the resource; i.e. it SHALL NOT be necessary to retrieve the list items in order to have a human-readable rendering of the content. In addition, if the List.text.status is "generated", then the narrative should not suggest the list contains items for which there are no corresponding List.item elements. If the list has flags, the representation should make clear use of visual hints (borders, lines, bullet marks, etc.) to ensure that human readers do not get confused about which flags belong with which item on space-poor displays (e.g. to prevent wrapping from separating the flags from the items).

Note that when a List resource is used in a Document, the narrative of the list is part of the attested content of the document.

In a dynamic environment, the narrative content of a list will be limited to the version of the linked resources at the time the list was last updated. It may be even earlier if the narrative isn't updated to reflect the most recent version of all referenced resources at each update. Best practice for 'working' lists is to update the narrative to reflect the most recent content of all list elements each time the list is revised. Lists should therefore not be relied on as a real-time view of the referenced content. There are a few possible approaches to work around this issue:

• Provide minimal information about the listed resources, possibly limited to only a link. (Not recommended as this severely limits the usefulness of the narrative and is particularly problematic for things like documents where the only attested content might be the List narrative.)

• Include only "generated" narrative, so the retriever can easily generate their own "current" view of the list by retrieving the referenced resources, ignoring the fixed narrative.

• The server hosting the list can subscribe to all referenced resources and auto-update the narrative each time one of the referenced resources changes (or at least on a semi-frequent basis).

2.39.9 Empty Reason

If a list is empty, there could be several different reasons why this is so. For example:

• There are no appropriate entries for the list (i.e. the patient has no known medications/allergies/history)

• The sender (human or system) deemed that these were not related to this context of patient care (usually for privacy related reasons)

• The source system doesn't support these types of entries

• The information to populate the list wasn't gathered - i.e. "Not asked"

Given these possibilities - especially the common and significant first case - for many kinds of lists, source systems SHOULD provide an empty reason if the list is empty. Because of the importance of the first case, the special value "nil-known" should be used when there are no (significant) entries in this context of care. Note that this concept is sometimes described differently, such as "patient denies taking medications", or "patient was unable to identify any relevant medical history".

When receiving a list, systems should not assume that the list is complete (some entries may have been withheld for a variety of reasons), unless there are specific trading partner arrangements in place or, if the list is empty, that there are actually nil known, unless the "nil-known" code is present.

If a list is empty, the narrative should contain text equivalent to the empty reason.

Note that there are also many kinds of lists that can be empty with no need for an empty reason (example)

2.39.10 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
code | token | What the purpose of this list is | List.code | 13 Resources
date | date | When the list was prepared | List.date | 17 Resources
empty-reason | token | Why list is empty | List.emptyReason
encounter | reference | Context in which list created | List.encounter (Encounter) | 12 Resources
identifier | token | Business identifier | List.identifier | 30 Resources
item | reference | Actual entry | List.entry.item (Any)
notes | string | The annotation - text content (as markdown) | List.note.text
patient | reference | If all resources have the same subject | List.subject.where(resolve() is Patient) (Patient) | 33 Resources
source | reference | Who and/or what defined the list contents (aka Author) | List.source (Practitioner, Device, Patient, PractitionerRole)
status | token | current | retired | entered-in-error | List.status
subject | reference | If all resources have the same subject | List.subject (Group, Device, Patient, Location)
title | string | Descriptive name for the list | List.title

2.39.11 Resource List - Examples

URL: https://hl7.org/fhir/R4/list-examples.html

2.39.11 Resource List - Examples

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Not Classified | Compartments: Device, Patient, Practitioner

Example Name | id | Format
General List Example | example | XML | JSON | Turtle
Empty List Example | example-empty | XML | JSON | Turtle
Empty List (no exception reason) | example-simple-empty | XML | JSON | Turtle
Example Medication Change List | med-list | XML | JSON | Turtle
Real-world patient example | f201 | XML | JSON | Turtle
Simple genetic family member history | genetic | XML | JSON | Turtle
Example for risk assessment | prognosis | XML | JSON | Turtle
Example of current allergies list | current-allergies | XML | JSON | Turtle
Simple genetic pedigree tree example with double cousin replationship | example-double-cousin-relationship | XML | JSON | Turtle
A long list with many entries for testing handling longer lists | long | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

2.39.12 Resource List - Detailed Descriptions

URL: https://hl7.org/fhir/R4/list-definitions.html

2.39.12 Resource List - Detailed Descriptions

FHIR Infrastructure Work Group | Maturity Level: 1 | Trial Use | Security Category: Not Classified | Compartments: Device, Patient, Practitioner

Detailed Descriptions for the elements in the List resource.

List
Element Id | List
Definition | A list is a curated collection of resources.
Cardinality | 0..*
Type | DomainResource
Alternate Names | Collection; WorkingList; Organizer
Invariants | Defined on this element lst-1 Rule A list can only have an emptyReason if it is empty emptyReason.empty() or entry.empty() lst-2 Rule The deleted flag can only be used if the mode of the list is "changes" mode = 'changes' or entry.deleted.empty() lst-3 Rule An entry date can only be used if the mode of the list is "working" mode = 'working' or entry.date.empty()
Defined on this element
lst-1 | Rule | A list can only have an emptyReason if it is empty | emptyReason.empty() or entry.empty()
lst-2 | Rule | The deleted flag can only be used if the mode of the list is "changes" | mode = 'changes' or entry.deleted.empty()
lst-3 | Rule | An entry date can only be used if the mode of the list is "working" | mode = 'working' or entry.date.empty()
List.identifier
Element Id | List.identifier
Definition | Identifier for the List assigned for business purposes outside the context of FHIR.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
List.status
Element Id | List.status
Definition | Indicates the current state of this list.
Cardinality | 1..1
Terminology Binding | ListStatus (Required)
Type | code
Is Modifier | true (Reason: This element is labeled as a modifier because it is a status element that contains status entered-in-error which means that the resource should not be treated as valid)
Summary | true
Comments | This element is labeled as a modifier because the status contains codes that mark the resource as not currently valid.
List.mode
Element Id | List.mode
Definition | How this list was prepared - whether it is a working list that is suitable for being maintained on an ongoing basis, or if it represents a snapshot of a list of items from another source, or whether it is a prepared list where items may be marked as added, modified or deleted.
Cardinality | 1..1
Terminology Binding | ListMode (Required)
Type | code
Is Modifier | true (Reason: If set to "changes", the list is considered incomplete, while the other two codes indicate the list is complete, which changes the understanding of the elements listed)
Requirements | Lists are used in various ways, and it must be known in what way it is safe to use them.
Summary | true
Comments | This element is labeled as a modifier because a change list must not be misunderstood as a complete list.
List.title
Element Id | List.title

Definition | A label for the list assigned by the author.
Cardinality | 0..1
Type | string
Requirements | Allows customization beyond just the code identifying the kind of list.
Summary | true
List.code
Element Id | List.code
Definition | This code defines the purpose of the list - why it was created.
Cardinality | 0..1
Terminology Binding | Example Use Codes for List (Example)
Type | CodeableConcept
Requirements | Lists often contain subsets of resources rather than an exhaustive list. The code identifies what type of subset is included.
Summary | true
Comments | If there is no code, the purpose of the list is implied where it is used, such as in a document section using Document.section.code.
List.subject
Element Id | List.subject
Definition | The common subject (or patient) of the resources that are in the list if there is one.
Cardinality | 0..1
Type | Reference(Patient | Group | Device | Location)
Requirements | The primary purpose of listing the subject explicitly is to help with finding the right list.
Summary | true
Comments | Some purely arbitrary lists do not have a common subject, so this is optional.
List.encounter
Element Id | List.encounter
Definition | The encounter that is the context in which this list was created.
Cardinality | 0..1
Type | Reference(Encounter)
List.date
Element Id | List.date
Definition | The date that the list was prepared.
Cardinality | 0..1
Type | dateTime
Requirements | Identifies how current the list is which affects relevance.
Summary | true
Comments | The actual important date is the date of currency of the resources that were summarized, but it is usually assumed that these are current when the preparation occurs.
List.source
Element Id | List.source
Definition | The entity responsible for deciding what the contents of the list were. Where the list was created by a human, this is the same as the author of the list.
Cardinality | 0..1
Type | Reference(Practitioner | PractitionerRole | Patient | Device)
Requirements | Allows follow-up as well as context.
Alternate Names | Author
Summary | true
Comments | The primary source is the entity that made the decisions what items are in the list. This may be software or user.
List.orderedBy
Element Id | List.orderedBy
Definition | What order applies to the items in the list.
Cardinality | 0..1
Terminology Binding | List Order Codes (Preferred)
Type | CodeableConcept
Requirements | Important for presentation and rendering. Lists may be sorted to place more important information first or to group related entries.
Comments | Applications SHOULD render ordered lists in the order provided, but MAY allow users to re-order based on their own preferences as well. If there is no order specified, the order is unknown, though there may still be some order.
List.note
Element Id | List.note
Definition | Comments that apply to the overall list.
Cardinality | 0..*
Type | Annotation
List.entry
Element Id | List.entry
Definition | Entries in this list.
Cardinality | 0..*
Comments | If there are no entries in the list, an emptyReason SHOULD be provided.
Invariants | Affect this element lst-1 Rule A list can only have an emptyReason if it is empty emptyReason.empty() or entry.empty()
Affect this element
lst-1 | Rule | A list can only have an emptyReason if it is empty | emptyReason.empty() or entry.empty()
List.entry.flag

Element Id | List.entry.flag
Definition | The flag allows the system constructing the list to indicate the role and significance of the item in the list.
Cardinality | 0..1
Terminology Binding | Patient Medicine Change Types (Example)
Type | CodeableConcept
Requirements | This field is present to support various clinical uses of lists, such as a discharge summary medication list, where flags specify whether the medication was added, modified, or deleted from the list.
Comments | The flag can only be understood in the context of the List.code. If the flag means that the entry has actually been deleted from the list, the deleted element SHALL be true. Deleted can only be used if the List.mode is "changes".
List.entry.deleted
Element Id | List.entry.deleted
Definition | True if this item is marked as deleted in the list.
Cardinality | 0..1
Type | boolean
Is Modifier | true (Reason: If deleted is true, then the item included in the list isn't actually part of the list anymore)
Meaning if Missing | List items are generally only treated as deleted when this element explicitly carries a value of true. Systems SHOULD always populate this value when mode is 'changes'
Requirements | The flag element may contain codes that an application processing the list does not understand. However there can be no ambiguity if a list item is actually marked as "deleted".
Comments | If the flag means that the entry has actually been deleted from the list, the deleted element SHALL be true. Both flag and deleted can only be used if the List.mode is "changes". A deleted entry should be displayed in narrative as deleted. This element is labeled as a modifier because it indicates that an item is (to be) no longer in the list.
Invariants | Affect this element lst-2 Rule The deleted flag can only be used if the mode of the list is "changes" mode = 'changes' or entry.deleted.empty()
Affect this element
lst-2 | Rule | The deleted flag can only be used if the mode of the list is "changes" | mode = 'changes' or entry.deleted.empty()
List.entry.date
Element Id | List.entry.date
Definition | When this item was added to the list.
Cardinality | 0..1
Type | dateTime
Requirements | The date may be significant for understanding the meaning of items in a working list.
List.entry.item
Element Id | List.entry.item
Definition | A reference to the actual resource from which data was derived.
Cardinality | 1..1
Type | Reference(Any)
List.emptyReason
Element Id | List.emptyReason
Definition | If the list is empty, why the list is empty.
Cardinality | 0..1
Terminology Binding | List Empty Reasons (Preferred)
Type | CodeableConcept
Requirements | Allows capturing things like "none exist" or "not asked" which can be important for most lists.
Comments | The various reasons for an empty list make a significant interpretation to its interpretation. Note that this code is for use when the entire list has been suppressed, and not for when individual items are omitted - implementers may consider using a text note or a flag on an entry in these cases.
Invariants | Affect this element lst-1 Rule A list can only have an emptyReason if it is empty emptyReason.empty() or entry.empty()
Affect this element
lst-1 | Rule | A list can only have an emptyReason if it is empty | emptyReason.empty() or entry.empty()

2.39.13 Resource List - Mappings

URL: https://hl7.org/fhir/R4/list-mappings.html

2.39.13 Resource List - Mappings

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Not Classified | Compartments: Device, Patient, Practitioner

Mappings:

FiveWs Pattern Mapping

RIM Mapping

Mappings for the list resource (see Mappings to Other Standards for further information & status).

2.39.13.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

List
identifier | FiveWs.identifier
status | FiveWs.status
mode | FiveWs.class
code | FiveWs.what[x]
subject | FiveWs.subject[x]

encounter | FiveWs.context
date | FiveWs.recorded
source | FiveWs.author

2.39.13.2 RIM Mapping (http://hl7.org/v3)

List | Act[classCode<ORG,moodCode=EVN]
identifier | .id
status | .status[current=active;retired=obsolete;entered-in-error=nullified]
mode | .outBoundRelationship[typeCode=COMP].target[classCode=OBS"].value
title | .title
code | .code
subject | .participation[typeCode<SUB].role (and sometimes .player)
encounter | inboundRelationship[typeCode=COMP].source[classCode=ENC, moodCode=EVN]
date | .participation[typeCode<AUT].time[type=TS]
source | .participation[typeCode<AUT].role[classCode=REL].player[classCode=PSN,determinerCode=INST] or .participation[typeCode<AUT].role[classCode=REL].player[classCode=DEV,determinerCode=INST]
orderedBy | .outboundRelationship[typeCode=COMP].sequenceNumber > 1
note | .inboundRelationship[typeCode=SUBJ].act.text
entry | .outboundRelationship[typeCode=COMP] or .participation[typeCode=SBJ]
flag | .outBoundRelationship[typeCode=COMP].target[classCode=OBS"].value
deleted | added: .updateMode=("A","AR") retained: .updateMode="NC" updated: .updateMode="R" deleted: .updateMode="D"
date | .availabilityTime
item | .target or .role or .role.entity
emptyReason | .inboundRelationship[typeCode=SUBJ,code<ListEmptyReason].value[type=CD]

2.39.14 Resource List - Extensions & Profiles

URL: https://hl7.org/fhir/R4/list-profiles.html

2.39.14 Resource List - Extensions & Profiles

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Not Classified | Compartments: Device, Patient, Practitioner

This table lists profiles and extensions for the List resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

2.39.14.1 Profiles

No Profiles defined for this resource

2.39.14.2 Extensions

list-changeBase | List | changeBase | for List HL7 Extensions
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

list-changeBase | changeBase | for List HL7 Extensions
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

2.39.14.3 Search Extensions

No Search Extensions defined for this resource

2.39.15 Resource List - Operations

URL: https://hl7.org/fhir/R4/list-operations.html

2.39.15 Resource List - Operations

FHIR Infrastructure Work Group | Maturity Level: 1 | Trial Use | Security Category: Not Classified | Compartments: Device, Patient, Practitioner

This resource has 1 operation operations associated with it:

$find | Find a functional list

For more information about operations, including how they are invoked, see Operations.

2.39.15.1 Operation $find on List

URL: https://hl7.org/fhir/R4/list-operation-find.html

2.39.15.1 Operation $find on List

FHIR Infrastructure Work Group | Maturity Level: 1 | Trial Use | Compartments: N/A

This operation allows a client to find an identified list for a particular function by its function. The operation takes two parameters, the identity of a patient, and the name of a functional list. The list of defined functional lists can be found at Current Resource Lists. Applications are not required to support all the lists, and may define additional lists of their own. If the system is able to locate a list that serves the identified purpose, it returns it as the body of the response with a 200 OK status. If the resource cannot be located, the server returns a 404 not found (optionally with an OperationOutcome resource).

The official URL for this operation definition is

```
http://hl7.org/fhir/OperationDefinition/List-find
```

Formal Definition (as a OperationDefinition).

URL: [base]/List/$find

This is an idempotent operation

In Parameters:

Name	Cardinality	Type	Binding	Profile	Documentation
patient	1..1	id			The id of a patient resource located on the server on which this operation is executed
name	1..1	code			The code for the functional list that is being found

Note that servers may support searching by a functional list, and not support this operation that allows clients to find the list directly

2.39.15.1.1 Examples

Request: Find the current allergy list for patient 'example'

```
GET /ehr/fhir/List/$find?patient=example&name=$current-allergies
```

Response: Return the current allergies list for the patient

```
HTTP/1.1 200 OK
[other headers]

<?xml version="1.0" encoding="UTF-8"?>
<List xmlns="http://hl7.org/fhir">
  <id value="current-allergies"/>
  <text>
    <status value="generated"/>
    <div xmlns="http://www.w3.org/1999/xhtml">
      <p>Patient Peter Chalmers, DOB = Dec 25, 1974, MRN = 12345 (Acme Healthcare) has the following allergies</p>
      <ul>
        <li>cashew nut allergenic extract Injectable Product (<b>High</b>)</li>
        <li>Allergenic extract, penicillin (high)</li>
      </ul>
    </div>
  </text>
  <status value="current"/>
  <mode value="working"/>
  <title value="Current Allergy List"/>
  <code>
    <coding>
      <system value="http://loinc.org"/>
      <code value="52472-8"/>
      <display value="Allergies and Adverse Drug Reactions"/>
    </coding>
    <text value="Current Allergy List"/>
  </code>
  <date value="2015-07-14T23:10:23+11:00"/>
  <source>
    <reference value="Patient/example"/>
  </source>
  <orderedBy>
    <coding>
      <system value="http://terminology.hl7.org/CodeSystem/list-order"/>
      <code value="entry-date"/>
    </coding>
  </orderedBy>
  <entry>
    <item>
      <reference value="AllergyIntolerance/example"/>
    </item>
  </entry>
  <entry>
    <item>
      <reference value="AllergyIntolerance/medication"/>
    </item>
  </entry>
</List>
```

For more information about operations, including how they are invoked, see Operations.

2.39.16 Resource List - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/list-version-maps.html

No validation errors - all conversions are clean

2.39.16 Resource List - R3/R4 Conversions

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Not Classified | Compartments: Device, Patient, Practitioner

R3 : R4 Conversion maps for List.

Functional status for this map: 9 tests that all execute ok. All tests pass round-trip testing and 5 r3 resources are invalid (0 errors). (see documentation)

2.39.16.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/List3to4" = "R3 to R4 Conversions for List"

uses "http://hl7.org/fhir/3.0/StructureDefinition/List" alias ListR3 as source
uses "http://hl7.org/fhir/StructureDefinition/List" alias List as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group List(source src : ListR3, target tgt : List) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.status -> tgt.status;
  src.mode -> tgt.mode;
  src.title -> tgt.title;
  src.code -> tgt.code;
  src.subject -> tgt.subject;
  src.encounter -> tgt.encounter;
  src.date -> tgt.date;
  src.source -> tgt.source;
  src.orderedBy -> tgt.orderedBy;
  src.note -> tgt.note;
  src.entry as s -> tgt.entry as t then entry(s, t);
  src.emptyReason -> tgt.emptyReason;
}

group entry(source src, target tgt) extends BackboneElement {
  src.flag -> tgt.flag;
  src.deleted -> tgt.deleted;
  src.date -> tgt.date;
  src.item -> tgt.item;
}
```

2.39.16.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/List4to3" = "R4 to R3 Conversion for List"

uses "http://hl7.org/fhir/StructureDefinition/List" alias List as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/List" alias ListR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group List(source src : ListR3, target tgt : List) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.status -> tgt.status;
  src.mode -> tgt.mode;
  src.title -> tgt.title;
  src.code -> tgt.code;
  src.subject -> tgt.subject;
  src.encounter -> tgt.encounter;
  src.date -> tgt.date;
  src.source -> tgt.source;
  src.orderedBy -> tgt.orderedBy;
  src.note -> tgt.note;
  src.entry as s -> tgt.entry as t then entry(s, t);
  src.emptyReason -> tgt.emptyReason;
}

group entry(source src, target tgt) extends BackboneElement {
  src.flag -> tgt.flag;
  src.deleted -> tgt.deleted;
  src.date -> tgt.date;
  src.item -> tgt.item;
}
```

2.39.16.3 R4 Validation Errors

2.39.17 List HL7 Extensions

URL: https://hl7.org/fhir/R4/list-extensions.html

Defines common extensions used with or related to the List resource

2.39.17 List HL7 Extensions

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative

2.39.17.1 Content

Extensions:
list-changeBase | changeBase : Reference to the List that a "change" list is asserting changes with respect to.

2.39.18 JSON Schema for List

URL: https://hl7.org/fhir/R4/list.schema.json.html

2.39.18 JSON Schema for List

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Device, Patient, Practitioner

This is a representation of the json schema for List, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/List",
  "$ref": "#/definitions/List",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "List": {
      "description": "A list is a curated collection of resources.",
      "properties": {
        "resourceType": {
          "description": "This is a List resource",
          "const": "List"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
          "description": "A human-readable narrative that contains a summary of the resource and can be used to
    represent the content of the resource to a human. The narrative need not encode all the structured data, but is
    required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
    Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
          "$ref": "Narrative.schema.json#/definitions/Narrative"
        },
        "contained": {
          "description": "These resources do not have an independent existence apart from the resource that contains
    them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
          "items": {
            "$ref": "ResourceList.schema.json#/definitions/ResourceList"
          },
          "type": "array"
        },
        "extension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
     definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
```

```
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "Identifier for the List assigned for business purposes outside the context of FHIR.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "status": {
      "description": "Indicates the current state of this list.",
      "enum": [
        "current",
        "retired",
        "entered-in-error"
      ]
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "mode": {
      "description": "How this list was prepared - whether it is a working list that is suitable for being
maintained on an ongoing basis, or if it represents a snapshot of a list of items from another source, or whether it
 is a prepared list where items may be marked as added, modified or deleted.",
      "enum": [
        "working",
        "snapshot",
        "changes"
      ]
    },
    "_mode": {
      "description": "Extensions for mode",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "title": {
      "description": "A label for the list assigned by the author.",
      "$ref": "#/definitions/string"
    },
    "_title": {
      "description": "Extensions for title",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "code": {
      "description": "This code defines the purpose of the list - why it was created.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "subject": {
      "description": "The common subject (or patient) of the resources that are in the list if there is one.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "encounter": {
      "description": "The encounter that is the context in which this list was created.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "date": {
      "description": "The date that the list was prepared.",
      "$ref": "#/definitions/dateTime"
    },
    "_date": {
      "description": "Extensions for date",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "source": {
      "description": "The entity responsible for deciding what the contents of the list were. Where the list was
created by a human, this is the same as the author of the list.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "orderedBy": {
      "description": "What order applies to the items in the list.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "note": {
      "description": "Comments that apply to the overall list.",
      "items": {
        "$ref": "Annotation.schema.json#/definitions/Annotation"
      },
```

```
        "type": "array"
      },
      "entry": {
        "description": "Entries in this list.",
        "items": {
          "$ref": "#/definitions/List_Entry"
        },
        "type": "array"
      },
      "emptyReason": {
        "description": "If the list is empty, why the list is empty.",
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      }
    },
    "additionalProperties": false,
    "required": [
      "resourceType"
    ]
  },
  "List_Entry": {
    "description": "A list is a curated collection of resources.",
    "properties": {
      "id": {
        "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
        "$ref": "string.schema.json#/definitions/string"
      },
      "extension": {
        "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
        "items": {
          "$ref": "Extension.schema.json#/definitions/Extension"
        },
        "type": "array"
      },
      "modifierExtension": {
        "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
        "items": {
          "$ref": "Extension.schema.json#/definitions/Extension"
        },
        "type": "array"
      },
      "flag": {
        "description": "The flag allows the system constructing the list to indicate the role and significance of the
item in the list.",
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "deleted": {
        "description": "True if this item is marked as deleted in the list.",
        "$ref": "#/definitions/boolean"
      },
      "_deleted": {
        "description": "Extensions for deleted",
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "date": {
        "description": "When this item was added to the list.",
        "$ref": "#/definitions/dateTime"
      },
      "_date": {
        "description": "Extensions for date",
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "item": {
        "description": "A reference to the actual resource from which data was derived.",
        "$ref": "Reference.schema.json#/definitions/Reference"
      }
    },
    "additionalProperties": false,
    "required": [
      "item"
    ]
  }
 }
}
```


2.39.19 Resource List - Maturity Dependency View

URL: https://hl7.org/fhir/R4/list-dependencies.html

2.39.19 Resource List - Maturity Dependency View

FHIR Infrastructure Work Group | Maturity Level: 1 | Trial Use | Compartments: Device, Patient, Practitioner

Dependency Graph for List FMM level 1

Name | Card. | Type | Dependency Analysis
List | | DomainResource
identifier | 0..* | Identifier
status | 1..1 | code
mode | 1..1 | code
title | 0..1 | string
code | 0..1 | CodeableConcept
subject | 0..1 | Reference(Patient | Group | Device | Location)
encounter | 0..1 | Reference(Encounter)
date | 0..1 | dateTime
source | 0..1 | Reference(Practitioner | PractitionerRole | Patient | Device)
orderedBy | 0..1 | CodeableConcept
note | 0..* | Annotation
entry | 0..* | BackboneElement
flag | 0..1 | CodeableConcept
deleted | 0..1 | boolean
date | 0..1 | dateTime
item | 1..1 | Reference(Any)
emptyReason | 0..1 | CodeableConcept

2.39.20 Value Set Sc-valueset-list-status - Status Code Mapping

URL: https://hl7.org/fhir/R4/sc-valueset-list-status.html

2.39.20 Value Set Sc-valueset-list-status - Status Code Mapping

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Device, Patient, Practitioner

This is the narrative for the resource. See also the XML, JSON or Turtle format.

2.39.21 ListStatusCanonicalMap (http://hl7.org/fhir/ConceptMap/sc-list-status)

Mapping from http://hl7.org/fhir/ValueSet/list-status to http://hl7.org/fhir/ValueSet/resource-status

DRAFT. Published on 01/11/2019 9:29:23 AM by HL7 (FHIR Project) (http://hl7.org/fhir , fhir@lists.hl7.org).

Canonical Mapping for "The current state of the list."

Source Code | Equivalence | Destination Code
entered-in-error | equivalent | error
current | equivalent | active
retired | equivalent | inactive

2.41 Resource Composition - Content

URL: https://hl7.org/fhir/R4/composition.html
Structured Documents Work Group | Maturity Level: 2 | Trial Use | Security Category: Not Classified | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

A set of healthcare-related information that is assembled together into a single logical package that provides a single coherent statement of meaning, establishes its own context and that has clinical attestation with regard to who is making the statement. A Composition defines the structure and narrative content necessary for a document. However, a Composition alone does not constitute a document. Rather, the Composition must be the first entry in a Bundle where Bundle.type=document, and any other resources referenced from Composition must be included as subsequent entries in the Bundle (for example Patient, Practitioner, Encounter, etc.).

2.41.1 Scope and Usage

A Composition is the basic structure from which FHIR Documents - immutable bundles with attested narrative - are built. A single logical composition may be associated with a series of derived documents, each of which is a frozen copy of the composition.

Note: EN 13606 uses the term "Composition" to refer to a single commit to an EHR system, and offers some common examples: a composition containing a consultation note, a progress note, a report or a letter, an investigation report, a prescription form or a set of bedside nursing observations. Using Composition for an attested EHR commit is a valid use of the Composition resource, but for FHIR purposes, it would be usual to make more granular updates with individual provenance statements.

The Clinical Document profile constrains Composition to specify a clinical document (matching CDA). See also the comparison with CDA.

2.41.2 Boundaries and Relationships

Composition is a structure for grouping information for purposes of persistence and attestability. There are several other grouping structures in FHIR with distinct purposes:

• The List resource - enumerates a flat collection of resources and provides features for managing the collection. While a particular List instance may represent a "snapshot", from a business process perspective, the notion of "list" is dynamic – items are added and removed over time. The List resource references other resources. Lists may be curated and have specific business meaning.

• The Group resource - defines a group of specific people, animals, devices, etc. by enumerating them, or by describing qualities that group members have. The Group resource refers to other resources, possibly implicitly. Groups are intended to be acted upon or observed as a whole (e.g., performing therapy on a group, calculating risk for a group, etc.). This resource will commonly be used for public health (e.g., describing an at-risk population), clinical trials (e.g., defining a test subject pool) and similar purposes.

• The Bundle resource - is an infrastructure container for a group of resources. It does not have narrative and is used to group collections of resources for transmission, persistence or processing (e.g., messages, documents, transactions, query responses, etc.). The content of bundles is typically algorithmically determined for a particular exchange or persistence purpose.

• The Composition resource - defines a set of healthcare-related information that is assembled together into a single logical document that provides a single coherent statement of meaning, establishes its own context and that has clinical attestation with regard to who is making the statement. The Composition resource provides the basic structure of a FHIR document. The full content of the document is expressed using a Bundle containing the Composition and its entries.

The Composition resource organizes clinical and administrative content into sections, each of which contains a narrative, and references other resources for supporting data. The narrative content of the various sections in a Composition are supported by the resources referenced in the section entries. The complete set of content to make up a document includes the Composition resource together with various resources pointed to or indirectly connected to the Composition, all gathered together into a Bundle for transport and persistence. Resources associated with the following list of Composition references SHALL be included in the Bundle:

• Composition.subject

• Composition.encounter

• Composition.author

• Composition.attester.party

• Composition.custodian

• Composition.event.detail

• Composition.section.author

• Composition.section.focus

• Composition.section.entry

Other resources referred to by those resources MAY be included in the Bundle at the discretion of the authoring system as documented in the system's operation definition (such as $document operation), or as specified by any applicable profiles.

2.41.3 Background and Context

2.41.3.1 Composition Status Codes

Every composition has a status element, which describes the status of the content of the composition, taken from this list of codes:

2.41.3.2 The workflow/clinical status of the composition.

preliminary | This is a preliminary composition or document (also known as initial or interim). The content may be incomplete or unverified.

final | This version of the composition is complete and verified by an appropriate person and no further work is planned. Any subsequent updates would be on a new version of the composition.

amended | The composition content or the referenced resources have been modified (edited or added to) subsequent to being released as "final" and the composition is complete and verified by an authorized person.

entered-in-error | The composition or document was originally created/issued in error, and this is an amendment that marks that the entire series should not be considered as valid.

Composition status generally only moves down through this list - it moves from preliminary to final and then it may progress to amended. Note that in many workflows, only final compositions are made available and the preliminary status is not used.

A very few compositions are created entirely in error in the workflow - usually the composition concerns the wrong patient or is written by the wrong author, and the error is only detected after the composition has been used or documents have been derived from it. To support resolution of this case, the composition is updated to be marked as entered-in-error and a new derived document can be created. This means that the entire series of derived documents is now considered to be created in error and systems receiving derived documents based on retracted compositions SHOULD remove data taken from earlier documents from routine use and/or take other appropriate actions. Systems are not required to provide this workflow or support documents derived from retracted compositions, but they SHALL NOT ignore a status of entered-in-error. Note that systems that handle compositions or derived documents and don't support the error status need to define some other way of handling compositions that are created in error; while this is not a common occurrence, some clinical systems have no provision for removing erroneous information from a patient's record, and there is no way for a user to know that it is not fit for use - this is not safe.

2.41.3.3 Note for CDA aware readers

Many users of this specification are familiar with the Clinical Document Architecture (CDA) and related specifications. CDA is a primary design input to the Composition resource (other principal inputs are other HL7 document specifications and EN13606). There are three important structural differences between CDA and the Composition resource:

• A composition is a logical construct- its identifier matches to the CDA ClinicalDocument.setId. Composition resources are wrapped into Document structures, for exchange of the whole package (the composition and its parts), and this wrapped, sealed entity is equivalent to a CDA document, where the Bundle.id is equivalent in function to ClinicalDocument.id (but it is not identical when interconverting, since it's a transform between them).

• The composition section defines a section (or sub-section) of the document, but unlike CDA, the section entries are actually references to other resources that hold the supporting data content for the section. This design means that the data can be reused in many other ways.

• Unlike CDA, the context defined in the Composition (the confidentiality, subject, author, event, event period and encounter) apply to the composition and do not specifically apply to the resources referenced from the section.entry. There is no context flow model in FHIR, so each resource referenced from within a Composition expresses its own individual context. In this way, clinical content can safely be extracted from the composition.

In addition, note that both the code lists (e.g., Composition.status) and the Composition resource are mapped to HL7 v3 and/or CDA.

This resource is referenced by itself, Contract and Procedure

2.41.4 Resource Content

• Structure

• UML

• XML

• JSON

• Turtle

• R3 Diff

• All

Structure

Name | Flags | Card. | Type | Description & Constraints
Composition | TU | | DomainResource | A set of resources composed into a single coherent clinical statement with clinical attestation Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..1 | Identifier | Version-independent identifier for the Composition
status | ?!Σ | 1..1 | code | preliminary | final | amended | entered-in-error CompositionStatus (Required)
type | Σ | 1..1 | CodeableConcept | Kind of composition (LOINC if possible) FHIR Document Type Codes (Preferred)
category | Σ | 0..* | CodeableConcept | Categorization of Composition Document Class Value Set (Example)
subject | Σ | 0..1 | Reference(Any) | Who and/or what the composition is about
encounter | Σ | 0..1 | Reference(Encounter) | Context of the Composition
date | Σ | 1..1 | dateTime | Composition editing time
author | Σ | 1..* | Reference(Practitioner | PractitionerRole | Device | Patient | RelatedPerson | Organization) | Who and/or what authored the composition
title | Σ | 1..1 | string | Human Readable name/title
confidentiality | Σ | 0..1 | code | As defined by affinity domain V3 Value SetConfidentialityClassification (Required)
attester | | 0..* | BackboneElement | Attests to accuracy of composition
mode | | 1..1 | code | personal | professional | legal | official CompositionAttestationMode (Required)
time | | 0..1 | dateTime | When the composition was attested
party | | 0..1 | Reference(Patient | RelatedPerson | Practitioner | PractitionerRole | Organization) | Who attested the composition
custodian | Σ | 0..1 | Reference(Organization) | Organization which maintains the composition
relatesTo | | 0..* | BackboneElement | Relationships to other compositions/documents

code | | 1..1 | code | replaces | transforms | signs | appends DocumentRelationshipType (Required)
target[x] | | 1..1 | | Target of the relationship
targetIdentifier | | | Identifier
targetReference | | | Reference(Composition)
event | Σ | 0..* | BackboneElement | The clinical service(s) being documented
code | Σ | 0..* | CodeableConcept | Code(s) that apply to the event being documented v3 Code System ActCode (Example)
period | Σ | 0..1 | Period | The period covered by the documentation
detail | Σ | 0..* | Reference(Any) | The event(s) being documented
section | I | 0..* | BackboneElement | Composition is broken into sections + Rule: A section must contain at least one of text, entries, or sub-sections + Rule: A section can only have an emptyReason if it is empty
title | | 0..1 | string | Label for section (e.g. for ToC)
code | | 0..1 | CodeableConcept | Classification of section (recommended) Document Section Codes (Example)
author | | 0..* | Reference(Practitioner | PractitionerRole | Device | Patient | RelatedPerson | Organization) | Who and/or what authored the section
focus | | 0..1 | Reference(Any) | Who/what the section is about, when it is not about the subject of composition
text | I | 0..1 | Narrative | Text summary of the section, for human interpretation
mode | | 0..1 | code | working | snapshot | changes ListMode (Required)
orderedBy | | 0..1 | CodeableConcept | Order of section entries List Order Codes (Preferred)
entry | I | 0..* | Reference(Any) | A reference to data that supports this section
emptyReason | I | 0..1 | CodeableConcept | Why the section is empty List Empty Reasons (Preferred)
section | I | 0..* | see section | Nested Section
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Composition xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..1 Identifier Version-independent identifier for the Composition --></identifier>
 <status value="[code]"/><!-- 1..1 preliminary | final | amended | entered-in-error -->
 <type><!-- 1..1 CodeableConcept Kind of composition (LOINC if possible) --></type>
 <category><!-- 0..* CodeableConcept Categorization of Composition --></category>
 <subject><!-- 0..1 Reference(Any) Who and/or what the composition is about --></subject>
 <encounter><!-- 0..1 Reference(Encounter) Context of the Composition --></encounter>
 <date value="[dateTime]"/><!-- 1..1 Composition editing time -->
 <author><!-- 1..* Reference(Practitioner|PractitionerRole|Device|Patient|
   RelatedPerson|Organization) Who and/or what authored the composition --></author>
 <title value="[string]"/><!-- 1..1 Human Readable name/title -->
 <confidentiality value="[code]"/><!-- 0..1 As defined by affinity domain -->
 <attester>  <!-- 0..* Attests to accuracy of composition -->
  <mode value="[code]"/><!-- 1..1 personal | professional | legal | official -->
  <time value="[dateTime]"/><!-- 0..1 When the composition was attested -->
  <party><!-- 0..1 Reference(Patient|RelatedPerson|Practitioner|PractitionerRole|
    Organization) Who attested the composition --></party>
 </attester>
 <custodian><!-- 0..1 Reference(Organization) Organization which maintains the composition --></custodian>
 <relatesTo>  <!-- 0..* Relationships to other compositions/documents -->
  <code value="[code]"/><!-- 1..1 replaces | transforms | signs | appends -->
  <target[x]><!-- 1..1 Identifier|Reference(Composition) Target of the relationship --></target[x]>
 </relatesTo>
 <event>  <!-- 0..* The clinical service(s) being documented -->
  <code><!-- 0..* CodeableConcept Code(s) that apply to the event being documented --></code>
  <period><!-- 0..1 Period The period covered by the documentation --></period>
  <detail><!-- 0..* Reference(Any) The event(s) being documented --></detail>
 </event>
 <section>  <!-- 0..* Composition is broken into sections -->
  <title value="[string]"/><!-- 0..1 Label for section (e.g. for ToC) -->
  <code><!-- 0..1 CodeableConcept Classification of section (recommended) --></code>
  <author><!-- 0..* Reference(Practitioner|PractitionerRole|Device|Patient|
    RelatedPerson|Organization) Who and/or what authored the section --></author>
  <focus><!-- 0..1 Reference(Any) Who/what the section is about, when it is not about the subject of composition
    --></focus>
  <text><!--  0..1 Narrative Text summary of the section, for human interpretation --></text>
  <mode value="[code]"/><!-- 0..1 working | snapshot | changes -->
  <orderedBy><!-- 0..1 CodeableConcept Order of section entries --></orderedBy>
  <entry><!--  0..* Reference(Any) A reference to data that supports this section --></entry>
  <emptyReason><!--  0..1 CodeableConcept Why the section is empty --></emptyReason>
  <section><!--  0..* Content as for Composition.section Nested Section --></section>
 </section>
</Composition>
```

JSON Template

```
{
  "resourceType" : "Composition",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : { Identifier }, // Version-independent identifier for the Composition
  "status" : "<code>", // R!  preliminary | final | amended | entered-in-error
  "type" : { CodeableConcept }, // R!  Kind of composition (LOINC if possible)
  "category" : [{ CodeableConcept }], // Categorization of Composition
  "subject" : { Reference(Any) }, // Who and/or what the composition is about
  "encounter" : { Reference(Encounter) }, // Context of the Composition
  "date" : "<dateTime>", // R!  Composition editing time
```

```
  "author" : [{ Reference(Practitioner|PractitionerRole|Device|Patient|
   RelatedPerson|Organization) }], // R!  Who and/or what authored the composition
  "title" : "<string>", // R!  Human Readable name/title
  "confidentiality" : "<code>", // As defined by affinity domain
  "attester" : [{ // Attests to accuracy of composition
    "mode" : "<code>", // R!  personal | professional | legal | official
    "time" : "<dateTime>", // When the composition was attested
    "party" : { Reference(Patient|RelatedPerson|Practitioner|PractitionerRole|
    Organization) } // Who attested the composition
  }],
  "custodian" : { Reference(Organization) }, // Organization which maintains the composition
  "relatesTo" : [{ // Relationships to other compositions/documents
    "code" : "<code>", // R!  replaces | transforms | signs | appends
    // target[x]: Target of the relationship. One of these 2:
    "targetIdentifier" : { Identifier }
    "targetReference" : { Reference(Composition) }
  }],
  "event" : [{ // The clinical service(s) being documented
    "code" : [{ CodeableConcept }], // Code(s) that apply to the event being documented
    "period" : { Period }, // The period covered by the documentation
    "detail" : [{ Reference(Any) }] // The event(s) being documented
  }],
  "section" : [{ // Composition is broken into sections
    "title" : "<string>", // Label for section (e.g. for ToC)
    "code" : { CodeableConcept }, // Classification of section (recommended)
    "author" : [{ Reference(Practitioner|PractitionerRole|Device|Patient|
    RelatedPerson|Organization) }], // Who and/or what authored the section
    "focus" : { Reference(Any) }, // Who/what the section is about, when it is not about the subject of composition
    "text" : { Narrative }, // C? Text summary of the section, for human interpretation
    "mode" : "<code>", // working | snapshot | changes
    "orderedBy" : { CodeableConcept }, // Order of section entries
    "entry" : [{ Reference(Any) }], // C? A reference to data that supports this section
    "emptyReason" : { CodeableConcept }, // C? Why the section is empty
    "section" : [{ Content as for Composition.section }] // C? Nested Section
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Composition;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Composition.identifier [ Identifier ]; # 0..1 Version-independent identifier for the Composition
  fhir:Composition.status [ code ]; # 1..1 preliminary | final | amended | entered-in-error
  fhir:Composition.type [ CodeableConcept ]; # 1..1 Kind of composition (LOINC if possible)
  fhir:Composition.category [ CodeableConcept ], ... ; # 0..* Categorization of Composition
  fhir:Composition.subject [ Reference(Any) ]; # 0..1 Who and/or what the composition is about
  fhir:Composition.encounter [ Reference(Encounter) ]; # 0..1 Context of the Composition
  fhir:Composition.date [ dateTime ]; # 1..1 Composition editing time
  fhir:Composition.author [ Reference(Practitioner|PractitionerRole|Device|Patient|RelatedPerson|Organization) ], ... ;
    # 1..* Who and/or what authored the composition
  fhir:Composition.title [ string ]; # 1..1 Human Readable name/title
  fhir:Composition.confidentiality [ code ]; # 0..1 As defined by affinity domain
  fhir:Composition.attester [ # 0..* Attests to accuracy of composition
    fhir:Composition.attester.mode [ code ]; # 1..1 personal | professional | legal | official
    fhir:Composition.attester.time [ dateTime ]; # 0..1 When the composition was attested
    fhir:Composition.attester.party [ Reference(Patient|RelatedPerson|Practitioner|PractitionerRole|Organization) ]; #
    0..1 Who attested the composition
  ], ...;
  fhir:Composition.custodian [ Reference(Organization) ]; # 0..1 Organization which maintains the composition
  fhir:Composition.relatesTo [ # 0..* Relationships to other compositions/documents
    fhir:Composition.relatesTo.code [ code ]; # 1..1 replaces | transforms | signs | appends
    # Composition.relatesTo.target[x] : 1..1 Target of the relationship. One of these 2
      fhir:Composition.relatesTo.targetIdentifier [ Identifier ]
      fhir:Composition.relatesTo.targetReference [ Reference(Composition) ]
  ], ...;
  fhir:Composition.event [ # 0..* The clinical service(s) being documented
    fhir:Composition.event.code [ CodeableConcept ], ... ; # 0..* Code(s) that apply to the event being documented
    fhir:Composition.event.period [ Period ]; # 0..1 The period covered by the documentation
    fhir:Composition.event.detail [ Reference(Any) ], ... ; # 0..* The event(s) being documented
  ], ...;
  fhir:Composition.section [ # 0..* Composition is broken into sections
    fhir:Composition.section.title [ string ]; # 0..1 Label for section (e.g. for ToC)
    fhir:Composition.section.code [ CodeableConcept ]; # 0..1 Classification of section (recommended)
    fhir:Composition.section.author [ Reference(Practitioner|PractitionerRole|Device|Patient|RelatedPerson|Organization)
     ], ... ; # 0..* Who and/or what authored the section
    fhir:Composition.section.focus [ Reference(Any) ]; # 0..1 Who/what the section is about, when it is not about the
    subject of composition
    fhir:Composition.section.text [ Narrative ]; # 0..1 Text summary of the section, for human interpretation
    fhir:Composition.section.mode [ code ]; # 0..1 working | snapshot | changes
    fhir:Composition.section.orderedBy [ CodeableConcept ]; # 0..1 Order of section entries
    fhir:Composition.section.entry [ Reference(Any) ], ... ; # 0..* A reference to data that supports this section
    fhir:Composition.section.emptyReason [ CodeableConcept ]; # 0..1 Why the section is empty
    fhir:Composition.section.section [ See Composition.section ], ... ; # 0..* Nested Section
  ], ...;
```

]

Changes since R3

Composition
Composition.status | Change value set from http://hl7.org/fhir/ValueSet/composition-status to http://hl7.org/fhir/ValueSet/composition-status|4.0.1
Composition.category | Renamed from class to category Max Cardinality changed from 1 to *
Composition.subject | Min Cardinality changed from 1 to 0
Composition.author | Type Reference: Added Target Types PractitionerRole, Organization
Composition.confidentiality | Change value set from http://terminology.hl7.org/ValueSet/v3-ConfidentialityClassification to http://terminology.hl7.org/ValueSet/v3-ConfidentialityClassification|2014-03-26 No longer marked as Modifier
Composition.attester.mode | Max Cardinality changed from * to 1 Change value set from http://hl7.org/fhir/ValueSet/composition-attestation-mode to http://hl7.org/fhir/ValueSet/composition-attestation-mode|4.0.1
Composition.attester.party | Type Reference: Added Target Types RelatedPerson, PractitionerRole
Composition.relatesTo.code | Change value set from http://hl7.org/fhir/ValueSet/document-relationship-type to http://hl7.org/fhir/ValueSet/document-relationship-type|4.0.1
Composition.section.author | Added Element
Composition.section.focus | Added Element
Composition.section.mode | Change value set from http://hl7.org/fhir/ValueSet/list-mode to http://hl7.org/fhir/ValueSet/list-mode|4.0.1 No longer marked as Modifier

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 1 test that all execute ok. All tests pass round-trip testing and all r3 resources are valid.)

Structure

Name | Flags | Card. | Type | Description & Constraints
Composition | TU | | DomainResource | A set of resources composed into a single coherent clinical statement with clinical attestation Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..1 | Identifier | Version-independent identifier for the Composition
status | ?!Σ | 1..1 | code | preliminary | final | amended | entered-in-error CompositionStatus (Required)
type | Σ | 1..1 | CodeableConcept | Kind of composition (LOINC if possible) FHIR Document Type Codes (Preferred)
category | Σ | 0..* | CodeableConcept | Categorization of Composition Document Class Value Set (Example)
subject | Σ | 0..1 | Reference(Any) | Who and/or what the composition is about
encounter | Σ | 0..1 | Reference(Encounter) | Context of the Composition
date | Σ | 1..1 | dateTime | Composition editing time
author | Σ | 1..* | Reference(Practitioner | PractitionerRole | Device | Patient | RelatedPerson | Organization) | Who and/or what authored the composition
title | Σ | 1..1 | string | Human Readable name/title
confidentiality | Σ | 0..1 | code | As defined by affinity domain V3 Value SetConfidentialityClassification (Required)
attester | | 0..* | BackboneElement | Attests to accuracy of composition
mode | | 1..1 | code | personal | professional | legal | official CompositionAttestationMode (Required)
time | | 0..1 | dateTime | When the composition was attested
party | | 0..1 | Reference(Patient | RelatedPerson | Practitioner | PractitionerRole | Organization) | Who attested the composition
custodian | Σ | 0..1 | Reference(Organization) | Organization which maintains the composition
relatesTo | | 0..* | BackboneElement | Relationships to other compositions/documents
code | | 1..1 | code | replaces | transforms | signs | appends DocumentRelationshipType (Required)
target[x] | | 1..1 | | Target of the relationship
targetIdentifier | | | Identifier
targetReference | | | Reference(Composition)
event | Σ | 0..* | BackboneElement | The clinical service(s) being documented
code | Σ | 0..* | CodeableConcept | Code(s) that apply to the event being documented v3 Code System ActCode (Example)
period | Σ | 0..1 | Period | The period covered by the documentation
detail | Σ | 0..* | Reference(Any) | The event(s) being documented
section | I | 0..* | BackboneElement | Composition is broken into sections + Rule: A section must contain at least one of text, entries, or sub-sections + Rule: A section can only have an emptyReason if it is empty
title | | 0..1 | string | Label for section (e.g. for ToC)
code | | 0..1 | CodeableConcept | Classification of section (recommended) Document Section Codes (Example)
author | | 0..* | Reference(Practitioner | PractitionerRole | Device | Patient | RelatedPerson | Organization) | Who and/or what authored the section
focus | | 0..1 | Reference(Any) | Who/what the section is about, when it is not about the subject of composition
text | I | 0..1 | Narrative | Text summary of the section, for human interpretation
mode | | 0..1 | code | working | snapshot | changes ListMode (Required)
orderedBy | | 0..1 | CodeableConcept | Order of section entries List Order Codes (Preferred)
entry | I | 0..* | Reference(Any) | A reference to data that supports this section
emptyReason | I | 0..1 | CodeableConcept | Why the section is empty List Empty Reasons (Preferred)
section | I | 0..* | see section | Nested Section
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Composition xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
```

```
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..1 Identifier Version-independent identifier for the Composition --></identifier>
 <status value="[code]"/><!-- 1..1 preliminary | final | amended | entered-in-error -->
 <type><!-- 1..1 CodeableConcept Kind of composition (LOINC if possible) --></type>
 <category><!-- 0..* CodeableConcept Categorization of Composition --></category>
 <subject><!-- 0..1 Reference(Any) Who and/or what the composition is about --></subject>
 <encounter><!-- 0..1 Reference(Encounter) Context of the Composition --></encounter>
 <date value="[dateTime]"/><!-- 1..1 Composition editing time -->
 <author><!-- 1..* Reference(Practitioner|PractitionerRole|Device|Patient|
   RelatedPerson|Organization) Who and/or what authored the composition --></author>
 <title value="[string]"/><!-- 1..1 Human Readable name/title -->
 <confidentiality value="[code]"/><!-- 0..1 As defined by affinity domain -->
 <attester>  <!-- 0..* Attests to accuracy of composition -->
  <mode value="[code]"/><!-- 1..1 personal | professional | legal | official -->
  <time value="[dateTime]"/><!-- 0..1 When the composition was attested -->
  <party><!-- 0..1 Reference(Patient|RelatedPerson|Practitioner|PractitionerRole|
    Organization) Who attested the composition --></party>
 </attester>
 <custodian><!-- 0..1 Reference(Organization) Organization which maintains the composition --></custodian>
 <relatesTo>  <!-- 0..* Relationships to other compositions/documents -->
  <code value="[code]"/><!-- 1..1 replaces | transforms | signs | appends -->
  <target[x]><!-- 1..1 Identifier|Reference(Composition) Target of the relationship --></target[x]>
 </relatesTo>
 <event>  <!-- 0..* The clinical service(s) being documented -->
  <code><!-- 0..* CodeableConcept Code(s) that apply to the event being documented --></code>
  <period><!-- 0..1 Period The period covered by the documentation --></period>
  <detail><!-- 0..* Reference(Any) The event(s) being documented --></detail>
 </event>
 <section>  <!-- 0..* Composition is broken into sections -->
  <title value="[string]"/><!-- 0..1 Label for section (e.g. for ToC) -->
  <code><!-- 0..1 CodeableConcept Classification of section (recommended) --></code>
  <author><!-- 0..* Reference(Practitioner|PractitionerRole|Device|Patient|
    RelatedPerson|Organization) Who and/or what authored the section --></author>
  <focus><!-- 0..1 Reference(Any) Who/what the section is about, when it is not about the subject of composition
    --></focus>
  <text><!--  0..1 Narrative Text summary of the section, for human interpretation --></text>
  <mode value="[code]"/><!-- 0..1 working | snapshot | changes -->
  <orderedBy><!-- 0..1 CodeableConcept Order of section entries --></orderedBy>
  <entry><!--  0..* Reference(Any) A reference to data that supports this section --></entry>
  <emptyReason><!--  0..1 CodeableConcept Why the section is empty --></emptyReason>
  <section><!--  0..* Content as for Composition.section Nested Section --></section>
 </section>
</Composition>
```

JSON Template

```
{
  "resourceType" : "Composition",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : { Identifier }, // Version-independent identifier for the Composition
  "status" : "<code>", // R!  preliminary | final | amended | entered-in-error
  "type" : { CodeableConcept }, // R!  Kind of composition (LOINC if possible)
  "category" : [{ CodeableConcept }], // Categorization of Composition
  "subject" : { Reference(Any) }, // Who and/or what the composition is about
  "encounter" : { Reference(Encounter) }, // Context of the Composition
  "date" : "<dateTime>", // R!  Composition editing time
  "author" : [{ Reference(Practitioner|PractitionerRole|Device|Patient|
   RelatedPerson|Organization) }], // R!  Who and/or what authored the composition
  "title" : "<string>", // R!  Human Readable name/title
  "confidentiality" : "<code>", // As defined by affinity domain
  "attester" : [{ // Attests to accuracy of composition
    "mode" : "<code>", // R!  personal | professional | legal | official
    "time" : "<dateTime>", // When the composition was attested
    "party" : { Reference(Patient|RelatedPerson|Practitioner|PractitionerRole|
    Organization) } // Who attested the composition
  }],
  "custodian" : { Reference(Organization) }, // Organization which maintains the composition
  "relatesTo" : [{ // Relationships to other compositions/documents
    "code" : "<code>", // R!  replaces | transforms | signs | appends
    // target[x]: Target of the relationship. One of these 2:
    "targetIdentifier" : { Identifier }
    "targetReference" : { Reference(Composition) }
  }],
  "event" : [{ // The clinical service(s) being documented
    "code" : [{ CodeableConcept }], // Code(s) that apply to the event being documented
    "period" : { Period }, // The period covered by the documentation
    "detail" : [{ Reference(Any) }] // The event(s) being documented
  }],
  "section" : [{ // Composition is broken into sections
    "title" : "<string>", // Label for section (e.g. for ToC)
    "code" : { CodeableConcept }, // Classification of section (recommended)
    "author" : [{ Reference(Practitioner|PractitionerRole|Device|Patient|
    RelatedPerson|Organization) }], // Who and/or what authored the section
    "focus" : { Reference(Any) }, // Who/what the section is about, when it is not about the subject of composition
    "text" : { Narrative }, // C? Text summary of the section, for human interpretation
    "mode" : "<code>", // working | snapshot | changes
    "orderedBy" : { CodeableConcept }, // Order of section entries
    "entry" : [{ Reference(Any) }], // C? A reference to data that supports this section
    "emptyReason" : { CodeableConcept }, // C? Why the section is empty
```

```
    "section" : [{ Content as for Composition.section }] // C? Nested Section
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Composition;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Composition.identifier [ Identifier ]; # 0..1 Version-independent identifier for the Composition
  fhir:Composition.status [ code ]; # 1..1 preliminary | final | amended | entered-in-error
  fhir:Composition.type [ CodeableConcept ]; # 1..1 Kind of composition (LOINC if possible)
  fhir:Composition.category [ CodeableConcept ], ... ; # 0..* Categorization of Composition
  fhir:Composition.subject [ Reference(Any) ]; # 0..1 Who and/or what the composition is about
  fhir:Composition.encounter [ Reference(Encounter) ]; # 0..1 Context of the Composition
  fhir:Composition.date [ dateTime ]; # 1..1 Composition editing time
  fhir:Composition.author [ Reference(Practitioner|PractitionerRole|Device|Patient|RelatedPerson|Organization) ], ... ;
    # 1..* Who and/or what authored the composition
  fhir:Composition.title [ string ]; # 1..1 Human Readable name/title
  fhir:Composition.confidentiality [ code ]; # 0..1 As defined by affinity domain
  fhir:Composition.attester [ # 0..* Attests to accuracy of composition
    fhir:Composition.attester.mode [ code ]; # 1..1 personal | professional | legal | official
    fhir:Composition.attester.time [ dateTime ]; # 0..1 When the composition was attested
    fhir:Composition.attester.party [ Reference(Patient|RelatedPerson|Practitioner|PractitionerRole|Organization) ]; #
    0..1 Who attested the composition
  ], ...;
  fhir:Composition.custodian [ Reference(Organization) ]; # 0..1 Organization which maintains the composition
  fhir:Composition.relatesTo [ # 0..* Relationships to other compositions/documents
    fhir:Composition.relatesTo.code [ code ]; # 1..1 replaces | transforms | signs | appends
    # Composition.relatesTo.target[x] : 1..1 Target of the relationship. One of these 2
      fhir:Composition.relatesTo.targetIdentifier [ Identifier ]
      fhir:Composition.relatesTo.targetReference [ Reference(Composition) ]
  ], ...;
  fhir:Composition.event [ # 0..* The clinical service(s) being documented
    fhir:Composition.event.code [ CodeableConcept ], ... ; # 0..* Code(s) that apply to the event being documented
    fhir:Composition.event.period [ Period ]; # 0..1 The period covered by the documentation
    fhir:Composition.event.detail [ Reference(Any) ], ... ; # 0..* The event(s) being documented
  ], ...;
  fhir:Composition.section [ # 0..* Composition is broken into sections
    fhir:Composition.section.title [ string ]; # 0..1 Label for section (e.g. for ToC)
    fhir:Composition.section.code [ CodeableConcept ]; # 0..1 Classification of section (recommended)
    fhir:Composition.section.author [ Reference(Practitioner|PractitionerRole|Device|Patient|RelatedPerson|Organization)
     ], ... ; # 0..* Who and/or what authored the section
    fhir:Composition.section.focus [ Reference(Any) ]; # 0..1 Who/what the section is about, when it is not about the
    subject of composition
    fhir:Composition.section.text [ Narrative ]; # 0..1 Text summary of the section, for human interpretation
    fhir:Composition.section.mode [ code ]; # 0..1 working | snapshot | changes
    fhir:Composition.section.orderedBy [ CodeableConcept ]; # 0..1 Order of section entries
    fhir:Composition.section.entry [ Reference(Any) ], ... ; # 0..* A reference to data that supports this section
    fhir:Composition.section.emptyReason [ CodeableConcept ]; # 0..1 Why the section is empty
    fhir:Composition.section.section [ See Composition.section ], ... ; # 0..* Nested Section
  ], ...;
]
```

Changes since Release 3

Composition
Composition.status | Change value set from http://hl7.org/fhir/ValueSet/composition-status to http://hl7.org/fhir/ValueSet/composition-status|4.0.1
Composition.category | Renamed from class to category Max Cardinality changed from 1 to *
Composition.subject | Min Cardinality changed from 1 to 0
Composition.author | Type Reference: Added Target Types PractitionerRole, Organization
Composition.confidentiality | Change value set from http://terminology.hl7.org/ValueSet/v3-ConfidentialityClassification to http://terminology.hl7.org/ValueSet/v3-ConfidentialityClassification|2014-03-26 No longer marked as Modifier
Composition.attester.mode | Max Cardinality changed from * to 1 Change value set from http://hl7.org/fhir/ValueSet/composition-attestation-mode to http://hl7.org/fhir/ValueSet/composition-attestation-mode|4.0.1
Composition.attester.party | Type Reference: Added Target Types RelatedPerson, PractitionerRole
Composition.relatesTo.code | Change value set from http://hl7.org/fhir/ValueSet/document-relationship-type to http://hl7.org/fhir/ValueSet/document-relationship-type|4.0.1
Composition.section.author | Added Element
Composition.section.focus | Added Element
Composition.section.mode | Change value set from http://hl7.org/fhir/ValueSet/list-mode to http://hl7.org/fhir/ValueSet/list-mode|4.0.1 No longer marked as Modifier

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 1 test that all execute ok. All tests pass round-trip testing and all r3 resources are valid.)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

2.41.4.1 Terminology Bindings

Path | Definition | Type | Reference
Composition.status | The workflow/clinical status of the composition. | Required | CompositionStatus
Composition.type | Type of a composition. | Preferred | FHIRDocumentTypeCodes
Composition.category | High-level kind of a clinical document at a macro level. | Example | DocumentClassValueSet
Composition.confidentiality | Codes specifying the level of confidentiality of the composition. | Required | v3.ConfidentialityClassification
Composition.attester.mode | The way in which a person authenticated a composition. | Required | CompositionAttestationMode
Composition.relatesTo.code | The type of relationship between documents. | Required | DocumentRelationshipType
Composition.event.code | This list of codes represents the main clinical acts being documented. | Example | v3.ActCode
Composition.section.code | Classification of a section of a composition/document. | Example | DocumentSectionCodes
Composition.section.mode | The processing mode that applies to this section. | Required | ListMode
Composition.section.orderedBy | What order applies to the items in the entry. | Preferred | ListOrderCodes
Composition.section.emptyReason | If a section is empty, why it is empty. | Preferred | ListEmptyReasons

2.41.4.2 Constraints

id | Level | Location | Description | Expression
cmp-1 | Rule | Composition.section | A section must contain at least one of text, entries, or sub-sections | text.exists() or entry.exists() or section.exists()
cmp-2 | Rule | Composition.section | A section can only have an emptyReason if it is empty | emptyReason.empty() or entry.empty()

2.41.5 Notes:

• The author and the attester are often the same person, but this might not be the case in some clinical workflows.

• The attester attests contents of the document resource, the subject resource and the resources referred to in the Composition.section.content references. Because documents are often derived Compositions and the attestation from the composition is held to apply to the document, the method for presenting a document should be used when/if attesters review the content of the composition.

• The custodian is responsible for the maintenance of the composition and any documents derived from it. With regard to the documents, they are responsible for the policy regarding persistence of the documents. Although they need not actually retain a copy of the document, they SHOULD do so.

• It is acceptable for a Composition to contain only narrative (Composition.section.text) and no entries (Composition.section.entry)

2.41.6 Compositions about multiple entities

Typically, a composition is made about the subject - e.g. a patient, or group of patients, location, or device - and the distinction between the subject and the composition describes the subject. Some kinds of documents contain data about other parties or entities that are relevant to the subject of the document. Some examples:

• A neonatal discharge summary that contains information about the mother

• A family history document that contains multiple sections for different family members

• A social health evaluation document that contains information about the patient's family members

• A procedure report that contains details about a device implanted in the patient

In all these cases, the subject of the document is a single patient, but some of the details are actually related to other persons or entities. When this happens, these other entities are detailed in the Composition.section.focus element. In the absence of a focus, it is assumed that the subject of the composition is the focus.

If a focus is specified, then the resources referred to from the section SHALL either:

• specify that their subject (however named e.g. patient) or focus element (if present) is the focus indicated

• not have a subject element

A few compositions genuinely cover multiple subjects - different sections are about different subjects. In such case, Composition.subject is omitted, and the extension section-subject is used on each section to indicate the subject.

Trial-Use Note: Feedback is welcome on two issues related to Composition:

• For many compositions, the title is the same as the text or a display name of Composition.type (e.g., a "consultation" or "progress note"). Note that CDA does not make title mandatory, but there are no known cases where it is useful for title to be omitted, so it is mandatory here during the trial use period.

• A client can ask a server to generate a fully bundled document from a Composition resource using the $snapshot operation. This operation definition does not resolve the question how document signatures are created. This is an open issue during the period of trial use, and feedback is requested regarding this question.

Feedback here .

2.41.7 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
attester | reference | Who attested the composition | Composition.attester.party (Practitioner, Organization, Patient, PractitionerRole, RelatedPerson)
author | reference | Who and/or what authored the composition | Composition.author (Practitioner, Organization, Device, Patient, PractitionerRole, RelatedPerson)
category | token | Categorization of Composition | Composition.category
confidentiality | token | As defined by affinity domain | Composition.confidentiality
context | token | Code(s) that apply to the event being documented | Composition.event.code
date | date | Composition editing time | Composition.date | 17 Resources
encounter | reference | Context of the Composition | Composition.encounter (Encounter) | 12 Resources
entry | reference | A reference to data that supports this section | Composition.section.entry (Any)
identifier | token | Version-independent identifier for the Composition | Composition.identifier | 30 Resources
patient | reference | Who and/or what the composition is about | Composition.subject.where(resolve() is Patient) (Patient) | 33 Resources
period | date | The period covered by the documentation | Composition.event.period
related-id | token | Target of the relationship | (Composition.relatesTo.target as Identifier)
related-ref | reference | Target of the relationship | (Composition.relatesTo.target as Reference) (Composition)
section | token | Classification of section (recommended) | Composition.section.code
status | token | preliminary | final | amended | entered-in-error | Composition.status
subject | reference | Who and/or what the composition is about | Composition.subject (Any)
title | string | Human Readable name/title | Composition.title
type | token | Kind of composition (LOINC if possible) | Composition.type | 5 Resources

2.41.8 Resource Composition - Examples

URL: https://hl7.org/fhir/R4/composition-examples.html

2.41.8 Resource Composition - Examples

Structured Documents Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Not Classified | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

Example Name | id | Format
Simple Example of a Composition | example | XML | JSON | Turtle
Example of a discharge summary | father | XML | JSON | Turtle
Example of a composition covering details from more than one patient | example-mixed | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

2.41.9 Resource Composition - Detailed Descriptions

URL: https://hl7.org/fhir/R4/composition-definitions.html

2.41.9 Resource Composition - Detailed Descriptions

Structured Documents Work Group | Maturity Level: 2 | Trial Use | Security Category: Not Classified | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

Detailed Descriptions for the elements in the Composition resource.

Composition
Element Id | Composition
Definition | A set of healthcare-related information that is assembled together into a single logical package that provides a single coherent statement of meaning, establishes its own context and that has clinical attestation with regard to who is making the statement. A Composition defines the structure and narrative content necessary for a document. However, a Composition alone does not constitute a document. Rather, the Composition must be the first entry in a Bundle where Bundle.type=document, and any other resources referenced from Composition must be included as subsequent entries in the Bundle (for example Patient, Practitioner, Encounter, etc.).
Cardinality | 0..*
Type | DomainResource
Requirements | To support documents, and also to capture the EN13606 notion of an attested commit to the patient EHR, and to allow a set of disparate resources at the information/engineering level to be gathered into a clinical statement.
Comments | While the focus of this specification is on patient-specific clinical statements, this resource can also apply to other healthcare-related statements such as study protocol designs, healthcare invoices and other activities that are not necessarily patient-specific or clinical.
Composition.identifier
Element Id | Composition.identifier
Definition | A version-independent identifier for the Composition. This identifier stays constant as the composition is changed over time.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..1
Type | Identifier

Summary | true
Comments | Similar to ClinicalDocument/setId in CDA. See discussion in resource definition for how these relate.
Composition.status
Element Id | Composition.status
Definition | The workflow/clinical status of this composition. The status is a marker for the clinical standing of the document.
Cardinality | 1..1
Terminology Binding | CompositionStatus (Required)
Type | code
Is Modifier | true (Reason: This element is labelled as a modifier because it is a status element that contains status entered-in-error which means that the resource should not be treated as valid)
Requirements | Need to be able to mark interim, amended, or withdrawn compositions or documents.
Summary | true
Comments | If a composition is marked as withdrawn, the compositions/documents in the series, or data from the composition or document series, should never be displayed to a user without being clearly marked as untrustworthy. The flag "entered-in-error" is why this element is labeled as a modifier of other elements. Some reporting work flows require that the original narrative of a final document never be altered; instead, only new narrative can be added. The composition resource has no explicit status for explicitly noting whether this business rule is in effect. This would be handled by an extension if required.
Composition.type
Element Id | Composition.type
Definition | Specifies the particular kind of composition (e.g. History and Physical, Discharge Summary, Progress Note). This usually equates to the purpose of making the composition.
Cardinality | 1..1
Terminology Binding | FHIR Document Type Codes (Preferred)
Type | CodeableConcept
Requirements | Key metadata element describing the composition, used in searching/filtering.
Summary | true
Comments | For Composition type, LOINC is ubiquitous and strongly endorsed by HL7. Most implementation guides will require a specific LOINC code, or use LOINC as an extensible binding.
Composition.category
Element Id | Composition.category
Definition | A categorization for the type of the composition - helps for indexing and searching. This may be implied by or derived from the code specified in the Composition Type.
Cardinality | 0..*
Terminology Binding | Document Class Value Set (Example)
Type | CodeableConcept
Requirements | Helps humans to assess whether the composition is of interest when viewing an index of compositions or documents.
Summary | true
Comments | This is a metadata field from XDS/MHD .
Composition.subject
Element Id | Composition.subject
Definition | Who or what the composition is about. The composition can be about a person, (patient or healthcare practitioner), a device (e.g. a machine) or even a group of subjects (such as a document about a herd of livestock, or a set of patients that share a common exposure).
Cardinality | 0..1
Type | Reference(Any)
Requirements | Essential metadata for searching for the composition. Identifies who and/or what the composition/document is about.
Summary | true
Comments | For clinical documents, this is usually the patient.
Composition.encounter
Element Id | Composition.encounter
Definition | Describes the clinical encounter or type of care this documentation is associated with.
Cardinality | 0..1
Type | Reference(Encounter)
Requirements | Provides context for the composition and supports searching.
Summary | true
Composition.date
Element Id | Composition.date
Definition | The composition editing time, when the composition was last logically changed by the author.
Cardinality | 1..1
Type | dateTime
Requirements | dateTime is used for tracking, organizing versions and searching. Note that this is the time of authoring. When packaged in a document, Bundle.timestamp is the date of packaging.
Summary | true
Comments | The Last Modified Date on the composition may be after the date of the document was attested without being changed.
Composition.author
Element Id | Composition.author
Definition | Identifies who is responsible for the information in the composition, not necessarily who typed it in.
Cardinality | 1..*
Type | Reference(Practitioner | PractitionerRole | Device | Patient | RelatedPerson | Organization)
Requirements | Identifies who is responsible for the content.

Summary | true
Composition.title
Element Id | Composition.title
Definition | Official human-readable label for the composition.
Cardinality | 1..1
Type | string
Summary | true
Comments | For many compositions, the title is the same as the text or a display name of Composition.type (e.g. a "consultation" or "progress note"). Note that CDA does not make title mandatory, but there are no known cases where it is useful for title to be omitted, so it is mandatory here. Feedback on this requirement is welcome during the trial use period.
Composition.confidentiality
Element Id | Composition.confidentiality
Definition | The code specifying the level of confidentiality of the Composition.
Cardinality | 0..1
Terminology Binding | V3 Value SetConfidentialityClassification (Required)
Type | code
Summary | true
Comments | The exact use of this element, and enforcement and issues related to highly sensitive documents are out of scope for the base specification, and delegated to implementation profiles (see security section). This element is labeled as a modifier because highly confidential documents must not be treated as if they are not.
Composition.attester
Element Id | Composition.attester
Definition | A participant who has attested to the accuracy of the composition/document.
Cardinality | 0..*
Requirements | Identifies responsibility for the accuracy of the composition content.
Comments | Only list each attester once.
Composition.attester.mode
Element Id | Composition.attester.mode
Definition | The type of attestation the authenticator offers.
Cardinality | 1..1
Terminology Binding | CompositionAttestationMode (Required)
Type | code
Requirements | Indicates the level of authority of the attestation.
Composition.attester.time
Element Id | Composition.attester.time
Definition | When the composition was attested by the party.
Cardinality | 0..1
Type | dateTime
Requirements | Identifies when the information in the composition was deemed accurate. (Things may have changed since then.).
Composition.attester.party
Element Id | Composition.attester.party
Definition | Who attested the composition in the specified way.
Cardinality | 0..1
Type | Reference(Patient | RelatedPerson | Practitioner | PractitionerRole | Organization)
Requirements | Identifies who has taken on the responsibility for accuracy of the composition content.
Composition.custodian
Element Id | Composition.custodian
Definition | Identifies the organization or group who is responsible for ongoing maintenance of and access to the composition/document information.
Cardinality | 0..1
Type | Reference(Organization)
Requirements | Identifies where to go to find the current version, where to report issues, etc.
Summary | true
Comments | This is useful when documents are derived from a composition - provides guidance for how to get the latest version of the document. This is optional because this is sometimes not known by the authoring system, and can be inferred by context. However, it is important that this information be known when working with a derived document, so providing a custodian is encouraged.
Composition.relatesTo
Element Id | Composition.relatesTo
Definition | Relationships that this composition has with other compositions or documents that already exist.
Cardinality | 0..*
Comments | A document is a version specific composition.
Composition.relatesTo.code
Element Id | Composition.relatesTo.code
Definition | The type of relationship that this composition has with anther composition or document.
Cardinality | 1..1
Terminology Binding | DocumentRelationshipType (Required)
Type | code
Comments | If this document appends another document, then the document cannot be fully understood without also accessing the referenced document.
Composition.relatesTo.target[x]
Element Id | Composition.relatesTo.target[x]
Definition | The target composition/document of this relationship.

Cardinality | 1..1
Type | Identifier|Reference(Composition)
[x] Note | See Choice of Data Types for further information about how to use [x]
Composition.event
Element Id | Composition.event
Definition | The clinical service, such as a colonoscopy or an appendectomy, being documented.
Cardinality | 0..*
Requirements | Provides context for the composition and creates a linkage between a resource describing an event and the composition created describing the event.
Summary | true
Comments | The event needs to be consistent with the type element, though can provide further information if desired.
Composition.event.code
Element Id | Composition.event.code
Definition | This list of codes represents the main clinical acts, such as a colonoscopy or an appendectomy, being documented. In some cases, the event is inherent in the typeCode, such as a "History and Physical Report" in which the procedure being documented is necessarily a "History and Physical" act.
Cardinality | 0..*
Terminology Binding | v3 Code System ActCode (Example)
Type | CodeableConcept
Summary | true
Comments | An event can further specialize the act inherent in the typeCode, such as where it is simply "Procedure Report" and the procedure was a "colonoscopy". If one or more eventCodes are included, they SHALL NOT conflict with the values inherent in the classCode, practiceSettingCode or typeCode, as such a conflict would create an ambiguous situation. This short list of codes is provided to be used as key words for certain types of queries.
Composition.event.period
Element Id | Composition.event.period
Definition | The period of time covered by the documentation. There is no assertion that the documentation is a complete representation for this period, only that it documents events during this time.
Cardinality | 0..1
Type | Period
Summary | true
Composition.event.detail
Element Id | Composition.event.detail
Definition | The description and/or reference of the event(s) being documented. For example, this could be used to document such a colonoscopy or an appendectomy.
Cardinality | 0..*
Type | Reference(Any)
Summary | true
Composition.section
Element Id | Composition.section
Definition | The root of the sections that make up the composition.
Cardinality | 0..*
Invariants | Defined on this element cmp-1 Rule A section must contain at least one of text, entries, or sub-sections text.exists() or entry.exists() or section.exists() cmp-2 Rule A section can only have an emptyReason if it is empty emptyReason.empty() or entry.empty()
Defined on this element
cmp-1 | Rule | A section must contain at least one of text, entries, or sub-sections | text.exists() or entry.exists() or section.exists()
cmp-2 | Rule | A section can only have an emptyReason if it is empty | emptyReason.empty() or entry.empty()
Composition.section.title
Element Id | Composition.section.title
Definition | The label for this particular section. This will be part of the rendered content for the document, and is often used to build a table of contents.
Cardinality | 0..1
Type | string
Requirements | Section headings are often standardized for different types of documents. They give guidance to humans on how the document is organized.
Alternate Names | header; label; caption
Comments | The title identifies the section for a human reader. The title must be consistent with the narrative of the resource that is the target of the section.content reference. Generally, sections SHOULD have titles, but in some documents, it is unnecessary or inappropriate. Typically, this is where a section has subsections that have their own adequately distinguishing title, or documents that only have a single section. Most Implementation Guides will make section title to be a required element.
Composition.section.code
Element Id | Composition.section.code
Definition | A code identifying the kind of content contained within the section. This must be consistent with the section title.
Cardinality | 0..1
Terminology Binding | Document Section Codes (Example)
Type | CodeableConcept
Requirements | Provides computable standardized labels to topics within the document.
Comments | The code identifies the section for an automated processor of the document. This is particularly relevant when using profiles to control the structure of the document. If the section has content (instead of sub-sections), the section.code does not change the meaning or interpretation of the resource that is the content of the section in the comments for the section.code.
Composition.section.author
Element Id | Composition.section.author
Definition | Identifies who is responsible for the information in this section, not necessarily who typed it in.

Cardinality | 0..*
Type | Reference(Practitioner | PractitionerRole | Device | Patient | RelatedPerson | Organization)
Requirements | Identifies who is responsible for the content.
Composition.section.focus
Element Id | Composition.section.focus
Definition | The actual focus of the section when it is not the subject of the composition, but instead represents something or someone associated with the subject such as (for a patient subject) a spouse, parent, fetus, or donor. If not focus is specified, the focus is assumed to be focus of the parent section, or, for a section in the Composition itself, the subject of the composition. Sections with a focus SHALL only include resources where the logical subject (patient, subject, focus, etc.) matches the section focus, or the resources have no logical subject (few resources).
Cardinality | 0..1
Type | Reference(Any)
Comments | Typically, sections in a doument are about the subject of the document, whether that is a patient, or group of patients, location, or device, or whatever. For some kind of documents, some sections actually contain data about related entities. Typical examples are a section in a newborn discharge summary concerning the mother, or family history documents, with a section about each family member, though there are many other examples.
Composition.section.text
Element Id | Composition.section.text
Definition | A human-readable narrative that contains the attested content of the section, used to represent the content of the resource to a human. The narrative need not encode all the structured data, but is required to contain sufficient detail to make it "clinically safe" for a human to just read the narrative.
Cardinality | 0..1
Type | Narrative
Comments | Document profiles may define what content should be represented in the narrative to ensure clinical safety.
Invariants | Affect this element cmp-1 Rule A section must contain at least one of text, entries, or sub-sections text.exists() or entry.exists() or section.exists()
Affect this element
cmp-1 | Rule | A section must contain at least one of text, entries, or sub-sections | text.exists() or entry.exists() or section.exists()
Composition.section.mode
Element Id | Composition.section.mode
Definition | How the entry list was prepared - whether it is a working list that is suitable for being maintained on an ongoing basis, or if it represents a snapshot of a list of items from another source, or whether it is a prepared list where items may be marked as added, modified or deleted.
Cardinality | 0..1
Terminology Binding | ListMode (Required)
Type | code
Requirements | Sections are used in various ways, and it must be known in what way it is safe to use the entries in them.
Comments | This element is labeled as a modifier because a change list must not be misunderstood as a complete list.
Composition.section.orderedBy
Element Id | Composition.section.orderedBy
Definition | Specifies the order applied to the items in the section entries.
Cardinality | 0..1
Terminology Binding | List Order Codes (Preferred)
Type | CodeableConcept
Requirements | Important for presentation and rendering. Lists may be sorted to place more important information first or to group related entries.
Comments | Applications SHOULD render ordered lists in the order provided, but MAY allow users to re-order based on their own preferences as well. If there is no order specified, the order is unknown, though there may still be some order.
Composition.section.entry
Element Id | Composition.section.entry
Definition | A reference to the actual resource from which the narrative in the section is derived.
Cardinality | 0..*
Type | Reference(Any)
Comments | If there are no entries in the list, an emptyReason SHOULD be provided.
Invariants | Affect this element cmp-2 Rule A section can only have an emptyReason if it is empty emptyReason.empty() or entry.empty()
Affect this element
cmp-2 | Rule | A section can only have an emptyReason if it is empty | emptyReason.empty() or entry.empty()
Composition.section.emptyReason
Element Id | Composition.section.emptyReason
Definition | If the section is empty, why the list is empty. An empty section typically has some text explaining the empty reason.
Cardinality | 0..1
Terminology Binding | List Empty Reasons (Preferred)
Type | CodeableConcept
Requirements | Allows capturing things like "none exist" or "not asked" which can be important for most lists.
Comments | The various reasons for an empty section make a significant interpretation to its interpretation. Note that this code is for use when the entire section content has been suppressed, and not for when individual items are omitted - implementers may consider using a text note or a flag on an entry in these cases.
Invariants | Affect this element cmp-2 Rule A section can only have an emptyReason if it is empty emptyReason.empty() or entry.empty()
Affect this element
cmp-2 | Rule | A section can only have an emptyReason if it is empty | emptyReason.empty() or entry.empty()
Composition.section.section
Element Id | Composition.section.section

Definition | A nested sub-section within this section.
Cardinality | 0..*
Type | See Composition.section
Comments | Nested sections are primarily used to help human readers navigate to particular portions of the document.
Invariants | Affect this element cmp-1 Rule A section must contain at least one of text, entries, or sub-sections text.exists() or entry.exists() or section.exists()
Affect this element
cmp-1 | Rule | A section must contain at least one of text, entries, or sub-sections | text.exists() or entry.exists() or section.exists()

2.41.10 Resource Composition - Mappings

URL: https://hl7.org/fhir/R4/composition-mappings.html

2.41.10 Resource Composition - Mappings

Structured Documents Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Not Classified | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

CDA (R2)

RIM Mapping

FHIR DocumentReference

Mappings for the composition resource (see Mappings to Other Standards for further information & status).

2.41.10.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

Composition
identifier | FiveWs.identifier
status | FiveWs.status
type | FiveWs.class
category | FiveWs.class
subject | FiveWs.subject[x]
encounter | FiveWs.context
date | FiveWs.done[x]
author | FiveWs.author
party | FiveWs.witness
author | FiveWs.author
mode | FiveWs.class

2.41.10.2 Workflow Pattern (http://hl7.org/fhir/workflow)

Composition | Event
identifier | Event.identifier
status | Event.status
type | Event.code
category | Event.subject
encounter | Event.context
date | Event.occurrence[x]
author | Event.performer

2.41.10.3 CDA (R2) (http://hl7.org/v3/cda)

Composition | ClinicalDocument
(meta.profile) | ClinicalDocument.templateId
(language) | ClinicalDocument.languageCode
identifier | .setId
status | n/a
type | .code
category | n/a
subject | .recordTarget
encounter | .componentOf.encompassingEncounter
date | .effectiveTime
author | .author.assignedAuthor
title | .title
confidentiality | .confidentialityCode
attester | .authenticator/.legalAuthenticator
mode | implied by .authenticator/.legalAuthenticator
time | .authenticator.time
party | .authenticator.assignedEnttty
custodian | .custodian.assignedCustodian
relatesTo | .relatedDocument

code | .relatedDocument.typeCode
target[x] | .relatedDocument.id
event | .documentationOf.serviceEvent
code | .code
period | .effectiveTime
detail | n/a
section | .component.structuredBody.component.section
title | .title
code | .code
author | .author.assignedAuthor
focus | .subject? (CDA did not differentiate between subject and focus)
text | .text
mode | n/a
orderedBy | n/a
entry | .entry
emptyReason | n/a
section | .component.section

2.41.10.4 RIM Mapping (http://hl7.org/v3)

Composition | Document[classCode="DOC" and moodCode="EVN" and isNormalAct()]
identifier | Document.id / Document.setId
status | interim: .completionCode="IN" & ./statusCode[isNormalDatatype()]="active"; final: .completionCode="AU" && ./statusCode[isNormalDatatype()]="complete" and not(./inboundRelationship[typeCode="SUBJ" and isNormalActRelationship()]/source[subsumesCode("ActClass#CACT") and moodCode="EVN" and domainMember("ReviseComposition", code) and isNormalAct()]); amended: .completionCode="AU" && ./statusCode[isNormalDatatype()]="complete" and ./inboundRelationship[typeCode="SUBJ" and isNormalActRelationship()]/source[subsumesCode("ActClass#CACT") and moodCode="EVN" and domainMember("ReviseComposition", code) and isNormalAct() and statusCode="completed"]; withdrawn : .completionCode=NI && ./statusCode[isNormalDatatype()]="obsolete"
type | ./code
category | .outboundRelationship[typeCode="COMP].target[classCode="LIST", moodCode="EVN"].code
subject | .participation[typeCode="SBJ"].role[typeCode="PAT"]
encounter | unique(highest(./outboundRelationship[typeCode="SUBJ" and isNormalActRelationship()], priorityNumber)/target[moodCode="EVN" and classCode=("ENC", "PCPR") and isNormalAct])
date | .effectiveTime[type="TS"]
author | .participation[typeCode="AUT"].role[classCode="ASSIGNED"]
title | ./title
confidentiality | .confidentialityCode
attester | .participation[typeCode="AUTHEN"].role[classCode="ASSIGNED"]
mode | unique(./modeCode)
time | ./time[type="TS" and isNormalDatatype()]
party | ./role[classCode="ASSIGNED" and isNormalRole]/player[determinerCode="INST" and classCode=("DEV", "PSN") and isNormalEntity()] or ./role[classCode="ASSIGNED" and isNormalRole and not(player)]/scoper[determinerCode="INST" and classCode="ORG" and isNormalEntity()]
custodian | .participation[typeCode="RCV"].role[classCode="CUST"].scoper[classCode="ORG" and determinerCode="INST"]
relatesTo | .outboundRelationship
code | .outboundRelationship.typeCode
target[x] | .target[classCode="DOC", moodCode="EVN"].id
event | outboundRelationship[typeCode="SUBJ"].target[classCode<'ACT']
code | .code
period | .effectiveTime
detail | .outboundRelationship[typeCode="SUBJ"].target
section | ./outboundRelationship[typeCode="COMP" and isNormalActRelationship()]/target[moodCode="EVN" and classCode="DOCSECT" and isNormalAct]
title | ./title
code | ./code
author | .participation[typeCode="AUT"].role[classCode="ASSIGNED"]
text | .text
mode | .outBoundRelationship[typeCode=COMP].target[classCode=OBS"].value
orderedBy | .outboundRelationship[typeCode=COMP].sequenceNumber > 1
entry | .outboundRelationship[typeCode=COMP] or .participation[typeCode=SBJ]
emptyReason | .inboundRelationship[typeCode=SUBJ,code<ListEmptyReason].value[type=CD]
section | ./outboundRelationship[typeCode="COMP" and isNormalActRelationship()]/target[moodCode="EVN" and classCode="DOCSECT" and isNormalAct]

2.41.10.5 FHIR DocumentReference (http://hl7.org/fhir/documentreference)

Composition | when described by DocumentReference
identifier | DocumentReference.masterIdentifier
status | DocumentReference.status
type | DocumentReference.type
category | DocumenttReference.category
subject | DocumentReference.subject
encounter | DocumentReference.context.encounter

date | DocumentReference.content.attachment.date, DocumentReference.created
author | DocumentReference.agent
title | DocumentReference.description, DocumentReference.content.attachment.title
confidentiality | DocumentReference.securityLabel
attester | DocumentReference.authenticator
mode
time
party
custodian | DocumentReference.custodian
relatesTo | DocumentReference.relatesTo
code | DocumentReference.relatesTo.code
target[x] | DocumentReference.relatesTo.target
event | DocumentReference.event
code | DocumentReference.event.code
period | DocumentReference.event.period
detail
section
title
code
author
focus
text
mode
orderedBy
entry
emptyReason
section

2.41.11 Resource Composition - Extensions & Profiles

URL: https://hl7.org/fhir/R4/composition-profiles.html

2.41.11 Resource Composition - Extensions & Profiles

Structured Documents Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Not Classified | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

This table lists profiles and extensions for the Composition resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

2.41.11.1 Profiles

Profile | Description | Context
Catalog | describes how the Composition resource is used to represent a Catalog | Profile for Catalog
ClinicalDocument | The Clinical Document profile constrains Composition to specify a clinical document (matching CDA). The base Composition is a general resource for compositions or documents about any kind of subject that might be encountered in healthcare including such things as guidelines, medicines, etc. A clinical document is focused on documents related to the provision of care process, where the subject is a patient, a group of patients, or a closely related concept. A clinical document has additional requirements around confidentiality that do not apply in the same way to other kinds of documents. | Clinical Document

2.41.11.2 Extensions

composition-clinicaldocument-otherConfidentiality | Composition | otherConfidentiality | for Clinical Document
composition-clinicaldocument-versionNumber | Composition | versionNumber | for Clinical Document
composition-section-subject | Composition.section | section-subject | for Composition HL7 Extensions
cqm-ValidityPeriod | Composition | ValidityPeriod | for Profile for Catalog
workflow-researchStudy | Composition | researchStudy | for Workflow Pattern HL7 Extensions
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

2.41.11.3 Search Extensions

No Search Extensions defined for this resource

2.41.12 Resource Composition - Operations

URL: https://hl7.org/fhir/R4/composition-operations.html

2.41.12 Resource Composition - Operations

Structured Documents Work Group | Maturity Level: 2 | Trial Use | Security Category: Not Classified | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

This resource has 1 operation operations associated with it:

$document | Generate a Document

For more information about operations, including how they are invoked, see Operations.

2.41.12.1 Operation $document on Composition

URL: https://hl7.org/fhir/R4/composition-operation-document.html

2.41.12.1 Operation $document on Composition

Structured Documents Work Group | Maturity Level: 2 | Trial Use | Compartments: N/A

A client can ask a server to generate a fully bundled document from a composition resource. The server takes the composition resource, locates all the referenced resources and other additional resources as configured or requested and either returns a full document bundle, or returns an error. Note that since this is a search operation, the document bundle is wrapped inside the search bundle. If some of the resources are located on other servers, it is at the discretion of the server whether to retrieve them or return an error. If the correct version of the document that would be generated already exists, then the server can return the existing one.

The official URL for this operation definition is

```
http://hl7.org/fhir/OperationDefinition/Composition-document
```

Formal Definition (as a OperationDefinition).

URL: [base]/Composition/$document

URL: [base]/Composition/[id]/$document

This is an idempotent operation

In Parameters:

Name	Cardinality	Type	Binding	Profile	Documentation
id	0..1	uri			Identifies the composition to use. This can either be a simple id, which identifies a composition, or it can be a full URL, which identifies a composition on another server. Notes: GET [base]/Composition/[id]/$document is identical in meaning to GET [base]/Composition/$document?id=[id] the id parameter SHALL NOT be used if the operation is requested on a particular composition (e.g. GET [base]/Composition/[id]/$document?id=[id] is not allowed) Servers are not required to support generating documents on Compositions located on another server
persist	0..1	boolean			Whether to store the document at the bundle end-point (/Bundle) or not once it is generated. Value = true or false (default is for the server to decide). If the document is stored, it's location can be inferred from the Bundle.id, but it SHOULD be provided explicitly in the HTTP Location header in the response
graph	0..1	uri			Canonical reference to a GraphDefinition. If a URL is provided, it is the canonical reference to a GraphDefinition that it controls what resources are to be added to the bundle when building the document. The GraphDefinition can also specify profiles that apply to the various resources

Note: this operation definition does not resolve the question how document signatures are created. This is an open issue during the period of trial use, and feedback is requested regarding this question

2.41.12.1.1 Examples

Request: Same, as a post request using the id parameter

```
GET http://fhir.healthintersections.com.au/open/Composition/180f219f-97a8-486d-99d9-ed631fe4fc57/$document?persist=true

POST http://fhir.healthintersections.com.au/open/Composition/$document

<Parameters xmlns="http://hl7.org/fhir">
  <parameter>
    <name value="persist"/>
    <valueBoolean value="true"/>
  </parameter>
  <parameter>
    <name value="id"/>
    <valueUri value="180f219f-97a8-486d-99d9-ed631fe4fc57"/>
  </parameter>
</Parameters>
```

Response:

```
HTTP/1.1 200 OK
[other headers]

<?xml version="1.0" encoding="UTF-8"?>

<Bundle>
 xsi:schemaLocation="http://hl7.org/fhir ../../schema/bundle.xsd"
```

```
 xmlns="http://hl7.org/fhir">

  <id value="father"/>
  <meta>
    <lastUpdated value="2013-05-28T22:12:21Z"/>
  </meta>
  <identifier>
    <system value="urn:ietf:rfc:3986"/>
    <value value="urn:uuid:0c3151bd-1cbf-4d64-b04d-cd9187a4c6e0"/>
  </identifier>
  <type value="document"/>
  <timestamp value="2013-05-28T22:12:21Z"/>
<!--   The Composition resource   -->
  <entry>
    <fullUrl value="http://fhir.healthintersections.com.au/open/Composition/180f219f-97a8-486d-99d9-ed631fe4fc57"/>
    <resource>
      <Composition>
        <id value="180f219f-97a8-486d-99d9-ed631fe4fc57"/>
        <meta>
          <lastUpdated value="2013-05-28T22:12:21Z"/>
        </meta>
        <status value="final"/>
        <type>
          <coding>
            <system value="http://loinc.org"/>
            <code value="28655-9"/>
          </coding>
          <text value="Discharge Summary from Responsible Clinician"/>
        </type>
        <subject>
          <reference value="http://fhir.healthintersections.com.au/open/Patient/d1"/>
          <display value="Eve Everywoman"/>
        </subject>
      <!--   The Encounter resource. Points directly to an Encounter resource. Contains the dates of admission,
    specialtyu etc.   -->
        <encounter>
          <reference value="http://fhir.healthintersections.com.au/open/Encounter/doc-example"/>
        </encounter>
        <date value="2013-02-01T12:30:02Z"/>
        <author>
          <reference value="Practitioner/example"/>
          <display value="Doctor Dave"/>
        </author>
        <title value="Discharge Summary"/>
        <confidentiality value="N"/>
        <section>
          <title value="Reason for admission"/>
          <code>
            <coding>
              <system value="http://loinc.org"/>
              <code value="29299-5"/>
              <display value="Reason for visit Narrative"/>
            </coding>
          </code>
          <text>
            <status value="additional"/>
            <div xmlns="http://www.w3.org/1999/xhtml">

              <table>

                <thead>

                  <tr>

                    <td>Details</td>

                    <td/>

                  </tr>

                </thead>

                <tbody>

                  <tr>

                    <td>Acute Asthmatic attack. Was wheezing for days prior to admission.</td>

                    <td/>

                  </tr>

                </tbody>

              </table>

            </div>
          </text>
          <entry>
```

```
      <reference value="urn:uuid:541a72a8-df75-4484-ac89-ac4923f03b81"/>
    </entry>
  </section>
<!--   Points to the  medications on discharge   -->
  <section>
    <title value="Medications on Discharge"/>
    <code>
      <coding>
        <system value="http://loinc.org"/>
        <code value="10183-2"/>
        <display value="Hospital discharge medications Narrative"/>
      </coding>
    </code>
    <text>
      <status value="additional"/>
      <div xmlns="http://www.w3.org/1999/xhtml">

        <table>

          <thead>

            <tr>

              <td>Medication</td>
              <td>Last Change</td>

              <td>Last ChangeReason</td>

            </tr>

          </thead>

          <tbody>

            <tr>

              <td>Theophylline 200mg BD after meals</td>

              <td>continued</td>

            </tr>

            <tr>

              <td>Ventolin Inhaler</td>

              <td>stopped</td>

              <td>Getting side effect of tremor</td>

            </tr>

          </tbody>

        </table>

      </div>
    </text>
    <mode value="working"/>
    <!--   This is an entry that refers to a MedicationRequest resource in this document.
          It is a new medication (as indicated by the flag)  -->
    <entry>
      <reference value="urn:uuid:124a6916-5d84-4b8c-b250-10cefb8e6e86"/>
    </entry>
    <entry>
      <reference value="urn:uuid:673f8db5-0ffd-4395-9657-6da00420bbc1"/>
    </entry>
  </section>
<!--   Points to the  allergies   -->
  <section>
    <title value="Known allergies"/>
    <code>
      <coding>
        <system value="http://loinc.org"/>
        <code value="48765-2"/>
        <display value="Allergies and adverse reactions Document"/>
      </coding>
    </code>
    <text>
      <status value="additional"/>
      <div xmlns="http://www.w3.org/1999/xhtml">

        <table>

          <thead>

            <tr>

              <td>Allergen</td>
```

```
                    <td>Reaction</td>
                  </tr>
                </thead>
                <tbody>
                  <tr>
                    <td>Doxycycline</td>
                    <td>Hives</td>
                  </tr>
                </tbody>
              </table>
            </div>
          </text>
          <entry>
            <reference value="urn:uuid:47600e0f-b6b5-4308-84b5-5dec157f7637"/>
          </entry>
        </section>
      </Composition>
    </resource>
  </entry>
<!--   The Practitioner Resource. In this document they are the author of the document
    (There is a reference from the document resource). Note that, strictly, it doesn't need to be within the document as
     the
    recipient knows where to go and get it if they need it - assuming it is available on-line of course.  -->
  <entry>
    <fullUrl value="http://fhir.healthintersections.com.au/open/Practitioner/example"/>
    <resource>
      <Practitioner>
        <id value="example"/>
        <meta>
          <lastUpdated value="2013-05-05T16:13:03Z"/>
        </meta>
        <text>
          <status value="generated"/>
          <div xmlns="http://www.w3.org/1999/xhtml">
            <p>Dr Adam Careful</p>
          </div>
        </text>
        <identifier>
          <system value="http://www.acme.org/practitioners"/>
          <value value="23"/>
        </identifier>
        <name>
          <family value="Careful"/>
          <given value="Adam"/>
          <prefix value="Dr"/>
        </name>
      </Practitioner>
    </resource>
  </entry>
<!--   The Patient who is the subject of the document. Same coments as practitioner.   -->
  <entry>
    <fullUrl value="http://fhir.healthintersections.com.au/open/Patient/d1"/>
    <resource>
      <Patient>
        <id value="d1"/>
        <text>
          <status value="generated"/>
          <div xmlns="http://www.w3.org/1999/xhtml">
            <h1>Eve Everywoman</h1>
          </div>
        </text>
        <active value="true"/>
        <name>
          <text value="Eve Everywoman"/>
          <family value="Everywoman1"/>
          <given value="Eve"/>
        </name>
        <telecom>
          <system value="phone"/>
          <value value="555-555-2003"/>
          <use value="work"/>
        </telecom>
        <gender value="female"/>
        <birthDate value="1955-01-06"/>
        <address>
```

```
          <use value="home"/>
          <line value="2222 Home Street"/>
        </address>
      </Patient>
    </resource>
  </entry>
<!--   The encounter that is being documented.  -->
  <entry>
    <fullUrl value="http://fhir.healthintersections.com.au/open/Encounter/doc-example"/>
    <resource>
      <Encounter>
        <id value="doc-example"/>
        <meta>
          <lastUpdated value="2013-05-05T16:13:03Z"/>
        </meta>
        <text>
          <status value="generated"/>
          <div xmlns="http://www.w3.org/1999/xhtml"> Admitted to Orthopedics Service,
                        Middlemore Hospital between Jan 20 and Feb ist 2013 </div>
        </text>
        <identifier>
          <value value="S100"/>
        </identifier>
        <status value="finished"/>
        <class>
            <system value="http://terminology.hl7.org/CodeSystem/v3-ActCode"/>
            <code value="IMP"/>
            <display value="inpatient encounter"/>
        </class>
        <type>
          <text value="Orthopedic Admission"/>
        </type>
        <subject>
          <reference value="Patient/d1"/>
        </subject>
        <period>
          <start value="2013-01-20T12:30:02Z"/>
          <end value="2013-02-01T12:30:02Z"/>
        </period>
        <hospitalization>
          <dischargeDisposition>
            <text value="Discharged to care of GP"/>
          </dischargeDisposition>
        </hospitalization>
      </Encounter>
    </resource>
  </entry>
  <entry>
    <fullUrl value="urn:uuid:541a72a8-df75-4484-ac89-ac4923f03b81"/>
    <resource>
      <Observation>
        <meta>
          <lastUpdated value="2013-05-05T16:13:03Z"/>
        </meta>
        <text>
          <status value="additional"/>
          <div xmlns="http://www.w3.org/1999/xhtml"> Acute Asthmatic attack. Was wheezing
                        for days prior to admission. </div>
        </text>
        <status value="final"/>
        <code>
          <coding>
            <system value="http://loinc.org"/>
            <code value="46241-6"/>
          </coding>
          <text value="Reason for admission"/>
        </code>
        <subject>
          <reference value="http://fhir.healthintersections.com.au/open/Patient/d1"/>
          <display value="Eve Everywoman"/>
        </subject>
        <encounter>
          <reference value="http://fhir.healthintersections.com.au/open/Encounter/doc-example"/>
        </encounter>
        <valueString value="Acute Asthmatic attack. Was wheezing for days prior to admission."/>
      </Observation>
    </resource>
  </entry>
<!--   The first medication in the medications list   -->
  <entry>
    <fullUrl value="urn:uuid:124a6916-5d84-4b8c-b250-10cefb8e6e86"/>
    <resource>
      <MedicationRequest>
        <meta>
          <lastUpdated value="2013-05-05T16:13:03Z"/>
        </meta>
      <!--   The Human readible version of the script   -->
        <text>
          <status value="generated"/>
```

```
          <div xmlns="http://www.w3.org/1999/xhtml">

            <p>Theophylline 200mg twice a day</p>

          </div>
        </text>
      <!--   The patient details will generally be in the Document resource in the Document bundle, but could just as
    easily
        point to a patient resource on a server  -->
        <status value='unknown'/>
        <intent value="order"/>
        <medicationCodeableConcept>
          <coding>
            <system value="http://snomed.info/sct"/>
            <code value="66493003"/>
          </coding>
          <text value="Theophylline 200mg"/>
        </medicationCodeableConcept>
     <subject>
          <reference value="http://fhir.healthintersections.com.au/open/Patient/d1"/>
          <display value="Peter Patient"/>
        </subject>
      <!--   The prescriber details could also be within the bundle, but has the same options as patient  -->
        <requester>
          <reference value="Practitioner/example"/>
          <display value="Peter Practitioner"/>
        </requester>
        <reasonCode>
          <text value="Management of Asthma"/>
        </reasonCode>
      <!--   a reference to the medication being prescribed. As described earlier, this could be contained (as is the
    example here), separately
        within the document bundle or simply a reference to a remote server. See comment in the contained resource  -->
            <dosageInstruction>
          <additionalInstruction>
            <text value="Take with Food"/>
          </additionalInstruction>
        <!--   twice a day   -->
          <timing>
            <repeat>
              <frequency value="2"/>
              <period value="1"/>
              <periodUnit value="d"/>
            </repeat>
          </timing>
        <!--   Orally   -->
          <route>
            <coding>
              <system value="http://snomed.info/sct"/>
              <code value="394899003"/>
              <display value="oral administration of treatment"/>
            </coding>
          </route>
               <doseAndRate>
                 <type>
                   <coding>
                     <system value="http://terminology.hl7.org/CodeSystem/dose-rate-type"/>
                     <code value="ordered"/>
                     <display value="Ordered"/>
                   </coding>
                 </type>
                 <doseQuantity>
                   <value value="1"/>
                   <unit value="tablet"/>
                   <system value="http://unitsofmeasure.org"/>
                   <code value="tbl"/>
                 </doseQuantity>
               </doseAndRate>

        </dosageInstruction>
      </MedicationRequest>
    </resource>
  </entry>
<!--   The second medication in the medications list   -->
  <entry>
    <fullUrl value="urn:uuid:673f8db5-0ffd-4395-9657-6da00420bbc1"/>
    <resource>
      <MedicationStatement>
      <!--   The Human readible version of the medication statement  -->
        <text>
          <status value="generated"/>
          <div xmlns="http://www.w3.org/1999/xhtml">

            <p>Ventolin inhaler discontinued</p>

          </div>
        </text>
      <!--   The patient details will generally be in the Document resource in the Document bundle, but could just as
    easily
```

```
        point to a patient resource on a server  -->
        <status value='active'/>
    <!--   a reference to the medication being prescribed. As described earlier, this could be contained (as is the
      example here), separately
        within the document bundle or simply a reference to a remote server. See comment in the contained resource  -->
        <statusReason>
          <text value="Management of Asthma"/>
        </statusReason>
        <medicationCodeableConcept>
          <text value="Ventolin Inhaler"/>
        </medicationCodeableConcept>
        <subject>
          <reference value="http://fhir.healthintersections.com.au/open/Patient/d1"/>
          <display value="Peter Patient"/>
        </subject>
        <dateAsserted value="2013-05-05T16:13:03Z"/>

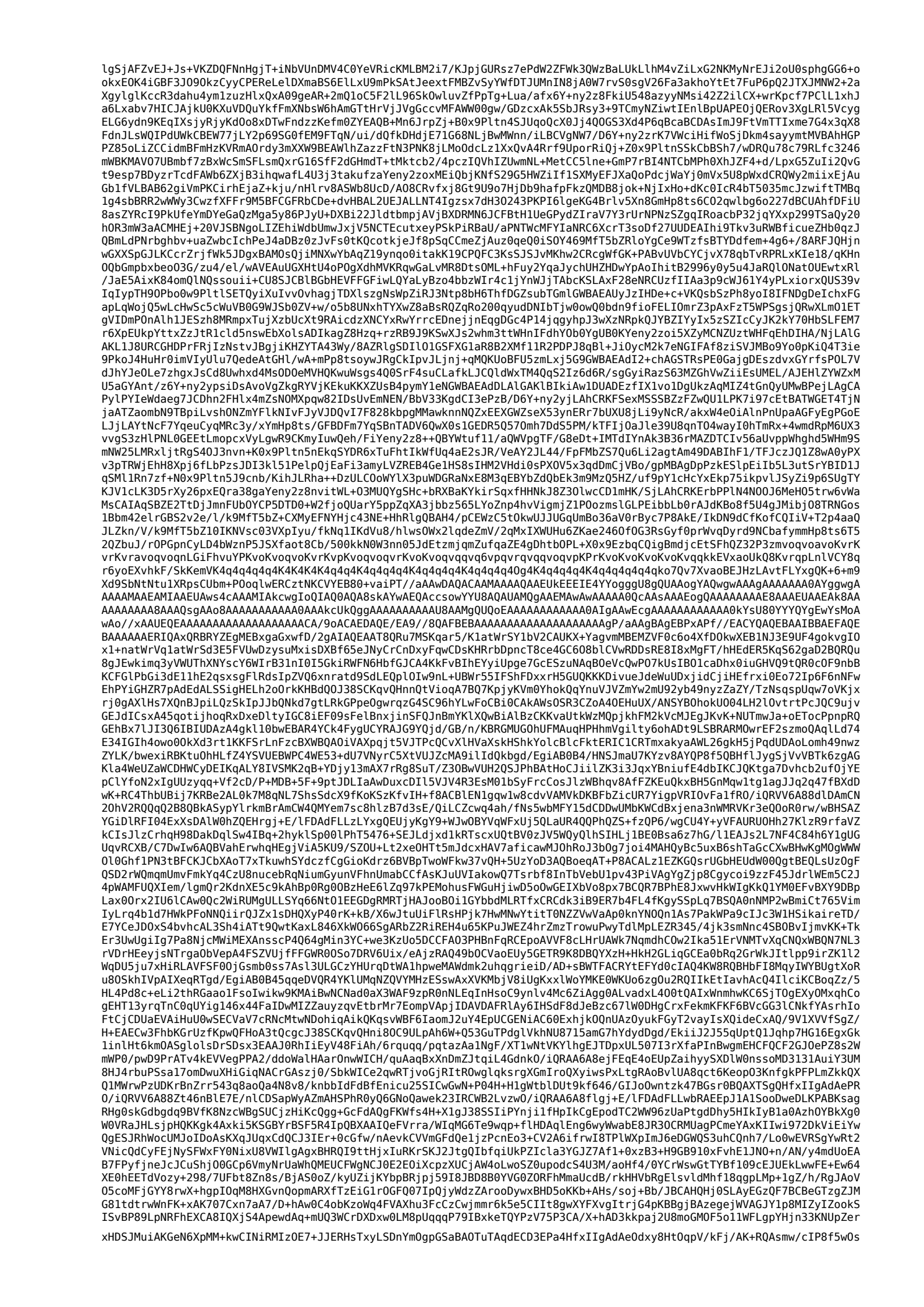



       </MedicationStatement>
      </resource>
    </entry>
  <!--   The Allergy.  -->
    <entry>
      <fullUrl value="urn:uuid:47600e0f-b6b5-4308-84b5-5dec157f7637"/>
      <resource>
        <AllergyIntolerance>
          <meta>
            <lastUpdated value="2013-05-05T16:13:03Z"/>
          </meta>
          <text>
            <status value="generated"/>
            <div xmlns="http://www.w3.org/1999/xhtml">Sensitivity to Doxycycline :
                        Hives</div>
          </text>
          <clinicalStatus>
        <coding>
          <system value="http://terminology.hl7.org/CodeSystem/allergyintolerance-clinical"/>
          <code value="active"/>
          <display value="Active"/>
        </coding>
        </clinicalStatus>

        <verificationStatus>
        <coding>
          <system value="http://terminology.hl7.org/CodeSystem/allergyintolerance-verification"/>
          <code value="confirmed"/>
          <display value="Confirmed"/>
        </coding>
        </verificationStatus>
          <type value="allergy"/>
          <criticality value="high"/>
          <code>
            <text value="Doxycycline"/>
          </code>
          <patient>
            <reference value="http://fhir.healthintersections.com.au/open/Patient/d1"/>
            <display value="Eve Everywoman"/>
          </patient>
          <recordedDate value="2012-09-17"/>
          <reaction>
            <manifestation>
              <coding>
                <system value="http://example.org/system"/>
                <code value="xxx"/>
                <display value="Hives"/>
              </coding>
              <text value="Hives"/>
            </manifestation>
          </reaction>
        </AllergyIntolerance>
      </resource>
    </entry>
    <signature>
      <type>
        <system value="urn:iso-astm:E1762-95:2013"/>
        <code value="1.2.840.10065.1.12.1.1"/>
        <display value="Author's Signature"/>
      </type>
      <when value="2015-08-31T07:42:33+10:00"/>
      <who>
        <reference value="Device/software"/>
      </who>
      <onBehalfOf>
        <reference value="Organization/example"/>
      </onBehalfOf>
      <!-- a real document would typically have an digital signature, but
        the publishing tools are not yet setup to produce a valid digital
```

```
  signature. For now, an image of a signature -->
<sigFormat value="image/jpg"/>
<data value="/9j/4AAQSkZJRgABAQAAAQABAAD/2wBDAAoHBwgHBgoICAgLCgoLDhgQDg0NDh0VFhEYIx8lJCIfIiEmKzcvJik0KSEiMEExNDk7Pj4
+JS5ESUM8SDc9Pjv/2wBDAQoLCw4NDhwQEBw7KCIoOzs7Ozs7Ozs7Ozs7Ozs7Ozs7Ozs7Ozs7Ozs7Ozs7Ozs7Ozs7Ozs7Ozs7Ozv/wgARCAD
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ipeZ4zKs06VB8DpWvyPF8ZTa9pJQRPkt5+H7I2zfLMdB+8/8AhewwKh6fmfoDmPTX9mL12/ZJ0/VQmTL3wZ7W0YsFyWe22ZSMm3FtAPCT5bTwx5NxPvZ
9Ro+/rQcppu/Bzvl+dgeGGqJfrvEqn9AfJ9A8/QAAAAAAEupLFKbSMm3FtEWtxZ0OGrQPP6+vkmxPwUb/AD/QCfQnmjQAAAB58+dIhUDaSiqRS0Alzzp
ECqagAJdSUftKZTMm3FtOdoSeoPON7eQ+Kf4QbOfQZLsewAAMXl5GboIdwGU5qlkrFKXQ4k7bn6kQuQ/b4LcejDK1DkunKEj3jnUJeMydXw/UFCVVln5
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fEpeXqPn6AAnyyhI8956VOaqjTvAAAAAAAACXUln5Tm0jJtxbQB5eoxeuiKWotqKWnl6gCfQwGbRQEudQinUT+c642AAAAAAAAAS6ksUplMybcW0AAY9
k8/c3pnKvvz9AoM+gYto8/TBvJu6D9n5YxVQAAAAAAAABLqSz8pzKZk24toAB583r3HvNp5Dx9qXOn3S1wi/5+kc0/WuCfPReXqAAAAAAAAAAJVWUflS
ZTMm3FtAGX24kodV8/Q8PcZefuzS/N2+pN8fmkaOaq5S0AAAAAAAAAABLqSxSm0jJtxbRln5CF3GoAMW3nzdP6LnzoMurzPTH4+RM6oAAAAAAAAAAAEu
pLPynNpEPLe8jnK+wY/3WMjYMf5t/DHK6H8Mv5sGRr/TH+bBka/0x/mwY2wY2sZGwZfzYMX7rGRrGP82ZAyfpqfuQ15XobqPx6H//xAAqEAACAgIBAgc
AAgIDAAAAAAADBAECAAUUFTQQERITIDJAIjAjJCEzNf/aAAgBAQABBQJRRayfDVzhq5w1c4aucNXHKpqhV1vvWNCCcl9wQuGrnDVzhq5w1c4aucNXOGr
nDVzhq5w1clRSsUprSWusmOi5NezlVkr5Cik4NdIw6315G6H15M9/XWCuhZiDHFUShlLLxweoUdQqqCv8zGRDlHAUX14/fDw1c4auOLgEuC/qrifY/Bh
ka1F1JsR129JoESWS7FCkpsLD0xSFYmYrFDDJRp0ClHXIFAzCLjRvTsuosGF7m0oblt+vVLgEts17kPRZr1KqujvX35qLmkPrdWRe/RLkwatVltfqyrN
RqyWW6Kp7R9bFk+kwDDiu4lfVBskvp1QfDY9kr9cT7HxMWoAqUluXGryStl9dFlysmBfXrVgsMCXpagvbaJNfMKnAZAadWQzYwiFjOvE0UqASWVVoqOd
arYsRFY+E0rNv7tj2Sv1xPsfHzJsmNg7CQESzSKIsmJCa8SNAUXoAQpwdIHW8iBCIb3f/AKIiI/FseyV+uJ9j4bAszkmAnSuss0WlKjrMecMsh14IvsH
irwwsHweJUaq9JEv+rY9kr9cT7HL3gdFDFYIqmNWvhMxWAebzQ1o2FlRxTa+Epjs1/Re9R1JsimmI23muaxPBnYBVPhGy8/wa2CymR1FyIVZSGBijFPj
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2zZSYgL2UXHBpBC2Mr7z1EhOOjQbGgRk2JZDmx7JX64n2NyDFAzncajUkNW62tRrzHzyFZwgzEdHAzLteG2NZYUtCrnL873YCKeSD0GfOyWBKa+tVjv
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l+qjMIv8AVseyV+uJ9jMxWNUCG3SEoEdRl25YiKxjgrXqOxxL9T8qD3orR10Ex1MjVFloi0efymYjD7S+VT2RTEbbXwc3kXxnWJ2LQYxR/TseyV+uJ9j
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YLjhY+TToE62nYt1qJLXxUrrFFiOrnol7l/x7HslfrifY/CKwMbAptgSwYeB/ybb4tXZ9Ky3+b3DWaquqpgmJa2V6D9jV3m+t/HseyV+uJ9j8jf6ruLf
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p0zqdM6pTOp0zqdM6nTOpVzqVc6nTOp0zqdM6nTOp0zqg86pTOp0zqdM6nTOpVxlvkgWr5VywonOPXOPXOPXOPXOPXOPXOPXOPXOPXONXOPXOPXOP
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iDNqPvnqSOCK8uFkJVD0/8GkxKWnxB4xc6L01/qZHYPHmZl0GYf8AFHe40S+3KLVnl6NnRfvrOueF7WzRHLpzVYcJ21dztY0ZDHtOYjMbE0ABYDYPFDF
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laVpZUszg8K6/0/fJusWQSX0VCL5jyVHGdqqAfzbesv1Ddg9mvy3GdjZVFzU80CXxMhtc8WNPv5VpwpDxpDtO6STYDaa8OxJyYaI/hZjas0sRjjA/BRR
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OHkae1g0ch+XNWZL6GxBFiDzeM3rL9Q3YPZr8txIAmZTeST1V5KCWsTYsOm1ZnYKo5Sa/02BfrlOXsoG2FHRwqaM+C2YWPGpMDLhmCnKoeLhX5TV8PLn
j5uasU9yFyrpz9PZ4+UcOVuLGNpp455Q/4dyEHBQ30HiTF5MmFw+mguS/2akKJvcOVd7XZz37dxVtwhCST7xROYm/JzVi8SwRgzb2Lc33amd9FjL8mwA
mppG1kkk1P3sqEMpUhALGkwuGNpHNs52DWnxGrSMxuWsbG5vaImQXK3Rc9d7bq4qNmjwysqqD6dyAd1vWX6huwezX5bjkaiSYQW5Qg1bcVpJ4
zla4twqtvvvymvwpVbo5aaB7B14ut7ryGoEKhiScqn0jst/urPEd4l/WnL189SrMuSURrcc+p1HR42QfiTckaamnxM73kI1JP+1afETeUxBzbeTk3WOH
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OmlxTyZ3LEhg1/fs23p1iYrDGcjHZmbmqFNeLc3rO+pPFXno4o8GNYFVjfYSazNqx4q89STRMOGMtjWGwUcO/GDUhW2v93ol0yOjZHHTuN6y/UN2D2a/
```

```
KryOqDnY2pThheUFmLNoBfTZ1Wr/W4uRyTfKp0FZpI41v+rhUFwGH3qHXLZPsU0++QGVrFH3vtvQaSDM68R8Pr7iOb/ihxSy65G4ynq3I8RHxw2X3fYp
A7jMzZODqM3NWVMPO+l75Mo7bVaSVEPMzWrPv8eW9s2YWoYfCI6Kw1kK62va4rMeFITcE6yOeiklxnAjXVYR/+vFNht21jI5ZVULa1ztGtNvKlMMguzN
pf75qMkuuIdbL/S3R300AmEjyjQD0QRr/AJoTYhs8uUKo/QKhwySpxuHzDr7a8HuolkIzAG9xz7Of720mDw6kC1idpa/JSQ5SpQa6VhsInCzTM0qtYMd
dp916NtvJQxGJe0smyK2oH+KiwPBbhZ5rcnRf721IXH4cR1v0V4GgvMmIzXHONK8HwqNHJ/7JTpbo6KsWsqasx5akjjRsqbH56b1l+obsHs1+VDFYvye
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UxeQPMQOGXO34c1RrA2VXjyzNf40EhxHAbymblpd6Xhq6tnY66GvCMM29Yi1r8h66/wBdKVReTTX4UI41yquwU3rL9Q3YPZr8tzGLFc4iaYqBzDnrhAP
IeM1Ek2A2mhvWaHDekTozf8VkQFem/G66ZZEIVRd025f6l/p6OSgQbg7DSl0ViuwkbKCyXGU3BXbuhDd3JtkTU0rSHeIf/lc328tCONcqrsH8uSL9Q06
6jSQWZBl281QSMxG8tfTl3JJDiEXJEEsTy3o73Ij225Tf+U3rL9Q3YPZr8qJJsBtNS4xxYBrgdJoySNlVdpoSy8DCLxVvxqAAsBsG4joUV4jmzvyVmwa
KY00fDsblSDrY1mfB4pbDX8PQUS0EgsNcuv3yUckUrNbZYVwpN6UtokIJkNRjD4dorXJfEIbn4dXLWvjam1EYLDvPY2z5brQnmZNDmETObA8myrTCF3Y
cBIblr9XNSmQWe3CHT4xkMN2JudTVo0VBzKLfym9ZfqG7B7NflUwUcl+2hFGwLjVuQ/fJW8nTDI2VrbXf9PV3UABYDYNwkmwG01oVdG94Nb6jS4UJl4d
8oYcnQaWDwhpZH/uvy8mm5aRFcczC9AAWA2DxrSPdv0LqaO9QJGh2O4OnfQlxDvO/9Z0+FE8GONatD+FCdsh4zerRcZnkba7m5P5JvWX6huwezX5UQRc
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IdzG/2fKt75bX6vFWH/6sF9209gO6MSDllXZoLe+sTHOLymTPdF23qRMLGRCq3c8tqjjjN1A28/5RvWX6huwezX5ePJF+oaddRSnaRr11Ou0yxq/VbSm
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o2V4TDHFKzDUOL5+o8leEYUNAzbUIsD0EclBlTIF0kDch5vv/AKGMh05JdNo56N4hvWXMJEbNf3bjOxsqi5qHpUMes61P7NvlVz6bEj5f4/Lt6y/UN2D
2a/Lxt5y53m4Kra9JFy+kec08d7Z1IvQTgrDJxehubcdNkWJG+Je/G5fvqogi4O0UcFM3E1h11ZddxcMpu0zZDl5By9lAAWA2Clw8Y4c5yjW1LGNiiw/
Lt6y/UN2D2a/LxTLKbAdteH4sXJ8kObdZL2uNDzdNJMvpCsLppm0J6dPvr3FyNlljOZGty1Z1yTLx0tsp8WNVQb2h5+c/43GbbHhRb+77+X5hvWX6huw
ezX5eJmdgqjlJrO4K4dLjTT7NAAWA2DxD7R/qNYZ1TM4nW1XGnODtHXub9hpN6ntYn9QpIwLZRTS6FvRBrM999kOZ8238w3rL9Q3YPZr8t0yymwHbSXV
hAzaWGi0IohYDt8SSa18o0qK7ZiwzE9etYKPkaYH4f97hiPBlAuV+9u5nla3MOU0MVi48sSj8NOQ/mW9ZfqG7B7Nflub2n4k52IK37+IPvshHF2Zfh4y
ol1iwz3ZrcZhybkMfAKwIXP38NwZlBsbi42VYTYhOlZT/AJozEvK/6pTe35pvWX6huxxvh5booBtl76thYWVuVmtpW+GGeWbW8ht315Cbs7683m7O+vN
5uzvrzebs7683m7O+vN5uzvryE3Z315Cbs76nxMkEmaTRcttF+NebzdnfXkJuzvrzebs7683m7O+vN5uzvryE3Z315Cbs7683m7O+vN5uzvryE3Z315C
bs768hN2d9eQm7O+vITdnfXkJuzvryE3Z315vN/t7683m7O+vN5uzvrzebs7683m7O+vN5uzvryE3Z315CXs768hN2d9eQm7O+vITdnfXkJuzvrzebs7
63pYZASRqbc/X+b2fy2O95g1rHTSl4JOi3F/jXDRjddOENOurSDhCtEIvm0BGh5KOXac3L06UhsxHLc7O3d//xAApEAEAAQMDAgYCAwEAAAAAAABEQA
hMUFRYRCBcZGhwfDxILFA0eEw/9oACAEBAAE/IUHAoisK+be1fNvavm3tXzb2r5t7VpBwo6dtqG3SQDOGrrHnsLZWSoWANPdnwq0aJMViUFjd48K+be1
fNvavm3tXzb2r5t7V829q+be1fNvavm3tXzb2oq0SgAUBxEUWlJwTAHamWR0Fa/Fad63RNOWzzQclSkmQs0UipD+qmumZBzPzUrWCAAGJhFpW7eU3gcC
RaHTMWnWrich3JYiCdy1O6UUAW2MwhOPeyiwBkdgNYx5kDpwOARbMaGfGgOfMtO5MRDUJFYnLcYzeQncwTQaYQVJeExZkh2tUEaYBZ6ExfltaKlgeUnk
drbNq+be1fNvam7lw5KIh6fLbPxITICOtIWa5qEEYG5mCU31ny5mdDgSDd89e/OumrFLiYKEBroTgtWUJ4Rdq4X0TltE02okHNmkm+cp7T9lCVuTfElB
```



```
    QA3xi8qf9WAvNRQYdyF9Cm330p6ojbgp2lp0BoiETga6q1llELAlmy8O6u0WA+sh4B2VX0AU+YNhQdIWeRRkWM9X+isGCBWCCvaeLp9ItbpWER9lPROM
    51stliM08pej9zA/x/lAiLCZcwq4ah/fLNhdgqkGzueQddg7+JBEADoDx+Pcj/wAcNRS+ip4wXri/IgwMe2AKfPois20fIjBQEQFIey4bpNMMapQtViq
    sLzqaFc4D5YVBqLTsHqaTSQhY726BRH9zAf1/jA6xADfGLyp/1YC8Pz1sgCssLqK4NHeICb5zeVP+BAD8I72cKLFSnQu2WcDdqnQXVWozfMvng3rUGAw
    fEp/B/fChhtLpS4BU6dUBg4V/EBLEy2qnwWqG8kNt3Cia1wrA10j+5Qf4/wAIHg8TB2p/rDV6DE+pqI3qorPEU+T8gfgJMqpE7Ipukp5nOwMXBNAMbE7
    vXp1leND1ToXE3hU1RNEcwTfa53zVU8GiKjw01Iibf3UD/P6VrDkgKntpyt7w0h4kK+V6PFaKe+YMbRRneldRn4f44+/otWj6jv8A89mImADWwrmOHYf
    hPvi9cfVf+/Xi0PVd9/P0uft6ffj9en7Kff5Lni549/jggIr/AByGWNEUeUhUQ1XHkMJjomvPbOMz0eue2Mgm1iel0cIh45OL0Hinj9c+L9c+P9c+H9c
    +L9c+L9cA8frinj9cV8frj7f658P658P658X658H658H64p4/XPi/XPi/XPi/XFPH64+1+uHt/rj7P658b64D4/XH2/1wPx+ufF+uLeP1z4f1z4f1z4f
    1xTx+ufH+uCDt05CIgMvgbXhSJgdQNEdlKDPfi+6kztQRTYSP43jrBIBGEBXH2a88GqaOjW4QWg9unwfWxa2hBcXPqXU4r3dySlRl7XqmoeSAen//2Q=
    ="/>
  </signature>
</Bundle>
```

For more information about operations, including how they are invoked, see Operations.

2.41.13 Resource Composition - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/composition-version-maps.html

No validation errors - all conversions are clean

2.41.13 Resource Composition - R3/R4 Conversions

Structured Documents Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Not Classified | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

R3 : R4 Conversion maps for Composition.

Functional status for this map: 1 test that all execute ok. All tests pass round-trip testing and all r3 resources are valid. (see documentation)

2.41.13.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/Composition3to4" = "R3 to R4 Conversions for Composition"

uses "http://hl7.org/fhir/3.0/StructureDefinition/Composition" alias CompositionR3 as source
uses "http://hl7.org/fhir/StructureDefinition/Composition" alias Composition as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group Composition(source src : CompositionR3, target tgt : Composition) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.status -> tgt.status;
  src.type -> tgt.type;
  src.class -> tgt.category;
  src.subject -> tgt.subject;
  src.encounter -> tgt.encounter;
  src.date -> tgt.date;
  src.author -> tgt.author;
  src.title -> tgt.title;
  src.confidentiality -> tgt.confidentiality;
  src.attester as vs0 -> tgt.attester as vt0 then attester(vs0, vt0);
  src.custodian -> tgt.custodian;
  src.relatesTo as vs0 -> tgt.relatesTo as vt0 then relatesTo(vs0, vt0);
  src.event as vs0 -> tgt.event as vt0 then event(vs0, vt0);
  src.section as vs0 -> tgt.section as vt0 then section(vs0, vt0);
}

group attester(source src, target tgt) extends BackboneElement {
  src.mode -> tgt.mode;
  src.time -> tgt.time;
  src.party -> tgt.party;
}

group relatesTo(source src, target tgt) extends BackboneElement {
  src.code -> tgt.code;
  src.target : Identifier as vs -> tgt.target = create('Identifier') as vt then Identifier(vs, vt);
  src.target : Reference as vs -> tgt.target = create('Reference') as vt then Reference(vs, vt);
}

group event(source src, target tgt) extends BackboneElement {
  src.code -> tgt.code;
  src.period -> tgt.period;
  src.detail -> tgt.detail;
}

group section(source src, target tgt) extends BackboneElement {
  src.title -> tgt.title;
  src.code -> tgt.code;
  src.text -> tgt.text;
  src.mode -> tgt.mode;
  src.orderedBy -> tgt.orderedBy;
  src.entry -> tgt.entry;
  src.emptyReason -> tgt.emptyReason;
  src.section as vs0 -> tgt.section as vt0 then section(vs0, vt0);
}
```

2.41.13.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/Composition4to3" = "R4 to R3 Conversion for Composition"
```

```
uses "http://hl7.org/fhir/StructureDefinition/Composition" alias Composition as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/Composition" alias CompositionR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group Composition(source src : Composition, target tgt : CompositionR3) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.status -> tgt.status;
  src.type -> tgt.type;
  src.category -> tgt.class;
  src.subject -> tgt.subject;
  src.encounter -> tgt.encounter;
  src.date -> tgt.date;
  src.author -> tgt.author;
  src.title -> tgt.title;
  src.confidentiality -> tgt.confidentiality;
  src.attester as vs0 -> tgt.attester as vt0 then attester(vs0, vt0);
  src.custodian -> tgt.custodian;
  src.relatesTo as vs0 -> tgt.relatesTo as vt0 then relatesTo(vs0, vt0);
  src.event as vs0 -> tgt.event as vt0 then event(vs0, vt0);
  src.section as vs0 -> tgt.section as vt0 then section(vs0, vt0);
}

group attester(source src, target tgt) extends BackboneElement {
  src.mode -> tgt.mode;
  src.time -> tgt.time;
  src.party -> tgt.party;
}

group relatesTo(source src, target tgt) extends BackboneElement {
  src.code -> tgt.code;
  src.target : Identifier as vs -> tgt.target = create('Identifier') as vt then Identifier(vs, vt);
  src.target : Reference as vs -> tgt.target = create('Reference') as vt then Reference(vs, vt);
}

group event(source src, target tgt) extends BackboneElement {
  src.code -> tgt.code;
  src.period -> tgt.period;
  src.detail -> tgt.detail;
}

group section(source src, target tgt) extends BackboneElement {
  src.title -> tgt.title;
  src.code -> tgt.code;
  src.text -> tgt.text;
  src.mode -> tgt.mode;
  src.orderedBy -> tgt.orderedBy;
  src.entry -> tgt.entry;
  src.emptyReason -> tgt.emptyReason;
  src.section as vs0 -> tgt.section as vt0 then section(vs0, vt0);
}
```

2.41.13.3 R4 Validation Errors

2.41.14 Clinical Document

URL: https://hl7.org/fhir/R4/composition-clinicaldocument.html

The Clinical Document profile constrains Composition to specify a clinical document (matching CDA). The base Composition is a general resource for compositions or documents about any kind of subject that might be encountered in healthcare including such things as guidelines, medicines, etc. A clinical document is focused on documents related to the provision of care process, where the subject is a patient, a group of patients, or a closely related concept. A clinical document has additional requirements around confidentiality that do not apply in the same way to other kinds of documents.

2.41.14 Clinical Document

Structured Documents Work Group | Maturity Level: N/A | Standards Status: Informative

2.41.14.1 Content

Profiles:

ClinicalDocument | The Clinical Document profile constrains Composition to specify a clinical document (matching CDA). The base Composition is a general resource for compositions or documents about any kind of subject that might be encountered in healthcare including such things as guidelines, medicines, etc. A clinical document is focused on documents related to the provision of care process, where the subject is a patient, a group of patients, or a closely related concept. A clinical document has additional requirements around confidentiality that do not apply in the same way to other kinds of documents.

Extensions:

composition-clinicaldocument-otherConfidentiality | otherConfidentiality : Carries additional confidentiality codes beyond the base fixed code specified in the CDA document.

composition-clinicaldocument-versionNumber | versionNumber : Version specific identifier for the composition, assigned when each version is created/updated.

2.41.14.2 StructureDefinition: ClinicalDocument

URL: https://hl7.org/fhir/R4/clinicaldocument.html

2.41.14.2 StructureDefinition: ClinicalDocument

Structured Documents Work Group | Maturity Level: 1 | Trial Use | Use Context: Any

The official URL for this profile is:

`http://hl7.org/fhir/StructureDefinition/clinicaldocument`

The Clinical Document profile constrains Composition to specify a clinical document (matching CDA). The base Composition is a general resource for compositions or documents about any kind of subject that might be encountered in healthcare including such things as guidelines, medicines, etc. A clinical document is focused on documents related to the provision of care process, where the subject is a patient, a group of patients, or a closely related concept. A clinical document has additional requirements around confidentiality that do not apply in the same way to other kinds of documents.

This profile was published on Tue, Mar 17, 2015 00:00+1100 as a draft by Health Level Seven, Inc. - Structured Documents WG.

2.41.14.2.1 Formal Views of Profile Content

Description of Profiles, Differentials, Snapshots, and how the XML and JSON presentations work.

• Text Summary
• Differential Table
• Snapshot Table
• XML Template
• JSON Template
• All

This structure is derived from Composition.

Summary

Extensions

This structure refers to these extensions:

• http://hl7.org/fhir/StructureDefinition/composition-clinicaldocument-versionNumber

This structure is derived from Composition.

Name | Flags | Card. | Type | Description & Constraints
Composition | | 0..*
composition-clinicaldocument-versionNumber | | 0..* | string | Version-specific identifier for composition URL: http://hl7.org/fhir/StructureDefinition/composition-clinicaldocument-versionNumber
subject | | 0..1 | Reference(Patient | Practitioner | Group | Device | Location)
Documentation for this format

Name | Flags | Card. | Type | Description & Constraints
Composition | I | 0..* | | A set of resources composed into a single coherent clinical statement with clinical attestation
id | Σ | 0..1 | string | Logical id of this artifact
meta | ΣI | 0..1 | Meta | Metadata about the resource
implicitRules | ?!ΣI | 0..1 | uri | A set of rules under which this content was created
language | I | 0..1 | code | Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | I | 0..* | Extension | Extension Slice: Unordered, Open by value:url
composition-clinicaldocument-versionNumber | I | 0..* | string | Version-specific identifier for composition URL: http://hl7.org/fhir/StructureDefinition/composition-clinicaldocument-versionNumber
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
identifier | ΣI | 0..1 | Identifier | Version-independent identifier for the Composition
status | ?!ΣI | 1..1 | code | preliminary | final | amended | entered-in-error Binding: CompositionStatus (required)
type | ΣI | 1..1 | CodeableConcept | Kind of composition (LOINC if possible) Binding: FHIR Document Type Codes (preferred)
category | ΣI | 0..* | CodeableConcept | Categorization of Composition Binding: Document Class Value Set (example)
subject | ΣI | 0..1 | Reference(Patient | Practitioner | Group | Device | Location) | Who and/or what the composition is about
encounter | ΣI | 0..1 | Reference(Encounter) | Context of the Composition
date | ΣI | 1..1 | dateTime | Composition editing time
author | ΣI | 1..* | Reference(Practitioner | PractitionerRole | Device | Patient | RelatedPerson | Organization) | Who and/or what authored the composition
title | ΣI | 1..1 | string | Human Readable name/title
confidentiality | ΣI | 0..1 | code | As defined by affinity domain Binding: ConfidentialityClassification (required)
attester | I | 0..* | BackboneElement | Attests to accuracy of composition
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
mode | I | 1..1 | code | personal | professional | legal | official Binding: CompositionAttestationMode (required)

time | I | 0..1 | dateTime | When the composition was attested
party | I | 0..1 | Reference(Patient | RelatedPerson | Practitioner | PractitionerRole | Organization) | Who attested the composition
custodian | ΣI | 0..1 | Reference(Organization) | Organization which maintains the composition
relatesTo | I | 0..* | BackboneElement | Relationships to other compositions/documents
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | I | 1..1 | code | replaces | transforms | signs | appends Binding: DocumentRelationshipType (required)
target[x] | I | 1..1 | | Target of the relationship
targetIdentifier | | | Identifier
targetReference | | | Reference(composition.html)
event | ΣI | 0..* | BackboneElement | The clinical service(s) being documented
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | ΣI | 0..* | CodeableConcept | Code(s) that apply to the event being documented Binding: ActCode (example)
period | ΣI | 0..1 | Period | The period covered by the documentation
detail | ΣI | 0..* | Reference(Resource) | The event(s) being documented
section | I | 0..* | BackboneElement | Composition is broken into sections cmp-1: A section must contain at least one of text, entries, or sub-sections cmp-2: A section can only have an emptyReason if it is empty
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
title | I | 0..1 | string | Label for section (e.g. for ToC)
code | I | 0..1 | CodeableConcept | Classification of section (recommended) Binding: Document Section Codes (example)
author | I | 0..* | Reference(Practitioner | PractitionerRole | Device | Patient | RelatedPerson | Organization) | Who and/or what authored the section
focus | I | 0..1 | Reference(Resource) | Who/what the section is about, when it is not about the subject of composition
text | I | 0..1 | Narrative | Text summary of the section, for human interpretation
mode | I | 0..1 | code | working | snapshot | changes Binding: ListMode (required)
orderedBy | I | 0..1 | CodeableConcept | Order of section entries Binding: List Order Codes (preferred)
entry | I | 0..* | Reference(Resource) | A reference to data that supports this section
emptyReason | I | 0..1 | CodeableConcept | Why the section is empty Binding: List Empty Reasons (preferred)
section | I | 0..* | | Unknown reference to #Composition.section Nested Section
Documentation for this format

```
<!-- Clinical Document -->

<Composition xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!--  0..1 Identifier
     Version-independent identifier for the Composition --></identifier>
 <status value="[code]"/><!--  1..1 preliminary | final | amended | entered-in-error   -->
 <type><!--  1..1 CodeableConcept Kind of composition (LOINC if possible)   --></type>
 <category><!--  0..* CodeableConcept Categorization of Composition   --></category>
 <subject><!--  0..1 Reference(Patient)
     Who and/or what the composition is about --></subject>
 <encounter><!--  0..1 Reference(Encounter) Context of the Composition --></encounter>
 <date value="[dateTime]"/><!--  1..1 Composition editing time -->
 <author><!--  1..* Reference(Practitioner)
     Who and/or what authored the composition --></author>
 <title value="[string]"/><!--  1..1 Human Readable name/title -->
 <confidentiality value="[code]"/><!--  0..1 As defined by affinity domain    -->
 <attester>  0..* BackboneElement  <!--  0..* Attests to accuracy of composition -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <mode value="[code]"/><!--  1..1 personal | professional | legal | official   -->
  <time value="[dateTime]"/><!--  0..1 When the composition was attested -->
  <party><!--  0..1 Reference(Patient)
      Who attested the composition --></party>
 </attester>
 <custodian><!--  0..1 Reference(Organization)
     Organization which maintains the composition --></custodian>
 <relatesTo>  0..* BackboneElement  <!--  0..* Relationships to other compositions/documents -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <code value="[code]"/><!--  1..1 replaces | transforms | signs | appends   -->
  <target[x]><!--  1..1 Identifier|Reference(Composition)
      Target of the relationship --></target[x]>
 </relatesTo>
 <event>  0..* BackboneElement  <!--  0..* The clinical service(s) being documented -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
```

```
  <code><!--  0..* CodeableConcept Code(s) that apply to the event being documented   --></code>
  <period><!--  0..1 Period
      The period covered by the documentation --></period>
  <detail><!--  0..* Reference(Resource)
      The event(s) being documented --></detail>
 </event>
 <section>  0..* BackboneElement  <!--  0..* Composition is broken into sections -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <title value="[string]"/><!--  0..1 Label for section (e.g. for ToC) -->
  <code><!--  0..1 CodeableConcept Classification of section (recommended)   --></code>
  <author><!--  0..* Reference(Practitioner)
      Who and/or what authored the section --></author>
  <focus><!--  0..1 Reference(Resource)
      Who/what the section is about, when it is not about the subject of composition --></focus>
  <text><!--  0..1 Narrative
      Text summary of the section, for human interpretation --></text>
  <mode value="[code]"/><!--  0..1 working | snapshot | changes   -->
  <orderedBy><!--  0..1 CodeableConcept Order of section entries   --></orderedBy>
  <entry><!--  0..* Reference(Resource)
      A reference to data that supports this section --></entry>
  <emptyReason><!--  0..1 CodeableConcept Why the section is empty   --></emptyReason>
  <section><!-- See #Composition.section  Nested Section --></section>
 </section>
</Composition>
```

```
{ // Clinical Document
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "extension" : [ //  sliced by value:url  in the specified order, Open
      { // Version-specific identifier for composition // C?
        // from Element: extension
      }
    ],
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "identifier" : { Identifier }, // C?Version-independent identifier for the Composition
    "status" : "<code>", // C? R! preliminary | final | amended | entered-in-error
    "type" : { CodeableConcept }, // C? R! Kind of composition (LOINC if possible)
    "category" : [{ CodeableConcept }], // C?Categorization of Composition
    "subject" : { Reference(Patient) }, // C?Who and/or what the composition is about
    "encounter" : { Reference(Encounter) }, // C?Context of the Composition
    "date" : "<dateTime>", // C? R! Composition editing time
    "author" : [{ Reference(Practitioner) }], // C? R! Who and/or what authored the composition
    "title" : "<string>", // C? R! Human Readable name/title
    "confidentiality" : "<code>", // C?As defined by affinity domain
    "attester" : [{ BackboneElement }], // C?Attests to accuracy of composition
    "custodian" : { Reference(Organization) }, // C?Organization which maintains the composition
    "relatesTo" : [{ BackboneElement }], // C?Relationships to other compositions/documents
    "event" : [{ BackboneElement }], // C?The clinical service(s) being documented
    "section" : [{ BackboneElement }], // C?Composition is broken into sections
  }
```

This structure is derived from Composition.

Summary

Extensions

This structure refers to these extensions:

• http://hl7.org/fhir/StructureDefinition/composition-clinicaldocument-versionNumber

Differential View

This structure is derived from Composition.

Name	Flags	Card.	Type	Description & Constraints
Composition		0..*		
composition-clinicaldocument-versionNumber		0..*	string	Version-specific identifier for composition URL: http://hl7.org/fhir/StructureDefinition/composition-clinicaldocument-versionNumber
subject		0..1	Reference(Patient | Practitioner | Group | Device | Location)	Documentation for this format

Snapshot View

Name	Flags	Card.	Type	Description & Constraints
Composition	I	0..*		A set of resources composed into a single coherent clinical statement with clinical attestation
id	Σ	0..1	string	Logical id of this artifact
meta	ΣI	0..1	Meta	Metadata about the resource
implicitRules	?!ΣI	0..1	uri	A set of rules under which this content was created
language	I	0..1	code	Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text	I	0..1	Narrative	Text summary of the resource, for human interpretation
contained		0..*	Resource	Contained, inline Resources
extension	I	0..*	Extension	Extension Slice: Unordered, Open by value:url
composition-clinicaldocument-versionNumber	I	0..*	string	Version-specific identifier for composition URL: http://hl7.org/fhir/StructureDefinition/composition-clinicaldocument-versionNumber

modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
identifier | ΣI | 0..1 | Identifier | Version-independent identifier for the Composition
status | ?!ΣI | 1..1 | code | preliminary | final | amended | entered-in-error Binding: CompositionStatus (required)
type | ΣI | 1..1 | CodeableConcept | Kind of composition (LOINC if possible) Binding: FHIR Document Type Codes (preferred)
category | ΣI | 0..* | CodeableConcept | Categorization of Composition Binding: Document Class Value Set (example)
subject | ΣI | 0..1 | Reference(Patient | Practitioner | Group | Device | Location) | Who and/or what the composition is about
encounter | ΣI | 0..1 | Reference(Encounter) | Context of the Composition
date | ΣI | 1..1 | dateTime | Composition editing time
author | ΣI | 1..* | Reference(Practitioner | PractitionerRole | Device | Patient | RelatedPerson | Organization) | Who and/or what authored the composition
title | ΣI | 1..1 | string | Human Readable name/title
confidentiality | ΣI | 0..1 | code | As defined by affinity domain Binding: ConfidentialityClassification (required)
attester | I | 0..* | BackboneElement | Attests to accuracy of composition
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
mode | I | 1..1 | code | personal | professional | legal | official Binding: CompositionAttestationMode (required)
time | I | 0..1 | dateTime | When the composition was attested
party | I | 0..1 | Reference(Patient | RelatedPerson | Practitioner | PractitionerRole | Organization) | Who attested the composition
custodian | ΣI | 0..1 | Reference(Organization) | Organization which maintains the composition
relatesTo | I | 0..* | BackboneElement | Relationships to other compositions/documents
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | I | 1..1 | code | replaces | transforms | signs | appends Binding: DocumentRelationshipType (required)
target[x] | I | 1..1 | | Target of the relationship
targetIdentifier | | | Identifier
targetReference | | | Reference(composition.html)
event | ΣI | 0..* | BackboneElement | The clinical service(s) being documented
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | ΣI | 0..* | CodeableConcept | Code(s) that apply to the event being documented Binding: ActCode (example)
period | ΣI | 0..1 | Period | The period covered by the documentation
detail | ΣI | 0..* | Reference(Resource) | The event(s) being documented
section | I | 0..* | BackboneElement | Composition is broken into sections cmp-1: A section must contain at least one of text, entries, or sub-sections cmp-2: A section can only have an emptyReason if it is empty
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
title | I | 0..1 | string | Label for section (e.g. for ToC)
code | I | 0..1 | CodeableConcept | Classification of section (recommended) Binding: Document Section Codes (example)
author | I | 0..* | Reference(Practitioner | PractitionerRole | Device | Patient | RelatedPerson | Organization) | Who and/or what authored the section
focus | I | 0..1 | Reference(Resource) | Who/what the section is about, when it is not about the subject of composition
text | I | 0..1 | Narrative | Text summary of the section, for human interpretation
mode | I | 0..1 | code | working | snapshot | changes Binding: ListMode (required)
orderedBy | I | 0..1 | CodeableConcept | Order of section entries Binding: List Order Codes (preferred)
entry | I | 0..* | Reference(Resource) | A reference to data that supports this section
emptyReason | I | 0..1 | CodeableConcept | Why the section is empty Binding: List Empty Reasons (preferred)
section | I | 0..* | | Unknown reference to #Composition.section Nested Section
Documentation for this format

XML Template

```
<!-- Clinical Document -->

<Composition xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!--  0..1 Identifier
     Version-independent identifier for the Composition --></identifier>
 <status value="[code]"/><!--  1..1 preliminary | final | amended | entered-in-error   -->
 <type><!--  1..1 CodeableConcept Kind of composition (LOINC if possible)   --></type>
 <category><!--  0..* CodeableConcept Categorization of Composition   --></category>
 <subject><!--  0..1 Reference(Patient)
     Who and/or what the composition is about --></subject>
 <encounter><!--  0..1 Reference(Encounter) Context of the Composition --></encounter>
 <date value="[dateTime]"/><!--  1..1 Composition editing time -->
 <author><!--  1..* Reference(Practitioner)
```

```
    Who and/or what authored the composition --></author>
 <title value="[string]"/><!--  1..1 Human Readable name/title -->
 <confidentiality value="[code]"/><!--  0..1 As defined by affinity domain    -->
 <attester>  0..* BackboneElement  <!--  0..* Attests to accuracy of composition -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <mode value="[code]"/><!--  1..1 personal | professional | legal | official   -->
  <time value="[dateTime]"/><!--  0..1 When the composition was attested -->
  <party><!--  0..1 Reference(Patient)
      Who attested the composition --></party>
 </attester>
 <custodian><!--  0..1 Reference(Organization)
     Organization which maintains the composition --></custodian>
 <relatesTo>  0..* BackboneElement  <!--  0..* Relationships to other compositions/documents -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <code value="[code]"/><!--  1..1 replaces | transforms | signs | appends   -->
  <target[x]><!--  1..1 Identifier|Reference(Composition)
      Target of the relationship --></target[x]>
 </relatesTo>
 <event>  0..* BackboneElement  <!--  0..* The clinical service(s) being documented -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <code><!--  0..* CodeableConcept Code(s) that apply to the event being documented   --></code>
  <period><!--  0..1 Period
      The period covered by the documentation --></period>
  <detail><!--  0..* Reference(Resource)
      The event(s) being documented --></detail>
 </event>
 <section>  0..* BackboneElement  <!--  0..* Composition is broken into sections -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <title value="[string]"/><!--  0..1 Label for section (e.g. for ToC) -->
  <code><!--  0..1 CodeableConcept Classification of section (recommended)   --></code>
  <author><!--  0..* Reference(Practitioner)
      Who and/or what authored the section --></author>
  <focus><!--  0..1 Reference(Resource)
      Who/what the section is about, when it is not about the subject of composition --></focus>
  <text><!--  0..1 Narrative
      Text summary of the section, for human interpretation --></text>
  <mode value="[code]"/><!--  0..1 working | snapshot | changes   -->
  <orderedBy><!--  0..1 CodeableConcept Order of section entries   --></orderedBy>
  <entry><!--  0..* Reference(Resource)
      A reference to data that supports this section --></entry>
  <emptyReason><!--  0..1 CodeableConcept Why the section is empty   --></emptyReason>
  <section><!-- See #Composition.section  Nested Section --></section>
 </section>
</Composition>
```

JSON Template

```
{ // Clinical Document
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "extension" : [ //  sliced by value:url  in the specified order, Open
      { // Version-specific identifier for composition // C?
        // from Element: extension
      }
    ],
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "identifier" : { Identifier }, // C?Version-independent identifier for the Composition
    "status" : "<code>", // C? R! preliminary | final | amended | entered-in-error
    "type" : { CodeableConcept }, // C? R! Kind of composition (LOINC if possible)
    "category" : [{ CodeableConcept }], // C?Categorization of Composition
    "subject" : { Reference(Patient) }, // C?Who and/or what the composition is about
    "encounter" : { Reference(Encounter) }, // C?Context of the Composition
    "date" : "<dateTime>", // C? R! Composition editing time
    "author" : [{ Reference(Practitioner) }], // C? R! Who and/or what authored the composition
    "title" : "<string>", // C? R! Human Readable name/title
    "confidentiality" : "<code>", // C?As defined by affinity domain
    "attester" : [{ BackboneElement }], // C?Attests to accuracy of composition
    "custodian" : { Reference(Organization) }, // C?Organization which maintains the composition
    "relatesTo" : [{ BackboneElement }], // C?Relationships to other compositions/documents
    "event" : [{ BackboneElement }], // C?The clinical service(s) being documented
    "section" : [{ BackboneElement }], // C?Composition is broken into sections
  }
```

Alternate definitions: Master Definition (XML, JSON), Schematron

2.41.14.2.2 Terminology Bindings

Path	Name	Conformance	ValueSet
Composition.language	Common Languages	preferred	Common Languages
Composition.status	CompositionStatus	required	CompositionStatus
Composition.type	FHIR Document Type Codes	preferred	FHIR Document Type Codes
Composition.category	Document Class Value Set	example	Document Class Value Set

Composition.confidentiality | V3 Value SetConfidentialityClassification | required | V3 Value SetConfidentialityClassification
Composition.attester.mode | CompositionAttestationMode | required | CompositionAttestationMode
Composition.relatesTo.code | DocumentRelationshipType | required | DocumentRelationshipType
Composition.event.code | v3 Code System ActCode | example | v3 Code System ActCode
Composition.section.code | Document Section Codes | example | Document Section Codes
Composition.section.mode | ListMode | required | ListMode
Composition.section.orderedBy | List Order Codes | preferred | List Order Codes
Composition.section.emptyReason | List Empty Reasons | preferred | List Empty Reasons

2.41.14.2.3 Constraints

Id | Path | Details | Requirements
cmp-1 | Composition.section | A section must contain at least one of text, entries, or sub-sections Expression : text.exists() or entry.exists() or section.exists()
cmp-2 | Composition.section | A section can only have an emptyReason if it is empty Expression : emptyReason.empty() or entry.empty()

2.41.15 Profile for Catalog

URL: https://hl7.org/fhir/R4/composition-catalog.html

2.41.15 Profile for Catalog

Structured Documents Work Group | Maturity Level: N/A | Standards Status: Informative

2.41.15.1 Introduction

This profile creates a document that represents a Catalog. The Catalog is a curated collection of related entries (CatalogEntry) which can point to devices, medications, specimenDefinitions, planDefinitions or any definitional resources.

The Definitional resources contain static, immutable attributes. The Catalog provides the curated, approved and context-specific attributes and relations.

2.41.15.2 Content

Profiles:
Catalog | A set of resources composed into a single coherent clinical statement with clinical attestation
Extensions:
cqm-ValidityPeriod | ValidityPeriod : The period in which the catalog is valid.

2.41.15.3 StructureDefinition: Catalog

URL: https://hl7.org/fhir/R4/catalog.html

2.41.15.3 StructureDefinition: Catalog

Structured Documents Work Group | Maturity Level: 1 | Trial Use | Use Context: Any

The official URL for this profile is:

`http://hl7.org/fhir/StructureDefinition/catalog`

A set of resources composed into a single coherent clinical statement with clinical attestation

This profile was published on Fri, Jan 31, 2014 00:00+1100 as a draft by Health Level Seven, Inc. - Clinical Quality Information WG.

2.41.15.3.1 Formal Views of Profile Content

Description of Profiles, Differentials, Snapshots, and how the XML and JSON presentations work.

• Text Summary
• Differential Table
• Snapshot Table
• XML Template
• JSON Template
• All

This structure is derived from Composition.

Summary

Mandatory: 4 elements (+1 nested mandatory element) Fixed Value: 1 element Prohibited: 1 element

Extensions

This structure refers to these extensions:

• http://hl7.org/fhir/StructureDefinition/cqm-ValidityPeriod

This structure is derived from Composition.

Name | Flags | Card. | Type | Description & Constraints
Composition | | 0..* | | A set of resources composed into a single coherent clinical statement with clinical attestation
cqm-ValidityPeriod | | 1..1 | dateTime | The validity of the catalog URL: http://hl7.org/fhir/StructureDefinition/cqm-ValidityPeriod
type | | 1..1 | CodeableConcept | The type of document - a Catalog Fixed Value: As shown
text | | 1..1 | string | Plain text representation of the concept Fixed Value: Catalog
category | | 1..1 | CodeableConcept | The Content of the section Binding: CatalogType (example)
subject | | 0..0
date:IssueDate | | 1..1 | dateTime | When the Catalog was created
section | | 0..* | BackboneElement
entry | | 1..* | Reference(CatalogEntry) | The Content of the section
Documentation for this format

Name | Flags | Card. | Type | Description & Constraints
Composition | I | 0..* | | A set of resources composed into a single coherent clinical statement with clinical attestation
id | Σ | 0..1 | string | Logical id of this artifact
meta | ΣI | 0..1 | Meta | Metadata about the resource
implicitRules | ?!ΣI | 0..1 | uri | A set of rules under which this content was created
language | I | 0..1 | code | Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | I | 0..* | Extension | Extension Slice: Unordered, Open by value:url
cqm-ValidityPeriod | I | 1..1 | dateTime | The validity of the catalog URL: http://hl7.org/fhir/StructureDefinition/cqm-ValidityPeriod
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
identifier | ΣI | 0..1 | Identifier | Version-independent identifier for the Composition
status | ?!ΣI | 1..1 | code | preliminary | final | amended | entered-in-error Binding: CompositionStatus (required)
type | ΣI | 1..1 | CodeableConcept | The type of document - a Catalog Binding: FHIR Document Type Codes (preferred) Fixed Value: As shown
id | | 0..0 | string | Unique id for inter-element referencing
extension | | 0..0 | Extension | Additional content defined by implementations
coding | | 0..0 | Coding | Code defined by a terminology system
text | | 1..1 | string | Plain text representation of the concept Fixed Value: Catalog
category | ΣI | 1..1 | CodeableConcept | The Content of the section Binding: CatalogType (example)
encounter | ΣI | 0..1 | Reference(Encounter) | Context of the Composition
date:IssueDate | ΣI | 1..1 | dateTime | When the Catalog was created
author | ΣI | 1..* | Reference(Practitioner | PractitionerRole | Device | Patient | RelatedPerson | Organization) | Who and/or what authored the composition
title | ΣI | 1..1 | string | Human Readable name/title
confidentiality | ΣI | 0..1 | code | As defined by affinity domain Binding: ConfidentialityClassification (required)
attester | I | 0..* | BackboneElement | Attests to accuracy of composition
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
mode | I | 1..1 | code | personal | professional | legal | official Binding: CompositionAttestationMode (required)
time | I | 0..1 | dateTime | When the composition was attested
party | I | 0..1 | Reference(Patient | RelatedPerson | Practitioner | PractitionerRole | Organization) | Who attested the composition
custodian | ΣI | 0..1 | Reference(Organization) | Organization which maintains the composition
relatesTo | I | 0..* | BackboneElement | Relationships to other compositions/documents
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | I | 1..1 | code | replaces | transforms | signs | appends Binding: DocumentRelationshipType (required)
target[x] | I | 1..1 | | Target of the relationship
targetIdentifier | | | Identifier
targetReference | | | Reference(composition.html)
event | ΣI | 0..* | BackboneElement | The clinical service(s) being documented
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | ΣI | 0..* | CodeableConcept | Code(s) that apply to the event being documented Binding: ActCode (example)
period | ΣI | 0..1 | Period | The period covered by the documentation
detail | ΣI | 0..* | Reference(Resource) | The event(s) being documented
section | I | 0..* | BackboneElement | Composition is broken into sections
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
title | I | 0..1 | string | Label for section (e.g. for ToC)
code | I | 0..1 | CodeableConcept | Classification of section (recommended) Binding: Document Section Codes (example)
author | I | 0..* | Reference(Practitioner | PractitionerRole | Device | Patient | RelatedPerson | Organization) | Who and/or what authored the section
focus | I | 0..1 | Reference(Resource) | Who/what the section is about, when it is not about the subject of composition
text | I | 0..1 | Narrative | Text summary of the section, for human interpretation
mode | I | 0..1 | code | working | snapshot | changes Binding: ListMode (required)
orderedBy | I | 0..1 | CodeableConcept | Order of section entries Binding: List Order Codes (preferred)

entry | I | 1..* | Reference(CatalogEntry) | The Content of the section
emptyReason | I | 0..1 | CodeableConcept | Why the section is empty Binding: List Empty Reasons (preferred)
section | I | 0..* | | Unknown reference to #Composition.section Nested Section
Documentation for this format

```
<!-- Profile for Catalog -->

<Composition xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!--  0..1 Identifier
     Version-independent identifier for the Composition --></identifier>
 <status value="[code]"/><!--  1..1 preliminary | final | amended | entered-in-error   -->
 <type><!--  1..1 CodeableConcept The type of document - a Catalog   --></type>
 <category><!--  1..1 CodeableConcept The Content of the section   --></category>
 <subject><!--  0..0 Reference(Resource)
     Who and/or what the composition is about --></subject>
 <encounter><!--  0..1 Reference(Encounter) Context of the Composition --></encounter>
 <date value="[dateTime]"/><!--  1..1 When the Catalog was created -->
 <author><!--  1..* Reference(Practitioner)
     Who and/or what authored the composition --></author>
 <title value="[string]"/><!--  1..1 Human Readable name/title -->
 <confidentiality value="[code]"/><!--  0..1 As defined by affinity domain    -->
 <attester>  0..* BackboneElement  <!--  0..* Attests to accuracy of composition -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <mode value="[code]"/><!--  1..1 personal | professional | legal | official   -->
  <time value="[dateTime]"/><!--  0..1 When the composition was attested -->
  <party><!--  0..1 Reference(Patient)
      Who attested the composition --></party>
 </attester>
 <custodian><!--  0..1 Reference(Organization)
     Organization which maintains the composition --></custodian>
 <relatesTo>  0..* BackboneElement  <!--  0..* Relationships to other compositions/documents -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <code value="[code]"/><!--  1..1 replaces | transforms | signs | appends   -->
  <target[x]><!--  1..1 Identifier|Reference(Composition)
      Target of the relationship --></target[x]>
 </relatesTo>
 <event>  0..* BackboneElement  <!--  0..* The clinical service(s) being documented -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <code><!--  0..* CodeableConcept Code(s) that apply to the event being documented   --></code>
  <period><!--  0..1 Period
      The period covered by the documentation --></period>
  <detail><!--  0..* Reference(Resource)
      The event(s) being documented --></detail>
 </event>
 <section>  0..* BackboneElement  <!--  0..* Composition is broken into sections -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <title value="[string]"/><!--  0..1 Label for section (e.g. for ToC) -->
  <code><!--  0..1 CodeableConcept Classification of section (recommended)   --></code>
  <author><!--  0..* Reference(Practitioner)
      Who and/or what authored the section --></author>
  <focus><!--  0..1 Reference(Resource)
      Who/what the section is about, when it is not about the subject of composition --></focus>
  <text><!--  0..1 Narrative
      Text summary of the section, for human interpretation --></text>
  <mode value="[code]"/><!--  0..1 working | snapshot | changes   -->
  <orderedBy><!--  0..1 CodeableConcept Order of section entries   --></orderedBy>
  <entry><!--  1..* Reference(CatalogEntry)
      The Content of the section --></entry>
  <emptyReason><!--  0..1 CodeableConcept Why the section is empty   --></emptyReason>
  <section><!-- See #Composition.section  Nested Section --></section>
 </section>
</Composition>
```

```
{ // Profile for Catalog
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "extension" : [ //  sliced by value:url  in the specified order, Open
      { // The validity of the catalog // C? R!
        // from Element: extension
      }
    ],
```

```
  "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
  "identifier" : { Identifier }, // C?Version-independent identifier for the Composition
  "status" : "<code>", // C? R! preliminary | final | amended | entered-in-error
  "type" : { CodeableConcept }, // C? R!
  "category" : { CodeableConcept }, // C? R! The Content of the section
  "encounter" : { Reference(Encounter) }, // C?Context of the Composition
  "date" : "<dateTime>", // C? R! When the Catalog was created
  "author" : [{ Reference(Practitioner) }], // C? R! Who and/or what authored the composition
  "title" : "<string>", // C? R! Human Readable name/title
  "confidentiality" : "<code>", // C?As defined by affinity domain
  "attester" : [{ BackboneElement }], // C?Attests to accuracy of composition
  "custodian" : { Reference(Organization) }, // C?Organization which maintains the composition
  "relatesTo" : [{ BackboneElement }], // C?Relationships to other compositions/documents
  "event" : [{ BackboneElement }], // C?The clinical service(s) being documented
  "section" : [{ BackboneElement }], // C?Composition is broken into sections
}
```

This structure is derived from Composition.

Summary

Mandatory: 4 elements (+1 nested mandatory element) Fixed Value: 1 element Prohibited: 1 element

Extensions

This structure refers to these extensions:

• http://hl7.org/fhir/StructureDefinition/cqm-ValidityPeriod

Differential View

This structure is derived from Composition.

Name	Flags	Card.	Type	Description & Constraints
Composition		0..*		A set of resources composed into a single coherent clinical statement with clinical attestation
cqm-ValidityPeriod		1..1	dateTime	The validity of the catalog URL: http://hl7.org/fhir/StructureDefinition/cqm-ValidityPeriod
type		1..1	CodeableConcept	The type of document - a Catalog Fixed Value: As shown
text		1..1	string	Plain text representation of the concept Fixed Value: Catalog
category		1..1	CodeableConcept	The Content of the section Binding: CatalogType (example)
subject		0..0		
date:IssueDate		1..1	dateTime	When the Catalog was created
section		0..*	BackboneElement	
entry		1..*	Reference(CatalogEntry)	The Content of the section

Documentation for this format

Snapshot View

Name	Flags	Card.	Type	Description & Constraints
Composition	I	0..*		A set of resources composed into a single coherent clinical statement with clinical attestation
id	Σ	0..1	string	Logical id of this artifact
meta	ΣI	0..1	Meta	Metadata about the resource
implicitRules	?!ΣI	0..1	uri	A set of rules under which this content was created
language	I	0..1	code	Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text	I	0..1	Narrative	Text summary of the resource, for human interpretation
contained		0..*	Resource	Contained, inline Resources
extension	I	0..*	Extension	Extension Slice: Unordered, Open by value:url
cqm-ValidityPeriod	I	1..1	dateTime	The validity of the catalog URL: http://hl7.org/fhir/StructureDefinition/cqm-ValidityPeriod
modifierExtension	?!I	0..*	Extension	Extensions that cannot be ignored
identifier	ΣI	0..1	Identifier	Version-independent identifier for the Composition
status	?!ΣI	1..1	code	preliminary | final | amended | entered-in-error Binding: CompositionStatus (required)
type	ΣI	1..1	CodeableConcept	The type of document - a Catalog Binding: FHIR Document Type Codes (preferred) Fixed Value: As shown
id		0..0	string	Unique id for inter-element referencing
extension		0..0	Extension	Additional content defined by implementations
coding		0..0	Coding	Code defined by a terminology system
text		1..1	string	Plain text representation of the concept Fixed Value: Catalog
category	ΣI	1..1	CodeableConcept	The Content of the section Binding: CatalogType (example)
encounter	ΣI	0..1	Reference(Encounter)	Context of the Composition
date:IssueDate	ΣI	1..1	dateTime	When the Catalog was created
author	ΣI	1..*	Reference(Practitioner | PractitionerRole | Device | Patient | RelatedPerson | Organization)	Who and/or what authored the composition
title	ΣI	1..1	string	Human Readable name/title
confidentiality	ΣI	0..1	code	As defined by affinity domain Binding: ConfidentialityClassification (required)
attester	I	0..*	BackboneElement	Attests to accuracy of composition
id		0..1	string	Unique id for inter-element referencing
extension	I	0..*	Extension	Additional content defined by implementations
modifierExtension	?!ΣI	0..*	Extension	Extensions that cannot be ignored even if unrecognized
mode	I	1..1	code	personal | professional | legal | official Binding: CompositionAttestationMode (required)
time	I	0..1	dateTime	When the composition was attested
party	I	0..1	Reference(Patient | RelatedPerson | Practitioner | PractitionerRole | Organization)	Who attested the composition
custodian	ΣI	0..1	Reference(Organization)	Organization which maintains the composition
relatesTo	I	0..*	BackboneElement	Relationships to other compositions/documents

id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | I | 1..1 | code | replaces | transforms | signs | appends Binding: DocumentRelationshipType (required)
target[x] | I | 1..1 | | Target of the relationship
targetIdentifier | | | Identifier
targetReference | | | Reference(composition.html)
event | ΣI | 0..* | BackboneElement | The clinical service(s) being documented
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | ΣI | 0..* | CodeableConcept | Code(s) that apply to the event being documented Binding: ActCode (example)
period | ΣI | 0..1 | Period | The period covered by the documentation
detail | ΣI | 0..* | Reference(Resource) | The event(s) being documented
section | I | 0..* | BackboneElement | Composition is broken into sections
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
title | I | 0..1 | string | Label for section (e.g. for ToC)
code | I | 0..1 | CodeableConcept | Classification of section (recommended) Binding: Document Section Codes (example)
author | I | 0..* | Reference(Practitioner | PractitionerRole | Device | Patient | RelatedPerson | Organization) | Who and/or what authored the section
focus | I | 0..1 | Reference(Resource) | Who/what the section is about, when it is not about the subject of composition
text | I | 0..1 | Narrative | Text summary of the section, for human interpretation
mode | I | 0..1 | code | working | snapshot | changes Binding: ListMode (required)
orderedBy | I | 0..1 | CodeableConcept | Order of section entries Binding: List Order Codes (preferred)
entry | I | 1..* | Reference(CatalogEntry) | The Content of the section
emptyReason | I | 0..1 | CodeableConcept | Why the section is empty Binding: List Empty Reasons (preferred)
section | I | 0..* | | Unknown reference to #Composition.section Nested Section
Documentation for this format

XML Template

```
<!-- Profile for Catalog -->

<Composition xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!--  0..1 Identifier
     Version-independent identifier for the Composition --></identifier>
 <status value="[code]"/><!--  1..1 preliminary | final | amended | entered-in-error   -->
 <type><!--  1..1 CodeableConcept The type of document - a Catalog   --></type>
 <category><!--  1..1 CodeableConcept The Content of the section   --></category>
 <subject><!--  0..0 Reference(Resource)
     Who and/or what the composition is about --></subject>
 <encounter><!--  0..1 Reference(Encounter) Context of the Composition --></encounter>
 <date value="[dateTime]"/><!--  1..1 When the Catalog was created -->
 <author><!--  1..* Reference(Practitioner)
     Who and/or what authored the composition --></author>
 <title value="[string]"/><!--  1..1 Human Readable name/title -->
 <confidentiality value="[code]"/><!--  0..1 As defined by affinity domain    -->
 <attester>  0..* BackboneElement  <!--  0..* Attests to accuracy of composition -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <mode value="[code]"/><!--  1..1 personal | professional | legal | official   -->
  <time value="[dateTime]"/><!--  0..1 When the composition was attested -->
  <party><!--  0..1 Reference(Patient)
      Who attested the composition --></party>
 </attester>
 <custodian><!--  0..1 Reference(Organization)
     Organization which maintains the composition --></custodian>
 <relatesTo>  0..* BackboneElement  <!--  0..* Relationships to other compositions/documents -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <code value="[code]"/><!--  1..1 replaces | transforms | signs | appends   -->
  <target[x]><!--  1..1 Identifier|Reference(Composition)
      Target of the relationship --></target[x]>
 </relatesTo>
 <event>  0..* BackboneElement  <!--  0..* The clinical service(s) being documented -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <code><!--  0..* CodeableConcept Code(s) that apply to the event being documented   --></code>
  <period><!--  0..1 Period
      The period covered by the documentation --></period>
  <detail><!--  0..* Reference(Resource)
```

```
      The event(s) being documented --></detail>
 </event>
 <section>  0..* BackboneElement  <!--  0..* Composition is broken into sections -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <title value="[string]"/><!--  0..1 Label for section (e.g. for ToC) -->
  <code><!--  0..1 CodeableConcept Classification of section (recommended)   --></code>
  <author><!--  0..* Reference(Practitioner)
      Who and/or what authored the section --></author>
  <focus><!--  0..1 Reference(Resource)
      Who/what the section is about, when it is not about the subject of composition --></focus>
  <text><!--  0..1 Narrative
      Text summary of the section, for human interpretation --></text>
  <mode value="[code]"/><!--  0..1 working | snapshot | changes   -->
  <orderedBy><!--  0..1 CodeableConcept Order of section entries   --></orderedBy>
  <entry><!--  1..* Reference(CatalogEntry)
      The Content of the section --></entry>
  <emptyReason><!--  0..1 CodeableConcept Why the section is empty   --></emptyReason>
  <section><!-- See #Composition.section  Nested Section --></section>
 </section>
</Composition>
```

JSON Template

```
{ // Profile for Catalog
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "extension" : [ //  sliced by value:url  in the specified order, Open
      { // The validity of the catalog // C? R!
        // from Element: extension
      }
    ],
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "identifier" : { Identifier }, // C?Version-independent identifier for the Composition
    "status" : "<code>", // C? R! preliminary | final | amended | entered-in-error
    "type" : { CodeableConcept }, // C? R!
    "category" : { CodeableConcept }, // C? R! The Content of the section
    "encounter" : { Reference(Encounter) }, // C?Context of the Composition
    "date" : "<dateTime>", // C? R! When the Catalog was created
    "author" : [{ Reference(Practitioner) }], // C? R! Who and/or what authored the composition
    "title" : "<string>", // C? R! Human Readable name/title
    "confidentiality" : "<code>", // C?As defined by affinity domain
    "attester" : [{ BackboneElement }], // C?Attests to accuracy of composition
    "custodian" : { Reference(Organization) }, // C?Organization which maintains the composition
    "relatesTo" : [{ BackboneElement }], // C?Relationships to other compositions/documents
    "event" : [{ BackboneElement }], // C?The clinical service(s) being documented
    "section" : [{ BackboneElement }], // C?Composition is broken into sections
  }
```

Alternate definitions: Master Definition (XML, JSON), Schematron

2.41.15.3.2 Terminology Bindings

Path | Name | Conformance | ValueSet
Composition.language | Common Languages | preferred | Common Languages
Composition.status | CompositionStatus | required | CompositionStatus
Composition.type | FHIR Document Type Codes | preferred | FHIR Document Type Codes
Composition.category | CatalogType | example | CatalogType
Composition.confidentiality | V3 Value SetConfidentialityClassification | required | V3 Value SetConfidentialityClassification
Composition.attester.mode | CompositionAttestationMode | required | CompositionAttestationMode
Composition.relatesTo.code | DocumentRelationshipType | required | DocumentRelationshipType
Composition.event.code | v3 Code System ActCode | example | v3 Code System ActCode
Composition.section.code | Document Section Codes | example | Document Section Codes
Composition.section.mode | ListMode | required | ListMode
Composition.section.orderedBy | List Order Codes | preferred | List Order Codes
Composition.section.emptyReason | List Empty Reasons | preferred | List Empty Reasons

2.41.16 Composition HL7 Extensions

URL: https://hl7.org/fhir/R4/composition-extensions.html

Defines common extensions used with or related to the Composition resource

2.41.16 Composition HL7 Extensions

Structured Documents Work Group | Maturity Level: N/A | Standards Status: Informative

2.41.16.1 Content

Extensions:

composition-section-subject | section-subject : Specifies that the section has a different subject that the Composition, or it's container section.

2.41.17 JSON Schema for Composition

URL: https://hl7.org/fhir/R4/composition.schema.json.html

2.41.17 JSON Schema for Composition

Structured Documents Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

This is a representation of the json schema for Composition, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/Composition",
  "$ref": "#/definitions/Composition",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "Composition": {
      "description": "A set of healthcare-related information that is assembled together into a single logical package
    that provides a single coherent statement of meaning, establishes its own context and that has clinical attestation
    with regard to who is making the statement. A Composition defines the structure and narrative content necessary for
    a document. However, a Composition alone does not constitute a document. Rather, the Composition must be the first
    entry in a Bundle where Bundle.type\u003ddocument, and any other resources referenced from Composition must be
    included as subsequent entries in the Bundle (for example Patient, Practitioner, Encounter, etc.).",
      "properties": {
        "resourceType": {
          "description": "This is a Composition resource",
          "const": "Composition"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
          "description": "A human-readable narrative that contains a summary of the resource and can be used to
    represent the content of the resource to a human. The narrative need not encode all the structured data, but is
    required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
    Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
          "$ref": "Narrative.schema.json#/definitions/Narrative"
        },
        "contained": {
          "description": "These resources do not have an independent existence apart from the resource that contains
    them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
          "items": {
            "$ref": "ResourceList.schema.json#/definitions/ResourceList"
          },
          "type": "array"
        },
        "extension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
     definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "modifierExtension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource and that modifies the understanding of the element that contains it and/or the understanding of the
    containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
    of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
    met as part of the definition of the extension. Applications processing a resource are required to check for
```

```
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "A version-independent identifier for the Composition. This identifier stays constant as the
composition is changed over time.",
      "$ref": "Identifier.schema.json#/definitions/Identifier"
    },
    "status": {
      "description": "The workflow/clinical status of this composition. The status is a marker for the clinical
standing of the document.",
      "enum": [
        "preliminary",
        "final",
        "amended",
        "entered-in-error"
      ]
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "type": {
      "description": "Specifies the particular kind of composition (e.g. History and Physical, Discharge Summary,
Progress Note). This usually equates to the purpose of making the composition.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "category": {
      "description": "A categorization for the type of the composition - helps for indexing and searching. This may
be implied by or derived from the code specified in the Composition Type.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "subject": {
      "description": "Who or what the composition is about. The composition can be about a person, (patient or
healthcare practitioner), a device (e.g. a machine) or even a group of subjects (such as a document about a herd of
livestock, or a set of patients that share a common exposure).",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "encounter": {
      "description": "Describes the clinical encounter or type of care this documentation is associated with.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "date": {
      "description": "The composition editing time, when the composition was last logically changed by the author.",
      "$ref": "#/definitions/dateTime"
    },
    "_date": {
      "description": "Extensions for date",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "author": {
      "description": "Identifies who is responsible for the information in the composition, not necessarily who
typed it in.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "title": {
      "description": "Official human-readable label for the composition.",
      "$ref": "#/definitions/string"
    },
    "_title": {
      "description": "Extensions for title",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "confidentiality": {
      "description": "The code specifying the level of confidentiality of the Composition.",
      "$ref": "#/definitions/code"
    },
    "_confidentiality": {
      "description": "Extensions for confidentiality",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "attester": {
      "description": "A participant who has attested to the accuracy of the composition/document.",
      "items": {
        "$ref": "#/definitions/Composition_Attester"
      },
      "type": "array"
    },
    "custodian": {
```

```
      "description": "Identifies the organization or group who is responsible for ongoing maintenance of and access to the composition/document information.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "relatesTo": {
      "description": "Relationships that this composition has with other compositions or documents that already exist.",
      "items": {
        "$ref": "#/definitions/Composition_RelatesTo"
      },
      "type": "array"
    },
    "event": {
      "description": "The clinical service, such as a colonoscopy or an appendectomy, being documented.",
      "items": {
        "$ref": "#/definitions/Composition_Event"
      },
      "type": "array"
    },
    "section": {
      "description": "The root of the sections that make up the composition.",
      "items": {
        "$ref": "#/definitions/Composition_Section"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "author",
    "type",
    "resourceType"
  ]
},
"Composition_Attester": {
  "description": "A set of healthcare-related information that is assembled together into a single logical package that provides a single coherent statement of meaning, establishes its own context and that has clinical attestation with regard to who is making the statement. A Composition defines the structure and narrative content necessary for a document. However, a Composition alone does not constitute a document. Rather, the Composition must be the first entry in a Bundle where Bundle.type\u003ddocument, and any other resources referenced from Composition must be included as subsequent entries in the Bundle (for example Patient, Practitioner, Encounter, etc.).",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element and that modifies the understanding of the element in which it is contained and/or the understanding of the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension. Applications processing a resource are required to check for modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "mode": {
      "description": "The type of attestation the authenticator offers.",
      "enum": [
        "personal",
        "professional",
        "legal",
        "official"
      ]
    },
    "_mode": {
      "description": "Extensions for mode",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "time": {
      "description": "When the composition was attested by the party.",
      "$ref": "#/definitions/dateTime"
    },
    "_time": {
```

```
      "description": "Extensions for time",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "party": {
      "description": "Who attested the composition in the specified way.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    }
  },
  "additionalProperties": false
},
"Composition_RelatesTo": {
  "description": "A set of healthcare-related information that is assembled together into a single logical package
that provides a single coherent statement of meaning, establishes its own context and that has clinical attestation
with regard to who is making the statement. A Composition defines the structure and narrative content necessary for
a document. However, a Composition alone does not constitute a document. Rather, the Composition must be the first
entry in a Bundle where Bundle.type\u003ddocument, and any other resources referenced from Composition must be
included as subsequent entries in the Bundle (for example Patient, Practitioner, Encounter, etc.).",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "code": {
      "description": "The type of relationship that this composition has with anther composition or document.",
      "$ref": "#/definitions/code"
    },
    "_code": {
      "description": "Extensions for code",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "targetIdentifier": {
      "description": "The target composition/document of this relationship.",
      "$ref": "Identifier.schema.json#/definitions/Identifier"
    },
    "targetReference": {
      "description": "The target composition/document of this relationship.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    }
  },
  "additionalProperties": false
},
"Composition_Event": {
  "description": "A set of healthcare-related information that is assembled together into a single logical package
that provides a single coherent statement of meaning, establishes its own context and that has clinical attestation
with regard to who is making the statement. A Composition defines the structure and narrative content necessary for
a document. However, a Composition alone does not constitute a document. Rather, the Composition must be the first
entry in a Bundle where Bundle.type\u003ddocument, and any other resources referenced from Composition must be
included as subsequent entries in the Bundle (for example Patient, Practitioner, Encounter, etc.).",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
```

```
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "code": {
      "description": "This list of codes represents the main clinical acts, such as a colonoscopy or an
appendectomy, being documented. In some cases, the event is inherent in the typeCode, such as a \"History and
Physical Report\" in which the procedure being documented is necessarily a \"History and Physical\" act.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "period": {
      "description": "The period of time covered by the documentation. There is no assertion that the documentation
is a complete representation for this period, only that it documents events during this time.",
      "$ref": "Period.schema.json#/definitions/Period"
    },
    "detail": {
      "description": "The description and/or reference of the event(s) being documented. For example, this could be
used to document such a colonoscopy or an appendectomy.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    }
  },
  "additionalProperties": false
},
"Composition_Section": {
  "description": "A set of healthcare-related information that is assembled together into a single logical package
that provides a single coherent statement of meaning, establishes its own context and that has clinical attestation
with regard to who is making the statement. A Composition defines the structure and narrative content necessary for
a document. However, a Composition alone does not constitute a document. Rather, the Composition must be the first
entry in a Bundle where Bundle.type\u003ddocument, and any other resources referenced from Composition must be
included as subsequent entries in the Bundle (for example Patient, Practitioner, Encounter, etc.).",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "title": {
      "description": "The label for this particular section.  This will be part of the rendered content for the
document, and is often used to build a table of contents.",
      "$ref": "#/definitions/string"
    },
    "_title": {
      "description": "Extensions for title",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "code": {
      "description": "A code identifying the kind of content contained within the section. This must be consistent
with the section title.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
```

```
        },
        "author": {
          "description": "Identifies who is responsible for the information in this section, not necessarily who typed
    it in.",
          "items": {
            "$ref": "Reference.schema.json#/definitions/Reference"
          },
          "type": "array"
        },
        "focus": {
          "description": "The actual focus of the section when it is not the subject of the composition, but instead
    represents something or someone associated with the subject such as (for a patient subject) a spouse, parent, fetus,
     or donor. If not focus is specified, the focus is assumed to be focus of the parent section, or, for a section in
    the Composition itself, the subject of the composition. Sections with a focus SHALL only include resources where the
     logical subject (patient, subject, focus, etc.) matches the section focus, or the resources have no logical subject
     (few resources).",
          "$ref": "Reference.schema.json#/definitions/Reference"
        },
        "text": {
          "description": "A human-readable narrative that contains the attested content of the section, used to
    represent the content of the resource to a human. The narrative need not encode all the structured data, but is
    required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.",
          "$ref": "Narrative.schema.json#/definitions/Narrative"
        },
        "mode": {
          "description": "How the entry list was prepared - whether it is a working list that is suitable for being
    maintained on an ongoing basis, or if it represents a snapshot of a list of items from another source, or whether it
     is a prepared list where items may be marked as added, modified or deleted.",
          "$ref": "#/definitions/code"
        },
        "_mode": {
          "description": "Extensions for mode",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "orderedBy": {
          "description": "Specifies the order applied to the items in the section entries.",
          "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
        },
        "entry": {
          "description": "A reference to the actual resource from which the narrative in the section is derived.",
          "items": {
            "$ref": "Reference.schema.json#/definitions/Reference"
          },
          "type": "array"
        },
        "emptyReason": {
          "description": "If the section is empty, why the list is empty. An empty section typically has some text
    explaining the empty reason.",
          "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
        },
        "section": {
          "description": "A nested sub-section within this section.",
          "items": {
            "$ref": "#/definitions/Composition_Section"
          },
          "type": "array"
        }
      },
      "additionalProperties": false
    }
  }
}
```


2.41.18 Resource Composition - Maturity Dependency View

URL: https://hl7.org/fhir/R4/composition-dependencies.html

2.41.18 Resource Composition - Maturity Dependency View

Structured Documents Work Group | Maturity Level: 2 | Trial Use | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

Dependency Graph for Composition FMM level 2

Name | Card. | Type | Dependency Analysis
Composition | | DomainResource
identifier | 0..1 | Identifier
status | 1..1 | code
type | 1..1 | CodeableConcept
category | 0..* | CodeableConcept
subject | 0..1 | Reference(Any) | Type Warning: (Any = FMM1-Trial Use vs. Element = FMM2-Trial Use)
encounter | 0..1 | Reference(Encounter)
date | 1..1 | dateTime
author | 1..* | Reference(Practitioner | PractitionerRole | Device | Patient | RelatedPerson | Organization)
title | 1..1 | string

confidentiality | 0..1 | code
attester | 0..* | BackboneElement
mode | 1..1 | code
time | 0..1 | dateTime
party | 0..1 | Reference(Patient | RelatedPerson | Practitioner | PractitionerRole | Organization)
custodian | 0..1 | Reference(Organization)
relatesTo | 0..* | BackboneElement
code | 1..1 | code
target[x] | 1..1
targetIdentifier | | Identifier
targetReference | | Reference(Composition
event | 0..* | BackboneElement
code | 0..* | CodeableConcept
period | 0..1 | Period
detail | 0..* | Reference(Any) | Type Warning: (Any = FMM1-Trial Use vs. Element = FMM2-Trial Use)
section | 0..* | BackboneElement
title | 0..1 | string
code | 0..1 | CodeableConcept
author | 0..* | Reference(Practitioner | PractitionerRole | Device | Patient | RelatedPerson | Organization)
focus | 0..1 | Reference(Any) | Type Warning: (Any = FMM1-Trial Use vs. Element = FMM2-Trial Use)
text | 0..1 | Narrative
mode | 0..1 | code
orderedBy | 0..1 | CodeableConcept
entry | 0..* | Reference(Any) | Type Warning: (Any = FMM1-Trial Use vs. Element = FMM2-Trial Use)
emptyReason | 0..1 | CodeableConcept
section | 0..* | see section

2.42 Resource DocumentReference - Content

URL: https://hl7.org/fhir/R4/documentreference.html
Structured Documents Work Group | Maturity Level: 3 | Trial Use | Security Category: Not Classified | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

A reference to a document of any kind for any purpose. Provides metadata about the document so that the document can be discovered and managed. The scope of a document is any seralized object with a mime-type, so includes formal patient centric documents (CDA), cliical notes, scanned paper, and non-patient specific documents like policy text.

2.42.1 Scope and Usage

A DocumentReference resource is used to index a document, clinical note, and other binary objects to make them available to a healthcare system. A document is some sequence of bytes that is identifiable, establishes its own context (e.g., what subject, author, etc. can be displayed to the user), and has defined update management. The DocumentReference resource can be used with any document format that has a recognized mime type and that conforms to this definition.

Typically, DocumentReference resources are used in document indexing systems, such as IHE XDS , such as profiled in IHE Mobile access to Health Documents .

DocumentReference is metadata describing a document such as:

• CDA documents in FHIR systems
• FHIR documents stored elsewhere (i.e. registry/repository following the XDS model)
• PDF documents , Scanned Paper, and digital records of faxes
• Clinical Notes in various forms
• Image files (e.g., JPEG, GIF, TIFF)
• Non-Standard formats (e.g., WORD)
• Other kinds of documents, such as records of prescriptions or immunizations

2.42.2 Boundaries and Relationships

FHIR defines both a document format and this document reference. FHIR documents are for documents that are authored and assembled in FHIR. This resource is mainly intended for general references to assembled documents.

The document that is a target of the reference can be a reference to a FHIR document served by another server, or the target can be stored in the special FHIR Binary Resource, or the target can be stored on some other server system. The document reference is also able to address documents that are retrieved by a service call such as an XDS.b RetrieveDocumentSet, or a DICOM exchange, or an HL7 v2 message query - though the way each of these service calls works must be specified in some external standard or other documentation.

A DocumentReference describes some other document. This means that there are two sets of provenance information relevant here: the provenance of the document, and the provenance of the document reference. Sometimes, the provenance information is closely related, as when the document producer also produces the document reference, but in other workflows, the document reference is generated later by other actors. In the DocumentReference resource, the meta content refers to the provenance of the reference itself, while the content described below concerns the document it references. Like all resources, there is overlap between the information in the resource directly, and in the general Provenance resource. This is discussed as part of the description of the Provenance resource.

This resource is referenced by AdverseEvent, CarePlan, Communication, CommunicationRequest, Consent, Contract, DeviceRequest, DeviceUseStatement, itself, FamilyMemberHistory, GuidanceResponse, ImagingStudy, MedicationKnowledge, MedicinalProduct, Observation, Procedure, RequestGroup, RiskAssessment, ServiceRequest, SubstanceReferenceInformation, SubstanceSpecification and SupplyRequest

2.42.3 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
DocumentReference | TU | | DomainResource | A reference to a document Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
masterIdentifier | Σ | 0..1 | Identifier | Master Version Specific Identifier
identifier | Σ | 0..* | Identifier | Other identifiers for the document
status | ?!Σ | 1..1 | code | current | superseded | entered-in-error DocumentReferenceStatus (Required)
docStatus | Σ | 0..1 | code | preliminary | final | amended | entered-in-error CompositionStatus (Required)
type | Σ | 0..1 | CodeableConcept | Kind of document (LOINC if possible) Document Type Value Set (Preferred)
category | Σ | 0..* | CodeableConcept | Categorization of document Document Class Value Set (Example)
subject | Σ | 0..1 | Reference(Patient | Practitioner | Group | Device) | Who/what is the subject of the document
date | Σ | 0..1 | instant | When this document reference was created
author | Σ | 0..* | Reference(Practitioner | PractitionerRole | Organization | Device | Patient | RelatedPerson) | Who and/or what authored the document
authenticator | | 0..1 | Reference(Practitioner | PractitionerRole | Organization) | Who/what authenticated the document
custodian | | 0..1 | Reference(Organization) | Organization which maintains the document
relatesTo | Σ | 0..* | BackboneElement | Relationships to other documents
code | Σ | 1..1 | code | replaces | transforms | signs | appends DocumentRelationshipType (Required)
target | Σ | 1..1 | Reference(DocumentReference) | Target of the relationship
description | Σ | 0..1 | string | Human-readable description
securityLabel | Σ | 0..* | CodeableConcept | Document security-tags SecurityLabels (Extensible)
content | Σ | 1..* | BackboneElement | Document referenced
attachment | Σ | 1..1 | Attachment | Where to access the document
format | Σ | 0..1 | Coding | Format/content rules for the document DocumentReference Format Code Set (Preferred)
context | Σ | 0..1 | BackboneElement | Clinical context of document
encounter | | 0..* | Reference(Encounter | EpisodeOfCare) | Context of the document content
event | | 0..* | CodeableConcept | Main clinical acts documented v3 Code System ActCode (Example)
period | Σ | 0..1 | Period | Time of service that is being documented
facilityType | | 0..1 | CodeableConcept | Kind of facility where patient was seen Facility Type Code Value Set (Example)
practiceSetting | | 0..1 | CodeableConcept | Additional details about where the content was created (e.g. clinical specialty) Practice Setting Code Value Set (Example)
sourcePatientInfo | | 0..1 | Reference(Patient) | Patient demographics from source
related | | 0..* | Reference(Any) | Related identifiers or resources
Documentation for this format

UML Diagram (Legend)

XML Template

```
<DocumentReference xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <masterIdentifier><!-- 0..1 Identifier Master Version Specific Identifier --></masterIdentifier>
 <identifier><!-- 0..* Identifier Other identifiers for the document --></identifier>
 <status value="[code]"/><!-- 1..1 current | superseded | entered-in-error -->
 <docStatus value="[code]"/><!-- 0..1 preliminary | final | amended | entered-in-error -->
 <type><!-- 0..1 CodeableConcept Kind of document (LOINC if possible) --></type>
 <category><!-- 0..* CodeableConcept Categorization of document --></category>
 <subject><!-- 0..1 Reference(Patient|Practitioner|Group|Device) Who/what is the subject of the document --></subject>
```

```
 <date value="[instant]"/><!-- 0..1 When this document reference was created -->
 <author><!-- 0..* Reference(Practitioner|PractitionerRole|Organization|Device|
   Patient|RelatedPerson) Who and/or what authored the document --></author>
 <authenticator><!-- 0..1 Reference(Practitioner|PractitionerRole|Organization) Who/what authenticated the document
    --></authenticator>
 <custodian><!-- 0..1 Reference(Organization) Organization which maintains the document --></custodian>
 <relatesTo>  <!-- 0..* Relationships to other documents -->
  <code value="[code]"/><!-- 1..1 replaces | transforms | signs | appends -->
  <target><!-- 1..1 Reference(DocumentReference) Target of the relationship --></target>
 </relatesTo>
 <description value="[string]"/><!-- 0..1 Human-readable description -->
 <securityLabel><!-- 0..* CodeableConcept Document security-tags --></securityLabel>
 <content>  <!-- 1..* Document referenced -->
  <attachment><!-- 1..1 Attachment Where to access the document --></attachment>
  <format><!-- 0..1 Coding Format/content rules for the document --></format>
 </content>
 <context>  <!-- 0..1 Clinical context of document -->
  <encounter><!-- 0..* Reference(Encounter|EpisodeOfCare) Context of the document  content --></encounter>
  <event><!-- 0..* CodeableConcept Main clinical acts documented --></event>
  <period><!-- 0..1 Period Time of service that is being documented --></period>
  <facilityType><!-- 0..1 CodeableConcept Kind of facility where patient was seen --></facilityType>
  <practiceSetting><!-- 0..1 CodeableConcept Additional details about where the content was created (e.g. clinical
    specialty) --></practiceSetting>
  <sourcePatientInfo><!-- 0..1 Reference(Patient) Patient demographics from source --></sourcePatientInfo>
  <related><!-- 0..* Reference(Any) Related identifiers or resources --></related>
 </context>
</DocumentReference>
```

JSON Template

```
{
  "resourceType" : "DocumentReference",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "masterIdentifier" : { Identifier }, // Master Version Specific Identifier
  "identifier" : [{ Identifier }], // Other identifiers for the document
  "status" : "<code>", // R!  current | superseded | entered-in-error
  "docStatus" : "<code>", // preliminary | final | amended | entered-in-error
  "type" : { CodeableConcept }, // Kind of document (LOINC if possible)
  "category" : [{ CodeableConcept }], // Categorization of document
  "subject" : { Reference(Patient|Practitioner|Group|Device) }, // Who/what is the subject of the document
  "date" : "<instant>", // When this document reference was created
  "author" : [{ Reference(Practitioner|PractitionerRole|Organization|Device|
   Patient|RelatedPerson) }], // Who and/or what authored the document
  "authenticator" : { Reference(Practitioner|PractitionerRole|Organization) }, // Who/what authenticated the document
  "custodian" : { Reference(Organization) }, // Organization which maintains the document
  "relatesTo" : [{ // Relationships to other documents
    "code" : "<code>", // R!  replaces | transforms | signs | appends
    "target" : { Reference(DocumentReference) } // R!  Target of the relationship
  }],
  "description" : "<string>", // Human-readable description
  "securityLabel" : [{ CodeableConcept }], // Document security-tags
  "content" : [{ // R!  Document referenced
    "attachment" : { Attachment }, // R!  Where to access the document
    "format" : { Coding } // Format/content rules for the document
  }],
  "context" : { // Clinical context of document
    "encounter" : [{ Reference(Encounter|EpisodeOfCare) }], // Context of the document  content
    "event" : [{ CodeableConcept }], // Main clinical acts documented
    "period" : { Period }, // Time of service that is being documented
    "facilityType" : { CodeableConcept }, // Kind of facility where patient was seen
    "practiceSetting" : { CodeableConcept }, // Additional details about where the content was created (e.g. clinical
    specialty)
    "sourcePatientInfo" : { Reference(Patient) }, // Patient demographics from source
    "related" : [{ Reference(Any) }] // Related identifiers or resources
  }
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:DocumentReference;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:DocumentReference.masterIdentifier [ Identifier ]; # 0..1 Master Version Specific Identifier
  fhir:DocumentReference.identifier [ Identifier ], ... ; # 0..* Other identifiers for the document
  fhir:DocumentReference.status [ code ]; # 1..1 current | superseded | entered-in-error
  fhir:DocumentReference.docStatus [ code ]; # 0..1 preliminary | final | amended | entered-in-error
  fhir:DocumentReference.type [ CodeableConcept ]; # 0..1 Kind of document (LOINC if possible)
  fhir:DocumentReference.category [ CodeableConcept ], ... ; # 0..* Categorization of document
  fhir:DocumentReference.subject [ Reference(Patient|Practitioner|Group|Device) ]; # 0..1 Who/what is the subject of the
     document
  fhir:DocumentReference.date [ instant ]; # 0..1 When this document reference was created
  fhir:DocumentReference.author [ Reference(Practitioner|PractitionerRole|Organization|Device|Patient|RelatedPerson) ],
    ... ; # 0..* Who and/or what authored the document
  fhir:DocumentReference.authenticator [ Reference(Practitioner|PractitionerRole|Organization) ]; # 0..1 Who/what
    authenticated the document
```

```
  fhir:DocumentReference.custodian [ Reference(Organization) ]; # 0..1 Organization which maintains the document
  fhir:DocumentReference.relatesTo [ # 0..* Relationships to other documents
    fhir:DocumentReference.relatesTo.code [ code ]; # 1..1 replaces | transforms | signs | appends
    fhir:DocumentReference.relatesTo.target [ Reference(DocumentReference) ]; # 1..1 Target of the relationship
  ], ...;
  fhir:DocumentReference.description [ string ]; # 0..1 Human-readable description
  fhir:DocumentReference.securityLabel [ CodeableConcept ], ... ; # 0..* Document security-tags
  fhir:DocumentReference.content [ # 1..* Document referenced
    fhir:DocumentReference.content.attachment [ Attachment ]; # 1..1 Where to access the document
    fhir:DocumentReference.content.format [ Coding ]; # 0..1 Format/content rules for the document
  ], ...;
  fhir:DocumentReference.context [ # 0..1 Clinical context of document
    fhir:DocumentReference.context.encounter [ Reference(Encounter|EpisodeOfCare) ], ... ; # 0..* Context of the
    document  content
    fhir:DocumentReference.context.event [ CodeableConcept ], ... ; # 0..* Main clinical acts documented
    fhir:DocumentReference.context.period [ Period ]; # 0..1 Time of service that is being documented
    fhir:DocumentReference.context.facilityType [ CodeableConcept ]; # 0..1 Kind of facility where patient was seen
    fhir:DocumentReference.context.practiceSetting [ CodeableConcept ]; # 0..1 Additional details about where the
    content was created (e.g. clinical specialty)
    fhir:DocumentReference.context.sourcePatientInfo [ Reference(Patient) ]; # 0..1 Patient demographics from source
    fhir:DocumentReference.context.related [ Reference(Any) ], ... ; # 0..* Related identifiers or resources
  ];
]
```

Changes since R3

DocumentReference
DocumentReference.status | Change value set from http://hl7.org/fhir/ValueSet/document-reference-status to http://hl7.org/fhir/ValueSet/document-reference-status|4.0.1
DocumentReference.docStatus | Change value set from http://hl7.org/fhir/ValueSet/composition-status to http://hl7.org/fhir/ValueSet/composition-status|4.0.1
DocumentReference.type | Min Cardinality changed from 1 to 0
DocumentReference.category | Renamed from class to category Max Cardinality changed from 1 to *
DocumentReference.date | Added Element
DocumentReference.author | Type Reference: Added Target Type PractitionerRole
DocumentReference.authenticator | Type Reference: Added Target Type PractitionerRole
DocumentReference.relatesTo | No longer marked as Modifier
DocumentReference.relatesTo.code | Change value set from http://hl7.org/fhir/ValueSet/document-relationship-type to http://hl7.org/fhir/ValueSet/document-relationship-type|4.0.1
DocumentReference.context.encounter | Max Cardinality changed from 1 to * Type Reference: Added Target Type EpisodeOfCare
DocumentReference.context.related | Type changed from BackboneElement to Reference(Resource)
DocumentReference.created | deleted
DocumentReference.indexed | deleted
DocumentReference.context.related.identifier | deleted
DocumentReference.context.related.ref | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 1 test of which 1 fail to execute.)

Structure

Name | Flags | Card. | Type | Description & Constraints
DocumentReference | TU | | DomainResource | A reference to a document Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
masterIdentifier | Σ | 0..1 | Identifier | Master Version Specific Identifier
identifier | Σ | 0..* | Identifier | Other identifiers for the document
status | ?!Σ | 1..1 | code | current | superseded | entered-in-error DocumentReferenceStatus (Required)
docStatus | Σ | 0..1 | code | preliminary | final | amended | entered-in-error CompositionStatus (Required)
type | Σ | 0..1 | CodeableConcept | Kind of document (LOINC if possible) Document Type Value Set (Preferred)
category | Σ | 0..* | CodeableConcept | Categorization of document Document Class Value Set (Example)
subject | Σ | 0..1 | Reference(Patient | Practitioner | Group | Device) | Who/what is the subject of the document
date | Σ | 0..1 | instant | When this document reference was created
author | Σ | 0..* | Reference(Practitioner | PractitionerRole | Organization | Device | Patient | RelatedPerson) | Who and/or what authored the document
authenticator | | 0..1 | Reference(Practitioner | PractitionerRole | Organization) | Who/what authenticated the document
custodian | | 0..1 | Reference(Organization) | Organization which maintains the document
relatesTo | Σ | 0..* | BackboneElement | Relationships to other documents
code | Σ | 1..1 | code | replaces | transforms | signs | appends DocumentRelationshipType (Required)
target | Σ | 1..1 | Reference(DocumentReference) | Target of the relationship
description | Σ | 0..1 | string | Human-readable description
securityLabel | Σ | 0..* | CodeableConcept | Document security-tags SecurityLabels (Extensible)
content | Σ | 1..* | BackboneElement | Document referenced
attachment | Σ | 1..1 | Attachment | Where to access the document
format | Σ | 0..1 | Coding | Format/content rules for the document DocumentReference Format Code Set (Preferred)
context | Σ | 0..1 | BackboneElement | Clinical context of document
encounter | | 0..* | Reference(Encounter | EpisodeOfCare) | Context of the document content
event | | 0..* | CodeableConcept | Main clinical acts documented v3 Code System ActCode (Example)
period | Σ | 0..1 | Period | Time of service that is being documented

facilityType | | 0..1 | CodeableConcept | Kind of facility where patient was seen Facility Type Code Value Set (Example)
practiceSetting | | 0..1 | CodeableConcept | Additional details about where the content was created (e.g. clinical specialty) Practice Setting Code Value Set (Example)
sourcePatientInfo | | 0..1 | Reference(Patient) | Patient demographics from source
related | | 0..* | Reference(Any) | Related identifiers or resources
Documentation for this format

UML Diagram (Legend)

XML Template

```
<DocumentReference xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <masterIdentifier><!-- 0..1 Identifier Master Version Specific Identifier --></masterIdentifier>
 <identifier><!-- 0..* Identifier Other identifiers for the document --></identifier>
 <status value="[code]"/><!-- 1..1 current | superseded | entered-in-error -->
 <docStatus value="[code]"/><!-- 0..1 preliminary | final | amended | entered-in-error -->
 <type><!-- 0..1 CodeableConcept Kind of document (LOINC if possible) --></type>
 <category><!-- 0..* CodeableConcept Categorization of document --></category>
 <subject><!-- 0..1 Reference(Patient|Practitioner|Group|Device) Who/what is the subject of the document --></subject>
 <date value="[instant]"/><!-- 0..1 When this document reference was created -->
 <author><!-- 0..* Reference(Practitioner|PractitionerRole|Organization|Device|
   Patient|RelatedPerson) Who and/or what authored the document --></author>
 <authenticator><!-- 0..1 Reference(Practitioner|PractitionerRole|Organization) Who/what authenticated the document
    --></authenticator>
 <custodian><!-- 0..1 Reference(Organization) Organization which maintains the document --></custodian>
 <relatesTo>  <!-- 0..* Relationships to other documents -->
  <code value="[code]"/><!-- 1..1 replaces | transforms | signs | appends -->
  <target><!-- 1..1 Reference(DocumentReference) Target of the relationship --></target>
 </relatesTo>
 <description value="[string]"/><!-- 0..1 Human-readable description -->
 <securityLabel><!-- 0..* CodeableConcept Document security-tags --></securityLabel>
 <content>  <!-- 1..* Document referenced -->
  <attachment><!-- 1..1 Attachment Where to access the document --></attachment>
  <format><!-- 0..1 Coding Format/content rules for the document --></format>
 </content>
 <context>  <!-- 0..1 Clinical context of document -->
  <encounter><!-- 0..* Reference(Encounter|EpisodeOfCare) Context of the document  content --></encounter>
  <event><!-- 0..* CodeableConcept Main clinical acts documented --></event>
  <period><!-- 0..1 Period Time of service that is being documented --></period>
  <facilityType><!-- 0..1 CodeableConcept Kind of facility where patient was seen --></facilityType>
  <practiceSetting><!-- 0..1 CodeableConcept Additional details about where the content was created (e.g. clinical
    specialty) --></practiceSetting>
  <sourcePatientInfo><!-- 0..1 Reference(Patient) Patient demographics from source --></sourcePatientInfo>
  <related><!-- 0..* Reference(Any) Related identifiers or resources --></related>
 </context>
</DocumentReference>
```

JSON Template

```
{
  "resourceType" : "DocumentReference",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "masterIdentifier" : { Identifier }, // Master Version Specific Identifier
  "identifier" : [{ Identifier }], // Other identifiers for the document
  "status" : "<code>", // R!  current | superseded | entered-in-error
  "docStatus" : "<code>", // preliminary | final | amended | entered-in-error
  "type" : { CodeableConcept }, // Kind of document (LOINC if possible)
  "category" : [{ CodeableConcept }], // Categorization of document
  "subject" : { Reference(Patient|Practitioner|Group|Device) }, // Who/what is the subject of the document
  "date" : "<instant>", // When this document reference was created
  "author" : [{ Reference(Practitioner|PractitionerRole|Organization|Device|
   Patient|RelatedPerson) }], // Who and/or what authored the document
  "authenticator" : { Reference(Practitioner|PractitionerRole|Organization) }, // Who/what authenticated the document
  "custodian" : { Reference(Organization) }, // Organization which maintains the document
  "relatesTo" : [{ // Relationships to other documents
    "code" : "<code>", // R!  replaces | transforms | signs | appends
    "target" : { Reference(DocumentReference) } // R!  Target of the relationship
  }],
  "description" : "<string>", // Human-readable description
  "securityLabel" : [{ CodeableConcept }], // Document security-tags
  "content" : [{ // R!  Document referenced
    "attachment" : { Attachment }, // R!  Where to access the document
    "format" : { Coding } // Format/content rules for the document
  }],
  "context" : { // Clinical context of document
    "encounter" : [{ Reference(Encounter|EpisodeOfCare) }], // Context of the document  content
    "event" : [{ CodeableConcept }], // Main clinical acts documented
    "period" : { Period }, // Time of service that is being documented
    "facilityType" : { CodeableConcept }, // Kind of facility where patient was seen
    "practiceSetting" : { CodeableConcept }, // Additional details about where the content was created (e.g. clinical
    specialty)
    "sourcePatientInfo" : { Reference(Patient) }, // Patient demographics from source
    "related" : [{ Reference(Any) }] // Related identifiers or resources
  }
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:DocumentReference;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:DocumentReference.masterIdentifier [ Identifier ]; # 0..1 Master Version Specific Identifier
  fhir:DocumentReference.identifier [ Identifier ], ... ; # 0..* Other identifiers for the document
  fhir:DocumentReference.status [ code ]; # 1..1 current | superseded | entered-in-error
  fhir:DocumentReference.docStatus [ code ]; # 0..1 preliminary | final | amended | entered-in-error
  fhir:DocumentReference.type [ CodeableConcept ]; # 0..1 Kind of document (LOINC if possible)
  fhir:DocumentReference.category [ CodeableConcept ], ... ; # 0..* Categorization of document
  fhir:DocumentReference.subject [ Reference(Patient|Practitioner|Group|Device) ]; # 0..1 Who/what is the subject of the
     document
  fhir:DocumentReference.date [ instant ]; # 0..1 When this document reference was created
  fhir:DocumentReference.author [ Reference(Practitioner|PractitionerRole|Organization|Device|Patient|RelatedPerson) ],
    ... ; # 0..* Who and/or what authored the document
  fhir:DocumentReference.authenticator [ Reference(Practitioner|PractitionerRole|Organization) ]; # 0..1 Who/what
    authenticated the document
  fhir:DocumentReference.custodian [ Reference(Organization) ]; # 0..1 Organization which maintains the document
  fhir:DocumentReference.relatesTo [ # 0..* Relationships to other documents
    fhir:DocumentReference.relatesTo.code [ code ]; # 1..1 replaces | transforms | signs | appends
    fhir:DocumentReference.relatesTo.target [ Reference(DocumentReference) ]; # 1..1 Target of the relationship
  ], ...;
  fhir:DocumentReference.description [ string ]; # 0..1 Human-readable description
  fhir:DocumentReference.securityLabel [ CodeableConcept ], ... ; # 0..* Document security-tags
  fhir:DocumentReference.content [ # 1..* Document referenced
    fhir:DocumentReference.content.attachment [ Attachment ]; # 1..1 Where to access the document
    fhir:DocumentReference.content.format [ Coding ]; # 0..1 Format/content rules for the document
  ], ...;
  fhir:DocumentReference.context [ # 0..1 Clinical context of document
    fhir:DocumentReference.context.encounter [ Reference(Encounter|EpisodeOfCare) ], ... ; # 0..* Context of the
    document  content
    fhir:DocumentReference.context.event [ CodeableConcept ], ... ; # 0..* Main clinical acts documented
    fhir:DocumentReference.context.period [ Period ]; # 0..1 Time of service that is being documented
    fhir:DocumentReference.context.facilityType [ CodeableConcept ]; # 0..1 Kind of facility where patient was seen
    fhir:DocumentReference.context.practiceSetting [ CodeableConcept ]; # 0..1 Additional details about where the
    content was created (e.g. clinical specialty)
    fhir:DocumentReference.context.sourcePatientInfo [ Reference(Patient) ]; # 0..1 Patient demographics from source
    fhir:DocumentReference.context.related [ Reference(Any) ], ... ; # 0..* Related identifiers or resources
  ];
]
```

Changes since Release 3

DocumentReference
DocumentReference.status | Change value set from http://hl7.org/fhir/ValueSet/document-reference-status to http://hl7.org/fhir/ValueSet/document-reference-status|4.0.1
DocumentReference.docStatus | Change value set from http://hl7.org/fhir/ValueSet/composition-status to http://hl7.org/fhir/ValueSet/composition-status|4.0.1
DocumentReference.type | Min Cardinality changed from 1 to 0
DocumentReference.category | Renamed from class to category Max Cardinality changed from 1 to *
DocumentReference.date | Added Element
DocumentReference.author | Type Reference: Added Target Type PractitionerRole
DocumentReference.authenticator | Type Reference: Added Target Type PractitionerRole
DocumentReference.relatesTo | No longer marked as Modifier
DocumentReference.relatesTo.code | Change value set from http://hl7.org/fhir/ValueSet/document-relationship-type to http://hl7.org/fhir/ValueSet/document-relationship-type|4.0.1
DocumentReference.context.encounter | Max Cardinality changed from 1 to * Type Reference: Added Target Type EpisodeOfCare
DocumentReference.context.related | Type changed from BackboneElement to Reference(Resource)
DocumentReference.created | deleted
DocumentReference.indexed | deleted
DocumentReference.context.related.identifier | deleted
DocumentReference.context.related.ref | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 1 test of which 1 fail to execute.)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

2.42.3.1 Terminology Bindings

Path | Definition | Type | Reference
DocumentReference.status | The status of the document reference. | Required | DocumentReferenceStatus
DocumentReference.docStatus | Status of the underlying document. | Required | CompositionStatus
DocumentReference.type | Precise type of clinical document. | Preferred | DocumentTypeValueSet
DocumentReference.category | High-level kind of a clinical document at a macro level. | Example | DocumentClassValueSet
DocumentReference.relatesTo.code | The type of relationship between documents. | Required | DocumentRelationshipType

DocumentReference.securityLabel | Security Labels from the Healthcare Privacy and Security Classification System. | Extensible | All Security Labels
DocumentReference.content.format | Document Format Codes. | Preferred | DocumentReferenceFormatCodeSet
DocumentReference.context.event | This list of codes represents the main clinical acts being documented. | Example | v3.ActCode
DocumentReference.context.facilityType | XDS Facility Type. | Example | FacilityTypeCodeValueSet
DocumentReference.context.practiceSetting | Additional details about where the content was created (e.g. clinical specialty). | Example | PracticeSettingCodeValueSet

2.42.4 Implementation Notes

• The use of the .docStatus codes is discussed in the Composition description

• The resources maintain one way relationships that point backwards - e.g., the document that replaces one document points towards the document that it replaced. The reverse relationships can be followed by using indexes built from the resources. Typically, this is done using the search parameters described below. Given that documents may have other documents that replace or append them, clients should always check these relationships when accessing documents

2.42.4.1 Generating a Document Reference

A client can ask a server to generate a document reference from a document. The server reads the existing document and generates a matching DocumentReference resource, or returns one it has previously generated. Servers may be able to return or generate document references for the following types of content:

Type | Comments
FHIR Documents | The uri refers to an existing Document
CDA Document | The uri is a reference to a Binary end-point that returns either a CDA document, or some kind of CDA Package that the server knows how to process (e.g., an IHE .zip)
Other | The server can be asked to generate a document reference for other kinds of documents. For some of these documents (e.g., PDF documents) a server could only provide a document reference if it already existed or the server had special knowledge of the document.

The server either returns a search result containing a single document reference, or it returns an error. If the URI refers to another server, it is at the discretion of the server whether to retrieve it or return an error.

The operation is initiated by a named query, using _query=generate on the /DocumentReference end-point:

```
GET [service-url]/DocumentReference/?_query=generate&uri=:url&...
```

The "uri" parameter is a relative or absolute reference to one of the document types described above. Other parameters may be supplied:

Name | Meaning
persist | Whether to store the document at the document end-point (/Document) or not, once it is generated. Value = true or false (default is for the server to decide).

2.42.5 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
authenticator | reference | Who/what authenticated the document | DocumentReference.authenticator (Practitioner, Organization, PractitionerRole)
author | reference | Who and/or what authored the document | DocumentReference.author (Practitioner, Organization, Device, Patient, PractitionerRole, RelatedPerson)
category | token | Categorization of document | DocumentReference.category
contenttype | token | Mime type of the content, with charset etc. | DocumentReference.content.attachment.contentType
custodian | reference | Organization which maintains the document | DocumentReference.custodian (Organization)
date | date | When this document reference was created | DocumentReference.date
description | string | Human-readable description | DocumentReference.description
encounter | reference | Context of the document content | DocumentReference.context.encounter (EpisodeOfCare, Encounter) | 12 Resources
event | token | Main clinical acts documented | DocumentReference.context.event
facility | token | Kind of facility where patient was seen | DocumentReference.context.facilityType
format | token | Format/content rules for the document | DocumentReference.content.format
identifier | token | Master Version Specific Identifier | DocumentReference.masterIdentifier | DocumentReference.identifier | 30 Resources
language | token | Human language of the content (BCP-47) | DocumentReference.content.attachment.language
location | uri | Uri where the data can be found | DocumentReference.content.attachment.url
patient | reference | Who/what is the subject of the document | DocumentReference.subject.where(resolve() is Patient) (Patient) | 33 Resources
period | date | Time of service that is being documented | DocumentReference.context.period
related | reference | Related identifiers or resources | DocumentReference.context.related (Any)
relatesto | reference | Target of the relationship | DocumentReference.relatesTo.target (DocumentReference)
relation | token | replaces | transforms | signs | appends | DocumentReference.relatesTo.code
relationship | composite | Combination of relation and relatesTo | On DocumentReference.relatesTo: relatesto: code relation: target
security-label | token | Document security-tags | DocumentReference.securityLabel

setting | token | Additional details about where the content was created (e.g. clinical specialty) | DocumentReference.context.practiceSetting
status | token | current | superseded | entered-in-error | DocumentReference.status
subject | reference | Who/what is the subject of the document | DocumentReference.subject (Practitioner, Group, Device, Patient)
type | token | Kind of document (LOINC if possible) | DocumentReference.type | 5 Resources

2.42.6 Resource DocumentReference - Examples

URL: https://hl7.org/fhir/R4/documentreference-examples.html

2.42.6 Resource DocumentReference - Examples

Structured Documents Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Not Classified | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

Example Name | id | Format
XDS example | xds | XML | JSON | Turtle
Generic example | example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

2.42.7 Resource DocumentReference - Detailed Descriptions

URL: https://hl7.org/fhir/R4/documentreference-definitions.html

2.42.7 Resource DocumentReference - Detailed Descriptions

Structured Documents Work Group | Maturity Level: 3 | Trial Use | Security Category: Not Classified | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

Detailed Descriptions for the elements in the DocumentReference resource.

DocumentReference
Element Id | DocumentReference
Definition | A reference to a document of any kind for any purpose. Provides metadata about the document so that the document can be discovered and managed. The scope of a document is any seralized object with a mime-type, so includes formal patient centric documents (CDA), cliical notes, scanned paper, and non-patient specific documents like policy text.
Cardinality | 0..*
Type | DomainResource
Comments | Usually, this is used for documents other than those defined by FHIR.
DocumentReference.masterIdentifier
Element Id | DocumentReference.masterIdentifier
Definition | Document identifier as assigned by the source of the document. This identifier is specific to this version of the document. This unique identifier may be used elsewhere to identify this version of the document.
Cardinality | 0..1
Type | Identifier
Requirements | The structure and format of this Id shall be consistent with the specification corresponding to the formatCode attribute. (e.g. for a DICOM standard document a 64-character numeric UID, for an HL7 CDA format a serialization of the CDA Document Id extension and root in the form "oid^extension", where OID is a 64 digits max, and the Id is a 16 UTF-8 char max. If the OID is coded without the extension then the '^' character shall not be included.).
Summary | true
Comments | CDA Document Id extension and root.
DocumentReference.identifier
Element Id | DocumentReference.identifier
Definition | Other identifiers associated with the document, including version independent identifiers.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Summary | true
DocumentReference.status
Element Id | DocumentReference.status
Definition | The status of this document reference.
Cardinality | 1..1
Terminology Binding | DocumentReferenceStatus (Required)
Type | code
Is Modifier | true (Reason: This element is labelled as a modifier because it is a status element that contains status entered-in-error which means that the resource should not be treated as valid)
Summary | true
Comments | This is the status of the DocumentReference object, which might be independent from the docStatus element. This element is labeled as a modifier because the status contains the codes that mark the document or reference as not currently valid.
DocumentReference.docStatus
Element Id | DocumentReference.docStatus
Definition | The status of the underlying document.
Cardinality | 0..1
Terminology Binding | CompositionStatus (Required)

Type | code
Summary | true
Comments | The document that is pointed to might be in various lifecycle states.
DocumentReference.type
Element Id | DocumentReference.type
Definition | Specifies the particular kind of document referenced (e.g. History and Physical, Discharge Summary, Progress Note). This usually equates to the purpose of making the document referenced.
Cardinality | 0..1
Terminology Binding | Document Type Value Set (Preferred)
Type | CodeableConcept
Summary | true
Comments | Key metadata element describing the document that describes he exact type of document. Helps humans to assess whether the document is of interest when viewing a list of documents.
DocumentReference.category
Element Id | DocumentReference.category
Definition | A categorization for the type of document referenced - helps for indexing and searching. This may be implied by or derived from the code specified in the DocumentReference.type.
Cardinality | 0..*
Terminology Binding | Document Class Value Set (Example)
Type | CodeableConcept
Alternate Names | claxs
Summary | true
Comments | Key metadata element describing the the category or classification of the document. This is a broader perspective that groups similar documents based on how they would be used. This is a primary key used in searching.
DocumentReference.subject
Element Id | DocumentReference.subject
Definition | Who or what the document is about. The document can be about a person, (patient or healthcare practitioner), a device (e.g. a machine) or even a group of subjects (such as a document about a herd of farm animals, or a set of patients that share a common exposure).
Cardinality | 0..1
Type | Reference(Patient | Practitioner | Group | Device)
Summary | true
DocumentReference.date
Element Id | DocumentReference.date
Definition | When the document reference was created.
Cardinality | 0..1
Type | instant
Alternate Names | indexed
Summary | true
Comments | Referencing/indexing time is used for tracking, organizing versions and searching.
DocumentReference.author
Element Id | DocumentReference.author
Definition | Identifies who is responsible for adding the information to the document.
Cardinality | 0..*
Type | Reference(Practitioner | PractitionerRole | Organization | Device | Patient | RelatedPerson)
Summary | true
Comments | Not necessarily who did the actual data entry (i.e. typist) or who was the source (informant).
DocumentReference.authenticator
Element Id | DocumentReference.authenticator
Definition | Which person or organization authenticates that this document is valid.
Cardinality | 0..1
Type | Reference(Practitioner | PractitionerRole | Organization)
Comments | Represents a participant within the author institution who has legally authenticated or attested the document. Legal authentication implies that a document has been signed manually or electronically by the legal Authenticator.
DocumentReference.custodian
Element Id | DocumentReference.custodian
Definition | Identifies the organization or group who is responsible for ongoing maintenance of and access to the document.
Cardinality | 0..1
Type | Reference(Organization)
Comments | Identifies the logical organization (software system, vendor, or department) to go to find the current version, where to report issues, etc. This is different from the physical location (URL, disk drive, or server) of the document, which is the technical location of the document, which host may be delegated to the management of some other organization.
DocumentReference.relatesTo
Element Id | DocumentReference.relatesTo
Definition | Relationships that this document has with other document references that already exist.
Cardinality | 0..*
Summary | true
Comments | This element is labeled as a modifier because documents that append to other documents are incomplete on their own.
DocumentReference.relatesTo.code
Element Id | DocumentReference.relatesTo.code
Definition | The type of relationship that this document has with anther document.
Cardinality | 1..1

Terminology Binding | DocumentRelationshipType (Required)
Type | code
Summary | true
Comments | If this document appends another document, then the document cannot be fully understood without also accessing the referenced document.
DocumentReference.relatesTo.target
Element Id | DocumentReference.relatesTo.target
Definition | The target document of this relationship.
Cardinality | 1..1
Type | Reference(DocumentReference)
Summary | true
DocumentReference.description
Element Id | DocumentReference.description
Definition | Human-readable description of the source document.
Cardinality | 0..1
Type | string
Requirements | Helps humans to assess whether the document is of interest.
Summary | true
Comments | What the document is about, a terse summary of the document.
DocumentReference.securityLabel
Element Id | DocumentReference.securityLabel
Definition | A set of Security-Tag codes specifying the level of privacy/security of the Document. Note that DocumentReference.meta.security contains the security labels of the "reference" to the document, while DocumentReference.securityLabel contains a snapshot of the security labels on the document the reference refers to.
Cardinality | 0..*
Terminology Binding | SecurityLabels (Extensible)
Type | CodeableConcept
Requirements | Use of the Health Care Privacy/Security Classification (HCS) system of security-tag use is recommended.
Summary | true
Comments | The confidentiality codes can carry multiple vocabulary items. HL7 has developed an understanding of security and privacy tags that might be desirable in a Document Sharing environment, called HL7 Healthcare Privacy and Security Classification System (HCS). The following specification is recommended but not mandated, as the vocabulary bindings are an administrative domain responsibility. The use of this method is up to the policy domain such as the XDS Affinity Domain or other Trust Domain where all parties including sender and recipients are trusted to appropriately tag and enforce. In the HL7 Healthcare Privacy and Security Classification (HCS) there are code systems specific to Confidentiality, Sensitivity, Integrity, and Handling Caveats. Some values would come from a local vocabulary as they are related to workflow roles and special projects.
DocumentReference.content
Element Id | DocumentReference.content
Definition | The document and format referenced. There may be multiple content element repetitions, each with a different format.
Cardinality | 1..*
Summary | true
DocumentReference.content.attachment
Element Id | DocumentReference.content.attachment
Definition | The document or URL of the document along with critical metadata to prove content has integrity.
Cardinality | 1..1
Type | Attachment
Summary | true
DocumentReference.content.format
Element Id | DocumentReference.content.format
Definition | An identifier of the document encoding, structure, and template that the document conforms to beyond the base format indicated in the mimeType.
Cardinality | 0..1
Terminology Binding | DocumentReference Format Code Set (Preferred)
Type | Coding
Summary | true
Comments | Note that while IHE mostly issues URNs for format types, not all documents can be identified by a URI.
DocumentReference.context
Element Id | DocumentReference.context
Definition | The clinical context in which the document was prepared.
Cardinality | 0..1
Summary | true
Comments | These values are primarily added to help with searching for interesting/relevant documents.
DocumentReference.context.encounter
Element Id | DocumentReference.context.encounter
Definition | Describes the clinical encounter or type of care that the document content is associated with.
Cardinality | 0..*
Type | Reference(Encounter | EpisodeOfCare)
DocumentReference.context.event
Element Id | DocumentReference.context.event
Definition | This list of codes represents the main clinical acts, such as a colonoscopy or an appendectomy, being documented. In some cases, the event is inherent in the type Code, such as a "History and Physical Report" in which the procedure being documented is necessarily a "History and Physical" act.

Cardinality | 0..*
Terminology Binding | v3 Code System ActCode (Example)
Type | CodeableConcept
Comments | An event can further specialize the act inherent in the type, such as where it is simply "Procedure Report" and the procedure was a "colonoscopy". If one or more event codes are included, they shall not conflict with the values inherent in the class or type elements as such a conflict would create an ambiguous situation.
DocumentReference.context.period
Element Id | DocumentReference.context.period
Definition | The time period over which the service that is described by the document was provided.
Cardinality | 0..1
Type | Period
Summary | true
DocumentReference.context.facilityType
Element Id | DocumentReference.context.facilityType
Definition | The kind of facility where the patient was seen.
Cardinality | 0..1
Terminology Binding | Facility Type Code Value Set (Example)
Type | CodeableConcept
DocumentReference.context.practiceSetting
Element Id | DocumentReference.context.practiceSetting
Definition | This property may convey specifics about the practice setting where the content was created, often reflecting the clinical specialty.
Cardinality | 0..1
Terminology Binding | Practice Setting Code Value Set (Example)
Type | CodeableConcept
Requirements | This is an important piece of metadata that providers often rely upon to quickly sort and/or filter out to find specific content.
Comments | This element should be based on a coarse classification system for the class of specialty practice. Recommend the use of the classification system for Practice Setting, such as that described by the Subject Matter Domain in LOINC.
DocumentReference.context.sourcePatientInfo
Element Id | DocumentReference.context.sourcePatientInfo
Definition | The Patient Information as known when the document was published. May be a reference to a version specific, or contained.
Cardinality | 0..1
Type | Reference(Patient)
DocumentReference.context.related
Element Id | DocumentReference.context.related
Definition | Related identifiers or resources associated with the DocumentReference.
Cardinality | 0..*
Type | Reference(Any)
Comments | May be identifiers or resources that caused the DocumentReference or referenced Document to be created.

2.42.8 Resource DocumentReference - Mappings

URL: https://hl7.org/fhir/R4/documentreference-mappings.html

2.42.8 Resource DocumentReference - Mappings

Structured Documents Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Not Classified | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

HL7 v2 Mapping

CDA (R2)

RIM Mapping

XDS metadata equivalent

FHIR Composition

Mappings for the documentreference resource (see Mappings to Other Standards for further information & status).

2.42.8.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

DocumentReference
masterIdentifier | FiveWs.identifier
identifier | FiveWs.identifier
status | FiveWs.status
docStatus | FiveWs.status
type | FiveWs.class
category | FiveWs.class

subject | FiveWs.subject[x]
date | FiveWs.recorded
authenticator | FiveWs.witness
encounter | FiveWs.context

2.42.8.2 Workflow Pattern (http://hl7.org/fhir/workflow)

DocumentReference | Event
masterIdentifier | Event.identifier
identifier | Event.identifier
status | Event.status
type | Event.code
subject | Event.subject
date | Event.occurrence[x]
author | Event.performer.actor
authenticator | Event.performer.actor
custodian | Event.performer.actor
encounter | Event.context

2.42.8.3 HL7 v2 Mapping (http://hl7.org/v2)

DocumentReference
masterIdentifier | TXA-12
identifier | TXA-16?
status | TXA-19
docStatus | TXA-17
type | TXA-2
category
subject | PID-3 (No standard way to define a Practitioner or Group subject in HL7 v2 MDM message)
date
author | TXA-9 (No standard way to indicate a Device in HL7 v2 MDM message)
authenticator | TXA-10
custodian
relatesTo
code
target
description | TXA-25
securityLabel | TXA-18
content
attachment | TXA-3 for mime type
format
context
encounter
event
period
facilityType
practiceSetting
sourcePatientInfo
related

2.42.8.4 CDA (R2) (http://hl7.org/v3/cda)

DocumentReference | when describing a CDA
masterIdentifier | ClinicalDocument/id
identifier
status
docStatus
type | ClinicalDocument/code/@code The typeCode should be mapped from the ClinicalDocument/code element to a set of document type codes configured in the affinity domain. One suggested coding system to use for typeCode is LOINC, in which case the mapping step can be omitted.
category | Derived from a mapping of /ClinicalDocument/code/@code to an Affinity Domain specified coded value to use and coding system. Affinity Domains are encouraged to use the appropriate value for Type of Service, based on the LOINC Type of Service (see Page 53 of the LOINC User's Manual). Must be consistent with /ClinicalDocument/code/@code
subject | ClinicalDocument/recordTarget/
date
author | ClinicalDocument/author
authenticator | ClinicalDocument/legalAuthenticator
custodian
relatesTo
code
target
description
securityLabel | ClinicalDocument/confidentialityCode/@code
content
attachment | ClinicalDocument/languageCode, ClinicalDocument/title, ClinicalDocument/date

format | derived from the IHE Profile or Implementation Guide templateID
context
encounter
event
period | ClinicalDocument/documentationOf/ serviceEvent/effectiveTime/low/ @value --> ClinicalDocument/documentationOf/ serviceEvent/effectiveTime/high/ @value
facilityType | usually a mapping to a local ValueSet. Must be consistent with /clinicalDocument/code
practiceSetting | usually from a mapping to a local ValueSet
sourcePatientInfo | ClinicalDocument/recordTarget/
related | ClinicalDocument/relatedDocument

2.42.8.5 RIM Mapping (http://hl7.org/v3)

DocumentReference | Document[classCode="DOC" and moodCode="EVN"]
masterIdentifier | .id
identifier | .id / .setId
status | interim: .completionCode="IN" & ./statusCode[isNormalDatatype()]="active"; final: .completionCode="AU" && ./statusCode[isNormalDatatype()]="complete" and not(./inboundRelationship[typeCode="SUBJ" and isNormalActRelationship()]/source[subsumesCode("ActClass#CACT") and moodCode="EVN" and domainMember("ReviseDocument", code) and isNormalAct()]); amended: .completionCode="AU" && ./statusCode[isNormalDatatype()]="complete" and ./inboundRelationship[typeCode="SUBJ" and isNormalActRelationship()]/source[subsumesCode("ActClass#CACT") and moodCode="EVN" and domainMember("ReviseDocument", code) and isNormalAct() and statusCode="completed"]; withdrawn : .completionCode=NI && ./statusCode[isNormalDatatype()]="obsolete"
docStatus | .statusCode
type | ./code
category | .outboundRelationship[typeCode="COMP].target[classCode="LIST", moodCode="EVN"].code
subject | .participation[typeCode="SBJ"].role[typeCode="PAT"]
date | .availabilityTime[type="TS"]
author | .participation[typeCode="AUT"].role[classCode="ASSIGNED"]
authenticator | .participation[typeCode="AUTHEN"].role[classCode="ASSIGNED"]
custodian | .participation[typeCode="RCV"].role[classCode="CUST"].scoper[classCode="ORG" and determinerCode="INST"]
relatesTo | .outboundRelationship
code | .outboundRelationship.typeCode
target | .target[classCode="DOC", moodCode="EVN"].id
description | .outboundRelationship[typeCode="SUBJ"].target.text
securityLabel | .confidentialityCode
content | document.text
attachment | document.text
format | document.text
context | outboundRelationship[typeCode="SUBJ"].target[classCode<'ACT']
encounter | unique(highest(./outboundRelationship[typeCode="SUBJ" and isNormalActRelationship()], priorityNumber)/target[moodCode="EVN" and classCode=("ENC", "PCPR") and isNormalAct])
event | .code
period | .effectiveTime
facilityType | .participation[typeCode="LOC"].role[classCode="DSDLOC"].code
practiceSetting | .participation[typeCode="LOC"].role[classCode="DSDLOC"].code
sourcePatientInfo | .participation[typeCode="SBJ"].role[typeCode="PAT"]
related | ./outboundRelationship[typeCode="PERT" and isNormalActRelationship()] / target[isNormalAct]

2.42.8.6 XDS metadata equivalent (http://ihe.net/xds)

DocumentReference
masterIdentifier | DocumentEntry.uniqueId
identifier | DocumentEntry.entryUUID
status | DocumentEntry.availabilityStatus
docStatus
type | DocumentEntry.type
category | DocumentEntry.class
subject | DocumentEntry.patientId
date
author | DocumentEntry.author
authenticator | DocumentEntry.legalAuthenticator
custodian
relatesTo | DocumentEntry Associations
code | DocumentEntry Associations type
target | DocumentEntry Associations reference
description | DocumentEntry.comments
securityLabel | DocumentEntry.confidentialityCode
content
attachment | DocumentEntry.mimeType, DocumentEntry.languageCode, DocumentEntry.URI, DocumentEntry.size, DocumentEntry.hash, DocumentEntry.title, DocumentEntry.creationTime
format | DocumentEntry.formatCode
context

encounter
event | DocumentEntry.eventCodeList
period | DocumentEntry.serviceStartTime, DocumentEntry.serviceStopTime
facilityType | DocumentEntry.healthcareFacilityTypeCode
practiceSetting | DocumentEntry.practiceSettingCode
sourcePatientInfo | DocumentEntry.sourcePatientInfo, DocumentEntry.sourcePatientId
related | DocumentEntry.referenceIdList

2.42.8.7 FHIR Composition (http://hl7.org/fhir/composition)

DocumentReference | when describing a Composition
masterIdentifier | Composition.identifier
identifier
status
docStatus | Composition.status
type | Composition.type
category | Composition.class
subject | Composition.subject
date | Composition.date
author | Composition.author
authenticator | Composition.attester
custodian | Composition.custodian
relatesTo | Composition.relatesTo
code | Composition.relatesTo.code
target | Composition.relatesTo.target
description
securityLabel | Composition.confidentiality, Composition.meta.security
content | Bundle(Composition+*)
attachment | Composition.language, Composition.title, Composition.date
format | Composition.meta.profile
context
encounter | Composition.encounter
event | Composition.event.code
period | Composition.event.period
facilityType | usually from a mapping to a local ValueSet
practiceSetting | usually from a mapping to a local ValueSet
sourcePatientInfo | Composition.subject
related | Composition.event.detail

2.42.9 Resource DocumentReference - Extensions & Profiles

URL: https://hl7.org/fhir/R4/documentreference-profiles.html

2.42.9 Resource DocumentReference - Extensions & Profiles

Structured Documents Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Not Classified | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

This table lists profiles and extensions for the DocumentReference resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

2.42.9.1 Profiles

No Profiles defined for this resource

2.42.9.2 Extensions

No Extensions defined for this resource
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

workflow-reasonReference | reasonReference | for Workflow Pattern HL7 Extensions
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

2.42.9.3 Search Extensions

No Search Extensions defined for this resource

2.42.10 Resource DocumentReference - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/documentreference-version-maps.html

No validation errors - all conversions are clean

2.42.10 Resource DocumentReference - R3/R4 Conversions

Structured Documents Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Not Classified | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

R3 : R4 Conversion maps for DocumentReference.

Functional status for this map: 1 test of which 1 fail to execute. (see documentation)

2.42.10.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/DocumentReference3to4" = "R3 to R4 Conversions for DocumentReference"

uses "http://hl7.org/fhir/3.0/StructureDefinition/DocumentReference" alias DocumentReferenceR3 as source
uses "http://hl7.org/fhir/StructureDefinition/DocumentReference" alias DocumentReference as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group DocumentReference(source src : DocumentReferenceR3, target tgt : DocumentReference) extends DomainResource
    <<type+>> {
  src.masterIdentifier -> tgt.masterIdentifier;
  src.identifier -> tgt.identifier;
  src.status -> tgt.status;
  src.docStatus -> tgt.docStatus;
  src.type -> tgt.type;
  src.class -> tgt.category;
  src.subject -> tgt.subject;
  src.indexed -> tgt.date;
  src.created as vs ->  tgt.extension as ext,  ext.url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    DocumentReference.created',  ext.value = vs;
  src.author as s -> tgt.agent as t then agent(s, t);
  src.authenticator -> tgt.authenticator;
  src.custodian -> tgt.custodian;
  src.relatesTo as vs0 -> tgt.relatesTo as vt0 then relatesTo(vs0, vt0);
  src.description -> tgt.description;
  src.securityLabel -> tgt.securityLabel;
  src.content as vs0 -> tgt.content as vt0 then content(vs0, vt0);
  src.context as vs0 -> tgt.context as vt0 then context(vs0, vt0);
}

group agent(source src, target tgt) extends BackboneElement {
  src ->  tgt.type as cc,  cc.coding as c,  c.system = 'http://terminology.hl7.org/CodeSystem/v3-ParticipationType',
    c.code = 'AUT' "author";
  src -> tgt.who as vt0 then Reference(src, vt0) "who";
}

group relatesTo(source src, target tgt) extends BackboneElement {
  src.code -> tgt.code;
  src.target -> tgt.target;
}

group content(source src, target tgt) extends BackboneElement {
  src.attachment -> tgt.attachment;
  src.format -> tgt.format;
}

group context(source src, target tgt) extends BackboneElement {
  src.encounter -> tgt.encounter;
  src.event -> tgt.event;
  src.period -> tgt.period;
  src.facilityType -> tgt.facilityType;
  src.practiceSetting -> tgt.practiceSetting;
  src.sourcePatientInfo -> tgt.sourcePatientInfo;
  src.related as vs0 -> tgt.related as vt0 then related(vs0, vt0);
}

group related(source src, target tgt) {
  src.identifier as vs0 then Identifier2Reference(vs0, tgt);
  src.ref as vs0 then Reference(vs0, tgt);
}
```

2.42.10.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/DocumentReference4to3" = "R4 to R3 Conversion for DocumentReference"

uses "http://hl7.org/fhir/StructureDefinition/DocumentReference" alias DocumentReference as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/DocumentReference" alias DocumentReferenceR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group DocumentReference(source src : DocumentReference, target tgt : DocumentReferenceR3) extends DomainResource
```

```
    <<type+>> {
  src.masterIdentifier -> tgt.masterIdentifier;
  src.identifier -> tgt.identifier;
  src.status -> tgt.status;
  src.docStatus -> tgt.docStatus;
  src.type -> tgt.type;
  src.category -> tgt.class;
  src.subject -> tgt.subject;
  src.date -> tgt.indexed;
  src.extension as ext where url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-DocumentReference.created'
    then {
    ext.value as vs0 -> tgt.created = vs0 "created2";
  } "created";
  src.agent as s where type.coding.where((system = 'http://terminology.hl7.org/CodeSystem/v3-ParticipationType') and
    (code = 'AUT')).exists() -> tgt.author as t then agent(s, t);
  src.authenticator -> tgt.authenticator;
  src.custodian -> tgt.custodian;
  src.relatesTo as vs0 -> tgt.relatesTo as vt0 then relatesTo(vs0, vt0);
  src.description -> tgt.description;
  src.securityLabel -> tgt.securityLabel;
  src.content as vs0 -> tgt.content as vt0 then content(vs0, vt0);
  src.context as vs0 -> tgt.context as vt0 then context(vs0, vt0);
}

group agent(source src, target tgt) extends Element {
  src.who as vs0 then Reference(vs0, tgt);
}

group relatesTo(source src, target tgt) extends BackboneElement {
  src.code -> tgt.code;
  src.target -> tgt.target;
}

group content(source src, target tgt) extends BackboneElement {
  src.attachment -> tgt.attachment;
  src.format -> tgt.format;
}

group context(source src, target tgt) extends BackboneElement {
  src.encounter -> tgt.encounter;
  src.event -> tgt.event;
  src.period -> tgt.period;
  src.facilityType -> tgt.facilityType;
  src.practiceSetting -> tgt.practiceSetting;
  src.sourcePatientInfo -> tgt.sourcePatientInfo;
  src.related as vs0 -> tgt.related as vt0 then related(vs0, vt0);
}

group related(source src, target tgt) {
  src where reference.exists() -> tgt.ref as vt0 then ReferenceNoIdentifier(src, vt0) "reference";
  src where identifier.exists() -> tgt.identifier as vt0 then Reference2Identifier(src, vt0) "identifier";
}
```

2.42.10.3 R4 Validation Errors

2.42.11 JSON Schema for DocumentReference

URL: https://hl7.org/fhir/R4/documentreference.schema.json.html

2.42.11 JSON Schema for DocumentReference

Structured Documents Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

This is a representation of the json schema for DocumentReference, which is just a part of the full JSON Schema.

```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/DocumentReference",
  "$ref": "#/definitions/DocumentReference",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "DocumentReference": {
      "description": "A reference to a document of any kind for any purpose. Provides metadata about the document so
    that the document can be discovered and managed. The scope of a document is any seralized object with a mime-type,
    so includes formal patient centric documents (CDA), cliical notes, scanned paper, and non-patient specific documents
     like policy text.",
      "properties": {
        "resourceType": {
          "description": "This is a DocumentReference resource",
          "const": "DocumentReference"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
```

```
    "meta": {
      "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
Changes to the content might not always be associated with version changes to the resource.",
      "$ref": "Meta.schema.json#/definitions/Meta"
    },
    "implicitRules": {
      "description": "A reference to a set of rules that were followed when the resource was constructed, and which
must be understood when processing the content. Often, this is a reference to an implementation guide that defines
the special rules along with other profiles etc.",
      "$ref": "#/definitions/uri"
    },
    "_implicitRules": {
      "description": "Extensions for implicitRules",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "language": {
      "description": "The base language in which the resource is written.",
      "$ref": "#/definitions/code"
    },
    "_language": {
      "description": "Extensions for language",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "text": {
      "description": "A human-readable narrative that contains a summary of the resource and can be used to
represent the content of the resource to a human. The narrative need not encode all the structured data, but is
required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
      "$ref": "Narrative.schema.json#/definitions/Narrative"
    },
    "contained": {
      "description": "These resources do not have an independent existence apart from the resource that contains
them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
 definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "masterIdentifier": {
      "description": "Document identifier as assigned by the source of the document. This identifier is specific to
this version of the document. This unique identifier may be used elsewhere to identify this version of the
document.",
      "$ref": "Identifier.schema.json#/definitions/Identifier"
    },
    "identifier": {
      "description": "Other identifiers associated with the document, including version independent identifiers.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "status": {
      "description": "The status of this document reference.",
      "enum": [
        "current",
        "superseded",
        "entered-in-error"
      ]
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "docStatus": {
      "description": "The status of the underlying document.",
```

```
      "$ref": "#/definitions/code"
    },
    "_docStatus": {
      "description": "Extensions for docStatus",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "type": {
      "description": "Specifies the particular kind of document referenced  (e.g. History and Physical, Discharge Summary, Progress Note). This usually equates to the purpose of making the document referenced.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "category": {
      "description": "A categorization for the type of document referenced - helps for indexing and searching. This may be implied by or derived from the code specified in the DocumentReference.type.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "subject": {
      "description": "Who or what the document is about. The document can be about a person, (patient or healthcare practitioner), a device (e.g. a machine) or even a group of subjects (such as a document about a herd of farm animals, or a set of patients that share a common exposure).",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "date": {
      "description": "When the document reference was created.",
      "$ref": "#/definitions/instant"
    },
    "_date": {
      "description": "Extensions for date",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "author": {
      "description": "Identifies who is responsible for adding the information to the document.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "authenticator": {
      "description": "Which person or organization authenticates that this document is valid.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "custodian": {
      "description": "Identifies the organization or group who is responsible for ongoing maintenance of and access to the document.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "relatesTo": {
      "description": "Relationships that this document has with other document references that already exist.",
      "items": {
        "$ref": "#/definitions/DocumentReference_RelatesTo"
      },
      "type": "array"
    },
    "description": {
      "description": "Human-readable description of the source document.",
      "$ref": "#/definitions/string"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "securityLabel": {
      "description": "A set of Security-Tag codes specifying the level of privacy/security of the Document. Note that DocumentReference.meta.security contains the security labels of the \"reference\" to the document, while DocumentReference.securityLabel contains a snapshot of the security labels on the document the reference refers to.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "content": {
      "description": "The document and format referenced. There may be multiple content element repetitions, each with a different format.",
      "items": {
        "$ref": "#/definitions/DocumentReference_Content"
      },
      "type": "array"
    },
    "context": {
      "description": "The clinical context in which the document was prepared.",
      "$ref": "#/definitions/DocumentReference_Context"
    }
  },
  "additionalProperties": false,
  "required": [
```

```
    "content",
    "resourceType"
  ]
},
"DocumentReference_RelatesTo": {
  "description": "A reference to a document of any kind for any purpose. Provides metadata about the document so
that the document can be discovered and managed. The scope of a document is any seralized object with a mime-type,
so includes formal patient centric documents (CDA), cliical notes, scanned paper, and non-patient specific documents
 like policy text.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "code": {
      "description": "The type of relationship that this document has with anther document.",
      "enum": [
        "replaces",
        "transforms",
        "signs",
        "appends"
      ]
    },
    "_code": {
      "description": "Extensions for code",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "target": {
      "description": "The target document of this relationship.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    }
  },
  "additionalProperties": false,
  "required": [
    "target"
  ]
},
"DocumentReference_Content": {
  "description": "A reference to a document of any kind for any purpose. Provides metadata about the document so
that the document can be discovered and managed. The scope of a document is any seralized object with a mime-type,
so includes formal patient centric documents (CDA), cliical notes, scanned paper, and non-patient specific documents
 like policy text.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
```

```
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "attachment": {
      "description": "The document or URL of the document along with critical metadata to prove content has
integrity.",
      "$ref": "Attachment.schema.json#/definitions/Attachment"
    },
    "format": {
      "description": "An identifier of the document encoding, structure, and template that the document conforms to
beyond the base format indicated in the mimeType.",
      "$ref": "Coding.schema.json#/definitions/Coding"
    }
  },
  "additionalProperties": false,
  "required": [
    "attachment"
  ]
},
"DocumentReference_Context": {
  "description": "A reference to a document of any kind for any purpose. Provides metadata about the document so
that the document can be discovered and managed. The scope of a document is any seralized object with a mime-type,
so includes formal patient centric documents (CDA), cliical notes, scanned paper, and non-patient specific documents
 like policy text.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "encounter": {
      "description": "Describes the clinical encounter or type of care that the document content is associated
with.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "event": {
      "description": "This list of codes represents the main clinical acts, such as a colonoscopy or an
appendectomy, being documented. In some cases, the event is inherent in the type Code, such as a \"History and
Physical Report\" in which the procedure being documented is necessarily a \"History and Physical\" act.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "period": {
      "description": "The time period over which the service that is described by the document was provided.",
      "$ref": "Period.schema.json#/definitions/Period"
    },
    "facilityType": {
      "description": "The kind of facility where the patient was seen.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "practiceSetting": {
      "description": "This property may convey specifics about the practice setting where the content was created,
often reflecting the clinical specialty.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
```

```
        "sourcePatientInfo": {
          "description": "The Patient Information as known when the document was published. May be a reference to a
    version specific, or contained.",
          "$ref": "Reference.schema.json#/definitions/Reference"
        },
        "related": {
          "description": "Related identifiers or resources associated with the DocumentReference.",
          "items": {
            "$ref": "Reference.schema.json#/definitions/Reference"
          },
          "type": "array"
        }
      },
      "additionalProperties": false
    }
  }
}
```


2.42.12 Resource DocumentReference - Maturity Dependency View

URL: https://hl7.org/fhir/R4/documentreference-dependencies.html

2.42.12 Resource DocumentReference - Maturity Dependency View

Structured Documents Work Group | Maturity Level: 3 | Trial Use | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

Dependency Graph for DocumentReference FMM level 3

Name | Card. | Type | Dependency Analysis
DocumentReference | | DomainResource
masterIdentifier | 0..1 | Identifier
identifier | 0..* | Identifier
status | 1..1 | code
docStatus | 0..1 | code
type | 0..1 | CodeableConcept
category | 0..* | CodeableConcept
subject | 0..1 | Reference(Patient | Practitioner | Group | Device) | Type Warning: (Group = FMM1-Trial Use vs. Element = FMM3-Trial Use) Type Warning: (Device = FMM2-Trial Use vs. Element = FMM3-Trial Use)
date | 0..1 | instant
author | 0..* | Reference(Practitioner | PractitionerRole | Organization | Device | Patient | RelatedPerson) | Type Warning: (PractitionerRole = FMM2-Trial Use vs. Element = FMM3-Trial Use) Type Warning: (Device = FMM2-Trial Use vs. Element = FMM3-Trial Use) Type Warning: (RelatedPerson = FMM2-Trial Use vs. Element = FMM3-Trial Use)
authenticator | 0..1 | Reference(Practitioner | PractitionerRole | Organization) | Type Warning: (PractitionerRole = FMM2-Trial Use vs. Element = FMM3-Trial Use)
custodian | 0..1 | Reference(Organization)
relatesTo | 0..* | BackboneElement
code | 1..1 | code
target | 1..1 | Reference(DocumentReference)
description | 0..1 | string
securityLabel | 0..* | CodeableConcept
content | 1..* | BackboneElement
attachment | 1..1 | Attachment
format | 0..1 | Coding
context | 0..1 | BackboneElement
encounter | 0..* | Reference(Encounter | EpisodeOfCare) | Type Warning: (Encounter = FMM2-Trial Use vs. Element = FMM3-Trial Use) Type Warning: (EpisodeOfCare = FMM2-Trial Use vs. Element = FMM3-Trial Use)
event | 0..* | CodeableConcept
period | 0..1 | Period
facilityType | 0..1 | CodeableConcept
practiceSetting | 0..1 | CodeableConcept
sourcePatientInfo | 0..1 | Reference(Patient)
related | 0..* | Reference(Any) | Type Warning: (Any = FMM1-Trial Use vs. Element = FMM3-Trial Use)

2.42.13 Value Set Sc-valueset-composition-status - Status Code Mapping

URL: https://hl7.org/fhir/R4/sc-valueset-composition-status.html

2.42.13 Value Set Sc-valueset-composition-status - Status Code Mapping

Structured Documents Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

This is the narrative for the resource. See also the XML, JSON or Turtle format.

2.42.14 CompositionStatusCanonicalMap (http://hl7.org/fhir/ConceptMap/sc-composition-status)

Mapping from http://hl7.org/fhir/ValueSet/composition-status to http://hl7.org/fhir/ValueSet/resource-status

DRAFT. Published on 01/11/2019 9:29:23 AM by HL7 (FHIR Project) (http://hl7.org/fhir , fhir@lists.hl7.org).

Canonical Mapping for "The workflow/clinical status of the composition."

Source Code | Equivalence | Destination Code
entered-in-error | equivalent | error
preliminary | equivalent | draft
amended | equivalent | replaced
final | equivalent | complete

2.43 Resource DocumentManifest - Content

URL: https://hl7.org/fhir/R4/documentmanifest.html
Structured Documents Work Group | Maturity Level: 2 | Trial Use | Security Category: Not Classified | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

A collection of documents compiled for a purpose together with metadata that applies to the collection.

2.43.1 Scope and Usage

A document manifest gathers a set of Resources into a single package with some clinical context that may be the subject of workflow such as targeted delivery. The usage of "document" in the case of DocumentManifest, covers the broader scope of "document" as defined in REST and HTTP. This use is not limited to the formal "Document" model used for FHIR documents or DocumentReference resources.

Typically, DocumentManifest resources are used in document indexing systems, such as IHE XDS . Such as profiled in IHE Mobile access to Health Documents . When used as a Manifest of "Documents" in this way, the DocumentManifest will point exclusively to DocumentReference resources.

A DocumentManifest resource could be used to bring clinical context grouping to a broader set of Resources.

2.43.2 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
DocumentManifest | TU | | DomainResource | A list that defines a set of documents Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
masterIdentifier | Σ | 0..1 | Identifier | Unique Identifier for the set of documents
identifier | Σ | 0..* | Identifier | Other identifiers for the manifest
status | ?!Σ | 1..1 | code | current | superseded | entered-in-error DocumentReferenceStatus (Required)
type | Σ | 0..1 | CodeableConcept | Kind of document set v3 Code System ActCode (Example)
subject | Σ | 0..1 | Reference(Patient | Practitioner | Group | Device) | The subject of the set of documents
created | | 0..1 | dateTime | When this document manifest created
author | Σ | 0..* | Reference(Practitioner | PractitionerRole | Organization | Device | Patient | RelatedPerson) | Who and/or what authored the DocumentManifest
recipient | | 0..* | Reference(Patient | Practitioner | PractitionerRole | RelatedPerson | Organization) | Intended to get notified about this set of documents
source | | 0..1 | uri | The source system/application/software
description | Σ | 0..1 | string | Human-readable description (title)
content | Σ | 1..* | Reference(Any) | Items in manifest
related | | 0..* | BackboneElement | Related things
identifier | | 0..1 | Identifier | Identifiers of things that are related
ref | | 0..1 | Reference(Any) | Related Resource
Documentation for this format

UML Diagram (Legend)

XML Template

```
<DocumentManifest xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <masterIdentifier><!-- 0..1 Identifier Unique Identifier for the set of documents --></masterIdentifier>
 <identifier><!-- 0..* Identifier Other identifiers for the manifest --></identifier>
```

```
 <status value="[code]"/><!-- 1..1 current | superseded | entered-in-error -->
 <type><!-- 0..1 CodeableConcept Kind of document set --></type>
 <subject><!-- 0..1 Reference(Patient|Practitioner|Group|Device) The subject of the set of documents --></subject>
 <created value="[dateTime]"/><!-- 0..1 When this document manifest created -->
 <author><!-- 0..* Reference(Practitioner|PractitionerRole|Organization|Device|
   Patient|RelatedPerson) Who and/or what authored the DocumentManifest --></author>
 <recipient><!-- 0..* Reference(Patient|Practitioner|PractitionerRole|
   RelatedPerson|Organization) Intended to get notified about this set of documents --></recipient>
 <source value="[uri]"/><!-- 0..1 The source system/application/software -->
 <description value="[string]"/><!-- 0..1 Human-readable description (title) -->
 <content><!-- 1..* Reference(Any) Items in manifest --></content>
 <related>  <!-- 0..* Related things -->
  <identifier><!-- 0..1 Identifier Identifiers of things that are related --></identifier>
  <ref><!-- 0..1 Reference(Any) Related Resource --></ref>
 </related>
</DocumentManifest>
```

JSON Template

```
{
  "resourceType" : "DocumentManifest",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "masterIdentifier" : { Identifier }, // Unique Identifier for the set of documents
  "identifier" : [{ Identifier }], // Other identifiers for the manifest
  "status" : "<code>", // R!  current | superseded | entered-in-error
  "type" : { CodeableConcept }, // Kind of document set
  "subject" : { Reference(Patient|Practitioner|Group|Device) }, // The subject of the set of documents
  "created" : "<dateTime>", // When this document manifest created
  "author" : [{ Reference(Practitioner|PractitionerRole|Organization|Device|
   Patient|RelatedPerson) }], // Who and/or what authored the DocumentManifest
  "recipient" : [{ Reference(Patient|Practitioner|PractitionerRole|
   RelatedPerson|Organization) }], // Intended to get notified about this set of documents
  "source" : "<uri>", // The source system/application/software
  "description" : "<string>", // Human-readable description (title)
  "content" : [{ Reference(Any) }], // R!  Items in manifest
  "related" : [{ // Related things
    "identifier" : { Identifier }, // Identifiers of things that are related
    "ref" : { Reference(Any) } // Related Resource
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:DocumentManifest;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:DocumentManifest.masterIdentifier [ Identifier ]; # 0..1 Unique Identifier for the set of documents
  fhir:DocumentManifest.identifier [ Identifier ], ... ; # 0..* Other identifiers for the manifest
  fhir:DocumentManifest.status [ code ]; # 1..1 current | superseded | entered-in-error
  fhir:DocumentManifest.type [ CodeableConcept ]; # 0..1 Kind of document set
  fhir:DocumentManifest.subject [ Reference(Patient|Practitioner|Group|Device) ]; # 0..1 The subject of the set of
    documents
  fhir:DocumentManifest.created [ dateTime ]; # 0..1 When this document manifest created
  fhir:DocumentManifest.author [ Reference(Practitioner|PractitionerRole|Organization|Device|Patient|RelatedPerson) ],
    ... ; # 0..* Who and/or what authored the DocumentManifest
  fhir:DocumentManifest.recipient [ Reference(Patient|Practitioner|PractitionerRole|RelatedPerson|Organization) ], ... ;
     # 0..* Intended to get notified about this set of documents
  fhir:DocumentManifest.source [ uri ]; # 0..1 The source system/application/software
  fhir:DocumentManifest.description [ string ]; # 0..1 Human-readable description (title)
  fhir:DocumentManifest.content [ Reference(Any) ], ... ; # 1..* Items in manifest
  fhir:DocumentManifest.related [ # 0..* Related things
    fhir:DocumentManifest.related.identifier [ Identifier ]; # 0..1 Identifiers of things that are related
    fhir:DocumentManifest.related.ref [ Reference(Any) ]; # 0..1 Related Resource
  ], ...;
]
```

Changes since R3

DocumentManifest

DocumentManifest.status | Change value set from http://hl7.org/fhir/ValueSet/document-reference-status to http://hl7.org/fhir/ValueSet/document-reference-status|4.0.1

DocumentManifest.author | Type Reference: Added Target Type PractitionerRole

DocumentManifest.recipient | Type Reference: Added Target Type PractitionerRole

DocumentManifest.content | Type changed from BackboneElement to Reference(Resource)

DocumentManifest.content.p[x] | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 2 tests of which 2 fail to execute.)

Structure

Name | Flags | Card. | Type | Description & Constraints

DocumentManifest | TU | | DomainResource | A list that defines a set of documents Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
masterIdentifier | Σ | 0..1 | Identifier | Unique Identifier for the set of documents
identifier | Σ | 0..* | Identifier | Other identifiers for the manifest
status | ?!Σ | 1..1 | code | current | superseded | entered-in-error DocumentReferenceStatus (Required)
type | Σ | 0..1 | CodeableConcept | Kind of document set v3 Code System ActCode (Example)
subject | Σ | 0..1 | Reference(Patient | Practitioner | Group | Device) | The subject of the set of documents
created | | 0..1 | dateTime | When this document manifest created
author | Σ | 0..* | Reference(Practitioner | PractitionerRole | Organization | Device | Patient | RelatedPerson) | Who and/or what authored the DocumentManifest
recipient | | 0..* | Reference(Patient | Practitioner | PractitionerRole | RelatedPerson | Organization) | Intended to get notified about this set of documents
source | | 0..1 | uri | The source system/application/software
description | Σ | 0..1 | string | Human-readable description (title)
content | Σ | 1..* | Reference(Any) | Items in manifest
related | | 0..* | BackboneElement | Related things
identifier | | 0..1 | Identifier | Identifiers of things that are related
ref | | 0..1 | Reference(Any) | Related Resource
Documentation for this format

UML Diagram (Legend)

XML Template

```
<DocumentManifest xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <masterIdentifier><!-- 0..1 Identifier Unique Identifier for the set of documents --></masterIdentifier>
 <identifier><!-- 0..* Identifier Other identifiers for the manifest --></identifier>
 <status value="[code]"/><!-- 1..1 current | superseded | entered-in-error -->
 <type><!-- 0..1 CodeableConcept Kind of document set --></type>
 <subject><!-- 0..1 Reference(Patient|Practitioner|Group|Device) The subject of the set of documents --></subject>
 <created value="[dateTime]"/><!-- 0..1 When this document manifest created -->
 <author><!-- 0..* Reference(Practitioner|PractitionerRole|Organization|Device|
   Patient|RelatedPerson) Who and/or what authored the DocumentManifest --></author>
 <recipient><!-- 0..* Reference(Patient|Practitioner|PractitionerRole|
   RelatedPerson|Organization) Intended to get notified about this set of documents --></recipient>
 <source value="[uri]"/><!-- 0..1 The source system/application/software -->
 <description value="[string]"/><!-- 0..1 Human-readable description (title) -->
 <content><!-- 1..* Reference(Any) Items in manifest --></content>
 <related>  <!-- 0..* Related things -->
  <identifier><!-- 0..1 Identifier Identifiers of things that are related --></identifier>
  <ref><!-- 0..1 Reference(Any) Related Resource --></ref>
 </related>
</DocumentManifest>
```

JSON Template

```
{
  "resourceType" : "DocumentManifest",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "masterIdentifier" : { Identifier }, // Unique Identifier for the set of documents
  "identifier" : [{ Identifier }], // Other identifiers for the manifest
  "status" : "<code>", // R!  current | superseded | entered-in-error
  "type" : { CodeableConcept }, // Kind of document set
  "subject" : { Reference(Patient|Practitioner|Group|Device) }, // The subject of the set of documents
  "created" : "<dateTime>", // When this document manifest created
  "author" : [{ Reference(Practitioner|PractitionerRole|Organization|Device|
   Patient|RelatedPerson) }], // Who and/or what authored the DocumentManifest
  "recipient" : [{ Reference(Patient|Practitioner|PractitionerRole|
   RelatedPerson|Organization) }], // Intended to get notified about this set of documents
  "source" : "<uri>", // The source system/application/software
  "description" : "<string>", // Human-readable description (title)
  "content" : [{ Reference(Any) }], // R!  Items in manifest
  "related" : [{ // Related things
    "identifier" : { Identifier }, // Identifiers of things that are related
    "ref" : { Reference(Any) } // Related Resource
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:DocumentManifest;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:DocumentManifest.masterIdentifier [ Identifier ]; # 0..1 Unique Identifier for the set of documents
  fhir:DocumentManifest.identifier [ Identifier ], ... ; # 0..* Other identifiers for the manifest
  fhir:DocumentManifest.status [ code ]; # 1..1 current | superseded | entered-in-error
  fhir:DocumentManifest.type [ CodeableConcept ]; # 0..1 Kind of document set
  fhir:DocumentManifest.subject [ Reference(Patient|Practitioner|Group|Device) ]; # 0..1 The subject of the set of
    documents
  fhir:DocumentManifest.created [ dateTime ]; # 0..1 When this document manifest created
```

```
  fhir:DocumentManifest.author [ Reference(Practitioner|PractitionerRole|Organization|Device|Patient|RelatedPerson) ],
    ... ; # 0..* Who and/or what authored the DocumentManifest
  fhir:DocumentManifest.recipient [ Reference(Patient|Practitioner|PractitionerRole|RelatedPerson|Organization) ], ... ;
     # 0..* Intended to get notified about this set of documents
  fhir:DocumentManifest.source [ uri ]; # 0..1 The source system/application/software
  fhir:DocumentManifest.description [ string ]; # 0..1 Human-readable description (title)
  fhir:DocumentManifest.content [ Reference(Any) ], ... ; # 1..* Items in manifest
  fhir:DocumentManifest.related [ # 0..* Related things
    fhir:DocumentManifest.related.identifier [ Identifier ]; # 0..1 Identifiers of things that are related
    fhir:DocumentManifest.related.ref [ Reference(Any) ]; # 0..1 Related Resource
  ], ...;
]
```

Changes since Release 3

DocumentManifest
DocumentManifest.status | Change value set from http://hl7.org/fhir/ValueSet/document-reference-status to http://hl7.org/fhir/ValueSet/document-reference-status|4.0.1
DocumentManifest.author | Type Reference: Added Target Type PractitionerRole
DocumentManifest.recipient | Type Reference: Added Target Type PractitionerRole
DocumentManifest.content | Type changed from BackboneElement to Reference(Resource)
DocumentManifest.content.p[x] | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 2 tests of which 2 fail to execute.)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

2.43.2.1 Terminology Bindings

Path | Definition | Type | Reference
DocumentManifest.status | The status of the document reference. | Required | DocumentReferenceStatus
DocumentManifest.type | The activity that caused the DocumentManifest to be created. | Example | v3.ActCode

2.43.3 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
author | reference | Who and/or what authored the DocumentManifest | DocumentManifest.author (Practitioner, Organization, Device, Patient, PractitionerRole, RelatedPerson)
created | date | When this document manifest created | DocumentManifest.created
description | string | Human-readable description (title) | DocumentManifest.description
identifier | token | Unique Identifier for the set of documents | DocumentManifest.masterIdentifier | DocumentManifest.identifier | 30 Resources
item | reference | Items in manifest | DocumentManifest.content (Any)
patient | reference | The subject of the set of documents | DocumentManifest.subject.where(resolve() is Patient) (Patient) | 33 Resources
recipient | reference | Intended to get notified about this set of documents | DocumentManifest.recipient (Practitioner, Organization, Patient, PractitionerRole, RelatedPerson)
related-id | token | Identifiers of things that are related | DocumentManifest.related.identifier
related-ref | reference | Related Resource | DocumentManifest.related.ref (Any)
source | uri | The source system/application/software | DocumentManifest.source
status | token | current | superseded | entered-in-error | DocumentManifest.status
subject | reference | The subject of the set of documents | DocumentManifest.subject (Practitioner, Group, Device, Patient)
type | token | Kind of document set | DocumentManifest.type | 5 Resources

2.43.4 Resource DocumentManifest - Examples

URL: https://hl7.org/fhir/R4/documentmanifest-examples.html

2.43.4 Resource DocumentManifest - Examples

Structured Documents Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Not Classified | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

Example Name | id | Format
XDS SubmissionSet example | example | XML | JSON | Turtle
Example of FM attachment | 654789 | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

2.43.5 Resource DocumentManifest - Detailed Descriptions

URL: https://hl7.org/fhir/R4/documentmanifest-definitions.html

2.43.5 Resource DocumentManifest - Detailed Descriptions

Structured Documents Work Group | Maturity Level: 2 | Trial Use | Security Category: Not Classified | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

Detailed Descriptions for the elements in the DocumentManifest resource.

DocumentManifest
Element Id | DocumentManifest
Definition | A collection of documents compiled for a purpose together with metadata that applies to the collection.
Cardinality | 0..*
Type | DomainResource
DocumentManifest.masterIdentifier
Element Id | DocumentManifest.masterIdentifier
Definition | A single identifier that uniquely identifies this manifest. Principally used to refer to the manifest in non-FHIR contexts.
Cardinality | 0..1
Type | Identifier
Summary | true
DocumentManifest.identifier
Element Id | DocumentManifest.identifier
Definition | Other identifiers associated with the document manifest, including version independent identifiers.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Summary | true
DocumentManifest.status
Element Id | DocumentManifest.status
Definition | The status of this document manifest.
Cardinality | 1..1
Terminology Binding | DocumentReferenceStatus (Required)
Type | code
Is Modifier | true (Reason: This element is labelled as a modifier because it is a status element that contains status entered-in-error which means that the resource should not be treated as valid)
Summary | true
Comments | This element is labeled as a modifier because the status contains the codes that mark the manifest as not currently valid.
DocumentManifest.type
Element Id | DocumentManifest.type
Definition | The code specifying the type of clinical activity that resulted in placing the associated content into the DocumentManifest.
Cardinality | 0..1
Terminology Binding | v3 Code System ActCode (Example)
Type | CodeableConcept
Summary | true
Comments | Specifies the kind of this set of documents (e.g. Patient Summary, Discharge Summary, Prescription, etc.). The type of a set of documents may be the same as one of the documents in it - especially if there is only one - but it may be wider.
DocumentManifest.subject
Element Id | DocumentManifest.subject
Definition | Who or what the set of documents is about. The documents can be about a person, (patient or healthcare practitioner), a device (i.e. machine) or even a group of subjects (such as a document about a herd of farm animals, or a set of patients that share a common exposure). If the documents cross more than one subject, then more than one subject is allowed here (unusual use case).
Cardinality | 0..1
Type | Reference(Patient | Practitioner | Group | Device)
Summary | true
DocumentManifest.created
Element Id | DocumentManifest.created
Definition | When the document manifest was created for submission to the server (not necessarily the same thing as the actual resource last modified time, since it may be modified, replicated, etc.).
Cardinality | 0..1
Type | dateTime
Comments | Creation time is used for tracking, organizing versions and searching. This is the creation time of the document set, not the documents on which it is based.
DocumentManifest.author
Element Id | DocumentManifest.author
Definition | Identifies who is the author of the manifest. Manifest author is not necessarly the author of the references included.
Cardinality | 0..*
Type | Reference(Practitioner | PractitionerRole | Organization | Device | Patient | RelatedPerson)
Summary | true
Comments | Not necessarily who did the actual data entry (i.e. typist) or who was the source (informant).
DocumentManifest.recipient
Element Id | DocumentManifest.recipient
Definition | A patient, practitioner, or organization for which this set of documents is intended.
Cardinality | 0..*
Type | Reference(Patient | Practitioner | PractitionerRole | RelatedPerson | Organization)

Comments | How the recipient receives the document set or is notified of it is up to the implementation. This element is just a statement of intent. If the recipient is a person, and it is not known whether the person is a patient or a practitioner, RelatedPerson would be the default choice.
DocumentManifest.source
Element Id | DocumentManifest.source
Definition | Identifies the source system, application, or software that produced the document manifest.
Cardinality | 0..1
Type | uri
DocumentManifest.description
Element Id | DocumentManifest.description
Definition | Human-readable description of the source document. This is sometimes known as the "title".
Cardinality | 0..1
Type | string
Requirements | Helps humans to assess whether the document is of interest.
Summary | true
Comments | What the document is about, rather than a terse summary of the document. It is commonly the case that records do not have a title and are collectively referred to by the display name of Record code (e.g. a "consultation" or "progress note").
DocumentManifest.content
Element Id | DocumentManifest.content
Definition | The list of Resources that consist of the parts of this manifest.
Cardinality | 1..*
Type | Reference(Any)
Requirements | Any Resources are allowed to be grouped in a Manifest. The manifest brings context to the group.
Summary | true
Comments | When used for XDS the intended focus of the DocumentManifest is for the reference to target to be a set of DocumentReference Resources. The reference is to "Any" to support EN 13606 usage, where an extract is DocumentManifest that references List and Composition resources.
DocumentManifest.related
Element Id | DocumentManifest.related
Definition | Related identifiers or resources associated with the DocumentManifest.
Cardinality | 0..*
Comments | May be identifiers or resources that caused the DocumentManifest to be created.
DocumentManifest.related.identifier
Element Id | DocumentManifest.related.identifier
Definition | Related identifier to this DocumentManifest. For example, Order numbers, accession numbers, XDW workflow numbers.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..1
Type | Identifier
Comments | If both identifier and ref elements are present they shall refer to the same thing.
DocumentManifest.related.ref
Element Id | DocumentManifest.related.ref
Definition | Related Resource to this DocumentManifest. For example, Order, ServiceRequest, Procedure, EligibilityRequest, etc.
Cardinality | 0..1
Type | Reference(Any)
Comments | If both identifier and ref elements are present they shall refer to the same thing.

2.43.6 Resource DocumentManifest - Mappings

URL: https://hl7.org/fhir/R4/documentmanifest-mappings.html

2.43.6 Resource DocumentManifest - Mappings

Structured Documents Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Not Classified | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

HL7 v2 Mapping

RIM Mapping

XDS metadata equivalent

Mappings for the documentmanifest resource (see Mappings to Other Standards for further information & status).

2.43.6.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

DocumentManifest
masterIdentifier | FiveWs.identifier
identifier | FiveWs.identifier
status | FiveWs.status
type | FiveWs.class

subject | FiveWs.subject[x]
created | FiveWs.done[x]
author | FiveWs.author
recipient | FiveWs.cause

2.43.6.2 Workflow Pattern (http://hl7.org/fhir/workflow)

DocumentManifest | Event
masterIdentifier | Event.identifier
identifier | Event.identifier
status | Event.status
type | Event.code
subject | Event.subject
created | Event.occurrence[x]
author | Event.performer.actor
recipient | Event.performer

2.43.6.3 HL7 v2 Mapping (http://hl7.org/v2)

DocumentManifest
masterIdentifier
identifier | TXA-16-Unique Document File Name?
status
type
subject
created
author
recipient
source
description
content
related
identifier
ref

2.43.6.4 RIM Mapping (http://hl7.org/v3)

DocumentManifest | Act[classCode=FOLDER, moodCode=EVN]
masterIdentifier | id [need a list semantic to indicate order matters]
identifier | .id / .setId
status | .statusCode
type | .code
subject | .participation[typeCode=SBJ].role
created | .participation[typeCode=AUT].time
author | .participation[typeCode=AUT].role
recipient | .participation[typeCode=IRCP].role
source | .participation[typeCode=PFM].time
description | .title
content | .outboundRelationship[typeCode=COMP].target
related | n/a

2.43.6.5 XDS metadata equivalent (http://ihe.net/xds)

DocumentManifest
masterIdentifier | SubmissionSet.uniqueId
identifier | SubmissionSet.entryUUID
status | SubmissionSet status
type | SubmissionSet.contentTypeCode
subject | SubmissionSet.patientId
created | SubmissionSet.submissionTime
author | SubmissionSet.author
recipient | SubmissionSet.intendedRecipient
source | SubmissionSet.sourceId
description | SubmissionSet.title
content | SubmissionSet DocumentEntry(s)
related
identifier
ref

2.43.7 Resource DocumentManifest - Extensions & Profiles

URL: https://hl7.org/fhir/R4/documentmanifest-profiles.html

2.43.7 Resource DocumentManifest - Extensions & Profiles

Structured Documents Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Not Classified | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

This table lists profiles and extensions for the DocumentManifest resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

2.43.7.1 Profiles

No Profiles defined for this resource

2.43.7.2 Extensions

No Extensions defined for this resource
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

2.43.7.3 Search Extensions

No Search Extensions defined for this resource

2.43.8 Resource DocumentManifest - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/documentmanifest-version-maps.html

No validation errors - all conversions are clean

2.43.8 Resource DocumentManifest - R3/R4 Conversions

Structured Documents Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Not Classified | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

R3 : R4 Conversion maps for DocumentManifest.

Functional status for this map: 2 tests of which 2 fail to execute. (see documentation)

2.43.8.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/DocumentManifest3to4" = "R3 to R4 Conversions for DocumentManifest"

uses "http://hl7.org/fhir/3.0/StructureDefinition/DocumentManifest" alias DocumentManifestR3 as source
uses "http://hl7.org/fhir/StructureDefinition/DocumentManifest" alias DocumentManifest as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group DocumentManifest(source src : DocumentManifestR3, target tgt : DocumentManifest) extends DomainResource <<type+>>
    {
  src.masterIdentifier -> tgt.masterIdentifier;
  src.identifier -> tgt.identifier;
  src.status -> tgt.status;
  src.type -> tgt.type;
  src.subject -> tgt.subject;
  src.created -> tgt.created;
  src.author as s -> tgt.agent as t then DocumentManifestAgent(s, t);
  src.recipient -> tgt.recipient;
  src.source -> tgt.source;
  src.description -> tgt.description;
  src.content as s -> tgt.content as t then content(s, t);
  src.related as s -> tgt.related as t then DocumentManifestRelated(s, t);
}

group DocumentManifestAgent(source src, target tgt) extends BackboneElement {
  src ->  tgt.type as cc,  cc.coding as c,  c.system = 'http://terminology.hl7.org/CodeSystem/v3-ParticipationType',
    c.code = 'AUT' "author";
  src -> tgt.who as vt0 then Reference(src, vt0) "who";
}

group content(source src, target tgt) extends Element {
  src.p : Attachment as s ->  tgt.extension as ext,  ext.url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    DocumentManifest.content.p_Attachment',  ext.value = s;
  src.p : Reference as s then Reference(s, tgt);
}

group DocumentManifestRelated(source src, target tgt) extends BackboneElement {
```

```
  src.identifier -> tgt.identifier;
  src.ref -> tgt.ref;
}
```

2.43.8.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/DocumentManifest4to3" = "R4 to R3 Conversion for DocumentManifest"

uses "http://hl7.org/fhir/StructureDefinition/DocumentManifest" alias DocumentManifest as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/DocumentManifest" alias DocumentManifestR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group DocumentManifest(source src : DocumentManifestR3, target tgt : DocumentManifest) extends DomainResource <<type+>>
    {
  src.masterIdentifier -> tgt.masterIdentifier;
  src.identifier -> tgt.identifier;
  src.status -> tgt.status;
  src.type -> tgt.type;
  src.subject -> tgt.subject;
  src.created -> tgt.created;
  src.agent as s where type.coding.where((system = 'http://terminology.hl7.org/CodeSystem/v3-ParticipationType') and
    (code = 'AUT')).exists() -> tgt.author as t then DocumentManifestAgent(s, t);
  src.recipient -> tgt.recipient;
  src.source -> tgt.source;
  src.description -> tgt.description;
  src.content as s -> tgt.content as t then content(s, t);
  src.related as s -> tgt.related as t then DocumentManifestRelated(s, t);
}

group DocumentManifestAgent(source src, target tgt) extends BackboneElement {
  src.who as vs0 then Reference(vs0, tgt);
}

group content(source src, target tgt) extends Element {
  src.extension as s where url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    DocumentManifest.content.p_Attachment' then contentAttachment(s, tgt);
  src where extension.where(url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    DocumentManifest.content.p_Attachment').empty() -> tgt.p = create('Reference') as p then Reference(src, p)
    "attachment";
}

group contentAttachment(source src, target tgt) extends Element {
  src.value : Attachment as s -> tgt.p = create('Attachment') as t then Attachment(s, t);
}

group DocumentManifestRelated(source src, target tgt) extends BackboneElement {
  src.identifier -> tgt.identifier;
  src.ref -> tgt.ref;
}
```

2.43.8.3 R4 Validation Errors

2.43.9 JSON Schema for DocumentManifest

URL: https://hl7.org/fhir/R4/documentmanifest.schema.json.html

2.43.9 JSON Schema for DocumentManifest

Structured Documents Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

This is a representation of the json schema for DocumentManifest, which is just a part of the full JSON Schema.

```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/DocumentManifest",
  "$ref": "#/definitions/DocumentManifest",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "DocumentManifest": {
      "description": "A collection of documents compiled for a purpose together with metadata that applies to the
    collection.",
      "properties": {
        "resourceType": {
          "description": "This is a DocumentManifest resource",
          "const": "DocumentManifest"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
```

```
    "implicitRules": {
      "description": "A reference to a set of rules that were followed when the resource was constructed, and which
must be understood when processing the content. Often, this is a reference to an implementation guide that defines
the special rules along with other profiles etc.",
      "$ref": "#/definitions/uri"
    },
    "_implicitRules": {
      "description": "Extensions for implicitRules",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "language": {
      "description": "The base language in which the resource is written.",
      "$ref": "#/definitions/code"
    },
    "_language": {
      "description": "Extensions for language",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "text": {
      "description": "A human-readable narrative that contains a summary of the resource and can be used to
represent the content of the resource to a human. The narrative need not encode all the structured data, but is
required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
      "$ref": "Narrative.schema.json#/definitions/Narrative"
    },
    "contained": {
      "description": "These resources do not have an independent existence apart from the resource that contains
them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
 definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "masterIdentifier": {
      "description": "A single identifier that uniquely identifies this manifest. Principally used to refer to the
manifest in non-FHIR contexts.",
      "$ref": "Identifier.schema.json#/definitions/Identifier"
    },
    "identifier": {
      "description": "Other identifiers associated with the document manifest, including version independent
identifiers.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "status": {
      "description": "The status of this document manifest.",
      "enum": [
        "current",
        "superseded",
        "entered-in-error"
      ]
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "type": {
      "description": "The code specifying the type of clinical activity that resulted in placing the associated
content into the DocumentManifest.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "subject": {
      "description": "Who or what the set of documents is about. The documents can be about a person, (patient or
```

```
healthcare practitioner), a device (i.e. machine) or even a group of subjects (such as a document about a herd of farm animals, or a set of patients that share a common exposure). If the documents cross more than one subject, then more than one subject is allowed here (unusual use case).",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "created": {
      "description": "When the document manifest was created for submission to the server (not necessarily the same thing as the actual resource last modified time, since it may be modified, replicated, etc.).",
      "$ref": "#/definitions/dateTime"
    },
    "_created": {
      "description": "Extensions for created",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "author": {
      "description": "Identifies who is the author of the manifest. Manifest author is not necessarly the author of the references included.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "recipient": {
      "description": "A patient, practitioner, or organization for which this set of documents is intended.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "source": {
      "description": "Identifies the source system, application, or software that produced the document manifest.",
      "$ref": "#/definitions/uri"
    },
    "_source": {
      "description": "Extensions for source",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "description": {
      "description": "Human-readable description of the source document. This is sometimes known as the \"title\".",
      "$ref": "#/definitions/string"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "content": {
      "description": "The list of Resources that consist of the parts of this manifest.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "related": {
      "description": "Related identifiers or resources associated with the DocumentManifest.",
      "items": {
        "$ref": "#/definitions/DocumentManifest_Related"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "content",
    "resourceType"
  ]
},
"DocumentManifest_Related": {
  "description": "A collection of documents compiled for a purpose together with metadata that applies to the collection.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element and that modifies the understanding of the element in which it is contained and/or the understanding of the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
```

```
    use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
     of the definition of the extension. Applications processing a resource are required to check for modifier
    extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
    (including cannot change the meaning of modifierExtension itself).",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "identifier": {
          "description": "Related identifier to this DocumentManifest.  For example, Order numbers, accession numbers,
    XDW workflow numbers.",
          "$ref": "Identifier.schema.json#/definitions/Identifier"
        },
        "ref": {
          "description": "Related Resource to this DocumentManifest. For example, Order, ServiceRequest,  Procedure,
    EligibilityRequest, etc.",
          "$ref": "Reference.schema.json#/definitions/Reference"
        }
      },
      "additionalProperties": false
    }
  }
}
```


2.43.10 Resource DocumentManifest - Maturity Dependency View

URL: https://hl7.org/fhir/R4/documentmanifest-dependencies.html

2.43.10 Resource DocumentManifest - Maturity Dependency View

Structured Documents Work Group | Maturity Level: 2 | Trial Use | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

Dependency Graph for DocumentManifest FMM level 2

Name | Card. | Type | Dependency Analysis
DocumentManifest | | DomainResource
masterIdentifier | 0..1 | Identifier
identifier | 0..* | Identifier
status | 1..1 | code
type | 0..1 | CodeableConcept
subject | 0..1 | Reference(Patient | Practitioner | Group | Device) | Type Warning: (Group = FMM1-Trial Use vs. Element = FMM2-Trial Use)
created | 0..1 | dateTime
author | 0..* | Reference(Practitioner | PractitionerRole | Organization | Device | Patient | RelatedPerson)
recipient | 0..* | Reference(Patient | Practitioner | PractitionerRole | RelatedPerson | Organization)
source | 0..1 | uri
description | 0..1 | string
content | 1..* | Reference(Any) | Type Warning: (Any = FMM1-Trial Use vs. Element = FMM2-Trial Use)
related | 0..* | BackboneElement
identifier | 0..1 | Identifier
ref | 0..1 | Reference(Any) | Type Warning: (Any = FMM1-Trial Use vs. Element = FMM2-Trial Use)

2.43.11 Value Set Sc-valueset-document-reference-status - Status Code Mapping

URL: https://hl7.org/fhir/R4/sc-valueset-document-reference-status.html

2.43.11 Value Set Sc-valueset-document-reference-status - Status Code Mapping

Structured Documents Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

This is the narrative for the resource. See also the XML, JSON or Turtle format.

2.43.12 DocumentReferenceStatusCanonicalMap (http://hl7.org/fhir/ConceptMap/sc-document-reference-status)

Mapping from http://hl7.org/fhir/ValueSet/document-reference-status to http://hl7.org/fhir/ValueSet/resource-status

DRAFT. Published on 01/11/2019 9:29:23 AM by HL7 (FHIR Project) (http://hl7.org/fhir , fhir@lists.hl7.org).

Canonical Mapping for "The status of the document reference."

Source Code | Equivalence | Destination Code

entered-in-error | equivalent | error
current | equivalent | active
superseded | equivalent | inactive

2.44 Resource OperationOutcome - Content

URL: https://hl7.org/fhir/R4/operationoutcome.html
FHIR Infrastructure Work Group | Maturity Level: N | Normative (from v4.0.0) | | Compartments: Not linked to any defined compartments

| This page has been approved as part of an ANSI standard. See the Infrastructure Package for further details.

A collection of error, warning, or information messages that result from a system action.

2.44.1 Scope and Usage

Operation outcomes are sets of error, warning and information messages that provide detailed information about the outcome of an attempted system operation. They are provided as a direct system response, or component of one, and provide information about the outcome of the operation.

The OperationOutcome resource is used in the following circumstances:

• When a RESTful interaction or operation fails
• As the response on a validation operation to provide information about the outcome
• As part of a message response, usually when the message has not been processed correctly
• As the response to a batch/transaction, when requested
• As part of a search's Bundle response containing information about the search

2.44.2 Boundaries and Relationships

This resource is not used for reporting clinical or workflow issues associated with a proposed or ongoing action; these would be handled using DetectedIssue or other resources. OperationOutcome is not designed to be persisted or referenced from other parts of the workflow.

It is possible to have both OperationOutcome and DetectedIssue together, where the OperationOutcome might indicate that a requested action was rejected due to a clinical issue and the DetectedIssue provides the details of the issue.

This resource is referenced by GuidanceResponse and MessageHeader

2.44.3 Resource Content

• Structure
• UML
• XML
• JSON
• Turtle
• R3 Diff
• All

Structure

Name | Flags | Card. | Type | Description & Constraints
OperationOutcome | ΣN | | DomainResource | Information about the success/failure of an action Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
issue | Σ | 1..* | BackboneElement | A single issue associated with the action
severity | Σ | 1..1 | code | fatal | error | warning | information IssueSeverity (Required)
code | Σ | 1..1 | code | Error or warning code IssueType (Required)
details | Σ | 0..1 | CodeableConcept | Additional details about the error Operation Outcome Codes (Example)
diagnostics | Σ | 0..1 | string | Additional diagnostic information about the issue
location | ΣXD | 0..* | string | Deprecated: Path of element(s) related to issue
expression | Σ | 0..* | string | FHIRPath of element(s) related to issue
Documentation for this format

UML Diagram (Legend)

XML Template

```
<OperationOutcome xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <issue>  <!-- 1..* A single issue associated with the action -->
  <severity value="[code]"/><!-- 1..1 fatal | error | warning | information -->
  <code value="[code]"/><!-- 1..1 Error or warning code -->
  <details><!-- 0..1 CodeableConcept Additional details about the error --></details>
  <diagnostics value="[string]"/><!-- 0..1 Additional diagnostic information about the issue -->
  <location value="[string]"/><!-- 0..* Deprecated: Path of element(s) related to issue -->
  <expression value="[string]"/><!-- 0..* FHIRPath of element(s) related to issue -->
```

```
 </issue>
</OperationOutcome>
```

JSON Template

```
{
  "resourceType" : "OperationOutcome",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "issue" : [{ // R!  A single issue associated with the action
    "severity" : "<code>", // R!  fatal | error | warning | information
    "code" : "<code>", // R!  Error or warning code
    "details" : { CodeableConcept }, // Additional details about the error
    "diagnostics" : "<string>", // Additional diagnostic information about the issue
    "location" : ["<string>"], // Deprecated: Path of element(s) related to issue
    "expression" : ["<string>"] // FHIRPath of element(s) related to issue
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:OperationOutcome;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:OperationOutcome.issue [ # 1..* A single issue associated with the action
    fhir:OperationOutcome.issue.severity [ code ]; # 1..1 fatal | error | warning | information
    fhir:OperationOutcome.issue.code [ code ]; # 1..1 Error or warning code
    fhir:OperationOutcome.issue.details [ CodeableConcept ]; # 0..1 Additional details about the error
    fhir:OperationOutcome.issue.diagnostics [ string ]; # 0..1 Additional diagnostic information about the issue
    fhir:OperationOutcome.issue.location [ string ], ... ; # 0..* Deprecated: Path of element(s) related to issue
    fhir:OperationOutcome.issue.expression [ string ], ... ; # 0..* FHIRPath of element(s) related to issue
  ], ...;
]
```

Changes since R3

OperationOutcome

OperationOutcome.issue.severity | Change value set from http://hl7.org/fhir/ValueSet/issue-severity to http://hl7.org/fhir/ValueSet/issue-severity|4.0.1 No longer marked as Modifier

OperationOutcome.issue.code | Change value set from http://hl7.org/fhir/ValueSet/issue-type to http://hl7.org/fhir/ValueSet/issue-type|4.0.1

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 6 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.)

Structure

Name | Flags | Card. | Type | Description & Constraints

OperationOutcome | ΣN | | DomainResource | Information about the success/failure of an action Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

issue | Σ | 1..* | BackboneElement | A single issue associated with the action

severity | Σ | 1..1 | code | fatal | error | warning | information IssueSeverity (Required)

code | Σ | 1..1 | code | Error or warning code IssueType (Required)

details | Σ | 0..1 | CodeableConcept | Additional details about the error Operation Outcome Codes (Example)

diagnostics | Σ | 0..1 | string | Additional diagnostic information about the issue

location | ΣXD | 0..* | string | Deprecated: Path of element(s) related to issue

expression | Σ | 0..* | string | FHIRPath of element(s) related to issue

Documentation for this format

UML Diagram (Legend)

XML Template

```
<OperationOutcome xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <issue>  <!-- 1..* A single issue associated with the action -->
  <severity value="[code]"/><!-- 1..1 fatal | error | warning | information -->
  <code value="[code]"/><!-- 1..1 Error or warning code -->
  <details><!-- 0..1 CodeableConcept Additional details about the error --></details>
  <diagnostics value="[string]"/><!-- 0..1 Additional diagnostic information about the issue -->
  <location value="[string]"/><!-- 0..* Deprecated: Path of element(s) related to issue -->
  <expression value="[string]"/><!-- 0..* FHIRPath of element(s) related to issue -->
 </issue>
</OperationOutcome>
```

JSON Template

```
{
  "resourceType" : "OperationOutcome",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "issue" : [{ // R!  A single issue associated with the action
```

```
    "severity" : "<code>", // R!  fatal | error | warning | information
    "code" : "<code>", // R!  Error or warning code
    "details" : { CodeableConcept }, // Additional details about the error
    "diagnostics" : "<string>", // Additional diagnostic information about the issue
    "location" : ["<string>"], // Deprecated: Path of element(s) related to issue
    "expression" : ["<string>"] // FHIRPath of element(s) related to issue
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:OperationOutcome;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:OperationOutcome.issue [ # 1..* A single issue associated with the action
    fhir:OperationOutcome.issue.severity [ code ]; # 1..1 fatal | error | warning | information
    fhir:OperationOutcome.issue.code [ code ]; # 1..1 Error or warning code
    fhir:OperationOutcome.issue.details [ CodeableConcept ]; # 0..1 Additional details about the error
    fhir:OperationOutcome.issue.diagnostics [ string ]; # 0..1 Additional diagnostic information about the issue
    fhir:OperationOutcome.issue.location [ string ], ... ; # 0..* Deprecated: Path of element(s) related to issue
    fhir:OperationOutcome.issue.expression [ string ], ... ; # 0..* FHIRPath of element(s) related to issue
  ], ...;
]
```

Changes since Release 3

OperationOutcome

OperationOutcome.issue.severity | Change value set from http://hl7.org/fhir/ValueSet/issue-severity to http://hl7.org/fhir/ValueSet/issue-severity|4.0.1 No longer marked as Modifier

OperationOutcome.issue.code | Change value set from http://hl7.org/fhir/ValueSet/issue-type to http://hl7.org/fhir/ValueSet/issue-type|4.0.1

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 6 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

2.44.3.1 Terminology Bindings

Path | Definition | Type | Reference

OperationOutcome.issue.severity | How the issue affects the success of the action. | Required | IssueSeverity

OperationOutcome.issue.code | A code that describes the type of issue. | Required | IssueType

OperationOutcome.issue.details | A code that provides details as the exact issue. | Example | OperationOutcomeCodes

2.44.4 Using Operation Outcome Resources

On the RESTful interface, operation outcome resources are only relevant when a level of computable detail is required that is more granular than that provided by the HTTP response codes. This granularity could include:

• more detail about the location of an issue

• the ability to identify multiple distinct issues

• provision of finer error codes that connect to known business failure states

Operation outcomes returned SHOULD be in alignment with the HTTP response code. For example, if the HTTP code indicates a failure (300+), at least one of the issues should have a severity of "error", indicating the reason for the failure.

Operation outcomes must include at least one issue. See the All OK example for a template for an operation that returns no errors.

2.44.5 Using the Expression

Each issue in an operation outcome SHOUD have an expression that reports a location for the issue. A correctly populated expression can allow client systems to:

• Connect return errors with the appropriate UI widget

• Route errors to the correct application or system log

• Develop more intelligent testing applications

In order to support these kinds of usages, applications need to use the expression element consistently. Applications can use the expression element to refer to a location inside a resource, or some location in the HTTP request (when appropriate).

Error locations are always reported using a simplified FHIRPath statement in the expression element. In addition to the simplified FHIRPath restrictions, the expression SHALL not contain a .resolve() function. These

examples are legal:

```
<expression value="Patient.identifier"/>
```

```
"expression" : "Patient.identifier[2].value"
```

but this is not:

```
"expression" : "Patient.identifier.where(system.value='http://example.com/mrn').value"
```

2.44.5.1 Reporting Errors in the HTTP Headers

Servers may also need to report errors in the HTTP headers - especially query parameters when processing searches. Errors are reported using a case sensitive expression that has two parts, a fixed "http" and the header or query parameter name separated by a ".". Some examples:

Expresson | Description
http.code | A reference to the search parameter "code"
http."name:exact" | A reference to the search parameter "name" with the modifier ":exact"
http.Authorization | A reference to the Authorization header - perhaps to indicate that it is missing, and some form of authentication is required.

This specification follows the convention that HTTP headers start with an uppercase character, and URL parameters start with a lowercase character. Implementations are encouraged to follow this convention, so that no name collisions occur.

2.44.6 Resource OperationOutcome - Examples

URL: https://hl7.org/fhir/R4/operationoutcome-examples.html

2.44.6 Resource OperationOutcome - Examples

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | | Compartments: Not linked to any defined compartments

Example Name | id | Format
General Outcome Example | 101 | XML | JSON | Turtle
An error that might be returned when an unexpected exception happens on a server | exception | XML | JSON | Turtle
An operation outcome that could be returned after a successful operation (no issues) | allok | XML | JSON | Turtle
A validator has found problems with a resource | validationfail | XML | JSON | Turtle
A search error (illegal modifier) | searchfail | XML | JSON | Turtle
A note that additional information might be available if Break-The-Glass is used | break-the-glass | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

2.44.7 Resource OperationOutcome - Detailed Descriptions

URL: https://hl7.org/fhir/R4/operationoutcome-definitions.html

2.44.7 Resource OperationOutcome - Detailed Descriptions

FHIR Infrastructure Work Group | Maturity Level: N | Normative (from v4.0.0) | | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the OperationOutcome resource.

OperationOutcome
Element Id | OperationOutcome
Definition | A collection of error, warning, or information messages that result from a system action.
Cardinality | 0..*
Type | DomainResource
Summary | true
Comments | Can result from the failure of a REST call or be part of the response message returned from a request message.
OperationOutcome.issue
Element Id | OperationOutcome.issue
Definition | An error, warning, or information message that results from a system action.
Cardinality | 1..*
Summary | true
OperationOutcome.issue.severity
Element Id | OperationOutcome.issue.severity
Definition | Indicates whether the issue indicates a variation from successful processing.
Cardinality | 1..1
Terminology Binding | IssueSeverity (Required)
Type | code
Requirements | Indicates how relevant the issue is to the overall success of the action.
Summary | true
Comments | This is labeled as "Is Modifier" because applications should not confuse hints and warnings with errors.
OperationOutcome.issue.code
Element Id | OperationOutcome.issue.code

Definition | Describes the type of the issue. The system that creates an OperationOutcome SHALL choose the most applicable code from the IssueType value set, and may additional provide its own code for the error in the details element.
Cardinality | 1..1
Terminology Binding | IssueType (Required)
Type | code
Requirements | Expresses the issue in a human and computer-friendly way, allowing the requesting system to behave differently based on the type of issue. The required element provides a base level of computable interoperability across all systems for processing an outcome, allowing systems to respond appropriately regardless of the internal system error codes returned.
Summary | true
OperationOutcome.issue.details
Element Id | OperationOutcome.issue.details
Definition | Additional details about the error. This may be a text description of the error or a system code that identifies the error.
Cardinality | 0..1
Terminology Binding | Operation Outcome Codes (Example)
Type | CodeableConcept
Summary | true
Comments | A human readable description of the error issue SHOULD be placed in details.text.
OperationOutcome.issue.diagnostics
Element Id | OperationOutcome.issue.diagnostics
Definition | Additional diagnostic information about the issue.
Cardinality | 0..1
Type | string
Summary | true
Comments | This may be a description of how a value is erroneous, a stack dump to help trace the issue or other troubleshooting information.
OperationOutcome.issue.location
Standards Status | This element has a standards status of "Deprecated" which is different from the status of the whole resource
Element Id | OperationOutcome.issue.location
Definition | This element is deprecated because it is XML specific. It is replaced by issue.expression, which is format independent, and simpler to parse. For resource issues, this will be a simple XPath limited to element names, repetition indicators and the default child accessor that identifies one of the elements in the resource that caused this issue to be raised. For HTTP errors, will be "http." + the parameter name.
Cardinality | 0..*
Type | string
Requirements | Allows systems to highlight or otherwise guide users to elements implicated in issues to allow them to be fixed more easily.
Summary | true
Comments | The root of the XPath is the resource or bundle that generated OperationOutcome. Each XPath SHALL resolve to a single node. This element is deprecated, and is being replaced by expression.
OperationOutcome.issue.expression
Element Id | OperationOutcome.issue.expression
Definition | A simple subset of FHIRPath limited to element names, repetition indicators and the default child accessor that identifies one of the elements in the resource that caused this issue to be raised.
Cardinality | 0..*
Type | string
Requirements | Allows systems to highlight or otherwise guide users to elements implicated in issues to allow them to be fixed more easily.
Summary | true
Comments | The root of the FHIRPath is the resource or bundle that generated OperationOutcome. Each FHIRPath SHALL resolve to a single node.

2.44.8 Resource OperationOutcome - Mappings

URL: https://hl7.org/fhir/R4/operationoutcome-mappings.html

2.44.8 Resource OperationOutcome - Mappings

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | | Compartments: Not linked to any defined compartments

Mappings:

HL7 v2 Mapping

RIM Mapping

Mappings for the operationoutcome resource (see Mappings to Other Standards for further information & status).

2.44.8.1 HL7 v2 Mapping (http://hl7.org/v2)

OperationOutcome
issue | ERR
severity | ERR-4
code | ERR-3
details | ERR-5
diagnostics | ERR-8

location | ERR-2
expression | ERR-2

2.44.8.2 RIM Mapping (http://hl7.org/v3)

OperationOutcome | Grouping is handled through multiple repetitions from whatever context references the outcomes (no specific grouper required)
issue | AcknowledgementDetail or Observation[classCode="ALRT" and moodCode="EVN"]
severity | ./typeCode[parent::AcknowledgmentDetail] or unique(./inboundRelationship[parent::Observation and typeCode='SUBJ' and isNormalActRelationship()]/source[classCode="OBS" and moodCode="EVN" and implies(code, ActCode#SEV and isNormalAct())/value[xsi:type="CD"]
code | ./code
details | ./code
diagnostics | ./text[parent::AcknowledgmentDetail]
location | ./location[parent::AcknowledgmentDetail]
expression | ./location[parent::AcknowledgmentDetail]

2.44.9 Resource OperationOutcome - Extensions & Profiles

URL: https://hl7.org/fhir/R4/operationoutcome-profiles.html

2.44.9 Resource OperationOutcome - Extensions & Profiles

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | | Compartments: Not linked to any defined compartments

This table lists profiles and extensions for the OperationOutcome resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

2.44.9.1 Profiles

No Profiles defined for this resource

2.44.9.2 Extensions

operationoutcome-authority | OperationOutcome.issue | authority | for Standard OperationOutcome Extensions
operationoutcome-detectedIssue | OperationOutcome.issue | detectedIssue | for Standard OperationOutcome Extensions
operationoutcome-issue-source | OperationOutcome.issue | issue-source | for Standard OperationOutcome Extensions
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

2.44.9.3 Search Extensions

No Search Extensions defined for this resource

2.44.10 Resource OperationOutcome - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/operationoutcome-version-maps.html

No validation errors - all conversions are clean

2.44.10 Resource OperationOutcome - R3/R4 Conversions

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | | Compartments: Not linked to any defined compartments

R3 : R4 Conversion maps for OperationOutcome.

Functional status for this map: 6 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid. (see documentation)

2.44.10.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/OperationOutcome3to4" = "R3 to R4 Conversions for OperationOutcome"

uses "http://hl7.org/fhir/3.0/StructureDefinition/OperationOutcome" alias OperationOutcomeR3 as source
uses "http://hl7.org/fhir/StructureDefinition/OperationOutcome" alias OperationOutcome as target
```

```
imports "http://hl7.org/fhir/StructureMap/*3to4"

group OperationOutcome(source src : OperationOutcomeR3, target tgt : OperationOutcome) extends DomainResource <<type+>>
    {
  src.issue as vs0 -> tgt.issue as vt0 then {
    vs0.severity -> vt0.severity;
    vs0.code -> vt0.code;
    vs0.details -> vt0.details;
    vs0.diagnostics -> vt0.diagnostics;
    vs0.location -> vt0.location;
    vs0.expression -> vt0.expression;
  };
}
```

2.44.10.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/OperationOutcome4to3" = "R4 to R3 Conversion for OperationOutcome"

uses "http://hl7.org/fhir/StructureDefinition/OperationOutcome" alias OperationOutcome as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/OperationOutcome" alias OperationOutcomeR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group OperationOutcome(source src : OperationOutcome, target tgt : OperationOutcomeR3) extends DomainResource <<type+>>
    {
  src.issue as vs0 -> tgt.issue as vt0 then {
    vs0.severity -> vt0.severity;
    vs0.code -> vt0.code;
    vs0.details -> vt0.details;
    vs0.diagnostics -> vt0.diagnostics;
    vs0.location -> vt0.location;
    vs0.expression -> vt0.expression;
  };
}
```

2.44.10.3 R4 Validation Errors

2.44.11 Standard OperationOutcome Extensions

URL: https://hl7.org/fhir/R4/operationoutcome-extensions.html

This profile describes common extensions that are used with OperationOutcomes

2.44.11 Standard OperationOutcome Extensions

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative

2.44.11.1 Content

Extensions:

operationoutcome-issue-source | issue-source : Helps a user track down the source of the problem.

operationoutcome-authority | authority : A reference to where the rule is defined (based on the authoritative URLs in the applicable conformance resources).

operationoutcome-detectedIssue | detectedIssue : A reference to a stored contraindication that is the basis for this issue. A recipient can expect that the item referenced in this extension is being retained for record keeping purposes.

2.44.12 JSON Schema for OperationOutcome

URL: https://hl7.org/fhir/R4/operationoutcome.schema.json.html

2.44.12 JSON Schema for OperationOutcome

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is a representation of the json schema for OperationOutcome, which is just a part of the full JSON Schema.

```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/OperationOutcome",
  "$ref": "#/definitions/OperationOutcome",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "OperationOutcome": {
      "description": "A collection of error, warning, or information messages that result from a system action.",
      "properties": {
        "resourceType": {
          "description": "This is a OperationOutcome resource",
          "const": "OperationOutcome"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
```

```
    "meta": {
      "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
Changes to the content might not always be associated with version changes to the resource.",
      "$ref": "Meta.schema.json#/definitions/Meta"
    },
    "implicitRules": {
      "description": "A reference to a set of rules that were followed when the resource was constructed, and which
must be understood when processing the content. Often, this is a reference to an implementation guide that defines
the special rules along with other profiles etc.",
      "$ref": "#/definitions/uri"
    },
    "_implicitRules": {
      "description": "Extensions for implicitRules",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "language": {
      "description": "The base language in which the resource is written.",
      "$ref": "#/definitions/code"
    },
    "_language": {
      "description": "Extensions for language",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "text": {
      "description": "A human-readable narrative that contains a summary of the resource and can be used to
represent the content of the resource to a human. The narrative need not encode all the structured data, but is
required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
      "$ref": "Narrative.schema.json#/definitions/Narrative"
    },
    "contained": {
      "description": "These resources do not have an independent existence apart from the resource that contains
them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
 definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "issue": {
      "description": "An error, warning, or information message that results from a system action.",
      "items": {
        "$ref": "#/definitions/OperationOutcome_Issue"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "issue",
    "resourceType"
  ]
},
"OperationOutcome_Issue": {
  "description": "A collection of error, warning, or information messages that result from a system action.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
```

```
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "severity": {
      "description": "Indicates whether the issue indicates a variation from successful processing.",
      "enum": [
        "fatal",
        "error",
        "warning",
        "information"
      ]
    },
    "_severity": {
      "description": "Extensions for severity",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "code": {
      "description": "Describes the type of the issue. The system that creates an OperationOutcome SHALL choose the
most applicable code from the IssueType value set, and may additional provide its own code for the error in the
details element.",
      "enum": [
        "invalid",
        "structure",
        "required",
        "value",
        "invariant",
        "security",
        "login",
        "unknown",
        "expired",
        "forbidden",
        "suppressed",
        "processing",
        "not-supported",
        "duplicate",
        "multiple-matches",
        "not-found",
        "deleted",
        "too-long",
        "code-invalid",
        "extension",
        "too-costly",
        "business-rule",
        "conflict",
        "transient",
        "lock-error",
        "no-store",
        "exception",
        "timeout",
        "incomplete",
        "throttled",
        "informational"
      ]
    },
    "_code": {
      "description": "Extensions for code",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "details": {
      "description": "Additional details about the error. This may be a text description of the error or a system
code that identifies the error.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "diagnostics": {
      "description": "Additional diagnostic information about the issue.",
      "$ref": "#/definitions/string"
    },
    "_diagnostics": {
      "description": "Extensions for diagnostics",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "location": {
      "description": "This element is deprecated because it is XML specific. It is replaced by issue.expression,
```

```
    which is format independent, and simpler to parse. \n\nFor resource issues, this will be a simple XPath limited to
    element names, repetition indicators and the default child accessor that identifies one of the elements in the
    resource that caused this issue to be raised.  For HTTP errors, will be \"http.\" + the parameter name.",
          "items": {
            "$ref": "#/definitions/string"
          },
          "type": "array"
        },
        "_location": {
          "description": "Extensions for location",
          "items": {
            "$ref": "Element.schema.json#/definitions/Element"
          },
          "type": "array"
        },
        "expression": {
          "description": "A [simple subset of FHIRPath](fhirpath.html#simple) limited to element names, repetition
    indicators and the default child accessor that identifies one of the elements in the resource that caused this issue
     to be raised.",
          "items": {
            "$ref": "#/definitions/string"
          },
          "type": "array"
        },
        "_expression": {
          "description": "Extensions for expression",
          "items": {
            "$ref": "Element.schema.json#/definitions/Element"
          },
          "type": "array"
        }
      },
      "additionalProperties": false
    }
  }
}
```


2.44.13 Resource OperationOutcome - Maturity Dependency View

URL: https://hl7.org/fhir/R4/operationoutcome-dependencies.html

2.44.13 Resource OperationOutcome - Maturity Dependency View

FHIR Infrastructure Work Group | Maturity Level: N | Normative (from v4.0.0) | Compartments: Not linked to any defined compartments

| This page has been approved as part of an ANSI standard. See the Infrastructure Package for further details.

Dependency Graph for OperationOutcome FMM level 5

Name | Card. | Type | Dependency Analysis
OperationOutcome | | DomainResource
issue | 1..* | BackboneElement
severity | 1..1 | code
code | 1..1 | code
details | 0..1 | CodeableConcept
diagnostics | 0..1 | string
location XD | 0..* | string | Type Error: (string = FMM5-Normative vs. Element = FMM5-Deprecated)
expression | 0..* | string

2.45 Resource Parameters - Content

URL: https://hl7.org/fhir/R4/parameters.html
FHIR Infrastructure Work Group | Maturity Level: N | Normative (from v4.0.0) | | Compartments: Not linked to any defined compartments

| This page has been approved as part of an ANSI standard. See the Infrastructure Package for further details.

This resource is a non-persisted resource used to pass information into and back from an operation. It has no other use, and there is no RESTful endpoint associated with it.

2.45.1 Scope and Usage

This resource is a non-persisted resource used to pass information into and back from an operation.

2.45.2 Boundaries and Relationships

This special resource has no other use than for operation parameters, and there is no RESTful end-point associated with it. For further information, see the operations page.

OperationDefinition defines constraints on the Parameters resource instances that are used to convey the inputs and outputs of the operation.

This resource is referenced by GuidanceResponse

2.45.3 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
Parameters | ΣN | | Resource | Operation Request or Response Elements defined in Ancestors: id, meta, implicitRules, language
parameter | ΣI | 0..* | BackboneElement | Operation Parameter + Rule: A parameter must have one and only one of (value, resource, part)
name | Σ | 1..1 | string | Name from the definition
value[x] | ΣI | 0..1 | * | If parameter is a data type
resource | ΣI | 0..1 | Resource | If parameter is a whole resource
part | Σ | 0..* | see parameter | Named part of a multi-part parameter
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Parameters xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <parameter>  <!-- 0..* Operation Parameter -->
  <name value="[string]"/><!-- 1..1 Name from the definition -->
  <value[x]><!--  0..1 * If parameter is a data type --></value[x]>
  <resource><!--  0..1 Resource If parameter is a whole resource --></resource>
  <part><!-- 0..* Content as for Parameters.parameter Named part of a multi-part parameter --></part>
 </parameter>
</Parameters>
```

JSON Template

```
{
  "resourceType" : "Parameters",
  // from Resource: id, meta, implicitRules, and language
  "parameter" : [{ // Operation Parameter
    "name" : "<string>", // R!  Name from the definition
    // value[x]: If parameter is a data type. One of these 50:
    "valueBase64Binary" : "<base64Binary>",
    "valueBoolean" : <boolean>,
    "valueCanonical" : "<canonical>",
    "valueCode" : "<code>",
    "valueDate" : "<date>",
    "valueDateTime" : "<dateTime>",
    "valueDecimal" : <decimal>,
    "valueId" : "<id>",
    "valueInstant" : "<instant>",
    "valueInteger" : <integer>,
    "valueMarkdown" : "<markdown>",
    "valueOid" : "<oid>",
    "valuePositiveInt" : "<positiveInt>",
    "valueString" : "<string>",
    "valueTime" : "<time>",
    "valueUnsignedInt" : "<unsignedInt>",
    "valueUri" : "<uri>",
    "valueUrl" : "<url>",
    "valueUuid" : "<uuid>",
    "valueAddress" : { Address },
    "valueAge" : { Age },
    "valueAnnotation" : { Annotation },
    "valueAttachment" : { Attachment },
    "valueCodeableConcept" : { CodeableConcept },
    "valueCoding" : { Coding },
    "valueContactPoint" : { ContactPoint },
    "valueCount" : { Count },
    "valueDistance" : { Distance },
    "valueDuration" : { Duration },
    "valueHumanName" : { HumanName },
    "valueIdentifier" : { Identifier },
    "valueMoney" : { Money },
    "valuePeriod" : { Period },
    "valueQuantity" : { Quantity },
    "valueRange" : { Range },
    "valueRatio" : { Ratio },
    "valueReference" : { Reference },
    "valueSampledData" : { SampledData },
    "valueSignature" : { Signature },
```

```
    "valueTiming" : { Timing },
    "valueContactDetail" : { ContactDetail },
    "valueContributor" : { Contributor },
    "valueDataRequirement" : { DataRequirement },
    "valueExpression" : { Expression },
    "valueParameterDefinition" : { ParameterDefinition },
    "valueRelatedArtifact" : { RelatedArtifact },
    "valueTriggerDefinition" : { TriggerDefinition },
    "valueUsageContext" : { UsageContext },
    "valueDosage" : { Dosage },
    "valueMeta" : { Meta },
    "resource" : { Resource }, // C? If parameter is a whole resource
    "part" : [{ Content as for Parameters.parameter }] // Named part of a multi-part parameter
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Parameters;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  fhir:Parameters.parameter [ # 0..* Operation Parameter
    fhir:Parameters.parameter.name [ string ]; # 1..1 Name from the definition
    # Parameters.parameter.value[x] : 0..1 If parameter is a data type. One of these 50
      fhir:Parameters.parameter.valueBase64Binary [ base64Binary ]
      fhir:Parameters.parameter.valueBoolean [ boolean ]
      fhir:Parameters.parameter.valueCanonical [ canonical ]
      fhir:Parameters.parameter.valueCode [ code ]
      fhir:Parameters.parameter.valueDate [ date ]
      fhir:Parameters.parameter.valueDateTime [ dateTime ]
      fhir:Parameters.parameter.valueDecimal [ decimal ]
      fhir:Parameters.parameter.valueId [ id ]
      fhir:Parameters.parameter.valueInstant [ instant ]
      fhir:Parameters.parameter.valueInteger [ integer ]
      fhir:Parameters.parameter.valueMarkdown [ markdown ]
      fhir:Parameters.parameter.valueOid [ oid ]
      fhir:Parameters.parameter.valuePositiveInt [ positiveInt ]
      fhir:Parameters.parameter.valueString [ string ]
      fhir:Parameters.parameter.valueTime [ time ]
      fhir:Parameters.parameter.valueUnsignedInt [ unsignedInt ]
      fhir:Parameters.parameter.valueUri [ uri ]
      fhir:Parameters.parameter.valueUrl [ url ]
      fhir:Parameters.parameter.valueUuid [ uuid ]
      fhir:Parameters.parameter.valueAddress [ Address ]
      fhir:Parameters.parameter.valueAge [ Age ]
      fhir:Parameters.parameter.valueAnnotation [ Annotation ]
      fhir:Parameters.parameter.valueAttachment [ Attachment ]
      fhir:Parameters.parameter.valueCodeableConcept [ CodeableConcept ]
      fhir:Parameters.parameter.valueCoding [ Coding ]
      fhir:Parameters.parameter.valueContactPoint [ ContactPoint ]
      fhir:Parameters.parameter.valueCount [ Count ]
      fhir:Parameters.parameter.valueDistance [ Distance ]
      fhir:Parameters.parameter.valueDuration [ Duration ]
      fhir:Parameters.parameter.valueHumanName [ HumanName ]
      fhir:Parameters.parameter.valueIdentifier [ Identifier ]
      fhir:Parameters.parameter.valueMoney [ Money ]
      fhir:Parameters.parameter.valuePeriod [ Period ]
      fhir:Parameters.parameter.valueQuantity [ Quantity ]
      fhir:Parameters.parameter.valueRange [ Range ]
      fhir:Parameters.parameter.valueRatio [ Ratio ]
      fhir:Parameters.parameter.valueReference [ Reference ]
      fhir:Parameters.parameter.valueSampledData [ SampledData ]
      fhir:Parameters.parameter.valueSignature [ Signature ]
      fhir:Parameters.parameter.valueTiming [ Timing ]
      fhir:Parameters.parameter.valueContactDetail [ ContactDetail ]
      fhir:Parameters.parameter.valueContributor [ Contributor ]
      fhir:Parameters.parameter.valueDataRequirement [ DataRequirement ]
      fhir:Parameters.parameter.valueExpression [ Expression ]
      fhir:Parameters.parameter.valueParameterDefinition [ ParameterDefinition ]
      fhir:Parameters.parameter.valueRelatedArtifact [ RelatedArtifact ]
      fhir:Parameters.parameter.valueTriggerDefinition [ TriggerDefinition ]
      fhir:Parameters.parameter.valueUsageContext [ UsageContext ]
      fhir:Parameters.parameter.valueDosage [ Dosage ]
      fhir:Parameters.parameter.valueMeta [ Meta ]
    fhir:Parameters.parameter.resource [ Resource ]; # 0..1 If parameter is a whole resource
    fhir:Parameters.parameter.part [ See Parameters.parameter ], ... ; # 0..* Named part of a multi-part parameter
  ], ...;
]
```

Changes since R3

Parameters

Parameters.parameter.value[x] | Add Types canonical, url, uuid, ContactDetail, Contributor, DataRequirement, Expression, ParameterDefinition, RelatedArtifact, TriggerDefinition, UsageContext, Dosage

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 1 test that all execute ok. 1 fail round-trip testing and 1 r3 resources are invalid (0 errors).)

Structure

Name | Flags | Card. | Type | Description & Constraints

Parameters | ΣN | | Resource | Operation Request or Response Elements defined in Ancestors: id, meta, implicitRules, language

parameter | ΣI | 0..* | BackboneElement | Operation Parameter + Rule: A parameter must have one and only one of (value, resource, part)

name | Σ | 1..1 | string | Name from the definition

value[x] | ΣI | 0..1 | * | If parameter is a data type

resource | ΣI | 0..1 | Resource | If parameter is a whole resource

part | Σ | 0..* | see parameter | Named part of a multi-part parameter

Documentation for this format

UML Diagram (Legend)

XML Template

```
<Parameters xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <parameter>  <!-- 0..* Operation Parameter -->
  <name value="[string]"/><!-- 1..1 Name from the definition -->
  <value[x]><!--  0..1 * If parameter is a data type --></value[x]>
  <resource><!--  0..1 Resource If parameter is a whole resource --></resource>
  <part><!-- 0..* Content as for Parameters.parameter Named part of a multi-part parameter --></part>
 </parameter>
</Parameters>
```

JSON Template

```
{
  "resourceType" : "Parameters",
  // from Resource: id, meta, implicitRules, and language
  "parameter" : [{ // Operation Parameter
    "name" : "<string>", // R!  Name from the definition
    // value[x]: If parameter is a data type. One of these 50:
    "valueBase64Binary" : "<base64Binary>",
    "valueBoolean" : <boolean>,
    "valueCanonical" : "<canonical>",
    "valueCode" : "<code>",
    "valueDate" : "<date>",
    "valueDateTime" : "<dateTime>",
    "valueDecimal" : <decimal>,
    "valueId" : "<id>",
    "valueInstant" : "<instant>",
    "valueInteger" : <integer>,
    "valueMarkdown" : "<markdown>",
    "valueOid" : "<oid>",
    "valuePositiveInt" : "<positiveInt>",
    "valueString" : "<string>",
    "valueTime" : "<time>",
    "valueUnsignedInt" : "<unsignedInt>",
    "valueUri" : "<uri>",
    "valueUrl" : "<url>",
    "valueUuid" : "<uuid>",
    "valueAddress" : { Address },
    "valueAge" : { Age },
    "valueAnnotation" : { Annotation },
    "valueAttachment" : { Attachment },
    "valueCodeableConcept" : { CodeableConcept },
    "valueCoding" : { Coding },
    "valueContactPoint" : { ContactPoint },
    "valueCount" : { Count },
    "valueDistance" : { Distance },
    "valueDuration" : { Duration },
    "valueHumanName" : { HumanName },
    "valueIdentifier" : { Identifier },
    "valueMoney" : { Money },
    "valuePeriod" : { Period },
    "valueQuantity" : { Quantity },
    "valueRange" : { Range },
    "valueRatio" : { Ratio },
    "valueReference" : { Reference },
    "valueSampledData" : { SampledData },
    "valueSignature" : { Signature },
    "valueTiming" : { Timing },
    "valueContactDetail" : { ContactDetail },
    "valueContributor" : { Contributor },
    "valueDataRequirement" : { DataRequirement },
    "valueExpression" : { Expression },
    "valueParameterDefinition" : { ParameterDefinition },
    "valueRelatedArtifact" : { RelatedArtifact },
    "valueTriggerDefinition" : { TriggerDefinition },
    "valueUsageContext" : { UsageContext },
    "valueDosage" : { Dosage },
    "valueMeta" : { Meta },
    "resource" : { Resource }, // C? If parameter is a whole resource
```

```
    "part" : [{ Content as for Parameters.parameter }] // Named part of a multi-part parameter
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Parameters;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  fhir:Parameters.parameter [ # 0..* Operation Parameter
    fhir:Parameters.parameter.name [ string ]; # 1..1 Name from the definition
    # Parameters.parameter.value[x] : 0..1 If parameter is a data type. One of these 50
      fhir:Parameters.parameter.valueBase64Binary [ base64Binary ]
      fhir:Parameters.parameter.valueBoolean [ boolean ]
      fhir:Parameters.parameter.valueCanonical [ canonical ]
      fhir:Parameters.parameter.valueCode [ code ]
      fhir:Parameters.parameter.valueDate [ date ]
      fhir:Parameters.parameter.valueDateTime [ dateTime ]
      fhir:Parameters.parameter.valueDecimal [ decimal ]
      fhir:Parameters.parameter.valueId [ id ]
      fhir:Parameters.parameter.valueInstant [ instant ]
      fhir:Parameters.parameter.valueInteger [ integer ]
      fhir:Parameters.parameter.valueMarkdown [ markdown ]
      fhir:Parameters.parameter.valueOid [ oid ]
      fhir:Parameters.parameter.valuePositiveInt [ positiveInt ]
      fhir:Parameters.parameter.valueString [ string ]
      fhir:Parameters.parameter.valueTime [ time ]
      fhir:Parameters.parameter.valueUnsignedInt [ unsignedInt ]
      fhir:Parameters.parameter.valueUri [ uri ]
      fhir:Parameters.parameter.valueUrl [ url ]
      fhir:Parameters.parameter.valueUuid [ uuid ]
      fhir:Parameters.parameter.valueAddress [ Address ]
      fhir:Parameters.parameter.valueAge [ Age ]
      fhir:Parameters.parameter.valueAnnotation [ Annotation ]
      fhir:Parameters.parameter.valueAttachment [ Attachment ]
      fhir:Parameters.parameter.valueCodeableConcept [ CodeableConcept ]
      fhir:Parameters.parameter.valueCoding [ Coding ]
      fhir:Parameters.parameter.valueContactPoint [ ContactPoint ]
      fhir:Parameters.parameter.valueCount [ Count ]
      fhir:Parameters.parameter.valueDistance [ Distance ]
      fhir:Parameters.parameter.valueDuration [ Duration ]
      fhir:Parameters.parameter.valueHumanName [ HumanName ]
      fhir:Parameters.parameter.valueIdentifier [ Identifier ]
      fhir:Parameters.parameter.valueMoney [ Money ]
      fhir:Parameters.parameter.valuePeriod [ Period ]
      fhir:Parameters.parameter.valueQuantity [ Quantity ]
      fhir:Parameters.parameter.valueRange [ Range ]
      fhir:Parameters.parameter.valueRatio [ Ratio ]
      fhir:Parameters.parameter.valueReference [ Reference ]
      fhir:Parameters.parameter.valueSampledData [ SampledData ]
      fhir:Parameters.parameter.valueSignature [ Signature ]
      fhir:Parameters.parameter.valueTiming [ Timing ]
      fhir:Parameters.parameter.valueContactDetail [ ContactDetail ]
      fhir:Parameters.parameter.valueContributor [ Contributor ]
      fhir:Parameters.parameter.valueDataRequirement [ DataRequirement ]
      fhir:Parameters.parameter.valueExpression [ Expression ]
      fhir:Parameters.parameter.valueParameterDefinition [ ParameterDefinition ]
      fhir:Parameters.parameter.valueRelatedArtifact [ RelatedArtifact ]
      fhir:Parameters.parameter.valueTriggerDefinition [ TriggerDefinition ]
      fhir:Parameters.parameter.valueUsageContext [ UsageContext ]
      fhir:Parameters.parameter.valueDosage [ Dosage ]
      fhir:Parameters.parameter.valueMeta [ Meta ]
    fhir:Parameters.parameter.resource [ Resource ]; # 0..1 If parameter is a whole resource
    fhir:Parameters.parameter.part [ See Parameters.parameter ], ... ; # 0..* Named part of a multi-part parameter
  ], ...;
]
```

Changes since Release 3

Parameters

Parameters.parameter.value[x] | Add Types canonical, url, uuid, ContactDetail, Contributor, DataRequirement, Expression, ParameterDefinition, RelatedArtifact, TriggerDefinition, UsageContext, Dosage

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 1 test that all execute ok. 1 fail round-trip testing and 1 r3 resources are invalid (0 errors).)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) & the dependency analysis

2.45.3.1 Constraints

id | Level | Location | Description | Expression

inv-1 | Rule | Parameters.parameter | A parameter must have one and only one of (value, resource, part) | (part.exists() and value.empty() and resource.empty()) or (part.empty() and (value.exists() xor resource.exists()))

Note: for technical compatibility reasons, the Parameters resource inherits from Resource, but since the parameter exchange format has no end-point and/or persistence, it never has an id, a versionId, or a lastUpdated. The other features of Resource (tags, profiles, security labels, language etc.) may have use when operations are executed.

2.45.3.2 Names and Identifiers

The following rules apply to name and identifier uniqueness within a Parameters resource:

Parameter Names | Parameter names can be repeated at any level. The meaning of duplicate parameter names - and whether it is valid to repeat any given parameter name - depends on the context. Typically, operation definitions define what parameters can be used, and whether they can repeat. The order of parameters with different names is not considered significant

Resources | Resources in parameter.resource do not need to be unique, though the interpretation of non-unique resources, or different versions of the same resource, may be ambiguous in cases where the parameter names do not clearly differentiate the purpose of the parameter. For additional commentary, see the notes about about resource uniqueness in Bundles, though be aware that those rules do not apply to the Parameters resource unless the fullUrl extension is present on parameters with resources.

Internal References | When internal references are resolved in a resource in a parameter.resource, the resolution stops at parameter.resource. This allows resources to appear more than once in a Parameters resource, and also means that resources cannot reference content in another resource using an internal reference (except for references inside contained resources)

References | When resolving references in resources, the applicable Operation Definition may specify how references may be resolved between parameters. If a reference cannot be resolved between the parameters, the application should fall back to it's general resource resolution methods

2.45.4 Resource Parameters - Examples

URL: https://hl7.org/fhir/R4/parameters-examples.html

2.45.4 Resource Parameters - Examples

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | | Compartments: Not linked to any defined compartments

In addition to this standard example, there are many examples of the Parameters resource in the operation examples:

- CapabilityStatement/$subset
- CapabilityStatement/$implements
- CapabilityStatement/$conforms
- CapabilityStatement/$versions
- CodeSystem/$lookup
- CodeSystem/$subsumes
- CodeSystem/$find-matches
- Composition/$document
- Observation/$stats

Example Name | id | Format
Parameters Example | example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

2.45.5 Resource Parameters - Detailed Descriptions

URL: https://hl7.org/fhir/R4/parameters-definitions.html

2.45.5 Resource Parameters - Detailed Descriptions

FHIR Infrastructure Work Group | Maturity Level: N | Normative (from v4.0.0) | | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the Parameters resource.

Parameters
Element Id | Parameters
Definition | This resource is a non-persisted resource used to pass information into and back from an operation. It has no other use, and there is no RESTful endpoint associated with it.
Cardinality | 0..*
Type | Resource
Summary | true
Comments | The parameters that may be used are defined by the OperationDefinition resource.
Parameters.parameter
Element Id | Parameters.parameter
Definition | A parameter passed to or received from the operation.

Cardinality | 0..*
Summary | true
Invariants | Defined on this element inv-1 Rule A parameter must have one and only one of (value, resource, part) (part.exists() and value.empty() and resource.empty()) or (part.empty() and (value.exists() xor resource.exists()))
Defined on this element
inv-1 | Rule | A parameter must have one and only one of (value, resource, part) | (part.exists() and value.empty() and resource.empty()) or (part.empty() and (value.exists() xor resource.exists()))
Parameters.parameter.name
Element Id | Parameters.parameter.name
Definition | The name of the parameter (reference to the operation definition).
Cardinality | 1..1
Type | string
Summary | true
Parameters.parameter.value[x]
Element Id | Parameters.parameter.value[x]
Definition | If the parameter is a data type.
Cardinality | 0..1
Type | *
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
Invariants | Affect this element inv-1 Rule A parameter must have one and only one of (value, resource, part) (part.exists() and value.empty() and resource.empty()) or (part.empty() and (value.exists() xor resource.exists()))
Affect this element
inv-1 | Rule | A parameter must have one and only one of (value, resource, part) | (part.exists() and value.empty() and resource.empty()) or (part.empty() and (value.exists() xor resource.exists()))
Parameters.parameter.resource
Element Id | Parameters.parameter.resource
Definition | If the parameter is a whole resource.
Cardinality | 0..1
Type | Resource
Summary | true
Comments | When resolving references in resources, the operation definition may specify how references may be resolved between parameters. If a reference cannot be resolved between the parameters, the application should fall back to it's general resource resolution methods.
Invariants | Affect this element inv-1 Rule A parameter must have one and only one of (value, resource, part) (part.exists() and value.empty() and resource.empty()) or (part.empty() and (value.exists() xor resource.exists()))
Affect this element
inv-1 | Rule | A parameter must have one and only one of (value, resource, part) | (part.exists() and value.empty() and resource.empty()) or (part.empty() and (value.exists() xor resource.exists()))
Parameters.parameter.part
Element Id | Parameters.parameter.part
Definition | A named part of a multi-part parameter.
Cardinality | 0..*
Type | See Parameters.parameter
Summary | true
Comments | Only one level of nested parameters is allowed.

2.45.6 Resource Parameters - Mappings

URL: https://hl7.org/fhir/R4/parameters-mappings.html

2.45.6 Resource Parameters - Mappings

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | | Compartments: Not linked to any defined compartments

Mappings:

HL7 v2 Mapping

RIM Mapping

Mappings for the parameters resource (see Mappings to Other Standards for further information & status).

2.45.6.1 HL7 v2 Mapping (http://hl7.org/v2)

Parameters | n/a

2.45.6.2 RIM Mapping (http://hl7.org/v3)

Parameters | n/a

2.45.7 Resource Parameters - Extensions & Profiles

URL: https://hl7.org/fhir/R4/parameters-profiles.html

2.45.7 Resource Parameters - Extensions & Profiles

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | | Compartments: Not linked to any defined compartments

This table lists profiles and extensions for the Parameters resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

2.45.7.1 Profiles

Resources of type Parameters do not have extensions at the root element, but extensions MAY be present on the elements in the resource

No Profiles defined for this resource

2.45.7.2 Extensions

parameters-fullUrl | Parameters.parameter | fullUrl | for Parameters HL7 Extensions
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

2.45.7.3 Search Extensions

No Search Extensions defined for this resource

2.45.8 Resource Parameters - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/parameters-version-maps.html

No validation errors - all conversions are clean

2.45.8 Resource Parameters - R3/R4 Conversions

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | | Compartments: Not linked to any defined compartments

R3 : R4 Conversion maps for Parameters.

Functional status for this map: 1 test that all execute ok. 1 fail round-trip testing and 1 r3 resources are invalid (0 errors). (see documentation)

2.45.8.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/Parameters3to4" = "R3 to R4 Conversions for Parameters"

uses "http://hl7.org/fhir/3.0/StructureDefinition/Parameters" alias ParametersR3 as source
uses "http://hl7.org/fhir/StructureDefinition/Parameters" alias Parameters as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group Parameters(source src : ParametersR3, target tgt : Parameters) extends Resource <<type+>> {
  src.parameter as vs0 -> tgt.parameter as vt0 then {
    vs0.name -> vt0.name;
    vs0.value : base64Binary as vs -> vt0.value = create('base64Binary') as vt then base64Binary(vs, vt);
    vs0.value : boolean as vs -> vt0.value = create('boolean') as vt then boolean(vs, vt);
    vs0.value : code as vs -> vt0.value = create('code') as vt then code(vs, vt);
    vs0.value : date as vs -> vt0.value = create('date') as vt then date(vs, vt);
    vs0.value : dateTime as vs -> vt0.value = create('dateTime') as vt then dateTime(vs, vt);
    vs0.value : decimal as vs -> vt0.value = create('decimal') as vt then decimal(vs, vt);
    vs0.value : id as vs -> vt0.value = create('id') as vt then id(vs, vt);
    vs0.value : instant as vs -> vt0.value = create('instant') as vt then instant(vs, vt);
    vs0.value : integer as vs -> vt0.value = create('integer') as vt then integer(vs, vt);
    vs0.value : markdown as vs -> vt0.value = create('markdown') as vt then markdown(vs, vt);
    vs0.value : oid as vs -> vt0.value = create('oid') as vt then oid(vs, vt);
    vs0.value : positiveInt as vs -> vt0.value = create('positiveInt') as vt then positiveInt(vs, vt);
    vs0.value : string as vs -> vt0.value = create('string') as vt then string(vs, vt);
    vs0.value : time as vs -> vt0.value = create('time') as vt then time(vs, vt);
    vs0.value : unsignedInt as vs -> vt0.value = create('unsignedInt') as vt then unsignedInt(vs, vt);
    vs0.value : uri as vs -> vt0.value = create('uri') as vt then uri(vs, vt);
    vs0.value : Address as vs -> vt0.value = create('Address') as vt then Address(vs, vt);
    vs0.value : Age as vs -> vt0.value = create('Age') as vt then Age(vs, vt);
    vs0.value : Annotation as vs -> vt0.value = create('Annotation') as vt then Annotation(vs, vt);
    vs0.value : Attachment as vs -> vt0.value = create('Attachment') as vt then Attachment(vs, vt);
    vs0.value : CodeableConcept as vs -> vt0.value = create('CodeableConcept') as vt then CodeableConcept(vs, vt);
```

```
    vs0.value : Coding as vs -> vt0.value = create('Coding') as vt then Coding(vs, vt);
    vs0.value : ContactPoint as vs -> vt0.value = create('ContactPoint') as vt then ContactPoint(vs, vt);
    vs0.value : Count as vs -> vt0.value = create('Count') as vt then Count(vs, vt);
    vs0.value : Distance as vs -> vt0.value = create('Distance') as vt then Distance(vs, vt);
    vs0.value : Duration as vs -> vt0.value = create('Duration') as vt then Duration(vs, vt);
    vs0.value : HumanName as vs -> vt0.value = create('HumanName') as vt then HumanName(vs, vt);
    vs0.value : Identifier as vs -> vt0.value = create('Identifier') as vt then Identifier(vs, vt);
    vs0.value : Money as vs -> vt0.value = create('Money') as vt then Money(vs, vt);
    vs0.value : Period as vs -> vt0.value = create('Period') as vt then Period(vs, vt);
    vs0.value : Quantity as vs -> vt0.value = create('Quantity') as vt then Quantity(vs, vt);
    vs0.value : Range as vs -> vt0.value = create('Range') as vt then Range(vs, vt);
    vs0.value : Ratio as vs -> vt0.value = create('Ratio') as vt then Ratio(vs, vt);
    vs0.value : Reference as vs -> vt0.value = create('Reference') as vt then Reference(vs, vt);
    vs0.value : SampledData as vs -> vt0.value = create('SampledData') as vt then SampledData(vs, vt);
    vs0.value : Signature as vs -> vt0.value = create('Signature') as vt then Signature(vs, vt);
    vs0.value : Timing as vs -> vt0.value = create('Timing') as vt then Timing(vs, vt);
  };
}
```

2.45.8.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/Parameters4to3" = "R4 to R3 Conversion for Parameters"

uses "http://hl7.org/fhir/StructureDefinition/Parameters" alias Parameters as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/Parameters" alias ParametersR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group Parameters(source src : Parameters, target tgt : ParametersR3) extends Resource <<type+>> {
  src.parameter as vs0 -> tgt.parameter as vt0 then {
    vs0.name -> vt0.name;
    vs0.value : base64Binary as vs -> vt0.value = create('base64Binary') as vt then base64Binary(vs, vt);
    vs0.value : boolean as vs -> vt0.value = create('boolean') as vt then boolean(vs, vt);
    vs0.value : code as vs -> vt0.value = create('code') as vt then code(vs, vt);
    vs0.value : date as vs -> vt0.value = create('date') as vt then date(vs, vt);
    vs0.value : dateTime as vs -> vt0.value = create('dateTime') as vt then dateTime(vs, vt);
    vs0.value : decimal as vs -> vt0.value = create('decimal') as vt then decimal(vs, vt);
    vs0.value : id as vs -> vt0.value = create('id') as vt then id(vs, vt);
    vs0.value : instant as vs -> vt0.value = create('instant') as vt then instant(vs, vt);
    vs0.value : integer as vs -> vt0.value = create('integer') as vt then integer(vs, vt);
    vs0.value : markdown as vs -> vt0.value = create('markdown') as vt then markdown(vs, vt);
    vs0.value : oid as vs -> vt0.value = create('oid') as vt then oid(vs, vt);
    vs0.value : positiveInt as vs -> vt0.value = create('positiveInt') as vt then positiveInt(vs, vt);
    vs0.value : string as vs -> vt0.value = create('string') as vt then string(vs, vt);
    vs0.value : time as vs -> vt0.value = create('time') as vt then time(vs, vt);
    vs0.value : unsignedInt as vs -> vt0.value = create('unsignedInt') as vt then unsignedInt(vs, vt);
    vs0.value : uri as vs -> vt0.value = create('uri') as vt then uri(vs, vt);
    vs0.value : Address as vs -> vt0.value = create('Address') as vt then Address(vs, vt);
    vs0.value : Age as vs -> vt0.value = create('Age') as vt then Age(vs, vt);
    vs0.value : Annotation as vs -> vt0.value = create('Annotation') as vt then Annotation(vs, vt);
    vs0.value : Attachment as vs -> vt0.value = create('Attachment') as vt then Attachment(vs, vt);
    vs0.value : CodeableConcept as vs -> vt0.value = create('CodeableConcept') as vt then CodeableConcept(vs, vt);
    vs0.value : Coding as vs -> vt0.value = create('Coding') as vt then Coding(vs, vt);
    vs0.value : ContactPoint as vs -> vt0.value = create('ContactPoint') as vt then ContactPoint(vs, vt);
    vs0.value : Count as vs -> vt0.value = create('Count') as vt then Count(vs, vt);
    vs0.value : Distance as vs -> vt0.value = create('Distance') as vt then Distance(vs, vt);
    vs0.value : Duration as vs -> vt0.value = create('Duration') as vt then Duration(vs, vt);
    vs0.value : HumanName as vs -> vt0.value = create('HumanName') as vt then HumanName(vs, vt);
    vs0.value : Identifier as vs -> vt0.value = create('Identifier') as vt then Identifier(vs, vt);
    vs0.value : Money as vs -> vt0.value = create('Money') as vt then Money(vs, vt);
    vs0.value : Period as vs -> vt0.value = create('Period') as vt then Period(vs, vt);
    vs0.value : Quantity as vs -> vt0.value = create('Quantity') as vt then Quantity(vs, vt);
    vs0.value : Range as vs -> vt0.value = create('Range') as vt then Range(vs, vt);
    vs0.value : Ratio as vs -> vt0.value = create('Ratio') as vt then Ratio(vs, vt);
    vs0.value : Reference as vs -> vt0.value = create('Reference') as vt then Reference(vs, vt);
    vs0.value : SampledData as vs -> vt0.value = create('SampledData') as vt then SampledData(vs, vt);
    vs0.value : Signature as vs -> vt0.value = create('Signature') as vt then Signature(vs, vt);
    vs0.value : Timing as vs -> vt0.value = create('Timing') as vt then Timing(vs, vt);
  };
}
```

2.45.8.3 R4 Validation Errors

2.45.9 Parameters HL7 Extensions

URL: https://hl7.org/fhir/R4/parameters-extensions.html

Defines common extensions used with or related to the Parameters resource

2.45.9 Parameters HL7 Extensions

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative

2.45.9.1 Content

Extensions:

parameters-fullUrl | fullUrl : This specifies the fullUrl for the resource in parameters.resource, if there is one. When fullUrl is provided, ithe resource resolution method described for Bundle.

2.45.10 JSON Schema for Parameters

URL: https://hl7.org/fhir/R4/parameters.schema.json.html

2.45.10 JSON Schema for Parameters

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is a representation of the json schema for Parameters, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/Parameters",
  "$ref": "#/definitions/Parameters",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "Parameters": {
      "description": "This resource is a non-persisted resource used to pass information into and back from an
    [operation](operations.html). It has no other use, and there is no RESTful endpoint associated with it.",
      "properties": {
        "resourceType": {
          "description": "This is a Parameters resource",
          "const": "Parameters"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "parameter": {
          "description": "A parameter passed to or received from the operation.",
          "items": {
            "$ref": "#/definitions/Parameters_Parameter"
          },
          "type": "array"
        }
      },
      "additionalProperties": false,
      "required": [
        "resourceType"
      ]
    },
    "Parameters_Parameter": {
      "description": "This resource is a non-persisted resource used to pass information into and back from an
    [operation](operations.html). It has no other use, and there is no RESTful endpoint associated with it.",
      "properties": {
        "id": {
          "description": "Unique id for the element within a resource (for internal references). This may be any string
    value that does not contain spaces.",
          "$ref": "string.schema.json#/definitions/string"
        },
        "extension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
    definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "modifierExtension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the element and that modifies the understanding of the element in which it is contained and/or the understanding of
    the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
```

```
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "name": {
      "description": "The name of the parameter (reference to the operation definition).",
      "$ref": "#/definitions/string"
    },
    "_name": {
      "description": "Extensions for name",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueBase64Binary": {
      "description": "If the parameter is a data type.",
      "pattern": "^(\\s*([0-9a-zA-Z\\+/\u003d]){4}\\s*)+$",
      "type": "string"
    },
    "_valueBase64Binary": {
      "description": "Extensions for valueBase64Binary",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueBoolean": {
      "description": "If the parameter is a data type.",
      "pattern": "^true|false$",
      "type": "boolean"
    },
    "_valueBoolean": {
      "description": "Extensions for valueBoolean",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueCanonical": {
      "description": "If the parameter is a data type.",
      "pattern": "^\\S*$",
      "type": "string"
    },
    "_valueCanonical": {
      "description": "Extensions for valueCanonical",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueCode": {
      "description": "If the parameter is a data type.",
      "pattern": "^[^\\s]+(\\s[^\\s]+)*$",
      "type": "string"
    },
    "_valueCode": {
      "description": "Extensions for valueCode",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueDate": {
      "description": "If the parameter is a data type.",
      "pattern":
"^([0-9]([0-9]([0-9][1-9]|[1-9]0)|[1-9]00)|[1-9]000)(-(0[1-9]|1[0-2])(-(0[1-9]|[1-2][0-9]|3[0-1]))?)?$",
      "type": "string"
    },
    "_valueDate": {
      "description": "Extensions for valueDate",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueDateTime": {
      "description": "If the parameter is a data type.",
      "pattern": "^([0-9]([0-9]([0-9][1-9]|[1-9]0)|[1-9]00)|[1-9]000)(-(0[1-9]|1[0-2])(-(0[1-9]|[1-2][0-9]|3[0-
1])(T([01][0-9]|2[0-3]):[0-5][0-9]:([0-5][0-9]|60)(\\.[0-9]+)?(Z|(\\+|-)((0[0-9]|1[0-3]):[0-5][0-9]|14:00)))?)?)?$",
      "type": "string"
    },
    "_valueDateTime": {
      "description": "Extensions for valueDateTime",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueDecimal": {
      "description": "If the parameter is a data type.",
      "pattern": "^-?(0|[1-9][0-9]*)(\\.[0-9]+)?([eE][+-]?[0-9]+)?$",
      "type": "number"
    },
    "_valueDecimal": {
      "description": "Extensions for valueDecimal",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueId": {
      "description": "If the parameter is a data type.",
      "pattern": "^[A-Za-z0-9\\-\\.]{1,64}$",
      "type": "string"
    },
    "_valueId": {
```

```
      "description": "Extensions for valueId",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueInstant": {
      "description": "If the parameter is a data type.",
      "pattern": "^([0-9]([0-9]([0-9][1-9]|[1-9]0)|[1-9]00)|[1-9]000)-(0[1-9]|1[0-2])-(0[1-9]|[1-2][0-9]|3[0-
1])T([01][0-9]|2[0-3]):[0-5][0-9]:([0-5][0-9]|60)(\\.[0-9]+)?(Z|(\\+|-)((0[0-9]|1[0-3]):[0-5][0-9]|14:00))$",
      "type": "string"
    },
    "_valueInstant": {
      "description": "Extensions for valueInstant",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueInteger": {
      "description": "If the parameter is a data type.",
      "pattern": "^-?([0]|([1-9][0-9]*))$",
      "type": "number"
    },
    "_valueInteger": {
      "description": "Extensions for valueInteger",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueMarkdown": {
      "description": "If the parameter is a data type.",
      "pattern": "^[ \\r\\n\\t\\S]+$",
      "type": "string"
    },
    "_valueMarkdown": {
      "description": "Extensions for valueMarkdown",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueOid": {
      "description": "If the parameter is a data type.",
      "pattern": "^urn:oid:[0-2](\\.(0|[1-9][0-9]*))+$",
      "type": "string"
    },
    "_valueOid": {
      "description": "Extensions for valueOid",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valuePositiveInt": {
      "description": "If the parameter is a data type.",
      "pattern": "^[1-9][0-9]*$",
      "type": "number"
    },
    "_valuePositiveInt": {
      "description": "Extensions for valuePositiveInt",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueString": {
      "description": "If the parameter is a data type.",
      "pattern": "^[ \\r\\n\\t\\S]+$",
      "type": "string"
    },
    "_valueString": {
      "description": "Extensions for valueString",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueTime": {
      "description": "If the parameter is a data type.",
      "pattern": "^([01][0-9]|2[0-3]):[0-5][0-9]:([0-5][0-9]|60)(\\.[0-9]+)?$",
      "type": "string"
    },
    "_valueTime": {
      "description": "Extensions for valueTime",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueUnsignedInt": {
      "description": "If the parameter is a data type.",
      "pattern": "^[0]|([1-9][0-9]*)$",
      "type": "number"
    },
    "_valueUnsignedInt": {
      "description": "Extensions for valueUnsignedInt",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueUri": {
      "description": "If the parameter is a data type.",
      "pattern": "^\\S*$",
      "type": "string"
    },
    "_valueUri": {
      "description": "Extensions for valueUri",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueUrl": {
      "description": "If the parameter is a data type.",
      "pattern": "^\\S*$",
      "type": "string"
```

```
},
"_valueUrl": {
  "description": "Extensions for valueUrl",
  "$ref": "Element.schema.json#/definitions/Element"
},
"valueUuid": {
  "description": "If the parameter is a data type.",
  "pattern": "^urn:uuid:[0-9a-f]{8}-[0-9a-f]{4}-[0-9a-f]{4}-[0-9a-f]{4}-[0-9a-f]{12}$",
  "type": "string"
},
"_valueUuid": {
  "description": "Extensions for valueUuid",
  "$ref": "Element.schema.json#/definitions/Element"
},
"valueAddress": {
  "description": "If the parameter is a data type.",
  "$ref": "Address.schema.json#/definitions/Address"
},
"valueAge": {
  "description": "If the parameter is a data type.",
  "$ref": "Age.schema.json#/definitions/Age"
},
"valueAnnotation": {
  "description": "If the parameter is a data type.",
  "$ref": "Annotation.schema.json#/definitions/Annotation"
},
"valueAttachment": {
  "description": "If the parameter is a data type.",
  "$ref": "Attachment.schema.json#/definitions/Attachment"
},
"valueCodeableConcept": {
  "description": "If the parameter is a data type.",
  "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
},
"valueCoding": {
  "description": "If the parameter is a data type.",
  "$ref": "Coding.schema.json#/definitions/Coding"
},
"valueContactPoint": {
  "description": "If the parameter is a data type.",
  "$ref": "ContactPoint.schema.json#/definitions/ContactPoint"
},
"valueCount": {
  "description": "If the parameter is a data type.",
  "$ref": "Count.schema.json#/definitions/Count"
},
"valueDistance": {
  "description": "If the parameter is a data type.",
  "$ref": "Distance.schema.json#/definitions/Distance"
},
"valueDuration": {
  "description": "If the parameter is a data type.",
  "$ref": "Duration.schema.json#/definitions/Duration"
},
"valueHumanName": {
  "description": "If the parameter is a data type.",
  "$ref": "HumanName.schema.json#/definitions/HumanName"
},
"valueIdentifier": {
  "description": "If the parameter is a data type.",
  "$ref": "Identifier.schema.json#/definitions/Identifier"
},
"valueMoney": {
  "description": "If the parameter is a data type.",
  "$ref": "Money.schema.json#/definitions/Money"
},
"valuePeriod": {
  "description": "If the parameter is a data type.",
  "$ref": "Period.schema.json#/definitions/Period"
},
"valueQuantity": {
  "description": "If the parameter is a data type.",
  "$ref": "Quantity.schema.json#/definitions/Quantity"
},
"valueRange": {
  "description": "If the parameter is a data type.",
  "$ref": "Range.schema.json#/definitions/Range"
},
"valueRatio": {
  "description": "If the parameter is a data type.",
  "$ref": "Ratio.schema.json#/definitions/Ratio"
},
"valueReference": {
  "description": "If the parameter is a data type.",
  "$ref": "Reference.schema.json#/definitions/Reference"
},
"valueSampledData": {
  "description": "If the parameter is a data type.",
  "$ref": "SampledData.schema.json#/definitions/SampledData"
```

```
        },
        "valueSignature": {
          "description": "If the parameter is a data type.",
          "$ref": "Signature.schema.json#/definitions/Signature"
        },
        "valueTiming": {
          "description": "If the parameter is a data type.",
          "$ref": "Timing.schema.json#/definitions/Timing"
        },
        "valueContactDetail": {
          "description": "If the parameter is a data type.",
          "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
        },
        "valueContributor": {
          "description": "If the parameter is a data type.",
          "$ref": "Contributor.schema.json#/definitions/Contributor"
        },
        "valueDataRequirement": {
          "description": "If the parameter is a data type.",
          "$ref": "DataRequirement.schema.json#/definitions/DataRequirement"
        },
        "valueExpression": {
          "description": "If the parameter is a data type.",
          "$ref": "Expression.schema.json#/definitions/Expression"
        },
        "valueParameterDefinition": {
          "description": "If the parameter is a data type.",
          "$ref": "ParameterDefinition.schema.json#/definitions/ParameterDefinition"
        },
        "valueRelatedArtifact": {
          "description": "If the parameter is a data type.",
          "$ref": "RelatedArtifact.schema.json#/definitions/RelatedArtifact"
        },
        "valueTriggerDefinition": {
          "description": "If the parameter is a data type.",
          "$ref": "TriggerDefinition.schema.json#/definitions/TriggerDefinition"
        },
        "valueUsageContext": {
          "description": "If the parameter is a data type.",
          "$ref": "UsageContext.schema.json#/definitions/UsageContext"
        },
        "valueDosage": {
          "description": "If the parameter is a data type.",
          "$ref": "Dosage.schema.json#/definitions/Dosage"
        },
        "valueMeta": {
          "description": "If the parameter is a data type.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "resource": {
          "description": "If the parameter is a whole resource.",
          "$ref": "ResourceList.schema.json#/definitions/ResourceList"
        },
        "part": {
          "description": "A named part of a multi-part parameter.",
          "items": {
            "$ref": "#/definitions/Parameters_Parameter"
          },
          "type": "array"
        }
      },
      "additionalProperties": false
    }
  }
}
```

2.45.11 Resource Parameters - Maturity Dependency View

URL: https://hl7.org/fhir/R4/parameters-dependencies.html

2.45.11 Resource Parameters - Maturity Dependency View

FHIR Infrastructure Work Group | Maturity Level: N | Normative (from v4.0.0) | Compartments: Not linked to any defined compartments

| This page has been approved as part of an ANSI standard. See the Infrastructure Package for further details.

Dependency Graph for Parameters FMM level 5

Name | Card. | Type | Dependency Analysis
Parameters | | Resource
parameter | 0..* | BackboneElement
name | 1..1 | string
value[x] | 0..1 | * | Type Error: (* = FMM2-Trial Use vs. Element = FMM5-Normative)
resource | 0..1 | Resource
part | 0..* | see parameter

2.46 Resource Subscription - Content

URL: https://hl7.org/fhir/R4/subscription.html
FHIR Infrastructure Work Group | Maturity Level: 3 | Trial Use | Security Category: Business | Compartments: Not linked to any defined compartments

The subscription resource is used to define a push-based subscription from a server to another system. Once a subscription is registered with the server, the server checks every resource that is created or updated, and if the resource matches the given criteria, it sends a message on the defined "channel" so that another system can take an appropriate action.

2.46.1 Scope and Usage

Once a subscription is created, any newly created or updated resources that meet the criteria in the resource cause a notification to be sent using the provided channel. The criteria are Search strings that have the same interpretation as if they were appended to the base URL and submitted using the REST API. Note that the search criteria are applied to the new value of the resource. The consequence of this is that there is no notification when a resource is deleted, or when a resource is updated so that it no longer meets the criteria.

The server is able to send notifications without any information about the matching resource, or with the entire resource.

Several different types of channels are supported:

• rest-hook: A post is made to the URL identified in the Subscription resource. If the subscription requests that the whole resource is included, the URL is interpreted as the service base

• websocket: A PING message is sent to the designated URI

• email/sms: A notification is sent to the nominated email address or SMS number

• message: The resource is sent to the application identified in the URI as a message

See below for further discussion of the various channels. Note that sending the entire resource creates security concerns that must be managed by the server.

Subscriptions are active resources; a server can only accept a subscription if it will execute the specified channel for any resources subsequently received. The subscription is no longer active once it is deleted from the server.

2.46.2 Boundaries and Relationships

As an alternative to subscriptions, the RESTful API describes a polling-based subscription method using bundles and the history operation. This method of polling allows for a much tighter relationship between the client and the server that doesn't involve missing updates and/or deletes.

When using the Subscription resource, the FHIR server combines the roles of publisher and information distributer. Other arrangements of the publish and subscribe pattern describe separate agents for the two roles. Implementers may implement the Subscription resource using an architecture with separate agents, or using any other pub/sub architectire (e.g. see FHIRCast , or, more generally, W3C Pub/Sub).

2.46.3 Resource Content

• Structure

• UML

• XML

• JSON

• Turtle

• R3 Diff

• All

Structure

Name | Flags | Card. | Type | Description & Constraints
Subscription | ΣTU | | DomainResource | Server push subscription criteria Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
status | ?!Σ | 1..1 | code | requested | active | error | off SubscriptionStatus (Required)
contact | Σ | 0..* | ContactPoint | Contact details for source (e.g. troubleshooting)
end | Σ | 0..1 | instant | When to automatically delete the subscription
reason | Σ | 1..1 | string | Description of why this subscription was created
criteria | Σ | 1..1 | string | Rule for server push
error | Σ | 0..1 | string | Latest error note
channel | Σ | 1..1 | BackboneElement | The channel on which to report matches to the criteria
type | Σ | 1..1 | code | rest-hook | websocket | email | sms | message SubscriptionChannelType (Required)
endpoint | Σ | 0..1 | url | Where the channel points to
payload | Σ | 0..1 | code | MIME type to send, or omit for no payload MimeType (Required)
header | Σ | 0..* | string | Usage depends on the channel type

Documentation for this format

UML Diagram (Legend)

XML Template

```
<Subscription xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <status value="[code]"/><!-- 1..1 requested | active | error | off -->
 <contact><!-- 0..* ContactPoint Contact details for source (e.g. troubleshooting) --></contact>
 <end value="[instant]"/><!-- 0..1 When to automatically delete the subscription -->
 <reason value="[string]"/><!-- 1..1 Description of why this subscription was created -->
 <criteria value="[string]"/><!-- 1..1 Rule for server push -->
 <error value="[string]"/><!-- 0..1 Latest error note -->
 <channel>  <!-- 1..1 The channel on which to report matches to the criteria -->
  <type value="[code]"/><!-- 1..1 rest-hook | websocket | email | sms | message -->
  <endpoint value="[url]"/><!-- 0..1 Where the channel points to -->
  <payload value="[code]"/><!-- 0..1 MIME type to send, or omit for no payload -->
  <header value="[string]"/><!-- 0..* Usage depends on the channel type -->
 </channel>
</Subscription>
```

JSON Template

```
{
  "resourceType" : "Subscription",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "status" : "<code>", // R!  requested | active | error | off
  "contact" : [{ ContactPoint }], // Contact details for source (e.g. troubleshooting)
  "end" : "<instant>", // When to automatically delete the subscription
  "reason" : "<string>", // R!  Description of why this subscription was created
  "criteria" : "<string>", // R!  Rule for server push
  "error" : "<string>", // Latest error note
  "channel" : { // R!  The channel on which to report matches to the criteria
    "type" : "<code>", // R!  rest-hook | websocket | email | sms | message
    "endpoint" : "<url>", // Where the channel points to
    "payload" : "<code>", // MIME type to send, or omit for no payload
    "header" : ["<string>"] // Usage depends on the channel type
  }
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Subscription;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Subscription.status [ code ]; # 1..1 requested | active | error | off
  fhir:Subscription.contact [ ContactPoint ], ... ; # 0..* Contact details for source (e.g. troubleshooting)
  fhir:Subscription.end [ instant ]; # 0..1 When to automatically delete the subscription
  fhir:Subscription.reason [ string ]; # 1..1 Description of why this subscription was created
  fhir:Subscription.criteria [ string ]; # 1..1 Rule for server push
  fhir:Subscription.error [ string ]; # 0..1 Latest error note
  fhir:Subscription.channel [ # 1..1 The channel on which to report matches to the criteria
    fhir:Subscription.channel.type [ code ]; # 1..1 rest-hook | websocket | email | sms | message
    fhir:Subscription.channel.endpoint [ url ]; # 0..1 Where the channel points to
    fhir:Subscription.channel.payload [ code ]; # 0..1 MIME type to send, or omit for no payload
    fhir:Subscription.channel.header [ string ], ... ; # 0..* Usage depends on the channel type
  ];
]
```

Changes since R3

Subscription

Subscription.status | Change value set from http://hl7.org/fhir/ValueSet/subscription-status to http://hl7.org/fhir/ValueSet/subscription-status|4.0.1

Subscription.channel.type | Change value set from http://hl7.org/fhir/ValueSet/subscription-channel-type to http://hl7.org/fhir/ValueSet/subscription-channel-type|4.0.1

Subscription.channel.endpoint | Type changed from uri to url

Subscription.channel.payload | Type changed from string to code Add Binding http://hl7.org/fhir/ValueSet/mimetypes|4.0.1 (required)

Subscription.tag | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 2 tests that all execute ok. 2 fail round-trip testing and all r3 resources are valid.)

Structure

Name | Flags | Card. | Type | Description & Constraints

Subscription | ΣTU | | DomainResource | Server push subscription criteria Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

status | ?!Σ | 1..1 | code | requested | active | error | off SubscriptionStatus (Required)

contact | Σ | 0..* | ContactPoint | Contact details for source (e.g. troubleshooting)
end | Σ | 0..1 | instant | When to automatically delete the subscription
reason | Σ | 1..1 | string | Description of why this subscription was created
criteria | Σ | 1..1 | string | Rule for server push
error | Σ | 0..1 | string | Latest error note
channel | Σ | 1..1 | BackboneElement | The channel on which to report matches to the criteria
type | Σ | 1..1 | code | rest-hook | websocket | email | sms | message SubscriptionChannelType (Required)
endpoint | Σ | 0..1 | url | Where the channel points to
payload | Σ | 0..1 | code | MIME type to send, or omit for no payload MimeType (Required)
header | Σ | 0..* | string | Usage depends on the channel type
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Subscription xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <status value="[code]"/><!-- 1..1 requested | active | error | off -->
 <contact><!-- 0..* ContactPoint Contact details for source (e.g. troubleshooting) --></contact>
 <end value="[instant]"/><!-- 0..1 When to automatically delete the subscription -->
 <reason value="[string]"/><!-- 1..1 Description of why this subscription was created -->
 <criteria value="[string]"/><!-- 1..1 Rule for server push -->
 <error value="[string]"/><!-- 0..1 Latest error note -->
 <channel>  <!-- 1..1 The channel on which to report matches to the criteria -->
  <type value="[code]"/><!-- 1..1 rest-hook | websocket | email | sms | message -->
  <endpoint value="[url]"/><!-- 0..1 Where the channel points to -->
  <payload value="[code]"/><!-- 0..1 MIME type to send, or omit for no payload -->
  <header value="[string]"/><!-- 0..* Usage depends on the channel type -->
 </channel>
</Subscription>
```

JSON Template

```
{
  "resourceType" : "Subscription",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "status" : "<code>", // R!  requested | active | error | off
  "contact" : [{ ContactPoint }], // Contact details for source (e.g. troubleshooting)
  "end" : "<instant>", // When to automatically delete the subscription
  "reason" : "<string>", // R!  Description of why this subscription was created
  "criteria" : "<string>", // R!  Rule for server push
  "error" : "<string>", // Latest error note
  "channel" : { // R!  The channel on which to report matches to the criteria
    "type" : "<code>", // R!  rest-hook | websocket | email | sms | message
    "endpoint" : "<url>", // Where the channel points to
    "payload" : "<code>", // MIME type to send, or omit for no payload
    "header" : ["<string>"] // Usage depends on the channel type
  }
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Subscription;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Subscription.status [ code ]; # 1..1 requested | active | error | off
  fhir:Subscription.contact [ ContactPoint ], ... ; # 0..* Contact details for source (e.g. troubleshooting)
  fhir:Subscription.end [ instant ]; # 0..1 When to automatically delete the subscription
  fhir:Subscription.reason [ string ]; # 1..1 Description of why this subscription was created
  fhir:Subscription.criteria [ string ]; # 1..1 Rule for server push
  fhir:Subscription.error [ string ]; # 0..1 Latest error note
  fhir:Subscription.channel [ # 1..1 The channel on which to report matches to the criteria
    fhir:Subscription.channel.type [ code ]; # 1..1 rest-hook | websocket | email | sms | message
    fhir:Subscription.channel.endpoint [ url ]; # 0..1 Where the channel points to
    fhir:Subscription.channel.payload [ code ]; # 0..1 MIME type to send, or omit for no payload
    fhir:Subscription.channel.header [ string ], ... ; # 0..* Usage depends on the channel type
  ];
]
```

Changes since Release 3

Subscription
Subscription.status | Change value set from http://hl7.org/fhir/ValueSet/subscription-status to http://hl7.org/fhir/ValueSet/subscription-status|4.0.1
Subscription.channel.type | Change value set from http://hl7.org/fhir/ValueSet/subscription-channel-type to http://hl7.org/fhir/ValueSet/subscription-channel-type|4.0.1
Subscription.channel.endpoint | Type changed from uri to url
Subscription.channel.payload | Type changed from string to code Add Binding http://hl7.org/fhir/ValueSet/mimetypes|4.0.1 (required)
Subscription.tag | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 2 tests that all execute ok. 2 fail round-trip testing and all r3 resources are valid.)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

2.46.3.1 Terminology Bindings

Path	Definition	Type	Reference
Subscription.status	The status of a subscription.	Required	SubscriptionStatus
Subscription.channel.type	The type of method used to execute a subscription.	Required	SubscriptionChannelType
Subscription.channel.payload	The mime type of an attachment. Any valid mime type is allowed.	Required	Mime Types

2.46.4 Safety and Security

Executing each of the channels documented below involves the server sending a communication that will reveal information about the client and server relationship, and, if the entire resource is sent, administrative or clinical information that may be quite sensitive and/or protected under law. Servers are responsible for ensuring appropriate security is employed for each channel. The subscription resource does not address these concerns directly - it is assumed that these are administered by other configuration processes. For instance, a server might maintain a whitelist of acceptable servers for the rest-create/rest-update methods.

Emails should generally be secured using some technique such as Direct .

2.46.5 Managing Subscriptions and Errors

A subscription is defined by creating the Subscription resource on the server. When the subscription is created by the client, it sets the status to "requested". After POSTing the subscription, the client parses the Location header and saves the new Subscription's logical id for use in subsequent operations.

When the server receives a subscription, it SHOULD check that it is prepared to accept/process the subscription. If it is, it sets the subscription to active, and then process it like a normal create. If it isn't, it SHOULD return an error with an OperationOutcome instead of processing the create.

The criteria are subject to the same limitations as the client that created it, such as access to patient compartments etc. Note that the subscription remains active after the client access tokens expire.

Once the server has activated the subscription, it sets the status to "active" (note: the server can do this as it accepts the resource if it wants).

An appropriately authorized client can use search and/or history operations to see what subscriptions are currently active on the server. Once the subscription is no longer desired, the client deletes the subscription from the server.

The server may retry the notification a fixed number of times and/or refer errors to its own alert logs. If the notification fails, the server should set the status to 'error' and mark the error in the resource. If the notification succeeds, the server should update the status to "active again. If a subscription fails consistently a server may choose set the subscription status to off and stop trying to send notifications.

If a subscription nominates a fixed end date, the server automatically deletes it at the specified time.

2.46.6 Tracking Subscription Notifications

Applications that wish to track whether notifications have been sent for particular resources (or versions of resources) can look at the AuditEvent resources. For example:

```
GET [base]/AuditEvent?entity=Patient/103
```

This search will return all the AuditEvent resources that are about Patient 103. At this time there is no deterministic way to tell say which of those AuditEvent resources come from the subscription sub-system that actually handles notifications. This is planned to be resolved in a future version of this specification. In the mean time, servers are encouraged to create AuditEvent records when performing notifications and document how clients can identify the relevant records when searching.

In addition, servers might also create Communication resources for some of the notifications that are sent in response to a subscription, such as when sending emails.

```
GET [base]/Communication?based-on=Subscription/103
```

This returns a list of communications sent by a subscription. TODO: search on payload....

2.46.7 Channels

2.46.7.1 REST Hook

This uses an empty POST message to alert the subscriber that new results are available - POST to [base]/Subscription:

```
{
```

```
  "resourceType": "Subscription",
  "criteria": "Observation?name=http://loinc.org|1975-2&_format=json",
  "channel": {
    "type": "rest-hook",
    "endpoint": "https://biliwatch.com/customers/mount-auburn-miu/on-result",
    "header": "Authorization: Bearer secret-token-abc-123"
  }
}
```

When a resource is created or updated that meets the criteria, the server sends a POST request with no body to the nominated URL.

When the subscriber receives a POST to https://biliwatch.com/customers/mount-auburn-miu/on-result, it re-issues the criteria as a query to the server, appending &_since=:last (where :last is replaced by the time at which the client last checked). In this way it can fetch all new relevant Observations.

Since payload is missing, the data in the resources is only available through the REST API, which helps consolidate authorization and authentication logic. The server must append the headers, if any are given, to the POST request that it makes to the client.

Alternatively, the server can be asked to send the entire resource to a nominated FHIR end-point. This is usually appropriate for defining routing rules within a managed eco-system such as a healthcare institution.

```
{
  "channel": {
    "type": "rest-hook",
    "endpoint": "https://internal.acme.com/research/saturn",
    "payload": "application/fhir+json"
  }
}
```

This requests that a server forward a copy of any matching resource in JSON format to the nominated server as an Update operation using the nominated URL as the service base. In order to execute this channel, the server must know how to authenticate appropriately with the destination server. This can be done by the subscription resource providing an authentication header for the server to use, or alternatively, the server may be specifically configured to be able to use the nominated server.

2.46.7.2 WebSockets

Subscriptions are created exclusively via the FHIR REST API. But notifications need not occur via REST. Indeed, some subscribers may be unable to expose an outward-facing HTTP server to receive triggered notifications. For example, a pure client-side Web app or mobile app may want to subscribe to a data feed without polling using the /history operation. This can be accomplished using a websocket notification channel.

A client can declare its intention to listen via Web Sockets:

```
{
  "channel": {
    "type": "websocket"
  }
}
```

The subscriber would then initiate a Web Socket connection to the server, at a URL advertised in the FHIR server's Capability statement (subscriptions/webSocketUrl (todo)). A simple protocol is used to listen for notifications:

• Client connects a secure Web Socket to the server's webSocketUrl (see websocket extension in the server's CapabilityStatement).

• Client authenticates to server using a server-specified Web socket protocol (e.g. OAuth bearer token presentation).

• Client sends a bind :id message over the socket (using the logical id of the subscription). For example, the client might issue: bind 123).

• Server responds with a "bound :id" message to acknowledge.

• Server sends a "ping :id" message to notify the client each time a new result is available

2.46.7.3 Email/SMS

A client can register for its user to receive notifications by email:

```
{
  "channel": {
    "type": "email",
    "endpoint": "mailto:mt-auburn-results@direct.biliwatch.com",
    "header": "A new bilirubin result has arrived!"
  }
}
```

The server would send a new message for each matching resource. The body of the email may be empty, or it may contain a reference to the search or the matching resource. It is at the discretion of the server as to how much information to provide. Subscription.channel.header sets the subject of the email. The email should be secured appropriately, such as using Direct, as specified by the rules of the applicable jurisdictions.

SMS works very similarly:

```
{
  "channel": {
    "type": "sms",
    "endpoint": "tel:+1555-345-5555"
  }
}
```

Note: SMS messages are extremely limited in size, so channel.payload will usually be omitted (signifying that no payload is to be sent). The recipient may be human, but this is not always the case. Irrespective of size, most servers will refuse to send payloads in SMS for security reasons, and may refuse to send emails unless encrypted.

A mime/type of text/plain can be useful for email and sms along with some extension describing how to convert resources to a text representation. This specification may provide supporting infrastructure for this in the future.

2.46.7.4 Messaging

A client can register for its user to receive notifications by messaging:

```
{
  "channel": {
    "type": "message",
    "endpoint": "http://ehr.example.org/fhir/$process-message"
  }
}
```

For each matching resource, a server will send a message to the nominated end-point. Most servers will require that the end-point is white-listed prior to allowing these kinds of subscriptions.

STU Note: The details of the message - mainly the event code - are still to be resolved during the trial use period.

Feedback is welcome here .

2.46.8 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
contact | token | Contact details for the subscription | Subscription.contact
criteria | string | The search rules used to determine when to send a notification | Subscription.criteria
payload | token | The mime-type of the notification payload | Subscription.channel.payload
status | token | The current state of the subscription | Subscription.status
type | token | The type of channel for the sent notifications | Subscription.channel.type
url | uri | The uri that will receive the notifications | Subscription.channel.endpoint

2.46.9 Resource Subscription - Examples

URL: https://hl7.org/fhir/R4/subscription-examples.html

2.46.9 Resource Subscription - Examples

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

Example Name | id | Format
Example Subscription | example | XML | JSON | Turtle
Example of subscription in error status | example-error | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

2.46.10 Resource Subscription - Detailed Descriptions

URL: https://hl7.org/fhir/R4/subscription-definitions.html

2.46.10 Resource Subscription - Detailed Descriptions

FHIR Infrastructure Work Group | Maturity Level: 3 | Trial Use | Security Category: Business | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the Subscription resource.

Subscription
Element Id | Subscription
Definition | The subscription resource is used to define a push-based subscription from a server to another system. Once a subscription is registered with the server, the server checks every resource that is created or updated, and if the resource matches the given criteria, it sends a message on the defined "channel" so that another system can take an appropriate action.
Cardinality | 0..*
Type | DomainResource
Alternate Names | WebHook; Hook; Routing Rule

Summary | true
Subscription.status
Element Id | Subscription.status
Definition | The status of the subscription, which marks the server state for managing the subscription.
Cardinality | 1..1
Terminology Binding | SubscriptionStatus (Required)
Type | code
Is Modifier | true (Reason: This element is labelled as a modifier because it is a status element that contains status entered-in-error which means that the resource should not be treated as valid)
Summary | true
Comments | A client can only submit subscription resources in the requested or off state. Only the server can move a subscription from requested to active, and then to error. Either the server or the client can turn a subscription off. This element is labeled as a modifier because the status contains codes that mark the resource as not currently valid.
Subscription.contact
Element Id | Subscription.contact
Definition | Contact details for a human to contact about the subscription. The primary use of this for system administrator troubleshooting.
Cardinality | 0..*
Type | ContactPoint
Summary | true
Subscription.end
Element Id | Subscription.end
Definition | The time for the server to turn the subscription off.
Cardinality | 0..1
Type | instant
Summary | true
Comments | The server is permitted to deviate from this time but should observe it.
Subscription.reason
Element Id | Subscription.reason
Definition | A description of why this subscription is defined.
Cardinality | 1..1
Type | string
Summary | true
Subscription.criteria
Element Id | Subscription.criteria
Definition | The rules that the server should use to determine when to generate notifications for this subscription.
Cardinality | 1..1
Type | string
Summary | true
Comments | The rules are search criteria (without the [base] part). Like Bundle.entry.request.url, it has no leading "/".
Subscription.error
Element Id | Subscription.error
Definition | A record of the last error that occurred when the server processed a notification.
Cardinality | 0..1
Type | string
Summary | true
Subscription.channel
Element Id | Subscription.channel
Definition | Details where to send notifications when resources are received that meet the criteria.
Cardinality | 1..1
Summary | true
Subscription.channel.type
Element Id | Subscription.channel.type
Definition | The type of channel to send notifications on.
Cardinality | 1..1
Terminology Binding | SubscriptionChannelType (Required)
Type | code
Summary | true
Subscription.channel.endpoint
Element Id | Subscription.channel.endpoint
Definition | The url that describes the actual end-point to send messages to.
Cardinality | 0..1
Type | url
Summary | true
Comments | For rest-hook, and websocket, the end-point must be an http: or https: URL; for email, a mailto: url, for sms, a tel: url, and for message the endpoint can be in any form of url the server understands (usually, http: or mllp:). The URI is allowed to be relative; in which case, it is relative to the server end-point (since there may be more than one, clients should avoid using relative URIs).
Subscription.channel.payload
Element Id | Subscription.channel.payload
Definition | The mime type to send the payload in - either application/fhir+xml, or application/fhir+json. If the payload is not present, then there is no payload in the notification, just a notification. The mime type "text/plain" may also be used for Email and SMS subscriptions.

Cardinality | 0..1
Terminology Binding | MimeType (Required)
Type | code
Summary | true
Comments | Sending the payload has obvious security implications. The server is responsible for ensuring that the content is appropriately secured.
Subscription.channel.header
Element Id | Subscription.channel.header
Definition | Additional headers / information to send as part of the notification.
Cardinality | 0..*
Type | string
Summary | true
Comments | Exactly what these mean depend on the channel type. They can convey additional information to the recipient and/or meet security requirements; for example, support of multiple headers in the outgoing notifications for rest-hook type subscriptions.

2.46.11 Resource Subscription - Mappings

URL: https://hl7.org/fhir/R4/subscription-mappings.html

2.46.11 Resource Subscription - Mappings

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

Mappings:

FiveWs Pattern Mapping

RIM Mapping

Mappings for the subscription resource (see Mappings to Other Standards for further information & status).

2.46.11.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

Subscription
status | FiveWs.status
contact | FiveWs.subject[x]
end | FiveWs.done[x]
reason | FiveWs.why[x]

2.46.11.2 RIM Mapping (http://hl7.org/v3)

Subscription | N/A

2.46.12 Resource Subscription - Extensions & Profiles

URL: https://hl7.org/fhir/R4/subscription-profiles.html

2.46.12 Resource Subscription - Extensions & Profiles

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

This table lists profiles and extensions for the Subscription resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

2.46.12.1 Profiles

No Profiles defined for this resource

2.46.12.2 Extensions

No Extensions defined for this resource
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

2.46.12.3 Search Extensions

No Search Extensions defined for this resource

2.46.13 Resource Subscription - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/subscription-version-maps.html

No validation errors - all conversions are clean

2.46.13 Resource Subscription - R3/R4 Conversions

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

R3 : R4 Conversion maps for Subscription.

Functional status for this map: 2 tests that all execute ok. 2 fail round-trip testing and all r3 resources are valid. (see documentation)

2.46.13.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/Subscription3to4" = "R3 to R4 Conversions for Subscription"

uses "http://hl7.org/fhir/3.0/StructureDefinition/Subscription" alias SubscriptionR3 as source
uses "http://hl7.org/fhir/StructureDefinition/Subscription" alias Subscription as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group Subscription(source src : SubscriptionR3, target tgt : Subscription) extends DomainResource <<type+>> {
  src.status -> tgt.status;
  src.contact -> tgt.contact;
  src.end -> tgt.end;
  src.reason -> tgt.reason;
  src.criteria -> tgt.criteria;
  src.error -> tgt.error;
  src.channel as vs0 -> tgt.channel as vt0 then channel(vs0, vt0);
}

group channel(source src, target tgt) extends BackboneElement {
  src.type -> tgt.type;
  src.endpoint -> tgt.endpoint;
  src.payload -> tgt.payload;
  src.header -> tgt.header;
}
```

2.46.13.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/Subscription4to3" = "R4 to R3 Conversion for Subscription"

uses "http://hl7.org/fhir/StructureDefinition/Subscription" alias Subscription as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/Subscription" alias SubscriptionR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group Subscription(source src : Subscription, target tgt : SubscriptionR3) extends DomainResource <<type+>> {
  src.status -> tgt.status;
  src.contact -> tgt.contact;
  src.end -> tgt.end;
  src.reason -> tgt.reason;
  src.criteria -> tgt.criteria;
  src.error -> tgt.error;
  src.channel as vs0 -> tgt.channel as vt0 then channel(vs0, vt0);
}

group channel(source src, target tgt) extends BackboneElement {
  src.type -> tgt.type;
  src.endpoint -> tgt.endpoint;
  src.payload -> tgt.payload;
  src.header -> tgt.header;
}
```

2.46.13.3 R4 Validation Errors

2.46.14 JSON Schema for Subscription

URL: https://hl7.org/fhir/R4/subscription.schema.json.html

2.46.14 JSON Schema for Subscription

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is a representation of the json schema for Subscription, which is just a part of the full JSON Schema.

```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/Subscription",
```

```
"$ref": "#/definitions/Subscription",
"description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
"definitions": {
  "Subscription": {
    "description": "The subscription resource is used to define a push-based subscription from a server to another
  system. Once a subscription is registered with the server, the server checks every resource that is created or
  updated, and if the resource matches the given criteria, it sends a message on the defined \"channel\" so that
  another system can take an appropriate action.",
    "properties": {
      "resourceType": {
        "description": "This is a Subscription resource",
        "const": "Subscription"
      },
      "id": {
        "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
   never changes.",
        "$ref": "id.schema.json#/definitions/id"
      },
      "meta": {
        "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
  Changes to the content might not always be associated with version changes to the resource.",
        "$ref": "Meta.schema.json#/definitions/Meta"
      },
      "implicitRules": {
        "description": "A reference to a set of rules that were followed when the resource was constructed, and which
  must be understood when processing the content. Often, this is a reference to an implementation guide that defines
  the special rules along with other profiles etc.",
        "$ref": "#/definitions/uri"
      },
      "_implicitRules": {
        "description": "Extensions for implicitRules",
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "language": {
        "description": "The base language in which the resource is written.",
        "$ref": "#/definitions/code"
      },
      "_language": {
        "description": "Extensions for language",
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "text": {
        "description": "A human-readable narrative that contains a summary of the resource and can be used to
  represent the content of the resource to a human. The narrative need not encode all the structured data, but is
  required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
  Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
        "$ref": "Narrative.schema.json#/definitions/Narrative"
      },
      "contained": {
        "description": "These resources do not have an independent existence apart from the resource that contains
  them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
        "items": {
          "$ref": "ResourceList.schema.json#/definitions/ResourceList"
        },
        "type": "array"
      },
      "extension": {
        "description": "May be used to represent additional information that is not part of the basic definition of
  the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
   definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
  that SHALL be met as part of the definition of the extension.",
        "items": {
          "$ref": "Extension.schema.json#/definitions/Extension"
        },
        "type": "array"
      },
      "modifierExtension": {
        "description": "May be used to represent additional information that is not part of the basic definition of
  the resource and that modifies the understanding of the element that contains it and/or the understanding of the
  containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
  of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
  extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
  met as part of the definition of the extension. Applications processing a resource are required to check for
  modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
  DomainResource (including cannot change the meaning of modifierExtension itself).",
        "items": {
          "$ref": "Extension.schema.json#/definitions/Extension"
        },
        "type": "array"
      },
      "status": {
        "description": "The status of the subscription, which marks the server state for managing the subscription.",
        "enum": [
          "requested",
          "active",
          "error",
          "off"
        ]
      },
```

```
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "contact": {
      "description": "Contact details for a human to contact about the subscription. The primary use of this for
system administrator troubleshooting.",
      "items": {
        "$ref": "ContactPoint.schema.json#/definitions/ContactPoint"
      },
      "type": "array"
    },
    "end": {
      "description": "The time for the server to turn the subscription off.",
      "$ref": "#/definitions/instant"
    },
    "_end": {
      "description": "Extensions for end",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "reason": {
      "description": "A description of why this subscription is defined.",
      "$ref": "#/definitions/string"
    },
    "_reason": {
      "description": "Extensions for reason",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "criteria": {
      "description": "The rules that the server should use to determine when to generate notifications for this
subscription.",
      "$ref": "#/definitions/string"
    },
    "_criteria": {
      "description": "Extensions for criteria",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "error": {
      "description": "A record of the last error that occurred when the server processed a notification.",
      "$ref": "#/definitions/string"
    },
    "_error": {
      "description": "Extensions for error",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "channel": {
      "description": "Details where to send notifications when resources are received that meet the criteria.",
      "$ref": "#/definitions/Subscription_Channel"
    }
  },
  "additionalProperties": false,
  "required": [
    "channel",
    "resourceType"
  ]
},
"Subscription_Channel": {
  "description": "The subscription resource is used to define a push-based subscription from a server to another
system. Once a subscription is registered with the server, the server checks every resource that is created or
updated, and if the resource matches the given criteria, it sends a message on the defined \"channel\" so that
another system can take an appropriate action.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
```

```
          "type": "array"
        },
        "type": {
          "description": "The type of channel to send notifications on.",
          "enum": [
            "rest-hook",
            "websocket",
            "email",
            "sms",
            "message"
          ]
        },
        "_type": {
          "description": "Extensions for type",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "endpoint": {
          "description": "The url that describes the actual end-point to send messages to.",
          "$ref": "#/definitions/url"
        },
        "_endpoint": {
          "description": "Extensions for endpoint",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "payload": {
          "description": "The mime type to send the payload in - either application/fhir+xml, or application/fhir+json.
    If the payload is not present, then there is no payload in the notification, just a notification. The mime type
    \"text/plain\" may also be used for Email and SMS subscriptions.",
          "$ref": "#/definitions/code"
        },
        "_payload": {
          "description": "Extensions for payload",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "header": {
          "description": "Additional headers / information to send as part of the notification.",
          "items": {
            "$ref": "#/definitions/string"
          },
          "type": "array"
        },
        "_header": {
          "description": "Extensions for header",
          "items": {
            "$ref": "Element.schema.json#/definitions/Element"
          },
          "type": "array"
        }
      },
      "additionalProperties": false
    }
  }
}
```

2.46.15 Resource Subscription - Maturity Dependency View

URL: https://hl7.org/fhir/R4/subscription-dependencies.html

2.46.15 Resource Subscription - Maturity Dependency View

FHIR Infrastructure Work Group | Maturity Level: 3 | Trial Use | Compartments: Not linked to any defined compartments

Dependency Graph for Subscription FMM level 3

Name | Card. | Type | Dependency Analysis
Subscription | | DomainResource
status | 1..1 | code
contact | 0..* | ContactPoint
end | 0..1 | instant
reason | 1..1 | string
criteria | 1..1 | string
error | 0..1 | string
channel | 1..1 | BackboneElement
type | 1..1 | code
endpoint | 0..1 | url
payload | 0..1 | code
header | 0..* | string

2.46.16 Value Set Sc-valueset-subscription-status - Status Code Mapping

URL: https://hl7.org/fhir/R4/sc-valueset-subscription-status.html

2.46.16 Value Set Sc-valueset-subscription-status - Status Code Mapping

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is the narrative for the resource. See also the XML, JSON or Turtle format.

2.46.17 SubscriptionStatusCanonicalMap (http://hl7.org/fhir/ConceptMap/sc-subscription-status)

Mapping from http://hl7.org/fhir/ValueSet/subscription-status to http://hl7.org/fhir/ValueSet/resource-status

DRAFT. Published on 01/11/2019 9:29:23 AM by HL7 (FHIR Project) (http://hl7.org/fhir , fhir@lists.hl7.org).

Canonical Mapping for "The status of a subscription."

Source Code	Equivalence	Destination Code
entered-in-error	equivalent	error
requested	equivalent	requested
active	equivalent	active
error	equivalent	failed
off	equivalent	inactive

2.47 Resource MessageHeader - Content

URL: https://hl7.org/fhir/R4/messageheader.html

Infrastructure And Messaging Work Group | Maturity Level: 4 | Trial Use | Security Category: Not Classified | Compartments: Device, Practitioner

The header for a message exchange that is either requesting or responding to an action. The reference(s) that are the subject of the action as well as other information related to the action are typically transmitted in a bundle in which the MessageHeader resource instance is the first resource in the bundle.

2.47.1 Scope and Usage

The MessageHeader resource is defined in order to support Messaging using FHIR resources. The principal usage of the MessageHeader resource is when messages are exchanged. However, as a resource that can be used with the RESTful framework, the MessageHeader resource has the normal resource end-point ([base-url]/MessageHeader), which is used to manage a set of static messages resources. This could be used to make an archive of past messages available. Creating or updating Message resources in this fashion does not represent the actual occurrence of any event, nor can it trigger any logic associated with the actual event. It is just for managing a set of message resources.

2.47.2 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name	Flags	Card.	Type	Description & Constraints
MessageHeader	ΣTU		DomainResource	A resource that describes a message that is exchanged between systems Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
event[x]	Σ	1..1		Code for the event this message represents or link to event definition MessageEvent (Example)
eventCoding			Coding	
eventUri			uri	
destination	Σ	0..*	BackboneElement	Message destination application(s)
name	Σ	0..1	string	Name of system
target	Σ	0..1	Reference(Device)	Particular delivery destination within the destination
endpoint	Σ	1..1	url	Actual destination address or id
receiver	Σ	0..1	Reference(Practitioner | PractitionerRole | Organization)	Intended "real-world" recipient for the data
sender	Σ	0..1	Reference(Practitioner | PractitionerRole | Organization)	Real world sender of the message
enterer	Σ	0..1	Reference(Practitioner | PractitionerRole)	The source of the data entry
author	Σ	0..1	Reference(Practitioner | PractitionerRole)	The source of the decision
source	Σ	1..1	BackboneElement	Message source application
name	Σ	0..1	string	Name of system
software	Σ	0..1	string	Name of software running the system
version	Σ	0..1	string	Version of software running
contact	Σ	0..1	ContactPoint	Human contact for problems
endpoint	Σ	1..1	url	Actual message source address or id

responsible | Σ | 0..1 | Reference(Practitioner | PractitionerRole | Organization) | Final responsibility for event
reason | Σ | 0..1 | CodeableConcept | Cause of event Example Message Reason Codes (Example)
response | Σ | 0..1 | BackboneElement | If this is a reply to prior message
identifier | Σ | 1..1 | id | Id of original message
code | Σ | 1..1 | code | ok | transient-error | fatal-error ResponseType (Required)
details | Σ | 0..1 | Reference(OperationOutcome) | Specific list of hints/warnings/errors
focus | Σ | 0..* | Reference(Any) | The actual content of the message
definition | Σ | 0..1 | canonical(MessageDefinition) | Link to the definition for this message
Documentation for this format

UML Diagram (Legend)

XML Template

```
<MessageHeader xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <event[x]><!-- 1..1 Coding|uri Code for the event this message represents or link to event definition --></event[x]>
 <destination>  <!-- 0..* Message destination application(s) -->
  <name value="[string]"/><!-- 0..1 Name of system -->
  <target><!-- 0..1 Reference(Device) Particular delivery destination within the destination --></target>
  <endpoint value="[url]"/><!-- 1..1 Actual destination address or id -->
  <receiver><!-- 0..1 Reference(Practitioner|PractitionerRole|Organization) Intended "real-world" recipient for the data
      --></receiver>
 </destination>
 <sender><!-- 0..1 Reference(Practitioner|PractitionerRole|Organization) Real world sender of the message --></sender>
 <enterer><!-- 0..1 Reference(Practitioner|PractitionerRole) The source of the data entry --></enterer>
 <author><!-- 0..1 Reference(Practitioner|PractitionerRole) The source of the decision --></author>
 <source>  <!-- 1..1 Message source application -->
  <name value="[string]"/><!-- 0..1 Name of system -->
  <software value="[string]"/><!-- 0..1 Name of software running the system -->
  <version value="[string]"/><!-- 0..1 Version of software running -->
  <contact><!-- 0..1 ContactPoint Human contact for problems --></contact>
  <endpoint value="[url]"/><!-- 1..1 Actual message source address or id -->
 </source>
 <responsible><!-- 0..1 Reference(Practitioner|PractitionerRole|Organization) Final responsibility for event
    --></responsible>
 <reason><!-- 0..1 CodeableConcept Cause of event --></reason>
 <response>  <!-- 0..1 If this is a reply to prior message -->
  <identifier value="[id]"/><!-- 1..1 Id of original message -->
  <code value="[code]"/><!-- 1..1 ok | transient-error | fatal-error -->
  <details><!-- 0..1 Reference(OperationOutcome) Specific list of hints/warnings/errors --></details>
 </response>
 <focus><!-- 0..* Reference(Any) The actual content of the message --></focus>
 <definition><!-- 0..1 canonical(MessageDefinition) Link to the definition for this message --></definition>
</MessageHeader>
```

JSON Template

```
{
  "resourceType" : "MessageHeader",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  // event[x]: Code for the event this message represents or link to event definition. One of these 2:
  "eventCoding" : { Coding },
  "eventUri" : "<uri>",
  "destination" : [{ // Message destination application(s)
    "name" : "<string>", // Name of system
    "target" : { Reference(Device) }, // Particular delivery destination within the destination
    "endpoint" : "<url>", // R!  Actual destination address or id
    "receiver" : { Reference(Practitioner|PractitionerRole|Organization) } // Intended "real-world" recipient for the
    data
  }],
  "sender" : { Reference(Practitioner|PractitionerRole|Organization) }, // Real world sender of the message
  "enterer" : { Reference(Practitioner|PractitionerRole) }, // The source of the data entry
  "author" : { Reference(Practitioner|PractitionerRole) }, // The source of the decision
  "source" : { // R!  Message source application
    "name" : "<string>", // Name of system
    "software" : "<string>", // Name of software running the system
    "version" : "<string>", // Version of software running
    "contact" : { ContactPoint }, // Human contact for problems
    "endpoint" : "<url>" // R!  Actual message source address or id
  },
  "responsible" : { Reference(Practitioner|PractitionerRole|Organization) }, // Final responsibility for event
  "reason" : { CodeableConcept }, // Cause of event
  "response" : { // If this is a reply to prior message
    "identifier" : "<id>", // R!  Id of original message
    "code" : "<code>", // R!  ok | transient-error | fatal-error
    "details" : { Reference(OperationOutcome) } // Specific list of hints/warnings/errors
  },
  "focus" : [{ Reference(Any) }], // The actual content of the message
  "definition" : { canonical(MessageDefinition) } // Link to the definition for this message
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:MessageHeader;
```

```
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  # MessageHeader.event[x] : 1..1 Code for the event this message represents or link to event definition. One of these 2
    fhir:MessageHeader.eventCoding [ Coding ]
    fhir:MessageHeader.eventUri [ uri ]
  fhir:MessageHeader.destination [ # 0..* Message destination application(s)
    fhir:MessageHeader.destination.name [ string ]; # 0..1 Name of system
    fhir:MessageHeader.destination.target [ Reference(Device) ]; # 0..1 Particular delivery destination within the
    destination
    fhir:MessageHeader.destination.endpoint [ url ]; # 1..1 Actual destination address or id
    fhir:MessageHeader.destination.receiver [ Reference(Practitioner|PractitionerRole|Organization) ]; # 0..1 Intended
    "real-world" recipient for the data
  ], ...;
  fhir:MessageHeader.sender [ Reference(Practitioner|PractitionerRole|Organization) ]; # 0..1 Real world sender of the
    message
  fhir:MessageHeader.enterer [ Reference(Practitioner|PractitionerRole) ]; # 0..1 The source of the data entry
  fhir:MessageHeader.author [ Reference(Practitioner|PractitionerRole) ]; # 0..1 The source of the decision
  fhir:MessageHeader.source [ # 1..1 Message source application
    fhir:MessageHeader.source.name [ string ]; # 0..1 Name of system
    fhir:MessageHeader.source.software [ string ]; # 0..1 Name of software running the system
    fhir:MessageHeader.source.version [ string ]; # 0..1 Version of software running
    fhir:MessageHeader.source.contact [ ContactPoint ]; # 0..1 Human contact for problems
    fhir:MessageHeader.source.endpoint [ url ]; # 1..1 Actual message source address or id
  ];
  fhir:MessageHeader.responsible [ Reference(Practitioner|PractitionerRole|Organization) ]; # 0..1 Final responsibility
    for event
  fhir:MessageHeader.reason [ CodeableConcept ]; # 0..1 Cause of event
  fhir:MessageHeader.response [ # 0..1 If this is a reply to prior message
    fhir:MessageHeader.response.identifier [ id ]; # 1..1 Id of original message
    fhir:MessageHeader.response.code [ code ]; # 1..1 ok | transient-error | fatal-error
    fhir:MessageHeader.response.details [ Reference(OperationOutcome) ]; # 0..1 Specific list of hints/warnings/errors
  ];
  fhir:MessageHeader.focus [ Reference(Any) ], ... ; # 0..* The actual content of the message
  fhir:MessageHeader.definition [ canonical(MessageDefinition) ]; # 0..1 Link to the definition for this message
]
```

Changes since R3

MessageHeader
MessageHeader.event[x] | Renamed from event to event[x] Add Type uri
MessageHeader.destination.endpoint | Type changed from uri to url
MessageHeader.destination.receiver | Moved from MessageHeader to MessageHeader.destination Type Reference: Added Target Type PractitionerRole
MessageHeader.sender | Type Reference: Added Target Type PractitionerRole
MessageHeader.enterer | Type Reference: Added Target Type PractitionerRole
MessageHeader.author | Type Reference: Added Target Type PractitionerRole
MessageHeader.source.endpoint | Type changed from uri to url
MessageHeader.responsible | Type Reference: Added Target Type PractitionerRole
MessageHeader.response.code | Change value set from http://hl7.org/fhir/ValueSet/response-code to http://hl7.org/fhir/ValueSet/response-code|4.0.1
MessageHeader.definition | Added Element
MessageHeader.timestamp | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 1 test that all execute ok. All tests pass round-trip testing and 1 r3 resources are invalid (0 errors).)

Structure

Name | Flags | Card. | Type | Description & Constraints
MessageHeader | ΣTU | | DomainResource | A resource that describes a message that is exchanged between systems Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
event[x] | Σ | 1..1 | | Code for the event this message represents or link to event definition MessageEvent (Example)
eventCoding | | | Coding
eventUri | | | uri
destination | Σ | 0..* | BackboneElement | Message destination application(s)
name | Σ | 0..1 | string | Name of system
target | Σ | 0..1 | Reference(Device) | Particular delivery destination within the destination
endpoint | Σ | 1..1 | url | Actual destination address or id
receiver | Σ | 0..1 | Reference(Practitioner | PractitionerRole | Organization) | Intended "real-world" recipient for the data
sender | Σ | 0..1 | Reference(Practitioner | PractitionerRole | Organization) | Real world sender of the message
enterer | Σ | 0..1 | Reference(Practitioner | PractitionerRole) | The source of the data entry
author | Σ | 0..1 | Reference(Practitioner | PractitionerRole) | The source of the decision
source | Σ | 1..1 | BackboneElement | Message source application
name | Σ | 0..1 | string | Name of system
software | Σ | 0..1 | string | Name of software running the system
version | Σ | 0..1 | string | Version of software running
contact | Σ | 0..1 | ContactPoint | Human contact for problems
endpoint | Σ | 1..1 | url | Actual message source address or id

responsible | Σ | 0..1 | Reference(Practitioner | PractitionerRole | Organization) | Final responsibility for event
reason | Σ | 0..1 | CodeableConcept | Cause of event Example Message Reason Codes (Example)
response | Σ | 0..1 | BackboneElement | If this is a reply to prior message
identifier | Σ | 1..1 | id | Id of original message
code | Σ | 1..1 | code | ok | transient-error | fatal-error ResponseType (Required)
details | Σ | 0..1 | Reference(OperationOutcome) | Specific list of hints/warnings/errors
focus | Σ | 0..* | Reference(Any) | The actual content of the message
definition | Σ | 0..1 | canonical(MessageDefinition) | Link to the definition for this message
Documentation for this format

UML Diagram (Legend)

XML Template

```
<MessageHeader xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <event[x]><!-- 1..1 Coding|uri Code for the event this message represents or link to event definition --></event[x]>
 <destination>  <!-- 0..* Message destination application(s) -->
  <name value="[string]"/><!-- 0..1 Name of system -->
  <target><!-- 0..1 Reference(Device) Particular delivery destination within the destination --></target>
  <endpoint value="[url]"/><!-- 1..1 Actual destination address or id -->
  <receiver><!-- 0..1 Reference(Practitioner|PractitionerRole|Organization) Intended "real-world" recipient for the data
      --></receiver>
 </destination>
 <sender><!-- 0..1 Reference(Practitioner|PractitionerRole|Organization) Real world sender of the message --></sender>
 <enterer><!-- 0..1 Reference(Practitioner|PractitionerRole) The source of the data entry --></enterer>
 <author><!-- 0..1 Reference(Practitioner|PractitionerRole) The source of the decision --></author>
 <source>  <!-- 1..1 Message source application -->
  <name value="[string]"/><!-- 0..1 Name of system -->
  <software value="[string]"/><!-- 0..1 Name of software running the system -->
  <version value="[string]"/><!-- 0..1 Version of software running -->
  <contact><!-- 0..1 ContactPoint Human contact for problems --></contact>
  <endpoint value="[url]"/><!-- 1..1 Actual message source address or id -->
 </source>
 <responsible><!-- 0..1 Reference(Practitioner|PractitionerRole|Organization) Final responsibility for event
    --></responsible>
 <reason><!-- 0..1 CodeableConcept Cause of event --></reason>
 <response>  <!-- 0..1 If this is a reply to prior message -->
  <identifier value="[id]"/><!-- 1..1 Id of original message -->
  <code value="[code]"/><!-- 1..1 ok | transient-error | fatal-error -->
  <details><!-- 0..1 Reference(OperationOutcome) Specific list of hints/warnings/errors --></details>
 </response>
 <focus><!-- 0..* Reference(Any) The actual content of the message --></focus>
 <definition><!-- 0..1 canonical(MessageDefinition) Link to the definition for this message --></definition>
</MessageHeader>
```

JSON Template

```
{
  "resourceType" : "MessageHeader",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  // event[x]: Code for the event this message represents or link to event definition. One of these 2:
  "eventCoding" : { Coding },
  "eventUri" : "<uri>",
  "destination" : [{ // Message destination application(s)
    "name" : "<string>", // Name of system
    "target" : { Reference(Device) }, // Particular delivery destination within the destination
    "endpoint" : "<url>", // R!  Actual destination address or id
    "receiver" : { Reference(Practitioner|PractitionerRole|Organization) } // Intended "real-world" recipient for the
    data
  }],
  "sender" : { Reference(Practitioner|PractitionerRole|Organization) }, // Real world sender of the message
  "enterer" : { Reference(Practitioner|PractitionerRole) }, // The source of the data entry
  "author" : { Reference(Practitioner|PractitionerRole) }, // The source of the decision
  "source" : { // R!  Message source application
    "name" : "<string>", // Name of system
    "software" : "<string>", // Name of software running the system
    "version" : "<string>", // Version of software running
    "contact" : { ContactPoint }, // Human contact for problems
    "endpoint" : "<url>" // R!  Actual message source address or id
  },
  "responsible" : { Reference(Practitioner|PractitionerRole|Organization) }, // Final responsibility for event
  "reason" : { CodeableConcept }, // Cause of event
  "response" : { // If this is a reply to prior message
    "identifier" : "<id>", // R!  Id of original message
    "code" : "<code>", // R!  ok | transient-error | fatal-error
    "details" : { Reference(OperationOutcome) } // Specific list of hints/warnings/errors
  },
  "focus" : [{ Reference(Any) }], // The actual content of the message
  "definition" : { canonical(MessageDefinition) } // Link to the definition for this message
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:MessageHeader;
```

```
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  # MessageHeader.event[x] : 1..1 Code for the event this message represents or link to event definition. One of these 2
    fhir:MessageHeader.eventCoding [ Coding ]
    fhir:MessageHeader.eventUri [ uri ]
  fhir:MessageHeader.destination [ # 0..* Message destination application(s)
    fhir:MessageHeader.destination.name [ string ]; # 0..1 Name of system
    fhir:MessageHeader.destination.target [ Reference(Device) ]; # 0..1 Particular delivery destination within the
    destination
    fhir:MessageHeader.destination.endpoint [ url ]; # 1..1 Actual destination address or id
    fhir:MessageHeader.destination.receiver [ Reference(Practitioner|PractitionerRole|Organization) ]; # 0..1 Intended
    "real-world" recipient for the data
  ], ...;
  fhir:MessageHeader.sender [ Reference(Practitioner|PractitionerRole|Organization) ]; # 0..1 Real world sender of the
    message
  fhir:MessageHeader.enterer [ Reference(Practitioner|PractitionerRole) ]; # 0..1 The source of the data entry
  fhir:MessageHeader.author [ Reference(Practitioner|PractitionerRole) ]; # 0..1 The source of the decision
  fhir:MessageHeader.source [ # 1..1 Message source application
    fhir:MessageHeader.source.name [ string ]; # 0..1 Name of system
    fhir:MessageHeader.source.software [ string ]; # 0..1 Name of software running the system
    fhir:MessageHeader.source.version [ string ]; # 0..1 Version of software running
    fhir:MessageHeader.source.contact [ ContactPoint ]; # 0..1 Human contact for problems
    fhir:MessageHeader.source.endpoint [ url ]; # 1..1 Actual message source address or id
  ];
  fhir:MessageHeader.responsible [ Reference(Practitioner|PractitionerRole|Organization) ]; # 0..1 Final responsibility
    for event
  fhir:MessageHeader.reason [ CodeableConcept ]; # 0..1 Cause of event
  fhir:MessageHeader.response [ # 0..1 If this is a reply to prior message
    fhir:MessageHeader.response.identifier [ id ]; # 1..1 Id of original message
    fhir:MessageHeader.response.code [ code ]; # 1..1 ok | transient-error | fatal-error
    fhir:MessageHeader.response.details [ Reference(OperationOutcome) ]; # 0..1 Specific list of hints/warnings/errors
  ];
  fhir:MessageHeader.focus [ Reference(Any) ], ... ; # 0..* The actual content of the message
  fhir:MessageHeader.definition [ canonical(MessageDefinition) ]; # 0..1 Link to the definition for this message
]
```

Changes since Release 3

MessageHeader
MessageHeader.event[x] | Renamed from event to event[x] Add Type uri
MessageHeader.destination.endpoint | Type changed from uri to url
MessageHeader.destination.receiver | Moved from MessageHeader to MessageHeader.destination Type Reference: Added Target Type PractitionerRole
MessageHeader.sender | Type Reference: Added Target Type PractitionerRole
MessageHeader.enterer | Type Reference: Added Target Type PractitionerRole
MessageHeader.author | Type Reference: Added Target Type PractitionerRole
MessageHeader.source.endpoint | Type changed from uri to url
MessageHeader.responsible | Type Reference: Added Target Type PractitionerRole
MessageHeader.response.code | Change value set from http://hl7.org/fhir/ValueSet/response-code to http://hl7.org/fhir/ValueSet/response-code|4.0.1
MessageHeader.definition | Added Element
MessageHeader.timestamp | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 1 test that all execute ok. All tests pass round-trip testing and 1 r3 resources are invalid (0 errors).)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

2.47.2.1 Terminology Bindings

Path | Definition | Type | Reference
MessageHeader.event[x] | One of the message events defined as part of this version of FHIR. | Example | Message Events
MessageHeader.reason | Reason for event occurrence. | Example | ExampleMessageReasonCodes
MessageHeader.response.code | The kind of response to a message. | Required | ResponseType

2.47.3 Notes:

• The resources referenced by the enterer, author and responsible elements may all be included in the message bundle or left out on the basis that the recipient (and any intermediaries) are able to locate/resolve the resources independently. The former would be suitable for loosely coupled systems, and the latter for tightly coupled systems. The messaging Capability statement for an application may reference a Structure Definition that describes how the bundling occurs

• The actual content of the focus resource is specified for each message event (see the list on the messaging page). Any resources referenced in the focus element are always included in the bundle

• If MessageHeader.source.endpoint and MessageHeader.destination.endpoint, are literal URLs, then they SHOULD identify the addresses to which messages can be delivered. If they are logical URIs (i.e.

non-dereferenceable), message delivery intermediaries must know how to deliver the message to the destination application.

• The time of the event will be found in the focus resource. The time of the message will be found in Bundle.timestamp

• The author and the receiver are not the actual technical systems - these are the human or organizations that make use of the technical systems

• A receiver is not obligated to reject messages which do not explicitly identify it as receiver (e.g. a tracker will get messages that are destined for some other system)

• The value set MessageEvent is populated by the authors of the resources declaring the events

2.47.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
author | reference | The source of the decision | MessageHeader.author (Practitioner, PractitionerRole)
code | token | ok | transient-error | fatal-error | MessageHeader.response.code
destination | string | Name of system | MessageHeader.destination.name
destination-uri | uri | Actual destination address or id | MessageHeader.destination.endpoint
enterer | reference | The source of the data entry | MessageHeader.enterer (Practitioner, PractitionerRole)
event | token | Code for the event this message represents or link to event definition | MessageHeader.event
focus | reference | The actual content of the message | MessageHeader.focus (Any)
receiver | reference | Intended "real-world" recipient for the data | MessageHeader.destination.receiver (Practitioner, Organization, PractitionerRole)
response-id | token | Id of original message | MessageHeader.response.identifier
responsible | reference | Final responsibility for event | MessageHeader.responsible (Practitioner, Organization, PractitionerRole)
sender | reference | Real world sender of the message | MessageHeader.sender (Practitioner, Organization, PractitionerRole)
source | string | Name of system | MessageHeader.source.name
source-uri | uri | Actual message source address or id | MessageHeader.source.endpoint
target | reference | Particular delivery destination within the destination | MessageHeader.destination.target (Device)

2.47.5 Resource MessageHeader - Examples

URL: https://hl7.org/fhir/R4/messageheader-examples.html

2.47.5 Resource MessageHeader - Examples

Infrastructure And Messaging Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Not Classified | Compartments: Device, Practitioner

Example Name | id | Format
A typical message header | 1cbdfb97-5859-48a4-8301-d54eab818d68 | XML | JSON | Turtle
A request to link two patients | 10bb101f-a121-4264-a920-67be9cb82c74 | XML | JSON | Turtle
A response to a request to link two patients | 3a0707d3-549e-4467-b8b8-5a2ab3800efe | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

2.47.6 Resource MessageHeader - Detailed Descriptions

URL: https://hl7.org/fhir/R4/messageheader-definitions.html

2.47.6 Resource MessageHeader - Detailed Descriptions

Infrastructure And Messaging Work Group | Maturity Level: 4 | Trial Use | Security Category: Not Classified | Compartments: Device, Practitioner

Detailed Descriptions for the elements in the MessageHeader resource.

MessageHeader
Element Id | MessageHeader
Definition | The header for a message exchange that is either requesting or responding to an action. The reference(s) that are the subject of the action as well as other information related to the action are typically transmitted in a bundle in which the MessageHeader resource instance is the first resource in the bundle.
Cardinality | 0..*
Type | DomainResource
Requirements | Many implementations are not prepared to use REST and need a messaging based infrastructure.
Summary | true
MessageHeader.event[x]
Element Id | MessageHeader.event[x]
Definition | Code that identifies the event this message represents and connects it with its definition. Events defined as part of the FHIR specification have the system value "http://terminology.hl7.org/CodeSystem/message-events". Alternatively uri to the EventDefinition.
Cardinality | 1..1

Terminology Binding | the Event List in the messaging framework
Type | Coding|uri
[x] Note | See Choice of Data Types for further information about how to use [x]
Requirements | Drives the behavior associated with this message.
Summary | true
Comments | The time of the event will be found in the focus resource. The time of the message will be found in Bundle.timestamp.
MessageHeader.destination
Element Id | MessageHeader.destination
Definition | The destination application which the message is intended for.
Cardinality | 0..*
Requirements | Indicates where message is to be sent for routing purposes. Allows verification of "am I the intended recipient".
Summary | true
Comments | There SHOULD be at least one destination, but in some circumstances, the source system is unaware of any particular destination system.
MessageHeader.destination.name
Element Id | MessageHeader.destination.name
Definition | Human-readable name for the target system.
Cardinality | 0..1
Type | string
Requirements | May be used for routing of response and/or to support audit.
Summary | true
MessageHeader.destination.target
Element Id | MessageHeader.destination.target
Definition | Identifies the target end system in situations where the initial message transmission is to an intermediary system.
Cardinality | 0..1
Type | Reference(Device)
Requirements | Supports multi-hop routing.
Summary | true
MessageHeader.destination.endpoint
Element Id | MessageHeader.destination.endpoint
Definition | Indicates where the message should be routed to.
Cardinality | 1..1
Type | url
Requirements | Identifies where to route the message.
Summary | true
Comments | The id may be a non-resolvable URI for systems that do not use standard network-based addresses.
MessageHeader.destination.receiver
Element Id | MessageHeader.destination.receiver
Definition | Allows data conveyed by a message to be addressed to a particular person or department when routing to a specific application isn't sufficient.
Cardinality | 0..1
Type | Reference(Practitioner | PractitionerRole | Organization)
Requirements | Allows routing beyond just the application level.
Summary | true
MessageHeader.sender
Element Id | MessageHeader.sender
Definition | Identifies the sending system to allow the use of a trust relationship.
Cardinality | 0..1
Type | Reference(Practitioner | PractitionerRole | Organization)
Requirements | Allows routing beyond just the application level.
Summary | true
Comments | Use case is for where a (trusted) sending system is responsible for multiple organizations, and therefore cannot differentiate based on source endpoint / authentication alone.
MessageHeader.enterer
Element Id | MessageHeader.enterer
Definition | The person or device that performed the data entry leading to this message. When there is more than one candidate, pick the most proximal to the message. Can provide other enterers in extensions.
Cardinality | 0..1
Type | Reference(Practitioner | PractitionerRole)
Requirements | Need to know for audit/traceback requirements and possibly for authorization.
Summary | true
Comments | Usually only for the request but can be used in a response.
MessageHeader.author
Element Id | MessageHeader.author
Definition | The logical author of the message - the person or device that decided the described event should happen. When there is more than one candidate, pick the most proximal to the MessageHeader. Can provide other authors in extensions.
Cardinality | 0..1
Type | Reference(Practitioner | PractitionerRole)
Requirements | Need to know for audit/traceback requirements and possibly for authorization.
Summary | true
Comments | Usually only for the request but can be used in a response.
MessageHeader.source
Element Id | MessageHeader.source

Definition | The source application from which this message originated.
Cardinality | 1..1
Requirements | Allows replies, supports audit.
Summary | true
MessageHeader.source.name
Element Id | MessageHeader.source.name
Definition | Human-readable name for the source system.
Cardinality | 0..1
Type | string
Requirements | May be used to support audit.
Summary | true
MessageHeader.source.software
Element Id | MessageHeader.source.software
Definition | May include configuration or other information useful in debugging.
Cardinality | 0..1
Type | string
Requirements | Supports audit and possibly interface engine behavior.
Summary | true
MessageHeader.source.version
Element Id | MessageHeader.source.version
Definition | Can convey versions of multiple systems in situations where a message passes through multiple hands.
Note | This is a business versionId, not a resource version id (see discussion)
Cardinality | 0..1
Type | string
Requirements | Supports audit and possibly interface engine behavior.
Summary | true
MessageHeader.source.contact
Element Id | MessageHeader.source.contact
Definition | An e-mail, phone, website or other contact point to use to resolve issues with message communications.
Cardinality | 0..1
Type | ContactPoint
Requirements | Allows escalation of technical issues.
Summary | true
MessageHeader.source.endpoint
Element Id | MessageHeader.source.endpoint
Definition | Identifies the routing target to send acknowledgements to.
Cardinality | 1..1
Type | url
Requirements | Identifies where to send responses, may influence security permissions.
Summary | true
Comments | The id may be a non-resolvable URI for systems that do not use standard network-based addresses.
MessageHeader.responsible
Element Id | MessageHeader.responsible
Definition | The person or organization that accepts overall responsibility for the contents of the message. The implication is that the message event happened under the policies of the responsible party.
Cardinality | 0..1
Type | Reference(Practitioner | PractitionerRole | Organization)
Requirements | Need to know for audit/traceback requirements and possibly for authorization.
Summary | true
Comments | Usually only for the request but can be used in a response.
MessageHeader.reason
Element Id | MessageHeader.reason
Definition | Coded indication of the cause for the event - indicates a reason for the occurrence of the event that is a focus of this message.
Cardinality | 0..1
Terminology Binding | Example Message Reason Codes (Example)
Type | CodeableConcept
Requirements | Need to be able to track why resources are being changed and report in the audit log/history of the resource. May affect authorization.
Summary | true
MessageHeader.response
Element Id | MessageHeader.response
Definition | Information about the message that this message is a response to. Only present if this message is a response.
Cardinality | 0..1
Summary | true
MessageHeader.response.identifier
Element Id | MessageHeader.response.identifier
Definition | The MessageHeader.id of the message to which this message is a response.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 1..1
Type | id
Requirements | Allows receiver to know what message is being responded to.

Summary | true
MessageHeader.response.code
Element Id | MessageHeader.response.code
Definition | Code that identifies the type of response to the message - whether it was successful or not, and whether it should be resent or not.
Cardinality | 1..1
Terminology Binding | ResponseType (Required)
Type | code
Requirements | Allows the sender of the acknowledge message to know if the request was successful or if action is needed.
Summary | true
Comments | This is a generic response to the request message. Specific data for the response will be found in MessageHeader.focus.
MessageHeader.response.details
Element Id | MessageHeader.response.details
Definition | Full details of any issues found in the message.
Cardinality | 0..1
Type | Reference(OperationOutcome)
Requirements | Allows the sender of the message to determine what the specific issues are.
Summary | true
Comments | This SHALL be contained in the bundle. If any of the issues are errors, the response code SHALL be an error.
MessageHeader.focus
Element Id | MessageHeader.focus
Definition | The actual data of the message - a reference to the root/focus class of the event.
Cardinality | 0..*
Type | Reference(Any)
Requirements | Every message event is about actual data, a single resource, that is identified in the definition of the event, and perhaps some or all linked resources.
Summary | true
Comments | The data is defined where the transaction type is defined. The transaction data is always included in the bundle that is the full message. Only the root resource is specified. The resources it references should be contained in the bundle but are not also listed here. Multiple repetitions are allowed to cater for merges and other situations with multiple focal targets.
MessageHeader.definition
Element Id | MessageHeader.definition
Definition | Permanent link to the MessageDefinition for this message.
Cardinality | 0..1
Type | canonical(MessageDefinition)
Requirements | Allows sender to define the expected contents of the message.
Summary | true

2.47.7 Resource MessageHeader - Mappings

URL: https://hl7.org/fhir/R4/messageheader-mappings.html

2.47.7 Resource MessageHeader - Mappings

Infrastructure And Messaging Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Not Classified | Compartments: Device, Practitioner

Mappings:

FiveWs Pattern Mapping

HL7 v2 Mapping

RIM Mapping

Mappings for the messageheader resource (see Mappings to Other Standards for further information & status).

2.47.7.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

MessageHeader
event[x] | FiveWs.what[x]
destination | FiveWs.subject[x]
receiver | FiveWs.subject[x]
sender | FiveWs.subject[x]
enterer | FiveWs.author
author | FiveWs.author
source | FiveWs.actor
responsible | FiveWs.witness
reason | FiveWs.why[x]

2.47.7.2 HL7 v2 Mapping (http://hl7.org/v2)

MessageHeader | MSH / MSA / ERR
event[x] | MSH-9.2
destination
name | MSH-5

target
endpoint | MSH-25 (or MSH-6)
receiver | PRT-5:PRT-4='WAYR' / PRT-8:PRT-4='WAYR'
sender | PRT-5:PRT-4='WAYR' / PRT-8:PRT-4='WAYR'
enterer | EVN-5 / ORC-10 / PRT-5:PRT-4='EP' / ROL where ROL.3 is EP or ORC.10
author | ORC-19 / PRT-5:PRT-4='AUT' / ROL where ROL.3 is IP or ORC.12
source
name | MSH-3
software | SFT-3 (+SFT-1)
version | SFT-2
contact | (MSH-22?)
endpoint | MSH-24
responsible | ORC-12 / PRT-5:PRT-4='OP' / PRT-8:PRT-4='OP' / ROL where ROL.3 is RO or ORC.11
reason | EVN.4 / ORC.16 / OBR-31-reason for study / BPO-13-BP indication for use / RXO-20-indication / RXE-27-give indication / RXD-21-indication / RXG-22-indication / RXA-19-indication
response | MSA
identifier | MSA-2
code | MSA-1
details | ERR
focus
definition

2.47.7.3 RIM Mapping (http://hl7.org/v3)

MessageHeader | Message
event[x] | ./payload[classCode="CACT" and moodCode="EVN" and isNormalAct()]/code[isNormalDatatype()]/code
destination | ./communicationFunction[typeCode="RCV"]
name | unique(./entity[determinerCode="INST" and (classCode=("DEV", "PSN", "ORG"]/name[type='ST' and isNormalDatatype()])
target | unique(./entity[determinerCode="INST" and (classCode=("DEV", "PSN", "ORG"]
endpoint | ./telecom
receiver | Not directly supported. Could be shared using attentionLine.value with an appropriate keyWordText
sender | Not directly supported. Could be shared using attentionLine.value with an appropriate keyWordText
enterer | unique(./controlAct[classCode="CACT" and classCode="EVN" and isNormalAct()]/participation[typeCode="ENT" and isNormalParticipation]/role[classCode="ASSIGNED" and isNormalRole()]/player[isNormalEntity() and classCode=("DEV", "PSN") and determinerCode="INST"])
author | unique(./controlAct[classCode="CACT" and classCode="EVN" and isNormalAct()]/participation[isHighest(priorityCode) and typeCode="AUT" and isNormalParticipation]/role[classCode="ASSIGNED" and isNormalRole()]/player[isNormalEntity() and classCode=("DEV", "PSN") and determinerCode="INST"])
source | unique(./communicationFunction[typeCode="SND"])
name | unique(./entity[determinerCode="INST" and classCode="DEV"]/name[type='ST' and isNormalDatatype()])
software | unique(./entity[determinerCode="INST" and classCode="DEV"]/softwareName[isNormalDatatype()])
version | (Not supported)
contact | unique(./entity[determinerCode="INST" and classCode="DEV"]/scopesRole[normalRole() and classCode="CON" and layer[classCode="PSN"]/telecom[isNormalDatatype()])
endpoint | ./telecom[isNormalDatatype()]
responsible | unique(./controlAct[classCode="CACT" and classCode="EVN" and isNormalAct()]/participation[isHighest(priorityCode) and typeCode="RESP" and isNormalParticipation]/role[classCode="ASSIGNED" and isNormalRole()]/player[isNormalEntity() and classCode="PSN" and determinerCode="INST"] or ./controlAct[classCode="CACT" and classCode="EVN" and isNormalAct()]/participation[isHighest(priorityCode) and typeCode="RESP" and isNormalParticipation]/role[classCode="ASSIGNED" and isNormalRole()]/scoper[isNormalEntity() and classCode="ORG" and determinerCode="INST"])
reason | unique(./controlAct[classCode="CACT" and classCode="EVN" and isNormalAct()]/reasonCode[isNormalDatatype()])
response | ./conveyedAcknowledgement
identifier | ./acknowledges.id[isNormalDatatype()]
code | ./typeCode[isNormalDatatype()]
details | AcknowledgementDetail or Observation[classCode="ALRT" and moodCode="EVN"]
focus | unique(./controlAct[classCode="CACT" and classCode="EVN" and isNormalAct()]/participation[typeCode="SUBJ" and isNormalParticipation]/role or ./controlAct[classCode="CACT" and classCode="EVN" and isNormalAct()]/outboundRelationship[typeCode="SBJ" and isNormalActRelatoinship]/target)
definition | Not directly supported.

2.47.8 Resource MessageHeader - Extensions & Profiles

URL: https://hl7.org/fhir/R4/messageheader-profiles.html

2.47.8 Resource MessageHeader - Extensions & Profiles

Infrastructure And Messaging Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Not Classified | Compartments: Device, Practitioner

This table lists profiles and extensions for the MessageHeader resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

2.47.8.1 Profiles

No Profiles defined for this resource

2.47.8.2 Extensions

messageheader-response-request | MessageHeader | messageheader-response-request | for Common extensions for message header
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

2.47.8.3 Search Extensions

No Search Extensions defined for this resource

2.47.9 Resource MessageHeader - Operations

URL: https://hl7.org/fhir/R4/messageheader-operations.html

2.47.9 Resource MessageHeader - Operations

Infrastructure And Messaging Work Group | Maturity Level: 4 | Trial Use | Security Category: Not Classified | Compartments: Device, Practitioner

This resource has 1 operation operations associated with it:

$process-message | Process Message

For more information about operations, including how they are invoked, see Operations.

2.47.9.1 Operation $process-message on MessageHeader

URL: https://hl7.org/fhir/R4/messageheader-operation-process-message.html

2.47.9.1 Operation $process-message on MessageHeader

Infrastructure And Messaging Work Group | Maturity Level: 4 | Trial Use | Compartments: N/A

This operation accepts a message, processes it according to the definition of the event in the message header, and returns one or more response messages.

In addition to processing the message event, a server may choose to retain all or some the resources and make them available on a RESTful interface, but is not required to do so.

The official URL for this operation definition is

```
http://hl7.org/fhir/OperationDefinition/MessageHeader-process-message
```

Formal Definition (as a OperationDefinition).

URL: [base]/$process-message

This is not an idempotent operation

In Parameters:
Name | Cardinality | Type | Binding | Profile | Documentation
content | 1..1 | Bundle | | | The message to process (or, if using asynchronous messaging, it may be a response message to accept)
async | 0..1 | boolean | | | If 'true' the message is processed using the asynchronous messaging pattern
response-url | 0..1 | url | | | A URL to submit response messages to, if asynchronous messaging is being used, and if the MessageHeader.source.endpoint is not the appropriate place to submit responses
Out Parameters:
Name | Cardinality | Type | Binding | Profile | Documentation
return | 0..1 | Bundle | | | A response message, if synchronous messaging is being used (mandatory in this case). For asynchronous messaging, there is no return value Note: as this is the only out parameter, it is a resource, and it has the name 'return', the result of this operation is returned directly as a resource

This operation does not use the parameters resource; the parameters "async" and "response-url" always go in the URL, if they are used, and the "content" parameter is always the body of the HTTP message.

When processing messages, a server may return one of several status codes:

• 200 OK: Indicates that the message has been fully processed. If an application-level response is expected for the submitted message, that response SHALL be returned as the body of the 200 response.

• 202 Accepted: Indicates that the receiving system has accepted custody of the message

• 204 No Content: Indicates that the message has been fully processed and would normally have had an application-level response, but because of instructions from the sender (e.g. the messageheader-response-request extension), no response is being provided

• 300+: Indicates that the message was not successfully processed. The server MAY return an OperationOutcome with additional information, and SHOULD do so if the response code is 400 or greater.
 The client SHALL interpret a 4xx response to indicate that there is no point resubmitting the unaltered message, and a 5xx response to indicate an unexpected error occurred on the part of the server, with the implication that it may be appropriate to resubmit the original message. Doing so SHOULD NOT result in a duplicate message response. Repeated failures indicate either a fatal problem with the submission or a problem with the receiving application.

The following rules apply when using $process-message:

• The operation only accepts POST transactions - any other HTTP method will result in an HTTP error

• The request content type submitted is always Bundle with type "message" containing a Message Header resource as the first resource

• The response content type returned is always Bundle with type "message" containing a Message Header resource as the first resource, or an HTTP error

• If the response is an error, the body SHOULD be an Errors ∓ Warning resource with full details

• The mailbox may be authenticated using standard HTTP authentication methods, including OAuth

The $process-message operation can be used by any HTTP end-point that accepts FHIR messages, not just FHIR RESTful servers.

In order to ensure consistency of processing, the logical rules regarding processing of Bundle.id and message id SHALL be followed when messages are processed using this operation.

The $process-message operation may be used synchronously, or asynchronously.

The following rules apply when using the $process-message operation synchronously:

• The URL (http://server/base/$process-message) has no parameters

• It is an error if the sender POSTs a message that requires multiple response messages

• Servers SHALL accept multiple concurrent message submissions and process them correctly (they are allowed to process them sequentially internally, but multiple concurrent submissions is not an error in its own right)

The following rules apply when using the $process-message operation asynchronously:

• The URL has at least one parameter: http://server/base/$process-message?async=true

• The server acknowledges the message with a 200 OK with no body, or returns an HTTP error if the message cannot be processed

• Accepting the message means that the server has understood the message enough to know where to respond

• An OperationOutcome SHOULD be returned in either case

• By default, the server responds by invoking the $process-message using the sender's stated end-point in the message: POST [MessageHeader.source.endpoint]/$process-messages]

• Since the source end-point may be manipulated by message transfer engines, an alternative response address may be specified using the parameter "response-url": http://server/base/$process-message?async=true&response-url=http://server2.com/base/anything. The endpoint at the specified URL SHALL implement the signature of the $process-message operation (parameter async=true, accept a Bundle, return a 200 OK or an error)

• The server submits response messages to the appropriate end-point with the parameter async=true. There is no response message for the response messages

2.47.9.1.1 Examples

Request: Request to link 2 patients, and respond asynchronously

```
POST /ehr/fhir/$process-message?aync=true&response-url=http://example.org/clients/ehr-lite
[other headers]

<?xml version="1.0" encoding="UTF-8"?>

<Bundle xmlns="http://hl7.org/fhir">
  <id value="10bb101f-a121-4264-a920-67be9cb82c74"/>
  <type value="message"/>
  <timestamp value="2015-07-14T11:15:33+10:00"/>
  <entry>
    <fullUrl value="urn:uuid:267b18ce-3d37-4581-9baa-6fada338038b"/>
```

```
  <resource>
    <MessageHeader>
      <!--
        Note about the id: according to the spec, the fullUrl must not
        disagree with the id in the resource. For instance, if the fullUrl
        was http://server/path/MessageHeader/267b18ce-3d37-4581-9baa-6fada338038b,
        then the id of the resource must be 267b18ce-3d37-4581-9baa-6fada338038b.

        However, in this case, a fullUrl of urn:uuid:[x] makes no claim about
        the id of the resource, and it can be different, or missing. Note, though
        that this doesn't really make sense; if it's present, there's less
        potential for confusion if they match, like in this case
      -->
      <id value="267b18ce-3d37-4581-9baa-6fada338038b"/>
      <text>
        <status value="generated"/>
        <div xmlns="http://www.w3.org/1999/xhtml">

          <p>This message is a request to link Patient records 654321 (Patient Donald DUCK @ Acme Healthcare, Inc) and
   123456 (Patient Donald D DUCK @ Acme Healthcare, Inc)</p>

        </div>
      </text>
      <eventCoding>
        <system value="http://example.org/fhir/message-events"/>
        <code value="patient-link"/>
      </eventCoding>
      <source>
        <endpoint value="http://example.org/clients/ehr-lite"/>
      </source>
      <responsible>
        <reference value="http://acme.com/ehr/fhir/Practitioner/2323-33-4"/>
      </responsible>
    <!--  this message is posted to http://acme.com/ehr/fhir,
           with an event 'link' to link 2 patient records, and nominates
           2 patients on the server  -->
      <focus>
        <reference value="http://acme.com/ehr/fhir/Patient/pat1"/>
      </focus>
      <focus>
        <reference value="http://acme.com/ehr/fhir/Patient/pat12"/>
      </focus>
    </MessageHeader>
  </resource>
</entry>
<entry>
  <fullUrl value="http://acme.com/ehr/fhir/Patient/pat1"/>
  <resource>
    <Patient>
      <id value="pat1"/>
      <text>
        <status value="generated"/>
        <div xmlns="http://www.w3.org/1999/xhtml">

          <p>Patient Donald DUCK @ Acme Healthcare, Inc. MR = 654321</p>

        </div>
      </text>
      <identifier>
        <use value="usual"/>
        <type>
          <coding>
            <system value="http://terminology.hl7.org/CodeSystem/v2-0203"/>
            <code value="MR"/>
          </coding>
        </type>
        <system value="urn:oid:0.1.2.3.4.5.6.7"/>
        <value value="654321"/>
      </identifier>
      <active value="true"/>
      <name>
        <use value="official"/>
        <family value="Donald"/>
        <given value="Duck"/>
      </name>
      <gender value="male"/>
      <contact>
        <relationship>
          <coding>
            <system value="http://example.org/fhir/CodeSystem/patient-contact-relationship"/>
            <code value="E"/>
          </coding>
        </relationship>
        <organization>
          <reference value="Organization/1"/>
          <display value="Walt Disney Corporation"/>
        </organization>
      </contact>
      <managingOrganization>
```

```
        <reference value="Organization/1"/>
        <display value="ACME Healthcare, Inc"/>
      </managingOrganization>
    </Patient>
  </resource>
</entry>
<entry>
  <fullUrl value="http://acme.com/ehr/fhir/Patient/pat12"/>
  <resource>
    <Patient>
      <id value="pat2"/>
      <text>
        <status value="generated"/>
        <div xmlns="http://www.w3.org/1999/xhtml">

          <p>Patient Donald D DUCK @ Acme Healthcare, Inc. MR = 123456</p>

        </div>
      </text>
      <identifier>
        <use value="usual"/>
        <type>
          <coding>
            <system value="http://terminology.hl7.org/CodeSystem/v2-0203"/>
            <code value="MR"/>
          </coding>
        </type>
        <system value="urn:oid:0.1.2.3.4.5.6.7"/>
        <value value="123456"/>
      </identifier>
      <active value="true"/>
      <name>
        <use value="official"/>
        <family value="Donald"/>
        <given value="Duck"/>
        <given value="D"/>
      </name>
      <gender value="other">
        <extension url="http://example.org/Profile/administrative-status">
          <valueCodeableConcept>
            <coding>
              <system value="http://terminology.hl7.org/CodeSystem/v2-0001"/>
              <code value="A"/>
              <display value="Ambiguous"/>
            </coding>
          </valueCodeableConcept>
        </extension>
      </gender>
      <managingOrganization>
        <reference value="Organization/1"/>
        <display value="ACME Healthcare, Inc"/>
      </managingOrganization>
    </Patient>
  </resource>
</entry>
</Bundle>
```

Response: Response - since this is asynchronous, the response is empty

```
200 OK
[other headers]
```

Response: Response - if the message couldn't be accepted (e.g. there'll be no asynchronous response

```
500 OK Internal Server Error
[other headers]

<?xml version="1.0" encoding="UTF-8"?>
<OperationOutcome xmlns="http://hl7.org/fhir">
  <id value="exception"/>
  <text>
    <status value="generated"/>
    <div xmlns="http://www.w3.org/1999/xhtml">
      <p>SQL Link Communication Error (dbx = 34234)</p>
    </div>
  </text>
  <issue>
    <severity value="error"/>
    <code value="exception"/>
    <details>
      <text value="SQL Link Communication Error (dbx = 34234)"/>
    </details>
  </issue>
</OperationOutcome>
```

For more information about operations, including how they are invoked, see Operations.

2.47.10 Resource MessageHeader - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/messageheader-version-maps.html

No validation errors - all conversions are clean

2.47.10 Resource MessageHeader - R3/R4 Conversions

Infrastructure And Messaging Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Not Classified | Compartments: Device, Practitioner

R3 : R4 Conversion maps for MessageHeader.

Functional status for this map: 1 test that all execute ok. All tests pass round-trip testing and 1 r3 resources are invalid (0 errors). (see documentation)

2.47.10.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/MessageHeader3to4" = "R3 to R4 Conversions for MessageHeader"

uses "http://hl7.org/fhir/3.0/StructureDefinition/MessageHeader" alias MessageHeaderR3 as source
uses "http://hl7.org/fhir/StructureDefinition/MessageHeader" alias MessageHeader as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group MessageHeader(source src : MessageHeaderR3, target tgt : MessageHeader) extends DomainResource <<type+>> {
  src.event -> tgt.event;
  src.destination as vs0 -> tgt.destination as vt0 then destination(vs0, vt0, src);
  src.sender -> tgt.sender;
  src.timestamp as vs ->  tgt.extension as ext,  ext.url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    MessageHeader.timestamp',  ext.value = vs;
  src.enterer -> tgt.enterer;
  src.author -> tgt.author;
  src.source as vs0 -> tgt.source as vt0 then source(vs0, vt0);
  src.responsible -> tgt.responsible;
  src.reason -> tgt.reason;
  src.response as vs0 -> tgt.response as vt0 then response(vs0, vt0);
  src.focus -> tgt.focus;
}

group destination(source src, target tgt, source msrc) extends BackboneElement {
  src.name -> tgt.name;
  src.target -> tgt.target;
  src.endpoint -> tgt.endpoint;
  msrc.receiver -> tgt.receiver;
}

group source(source src, target tgt) extends BackboneElement {
  src.name -> tgt.name;
  src.software -> tgt.software;
  src.version -> tgt.version;
  src.contact -> tgt.contact;
  src.endpoint -> tgt.endpoint;
}

group response(source src, target tgt) extends BackboneElement {
  src.identifier -> tgt.identifier;
  src.code -> tgt.code;
  src.details -> tgt.details;
}
```

2.47.10.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/MessageHeader4to3" = "R4 to R3 Conversion for MessageHeader"

uses "http://hl7.org/fhir/StructureDefinition/MessageHeader" alias MessageHeader as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/MessageHeader" alias MessageHeaderR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group MessageHeader(source src : MessageHeader, target tgt : MessageHeaderR3) extends DomainResource <<type+>> {
  src.event -> tgt.event;
  src.destination as vs0 -> tgt.destination as vt0 then destination(vs0, vt0, tgt);
  src.sender -> tgt.sender;
  src.extension as ext where url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-MessageHeader.timestamp' then
    {
    ext.value -> tgt.timestamp "timestamp2";
  } "timestamp";
  src.enterer -> tgt.enterer;
  src.author -> tgt.author;
  src.source as vs0 -> tgt.source as vt0 then source(vs0, vt0);
  src.responsible -> tgt.responsible;
  src.reason -> tgt.reason;
  src.response as vs0 -> tgt.response as vt0 then response(vs0, vt0);
  src.focus -> tgt.focus;
}

group destination(source src, target tgt, target mtgt) extends BackboneElement {
  src.name -> tgt.name;
  src.target -> tgt.target;
  src.endpoint -> tgt.endpoint;
  src.receiver -> mtgt.receiver;
```

```
}

group source(source src, target tgt) extends BackboneElement {
  src.name -> tgt.name;
  src.software -> tgt.software;
  src.version -> tgt.version;
  src.contact -> tgt.contact;
  src.endpoint -> tgt.endpoint;
}

group response(source src, target tgt) extends BackboneElement {
  src.identifier -> tgt.identifier;
  src.code -> tgt.code;
  src.details -> tgt.details;
}
```

2.47.10.3 R4 Validation Errors

2.47.11 Common extensions for message header

URL: https://hl7.org/fhir/R4/messageheader-extensions.html

Defines "common" extensions for use with the messageheader resource and contained elements

2.47.11 Common extensions for message header

Infrastructure And Messaging Work Group | Maturity Level: N/A | Standards Status: Informative

2.47.11.1 Content

Extensions:

messageheader-response-request | messageheader-response-request : This extension enables the capability currently available through MSH-16 (Application Level acknowledgement) in HL7 Version 2 to declare at a message instance level whether a response is required or only upon error or success, or never.

2.47.12 JSON Schema for MessageHeader

URL: https://hl7.org/fhir/R4/messageheader.schema.json.html

2.47.12 JSON Schema for MessageHeader

Infrastructure And Messaging Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Device, Practitioner

This is a representation of the json schema for MessageHeader, which is just a part of the full JSON Schema.

```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/MessageHeader",
  "$ref": "#/definitions/MessageHeader",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "MessageHeader": {
      "description": "The header for a message exchange that is either requesting or responding to an action.  The
    reference(s) that are the subject of the action as well as other information related to the action are typically
    transmitted in a bundle in which the MessageHeader resource instance is the first resource in the bundle.",
      "properties": {
        "resourceType": {
          "description": "This is a MessageHeader resource",
          "const": "MessageHeader"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
```

```
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "text": {
      "description": "A human-readable narrative that contains a summary of the resource and can be used to
represent the content of the resource to a human. The narrative need not encode all the structured data, but is
required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
      "$ref": "Narrative.schema.json#/definitions/Narrative"
    },
    "contained": {
      "description": "These resources do not have an independent existence apart from the resource that contains
them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
 definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "eventCoding": {
      "description": "Code that identifies the event this message represents and connects it with its definition.
Events defined as part of the FHIR specification have the system value
\"http://terminology.hl7.org/CodeSystem/message-events\".  Alternatively uri to the EventDefinition.",
      "$ref": "Coding.schema.json#/definitions/Coding"
    },
    "eventUri": {
      "description": "Code that identifies the event this message represents and connects it with its definition.
Events defined as part of the FHIR specification have the system value
\"http://terminology.hl7.org/CodeSystem/message-events\".  Alternatively uri to the EventDefinition.",
      "pattern": "^\\S*$",
      "type": "string"
    },
    "_eventUri": {
      "description": "Extensions for eventUri",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "destination": {
      "description": "The destination application which the message is intended for.",
      "items": {
        "$ref": "#/definitions/MessageHeader_Destination"
      },
      "type": "array"
    },
    "sender": {
      "description": "Identifies the sending system to allow the use of a trust relationship.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "enterer": {
      "description": "The person or device that performed the data entry leading to this message. When there is more
 than one candidate, pick the most proximal to the message. Can provide other enterers in extensions.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "author": {
      "description": "The logical author of the message - the person or device that decided the described event
should happen. When there is more than one candidate, pick the most proximal to the MessageHeader. Can provide other
 authors in extensions.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "source": {
      "description": "The source application from which this message originated.",
      "$ref": "#/definitions/MessageHeader_Source"
    },
    "responsible": {
      "description": "The person or organization that accepts overall responsibility for the contents of the
message. The implication is that the message event happened under the policies of the responsible party.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
```

```
    "reason": {
      "description": "Coded indication of the cause for the event - indicates  a reason for the occurrence of the
event that is a focus of this message.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "response": {
      "description": "Information about the message that this message is a response to.  Only present if this
message is a response.",
      "$ref": "#/definitions/MessageHeader_Response"
    },
    "focus": {
      "description": "The actual data of the message - a reference to the root/focus class of the event.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "definition": {
      "description": "Permanent link to the MessageDefinition for this message.",
      "$ref": "canonical.schema.json#/definitions/canonical"
    }
  },
  "additionalProperties": false,
  "required": [
    "source",
    "resourceType"
  ]
},
"MessageHeader_Destination": {
  "description": "The header for a message exchange that is either requesting or responding to an action.  The
reference(s) that are the subject of the action as well as other information related to the action are typically
transmitted in a bundle in which the MessageHeader resource instance is the first resource in the bundle.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "name": {
      "description": "Human-readable name for the target system.",
      "$ref": "#/definitions/string"
    },
    "_name": {
      "description": "Extensions for name",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "target": {
      "description": "Identifies the target end system in situations where the initial message transmission is to an
 intermediary system.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "endpoint": {
      "description": "Indicates where the message should be routed to.",
      "$ref": "#/definitions/url"
    },
    "_endpoint": {
      "description": "Extensions for endpoint",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "receiver": {
      "description": "Allows data conveyed by a message to be addressed to a particular person or department when
routing to a specific application isn\u0027t sufficient.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    }
  },
```

```
  "additionalProperties": false
},
"MessageHeader_Source": {
  "description": "The header for a message exchange that is either requesting or responding to an action.  The
reference(s) that are the subject of the action as well as other information related to the action are typically
transmitted in a bundle in which the MessageHeader resource instance is the first resource in the bundle.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "name": {
      "description": "Human-readable name for the source system.",
      "$ref": "#/definitions/string"
    },
    "_name": {
      "description": "Extensions for name",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "software": {
      "description": "May include configuration or other information useful in debugging.",
      "$ref": "#/definitions/string"
    },
    "_software": {
      "description": "Extensions for software",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "version": {
      "description": "Can convey versions of multiple systems in situations where a message passes through multiple
hands.",
      "$ref": "#/definitions/string"
    },
    "_version": {
      "description": "Extensions for version",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "contact": {
      "description": "An e-mail, phone, website or other contact point to use to resolve issues with message
communications.",
      "$ref": "ContactPoint.schema.json#/definitions/ContactPoint"
    },
    "endpoint": {
      "description": "Identifies the routing target to send acknowledgements to.",
      "$ref": "#/definitions/url"
    },
    "_endpoint": {
      "description": "Extensions for endpoint",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
},
"MessageHeader_Response": {
  "description": "The header for a message exchange that is either requesting or responding to an action.  The
reference(s) that are the subject of the action as well as other information related to the action are typically
transmitted in a bundle in which the MessageHeader resource instance is the first resource in the bundle.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
```

```
    the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
    definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "modifierExtension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the element and that modifies the understanding of the element in which it is contained and/or the understanding of
    the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
    use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
     of the definition of the extension. Applications processing a resource are required to check for modifier
    extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
    (including cannot change the meaning of modifierExtension itself).",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "identifier": {
          "description": "The MessageHeader.id of the message to which this message is a response.",
          "$ref": "#/definitions/id"
        },
        "_identifier": {
          "description": "Extensions for identifier",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "code": {
          "description": "Code that identifies the type of response to the message - whether it was successful or not,
    and whether it should be resent or not.",
          "enum": [
            "ok",
            "transient-error",
            "fatal-error"
          ]
        },
        "_code": {
          "description": "Extensions for code",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "details": {
          "description": "Full details of any issues found in the message.",
          "$ref": "Reference.schema.json#/definitions/Reference"
        }
      },
      "additionalProperties": false
    }
  }
}
```


2.47.13 Resource MessageHeader - Maturity Dependency View

URL: https://hl7.org/fhir/R4/messageheader-dependencies.html

2.47.13 Resource MessageHeader - Maturity Dependency View

Infrastructure And Messaging Work Group | Maturity Level: 4 | Trial Use | Compartments: Device, Practitioner

Dependency Graph for MessageHeader FMM level 4

Name | Card. | Type | Dependency Analysis
MessageHeader | | DomainResource
event[x] | 1..1
eventCoding | | Coding
eventUri | | uri
destination | 0..* | BackboneElement
name | 0..1 | string
target | 0..1 | Reference(Device) | Type Warning: (Device = FMM2-Trial Use vs. Element = FMM4-Trial Use)
endpoint | 1..1 | url
receiver | 0..1 | Reference(Practitioner | PractitionerRole | Organization) | Type Warning: (Practitioner = FMM3-Trial Use vs. Element = FMM4-Trial Use) Type Warning: (PractitionerRole = FMM2-Trial Use vs. Element = FMM4-Trial Use) Type Warning: (Organization = FMM3-Trial Use vs. Element = FMM4-Trial Use)
sender | 0..1 | Reference(Practitioner | PractitionerRole | Organization) | Type Warning: (Practitioner = FMM3-Trial Use vs. Element = FMM4-Trial Use) Type Warning: (PractitionerRole = FMM2-Trial Use vs. Element = FMM4-Trial Use) Type Warning: (Organization = FMM3-Trial Use vs. Element = FMM4-Trial Use)
enterer | 0..1 | Reference(Practitioner | PractitionerRole) | Type Warning: (Practitioner = FMM3-Trial Use vs. Element = FMM4-Trial Use) Type Warning: (PractitionerRole = FMM2-Trial Use vs. Element = FMM4-Trial Use)
author | 0..1 | Reference(Practitioner | PractitionerRole) | Type Warning: (Practitioner = FMM3-Trial Use vs. Element = FMM4-Trial Use) Type Warning: (PractitionerRole = FMM2-Trial Use vs. Element = FMM4-Trial Use)
source | 1..1 | BackboneElement
name | 0..1 | string

software | 0..1 | string
version | 0..1 | string
contact | 0..1 | ContactPoint
endpoint | 1..1 | url
responsible | 0..1 | Reference(Practitioner | PractitionerRole | Organization) | Type Warning: (Practitioner = FMM3-Trial Use vs. Element = FMM4-Trial Use) Type Warning: (PractitionerRole = FMM2-Trial Use vs. Element = FMM4-Trial Use) Type Warning: (Organization = FMM3-Trial Use vs. Element = FMM4-Trial Use)
reason | 0..1 | CodeableConcept
response | 0..1 | BackboneElement
identifier | 1..1 | id
code | 1..1 | code
details | 0..1 | Reference(OperationOutcome)
focus | 0..* | Reference(Any) | Type Warning: (Any = FMM1-Trial Use vs. Element = FMM4-Trial Use)
definition | 0..1 | canonical(MessageDefinition) | Type Warning: (MessageDefinition = FMM1-Trial Use vs. Element = FMM4-Trial Use)

2.48 Resource MessageDefinition - Content

URL: https://hl7.org/fhir/R4/messagedefinition.html
Infrastructure And Messaging Work Group | Maturity Level: 1 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Defines the characteristics of a message that can be shared between systems, including the type of event that initiates the message, the content to be transmitted and what response(s), if any, are permitted.

2.48.1 Scope and Usage

Implementers should be familiar with the messaging concepts and descriptions found in Messaging using FHIR Resources before working with this resource.

This resource is referenced by CapabilityStatement, itself and MessageHeader

2.48.2 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
MessageDefinition | ITU | | DomainResource | A resource that defines a type of message that can be exchanged between systems + Warning: Name should be usable as an identifier for the module by machine processing applications such as code generation Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
url | Σ | 0..1 | uri | Business Identifier for a given MessageDefinition
identifier | Σ | 0..* | Identifier | Primary key for the message definition on a given server
version | Σ | 0..1 | string | Business version of the message definition
name | ΣI | 0..1 | string | Name for this message definition (computer friendly)
title | Σ | 0..1 | string | Name for this message definition (human friendly)
replaces | Σ | 0..* | canonical(MessageDefinition) | Takes the place of
status | ?!Σ | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)
experimental | Σ | 0..1 | boolean | For testing purposes, not real usage
date | Σ | 1..1 | dateTime | Date last changed
publisher | Σ | 0..1 | string | Name of the publisher (organization or individual)
contact | Σ | 0..* | ContactDetail | Contact details for the publisher
description | Σ | 0..1 | markdown | Natural language description of the message definition
useContext | Σ | 0..* | UsageContext | The context that the content is intended to support
jurisdiction | Σ | 0..* | CodeableConcept | Intended jurisdiction for message definition (if applicable) Jurisdiction (Extensible)
purpose | Σ | 0..1 | markdown | Why this message definition is defined
copyright | | 0..1 | markdown | Use and/or publishing restrictions
base | Σ | 0..1 | canonical(MessageDefinition) | Definition this one is based on
parent | Σ | 0..* | canonical(ActivityDefinition | PlanDefinition) | Protocol/workflow this is part of
event[x] | Σ | 1..1 | | Event code or link to the EventDefinition MessageEvent (Example)
eventCoding | | | Coding
eventUri | | | uri
category | Σ | 0..1 | code | consequence | currency | notification MessageSignificanceCategory (Required)
focus | ΣI | 0..* | BackboneElement | Resource(s) that are the subject of the event + Rule: Max must be postive int or *
code | Σ | 1..1 | code | Type of resource ResourceType (Required)
profile | | 0..1 | canonical(StructureDefinition) | Profile that must be adhered to by focus

min | Σ | 1..1 | unsignedInt | Minimum number of focuses of this type
max | I | 0..1 | string | Maximum number of focuses of this type
responseRequired | | 0..1 | code | always | on-error | never | on-success messageheader-response-request (Required)
allowedResponse | | 0..* | BackboneElement | Responses to this message
message | | 1..1 | canonical(MessageDefinition) | Reference to allowed message definition response
situation | | 0..1 | markdown | When should this response be used
graph | | 0..* | canonical(GraphDefinition) | Canonical reference to a GraphDefinition
Documentation for this format

UML Diagram (Legend)

XML Template

```
<MessageDefinition xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 0..1 Business Identifier for a given MessageDefinition -->
 <identifier><!-- 0..* Identifier Primary key for the message definition on a given server --></identifier>
 <version value="[string]"/><!-- 0..1 Business version of the message definition -->
 <name value="[string]"/><!--  0..1 Name for this message definition (computer friendly) -->
 <title value="[string]"/><!-- 0..1 Name for this message definition (human friendly) -->
 <replaces><!-- 0..* canonical(MessageDefinition) Takes the place of --></replaces>
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
 <date value="[dateTime]"/><!-- 1..1 Date last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!-- 0..1 Natural language description of the message definition -->
 <useContext><!-- 0..* UsageContext The context that the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for message definition (if applicable) --></jurisdiction>
 <purpose value="[markdown]"/><!-- 0..1 Why this message definition is defined -->
 <copyright value="[markdown]"/><!-- 0..1 Use and/or publishing restrictions -->
 <base><!-- 0..1 canonical(MessageDefinition) Definition this one is based on --></base>
 <parent><!-- 0..* canonical(ActivityDefinition|PlanDefinition) Protocol/workflow this is part of --></parent>
 <event[x]><!-- 1..1 Coding|uri Event code  or link to the EventDefinition --></event[x]>
 <category value="[code]"/><!-- 0..1 consequence | currency | notification -->
 <focus>  <!-- 0..* Resource(s) that are the subject of the event -->
  <code value="[code]"/><!-- 1..1 Type of resource -->
  <profile><!-- 0..1 canonical(StructureDefinition) Profile that must be adhered to by focus --></profile>
  <min value="[unsignedInt]"/><!-- 1..1 Minimum number of focuses of this type -->
  <max value="[string]"/><!--  0..1 Maximum number of focuses of this type -->
 </focus>
 <responseRequired value="[code]"/><!-- 0..1 always | on-error | never | on-success -->
 <allowedResponse>  <!-- 0..* Responses to this message -->
  <message><!-- 1..1 canonical(MessageDefinition) Reference to allowed message definition response --></message>
  <situation value="[markdown]"/><!-- 0..1 When should this response be used -->
 </allowedResponse>
 <graph><!-- 0..* canonical(GraphDefinition) Canonical reference to a GraphDefinition --></graph>
</MessageDefinition>
```

JSON Template

```
{
  "resourceType" : "MessageDefinition",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // Business Identifier for a given MessageDefinition
  "identifier" : [{ Identifier }], // Primary key for the message definition on a given server
  "version" : "<string>", // Business version of the message definition
  "name" : "<string>", // C? Name for this message definition (computer friendly)
  "title" : "<string>", // Name for this message definition (human friendly)
  "replaces" : [{ canonical(MessageDefinition) }], // Takes the place of
  "status" : "<code>", // R!  draft | active | retired | unknown
  "experimental" : <boolean>, // For testing purposes, not real usage
  "date" : "<dateTime>", // R!  Date last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "description" : "<markdown>", // Natural language description of the message definition
  "useContext" : [{ UsageContext }], // The context that the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for message definition (if applicable)
  "purpose" : "<markdown>", // Why this message definition is defined
  "copyright" : "<markdown>", // Use and/or publishing restrictions
  "base" : { canonical(MessageDefinition) }, // Definition this one is based on
  "parent" : [{ canonical(ActivityDefinition|PlanDefinition) }], // Protocol/workflow this is part of
  // event[x]: Event code  or link to the EventDefinition. One of these 2:
  "eventCoding" : { Coding },
  "eventUri" : "<uri>",
  "category" : "<code>", // consequence | currency | notification
  "focus" : [{ // Resource(s) that are the subject of the event
    "code" : "<code>", // R!  Type of resource
    "profile" : { canonical(StructureDefinition) }, // Profile that must be adhered to by focus
    "min" : "<unsignedInt>", // R!  Minimum number of focuses of this type
    "max" : "<string>" // C? Maximum number of focuses of this type
  }],
  "responseRequired" : "<code>", // always | on-error | never | on-success
  "allowedResponse" : [{ // Responses to this message
    "message" : { canonical(MessageDefinition) }, // R!  Reference to allowed message definition response
    "situation" : "<markdown>" // When should this response be used
  }],
```

```
  "graph" : [{ canonical(GraphDefinition) }] // Canonical reference to a GraphDefinition
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:MessageDefinition;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:MessageDefinition.url [ uri ]; # 0..1 Business Identifier for a given MessageDefinition
  fhir:MessageDefinition.identifier [ Identifier ], ... ; # 0..* Primary key for the message definition on a given
    server
  fhir:MessageDefinition.version [ string ]; # 0..1 Business version of the message definition
  fhir:MessageDefinition.name [ string ]; # 0..1 Name for this message definition (computer friendly)
  fhir:MessageDefinition.title [ string ]; # 0..1 Name for this message definition (human friendly)
  fhir:MessageDefinition.replaces [ canonical(MessageDefinition) ], ... ; # 0..* Takes the place of
  fhir:MessageDefinition.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:MessageDefinition.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
  fhir:MessageDefinition.date [ dateTime ]; # 1..1 Date last changed
  fhir:MessageDefinition.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:MessageDefinition.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:MessageDefinition.description [ markdown ]; # 0..1 Natural language description of the message definition
  fhir:MessageDefinition.useContext [ UsageContext ], ... ; # 0..* The context that the content is intended to support
  fhir:MessageDefinition.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for message definition (if
     applicable)
  fhir:MessageDefinition.purpose [ markdown ]; # 0..1 Why this message definition is defined
  fhir:MessageDefinition.copyright [ markdown ]; # 0..1 Use and/or publishing restrictions
  fhir:MessageDefinition.base [ canonical(MessageDefinition) ]; # 0..1 Definition this one is based on
  fhir:MessageDefinition.parent [ canonical(ActivityDefinition|PlanDefinition) ], ... ; # 0..* Protocol/workflow this is
     part of
  # MessageDefinition.event[x] : 1..1 Event code  or link to the EventDefinition. One of these 2
    fhir:MessageDefinition.eventCoding [ Coding ]
    fhir:MessageDefinition.eventUri [ uri ]
  fhir:MessageDefinition.category [ code ]; # 0..1 consequence | currency | notification
  fhir:MessageDefinition.focus [ # 0..* Resource(s) that are the subject of the event
    fhir:MessageDefinition.focus.code [ code ]; # 1..1 Type of resource
    fhir:MessageDefinition.focus.profile [ canonical(StructureDefinition) ]; # 0..1 Profile that must be adhered to by
    focus
    fhir:MessageDefinition.focus.min [ unsignedInt ]; # 1..1 Minimum number of focuses of this type
    fhir:MessageDefinition.focus.max [ string ]; # 0..1 Maximum number of focuses of this type
  ], ...;
  fhir:MessageDefinition.responseRequired [ code ]; # 0..1 always | on-error | never | on-success
  fhir:MessageDefinition.allowedResponse [ # 0..* Responses to this message
    fhir:MessageDefinition.allowedResponse.message [ canonical(MessageDefinition) ]; # 1..1 Reference to allowed message
     definition response
    fhir:MessageDefinition.allowedResponse.situation [ markdown ]; # 0..1 When should this response be used
  ], ...;
  fhir:MessageDefinition.graph [ canonical(GraphDefinition) ], ... ; # 0..* Canonical reference to a GraphDefinition
]
```

Changes since R3

MessageDefinition
MessageDefinition.identifier | Max Cardinality changed from 1 to *
MessageDefinition.replaces | Type changed from Reference(MessageDefinition) to canonical(MessageDefinition)
MessageDefinition.status | Change value set from http://hl7.org/fhir/ValueSet/publication-status to http://hl7.org/fhir/ValueSet/publication-status|4.0.1
MessageDefinition.experimental | No longer marked as Modifier
MessageDefinition.base | Type changed from Reference(MessageDefinition) to canonical(MessageDefinition)
MessageDefinition.parent | Type changed from Reference(ActivityDefinition | PlanDefinition) to canonical(ActivityDefinition | PlanDefinition)
MessageDefinition.event[x] | Renamed from event to event[x] Add Type uri
MessageDefinition.category | Change value set from http://hl7.org/fhir/ValueSet/message-significance-category to http://hl7.org/fhir/ValueSet/message-significance-category|4.0.1
MessageDefinition.focus.code | Change value set from http://hl7.org/fhir/ValueSet/resource-types to http://hl7.org/fhir/ValueSet/resource-types|4.0.1
MessageDefinition.focus.profile | Type changed from Reference(StructureDefinition) to canonical(StructureDefinition)
MessageDefinition.focus.min | Min Cardinality changed from 0 to 1 Default Value "0" removed
MessageDefinition.responseRequired | Type changed from boolean to code Add Binding http://hl7.org/fhir/ValueSet/messageheader-response-request|4.0.1 (required) Default Value "false" removed
MessageDefinition.allowedResponse.message | Type changed from Reference(MessageDefinition) to canonical(MessageDefinition)
MessageDefinition.graph | Added Element

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 3 tests that all execute ok. All tests pass round-trip testing and 2 r3 resources are invalid (0 errors).)

Structure

Name | Flags | Card. | Type | Description & Constraints

MessageDefinition | ITU | | DomainResource | A resource that defines a type of message that can be exchanged between systems + Warning: Name should be usable as an identifier for the module by machine processing applications such as code generation Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
url | Σ | 0..1 | uri | Business Identifier for a given MessageDefinition
identifier | Σ | 0..* | Identifier | Primary key for the message definition on a given server
version | Σ | 0..1 | string | Business version of the message definition
name | ΣI | 0..1 | string | Name for this message definition (computer friendly)
title | Σ | 0..1 | string | Name for this message definition (human friendly)
replaces | Σ | 0..* | canonical(MessageDefinition) | Takes the place of
status | ?!Σ | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)
experimental | Σ | 0..1 | boolean | For testing purposes, not real usage
date | Σ | 1..1 | dateTime | Date last changed
publisher | Σ | 0..1 | string | Name of the publisher (organization or individual)
contact | Σ | 0..* | ContactDetail | Contact details for the publisher
description | Σ | 0..1 | markdown | Natural language description of the message definition
useContext | Σ | 0..* | UsageContext | The context that the content is intended to support
jurisdiction | Σ | 0..* | CodeableConcept | Intended jurisdiction for message definition (if applicable) Jurisdiction (Extensible)
purpose | Σ | 0..1 | markdown | Why this message definition is defined
copyright | | 0..1 | markdown | Use and/or publishing restrictions
base | Σ | 0..1 | canonical(MessageDefinition) | Definition this one is based on
parent | Σ | 0..* | canonical(ActivityDefinition | PlanDefinition) | Protocol/workflow this is part of
event[x] | Σ | 1..1 | | Event code or link to the EventDefinition MessageEvent (Example)
eventCoding | | | Coding
eventUri | | | uri
category | Σ | 0..1 | code | consequence | currency | notification MessageSignificanceCategory (Required)
focus | ΣI | 0..* | BackboneElement | Resource(s) that are the subject of the event + Rule: Max must be postive int or *
code | Σ | 1..1 | code | Type of resource ResourceType (Required)
profile | | 0..1 | canonical(StructureDefinition) | Profile that must be adhered to by focus
min | Σ | 1..1 | unsignedInt | Minimum number of focuses of this type
max | I | 0..1 | string | Maximum number of focuses of this type
responseRequired | | 0..1 | code | always | on-error | never | on-success messageheader-response-request (Required)
allowedResponse | | 0..* | BackboneElement | Responses to this message
message | | 1..1 | canonical(MessageDefinition) | Reference to allowed message definition response
situation | | 0..1 | markdown | When should this response be used
graph | | 0..* | canonical(GraphDefinition) | Canonical reference to a GraphDefinition
Documentation for this format

UML Diagram (Legend)

XML Template

```
<MessageDefinition xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 0..1 Business Identifier for a given MessageDefinition -->
 <identifier><!-- 0..* Identifier Primary key for the message definition on a given server --></identifier>
 <version value="[string]"/><!-- 0..1 Business version of the message definition -->
 <name value="[string]"/><!--  0..1 Name for this message definition (computer friendly) -->
 <title value="[string]"/><!-- 0..1 Name for this message definition (human friendly) -->
 <replaces><!-- 0..* canonical(MessageDefinition) Takes the place of --></replaces>
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
 <date value="[dateTime]"/><!-- 1..1 Date last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!-- 0..1 Natural language description of the message definition -->
 <useContext><!-- 0..* UsageContext The context that the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for message definition (if applicable) --></jurisdiction>
 <purpose value="[markdown]"/><!-- 0..1 Why this message definition is defined -->
 <copyright value="[markdown]"/><!-- 0..1 Use and/or publishing restrictions -->
 <base><!-- 0..1 canonical(MessageDefinition) Definition this one is based on --></base>
 <parent><!-- 0..* canonical(ActivityDefinition|PlanDefinition) Protocol/workflow this is part of --></parent>
 <event[x]><!-- 1..1 Coding|uri Event code  or link to the EventDefinition --></event[x]>
 <category value="[code]"/><!-- 0..1 consequence | currency | notification -->
 <focus>  <!-- 0..* Resource(s) that are the subject of the event -->
  <code value="[code]"/><!-- 1..1 Type of resource -->
  <profile><!-- 0..1 canonical(StructureDefinition) Profile that must be adhered to by focus --></profile>
  <min value="[unsignedInt]"/><!-- 1..1 Minimum number of focuses of this type -->
  <max value="[string]"/><!--  0..1 Maximum number of focuses of this type -->
 </focus>
 <responseRequired value="[code]"/><!-- 0..1 always | on-error | never | on-success -->
 <allowedResponse>  <!-- 0..* Responses to this message -->
  <message><!-- 1..1 canonical(MessageDefinition) Reference to allowed message definition response --></message>
  <situation value="[markdown]"/><!-- 0..1 When should this response be used -->
 </allowedResponse>
 <graph><!-- 0..* canonical(GraphDefinition) Canonical reference to a GraphDefinition --></graph>
</MessageDefinition>
```

JSON Template

```
{
  "resourceType" : "MessageDefinition",
  // from Resource: id, meta, implicitRules, and language
```

```
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // Business Identifier for a given MessageDefinition
  "identifier" : [{ Identifier }], // Primary key for the message definition on a given server
  "version" : "<string>", // Business version of the message definition
  "name" : "<string>", // C? Name for this message definition (computer friendly)
  "title" : "<string>", // Name for this message definition (human friendly)
  "replaces" : [{ canonical(MessageDefinition) }], // Takes the place of
  "status" : "<code>", // R!  draft | active | retired | unknown
  "experimental" : <boolean>, // For testing purposes, not real usage
  "date" : "<dateTime>", // R!  Date last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "description" : "<markdown>", // Natural language description of the message definition
  "useContext" : [{ UsageContext }], // The context that the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for message definition (if applicable)
  "purpose" : "<markdown>", // Why this message definition is defined
  "copyright" : "<markdown>", // Use and/or publishing restrictions
  "base" : { canonical(MessageDefinition) }, // Definition this one is based on
  "parent" : [{ canonical(ActivityDefinition|PlanDefinition) }], // Protocol/workflow this is part of
  // event[x]: Event code  or link to the EventDefinition. One of these 2:
  "eventCoding" : { Coding },
  "eventUri" : "<uri>",
  "category" : "<code>", // consequence | currency | notification
  "focus" : [{ // Resource(s) that are the subject of the event
    "code" : "<code>", // R!  Type of resource
    "profile" : { canonical(StructureDefinition) }, // Profile that must be adhered to by focus
    "min" : "<unsignedInt>", // R!  Minimum number of focuses of this type
    "max" : "<string>" // C? Maximum number of focuses of this type
  }],
  "responseRequired" : "<code>", // always | on-error | never | on-success
  "allowedResponse" : [{ // Responses to this message
    "message" : { canonical(MessageDefinition) }, // R!  Reference to allowed message definition response
    "situation" : "<markdown>" // When should this response be used
  }],
  "graph" : [{ canonical(GraphDefinition) }] // Canonical reference to a GraphDefinition
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:MessageDefinition;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:MessageDefinition.url [ uri ]; # 0..1 Business Identifier for a given MessageDefinition
  fhir:MessageDefinition.identifier [ Identifier ], ... ; # 0..* Primary key for the message definition on a given
    server
  fhir:MessageDefinition.version [ string ]; # 0..1 Business version of the message definition
  fhir:MessageDefinition.name [ string ]; # 0..1 Name for this message definition (computer friendly)
  fhir:MessageDefinition.title [ string ]; # 0..1 Name for this message definition (human friendly)
  fhir:MessageDefinition.replaces [ canonical(MessageDefinition) ], ... ; # 0..* Takes the place of
  fhir:MessageDefinition.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:MessageDefinition.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
  fhir:MessageDefinition.date [ dateTime ]; # 1..1 Date last changed
  fhir:MessageDefinition.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:MessageDefinition.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:MessageDefinition.description [ markdown ]; # 0..1 Natural language description of the message definition
  fhir:MessageDefinition.useContext [ UsageContext ], ... ; # 0..* The context that the content is intended to support
  fhir:MessageDefinition.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for message definition (if
     applicable)
  fhir:MessageDefinition.purpose [ markdown ]; # 0..1 Why this message definition is defined
  fhir:MessageDefinition.copyright [ markdown ]; # 0..1 Use and/or publishing restrictions
  fhir:MessageDefinition.base [ canonical(MessageDefinition) ]; # 0..1 Definition this one is based on
  fhir:MessageDefinition.parent [ canonical(ActivityDefinition|PlanDefinition) ], ... ; # 0..* Protocol/workflow this is
     part of
  # MessageDefinition.event[x] : 1..1 Event code  or link to the EventDefinition. One of these 2
    fhir:MessageDefinition.eventCoding [ Coding ]
    fhir:MessageDefinition.eventUri [ uri ]
  fhir:MessageDefinition.category [ code ]; # 0..1 consequence | currency | notification
  fhir:MessageDefinition.focus [ # 0..* Resource(s) that are the subject of the event
    fhir:MessageDefinition.focus.code [ code ]; # 1..1 Type of resource
    fhir:MessageDefinition.focus.profile [ canonical(StructureDefinition) ]; # 0..1 Profile that must be adhered to by
    focus
    fhir:MessageDefinition.focus.min [ unsignedInt ]; # 1..1 Minimum number of focuses of this type
    fhir:MessageDefinition.focus.max [ string ]; # 0..1 Maximum number of focuses of this type
  ], ...;
  fhir:MessageDefinition.responseRequired [ code ]; # 0..1 always | on-error | never | on-success
  fhir:MessageDefinition.allowedResponse [ # 0..* Responses to this message
    fhir:MessageDefinition.allowedResponse.message [ canonical(MessageDefinition) ]; # 1..1 Reference to allowed message
     definition response
    fhir:MessageDefinition.allowedResponse.situation [ markdown ]; # 0..1 When should this response be used
  ], ...;
  fhir:MessageDefinition.graph [ canonical(GraphDefinition) ], ... ; # 0..* Canonical reference to a GraphDefinition
]
```

Changes since Release 3

MessageDefinition

MessageDefinition.identifier | Max Cardinality changed from 1 to *
MessageDefinition.replaces | Type changed from Reference(MessageDefinition) to canonical(MessageDefinition)
MessageDefinition.status | Change value set from http://hl7.org/fhir/ValueSet/publication-status to http://hl7.org/fhir/ValueSet/publication-status|4.0.1
MessageDefinition.experimental | No longer marked as Modifier
MessageDefinition.base | Type changed from Reference(MessageDefinition) to canonical(MessageDefinition)
MessageDefinition.parent | Type changed from Reference(ActivityDefinition | PlanDefinition) to canonical(ActivityDefinition | PlanDefinition)
MessageDefinition.event[x] | Renamed from event to event[x] Add Type uri
MessageDefinition.category | Change value set from http://hl7.org/fhir/ValueSet/message-significance-category to http://hl7.org/fhir/ValueSet/message-significance-category|4.0.1
MessageDefinition.focus.code | Change value set from http://hl7.org/fhir/ValueSet/resource-types to http://hl7.org/fhir/ValueSet/resource-types|4.0.1
MessageDefinition.focus.profile | Type changed from Reference(StructureDefinition) to canonical(StructureDefinition)
MessageDefinition.focus.min | Min Cardinality changed from 0 to 1 Default Value "0" removed
MessageDefinition.responseRequired | Type changed from boolean to code Add Binding http://hl7.org/fhir/ValueSet/messageheader-response-request|4.0.1 (required) Default Value "false" removed
MessageDefinition.allowedResponse.message | Type changed from Reference(MessageDefinition) to canonical(MessageDefinition)
MessageDefinition.graph | Added Element

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 3 tests that all execute ok. All tests pass round-trip testing and 2 r3 resources are invalid (0 errors).)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

2.48.2.1 Terminology Bindings

Path | Definition | Type | Reference
MessageDefinition.status | The lifecycle status of an artifact. | Required | PublicationStatus
MessageDefinition.jurisdiction | Countries and regions within which this artifact is targeted for use. | Extensible | Jurisdiction ValueSet
MessageDefinition.event[x] | One of the message events defined as part of this version of FHIR. | Example | Message Events
MessageDefinition.category | The impact of the content of a message. | Required | MessageSignificanceCategory
MessageDefinition.focus.code | One of the resource types defined as part of this version of FHIR. | Required | Resource Types
MessageDefinition.responseRequired | HL7-defined table of codes which identify conditions under which acknowledgments are required to be returned in response to a message. | Required | messageheader-response-request

2.48.2.2 Constraints

id | Level | Location | Description | Expression
msd-0 | Warning | (base) | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
md-1 | Rule | MessageDefinition.focus | Max must be postive int or * | max='*' or (max.toInteger() > 0)

2.48.3 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
category | token | The behavior associated with the message | MessageDefinition.category
context | token | A use context assigned to the message definition | (MessageDefinition.useContext.value as CodeableConcept)
context-quantity | quantity | A quantity- or range-valued use context assigned to the message definition | (MessageDefinition.useContext.value as Quantity) | (MessageDefinition.useContext.value as Range)
context-type | token | A type of use context assigned to the message definition | MessageDefinition.useContext.code
context-type-quantity | composite | A use context type and quantity- or range-based value assigned to the message definition | On MessageDefinition.useContext: context-type: code context-quantity: value.as(Quantity) | value.as(Range)
context-type-value | composite | A use context type and value assigned to the message definition | On MessageDefinition.useContext: context-type: code context: value.as(CodeableConcept)
date | date | The message definition publication date | MessageDefinition.date
description | string | The description of the message definition | MessageDefinition.description
event | token | The event that triggers the message or link to the event definition. | MessageDefinition.event
focus | token | A resource that is a permitted focus of the message | MessageDefinition.focus.code
identifier | token | External identifier for the message definition | MessageDefinition.identifier
jurisdiction | token | Intended jurisdiction for the message definition | MessageDefinition.jurisdiction
name | string | Computationally friendly name of the message definition | MessageDefinition.name
parent | reference | A resource that is the parent of the definition | MessageDefinition.parent (PlanDefinition, ActivityDefinition)
publisher | string | Name of the publisher of the message definition | MessageDefinition.publisher
status | token | The current status of the message definition | MessageDefinition.status
title | string | The human-friendly name of the message definition | MessageDefinition.title
url | uri | The uri that identifies the message definition | MessageDefinition.url
version | token | The business version of the message definition | MessageDefinition.version

2.48.4 Resource MessageDefinition - Examples

URL: https://hl7.org/fhir/R4/messagedefinition-examples.html

2.48.4 Resource MessageDefinition - Examples

Infrastructure And Messaging Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Example Name | id | Format
A typical message definition | example | XML | JSON | Turtle
An example of a patient link notification | patient-link-notification | XML | JSON | Turtle
An example of a patient link response | patient-link-response | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

2.48.5 Resource MessageDefinition - Detailed Descriptions

URL: https://hl7.org/fhir/R4/messagedefinition-definitions.html

2.48.5 Resource MessageDefinition - Detailed Descriptions

Infrastructure And Messaging Work Group | Maturity Level: 1 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the MessageDefinition resource.

MessageDefinition
Element Id | MessageDefinition
Definition | Defines the characteristics of a message that can be shared between systems, including the type of event that initiates the message, the content to be transmitted and what response(s), if any, are permitted.
Cardinality | 0..*
Type | DomainResource
Requirements | Allows messages to be defined once and re-used across systems.
Comments | This would be a MIF-level artifact.
Invariants | Defined on this element msd-0 Warning Name should be usable as an identifier for the module by machine processing applications such as code generation name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
Defined on this element
msd-0 | Warning | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
MessageDefinition.url
Element Id | MessageDefinition.url
Definition | The business identifier that is used to reference the MessageDefinition and is expected to be consistent from server to server.
Cardinality | 0..1
Type | uri
Requirements | Allows the message definition to be referenced by a single globally unique identifier.
Summary | true
Comments | Can be a urn:uuid: or a urn:oid: but real http: addresses are preferred. Multiple instances may share the same URL if they have a distinct version. The determination of when to create a new version of a resource (same url, new version) vs. defining a new artifact is up to the author. Considerations for making this decision are found in Technical and Business Versions. In some cases, the resource can no longer be found at the stated url, but the url itself cannot change. Implementations can use the meta.source element to indicate where the current master source of the resource can be found.
MessageDefinition.identifier
Element Id | MessageDefinition.identifier
Definition | A formal identifier that is used to identify this message definition when it is represented in other formats, or referenced in a specification, model, design or an instance.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Requirements | Allows externally provided and/or usable business identifiers to be easily associated with the module.
Summary | true
Comments | Typically, this is used for identifiers that can go in an HL7 V3 II (instance identifier) data type, and can then identify this message definition outside of FHIR, where it is not possible to use the logical URI.
MessageDefinition.version
Element Id | MessageDefinition.version
Definition | The identifier that is used to identify this version of the message definition when it is referenced in a specification, model, design or instance. This is an arbitrary value managed by the message definition author and is not expected to be globally unique. For example, it might be a timestamp (e.g. yyyymmdd) if a managed version is not available. There is also no expectation that versions can be placed in a lexicographical sequence.
Note | This is a business versionId, not a resource version id (see discussion)
Cardinality | 0..1
Type | string
Summary | true
Comments | There may be different message definition instances that have the same identifier but different versions. The version can be appended to the url in a reference to allow a reference to a particular business version of the message definition with the

format [url]|[version].
MessageDefinition.name
Element Id | MessageDefinition.name
Definition | A natural language name identifying the message definition. This name should be usable as an identifier for the module by machine processing applications such as code generation.
Cardinality | 0..1
Type | string
Requirements | Support human navigation and code generation.
Summary | true
Comments | The name is not expected to be globally unique. The name should be a simple alphanumeric type name to ensure that it is machine-processing friendly.
Invariants | Affect this element inv-0 Warning Name should be usable as an identifier for the module by machine processing applications such as code generation name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
Affect this element
inv-0 | Warning | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
MessageDefinition.title
Element Id | MessageDefinition.title
Definition | A short, descriptive, user-friendly title for the message definition.
Cardinality | 0..1
Type | string
Summary | true
Comments | This name does not need to be machine-processing friendly and may contain punctuation, white-space, etc.
MessageDefinition.replaces
Element Id | MessageDefinition.replaces
Definition | A MessageDefinition that is superseded by this definition.
Cardinality | 0..*
Type | canonical(MessageDefinition)
Summary | true
MessageDefinition.status
Element Id | MessageDefinition.status
Definition | The status of this message definition. Enables tracking the life-cycle of the content.
Cardinality | 1..1
Terminology Binding | PublicationStatus (Required)
Type | code
Is Modifier | true (Reason: This is labeled as "Is Modifier" because applications should not use a retired {{title}} without due consideration)
Summary | true
Comments | Allows filtering of message definitions that are appropriate for use versus not.
MessageDefinition.experimental
Element Id | MessageDefinition.experimental
Definition | A Boolean value to indicate that this message definition is authored for testing purposes (or education/evaluation/marketing) and is not intended to be used for genuine usage.
Cardinality | 0..1
Type | boolean
Requirements | Enables experimental content to be developed following the same lifecycle that would be used for a production-level message definition.
Summary | true
Comments | Allows filtering of message definitions that are appropriate for use versus not.
MessageDefinition.date
Element Id | MessageDefinition.date
Definition | The date (and optionally time) when the message definition was published. The date must change when the business version changes and it must change if the status code changes. In addition, it should change when the substantive content of the message definition changes.
Cardinality | 1..1
Type | dateTime
Alternate Names | Revision Date
Summary | true
Comments | Note that this is not the same as the resource last-modified-date, since the resource may be a secondary representation of the message definition. Additional specific dates may be added as extensions or be found by consulting Provenances associated with past versions of the resource.
MessageDefinition.publisher
Element Id | MessageDefinition.publisher
Definition | The name of the organization or individual that published the message definition.
Cardinality | 0..1
Type | string
Requirements | Helps establish the "authority/credibility" of the message definition. May also allow for contact.
Summary | true
Comments | Usually an organization but may be an individual. The publisher (or steward) of the message definition is the organization or individual primarily responsible for the maintenance and upkeep of the message definition. This is not necessarily the same individual or organization that developed and initially authored the content. The publisher is the primary point of contact for questions or issues with the message definition. This item SHOULD be populated unless the information is available from context.
MessageDefinition.contact

Element Id | MessageDefinition.contact
Definition | Contact details to assist a user in finding and communicating with the publisher.
Cardinality | 0..*
Type | ContactDetail
Summary | true
Comments | May be a web site, an email address, a telephone number, etc.
MessageDefinition.description
Element Id | MessageDefinition.description
Definition | A free text natural language description of the message definition from a consumer's perspective.
Cardinality | 0..1
Type | markdown
Summary | true
Comments | This description can be used to capture details such as why the message definition was built, comments about misuse, instructions for clinical use and interpretation, literature references, examples from the paper world, etc. It is not a rendering of the message definition as conveyed in the 'text' field of the resource itself. This item SHOULD be populated unless the information is available from context (e.g. the language of the message definition is presumed to be the predominant language in the place the message definition was created).
MessageDefinition.useContext
Element Id | MessageDefinition.useContext
Definition | The content was developed with a focus and intent of supporting the contexts that are listed. These contexts may be general categories (gender, age, ...) or may be references to specific programs (insurance plans, studies, ...) and may be used to assist with indexing and searching for appropriate message definition instances.
Cardinality | 0..*
Type | UsageContext
Requirements | Assist in searching for appropriate content.
Summary | true
Comments | When multiple useContexts are specified, there is no expectation that all or any of the contexts apply.
MessageDefinition.jurisdiction
Element Id | MessageDefinition.jurisdiction
Definition | A legal or geographic region in which the message definition is intended to be used.
Cardinality | 0..*
Terminology Binding | Jurisdiction (Extensible)
Type | CodeableConcept
Summary | true
Comments | It may be possible for the message definition to be used in jurisdictions other than those for which it was originally designed or intended.
MessageDefinition.purpose
Element Id | MessageDefinition.purpose
Definition | Explanation of why this message definition is needed and why it has been designed as it has.
Cardinality | 0..1
Type | markdown
Summary | true
Comments | This element does not describe the usage of the message definition. Instead, it provides traceability of ''why'' the resource is either needed or ''why'' it is defined as it is. This may be used to point to source materials or specifications that drove the structure of this message definition.
MessageDefinition.copyright
Element Id | MessageDefinition.copyright
Definition | A copyright statement relating to the message definition and/or its contents. Copyright statements are generally legal restrictions on the use and publishing of the message definition.
Cardinality | 0..1
Type | markdown
Requirements | Consumers must be able to determine any legal restrictions on the use of the message definition and/or its content.
Alternate Names | License; Restrictions
MessageDefinition.base
Element Id | MessageDefinition.base
Definition | The MessageDefinition that is the basis for the contents of this resource.
Cardinality | 0..1
Type | canonical(MessageDefinition)
Summary | true
MessageDefinition.parent
Element Id | MessageDefinition.parent
Definition | Identifies a protocol or workflow that this MessageDefinition represents a step in.
Cardinality | 0..*
Type | canonical(ActivityDefinition | PlanDefinition)
Summary | true
Comments | It should be possible to use MessageDefinition to describe a message to be used by certain steps in a particular protocol as part of a PlanDefinition or ActivityDefinition.
MessageDefinition.event[x]
Element Id | MessageDefinition.event[x]
Definition | Event code or link to the EventDefinition.
Cardinality | 1..1
Terminology Binding | the Event List in the messaging framework

Type | Coding|uri
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
MessageDefinition.category
Element Id | MessageDefinition.category
Definition | The impact of the content of the message.
Cardinality | 0..1
Terminology Binding | MessageSignificanceCategory (Required)
Type | code
Summary | true
MessageDefinition.focus
Element Id | MessageDefinition.focus
Definition | Identifies the resource (or resources) that are being addressed by the event. For example, the Encounter for an admit message or two Account records for a merge.
Cardinality | 0..*
Summary | true
Invariants | Defined on this element md-1 Rule Max must be postive int or * max='*' or (max.toInteger() > 0)
Defined on this element
md-1 | Rule | Max must be postive int or * | max='*' or (max.toInteger() > 0)
MessageDefinition.focus.code
Element Id | MessageDefinition.focus.code
Definition | The kind of resource that must be the focus for this message.
Cardinality | 1..1
Terminology Binding | Any defined Resource Type name
Type | code
Summary | true
Comments | Multiple focuses addressing different resources may occasionally occur. E.g. to link or unlink a resource from a particular account or encounter, etc.
MessageDefinition.focus.profile
Element Id | MessageDefinition.focus.profile
Definition | A profile that reflects constraints for the focal resource (and potentially for related resources).
Cardinality | 0..1
Type | canonical(StructureDefinition)
Requirements | This profile allows setting boundaries for what content must appear within the message bundle vs. outside based on the declared aggregation constraints on referenced resources on this and related profiles.
Comments | This should be present for most message definitions. However, if the message focus is only a single resource and there is no need to include referenced resources or otherwise enforce the presence of particular elements, it can be omitted.
MessageDefinition.focus.min
Element Id | MessageDefinition.focus.min
Definition | Identifies the minimum number of resources of this type that must be pointed to by a message in order for it to be valid against this MessageDefinition.
Cardinality | 1..1
Type | unsignedInt
Summary | true
MessageDefinition.focus.max
Element Id | MessageDefinition.focus.max
Definition | Identifies the maximum number of resources of this type that must be pointed to by a message in order for it to be valid against this MessageDefinition.
Cardinality | 0..1
Type | string
Invariants | Affect this element md-1 Rule Max must be postive int or * max='*' or (max.toInteger() > 0)
Affect this element
md-1 | Rule | Max must be postive int or * | max='*' or (max.toInteger() > 0)
MessageDefinition.responseRequired
Element Id | MessageDefinition.responseRequired
Definition | Declare at a message definition level whether a response is required or only upon error or success, or never.
Cardinality | 0..1
Terminology Binding | messageheader-response-request (Required)
Type | code
Comments | This enables the capability currently available through MSH-16 (Application Level acknowledgement) in HL7 Version 2 to declare at a message instance level whether a response is required or only upon error or success, or never.
MessageDefinition.allowedResponse
Element Id | MessageDefinition.allowedResponse
Definition | Indicates what types of messages may be sent as an application-level response to this message.
Cardinality | 0..*
Comments | This indicates an application level response to "close" a transaction implicit in a particular request message. To define a complete workflow scenario, look to the [[PlanDefinition]] resource which allows the definition of complex orchestrations, conditionality, etc.
MessageDefinition.allowedResponse.message
Element Id | MessageDefinition.allowedResponse.message
Definition | A reference to the message definition that must be adhered to by this supported response.
Cardinality | 1..1
Type | canonical(MessageDefinition)

MessageDefinition.allowedResponse.situation
Element Id | MessageDefinition.allowedResponse.situation
Definition | Provides a description of the circumstances in which this response should be used (as opposed to one of the alternative responses).
Cardinality | 0..1
Type | markdown
MessageDefinition.graph
Element Id | MessageDefinition.graph
Definition | Canonical reference to a GraphDefinition. If a URL is provided, it is the canonical reference to a GraphDefinition that it controls what resources are to be added to the bundle when building the document. The GraphDefinition can also specify profiles that apply to the various resources.
Cardinality | 0..*
Type | canonical(GraphDefinition)

2.48.6 Resource MessageDefinition - Mappings

URL: https://hl7.org/fhir/R4/messagedefinition-mappings.html

2.48.6 Resource MessageDefinition - Mappings

Infrastructure And Messaging Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

HL7 v2 Mapping

RIM Mapping

Mappings for the messagedefinition resource (see Mappings to Other Standards for further information & status).

2.48.6.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

MessageDefinition
url | FiveWs.identifier
identifier | FiveWs.identifier
version | FiveWs.version
status | FiveWs.status
experimental | FiveWs.class
date | FiveWs.recorded
publisher | FiveWs.witness
purpose | FiveWs.why[x]

2.48.6.2 Workflow Pattern (http://hl7.org/fhir/workflow)

MessageDefinition
url | Definition.url
identifier | Definition.identifier
version | Definition.version
title | Definition.title
status | Definition.status
experimental | Definition.experimental
date | Definition.date
publisher | Definition.publisher
contact | Definition.contact
description | Definition.description
useContext | Definition.useContext
jurisdiction | Definition.jurisdiction
purpose | Definition.purpose
copyright | Definition.copyright

2.48.6.3 HL7 v2 Mapping (http://hl7.org/v2)

MessageDefinition | N/A

2.48.6.4 RIM Mapping (http://hl7.org/v3)

MessageDefinition | N/A

2.48.7 Resource MessageDefinition - Extensions & Profiles

URL: https://hl7.org/fhir/R4/messagedefinition-profiles.html

2.48.7 Resource MessageDefinition - Extensions & Profiles

Infrastructure And Messaging Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

This table lists profiles and extensions for the MessageDefinition resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

2.48.7.1 Profiles

No Profiles defined for this resource

2.48.7.2 Extensions

replaces | MessageDefinition | replaces | for General Extensions for use by FHIR Implementers
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

2.48.7.3 Search Extensions

No Search Extensions defined for this resource

2.48.8 Resource MessageDefinition - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/messagedefinition-version-maps.html

No validation errors - all conversions are clean

2.48.8 Resource MessageDefinition - R3/R4 Conversions

Infrastructure And Messaging Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

R3 : R4 Conversion maps for MessageDefinition.

Functional status for this map: 3 tests that all execute ok. All tests pass round-trip testing and 2 r3 resources are invalid (0 errors). (see documentation)

2.48.8.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/MessageDefinition3to4" = "R3 to R4 Conversions for MessageDefinition"

conceptmap "MessageSignificanceCategory" {
  prefix s = "http://hl7.org/fhir/message-significance-category"
  prefix t = "http://hl7.org/fhir/message-significance-category"

  s:Consequence - t:consequence
  s:Currency - t:currency
  s:Notification - t:notification
}

uses "http://hl7.org/fhir/3.0/StructureDefinition/MessageDefinition" alias MessageDefinitionR3 as source
uses "http://hl7.org/fhir/StructureDefinition/MessageDefinition" alias MessageDefinition as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group MessageDefinition(source src : MessageDefinitionR3, target tgt : MessageDefinition) extends DomainResource
    <<type+>> {
  src.url -> tgt.url;
  src.identifier -> tgt.identifier;
  src.version -> tgt.version;
  src.name -> tgt.name;
  src.title -> tgt.title;
  src.status -> tgt.status;
  src.experimental -> tgt.experimental;
  src.date -> tgt.date;
  src.publisher -> tgt.publisher;
  src.contact -> tgt.contact;
  src.description -> tgt.description;
  src.useContext -> tgt.useContext;
  src.jurisdiction -> tgt.jurisdiction;
  src.purpose -> tgt.purpose;
  src.copyright -> tgt.copyright;
  src.base -> tgt.base;
```

```
  src.parent -> tgt.parent;
  src.replaces -> tgt.replaces;
  src.event as vs0 -> tgt.event = create('Coding') as vt0 then Coding(vs0, vt0);
  src.category as v -> tgt.category = translate(v, '#MessageSignificanceCategory', 'code');
  src.focus as vs0 -> tgt.focus as vt0 then focus(vs0, vt0);
  src.responseRequired where value = true -> tgt.responseRequired = 'always';
  src.responseRequired where value = false -> tgt.responseRequired = 'never';
  src.allowedResponse as vs0 -> tgt.allowedResponse as vt0 then allowedResponse(vs0, vt0);
}

group focus(source src, target tgt) extends BackboneElement {
  src.code -> tgt.code;
  src.profile -> tgt.profile;
  src.min -> tgt.min;
  src.max -> tgt.max;
}

group allowedResponse(source src, target tgt) extends BackboneElement {
  src.message -> tgt.message;
  src.situation -> tgt.situation;
}
```

2.48.8.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/MessageDefinition4to3" = "R4 to R3 Conversion for MessageDefinition"

conceptmap "MessageSignificanceCategory" {
  prefix s = "http://hl7.org/fhir/message-significance-category"
  prefix t = "http://hl7.org/fhir/message-significance-category"

  s:consequence - t:Consequence
  s:currency - t:Currency
  s:notification - t:Notification
}

uses "http://hl7.org/fhir/StructureDefinition/MessageDefinition" alias MessageDefinition as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/MessageDefinition" alias MessageDefinitionR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group MessageDefinition(source src : MessageDefinition, target tgt : MessageDefinitionR3) extends DomainResource
    <<type+>> {
  src.url -> tgt.url;
  src.identifier -> tgt.identifier;
  src.version -> tgt.version;
  src.name -> tgt.name;
  src.title -> tgt.title;
  src.status -> tgt.status;
  src.experimental -> tgt.experimental;
  src.date -> tgt.date;
  src.publisher -> tgt.publisher;
  src.contact -> tgt.contact;
  src.description -> tgt.description;
  src.useContext -> tgt.useContext;
  src.jurisdiction -> tgt.jurisdiction;
  src.purpose -> tgt.purpose;
  src.copyright -> tgt.copyright;
  src.base -> tgt.base;
  src.parent -> tgt.parent;
  src.replaces -> tgt.replaces;
  src.event as vs0 -> tgt.event = create('Coding') as vt0 then Coding(vs0, vt0);
  src.category as v -> tgt.category = translate(v, '#MessageSignificanceCategory', 'code');
  src.focus as vs0 -> tgt.focus as vt0 then focus(vs0, vt0);
  src.responseRequired where value = 'always' -> tgt.responseRequired = true;
  src.responseRequired where value = 'never' -> tgt.responseRequired = false;
  src.allowedResponse as vs0 -> tgt.allowedResponse as vt0 then allowedResponse(vs0, vt0);
}

group focus(source src, target tgt) extends BackboneElement {
  src.code -> tgt.code;
  src.profile -> tgt.profile;
  src.min -> tgt.min;
  src.max -> tgt.max;
}

group allowedResponse(source src, target tgt) extends BackboneElement {
  src.message -> tgt.message;
  src.situation -> tgt.situation;
}
```

2.48.8.3 R4 Validation Errors

2.48.9 JSON Schema for MessageDefinition

URL: https://hl7.org/fhir/R4/messagedefinition.schema.json.html

2.48.9 JSON Schema for MessageDefinition

This is a representation of the json schema for MessageDefinition, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/MessageDefinition",
  "$ref": "#/definitions/MessageDefinition",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "MessageDefinition": {
      "description": "Defines the characteristics of a message that can be shared between systems, including the type of
     event that initiates the message, the content to be transmitted and what response(s), if any, are permitted.",
      "properties": {
        "resourceType": {
          "description": "This is a MessageDefinition resource",
          "const": "MessageDefinition"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
          "description": "A human-readable narrative that contains a summary of the resource and can be used to
    represent the content of the resource to a human. The narrative need not encode all the structured data, but is
    required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
    Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
          "$ref": "Narrative.schema.json#/definitions/Narrative"
        },
        "contained": {
          "description": "These resources do not have an independent existence apart from the resource that contains
    them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
          "items": {
            "$ref": "ResourceList.schema.json#/definitions/ResourceList"
          },
          "type": "array"
        },
        "extension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
     definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "modifierExtension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource and that modifies the understanding of the element that contains it and/or the understanding of the
    containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
    of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
    met as part of the definition of the extension. Applications processing a resource are required to check for
    modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
    DomainResource (including cannot change the meaning of modifierExtension itself).",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "url": {
          "description": "The business identifier that is used to reference the MessageDefinition and *is* expected to
    be consistent from server to server.",
          "$ref": "#/definitions/uri"
```

```
    },
    "_url": {
      "description": "Extensions for url",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "identifier": {
      "description": "A formal identifier that is used to identify this message definition when it is represented in
 other formats, or referenced in a specification, model, design or an instance.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "version": {
      "description": "The identifier that is used to identify this version of the message definition when it is
referenced in a specification, model, design or instance. This is an arbitrary value managed by the message
definition author and is not expected to be globally unique. For example, it might be a timestamp (e.g. yyyymmdd) if
 a managed version is not available. There is also no expectation that versions can be placed in a lexicographical
sequence.",
      "$ref": "#/definitions/string"
    },
    "_version": {
      "description": "Extensions for version",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "name": {
      "description": "A natural language name identifying the message definition. This name should be usable as an
identifier for the module by machine processing applications such as code generation.",
      "$ref": "#/definitions/string"
    },
    "_name": {
      "description": "Extensions for name",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "title": {
      "description": "A short, descriptive, user-friendly title for the message definition.",
      "$ref": "#/definitions/string"
    },
    "_title": {
      "description": "Extensions for title",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "replaces": {
      "description": "A MessageDefinition that is superseded by this definition.",
      "items": {
        "$ref": "canonical.schema.json#/definitions/canonical"
      },
      "type": "array"
    },
    "status": {
      "description": "The status of this message definition. Enables tracking the life-cycle of the content.",
      "enum": [
        "draft",
        "active",
        "retired",
        "unknown"
      ]
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "experimental": {
      "description": "A Boolean value to indicate that this message definition is authored for testing purposes (or
education/evaluation/marketing) and is not intended to be used for genuine usage.",
      "$ref": "#/definitions/boolean"
    },
    "_experimental": {
      "description": "Extensions for experimental",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "date": {
      "description": "The date  (and optionally time) when the message definition was published. The date must
change when the business version changes and it must change if the status code changes. In addition, it should
change when the substantive content of the message definition changes.",
      "$ref": "#/definitions/dateTime"
    },
    "_date": {
      "description": "Extensions for date",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "publisher": {
      "description": "The name of the organization or individual that published the message definition.",
      "$ref": "#/definitions/string"
    },
    "_publisher": {
      "description": "Extensions for publisher",
      "$ref": "Element.schema.json#/definitions/Element"
    },
```

```
    "contact": {
      "description": "Contact details to assist a user in finding and communicating with the publisher.",
      "items": {
        "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
      },
      "type": "array"
    },
    "description": {
      "description": "A free text natural language description of the message definition from a consumer\u0027s
perspective.",
      "$ref": "#/definitions/markdown"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "useContext": {
      "description": "The content was developed with a focus and intent of supporting the contexts that are listed.
These contexts may be general categories (gender, age, ...) or may be references to specific programs (insurance
plans, studies, ...) and may be used to assist with indexing and searching for appropriate message definition
instances.",
      "items": {
        "$ref": "UsageContext.schema.json#/definitions/UsageContext"
      },
      "type": "array"
    },
    "jurisdiction": {
      "description": "A legal or geographic region in which the message definition is intended to be used.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "purpose": {
      "description": "Explanation of why this message definition is needed and why it has been designed as it has.",
      "$ref": "#/definitions/markdown"
    },
    "_purpose": {
      "description": "Extensions for purpose",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "copyright": {
      "description": "A copyright statement relating to the message definition and/or its contents. Copyright
statements are generally legal restrictions on the use and publishing of the message definition.",
      "$ref": "#/definitions/markdown"
    },
    "_copyright": {
      "description": "Extensions for copyright",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "base": {
      "description": "The MessageDefinition that is the basis for the contents of this resource.",
      "$ref": "canonical.schema.json#/definitions/canonical"
    },
    "parent": {
      "description": "Identifies a protocol or workflow that this MessageDefinition represents a step in.",
      "items": {
        "$ref": "canonical.schema.json#/definitions/canonical"
      },
      "type": "array"
    },
    "eventCoding": {
      "description": "Event code or link to the EventDefinition.",
      "$ref": "Coding.schema.json#/definitions/Coding"
    },
    "eventUri": {
      "description": "Event code or link to the EventDefinition.",
      "pattern": "^\\S*$",
      "type": "string"
    },
    "_eventUri": {
      "description": "Extensions for eventUri",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "category": {
      "description": "The impact of the content of the message.",
      "enum": [
        "consequence",
        "currency",
        "notification"
      ]
    },
    "_category": {
      "description": "Extensions for category",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "focus": {
      "description": "Identifies the resource (or resources) that are being addressed by the event.  For example,
the Encounter for an admit message or two Account records for a merge.",
```

```
      "items": {
        "$ref": "#/definitions/MessageDefinition_Focus"
      },
      "type": "array"
    },
    "responseRequired": {
      "description": "Declare at a message definition level whether a response is required or only upon error or
success, or never.",
      "enum": [
        "always",
        "on-error",
        "never",
        "on-success"
      ]
    },
    "_responseRequired": {
      "description": "Extensions for responseRequired",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "allowedResponse": {
      "description": "Indicates what types of messages may be sent as an application-level response to this
message.",
      "items": {
        "$ref": "#/definitions/MessageDefinition_AllowedResponse"
      },
      "type": "array"
    },
    "graph": {
      "description": "Canonical reference to a GraphDefinition. If a URL is provided, it is the canonical reference
to a [[[GraphDefinition]]] that it controls what resources are to be added to the bundle when building the document.
 The GraphDefinition can also specify profiles that apply to the various resources.",
      "items": {
        "$ref": "canonical.schema.json#/definitions/canonical"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "resourceType"
  ]
},
"MessageDefinition_Focus": {
  "description": "Defines the characteristics of a message that can be shared between systems, including the type of
 event that initiates the message, the content to be transmitted and what response(s), if any, are permitted.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "code": {
      "description": "The kind of resource that must be the focus for this message.",
      "$ref": "#/definitions/code"
    },
    "_code": {
      "description": "Extensions for code",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "profile": {
      "description": "A profile that reflects constraints for the focal resource (and potentially for related
resources).",
      "$ref": "canonical.schema.json#/definitions/canonical"
    },
    "min": {
```

```
        "description": "Identifies the minimum number of resources of this type that must be pointed to by a message
    in order for it to be valid against this MessageDefinition.",
        "$ref": "#/definitions/unsignedInt"
      },
      "_min": {
        "description": "Extensions for min",
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "max": {
        "description": "Identifies the maximum number of resources of this type that must be pointed to by a message
    in order for it to be valid against this MessageDefinition.",
        "$ref": "#/definitions/string"
      },
      "_max": {
        "description": "Extensions for max",
        "$ref": "Element.schema.json#/definitions/Element"
      }
    },
    "additionalProperties": false
  },
  "MessageDefinition_AllowedResponse": {
    "description": "Defines the characteristics of a message that can be shared between systems, including the type of
     event that initiates the message, the content to be transmitted and what response(s), if any, are permitted.",
    "properties": {
      "id": {
        "description": "Unique id for the element within a resource (for internal references). This may be any string
    value that does not contain spaces.",
        "$ref": "string.schema.json#/definitions/string"
      },
      "extension": {
        "description": "May be used to represent additional information that is not part of the basic definition of
    the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
    definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
        "items": {
          "$ref": "Extension.schema.json#/definitions/Extension"
        },
        "type": "array"
      },
      "modifierExtension": {
        "description": "May be used to represent additional information that is not part of the basic definition of
    the element and that modifies the understanding of the element in which it is contained and/or the understanding of
    the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
    use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
     of the definition of the extension. Applications processing a resource are required to check for modifier
    extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
    (including cannot change the meaning of modifierExtension itself).",
        "items": {
          "$ref": "Extension.schema.json#/definitions/Extension"
        },
        "type": "array"
      },
      "message": {
        "description": "A reference to the message definition that must be adhered to by this supported response.",
        "$ref": "canonical.schema.json#/definitions/canonical"
      },
      "situation": {
        "description": "Provides a description of the circumstances in which this response should be used (as opposed
    to one of the alternative responses).",
        "$ref": "#/definitions/markdown"
      },
      "_situation": {
        "description": "Extensions for situation",
        "$ref": "Element.schema.json#/definitions/Element"
      }
    },
    "additionalProperties": false,
    "required": [
      "message"
    ]
  }
 }
}
```


2.48.10 Resource MessageDefinition - Maturity Dependency View

URL: https://hl7.org/fhir/R4/messagedefinition-dependencies.html

2.48.10 Resource MessageDefinition - Maturity Dependency View

Infrastructure And Messaging Work Group | Maturity Level: 1 | Trial Use | Compartments: Not linked to any defined compartments

Dependency Graph for MessageDefinition FMM level 1

Name | Card. | Type | Dependency Analysis

MessageDefinition | | DomainResource
url | 0..1 | uri
identifier | 0..* | Identifier
version | 0..1 | string
name | 0..1 | string
title | 0..1 | string
replaces | 0..* | canonical(MessageDefinition)
status | 1..1 | code
experimental | 0..1 | boolean
date | 1..1 | dateTime
publisher | 0..1 | string
contact | 0..* | ContactDetail
description | 0..1 | markdown
useContext | 0..* | UsageContext
jurisdiction | 0..* | CodeableConcept
purpose | 0..1 | markdown
copyright | 0..1 | markdown
base | 0..1 | canonical(MessageDefinition)
parent | 0..* | canonical(ActivityDefinition | PlanDefinition)
event[x] | 1..1
eventCoding | | Coding
eventUri | | uri
category | 0..1 | code
focus | 0..* | BackboneElement
code | 1..1 | code
profile | 0..1 | canonical(StructureDefinition)
min | 1..1 | unsignedInt
max | 0..1 | string
responseRequired | 0..1 | code
allowedResponse | 0..* | BackboneElement
message | 1..1 | canonical(MessageDefinition)
situation | 0..1 | markdown
graph | 0..* | canonical(GraphDefinition)